

**VP28.5.4** Two straight, parallel, superconducting wires are 0.900 m apart. Both wires are oriented straight up and down. One wire carries a current of  $1.30 \times 10^4$  A upward and experiences a magnetic force per unit length of 95.0 N/m that attracts it to the other wire. What are the magnitude and direction of the current in the other wire?

**Be sure to review EXAMPLES 28.8, 28.9, and 28.10 (Section 28.7) before attempting these problems.**

**VP28.10.1** A solid cylindrical conductor with radius 4.50 cm carries a current of 2.00 A. The current is uniformly distributed over the cross-sectional area of the conductor. Find the magnitude of the magnetic field at the following distances from the conductor axis: (a) 2.50 cm; (b) 4.50 cm; (c) 6.00 cm.

**VP28.10.2** A long, hollow cylinder with inner radius  $R_1$  and outer radius  $R_2$  carries current along its length. The current is uniformly distributed over the cross-sectional area of the cylinder and has current

density  $J$ . (a) Find the magnetic-field magnitude  $B$  as a function of the distance  $r$  from the conductor axis for points inside the hollow interior ( $r < R_1$ ), inside the solid conductor ( $R_2 > r > R_1$ ), and outside the conductor ( $r > R_2$ ). (b) For which value of  $r$  is  $B$  greatest?

**VP28.10.3** A very long solenoid with tightly wound coils has  $8.00 \times 10^3$  turns of wire per meter of length. What must the current be in the solenoid in order for the field inside the solenoid to have magnitude 0.0320 T?

**VP28.10.4** A copper wire carries a current of 0.840 A. You wrap this wire around a hollow toroidal core, forming a toroidal solenoid. The inner and outer radii of the core are 6.00 cm and 8.00 cm, respectively. (a) How many turns of wire around the core are required to produce a magnetic field of magnitude  $2.00 \times 10^{-3}$  T at a distance of 7.00 cm from the toroid axis? (b) With this number of turns, what are the maximum and minimum field magnitudes in the interior of the core?

## BRIDGING PROBLEM Magnetic Field of a Charged, Rotating Dielectric Disk

A thin dielectric disk with radius  $a$  has a total charge  $+Q$  distributed uniformly over its surface (**Fig. 28.30**). It rotates  $n$  times per second about an axis perpendicular to the surface of the disk and passing through its center. Find the magnetic field at the center of the disk.

### SOLUTION GUIDE

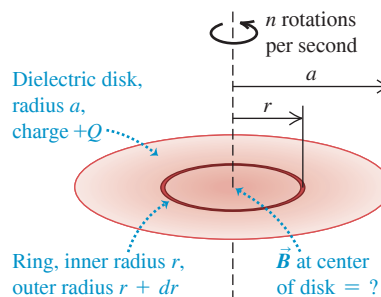
#### IDENTIFY and SET UP

1. Think of the rotating disk as a series of concentric rotating rings. Each ring acts as a circular current loop that produces a magnetic field at the center of the disk.
2. Use the results of Section 28.5 to find the magnetic field due to a single ring. Then integrate over all rings to find the total field.

#### EXECUTE

3. Find the charge on a ring with inner radius  $r$  and outer radius  $r + dr$  (**Fig. 28.30**).
4. How long does it take the charge found in step 3 to make a complete trip around the rotating ring? Use this to find the current of the rotating ring.
5. Use a result from Section 28.5 to determine the magnetic field that this ring produces at the center of the disk.

**Figure 28.30** Finding the  $\vec{B}$  field at the center of a uniformly charged, rotating disk.



6. Integrate your result from step 5 to find the total magnetic field from all rings with radii from  $r = 0$  to  $r = a$ .

#### EVALUATE

7. Does your answer have the correct units?
8. Suppose all of the charge were concentrated at the rim of the disk (at  $r = a$ ). Would this increase or decrease the field at the center of the disk?

## PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

### DISCUSSION QUESTIONS

**Q28.1** A topic of current interest in physics research is the search (thus far unsuccessful) for an isolated magnetic pole, or magnetic *monopole*. If such an entity were found, how could it be recognized? What would its properties be?

**Q28.2** Streams of charged particles emitted from the sun during periods of solar activity create a disturbance in the earth's magnetic field. How does this happen?

**Q28.3** The text discussed the magnetic field of an infinitely long, straight conductor carrying a current. Of course, there is no such thing

as an infinitely long *anything*. How do you decide whether a particular wire is long enough to be considered infinite?

**Q28.4** Two parallel conductors carrying current in the same direction attract each other. If they are permitted to move toward each other, the forces of attraction do work. From where does the energy come? Does this contradict the assertion in Chapter 27 that magnetic forces on moving charges do no work? Explain.

**Q28.5** Pairs of conductors carrying current into or out of the power-supply components of electronic equipment are sometimes twisted together to reduce magnetic-field effects. Why does this help?

**Q28.6** Suppose you have three long, parallel wires arranged so that in cross section they are at the corners of an equilateral triangle. Is there any way to arrange the currents so that all three wires attract each other? So that all three wires repel each other? Explain.

**Q28.7** In deriving the force on one of the long, current-carrying conductors in Section 28.4, why did we use the magnetic field due to only one of the conductors? That is, why didn't we use the *total* magnetic field due to *both* conductors?

**Q28.8** Two concentric, coplanar, circular loops of wire of different diameter carry currents in the same direction. Describe the nature of the force exerted on the inner loop by the outer loop and on the outer loop by the inner loop.

**Q28.9** A current was sent through a helical coil spring. The spring contracted, as though it had been compressed. Why?

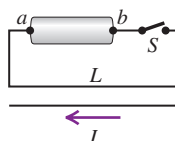
**Q28.10** What are the relative advantages and disadvantages of Ampere's law and the law of Biot and Savart for practical calculations of magnetic fields?

**Q28.11** Magnetic field lines never have a beginning or an end. Use this to explain why it is reasonable for the field of an ideal toroidal solenoid to be confined entirely to its interior, while a straight solenoid *must* have some field outside.

**Q28.12** Two very long, parallel wires carry equal currents in opposite directions. (a) Is there any place that their magnetic fields completely cancel? If so, where? If not, why not? (b) How would the answer to part (a) change if the currents were in the same direction?

**Q28.13** In the circuit shown in Fig. Q28.13, Figure Q28.13

when switch  $S$  is suddenly closed, the wire  $L$  is pulled toward the lower wire carrying current  $I$ . Which ( $a$  or  $b$ ) is the positive terminal of the battery? How do you know?



**Q28.14** A metal ring carries a current that causes a magnetic field  $B_0$  at the center of the ring and a field  $B$  at point  $P$  a distance  $x$  from the center along the axis of the ring. If the radius of the ring is doubled, find the magnetic field at the center. Will the field at point  $P$  change by the same factor? Why?

**Q28.15** Show that the units  $A \cdot m^2$  and  $J/T$  for the Bohr magneton are equivalent.

**Q28.16** Why should the permeability of a paramagnetic material be expected to decrease with increasing temperature?

**Q28.17** If a magnet is suspended over a container of liquid air, it attracts droplets to its poles. The droplets contain only liquid oxygen; even though nitrogen is the primary constituent of air, it is not attracted to the magnet. Explain what this tells you about the magnetic susceptibilities of oxygen and nitrogen, and explain why a magnet in ordinary, room-temperature air doesn't attract molecules of oxygen *gas* to its poles.

**Q28.18** What features of atomic structure determine whether an element is diamagnetic or paramagnetic? Explain.

**Q28.19** The magnetic susceptibility of paramagnetic materials is quite strongly temperature dependent, but that of diamagnetic materials is nearly independent of temperature. Why the difference?

**Q28.20** A cylinder of iron is placed so that it is free to rotate around its axis. Initially the cylinder is at rest, and a magnetic field is applied to the cylinder so that it is magnetized in a direction parallel to its axis. If the direction of the *external* field is suddenly reversed, the direction of magnetization will also reverse and the cylinder will begin rotating around its axis. (This is called the *Einstein-de Haas effect*.) Explain why the cylinder begins to rotate.

## EXERCISES

### Section 28.1 Magnetic Field of a Moving Charge

**28.1** •• A  $+6.00 \mu C$  point charge is moving at a constant  $8.00 \times 10^6$  m/s in the  $+y$ -direction, relative to a reference frame. At the instant when the point charge is at the origin of this reference frame, what is the magnetic-field vector  $\vec{B}$  it produces at the following points: (a)  $x = 0.500$  m,  $y = 0$ ,  $z = 0$ ; (b)  $x = 0$ ,  $y = -0.500$  m,  $z = 0$ ; (c)  $x = 0$ ,  $y = 0$ ,  $z = +0.500$  m; (d)  $x = 0$ ,  $y = -0.500$  m,  $z = +0.500$  m?

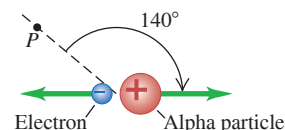
**28.2** • **Fields Within the Atom.** In the Bohr model of the hydrogen atom, the electron moves in a circular orbit of radius  $5.3 \times 10^{-11}$  m with a speed of  $2.2 \times 10^6$  m/s. If we are viewing the atom in such a way that the electron's orbit is in the plane of the paper with the electron moving clockwise, find the magnitude and direction of the electric and magnetic fields that the electron produces at the location of the nucleus (treated as a point).

**28.3** • An electron moves at  $0.100c$  as shown Figure E28.3

in Fig. E28.3. Find the magnitude and direction of the magnetic field this electron produces at the following points, each  $2.00 \mu m$  from the electron: (a) points  $A$  and  $B$ ; (b) point  $C$ ; (c) point  $D$ .

**28.4** •• An alpha particle (charge  $+2e$ ) and an electron move in opposite directions from the same point, each with the speed of  $2.50 \times 10^5$  m/s (Fig. E28.4). Find the magnitude and direction of the total magnetic field these charges produce at point  $P$ , which is  $8.65$  nm from each charge.

Figure E28.4

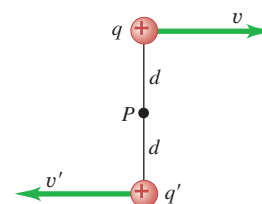
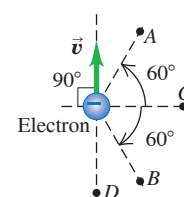


**28.5** • A  $-4.80 \mu C$  charge is moving at a constant speed of  $6.80 \times 10^5$  m/s in the  $+x$ -direction relative to a reference frame. At the instant when the point charge is at the origin, what is the magnetic-field vector it produces at the following points: (a)  $x = 0.500$  m,  $y = 0$ ,  $z = 0$ ; (b)  $x = 0$ ,  $y = 0.500$  m,  $z = 0$ ; (c)  $x = 0.500$  m,  $y = 0.500$  m,  $z = 0$ ; (d)  $x = 0$ ,  $y = 0$ ,  $z = 0.500$  m?

**28.6** • Positive point charges Figure E28.6

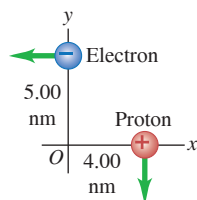
$q = +8.00 \mu C$  and  $q' = +3.00 \mu C$  are moving relative to an observer at point  $P$ , as shown in Fig. E28.6. The distance  $d$  is  $0.120$  m,  $v = 4.50 \times 10^6$  m/s, and  $v' = 9.00 \times 10^6$  m/s. (a) When the two charges are at the locations shown in the figure, what are the magnitude and direction of the net magnetic field they produce at point  $P$ ? (b) What are the magnitude and direction of the electric and magnetic forces that each charge exerts on the other, and what is the ratio of the magnitude of the electric force to the magnitude of the magnetic force? (c) If the direction of  $\vec{v}'$  is reversed, so both charges are moving in the same direction, what are the magnitude and direction of the magnetic forces that the two charges exert on each other?

**28.7** • At one instant, point  $P$  is  $6.60 \mu m$  to the left of a proton that is moving at  $3.30$  km/s in vacuum. (a) What is the direction of the unit vector in Eq. (28.2)? (b) What is the magnetic field caused by this proton if it is moving to the right or to the left? (c) What is the magnetic field (magnitude and direction) caused by this proton if it is moving toward the top of the page? (d) What is the answer to part (c) for an electron instead of a proton?



**28.8** • An electron and a proton are each moving at 735 km/s in perpendicular paths as shown in **Fig. E28.8**. At the instant when they are at the positions shown, find the magnitude and direction of (a) the total magnetic field they produce at the origin; (b) the magnetic field the electron produces at the location of the proton; (c) the total electric force and the total magnetic force that the electron exerts on the proton.

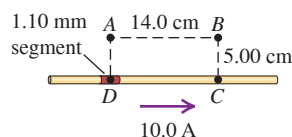
Figure E28.8



### Section 28.2 Magnetic Field of a Current Element

**28.9** • A straight wire carries a 10.0 A current (**Fig. E28.9**).  $ABCD$  is a rectangle with point  $D$  in the middle of a 1.10 mm segment of the wire and point  $C$  in the wire. Find the magnitude and direction of the magnetic field due to this segment at (a) point  $A$ ; (b) point  $B$ ; (c) point  $C$ .

Figure E28.9



**28.10** • A short current element  $d\vec{l} = (0.500 \text{ mm})\hat{j}$  carries a current of 5.40 A in the same direction as  $d\vec{l}$ . Point  $P$  is located at  $\vec{r} = (-0.730 \text{ m})\hat{i} + (0.390 \text{ m})\hat{k}$ . Use unit vectors to express the magnetic field at  $P$  produced by this current element.

**28.11** • A long, straight wire lies along the  $z$ -axis and carries a 4.00 A current in the  $+z$ -direction. Find the magnetic field (magnitude and direction) produced at the following points by a 0.500 mm segment of the wire centered at the origin: (a)  $x = 2.00 \text{ m}$ ,  $y = 0$ ,  $z = 0$ ; (b)  $x = 0$ ,  $y = 2.00 \text{ m}$ ,  $z = 0$ ; (c)  $x = 2.00 \text{ m}$ ,  $y = 2.00 \text{ m}$ ,  $z = 0$ ; (d)  $x = 0$ ,  $y = 0$ ,  $z = 2.00 \text{ m}$ .

**28.12** • Two parallel wires are 5.00 cm apart and carry currents in opposite directions, as shown in **Fig. E28.12**. Find the magnitude and direction of the magnetic field at point  $P$  due to two 1.50 mm segments of wire that are opposite each other and each 8.00 cm from  $P$ .

Figure E28.12

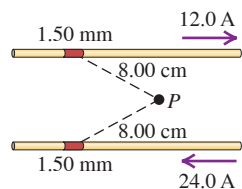
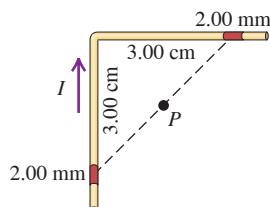


Figure E28.13



**28.13** • A wire carrying a 28.0 A current bends through a right angle. Consider two 2.00 mm segments of wire, each 3.00 cm from the bend (**Fig. E28.13**). Find the magnitude and direction of the magnetic field these two segments produce at point  $P$ , which is midway between them.

### Section 28.3 Magnetic Field of a Straight Current-Carrying Conductor

**28.14** • A long, straight wire lies along the  $x$ -axis and carries current  $I = 60.0 \text{ A}$  in the  $+x$ -direction. A small particle with mass  $3.00 \times 10^{-6} \text{ kg}$  and charge  $8.00 \times 10^{-3} \text{ C}$  is traveling in the vicinity of the wire. At an instant when the particle is on the  $y$ -axis at  $y = 8.00 \text{ cm}$ , its acceleration has components  $a_x = -5.00 \times 10^3 \text{ m/s}^2$  and  $a_y = +9.00 \times 10^3 \text{ m/s}^2$ . At that instant what are the  $x$ - and  $y$ -components of the velocity of the particle?

**28.15** • **The Magnetic Field from a Lightning Bolt.** Lightning bolts can carry currents up to approximately 20 kA. We can model such a current as the equivalent of a very long, straight wire. (a) If you were unfortunate enough to be 5.0 m away from such a lightning bolt, how large a magnetic field would you experience? (b) How does this field compare to one you would experience by being 5.0 cm from a long, straight household current of 10 A?

**28.16** • A very long, straight horizontal wire carries a current such that  $8.20 \times 10^{18}$  electrons per second pass any given point going from west to east. What are the magnitude and direction of the magnetic field this wire produces at a point 4.00 cm directly above it?

**28.17** • **BIO Currents in the Heart.** The body contains many small currents caused by the motion of ions in the organs and cells. Measurements of the magnetic field around the chest due to currents in the heart give values of about  $10 \mu\text{G}$ . Although the actual currents are rather complicated, we can gain a rough understanding of their magnitude if we model them as a long, straight wire. If the surface of the chest is 5.0 cm from this current, how large is the current in the heart?

**28.18** • Two long, straight wires, one above the other, are separated by a distance  $2a$  and are parallel to the  $x$ -axis. Let the  $+y$ -axis be in the plane of the wires in the direction from the lower wire to the upper wire. Each wire carries current  $I$  in the  $+x$ -direction. What are the magnitude and direction of the net magnetic field of the two wires at a point in the plane of the wires (a) midway between them; (b) at a distance  $a$  above the upper wire; (c) at a distance  $a$  below the lower wire?

**28.19** • A long, straight wire lies along the  $y$ -axis and carries a current  $I = 8.00 \text{ A}$  in the  $-y$ -direction (**Fig. E28.19**). In addition to the magnetic field due to the current in the wire, a uniform magnetic field  $\vec{B}_0$  with magnitude  $1.50 \times 10^{-6} \text{ T}$  is in the  $+x$ -direction. What is the total field (magnitude and direction) at the following points in the  $xz$ -plane:

(a)  $x = 0$ ,  $z = 1.00 \text{ m}$ ; (b)  $x = 1.00 \text{ m}$ ,  $z = 0$ ; (c)  $x = 0$ ,  $z = -0.25 \text{ m}$ ?

**28.20** • **BIO Transmission Lines and Health.** Currents in dc transmission lines can be 100 A or higher.

Some people are concerned that the electromagnetic fields from such lines near their homes could pose health dangers. For a line that has current 150 A and a height of 8.0 m above the ground, what magnetic field does the line produce at ground level? Express your answer in teslas and as a percentage of the earth's magnetic field, which is 0.50 G. Is this value cause for worry?

**28.21** • Two long, straight, parallel wires, 10.0 cm apart, carry equal 4.00 A currents in the same direction, as shown in **Fig. E28.21**. Find the magnitude and direction of the magnetic field at (a) point  $P_1$ , midway between the wires; (b) point  $P_2$ , 25.0 cm to the right of  $P_1$ ; (c) point  $P_3$ , 20.0 cm directly above  $P_1$ .

**28.22** • A rectangular loop with dimensions 4.20 cm by 9.50 cm carries current  $I$ . The current in the loop produces a magnetic field at the center of the loop that has magnitude  $5.50 \times 10^{-5} \text{ T}$  and direction away from you as you view the plane of the loop. What are the magnitude and direction (clockwise or counterclockwise) of the current in the loop?

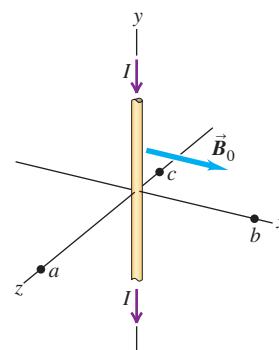
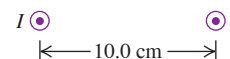
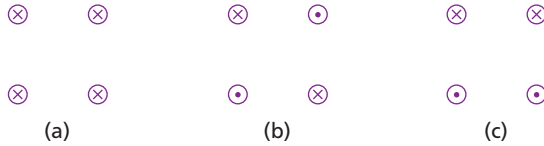


Figure E28.21



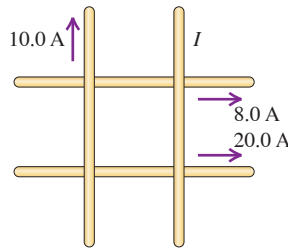
**28.23 •** Four long, parallel power lines each carry 100 A currents. A cross-sectional diagram of these lines is a square, 20.0 cm on each side. For each of the three cases shown in **Fig. E28.23**, calculate the magnetic field at the center of the square.

Figure E28.23



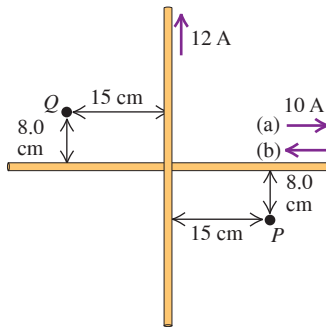
**28.24 •** Four very long, current-carrying wires in the same plane intersect to form a square 40.0 cm on each side, as shown in **Fig. E28.24**. Find the magnitude and direction of the current  $I$  so that the magnetic field at the center of the square is zero.

Figure E28.24



**28.25 ••** Two very long insulated wires perpendicular to each other in the same plane carry currents as shown in **Fig. E28.25**. Find the magnitude of the *net* magnetic field these wires produce at points  $P$  and  $Q$  if the 10.0 A current is (a) to the right or (b) to the left.

Figure E28.25

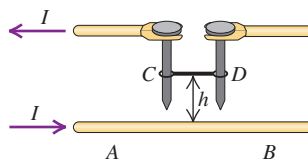


**28.26 •** A wire of length 20.0 cm lies along the  $x$ -axis with the center of the wire at the origin. The wire carries current  $I = 8.00$  A in the  $-x$ -direction. (a) What is the magnitude  $B$  of the magnetic field of the wire at the point  $y = 5.00$  cm on the  $y$ -axis? (b) What is the percent difference between the answer in (a) and the value you obtain if you assume the wire is infinitely long and use Eq. (28.9) to calculate  $B$ ?

#### Section 28.4 Force Between Parallel Conductors

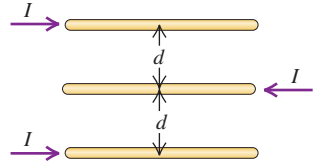
**28.27 •** A long, horizontal wire  $AB$  rests on the surface of a table and carries a current  $I$ . Horizontal wire  $CD$  is vertically above wire  $AB$  and is free to slide up and down on the two vertical metal guides  $C$  and  $D$  (**Fig. E28.27**). Wire  $CD$  is connected through the sliding contacts to another wire that also carries a current  $I$ , opposite in direction to the current in wire  $AB$ . The mass per unit length of the wire  $CD$  is  $\lambda$ . To what equilibrium height  $h$  will the wire  $CD$  rise, assuming that the magnetic force on it is due entirely to the current in the wire  $AB$ ?

Figure E28.27



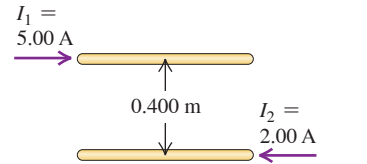
**28.28 •** Three very long parallel wires each carry current  $I$  in the directions shown in **Fig. E28.28**. If the separation between adjacent wires is  $d$ , calculate the magnitude and direction of the net magnetic force per unit length on each wire.

Figure E28.28



**28.29 •** Two long, parallel wires are separated by a distance of 0.400 m (**Fig. E28.29**). The currents  $I_1$  and  $I_2$  have the directions shown. (a) Calculate the magnitude of the force exerted by each wire on a 1.20 m length of the other. Is the force attractive or repulsive? (b) Each current is doubled, so that  $I_1$  becomes 10.0 A and  $I_2$  becomes 4.00 A. Now what is the magnitude of the force that each wire exerts on a 1.20 m length of the other?

Figure E28.29



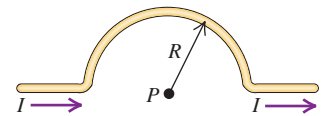
**28.30 •** Two long, parallel wires are separated by a distance of 2.50 cm. The force per unit length that each wire exerts on the other is  $4.00 \times 10^{-5}$  N/m, and the wires repel each other. The current in one wire is 0.600 A. (a) What is the current in the second wire? (b) Are the two currents in the same direction or in opposite directions?

#### Section 28.5 Magnetic Field of a Circular Current Loop

**28.31 • BIO Currents in the Brain.** The magnetic field around the head has been measured to be approximately  $3.0 \times 10^{-8}$  G. Although the currents that cause this field are quite complicated, we can get a rough estimate of their size by modeling them as a single circular current loop 16 cm (the width of a typical head) in diameter. What is the current needed to produce such a field at the center of the loop?

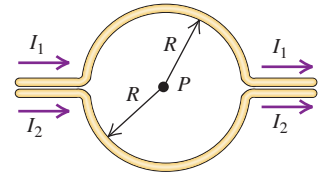
**28.32 •** Calculate the magnitude and direction of the magnetic field at point  $P$  due to the current in the semicircular section of wire shown in **Fig. E28.32**. (*Hint:* Does the current in the long, straight section of the wire produce any field at  $P$ ?)

Figure E28.32



**28.33 ••** Calculate the magnitude of the magnetic field at point  $P$  of **Fig. E28.33** in terms of  $R$ ,  $I_1$ , and  $I_2$ . What does your expression give when  $I_1 = I_2$ ?

Figure E28.33



**28.34 ••** A closely wound, circular coil with radius 2.40 cm has 800 turns. (a) What must the current in the coil be if the magnetic field at the center of the coil is 0.0770 T? (b) At what distance  $x$  from the center of the coil, on the axis of the coil, is the magnetic field half its value at the center?

**28.35 ••** Two concentric circular loops of wire lie on a tabletop, one inside the other. The inner wire has a diameter of 20.0 cm and carries a clockwise current of 12.0 A, as viewed from above, and the outer wire has a diameter of 30.0 cm. What must be the magnitude and direction (as viewed from above) of the current in the outer wire so that the net magnetic field due to this combination of wires is zero at the common center of the wires?

**28.36 ••** A closely wound coil has a radius of 6.00 cm and carries a current of 2.50 A. How many turns must it have if, at a point on the coil axis 6.00 cm from the center of the coil, the magnetic field is  $6.39 \times 10^{-4}$  T?

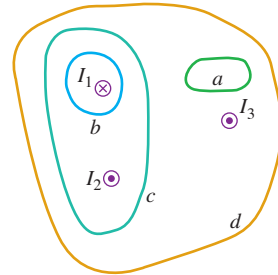


## Section 28.6 Ampere's Law

**28.37** • A closed curve encircles several conductors. The line integral  $\oint \vec{B} \cdot d\vec{l}$  around this curve is  $3.83 \times 10^{-4} \text{ T} \cdot \text{m}$ . (a) What is the net current in the conductors? (b) If you were to integrate around the curve in the opposite direction, what would be the value of the line integral? Explain.

**28.38** • Figure E28.38 shows, in cross section, several conductors that carry currents through the plane of the figure. The currents have the magnitudes  $I_1 = 4.0 \text{ A}$ ,  $I_2 = 6.0 \text{ A}$ , and  $I_3 = 2.0 \text{ A}$ , and the directions shown. Four paths, labeled  $a$  through  $d$ , are shown. What is the line integral  $\oint \vec{B} \cdot d\vec{l}$  for each path? Each integral involves going around the path in the counterclockwise direction. Explain your answers.

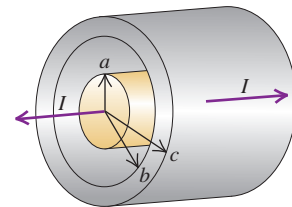
Figure E28.38



## Section 28.7 Applications of Ampere's Law

**28.39** • **Coaxial Cable.** A solid conductor with radius  $a$  is supported by insulating disks on the axis of a conducting tube with inner radius  $b$  and outer radius  $c$  (Fig. E28.39). The central conductor and tube carry equal currents  $I$  in opposite directions. The currents are distributed uniformly over the cross sections of each conductor. Derive an expression for the magnitude of the magnetic field (a) at points outside the central, solid conductor but inside the tube ( $a < r < b$ ) and (b) at points outside the tube ( $r > c$ ).

Figure E28.39



**28.40** • As a new electrical technician, you are designing a large solenoid to produce a uniform  $0.150 \text{ T}$  magnetic field near the center of the solenoid. You have enough wire for 4000 circular turns. This solenoid must be  $55.0 \text{ cm}$  long and  $2.80 \text{ cm}$  in diameter. What current will you need to produce the necessary field?

**28.41** • Repeat Exercise 28.39 for the case in which the current in the central, solid conductor is  $I_1$ , the current in the tube is  $I_2$ , and these currents are in the same direction rather than in opposite directions.

**28.42** • A  $15.0\text{-cm}$ -long solenoid with radius  $0.750 \text{ cm}$  is closely wound with 600 turns of wire. The current in the windings is  $8.00 \text{ A}$ . Compute the magnetic field at a point near the center of the solenoid.

**28.43** • A solenoid is designed to produce a magnetic field of  $0.0270 \text{ T}$  at its center. It has radius  $1.40 \text{ cm}$  and length  $40.0 \text{ cm}$ , and the wire can carry a maximum current of  $12.0 \text{ A}$ . (a) What minimum number of turns per unit length must the solenoid have? (b) What total length of wire is required?

**28.44** • An ideal toroidal solenoid (see Example 28.10) has inner radius  $r_1 = 15.0 \text{ cm}$  and outer radius  $r_2 = 18.0 \text{ cm}$ . The solenoid has 250 turns and carries a current of  $8.50 \text{ A}$ . What is the magnitude of the magnetic field at the following distances from the center of the torus: (a)  $12.0 \text{ cm}$ ; (b)  $16.0 \text{ cm}$ ; (c)  $20.0 \text{ cm}$ ?

**28.45** • A magnetic field of  $37.2 \text{ T}$  has been achieved at the MIT Francis Bitter Magnet Laboratory. Find the current needed to achieve such a field (a)  $2.00 \text{ cm}$  from a long, straight wire; (b) at the center of a circular coil of radius  $42.0 \text{ cm}$  that has 100 turns; (c) near the center of a solenoid with radius  $2.40 \text{ cm}$ , length  $32.0 \text{ cm}$ , and 40,000 turns.

## Section 28.8 Magnetic Materials

**28.46** • A toroidal solenoid with 400 turns of wire and a mean radius of  $6.0 \text{ cm}$  carries a current of  $0.25 \text{ A}$ . The relative permeability of the core is 80. (a) What is the magnetic field in the core? (b) What part of the magnetic field is due to the magnetic moments of the atoms in the core?

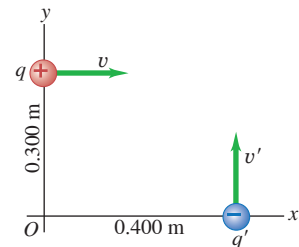
**28.47** • A long solenoid with 60 turns of wire per centimeter carries a current of  $0.15 \text{ A}$ . The wire that makes up the solenoid is wrapped around a solid core of silicon steel ( $K_m = 5200$ ). (The wire of the solenoid is jacketed with an insulator so that none of the current flows into the core.) (a) For a point inside the core, find the magnitudes of (i) the magnetic field  $\vec{B}_0$  due to the solenoid current; (ii) the magnetization  $\vec{M}$ ; (iii) the total magnetic field  $\vec{B}$ . (b) In a sketch of the solenoid and core, show the directions of the vectors  $\vec{B}$ ,  $\vec{B}_0$ , and  $\vec{M}$  inside the core.

**28.48** • The current in the windings of a toroidal solenoid is  $2.400 \text{ A}$ . There are 500 turns, and the mean radius is  $25.00 \text{ cm}$ . The toroidal solenoid is filled with a magnetic material. The magnetic field inside the windings is found to be  $1.940 \text{ T}$ . Calculate (a) the relative permeability and (b) the magnetic susceptibility of the material that fills the toroid.

## PROBLEMS

**28.49** • A pair of point charges,  $q = +8.00 \mu\text{C}$  and  $q' = -5.00 \mu\text{C}$ , are moving as shown in Fig. P28.49 with speeds  $v = 9.00 \times 10^4 \text{ m/s}$  and  $v' = 6.50 \times 10^4 \text{ m/s}$ . When the charges are at the locations shown in the figure, what are the magnitude and direction of (a) the magnetic field produced at the origin and (b) the magnetic force that  $q'$  exerts on  $q$ ?

Figure P28.49



**28.50** • At a particular instant, charge  $q_1 = +4.80 \times 10^{-6} \text{ C}$  is at the point  $(0, 0.250 \text{ m}, 0)$  and has velocity  $\vec{v}_1 = (9.20 \times 10^5 \text{ m/s})\hat{i}$ . Charge  $q_2 = -2.90 \times 10^{-6} \text{ C}$  is at the point  $(0.150 \text{ m}, 0, 0)$  and has velocity  $\vec{v}_2 = (-5.30 \times 10^5 \text{ m/s})\hat{j}$ . At this instant, what are the magnitude and direction of the magnetic force that  $q_1$  exerts on  $q_2$ ?

**28.51** • A long, straight wire lies along the  $x$ -axis and carries current  $I_1 = 2.00 \text{ A}$  in the  $+\hat{x}$ -direction. A second wire lies in the  $xy$ -plane and is parallel to the  $x$ -axis at  $y = +0.800 \text{ m}$ . It carries current  $I_2 = 6.00 \text{ A}$ , also in the  $+\hat{x}$ -direction. In addition to  $y \rightarrow \pm\infty$ , at what point on the  $y$ -axis is the resultant magnetic field of the two wires equal to zero?

**28.52** • Repeat Problem 28.51 for  $I_2$  in the  $-\hat{x}$ -direction, with all the other quantities the same.

**28.53** • We can estimate the strength of the magnetic field of a refrigerator magnet in the following way: Imagine the magnet as a collection of current-loop magnetic dipoles. (a) Derive the force between two current loops with radius  $R$  and current  $I$  separated by distance  $d \ll R$ . Very close to the wire its magnetic field is about the same as for an infinitely long wire, and Eq.(28.11) can be used. (b) Using Eq. (28.17), express the current  $I$  in terms of the magnetic field at the middle of the loop, and express the radius  $R$  in terms of the area of the loop. In this way, derive an expression for the force  $F$  between two identical current loops separated by a small distance  $d$  in terms of their mutual area  $A$  and center magnetic field  $B$ . (c) Rearrange your result to obtain an expression for the magnetic field of a dipole with area  $A$  in terms of the force  $F$  from an identical dipole separated by a small distance  $d$ . (d) Now notice that the force it takes to separate one magnet from your refrigerator is nearly the same as the force it takes to separate two magnets stuck together. Estimate that force  $F$ . (e) Estimate the area of a refrigerator magnet. (f) Assume that when these magnets are stuck together or to the refrigerator, they are separated by an effective distance  $d = 25 \mu\text{m}$ . Use the formula derived above to estimate the magnetic field strength of the magnet.

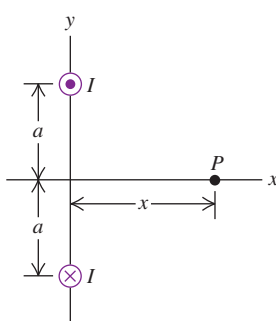
**28.54 •** A long, straight wire carries a current of 8.60 A. An electron is traveling in the vicinity of the wire. At the instant when the electron is 4.50 cm from the wire and traveling at a speed of  $6.00 \times 10^4$  m/s directly toward the wire, what are the magnitude and direction (relative to the direction of the current) of the force that the magnetic field of the current exerts on the electron?

**28.55 • CP** A long, straight wire carries a 13.0 A current. An electron is fired parallel to this wire with a velocity of 250 km/s in the same direction as the current, 2.00 cm from the wire. (a) Find the magnitude and direction of the electron's initial acceleration. (b) What should be the magnitude and direction of a uniform electric field that will allow the electron to continue to travel parallel to the wire? (c) Is it necessary to include the effects of gravity? Justify your answer.

**28.56 ••** An electron is moving in the vicinity of a long, straight wire that lies along the  $x$ -axis. The wire has a constant current of 9.00 A in the  $-x$ -direction. At an instant when the electron is at point  $(0, 0.200 \text{ m}, 0)$  and the electron's velocity is  $\vec{v} = (5.00 \times 10^4 \text{ m/s})\hat{i} - (3.00 \times 10^4 \text{ m/s})\hat{j}$ , what is the force that the wire exerts on the electron? Express the force in terms of unit vectors, and calculate its magnitude.

**28.57 •• CP** (a) Determine the transmission power  $P$  of your cell phone. (This information is available online.) (b) A typical cell phone battery supplies a 1.5 V potential. If your phone battery supplies the power  $P$ , what is a good estimate of the current supplied by the battery? (c) Estimate the width of your head. (d) Estimate the diameter of the phone speaker that goes next to your ear. Model the current in the speaker as a current loop with the same diameter as the speaker. Use these values to estimate the magnetic field generated by your phone midway between the ears when it is held near one ear. (e) How does your answer compare to the earth's field, which is about  $50 \mu\text{T}$ ?

**28.58 •• Figure P28.58** shows an end view of two long, parallel wires perpendicular to the  $xy$ -plane, each carrying a current  $I$  but in opposite directions. (a) Copy the diagram, and draw vectors to show the  $\vec{B}$  field of each wire and the net  $\vec{B}$  field at point  $P$ . (b) Derive the expression for the magnitude of  $\vec{B}$  at any point on the  $x$ -axis in terms of the  $x$ -coordinate of the point. What is the direction of  $\vec{B}$ ? (c) Graph the magnitude of  $\vec{B}$  at points on the  $x$ -axis. (d) At what value of  $x$  is the magnitude of  $\vec{B}$  a maximum? (e) What is the magnitude of  $\vec{B}$  when  $x \gg a$ ?



**28.59 •** Two long, straight, parallel wires are 1.00 m apart (Fig. P28.59). The wire on the left carries a current  $I_1$  of 6.00 A into the plane of the paper. (a) What must the magnitude and direction of the current  $I_2$  be for the net field at point  $P$  to be zero? (b) Then what are the magnitude and direction of the net field at  $Q$ ? (c) Then what is the magnitude of the net field at  $S$ ?

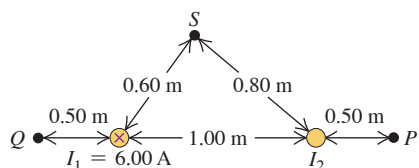
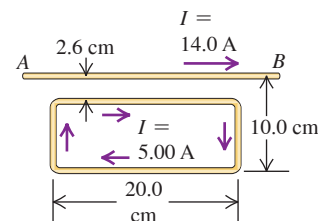


Figure P28.59

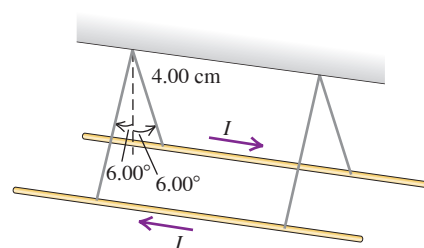
**28.60 •** The long, straight wire  $AB$  shown in Fig. P28.60 carries a current of 14.0 A. The rectangular loop whose long edges are parallel to the wire carries a current of 5.00 A. Find the magnitude and direction of the net force exerted on the loop by the magnetic field of the wire.

Figure P28.60



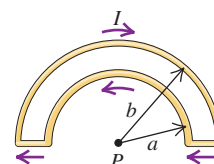
**28.61 ••• CP** Two long, parallel wires hang by 4.00-cm-long cords from a common axis (Fig. P28.61). The wires have a mass per unit length of 0.0125 kg/m and carry the same current in opposite directions. What is the current in each wire if the cords hang at an angle of  $6.00^\circ$  with the vertical?

Figure P28.61



**28.62 •** The wire semicircles shown in Fig. P28.62 have radii  $a$  and  $b$ . Calculate the net magnetic field (magnitude and direction) that the current in the wires produces at point  $P$ .

Figure P28.62



**28.63 •** A long, straight, solid cylinder, oriented with its axis in the  $z$ -direction, carries a current whose current density is  $\vec{J}$ . The current density, although symmetric about the cylinder axis, is not constant and varies according to the relationship

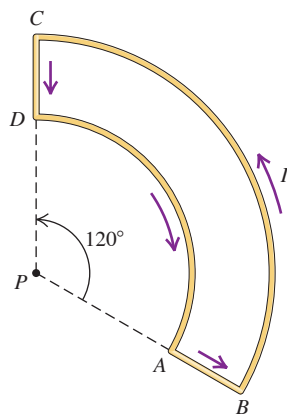
$$\vec{J} = \left(\frac{b}{r}\right)e^{(r-a)/\delta}\hat{k} \quad \text{for } r \leq a$$

$$= 0 \quad \text{for } r \geq a$$

where the radius of the cylinder is  $a = 5.00$  cm,  $r$  is the radial distance from the cylinder axis,  $b$  is a constant equal to 600 A/m, and  $\delta$  is a constant equal to 2.50 cm. (a) Let  $I_0$  be the total current passing through the entire cross section of the wire. Obtain an expression for  $I_0$  in terms of  $b$ ,  $\delta$ , and  $a$ . Evaluate your expression to obtain a numerical value for  $I_0$ . (b) Using Ampere's law, derive an expression for the magnetic field  $\vec{B}$  in the region  $r \geq a$ . Express your answer in terms of  $I_0$  rather than  $b$ . (c) Obtain an expression for the current  $I$  contained in a circular cross section of radius  $r \leq a$  and centered at the cylinder axis. Express your answer in terms of  $I_0$  rather than  $b$ . (d) Using Ampere's law, derive an expression for the magnetic field  $\vec{B}$  in the region  $r \leq a$ . (e) Evaluate the magnitude of the magnetic field at  $r = \delta$ ,  $r = a$ , and  $r = 2a$ .

**28.64 ••** Calculate the magnetic field (magnitude and direction) at a point  $P$  due to a current  $I = 12.0$  A in the wire shown in **Fig. P28.64**. Segment  $BC$  is an arc of a circle with radius  $30.0$  cm, and point  $P$  is at the center of curvature of the arc. Segment  $DA$  is an arc of a circle with radius  $20.0$  cm, and point  $P$  is at its center of curvature. Segments  $CD$  and  $AB$  are straight lines of length  $10.0$  cm each.

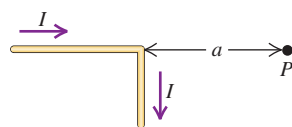
Figure P28.64



**28.65 • CALC** A long, straight wire with a circular cross section of radius  $R$  carries a current  $I$ . Assume that the current density is not constant across the cross section of the wire, but rather varies as  $J = ar$ , where  $a$  is a constant. (a) By the requirement that  $J$  integrated over the cross section of the wire gives the total current  $I$ , calculate the constant  $a$  in terms of  $I$  and  $R$ . (b) Use Ampere's law to calculate the magnetic field  $B(r)$  for (i)  $r \leq R$  and (ii)  $r \geq R$ . Express your answers in terms of  $I$ .

**28.66 • CALC** The wire shown in **Fig. P28.66** is infinitely long and carries a current  $I$ . Calculate the magnitude and direction of the magnetic field that this current produces at point  $P$ .

Figure P28.66



**28.67 • CALC** A long, straight, solid cylinder, oriented with its axis in the  $z$ -direction, carries a current whose current density is  $\vec{J}$ . The current density, although symmetric about the cylinder axis, is not constant but varies according to the relationship

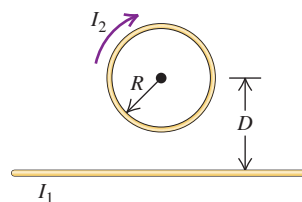
$$\vec{J} = \frac{2I_0}{\pi a^2} \left[ 1 - \left( \frac{r}{a} \right)^2 \right] \hat{k} \quad \text{for } r \leq a$$

$$= \mathbf{0} \quad \text{for } r \geq a$$

where  $a$  is the radius of the cylinder,  $r$  is the radial distance from the cylinder axis, and  $I_0$  is a constant having units of amperes. (a) Show that  $I_0$  is the total current passing through the entire cross section of the wire. (b) Using Ampere's law, derive an expression for the magnitude of the magnetic field  $\vec{B}$  in the region  $r \geq a$ . (c) Obtain an expression for the current  $I$  contained in a circular cross section of radius  $r \leq a$  and centered at the cylinder axis. (d) Using Ampere's law, derive an expression for the magnitude of the magnetic field  $\vec{B}$  in the region  $r \leq a$ . How do your results in parts (b) and (d) compare for  $r = a$ ?

**28.68 •** A circular loop has radius  $R$  and carries current  $I_2$  in a clockwise direction (**Fig. P28.68**). The center of the loop is a distance  $D$  above a long, straight wire. What are the magnitude and direction of the current  $I_1$  in the wire if the magnetic field at the center of the loop is zero?

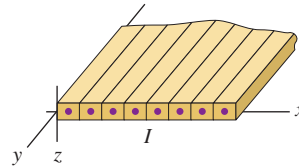
Figure P28.68



**28.69 • An Infinite Current Sheet.** Long, straight conductors with square cross sections and each carrying current  $I$  are laid side by side to form an infinite current sheet (**Fig. P28.69**). The conductors lie in the  $xy$ -plane, are parallel to the  $y$ -axis, and carry current in the  $+y$ -direction. There are  $n$  conductors per unit length measured along the  $x$ -axis.

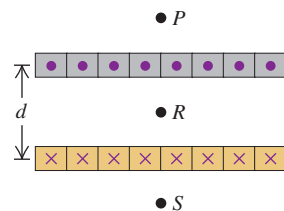
(a) What are the magnitude and direction of the magnetic field a distance  $a$  below the current sheet? (b) What are the magnitude and direction of the magnetic field a distance  $a$  above the current sheet?

Figure P28.69



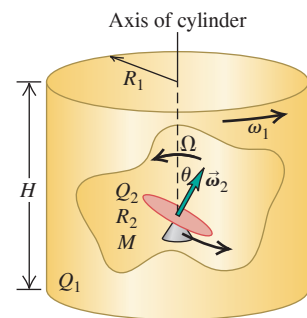
**28.70 •** Long, straight conductors with square cross section, each carrying current  $I$ , are laid side by side to form an infinite current sheet with current directed out of the plane of the page (**Fig. P28.70**). A second infinite current sheet is a distance  $d$  below the first and is parallel to it. The second sheet carries current into the plane of the page. Each sheet has  $n$  conductors per unit length. (Refer to Problem 28.69.) Calculate the magnitude and direction of the net magnetic field at (a) point  $P$  (above the upper sheet); (b) point  $R$  (midway between the two sheets); (c) point  $S$  (below the lower sheet).

Figure P28.70



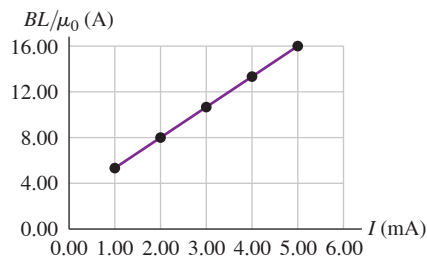
**28.71 ••• CP** A cylindrical shell with radius  $R_1$  and height  $H$  has charge  $Q_1$  and rotates around its axis with angular speed  $\omega_1$ , as shown in **Fig. P28.71**. Inside the cylinder, far from its edges, sits a very small disk with radius  $R_2$ , mass  $M$ , and charge  $Q_2$  mounted on a pivot, spinning with a large angular velocity  $\vec{\omega}_2$  and oriented at angle  $\theta$  with respect to the axis of the cylinder. The center of the disk is on the axis of the cylinder. The magnetic interaction between the cylinder and the disk causes a precession of the axis of the disk. (a) What is the magnitude of the enclosed current  $I_{\text{encl}}$  surrounded by a loop that has one vertical side that is along the axis of the cylinder and extends beyond the top and bottom of the cylinder? The other vertical side of the loop is very far outside the cylinder. (b) Assume the field is uniform within the cylinder and use Ampere's law to find the magnetic field at the center of the disk. (c) The magnetic moment of the disk has magnitude  $\mu = \frac{1}{4}Q_2\omega_2 R_2^2$ . What is the magnitude of the torque exerted on the disk? (d) What is the magnitude of the angular momentum of the disk?

Figure P28.71



**28.72 •• DATA** As a summer intern at a research lab, you are given a long solenoid that has two separate windings that are wound close together, in the same direction, on the same hollow cylindrical form. You must determine the number of turns in each winding. The solenoid has length  $L = 40.0$  cm and diameter  $2.80$  cm. You let a  $2.00$  mA current flow in winding 1 and vary the current  $I$  in winding 2; both currents flow in the same direction. Then you measure the magnetic-field magnitude  $B$  at the center of the solenoid as a function of  $I$ . You plot your results as  $BL/\mu_0$  versus  $I$ . The graph in **Fig. P28.72** shows the best-fit straight line to your data. (a) Explain why the data plotted in this way should fall close to a straight line. (b) Use **Fig. P28.72** to calculate  $N_1$  and  $N_2$ , the number of turns in windings 1 and 2. (c) If the current in winding 1 remains  $2.00$  mA in its original direction and winding 2 has  $I = 5.00$  mA in the opposite direction, what is  $B$  at the center of the solenoid?

Figure P28.72



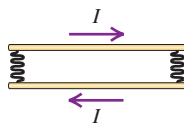
**28.73 •• DATA** You use a teslameter (a Hall-effect device) to measure the magnitude of the magnetic field at various distances from a long, straight, thick cylindrical copper cable that is carrying a large constant current. To exclude the earth's magnetic field from the measurement, you first set the meter to zero. You then measure the magnetic field  $B$  at distances  $x$  from the surface of the cable and obtain these data:

|          |       |       |       |       |       |
|----------|-------|-------|-------|-------|-------|
| $x$ (cm) | 2.0   | 4.0   | 6.0   | 8.0   | 10.0  |
| $B$ (mT) | 0.406 | 0.250 | 0.181 | 0.141 | 0.116 |

(a) You think you remember from your physics course that the magnetic field of a wire is inversely proportional to the distance from the wire. Therefore, you expect that the quantity  $Bx$  from your data will be constant. Calculate  $Bx$  for each data point in the table. Is  $Bx$  constant for this set of measurements? Explain. (b) Graph the data as  $x$  versus  $1/B$ . Explain why such a plot lies close to a straight line. (c) Use the graph in part (b) to calculate the current  $I$  in the cable and the radius  $R$  of the cable.

**28.74 ••• DATA** A pair of long, rigid metal rods, each of length 0.50 m, lie parallel to each other on a frictionless table. Their ends are connected by identical, very lightweight conducting springs with unstretched length  $l_0$  and force constant  $k$  (Fig. P28.74). When a current  $I$  runs through the circuit consisting of the rods and springs, the springs stretch. You measure the distance  $x$  each spring stretches for certain values of  $I$ . When  $I = 8.05$  A, you measure that  $x = 0.40$  cm. When  $I = 13.1$  A, you find  $x = 0.80$  cm. In both cases the rods are much longer than the stretched springs, so it is accurate to use Eq. (28.11) for two infinitely long, parallel conductors. (a) From these two measurements, calculate  $l_0$  and  $k$ . (b) If  $I = 12.0$  A, what distance  $x$  will each spring stretch? (c) What current is required for each spring to stretch 1.00 cm?

Figure P28.74

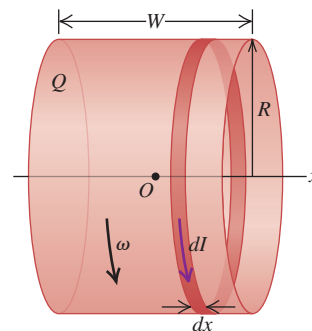


**28.75 ••• CP CALC** A plasma is a gas of ionized (charged) particles. When plasma is in motion, magnetic effects “squeeze” its volume, inducing inward pressure known as a pinch. Consider a cylindrical tube of plasma with radius  $R$  and length  $L$  moving with velocity  $\vec{v}$  along its axis. If there are  $n$  ions per unit volume and each ion has charge  $q$ , we can determine the pressure felt by the walls of the cylinder. (a) What is the volume charge density  $\rho$  in terms of  $n$  and  $q$ ? (b) The thickness of the cylinder “surface” is  $n^{-1/3}$ . What is the surface charge density  $\sigma$  in terms of  $n$  and  $q$ ? (c) The current density inside the cylinder is  $\vec{J} = \rho\vec{v}$ . Use this result along with Ampere’s law to determine the magnetic field on the surface of the cylinder. Denote the circumferential unit vector as  $\hat{\phi}$ . (d) The width of a differential strip of surface current is  $R d\phi$ . What is the differential current  $dI_{\text{surface}}$  that flows along this strip? (e) What differential force is felt by this strip due to the magnetic field generated by the volume current? (f) Integrate to determine the total force on the walls of the cylinder; then divide by the wall area to obtain the pressure in terms of  $n$ ,  $q$ ,  $R$ , and  $v$ . (g) If a plasma cylinder with radius 2.0 cm has a charge density of  $8.0 \times 10^{16}$  ions/cm<sup>3</sup>, where each ion has a charge of  $e = 1.6 \times 10^{-19}$  C and is moving axially with a speed of 20.0 m/s, what is its pinch pressure?

## CHALLENGE PROBLEMS

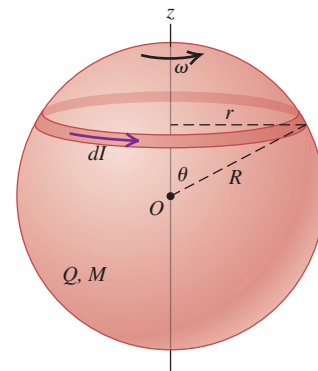
**28.76 ••• CALC** A cylindrical shell with radius  $R$  and length  $W$  carries a uniform charge  $Q$  and rotates about its axis with angular speed  $\omega$ . The center of the cylinder lies at the origin  $O$  and its axis is coincident with the  $x$ -axis, as shown in Fig. P28.76. (a) What is the charge density  $\sigma$ ? (b) What is the differential current  $dI$  on a circular strip of the cylinder centered at  $x$  and with width  $dx$ ? (c) Use Eq. (28.15) to write an expression for the differential magnetic field  $d\vec{B}$  at the origin due to this strip. (d) Integrate to determine the magnetic field at the origin.

Figure P28.76



**28.77 ••• CALC** When a rigid charge distribution with charge  $Q$  and mass  $M$  rotates about an axis, its magnetic moment  $\vec{\mu}$  is linearly proportional to its angular momentum  $\vec{L}$ , with  $\vec{\mu} = \gamma\vec{L}$ . The constant of proportionality  $\gamma$  is called the gyromagnetic ratio of the object. We can write  $\gamma = g(Q/2M)$ , where  $g$  is a dimensionless number called the  $g$ -factor of the object. Consider a spherical shell with mass  $M$  and uniformly distributed charge  $Q$  centered on the origin  $O$  and rotating about the  $z$ -axis with angular speed  $\omega$ . (a) A thin slice with latitude  $\theta$  measured with respect to the positive  $z$ -axis describes a current loop with width  $R d\theta$  and radius  $r = R \sin\theta$ , as shown in Fig. P28.77. What is the differential current  $dI$  carried by this loop, in terms of  $Q$ ,  $\omega$ ,  $R$ ,  $\theta$ , and  $d\theta$ ? (b) The differential magnetic moment contributed by that slice is  $d\vec{\mu} = A dI$ , where  $A = \pi r^2$  is the area enclosed by the loop. Express the differential magnetic moment in terms of  $Q$ ,  $\omega$ ,  $R$ ,  $\theta$ , and  $d\theta$ . (c) Integrate over  $\theta$  to determine the magnetic moment  $\vec{\mu}$ . (d) What is the magnitude of the angular momentum  $\vec{L}$ ? (e) Determine the gyromagnetic ratio  $\gamma$ . (f) What is the  $g$ -factor for a spherical shell?

Figure P28.77



**28.78 ••• CALC** The law of Biot and Savart in Eq. (28.7) generalizes to the case of surface currents as

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{\sigma \vec{v} \times \hat{r}}{r^2} da$$

where  $\sigma$  is the local charge density,  $\vec{v}$  is the local velocity, and  $da$  is a differential area element. Re-visit Challenge Problem 28.76 and use the above equation as an alternative means to derive the magnetic field at the center of the cylinder. Use the following steps: (a) Write the charge density  $\sigma$ . (b) The origin is at the center of the cylinder. What is the vector  $\vec{r}$  that points from the element with coordinates  $(x, y, z) = (x, R \cos\phi, R \sin\phi)$  to the origin? (c) What is the velocity  $\vec{v}$  of the element? (d) What is the vector product  $\vec{v} \times \hat{r}$ ? (e) An area element on the cylinder may be written as  $da = R dx d\phi$ . Use this and the previously established information to write the generalized law of Biot and Savart as a double integral. Evaluate the integral to determine the magnetic field  $\vec{B}$  at the center of the cylinder. (f) Is your result consistent with your result in Challenge Problem 28.76?



**28.79 ••• CP** Two long, straight

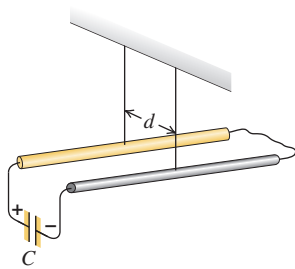
conducting wires with linear mass density  $\lambda$  are suspended from cords so that they are each horizontal, parallel to each other, and a distance  $d$  apart. The back ends of the wires are connected to each other by a slack, low-resistance connecting wire. A charged capacitor (capacitance  $C$ ) is now added to the system; the positive plate of the capacitor (initial charge  $+Q_0$ ) is connected to the front end of one of the wires, and the negative plate of the capacitor (initial charge  $-Q_0$ ) is connected to the front end of the other wire (**Fig. P28.79**). Both of these connections are also made by slack, low-resistance wires. When the connection is made, the wires are pushed aside by the repulsive force between the wires, and each wire has an initial horizontal velocity of magnitude  $v_0$ . Assume that the time constant for the capacitor to discharge is negligible compared to the time it takes for any appreciable displacement in the position of the wires to occur. (a) Show that the initial speed  $v_0$  of either wire is given by

$$v_0 = \frac{\mu_0 Q_0^2}{4\pi\lambda R C d}$$

where  $R$  is the total resistance of the circuit. (b) To what height  $h$  will each wire rise as a result of the circuit connection?

**28.80 •••** A wide, long, insulating belt has a uniform positive charge per unit area  $\sigma$  on its upper surface. Rollers at each end move the belt to the right at a constant speed  $v$ . Calculate the magnitude and direction of the magnetic field produced by the moving belt at a point just above its surface. (*Hint:* At points near the surface and far from its edges or ends, the moving belt can be considered to be an infinite current sheet like that in Problem 28.69.)

Figure P28.79

**MCAT-STYLE PASSAGE PROBLEMS**

**BIO Studying Magnetic Bacteria.** Some types of bacteria contain chains of ferromagnetic particles parallel to their long axis. The chains act like small bar magnets that align these *magnetotactic* bacteria with the earth's magnetic field. In one experiment to study the response of such bacteria to magnetic fields, a solenoid is constructed with copper wire 1.0 mm in diameter, evenly wound in a single layer to form a helical coil of length 40 cm and diameter 12 cm. The wire has a very thin layer of insulation, and the coil is wound so that adjacent turns are just touching. The solenoid, which generates a magnetic field, is in an enclosure that shields it from other magnetic fields. A sample of magnetotactic bacteria is placed inside the solenoid. The torque on an individual bacterium in the solenoid's magnetic field is proportional to the magnitude of the magnetic field and to the sine of the angle between the long axis of the bacterium and the magnetic-field direction.

**28.81** What current is needed in the wire so that the magnetic field experienced by the bacteria has a magnitude of  $150 \mu\text{T}$ ? (a)  $0.095 \text{ A}$ ; (b)  $0.12 \text{ A}$ ; (c)  $0.30 \text{ A}$ ; (d)  $14 \text{ A}$ .

**28.82** To use a larger sample, the experimenters construct a solenoid that has the same length, type of wire, and loop spacing but twice the diameter of the original. How does the maximum possible magnetic torque on a bacterium in this new solenoid compare with the torque the bacterium would have experienced in the original solenoid? Assume that the currents in the solenoids are the same. The maximum torque in the new solenoid is (a) twice that in the original one; (b) half that in the original one; (c) the same as that in the original one; (d) one-quarter that in the original one.

**28.83** The solenoid is removed from the enclosure and then used in a location where the earth's magnetic field is  $50 \mu\text{T}$  and points horizontally. A sample of bacteria is placed in the center of the solenoid, and the same current is applied that produced a magnetic field of  $150 \mu\text{T}$  in the lab. Describe the field experienced by the bacteria: The field (a) is still  $150 \mu\text{T}$ ; (b) is now  $200 \mu\text{T}$ ; (c) is between  $100$  and  $200 \mu\text{T}$ , depending on how the solenoid is oriented; (d) is between  $50$  and  $150 \mu\text{T}$ , depending on how the solenoid is oriented.

**ANSWERS****Chapter Opening Question?**

(iv) There would be *no* change in the magnetic field strength. From Example 28.9 (Section 28.7), the field inside a solenoid has magnitude  $B = \mu_0 n I$ , where  $n$  is the number of turns of wire per unit length. Joining two solenoids end to end doubles both the number of turns and the length, so the number of turns per unit length is unchanged.

**Key Example VARIATION Problems**

**VP28.2.1** (a)  $(3.20 \times 10^{-15} \text{ T})\hat{k}$  (b)  $(-8.00 \times 10^{-16} \text{ T})\hat{j}$  (c) zero (d)  $(1.13 \times 10^{-15} \text{ T})\hat{k}$

**VP28.2.2** (a)  $1.13 \times 10^{-28} \text{ N}$  in the  $-x$ -direction (toward the proton) (b)  $1.13 \times 10^{-28} \text{ N}$  in the  $+x$ -direction (toward the electron)

**VP28.2.3** (a)  $(0.600)\hat{i} + (0.800)\hat{j}$  (b)  $(-2.88 \times 10^{-11} \text{ T})\hat{k}$

**VP28.2.4**  $(9.83 \times 10^{-23} \text{ N})\hat{k}$

**VP28.5.1**  $2.24 \text{ A}$

**VP28.5.2** (a)  $(-6.00 \times 10^{-5} \text{ T})\hat{i}$  (b)  $(-4.67 \times 10^{-5} \text{ T})\hat{i}$  (c)  $(8.67 \times 10^{-5} \text{ T})\hat{i}$

**VP28.5.3** (a)  $5.00 \text{ A}$  (b)  $1.00 \text{ A}$

**VP28.5.4**  $3.29 \times 10^4 \text{ A}$ , upward

**VP28.10.1** (a)  $4.94 \times 10^{-6} \text{ T}$  (b)  $8.89 \times 10^{-6} \text{ T}$  (c)  $6.67 \times 10^{-6} \text{ T}$

**VP28.10.2** (a)  $B = 0$  for  $r < R_1$ ,  $B = (\mu_0 J/2r)(r^2 - R_1^2)$  for  $R_2 > r > R_1$ ,  $B = (\mu_0 J/2r)(R_2^2 - R_1^2)$  for  $r > R_2$  (b)  $r = R_2$

**VP28.10.3**  $3.18 \text{ A}$

**VP28.10.4** (a) 833 (b)  $2.33 \times 10^{-3} \text{ T}$  maximum,  $1.75 \times 10^{-3} \text{ T}$  minimum

**Bridging Problem**

$$B = \frac{\mu_0 n Q}{a}$$



**?** The card reader at your bank's cash machine scans the information that is coded in a magnetic pattern on the back of your card. Why must you remove the card quickly rather than hold it motionless in the card reader's slot? (i) To maximize the magnetic force on the card; (ii) to maximize the magnetic force on the mobile charges in the card reader; (iii) to generate an electric force on the card; (iv) to generate an electric force on the mobile charges in the card reader.

# 29 Electromagnetic Induction

**A**lmost every modern device or machine, from a computer to a washing machine to a power drill, has electric circuits at its heart. We learned in Chapter 25 that an electromotive force (emf) is required for a current to flow in a circuit; in Chapters 25 and 26 we almost always took the source of emf to be a battery. But for most devices that you plug into a wall socket, the source of emf is *not* a battery but an electric generating station. Such a station produces electrical energy by converting other forms of energy: gravitational potential energy at a hydroelectric plant, chemical energy in a coal-, gas-, or oil-fired plant, nuclear energy at a nuclear plant. But how is this energy conversion done?

The answer is a phenomenon known as *electromagnetic induction*: If the magnetic flux through a circuit changes, an emf and a current are induced in the circuit. In a power-generating station, magnets move relative to coils of wire to produce a changing magnetic flux in the coils and hence an emf.

The central principle of electromagnetic induction is *Faraday's law*. This law relates induced emf to changing magnetic flux in any loop, including a closed circuit. We also discuss Lenz's law, which helps us to predict the directions of induced emfs and currents. These principles will allow us to understand electrical energy-conversion devices such as motors, generators, and transformers.

Electromagnetic induction tells us that a time-varying magnetic field can act as a source of electric field. We'll also see how a time-varying *electric* field can act as a source of *magnetic* field. These remarkable results form part of a neat package of formulas, called *Maxwell's equations*, that describe the behavior of electric and magnetic fields in general. Maxwell's equations pave the way toward an understanding of electromagnetic waves, the topic of Chapter 32.

## LEARNING OUTCOMES

### In this chapter, you'll learn...

- 29.1** The experimental evidence that a changing magnetic field induces an emf.
- 29.2** How Faraday's law relates the induced emf in a loop to the change in magnetic flux through the loop.
- 29.3** How to determine the direction of an induced emf.
- 29.4** How to calculate the emf induced in a conductor moving through a magnetic field.
- 29.5** How a changing magnetic flux generates a circulating electric field.
- 29.6** How eddy currents arise in a metal that moves in a magnetic field.
- 29.7** The four fundamental equations that completely describe both electricity and magnetism.
- 29.8** The remarkable electric and magnetic properties of superconductors.

### You'll need to review...

- 23.1** Conservative electric fields.
- 25.4** Electromotive force (emf).
- 27.3, 27.8, 27.9** Magnetic flux; direct-current motors; Hall effect.
- 28.5–28.7** Magnetic field of a current loop and solenoid; Ampere's law.

**BIO APPLICATION Exploring the Brain with Induced emfs** Transcranial magnetic stimulation (TMS) is a technique for studying the function of various parts of the brain. A coil held to the subject's head carries a varying electric current, and so produces a varying magnetic field. This field causes an induced emf, and that triggers electric activity in the region of the brain underneath the coil. By observing how the TMS subject responds (for instance, which muscles move as a result of stimulating a certain part of the brain), a physician can test for various neurological conditions.



## 29.1 INDUCTION EXPERIMENTS

During the 1830s, several pioneering experiments with magnetically induced emf were carried out in England by Michael Faraday and in the United States by Joseph Henry (1797–1878). **Figure 29.1** shows several examples. In Fig. 29.1a, a coil of wire is connected to a galvanometer. When the nearby magnet is stationary, the meter shows no current. This isn't surprising; there is no source of emf in the circuit. But when we *move* the magnet either toward or away from the coil, the meter shows current in the circuit, but *only* while the magnet is moving (Fig. 29.1b). If we keep the magnet stationary and move the coil, we again detect a current during the motion. We call this an **induced current**, and the corresponding emf required to cause this current is called an **induced emf**.

In Fig. 29.1c we replace the magnet with a second coil connected to a battery. When the second coil is stationary, there is no current in the first coil. However, when we move the second coil toward or away from the first or move the first toward or away from the second, there is current in the first coil, but again *only* while one coil is moving relative to the other.

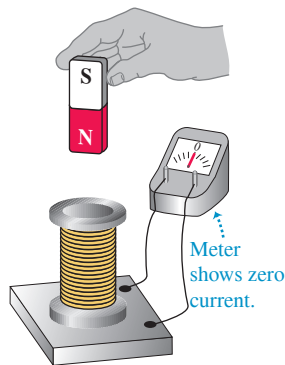
Finally, using the two-coil setup in Fig. 29.1d, we keep both coils stationary and vary the current in the second coil by opening and closing the switch. As we open or close the switch, there is a momentary current pulse in the first coil. The induced current in the first coil is present only while the current in the second coil is changing.

To explore further the common elements in these observations, let's consider a more detailed series of experiments (**Fig. 29.2**). We connect a coil of wire to a galvanometer and then place the coil between the poles of an electromagnet whose magnetic field we can vary. Here's what we observe:

1. When there is no current in the electromagnet, so that  $\vec{B} = 0$ , the galvanometer shows no current.
2. When the electromagnet is turned on, there is a momentary current through the meter as  $\vec{B}$  increases.
3. When  $\vec{B}$  levels off at a steady value, the current drops to zero.
4. With the coil in a horizontal plane, we squeeze it so as to decrease the cross-sectional area of the coil. The meter detects current only *during* the deformation, not before or after. When we increase the area to return the coil to its original shape, there is current in the opposite direction, but only while the area of the coil is changing.

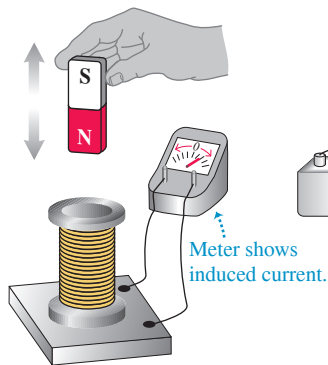
Figure 29.1 Demonstrating the phenomenon of induced current.

(a) A stationary magnet does NOT induce a current in a coil.

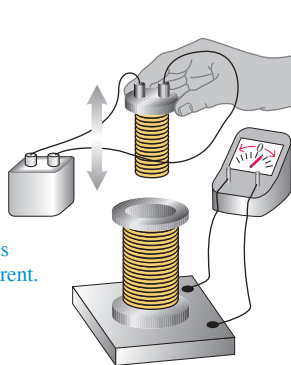


All these actions DO induce a current in the coil. What do they have in common?\*

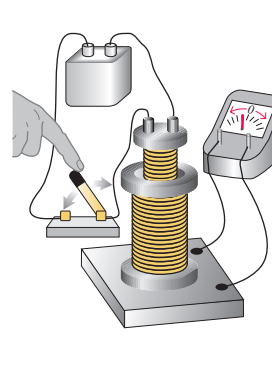
(b) Moving the magnet toward or away from the coil



(c) Moving a second, current-carrying coil toward or away from the coil



(d) Varying the current in the second coil (by closing or opening a switch)



\*Answer: They cause the magnetic field through the coil to *change*.

5. If we rotate the coil a few degrees about a horizontal axis, the meter detects current during the rotation, in the same direction as when we decreased the area. When we rotate the coil back, there is a current in the opposite direction during this rotation.
6. If we jerk the coil out of the magnetic field, there is a current during the motion, in the same direction as when we decreased the area.
7. If we decrease the number of turns in the coil by unwinding one or more turns, there is a current during the unwinding, in the same direction as when we decreased the area. If we wind more turns onto the coil, there is a current in the opposite direction during the winding.
8. When the magnet is turned off, there is a momentary current in the direction opposite to the current when it was turned on.
9. The faster we carry out any of these changes, the greater the current.
10. If all these experiments are repeated with a coil that has the same shape but different material and different resistance, the current in each case is inversely proportional to the total circuit resistance. This shows that the induced emfs that are causing the current do not depend on the material of the coil but only on its shape and the magnetic field.

The common element in these experiments is changing *magnetic flux*  $\Phi_B$  through the coil connected to the galvanometer. In each case the flux changes either because the magnetic field changes with time or because the coil is moving through a nonuniform magnetic field. What's more, in each case the induced emf is proportional to the *rate of change* of magnetic flux  $\Phi_B$  through the coil. The *direction* of the induced emf depends on whether the flux is increasing or decreasing. If the flux is constant, there is no induced emf.

Induced emfs have a tremendous number of practical applications. If you are reading these words indoors, you are making use of induced emfs right now! At the power plant that supplies your neighborhood, an electric generator produces an emf by varying the magnetic flux through coils of wire. (In the next section we'll see how this is done.) This emf supplies the voltage between the terminals of the wall sockets in your home, and this voltage supplies the power to your reading lamp.

Magnetically induced emfs, just like the emfs discussed in Section 25.4, are the result of *nonelectrostatic* forces. We have to distinguish carefully between the electrostatic electric fields produced by charges (according to Coulomb's law) and the nonelectrostatic electric fields produced by changing magnetic fields. We'll return to this distinction later in this chapter and the next.

## 29.2 FARADAY'S LAW

The common element in all induction effects is changing magnetic flux through a circuit. Before stating the simple physical law that summarizes all of the kinds of experiments described in Section 29.1, let's first review the concept of magnetic flux  $\Phi_B$  (which we introduced in Section 27.3). For an infinitesimal-area element  $d\vec{A}$  in a magnetic field  $\vec{B}$  (Fig. 29.3), the magnetic flux  $d\Phi_B$  through the area is

$$d\Phi_B = \vec{B} \cdot d\vec{A} = B_{\perp} dA = B dA \cos \phi$$

where  $B_{\perp}$  is the component of  $\vec{B}$  perpendicular to the surface of the area element and  $\phi$  is the angle between  $\vec{B}$  and  $d\vec{A}$ . (As in Chapter 27, be careful to distinguish between two quantities named "phi,"  $\phi$  and  $\Phi_B$ .) The total magnetic flux  $\Phi_B$  through a finite area is the integral of this expression over the area:

$$\Phi_B = \int \vec{B} \cdot d\vec{A} = \int B dA \cos \phi \quad (29.1)$$

Figure 29.2 A coil in a magnetic field. When the  $\vec{B}$  field is constant and the shape, location, and orientation of the coil do not change, no current is induced in the coil. A current is induced when any of these factors change.

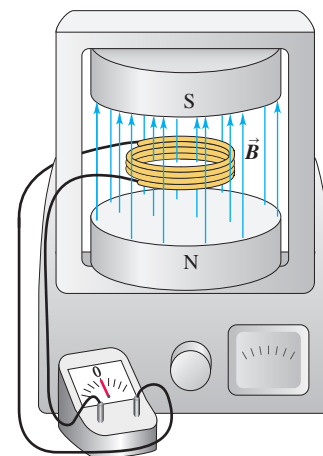
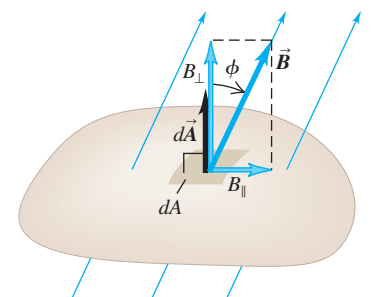


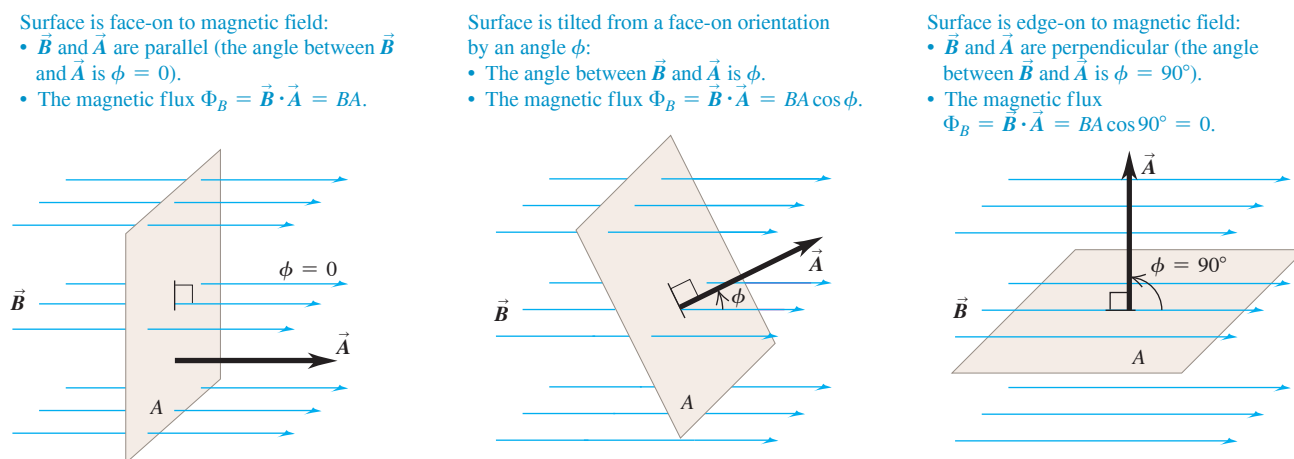
Figure 29.3 Calculating the magnetic flux through an area element.



Magnetic flux through element of area  $d\vec{A}$ :  
 $d\Phi_B = \vec{B} \cdot d\vec{A} = B_{\perp} dA = B dA \cos \phi$



Figure 29.4 Calculating the flux of a uniform magnetic field through a flat area. (Compare to Fig. 22.6, which shows the rules for calculating the flux of a uniform *electric* field.)



If  $\vec{B}$  is uniform over a flat area  $\vec{A}$ , then

$$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \phi \quad (29.2)$$

Figure 29.4 reviews the rules for using Eq. (29.2).

**CAUTION** Choosing the direction of  $d\vec{A}$  or  $\vec{A}$  In Eqs. (29.1) and (29.2) we must define the direction of the vector area  $d\vec{A}$  or  $\vec{A}$  unambiguously. There are always two directions perpendicular to any given area, and the sign of the magnetic flux through the area depends on which one we choose. For example, in Fig. 29.3 we chose  $d\vec{A}$  to point upward, so  $\phi$  is less than  $90^\circ$  and  $\vec{B} \cdot d\vec{A}$  is positive. We could have chosen  $d\vec{A}$  to point downward, in which case  $\phi$  would have been greater than  $90^\circ$  and  $\vec{B} \cdot d\vec{A}$  would have been negative. Both choices are equally good, but once we make a choice we must stick with it.

Faraday's law of induction states:

Faraday's law:  
The induced emf  $\mathcal{E}$  in a closed loop ... equals the negative of the time rate of change of magnetic flux through the loop.

$$\mathcal{E} = -\frac{d\Phi_B}{dt} \quad (29.3)$$

To understand the negative sign, we have to introduce a sign convention for the induced emf  $\mathcal{E}$ . But first let's look at a simple example of this law in action.

### EXAMPLE 29.1 Emf and current induced in a loop

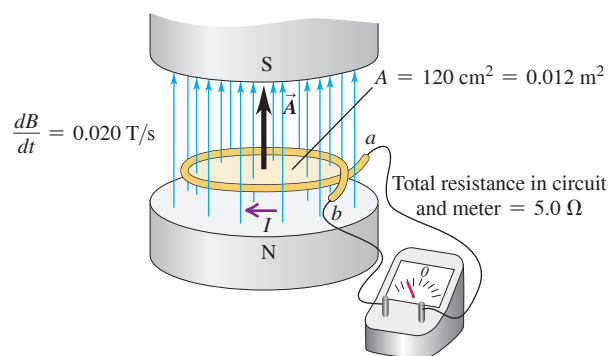
### WITH VARIATION PROBLEMS

The magnetic field between the poles of the electromagnet in Fig. 29.5 is uniform at any time, but its magnitude is increasing at the rate of  $0.020 \text{ T/s}$ . The area of the conducting loop in the field is  $120 \text{ cm}^2$ , and the total circuit resistance, including the meter, is  $5.0 \Omega$ . (a) Find the induced emf and the induced current in the circuit. (b) If the loop is replaced by one made of an insulator, what effect does this have on the induced emf and induced current?

**IDENTIFY and SET UP** The magnetic flux  $\Phi_B$  through the loop changes as the magnetic field changes. Hence there will be an induced emf  $\mathcal{E}$  and an induced current  $I$  in the loop. We calculate  $\Phi_B$  from Eq. (29.2), then find  $\mathcal{E}$  by using Faraday's law. Finally, we calculate  $I$  from  $\mathcal{E} = IR$ , where  $R$  is the total resistance of the circuit that includes the loop.

**EXECUTE** (a) The area vector  $\vec{A}$  for the loop is perpendicular to the plane of the loop; we take  $\vec{A}$  to be vertically upward. Then  $\vec{A}$  and  $\vec{B}$  are

Figure 29.5 A stationary conducting loop in an increasing magnetic field.



parallel, and because  $\vec{B}$  is uniform the magnetic flux through the loop is  $\Phi_B = \vec{B} \cdot \vec{A} = BA \cos 0 = BA$ . The area  $A = 0.012 \text{ m}^2$  is constant, so the rate of change of magnetic flux is

$$\begin{aligned} \frac{d\Phi_B}{dt} &= \frac{d(BA)}{dt} = \frac{dB}{dt}A = (0.020 \text{ T/s})(0.012 \text{ m}^2) \\ &= 2.4 \times 10^{-4} \text{ V} = 0.24 \text{ mV} \end{aligned}$$

This, apart from a sign that we haven't discussed yet, is the induced emf  $\mathcal{E}$ . The corresponding induced current is

$$I = \frac{\mathcal{E}}{R} = \frac{2.4 \times 10^{-4} \text{ V}}{5.0 \, \Omega} = 4.8 \times 10^{-5} \text{ A} = 0.048 \text{ mA}$$

(b) By changing to an insulating loop, we've made the resistance of the loop very high. Faraday's law, Eq. (29.3), does not involve the resistance of the circuit in any way, so the induced *emf* does not change. But the *current* will be smaller, as given by the equation  $I = \mathcal{E}/R$ .

If the loop is made of a perfect insulator with infinite resistance, the induced current is zero. This situation is analogous to an isolated battery whose terminals aren't connected to anything: An emf is present, but no current flows.

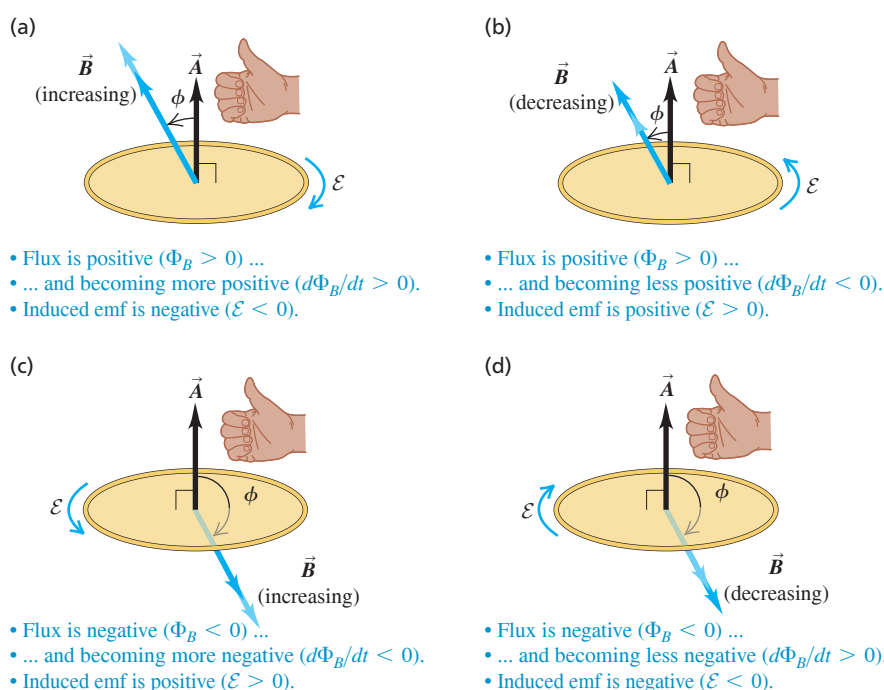
**EVALUATE** We can verify unit consistency in this calculation by noting that the magnetic-force relationship  $\vec{F} = q\vec{v} \times \vec{B}$  implies that the units of  $\vec{B}$  are the units of force divided by the units of (charge times velocity):  $1 \text{ T} = (1 \text{ N})/(1 \text{ C} \cdot \text{m/s})$ . The units of magnetic flux are then  $(1 \text{ T})(1 \text{ m}^2) = 1 \text{ N} \cdot \text{s} \cdot \text{m/C}$ , and the rate of change of magnetic flux is  $1 \text{ N} \cdot \text{m/C} = 1 \text{ J/C} = 1 \text{ V}$ . Thus the unit of  $d\Phi_B/dt$  is the volt, as required by Eq. (29.3). Also recall from Section 27.3 that the unit of magnetic flux is the weber (Wb):  $1 \text{ T} \cdot \text{m}^2 = 1 \text{ Wb}$ , so  $1 \text{ V} = 1 \text{ Wb/s}$ .

**KEYCONCEPT** When there is a change in the magnetic flux through a curve in space that forms a closed loop, an emf is induced in that loop. If the loop is made of a conductor, an induced current flows in response to the emf.

## Direction of Induced emf

We can find the direction of an induced emf or current by using Eq. (29.3) together with some simple sign rules. Here's the procedure:

1. Define a positive direction for the vector area  $\vec{A}$ .
2. From the directions of  $\vec{A}$  and the magnetic field  $\vec{B}$ , determine the sign of the magnetic flux  $\Phi_B$  and its rate of change  $d\Phi_B/dt$ . **Figure 29.6** shows several examples.
3. Determine the sign of the induced emf or current. If the flux is increasing, so  $d\Phi_B/dt$  is positive, then the induced emf or current is negative; if the flux is decreasing,  $d\Phi_B/dt$  is negative and the induced emf or current is positive.
4. Finally, use your right hand to determine the direction of the induced emf or current. Curl the fingers of your right hand around the  $\vec{A}$  vector, with your right thumb in the direction of  $\vec{A}$ . If the induced emf or current in the circuit is *positive*, it is in the same direction as your curled fingers; if the induced emf or current is *negative*, it is in the opposite direction.



**Figure 29.6** The magnetic flux is becoming (a) more positive, (b) less positive, (c) more negative, and (d) less negative. Therefore  $\Phi_B$  is increasing in (a) and (d) and decreasing in (b) and (c). In (a) and (d) the emfs are negative (they are opposite to the direction of the curled fingers of your right hand when your right thumb points along  $\vec{A}$ ). In (b) and (c) the emfs are positive (in the same direction as the curled fingers).

**CAUTION** Induced emfs are caused by changes in flux It can be tempting to think that *flux* is the cause of induced emf and that an induced emf will appear in a circuit whenever there is a magnetic field in the region bordered by the circuit. But Eq. (29.3) shows that only a *change* in flux through a circuit, not flux itself, can induce an emf in a circuit. If the flux through a circuit has a constant value, whether positive, negative, or zero, there is no induced emf. Note also that an induced emf will appear in a circuit if the flux is changed for any reason, including rotating the circuit relative to the magnetic field or changing the circuit's shape. **|**

In Example 29.1, in which  $\vec{A}$  is upward, a positive  $\mathcal{E}$  would be directed counterclockwise around the loop, as seen from above. Both  $\vec{A}$  and  $\vec{B}$  are upward in this example, so  $\Phi_B$  is positive; the magnitude  $B$  is increasing, so  $d\Phi_B/dt$  is positive. Hence by Eq. (29.3),  $\mathcal{E}$  in Example 29.1 is *negative*—that is, *clockwise* around the loop, as seen from above.

If the loop in Fig. 29.5 is a conductor, the clockwise induced emf causes a clockwise induced current. This induced current produces an additional magnetic field through the loop, and the right-hand rule described in Section 28.5 shows that this field is *opposite* in direction to the increasing field produced by the electromagnet. This is an example of a general rule called *Lenz's law*, which says that any induction effect tends to oppose the change that caused it; in this case the change is the increase in the flux of the electromagnet's field through the loop. (We'll study Lenz's law in detail in the next section.)

You should check out the signs of the induced emfs and currents for the list of experiments in Section 29.1. For example, when the loop in Fig. 29.2 is in a constant field and we tilt it or squeeze it to *decrease* the flux through it, the induced emf and current are counterclockwise, as seen from above.

If a coil has  $N$  identical turns and if the flux varies at the same rate through each turn, the *total* rate of change through all turns is  $N$  times that for a single turn. If  $\Phi_B$  is the flux through each turn, the total emf in a coil with  $N$  turns is

$$\mathcal{E} = -N \frac{d\Phi_B}{dt} \quad (29.4)$$

As we discussed in this chapter's introduction, induced emfs play an essential role in the generation of electrical power for commercial use. Several of the following examples explore different methods of producing emfs by changing the flux through a circuit.

### PROBLEM-SOLVING STRATEGY 29.1 Faraday's Law

**IDENTIFY** the relevant concepts: Faraday's law applies when a magnetic flux is changing. To use the law, identify an area through which there is a flux of magnetic field. This will usually be the area enclosed by a loop made of a conducting material (though not always—see part (b) of Example 29.1). Identify the target variables.

**SET UP** the problem using the following steps:

1. Faraday's law relates the induced emf to the rate of change of magnetic flux. To calculate this rate of change, you first have to understand what is making the flux change. Is the conductor moving or changing orientation? Is the magnetic field changing?
2. The area vector  $\vec{A}$  (or  $d\vec{A}$ ) must be perpendicular to the plane of the area. You always have two choices of its direction; for example, if the area is in a horizontal plane,  $\vec{A}$  could point up or down. Choose a direction and use it throughout the problem.

**EXECUTE** the solution as follows:

1. Calculate the magnetic flux from Eq. (29.2) if  $\vec{B}$  is uniform over the area of the loop or Eq. (29.1) if it isn't uniform. Remember the direction you chose for the area vector.
2. Calculate the induced emf from Eq. (29.3) or (if your conductor has  $N$  turns in a coil) Eq. (29.4). Apply the sign rule (described just after Example 29.1) to determine the positive direction of emf.
3. If the circuit resistance is known, you can calculate the magnitude of the induced current  $I$  by using  $\mathcal{E} = IR$ .

**EVALUATE** your answer: Check your results for the proper units, and double-check that you have properly implemented the sign rules for magnetic flux and induced emf.

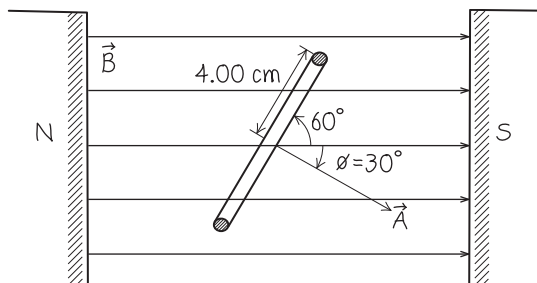
### EXAMPLE 29.2 Magnitude and direction of an induced emf

WITH  **ARIATION PROBLEMS**

A 500-loop circular wire coil with radius 4.00 cm is placed between the poles of a large electromagnet. The magnetic field is uniform and makes an angle of  $60^\circ$  with the plane of the coil; it decreases at 0.200 T/s. What are the magnitude and direction of the induced emf?

**IDENTIFY and SET UP** Our target variable is the emf induced by a varying magnetic flux through the coil. The flux varies because the magnetic field decreases in amplitude. We choose the area vector  $\vec{A}$  to be in the direction shown in Fig. 29.7. With this choice, the geometry is similar to that of Fig. 29.6b. That figure will help us determine the direction of the induced emf.

Figure 29.7 Our sketch for this problem.



**EXECUTE** The magnetic field is uniform over the loop, so we can calculate the flux from Eq. (29.2):  $\Phi_B = BA \cos \phi$ , where  $\phi = 30^\circ$ . In this expression, the only quantity that changes with time is the magnitude  $B$  of the field, so  $d\Phi_B/dt = (dB/dt)A \cos \phi$ .

**CAUTION Remember how  $\phi$  is defined** You may have been tempted to say that  $\phi = 60^\circ$  in this problem. If so, remember that  $\phi$  is the angle between  $\vec{A}$  and  $\vec{B}$ , *not* the angle between  $\vec{B}$  and the plane of the loop. 

From Eq. (29.4), the induced emf in the coil of  $N = 500$  turns is

$$\begin{aligned}\mathcal{E} &= -N \frac{d\Phi_B}{dt} = -N \frac{dB}{dt} A \cos \phi \\ &= -500(-0.200 \text{ T/s})\pi(0.0400 \text{ m})^2(\cos 30^\circ) = 0.435 \text{ V}\end{aligned}$$

The positive answer means that when you point your right thumb in the direction of area vector  $\vec{A}$  ( $30^\circ$  below field  $\vec{B}$  in Fig. 29.7), the positive direction for  $\mathcal{E}$  is in the direction of the curled fingers of your right hand. If you viewed the coil from the left in Fig. 29.7 and looked in the direction of  $\vec{A}$ , the emf would be clockwise.

**EVALUATE** If the ends of the wire are connected, the direction of current in the coil is in the same direction as the emf—that is, clockwise as seen from the left side of the coil. A clockwise current increases the magnetic flux through the coil, and therefore tends to oppose the decrease in total flux. This is an example of Lenz's law, which we'll discuss in Section 29.3.

**KEYCONCEPT** Faraday's law states that the induced emf in a coil is proportional to the rate of change of the magnetic flux through the coil and to the number of turns in the coil. The sign of the induced emf tells you its direction.

### EXAMPLE 29.3 Generator I: A simple alternator

### WITH VARIATION PROBLEMS

**Figure 29.8a** shows a simple *alternator*, a device that generates an emf. A rectangular loop is rotated with constant angular speed  $\omega$  about the axis shown. The magnetic field  $\vec{B}$  is uniform and constant. At time  $t = 0$ ,  $\phi = 0$ . Determine the induced emf.

**IDENTIFY and SET UP** The magnetic field  $\vec{B}$  and the loop area  $A$  are constant, but the flux through the loop varies because the loop rotates and so the angle  $\phi$  between  $\vec{B}$  and the area vector  $\vec{A}$  changes (Fig. 29.8a). Because the angular speed is constant and  $\phi = 0$  at  $t = 0$ , the angle as a function of time is  $\phi = \omega t$ .

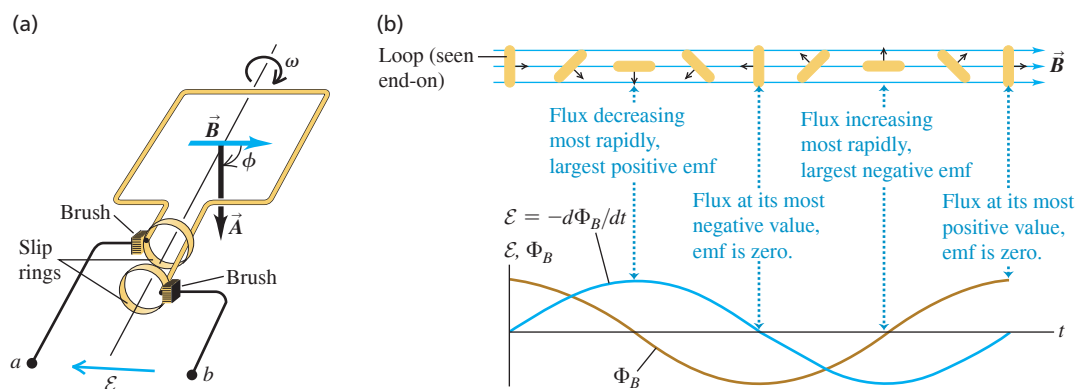
**EXECUTE** The magnetic field is uniform over the loop, so the magnetic flux is  $\Phi_B = BA \cos \phi = BA \cos \omega t$ . Hence, by Faraday's law [Eq. (29.3)] the induced emf is

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}(BA \cos \omega t) = \omega BA \sin \omega t$$

**EVALUATE** The induced emf  $\mathcal{E}$  varies sinusoidally with time (see Fig. 29.8b). When the plane of the loop is perpendicular to  $\vec{B}$  ( $\phi = 0$  or  $180^\circ$ ),  $\Phi_B$  reaches its maximum and minimum values. At these times, its instantaneous rate of change is zero and  $\mathcal{E}$  is zero. Conversely,  $\mathcal{E}$  reaches its maximum and minimum values when the plane of the loop is parallel to  $\vec{B}$  ( $\phi = 90^\circ$  or  $270^\circ$ ) and  $\Phi_B$  is changing most rapidly. We note that the induced emf does not depend on the *shape* of the loop, but only on its area.

We can use the alternator as a source of emf in an external circuit by using two *slip rings* that rotate with the loop, as shown in Fig. 29.8a. The rings slide against stationary contacts called *brushes*, which are connected to the output terminals  $a$  and  $b$ . Since the emf varies sinusoidally, the current that results in the circuit is an *alternating* current that also

**Figure 29.8** (a) Schematic diagram of an alternator. A conducting loop rotates in a magnetic field, producing an emf. Connections from each end of the loop to the external circuit are made by means of that end's slip ring. The system is shown at the time when the angle  $\phi = \omega t = 90^\circ$ . (b) Graph of the flux through the loop and the resulting emf between terminals  $a$  and  $b$ , along with the corresponding positions of the loop during one complete rotation.



*Continued*



varies sinusoidally in magnitude and direction. The amplitude of the emf can be increased by increasing the rotation speed, the field magnitude, or the loop area or by using  $N$  loops instead of one, as in Eq. (29.4).

Alternators are used in automobiles to generate the currents in the ignition, the lights, and the entertainment system. The arrangement is a little different than in this example; rather than having a rotating loop in a magnetic field, the loop stays fixed and an electromagnet rotates. (The rotation is provided by a mechanical connection between the alternator and the engine.) But the result is the same; the flux through the loop varies sinusoidally, producing a sinusoidally varying emf. Larger alternators of this same type are used in electric power plants (**Fig. 29.9**).

**KEYCONCEPT** In an alternator, a coil rotates relative to a magnetic field. The flux through the coil varies sinusoidally with time, as does the emf induced in the coil. The more rapid the rotation, the greater the magnitudes of the induced emf and current.

**Figure 29.9** A commercial alternator uses many loops of wire wound around a barrel-like structure called an armature. The armature and wire remain stationary while electromagnets rotate on a shaft (not shown) through the center of the armature. The resulting induced emf is far larger than would be possible with a single loop of wire.



### EXAMPLE 29.4 Generator II: A dc generator and back emf in a motor

The alternator in Example 29.3 produces a sinusoidally varying emf and hence an alternating current. **Figure 29.10a** shows a *direct-current* (dc) *generator* that produces an emf that always has the same sign. The arrangement of split rings, called a *commutator*, reverses the connections to the external circuit at angular positions at which the emf reverses. **Figure 29.10b** shows the resulting emf. Commercial dc generators have a large number of coils and commutator segments, smoothing out the bumps in the emf so that the terminal voltage is not only one-directional but also practically constant. This brush-and-commutator arrangement is the same as that in the direct-current motor discussed in Section 27.8. The motor's *back emf* is just the emf induced by the changing magnetic flux through its rotating coil. Consider a motor with a square, 500-turn coil 10.0 cm on a side. If the magnetic field has magnitude 0.200 T, at what rotation speed is the *average* back emf of the motor equal to 112 V?

**IDENTIFY and SET UP** As far as the rotating loop is concerned, the situation is the same as in Example 29.3 except that we now have  $N$  turns of wire. Without the commutator, the emf would alternate between positive and negative values and have an average value of zero (see Fig. 29.8b). With the commutator, the emf is never negative and its average value is positive (Fig. 29.10b). We'll use our result from Example 29.3 to obtain an expression for this average value and solve it for the rotational speed  $\omega$ .

**EXECUTE** Comparison of Figs. 29.8b and 29.10b shows that the back emf of the motor is just  $N$  times the absolute value of the emf found for an alternator in Example 29.3, as in Eq. (29.4):

$|\mathcal{E}| = N\omega BA |\sin \omega t|$ . To find the *average* back emf, we must replace  $|\sin \omega t|$  by its average value. We find this by integrating  $|\sin \omega t|$  over half a cycle, from  $t = 0$  to  $t = T/2 = \pi/\omega$ , and dividing by the elapsed time  $\pi/\omega$ . During this half cycle, the sine function is positive, so  $|\sin \omega t| = \sin \omega t$ , and we find

$$(|\sin \omega t|)_{\text{av}} = \frac{\int_0^{\pi/\omega} \sin \omega t \, dt}{\pi/\omega} = \frac{2}{\pi}$$

The average back emf is then

$$\mathcal{E}_{\text{av}} = \frac{2N\omega BA}{\pi}$$

Solving for  $\omega$ , we obtain

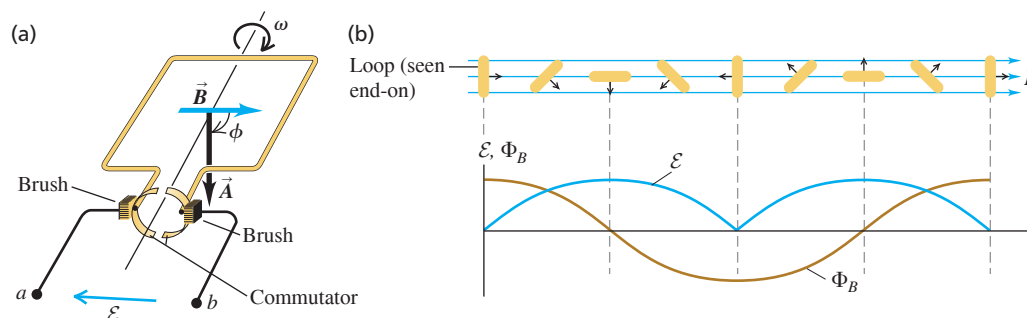
$$\omega = \frac{\pi \mathcal{E}_{\text{av}}}{2NBA} = \frac{\pi(112 \text{ V})}{2(500)(0.200 \text{ T})(0.100 \text{ m})^2} = 176 \text{ rad/s}$$

(Recall from Example 29.1 that  $1 \text{ V} = 1 \text{ Wb/s} = 1 \text{ T} \cdot \text{m}^2/\text{s}$ .)

**EVALUATE** The average back emf is directly proportional to  $\omega$ . Hence the slower the rotation speed, the less the back emf and the greater the possibility of burning out the motor, as we described in Example 27.11 (Section 27.8).

**KEYCONCEPT** The back emf in a motor is a result of Faraday's law. The average value of the back emf is proportional to the rotation speed of the motor.

**Figure 29.10** (a) Schematic diagram of a dc generator, using a split-ring commutator. The ring halves are attached to the loop and rotate with it. (b) Graph of the resulting induced emf between terminals  $a$  and  $b$ . Compare to Fig. 29.8b.



**EXAMPLE 29.5 Generator III: The slidewire generator**

**Figure 29.11** shows a U-shaped conductor in a uniform magnetic field  $\vec{B}$  perpendicular to the plane of the figure and directed *into* the page. We lay a metal rod (the “slidewire”) with length  $L$  across the two arms of the conductor, forming a circuit, and move it to the right with constant velocity  $\vec{v}$ . This induces an emf and a current, which is why this device is called a *slidewire generator*. Find the magnitude and direction of the resulting induced emf.

**IDENTIFY and SET UP** The magnetic flux changes because the area of the loop—bounded on the right by the moving rod—is increasing. Our target variable is the emf  $\mathcal{E}$  induced in this expanding loop. The magnetic field is uniform over the area of the loop, so we can find the flux from  $\Phi_B = BA \cos \phi$ . We choose the area vector  $\vec{A}$  to point straight into the page, in the same direction as  $\vec{B}$ . With this choice a positive emf will be one that is directed clockwise around the loop. (You can check this with the right-hand rule: Using your right hand, point your thumb into the page and curl your fingers as in Fig. 29.6.)

**EXECUTE** Since  $\vec{B}$  and  $\vec{A}$  point in the same direction, the angle  $\phi = 0$  and  $\Phi_B = BA$ . The magnetic-field magnitude  $B$  is constant, so the induced emf is

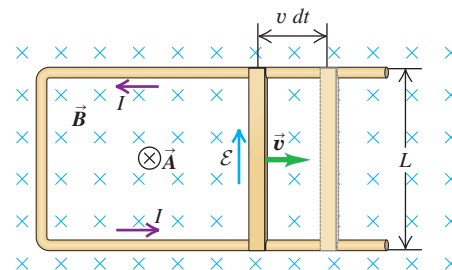
$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -B \frac{dA}{dt}$$

To calculate  $dA/dt$ , note that in a time  $dt$  the sliding rod moves a distance  $v dt$  (Fig. 29.11) and the loop area increases by an amount  $dA = Lv dt$ . Hence the induced emf is

$$\mathcal{E} = -B \frac{Lv dt}{dt} = -BLv$$

The minus sign tells us that the emf is directed *counterclockwise* around the loop. The induced current is also counterclockwise, as shown in the figure.

**Figure 29.11** A slidewire generator. The magnetic field  $\vec{B}$  and the vector area  $\vec{A}$  are both directed into the figure. The increase in magnetic flux (caused by an increase in area) induces the emf and current.



**EVALUATE** The emf of a slidewire generator is constant if  $\vec{v}$  is constant. Hence the slidewire generator is a *direct-current* generator. It's not a very practical device because the rod eventually moves beyond the U-shaped conductor and loses contact, after which the current stops.

**KEYCONCEPT** An induced emf can be produced if the area of a circuit in a magnetic field changes. The magnitude of the emf depends on the speed at which the area of the circuit changes.

**EXAMPLE 29.6 Work and power in the slidewire generator**

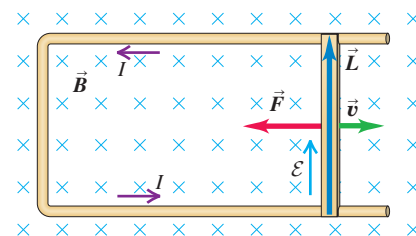
In the slidewire generator of Example 29.5, energy is dissipated in the circuit owing to its resistance. Let the resistance of the circuit (made up of the moving slidewire and the U-shaped conductor that connects the ends of the slidewire) at a given point in the slidewire's motion be  $R$ . Find the rate at which energy is dissipated in the circuit and the rate at which work must be done to move the rod through the magnetic field.

**IDENTIFY and SET UP** Our target variables are the *rates* at which energy is dissipated and at which work is done. Energy is dissipated in the circuit at the rate  $P_{\text{dissipated}} = I^2 R$ . The current  $I$  in the circuit equals  $|\mathcal{E}|/R$ ; we found an expression for the induced emf  $\mathcal{E}$  in this circuit in Example 29.5. There is a magnetic force  $\vec{F} = I\vec{L} \times \vec{B}$  on the rod, where  $\vec{L}$  points along the rod in the direction of the current. **Figure 29.12** shows that this force is opposite to the rod velocity  $\vec{v}$ ; to maintain the motion, whoever is pushing the rod must apply a force of equal magnitude in the direction of  $\vec{v}$ . This force does work at the rate  $P_{\text{applied}} = Fv$ .

**EXECUTE** First we'll calculate  $P_{\text{dissipated}}$ . From Example 29.5,  $\mathcal{E} = -BLv$ , so the current in the rod is  $I = |\mathcal{E}|/R = BLv/R$ . Hence

$$P_{\text{dissipated}} = I^2 R = \left(\frac{BLv}{R}\right)^2 R = \frac{B^2 L^2 v^2}{R}$$

**Figure 29.12** The magnetic force  $\vec{F} = I\vec{L} \times \vec{B}$  that acts on the rod due to the induced current is to the left, opposite to  $\vec{v}$ .



To calculate  $P_{\text{applied}}$ , we first calculate the magnitude of  $\vec{F} = I\vec{L} \times \vec{B}$ . Since  $\vec{L}$  and  $\vec{B}$  are perpendicular, this magnitude is

$$F = ILB = \frac{BLv}{R} LB = \frac{B^2 L^2 v}{R}$$

The applied force has the same magnitude and does work at the rate

$$P_{\text{applied}} = Fv = \frac{B^2 L^2 v^2}{R}$$

*Continued*

**CAUTION** You can't violate energy conservation You might think that reversing the direction of  $\vec{B}$  or of  $\vec{v}$  would allow the magnetic force  $\vec{F} = I\vec{L} \times \vec{B}$  to be in the *same* direction as  $\vec{v}$ . This would be a neat trick. Once the rod was moving, the changing magnetic flux would induce an emf and a current, and the magnetic force on the rod would make it move even faster, increasing the emf and current until the rod was moving at tremendous speed and producing electrical power at a prodigious rate. If this seems too good to be true and a violation of energy conservation, that's because it is. Reversing  $\vec{B}$  also reverses the sign of the induced emf and current and hence the direction of  $\vec{L}$ , so the magnetic force still opposes the motion of the rod; a similar result holds true if we reverse  $\vec{v}$ .

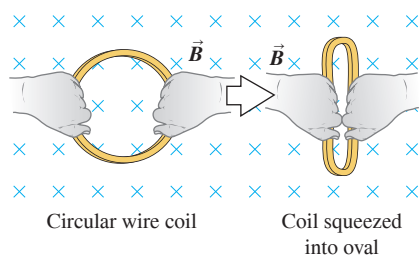
**EVALUATE** The rate at which work is done is exactly *equal* to the rate at which energy is dissipated in the resistance.

**KEYCONCEPT** You must do work to move any part of a circuit in a magnetic field. The work that you do is used to make the induced current flow in the presence of the circuit's resistance.

## Generators as Energy Converters

Example 29.6 shows that the slidewire generator doesn't produce electrical energy out of nowhere; the energy is supplied by whatever object exerts the force that keeps the rod moving. All that the generator does is *convert* that energy into a different form. The equality between the rate at which *mechanical* energy is supplied to a generator and the rate at which *electrical* energy is generated holds for all types of generators, including the alternator described in Example 29.3. (We are ignoring the effects of friction in the bearings of an alternator or between the rod and the U-shaped conductor of a slidewire generator. The energy lost to friction is not available for conversion to electrical energy, so in real generators the friction is kept to a minimum.)

In Chapter 27 we stated that the magnetic force on moving charges can never do work. You might think, however, that the magnetic force  $\vec{F} = I\vec{L} \times \vec{B}$  in Example 29.6 is doing (negative) work on the current-carrying rod as it moves, contradicting our earlier statement. In fact, the work done by the magnetic force is zero. The moving charges that make up the current in the rod in Fig. 29.12 have a vertical component of velocity, causing a horizontal component of force on these charges. As a result, there is a horizontal displacement of charge within the rod, the left side acquiring a net positive charge and the right side a net negative charge. The result is a horizontal component of electric field, perpendicular to the length of the rod (analogous to the Hall effect, described in Section 27.9). It is this field, in the direction of motion of the rod, that does work on the mobile charges in the rod and hence indirectly on the atoms making up the rod.



**TEST YOUR UNDERSTANDING OF SECTION 29.2** The accompanying figure shows a wire coil being squeezed in a uniform magnetic field. (a) While the coil is being squeezed, is the induced emf in the coil (i) clockwise, (ii) counterclockwise, or (iii) zero? (b) Once the coil has reached its final squeezed shape, is the induced emf in the coil (i) clockwise, (ii) counterclockwise, or (iii) zero?

### ANSWER

(a) (i) Initially there is magnetic flux into the plane of the page, which we call positive. While the loop is being squeezed, the flux is becoming less positive ( $d\Phi_B/dt < 0$ ) and so the induced emf is positive as in Fig. 29.6b ( $\mathcal{E} = -d\Phi_B/dt > 0$ ). If you point the thumb of your right hand into the page, your fingers curl clockwise, so this is the direction of positive induced emf. In (b) 'since the coil's shape is no longer changing, the magnetic flux is not changing and there is no induced emf.'

## 29.3 LENZ'S LAW

**Lenz's law** is a convenient alternative method for determining the direction of an induced current or emf. Lenz's law, named for the Russian physicist H. F. E. Lenz (1804–1865), is not an independent principle; it can be derived from Faraday's law. It always gives the same results as the sign rules we introduced in connection with Faraday's law, but it is often easier to use. Lenz's law also helps us gain intuitive understanding of various induction effects and of the role of energy conservation.

**LENZ'S LAW** The direction of any magnetic induction effect is such as to oppose the cause of the effect.

The “cause” may be changing flux through a stationary circuit due to a varying magnetic field, changing flux due to motion of the conductors that make up the circuit, or any combination. If the flux in a stationary circuit changes, as in Examples 29.1 and 29.2, the induced current sets up a magnetic field of its own. Within the area bounded by the circuit, this field is *opposite* to the original field if the original field is *increasing* but is in the *same* direction as the original field if the latter is *decreasing*. That is, the induced current opposes the *change in flux* through the circuit (*not* the flux itself).

If the flux change is due to motion of the conductors, as in Examples 29.3 through 29.6, the direction of the induced current in the moving conductor is such that the direction of the magnetic-field force on the conductor is opposite in direction to its motion. Thus the motion of the conductor, which caused the induced current, is opposed. We saw this explicitly for the slidewire generator in Example 29.6. In all these cases the induced current tries to preserve the *status quo* by opposing motion or a change of flux.

Lenz's law is also directly related to energy conservation. If the induced current in Example 29.6 were in the direction opposite to that given by Lenz's law, the magnetic force on the rod would accelerate it to ever-increasing speed with no external energy source, even though electrical energy is being dissipated in the circuit. This would be a clear violation of energy conservation and doesn't happen in nature.

### CONCEPTUAL EXAMPLE 29.7 Lenz's law and the slidewire generator

WITH  **ARIATION PROBLEMS**

In Fig. 29.11, the induced current in the loop causes an additional magnetic field in the area bounded by the loop. The direction of the induced current is counterclockwise, so from the discussion of Section 28.5, this additional magnetic field is directed *out of* the plane of the figure. That direction is opposite that of the original magnetic field, so it tends to cancel the effect of that field. This is just what Lenz's law predicts.

**KEYCONCEPT** A change in the magnetic flux through a circuit induces a current that produces an additional magnetic field of its own. This induced field always opposes the change in the magnetic flux (Lenz's law).

### CONCEPTUAL EXAMPLE 29.8 Lenz's law and the direction of induced current

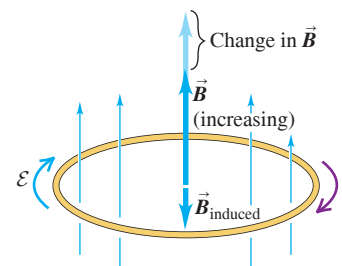
WITH  **ARIATION PROBLEMS**

In Fig. 29.13 there is a uniform magnetic field  $\vec{B}$  through the coil. The magnitude of the field is increasing, so there is an induced emf. Use Lenz's law to determine the direction of the resulting induced current.

**SOLUTION** This situation is the same as in Example 29.1 (Section 29.2). By Lenz's law the induced current must produce a magnetic field  $\vec{B}_{\text{induced}}$  inside the coil that is downward, opposing the change in flux. From the right-hand rule we described in Section 28.5 for the direction of the magnetic field produced by a circular loop,  $\vec{B}_{\text{induced}}$  will be in the desired direction if the induced current flows as shown in Fig. 29.13.

**Figure 29.14** (next page) shows several applications of Lenz's law to the similar situation of a magnet moving near a conducting loop. In each case, the induced current produces a magnetic field whose direction opposes the change in flux through the loop due to the magnet's motion.

**Figure 29.13** The induced current due to the change in  $\vec{B}$  is clockwise, as seen from above the loop. The added field  $\vec{B}_{\text{induced}}$  that it causes is downward, opposing the change in the upward field  $\vec{B}$ .

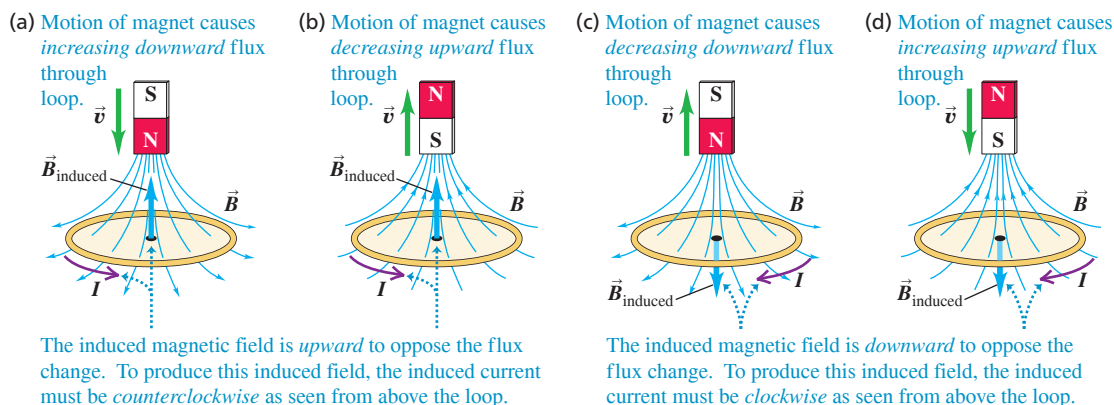


**KEYCONCEPT** Whenever there is a changing magnetic flux through a circuit, you can use Lenz's law to decide what the induced current direction will be: The current must produce an induced magnetic field that opposes the change in the flux.

*Continued*



Figure 29.14 Directions of induced currents as a bar magnet moves along the axis of a conducting loop. If the bar magnet is stationary, there is no induced current.



### Lenz's Law and the Response to Flux Changes

Since an induced current always opposes any change in magnetic flux through a circuit, how is it possible for the flux to change at all? The answer is that Lenz's law gives only the *direction* of an induced current; the *magnitude* of the current depends on the resistance of the circuit. The greater the circuit resistance, the less the induced current that appears to oppose any change in flux and the easier it is for a flux change to take effect. If the loop in Fig. 29.14 were made out of wood (an insulator), there would be almost no induced current in response to changes in the flux through the loop.

Conversely, the less the circuit resistance, the greater the induced current and the more difficult it is to change the flux through the circuit. If the loop in Fig. 29.14 is a good conductor, an induced current flows as long as the magnet moves relative to the loop. Once the magnet and loop are no longer in relative motion, the induced current very quickly decreases to zero because of the nonzero resistance in the loop.

The extreme case occurs when the resistance of the circuit is *zero*. Then the induced current in Fig. 29.14 will continue to flow even after the induced emf has disappeared—that is, even after the magnet has stopped moving relative to the loop. Thanks to this *persistent current*, it turns out that the flux through the loop is exactly the same as it was before the magnet started to move, so the flux through a loop of zero resistance *never* changes. Exotic materials called *superconductors* do indeed have zero resistance; we discuss these further in Section 29.8.

**TEST YOUR UNDERSTANDING OF SECTION 29.3** (a) Suppose the magnet in Fig. 29.14a were stationary and the loop of wire moved upward. Would the induced current in the loop be (i) in the same direction as shown in Fig. 29.14a, (ii) in the direction opposite to that shown in Fig. 29.14a, or (iii) zero? (b) Suppose the magnet and loop of wire in Fig. 29.14a both moved downward at the same velocity. Would the induced current in the loop be (i) in the same direction as shown in Fig. 29.14a, (ii) in the direction opposite to that shown in Fig. 29.14a, or (iii) zero?

### ANSWER

(a) (i) In (a), as in the original situation, the magnet and loop are approaching each other and the downward flux through the loop is increasing. Hence the induced emf and induced current are the same. In (b), since the magnet and loop are moving together, the flux through the loop is not changing and no emf is induced.

## 29.4 MOTIONAL EMF

We've seen several situations in which a conductor moves in a magnetic field, as in the generators discussed in Examples 29.3 through 29.6. We can gain additional insight into the origin of the induced emf in these situations by considering the magnetic forces on mobile charges in the conductor. **Figure 29.15a** shows the same moving rod that we discussed in Example 29.5, separated for the moment from the U-shaped conductor. The magnetic field  $\vec{B}$  is uniform and directed into the page, and we move the rod to the right at a constant velocity  $\vec{v}$ . A charged particle  $q$  in the rod then experiences a magnetic force  $\vec{F} = q\vec{v} \times \vec{B}$  with magnitude  $F = |q|vB$ . We'll assume in the following discussion that  $q$  is positive; in that case the direction of this force is upward along the rod, from  $b$  toward  $a$ .

This magnetic force causes the free charges in the rod to move, creating an excess of positive charge at the upper end  $a$  and negative charge at the lower end  $b$ . This in turn creates an electric field  $\vec{E}$  within the rod, in the direction from  $a$  toward  $b$  (opposite to the magnetic force). Charge continues to accumulate at the ends of the rod until  $\vec{E}$  becomes large enough for the downward electric force (with magnitude  $qE$ ) to cancel exactly the upward magnetic force (with magnitude  $qvB$ ). Then  $qE = qvB$  and the charges are in equilibrium.

The magnitude of the potential difference  $V_{ab} = V_a - V_b$  is equal to the electric-field magnitude  $E$  multiplied by the length  $L$  of the rod. From the above discussion,  $E = vB$ , so

$$V_{ab} = EL = vBL \quad (29.5)$$

with point  $a$  at higher potential than point  $b$ .

Now suppose the moving rod slides along a stationary U-shaped conductor, forming a complete circuit (Fig. 29.15b). No magnetic force acts on the charges in the stationary U-shaped conductor, but the charge that was near points  $a$  and  $b$  redistributes itself along the stationary conductor, creating an electric field within it. This field establishes a current in the direction shown. The moving rod has become a source of emf; within it, charge moves from lower to higher potential, and in the remainder of the circuit, charge moves from higher to lower potential. We call this emf a **motional emf**, denoted by  $\mathcal{E}$ . From the above discussion, the magnitude of this emf is

$$\text{Motional emf, } \mathcal{E} = vBL \quad (29.6)$$

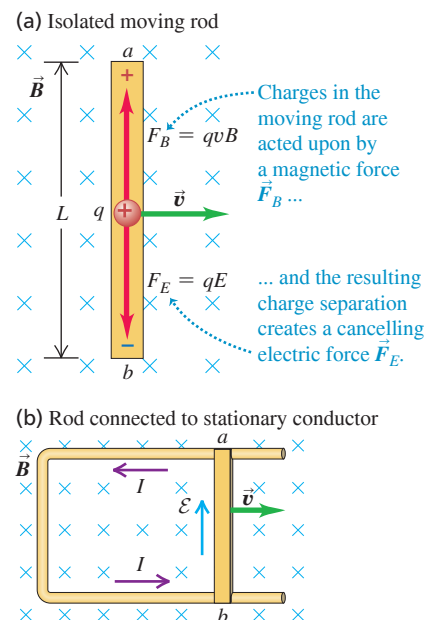
conductor length and velocity perpendicular to uniform  $\vec{B}$  Conductor speed Conductor length Magnitude of uniform magnetic field

This corresponds to a force per unit charge of magnitude  $vB$  acting for a distance  $L$  along the moving rod. If the total circuit resistance of the U-shaped conductor and the sliding rod is  $R$ , the induced current  $I$  in the circuit is given by  $vBL = IR$ . This is the same result we obtained in Section 29.2 by using Faraday's law, and indeed motional emf is a particular case of Faraday's law. Verify that if we express  $v$  in meters per second,  $B$  in teslas, and  $L$  in meters, then  $\mathcal{E}$  is in volts. (Recall that  $1 \text{ V} = 1 \text{ J/C} = 1 \text{ T} \cdot \text{m}^2/\text{s}$ .)

The emf associated with the moving rod in Fig. 29.15b is analogous to that of a battery with its positive terminal at  $a$  and its negative terminal at  $b$ , although the origins of the two emfs are quite different. In each case a nonelectrostatic force acts on the charges in the device, in the direction from  $b$  to  $a$ , and the emf is the work per unit charge done by this force when a charge moves from  $b$  to  $a$  in the device. When the device is connected to an external circuit, the direction of current is from  $b$  to  $a$  in the device and from  $a$  to  $b$  in the external circuit. Note that a motional emf is also present in the isolated moving rod in Fig. 29.15a, in the same way that a battery has an emf even when it's not part of a circuit.

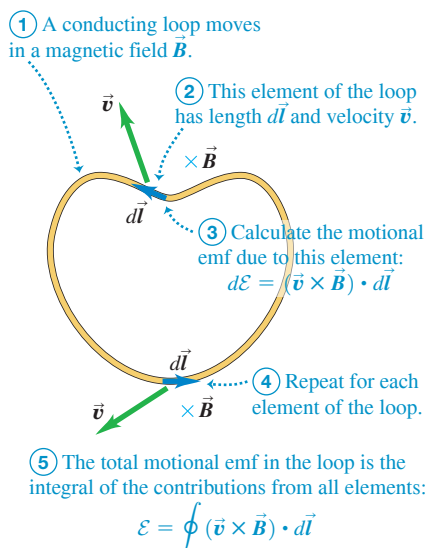
You can determine the direction of the induced emf in Fig. 29.15 by using Lenz's law, even if (as in Fig. 29.15a) the conductor does not form a complete circuit. In this case we can mentally complete the circuit between the ends of the conductor and use Lenz's law to determine the direction of the current. From this we can deduce the polarity of the ends of the open-circuit conductor. The direction from the  $-$  end to the  $+$  end within the conductor is the direction the current would have if the circuit were complete.

**Figure 29.15** A conducting rod moving in a uniform magnetic field. (a) The rod, the velocity, and the field are mutually perpendicular. (b) Direction of induced current in the circuit.



The motional emf  $\mathcal{E}$  in the moving rod creates an electric field in the stationary conductor.

Figure 29.16 Calculating the motional emf for a moving current loop. The velocity  $\vec{v}$  can be different for different elements if the loop is rotating or changing shape. The magnetic field  $\vec{B}$  can also have different values at different points around the loop.



## Motional emf: General Form

We can generalize the concept of motional emf for a conductor with *any* shape, moving in any magnetic field, uniform or not, if we assume that the magnetic field at each point does not vary with time (Fig. 29.16). For an element  $d\vec{l}$  of the conductor, the contribution  $d\mathcal{E}$  to the emf is the magnitude  $dl$  multiplied by the component of  $\vec{v} \times \vec{B}$  (the magnetic force per unit charge) parallel to  $d\vec{l}$ ; that is,

$$d\mathcal{E} = (\vec{v} \times \vec{B}) \cdot d\vec{l}$$

For any closed conducting loop, the total emf is

$$\text{Motional emf, general case} \quad \mathcal{E} = \oint (\vec{v} \times \vec{B}) \cdot d\vec{l} \quad (29.7)$$

Line integral over all elements of closed conducting loop

Length vector of conductor element

Velocity of conductor element

Magnetic field at position of element

This expression looks very different from our original statement of Faraday's law,  $\mathcal{E} = -d\Phi_B/dt$  [Eq. (29.3)]. In fact, though, the two statements are equivalent. It can be shown that the rate of change of magnetic flux through a moving conducting loop is always given by the negative of the expression in Eq. (29.7). Thus this equation gives us an alternative formulation of Faraday's law that is often convenient in problems with *moving* conductors. But when we have *stationary* conductors in changing magnetic fields, Eq. (29.7) *cannot* be used; in this case,  $\mathcal{E} = -d\Phi_B/dt$  is the only correct way to express Faraday's law.

### EXAMPLE 29.9 Motional emf in the slidewire generator

### WITH VARIATION PROBLEMS

Suppose the moving rod in Fig. 29.15b is 0.10 m long, the velocity  $v$  is 2.5 m/s, the total resistance of the loop is  $0.030 \, \Omega$ , and  $B$  is 0.60 T. Find the motional emf, the induced current, and the force acting on the rod.

**IDENTIFY and SET UP** We'll find the motional emf  $\mathcal{E}$  from Eq. (29.6) and the current from the values of  $\mathcal{E}$  and the resistance  $R$ . The force on the rod is a *magnetic* force, exerted by  $\vec{B}$  on the current in the rod; we'll find this force by using  $\vec{F} = I\vec{L} \times \vec{B}$ .

**EXECUTE** From Eq. (29.6) the motional emf is

$$\mathcal{E} = vBL = (2.5 \text{ m/s})(0.60 \text{ T})(0.10 \text{ m}) = 0.15 \text{ V}$$

The induced current in the loop is

$$I = \frac{\mathcal{E}}{R} = \frac{0.15 \text{ V}}{0.030 \, \Omega} = 5.0 \text{ A}$$

In the expression for the magnetic force,  $\vec{F} = I\vec{L} \times \vec{B}$ , the vector  $\vec{L}$  points in the same direction as the induced current in the rod

(from  $b$  to  $a$  in Fig. 29.15). The right-hand rule for vector products shows that this force is directed *opposite* to the rod's motion. Since  $\vec{L}$  and  $\vec{B}$  are perpendicular, the force has magnitude

$$F = ILB = (5.0 \text{ A})(0.10 \text{ m})(0.60 \text{ T}) = 0.30 \text{ N}$$

**EVALUATE** We can check our answer for the direction of  $\vec{F}$  by using Lenz's law. If we take the area vector  $\vec{A}$  to point into the plane of the loop, the magnetic flux is positive and increasing as the rod moves to the right and increases the area of the loop. Lenz's law tells us that a force appears to oppose this increase in flux. Hence the force on the rod is to the left, opposite its motion.

**KEYCONCEPT** When a straight conductor moves through a magnetic field, the magnetic forces on the mobile charges give rise to a motional emf. This emf is proportional to the field strength and to the speed and length of the conductor.

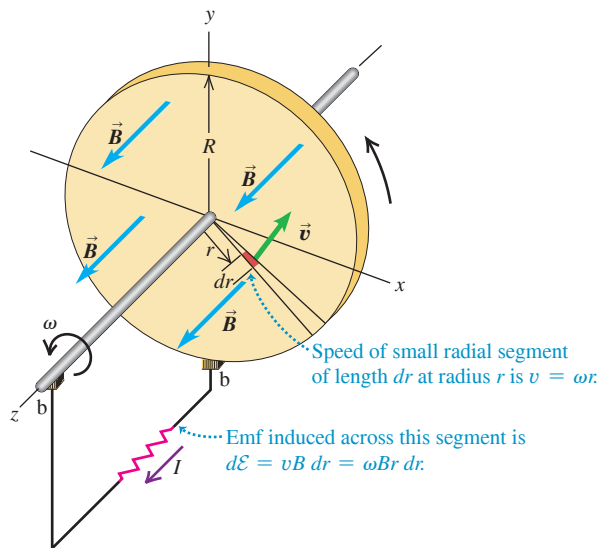
### EXAMPLE 29.10 The Faraday disk dynamo

Figure 29.17 shows a conducting disk with radius  $R$  that lies in the  $xy$ -plane and rotates with constant angular velocity  $\omega$  about the  $z$ -axis. The disk is in a uniform, constant  $\vec{B}$  field in the  $z$ -direction. Find the induced emf between the center and the rim of the disk.

**IDENTIFY and SET UP** A motional emf arises because the conducting disk moves relative to  $\vec{B}$ . The complication is that different parts of the disk move at different speeds  $v$ , depending on their distance from the rotation axis. We'll address this by considering small segments of the disk

and integrating their contributions to determine our target variable, the emf between the center and the rim. Consider the small segment of the disk shown in red in Fig. 29.17 and labeled by its velocity vector  $\vec{v}$ . The magnetic force per unit charge on this segment is  $\vec{v} \times \vec{B}$ , which points radially outward from the center of the disk. Hence the induced emf tends to make a current flow radially outward, which tells us that the moving conducting path to think about here is a straight line from the center to the rim. We can find the emf from each small disk segment along this line by using  $d\mathcal{E} = (\vec{v} \times \vec{B}) \cdot d\vec{l}$  and then integrate to find the total emf.

Figure 29.17 A conducting disk with radius  $R$  rotating at an angular speed  $\omega$  in a magnetic field  $\vec{B}$ . The emf is induced along radial lines of the disk and is applied to an external circuit through the two sliding contacts labeled b.



**EXECUTE** The length vector  $d\vec{l}$  (of length  $dr$ ) associated with the segment points radially outward, in the same direction as  $\vec{v} \times \vec{B}$ . The vectors  $\vec{v}$  and  $\vec{B}$  are perpendicular, and the magnitude of  $\vec{v}$  is  $v = \omega r$ . The emf from the segment is then  $d\mathcal{E} = \omega r B dr$ . The total emf is the integral of  $d\mathcal{E}$  from the center ( $r = 0$ ) to the rim ( $r = R$ ):

$$\mathcal{E} = \int_0^R \omega B r dr = \frac{1}{2} \omega B R^2$$

**EVALUATE** We can use this device as a source of emf in a circuit by completing the circuit through two stationary brushes (labeled b in the figure) that contact the disk and its conducting shaft as shown. Such a disk is called a *Faraday disk dynamo* or a *homopolar generator*. Unlike the alternator in Example 29.3, the Faraday disk dynamo is a direct-current generator; it produces an emf that is constant in time. Can you use Lenz's law to show that for the direction of rotation in Fig. 29.17, the current in the external circuit must be in the direction shown?

**KEYCONCEPT** In general, finding the induced emf in a conductor requires doing an integral. This is because the magnetic force on a mobile charge can be different at different locations in the conductor.

**TEST YOUR UNDERSTANDING OF SECTION 29.4** The earth's magnetic field points toward (magnetic) north. For simplicity, assume that the field has no vertical component (as is the case near the earth's equator). (a) If you hold a metal rod in your hand and walk toward the east, how should you orient the rod to get the maximum motional emf between its ends? (i) East-west; (ii) north-south; (iii) up-down; (iv) you get the same motional emf with all of these orientations. (b) How should you hold it to get *zero* emf as you walk toward the east? (i) East-west; (ii) north-south; (iii) up-down; (iv) none of these. (c) In which direction should you travel so that the motional emf across the rod is zero no matter how the rod is oriented? (i) West; (ii) north; (iii) south; (iv) straight up; (v) straight down.

### ANSWER

(a) (iii); (b) (i) or (ii); (c) (iii) or (iv). You'll get the maximum motional emf if you hold the rod vertically, so that its length is perpendicular to both the magnetic field and the direction of motion. With this orientation,  $\vec{L}$  is parallel to  $\vec{v} \times \vec{B}$ . If you hold the rod in any horizontal orientation,  $\vec{L}$  will be perpendicular to  $\vec{v} \times \vec{B}$  and no emf will be induced. If you walk due north or south,  $\vec{v} \times \vec{B}$  and no emf will be induced for any orientation of the rod.

## 29.5 INDUCED ELECTRIC FIELDS

When a conductor moves in a magnetic field, we can understand the induced emf on the basis of magnetic forces on charges in the conductor, as described in Section 29.4. But an induced emf also occurs when there is a changing flux through a stationary conductor. What is it that pushes the charges around the circuit in this type of situation?

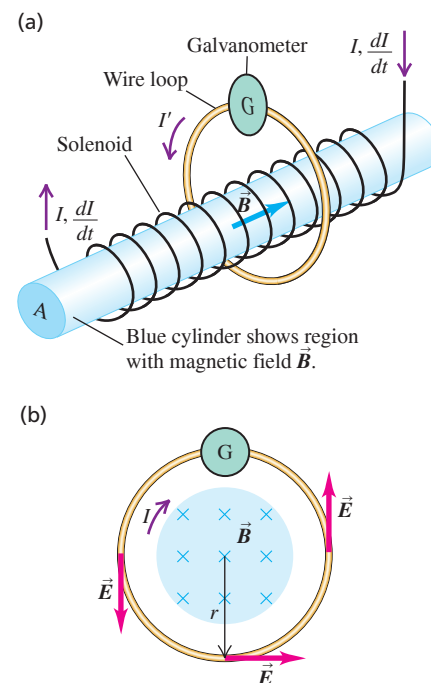
Let's consider the situation shown in **Fig. 29.18**. A long, thin solenoid with cross-sectional area  $A$  and  $n$  turns per unit length is encircled at its center by a circular conducting loop. The galvanometer  $G$  measures the current in the loop. A current  $I$  in the winding of the solenoid sets up a magnetic field  $\vec{B}$  along the solenoid axis, as shown, with magnitude  $B$  as calculated in Section 28.7:  $B = \mu_0 n I$ , where  $n$  is the number of turns per unit length. If we ignore the small field outside the solenoid and take the area vector  $\vec{A}$  to point in the same direction as  $\vec{B}$ , then the magnetic flux  $\Phi_B$  through the loop is

$$\Phi_B = BA = \mu_0 n I A$$

When the solenoid current  $I$  changes with time, the magnetic flux  $\Phi_B$  also changes, and according to Faraday's law the induced emf in the loop is given by

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\mu_0 n A \frac{dI}{dt} \quad (29.8)$$

Figure 29.18 (a) The windings of a long solenoid carry a current  $I$  that is increasing at a rate  $dI/dt$ . The magnetic flux in the solenoid is increasing at a rate  $d\Phi_B/dt$ , and this changing flux passes through a wire loop. An emf  $\mathcal{E} = -d\Phi_B/dt$  is induced in the loop, inducing a current  $I'$  that is measured by the galvanometer  $G$ . (b) Cross-sectional view.





If the total resistance of the loop is  $R$ , the induced current in the loop, which we may call  $I'$ , is  $I' = \mathcal{E}/R$ .

But what *force* makes the charges move around the wire loop? It can't be a magnetic force because the loop isn't even *in* a magnetic field. We are forced to conclude that there has to be an **induced electric field** in the conductor *caused by the changing magnetic flux*. Induced electric fields are *very* different from the electric fields caused by charges, which we discussed in Chapter 23. To see this, note that when a charge  $q$  goes once around the loop, the total work done on it by the electric field must be equal to  $q$  times the emf  $\mathcal{E}$ . That is, the electric field in the loop is *not conservative*, as we used the term in Section 23.1, because the line integral of  $\vec{E}$  around a closed path is not zero. Indeed, this line integral, representing the work done by the induced  $\vec{E}$  field per unit charge, is equal to the induced emf  $\mathcal{E}$ :

$$\oint \vec{E} \cdot d\vec{l} = \mathcal{E} \quad (29.9)$$

From Faraday's law the emf  $\mathcal{E}$  is also the negative of the rate of change of magnetic flux through the loop. Thus for this case we can restate Faraday's law as

|                                                                 |                                                                                                                                                    |                                                                                  |                |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------|
| <p>Faraday's law<br/>for a stationary<br/>integration path:</p> | <p style="color: #0070C0; font-size: small;">Line integral of electric field around path</p> $\oint \vec{E} \cdot d\vec{l} = - \frac{d\Phi_B}{dt}$ | <p>Negative of the time<br/>rate of change of<br/>magnetic flux through path</p> | <p>(29.10)</p> |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------|

Note that Faraday's law is *always* true in the form  $\mathcal{E} = -d\Phi_B/dt$ ; the form given in Eq. (29.10) is valid *only* if the path around which we integrate is *stationary*.

Let's apply Eq. (29.10) to the stationary circular loop in Fig. 29.18b, which we take to have radius  $r$ . Because of cylindrical symmetry,  $\vec{E}$  has the same magnitude at every point on the circle and is tangent to it at each point. (Symmetry would also permit the field to be *radial*, but then Gauss's law would require the presence of a net charge inside the circle, and there is none.) The line integral in Eq. (29.10) becomes simply the magnitude  $E$  times the circumference  $2\pi r$  of the loop,  $\oint \vec{E} \cdot d\vec{l} = 2\pi rE$ , and Eq. (29.10) gives

$$E = \frac{1}{2\pi r} \left| \frac{d\Phi_B}{dt} \right| \quad (29.11)$$

The directions of  $\vec{E}$  at points on the loop are shown in Fig. 29.18b. We know that  $\vec{E}$  has to have the direction shown when  $\vec{B}$  in the solenoid is increasing, because  $\oint \vec{E} \cdot d\vec{l}$  has to be negative when  $d\Phi_B/dt$  is positive. The same approach can be used to find the induced electric field *inside* the solenoid when the solenoid  $\vec{B}$  field is changing; we leave the details to you.

## Nonelectrostatic Electric Fields

We've learned that Faraday's law, Eq. (29.3), is valid for two rather different situations. In one, an emf is induced by magnetic forces on charges when a conductor moves through a magnetic field. In the other, a time-varying magnetic field induces an electric field and hence an emf; the  $\vec{E}$  field is induced even when no conductor is present. This  $\vec{E}$  field differs from an *electrostatic* field in an important way. It is *nonconservative*; the line integral  $\oint \vec{E} \cdot d\vec{l}$  around a closed path is not zero, and when a charge moves around a closed path, the field does a nonzero amount of work on it. It follows that for such a field the concept of *potential* has no meaning. We call such a field a **nonelectrostatic field**. In contrast, an *electrostatic* field is *always* conservative, as we discussed in Section 23.1, and always has an associated potential function. Despite this difference, the fundamental effect of *any* electric field is to exert a force  $\vec{F} = q\vec{E}$  on a charge  $q$ . This relationship is valid whether  $\vec{E}$  is conservative and produced by charges or nonconservative and produced by changing magnetic flux.

So a changing magnetic field acts as a source of electric field of a sort that we *cannot* produce with any static charge distribution. What's more, we'll see in Section 29.7 that a changing *electric* field acts as a source of *magnetic* field. We'll explore this symmetry between the two fields in our study of electromagnetic waves in Chapter 32.

If any doubt remains in your mind about the reality of magnetically induced electric fields, consider a few of the many practical applications (**Fig. 29.19**). Pickups in electric

Figure 29.19 Applications of induced electric fields. (a) This car is powered by an electric motor. As the car comes to a halt, the spinning wheels run the motor backward so that it acts as a generator. The resulting induced current is used to recharge the car's batteries. (b) The rotating crankshaft of a piston-engine airplane spins a magnet, inducing an emf in an adjacent coil and generating the spark that ignites fuel in the engine cylinders. This keeps the engine running even if the airplane's other electrical systems fail.



guitars use currents induced in stationary pickup coils by the vibration of nearby ferromagnetic strings. Alternators in most cars use rotating magnets to induce currents in stationary coils. Whether we realize it or not, magnetically induced electric fields play an important role in everyday life.

### EXAMPLE 29.11 Induced electric fields

### WITH VARIATION PROBLEMS

Suppose the long solenoid in Fig. 29.18a has 500 turns per meter and cross-sectional area  $4.0 \text{ cm}^2$ . The current in its windings is increasing at  $100 \text{ A/s}$ . (a) Find the magnitude of the induced emf in the wire loop outside the solenoid. (b) Find the magnitude of the induced electric field within the loop if its radius is  $2.0 \text{ cm}$ .

**IDENTIFY and SET UP** As in Fig. 29.18b, the increasing magnetic field inside the solenoid causes a change in the magnetic flux through the wire loop and hence induces an electric field  $\vec{E}$  around the loop. Our target variables are the induced emf  $\mathcal{E}$  and the electric-field magnitude  $E$ . We use Eq. (29.8) to determine the emf. The loop and the solenoid share the same central axis. Hence, by symmetry, the electric field is tangent to the loop and has the same magnitude  $E$  all the way around its circumference. We can therefore use Eq. (29.9) to find  $E$ .

**EXECUTE** (a) From Eq. (29.8), the induced emf is

$$\begin{aligned}\mathcal{E} &= -\frac{d\Phi_B}{dt} = -\mu_0 n A \frac{dI}{dt} \\ &= -(4\pi \times 10^{-7} \text{ Wb/A} \cdot \text{m})(500 \text{ turns/m})(4.0 \times 10^{-4} \text{ m}^2)(100 \text{ A/s}) \\ &= -25 \times 10^{-6} \text{ Wb/s} = -25 \times 10^{-6} \text{ V} = -25 \mu\text{V}\end{aligned}$$

(b) By symmetry the line integral  $\oint \vec{E} \cdot d\vec{l}$  has absolute value  $2\pi rE$  no matter which direction we integrate around the loop. This is equal to the absolute value of the emf, so

$$E = \frac{|\mathcal{E}|}{2\pi r} = \frac{25 \times 10^{-6} \text{ V}}{2\pi(2.0 \times 10^{-2} \text{ m})} = 2.0 \times 10^{-4} \text{ V/m}$$

**EVALUATE** In Fig. 29.18b the magnetic flux *into* the plane of the figure is increasing. According to the right-hand rule for induced emf (Fig. 29.6), a positive emf would be clockwise around the loop; the negative sign of  $\mathcal{E}$  shows that the emf is in the counterclockwise direction. Can you also show this by using Lenz's law?

**KEYCONCEPT** A changing magnetic flux through a closed loop produces an induced electric field  $\vec{E}$ . The line integral of  $\vec{E}$  around the loop is equal to the induced emf in the loop.

**TEST YOUR UNDERSTANDING OF SECTION 29.5** If you wiggle a magnet back and forth in your hand, are you generating an electric field? If so, is this field conservative?

**ANSWER** **yes, no** The magnetic field at a fixed position changes as you move the magnet, which induces an electric field. Such induced electric fields are *not* conservative.

## 29.6 EDDY CURRENTS

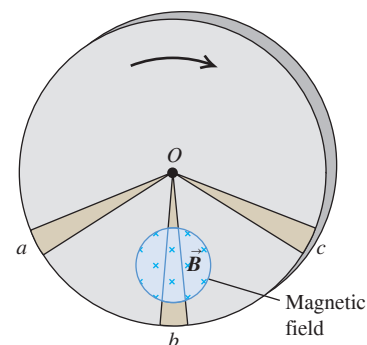
In the examples of induction effects that we have studied, the induced currents have been confined to well-defined paths in conductors and other components forming a circuit. However, many pieces of electrical equipment contain masses of metal moving in magnetic fields or located in changing magnetic fields. In situations like these we can have induced currents that circulate throughout the volume of a material. Because their flow patterns resemble swirling eddies in a river, we call these **eddy currents**.

As an example, consider a metallic disk rotating in a magnetic field perpendicular to the plane of the disk but confined to a limited portion of the disk's area, as shown in Fig. 29.20a. Sector  $Ob$  is moving across the field and has an emf induced in it. Sectors  $Oa$  and  $Oc$  are not in the field, but they provide return conducting paths for charges displaced along  $Ob$  to return from  $b$  to  $O$ . The result is a circulation of eddy currents in the disk, somewhat as sketched in Fig. 29.20b.

We can use Lenz's law to decide on the direction of the induced current in the neighborhood of sector  $Ob$ . This current must experience a magnetic force  $\vec{F} = I\vec{L} \times \vec{B}$  that *opposes* the rotation of the disk, and so this force must be to the right in Fig. 29.20b. Since  $\vec{B}$  is directed into the plane of the disk, the current and hence  $\vec{L}$  have downward components. The return currents lie outside the field, so they do not experience magnetic forces. The interaction between the eddy currents and the field causes a braking action on the disk. Such effects can be used to stop the rotation of a circular saw quickly when

Figure 29.20 Eddy currents induced in a rotating metal disk.

(a) Metal disk rotating through a magnetic field



(b) Resulting eddy currents and braking force

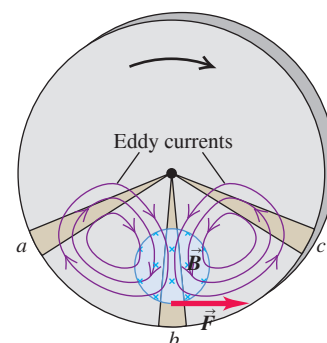
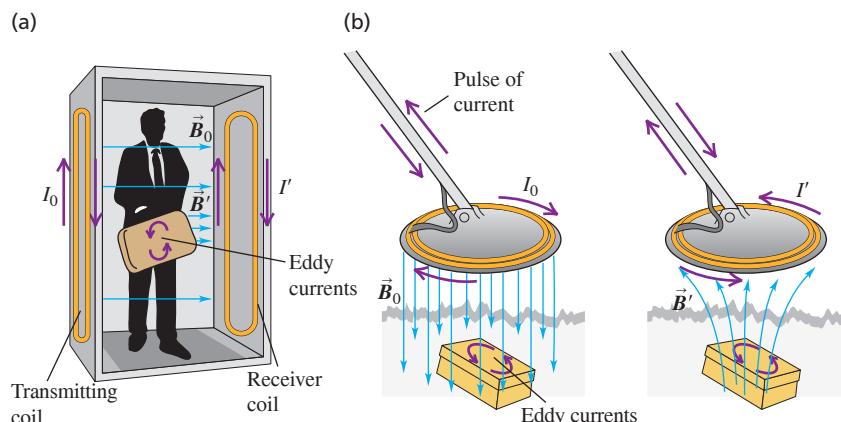
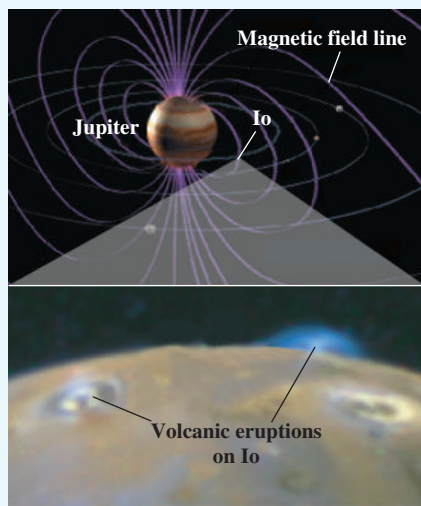


Figure 29.21 (a) A metal detector at an airport security checkpoint generates an alternating magnetic field  $\vec{B}_0$ . This induces eddy currents in a conducting object carried through the detector. The eddy currents in turn produce an alternating magnetic field  $\vec{B}'$ , which induces a current in the detector's receiver coil. (b) Portable metal detectors work on the same principle.



### APPLICATION Eddy Currents Help Power Io's Volcanoes

Jupiter's moon Io is slightly larger than the earth's moon. It moves at more than 60,000 km/h through Jupiter's intense magnetic field (about ten times stronger than the earth's field), which sets up strong eddy currents within Io that dissipate energy at a rate of  $10^{12}$  W. This dissipated energy helps to heat Io's interior and causes volcanic eruptions on its surface, as shown in the lower close-up image. (Gravitational effects from Jupiter cause even more heating.)



the power is turned off. Eddy current braking is used on some electrically powered rapid-transit vehicles. Electromagnets mounted in the cars induce eddy currents in the rails; the resulting magnetic fields cause braking forces on the electromagnets and thus on the cars.

Eddy currents have many other practical uses. In induction furnaces, eddy currents are used to heat materials in completely sealed containers for processes in which it is essential to avoid the slightest contamination of the materials. The metal detectors used at airport security checkpoints (Fig. 29.21a) operate by detecting eddy currents induced in metallic objects. Similar devices (Fig. 29.21b) are used to find buried treasure such as bottlecaps and lost pennies.

Eddy currents also have undesirable effects. In an alternating-current transformer, coils wrapped around an iron core carry a sinusoidally varying current. The resulting eddy currents in the core waste energy through  $I^2R$  heating and set up an unwanted opposing emf in the coils. To minimize these effects, the core is designed so that the paths for eddy currents are as narrow as possible. We'll describe how this is done when we discuss transformers in Section 31.6.

**TEST YOUR UNDERSTANDING OF SECTION 29.6** Suppose that the magnetic field in Fig. 29.20 were directed out of the plane of the figure and the disk were rotating counterclockwise. Compared to the directions of the force  $\vec{F}$  and the eddy currents shown in Fig. 29.20b, what would the new directions be? (i) The force  $\vec{F}$  and the eddy currents would both be in the same direction; (ii) the force  $\vec{F}$  would be in the same direction, but the eddy currents would be in the opposite direction; (iii) the force  $\vec{F}$  would be in the opposite direction, but the eddy currents would be in the same direction; (iv) the force  $\vec{F}$  and the eddy currents would be in the opposite directions.

### ANSWER

(iii) By Lenz's law, the force must oppose the motion of the disk through the magnetic field. Since the disk material is now moving to the right through the field region, the force  $\vec{F}$  is to the left—that is, in the opposite direction to that shown in Fig. 29.20b. To produce a leftward magnetic force  $\vec{F} = I\vec{L} \times \vec{B}$  on currents moving through a magnetic field  $\vec{B}$  directed out of the plane of the figure, the eddy currents must be moving downward in the figure—that is, in the same direction shown in Fig. 29.20b.

## 29.7 DISPLACEMENT CURRENT AND MAXWELL'S EQUATIONS

We have seen that a varying magnetic field gives rise to an induced electric field. In one of the more remarkable examples of the symmetry of nature, it turns out that a varying electric field gives rise to a magnetic field. This effect is of tremendous importance, for it turns out to explain the existence of radio waves, gamma rays, and visible light, as well as all other forms of electromagnetic waves.

## Generalizing Ampere's Law

To see the origin of the relationship between varying electric fields and magnetic fields, let's return to Ampere's law as given in Section 28.6, Eq. (28.20):

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$$

The problem with Ampere's law in this form is that it is *incomplete*. To see why, let's consider the process of charging a capacitor (**Fig. 29.22**). Conducting wires lead current  $i_C$  into one plate and out of the other; the charge  $Q$  increases, and the electric field  $\vec{E}$  between the plates increases. The notation  $i_C$  indicates *conduction* current to distinguish it from another kind of current we are about to encounter, called *displacement* current  $i_D$ . We use lowercase  $i$ 's and  $v$ 's to denote instantaneous values of currents and potential differences, respectively, that may vary with time.

Let's apply Ampere's law to the circular path shown. The integral  $\oint \vec{B} \cdot d\vec{l}$  around this path equals  $\mu_0 I_{\text{encl}}$ . For the plane circular area bounded by the circle,  $I_{\text{encl}}$  is just the current  $i_C$  in the left conductor. But the surface that bulges out to the right is bounded by the same circle, and the current through that surface is zero. So  $\oint \vec{B} \cdot d\vec{l}$  is equal to  $\mu_0 i_C$ , and at the same time it is equal to zero! This is a clear contradiction.

However, something else is happening on the bulged-out surface. As the capacitor charges, the electric field  $\vec{E}$  and the electric flux  $\Phi_E$  through the surface are increasing. We can determine their rates of change in terms of the charge and current. The instantaneous charge is  $q = Cv$ , where  $C$  is the capacitance and  $v$  is the instantaneous potential difference. For a parallel-plate capacitor,  $C = \epsilon_0 A/d$ , where  $A$  is the plate area and  $d$  is the spacing. The potential difference  $v$  between plates is  $v = Ed$ , where  $E$  is the electric-field magnitude between plates. (We ignore fringing and assume that  $\vec{E}$  is uniform in the region between the plates.) If this region is filled with a material with permittivity  $\epsilon$ , we replace  $\epsilon_0$  by  $\epsilon$  everywhere; we'll use  $\epsilon$  in the following discussion.

Substituting these expressions for  $C$  and  $v$  into  $q = Cv$ , we can express the capacitor charge  $q$  in terms of the electric flux  $\Phi_E = EA$  through the surface:

$$q = Cv = \frac{\epsilon A}{d}(Ed) = \epsilon EA = \epsilon \Phi_E \quad (29.12)$$

As the capacitor charges, the rate of change of  $q$  is the conduction current,  $i_C = dq/dt$ . Taking the derivative of Eq. (29.12) with respect to time, we get

$$i_C = \frac{dq}{dt} = \epsilon \frac{d\Phi_E}{dt} \quad (29.13)$$

Stretching our imagination a bit, we invent a fictitious **displacement current**  $i_D$  in the region between the plates, defined as

$$i_D = \epsilon \frac{d\Phi_E}{dt} \quad (29.14)$$

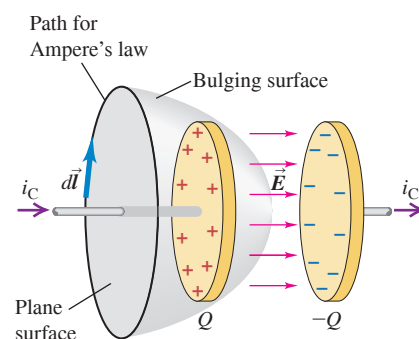
Displacement current through an area Time rate of change of electric flux through area  
Permittivity of material in area

That is, we imagine that the changing flux through the curved (bulged-out) surface in Fig. 29.22 is equivalent, in Ampere's law, to a conduction current through that surface. We include this fictitious current, along with the real conduction current  $i_C$ , in Ampere's law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i_C + i_D)_{\text{encl}} \quad (\text{generalized Ampere's law}) \quad (29.15)$$

Ampere's law in this form is obeyed no matter which surface we use in Fig. 29.22. For the flat surface,  $i_D$  is zero; for the curved surface,  $i_C$  is zero; and  $i_C$  for the flat surface equals  $i_D$  for the curved surface. Equation (29.15) remains valid in a magnetic material, provided that the magnetization is proportional to the external field and we replace  $\mu_0$  by  $\mu$ .

Figure 29.22 Parallel-plate capacitor being charged. The conduction current through the plane surface is  $i_C$ , but there is no conduction current through the surface that bulges out to pass between the plates. The two surfaces have a common boundary, so this difference in  $I_{\text{encl}}$  leads to an apparent contradiction in applying Ampere's law.





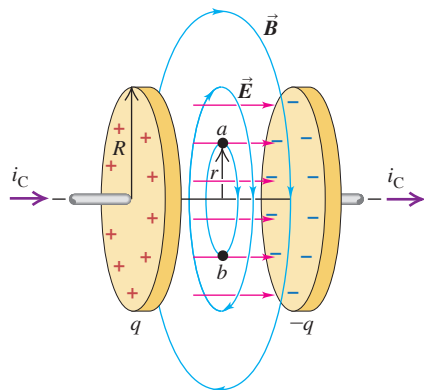
The fictitious displacement current  $i_D$  was invented in 1865 by the Scottish physicist James Clerk Maxwell. There is a corresponding *displacement current density*  $j_D = i_D/A$ ; using  $\Phi_E = EA$  and dividing Eq. (29.14) by  $A$ , we find

$$j_D = \epsilon \frac{dE}{dt} \quad (29.16)$$

We have pulled the concept out of thin air, as Maxwell did, but we see that it enables us to save Ampere's law in situations such as that in Fig. 29.22.

Another benefit of displacement current is that it lets us generalize Kirchhoff's junction rule, discussed in Section 26.2. Considering the left plate of the capacitor, we have conduction current into it but none out of it. But when we include the displacement current, we have conduction current coming in on one side and an equal displacement current coming out the other side. With this generalized meaning of the term "current," we can speak of current going *through* the capacitor.

Figure 29.23 A capacitor being charged by a current  $i_C$  has a displacement current equal to  $i_C$  between the plates, with displacement-current density  $j_D = \epsilon dE/dt$ . This can be regarded as the source of the magnetic field between the plates.



### The Reality of Displacement Current

You might well ask at this point whether displacement current has any real physical significance or whether it is just a ruse to satisfy Ampere's law and Kirchhoff's junction rule. Here's a fundamental experiment that helps to answer that question. We take a plane circular area between the capacitor plates (**Fig. 29.23**). If displacement current really plays the role in Ampere's law that we have claimed, then there ought to be a magnetic field in the region between the plates while the capacitor is charging. We can use our generalized Ampere's law, including displacement current, to predict what this field should be.

To be specific, let's picture round capacitor plates with radius  $R$ . To find the magnetic field at a point in the region between the plates at a distance  $r$  from the axis, we apply Ampere's law to a circle of radius  $r$  passing through the point, with  $r < R$ . This circle passes through points  $a$  and  $b$  in Fig. 29.23. The total current enclosed by the circle is  $j_D$  times its area, or  $(i_D/\pi R^2)(\pi r^2)$ . The integral  $\oint \vec{B} \cdot d\vec{l}$  in Ampere's law is just  $B$  times the circumference  $2\pi r$  of the circle, and because  $i_D = i_C$  for the charging capacitor, Ampere's law becomes

$$\begin{aligned} \oint \vec{B} \cdot d\vec{l} &= 2\pi r B = \mu_0 \frac{r^2}{R^2} i_C \quad \text{or} \\ B &= \frac{\mu_0}{2\pi} \frac{r}{R^2} i_C \end{aligned} \quad (29.17)$$

This result predicts that in the region between the plates  $\vec{B}$  is zero at the axis and increases linearly with distance from the axis. A similar calculation shows that *outside* the region between the plates (that is, for  $r > R$ ),  $\vec{B}$  is the same as though the wire were continuous and the plates not present at all.

When we *measure* the magnetic field in this region, we find that it really is there and that it behaves just as Eq. (29.17) predicts. This confirms directly the role of displacement current as a source of magnetic field. It is now established beyond reasonable doubt that Maxwell's displacement current, far from being just an artifice, is a fundamental fact of nature.

### Maxwell's Equations of Electromagnetism

We are now in a position to wrap up in a single package *all* of the relationships between electric and magnetic fields and their sources. This package consists of four equations, called **Maxwell's equations**. Maxwell did not discover all of these equations single-handedly (though he did develop the concept of displacement current). But he did put them together and recognized their significance, particularly in predicting the existence of electromagnetic waves.

For now we'll state Maxwell's equations in their simplest form, for the case in which we have charges and currents in otherwise empty space. In Chapter 32 we'll discuss how to modify these equations if a dielectric or a magnetic material is present.

Two of Maxwell's equations involve an integral of  $\vec{E}$  or  $\vec{B}$  over a closed surface. The first is simply Gauss's law for electric fields, Eq. (22.8):

**Gauss's law for  $\vec{E}$ :**

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (29.18)$$

Flux of electric field through a closed surface  
Charge enclosed by surface  
Electric constant

The second is the analogous relationship for *magnetic* fields, Eq. (27.8):

**Gauss's law for  $\vec{B}$ :**

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (29.19)$$

Flux of magnetic field through any closed surface ...  
... equals zero.

This statement means, among other things, that there are no magnetic monopoles (single magnetic charges) to act as sources of magnetic field.

The third and fourth equations involve a line integral of  $\vec{E}$  or  $\vec{B}$  around a closed path. Faraday's law states that a changing magnetic flux acts as a source of electric field:

**Faraday's law for a stationary integration path:**

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad (29.20)$$

Line integral of electric field around path  
Negative of the time rate of change of magnetic flux through path

If there is a changing magnetic field, the line integral in Eq. (29.20)—which must be carried out over a *stationary* closed path—is not zero. Thus the  $\vec{E}$  field produced by a changing  $\vec{B}$  is not conservative.

The fourth and final equation is Ampere's law including displacement current. It states that both a conduction current and a changing electric flux act as sources of magnetic field:

**Ampere's law for a stationary integration path:**

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left( i_C + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}} \quad (29.21)$$

Line integral of magnetic field around path  
Electric constant  
Time rate of change of electric flux through path  
Magnetic constant  
Conduction current through path  
Displacement current through path

It's worthwhile to look more carefully at the electric field  $\vec{E}$  and its role in Maxwell's equations. In general, the total  $\vec{E}$  field at a point in space can be the superposition of an electrostatic field  $\vec{E}_c$  caused by a distribution of charges at rest and a magnetically induced, nonelectrostatic field  $\vec{E}_n$ . That is,

$$\vec{E} = \vec{E}_c + \vec{E}_n$$

The electrostatic part  $\vec{E}_c$  is *always* conservative, so  $\oint \vec{E}_c \cdot d\vec{l} = 0$ . This conservative part of the field does not contribute to the integral in Faraday's law, so we can take  $\vec{E}$  in Eq. (29.20) to be the *total* electric field  $\vec{E}$ , including both the part  $\vec{E}_c$  due to charges and the magnetically induced part  $\vec{E}_n$ . Similarly, the nonconservative part  $\vec{E}_n$  of the  $\vec{E}$  field does not contribute to the integral in Gauss's law, because this part of the field is not caused by static charges. Hence  $\oint \vec{E}_n \cdot d\vec{A}$  is always zero. We conclude that in all the Maxwell equations,  $\vec{E}$  is the total electric field; these equations don't distinguish between conservative and nonconservative fields.

Figure 29.24 Maxwell's equations in empty space are highly symmetric.

In empty space there are no charges, so the fluxes of  $\vec{E}$  and  $\vec{B}$  through any closed surface are equal to zero.

$$\begin{aligned}\oint \vec{E} \cdot d\vec{A} &= 0 \\ \oint \vec{B} \cdot d\vec{A} &= 0 \\ \oint \vec{E} \cdot d\vec{l} &= -\frac{d\Phi_B}{dt} \\ \oint \vec{B} \cdot d\vec{l} &= \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}\end{aligned}$$

In empty space there are no conduction currents, so the line integrals of  $\vec{E}$  and  $\vec{B}$  around any closed path are related to the rate of change of flux of the other field.

## Symmetry in Maxwell's Equations

There is a remarkable symmetry in Maxwell's four equations. In empty space where there is no charge, the first two equations [Eqs. (29.18) and (29.19)] are identical in form, one containing  $\vec{E}$  and the other containing  $\vec{B}$  (Fig. 29.24). When we compare the second two equations, Eq. (29.20) says that a changing magnetic flux creates an electric field, and Eq. (29.21) says that a changing electric flux creates a magnetic field. In empty space, where there is no conduction current,  $i_C = 0$  and the two equations have the same form, apart from a numerical constant and a negative sign, with the roles of  $\vec{E}$  and  $\vec{B}$  exchanged.

We can rewrite Eqs. (29.20) and (29.21) in a different but equivalent form by introducing the definitions of magnetic flux,  $\Phi_B = \int \vec{B} \cdot d\vec{A}$ , and electric flux,  $\Phi_E = \int \vec{E} \cdot d\vec{A}$ , respectively. In empty space, where there is no charge or conduction current,  $i_C = 0$  and  $Q_{\text{encl}} = 0$ , and we have

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A} \quad (29.22)$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d}{dt} \int \vec{E} \cdot d\vec{A} \quad (29.23)$$

Again we notice the symmetry between the roles of  $\vec{E}$  and  $\vec{B}$  in these expressions.

The most remarkable feature of these equations is that a time-varying field of *either* kind induces a field of the other kind in neighboring regions of space. Maxwell recognized that these relationships predict the existence of electromagnetic disturbances consisting of time-varying electric and magnetic fields that travel or *propagate* from one region of space to another, even if no matter is present in the intervening space. Such disturbances, called *electromagnetic waves*, provide the physical basis for light, radio and television waves, infrared, ultraviolet, and x rays. We'll return to this vitally important topic in Chapter 32.

Although it may not be obvious, *all* the basic relationships between fields and their sources are contained in Maxwell's equations. We can derive Coulomb's law from Gauss's law, we can derive the law of Biot and Savart from Ampere's law, and so on. When we add the equation that defines the  $\vec{E}$  and  $\vec{B}$  fields in terms of the forces that they exert on a charge  $q$ , namely,

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \quad (29.24)$$

we have *all* the fundamental relationships of electromagnetism!

Maxwell's equations would have even greater symmetry between the  $\vec{E}$  and  $\vec{B}$  fields if single magnetic charges (magnetic monopoles) existed. The right side of Eq. (29.19) would contain the total *magnetic* charge enclosed by the surface, and the right side of Eq. (29.20) would include a magnetic monopole current term. However, no magnetic monopoles have yet been found.

In conciseness and generality, Maxwell's equations are in the same league with Newton's laws of motion and the laws of thermodynamics. Indeed, a major goal of science is learning how to express very broad and general relationships in a concise and compact form. Maxwell's synthesis of electromagnetism stands as a towering intellectual achievement, comparable to the Newtonian synthesis we described at the end of Section 13.5 and to the development of relativity and quantum mechanics in the 20th century.

**TEST YOUR UNDERSTANDING OF SECTION 29.7** (a) Which of Maxwell's equations explains how a credit card reader works? (b) Which one describes how a wire carrying a steady current generates a magnetic field?

### ANSWER

(a) **Faraday's law.** A credit card reader works by inducing currents in the reader's coils as the card's magnetized stripe is swiped (see the answer to the chapter opening question). Ampere's law describes how currents of all kinds (both conduction currents and displacement currents) give rise to magnetic fields.

## 29.8 SUPERCONDUCTIVITY

The most familiar property of a superconductor is the sudden disappearance of all electrical resistance when the material is cooled below a temperature called the *critical temperature*, denoted by  $T_c$ . We discussed this behavior and the circumstances of its discovery in Section 25.2. But superconductivity is far more than just the absence of measurable resistance. As we'll see in this section, superconductors also have extraordinary *magnetic* properties.

The first hint of unusual magnetic properties was the discovery that for any superconducting material the critical temperature  $T_c$  changes when the material is placed in an externally produced magnetic field  $\vec{B}_0$ . **Figure 29.25** shows this dependence for mercury, the first element in which superconductivity was observed. As the external field magnitude  $B_0$  increases, the superconducting transition occurs at lower and lower temperature. When  $B_0$  is greater than 0.0412 T, *no* superconducting transition occurs. The minimum magnitude of magnetic field that is needed to eliminate superconductivity at a temperature below  $T_c$  is called the *critical field*, denoted by  $B_c$ .

### The Meissner Effect

Another aspect of the magnetic behavior of superconductors appears if we place a homogeneous sphere of a superconducting material in a uniform applied magnetic field  $\vec{B}_0$  at a temperature  $T$  greater than  $T_c$ . The material is then in the normal phase, not the superconducting phase (**Fig. 29.26a**). Now we lower the temperature until the superconducting transition occurs. (We assume that the magnitude of  $\vec{B}_0$  is not large enough to prevent the phase transition.) What happens to the field?

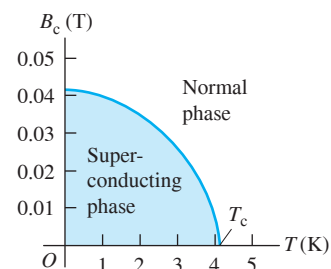
Measurements of the field outside the sphere show that the field lines become distorted as in **Fig. 29.26b**. There is no longer any field inside the material, except possibly in a very thin surface layer a hundred or so atoms thick. If a coil is wrapped around the sphere, the emf induced in the coil shows that during the superconducting transition the magnetic flux through the coil decreases from its initial value to zero; this is consistent with the absence of field inside the material. Finally, if the field is now turned off while the material is still in its superconducting phase, no emf is induced in the coil, and measurements show no field outside the sphere (**Fig. 29.26c**).

We conclude that during a superconducting transition in the presence of the field  $\vec{B}_0$ , all of the magnetic flux is expelled from the bulk of the sphere, and the magnetic flux  $\Phi_B$  through the coil becomes zero. This expulsion of magnetic flux is called the *Meissner effect*. As **Fig. 29.26b** shows, this expulsion crowds the magnetic field lines closer together to the side of the sphere, increasing  $\vec{B}$  there.

### Superconductor Levitation and Other Applications

The diamagnetic nature of a superconductor has some interesting *mechanical* consequences. A paramagnetic or ferromagnetic material is attracted by a permanent magnet because the magnetic dipoles in the material align with the nonuniform magnetic field of the permanent magnet. (We discussed this in Section 27.7.) For a diamagnetic material

**Figure 29.25** Phase diagram for pure mercury, showing the critical magnetic field  $B_c$  and its dependence on temperature. Superconductivity is impossible above the critical temperature  $T_c$ . The curves for other superconducting materials are similar but with different numerical values.



**Figure 29.26** A superconducting material (a) above the critical temperature and (b), (c) below the critical temperature.

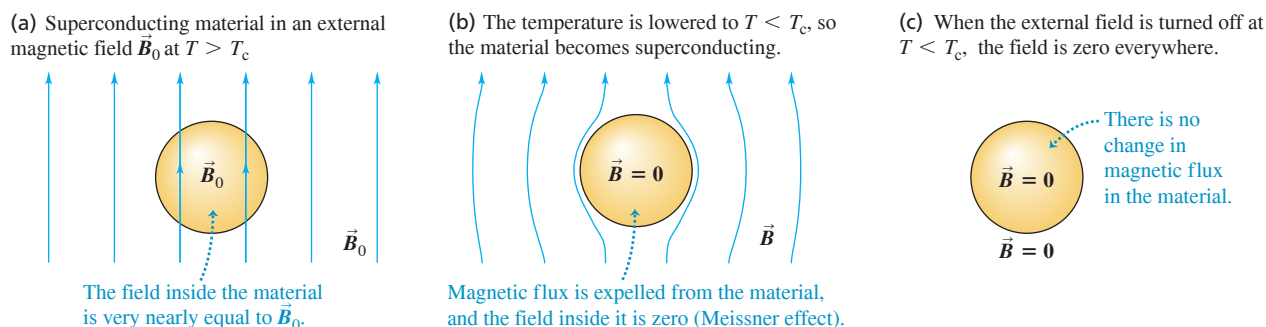
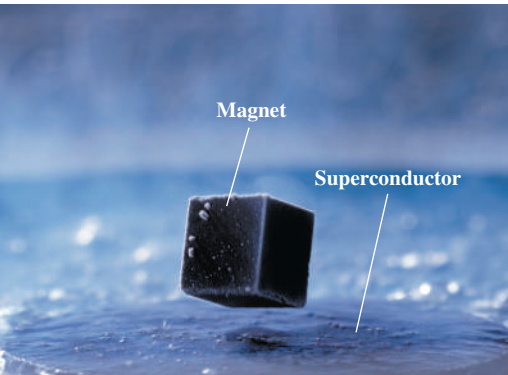




Figure 29.27 A superconductor exerts a repulsive force on a magnet, supporting the magnet in midair.



the magnetization is in the opposite sense, and a diamagnetic material is *repelled* by a permanent magnet. By Newton's third law the magnet is also repelled by the diamagnetic material. **Figure 29.27** shows the repulsion between a specimen of a high-temperature superconductor and a magnet; the magnet is supported ("levitated") by this repulsive magnetic force.

The behavior we have described is characteristic of what are called *type-I superconductors*. There is another class of superconducting materials called *type-II superconductors*. When such a material in the superconducting phase is placed in a magnetic field, the bulk of the material remains superconducting, but thin filaments of material, running parallel to the field, may return to the normal phase. Currents circulate around the boundaries of these filaments, and there *is* magnetic flux inside them. Type-II superconductors are used for electromagnets because they usually have much larger values of  $B_c$  than do type-I materials, permitting much larger magnetic fields without destroying the superconducting state. Type-II superconductors have *two* critical magnetic fields: The first,  $B_{c1}$ , is the field at which magnetic flux begins to enter the material, forming the filaments just described, and the second,  $B_{c2}$ , is the field at which the material becomes normal.

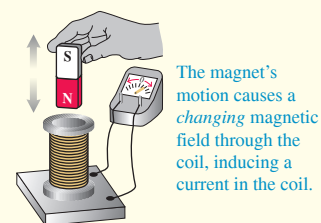
Superconducting electromagnets are in everyday use not only in research laboratories but also in medical MRI (magnetic resonance imaging) scanners. As we described in Section 27.7, scanning a patient through MRI requires a strong magnetic field to align the magnetic dipoles of the patient's atomic nuclei. A steady field of 1.5 T or more is needed, which is very difficult to produce with a conventional electromagnet, since this would require very high currents and hence very high energy losses due to resistance in the electromagnet coils. But with a superconducting electromagnet there is no resistive energy loss, and fields up to 10 T can routinely be attained.

Very sensitive measurements of magnetic fields can be made with superconducting quantum interference devices (SQUIDS), which can detect changes in magnetic flux of less than  $10^{-14}$  Wb; these devices have applications in medicine, geology, and other fields. The number of potential uses for superconductors has increased greatly since the discovery in 1987 of high-temperature superconductors. These materials have critical temperatures that are above the temperature of liquid nitrogen (about 77 K) and so are comparatively easy to attain. Development of practical applications of superconductor science promises to be an exciting chapter in contemporary technology.

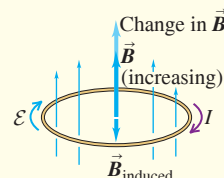
## CHAPTER 29 SUMMARY

**Faraday's law:** Faraday's law states that the induced emf in a closed loop equals the negative of the time rate of change of magnetic flux through the loop. This relationship is valid whether the flux change is caused by a changing magnetic field, motion of the loop, or both. (See Examples 29.1–29.6.)

$$\mathcal{E} = -\frac{d\Phi_B}{dt} \quad (29.3)$$



**Lenz's law:** Lenz's law states that an induced current or emf always tends to oppose or cancel out the change that caused it. Lenz's law can be derived from Faraday's law and is often easier to use. (See Examples 29.7 and 29.8.)



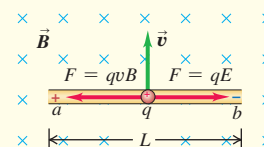
**Motional emf:** If a conductor moves in a magnetic field, a motional emf is induced. (See Examples 29.9 and 29.10.)

$$\mathcal{E} = vBL \quad (29.6)$$

(conductor with length  $L$  moves in uniform  $\vec{B}$  field,  $\vec{L}$  and  $\vec{v}$  both perpendicular to  $\vec{B}$  and to each other)

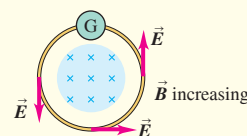
$$\mathcal{E} = \oint (\vec{v} \times \vec{B}) \cdot d\vec{l} \quad (29.7)$$

(all or part of a closed loop moves in a  $\vec{B}$  field)



**Induced electric fields:** When an emf is induced by a changing magnetic flux through a stationary conductor, there is an induced electric field  $\vec{E}$  of nonelectrostatic origin. This field is nonconservative and cannot be associated with a potential. (See Example 29.11.)

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad (29.10)$$



**Displacement current and Maxwell's equations:** A time-varying electric field generates displacement current  $i_D$ , which acts as a source of magnetic field in exactly the same way as conduction current. The relationships between electric and magnetic fields and their sources can be stated compactly in four equations, called Maxwell's equations. Together they form a complete basis for the relationship of  $\vec{E}$  and  $\vec{B}$  fields to their sources.

$$i_D = \epsilon \frac{d\Phi_E}{dt} \quad (29.14)$$

(displacement current)

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (29.18)$$

(Gauss's law for  $\vec{E}$  fields)

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (29.19)$$

(Gauss's law for  $\vec{B}$  fields)

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad (29.20)$$

(Faraday's law)

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left( i_C + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}} \quad (29.21)$$

(Ampere's law including displacement current)



## GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

## KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 29.1, 29.2, and 29.3** (Section 29.2) and **CONCEPTUAL EXAMPLES 29.7 and 29.8** (Section 29.3) before attempting these problems.

**VP29.8.1** A single circular loop of wire with radius 2.40 cm lies in the  $xy$ -plane. There is a uniform magnetic field that changes at a steady rate from 0.140 T in the  $+z$ -direction at  $t = 0$  to 0.110 T in the  $-z$ -direction at  $t = 2.00$  s. Take the area vector for the loop to be in the  $+z$ -direction. Find (a) the magnetic flux through the loop at  $t = 0$  and  $t = 2.00$  s, and (b) the magnitude of the induced emf in the loop while the field is changing.

**VP29.8.2** A circular coil with 455 turns of wire has radius 5.00 cm and resistance 14.5  $\Omega$ . The area vector of the coil points in the  $+z$ -direction. The coil is in a region of uniform magnetic field that points  $20.0^\circ$  from the  $+z$ -direction. The magnitude of the field is initially 0.0600 T and decreases at a rate of  $-3.00 \times 10^{-3}$  T/s. Find the magnitude of (a) the induced emf and (b) the induced current in the coil while the field is changing.

**VP29.8.3** A square single loop of wire 4.00 cm on a side with 875 turns lies in the  $xy$ -plane. The loop is in a uniform magnetic field that changes at a steady rate from  $\vec{B} = (0.200 \text{ T})\hat{i} + (0.150 \text{ T})\hat{k}$  at  $t = 0$  to  $\vec{B} = (0.300 \text{ T})\hat{i} + (-0.200 \text{ T})\hat{k}$  at  $t = 3.00$  s. At the time  $t = 2.00$  s, find (a) the magnitude of the induced emf and (b) the direction of the induced current in the coil as seen from a point on the  $+z$ -axis.

**VP29.8.4** A coil with area  $A$  and  $N$  turns of wire lies in the  $xy$ -plane. Its area vector points in the  $+z$ -direction. It is exposed to a uniform time-varying magnetic field  $\vec{B} = (B_0 \sin \omega t)\hat{k}$ , where  $\omega$  is a positive constant. Find (a) the induced emf in the coil and (b) the direction of the induced current in the coil as seen from a point on the  $+z$ -axis at  $t = 0$  and at  $t = \pi/\omega$ .

Be sure to review **EXAMPLE 29.9** (Section 29.4) before attempting these problems.

**VP29.9.1** An isolated conducting rod of length 8.00 cm is oriented parallel to the  $x$ -axis. It moves in the  $+y$ -direction at 3.90 m/s in the presence of a uniform magnetic field of magnitude 0.600 T that points in the  $-z$ -direction. Find (a) the magnitude of the motional emf in the rod and (b) the magnitude and direction of the electric field produced in the rod.

**VP29.9.2** In the slidewire generator of Fig. 29.15b, you reverse the rod's motion so that it moves to the left rather than to the right. The moving rod is 5.00 cm long and moves at 2.40 m/s, and the uniform

magnetic field has magnitude 0.150 T. If the resistance of the circuit at a given instant is 0.0200  $\Omega$ , find the current in the circuit and its direction around the circuit.

**VP29.9.3** An isolated conducting rod of length 8.00 cm is oriented parallel to the  $x$ -axis. At  $t = 0$ , end  $a$  is at  $x = y = z = 0$  and end  $b$  is at  $x = 8.00$  cm,  $y = z = 0$ . The rod moves with constant velocity  $\vec{v} = (1.50 \text{ m/s})\hat{i} + (2.00 \text{ m/s})\hat{j}$  in a uniform magnetic field  $\vec{B} = (-0.120 \text{ T})\hat{i} + (0.150 \text{ T})\hat{k}$ . (a) Find the magnitude of the induced emf in the rod. (b) Which end of the rod,  $a$  or  $b$ , is at higher potential?

**VP29.9.4** The magnetic force on the moving rod in a slidewire generator (Fig. 29.15b) has magnitude 0.0600 N at an instant when the resistance of the circuit is 0.750  $\Omega$ . The rod is 0.190 m long and the magnetic field (which is perpendicular to the plane of the generator) has magnitude 0.550 T. Find the speed of the rod.

Be sure to review **EXAMPLE 29.11** (Section 29.5) before attempting these problems.

**VP29.11.1** A long solenoid has a cross-sectional area of 3.00 cm<sup>2</sup>. The current through the windings is decreasing at a rate of 22.0 A/s. A wire loop of radius 3.10 cm is around the solenoid, parallel with its coils, and centered on the axis of the solenoid. The magnitude of the induced emf is 15.0  $\mu$ V. Find (a) the number of turns per meter in the solenoid and (b) the magnitude of the induced electric field within the loop.

**VP29.11.2** A circular wire loop of radius 0.360 cm lies in the  $xz$ -plane. There is a uniform magnetic field in the  $y$ -direction that decreases at 0.0150 T/s. Find the magnitude of the induced electric field in the wire.

**VP29.11.3** A long solenoid with cross-sectional area 4.00 cm<sup>2</sup> and 965 turns per meter is oriented with its axis along the  $z$ -axis. The field inside the solenoid points in the  $+z$ -direction. A wire loop of radius 5.00 cm is around the solenoid, parallel with its coils, centered on the axis of the solenoid, and lying in the  $xy$ -plane. Find the rate of change of the current in the solenoid if the electric field in the loop at the point  $x = 5.00$  cm,  $y = 0$ ,  $z = 0$  is (a)  $\vec{E} = (+1.20 \times 10^{-5} \text{ V/m})\hat{j}$ ; (b)  $\vec{E} = (-1.80 \times 10^{-5} \text{ V/m})\hat{j}$ .

**VP29.11.4** A long solenoid has 585 turns per meter and a cross-sectional area of 2.70 cm<sup>2</sup>. The current through the windings as a function of time is  $(0.600 \text{ A/s}^2)t^2$ . A wire loop of resistance 0.600  $\Omega$  is around the solenoid, parallel with its coils, and centered on the axis of the solenoid. What is the current induced in the loop at  $t = 13.9$  s?

## BRIDGING PROBLEM A Falling Square Loop

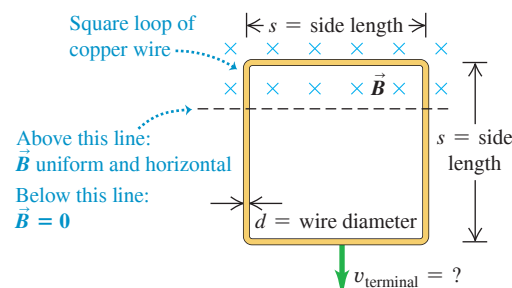
A vertically oriented square loop of copper wire falls from rest in a region in which the field  $\vec{B}$  is horizontal, uniform, and perpendicular to the plane of the loop, into a field-free region (Fig. 29.28). The side length of the loop is  $s$  and the wire diameter is  $d$ . The resistivity of copper is  $\rho_R$  and the density of copper is  $\rho_m$ . If the loop reaches its terminal speed while its upper segment is still in the magnetic-field region, find an expression for the terminal speed.

## SOLUTION GUIDE

## IDENTIFY and SET UP

1. The motion of the loop through the magnetic field induces an emf and a current in the loop. The field then gives rise to a magnetic

Figure 29.28 A wire loop falling in a horizontal magnetic field  $\vec{B}$ . The plane of the loop is perpendicular to  $\vec{B}$ .



force on this current that opposes the downward force of gravity. The loop reaches terminal speed (it no longer accelerates) when the upward magnetic force balances the downward force of gravity.

2. Consider the case in which the entire loop is in the magnetic-field region. Is there an induced emf in this case? If so, what is its direction?
3. Consider the case in which only the upper segment of the loop is in the magnetic-field region. Is there an induced emf in this case? If so, what is its direction?
4. For the case in which there is an induced emf and hence an induced current, what is the direction of the magnetic force on each of the four sides of the loop? What is the direction of the *net* magnetic force on the loop?

## PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

### DISCUSSION QUESTIONS

**Q29.1** A sheet of copper is placed between the poles of an electromagnet with the magnetic field perpendicular to the sheet. When the sheet is pulled out, a considerable force is required, and the force required increases with speed. Explain. Is a force required also when the sheet is inserted between the poles? Explain.

**Q29.2** In Fig. 29.8, if the angular speed  $\omega$  of the loop is doubled, then the frequency with which the induced current changes direction doubles, and the maximum emf also doubles. Why? Does the torque required to turn the loop change? Explain.

**Q29.3** Two circular loops lie side by side in the same plane. One is connected to a source that supplies an increasing current; the other is a simple closed ring. Is the induced current in the ring in the same direction as the current in the loop connected to the source, or opposite? What if the current in the first loop is decreasing? Explain.

**Q29.4** For Eq. (29.6), show that if  $v$  is in meters per second,  $B$  in teslas, and  $L$  in meters, then the units of the right-hand side of the equation are joules per coulomb or volts (the correct SI units for  $\mathcal{E}$ ).

**Q29.5** A long, straight conductor passes through the center of a metal ring, perpendicular to its plane. If the current in the conductor increases, is a current induced in the ring? Explain.

**Q29.6** A student asserted that if a permanent magnet is dropped down a vertical copper pipe, it eventually reaches a terminal velocity even if there is no air resistance. Why should this be? Or should it?

**Q29.7** An airplane is in level flight over Antarctica, where the magnetic field of the earth is mostly directed upward away from the ground. As viewed by a passenger facing toward the front of the plane, is the left or the right wingtip at higher potential? Does your answer depend on the direction the plane is flying?

**Q29.8** Consider the situation in Exercise 29.21. In part (a), find the direction of the force that the large circuit exerts on the small one. Explain how this result is consistent with Lenz's law.

**Q29.9** A metal rectangle is close to a long, straight, current-carrying wire, with two of its sides parallel to the wire. If the current in the long wire is decreasing, is the rectangle repelled by or attracted to the wire? Explain why this result is consistent with Lenz's law.

**Q29.10** A square conducting loop is in a region of uniform, constant magnetic field. Can the loop be rotated about an axis along one side and no emf be induced in the loop? Discuss, in terms of the orientation of the rotation axis relative to the magnetic-field direction.

### EXECUTE

5. For the case in which the loop is falling at speed  $v$  and there is an induced emf, find (i) the emf, (ii) the induced current, and (iii) the magnetic force on the loop in terms of its resistance  $R$ .
6. Find  $R$  and the mass of the loop in terms of the given information about the loop.
7. Use your results from steps 5 and 6 to find an expression for the terminal speed.

### EVALUATE

8. How does the terminal speed depend on the magnetic-field magnitude  $B$ ? Explain why this makes sense.

**Q29.11** Example 29.6 discusses the external force that must be applied to the slidewire to move it at constant speed. If there were a break in the left-hand end of the U-shaped conductor, how much force would be needed to move the slidewire at constant speed? As in the example, you can ignore friction.

**Q29.12** In the situation shown in Fig. 29.18, would it be appropriate to ask how much *energy* an electron gains during a complete trip around the wire loop with current  $I$ ? Would it be appropriate to ask what *potential difference* the electron moves through during such a complete trip? Explain your answers.

**Q29.13** A metal ring is oriented with the plane of its area perpendicular to a spatially uniform magnetic field that increases at a steady rate. If the radius of the ring is doubled, by what factor do (a) the emf induced in the ring and (b) the electric field induced in the ring change?

**Q29.14** Small one-cylinder gasoline engines sometimes use a device called a *magneto* to supply current to the spark plug. A permanent magnet is attached to the flywheel, and a stationary coil is mounted adjacent to it. Explain how this device is able to generate current. What happens when the magnet passes the coil?

**Q29.15** Does Lenz's law say that the induced current in a metal loop always flows to oppose the magnetic flux through that loop? Explain.

**Q29.16** Does Faraday's law say that a large magnetic flux induces a large emf in a coil? Explain.

**Q29.17** Can one have a displacement current as well as a conduction current within a conductor? Explain.

**Q29.18** Your physics study partner asks you to consider a parallel-plate capacitor that has a dielectric completely filling the volume between the plates. He then claims that Eqs. (29.13) and (29.14) show that the conduction current in the dielectric equals the displacement current in the dielectric. Do you agree? Explain.

**Q29.19** Match the mathematical statements of Maxwell's equations as given in Section 29.7 to these verbal statements. (a) Closed electric field lines are evidently produced only by changing magnetic flux. (b) Closed magnetic field lines are produced both by the motion of electric charge and by changing electric flux. (c) Electric field lines can start on positive charges and end on negative charges. (d) Evidently there are no magnetic monopoles on which to start and end magnetic field lines.

**Q29.20** If magnetic monopoles existed, the right-hand side of Eq. (29.20) would include a term proportional to the current of magnetic monopoles. Suppose a steady monopole current is moving in a long straight wire. Sketch the *electric* field lines that such a current would produce.



**Q29.21** A type-II superconductor in an external field between  $B_{c1}$  and  $B_{c2}$  has regions that contain magnetic flux and have resistance, and also has superconducting regions. What is the resistance of a long, thin cylinder of such material?

## EXERCISES

### Section 29.2 Faraday's Law

**29.1** • A single loop of wire with an area of  $0.0900 \text{ m}^2$  is in a uniform magnetic field that has an initial value of  $3.80 \text{ T}$ , is perpendicular to the plane of the loop, and is decreasing at a constant rate of  $0.190 \text{ T/s}$ . (a) What emf is induced in this loop? (b) If the loop has a resistance of  $0.600 \Omega$ , find the current induced in the loop.

**29.2** •• In a physics laboratory experiment, a coil with 200 turns enclosing an area of  $12 \text{ cm}^2$  is rotated in  $0.040 \text{ s}$  from a position where its plane is perpendicular to the earth's magnetic field to a position where its plane is parallel to the field. The earth's magnetic field at the lab location is  $6.0 \times 10^{-5} \text{ T}$ . (a) What is the magnetic flux through each turn of the coil before it is rotated? After it is rotated? (b) What is the average emf induced in the coil?

**29.3** • The magnetic flux through a coil is given by  $\Phi_B = \alpha t - \beta t^3$ , where  $\alpha$  and  $\beta$  are constants. (a) What are the units of  $\alpha$  and  $\beta$ ? (b) If the induced emf is zero at  $t = 0.500 \text{ s}$ , how is  $\alpha$  related to  $\beta$ ? (c) If the emf at  $t = 0$  is  $-1.60 \text{ V}$ , what is the emf at  $t = 0.250 \text{ s}$ ?

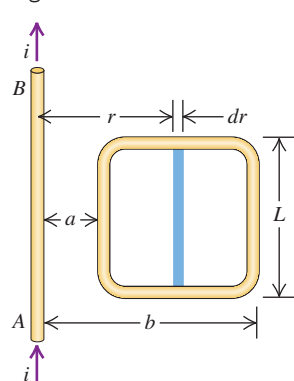
**29.4** •• A small, closely wound coil has  $N$  turns, area  $A$ , and resistance  $R$ . The coil is initially in a uniform magnetic field that has magnitude  $B$  and a direction perpendicular to the plane of the loop. The coil is then rapidly pulled out of the field so that the flux through the coil is reduced to zero in time  $\Delta t$ . (a) What are the magnitude of the average emf  $\mathcal{E}_{\text{av}}$  and average current  $I_{\text{av}}$  induced in the coil? (b) The total charge  $Q$  that flows through the coil is given by  $Q = I_{\text{av}}\Delta t$ . Derive an expression for  $Q$  in terms of  $N$ ,  $A$ ,  $B$ , and  $R$ . Note that  $Q$  does not depend on  $\Delta t$ . (c) What is  $Q$  if  $N = 150$  turns,  $A = 4.50 \text{ cm}^2$ ,  $R = 30.0 \Omega$ , and  $B = 0.200 \text{ T}$ ?

**29.5** • A circular loop of wire with a radius of  $12.0 \text{ cm}$  and oriented in the horizontal  $xy$ -plane is located in a region of uniform magnetic field. A field of  $1.5 \text{ T}$  is directed along the positive  $z$ -direction, which is upward. (a) If the loop is removed from the field region in a time interval of  $2.0 \text{ ms}$ , find the average emf that will be induced in the wire loop during the extraction process. (b) If the coil is viewed looking down on it from above, is the induced current in the loop clockwise or counterclockwise?

**29.6** • CALC A coil  $4.00 \text{ cm}$  in radius, containing 500 turns, is placed in a uniform magnetic field that varies with time according to  $B = (0.0120 \text{ T/s})t + (3.00 \times 10^{-5} \text{ T/s}^4)t^4$ . The coil is connected to a  $600 \Omega$  resistor, and its plane is perpendicular to the magnetic field. You can ignore the resistance of the coil. (a) Find the magnitude of the induced emf in the coil as a function of time. (b) What is the current in the resistor at time  $t = 5.00 \text{ s}$ ?

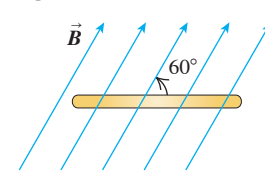
**29.7** • CALC The current in the long, straight wire  $AB$  shown in Fig. E29.7 is upward and is increasing steadily at a rate  $di/dt$ . (a) At an instant when the current is  $i$ , what are the magnitude and direction of the field  $\vec{B}$  at a distance  $r$  to the right of the wire? (b) What is the flux  $d\Phi_B$  through the narrow, shaded strip? (c) What is the total flux through the loop? (d) What is the induced emf in the loop? (e) Evaluate the numerical value of the induced emf if  $a = 12.0 \text{ cm}$ ,  $b = 36.0 \text{ cm}$ ,  $L = 24.0 \text{ cm}$ , and  $di/dt = 9.60 \text{ A/s}$ .

Figure E29.7



**29.8** • CALC A flat, circular, steel loop of radius  $75 \text{ cm}$  is at rest in a uniform magnetic field, as shown in an edge-on view in Fig. E29.8. The field is changing with time, according to  $B(t) = (1.4 \text{ T})e^{-(0.057 \text{ s}^{-1})t}$ . (a) Find the emf induced in the loop as a function of time. (b) When is the induced emf equal to  $\frac{1}{10}$  of its initial value? (c) Find the direction of the current induced in the loop, as viewed from above the loop.

Figure E29.8



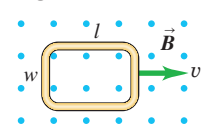
**29.9** • Shrinking Loop. A circular loop of flexible iron wire has an initial circumference of  $165.0 \text{ cm}$ , but its circumference is decreasing at a constant rate of  $12.0 \text{ cm/s}$  due to a tangential pull on the wire. The loop is in a constant, uniform magnetic field oriented perpendicular to the plane of the loop and with magnitude  $0.500 \text{ T}$ . (a) Find the emf induced in the loop at the instant when  $9.0 \text{ s}$  have passed. (b) Find the direction of the induced current in the loop as viewed looking along the direction of the magnetic field.

**29.10** • A closely wound rectangular coil of 80 turns has dimensions of  $25.0 \text{ cm}$  by  $40.0 \text{ cm}$ . The plane of the coil is rotated from a position where it makes an angle of  $37.0^\circ$  with a magnetic field of  $1.70 \text{ T}$  to a position perpendicular to the field. The rotation takes  $0.0600 \text{ s}$ . What is the average emf induced in the coil?

**29.11** •• CALC A circular loop of wire with radius  $2.00 \text{ cm}$  and resistance  $0.600 \Omega$  is in a region of a spatially uniform magnetic field  $\vec{B}$  that is perpendicular to the plane of the loop. At  $t = 0$  the magnetic field has magnitude  $B_0 = 3.00 \text{ T}$ . The magnetic field then decreases according to the equation  $B(t) = B_0 e^{-t/\tau}$ , where  $\tau = 0.500 \text{ s}$ . (a) What is the maximum magnitude of the current  $I$  induced in the loop? (b) What is the induced current  $I$  when  $t = 1.50 \text{ s}$ ?

**29.12** • A flat, rectangular coil of dimensions  $l$  and  $w$  is pulled with uniform speed  $v$  through a uniform magnetic field  $B$  with the plane of its area perpendicular to the field (Fig. E29.12). (a) Find the emf induced in this coil. (b) If the speed and magnetic field are both tripled, what is the induced emf?

Figure E29.12

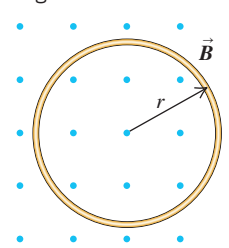


**29.13** •• The armature of a small generator consists of a flat, square coil with 120 turns and sides with a length of  $1.60 \text{ cm}$ . The coil rotates in a magnetic field of  $0.0750 \text{ T}$ . What is the angular speed of the coil if the maximum emf produced is  $24.0 \text{ mV}$ ?

### Section 29.3 Lenz's Law

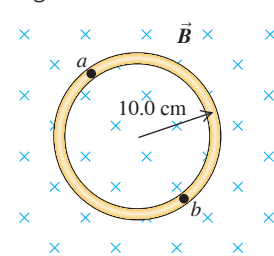
**29.14** • A circular loop of wire with radius  $r = 0.0480 \text{ m}$  and resistance  $R = 0.160 \Omega$  is in a region of spatially uniform magnetic field, as shown in Fig. E29.14. The magnetic field is directed out of the plane of the figure. The magnetic field has an initial value of  $8.00 \text{ T}$  and is decreasing at a rate of  $dB/dt = -0.680 \text{ T/s}$ . (a) Is the induced current in the loop clockwise or counterclockwise? (b) What is the rate at which electrical energy is being dissipated by the resistance of the loop?

Figure E29.14



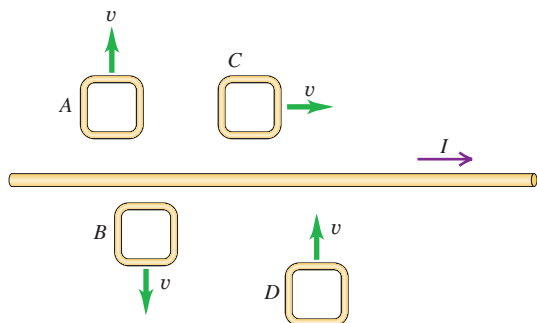
**29.15** • A circular loop of wire is in a region of spatially uniform magnetic field, as shown in Fig. E29.15. The magnetic field is directed into the plane of the figure. Determine the direction (clockwise or counterclockwise) of the induced current in the loop when (a)  $B$  is increasing; (b)  $B$  is decreasing; (c)  $B$  is constant with value  $B_0$ . Explain your reasoning.

Figure E29.15



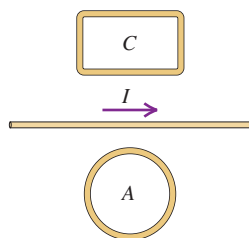
**29.16** • The current  $I$  in a long, straight wire is constant and is directed toward the right as in **Fig. E29.16**. Conducting loops  $A$ ,  $B$ ,  $C$ , and  $D$  are moving, in the directions shown, near the wire. (a) For each loop, is the direction of the induced current clockwise or counterclockwise, or is the induced current zero? (b) For each loop, what is the direction of the net force that the wire exerts on the loop? Give your reasoning for each answer.

Figure E29.16



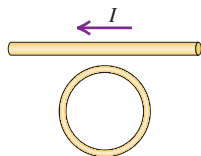
**29.17** • Two closed loops  $A$  and  $C$  are close to a long wire carrying a current  $I$  (**Fig. E29.17**). (a) Find the direction (clockwise or counterclockwise) of the current induced in each loop if  $I$  is steadily decreasing. (b) While  $I$  is decreasing, what is the direction of the net force that the wire exerts on each loop? Explain how you obtain your answer.

Figure E29.17



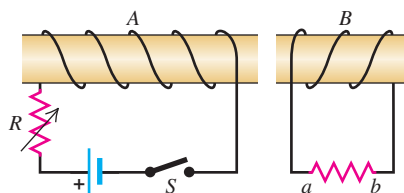
**29.18** • The current in **Fig. E29.18** obeys the equation  $I(t) = I_0 e^{-bt}$ , where  $b > 0$ . Find the direction (clockwise or counterclockwise) of the current induced in the round coil for  $t > 0$ .

Figure E29.18



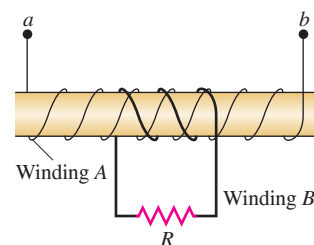
**29.19** • Using Lenz's law, determine the direction of the current in resistor  $ab$  of **Fig. E29.19** when (a) switch  $S$  is opened after having been closed for several minutes; (b) coil  $B$  is brought closer to coil  $A$  with the switch closed; (c) the resistance of  $R$  is decreased while the switch remains closed.

Figure E29.19



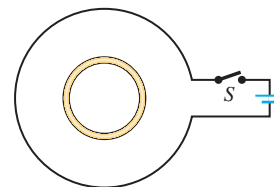
**29.20** • A cardboard tube is wrapped with two windings of insulated wire wound in opposite directions, as shown in **Fig. E29.20**. Terminals  $a$  and  $b$  of winding  $A$  may be connected to a battery through a reversing switch. State whether the induced current in the resistor  $R$  is from left to right or from right to left in the following circumstances: (a) the current in winding  $A$  is from  $a$  to  $b$  and is increasing; (b) the current in winding  $A$  is from  $b$  to  $a$  and is decreasing; (c) the current in winding  $A$  is from  $b$  to  $a$  and is increasing.

Figure E29.20



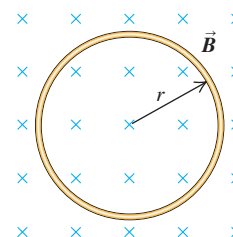
**29.21** • A small, circular ring is inside a larger loop that is connected to a battery and a switch (**Fig. E29.21**). Use Lenz's law to find the direction of the current induced in the small ring (a) just after switch  $S$  is closed; (b) after  $S$  has been closed a long time; (c) just after  $S$  has been reopened after it was closed for a long time.

Figure E29.21



**29.22** • **CALC** A circular loop of wire with radius  $r = 0.0250$  m and resistance  $R = 0.390 \Omega$  is in a region of spatially uniform magnetic field, as shown in **Fig. E29.22**. The magnetic field is directed into the plane of the figure. At  $t = 0$ ,  $B = 0$ . The magnetic field then begins increasing, with  $B(t) = (0.380 \text{ T/s}^3)t^3$ . What is the current in the loop (magnitude and direction) at the instant when  $B = 1.33$  T?

Figure E29.22



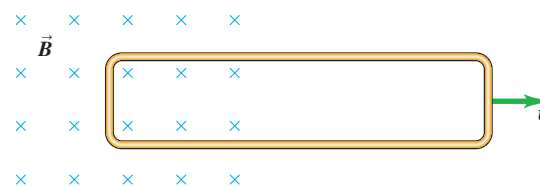
### Section 29.4 Motional Electromotive Force

**29.23** • A magnetic field of  $0.080$  T is in the  $y$ -direction. The velocity of wire segment  $S$  has a magnitude of  $78$  m/s and components of  $18$  m/s in the  $x$ -direction,  $24$  m/s in the  $y$ -direction, and  $72$  m/s in the  $z$ -direction. The segment has length  $0.50$  m and is parallel to the  $z$ -axis as it moves. (a) Find the motional emf induced between the ends of the segment. (b) What would the motional emf be if the wire segment was parallel to the  $y$ -axis?

**29.24** • A rectangular loop of wire with dimensions  $1.50$  cm by  $8.00$  cm and resistance  $R = 0.600 \Omega$  is being pulled to the right out of a region of uniform magnetic field. The magnetic field has magnitude  $B = 2.40$  T and is directed into the plane of **Fig. E29.24**. At the instant when the speed of the loop is  $3.00$  m/s and it is still partially in the field region, what force (magnitude and direction) does the magnetic field exert on the loop?

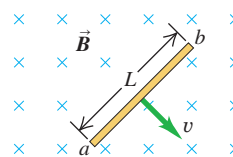
**29.25** • In **Fig. E29.25** a conducting rod of length  $L = 30.0$  cm moves in a magnetic field  $\vec{B}$  of magnitude  $0.450$  T directed into the plane of the figure.

Figure E29.24

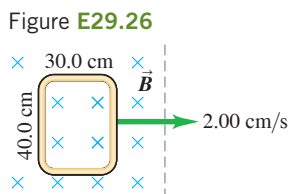


The rod moves with speed  $v = 5.00$  m/s in the direction shown. (a) What is the potential difference between the ends of the rod? (b) Which point,  $a$  or  $b$ , is at higher potential? (c) When the charges in the rod are in equilibrium, what are the magnitude and direction of the electric field within the rod? (d) When the charges in the rod are in equilibrium, which point,  $a$  or  $b$ , has an excess of positive charge? (e) What is the potential difference across the rod if it moves (i) parallel to  $ab$  and (ii) directly out of the page?

Figure E29.25

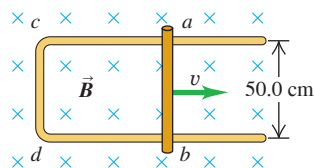


**29.26** • A rectangle measuring 30.0 cm by 40.0 cm is located inside a region of a spatially uniform magnetic field of 1.25 T, with the field perpendicular to the plane of the coil (**Fig. E29.26**). The coil is pulled out at a steady rate of 2.00 cm/s traveling perpendicular to the field lines. The region of the field ends abruptly as shown. Find the emf induced in this coil when it is (a) all inside the field; (b) partly inside the field; (c) all outside the field.



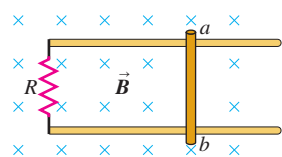
**29.27** • The conducting rod  $ab$  shown in **Fig. E29.27** makes contact with metal rails  $ca$  and  $db$ . The apparatus is in a uniform magnetic field of 0.800 T, perpendicular to the plane of the figure. (a) Find the magnitude of the emf induced in the rod when it is moving toward the right with a speed 7.50 m/s. (b) In what direction does the current flow in the rod? (c) If the resistance of the circuit  $abdc$  is  $1.50\ \Omega$  (assumed to be constant), find the force (magnitude and direction) required to keep the rod moving to the right with a constant speed of 7.50 m/s. You can ignore friction. (d) Compare the rate at which mechanical work is done by the force ( $Fv$ ) with the rate at which thermal energy is developed in the circuit ( $I^2R$ ).

Figure E29.27



**29.28** • A 0.650-m-long metal bar is pulled to the right at a steady 5.0 m/s perpendicular to a uniform, 0.750 T magnetic field. The bar rides on parallel metal rails connected through a  $25.0\ \Omega$  resistor (**Fig. E29.28**), so the apparatus makes a complete circuit. Ignore the resistance of the bar and the rails.

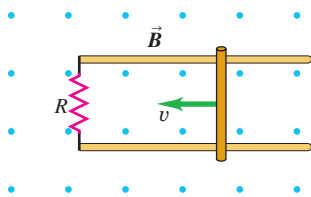
Figure E29.28



(a) Calculate the magnitude of the emf induced in the circuit. (b) Find the direction of the current induced in the circuit by using (i) the magnetic force on the charges in the moving bar; (ii) Faraday's law; (iii) Lenz's law. (c) Calculate the current through the resistor.

**29.29** • A 0.360-m-long metal bar is pulled to the left by an applied force  $F$ . The bar rides on parallel metal rails connected through a  $45.0\ \Omega$  resistor, as shown in **Fig. E29.29**, so the apparatus makes a complete circuit. You can ignore the resistance of the bar and rails. The circuit is in a uniform 0.650 T magnetic field that is directed out of the plane of the figure. At the instant when the bar is moving to the left at 5.90 m/s, (a) is the induced current in the circuit clockwise or counterclockwise and (b) what is the rate at which the applied force is doing work on the bar?

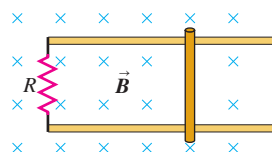
Figure E29.29



**29.30** • Consider the circuit shown in **Fig. E29.29**, but with the bar moving to the right with speed  $v$ . As in Exercise 29.29, the bar has length 0.360 m,  $R = 45.0\ \Omega$ , and  $B = 0.650\ \text{T}$ . (a) Is the induced current in the circuit clockwise or counterclockwise? (b) At an instant when the  $45.0\ \Omega$  resistor is dissipating electrical energy at a rate of  $0.840\ \text{J/s}$ , what is the speed of the bar?

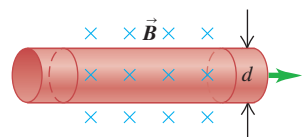
**29.31** • A 0.250-m-long bar moves on parallel rails that are connected through a  $6.00\ \Omega$  resistor, as shown in **Fig. E29.31**, so the apparatus makes a complete circuit. You can ignore the resistance of the bar and rails. The circuit is in a uniform magnetic field  $B = 1.20\ \text{T}$  that is directed into the plane of the figure. At an instant when the induced current in the circuit is counterclockwise and equal to 1.75 A, what is the velocity of the bar (magnitude and direction)?

Figure E29.31



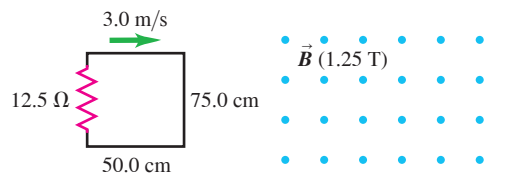
**29.32** • **BIO Measuring Blood Flow.** Blood contains positive and negative ions and thus is a conductor. A blood vessel, therefore, can be viewed as an electrical wire. We can even picture the flowing blood as a series of parallel conducting slabs whose thickness is the diameter  $d$  of the vessel moving with speed  $v$ . (See **Fig. E29.32**.) (a) If the blood vessel is placed in a magnetic field  $B$  perpendicular to the vessel, as in the figure, show that the motional potential difference induced across it is  $\mathcal{E} = vBd$ . (b) If you expect that the blood will be flowing at 15 cm/s for a vessel 5.0 mm in diameter, what strength of magnetic field will you need to produce a potential difference of 1.0 mV? (c) Show that the volume rate of flow ( $R$ ) of the blood is equal to  $R = \pi \mathcal{E} d / 4B$ . (Note: Although the method developed here is useful in measuring the rate of blood flow in a vessel, it is limited to use in surgery because measurement of the potential  $\mathcal{E}$  must be made directly across the vessel.)

Figure E29.32



**29.33** • A rectangular circuit is moved at a constant velocity of 3.0 m/s into, through, and then out of a uniform 1.25 T magnetic field, as shown in **Fig. E29.33**. The magnetic-field region is considerably wider than 50.0 cm. Find the magnitude and direction (clockwise or counterclockwise) of the current induced in the circuit as it is (a) going into the magnetic field; (b) totally within the magnetic field, but still moving; and (c) moving out of the field. (d) Sketch a graph of the current in this circuit as a function of time, including the preceding three cases.

Figure E29.33



### Section 29.5 Induced Electric Fields

**29.34** • A metal ring 4.50 cm in diameter is placed between the north and south poles of large magnets with the plane of its area perpendicular to the magnetic field. These magnets produce an initial uniform field of 1.12 T between them but are gradually pulled apart, causing this field to remain uniform but decrease steadily at  $0.250\ \text{T/s}$ . (a) What is the magnitude of the electric field induced in the ring? (b) In which direction (clockwise or counterclockwise) does the current flow as viewed by someone on the south pole of the magnet?

**29.35 ••** A long, thin solenoid has 400 turns per meter and radius 1.10 cm. The current in the solenoid is increasing at a uniform rate  $di/dt$ . The induced electric field at a point near the center of the solenoid and 3.50 cm from its axis is  $8.00 \times 10^{-6}$  V/m. Calculate  $di/dt$ .

**29.36 ••** A long, thin solenoid has 900 turns per meter and radius 2.50 cm. The current in the solenoid is increasing at a uniform rate of 36.0 A/s. What is the magnitude of the induced electric field at a point near the center of the solenoid and (a) 0.500 cm from the axis of the solenoid; (b) 1.00 cm from the axis of the solenoid?

**29.37 •** A long, straight solenoid with a cross-sectional area of  $8.00 \text{ cm}^2$  is wound with 90 turns of wire per centimeter, and the windings carry a current of 0.350 A. A second winding of 12 turns encircles the solenoid at its center. The current in the solenoid is turned off such that the magnetic field of the solenoid becomes zero in 0.0400 s. What is the average induced emf in the second winding?

### Section 29.7 Displacement Current and Maxwell's Equations

**29.38 •** A parallel-plate, air-filled capacitor is being charged as in Fig. 29.23. The circular plates have radius 4.00 cm, and at a particular instant the conduction current in the wires is 0.520 A. (a) What is the displacement current density  $j_D$  in the air space between the plates? (b) What is the rate at which the electric field between the plates is changing? (c) What is the induced magnetic field between the plates at a distance of 2.00 cm from the axis? (d) At 1.00 cm from the axis?

**29.39 • Displacement Current in a Dielectric.** Suppose that the parallel plates in Fig. 29.23 have an area of  $3.00 \text{ cm}^2$  and are separated by a 2.50-mm-thick sheet of dielectric that completely fills the volume between the plates. The dielectric has dielectric constant 4.70. (You can ignore fringing effects.) At a certain instant, the potential difference between the plates is 120 V and the conduction current  $i_C$  equals 6.00 mA. At this instant, what are (a) the charge  $q$  on each plate; (b) the rate of change of charge on the plates; (c) the displacement current in the dielectric?

**29.40 • CALC** In Fig. 29.23 the capacitor plates have area  $5.00 \text{ cm}^2$  and separation 2.00 mm. The plates are in vacuum. The charging current  $i_C$  has a constant value of 1.80 mA. At  $t = 0$  the charge on the plates is zero. (a) Calculate the charge on the plates, the electric field between the plates, and the potential difference between the plates when  $t = 0.500 \mu\text{s}$ . (b) Calculate  $dE/dt$ , the time rate of change of the electric field between the plates. Does  $dE/dt$  vary in time? (c) Calculate the displacement current density  $j_D$  between the plates, and from this the total displacement current  $i_D$ . How do  $i_C$  and  $i_D$  compare?

**29.41 • CALC** The electric flux is  $(4.0 \text{ V} \cdot \text{m/s}^5)t^5$  through a certain area of a dielectric that has dielectric constant 2.5. (a) Find the displacement current through that area at  $t = 1.5 \text{ s}$ . (b) At what time was the displacement current  $\frac{1}{6}$  as much?

### Section 29.8 Superconductivity

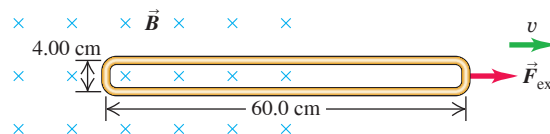
**29.42 •** At temperatures near absolute zero,  $B_c$  approaches 0.142 T for vanadium, a type-I superconductor. The normal phase of vanadium has a magnetic susceptibility close to zero. Consider a long, thin vanadium cylinder with its axis parallel to an external magnetic field  $\vec{B}_0$  in the  $+x$ -direction. At points far from the ends of the cylinder, by symmetry, all the magnetic vectors are parallel to the  $x$ -axis. At temperatures near absolute zero, what are the resultant magnetic field  $\vec{B}$  and the magnetization  $\vec{M}$  inside and outside the cylinder (far from the ends) for (a)  $\vec{B}_0 = (0.130 \text{ T})\hat{i}$  and (b)  $\vec{B}_0 = (0.260 \text{ T})\hat{i}$ ?

## PROBLEMS

**29.43 •• CP** A motor vehicle generates electrical power using an alternator, which employs electromagnetic induction to convert mechanical energy to electrical energy. The alternator acts as a dc generator (Example 29.4). The alternator maintains and replenishes charge on the car's battery and operates headlights, radiator fans, windshield wipers, power windows, computer systems, sensors, sound systems, and other components. (a) A typical car battery provides 70 amp-hours of charge. How many coulombs is that? (b) If headlights each draw 20 A of current, a radiator fan draws 10 A, and windshield wipers each draw 5 A, estimate the peak current needed for a car to operate on a rainy night. (c) A car's alternator supplies an average emf of 14 V as emf induced in a sequence of stator coils in the presence of a magnetic field created by rotor coil electromagnets turned by a pulley system. A stator coil may have 42 windings and a cross-sectional diameter of 5.0 cm, and it rotates at 400 Hz. Estimate the strength of the magnetic field generated by a rotor coil.

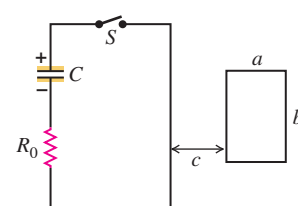
**29.44 ••** A very long, rectangular loop of wire can slide without friction on a horizontal surface. Initially the loop has part of its area in a region of uniform magnetic field that has magnitude  $B = 2.90 \text{ T}$  and is perpendicular to the plane of the loop. The loop has dimensions 4.00 cm by 60.0 cm, mass 24.0 g, and resistance  $R = 5.00 \times 10^{-3} \Omega$ . The loop is initially at rest; then a constant force  $F_{\text{ext}} = 0.180 \text{ N}$  is applied to the loop to pull it out of the field (Fig. P29.44). (a) What is the acceleration of the loop when  $v = 3.00 \text{ cm/s}$ ? (b) What are the loop's terminal speed and acceleration when the loop is moving at that terminal speed? (c) What is the acceleration of the loop when it is completely out of the magnetic field?

Figure P29.44



**29.45 •• CP CALC** In the circuit shown in Fig. P29.45, the capacitor has capacitance  $C = 20 \mu\text{F}$  and is initially charged to 100 V with the polarity shown. The resistor  $R_0$  has resistance  $10 \Omega$ . At time  $t = 0$  the switch  $S$  is closed. The small circuit is not connected in any way to the large one. The wire of the small circuit has a resistance of  $1.0 \Omega/\text{m}$  and contains 25 loops. The large circuit is a rectangle 2.0 m by 4.0 m, while the small one has dimensions  $a = 10.0 \text{ cm}$  and  $b = 20.0 \text{ cm}$ . The distance  $c$  is 5.0 cm. (The figure is not drawn to scale.) Both circuits are held stationary. Assume that only the wire nearest the small circuit produces an appreciable magnetic field through it. (a) Find the current in the large circuit 200  $\mu\text{s}$  after  $S$  is closed. (b) Find the current in the small circuit 200  $\mu\text{s}$  after  $S$  is closed. (Hint: See Exercise 29.7.) (c) Find the direction of the current in the small circuit. (d) Justify why we can ignore the magnetic field from all the wires of the large circuit except for the wire closest to the small circuit.

Figure P29.45



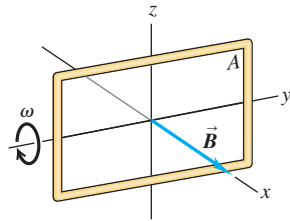
**29.46 •• CP CALC** In the circuit in Fig. P29.45, an emf of 90.0 V is added in series with the capacitor and the resistor, and the capacitor is initially uncharged. The emf is placed between the capacitor and switch  $S$ , with the positive terminal of the emf adjacent to the capacitor. Otherwise, the two circuits are the same as in Problem 29.45. The switch is closed at  $t = 0$ . When the current in the large circuit is 5.00 A, what are the magnitude and direction of the induced current in the small circuit?



**29.47 •• CALC** A very long, straight solenoid with a cross-sectional area of  $2.00 \text{ cm}^2$  is wound with 90.0 turns of wire per centimeter. Starting at  $t = 0$ , the current in the solenoid is increasing according to  $i(t) = (0.160 \text{ A/s}^2)t^2$ . A secondary winding of 5 turns encircles the solenoid at its center, such that the secondary winding has the same cross-sectional area as the solenoid. What is the magnitude of the emf induced in the secondary winding at the instant that the current in the solenoid is 3.20 A?

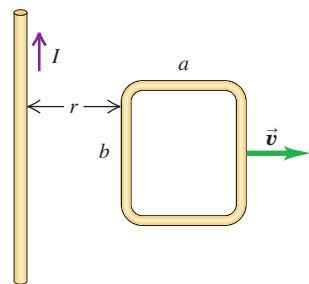
**29.48 •** Suppose the loop in **Fig. P29.48** Figure P29.48

is (a) rotated about the  $y$ -axis; (b) rotated about the  $x$ -axis; (c) rotated about an edge parallel to the  $z$ -axis. What is the maximum induced emf in each case if  $A = 600 \text{ cm}^2$ ,  $\omega = 35.0 \text{ rad/s}$ , and  $B = 0.320 \text{ T}$ ?



**29.49 •** In **Fig. P29.49** the loop is being pulled to the right at constant speed  $v$ . A constant current  $I$  flows in the long wire, in the direction shown. (a) Calculate the magnitude of the net emf  $\mathcal{E}$  induced in the loop. Do this two ways: (i) by using Faraday's law of induction (*Hint*: See Exercise 29.7) and (ii) by looking at the emf induced in each segment of the loop due to its motion. (b) Find the direction (clockwise or counterclockwise) of the current induced in the loop. Do this two ways: (i) using Lenz's law and (ii) using the magnetic force on charges in the loop. (c) Check your answer for the emf in part (a) in the following special cases to see whether it is physically reasonable: (i) The loop is stationary; (ii) the loop is very thin, so  $a \rightarrow 0$ ; (iii) the loop gets very far from the wire.

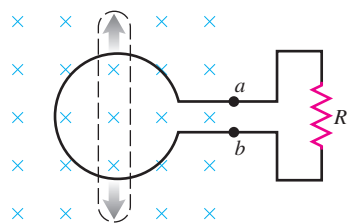
Figure P29.49



**29.50 ••** If you secure a refrigerator magnet about 2 mm from the metallic surface of a refrigerator door and then move the magnet sideways, you can feel a resistive force, indicating the presence of eddy currents in the surface. (a) Estimate the magnetic field strength  $B$  of the magnet to be 5 mT (Problem 28.53) and assume the magnet is rectangular with dimensions 4 cm wide by 2 cm high, so its area  $A$  is  $8 \text{ cm}^2$ . Now estimate the magnetic flux  $\Phi_B$  into the refrigerator door behind the magnet. (b) If you move the magnet sideways at a speed of 2 cm/s, what is a corresponding estimate of the time rate at which the magnetic flux through an area  $A$  fixed on the refrigerator is changing as the magnet passes over? Use this estimate to estimate the emf induced under the rectangle on the door's surface.

**29.51 •** A flexible circular loop Figure P29.51

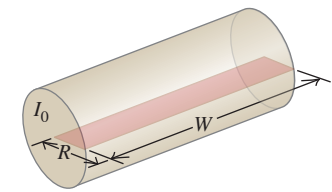
6.50 cm in diameter lies in a magnetic field with magnitude 1.35 T, directed into the plane of the page as shown in **Fig. P29.51**. The loop is pulled at the points indicated by the arrows, forming a loop of zero area in 0.250 s. (a) Find the average induced emf in the circuit. (b) What is the direction of the current in  $R$ : from  $a$  to  $b$  or from  $b$  to  $a$ ? Explain your reasoning.



**29.52 ••• CALC** A conducting rod with length  $L = 0.200 \text{ m}$ , mass  $m = 0.120 \text{ kg}$ , and resistance  $R = 80.0 \Omega$  moves without friction on metal rails as shown in **Fig. 29.11**. A uniform magnetic field with magnitude  $B = 1.50 \text{ T}$  is directed into the plane of the figure. The rod is initially at rest, and then a constant force with magnitude  $F = 1.90 \text{ N}$  and directed to the right is applied to the rod. How many seconds after the force is applied does the rod reach a speed of 25.0 m/s?

**29.53 •• CALC** A very long, cylindrical wire of radius  $R$  carries a current  $I_0$  uniformly distributed across the cross section of the wire. Calculate the magnetic flux through a rectangle that has one side of length  $W$  running down the center of the wire and another side of length  $R$ , as shown in **Fig. P29.53** (see Exercise 29.7).

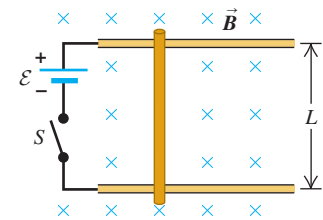
Figure P29.53



**29.54 •• CP CALC Terminal Speed.** A bar of length

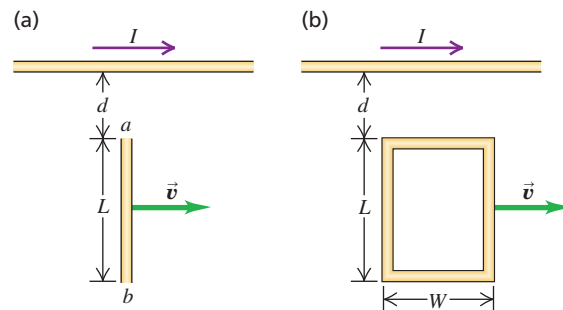
$L = 0.36 \text{ m}$  is free to slide without friction on horizontal rails as shown in **Fig. P29.54**. A uniform magnetic field  $B = 2.4 \text{ T}$  is directed into the plane of the figure. At one end of the rails there is a battery with emf  $\mathcal{E} = 12 \text{ V}$  and a switch  $S$ . The bar has mass 0.90 kg and resistance  $5.0 \Omega$ ; ignore all other resistance in the circuit. The switch is closed at time  $t = 0$ . (a) Sketch the bar's speed as a function of time. (b) Just after the switch is closed, what is the acceleration of the bar? (c) What is the acceleration of the bar when its speed is 2.0 m/s? (d) What is the bar's terminal speed?

Figure P29.54



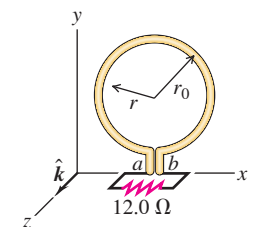
**29.55 • CALC** The long, straight wire shown in **Fig. P29.55a** carries constant current  $I$ . A metal bar with length  $L$  is moving at constant velocity  $\vec{v}$ , as shown in the figure. Point  $a$  is a distance  $d$  from the wire. (a) Calculate the emf induced in the bar. (b) Which point,  $a$  or  $b$ , is at higher potential? (c) If the bar is replaced by a rectangular wire loop of resistance  $R$  (**Fig. P29.55b**), what is the magnitude of the current induced in the loop?

Figure P29.55



**29.56 • CALC** A circular conducting ring with radius  $r_0 = 0.0420 \text{ m}$  lies in the  $xy$ -plane in a region of uniform magnetic field  $\vec{B} = B_0[1 - 3(t/t_0)^2 + 2(t/t_0)^3]\hat{k}$ . In this expression,  $t_0 = 0.0100 \text{ s}$  and is constant,  $t$  is time,  $\hat{k}$  is the unit vector in the  $+z$ -direction, and  $B_0 = 0.0800 \text{ T}$  and is constant. At points  $a$  and  $b$  (**Fig. P29.56**) there is a small gap in the ring with wires leading to an external circuit of resistance  $R = 12.0 \Omega$ . There is no magnetic field at the location of the external circuit. (a) Derive an expression, as a function of time, for the total magnetic flux  $\Phi_B$  through the ring. (b) Determine the emf induced in the ring at time  $t = 5.00 \times 10^{-3} \text{ s}$ . What is the polarity of the emf? (c) Because of the internal resistance of the ring, the current through  $R$  at the time given in part (b) is only 3.00 mA. Determine the internal resistance of the ring. (d) Determine the emf in the ring at a time  $t = 1.21 \times 10^{-2} \text{ s}$ . What is the polarity of the emf? (e) Determine the time at which the current through  $R$  reverses its direction.

Figure P29.56



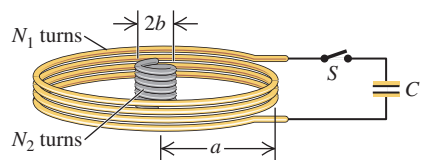
**29.57 • CALC** A slender rod, 0.240 m long, rotates with an angular speed of 8.80 rad/s about an axis through one end and perpendicular to the rod. The plane of rotation of the rod is perpendicular to a uniform magnetic field with a magnitude of 0.650 T. (a) What is the induced emf in the rod? (b) What is the potential difference between its ends? (c) Suppose instead the rod rotates at 8.80 rad/s about an axis through its center and perpendicular to the rod. In this case, what is the potential difference between the ends of the rod? Between the center of the rod and one end?

**29.58 ••** A 25.0-cm-long metal rod lies in the  $xy$ -plane and makes an angle of  $36.9^\circ$  with the positive  $x$ -axis and an angle of  $53.1^\circ$  with the positive  $y$ -axis. The rod is moving in the  $+x$ -direction with a speed of 6.80 m/s. The rod is in a uniform magnetic field  $\vec{B} = (0.120 \text{ T})\hat{i} - (0.220 \text{ T})\hat{j} - (0.0900 \text{ T})\hat{k}$ . (a) What is the magnitude of the emf induced in the rod? (b) Indicate in a sketch which end of the rod is at higher potential.

**29.59 •• CP CALC** A rectangular loop with width  $L$  and a slidewire with mass  $m$  are as shown in **Fig. P29.59**. A uniform magnetic field  $\vec{B}$  is directed perpendicular to the plane of the loop into the plane of the figure. The slidewire is given an initial speed of  $v_0$  and then released. There is no friction between the slidewire and the loop, and the resistance of the loop is negligible in comparison to the resistance  $R$  of the slidewire. (a) Obtain an expression for  $F$ , the magnitude of the force exerted on the wire while it is moving at speed  $v$ . (b) Show that the distance  $x$  that the wire moves before coming to rest is  $x = mv_0 R / L^2 B^2$ .

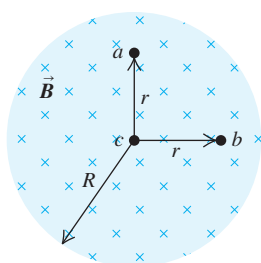
**29.60 ••• CP** A circular coil with  $N_1 = 5000$  turns is made of a conducting material with resistance  $0.0100 \Omega/\text{m}$  and radius  $a = 40.0 \text{ cm}$ . The coil is attached to a  $C = 10.00 \mu\text{F}$  capacitor as shown in **Fig. P29.60**. A second coil with radius  $b = 4.00 \text{ cm}$ , made of the same wire, with  $N_2 = 100$  turns, is concentric with the first coil and parallel to it. The capacitor has a charge of  $+100 \mu\text{C}$  on its upper plate, and the switch  $S$  is open. At time  $t = 0$  the switch is closed. (a) What is the magnitude of the current in the larger coil immediately after the switch is closed? (b) What is the magnetic flux through each turn of the smaller coil immediately after the switch is closed? (Since  $b \ll a$ , we may treat the magnetic field in the smaller coil due to the larger coil as uniform.) (c) What is the direction of the current in the smaller coil immediately after the switch is closed? (d) What is the direction of the current in the smaller coil at  $t = 1.26 \text{ ms}$ ? (e) What is the magnitude of the current in the smaller coil at  $t = 1.26 \text{ ms}$ ?

Figure P29.60



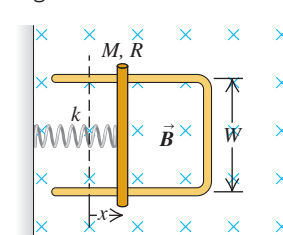
**29.61 •** The magnetic field  $\vec{B}$ , at all points within a circular region of radius  $R$ , is uniform in space and directed into the plane of the page as shown in **Fig. P29.61**. (The region could be a cross section inside the windings of a long, straight solenoid.) If the magnetic field is increasing at a rate  $dB/dt$ , what are the magnitude and direction of the force on a stationary positive point charge  $q$  located at points  $a$ ,  $b$ , and  $c$ ? (Point  $a$  is a distance  $r$  above the center of the region, point  $b$  is a distance  $r$  to the right of the center, and point  $c$  is at the center of the region.)

Figure P29.61



**29.62 ••• CP** A bar with mass  $M = 1.20 \text{ kg}$  and resistance  $R = 0.500 \Omega$  slides without friction on a horizontal U-shaped rail with width  $W = 40.0 \text{ cm}$  and negligible resistance. The bar is attached to a spring with spring constant  $k = 90.0 \text{ N/m}$ , as shown in **Fig. P29.62**. A constant magnetic field with magnitude 1.00 T points into the plane everywhere in the vicinity. At time  $t = 0$  the bar is stretched beyond its equilibrium position by an amount  $x = 10.0 \text{ cm}$  and released from rest.

Figure P29.62

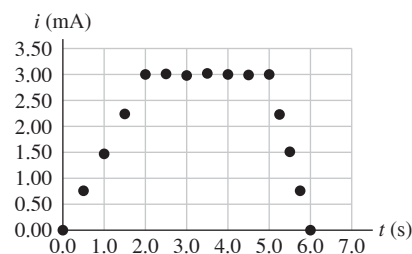


(a) This system behaves like a damped oscillator, described by Eq. (14.41). What is the damping coefficient  $b$ ? (b) With what frequency does the bar oscillate around its equilibrium position? (c) What is the amplitude of the motion at time  $t = 5.00 \text{ s}$ ? (d) What is the magnitude of the current in the bar when it passes the equilibrium position for the first time? (e) What is the direction of that current?

**29.63 ••• CALC** A dielectric of permittivity  $3.5 \times 10^{-11} \text{ F/m}$  completely fills the volume between two capacitor plates. For  $t > 0$  the electric flux through the dielectric is  $(8.0 \times 10^3 \text{ V} \cdot \text{m/s}^3)t^3$ . The dielectric is ideal and nonmagnetic; the conduction current in the dielectric is zero. At what time does the displacement current in the dielectric equal  $21 \mu\text{A}$ ?

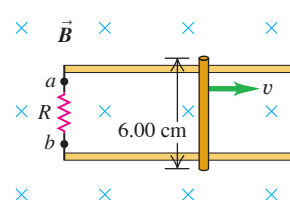
**29.64 •• DATA** You are evaluating the performance of a large electromagnet. The magnetic field of the electromagnet is zero at  $t = 0$  and increases as the current through the windings of the electromagnet is increased. You determine the magnetic field as a function of time by measuring the time dependence of the current induced in a small coil that you insert between the poles of the electromagnet, with the plane of the coil parallel to the pole faces as in **Fig. 29.5**. The coil has 4 turns, a radius of  $0.800 \text{ cm}$ , and a resistance of  $0.250 \Omega$ . You measure the current  $i$  in the coil as a function of time  $t$ . Your results are shown in **Fig. P29.64**. Throughout your measurements, the current induced in the coil remains in the same direction. Calculate the magnetic field at the location of the coil for (a)  $t = 2.00 \text{ s}$ , (b)  $t = 5.00 \text{ s}$ , and (c)  $t = 6.00 \text{ s}$ .

Figure P29.64



**29.65 •• DATA** You are conducting an experiment in which a metal bar of length  $6.00 \text{ cm}$  and mass  $0.200 \text{ kg}$  slides without friction on two parallel metal rails (**Fig. P29.65**). A resistor with resistance  $R = 0.800 \Omega$  is connected across one end of the rails so that the bar, rails, and resistor form a complete conducting path. The resistances of the rails and of the bar are much less than  $R$  and can be ignored. The entire apparatus is in a uniform magnetic field  $\vec{B}$  that is directed into the plane of the figure. You give the bar an initial velocity  $v = 20.0 \text{ cm/s}$  to the right and then release it, so that the only force on the bar then is the force exerted by the magnetic field. Using high-speed photography, you measure the magnitude of the acceleration of the bar as a function of its speed. Your results are given in the table (next page):

Figure P29.65

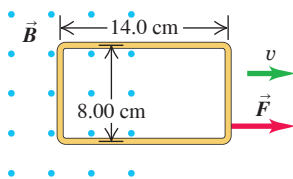


|                          |      |      |      |      |      |     |
|--------------------------|------|------|------|------|------|-----|
| $v$ (cm/s)               | 20.0 | 16.0 | 14.0 | 12.0 | 10.0 | 8.0 |
| $a$ (cm/s <sup>2</sup> ) | 6.2  | 4.9  | 4.3  | 3.7  | 3.1  | 2.5 |

(a) Plot the data as a graph of  $a$  versus  $v$ . Explain why the data points plotted this way lie close to a straight line, and determine the slope of the best-fit straight line for the data. (b) Use your graph from part (a) to calculate the magnitude  $B$  of the magnetic field. (c) While the bar is moving, which end of the resistor,  $a$  or  $b$ , is at higher potential? (d) How many seconds does it take the speed of the bar to decrease from 20.0 cm/s to 10.0 cm/s?

**29.66 ••• DATA** You measure the magnitude of the external force  $\vec{F}$  that must be applied to a rectangular conducting loop to pull it at constant speed  $v$  out of a region of uniform magnetic field  $\vec{B}$  that is directed out of the plane of **Fig. P29.66**. The loop has dimensions 14.0 cm by 8.00 cm and resistance  $4.00 \times 10^{-3} \Omega$ ; it does not change shape as it moves. The measurements you collect are listed in the table.

Figure P29.66



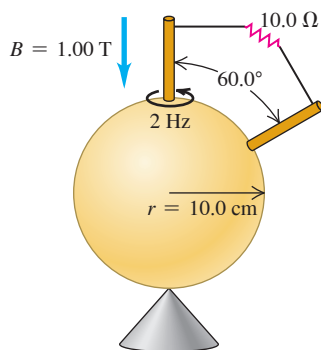
|            |      |      |      |      |      |
|------------|------|------|------|------|------|
| $F$ (N)    | 0.10 | 0.21 | 0.31 | 0.41 | 0.52 |
| $v$ (cm/s) | 2.0  | 4.0  | 6.0  | 8.0  | 10.0 |

(a) Plot the data as a graph of  $F$  versus  $v$ . Explain why the data points plotted this way lie close to a straight line, and determine the slope of the best-fit straight line for the data. (b) Use your graph from part (a) to calculate the magnitude  $B$  of the uniform magnetic field. (c) In **Fig. P29.66**, is the current induced in the loop clockwise or counterclockwise? (d) At what rate is electrical energy being dissipated in the loop when the speed of the loop is 5.00 cm/s?

### CHALLENGE PROBLEMS

**29.67 ••• CP** A conducting spherical shell with radius 10.0 cm spins about a vertical axis twice every second in the presence of a constant magnetic field  $\vec{B}$  with magnitude 1.00 T that points downward. Two conducting rods supported by a frame contact the sphere with conducting brushes and extend away from the sphere radially. One rod extends from the top of the sphere and the other forms a  $60.0^\circ$  angle with vertical, as shown in **Fig. P29.67**.

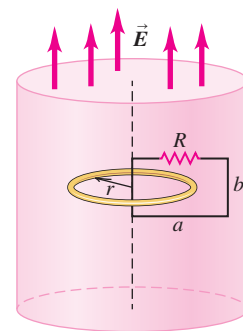
Figure P29.67



The outer ends of the rods are connected to each other by a conducting wire that includes a  $10.0 \Omega$  resistor. (a) Construct a Cartesian coordinate system with the origin at the center of the sphere, the  $z$ -axis pointing upward, and the  $y$ -axis pointing rightward so that both rods lie in the  $yz$ -plane. What is the velocity  $\vec{v}$  of a point on the sphere in the  $yz$ -plane at angle  $\theta$  measured from the positive  $z$ -axis? (b) What is the vector product  $\vec{v} \times \vec{B}$ ? (c) What is the line element  $d\vec{l}$  along the shortest path on the sphere from the upper rod to the angled rod at angle  $\theta$ ? (d) What is the magnitude of the current in the wire? (e) What direction does the current flow in the vertical rod?

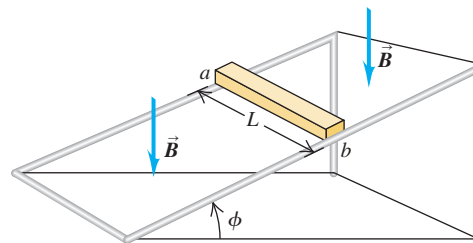
**29.68 ••• CP** A uniform electric field is directed axially in a cylindrical region that includes a rectangular loop of wire with total resistance  $R$ . This loop has radially oriented width  $a$  and axially oriented length  $b$ , and sits tight against the cylinder axis, as shown in **Fig. P29.68**. The electric field is zero at time  $t = 0$  and then increases in time according to  $\vec{E} = \eta t^2 \hat{k}$ , where  $\eta$  is a constant with units of  $\text{V}/(\text{m} \cdot \text{s}^2)$ . (a) What is the magnitude of the displacement current through a circular loop centered on the cylinder axis with radius  $r \leq a$ , at time  $t$ ? (b) Use Ampere's law to determine the magnitude of the magnetic field a distance  $r \leq a$  from the cylinder axis at time  $t$ . (c) What is the magnetic flux at time  $t$  through the rectangular wire loop? (d) What magnitude of current flows in the wire? (e) Does the current flow clockwise or counterclockwise from the perspective shown in the figure?

Figure P29.68



**29.69 •••** A metal bar with length  $L$ , mass  $m$ , and resistance  $R$  is placed on frictionless metal rails that are inclined at an angle  $\phi$  above the horizontal. The rails have negligible resistance. A uniform magnetic field of magnitude  $B$  is directed downward as shown in **Fig. P29.69**. The bar is released from rest and slides down the rails. (a) Is the direction of the current induced in the bar from  $a$  to  $b$  or from  $b$  to  $a$ ? (b) What is the terminal speed of the bar? (c) What is the induced current in the bar when the terminal speed has been reached? (d) After the terminal speed has been reached, at what rate is electrical energy being converted to thermal energy in the resistance of the bar? (e) After the terminal speed has been reached, at what rate is work being done on the bar by gravity? Compare your answer to that in part (d).

Figure P29.69

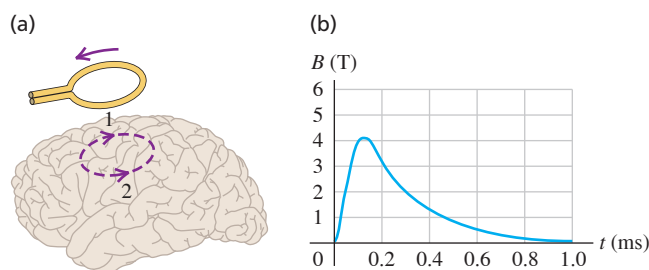


**29.70 ••• CP CALC** A square, conducting, wire loop of side  $L$ , total mass  $m$ , and total resistance  $R$  initially lies in the horizontal  $xy$ -plane, with corners at  $(x, y, z) = (0, 0, 0)$ ,  $(0, L, 0)$ ,  $(L, 0, 0)$ , and  $(L, L, 0)$ . There is a uniform, upward magnetic field  $\vec{B} = B\hat{k}$  in the space within and around the loop. The side of the loop that extends from  $(0, 0, 0)$  to  $(L, 0, 0)$  is held in place on the  $x$ -axis; the rest of the loop is free to pivot around this axis. When the loop is released, it begins to rotate due to the gravitational torque. (a) Find the *net* torque (magnitude and direction) that acts on the loop when it has rotated through an angle  $\phi$  from its original orientation and is rotating downward at an angular speed  $\omega$ . (b) Find the angular acceleration of the loop at the instant described in part (a). (c) Compared to the case with zero magnetic field, does it take the loop a longer or shorter time to rotate through  $90^\circ$ ? Explain. (d) Is mechanical energy conserved as the loop rotates downward? Explain.

# MCAT-STYLE PASSAGE PROBLEMS

**BIO Stimulating the Brain.** Communication in the nervous system is based on the propagation of electrical signals called *action potentials* along axons, which are extensions of nerve cells (see the MCAT-style Passage Problems in Chapter 26). Action potentials are generated when the electric potential difference across the membrane of the nerve cell changes: Specifically, the inside of the cell becomes more positive. Researchers in clinical medicine and neurobiology cannot stimulate nerves (even noninvasively) at specific locations in conscious human subjects. Using electrodes to apply current to the skin is painful and requires large currents, which could be dangerous.

Anthony Barker and colleagues at the University of Sheffield in England developed a technique called *transcranial magnetic stimulation* (TMS). In this widely used procedure, a coil positioned near the skull produces a time-varying magnetic field that induces in the conductive tissue of the brain (see part (a) of the figure) electric currents that are sufficient to cause action potentials in nerve cells. For example, if the coil is placed near the motor cortex (the region of the brain that controls voluntary movement), scientists can monitor muscle contraction and assess the connections between the brain and the muscles. Part (b) of the figure is a graph of the typical dependence on time  $t$  of the magnetic field  $B$  produced by the coil.



**29.71** In part (a) of the figure, a current pulse increases to a peak and then decreases to zero in the direction shown in the stimulating coil. What will be the direction of the induced current (dashed line) in the brain tissue? (a) 1; (b) 2; (c) 1 while the current increases in the stimulating coil, 2 while the current decreases; (d) 2 while the current increases in the stimulating coil, 1 while the current decreases.

# ANSWERS

## Chapter Opening Question ?

(iv) As the magnetic stripe moves through the card reader, the coded pattern of magnetization in the stripe causes a varying magnetic flux. An electric field is induced, which causes a current in the reader's circuits. If the card does not move, there is no induced current and none of the credit card's information is read.

## Key Example VARIATION Problems

**VP29.8.1** (a)  $+2.53 \times 10^{-4} \text{ T} \cdot \text{m}^2$  at  $t = 0$ ,  $-1.99 \times 10^{-4} \text{ T} \cdot \text{m}^2$  at  $t = 2.00 \text{ s}$  (b)  $2.26 \times 10^{-4} \text{ V}$

**VP29.8.2** (a)  $1.01 \times 10^{-2} \text{ V}$  (b)  $6.95 \times 10^{-4} \text{ A}$

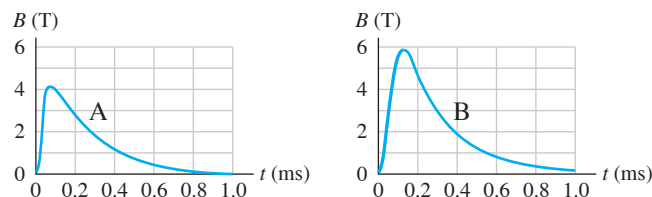
**VP29.8.3** (a)  $0.163 \text{ V}$  (b) counterclockwise

**VP29.8.4** (a)  $-\omega NB_0 A \cos \omega t$  (b) clockwise at  $t = 0$ , counterclockwise at  $t = \pi/\omega$

**29.72** Consider the brain tissue at the level of the dashed line to be a series of concentric circles, each behaving independently of the others. Where will the induced emf be the greatest? (a) At the center of the dashed line; (b) at the periphery of the dashed line; (c) nowhere—it will be the same in all concentric circles; (d) at the center while the stimulating current increases, at the periphery while the current decreases.

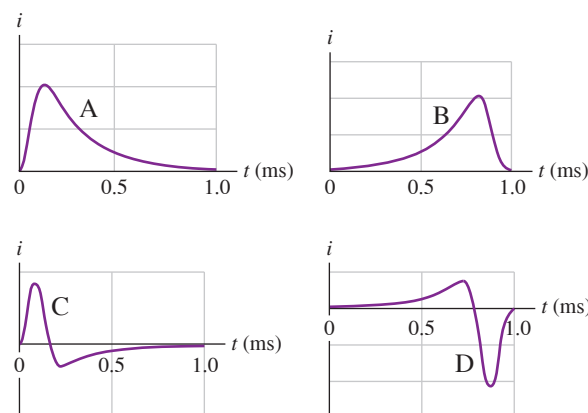
**29.73** It may be desirable to increase the maximum induced current in the brain tissue. In **Fig. P29.73**, which time-dependent graph of the magnetic field  $B$  in the coil achieves that goal? Assume that everything else remains constant. (a) A; (b) B; (c) either A or B; (d) neither A nor B.

Figure P29.73



**29.74** Which graph in **Fig. P29.74** best represents the time  $t$  dependence of the current  $i$  induced in the brain tissue, assuming that this tissue can be modeled as a resistive circuit? (The units of  $i$  are arbitrary.) (a) A; (b) B; (c) C; (d) D.

Figure P29.74



**VP29.9.1** (a)  $0.187 \text{ V}$  (b)  $2.34 \text{ V/m}$ ,  $+x$ -direction

**VP29.9.2**  $0.900 \text{ A}$ , clockwise

**VP29.9.3** (a)  $2.40 \times 10^{-2} \text{ V}$  (b) end b

**VP29.9.4**  $4.12 \text{ m/s}$

**VP29.11.1** (a)  $1.81 \times 10^3 \text{ turns/meter}$  (b)  $7.70 \times 10^{-5} \text{ V/m}$

**VP29.11.2**  $2.70 \times 10^{-5} \text{ V/m}$

**VP29.11.3** (a)  $-7.77 \text{ A/s}$  (b)  $+11.7 \text{ A/s}$

**VP29.11.4**  $5.52 \times 10^{-6} \text{ A}$

## Bridging Problem

$$v_{\text{terminal}} = 16\rho_m\rho_R g/B^2$$



? Many traffic lights change when a car rolls up to the intersection. This process works because the car contains (i) conducting material; (ii) insulating material that carries a net electric charge; (iii) ferromagnetic material; (iv) ferromagnetic material that is already magnetized.



# 30 Inductance

## LEARNING OUTCOMES

### In this chapter, you'll learn...

- 30.1 How a time-varying current in one coil can induce an emf in a second, unconnected coil.
- 30.2 How to relate the induced emf in a circuit to the rate of change of current in the same circuit.
- 30.3 How to calculate the energy stored in a magnetic field.
- 30.4 How to analyze circuits that include both a resistor and an inductor (coil).
- 30.5 Why electrical oscillations occur in circuits that include both an inductor and a capacitor.
- 30.6 Why oscillations decay in circuits with an inductor, a resistor, and a capacitor.

### You'll need to review...

- 14.2, 14.3, 14.7 Simple harmonic motion, damped oscillations.
- 24.1, 24.3 Capacitance, electric-field energy.
- 26.2, 26.4 Kirchhoff's rules,  $R$ - $C$  circuits.
- 28.4, 28.7, 28.8 Magnetic forces between conductors; field of a solenoid; permeability.
- 29.2, 29.3, 29.7 Faraday's law; Lenz's law; conservative and nonconservative electric fields.

Take a length of copper wire and wrap it around a pencil to form a coil. If you put this coil in a circuit, the coil behaves quite differently than a straight piece of wire. In an ordinary gasoline-powered car, a coil of this kind makes it possible for the 12 volt car battery to provide thousands of volts to the spark plugs in order for the plugs to fire and make the engine run. Other coils are used to keep fluorescent light fixtures shining. Larger coils placed under city streets are used to control the operation of traffic signals. All of these applications, and many others, involve the *induction* effects that we studied in Chapter 29.

A changing current in a coil induces an emf in an adjacent coil. The coupling between the coils is described by their *mutual inductance*. A changing current in a coil also induces an emf in that same coil. Such a coil is called an *inductor*, and the relationship of current to emf is described by the *inductance* (also called *self-inductance*) of the coil. If a coil is initially carrying a current, energy is released when the current decreases; this principle is used in automotive ignition systems. We'll find that this released energy was stored in the magnetic field caused by the current that was initially in the coil, and we'll look at some of the practical applications of magnetic-field energy.

We'll also take a first look at what happens when an inductor is part of a circuit. In Chapter 31 we'll go on to study how inductors behave in alternating-current circuits, and we'll learn why inductors play an essential role in modern electronics.

## 30.1 MUTUAL INDUCTANCE

In Section 28.4 we considered the magnetic interaction between two wires carrying *steady* currents; the current in one wire causes a magnetic field, which exerts a force on the current in the second wire. But an additional interaction arises between two circuits when there is a *changing* current in one of the circuits. Consider two neighboring coils of wire, as in **Fig. 30.1**. A current flowing in coil 1 produces a magnetic field  $\mathbf{B}$  and hence a magnetic flux through coil 2. If the current in coil 1 changes, the flux through coil 2 changes as well; according to Faraday's law (Section 29.2), this induces an emf in coil 2. In this way, a change in the current in one circuit can induce a current in a second circuit.

Let's analyze the situation shown in Fig. 30.1 in more detail. We'll use lowercase letters to represent quantities that vary with time; for example, a time-varying current is  $i$ , often with a subscript to identify the circuit. In Fig. 30.1 a current  $i_1$  in coil 1 sets up a magnetic field  $\vec{B}$ , and some of the (blue) field lines pass through coil 2. We denote the magnetic flux through *each* turn of coil 2, caused by the current  $i_1$  in coil 1, as  $\Phi_{B2}$ . (If the flux is different through different turns of the coil, then  $\Phi_{B2}$  denotes the *average* flux.) The magnetic field is proportional to  $i_1$ , so  $\Phi_{B2}$  is also proportional to  $i_1$ . When  $i_1$  changes,  $\Phi_{B2}$  changes; this changing flux induces an emf  $\mathcal{E}_2$  in coil 2, given by

$$\mathcal{E}_2 = -N_2 \frac{d\Phi_{B2}}{dt} \quad (30.1)$$

We could represent the proportionality of  $\Phi_{B2}$  and  $i_1$  in the form  $\Phi_{B2} = (\text{constant})i_1$ , but instead it is more convenient to include the number of turns  $N_2$  in the relationship. Introducing a proportionality constant  $M_{21}$ , called the **mutual inductance** of the two coils, we write

$$N_2\Phi_{B2} = M_{21}i_1 \quad (30.2)$$

where  $\Phi_{B2}$  is the flux through a *single* turn of coil 2. From this,

$$N_2 \frac{d\Phi_{B2}}{dt} = M_{21} \frac{di_1}{dt}$$

and we can rewrite Eq. (30.1) as

$$\mathcal{E}_2 = -M_{21} \frac{di_1}{dt} \quad (30.3)$$

That is, a change in the current  $i_1$  in coil 1 induces an emf in coil 2 that is directly proportional to the rate of change of  $i_1$  (**Fig. 30.2**).

We may also write the definition of mutual inductance, Eq. (30.2), as

$$M_{21} = \frac{N_2\Phi_{B2}}{i_1}$$

If the coils are in vacuum, the flux  $\Phi_{B2}$  through each turn of coil 2 is directly proportional to the current  $i_1$ . Then the mutual inductance  $M_{21}$  is a constant that depends only on the geometry of the two coils (the size, shape, number of turns, and orientation of each coil and the separation between the coils). If a magnetic material is present,  $M_{21}$  also depends on the magnetic properties of the material. If the material has nonlinear magnetic properties—that is, if the relative permeability  $K_m$  (defined in Section 28.8) is not constant and magnetization is not proportional to magnetic field—then  $\Phi_{B2}$  is no longer directly proportional to  $i_1$ . In that case the mutual inductance also depends on the value of  $i_1$ . In this discussion we'll assume that any magnetic material present has constant  $K_m$  so that flux *is* directly proportional to current and  $M_{21}$  depends on geometry only.

We can repeat our discussion for the opposite case in which a changing current  $i_2$  in coil 2 causes a changing flux  $\Phi_{B1}$  and an emf  $\mathcal{E}_1$  in coil 1. It turns out that the corresponding constant  $M_{12}$  is *always* equal to  $M_{21}$ , even though in general the two coils are not identical and the flux through them is not the same. We call this common value simply the mutual inductance, denoted by the symbol  $M$  without subscripts; it characterizes completely the induced-emf interaction of two coils. Then

$$\text{Mutually induced emfs: } \mathcal{E}_2 = -M \frac{di_1}{dt} \quad \text{and} \quad \mathcal{E}_1 = -M \frac{di_2}{dt} \quad (30.4)$$

Rate of change of current in coil 1      Rate of change of current in coil 2  
Induced emf in coil 2      Induced emf in coil 1  
Mutual inductance of coils 1 and 2

Figure 30.1 A current  $i_1$  in coil 1 gives rise to a magnetic flux through coil 2. (For clarity, only one-half of each coil is shown.)

**Mutual inductance:** If the current in coil 1 is changing, the changing flux through coil 2 induces an emf in coil 2.

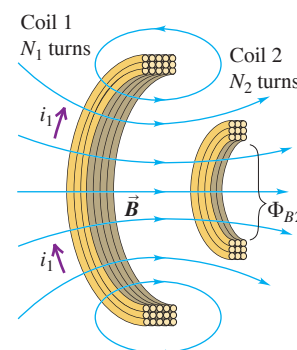
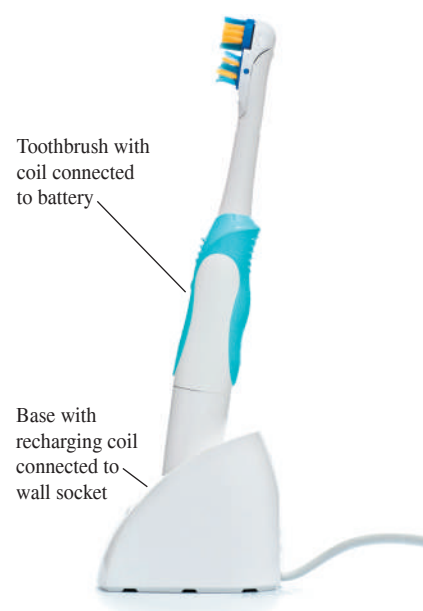


Figure 30.2 This electric toothbrush makes use of mutual inductance. The base contains a coil that is supplied with alternating current from a wall socket. Even though there is no direct electrical contact between the base and the toothbrush, this varying current induces an emf in a coil within the toothbrush itself, recharging the toothbrush battery.



The negative signs in Eqs. (30.4) reflect Lenz's law (Section 29.3). The first equation says that a current change in coil 1 causes a flux change through coil 2, inducing an emf in coil 2 that opposes the flux change; in the second equation, the roles of the two coils are interchanged. The mutual inductance  $M$  is

$$\text{Mutual inductance of coils 1 and 2} \quad M = \frac{N_2 \Phi_{B2}}{i_1} = \frac{N_1 \Phi_{B1}}{i_2} \quad (30.5)$$

Labels in diagram:  
 Turns in coil 2 (points to  $N_2$ )  
 Magnetic flux through each turn of coil 2 (points to  $\Phi_{B2}$ )  
 Turns in coil 1 (points to  $N_1$ )  
 Magnetic flux through each turn of coil 1 (points to  $\Phi_{B1}$ )  
 Current in coil 1 (causes flux through coil 2) (points to  $i_1$ )  
 Current in coil 2 (causes flux through coil 1) (points to  $i_2$ )

**CAUTION** Only a time-varying current induces an emf Only a *time-varying* current in a coil can induce an emf and hence a current in a second coil. Equations (30.4) show that the induced emf in each coil is directly proportional to the *rate of change* of the current in the other coil, not to the *value* of the current. A steady current in one coil, no matter how strong, cannot induce a current in a neighboring coil. **|**

The SI unit of mutual inductance is called the **henry** (1 H), in honor of the American physicist Joseph Henry (1797–1878), one of the discoverers of electromagnetic induction. From Eq. (30.5), one henry is equal to one weber per ampere. Other equivalent units, obtained by using Eqs. (30.4), are

$$1 \text{ H} = 1 \text{ Wb/A} = 1 \text{ V} \cdot \text{s/A} = 1 \Omega \cdot \text{s} = 1 \text{ J/A}^2$$

Just as the farad is a rather large unit of capacitance (see Section 24.1), the henry is a rather large unit of mutual inductance. Typical values of mutual inductance can be in the millihenry (mH) or microhenry ( $\mu\text{H}$ ) range.

### Drawbacks and Uses of Mutual Inductance

Mutual inductance can be a nuisance in electric circuits, since variations in current in one circuit can induce unwanted emfs in other nearby circuits. To minimize these effects, multiple-circuit systems must be designed so that  $M$  is as small as possible; for example, two coils would be placed far apart.

Happily, mutual inductance also has many useful applications. A *transformer*, used in alternating-current circuits to raise or lower voltages, is fundamentally no different from the two coils shown in Fig. 30.1. A time-varying alternating current in one coil of the transformer produces an alternating emf in the other coil; the value of  $M$ , which depends on the geometry of the coils, determines the amplitude of the induced emf in the second coil and hence the amplitude of the output voltage. (We'll describe transformers in more detail in Chapter 31.)

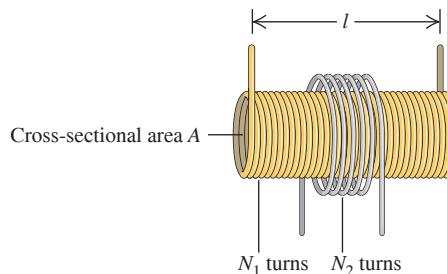
### EXAMPLE 30.1 Calculating mutual inductance

### WITH VARIATION PROBLEMS

In one form of Tesla coil (a high-voltage generator popular in science museums), a long solenoid with length  $l$  and cross-sectional area  $A$  is closely wound with  $N_1$  turns of wire. A coil with  $N_2$  turns surrounds it at its center (Fig. 30.3). Find the mutual inductance  $M$ .

**IDENTIFY and SET UP** Mutual inductance occurs here because a current in either coil sets up a magnetic field that causes a flux through the other coil. From Example 28.9 (Section 28.7) we have an expression [Eq. (28.23)] for the field magnitude  $B_1$  at the center of the solenoid (coil 1) in terms of the solenoid current  $i_1$ . This allows us to determine the flux through a cross section of the solenoid. Since there is almost no magnetic field outside a very long solenoid, this is also equal to the flux  $\Phi_{B2}$  through each turn of the *outer* coil (2). We then use Eq. (30.5), in the form  $M = N_2 \Phi_{B2} / i_1$ , to determine  $M$ .

Figure 30.3 A long solenoid with cross-sectional area  $A$  and  $N_1$  turns is surrounded at its center by a coil with  $N_2$  turns.



**EXECUTE** Equation (28.23) is expressed in terms of the number of turns per unit length, which for solenoid (1) is  $n_1 = N_1/L$ . So

$$B_1 = \mu_0 n_1 i_1 = \frac{\mu_0 N_1 i_1}{l}$$

The flux through a cross section of the solenoid equals  $B_1 A$ . As we mentioned above, this also equals the flux  $\Phi_{B2}$  through each turn of the outer coil, independent of its cross-sectional area. From Eq. (30.5), the mutual inductance  $M$  is then

$$M = \frac{N_2 \Phi_{B2}}{i_1} = \frac{N_2 B_1 A}{i_1} = \frac{N_2}{i_1} \frac{\mu_0 N_1 i_1}{l} A = \frac{\mu_0 A N_1 N_2}{l}$$

**EVALUATE** The mutual inductance  $M$  of any two coils is proportional to the product  $N_1 N_2$  of their numbers of turns. Notice that  $M$  depends only on the geometry of the two coils, not on the current.

Here's a numerical example to give you an idea of magnitudes. Suppose  $l = 0.50$  m,  $A = 10$  cm<sup>2</sup> =  $1.0 \times 10^{-3}$  m<sup>2</sup>,  $N_1 = 1000$  turns, and  $N_2 = 10$  turns. Then

$$\begin{aligned} M &= \frac{(4\pi \times 10^{-7} \text{ Wb/A} \cdot \text{m})(1.0 \times 10^{-3} \text{ m}^2)(1000)(10)}{0.50 \text{ m}} \\ &= 25 \times 10^{-6} \text{ Wb/A} = 25 \times 10^{-6} \text{ H} = 25 \mu\text{H} \end{aligned}$$

**KEYCONCEPT** A current in one coil causes a magnetic flux through an adjacent coil. How much flux is caused in the other coil by a given current in the first coil is determined by the mutual inductance  $M$  of the two coils. The value of  $M$  doesn't depend on the currents in the coils, only on their geometry.

### EXAMPLE 30.2 Emf due to mutual inductance

### WITH VARIATION PROBLEMS

In Example 30.1, suppose the current  $i_2$  in the outer coil is given by  $i_2 = (2.0 \times 10^6 \text{ A/s})t$ . (Currents in wires can indeed increase this rapidly for brief periods.) (a) At  $t = 3.0 \mu\text{s}$ , what is the average magnetic flux through each turn of the solenoid (coil 1) due to the current in the outer coil? (b) What is the induced emf in the solenoid?

**IDENTIFY and SET UP** In Example 30.1 we found the mutual inductance by relating the current in the solenoid to the flux produced in the outer coil; to do that, we used Eq. (30.5) in the form  $M = N_2 \Phi_{B2}/i_1$ . Here we are given the current  $i_2$  in the outer coil and want to find the resulting flux  $\Phi_1$  in the solenoid. The mutual inductance is the *same* in either case, and we have  $M = 25 \mu\text{H}$  from Example 30.1. We use Eq. (30.5) in the form  $M = N_1 \Phi_{B1}/i_2$  to determine the average flux  $\Phi_{B1}$  through each turn of the solenoid caused by current  $i_2$  in the outer coil. We then use Eqs. (30.4) to find the emf induced in the solenoid by the time variation of  $i_2$ .

**EXECUTE** (a) At  $t = 3.0 \mu\text{s} = 3.0 \times 10^{-6}$  s, the current in the outer coil is  $i_2 = (2.0 \times 10^6 \text{ A/s})(3.0 \times 10^{-6} \text{ s}) = 6.0$  A. We solve Eq. (30.5) for the flux  $\Phi_{B1}$  through each turn of coil 1:

$$\Phi_{B1} = \frac{M i_2}{N_1} = \frac{(25 \times 10^{-6} \text{ H})(6.0 \text{ A})}{1000} = 1.5 \times 10^{-7} \text{ Wb}$$

We emphasize that this is an *average* value; the flux can vary considerably between the center and the ends of the solenoid.

(b) We are given  $i_2 = (2.0 \times 10^6 \text{ A/s})t$ , so  $di_2/dt = 2.0 \times 10^6 \text{ A/s}$ ; then, from Eqs. (30.4), the induced emf in the solenoid is

$$\mathcal{E}_1 = -M \frac{di_2}{dt} = -(25 \times 10^{-6} \text{ H})(2.0 \times 10^6 \text{ A/s}) = -50 \text{ V}$$

**EVALUATE** This is a substantial induced emf in response to a very rapid current change. In an operating Tesla coil, there is a high-frequency alternating current rather than a continuously increasing current as in this example; both  $di_2/dt$  and  $\mathcal{E}_1$  alternate as well, with amplitudes that can be thousands of times larger than in this example.

**KEYCONCEPT** When the current in a coil changes, it causes a changing magnetic flux and hence an induced emf in an adjacent coil. The mutual inductance of the two coils determines how much emf is induced in one coil for a given rate of change of the current in the other coil.

**TEST YOUR UNDERSTANDING OF SECTION 30.1** Consider the Tesla coil described in Example 30.1. If you make the solenoid out of twice as much wire, so that it has twice as many turns and is twice as long, how much larger is the mutual inductance? (i)  $M$  is four times greater; (ii)  $M$  is twice as great; (iii)  $M$  is unchanged; (iv)  $M$  is  $\frac{1}{2}$  as great; (v)  $M$  is  $\frac{1}{4}$  as great.

### ANSWER

(iii) Doubling both the length of the solenoid ( $l$ ) and the number of turns of wire in the solenoid ( $N_1$ ) would have *no* effect on the mutual inductance  $M$ . Example 30.1 shows that  $M$  depends on the ratio of these quantities, which would remain unchanged. This is because the magnetic field produced by the solenoid depends on the number of turns *per unit length*, and the proposed change has no effect on this quantity.



## 30.2 SELF-INDUCTANCE AND INDUCTORS

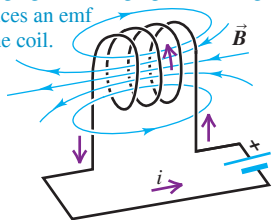
In our discussion of mutual inductance we considered two separate, independent circuits: A current in one circuit creates a magnetic field that gives rise to a flux through the second circuit. If the current in the first circuit changes, the flux through the second circuit changes and an emf is induced in the second circuit.

An important related effect occurs in a *single* isolated circuit. A current in a circuit sets up a magnetic field that causes a magnetic flux through the *same* circuit; this flux changes when the current changes. Thus any circuit that carries a varying current has an emf induced in it by the variation in *its own* magnetic field. Such an emf is called a **self-induced emf**. By Lenz's law, a self-induced emf opposes the change in the current that caused the emf and so tends to make it more difficult for variations in current to occur. Hence self-induced emfs can be of great importance whenever there is a varying current.

Self-induced emfs can occur in *any* circuit, since there is always some magnetic flux through the closed loop of a current-carrying circuit. But the effect is greatly enhanced if the circuit includes a coil with  $N$  turns of wire (Fig. 30.4). As a result of the current  $i$ , there is an average magnetic flux  $\Phi_B$  through each turn of the coil. In analogy to Eq. (30.5) we define the **self-inductance**  $L$  of the circuit as

Figure 30.4 The current  $i$  in the circuit causes a magnetic field  $\vec{B}$  in the coil and hence a flux through the coil.

**Self-inductance:** If the current  $i$  in the coil is changing, the changing flux through the coil induces an emf in the coil.



$$\text{Self-inductance (or inductance) of a coil} \quad L = \frac{N\Phi_B}{i} \quad (30.6)$$

Number of turns in coil  
Flux due to current through each turn of coil  
Current in coil

When there is no danger of confusion with mutual inductance, the self-inductance is called simply the **inductance**. Comparing Eqs. (30.5) and (30.6), we see that the units of self-inductance are the same as those of mutual inductance; the SI unit of self-inductance is the henry.

If the current  $i$  changes, so does the flux  $\Phi_B$ ; after we rearrange Eq. (30.6) and differentiate with respect to time, the rates of change are related by

$$N \frac{d\Phi_B}{dt} = L \frac{di}{dt}$$

From Faraday's law for a coil with  $N$  turns, Eq. (29.4), the self-induced emf is  $\mathcal{E} = -N d\Phi_B/dt$ , so it follows that

$$\text{Self-induced emf in a circuit} \quad \mathcal{E} = -L \frac{di}{dt} \quad (30.7)$$

Inductance of circuit  
Rate of change of current in circuit

The minus sign in Eq. (30.7) is a reflection of Lenz's law; it says that the self-induced emf in a circuit opposes any change in the current in that circuit.

Equation (30.7) states that the self-inductance of a circuit is the magnitude of the self-induced emf per unit rate of change of current. This relationship makes it possible to measure an unknown self-inductance: Change the current at a known rate  $di/dt$ , measure the induced emf, and take the ratio to determine  $L$ .

### Inductors as Circuit Elements

A circuit device that is designed to have a particular inductance is called an **inductor**, or a *choke*. The usual circuit symbol for an inductor is



Like resistors and capacitors, inductors are among the indispensable circuit elements of modern electronics. Their purpose is to oppose any variations in the current through the

### APPLICATION Inductors, Power Transmission, and Lightning Strikes

If lightning strikes part of an electrical power transmission system, it causes a sudden spike in voltage that can damage the components of the system as well as anything connected to that system (for example, home appliances). To minimize these effects, large inductors are incorporated into the transmission system. These use the principle that an inductor opposes and suppresses any rapid changes in the current.



circuit. An inductor in a direct-current circuit helps to maintain a steady current despite any fluctuations in the applied emf; in an alternating-current circuit, an inductor tends to suppress variations of the current that are more rapid than desired.

To understand the behavior of circuits containing inductors, we need to develop a general principle analogous to Kirchhoff's loop rule (discussed in Section 26.2). To apply that rule, we go around a conducting loop, measuring potential differences across successive circuit elements as we go. The algebraic sum of these differences around any closed loop must be zero because the electric field produced by charges distributed around the circuit is *conservative*. In Section 29.7 we denoted such a conservative field as  $\vec{E}_c$ .

When an inductor is included in the circuit, the situation changes. The magnetically induced electric field within the coils of the inductor is *not* conservative; as in Section 29.7, we'll denote it by  $\vec{E}_n$ . We need to think very carefully about the roles of the various fields. Let's assume we are dealing with an inductor whose coils have negligible resistance. Then a negligibly small electric field is required to make charge move through the coils, so the *total* electric field  $\vec{E}_c + \vec{E}_n$  within the coils must be zero, even though neither field is individually zero. Because  $\vec{E}_c$  is nonzero, there have to be accumulations of charge on the terminals of the inductor and the surfaces of its conductors to produce this field.

Consider the circuit shown in **Fig. 30.5**; the box contains some combination of batteries and variable resistors that enables us to control the current  $i$  in the circuit. According to Faraday's law, Eq. (29.10), the line integral of  $\vec{E}_n$  around the circuit is the negative of the rate of change of flux through the circuit, which in turn is given by Eq. (30.7). Combining these two relationships, we get

$$\oint \vec{E}_n \cdot d\vec{l} = -L \frac{di}{dt}$$

where we integrate clockwise around the loop (the direction of the assumed current). But  $\vec{E}_n$  is different from zero only within the inductor. Therefore the integral of  $\vec{E}_n$  around the whole loop can be replaced by its integral only from  $a$  to  $b$  through the inductor; that is,

$$\int_a^b \vec{E}_n \cdot d\vec{l} = -L \frac{di}{dt}$$

Next, because  $\vec{E}_c + \vec{E}_n = \mathbf{0}$  at each point within the inductor coils,  $\vec{E}_n = -\vec{E}_c$ . So we can rewrite the above equation as

$$\int_a^b \vec{E}_c \cdot d\vec{l} = L \frac{di}{dt}$$

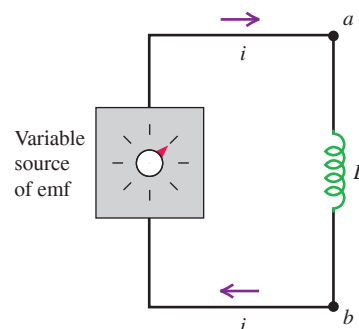
But this integral is just the potential  $V_{ab}$  of point  $a$  with respect to point  $b$ , so

$$V_{ab} = V_a - V_b = L \frac{di}{dt} \quad (30.8)$$

We conclude that there is a genuine potential difference between the terminals of the inductor, associated with conservative, electrostatic forces, even though the electric field associated with magnetic induction is nonconservative. Thus we are justified in using Kirchhoff's loop rule to analyze circuits that include inductors. Equation (30.8) gives the potential difference across an inductor in a circuit.

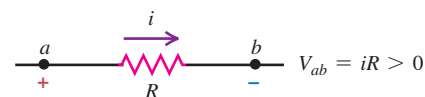
**CAUTION Self-induced emf opposes changes in current** Note that the self-induced emf does not oppose the current  $i$  itself; rather, it opposes any *change* ( $di/dt$ ) in the current. Thus the circuit behavior of an inductor is quite different from that of a resistor. **Figure 30.6** compares the behaviors of a resistor and an inductor and summarizes the sign relationships. ■

**Figure 30.5** A circuit containing a source of emf and an ideal inductor. The source is variable, so the current  $i$  and its rate of change  $di/dt$  can be varied.

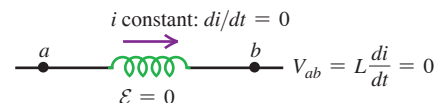


**Figure 30.6** (a) The potential difference across a resistor depends on the current, whereas (b), (c), (d) the potential difference across an inductor depends on the rate of change of the current.

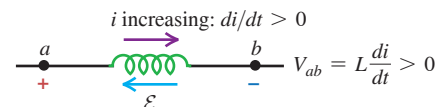
(a) Resistor with current  $i$  flowing from  $a$  to  $b$ : potential drops from  $a$  to  $b$ .



(b) Inductor with *constant* current  $i$  flowing from  $a$  to  $b$ : no potential difference.



(c) Inductor with *increasing* current  $i$  flowing from  $a$  to  $b$ : potential drops from  $a$  to  $b$ .



(d) Inductor with *decreasing* current  $i$  flowing from  $a$  to  $b$ : potential increases from  $a$  to  $b$ .

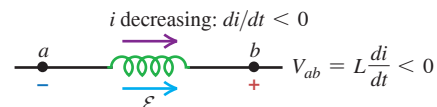


Figure 30.7 These fluorescent light tubes are wired in series with an inductor, or ballast, that helps to sustain the current flowing through the tubes.



## Applications of Inductors

Because an inductor opposes changes in current, it plays an important role in fluorescent light fixtures (**Fig. 30.7**). In such fixtures, current flows from the wiring into the gas that fills the tube, ionizing the gas and causing it to glow. However, an ionized gas or *plasma* is a highly nonohmic conductor: The greater the current, the more highly ionized the plasma becomes and the lower its resistance. If a sufficiently large voltage is applied to the plasma, the current can grow so much that it damages the circuitry outside the fluorescent tube. To prevent this problem, an inductor or *magnetic ballast* is put in series with the fluorescent tube to keep the current from growing out of bounds.

The ballast also makes it possible for the fluorescent tube to work with the alternating voltage provided by household wiring. This voltage oscillates sinusoidally with a frequency of 60 Hz, so that it goes momentarily to zero 120 times per second. If there were no ballast, the plasma in the fluorescent tube would rapidly deionize when the voltage went to zero and the tube would shut off. With a ballast present, a self-induced emf sustains the current and keeps the tube lit. Magnetic ballasts are also used for this purpose in streetlights (which obtain their light from a glowing mercury or sodium vapor) and in neon lights. (In compact fluorescent lamps, the magnetic ballast is replaced by a more complicated scheme that utilizes transistors, discussed in Chapter 42.)

The self-inductance of a circuit depends on its size, shape, and number of turns. For  $N$  turns close together, it is always proportional to  $N^2$ . It also depends on the magnetic properties of the material enclosed by the circuit. In the following examples we'll assume that the circuit encloses only vacuum (or air, which from the standpoint of magnetism is essentially vacuum). If, however, the flux is concentrated in a region containing a magnetic material with permeability  $\mu$ , then in the expression for  $B$  we must replace  $\mu_0$  (the permeability of vacuum) by  $\mu = K_m \mu_0$ , as discussed in Section 28.8. If the material is diamagnetic or paramagnetic, this replacement makes very little difference, since  $K_m$  is very close to 1. If the material is *ferromagnetic*, however, the difference is of crucial importance. A solenoid wound on a soft iron core having  $K_m = 5000$  can have an inductance approximately 5000 times as great as that of the same solenoid with an air core. Ferromagnetic-core inductors are very widely used in a variety of electronic and electric-power applications.

With ferromagnetic materials, the magnetization is in general not a linear function of magnetizing current, especially as saturation is approached. As a result, the inductance is not constant but can depend on current in a fairly complicated way. In our discussion we'll ignore this complication and assume always that the inductance is constant. This is a reasonable assumption even for a ferromagnetic material if the magnetization remains well below the saturation level.

Because automobiles contain steel, a ferromagnetic material, driving an automobile over a coil causes an appreciable increase in the coil's inductance. This effect is used in traffic light sensors, which use a large, current-carrying coil embedded under the road surface near an intersection. The circuitry connected to the coil detects the inductance change as a car drives over. When a preprogrammed number of cars have passed over the coil, the light changes to green to allow the cars through the intersection.

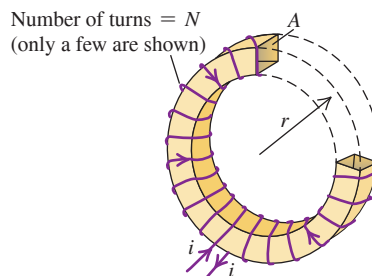
### EXAMPLE 30.3 Calculating self-inductance

### WITH VARIATION PROBLEMS

Determine the self-inductance of a toroidal solenoid with cross-sectional area  $A$  and mean radius  $r$ , closely wound with  $N$  turns of wire on a nonmagnetic core (**Fig. 30.8**). Assume that  $B$  is uniform across a cross section (that is, neglect the variation of  $B$  with distance from the toroid axis).

**IDENTIFY and SET UP** Our target variable is the self-inductance  $L$  of the toroidal solenoid. We can find  $L$  by using Eq. (30.6), which requires knowing the flux  $\Phi_B$  through each turn and the current  $i$  in the coil. For this, we use the results of Example 28.10 (Section 28.7), in which we found the magnetic field in the interior of a toroidal solenoid as a function of the current.

Figure 30.8 Determining the self-inductance of a closely wound toroidal solenoid. For clarity, only a few turns of the winding are shown. Part of the toroid has been cut away to show the cross-sectional area  $A$  and radius  $r$ .



**EXECUTE** From Eq. (30.6), the self-inductance is  $L = N\Phi_B/i$ . From Example 28.10, the field magnitude at a distance  $r$  from the toroid axis is  $B = \mu_0 Ni/2\pi r$ . If we assume that the field has this magnitude over the entire cross-sectional area  $A$ , then

$$\Phi_B = BA = \frac{\mu_0 NiA}{2\pi r}$$

The flux  $\Phi_B$  is the same through each turn, and

$$L = \frac{N\Phi_B}{i} = \frac{\mu_0 N^2 A}{2\pi r} \quad (\text{self-inductance of a toroidal solenoid})$$

**EVALUATE** Suppose  $N = 200$  turns,  $A = 5.0 \text{ cm}^2 = 5.0 \times 10^{-4} \text{ m}^2$ , and  $r = 0.10 \text{ m}$ ; then

$$\begin{aligned} L &= \frac{(4\pi \times 10^{-7} \text{ Wb/A} \cdot \text{m})(200)^2(5.0 \times 10^{-4} \text{ m}^2)}{2\pi(0.10 \text{ m})} \\ &= 40 \times 10^{-6} \text{ H} = 40 \mu\text{H} \end{aligned}$$

**KEYCONCEPT** The self-inductance  $L$  of a coil (or inductor) tells you how much magnetic flux is produced through that coil by the current that flows in it. Like mutual inductance, the value of  $L$  depends only on the coil's geometry, not on how much current it carries.

### EXAMPLE 30.4 Calculating self-induced emf

### WITH VARIATION PROBLEMS

If the current in the toroidal solenoid in Example 30.3 increases uniformly from 0 to 6.0 A in 3.0  $\mu\text{s}$ , find the magnitude and direction of the self-induced emf.

**IDENTIFY and SET UP** We are given  $L$ , the self-inductance, and  $di/dt$ , the rate of change of the solenoid current. We find the magnitude of the self-induced emf  $\mathcal{E}$  by using Eq. (30.7) and its direction by using Lenz's law.

**EXECUTE** We have  $di/dt = (6.0 \text{ A})/(3.0 \times 10^{-6} \text{ s}) = 2.0 \times 10^6 \text{ A/s}$ . From Eq. (30.7), the magnitude of the induced emf is

$$\begin{aligned} |\mathcal{E}| &= L \left| \frac{di}{dt} \right| \\ &= (40 \times 10^{-6} \text{ H})(2.0 \times 10^6 \text{ A/s}) = 80 \text{ V} \end{aligned}$$

The current is increasing, so according to Lenz's law the direction of the emf is opposite to that of the current. This corresponds to the situation in Fig. 30.6c; the emf is in the direction from  $b$  to  $a$ , like a battery with  $a$  as the  $+$  terminal and  $b$  the  $-$  terminal, tending to oppose the current increase from the external circuit.

**EVALUATE** This example shows that even a small inductance  $L$  can give rise to a substantial induced emf if the current changes rapidly.

**KEYCONCEPT** If the current in a coil changes, the resulting change in magnetic flux through the coil produces an induced emf. The self-inductance of the coil determines how much emf is induced in the coil for a given rate of change of the current in the coil.

**TEST YOUR UNDERSTANDING OF SECTION 30.2** Rank the following inductors in order of the potential difference  $V_{ab}$ , from most positive to most negative. In each case the inductor has zero resistance and the current flows from point  $a$  through the inductor to point  $b$ . (i) The current through a 2.0  $\mu\text{H}$  inductor increases from 1.0 A to 2.0 A in 0.50 s; (ii) the current through a 4.0  $\mu\text{H}$  inductor decreases from 3.0 A to 0 in 2.0 s; (iii) the current through a 1.0  $\mu\text{H}$  inductor remains constant at 4.0 A; (iv) the current through a 1.0  $\mu\text{H}$  inductor increases from 0 to 4.0 A in 0.25 s.

#### ANSWER

For the four cases we find (i)  $V_{ab} = (2.0 \mu\text{H})(2.0 \text{ A})/(0.50 \text{ s}) = 4.0 \mu\text{V}$ ; (ii)  $V_{ab} = -6.0 \mu\text{V}$ ; (iii)  $V_{ab} = 0$  because the rate of change of current is zero; and (iv)  $V_{ab} = (1.0 \mu\text{H})(4.0 \text{ A})/(0.25 \text{ s}) = 4.0 \mu\text{V}$ . **| (iv), (i), (iii), (ii)** From Eq. (30.8), the potential difference across the inductor is  $V_{ab} = L di/dt$ .

## 30.3 MAGNETIC-FIELD ENERGY

Establishing a current in an inductor requires an input of energy, and an inductor carrying a current has energy stored in it. Let's see how this comes about. In Fig. 30.5, an increasing current  $i$  in the inductor causes an emf  $\mathcal{E}$  between its terminals and a corresponding potential difference  $V_{ab}$  between the terminals of the source, with point  $a$  at higher potential than point  $b$ . Thus the source must be adding energy to the inductor, and the instantaneous power  $P$  (rate of transfer of energy into the inductor) is  $P = V_{ab}i$ .



## Energy Stored in an Inductor

We can calculate the total energy input  $U$  needed to establish a final current  $I$  in an inductor with inductance  $L$  if the initial current is zero. We assume that the inductor has zero resistance, so no energy is dissipated within the inductor. Let the rate of change of the current  $i$  at some instant be  $di/dt$ ; the current is increasing, so  $di/dt > 0$ . The voltage between the terminals  $a$  and  $b$  of the inductor at this instant is  $V_{ab} = L di/dt$ , and the rate  $P$  at which energy is being delivered to the inductor (equal to the instantaneous power supplied by the external source) is

$$P = V_{ab}i = Li \frac{di}{dt}$$

The energy  $dU$  supplied to the inductor during an infinitesimal time interval  $dt$  is  $dU = P dt$ , so

$$dU = Li di$$

The total energy  $U$  supplied while the current increases from zero to a final value  $I$  is

$$\text{Energy stored in an inductor} \rightarrow U = L \int_0^I i di = \frac{1}{2} LI^2 \quad (30.9)$$

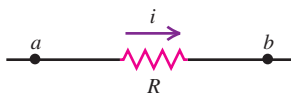
Inductance  
Final current  
Integral from initial (zero) value of instantaneous current to final value

After the current has reached its final steady value  $I$ ,  $di/dt = 0$  and no more energy is input to the inductor. When there is no current, the stored energy  $U$  is zero; when the current is  $I$ , the energy is  $\frac{1}{2} LI^2$ .

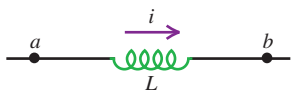
When the current decreases from  $I$  to zero, the inductor acts as a source that supplies a total amount of energy  $\frac{1}{2} LI^2$  to the external circuit. If we interrupt the circuit suddenly by opening a switch, the current decreases very rapidly, the induced emf is very large, and the energy may be dissipated in an arc across the switch contacts.

Figure 30.9 A resistor is a device in which energy is irrecoverably dissipated. By contrast, energy stored in a current-carrying inductor can be recovered when the current decreases to zero.

Resistor with current  $i$ : energy is dissipated.



Inductor with current  $i$ : energy is stored.



**CAUTION Energy, resistors, and inductors** Don't confuse the behavior of resistors and inductors where energy is concerned (Fig. 30.9). Energy flows into a resistor whenever a current passes through it, whether the current is steady or varying; this energy is dissipated in the form of heat. By contrast, energy flows into an ideal, zero-resistance inductor only when the current in the inductor *increases*. This energy is not dissipated; it is stored in the inductor and released when the current *decreases*. When a steady current flows through an inductor, there is no energy flow in or out. ■

## Magnetic Energy Density

The energy in an inductor is actually stored in the magnetic field of the coil, just as the energy of a capacitor is stored in the electric field between its plates. We can develop relationships for magnetic-field energy analogous to those we obtained for electric-field energy in Section 24.3 [Eqs. (24.9) and (24.11)]. We'll concentrate on one simple case, the ideal toroidal solenoid. This system has the advantage that its magnetic field is confined completely to a finite region of space within its core. As in Example 30.3, we assume that the cross-sectional area  $A$  is small enough that we can pretend the magnetic field is uniform over the area. The volume  $V$  enclosed by the toroidal solenoid is approximately equal to the circumference  $2\pi r$  multiplied by the area  $A$ :  $V = 2\pi rA$ . From Example 30.3, the self-inductance of the toroidal solenoid with vacuum within its coils is

$$L = \frac{\mu_0 N^2 A}{2\pi r}$$

From Eq. (30.9), the energy  $U$  stored in the toroidal solenoid when the current is  $I$  is

$$U = \frac{1}{2}LI^2 = \frac{1}{2} \frac{\mu_0 N^2 A}{2\pi r} I^2$$

The magnetic field and therefore this energy are localized in the volume  $V = 2\pi rA$  enclosed by the windings. The energy *per unit volume*, or *magnetic energy density*, is  $u = U/V$ :

$$u = \frac{U}{2\pi rA} = \frac{1}{2} \mu_0 \frac{N^2 I^2}{(2\pi r)^2}$$

We can express this in terms of the magnitude  $B$  of the magnetic field inside the toroidal solenoid. From Eq. (28.24) in Example 28.10 (Section 28.7), this is

$$B = \frac{\mu_0 NI}{2\pi r}$$

and so

$$\frac{N^2 I^2}{(2\pi r)^2} = \frac{B^2}{\mu_0^2}$$

When we substitute this into the above equation for  $u$ , we finally find the expression for **magnetic energy density** in vacuum:

$$\text{Magnetic energy density in vacuum} \quad u = \frac{B^2}{2\mu_0} \quad \text{Magnetic-field magnitude} \quad \text{Magnetic constant} \quad (30.10)$$

This is the magnetic analog of the energy per unit volume in an *electric* field in vacuum,  $u = \frac{1}{2}\epsilon_0 E^2$ , which we derived in Section 24.3. As an example, the energy density in the 1.5 T magnetic field of an MRI scanner (see Section 27.7) is  $u = B^2/2\mu_0 = (1.5 \text{ T})^2/(2 \times 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}) = 9.0 \times 10^5 \text{ J/m}^3$ .

When the material inside the toroid is not vacuum but a material with (constant) magnetic permeability  $\mu = K_m \mu_0$ , we replace  $\mu_0$  by  $\mu$  in Eq. (30.10):

$$\text{Magnetic energy density in a material} \quad u = \frac{B^2}{2\mu} \quad \text{Magnetic-field magnitude} \quad \text{Permeability of material} \quad (30.11)$$

Although we have derived Eq. (30.11) for only one special situation, it turns out to be the correct expression for the energy per unit volume associated with *any* magnetic-field configuration in a material with constant permeability. For vacuum, Eq. (30.11) reduces to Eq. (30.10). We'll use the expressions for electric-field and magnetic-field energy in Chapter 32 when we study the energy associated with electromagnetic waves.

Magnetic-field energy plays an important role in the ignition systems of gasoline-powered automobiles. A primary coil of about 250 turns is connected to the car's battery and produces a strong magnetic field. This coil is surrounded by a secondary coil with some 25,000 turns of very fine wire. When it is time for a spark plug to fire (see Fig. 20.5 in Section 20.3), the current to the primary coil is interrupted, the magnetic field quickly drops to zero, and an emf of tens of thousands of volts is induced in the secondary coil. The energy stored in the magnetic field thus goes into a powerful pulse of current that travels through the secondary coil to the spark plug, generating the spark that ignites the fuel-air mixture in the engine's cylinders (**Fig. 30.10**).

### APPLICATION A Magnetic Eruption on the Sun

This composite of two images of the sun shows a coronal mass ejection, a dramatic event in which about  $10^{12}$  kg (a billion tons) of material from the sun's outer atmosphere is ejected into space at speeds of 500 km/s or faster. Such ejections happen at intervals of a few hours to a few days. These immense eruptions are powered by the energy stored in the sun's magnetic field. Unlike the earth's relatively steady magnetic field, the sun's field is constantly changing, and regions of unusually strong field (and hence unusually high magnetic energy density) frequently form. A coronal mass ejection occurs when the energy stored in such a region is suddenly released.

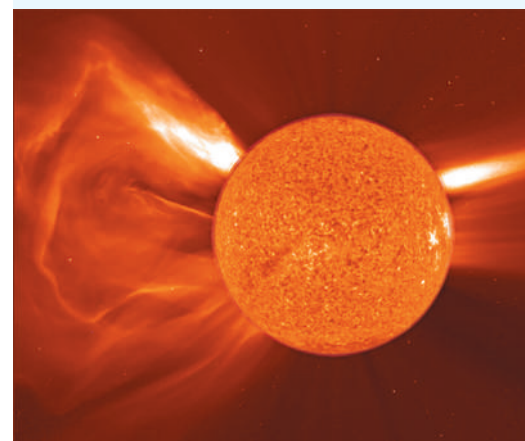


Figure 30.10 The energy required to fire an automobile spark plug is derived from magnetic-field energy stored in the ignition coil.



**EXAMPLE 30.5** Storing energy in an inductor

The electric-power industry would like to find efficient ways to store electrical energy generated during low-demand hours to help meet customer requirements during high-demand hours. Could a large inductor be used? What inductance would be needed to store  $1.00 \text{ kW} \cdot \text{h}$  of energy in a coil carrying a  $200 \text{ A}$  current?

**IDENTIFY and SET UP** We are given the required amount of stored energy  $U$  and the current  $I = 200 \text{ A}$ . We use Eq. (30.9) to find the self-inductance  $L$ .

**EXECUTE** Here we have  $I = 200 \text{ A}$  and  $U = 1.00 \text{ kW} \cdot \text{h} = (1.00 \times 10^3 \text{ W})(3600 \text{ s}) = 3.60 \times 10^6 \text{ J}$ . Solving Eq. (30.9) for  $L$ , we find

$$L = \frac{2U}{I^2} = \frac{2(3.60 \times 10^6 \text{ J})}{(200 \text{ A})^2} = 180 \text{ H}$$

**EVALUATE** The required inductance is more than a million times greater than the self-inductance of the toroidal solenoid of Example 30.3. Conventional wires that are to carry  $200 \text{ A}$  would have to be of large diameter to keep the resistance low and avoid unacceptable energy losses due to  $I^2R$  heating. As a result, a  $180 \text{ H}$  inductor using conventional wire would be very large (room-size). A superconducting inductor can be much smaller, since the resistance of a superconductor is zero and much thinner wires can be used. However, such wires must be kept at low temperature to remain superconducting, and maintaining this temperature itself requires energy. Despite these limitations, superconducting inductors are very efficient for energy storage and are presently in small-scale use by several electric utilities.

**KEYCONCEPT** Energy is stored in any magnetic field, including the field of a current-carrying inductor. The magnetic energy of an inductor depends on its self-inductance and current.

**TEST YOUR UNDERSTANDING OF SECTION 30.3** The current in a solenoid is reversed in direction while keeping the same magnitude. (a) Does this change the magnetic field within the solenoid? (b) Does this change the magnetic energy density in the solenoid?

**ANSWER** (a) **no** Reversing the direction of the current has no effect on the magnetic-field magnitude, but it causes the direction of the magnetic field to reverse. It has no effect on the magnetic-field energy density, which is proportional to the square of the magnitude of the magnetic field. (b) **yes**, (a) **yes**, (b) **no**

**30.4** THE *R-L* CIRCUIT

Let's look at some examples of the circuit behavior of an inductor. One thing is clear already; an inductor in a circuit makes it difficult for rapid changes in current to occur, thanks to the effects of self-induced emf. Equation (30.7) shows that the greater the rate of change of current  $di/dt$ , the greater the self-induced emf and the greater the potential difference between the inductor terminals. This equation, together with Kirchhoff's rules (see Section 26.2), gives us the principles we need to analyze circuits containing inductors.

**PROBLEM-SOLVING STRATEGY 30.1** Inductors in Circuits

**IDENTIFY** *the relevant concepts:* An inductor is just another circuit element, like a source of emf, a resistor, or a capacitor. One key difference is that when an inductor is included in a circuit, all the voltages, currents, and capacitor charges are in general functions of time, not constants as they have been in most of our previous circuit analysis. But even when the voltages and currents vary with time, Kirchhoff's rules (see Section 26.2) hold at each instant of time.

**SET UP** *the problem* using the following steps:

1. Follow the procedure given in Problem-Solving Strategy 26.2 (Section 26.2). Draw a circuit diagram and label all quantities, known and unknown. Apply the junction rule immediately to express the currents in terms of as few quantities as possible.
2. Determine which quantities are the target variables.

**EXECUTE** *the solution* as follows:

1. As in Problem-Solving Strategy 26.2, apply Kirchhoff's loop rule to each loop in the circuit.

2. Review the sign rules given in Problem-Solving Strategy 26.2. To get the correct sign for the potential difference between the terminals of an inductor, apply Lenz's law and the sign rule described in Section 30.2 in connection with Eq. (30.7) and Fig. 30.6. In Kirchhoff's loop rule, when we go through an inductor in the *same* direction as the assumed current, we encounter a voltage *drop* equal to  $L di/dt$ , so the corresponding term in the loop equation is  $-L di/dt$ . When we go through an inductor in the *opposite* direction from the assumed current, the potential difference is reversed and the term to use in the loop equation is  $+L di/dt$ .
3. Solve for the target variables.

**EVALUATE** *your answer:* Check whether your answer is consistent with the behavior of inductors. By Lenz's law, if the current through an inductor is changing, your result should indicate that the potential difference across the inductor opposes the change.

## Current Growth in an R-L Circuit

We can learn a great deal about inductor behavior by analyzing the circuit of **Fig. 30.11**. A circuit that includes both a resistor and an inductor, and possibly a source of emf, is called an **R-L circuit**. The inductor helps to prevent rapid changes in current, which can be useful if a steady current is required but the source has a fluctuating emf. The resistor  $R$  may be a separate circuit element, or it may be the resistance of the inductor windings; every real-life inductor has some resistance unless it is made of superconducting wire. By closing switch  $S_1$ , we can connect the R-L combination to a source with constant emf  $\mathcal{E}$ . (We assume that the source has zero internal resistance, so the terminal voltage equals  $\mathcal{E}$ .)

Suppose both switches are open to begin with, and then at some initial time  $t = 0$  we close switch  $S_1$ . The current cannot change suddenly from zero to some final value, since  $di/dt$  and the induced emf in the inductor would both be infinite. Instead, the current begins to grow at a rate that depends on the value of  $L$  in the circuit.

Let  $i$  be the current at some time  $t$  after switch  $S_1$  is closed, and let  $di/dt$  be its rate of change at that time. The potential differences  $v_{ab}$  (across the resistor) and  $v_{bc}$  (across the inductor) are

$$v_{ab} = iR \quad \text{and} \quad v_{bc} = L \frac{di}{dt}$$

Note that if the current is in the direction shown in Fig. 30.11 and is increasing, then both  $v_{ab}$  and  $v_{bc}$  are positive;  $a$  is at a higher potential than  $b$ , which in turn is at a higher potential than  $c$ . (Compare to Figs. 30.6a and 30.6c.) We apply Kirchhoff's loop rule, starting at the negative terminal and proceeding counterclockwise around the loop:

$$\mathcal{E} - iR - L \frac{di}{dt} = 0 \quad (30.12)$$

Solving this for  $di/dt$ , we find that the rate of increase of current is

$$\frac{di}{dt} = \frac{\mathcal{E} - iR}{L} = \frac{\mathcal{E}}{L} - \frac{R}{L}i \quad (30.13)$$

At the instant that switch  $S_1$  is first closed,  $i = 0$  and the potential drop across  $R$  is zero. The initial rate of change of current is

$$\left( \frac{di}{dt} \right)_{\text{initial}} = \frac{\mathcal{E}}{L}$$

The greater the inductance  $L$ , the more slowly the current increases.

As the current increases, the term  $(R/L)i$  in Eq. (30.13) also increases, and the *rate* of increase of current given by Eq. (30.13) becomes smaller and smaller. This means that the current is approaching a final, steady-state value  $I$ . When the current reaches this value, its rate of increase is zero. Then Eq. (30.13) becomes

$$\begin{aligned} \left( \frac{di}{dt} \right)_{\text{final}} = 0 &= \frac{\mathcal{E}}{L} - \frac{R}{L}I \quad \text{and} \\ I &= \frac{\mathcal{E}}{R} \end{aligned}$$

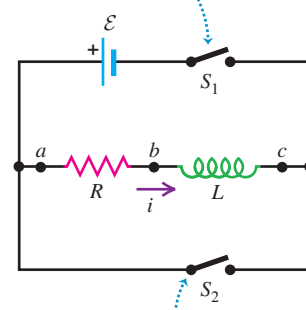
The *final* current  $I$  does not depend on the inductance  $L$ ; it is the same as it would be if the resistance  $R$  alone were connected to the source with emf  $\mathcal{E}$ .

**Figure 30.12** shows the behavior of the current as a function of time. To derive the equation for this curve (that is, an expression for current as a function of time), we proceed just as we did for the charging capacitor in Section 26.4. First we rearrange Eq. (30.13) to the form

$$\frac{di}{i - (\mathcal{E}/R)} = -\frac{R}{L}dt$$

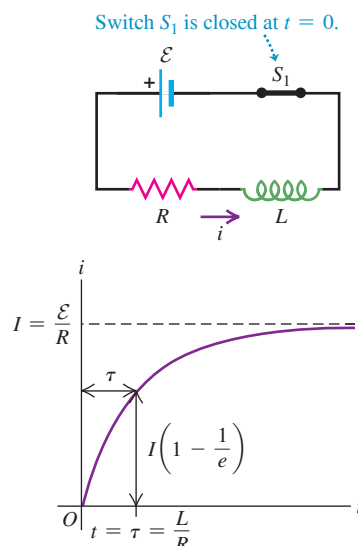
Figure 30.11 An R-L circuit.

Closing switch  $S_1$  connects the R-L combination in series with a source of emf  $\mathcal{E}$ .



Closing switch  $S_2$  while opening switch  $S_1$  disconnects the combination from the source.

Figure 30.12 Graph of  $i$  versus  $t$  for growth of current in an R-L circuit with an emf in series. The final current is  $I = \mathcal{E}/R$ ; after one time constant  $\tau$ , the current is  $1 - 1/e$  of this value.





This separates the variables, with  $i$  on the left side and  $t$  on the right. Then we integrate both sides, renaming the integration variables  $i'$  and  $t'$  so that we can use  $i$  and  $t$  as the upper limits. (The lower limit for each integral is zero, corresponding to zero current at the initial time  $t = 0$ .) We get

$$\int_0^i \frac{di'}{i' - (\mathcal{E}/R)} = - \int_0^t \frac{R}{L} dt'$$

$$\ln\left(\frac{i - (\mathcal{E}/R)}{-\mathcal{E}/R}\right) = -\frac{R}{L}t$$

Now we take exponentials of both sides and solve for  $i$ . We leave the details for you to work out; the final result is the equation of the curve in Fig. 30.12:

$$i = \frac{\mathcal{E}}{R}(1 - e^{-(R/L)t}) \quad (\text{current in an } R\text{-}L \text{ circuit with emf}) \quad (30.14)$$

Taking the derivative of Eq. (30.14), we find

$$\frac{di}{dt} = \frac{\mathcal{E}}{L}e^{-(R/L)t} \quad (30.15)$$

At time  $t = 0$ ,  $i = 0$  and  $di/dt = \mathcal{E}/L$ . As  $t \rightarrow \infty$ ,  $i \rightarrow \mathcal{E}/R$  and  $di/dt \rightarrow 0$ , as we predicted.

As Fig. 30.12 shows, the instantaneous current  $i$  first rises rapidly, then increases more slowly and approaches the final value  $I = \mathcal{E}/R$  asymptotically. At a time equal to  $L/R$ , the current has risen to  $(1 - 1/e)$ , or about 63%, of its final value. The quantity  $L/R$  is therefore a measure of how quickly the current builds toward its final value; this quantity is called the **time constant** for the circuit, denoted by  $\tau$ :

$$\text{Time constant } \tau = \frac{L}{R} \quad \begin{array}{l} \text{Inductance} \\ \text{Resistance} \end{array} \quad (30.16)$$

In a time equal to  $2\tau$ , the current reaches 86% of its final value; in  $5\tau$ , 99.3%; and in  $10\tau$ , 99.995%. (Compare the discussion in Section 26.4 of charging a capacitor of capacitance  $C$  that was in series with a resistor of resistance  $R$ ; the time constant for that situation was the product  $RC$ .)

The graphs of  $i$  versus  $t$  have the same general shape for all values of  $L$ . For a given value of  $R$ , the time constant  $\tau$  is greater for greater values of  $L$ . When  $L$  is small, the current rises rapidly to its final value; when  $L$  is large, it rises more slowly. For example, if  $R = 100 \, \Omega$  and  $L = 10 \, \text{H}$ ,

$$\tau = \frac{L}{R} = \frac{10 \, \text{H}}{100 \, \Omega} = 0.10 \, \text{s}$$

and the current increases to about 63% of its final value in 0.10 s. (Recall that  $1 \, \text{H} = 1 \, \Omega \cdot \text{s}$ .) But if  $L = 0.010 \, \text{H}$ ,  $\tau = 1.0 \times 10^{-4} \, \text{s} = 0.10 \, \text{ms}$ , and the rise is much more rapid.

Energy considerations offer us additional insight into the behavior of an  $R$ - $L$  circuit. The instantaneous rate at which the source delivers energy to the circuit is  $P = \mathcal{E}i$ . The instantaneous rate at which energy is dissipated in the resistor is  $i^2R$ , and the rate at which energy is stored in the inductor is  $i v_{bc} = Li \, di/dt$  [or, equivalently,  $(d/dt)(\frac{1}{2}Li^2) = Li \, di/dt$ ]. When we multiply Eq. (30.12) by  $i$  and rearrange, we find

$$\mathcal{E}i = i^2R + Li \frac{di}{dt} \quad (30.17)$$

Of the power  $\mathcal{E}i$  supplied by the source, part ( $i^2R$ ) is dissipated in the resistor and part ( $Li \, di/dt$ ) goes to store energy in the inductor. This discussion is analogous to our power analysis for a charging capacitor, given at the end of Section 26.4.

**CAUTION** **Current and its rate of change in an inductor** Note that the current  $i$  cannot change abruptly in a circuit that contains an inductor, so  $i$  must be a continuous function of time  $t$ . However, the rate of change of the current  $di/dt$  can change abruptly (for example, when the switch in Fig. 30.12a is closed and the current suddenly starts to increase, as shown at  $t = 0$  in Fig. 30.12b). ■

**EXAMPLE 30.6 Analyzing an R-L circuit****WITH VARIATION PROBLEMS**

A sensitive electronic device of resistance  $R = 175\ \Omega$  is to be connected to a source of emf (of negligible internal resistance) by a switch. The device is designed to operate with a 36 mA current, but to avoid damage to the device, the current can rise to no more than 4.9 mA in the first  $58\ \mu\text{s}$  after the switch is closed. An inductor is therefore connected in series with the device, as in Fig. 30.11; the switch in question is  $S_1$ . (a) What is the required source emf  $\mathcal{E}$ ? (b) What is the required inductance  $L$ ? (c) What is the  $R$ - $L$  time constant  $\tau$ ?

**IDENTIFY and SET UP** This problem concerns current and current growth in an  $R$ - $L$  circuit, so we can use the ideas of this section. Figure 30.12 shows the current  $i$  versus the time  $t$  that has elapsed since closing  $S_1$ . The graph shows that the final current is  $I = \mathcal{E}/R$ ; we are given  $R = 175\ \Omega$ , so the emf is determined by the requirement that the final current be  $I = 36\ \text{mA}$ . The other requirement is that the current be no more than  $i = 4.9\ \text{mA}$  at  $t = 58\ \mu\text{s}$ ; to satisfy this, we use Eq. (30.14) for the current as a function of time and solve for the inductance, which is the only unknown quantity. Equation (30.16) then tells us the time constant.

**EXECUTE** (a) We solve  $I = \mathcal{E}/R$  for  $\mathcal{E}$ :

$$\mathcal{E} = IR = (0.036\ \text{A})(175\ \Omega) = 6.3\ \text{V}$$

(b) To find the required inductance, we solve Eq. (30.14) for  $L$ . First we multiply through by  $(-R/\mathcal{E})$  and add 1 to both sides:

$$1 - \frac{iR}{\mathcal{E}} = e^{-(R/L)t}$$

Then we take natural logs of both sides, solve for  $L$ , and substitute:

$$\begin{aligned} L &= \frac{-Rt}{\ln(1 - iR/\mathcal{E})} \\ &= \frac{-(175\ \Omega)(58 \times 10^{-6}\ \text{s})}{\ln[1 - (4.9 \times 10^{-3}\ \text{A})(175\ \Omega)/(6.3\ \text{V})]} = 69\ \text{mH} \end{aligned}$$

(c) From Eq. (30.16),

$$\tau = \frac{L}{R} = \frac{69 \times 10^{-3}\ \text{H}}{175\ \Omega} = 3.9 \times 10^{-4}\ \text{s} = 390\ \mu\text{s}$$

**EVALUATE** Note that  $58\ \mu\text{s}$  is much less than the time constant. In  $58\ \mu\text{s}$  the current builds up from zero to 4.9 mA, a small fraction of its final value of 36 mA; after  $390\ \mu\text{s}$  the current equals  $(1 - 1/e)$  of its final value, or about  $(0.63)(36\ \text{mA}) = 23\ \text{mA}$ .

**KEYCONCEPT** The presence of an inductor in a circuit that contains a source of emf and a resistor slows the growth of the current when the circuit is completed. The equilibrium value of the current after a very long time is the same as if the inductor were not present.

**Current Decay in an R-L Circuit**

Now suppose switch  $S_1$  in the circuit of Fig. 30.11 has been closed for a while and the current has reached the value  $I_0$ . Resetting our stopwatch to redefine the initial time, we close switch  $S_2$  at time  $t = 0$ , bypassing the battery. (At the same time we should open  $S_1$  to protect the battery.) The current through  $R$  and  $L$  does not instantaneously go to zero but decays smoothly, as shown in **Fig. 30.13**. The Kirchhoff's-rule loop equation is obtained from Eq. (30.12) by omitting the  $\mathcal{E}$  term. We challenge you to retrace the steps in the above analysis and show that the current  $i$  varies with time according to

$$i = I_0 e^{-(R/L)t} \quad (30.18)$$

where  $I_0$  is the initial current at time  $t = 0$ . The time constant,  $\tau = L/R$ , is the time for current to decrease to  $1/e$ , or about 37%, of its original value. In time  $2\tau$  it has dropped to 13.5%, in time  $5\tau$  to 0.67%, and in  $10\tau$  to 0.0045%.

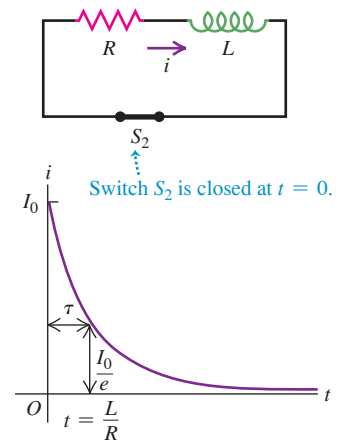
The energy that is needed to maintain the current during this decay is provided by the energy stored in the magnetic field of the inductor. The detailed energy analysis is simpler this time. In place of Eq. (30.17) we have

$$0 = i^2 R + Li \frac{di}{dt} \quad (30.19)$$

Now  $Li\,di/dt$  is negative; Eq. (30.19) shows that the energy stored in the inductor *decreases* at the same rate  $i^2 R$  at which energy is dissipated in the resistor.

This entire discussion should look familiar; the situation is very similar to that of a charging and discharging capacitor, analyzed in Section 26.4. It would be a good idea to compare that section with our discussion of the  $R$ - $L$  circuit.

**Figure 30.13** Graph of  $i$  versus  $t$  for decay of current in an  $R$ - $L$  circuit. After one time constant  $\tau$ , the current is  $1/e$  of its initial value.



**EXAMPLE 30.7** Energy in an  $R$ - $L$  circuit**WITH VARIATION PROBLEMS**

When the current in an  $R$ - $L$  circuit is decaying, what fraction of the original energy stored in the inductor has been dissipated after 2.3 time constants?

**IDENTIFY and SET UP** This problem concerns current decay in an  $R$ - $L$  circuit as well as the relationship between the current in an inductor and the amount of stored energy. The current  $i$  at any time  $t$  is given by Eq. (30.18); the stored energy associated with this current is given by Eq. (30.9),  $U = \frac{1}{2}Li^2$ .

**EXECUTE** From Eq. (30.18), the current  $i$  at any time  $t$  is

$$i = I_0 e^{-(R/L)t}$$

We substitute this into  $U = \frac{1}{2}Li^2$  to obtain an expression for the stored energy at any time:

$$U = \frac{1}{2}LI_0^2 e^{-2(R/L)t} = U_0 e^{-2(R/L)t}$$

where  $U_0 = \frac{1}{2}LI_0^2$  is the energy at the initial time  $t = 0$ . When  $t = 2.3\tau = 2.3L/R$ , we have

$$U = U_0 e^{-2(2.3)} = U_0 e^{-4.6} = 0.010U_0$$

That is, only 0.010 or 1.0% of the energy initially stored in the inductor remains, so 99.0% has been dissipated in the resistor.

**EVALUATE** To get a sense of what this result means, consider the  $R$ - $L$  circuit we analyzed in Example 30.6, for which  $\tau = 390 \mu\text{s}$ . With  $L = 69 \text{ mH}$  and  $I_0 = 36 \text{ mA}$ , we have

$$\begin{aligned} U_0 &= \frac{1}{2}LI_0^2 = \frac{1}{2}(0.069 \text{ H})(0.036 \text{ A})^2 \\ &= 4.5 \times 10^{-5} \text{ J} \end{aligned}$$

Of this, 99.0% or  $4.4 \times 10^{-5} \text{ J}$  is dissipated in  $2.3(390 \mu\text{s}) = 9.0 \times 10^{-4} \text{ s} = 0.90 \text{ ms}$ . In other words, this circuit can be almost completely powered off (or powered on) in 0.90 ms, so the minimum time for a complete on-off cycle is 1.8 ms. Even shorter cycle times are required for many purposes, such as in fast switching networks for telecommunications. In such cases a smaller time constant  $\tau = L/R$  is needed.

**KEYCONCEPT** The presence of an inductor in a circuit that contains a resistor slows the decay of the current when the source of emf is removed. As the current decays, the magnetic energy stored in the inductor is dissipated in the resistor.

**TEST YOUR UNDERSTANDING OF SECTION 30.4** (a) In Fig. 30.11, what are the algebraic signs of the potential differences  $v_{ab}$  and  $v_{bc}$  when switch  $S_1$  is closed and switch  $S_2$  is open? (i)  $v_{ab} > 0$ ,  $v_{bc} > 0$ ; (ii)  $v_{ab} > 0$ ,  $v_{bc} < 0$ ; (iii)  $v_{ab} < 0$ ,  $v_{bc} > 0$ ; (iv)  $v_{ab} < 0$ ,  $v_{bc} < 0$ . (b) What are the signs of  $v_{ab}$  and  $v_{bc}$  when  $S_1$  is open,  $S_2$  is closed, and current is flowing in the direction shown? (i)  $v_{ab} > 0$ ,  $v_{bc} > 0$ ; (ii)  $v_{ab} > 0$ ,  $v_{bc} < 0$ ; (iii)  $v_{ab} < 0$ ,  $v_{bc} > 0$ ; (iv)  $v_{ab} < 0$ ,  $v_{bc} < 0$ .

**ANSWER**

(a) (i) Recall that  $v_{ab}$  is the potential at  $a$  minus the potential at  $b$ , and similarly for  $v_{bc}$ . For either arrangement of the switches, current flows through the resistor from  $a$  to  $b$ . The upstream end of the resistor is always at the higher potential, so  $v_{ab}$  is positive. With  $S_1$  closed and  $S_2$  open, the current through the inductor flows from  $b$  to  $c$  and is increasing. The self-induced emf opposes this increase and is therefore directed from  $c$  toward  $b$ , which means that  $b$  is at the higher potential. Hence  $v_{bc}$  is positive. With  $S_1$  open and  $S_2$  closed, the inductor current again flows from  $b$  to  $c$  but is now decreasing. The self-induced emf is directed from  $b$  to  $c$  in an effort to sustain the decaying current, so  $c$  is at the higher potential and  $v_{bc}$  is negative.

**30.5 THE  $L$ - $C$  CIRCUIT**

A circuit containing an inductor and a capacitor shows an entirely new mode of behavior, characterized by *oscillating* current and charge. This is in sharp contrast to the *exponential* approach to a steady-state situation that we have seen with both  $R$ - $C$  and  $R$ - $L$  circuits. In the  **$L$ - $C$  circuit** in **Fig. 30.14a** we charge the capacitor to a potential difference  $V_m$  and initial charge  $Q_m = CV_m$  on its left-hand plate and then close the switch. What happens?

The capacitor begins to discharge through the inductor. Because of the induced emf in the inductor, the current cannot change instantaneously; it starts at zero and eventually builds up to a maximum value  $I_m$ . During this buildup the capacitor is discharging. At each instant the capacitor potential equals the induced emf, so as the capacitor discharges, the *rate of change* of current decreases. When the capacitor potential becomes zero, the induced emf is also zero, and the current has leveled off at its maximum value  $I_m$ .

Figure 30.14 In an oscillating L-C circuit, the charge on the capacitor and the current through the inductor both vary sinusoidally with time. Energy is transferred between magnetic energy in the inductor ( $U_B$ ) and electrical energy in the capacitor ( $U_E$ ). As in simple harmonic motion, the total energy  $E$  remains constant. (Compare Fig. 14.14 in Section 14.3.)

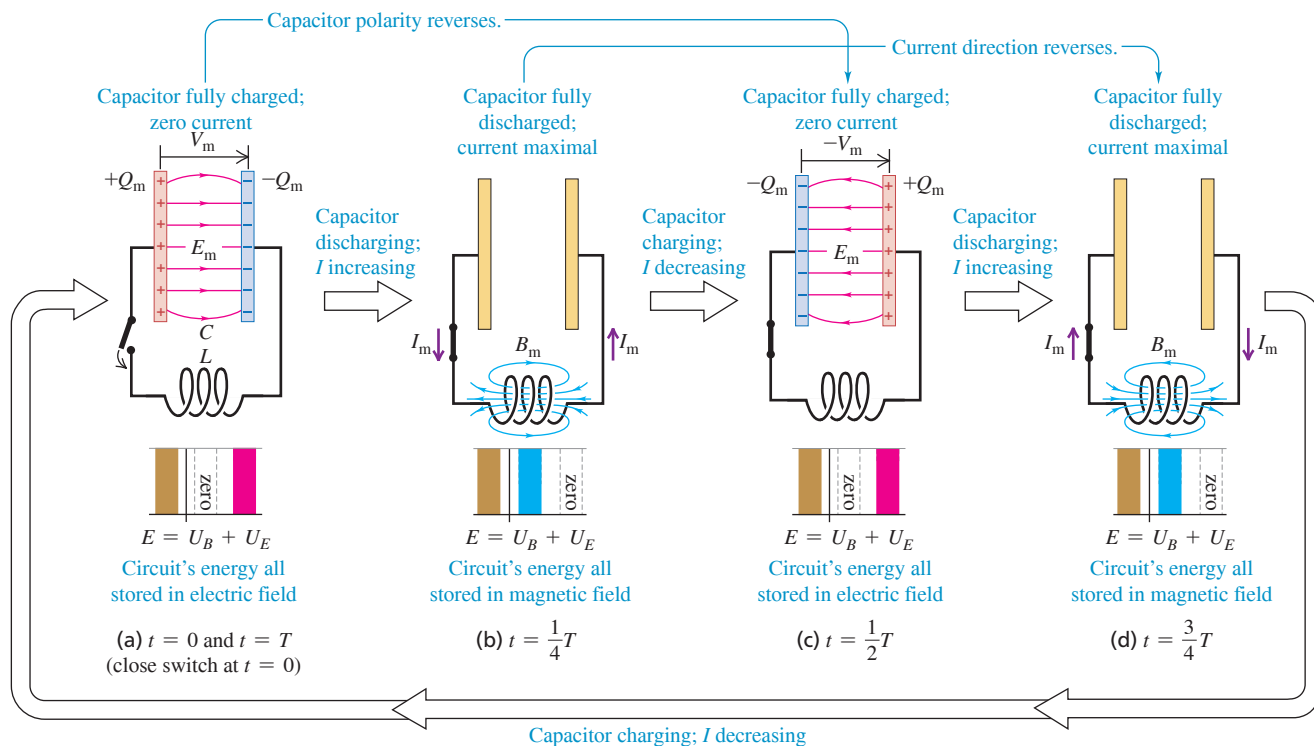


Figure 30.14b shows this situation; the capacitor has completely discharged. The potential difference between its terminals (and those of the inductor) has decreased to zero, and the current has reached its maximum value  $I_m$ .

During the discharge of the capacitor, the increasing current in the inductor has established a magnetic field in the space around it, and the energy that was initially stored in the capacitor's electric field is now stored in the inductor's magnetic field.

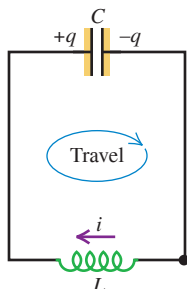
Although the capacitor is completely discharged in Fig. 30.14b, the current persists (it cannot change instantaneously), and the capacitor begins to charge with polarity opposite to that in the initial state. As the current decreases, the magnetic field also decreases, inducing an emf in the inductor in the *same* direction as the current; this slows down the decrease of the current. Eventually, the current and the magnetic field reach zero, and the capacitor has been charged in the sense *opposite* to its initial polarity (Fig. 30.14c), with potential difference  $-V_m$  and charge  $-Q_m$  on its left-hand plate.

The process now repeats in the reverse direction; a little later, the capacitor has again discharged, and there is a current in the inductor in the opposite direction (Fig. 30.14d). Still later, the capacitor charge returns to its original value (Fig. 30.14a), and the whole process repeats. If there are no energy losses, the charges on the capacitor continue to oscillate back and forth indefinitely. This process is called an **electrical oscillation**. (Before you read further, review the analogous case of *mechanical* oscillation in Sections 14.2 and 14.3.)

From an energy standpoint the oscillations of an electric circuit transfer energy from the capacitor's electric field to the inductor's magnetic field and back. The *total* energy associated with the circuit is constant. This is analogous to the transfer of energy in an oscillating mechanical system from potential energy to kinetic energy and back, with constant total energy (Section 14.3). As we'll see, this analogy goes much further.



Figure 30.15 Applying Kirchhoff's loop rule to the  $L$ - $C$  circuit. The direction of travel around the loop in the loop equation is shown. Just after the circuit is completed and the capacitor first begins to discharge, as in Fig. 30.14a, the current is negative (opposite to the direction shown).



## Electrical Oscillations in an $L$ - $C$ Circuit

To study the flow of charge in detail, we proceed just as we did for the  $R$ - $L$  circuit. **Figure 30.15** shows our definitions of  $q$  and  $i$ .

**CAUTION Positive current in an  $L$ - $C$  circuit** After you have examined Fig. 30.14, the positive direction for current in Fig. 30.15 may seem backward. In fact, we chose this direction to simplify the relationship between current and capacitor charge. We define the current at each instant to be  $i = dq/dt$ , the rate of change of the charge on the left-hand capacitor plate. If the capacitor is initially charged and begins to discharge as in Figs. 30.14a and 30.14b, then  $dq/dt < 0$  and the initial current  $i$  is negative; the current direction is opposite to the (positive) direction shown in Fig. 30.15. ▮

We apply Kirchhoff's loop rule to the circuit in Fig. 30.15. Starting at the lower-right corner of the circuit and adding voltages as we go clockwise around the loop, we obtain

$$-L \frac{di}{dt} - \frac{q}{C} = 0$$

Since  $i = dq/dt$ , it follows that  $di/dt = d^2q/dt^2$ . We substitute this expression into the above equation and divide by  $-L$  to obtain

$$\frac{d^2q}{dt^2} + \frac{1}{LC}q = 0 \quad (L\text{-}C \text{ circuit}) \quad (30.20)$$

Equation (30.20) has exactly the same form as the equation we derived for simple harmonic motion in Section 14.2, Eq. (14.4):  $d^2x/dt^2 = -(k/m)x$ , or

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

In an  $L$ - $C$  circuit the capacitor charge  $q$  plays the role of the displacement  $x$ , and the current  $i = dq/dt$  is analogous to the particle's velocity  $v_x = dx/dt$ . The inductance  $L$  is analogous to the mass  $m$ , and the reciprocal of the capacitance,  $1/C$ , is analogous to the force constant  $k$ .

Pursuing this analogy, we recall that the angular frequency  $\omega = 2\pi f$  of the harmonic oscillator is equal to  $(k/m)^{1/2}$  [Eq. (14.10)], and the position is given as a function of time by Eq. (14.13),

$$x = A \cos(\omega t + \phi)$$

where the amplitude  $A$  and the phase angle  $\phi$  depend on the initial conditions. In the analogous electrical situation, the capacitor charge  $q$  is given by

$$q = Q \cos(\omega t + \phi) \quad (30.21)$$

and the angular frequency  $\omega$  of oscillation is given by

$$\text{Angular frequency of oscillation in an } L\text{-}C \text{ circuit} \quad \omega = \sqrt{\frac{1}{LC}} \quad \begin{array}{l} \text{Capacitance} \\ \text{Inductance} \end{array} \quad (30.22)$$

Verify that Eq. (30.21) satisfies the loop equation, Eq. (30.20), when  $\omega$  has the value given by Eq. (30.22). In doing this, you'll find that the instantaneous current  $i = dq/dt$  is

$$i = -\omega Q \sin(\omega t + \phi) \quad (30.23)$$

Thus the charge and current in an  $L$ - $C$  circuit oscillate sinusoidally with time, with an angular frequency determined by the values of  $L$  and  $C$ . The ordinary frequency  $f$ , the number of cycles per second, is equal to  $\omega/2\pi$ . The constants  $Q$  and  $\phi$  in Eqs. (30.21) and (30.23) are determined by the initial conditions. If at time  $t = 0$  the left-hand capacitor plate in Fig. 30.15 has its maximum charge  $Q$  and the current  $i$  is zero, then  $\phi = 0$ . If  $q = 0$  at  $t = 0$ , then  $\phi = \pm\pi/2$  rad.

## Energy in an $L$ - $C$ Circuit

We can also analyze the  $L$ - $C$  circuit by using an energy approach. The analogy to simple harmonic motion is equally useful here. In the mechanical problem an object with mass  $m$  is attached to a spring with force constant  $k$ . Suppose we displace the object a distance  $A$  from its equilibrium position and release it from rest at time  $t = 0$ . The kinetic energy of the system at any later time is  $\frac{1}{2}mv_x^2$ , and its elastic potential energy is  $\frac{1}{2}kx^2$ . Because the system is conservative, the sum of these energies equals the initial energy of the system,  $\frac{1}{2}kA^2$ . We find the velocity  $v_x$  at any position  $x$  just as we did in Section 14.3, Eq. (14.22):

$$v_x = \pm \sqrt{\frac{k}{m}} \sqrt{A^2 - x^2} \quad (30.24)$$

The  $L$ - $C$  circuit is also a conservative system. Again let  $Q$  be the maximum capacitor charge. The magnetic-field energy  $\frac{1}{2}Li^2$  in the inductor at any time corresponds to the kinetic energy  $\frac{1}{2}mv^2$  of the oscillating object, and the electric-field energy  $q^2/2C$  in the capacitor corresponds to the elastic potential energy  $\frac{1}{2}kx^2$  of the spring. The sum of these energies is the total energy  $Q^2/2C$  of the system:

$$\frac{1}{2}Li^2 + \frac{q^2}{2C} = \frac{Q^2}{2C} \quad (30.25)$$

The total energy in the  $L$ - $C$  circuit is *constant*; it oscillates between the magnetic and the electric forms, just as the constant total mechanical energy in simple harmonic motion is constant and oscillates between the kinetic and potential forms.

Solving Eq. (30.25) for  $i$ , we find that when the charge on the capacitor is  $q$ , the current  $i$  is

$$i = \pm \sqrt{\frac{1}{LC}} \sqrt{Q^2 - q^2} \quad (30.26)$$

Verify this equation by substituting  $q$  from Eq. (30.21) and  $i$  from Eq. (30.23). Comparing Eqs. (30.24) and (30.26), we see that current  $i = dq/dt$  and charge  $q$  are related in the same way as are velocity  $v_x = dx/dt$  and position  $x$  in the mechanical problem.

**Table 30.1** summarizes the analogies between simple harmonic motion and  $L$ - $C$  circuit oscillations. The striking parallels shown there are so close that we can solve complicated mechanical problems by setting up analogous electric circuits and measuring the currents and voltages that correspond to the mechanical quantities to be determined. This is the basic principle of many analog computers. This analogy can be extended to *damped* oscillations, which we consider in the next section. In Chapter 31 we'll extend the analogy further to include *forced* electrical oscillations, which occur in all alternating-current circuits.

**TABLE 30.1** Oscillation of a Mass-Spring System Compared with Electrical Oscillation in an  $L$ - $C$  Circuit

### Mass-Spring System

$$\text{Kinetic energy} = \frac{1}{2}mv_x^2$$

$$\text{Potential energy} = \frac{1}{2}kx^2$$

$$\frac{1}{2}mv_x^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$$

$$v_x = \pm \sqrt{k/m} \sqrt{A^2 - x^2}$$

$$v_x = dx/dt$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$x = A \cos(\omega t + \phi)$$

### Inductor-Capacitor Circuit

$$\text{Magnetic energy} = \frac{1}{2}Li^2$$

$$\text{Electrical energy} = q^2/2C$$

$$\frac{1}{2}Li^2 + q^2/2C = Q^2/2C$$

$$i = \pm \sqrt{1/LC} \sqrt{Q^2 - q^2}$$

$$i = dq/dt$$

$$\omega = \sqrt{\frac{1}{LC}}$$

$$q = Q \cos(\omega t + \phi)$$

## EXAMPLE 30.8 An oscillating circuit

A 300 V dc power supply is used to charge a 25  $\mu\text{F}$  capacitor. After the capacitor is fully charged, it is disconnected from the power supply and connected across a 10 mH inductor. The resistance in the circuit is negligible. (a) Find the frequency and period of oscillation of the circuit. (b) Find the capacitor charge and the circuit current 1.2 ms after the inductor and capacitor are connected.

**IDENTIFY and SET UP** Our target variables are the oscillation frequency  $f$  and period  $T$ , as well as the charge  $q$  and current  $i$  at a particular time  $t$ . We are given the capacitance  $C$  and the inductance  $L$ , with which we can calculate the frequency and period from Eq. (30.22). We find the charge and current from Eqs. (30.21) and (30.23). Initially the capacitor is fully charged and the current is zero, as in Fig. 30.14a, so the phase angle is  $\phi = 0$  [see the discussion that follows Eq. (30.23)].

*Continued*

**EXECUTE** (a) The natural angular frequency is

$$\omega = \sqrt{\frac{1}{LC}} = \sqrt{\frac{1}{(10 \times 10^{-3} \text{ H})(25 \times 10^{-6} \text{ F})}} = 2.0 \times 10^3 \text{ rad/s}$$

The frequency  $f$  and period  $T$  are then

$$f = \frac{\omega}{2\pi} = \frac{2.0 \times 10^3 \text{ rad/s}}{2\pi \text{ rad/cycle}} = 320 \text{ Hz}$$

$$T = \frac{1}{f} = \frac{1}{320 \text{ Hz}} = 3.1 \times 10^{-3} \text{ s} = 3.1 \text{ ms}$$

(b) Since the period of the oscillation is  $T = 3.1 \text{ ms}$ ,  $t = 1.2 \text{ ms}$  equals  $0.38T$ ; this corresponds to a situation intermediate between Fig. 30.14b ( $t = T/4$ ) and Fig. 30.14c ( $t = T/2$ ). Comparing those figures with Fig. 30.15, we expect the capacitor charge  $q$  to be negative (that is, there will be negative charge on the left-hand plate of the capacitor) and the current  $i$  to be negative as well (that is, current will flow counterclockwise).

To find  $q$ , we use Eq. (30.21),  $q = Q \cos(\omega t + \phi)$ . The charge is maximum at  $t = 0$ , so  $\phi = 0$  and  $Q = C\mathcal{E} = (25 \times 10^{-6} \text{ F})(300 \text{ V}) = 7.5 \times 10^{-3} \text{ C}$ . Hence Eq. (30.21) becomes

$$q = (7.5 \times 10^{-3} \text{ C}) \cos \omega t$$

At time  $t = 1.2 \times 10^{-3} \text{ s}$ ,

$$\omega t = (2.0 \times 10^3 \text{ rad/s})(1.2 \times 10^{-3} \text{ s}) = 2.4 \text{ rad}$$

$$q = (7.5 \times 10^{-3} \text{ C}) \cos(2.4 \text{ rad}) = -5.5 \times 10^{-3} \text{ C}$$

From Eq. (30.23), the current  $i$  at any time is  $i = -\omega Q \sin \omega t$ . At  $t = 1.2 \times 10^{-3} \text{ s}$ ,

$$i = -(2.0 \times 10^3 \text{ rad/s})(7.5 \times 10^{-3} \text{ C}) \sin(2.4 \text{ rad}) = -10 \text{ A}$$

**EVALUATE** The signs of both  $q$  and  $i$  are negative, as predicted.

**KEYCONCEPT** An  $L$ - $C$  circuit that contains an inductor and a capacitor (and no resistance) is analogous to an ideal mass-spring system. The capacitor charge and the current in the circuit both oscillate sinusoidally, just like the displacement and the velocity for a mass on a spring.

### EXAMPLE 30.9 Energy in an oscillating circuit

### WITH VARIATION PROBLEMS

For the  $L$ - $C$  circuit of Example 30.8, find the magnetic and electrical energies (a) at  $t = 0$  and (b) at  $t = 1.2 \text{ ms}$ .

**IDENTIFY and SET UP** We must calculate the magnetic energy  $U_B$  (stored in the inductor) and the electrical energy  $U_E$  (stored in the capacitor) at two times during the  $L$ - $C$  circuit oscillation. From Example 30.8 we know the values of the capacitor charge  $q$  and circuit current  $i$  for both times. We use them to calculate  $U_B = \frac{1}{2}Li^2$  and  $U_E = q^2/2C$ .

**EXECUTE** (a) At  $t = 0$  there is no current and  $q = Q$ . Hence there is no magnetic energy, and all the energy in the circuit is in the form of electrical energy in the capacitor:

$$U_B = \frac{1}{2}Li^2 = 0 \quad U_E = \frac{Q^2}{2C} = \frac{(7.5 \times 10^{-3} \text{ C})^2}{2(25 \times 10^{-6} \text{ F})} = 1.1 \text{ J}$$

(b) From Example 30.8, at  $t = 1.2 \text{ ms}$  we have  $i = -10 \text{ A}$  and  $q = -5.5 \times 10^{-3} \text{ C}$ . Hence

$$U_B = \frac{1}{2}Li^2 = \frac{1}{2}(10 \times 10^{-3} \text{ H})(-10 \text{ A})^2 = 0.5 \text{ J}$$

$$U_E = \frac{q^2}{2C} = \frac{(-5.5 \times 10^{-3} \text{ C})^2}{2(25 \times 10^{-6} \text{ F})} = 0.6 \text{ J}$$

**EVALUATE** The magnetic and electrical energies are the same at  $t = 3T/8 = 0.375T$ , halfway between the situations in Figs. 30.14b and 30.14c. We saw in Example 30.8 that the time considered in part (b),  $t = 1.2 \text{ ms}$ , equals  $0.38T$ ; this is slightly later than  $0.375T$ , so  $U_B$  is slightly less than  $U_E$ . At all times the total energy  $E = U_B + U_E$  has the same value, 1.1 J. An  $L$ - $C$  circuit without resistance is a conservative system; no energy is dissipated.

**KEYCONCEPT** In an  $L$ - $C$  circuit, the inductor is analogous to the mass in an ideal mass-spring system and the capacitor is analogous to the spring. Magnetic energy in the inductor behaves like the kinetic energy of the mass; electrical energy in the capacitor behaves like the potential energy of the spring.

**TEST YOUR UNDERSTANDING OF SECTION 30.5** One way to think about the energy stored in an  $L$ - $C$  circuit is to say that the circuit elements do positive or negative work on the charges that move back and forth through the circuit. (a) Between stages (a) and (b) in Fig. 30.14, does the capacitor do positive or negative work on the charges? (b) What kind of force (electric or magnetic) does the capacitor exert on the charges to do this work? (c) During this process, does the inductor do positive or negative work on the charges? (d) What kind of force (electric or magnetic) does the inductor exert on the charges?

### ANSWER

**(a) positive, (b) electric, (c) negative, (d) electric** The capacitor loses energy between stages (a) and (b), so it does positive work on the charges. It does this by exerting an electric force that pushes current away from the positively charged left-hand capacitor plate and toward the negatively charged right-hand plate. At the same time, the inductor gains energy and does negative work on the moving charges. Although the inductor stores magnetic energy, the force that the inductor exerts is *electric*. This force comes about from the inductor's self-induced emf (see Section 30.2).

## 30.6 THE L-R-C SERIES CIRCUIT

In our discussion of the  $L$ - $C$  circuit we assumed that there was no *resistance* in the circuit. This is an idealization, of course; every real inductor has resistance in its windings, and there may also be resistance in the connecting wires. Because of resistance, the electromagnetic energy in the circuit is dissipated and converted to other forms, such as internal energy of the circuit materials. Resistance in an electric circuit is analogous to friction in a mechanical system.

Suppose an inductor with inductance  $L$  and a resistor of resistance  $R$  are connected in series across the terminals of a charged capacitor, forming an  **$L$ - $R$ - $C$  series circuit**. As before, the capacitor starts to discharge as soon as the circuit is completed. But due to  $i^2R$  losses in the resistor, the magnetic-field energy that the inductor acquires when the capacitor is completely discharged is *less* than the original electric-field energy of the capacitor. In the same way, the energy of the capacitor when the magnetic field has decreased to zero is still less and so on.

If the resistance  $R$  of the resistor is relatively small, the circuit still oscillates, but with **damped harmonic motion** (Fig. 30.16a), and we say that the circuit is **underdamped**. If we increase  $R$ , the oscillations die out more rapidly. When  $R$  reaches a certain value, the circuit no longer oscillates; it is **critically damped** (Fig. 30.16b). For still larger values of  $R$ , the circuit is **overdamped** (Fig. 30.16c), and the capacitor charge approaches zero even more slowly. We used these same terms to describe the behavior of the analogous mechanical system, the damped harmonic oscillator, in Section 14.7.

### Analyzing an L-R-C Series Circuit

To analyze  $L$ - $R$ - $C$  series circuit behavior in detail, consider the circuit shown in Fig. 30.17. It is like the  $L$ - $C$  circuit of Fig. 30.15 except for the added resistor  $R$ ; we also show the source that charges the capacitor initially. The labeling of the positive senses of  $q$  and  $i$  is the same as for the  $L$ - $C$  circuit.

First we close the switch in the upward position, connecting the capacitor to a source of emf  $\mathcal{E}$  for a long enough time to ensure that the capacitor acquires its final charge  $Q = C\mathcal{E}$  and any initial oscillations have died out. Then at time  $t = 0$  we flip the switch to the downward position, removing the source from the circuit and placing the capacitor in series with the resistor and inductor. Note that the initial current is negative, opposite to the direction of  $i$  shown in Fig. 30.17.

To find how  $q$  and  $i$  vary with time, we apply Kirchhoff's loop rule. Starting at point  $a$  and going around the loop in the direction  $abcda$ , we obtain

$$-iR - L \frac{di}{dt} - \frac{q}{C} = 0$$

Replacing  $i$  with  $dq/dt$  and rearranging, we get

$$\frac{d^2q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{1}{LC}q = 0 \quad (30.27)$$

Note that when  $R = 0$ , this reduces to Eq. (30.20) for an  $L$ - $C$  circuit.

There are general methods for obtaining solutions of Eq. (30.27). The form of the solution is different for the underdamped (small  $R$ ) and overdamped (large  $R$ ) cases. When  $R^2$  is less than  $4L/C$ , the solution has the form

$$q = Ae^{-(R/2L)t} \cos\left(\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}t + \phi\right) \quad (30.28)$$

where  $A$  and  $\phi$  are constants. You can take the first and second derivatives of this function and show by direct substitution that it does satisfy Eq. (30.27).

This solution corresponds to the *underdamped* behavior shown in Fig. 30.16a; the function represents a sinusoidal oscillation with an exponentially decaying amplitude. (Note that the exponential factor  $e^{-(R/2L)t}$  is *not* the same as the factor  $e^{-(R/L)t}$  that we encountered in describing the  $R$ - $L$  circuit in Section 30.4.) When  $R = 0$ , Eq. (30.28) reduces to Eq. (30.21)

Figure 30.16 Graphs of capacitor charge as a function of time in an  $L$ - $R$ - $C$  series circuit with initial charge  $Q$ .

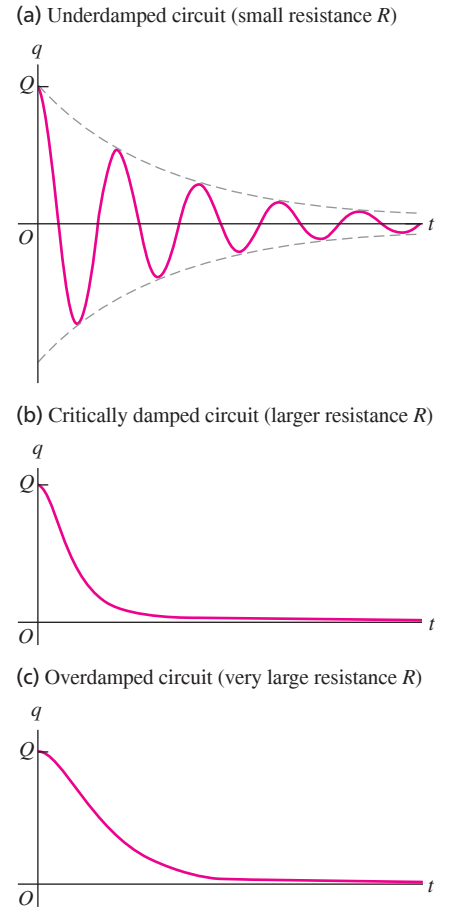
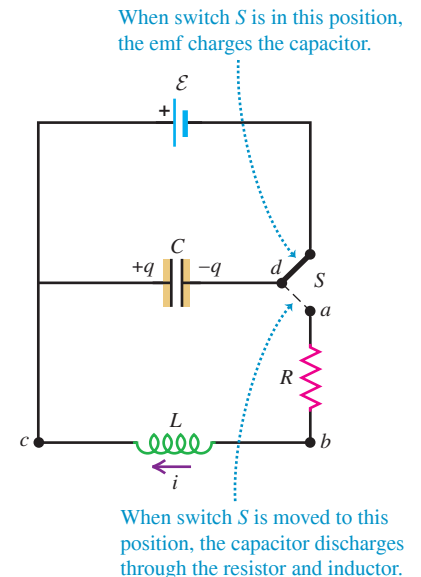


Figure 30.17 An  $L$ - $R$ - $C$  series circuit.





for the oscillations in an  $L$ - $C$  circuit. If  $R$  is not zero, the angular frequency of the oscillation is *less* than  $1/(LC)^{1/2}$  because of the term containing  $R$ . The angular frequency  $\omega'$  of the damped oscillations is given by

$$\text{Angular frequency of underdamped oscillations in an } L\text{-}R\text{-}C \text{ series circuit} \rightarrow \omega' = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \quad (30.29)$$

Inductance      Capacitance      Resistance

When  $R = 0$ , this reduces to Eq. (30.22),  $\omega = (1/LC)^{1/2}$ . As  $R$  increases,  $\omega'$  becomes smaller and smaller. When  $R^2 = 4L/C$ , the quantity under the radical becomes zero; the system no longer oscillates, and the case of *critical damping* (Fig. 30.16b) has been reached. For still larger values of  $R$  the system behaves as in Fig. 30.16c. In this case the circuit is *overdamped*, and  $q$  is given as a function of time by the sum of two decreasing exponential functions.

In the *underdamped* case the phase constant  $\phi$  in the cosine function of Eq. (30.28) provides for the possibility of both an initial charge and an initial current at time  $t = 0$ , analogous to an underdamped harmonic oscillator given both an initial displacement and an initial velocity (see Exercise 30.38).

We emphasize once more that the behavior of the  $L$ - $R$ - $C$  series circuit is completely analogous to that of the damped harmonic oscillator (see Section 14.7). We invite you to verify, for example, that if you start with Eq. (14.41) and substitute  $q$  for  $x$ ,  $L$  for  $m$ ,  $1/C$  for  $k$ , and  $R$  for the damping constant  $b$ , the result is Eq. (30.27). Similarly, the cross-over point between underdamping and overdamping occurs at  $b^2 = 4km$  for the mechanical system and at  $R^2 = 4L/C$  for the electrical one. Can you find still other aspects of this analogy?

The practical applications of the  $L$ - $R$ - $C$  series circuit emerge when we include a sinusoidally varying source of emf in the circuit. This is analogous to the *forced oscillations* that we discussed in Section 14.7, and there are analogous *resonance* effects. Such a circuit is called an *alternating-current (ac) circuit*. The analysis of ac circuits is the principal topic of the next chapter.

### EXAMPLE 30.10 An underdamped $L$ - $R$ - $C$ series circuit

### WITH VARIATION PROBLEMS

What resistance  $R$  is required (in terms of  $L$  and  $C$ ) to give an  $L$ - $R$ - $C$  series circuit a frequency that is one-half the undamped frequency?

**IDENTIFY and SET UP** This problem concerns an underdamped  $L$ - $R$ - $C$  series circuit (Fig. 30.16a). We want just enough resistance to reduce the oscillation frequency to one-half of the undamped value. Equation (30.29) gives the angular frequency  $\omega'$  of an underdamped  $L$ - $R$ - $C$  series circuit; Eq. (30.22) gives the angular frequency  $\omega$  of an undamped  $L$ - $C$  circuit. We use these two equations to solve for  $R$ .

**EXECUTE** From Eqs. (30.29) and (30.22), the requirement  $\omega' = \omega/2$  yields

$$\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = \frac{1}{2} \sqrt{\frac{1}{LC}}$$

We square both sides and solve for  $R$ :

$$R = \sqrt{\frac{3L}{C}}$$

For example, adding  $35 \, \Omega$  to the circuit of Example 30.8 ( $L = 10 \, \text{mH}$ ,  $C = 25 \, \mu\text{F}$ ) would reduce the frequency from 320 Hz to 160 Hz.

**EVALUATE** The circuit becomes critically damped with no oscillations when  $R = \sqrt{4L/C}$ . Our result for  $R$  is smaller than that, as it should be; we want the circuit to be *underdamped*.

**KEYCONCEPT** A series  $L$ - $R$ - $C$  circuit is analogous to a mass-spring system with damping. Increasing the resistance reduces the oscillation frequency of the charge and current; if the resistance is too great, the circuit is critically damped or overdamped and there is no oscillation.

**TEST YOUR UNDERSTANDING OF SECTION 30.6** An  $L$ - $R$ - $C$  series circuit includes a  $2.0 \, \Omega$  resistor. At  $t = 0$  the capacitor charge is  $2.0 \, \mu\text{C}$ . For which of the following values of the inductance and capacitance will the charge on the capacitor *not* oscillate? (i)  $L = 3.0 \, \mu\text{H}$ ,  $C = 6.0 \, \mu\text{F}$ ; (ii)  $L = 6.0 \, \mu\text{H}$ ,  $C = 3.0 \, \mu\text{F}$ ; (iii)  $L = 3.0 \, \mu\text{H}$ ,  $C = 3.0 \, \mu\text{F}$ .

### ANSWER

(i) and (iii) There are no oscillations if  $R^2 \geq 4L/C$ . In each case  $R^2 = (2.0 \, \Omega)^2 = 4.0 \, \Omega^2$ . In case (i)  $4L/C = 4(3.0 \, \mu\text{H})/(6.0 \, \mu\text{F}) = 2.0 \, \Omega^2$ , so there are no oscillations (the system is overdamped); in case (ii)  $4L/C = 4(6.0 \, \mu\text{H})/(3.0 \, \mu\text{F}) = 8.0 \, \Omega^2$ , so there are oscillations (the system is underdamped); and in case (iii)  $4L/C = 4(3.0 \, \mu\text{H})/(3.0 \, \mu\text{F}) = 4.0 \, \Omega^2$ , so there are no oscillations (the system is critically damped).

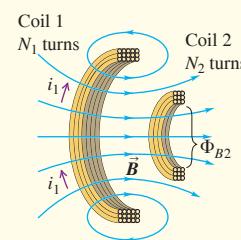
## CHAPTER 30 SUMMARY

**Mutual inductance:** When a changing current  $i_1$  in one circuit causes a changing magnetic flux in a second circuit, an emf  $\mathcal{E}_2$  is induced in the second circuit. Likewise, a changing current  $i_2$  in the second circuit induces an emf  $\mathcal{E}_1$  in the first circuit. If the circuits are coils of wire with  $N_1$  and  $N_2$  turns, the mutual inductance  $M$  can be expressed in terms of the average flux  $\Phi_{B2}$  through each turn of coil 2 caused by the current  $i_1$  in coil 1, or in terms of the average flux  $\Phi_{B1}$  through each turn of coil 1 caused by the current  $i_2$  in coil 2. The SI unit of mutual inductance is the henry, abbreviated H. (See Examples 30.1 and 30.2.)

$$\mathcal{E}_2 = -M \frac{di_1}{dt} \quad \text{and} \quad (30.4)$$

$$\mathcal{E}_1 = -M \frac{di_2}{dt} \quad (30.5)$$

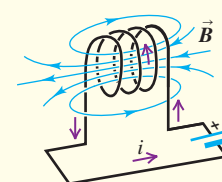
$$M = \frac{N_2 \Phi_{B2}}{i_1} = \frac{N_1 \Phi_{B1}}{i_2} \quad (30.6)$$



**Self-inductance:** A changing current  $i$  in any circuit causes a self-induced emf  $\mathcal{E}$ . The inductance (or self-inductance)  $L$  depends on the geometry of the circuit and the material surrounding it. The inductance of a coil of  $N$  turns is related to the average flux  $\Phi_B$  through each turn caused by the current  $i$  in the coil. An inductor is a circuit device, usually including a coil of wire, intended to have a substantial inductance. (See Examples 30.3 and 30.4.)

$$\mathcal{E} = -L \frac{di}{dt} \quad (30.7)$$

$$L = \frac{N\Phi_B}{i} \quad (30.8)$$

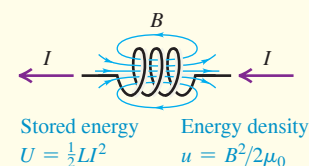


**Magnetic-field energy:** An inductor with inductance  $L$  carrying current  $I$  has energy  $U$  associated with the inductor's magnetic field. The magnetic energy density  $u$  (energy per unit volume) is proportional to the square of the magnetic-field magnitude. (See Example 30.5.)

$$U = \frac{1}{2} LI^2 \quad (30.9)$$

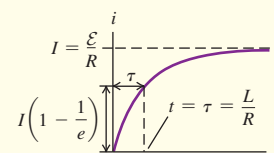
$$u = \frac{B^2}{2\mu_0} \quad (\text{in vacuum}) \quad (30.10)$$

$$u = \frac{B^2}{2\mu} \quad (\text{in a material with magnetic permeability } \mu) \quad (30.11)$$



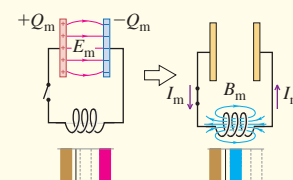
**R-L circuits:** In a circuit containing a resistor  $R$ , an inductor  $L$ , and a source of emf, the growth and decay of current are exponential. The time constant  $\tau$  is the time required for the current to approach within a fraction  $1/e$  of its final value. (See Examples 30.6 and 30.7.)

$$\tau = \frac{L}{R} \quad (30.16)$$



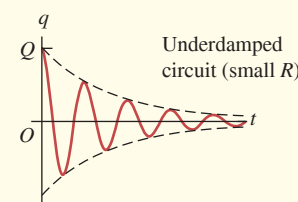
**L-C circuits:** A circuit that contains inductance  $L$  and capacitance  $C$  undergoes electrical oscillations with an angular frequency  $\omega$  that depends on  $L$  and  $C$ . This is analogous to a mechanical harmonic oscillator, with inductance  $L$  analogous to mass  $m$ , the reciprocal of capacitance  $1/C$  to force constant  $k$ , charge  $q$  to displacement  $x$ , and current  $i$  to velocity  $v_x$ . (See Examples 30.8 and 30.9.)

$$\omega = \sqrt{\frac{1}{LC}} \quad (30.22)$$



**L-R-C series circuits:** A circuit that contains inductance, resistance, and capacitance undergoes damped oscillations for sufficiently small resistance. The frequency  $\omega'$  of damped oscillations depends on the values of  $L$ ,  $R$ , and  $C$ . As  $R$  increases, the damping increases; if  $R$  is greater than a certain value, the behavior becomes overdamped and no longer oscillates. (See Example 30.10.)

$$\omega' = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \quad (30.29)$$





## GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

### KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 30.1 and 30.2** (Section 30.1) and **EXAMPLES 30.3 and 30.4** (Section 30.2) before attempting these problems.

**VP30.4.1** A long solenoid (coil 1) is surrounded at its center by a shorter coil (coil 2) as in Fig. 30.3. At an instant when the current in coil 1 is 24.0 A and increasing at a rate of 2.00 A/s, the induced emf in coil 2 is  $1.30 \times 10^{-4}$  V. (a) What is the mutual inductance of the two coils? (b) At this instant, what is the total magnetic flux through all the turns of coil 2 due to the current in coil 1?

**VP30.4.2** The mutual inductance of two coils (1 and 2) is 48.0  $\mu$ H. Find the magnitude of the emf induced in coil 2 if the current in coil 1 is (a) 35.0 A and increasing at 7.00 A/s; (b) 28.0 A and constant; (c) 26.0 A and decreasing at 3.00 A/s.

**VP30.4.3** A toroidal solenoid with self-inductance 76.0  $\mu$ H has 465 turns of wire. Find (a) the magnetic flux through each turn when the current in the solenoid is 12.0 A and (b) the magnitude of the induced emf in the solenoid when the current is 12.0 A and decreasing at 55.0 A/s.

**VP30.4.4** When a coil is carrying a current of 25.0 A that is increasing at 145 A/s, the induced emf in the coil has magnitude 3.70 mV. Find (a) the self-inductance of the coil and (b) the magnitude and direction of the induced emf at  $t = 2.00$  s if the current is  $i(t) = (225 \text{ A/s}^2)t^2$ .

Be sure to review **EXAMPLES 30.6 and 30.7** (Section 30.4) before attempting these problems.

**VP30.7.1** In the circuit shown in Fig. 30.11, the inductance is 4.90 mH and the emf is 24.0 V. Initially both switches are open. You want the current to reach 75.0% of its maximum value 80.0  $\mu$ s after switch  $S_1$  is closed. Find (a) the required resistance and (b) the current 80.0  $\mu$ s after switch  $S_1$  is closed.

**VP30.7.2** In the circuit shown in Fig. 30.11,  $\mathcal{E} = 12.0$  V,  $R = 18.0 \Omega$ , and  $L = 37.5$  mH. Initially both switches are open. You close switch  $S_1$

at  $t = 0$ . At  $t = 0.700$  ms, find (a) the current, (b) the rate of change of the current, and (c) the amount of energy stored in the inductor.

**VP30.7.3** In the circuit shown in Fig. 30.11,  $\mathcal{E} = 12.0$  V,  $R = 18.0 \Omega$ , and  $L = 37.5$  mH. Switch  $S_1$  has been closed for a long time and switch  $S_2$  is open. At  $t = 0$  you open switch  $S_1$  and close switch  $S_2$ . At  $t = 0.700$  ms, find (a) the current, (b) the rate of change of the current, and (c) the amount of energy stored in the inductor.

**VP30.7.4** For the  $R$ - $L$  circuit with an increasing current shown in Fig. 30.12, find (a) the time when the voltage across the inductor has the same magnitude as the voltage across the resistor, (b) the current at this time, and (c) the rate of change of the current at this time.

Be sure to review **EXAMPLES 30.8 and 30.9** (Section 30.5) and **EXAMPLE 30.10** (Section 30.6) before attempting these problems.

**VP30.10.1** The  $L$ - $C$  circuit in Fig. 30.15 contains an inductor with  $L = 25.0$   $\mu$ H. The charge and current in this circuit oscillate with frequency 445 Hz, and the maximum capacitor charge is 5.00 mC. Find (a) the capacitance of the capacitor and (b) the maximum current in the circuit.

**VP30.10.2** You construct an  $L$ - $C$  circuit that contains a 30.0  $\mu$ H inductor and a 655  $\mu$ F capacitor. At  $t = 0$  the capacitor has its maximum charge of 0.400 mC. Find (a) the first time after  $t = 0$  when the capacitor charge is zero and (b) the magnitude of the current in the circuit at this time.

**VP30.10.3** The sum of the electrical and magnetic energies in an  $L$ - $C$  circuit is 0.800 J. At a certain instant the energy is exactly half electrical and half magnetic, the capacitor charge is 5.30 mC, and the current is 8.00 A. Find (a) the capacitance, (b) the inductance, and (c) the angular frequency of oscillation.

**VP30.10.4** An  $L$ - $R$ - $C$  series circuit has inductance 42.0 mH, capacitance  $C$ , and resistance  $R$ . Without the resistor, the angular frequency of oscillation is 624 rad/s. With the resistor, the angular frequency is 208 rad/s. Find the values of (a)  $C$  and (b)  $R$ .

### BRIDGING PROBLEM Analyzing an $L$ - $C$ Circuit

An  $L$ - $C$  circuit like that shown in Fig. 30.14 consists of a 60.0 mH inductor and a 250  $\mu$ F capacitor. The initial charge on the capacitor is 6.00  $\mu$ C, and the initial current in the inductor is 0.400 mA. (a) What is the maximum energy stored in the inductor? (b) What is the maximum current in the inductor? (c) What is the maximum voltage across the capacitor? (d) When the current in the inductor has half its maximum value, what are the energy stored in the inductor and the voltage across the capacitor?

#### SOLUTION GUIDE

#### IDENTIFY and SET UP

1. An  $L$ - $C$  circuit is a conservative system—there is no resistance to dissipate energy. The energy oscillates between electrical energy in the capacitor and magnetic energy stored in the inductor.
2. Oscillations in an  $L$ - $C$  circuit are analogous to the mechanical oscillations of a particle at the end of an ideal spring (see Table 30.1). Compare this problem to the analogous mechanical problem (see Example 14.3 in Section 14.2 and Example 14.4 in Section 14.3).

3. Which key equations are needed to describe the capacitor? To describe the inductor?

#### EXECUTE

4. Find the initial total energy in the circuit. Use it to determine the maximum energy stored in the inductor during the oscillation.
5. Use the result of step 4 to find the maximum current in the inductor.
6. Use the result of step 4 to find the maximum energy stored in the capacitor during the oscillation. Then use this to find the maximum capacitor voltage.
7. Find the energy in the inductor and the capacitor charge when the current has half the value that you found in step 5.

#### EVALUATE

8. Initially, what fraction of the total energy is in the inductor? Is it possible to tell whether this is initially increasing or decreasing?

## PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

### DISCUSSION QUESTIONS

**Q30.1** In an electric trolley or bus system, the vehicle's motor draws current from an overhead wire by means of a long arm with an attachment at the end that slides along the overhead wire. A brilliant electric spark is often seen when the attachment crosses a junction in the wires where contact is momentarily lost. Explain this phenomenon.

**Q30.2** From Eq. (30.5)  $1 \text{ H} = 1 \text{ Wb/A}$ , and from Eqs. (30.4)  $1 \text{ H} = 1 \Omega \cdot \text{s}$ . Show that these two definitions are equivalent.

**Q30.3** In Fig. 30.1, if coil 2 is turned  $90^\circ$  so that its axis is vertical, does the mutual inductance increase or decrease? Explain.

**Q30.4** The tightly wound toroidal solenoid is one of the few configurations for which it is easy to calculate self-inductance. What features of the toroidal solenoid give it this simplicity?

**Q30.5** Two identical, closely wound, circular coils, each having self-inductance  $L$ , are placed next to each other, so that they are coaxial and almost touching. If they are connected in series, what is the self-inductance of the combination? What if they are connected in parallel? Can they be connected so that the total inductance is zero? Explain.

**Q30.6** Two closely wound circular coils have the same number of turns, but one has twice the radius of the other. How are the self-inductances of the two coils related? Explain your reasoning.

**Q30.7** You are to make a resistor by winding a wire around a cylindrical form. To make the inductance as small as possible, it is proposed that you wind half the wire in one direction and the other half in the opposite direction. Would this achieve the desired result? Why or why not?

**Q30.8** For the same magnetic field strength  $B$ , is the energy density greater in vacuum or in a magnetic material? Explain. Does Eq. (30.11) imply that for a long solenoid in which the current is  $I$  the energy stored is proportional to  $1/\mu$ ? And does this mean that for the same current less energy is stored when the solenoid is filled with a ferromagnetic material rather than with air? Explain.

**Q30.9** In an  $R$ - $C$  circuit, a resistor, an uncharged capacitor, a dc battery, and an open switch are in series. In an  $R$ - $L$  circuit, a resistor, an inductor, a dc battery, and an open switch are in series. Compare the behavior of the current in these circuits (a) just after the switch is closed and (b) long after the switch has been closed. In other words, compare the way in which a capacitor and an inductor affect a circuit.

**Q30.10 A Differentiating Circuit.** Figure **Q30.10**

The current in a resistanceless inductor is caused to vary with time as shown in the graph of **Fig. Q30.10**. (a) Sketch the pattern that would be observed on the screen of an oscilloscope connected to the terminals of the inductor. (The oscilloscope spot sweeps horizontally across the screen at a constant speed, and its vertical deflection is proportional to the potential difference between the inductor terminals.) (b) Explain why a circuit with an inductor can be described as a "differentiating circuit."

**Q30.11** In Section 30.5 Kirchhoff's loop rule is applied to an  $L$ - $C$  circuit where the capacitor is initially fully charged and the equation  $-L(di/dt) - (q/C) = 0$  is derived. But as the capacitor starts to discharge, the current increases from zero. The equation says  $L di/dt = -q/C$ , so it says  $L di/dt$  is negative. Explain how  $L di/dt$  can be negative when the current is increasing.

**Q30.12** In Section 30.5 the relationship  $i = dq/dt$  is used in deriving Eq. (30.20). But a flow of current corresponds to a decrease in the charge on the capacitor. Explain, therefore, why this is the correct equation to use in the derivation, rather than  $i = -dq/dt$ .

**Q30.13** In the  $R$ - $L$  circuit shown in Fig. 30.11, when switch  $S_1$  is closed, the potential  $v_{ac}$  changes suddenly and discontinuously, but the current does not. Explain why the voltage can change suddenly but the current can't.

**Q30.14** In the  $R$ - $L$  circuit shown in Fig. 30.11, is the current in the resistor always the same as the current in the inductor? How do you know?

**Q30.15** Suppose there is a steady current in an inductor. If you attempt to reduce the current to zero instantaneously by quickly opening a switch, an arc can appear at the switch contacts. Why? Is it physically possible to stop the current instantaneously? Explain.

**Q30.16** In an  $L$ - $R$ - $C$  series circuit, what criteria could be used to decide whether the system is overdamped or underdamped? For example, could we compare the maximum energy stored during one cycle to the energy dissipated during one cycle? Explain.

### EXERCISES

#### Section 30.1 Mutual Inductance

**30.1 •** Two coils have mutual inductance  $M = 3.25 \times 10^{-4} \text{ H}$ . The current  $i_1$  in the first coil increases at a uniform rate of  $830 \text{ A/s}$ . (a) What is the magnitude of the induced emf in the second coil? Is it constant? (b) Suppose that the current described is in the second coil rather than the first. What is the magnitude of the induced emf in the first coil?

**30.2 •** Two coils are wound around the same cylindrical form, like the coils in Example 30.1. When the current in the first coil is decreasing at a rate of  $-0.242 \text{ A/s}$ , the induced emf in the second coil has magnitude  $1.65 \times 10^{-3} \text{ V}$ . (a) What is the mutual inductance of the pair of coils? (b) If the second coil has 25 turns, what is the flux through each turn when the current in the first coil equals  $1.20 \text{ A}$ ? (c) If the current in the second coil increases at a rate of  $0.360 \text{ A/s}$ , what is the magnitude of the induced emf in the first coil?

**30.3 •** Two toroidal solenoids are wound around the same form so that the magnetic field of one passes through the turns of the other. Solenoid 1 has 700 turns, and solenoid 2 has 400 turns. When the current in solenoid 1 is  $6.52 \text{ A}$ , the average flux through each turn of solenoid 2 is  $0.0320 \text{ Wb}$ . (a) What is the mutual inductance of the pair of solenoids? (b) When the current in solenoid 2 is  $2.54 \text{ A}$ , what is the average flux through each turn of solenoid 1?

**30.4 •** A solenoidal coil with 25 turns of wire is wound tightly around another coil with 300 turns (see Example 30.1). The inner solenoid is  $25.0 \text{ cm}$  long and has a diameter of  $2.00 \text{ cm}$ . At a certain time, the current in the inner solenoid is  $0.120 \text{ A}$  and is increasing at a rate of  $1.75 \times 10^3 \text{ A/s}$ . For this time, calculate: (a) the average magnetic flux through each turn of the inner solenoid; (b) the mutual inductance of the two solenoids; (c) the emf induced in the outer solenoid by the changing current in the inner solenoid.

#### Section 30.2 Self-Inductance and Inductors

**30.5 •** A  $2.50 \text{ mH}$  toroidal solenoid has an average radius of  $6.00 \text{ cm}$  and a cross-sectional area of  $2.00 \text{ cm}^2$ . (a) How many coils does it have? (Make the same assumption as in Example 30.3.) (b) At what rate must the current through it change so that a potential difference of  $2.00 \text{ V}$  is developed across its ends?



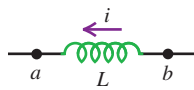
**30.6 •** A toroidal solenoid has 500 turns, cross-sectional area  $6.25 \text{ cm}^2$ , and mean radius  $4.00 \text{ cm}$ . (a) Calculate the coil's self-inductance. (b) If the current decreases uniformly from  $5.00 \text{ A}$  to  $2.00 \text{ A}$  in  $3.00 \text{ ms}$ , calculate the self-induced emf in the coil. (c) The current is directed from terminal  $a$  of the coil to terminal  $b$ . Is the direction of the induced emf from  $a$  to  $b$  or from  $b$  to  $a$ ?

**30.7 •** At the instant when the current in an inductor is increasing at a rate of  $0.0640 \text{ A/s}$ , the magnitude of the self-induced emf is  $0.0160 \text{ V}$ . (a) What is the inductance of the inductor? (b) If the inductor is a solenoid with 400 turns, what is the average magnetic flux through each turn when the current is  $0.720 \text{ A}$ ?

**30.8 ••** When the current in a toroidal solenoid is changing at a rate of  $0.0260 \text{ A/s}$ , the magnitude of the induced emf is  $12.6 \text{ mV}$ . When the current equals  $1.40 \text{ A}$ , the average flux through each turn of the solenoid is  $0.00285 \text{ Wb}$ . How many turns does the solenoid have?

**30.9 •** The inductor in **Fig. E30.9** has inductance  $0.260 \text{ H}$  and carries a current in the direction shown that is decreasing at a uniform rate,  $di/dt = -0.0180 \text{ A/s}$ . (a) Find the self-induced emf. (b) Which end of the inductor,  $a$  or  $b$ , is at a higher potential?

Figure E30.9



**30.10 •** The inductor shown in **Fig. E30.9** has inductance  $0.260 \text{ H}$  and carries a current in the direction shown. The current is changing at a constant rate. (a) The potential between points  $a$  and  $b$  is  $V_{ab} = 1.04 \text{ V}$ , with point  $a$  at higher potential. Is the current increasing or decreasing? (b) If the current at  $t = 0$  is  $12.0 \text{ A}$ , what is the current at  $t = 2.00 \text{ s}$ ?

**30.11 •• Inductance of a Solenoid.** (a) A long, straight solenoid has  $N$  turns, uniform cross-sectional area  $A$ , and length  $l$ . Show that the inductance of this solenoid is given by the equation  $L = \mu_0 AN^2/l$ . Assume that the magnetic field is uniform inside the solenoid and zero outside. (Your answer is approximate because  $B$  is actually smaller at the ends than at the center. For this reason, your answer is actually an upper limit on the inductance.) (b) A metallic laboratory spring is typically  $5.00 \text{ cm}$  long and  $0.150 \text{ cm}$  in diameter and has 50 coils. If you connect such a spring in an electric circuit, how much self-inductance must you include for it if you model it as an ideal solenoid?

**30.12 •** A long, straight solenoid has 800 turns. When the current in the solenoid is  $2.90 \text{ A}$ , the average flux through each turn of the solenoid is  $3.25 \times 10^{-3} \text{ Wb}$ . What must be the magnitude of the rate of change of the current in order for the self-induced emf to equal  $6.20 \text{ mV}$ ?

### Section 30.3 Magnetic-Field Energy

**30.13 ••** When the current in a long, straight, air-filled solenoid is changing at the rate of  $2000 \text{ A/s}$ , the voltage across the solenoid is  $0.600 \text{ V}$ . The solenoid has 1200 turns and uniform cross-sectional area  $25.0 \text{ mm}^2$ . Assume that the magnetic field is uniform inside the solenoid and zero outside, so the result  $L = \mu_0 AN^2/l$  (see Exercise 30.11) applies. What is the magnitude  $B$  of the magnetic field in the interior of the solenoid when the current in the solenoid is  $3.00 \text{ A}$ ?

**30.14 •** An inductor used in a dc power supply has an inductance of  $12.0 \text{ H}$  and a resistance of  $180 \Omega$ . It carries a current of  $0.500 \text{ A}$ . (a) What is the energy stored in the magnetic field? (b) At what rate is thermal energy developed in the inductor? (c) Does your answer to part (b) mean that the magnetic-field energy is decreasing with time? Explain.

**30.15 •** An air-filled toroidal solenoid has a mean radius of  $15.0 \text{ cm}$  and a cross-sectional area of  $5.00 \text{ cm}^2$ . When the current is  $12.0 \text{ A}$ , the energy stored is  $0.390 \text{ J}$ . How many turns does the winding have?

**30.16 •** An air-filled toroidal solenoid has 300 turns of wire, a mean radius of  $12.0 \text{ cm}$ , and a cross-sectional area of  $4.00 \text{ cm}^2$ . If the current is  $5.00 \text{ A}$ , calculate: (a) the magnetic field in the solenoid; (b) the self-inductance of the solenoid; (c) the energy stored in the magnetic field; (d) the energy density in the magnetic field. (e) Check your answer for part (d) by dividing your answer to part (c) by the volume of the solenoid.

**30.17 ••** A solenoid  $25.0 \text{ cm}$  long and with a cross-sectional area of  $0.500 \text{ cm}^2$  contains 400 turns of wire and carries a current of  $80.0 \text{ A}$ . Calculate: (a) the magnetic field in the solenoid; (b) the energy density in the magnetic field if the solenoid is filled with air; (c) the total energy contained in the coil's magnetic field (assume the field is uniform); (d) the inductance of the solenoid.

**30.18 •** It has been proposed to use large inductors as energy storage devices. (a) How much electrical energy is converted to light and thermal energy by a  $150 \text{ W}$  light bulb in one day? (b) If the amount of energy calculated in part (a) is stored in an inductor in which the current is  $80.0 \text{ A}$ , what is the inductance?

**30.19 ••** In a proton accelerator used in elementary particle physics experiments, the trajectories of protons are controlled by bending magnets that produce a magnetic field of  $4.80 \text{ T}$ . What is the magnetic-field energy in a  $10.0 \text{ cm}^3$  volume of space where  $B = 4.80 \text{ T}$ ?

**30.20 •** A region of vacuum contains both a uniform electric field with magnitude  $E$  and a uniform magnetic field with magnitude  $B$ . (a) What is the ratio  $E/B$  if the energy density for the magnetic field equals the energy density for the electric field? (b) If  $E = 500 \text{ V/m}$ , what is  $B$ , in teslas, if the magnetic-field and electric-field energy densities are equal?

### Section 30.4 The R-L Circuit

**30.21 •** An inductor with an inductance of  $2.50 \text{ H}$  and a resistance of  $8.00 \Omega$  is connected to the terminals of a battery with an emf of  $6.00 \text{ V}$  and negligible internal resistance. Find (a) the initial rate of increase of current in the circuit; (b) the rate of increase of current at the instant when the current is  $0.500 \text{ A}$ ; (c) the current  $0.250 \text{ s}$  after the circuit is closed; (d) the final steady-state current.

**30.22 •** In **Fig. 30.11**,  $R = 15.0 \Omega$  and the battery emf is  $6.30 \text{ V}$ . With switch  $S_2$  open, switch  $S_1$  is closed. After several minutes,  $S_1$  is opened and  $S_2$  is closed. (a) At  $2.00 \text{ ms}$  after  $S_1$  is opened, the current has decayed to  $0.280 \text{ A}$ . Calculate the inductance of the coil. (b) How long after  $S_1$  is opened will the current reach  $1.00\%$  of its original value?

**30.23 •** A  $35.0 \text{ V}$  battery with negligible internal resistance, a  $50.0 \Omega$  resistor, and a  $1.25 \text{ mH}$  inductor with negligible resistance are all connected in series with an open switch. The switch is suddenly closed. (a) How long after closing the switch will the current through the inductor reach one-half of its maximum value? (b) How long after closing the switch will the energy stored in the inductor reach one-half of its maximum value?

**30.24 ••** A resistor and an inductor are connected in series to a battery with emf  $240 \text{ V}$  and negligible internal resistance. The circuit is completed at time  $t = 0$ . At a later time  $t = T$  the current is  $5.00 \text{ A}$  and is increasing at a rate of  $20.0 \text{ A/s}$ . After a long time the current in the circuit is  $15.0 \text{ A}$ . What is the value of  $T$ , the time when the current is  $5.00 \text{ A}$ ?

**30.25 •** A resistor with  $R = 30.0 \Omega$  and an inductor with  $L = 0.600 \text{ H}$  are connected in series to a battery that has emf  $50.0 \text{ V}$  and negligible internal resistance. At time  $t$  after the circuit is completed, the energy stored in the inductor is  $0.400 \text{ J}$ . At this instant, what is the voltage across the inductor?

**30.26 •** In the circuit shown in **Fig. 30.11** switch  $S_1$  has been closed a long time while switch  $S_2$  has been left open. Then  $S_2$  is closed at the same instant when  $S_1$  is opened. Just after  $S_2$  is closed, the current through the resistor is  $12.0 \text{ A}$  and its rate of decrease is  $di/dt = -36.0 \text{ A/s}$ . How long does it take the current to decrease to  $6.00 \text{ A}$ , one-half its initial value?

**30.27 •** In **Fig. 30.11**, suppose that  $\mathcal{E} = 60.0 \text{ V}$ ,  $R = 240 \Omega$ , and  $L = 0.160 \text{ H}$ . With switch  $S_2$  open, switch  $S_1$  is left closed until a constant current is established. Then  $S_2$  is closed and  $S_1$  is opened, taking the battery out of the circuit. (a) What is the initial current in the resistor, just after  $S_2$  is closed and  $S_1$  is opened? (b) What is the current in the resistor at  $t = 4.00 \times 10^{-4} \text{ s}$ ? (c) What is the potential difference between points  $b$  and  $c$  at  $t = 4.00 \times 10^{-4} \text{ s}$ ? Which point is at a higher potential? (d) How long does it take the current to decrease to half its initial value?

**30.28 •** In Fig. 30.11, suppose that  $\mathcal{E} = 60.0 \text{ V}$ ,  $R = 240 \, \Omega$ , and  $L = 0.160 \text{ H}$ . Initially there is no current in the circuit. Switch  $S_2$  is left open, and switch  $S_1$  is closed. (a) Just after  $S_1$  is closed, what are the potential differences  $v_{ab}$  and  $v_{bc}$ ? (b) A long time (many time constants) after  $S_1$  is closed, what are  $v_{ab}$  and  $v_{bc}$ ? (c) What are  $v_{ab}$  and  $v_{bc}$  at an intermediate time when  $i = 0.150 \text{ A}$ ?

**30.29 •** In Fig. 30.11 switch  $S_1$  is closed while switch  $S_2$  is kept open. The inductance is  $L = 0.380 \text{ H}$ , the resistance is  $R = 48.0 \, \Omega$ , and the emf of the battery is  $18.0 \text{ V}$ . At time  $t$  after  $S_1$  is closed, the current in the circuit is increasing at a rate of  $di/dt = 7.20 \text{ A/s}$ . At this instant what is  $v_{ab}$ , the voltage across the resistor?

**30.30 ••** Consider the circuit in Exercise 30.21. (a) Just after the circuit is completed, at what rate is the battery supplying electrical energy to the circuit? (b) When the current has reached its final steady-state value, how much energy is stored in the inductor? What is the rate at which electrical energy is being dissipated in the resistance of the inductor? What is the rate at which the battery is supplying electrical energy to the circuit?

**30.31 •** Consider the rate  $P_L$  at which energy is being stored in the  $R$ - $L$  circuit of Fig. 30.12. Answer these questions, in terms of  $\mathcal{E}$ ,  $R$ , and  $L$  as needed: (a) What is  $P_L$  at  $t = 0$ , just after the circuit is completed? (b) What is  $P_L$  at  $t \rightarrow \infty$ , a long time after the circuit is completed? (c) What is  $P_L$  at the instant when  $i = \mathcal{E}/2R$ , one-half the current's final value?

### Section 30.5 The $L$ - $C$ Circuit

**30.32 ••** A  $15.0 \, \mu\text{F}$  capacitor is charged by a  $150.0 \text{ V}$  power supply, then disconnected from the power and connected in series with a  $0.280 \text{ mH}$  inductor. Calculate: (a) the oscillation frequency of the circuit; (b) the energy stored in the capacitor at time  $t = 0 \text{ ms}$  (the moment of connection with the inductor); (c) the energy stored in the inductor at  $t = 1.30 \text{ ms}$ .

**30.33 •** In an  $L$ - $C$  circuit,  $L = 85.0 \text{ mH}$  and  $C = 3.20 \, \mu\text{F}$ . During the oscillations the maximum current in the inductor is  $0.850 \text{ mA}$ . (a) What is the maximum charge on the capacitor? (b) What is the magnitude of the charge on the capacitor at an instant when the current in the inductor has magnitude  $0.500 \text{ mA}$ ?

**30.34 •** A  $7.50 \text{ nF}$  capacitor is charged to  $12.0 \text{ V}$ , then disconnected from the power supply and connected in series through a coil. The period of oscillation of the circuit is then measured to be  $8.60 \times 10^{-5} \text{ s}$ . Calculate: (a) the inductance of the coil; (b) the maximum charge on the capacitor; (c) the total energy of the circuit; (d) the maximum current in the circuit.

**30.35 •  $L$ - $C$  Oscillations.** A capacitor with capacitance  $6.00 \times 10^{-5} \text{ F}$  is charged by connecting it to a  $12.0 \text{ V}$  battery. The capacitor is disconnected from the battery and connected across an inductor with  $L = 1.50 \text{ H}$ . (a) What are the angular frequency  $\omega$  of the electrical oscillations and the period of these oscillations (the time for one oscillation)? (b) What is the initial charge on the capacitor? (c) How much energy is initially stored in the capacitor? (d) What is the charge on the capacitor  $0.0230 \text{ s}$  after the connection to the inductor is made? Interpret the sign of your answer. (e) At the time given in part (d), what is the current in the inductor? Interpret the sign of your answer. (f) At the time given in part (d), how much electrical energy is stored in the capacitor and how much is stored in the inductor?

**30.36 • A Radio Tuning Circuit.** The minimum capacitance of a variable capacitor in a radio is  $4.18 \text{ pF}$ . (a) What is the inductance of a coil connected to this capacitor if the oscillation frequency of the  $L$ - $C$  circuit is  $1600 \times 10^3 \text{ Hz}$ , corresponding to one end of the AM radio broadcast band, when the capacitor is set to its minimum capacitance? (b) The frequency at the other end of the broadcast band is  $540 \times 10^3 \text{ Hz}$ . What is the maximum capacitance of the capacitor if the oscillation frequency is adjustable over the range of the broadcast band?

**30.37 ••** An  $L$ - $C$  circuit containing an  $80.0 \text{ mH}$  inductor and a  $1.25 \text{ nF}$  capacitor oscillates with a maximum current of  $0.750 \text{ A}$ . Calculate: (a) the maximum charge on the capacitor and (b) the oscillation frequency of the circuit. (c) Assuming the capacitor had its maximum charge at time  $t = 0$ , calculate the energy stored in the inductor after  $2.50 \text{ ms}$  of oscillation.

### Section 30.6 The $L$ - $R$ - $C$ Series Circuit

**30.38 •** For the circuit of Fig. 30.17, let  $C = 15.0 \text{ nF}$ ,  $L = 22 \text{ mH}$ , and  $R = 75.0 \, \Omega$ . (a) Calculate the oscillation frequency of the circuit once the capacitor has been charged and the switch has been connected to point  $a$ . (b) How long will it take for the amplitude of the oscillation to decay to  $10.0\%$  of its original value? (c) What value of  $R$  would result in a critically damped circuit?

**30.39 •** An  $L$ - $R$ - $C$  series circuit has  $L = 0.450 \text{ H}$ ,  $C = 2.50 \times 10^{-5} \text{ F}$ , and resistance  $R$ . (a) What is the angular frequency of the circuit when  $R = 0$ ? (b) What value must  $R$  have to give a  $5.0\%$  decrease in angular frequency compared to the value calculated in part (a)?

**30.40 •** An  $L$ - $R$ - $C$  series circuit has  $L = 0.400 \text{ H}$ ,  $C = 7.00 \, \mu\text{F}$ , and  $R = 320 \, \Omega$ . At  $t = 0$  the current is zero and the initial charge on the capacitor is  $2.80 \times 10^{-4} \text{ C}$ . (a) What are the values of the constants  $A$  and  $\phi$  in Eq. (30.28)? (b) How much time does it take for each complete current oscillation after the switch in this circuit is closed? (c) What is the charge on the capacitor after the first complete current oscillation?

## PROBLEMS

**30.41 ••** It is possible to make your own inductor by winding wire around a cylinder, such as a pencil. Assume you have a spool of AWG 20 copper wire, which has a diameter of  $0.812 \text{ mm}$ . (a) Estimate the diameter of a pencil. (b) Estimate how many times can you tightly wrap AWG 20 copper wire around a pencil to form a solenoid with a length of  $4.0 \text{ cm}$ . (c) Estimate the inductance of this solenoid by assuming the magnetic field inside is constant. (d) If a current of  $1.0 \text{ A}$  flows through this solenoid, how much magnetic energy will be stored inside?

**30.42 •** An inductor is connected to the terminals of a battery that has an emf of  $16.0 \text{ V}$  and negligible internal resistance. The current is  $4.86 \text{ mA}$  at  $0.940 \text{ ms}$  after the connection is completed. After a long time, the current is  $6.45 \text{ mA}$ . What are (a) the resistance  $R$  of the inductor and (b) the inductance  $L$  of the inductor?

**30.43 •• CP** Consider a coil of wire that has radius  $3.00 \text{ cm}$  and carries a sinusoidal current given by  $i(t) = I_0 \sin(2\pi ft)$ , where the frequency  $f = 60.0 \text{ Hz}$  and the initial current  $I_0 = 1.20 \text{ A}$ . (a) Estimate the magnetic flux through this coil as the product of the magnetic field at the center of the coil and the area of the coil. Use this magnetic flux to estimate the self-inductance  $L$  of the coil. (b) Use the value of  $L$  that you estimated in part (a) to calculate the magnitude of the maximum emf induced in the coil.

**30.44 •• CALC** A coil has 400 turns and self-inductance  $7.50 \text{ mH}$ . The current in the coil varies with time according to  $i = (680 \text{ mA}) \cos(\pi t / 0.0250 \text{ s})$ . (a) What is the maximum emf induced in the coil? (b) What is the maximum average flux through each turn of the coil? (c) At  $t = 0.0180 \text{ s}$ , what is the magnitude of the induced emf?

**30.45 •• Solar Magnetic Energy.** Magnetic fields within a sunspot can be as strong as  $0.4 \text{ T}$ . (By comparison, the earth's magnetic field is about  $1/10,000$  as strong.) Sunspots can be as large as  $25,000 \text{ km}$  in radius. The material in a sunspot has a density of about  $3 \times 10^{-4} \text{ kg/m}^3$ . Assume  $\mu$  for the sunspot material is  $\mu_0$ . If  $100\%$  of the magnetic-field energy stored in a sunspot could be used to eject the sunspot's material away from the sun's surface, at what speed would that material be ejected? Compare to the sun's escape speed, which is about  $6 \times 10^5 \text{ m/s}$ . (Hint: Calculate the kinetic energy the magnetic field could supply to  $1 \text{ m}^3$  of sunspot material.)

**30.46 •• CP CALC A Coaxial Cable.** A small solid conductor with radius  $a$  is supported by insulating, nonmagnetic disks on the axis of a thin-walled tube with inner radius  $b$ . The inner and outer conductors carry equal currents  $i$  in opposite directions. (a) Use Ampere's law to find the magnetic field at any point in the volume between the conductors. (b) Write the expression for the flux  $d\Phi_B$  through a narrow strip of length  $l$  parallel to the axis, of width  $dr$ , at a distance  $r$  from the axis of the cable and lying in a plane containing the axis. (c) Integrate your expression from part (b) over the volume between the two conductors to find the total flux produced by a current  $i$  in the central conductor. (d) Show that the inductance of a length  $l$  of the cable is

$$L = l \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right)$$

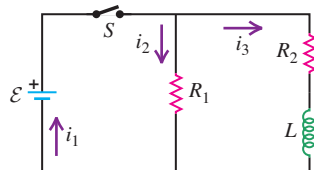
(e) Use Eq. (30.9) to calculate the energy stored in the magnetic field for a length  $l$  of the cable.

**30.47 •** (a) What would have to be the self-inductance of a solenoid for it to store 10.0 J of energy when a 2.00 A current runs through it? (b) If this solenoid's cross-sectional diameter is 4.00 cm, and if you could wrap its coils to a density of 10 coils/mm, how long would the solenoid be? (See Exercise 30.11.) Is this a realistic length for ordinary laboratory use?

**30.48 •• CALC** Consider the circuit in Fig. 30.11 with both switches open. At  $t = 0$  switch  $S_1$  is closed while switch  $S_2$  is left open. (a) Use Eq. (30.14) to derive an equation for the rate  $P_R$  at which electrical energy is being consumed in the resistor. In terms of  $\mathcal{E}$ ,  $R$ , and  $L$ , at what value of  $t$  is  $P_R$  a maximum? What is that maximum value? (b) Use Eqs. (30.14) and (30.15) to derive an equation for  $P_L$ , the rate at which energy is being stored in the inductor. (c) What is  $P_L$  at  $t = 0$  and as  $t \rightarrow \infty$ ? (d) In terms of  $\mathcal{E}$ ,  $R$ , and  $L$ , at what value of  $t$  is  $P_L$  a maximum? What is that maximum value? (e) Obtain an expression for  $P_{\mathcal{E}}$ , the rate at which the battery is supplying electrical energy to the circuit. In terms of  $\mathcal{E}$ ,  $R$ , and  $L$ , at what value of  $t$  is  $P_{\mathcal{E}}$  a maximum? What is that maximum value?

**30.49 • An Electromagnetic Car Alarm.** Your latest invention is a car alarm that produces sound at a particularly annoying frequency of 3500 Hz. To do this, the car-alarm circuitry must produce an alternating electric current of the same frequency. That's why your design includes an inductor and a capacitor in series. The maximum voltage across the capacitor is to be 12.0 V. To produce a sufficiently loud sound, the capacitor must store 0.0160 J of energy. What values of capacitance and inductance should you choose for your car-alarm circuit?

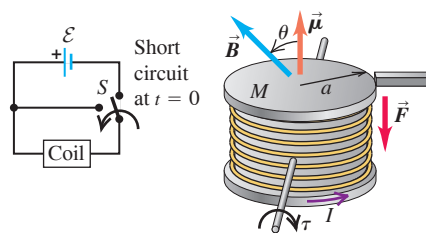
**30.50 •• CALC** An inductor with inductance  $L = 0.300$  H and negligible resistance is connected to a battery, a switch  $S$ , and two resistors,  $R_1 = 12.0 \Omega$  and  $R_2 = 16.0 \Omega$  (Fig. P30.50). The battery has emf 96.0 V and negligible internal resistance.  $S$  is closed at  $t = 0$ . (a) What are the currents  $i_1$ ,  $i_2$ , and  $i_3$  just after  $S$  is closed? (b) What are  $i_1$ ,  $i_2$ , and  $i_3$  after  $S$  has been closed a long time? (c) What is the value of  $t$  for which  $i_3$  has half of the final value that you calculated in part (b)? (d) When  $i_3$  has half of its final value, what are  $i_1$  and  $i_2$ ?



**30.51 ••• CP** An alternating-current electric motor includes a thin, hollow, cylindrical spool (similar to a ring) with mass  $M = 1.11$  kg and radius  $a = 5.00$  cm wrapped  $N = 500$  times with a copper wire with resistance  $R = 5.00 \Omega$  and inductance  $L = 77.0$  mH. Within the spool is a battery that supplies current  $I = 1.00$  A, which makes the spool a magnetic dipole with dipole moment  $\vec{\mu}$  parallel to the cylinder axis. A constant magnetic field with magnitude  $B = 2.00$  T is supplied by an external stator magnet, while the spool turns freely on an axis perpendicular to its own axis. At a certain time, a bar is inserted, stopping the spool's motion (Fig. P30.51). At that instant the angle between the spool

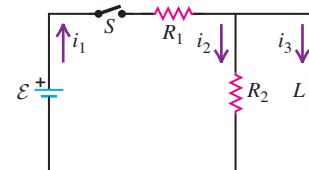
axis and the magnetic field is  $\theta = 45^\circ$ . (a) What is the magnitude of the downward force  $\vec{F}$  applied by the bar onto the spool immediately after the bar is inserted? (b) Later, at time  $t = 0$  with spool still at rest, the coil is short-circuited and a constant counter-torque  $\tau = 0.500$  N·m is applied. The current subsides, and the magnetic torque decreases exponentially. At what time  $t$  does the force applied by the bar vanish? (Hint: Determine when the magnetic torque balances the counter-torque.) (c) After the spool rotates  $180^\circ$  it becomes stuck on the top side of the bar. The counter-torque is no longer applied, and the switch is returned to its original position. After a long time, the bar is removed. What is the angular acceleration of the spool immediately after the bar is removed? The moment of inertia of the spool for an axis along its diameter is  $I = \frac{1}{2}Ma^2$ .

Figure P30.51



**30.52 ••• CALC** An inductor with inductance  $L = 0.200$  H and negligible resistance is connected to a battery, a switch  $S$ , and two resistors,  $R_1 = 8.00 \Omega$  and  $R_2 = 6.00 \Omega$  (Fig. P30.52). The battery has emf 48.0 V and negligible internal resistance.  $S$  is closed at  $t = 0$ . (a) What are the currents  $i_1$ ,  $i_2$ , and  $i_3$  just after  $S$  is closed? (b) What are  $i_1$ ,  $i_2$ , and  $i_3$  after  $S$  has been closed a long time? (c) Apply Kirchhoff's rules to the circuit and obtain a differential equation for  $i_3(t)$ . Integrate this equation to obtain an equation for  $i_3$  as a function of the time  $t$  that has elapsed since  $S$  was closed. (d) Use the equation that you derived in part (c) to calculate the value of  $t$  for which  $i_3$  has half of the final value that you calculated in part (b). (e) When  $i_3$  has half of its final value, what are  $i_1$  and  $i_2$ ?

Figure P30.52



**30.53 ••• CP CALC** A cylindrical solenoid with radius 1.00 cm and length 10.0 cm consists of 300 windings of AWG 20 copper wire, which has a resistance per length of  $0.0333 \Omega/\text{m}$ . This solenoid is connected in series with a  $10.0 \mu\text{F}$  capacitor, which is initially uncharged. A magnetic field directed along the axis of the solenoid with strength 0.100 T is switched on abruptly. (a) The solenoid may be considered an inductor and a resistor in series. Use Faraday's law to determine the average emf across the solenoid during the brief switch-on interval, and determine the net charge initially deposited on the capacitor. (See Exercise 29.4.) (b) At time  $t = 0$  the capacitor is fully charged and there is no current. How much time does it take for the capacitor to fully discharge the first time? (c) What is the frequency with which the current oscillates? (d) How much energy is stored in the capacitor at  $t = 0$ ? (e) How long does it take for the total energy stored in the circuit to drop to 10% of that value?

**30.54 ••** A  $6.40$  nF capacitor is charged to 24.0 V and then disconnected from the battery in the circuit and connected in series with a coil that has  $L = 0.0660$  H and negligible resistance. After the circuit has been completed, there are current oscillations. (a) At an instant when the charge of the capacitor is  $0.0800 \mu\text{C}$ , how much energy is stored in the capacitor and in the inductor, and what is the current in the inductor? (b) At the instant when the charge on the capacitor is  $0.0800 \mu\text{C}$ , what are the voltages across the capacitor and across the inductor, and what is the rate at which current in the inductor is changing?



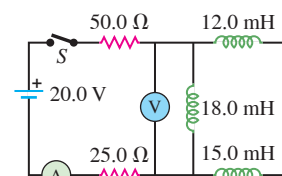
**30.55 •** An  $L$ - $C$  circuit consists of a  $60.0\text{ mH}$  inductor and a  $250\text{ }\mu\text{F}$  capacitor. The initial charge on the capacitor is  $6.00\text{ }\mu\text{C}$ , and the initial current in the inductor is zero. (a) What is the maximum voltage across the capacitor? (b) What is the maximum current in the inductor? (c) What is the maximum energy stored in the inductor? (d) When the current in the inductor has half its maximum value, what is the charge on the capacitor and what is the energy stored in the inductor?

**30.56 ••** A charged capacitor with  $C = 590\text{ }\mu\text{F}$  is connected in series to an inductor that has  $L = 0.330\text{ H}$  and negligible resistance. At an instant when the current in the inductor is  $i = 2.50\text{ A}$ , the current is increasing at a rate of  $di/dt = 73.0\text{ A/s}$ . During the current oscillations, what is the maximum voltage across the capacitor?

**30.57 ••• CP** In the circuit shown in Figure P30.57

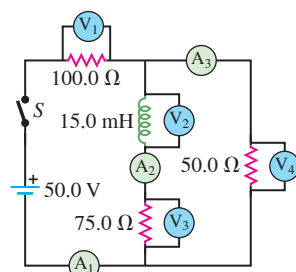
Fig. P30.57, the switch  $S$  has been open for a long time and is suddenly closed. Neither the battery nor the inductors have any appreciable resistance. What do the ammeter and the voltmeter read (a) just after  $S$  is closed; (b) after  $S$  has been closed a very long time; (c)  $0.115\text{ ms}$  after  $S$  is closed?

Figure P30.57



**30.58 •• CP** In the circuit shown in Fig. P30.58, find the reading in each ammeter and voltmeter (a) just after switch  $S$  is closed and (b) after  $S$  has been closed a very long time.

Figure P30.58



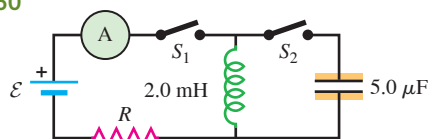
**30.59 •• DATA** To investigate the properties of a large industrial solenoid, you connect the solenoid and a resistor in series with a battery. Switches allow the battery to be replaced by a short circuit across the solenoid and resistor. Therefore Fig. 30.11 applies, with  $R = R_{\text{ext}} + R_L$ , where  $R_L$  is the resistance of the solenoid and  $R_{\text{ext}}$  is the resistance of the series resistor. With switch  $S_2$  open, you close switch  $S_1$  and keep it closed until the current  $i$  in the solenoid is constant (Fig. 30.11). Then you close  $S_2$  and open  $S_1$  simultaneously, using a rapid-response switching mechanism. With high-speed electronics you measure the time  $t_{\text{half}}$  that it takes for the current to decrease to half of its initial value. You repeat this measurement for several values of  $R_{\text{ext}}$  and obtain these results:

| $R_{\text{ext}}\text{ (}\Omega\text{)}$ | 3.0   | 4.0   | 5.0   | 6.0   | 7.0   | 8.0   | 10.0  | 12.0  |
|-----------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| $t_{\text{half}}\text{ (s)}$            | 0.735 | 0.654 | 0.589 | 0.536 | 0.491 | 0.453 | 0.393 | 0.347 |

(a) Graph your data in the form of  $1/t_{\text{half}}$  versus  $R_{\text{ext}}$ . Explain why the data points plotted this way fall close to a straight line. (b) Use your graph from part (a) to calculate the resistance  $R_L$  and inductance  $L$  of the solenoid. (c) If the current in the solenoid is  $20.0\text{ A}$ , how much energy is stored there? At what rate is electrical energy being dissipated in the resistance of the solenoid?

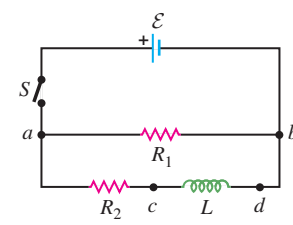
**30.60 ••** In the circuit shown in Fig. P30.60, switch  $S_1$  has been closed for a long enough time so that the current reads a steady  $3.50\text{ A}$ . Suddenly, switch  $S_2$  is closed and  $S_1$  is opened at the same instant. (a) What is the maximum charge that the capacitor will receive? (b) What is the current in the inductor at this time?

Figure P30.60



**30.61 •• CP** In the circuit shown in Fig. P30.61,  $\mathcal{E} = 60.0\text{ V}$ ,  $R_1 = 40.0\text{ }\Omega$ ,  $R_2 = 25.0\text{ }\Omega$ , and  $L = 0.300\text{ H}$ . Switch  $S$  is closed at  $t = 0$ . Just after the switch is closed, (a) what is the potential difference  $v_{ab}$  across the resistor  $R_1$ ; (b) which point,  $a$  or  $b$ , is at a higher potential; (c) what is the potential difference  $v_{cd}$  across the inductor  $L$ ; (d) which point,  $c$  or  $d$ , is at a higher potential? The switch is left closed a long time and then opened. Just after the switch is opened, (e) what is the potential difference  $v_{ab}$  across the resistor  $R_1$ ; (f) which point,  $a$  or  $b$ , is at a higher potential; (g) what is the potential difference  $v_{cd}$  across the inductor  $L$ ; (h) which point,  $c$  or  $d$ , is at a higher potential?

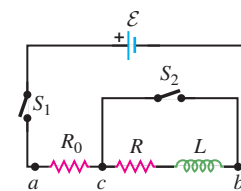
Figure P30.61



**30.62 •• CP** In the circuit shown in Fig. P30.61,  $\mathcal{E} = 60.0\text{ V}$ ,  $R_1 = 40.0\text{ }\Omega$ ,  $R_2 = 25.0\text{ }\Omega$ , and  $L = 0.300\text{ H}$ . (a) Switch  $S$  is closed. At some time  $t$  afterward, the current in the inductor is increasing at a rate of  $di/dt = 50.0\text{ A/s}$ . At this instant, what are the current  $i_1$  through  $R_1$  and the current  $i_2$  through  $R_2$ ? (Hint: Analyze two separate loops: one containing  $\mathcal{E}$  and  $R_1$  and the other containing  $\mathcal{E}$ ,  $R_2$ , and  $L$ .) (b) After the switch has been closed a long time, it is opened again. Just after it is opened, what is the current through  $R_1$ ?

**30.63 •• CALC** Consider the circuit shown in Fig. P30.63. Let  $\mathcal{E} = 36.0\text{ V}$ ,  $R_0 = 50.0\text{ }\Omega$ ,  $R = 150\text{ }\Omega$ , and  $L = 4.00\text{ H}$ . (a) Switch  $S_1$  is closed and switch  $S_2$  is left open. Just after  $S_1$  is closed, what are the current  $i_0$  through  $R_0$  and the potential differences  $v_{ac}$  and  $v_{cb}$ ?

Figure P30.63



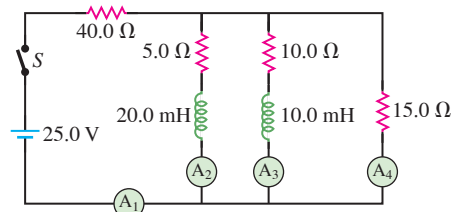
(b) After  $S_1$  has been closed a long time ( $S_2$  is still open) so that the current has reached its final, steady value, what are  $i_0$ ,  $v_{ac}$ , and  $v_{cb}$ ?

(c) Find the expressions for  $i_0$ ,  $v_{ac}$ , and  $v_{cb}$  as functions of the time  $t$  since  $S_1$  was closed. Your results should agree with part (a) when  $t = 0$  and with part (b) when  $t \rightarrow \infty$ . Graph  $i_0$ ,  $v_{ac}$ , and  $v_{cb}$  versus time.

**30.64 ••** After the current in the circuit of Fig. P30.63 has reached its final, steady value with switch  $S_1$  closed and  $S_2$  open, switch  $S_2$  is closed, thus short-circuiting the inductor. (Switch  $S_1$  remains closed. See Problem 30.63 for numerical values of the circuit elements.) (a) Just after  $S_2$  is closed, what are  $v_{ac}$  and  $v_{cb}$ , and what are the currents through  $R_0$ ,  $R$ , and  $S_2$ ? (b) A long time after  $S_2$  is closed, what are  $v_{ac}$  and  $v_{cb}$ , and what are the currents through  $R_0$ ,  $R$ , and  $S_2$ ? (c) Derive expressions for the currents through  $R_0$ ,  $R$ , and  $S_2$  as functions of the time  $t$  that has elapsed since  $S_2$  was closed. Your results should agree with part (a) when  $t = 0$  and with part (b) when  $t \rightarrow \infty$ . Graph these three currents versus time.

**30.65 •• CP** In the circuit shown in Fig. P30.65, switch  $S$  is closed at time  $t = 0$ . (a) Find the reading of each meter just after  $S$  is closed. (b) What does each meter read long after  $S$  is closed?

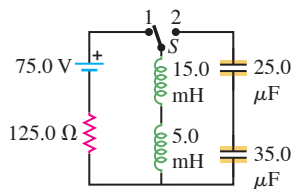
Figure P30.65





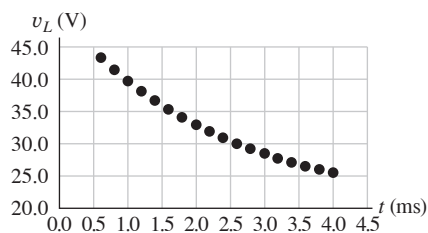
**30.66 •• CP** In the circuit shown in **Fig. P30.66**, neither the battery nor the inductors have any appreciable resistance, the capacitors are initially uncharged, and the switch  $S$  has been in position 1 for a very long time. (a) What is the current in the circuit? (b) The switch is now suddenly flipped to position 2. Find the maximum charge that each capacitor will receive, and how much time after the switch is flipped it will take them to acquire this charge.

Figure P30.66



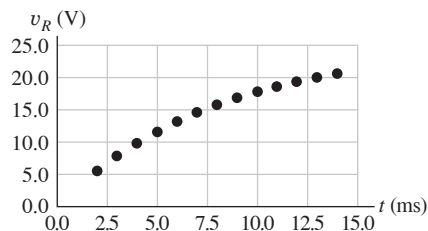
**30.67 •• DATA** During a summer internship as an electronics technician, you are asked to measure the self-inductance  $L$  of a solenoid. You connect the solenoid in series with a  $10.0\ \Omega$  resistor, a battery that has negligible internal resistance, and a switch. Using an ideal voltmeter, you measure and digitally record the voltage  $v_L$  across the solenoid as a function of the time  $t$  that has elapsed since the switch is closed. Your measured values are shown in **Fig. P30.67**, where  $v_L$  is plotted versus  $t$ . In addition, you measure that  $v_L = 50.0\ \text{V}$  just after the switch is closed and  $v_L = 20.0\ \text{V}$  a long time after it is closed. (a) Apply the loop rule to the circuit and obtain an equation for  $v_L$  as a function of  $t$ . [Hint: Use an analysis similar to that used to derive Eq. (30.15).] (b) What is the emf  $\mathcal{E}$  of the battery? (c) According to your measurements, what is the voltage amplitude across the  $10.0\ \Omega$  resistor as  $t \rightarrow \infty$ ? Use this result to calculate the current in the circuit as  $t \rightarrow \infty$ . (d) What is the resistance  $R_L$  of the solenoid? (e) Use the theoretical equation from part (a), **Fig. P30.67**, and the values of  $\mathcal{E}$  and  $R_L$  from parts (b) and (d) to calculate  $L$ . (Hint: According to the equation, what is  $v_L$  when  $t = \tau$ , one time constant? Use **Fig. P30.67** to estimate the value of  $t = \tau$ .)

Figure P30.67



**30.68 •• DATA** You are studying a solenoid of unknown resistance and inductance. You connect it in series with a  $50.0\ \Omega$  resistor, a  $25.0\ \text{V}$  battery that has negligible internal resistance, and a switch. Using an ideal voltmeter, you measure and digitally record the voltage  $v_R$  across the resistor as a function of the time  $t$  that has elapsed after the switch is closed. Your measured values are shown in **Fig. P30.68**, where  $v_R$  is plotted versus  $t$ . In addition, you measure that  $v_R = 0$  just after the switch is closed and  $v_R = 25.0\ \text{V}$  a long time after it is closed. (a) What is the resistance  $R_L$  of the solenoid? (b) Apply the loop rule to the circuit and obtain an equation for  $v_R$  as a function of  $t$ . (c) According to the equation that you derived in part (b), what is  $v_R$  when  $t = \tau$ , one time constant? Use **Fig. P30.68** to estimate the value of  $t = \tau$ . What is the inductance of the solenoid? (d) How much energy is stored in the inductor a long time after the switch is closed?

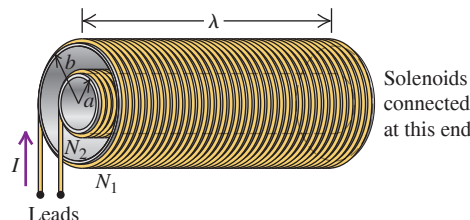
Figure P30.68



## CHALLENGE PROBLEMS

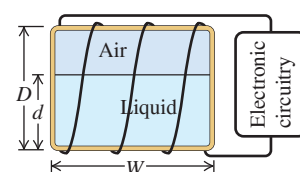
**30.69 ••• A** A long solenoid with  $N_1$  windings and radius  $b$  surrounds a coaxial, narrower solenoid with  $N_2$  windings and radius  $a < b$ , as shown in **Fig. P30.69**. At the far end, the outer solenoid is attached to the inner solenoid by a wire, so that current flows down the outer coil and then back through the inner coil as shown. (a) The two leads are attached to a supply circuit that includes a battery, supplying current  $I$  as indicated. What is the magnetic flux through each turn of the inner coil, taking rightward as the positive direction? (b) What is the magnetic flux through each turn of the outer coil? (c) What is the inductance as seen by the two leads? (d) What would be the inductance if the sense of the inner coil were reversed? (e) In the original configuration, what would be the inductance if  $\lambda = 20.0\ \text{cm}$ ,  $a = 1.00\ \text{cm}$ ,  $b = 2.00\ \text{cm}$ ,  $N_1 = 1200$ , and  $N_2 = 750$ ? (f) Using the values from part (e), what would be the inductance in the configuration of part (d)?

Figure P30.69



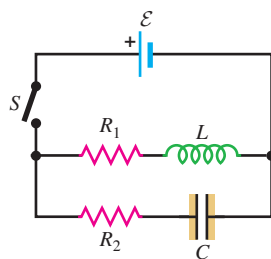
**30.70 ••• CP A Volume Gauge.** A tank containing a liquid has turns of wire wrapped around it, causing it to act like an inductor. The liquid content of the tank can be measured by using its inductance to determine the height of the liquid in the tank. The inductance of the tank changes from a value of  $L_0$  corresponding to a relative permeability of 1 when the tank is empty to a value of  $L_f$  corresponding to a relative permeability of  $K_m$  (the relative permeability of the liquid) when the tank is full. The appropriate electronic circuitry can determine the inductance to five significant figures and thus the effective relative permeability of the combined air and liquid within the rectangular cavity of the tank. The four sides of the tank each have width  $W$  and height  $D$  (**Fig. P30.70**). The height of the liquid in the tank is  $d$ . You can ignore any fringing effects and assume that the relative permeability of the material of which the tank is made can be ignored. (a) Derive an expression for  $d$  as a function of  $L$ , the inductance corresponding to a certain fluid height,  $L_0$ ,  $L_f$ , and  $D$ . (b) What is the inductance (to five significant figures) for a tank  $\frac{1}{4}$  full,  $\frac{1}{2}$  full,  $\frac{3}{4}$  full, and completely full if the tank contains liquid oxygen? Take  $L_0 = 0.63000\ \text{H}$ . The magnetic susceptibility of liquid oxygen is  $\chi_m = 1.52 \times 10^{-3}$ . (c) Repeat part (b) for mercury. The magnetic susceptibility of mercury is given in Table 28.1. (d) For which material is this volume gauge more practical?

Figure P30.70



**30.71 ••• CP CALC** Consider the circuit shown in **Fig. P30.71**. Switch  $S$  is closed at time  $t = 0$ , causing a current  $i_1$  through the inductive branch and a current  $i_2$  through the capacitive branch. The initial charge on the capacitor is zero, and the charge at time  $t$  is  $q_2$ . (a) Derive expressions for  $i_1$ ,  $i_2$ , and  $q_2$  as functions of time. Express your answers in terms of  $\mathcal{E}$ ,  $L$ ,  $C$ ,  $R_1$ ,  $R_2$ , and  $t$ . For the remainder of the problem let the circuit elements have the following values:  $\mathcal{E} = 48 \text{ V}$ ,  $L = 8.0 \text{ H}$ ,  $C = 20 \mu\text{F}$ ,  $R_1 = 25 \Omega$ , and  $R_2 = 5000 \Omega$ . (b) What is the initial current through the inductive branch? What is the initial current through the capacitive branch? (c) What are the currents through the inductive and capacitive branches a long time after the switch has been closed? How long is a “long time”? Explain. (d) At what time  $t_1$  (accurate to two significant figures) will the currents  $i_1$  and  $i_2$  be equal? (*Hint*: You might consider using series expansions for the exponentials.) (e) For the conditions given in part (d), determine  $i_1$ . (f) The total current through the battery is  $i = i_1 + i_2$ . At what time  $t_2$  (accurate to two significant figures) will  $i$  equal one-half of its final value? (*Hint*: The numerical work is greatly simplified if one makes suitable approximations. A sketch of  $i_1$  and  $i_2$  versus  $t$  may help you decide what approximations are valid.)

Figure P30.71



of superconducting wire cooled by liquid helium at a temperature just under its boiling point (4.2 K). After a current is established in the wire, the power supply is disconnected and the magnet leads are shorted together through a piece of superconductor so that the current flows without resistance as long as the liquid helium keeps the magnet cold.

Under rare circumstances, a small segment of the magnet's wire may lose its superconducting properties and develop resistance. In this segment, electrical energy is converted to thermal energy, which can boil off some of the liquid helium. More of the wire then warms up and loses its superconducting properties, thus dissipating even more energy as heat. Because the latent heat of vaporization of liquid helium is quite low (20.9 kJ/kg), once the wire begins to warm up, all of the liquid helium may boil off rapidly. This event, called a *quench*, can damage the magnet. Also, a large volume of helium gas is generated as the liquid helium boils off, causing an asphyxiation hazard, and the resulting rapid pressure buildup can lead to an explosion. You can see how important it is to keep the wire resistance in an MRI magnet at zero and to have devices that detect a quench and shut down the current immediately.

**30.72** How many turns does this typical MRI magnet have? (a) 1100; (b) 3000; (c) 4000; (d) 22,000.

**30.73** If a small part of this magnet loses its superconducting properties and the resistance of the magnet wire suddenly rises from 0 to a constant  $0.005 \Omega$ , how much time will it take for the current to decrease to half of its initial value? (a) 4.7 min; (b) 10 min; (c) 15 min; (d) 30 min.

**30.74** If part of the magnet develops resistance and liquid helium boils away, rendering more and more of the magnet nonsuperconducting, how will this quench affect the time for the current to drop to half of its initial value? (a) The time will be shorter because the resistance will increase; (b) the time will be longer because the resistance will increase; (c) the time will be the same; (d) not enough information is given.

**30.75** If all of the magnetic energy stored in this MRI magnet is converted to thermal energy, how much liquid helium will boil off? (a) 27 kg; (b) 38 kg; (c) 60 kg; (d) 110 kg.

## MCAT-STYLE PASSAGE PROBLEMS

**BIO Quenching an MRI Magnet.** Magnets carrying very large currents are used to produce the uniform, large-magnitude magnetic fields that are required for *magnetic resonance imaging* (MRI). A typical MRI magnet may be a solenoid that is 2.0 m long and 1.0 m in diameter, has a self-inductance of 4.4 H, and carries a current of 750 A. A normal wire carrying that much current would dissipate a great deal of electrical power as heat, so most MRI magnets are made with coils

## ANSWERS

### Chapter Opening Question ?

(iii) As explained in Section 30.2, traffic light sensors work by measuring the change in inductance of a coil embedded under the road surface when a car (which contains ferromagnetic material) drives over it.

### Key Example VARIATION Problems

**VP30.4.1** (a)  $6.50 \times 10^{-5} \text{ H}$  (b)  $1.56 \times 10^{-3} \text{ T} \cdot \text{m}^2$

**VP30.4.2** (a)  $3.36 \times 10^{-4} \text{ V}$  (b) zero (c)  $1.44 \times 10^{-4} \text{ V}$

**VP30.4.3** (a)  $1.96 \times 10^{-6} \text{ T} \cdot \text{m}^2$  (b)  $4.18 \times 10^{-3} \text{ V}$

**VP30.4.4** (a) 25.5  $\mu\text{H}$  (b) 23.0 mV, opposite to the current

**VP30.7.1** (a) 84.9  $\Omega$  (b) 0.212 A

**VP30.7.2** (a) 0.190 A (b) +229 A/s (c)  $6.79 \times 10^{-4} \text{ J}$

**VP30.7.3** (a) 0.476 A (b) -229 A/s (c)  $4.26 \times 10^{-3} \text{ J}$

**VP30.7.4** (a)  $(L/R)\ln 2 = 0.693(L/R)$  (b)  $\mathcal{E}/2R$  (c)  $\mathcal{E}/2L$

**VP30.10.1** (a)  $5.12 \times 10^{-3} \text{ F}$  (b) 14.0 A

**VP30.10.2** (a)  $2.20 \times 10^{-4} \text{ s}$  (b) 2.85 A

**VP30.10.3** (a) 35.1  $\mu\text{F}$  (b) 0.0125 H (c)  $1.51 \times 10^3 \text{ rad/s}$

**VP30.10.4** (a) 61.1  $\mu\text{F}$  (b) 49.4  $\Omega$

### Bridging Problem

(a)  $7.68 \times 10^{-8} \text{ J}$

(b) 1.60 mA

(c) 24.8 mV

(d)  $1.92 \times 10^{-8} \text{ J}$ , 21.5 mV

? Waves from a broadcasting station produce an alternating current in the circuits of a radio (like the one in this classic car). If a radio is tuned to a station at a frequency of 1000 kHz, it will also detect the transmissions from a station broadcasting at (i) 600 kHz; (ii) 800 kHz; (iii) 1200 kHz; (iv) all of these; (v) none of these.



# 31 Alternating Current

## LEARNING OUTCOMES

### In this chapter, you'll learn...

- 31.1 How phasors make it easy to describe sinusoidally varying quantities.
- 31.2 How to use reactance to describe the voltage across a circuit element that carries an alternating current.
- 31.3 How to analyze an  $L$ - $R$ - $C$  series circuit with a sinusoidal emf.
- 31.4 What determines the amount of power flowing into or out of an alternating-current circuit.
- 31.5 How an  $L$ - $R$ - $C$  series circuit responds to sinusoidal emfs of different frequencies.
- 31.6 Why transformers are useful, and how they work.

### You'll need to review...

- 14.2, 14.8 Simple harmonic motion, resonance.
- 16.5 Resonance and sound.
- 18.3 Root-mean-square (rms) values.
- 25.3 Diodes.
- 26.3 Galvanometers.
- 28.8 Hysteresis in magnetic materials.
- 29.2, 29.6, 29.7 Alternating-current generators; eddy currents; displacement current.
- 30.1, 30.2, 30.5, 30.6 Mutual inductance; voltage across an inductor;  $L$ - $C$  circuits;  $L$ - $R$ - $C$  series circuits.

During the 1880s in the United States there was a heated and acrimonious debate between two inventors over the best method of electric-power distribution. Thomas Edison favored direct current (dc)—that is, steady current that does not vary with time. George Westinghouse favored **alternating current (ac)**, with sinusoidally varying voltages and currents. He argued that transformers (which we'll study in this chapter) can be used to step the voltage up and down with ac but not with dc; low voltages are safer for consumer use, but high voltages and correspondingly low currents are best for long-distance power transmission to minimize  $i^2R$  losses in the cables.

Eventually, Westinghouse prevailed, and most present-day household and industrial power distribution systems operate with alternating current. Any appliance that you plug into a wall outlet uses ac. Circuits in modern communication equipment also make extensive use of ac.

In this chapter we'll learn how resistors, inductors, and capacitors behave in **ac circuits**—that is, circuits with sinusoidally varying voltages and currents. Many of the principles that we found useful in Chapter 30 are applicable, along with several new concepts related to the circuit behavior of inductors and capacitors. A key concept in this discussion is *resonance*, which we studied in Chapter 14 for mechanical systems.

## 31.1 PHASORS AND ALTERNATING CURRENTS

To supply an alternating current to a circuit, a source of alternating emf or voltage is required. An example of such a source is a coil of wire rotating with constant angular velocity in a magnetic field, which we discussed in Example 29.3 (Section 29.2). This develops a sinusoidal alternating emf and is the prototype of the commercial alternating-current generator or *alternator* (see Fig. 29.8).

We use the term **ac source** for any device that supplies a sinusoidally varying voltage (potential difference)  $v$  or current  $i$ . The usual circuit-diagram symbol for an ac source is



A sinusoidal voltage might be described by a function such as

$$v = V \cos \omega t \quad (31.1)$$

In this expression, (lowercase)  $v$  is the *instantaneous* potential difference; (uppercase)  $V$  is the maximum potential difference, which we call the **voltage amplitude**; and  $\omega$  is the *angular frequency*, equal to  $2\pi$  times the frequency  $f$  (**Fig. 31.1**).

In the United States and Canada, commercial electric-power distribution systems use a frequency  $f = 60$  Hz, corresponding to  $\omega = (2\pi \text{ rad})(60 \text{ s}^{-1}) = 377 \text{ rad/s}$ ; in much of the rest of the world,  $f = 50$  Hz ( $\omega = 314 \text{ rad/s}$ ) is used. Similarly, a sinusoidal current with a maximum value, or **current amplitude**, of  $I$  might be described as

$$\text{Sinusoidal alternating current: } i = I \cos \omega t \quad (31.2)$$

Instantaneous current  $i$       Angular frequency  $\omega$       Time  $t$   
 Current amplitude (maximum current)  $I$

## Phasor Diagrams

To represent sinusoidally varying voltages and currents in ac circuits, we'll use rotating vector diagrams similar to those we used in the study of simple harmonic motion in Section 14.2 (see Figs. 14.5b and 14.6). In these diagrams the instantaneous value of a quantity that varies sinusoidally with time is represented by the *projection* onto a horizontal axis of a vector with a length equal to the amplitude of the quantity. The vector rotates counterclockwise with constant angular speed  $\omega$ . These rotating vectors are called **phasors**, and diagrams containing them are called **phasor diagrams**. **Figure 31.2** shows a phasor diagram for the sinusoidal current described by Eq. (31.2). The projection of the phasor onto the horizontal axis at time  $t$  is  $I \cos \omega t$ ; this is why we chose to use the cosine function rather than the sine in Eq. (31.2).

**CAUTION** Just what is a phasor? A phasor isn't a real physical quantity with a direction in space, such as velocity or electric field. Rather, it's a *geometric* entity that helps us describe physical quantities that vary sinusoidally with time. In Section 14.2 we used a single phasor to represent the position of a particle undergoing simple harmonic motion. Here we'll use phasors to *add* sinusoidal voltages and currents. Combining sinusoidal quantities with phase differences then involves vector addition. We'll use phasors in a similar way in Chapters 35 and 36 in our study of interference effects with light. ■

## Rectified Alternating Current

How do we measure a sinusoidally varying current? In Section 26.3 we used a d'Arsonval galvanometer to measure steady currents. But if we pass a *sinusoidal* current through a d'Arsonval meter, the torque on the moving coil varies sinusoidally, with one direction half the time and the opposite direction the other half. The needle may wiggle a little if the frequency is low enough, but its average deflection is zero. Hence a d'Arsonval meter by itself isn't very useful for measuring alternating currents.

To get a measurable one-way current through the meter, we can use *diodes*, which we described in Section 25.3. A diode is a device that conducts better in one direction than in the other; an ideal diode has zero resistance for one direction of current and infinite resistance for the other. **Figure 31.3a** (next page) shows one possible arrangement, called

Figure 31.1 The voltage across a sinusoidal ac source.

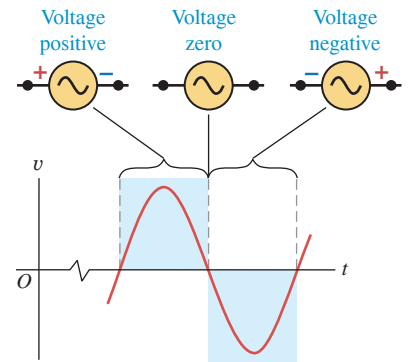


Figure 31.2 A phasor diagram.

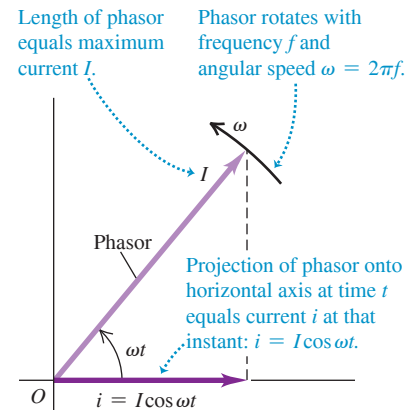
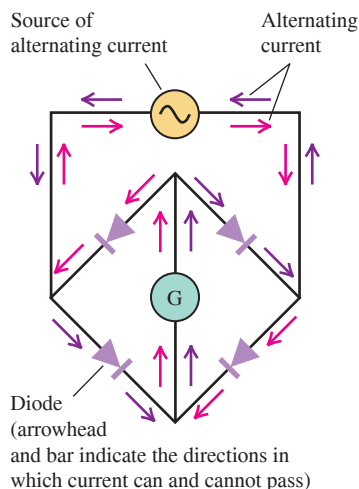




Figure 31.3 (a) A full-wave rectifier circuit. (b) Graph of the resulting current through the galvanometer G.

(a) A full-wave rectifier circuit



(b) Graph of the full-wave rectified current and its average value, the rectified average current  $I_{\text{rav}}$

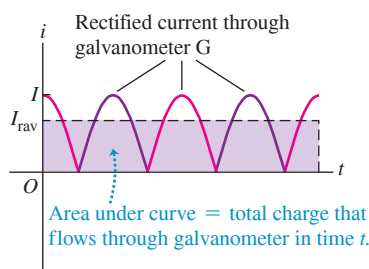
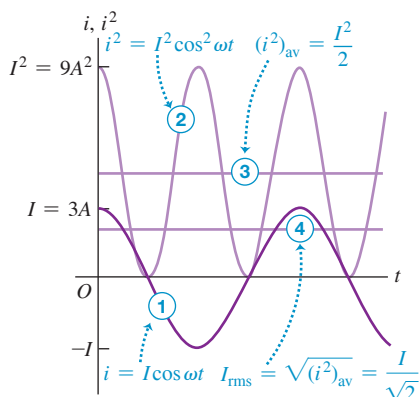


Figure 31.4 Calculating the root-mean-square (rms) value of an alternating current.

**Meaning of the rms value** of a sinusoidal quantity (here, ac current with  $I = 3 \text{ A}$ ):

- ① Graph current  $i$  versus time.
- ② Square the instantaneous current  $i$ .
- ③ Take the *average* (mean) value of  $i^2$ .
- ④ Take the *square root* of that average.



a *full-wave rectifier circuit*. The current through the galvanometer G is always upward, regardless of the direction of the current from the ac source (i.e., which part of the cycle the source is in). The graph in Fig. 31.3b shows the current through G: It pulsates but always has the same direction, and the average meter deflection is *not* zero.

The **rectified average current**  $I_{\text{rav}}$  is defined so that during any whole number of cycles, the total charge that flows is the same as though the current were constant with a value equal to  $I_{\text{rav}}$ . The notation  $I_{\text{rav}}$  and the name *rectified average* current emphasize that this is *not* the average of the original sinusoidal current. In Fig. 31.3b the total charge that flows in time  $t$  corresponds to the area under the curve of  $i$  versus  $t$  (recall that  $i = dq/dt$ , so  $q$  is the integral of  $t$ ); this area must equal the rectangular area with height  $I_{\text{rav}}$ . We see that  $I_{\text{rav}}$  is less than the maximum current  $I$ ; the two are related by

$$\text{Rectified average value of a sinusoidal current} \quad I_{\text{rav}} = \frac{2}{\pi} I = 0.637 I \quad \text{Current amplitude} \quad (31.3)$$

(The factor of  $2/\pi$  is the average value of  $|\cos \omega t|$  or of  $|\sin \omega t|$ ; see Example 29.4 in Section 29.2.) The galvanometer deflection is proportional to  $I_{\text{rav}}$ . The galvanometer scale can be calibrated to read  $I$ ,  $I_{\text{rav}}$ , or, most commonly,  $I_{\text{rms}}$  (discussed below).

## Root-Mean-Square (rms) Values

A more useful way to describe a quantity that can be either positive or negative is the *root-mean-square (rms) value*. We used rms values in Section 18.3 in connection with the speeds of molecules in a gas. We *square* the instantaneous current  $i$ , take the *average* (mean) value of  $i^2$ , and finally take the *square root* of that average. This procedure defines the **root-mean-square current**, denoted as  $I_{\text{rms}}$  (**Fig. 31.4**). Even when  $i$  is negative,  $i^2$  is always positive, so  $I_{\text{rms}}$  is never zero (unless  $i$  is zero at every instant).

Here's how we obtain  $I_{\text{rms}}$  for a sinusoidal current, like that shown in Fig. 31.4. If the instantaneous current is given by  $i = I \cos \omega t$ , then

$$i^2 = I^2 \cos^2 \omega t$$

Using a double-angle formula from trigonometry,

$$\cos^2 A = \frac{1}{2}(1 + \cos 2A)$$

we find

$$i^2 = I^2 \frac{1}{2}(1 + \cos 2\omega t) = \frac{1}{2} I^2 + \frac{1}{2} I^2 \cos 2\omega t$$

The average of  $\cos 2\omega t$  is zero because it is positive half the time and negative half the time. Thus the average of  $i^2$  is simply  $I^2/2$ . The square root of this is  $I_{\text{rms}}$ :

$$\text{Root-mean-square (rms) value of a sinusoidal current} \quad I_{\text{rms}} = \frac{I}{\sqrt{2}} \quad \text{Current amplitude} \quad (31.4)$$

In the same way, the root-mean-square value of a sinusoidal voltage is

$$\text{Root-mean-square (rms) value of a sinusoidal voltage} \quad V_{\text{rms}} = \frac{V}{\sqrt{2}} \quad \text{Voltage amplitude (maximum value)} \quad (31.5)$$

We can convert a rectifying ammeter into a voltmeter by adding a series resistor, just as for the dc case discussed in Section 26.3. Meters used for ac voltage and current measurements are nearly always calibrated to read rms values, not maximum or rectified average. Voltages and currents in power distribution systems are always described in terms of their rms values. The usual household power supply, “120 volt ac,” has an rms voltage of 120 V (Fig. 31.5). The voltage amplitude is

$$\begin{aligned} V &= \sqrt{2} V_{\text{rms}} \\ &= \sqrt{2}(120 \text{ V}) = 170 \text{ V} \end{aligned}$$

Sixty times per second, the instantaneous voltage across a socket’s terminals varies from 170 V to  $-170$  V and back again.

Figure 31.5 This wall socket delivers a root-mean-square (rms) voltage of 120 V.



### EXAMPLE 31.1 Current in a desktop computer

The plate on the back of a desktop computer says that it draws 2.7 A from a 120 V, 60 Hz line. For this computer, what are (a) the average current, (b) the average of the square of the current, and (c) the current amplitude?

**IDENTIFY and SET UP** This example is about alternating current. In part (a) we find the average, over a complete cycle, of the alternating current. In part (b) we recognize that the 2.7 A current draw of the computer is the rms value  $I_{\text{rms}}$ —that is, the *square root of the mean* (average) of the *square* of the current,  $(i^2)_{\text{av}}$ . In part (c) we use Eq. (31.4) to relate  $I_{\text{rms}}$  to the current amplitude.

**EXECUTE** (a) The average of *any* sinusoidally varying quantity, over any whole number of cycles, is zero.

(b) We are given  $I_{\text{rms}} = 2.7 \text{ A}$ . From the definition of rms value,

$$I_{\text{rms}} = \sqrt{(i^2)_{\text{av}}} \text{ so } (i^2)_{\text{av}} = (I_{\text{rms}})^2 = (2.7 \text{ A})^2 = 7.3 \text{ A}^2$$

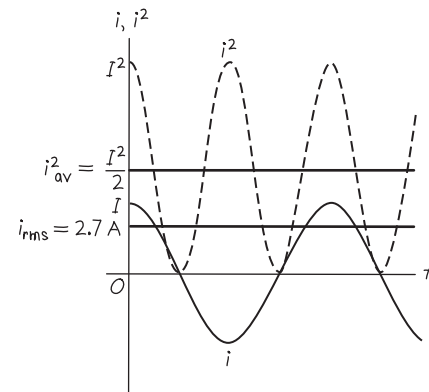
(c) From Eq. (31.4), the current amplitude  $I$  is

$$I = \sqrt{2} I_{\text{rms}} = \sqrt{2}(2.7 \text{ A}) = 3.8 \text{ A}$$

Figure 31.6 shows graphs of  $i$  and  $i^2$  versus time  $t$ .

**EVALUATE** Why would we be interested in the average of the square of the current? Recall that the rate at which energy is dissipated in a resistor  $R$

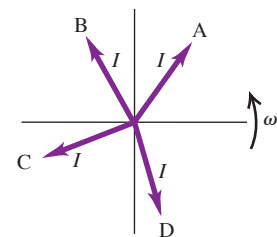
Figure 31.6 Our graphs of the current  $i$  and the square of the current  $i^2$  versus time  $t$ .



equals  $i^2 R$ . This rate varies if the current is alternating, so it is best described by its average value  $(i^2)_{\text{av}} R = I_{\text{rms}}^2 R$ . We’ll use this idea in Section 31.4.

**KEYCONCEPT** The root-mean-square (rms) value of a time-varying quantity is the square root of the average value of the square of that quantity. To find the rms value of a quantity that varies sinusoidally with time, like the current or voltage in an ac circuit, divide its amplitude by  $\sqrt{2}$ .

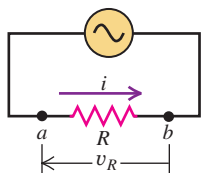
**TEST YOUR UNDERSTANDING OF SECTION 31.1** The accompanying figure shows four different current phasors with the same angular frequency  $\omega$ . At the time shown, which phasor corresponds to (a) a positive current that is becoming more positive; (b) a positive current that is decreasing toward zero; (c) a negative current that is becoming more negative; (d) a negative current that is decreasing in magnitude toward zero?



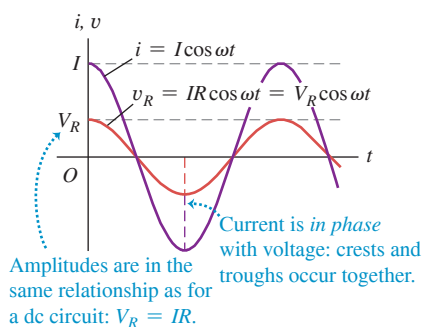
**ANSWER** (a) D; (b) A; (c) C; (d) B. For each phasor, the actual current is represented by the projection of that phasor onto the horizontal axis. The phasors all rotate counterclockwise around the origin with angular speed  $\omega$ , so at the instant shown the projection of phasor A is positive but trending toward zero; the projection of phasor B is negative and becoming more negative; the projection of phasor C is negative but trending toward zero; and the projection of phasor D is positive and becoming more positive.

Figure 31.7 Resistance  $R$  connected across an ac source.

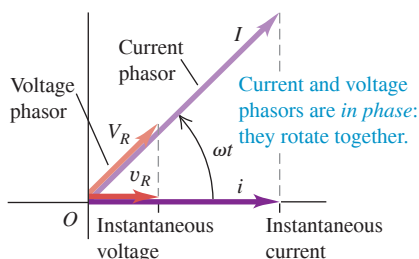
(a) Circuit with ac source and resistor



(b) Graphs of current and voltage versus time



(c) Phasor diagram



## 31.2 RESISTANCE AND REACTANCE

In this section we'll derive voltage–current relationships for individual circuit elements—resistors, inductors, and capacitors—carrying a sinusoidal current.

### Resistor in an ac Circuit

First let's consider a resistor with resistance  $R$  through which there is a sinusoidal current given by Eq. (31.2):  $i = I \cos \omega t$ . The positive direction of current is counterclockwise around the circuit (**Fig. 31.7a**). The current amplitude (maximum current) is  $I$ . From Ohm's law the instantaneous potential  $v_R$  of point  $a$  with respect to point  $b$  (that is, the instantaneous voltage across the resistor) is

$$v_R = iR = (IR) \cos \omega t \quad (31.6)$$

The maximum value of the voltage  $v_R$  is  $V_R$ , the *voltage amplitude*:

$$V_R = IR \quad \text{Amplitude of voltage across a resistor, ac circuit} = \frac{\text{Current amplitude}}{\text{Resistance}} \quad (31.7)$$

Hence we can also write

$$v_R = V_R \cos \omega t \quad (31.8)$$

Both the current  $i$  and the voltage  $v_R$  are proportional to  $\cos \omega t$ , so the current is *in phase* with the voltage. Equation (31.7) shows that the current and voltage amplitudes are related in the same way as in a dc circuit.

Figure 31.7b shows graphs of  $i$  and  $v_R$  as functions of time. The vertical scales for current and voltage are different, so the relative heights of the two curves are not significant. The corresponding phasor diagram is given in Fig. 31.7c. Because  $i$  and  $v_R$  are *in phase* and have the same frequency, the current and voltage phasors rotate together; they are parallel at each instant. Their projections on the horizontal axis represent the instantaneous current and voltage, respectively.

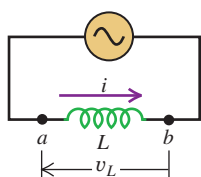
### Inductor in an ac Circuit

Now we replace the resistor in Fig. 31.7 with a pure inductor with self-inductance  $L$  and zero resistance (**Fig. 31.8a**). Again the current is  $i = I \cos \omega t$ , and the positive direction of current is counterclockwise around the circuit.

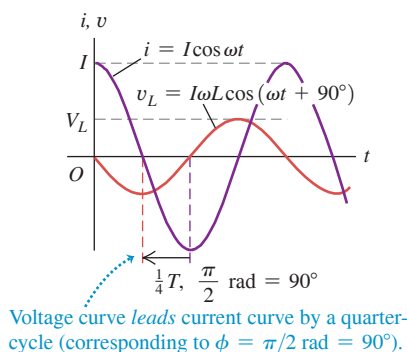
Although there is no resistance, there is a potential difference  $v_L$  between the inductor terminals  $a$  and  $b$  because the current varies with time, giving rise to a self-induced emf. The induced emf in the direction of  $i$  is given by Eq. (30.7),  $\mathcal{E} = -L di/dt$ ; however, the voltage  $v_L$  is *not* simply equal to  $\mathcal{E}$ . To see why, notice that if the current in the inductor is in the

Figure 31.8 Inductance  $L$  connected across an ac source.

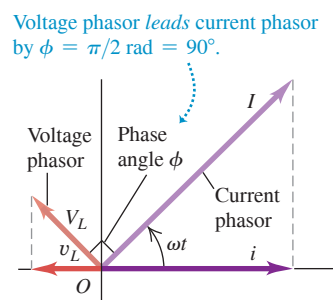
(a) Circuit with ac source and inductor



(b) Graphs of current and voltage versus time



(c) Phasor diagram



positive (counterclockwise) direction from  $a$  to  $b$  and is increasing, then  $di/dt$  is positive and the induced emf is directed to the left to oppose the increase in current; hence point  $a$  is at higher potential than is point  $b$ . Thus the potential of point  $a$  with respect to point  $b$  is positive and is given by  $v_L = +L di/dt$ , the *negative* of the induced emf. (Convince yourself that this expression gives the correct sign of  $v_L$  in *all* cases, including  $i$  counterclockwise and decreasing,  $i$  clockwise and increasing, and  $i$  clockwise and decreasing; also review Section 30.2.) So

$$v_L = L \frac{di}{dt} = L \frac{d}{dt}(I \cos \omega t) = -I\omega L \sin \omega t \quad (31.9)$$

The voltage  $v_L$  across the inductor at any instant is proportional to the *rate of change* of the current. The points of maximum voltage on the graph correspond to maximum steepness of the current curve, and the points of zero voltage are the points where the current curve has its maximum and minimum values (Fig. 31.8b). The voltage and current are *out of phase* by a quarter-cycle. Since the voltage peaks occur a quarter-cycle earlier than the current peaks, we say that the voltage *leads* the current by  $90^\circ$ . The phasor diagram in Fig. 31.8c also shows this relationship; the voltage phasor is ahead of the current phasor by  $90^\circ$ .

We can also obtain this phase relationship by rewriting Eq. (31.9) with the identity  $\cos(A + 90^\circ) = -\sin A$ :

$$v_L = I\omega L \cos(\omega t + 90^\circ) \quad (31.10)$$

This result shows that the voltage can be viewed as a cosine function with a “head start” of  $90^\circ$  relative to the current.

As we have done in Eq. (31.10), we’ll usually describe the phase of the *voltage* relative to the *current*, not the reverse. Thus if the current  $i$  in a circuit is

$$i = I \cos \omega t$$

and the voltage  $v$  of one point with respect to another is

$$v = V \cos(\omega t + \phi)$$

we call  $\phi$  the **phase angle**; it gives the phase of the *voltage* relative to the *current*. For a pure resistor,  $\phi = 0$ , and for a pure inductor,  $\phi = 90^\circ$ .

From Eq. (31.9) or (31.10) the amplitude  $V_L$  of the inductor voltage is

$$V_L = I\omega L \quad (31.11)$$

We define the **inductive reactance**  $X_L$  of an inductor as

$$X_L = \omega L \quad (\text{inductive reactance}) \quad (31.12)$$

Using  $X_L$ , we can write Eq. (31.11) in a form similar to Eq. (31.7) for a resistor:

$$\text{Amplitude of voltage across an inductor, ac circuit} \quad V_L = IX_L \quad \text{Current amplitude} \quad \text{Inductive reactance} \quad (31.13)$$

Because  $X_L$  is the ratio of a voltage and a current, its SI unit is the ohm, the same as for resistance.

## The Meaning of Inductive Reactance

The inductive reactance  $X_L$  is really a description of the self-induced emf that opposes any change in the current through the inductor. From Eq. (31.13), for a given current amplitude  $I$  the voltage  $v_L = +L di/dt$  across the inductor and the self-induced emf  $\mathcal{E} = -L di/dt$  both have an amplitude  $V_L$  that is directly proportional to  $X_L$ . According to Eq. (31.12), the inductive reactance and self-induced emf increase with more rapid variation in current (that is, increasing angular frequency  $\omega$ ) and increasing inductance  $L$ .

If an oscillating voltage of a given amplitude  $V_L$  is applied across the inductor terminals, the resulting current will have a smaller amplitude  $I$  for larger values of  $X_L$ . Since  $X_L$  is proportional to frequency, a high-frequency voltage applied to the inductor gives only a small current, while a lower-frequency voltage of the same amplitude gives rise to a larger

**CAUTION Inductor voltage and current are not in phase** Equation (31.13) relates the *amplitudes* of the oscillating voltage and current for the inductor in Fig. 31.8a. It does *not* say that the voltage at any instant is equal to the current at that instant multiplied by  $X_L$ . As Fig. 31.8b shows, the voltage and current are  $90^\circ$  out of phase. Voltage and current are in phase only for resistors, as in Eq. (31.6). **I**



current. Inductors are used in some circuit applications, such as power supplies and radio-interference filters, to block high frequencies while permitting lower frequencies or dc to pass through. A circuit device that uses an inductor for this purpose is called a *low-pass filter* (see Problem 31.48).

### EXAMPLE 31.2 An inductor in an ac circuit

### WITH VARIATION PROBLEMS

The current amplitude in a pure inductor in a radio receiver is to be  $250\ \mu\text{A}$  when the voltage amplitude is  $3.60\ \text{V}$  at a frequency of  $1.60\ \text{MHz}$  (at the upper end of the AM broadcast band). (a) What inductive reactance is needed? What inductance? (b) If the voltage amplitude is kept constant, what will be the current amplitude through this inductor at  $16.0\ \text{MHz}$ ? At  $160\ \text{kHz}$ ?

**IDENTIFY and SET UP** The circuit may have other elements, but in this example we don't care: All they do is provide the inductor with an oscillating voltage, so the other elements are lumped into the ac source shown in Fig. 31.8a. We are given the current amplitude  $I$  and the voltage amplitude  $V$ . Our target variables in part (a) are the inductive reactance  $X_L$  at  $1.60\ \text{MHz}$  and the inductance  $L$ , which we find from Eqs. (31.13) and (31.12). Knowing  $L$ , we use these equations in part (b) to find  $X_L$  and  $I$  at any frequency.

**EXECUTE** (a) From Eq. (31.13),

$$X_L = \frac{V_L}{I} = \frac{3.60\ \text{V}}{250 \times 10^{-6}\ \text{A}} = 1.44 \times 10^4\ \Omega = 14.4\ \text{k}\Omega$$

From Eq. (31.12), with  $\omega = 2\pi f$ ,

$$L = \frac{X_L}{2\pi f} = \frac{1.44 \times 10^4\ \Omega}{2\pi(1.60 \times 10^6\ \text{Hz})} = 1.43 \times 10^{-3}\ \text{H} = 1.43\ \text{mH}$$

(b) Combining Eqs. (31.12) and (31.13), we find  $I = V_L/X_L = V_L/\omega L = V_L/2\pi fL$ . Thus the current amplitude is inversely proportional to the frequency  $f$ . Since  $I = 250\ \mu\text{A}$  at  $f = 1.60\ \text{MHz}$ , the current amplitudes at  $16.0\ \text{MHz}$  ( $10f$ ) and  $160\ \text{kHz} = 0.160\ \text{MHz}$  ( $f/10$ ) will be, respectively, one-tenth as great ( $25.0\ \mu\text{A}$ ) and ten times as great ( $2500\ \mu\text{A} = 2.50\ \text{mA}$ ).

**EVALUATE** In general, the lower the frequency of an oscillating voltage applied across an inductor, the greater the amplitude of the resulting oscillating current.

**KEYCONCEPT** For any component in an ac circuit, the reactance is the ratio of the amplitude of the voltage across the component to the amplitude of the current through the component. If the component is an inductor, the reactance equals the product of the inductance and the angular frequency of the current.

**CAUTION** **Alternating current through a capacitor** Charge can't really move through the capacitor because its two plates are insulated from each other. But as the capacitor charges and discharges, there is at each instant a current  $i$  into one plate, an equal current out of the other plate, and an equal *displacement* current between the plates. (You should review the discussion of displacement current in Section 29.7.) Thus we often speak about alternating current *through* a capacitor. ■

### Capacitor in an ac Circuit

Finally, we connect a capacitor with capacitance  $C$  to the source, as in **Fig. 31.9a**, producing a current  $i = I\cos\omega t$  through the capacitor. Again, the positive direction of current is counterclockwise around the circuit.

To find the instantaneous voltage  $v_C$  across the capacitor—that is, the potential of point  $a$  with respect to point  $b$ —we first let  $q$  denote the charge on the left-hand plate of the capacitor in Fig. 31.9a (so  $-q$  is the charge on the right-hand plate). The current  $i$  is related to  $q$  by  $i = dq/dt$ ; with this definition, positive current corresponds to an increasing charge on the left-hand capacitor plate. Then

$$i = \frac{dq}{dt} = I\cos\omega t$$

Integrating this, we get

$$q = \frac{I}{\omega}\sin\omega t \quad (31.14)$$

Also, from Eq. (24.1) the charge  $q$  equals the voltage  $v_C$  multiplied by the capacitance,  $q = Cv_C$ . Using this in Eq. (31.14), we find

$$v_C = \frac{I}{\omega C}\sin\omega t \quad (31.15)$$

The instantaneous current  $i$  is equal to the rate of change  $dq/dt$  of the capacitor charge  $q$ ; since  $q = Cv_C$ ,  $i$  is also proportional to the rate of change of voltage. (Compare to an inductor, for which the situation is reversed and  $v_L$  is proportional to the rate of change of  $i$ .) Figure 31.9b shows  $v_C$  and  $i$  as functions of  $t$ . Because  $i = dq/dt = C dv_C/dt$ , the current has its greatest magnitude when the  $v_C$  curve is rising or falling most steeply and is zero when the  $v_C$  curve instantaneously levels off at its maximum and minimum values.

The peaks of capacitor voltage occur a quarter-cycle *after* the corresponding current peaks, and we say that the voltage *lags* the current by  $90^\circ$ . The phasor diagram in Fig. 31.9c shows this relationship; the voltage phasor is behind the current phasor by a quarter-cycle, or  $90^\circ$ .

We can also derive this phase difference by rewriting Eq. (31.15) with the identity  $\cos(A - 90^\circ) = \sin A$ :

$$v_C = \frac{I}{\omega C} \cos(\omega t - 90^\circ) \quad (31.16)$$

This corresponds to a phase angle  $\phi = -90^\circ$ . This cosine function has a “late start” of  $90^\circ$  compared with the current  $i = I \cos \omega t$ .

Equations (31.15) and (31.16) show that the voltage *amplitude*  $V_C$  is

$$V_C = \frac{I}{\omega C} \quad (31.17)$$

To put this expression in a form similar to Eq. (31.7) for a resistor,  $V_R = IR$ , we define a quantity  $X_C$ , called the **capacitive reactance** of the capacitor, as

$$X_C = \frac{1}{\omega C} \quad (\text{capacitive reactance}) \quad (31.18)$$

Then

$$\text{Amplitude of voltage across a capacitor, ac circuit} \quad V_C = IX_C \quad \text{Current amplitude} \quad \text{Capacitive reactance} \quad (31.19)$$

The SI unit of  $X_C$  is the ohm, the same as for resistance and inductive reactance, because  $X_C$  is the ratio of a voltage and a current.

**CAUTION** **Capacitor voltage and current are not in phase** Remember that Eq. (31.19) for a capacitor, like Eq. (31.13) for an inductor, is *not* a statement about the instantaneous values of voltage and current. The instantaneous values are  $90^\circ$  out of phase, as Fig. 31.9b shows. Rather, Eq. (31.19) relates the *amplitudes* of voltage and current. ■

## The Meaning of Capacitive Reactance

The capacitive reactance of a capacitor is inversely proportional both to the capacitance  $C$  and to the angular frequency  $\omega$ ; the greater the capacitance and the higher the frequency, the *smaller* the capacitive reactance  $X_C$ . Capacitors tend to pass high-frequency current and to block low-frequency currents and dc, just the opposite of inductors. A device that preferentially passes signals of high frequency is called a *high-pass filter* (see Problem 31.47).

### EXAMPLE 31.3 A resistor and a capacitor in an ac circuit

A  $200 \, \Omega$  resistor is connected in series with a  $5.0 \, \mu\text{F}$  capacitor. The voltage across the resistor is  $v_R = (1.20 \, \text{V}) \cos[(2500 \, \text{rad/s})t]$  (Fig. 31.10). (a) Derive an expression for the circuit current. (b) Determine the capacitive reactance of the capacitor. (c) Derive an expression for the voltage across the capacitor.

**IDENTIFY and SET UP** Since this is a series circuit, the current is the same through the capacitor as through the resistor. Our target variables are the current  $i$ , the capacitive reactance  $X_C$ , and the capacitor voltage  $v_C$ . We use Eq. (31.6) to find an expression for  $i$  in terms of the angular frequency  $\omega = 2500 \, \text{rad/s}$ , Eq. (31.18) to find  $X_C$ , Eq. (31.19) to find the capacitor voltage amplitude  $V_C$ , and Eq. (31.16) to write an expression for  $v_C$ .

**EXECUTE** (a) From Eq. (31.6),  $v_R = iR$ , we find

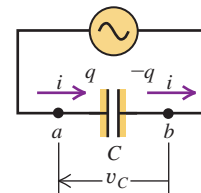
$$i = \frac{v_R}{R} = \frac{(1.20 \, \text{V}) \cos[(2500 \, \text{rad/s})t]}{200 \, \Omega} = (6.0 \times 10^{-3} \, \text{A}) \cos[(2500 \, \text{rad/s})t]$$

(b) From Eq. (31.18), the capacitive reactance at  $\omega = 2500 \, \text{rad/s}$  is

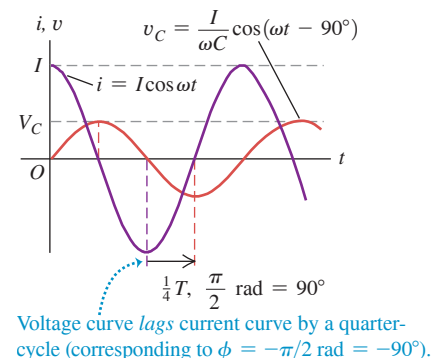
$$X_C = \frac{1}{\omega C} = \frac{1}{(2500 \, \text{rad/s})(5.0 \times 10^{-6} \, \text{F})} = 80 \, \Omega$$

Figure 31.9 Capacitor  $C$  connected across an ac source.

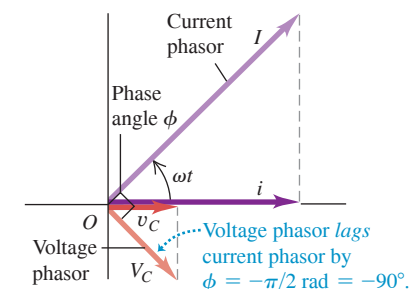
(a) Circuit with ac source and capacitor



(b) Graphs of current and voltage versus time

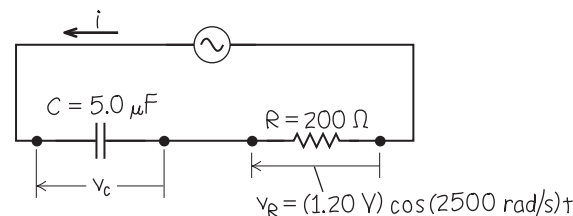


(c) Phasor diagram



### WITH VARIATION PROBLEMS

Figure 31.10 Our sketch for this problem.



(c) From Eq. (31.19), the capacitor voltage amplitude is

$$V_C = IX_C = (6.0 \times 10^{-3} \, \text{A})(80 \, \Omega) = 0.48 \, \text{V}$$

(The  $80 \, \Omega$  reactance of the capacitor is 40% of the resistor's  $200 \, \Omega$  resistance, so  $V_C$  is 40% of  $V_R$ .) The instantaneous capacitor voltage is given by Eq. (31.16):

$$v_C = V_C \cos(\omega t - 90^\circ) = (0.48 \, \text{V}) \cos[(2500 \, \text{rad/s})t - \pi/2 \, \text{rad}]$$

Continued

**EVALUATE** Although the same *current* passes through both the capacitor and the resistor, the *voltages* across them are different in both amplitude and phase. Note that in the expression for  $v_C$  we converted the  $90^\circ$  to  $\pi/2$  rad so that all the angular quantities have the same units. In ac circuit analysis, phase angles are often given in degrees, so be careful to convert to radians when necessary.

**KEYCONCEPT** The reactance (ratio of voltage amplitude to current amplitude) for a resistor in an ac circuit is equal to its resistance. For a capacitor, the reactance equals the reciprocal of the product of the capacitance and the angular frequency of the current.

Figure 31.11 Graphs of  $R$ ,  $X_L$ , and  $X_C$  as functions of angular frequency  $\omega$ .

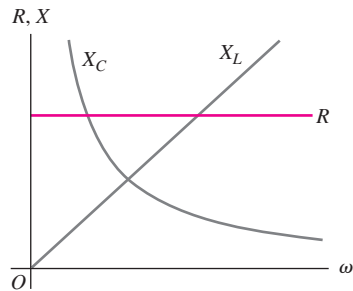
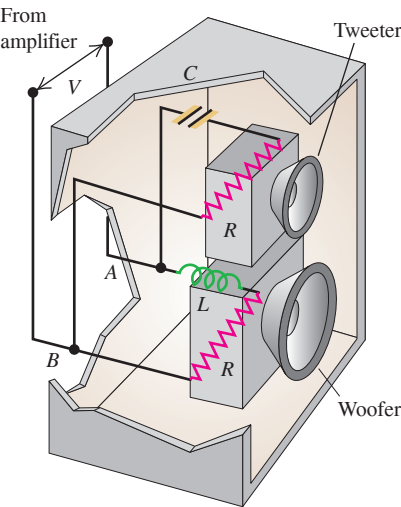
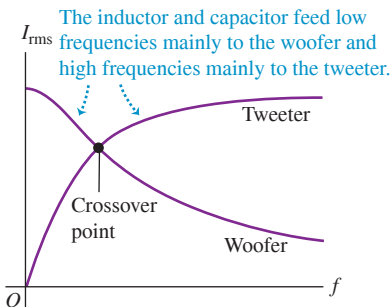


Figure 31.12 (a) The two speakers in this loudspeaker system are connected in parallel to the amplifier. (b) Graphs of current amplitude in the tweeter and woofer as functions of frequency for a given amplifier voltage amplitude.

(a) A crossover network in a loudspeaker system



(b) Graphs of rms current as functions of frequency for a given amplifier voltage



Comparing ac Circuit Elements

Table 31.1 summarizes the relationships of voltage and current amplitudes for the three circuit elements we have discussed. Note again that *instantaneous* voltage and current are proportional in a resistor, where there is zero phase difference between  $v_R$  and  $i$  (see Fig. 31.7b). The instantaneous voltage and current are *not* proportional in an inductor or capacitor, because there is a  $90^\circ$  phase difference in both cases (see Figs. 31.8b and 31.9b).

TABLE 31.1 Circuit Elements with Alternating Current

| Circuit Element | Amplitude Relationship | Circuit Quantity   | Phase of $v$            |
|-----------------|------------------------|--------------------|-------------------------|
| Resistor        | $V_R = IR$             | $R$                | In phase with $i$       |
| Inductor        | $V_L = IX_L$           | $X_L = \omega L$   | Leads $i$ by $90^\circ$ |
| Capacitor       | $V_C = IX_C$           | $X_C = 1/\omega C$ | Lags $i$ by $90^\circ$  |

Figure 31.11 shows how the resistance of a resistor and the reactances of an inductor and a capacitor vary with angular frequency  $\omega$ . Resistance  $R$  is independent of frequency, while the reactances  $X_L$  and  $X_C$  are not. If  $\omega = 0$ , corresponding to a dc circuit, there is *no* current through a capacitor because  $X_C \rightarrow \infty$ , and there is no inductive effect because  $X_L = 0$ . In the limit  $\omega \rightarrow \infty$ ,  $X_L$  also approaches infinity, and the current through an inductor becomes vanishingly small; recall that the self-induced emf opposes rapid changes in current. In this same limit,  $X_C$  and the voltage across a capacitor both approach zero; the current changes direction so rapidly that no charge can build up on either plate.

Figure 31.12 shows an application of the above discussion to a loudspeaker system. Low-frequency sounds are produced by the *woofer*, which is a speaker with large diameter; the *tweeter*, a speaker with smaller diameter, produces high-frequency sounds. In order to route signals of different frequency to the appropriate speaker, the woofer and tweeter are connected in parallel across the amplifier output. The capacitor in the tweeter branch blocks the low-frequency components of sound but passes the higher frequencies; the inductor in the woofer branch does the opposite.

**TEST YOUR UNDERSTANDING OF SECTION 31.2** An oscillating voltage of fixed amplitude is applied across a circuit element. If the frequency of this voltage is increased, will the amplitude of the current through the element (i) increase, (ii) decrease, or (iii) remain the same if it is (a) a resistor, (b) an inductor, or (c) a capacitor?

ANSWER

(a) (iii); (b) (ii); (c) (i) For a resistor,  $V_R = IR$ , so  $I = V_R/R$ . The voltage amplitude  $V_R$  and resistance  $R$  do not change with frequency, so the current amplitude  $I$  remains constant. For an inductor,  $V_L = IX_L$ , so  $I = V_L/X_L = V_L/\omega L$ . The voltage amplitude  $V_L$  and inductance  $L$  are constant, so the current amplitude  $I$  decreases as the frequency increases. For a capacitor,  $V_C = I/X_C$ , so  $I = V_C/X_C = V_C\omega C$ . The voltage amplitude  $V_C$  and capacitance  $C$  are constant, so the current amplitude  $I$  increases as the frequency increases.

### 31.3 THE L-R-C SERIES CIRCUIT

Many ac circuits used in practical electronic systems involve resistance, inductive reactance, and capacitive reactance. **Figure 31.13a** shows a simple example: a series circuit containing a resistor, an inductor, a capacitor, and an ac source. (In Section 30.6 we studied an L-R-C series circuit *without* a source.)

To analyze this circuit, we'll use a phasor diagram that includes the voltage and current phasors for each of the components. Because of Kirchhoff's loop rule, the instantaneous *total* voltage  $v_{ad}$  across all three components is equal to the source voltage at that instant. We'll show that the phasor representing this total voltage is the *vector sum* of the phasors for the individual voltages.

Figures 31.13b and 31.13c show complete phasor diagrams for the circuit of Fig. 31.13a. We assume that the source supplies a current  $i$  given by  $i = I \cos \omega t$ . Because the circuit elements are connected in series, the current at any instant is the same at every point in the circuit. Thus a *single phasor*  $I$ , with length proportional to the current amplitude, represents the current in *all* circuit elements.

As in Section 31.2, we use the symbols  $v_R$ ,  $v_L$ , and  $v_C$  for the instantaneous voltages across  $R$ ,  $L$ , and  $C$ , and the symbols  $V_R$ ,  $V_L$ , and  $V_C$  for the maximum voltages. We denote the instantaneous and maximum *source* voltages by  $v$  and  $V$ . Then, in Fig. 31.13a,  $v = v_{ad}$ ,  $v_R = v_{ab}$ ,  $v_L = v_{bc}$ , and  $v_C = v_{cd}$ .

The potential difference between the terminals of a resistor is *in phase* with the current in the resistor. Its maximum value  $V_R$  is given by Eq. (31.7):

$$V_R = IR$$

The phasor  $V_R$  in Fig. 31.13b, in phase with the current phasor  $I$ , represents the voltage across the resistor. Its projection onto the horizontal axis at any instant gives the instantaneous potential difference  $v_R$ .

The voltage across an inductor *leads* the current by  $90^\circ$ . Its voltage amplitude is given by Eq. (31.13):

$$V_L = IX_L$$

The phasor  $V_L$  in Fig. 31.13b represents the voltage across the inductor, and its projection onto the horizontal axis at any instant equals  $v_L$ .

The voltage across a capacitor *lags* the current by  $90^\circ$ . Its voltage amplitude is given by Eq. (31.19):

$$V_C = IX_C$$

The phasor  $V_C$  in Fig. 31.13b represents the voltage across the capacitor, and its projection onto the horizontal axis at any instant equals  $v_C$ .

The instantaneous potential difference  $v$  between terminals  $a$  and  $d$  is equal at every instant to the (algebraic) sum of the potential differences  $v_R$ ,  $v_L$ , and  $v_C$ . That is, it equals the sum of the *projections* of the phasors  $V_R$ ,  $V_L$ , and  $V_C$ . But the sum of the projections of these phasors is equal to the *projection* of their *vector sum*. So the vector sum  $V$  must be the phasor that represents the source voltage  $v$  and the instantaneous total voltage  $v_{ad}$  across the series of elements.

To form this vector sum, we first subtract the phasor  $V_C$  from the phasor  $V_L$ . (These two phasors always lie along the same line, with opposite directions.) This gives the phasor  $V_L - V_C$ . This is always at right angles to the phasor  $V_R$ , so from the Pythagorean theorem the magnitude of the phasor  $V$  is

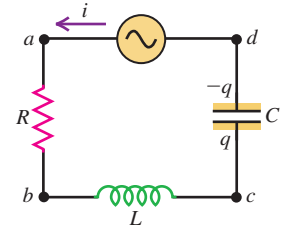
$$\begin{aligned} V &= \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{(IR)^2 + (IX_L - IX_C)^2} \quad \text{or} \\ V &= I\sqrt{R^2 + (X_L - X_C)^2} \end{aligned} \quad (31.20)$$

We define the **impedance**  $Z$  of an ac circuit as the ratio of the voltage amplitude across the circuit to the current amplitude in the circuit. From Eq. (31.20) the impedance of the L-R-C series circuit is

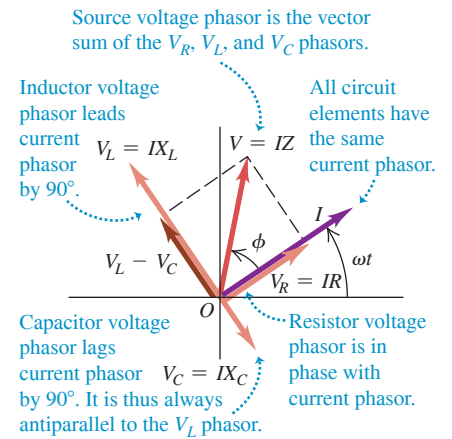
$$Z = \sqrt{R^2 + (X_L - X_C)^2} \quad (31.21)$$

Figure 31.13 An L-R-C series circuit with an ac source.

(a) L-R-C series circuit



(b) Phasor diagram for the case  $X_L > X_C$



(c) Phasor diagram for the case  $X_L < X_C$

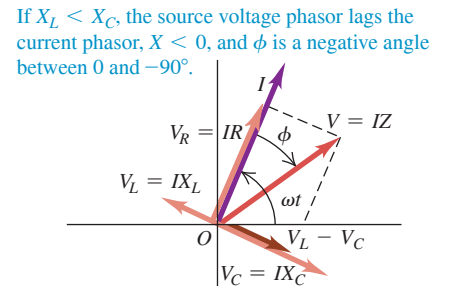


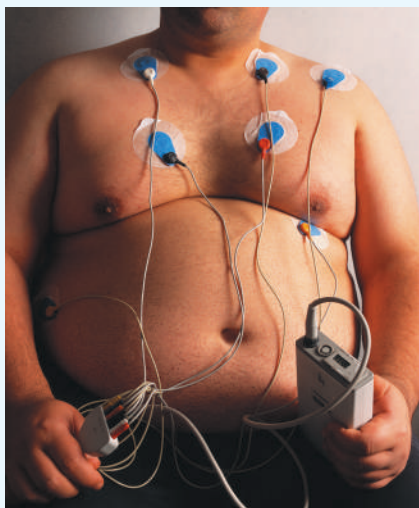


Figure 31.14 This gas-filled glass sphere has an alternating voltage between its surface and the electrode at its center. The glowing streamers show the resulting alternating current that passes through the gas. When you touch the outside of the sphere, your fingertips and the inner surface of the sphere act as the plates of a capacitor, and the sphere and your body together form an  $L$ - $R$ - $C$  series circuit. The current (which is low enough to be harmless) is drawn to your fingers because the path through your body has a low impedance.



### BIO APPLICATION Measuring Body Fat by Bioelectric Impedance Analysis

The electrodes attached to this overweight patient's chest are applying a small ac voltage of frequency 50 kHz. The attached instrumentation measures the amplitude and phase angle of the resulting current through the patient's body. These depend on the relative amounts of water and fat along the path followed by the current, and so provide a sensitive measure of body composition.



so we can rewrite Eq. (31.20) as

$$\text{Amplitude of voltage across an ac circuit} \quad V = IZ \quad \text{Current amplitude} \quad \text{Impedance of circuit} \quad (31.22)$$

While Eq. (31.21) is valid only for an  $L$ - $R$ - $C$  series circuit, we can use Eq. (31.22) to define the impedance of *any* network of resistors, inductors, and capacitors as the ratio of the amplitude of the voltage across the network to the current amplitude. The SI unit of impedance is the ohm.

### The Meaning of Impedance and Phase Angle

Equation (31.22) has a form similar to  $V = IR$ , with impedance  $Z$  in an ac circuit playing the role of resistance  $R$  in a dc circuit. Just as direct current tends to follow the path of least resistance, so alternating current tends to follow the path of lowest impedance (Fig. 31.14). Note, however, that impedance is actually a function of  $R$ ,  $L$ , and  $C$ , as well as of the angular frequency  $\omega$ . We can see this by substituting Eq. (31.12) for  $X_L$  and Eq. (31.18) for  $X_C$  into Eq. (31.21), giving the following complete expression for  $Z$  for a series circuit:

$$\text{Impedance of an } L\text{-}R\text{-}C \text{ series circuit} \quad Z = \sqrt{R^2 + [\omega L - (1/\omega C)]^2} \quad \begin{array}{l} \text{Resistance} \quad \text{Inductance} \quad \text{Capacitance} \\ \text{Angular frequency} \end{array} \quad (31.23)$$

Hence for a given amplitude  $V$  of the source voltage applied to the circuit, the amplitude  $I = V/Z$  of the resulting current will be different at different frequencies. We'll explore this frequency dependence in detail in Section 31.5.

In Fig. 31.13b, the angle  $\phi$  between the voltage and current phasors is the phase angle of the source voltage  $v$  with respect to the current  $i$ ; that is, it is the angle by which the source voltage leads the current. From the diagram,

$$\tan \phi = \frac{V_L - V_C}{V_R} = \frac{I(X_L - X_C)}{IR} = \frac{X_L - X_C}{R}$$

$$\text{Phase angle of voltage with respect to current in an } L\text{-}R\text{-}C \text{ series circuit} \quad \tan \phi = \frac{\omega L - 1/\omega C}{R} \quad \begin{array}{l} \text{Inductance} \quad \text{Angular frequency} \\ \text{Capacitance} \\ \text{Resistance} \end{array} \quad (31.24)$$

If the current is  $i = I \cos \omega t$ , then the source voltage  $v$  is

$$v = V \cos(\omega t + \phi) \quad (31.25)$$

Figure 31.13b shows the behavior of an  $L$ - $R$ - $C$  series circuit in which  $X_L > X_C$ . Figure 31.13c shows the behavior when  $X_L < X_C$ ; the voltage phasor  $V$  lies on the opposite side of the current phasor  $I$  and the voltage *lags* the current. In this case,  $X_L - X_C$  is *negative*,  $\tan \phi$  is negative, and  $\phi$  is a negative angle between  $0^\circ$  and  $-90^\circ$ . Since  $X_L$  and  $X_C$  depend on frequency, the phase angle  $\phi$  depends on frequency as well. We'll examine the consequences of this in Section 31.5.

All of the expressions that we've developed for an  $L$ - $R$ - $C$  series circuit are still valid if one of the circuit elements is missing. If the resistor is missing, we set  $R = 0$ ; if the inductor is missing, we set  $L = 0$ . But if the capacitor is missing, we set  $C = \infty$ , corresponding to the absence of any potential difference ( $v_C = q/C = 0$ ) or any capacitive reactance ( $X_C = 1/\omega C = 0$ ).

In this entire discussion we have described magnitudes of voltages and currents in terms of their *maximum* values, the voltage and current *amplitudes*. But we remarked at the end of Section 31.1 that these quantities are usually described in terms of rms values, not amplitudes. For any sinusoidally varying quantity, the rms value is always  $1/\sqrt{2}$  times the amplitude. All the relationships between voltage and current that we have derived in this and the preceding sections are still valid if we use rms quantities throughout instead of amplitudes. For example, if we divide Eq. (31.22) by  $\sqrt{2}$ , we get

$$\frac{V}{\sqrt{2}} = \frac{I}{\sqrt{2}}Z$$

which we can rewrite as

$$V_{\text{rms}} = I_{\text{rms}}Z \quad (31.26)$$

We can translate Eqs. (31.7), (31.13), and (31.19) in exactly the same way.

We have considered only ac circuits in which an inductor, a resistor, and a capacitor are in series. You can do a similar analysis for an  $L$ - $R$ - $C$  *parallel* circuit; see Problem 31.54.

### PROBLEM-SOLVING STRATEGY 31.1 Alternating-Current Circuits

**IDENTIFY** the relevant concepts: In analyzing ac circuits, we can apply all of the concepts used to analyze direct-current circuits, particularly those in Problem-Solving Strategies 26.1 and 26.2. But now we must distinguish between the amplitudes of alternating currents and voltages and their instantaneous values, and among resistance (for resistors), reactance (for inductors or capacitors), and impedance (for composite circuits).

**SET UP** the problem using the following steps:

1. Draw a diagram of the circuit and label all known and unknown quantities.
2. Identify the target variables.

**EXECUTE** the solution as follows:

1. Use the relationships derived in Sections 31.2 and 31.3 to solve for the target variables, using the following hints.
2. It's almost always easiest to work with angular frequency  $\omega = 2\pi f$  rather than ordinary frequency  $f$ .
3. Keep in mind the following phase relationships: For a resistor, voltage and current are *in phase*, so the corresponding phasors always point in the same direction. For an inductor, the voltage *leads* the current by  $90^\circ$  (i.e.,  $\phi = +90^\circ = \pi/2$  radians), so the voltage phasor points  $90^\circ$  counterclockwise from the current phasor. For a capacitor, the voltage *lags* the current by  $90^\circ$

(i.e.,  $\phi = -90^\circ = -\pi/2$  radians), so the voltage phasor points  $90^\circ$  clockwise from the current phasor.

4. Kirchhoff's rules hold *at each instant*. For example, in a series circuit, the instantaneous current is the same in all circuit elements; in a parallel circuit, the instantaneous potential difference is the same across all circuit elements.
5. Inductive reactance, capacitive reactance, and impedance are analogous to resistance; each represents the ratio of voltage amplitude  $V$  to current amplitude  $I$  in a circuit element or combination of elements. However, phase relationships are crucial. In applying Kirchhoff's loop rule, you must combine the effects of resistance and reactance by *vector* addition of the corresponding voltage phasors, as in Figs. 31.13b and 31.13c. When several circuit elements are in series, for example, you can't *add* all the numerical values of resistance and reactance to get the impedance; that would ignore the phase relationships.

**EVALUATE** your answer: When working with an  $L$ - $R$ - $C$  *series* circuit, you can check your results by comparing the values of the inductive and capacitive reactances  $X_L$  and  $X_C$ . If  $X_L > X_C$ , then the voltage amplitude across the inductor is greater than that across the capacitor and the phase angle  $\phi$  is positive (between  $0^\circ$  and  $90^\circ$ ). If  $X_L < X_C$ , then the voltage amplitude across the inductor is less than that across the capacitor and the phase angle  $\phi$  is negative (between  $0^\circ$  and  $-90^\circ$ ).

### EXAMPLE 31.4 An $L$ - $R$ - $C$ series circuit I

#### WITH VARIATION PROBLEMS

In the series circuit of Fig. 31.13a, suppose  $R = 300 \, \Omega$ ,  $L = 60 \, \text{mH}$ ,  $C = 0.50 \, \mu\text{F}$ ,  $V = 50 \, \text{V}$ , and  $\omega = 10,000 \, \text{rad/s}$ . Find the reactances  $X_L$  and  $X_C$ , the impedance  $Z$ , the current amplitude  $I$ , the phase angle  $\phi$ , and the voltage amplitude across each circuit element.

**IDENTIFY and SET UP** This problem uses the ideas developed in Section 31.2 and this section about the behavior of circuit elements in an ac circuit. We use Eqs. (31.12) and (31.18) to determine  $X_L$  and  $X_C$ , and Eq. (31.23) to find  $Z$ . We then use Eq. (31.22) to find the current amplitude and Eq. (31.24) to find the phase angle. The relationships in Table 31.1 then yield the voltage amplitudes.

**EXECUTE** The inductive and capacitive reactances are

$$X_L = \omega L = (10,000 \, \text{rad/s})(60 \, \text{mH}) = 600 \, \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{(10,000 \, \text{rad/s})(0.50 \times 10^{-6} \, \text{F})} = 200 \, \Omega$$

The impedance  $Z$  of the circuit is then

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{(300 \, \Omega)^2 + (600 \, \Omega - 200 \, \Omega)^2} = 500 \, \Omega$$

*Continued*

With source voltage amplitude  $V = 50\text{ V}$ , the current amplitude  $I$  and phase angle  $\phi$  are

$$I = \frac{V}{Z} = \frac{50\text{ V}}{500\ \Omega} = 0.10\text{ A}$$

$$\phi = \arctan \frac{X_L - X_C}{R} = \arctan \frac{400\ \Omega}{300\ \Omega} = 53^\circ$$

From Table 31.1, the voltage amplitudes  $V_R$ ,  $V_L$ , and  $V_C$  across the resistor, inductor, and capacitor, respectively, are

$$V_R = IR = (0.10\text{ A})(300\ \Omega) = 30\text{ V}$$

$$V_L = IX_L = (0.10\text{ A})(600\ \Omega) = 60\text{ V}$$

$$V_C = IX_C = (0.10\text{ A})(200\ \Omega) = 20\text{ V}$$

### EXAMPLE 31.5 An $L$ - $R$ - $C$ series circuit II

### WITH VARIATION PROBLEMS

For the  $L$ - $R$ - $C$  series circuit of Example 31.4, find expressions for the time dependence of the instantaneous current  $i$  and the instantaneous voltages across the resistor ( $v_R$ ), inductor ( $v_L$ ), capacitor ( $v_C$ ), and ac source ( $v$ ).

**IDENTIFY and SET UP** We describe the current by using Eq. (31.2), which assumes that the current is maximum at  $t = 0$ . The voltages are then given by Eq. (31.8) for the resistor, Eq. (31.10) for the inductor, Eq. (31.16) for the capacitor, and Eq. (31.25) for the source.

**EXECUTE** The current and the voltages all oscillate with the same angular frequency,  $\omega = 10,000\text{ rad/s}$ , and hence with the same period,  $2\pi/\omega = 2\pi/(10,000\text{ rad/s}) = 6.3 \times 10^{-4}\text{ s} = 0.63\text{ ms}$ . From Eq. (31.2), the current is

$$i = I \cos \omega t = (0.10\text{ A}) \cos [(10,000\text{ rad/s})t]$$

The resistor voltage is *in phase* with the current, so

$$v_R = V_R \cos \omega t = (30\text{ V}) \cos [(10,000\text{ rad/s})t]$$

The inductor voltage *leads* the current by  $90^\circ$ , so

$$v_L = V_L \cos(\omega t + 90^\circ) = -V_L \sin \omega t = -(60\text{ V}) \sin [(10,000\text{ rad/s})t]$$

The capacitor voltage *lags* the current by  $90^\circ$ , so

$$v_C = V_C \cos(\omega t - 90^\circ) = V_C \sin \omega t = (20\text{ V}) \sin [(10,000\text{ rad/s})t]$$

We found in Example 31.4 that the source voltage (equal to the voltage across the entire combination of resistor, inductor, and capacitor) *leads* the current by  $\phi = 53^\circ$ , so

$$v = V \cos(\omega t + \phi)$$

$$= (50\text{ V}) \cos \left[ (10,000\text{ rad/s})t + \left( \frac{2\pi\text{ rad}}{360^\circ} \right) (53^\circ) \right]$$

$$= (50\text{ V}) \cos [(10,000\text{ rad/s})t + 0.93\text{ rad}]$$

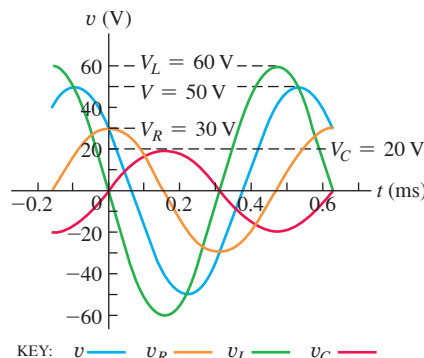
**EVALUATE** As in Fig. 31.13b,  $X_L > X_C$ ; hence the voltage amplitude across the inductor is greater than that across the capacitor and  $\phi$  is positive. The value  $\phi = 53^\circ$  means that the voltage *leads* the current by  $53^\circ$ .

Note that the source voltage amplitude  $V = 50\text{ V}$  is *not* equal to the sum of the voltage amplitudes across the separate circuit elements:  $50\text{ V} \neq 30\text{ V} + 60\text{ V} + 20\text{ V}$ . Instead,  $V$  is the *vector sum* of the  $V_R$ ,  $V_L$ , and  $V_C$  phasors. If you draw the phasor diagram like Fig. 31.13b for this particular situation, you'll see that  $V_R$ ,  $V_L - V_C$ , and  $V$  constitute a 3-4-5 right triangle.

**KEYCONCEPT** For an ac circuit that includes an inductor  $L$ , resistor  $R$ , and capacitor  $C$  in series, the impedance—the ratio of the amplitude of the voltage across the  $L$ - $R$ - $C$  combination to the amplitude of the current—depends on the values of  $L$ ,  $R$ ,  $C$ , and the angular frequency of the current.

**EVALUATE** Figure 31.15 graphs the four voltages versus time. The inductor voltage has a larger amplitude than the capacitor voltage because  $X_L > X_C$ . The *instantaneous* source voltage  $v$  is always equal to the sum of the instantaneous voltages  $v_R$ ,  $v_L$ , and  $v_C$ . You should verify this by measuring the values of the voltages shown in the graph at different values of the time  $t$ .

Figure 31.15 Graphs of the source voltage  $v$ , resistor voltage  $v_R$ , inductor voltage  $v_L$ , and capacitor voltage  $v_C$  as functions of time for the situation of Example 31.4. The current, which is not shown, is in phase with the resistor voltage.



**KEYCONCEPT** In an  $L$ - $R$ - $C$  series circuit that includes an ac source, the instantaneous voltage across the  $L$ - $R$ - $C$  combination is the sum of the instantaneous voltages across each of the three components. Because these voltages are not in phase with one another, the amplitude of the voltage across the  $L$ - $R$ - $C$  combination is *not* equal to the sum of the individual voltage amplitudes.

**TEST YOUR UNDERSTANDING OF SECTION 31.3** Rank the following ac circuits in order of their current amplitude, from highest to lowest value. (i) The circuit in Example 31.4; (ii) the circuit in Example 31.4 with both the capacitor and inductor removed; (iii) the circuit in Example 31.4 with both the resistor and capacitor removed; (iv) the circuit in Example 31.4 with both the resistor and inductor removed.

### ANSWER

(iv), (iii), (ii), (i). For the circuit in Example 31.4,  $I = (50\text{ V})/(500\ \Omega) = 0.10\text{ A}$ . If the capacitor and inductor are removed so that only the ac source and resistor remain, the circuit is like that shown in Fig. 31.7a; then  $I = (50\text{ V})/(300\ \Omega) = 0.17\text{ A}$ . If the resistor and capacitor are removed so that only the ac source and inductor remain, the circuit is like that shown in Fig. 31.8a; then  $I = (50\text{ V})/(600\ \Omega) = 0.083\text{ A}$ . Finally, if the resistor and inductor are removed so that only the ac source and capacitor remain, the circuit is like that shown in Fig. 31.9a; then  $I = (50\text{ V})/(200\ \Omega) = 0.25\text{ A}$ .

## 31.4 POWER IN ALTERNATING-CURRENT CIRCUITS

Alternating currents play a central role in systems for distributing, converting, and using electrical energy, so it's important to look at power relationships in ac circuits. For an ac circuit with instantaneous current  $i$  and current amplitude  $I$ , we'll consider an element of that circuit across which the instantaneous potential difference is  $v$  with voltage amplitude  $V$ . The instantaneous power  $p$  delivered to this circuit element is

$$p = vi$$

Let's first see what this means for individual circuit elements. We'll assume in each case that  $i = I \cos \omega t$ .

### Power in a Resistor

Suppose first that the circuit element is a *pure resistor*  $R$ , as in Fig. 31.7a; then  $v = v_R$  and  $i$  are *in phase*. We obtain the graph representing  $p$  by multiplying the heights of the graphs of  $v$  and  $i$  in Fig. 31.7b at each instant. The result is the black curve in **Fig. 31.16a**. The product  $vi$  is always positive because  $v$  and  $i$  are always either both positive or both negative. Hence energy is supplied *to* the resistor at every instant for both directions of  $i$ , although the power is not constant.

The power curve for a pure resistor is symmetric about a value equal to one-half its maximum value  $VI$ , so the *average power*  $P_{\text{av}}$  is

$$P_{\text{av}} = \frac{1}{2} VI \quad (\text{for a pure resistor}) \quad (31.27)$$

An equivalent expression is

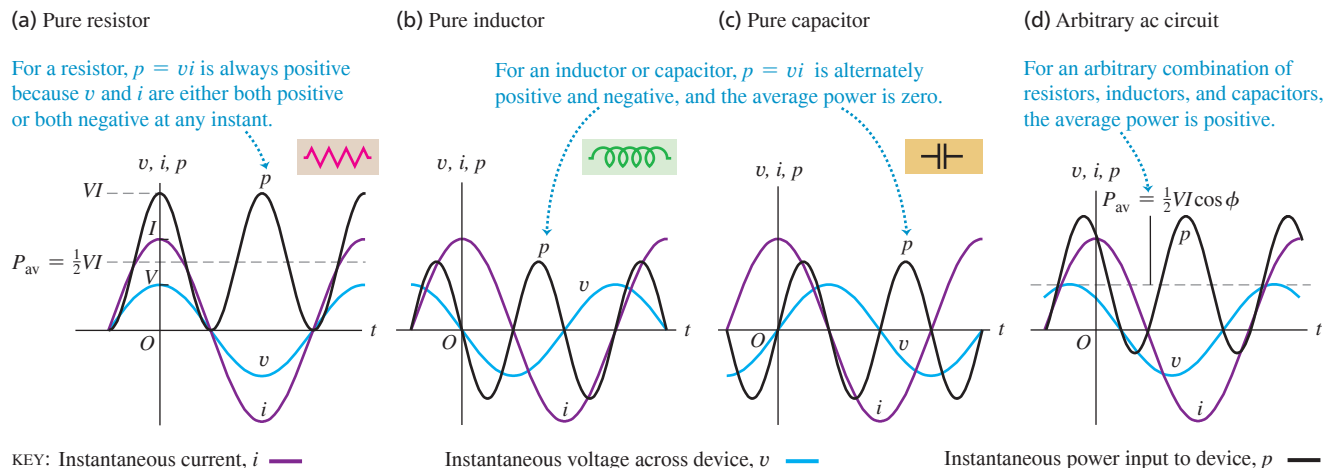
$$P_{\text{av}} = \frac{V}{\sqrt{2}} \frac{I}{\sqrt{2}} = V_{\text{rms}} I_{\text{rms}} \quad (\text{for a pure resistor}) \quad (31.28)$$

Also,  $V_{\text{rms}} = I_{\text{rms}} R$ , so we can express  $P_{\text{av}}$  by any of the equivalent forms

$$P_{\text{av}} = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R} = V_{\text{rms}} I_{\text{rms}} \quad (\text{for a pure resistor}) \quad (31.29)$$

Note that the expressions in Eq. (31.29) have the same form as the corresponding relationships for a dc circuit, Eq. (25.18). Also note that they are valid only for pure resistors, not for more complicated combinations of circuit elements.

Figure 31.16 Graphs of current, voltage, and power as functions of time for (a) a pure resistor, (b) a pure inductor, (c) a pure capacitor, and (d) an arbitrary ac circuit that can have resistance, inductance, and capacitance.





## Power in an Inductor

Next we connect the source to a pure inductor  $L$ , as in Fig. 31.8a. The voltage  $v = v_L$  leads the current  $i$  by  $90^\circ$ . When we multiply the curves of  $v$  and  $i$ , the product  $vi$  is *negative* during the half of the cycle when  $v$  and  $i$  have *opposite* signs. The power curve, shown in Fig. 31.16b, is symmetric about the horizontal axis; it is positive half the time and negative the other half, and the average power is zero. When  $p$  is positive, energy is being supplied to set up the magnetic field in the inductor; when  $p$  is negative, the field is collapsing and the inductor is returning energy to the source. The net energy transfer over one cycle is zero.

## Power in a Capacitor

Finally, we connect the source to a pure capacitor  $C$ , as in Fig. 31.9a. The voltage  $v = v_C$  lags the current  $i$  by  $90^\circ$ . Figure 31.16c shows the power curve; the average power is again zero. Energy is supplied to charge the capacitor and is returned to the source when the capacitor discharges. The net energy transfer over one cycle is again zero.

## Power in a General ac Circuit

In *any* ac circuit, with any combination of resistors, capacitors, and inductors, the voltage  $v$  across the entire circuit has some phase angle  $\phi$  with respect to the current  $i$ . Then the instantaneous power  $p$  is given by

$$p = vi = [V \cos(\omega t + \phi)] [I \cos \omega t] \quad (31.30)$$

The instantaneous power curve has the form shown in Fig. 31.16d. The area between the positive loops and the horizontal axis is greater than the area between the negative loops and the horizontal axis, and the average power is positive.

We can derive from Eq. (31.30) an expression for the *average* power  $P_{\text{av}}$  by using the identity for the cosine of the sum of two angles:

$$\begin{aligned} p &= [V(\cos \omega t \cos \phi - \sin \omega t \sin \phi)] [I \cos \omega t] \\ &= VI \cos \phi \cos^2 \omega t - VI \sin \phi \cos \omega t \sin \omega t \end{aligned}$$

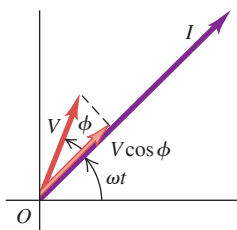
From the discussion in Section 31.1 that led to Eq. (31.4), the average value of  $\cos^2 \omega t$  (over one cycle) is  $\frac{1}{2}$ . Furthermore,  $\cos \omega t \sin \omega t$  is equal to  $\frac{1}{2} \sin 2\omega t$ , whose average over a cycle is zero. So the average power  $P_{\text{av}}$  is

$$\begin{aligned} \text{Average power into a general ac circuit} \quad P_{\text{av}} &= \frac{1}{2} VI \cos \phi = V_{\text{rms}} I_{\text{rms}} \cos \phi \end{aligned} \quad (31.31)$$

Phase angle of voltage with respect to current  
Voltage amplitude Current amplitude rms voltage rms current

Figure 31.17 Using phasors to calculate the average power for an arbitrary ac circuit.

Average power  $= \frac{1}{2} I(V \cos \phi)$ , where  $V \cos \phi$  is the component of  $V$  in phase with  $I$ .



**Figure 31.17** shows the general relationship of the current and voltage phasors. When  $v$  and  $i$  are in phase, so  $\phi = 0$ , the average power equals  $\frac{1}{2} VI = V_{\text{rms}} I_{\text{rms}}$ ; when  $v$  and  $i$  are  $90^\circ$  out of phase, the average power is zero. In the general case, when  $v$  has a phase angle  $\phi$  with respect to  $i$ , the average power equals  $\frac{1}{2} I$  multiplied by  $V \cos \phi$ , the component of the voltage phasor that is *in phase* with the current phasor. For the  $L$ - $R$ - $C$  series circuit, Figs. 31.13b and 31.13c show that  $V \cos \phi$  equals the voltage amplitude  $V_R$  for the resistor; hence Eq. (31.31) is the average power dissipated in the resistor. On average there is no energy flow into or out of the inductor or capacitor, so none of  $P_{\text{av}}$  goes into either of these circuit elements.

The factor  $\cos \phi$  is called the **power factor** of the circuit. For a pure resistance,  $\phi = 0$ ,  $\cos \phi = 1$ , and  $P_{\text{av}} = V_{\text{rms}} I_{\text{rms}}$ . For a pure inductor or capacitor,  $\phi = \pm 90^\circ$ ,  $\cos \phi = 0$ , and  $P_{\text{av}} = 0$ . For an  $L$ - $R$ - $C$  series circuit the power factor is equal to  $R/Z$ ; we leave the proof of this statement to you.

A low power factor (large angle  $\phi$  of lag or lead) is usually undesirable in power circuits. The reason is that for a given potential difference, a large current is needed to

supply a given amount of power. This results in large  $i^2R$  losses in the transmission lines. Many types of ac machinery draw a *lagging* current; that is, the current drawn by the machinery lags the applied voltage. Hence the voltage leads the current, so  $\phi > 0$  and  $\cos \phi < 1$ . The power factor can be corrected toward the ideal value of 1 by connecting a capacitor in parallel with the load. The current drawn by the capacitor *leads* the voltage (that is, the voltage across the capacitor lags the current), which compensates for the lagging current in the other branch of the circuit. The capacitor itself absorbs no net power from the line.

### EXAMPLE 31.6 Power in a hair dryer

### WITH VARIATION PROBLEMS

An electric hair dryer is rated at 1500 W (the *average* power) at 120 V (the *rms* voltage). Calculate (a) the resistance, (b) the rms current, and (c) the maximum instantaneous power. Assume that the dryer is a pure resistor. (The heating element acts as a resistor.)

**IDENTIFY and SET UP** We are given  $P_{\text{av}} = 1500 \text{ W}$  and  $V_{\text{rms}} = 120 \text{ V}$ . Our target variables are the resistance  $R$ , the rms current  $I_{\text{rms}}$ , and the maximum value  $p_{\text{max}}$  of the instantaneous power  $p$ . We solve Eq. (31.29) to find  $R$ , Eq. (31.28) to find  $I_{\text{rms}}$  from  $V_{\text{rms}}$  and  $P_{\text{av}}$ , and Eq. (31.30) to find  $p_{\text{max}}$ .

**EXECUTE** (a) From Eq. (31.29), the resistance is

$$R = \frac{V_{\text{rms}}^2}{P_{\text{av}}} = \frac{(120 \text{ V})^2}{1500 \text{ W}} = 9.6 \, \Omega$$

(b) From Eq. (31.28),

$$I_{\text{rms}} = \frac{P_{\text{av}}}{V_{\text{rms}}} = \frac{1500 \text{ W}}{120 \text{ V}} = 12.5 \text{ A}$$

(c) For a pure resistor, the voltage and current are in phase and the phase angle  $\phi$  is zero. Hence from Eq. (31.30), the instantaneous power is  $p = VI \cos^2 \omega t$  and the maximum instantaneous power is  $p_{\text{max}} = VI$ . From Eq. (31.27), this is twice the average power  $P_{\text{av}}$ , so

$$p_{\text{max}} = VI = 2P_{\text{av}} = 2(1500 \text{ W}) = 3000 \text{ W}$$

**EVALUATE** We can use Eq. (31.7) to confirm our result in part (b):  $I_{\text{rms}} = V_{\text{rms}}/R = (120 \text{ V})/(9.6 \, \Omega) = 12.5 \text{ A}$ . Note that some unscrupulous manufacturers of stereo amplifiers advertise the *peak* power output rather than the lower average value.

**KEYCONCEPT** The power delivered to a resistor in an ac circuit is not constant, but oscillates between zero and a maximum value. The average power delivered to the resistor is one-half of this maximum, and equals the rms current through the resistor times the rms voltage across the resistor.

### EXAMPLE 31.7 Power in an $L$ - $R$ - $C$ series circuit

### WITH VARIATION PROBLEMS

For the  $L$ - $R$ - $C$  series circuit of Example 31.4, (a) calculate the power factor and (b) calculate the average power delivered to the entire circuit and to each circuit element.

**IDENTIFY and SET UP** We can use the results of Example 31.4. The power factor is the cosine of the phase angle  $\phi$ , and we use Eq. (31.31) to find the average power delivered in terms of  $\phi$  and the amplitudes of voltage and current.

**EXECUTE** (a) The power factor is  $\cos \phi = \cos 53^\circ = 0.60$ .

(b) From Eq. (31.31),

$$P_{\text{av}} = \frac{1}{2} VI \cos \phi = \frac{1}{2} (50 \text{ V})(0.10 \text{ A})(0.60) = 1.5 \text{ W}$$

**EVALUATE** Although  $P_{\text{av}}$  is the average power delivered to the  $L$ - $R$ - $C$  combination, all of this power is dissipated in the *resistor*. As Figs. 31.16b and 31.16c show, the average power delivered to a pure inductor or pure capacitor is always zero.

**KEYCONCEPT** The power factor of an ac circuit equals the cosine of the phase angle  $\phi$  between the oscillating voltage across the circuit and the oscillating current in the circuit. The greater the power factor, the less current is required to supply a given average power for a given voltage amplitude.

**TEST YOUR UNDERSTANDING OF SECTION 31.4** Figure 31.16d shows that during part of a cycle of oscillation, the instantaneous power delivered to the circuit is negative. This means that energy is being extracted from the circuit. (a) Where is the energy extracted from? (i) The resistor; (ii) the inductor; (iii) the capacitor; (iv) the ac source; (v) more than one of these. (b) Where does the energy go? (i) The resistor; (ii) the inductor; (iii) the capacitor; (iv) the ac source; (v) more than one of these.

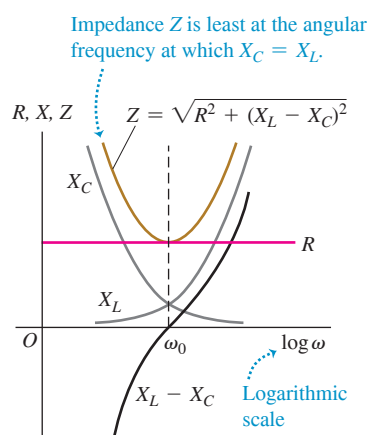
### ANSWER

implies that energy is being transferred back into the source. power means that energy is being transferred from the ac source to the circuit, so *negative* power (which stores magnetic-field energy) or the capacitor (which stores electric-field energy). Positive resistor and cannot be recovered. Instead, the energy must be extracted from either the inductor (a) (v); (b) (v); (c) (v); (d) (v); (e) (v); (f) (v); (g) (v); (h) (v); (i) (v); (j) (v); (k) (v); (l) (v); (m) (v); (n) (v); (o) (v); (p) (v); (q) (v); (r) (v); (s) (v); (t) (v); (u) (v); (v) (v); (w) (v); (x) (v); (y) (v); (z) (v); (aa) (v); (ab) (v); (ac) (v); (ad) (v); (ae) (v); (af) (v); (ag) (v); (ah) (v); (ai) (v); (aj) (v); (ak) (v); (al) (v); (am) (v); (an) (v); (ao) (v); (ap) (v); (aq) (v); (ar) (v); (as) (v); (at) (v); (au) (v); (av) (v); (aw) (v); (ax) (v); (ay) (v); (az) (v); (ba) (v); (bb) (v); (bc) (v); (bd) (v); (be) (v); (bf) (v); (bg) (v); (bh) (v); (bi) (v); (bj) (v); (bk) (v); (bl) (v); (bm) (v); (bn) (v); (bo) (v); (bp) (v); (bq) (v); (br) (v); (bs) (v); (bt) (v); (bu) (v); (bv) (v); (bw) (v); (bx) (v); (by) (v); (bz) (v); (ca) (v); (cb) (v); (cc) (v); (cd) (v); (ce) (v); (cf) (v); (cg) (v); (ch) (v); (ci) (v); (cj) (v); (ck) (v); (cl) (v); (cm) (v); (cn) (v); (co) (v); (cp) (v); (cq) (v); (cr) (v); (cs) (v); 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(G) (v); (H) (v); (I) (v); (J) (v); (K) (v); (L) (v); (M) (v); (N) (v); (O) (v); (P) (v); (Q) (v); (R) (v); (S) (v); (T) (v); (U) (v); (V) (v); (W) (v); (X) (v); (Y) (v); (Z) (v); (a) (v); (b) (v); (c) (v); (d) (v); (e) (v); (f) (v); (g) (v); (h) (v); (i) (v); (j) (v); (k) (

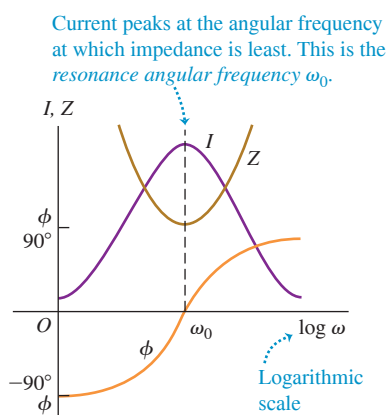
### 31.5 RESONANCE IN ALTERNATING-CURRENT CIRCUITS

Figure 31.18 How variations in the angular frequency of an ac circuit affect (a) reactance, resistance, and impedance, and (b) impedance, current amplitude, and phase angle.

(a) Reactance, resistance, and impedance as functions of angular frequency



(b) Impedance, current, and phase angle as functions of angular frequency



Much of the practical importance of  $L$ - $R$ - $C$  series circuits arises from the way in which such circuits respond to sources of different angular frequency  $\omega$ . For example, one type of tuning circuit used in radio receivers is simply an  $L$ - $R$ - $C$  series circuit. A radio signal of any given frequency produces a current of the same frequency in the receiver circuit, but the amplitude of the current is *greatest* if the signal frequency equals the particular frequency to which the receiver circuit is “tuned.” This effect is called *resonance*. The circuit is designed so that signals at other than the tuned frequency produce currents that are too small to make an audible sound come out of the radio’s speakers.

To see how an  $L$ - $R$ - $C$  series circuit can be used in this way, suppose we connect an ac source with constant voltage amplitude  $V$  but adjustable angular frequency  $\omega$  across an  $L$ - $R$ - $C$  series circuit. The current that appears in the circuit has the same angular frequency as the source and a current amplitude  $I = V/Z$ , where  $Z$  is the impedance of the  $L$ - $R$ - $C$  series circuit. This impedance depends on the frequency, as Eq. (31.23) shows.

**Figure 31.18a** shows graphs of  $R$ ,  $X_L$ ,  $X_C$ , and  $Z$  as functions of  $\omega$ . We have used a logarithmic angular frequency scale so that we can cover a wide range of frequencies. As the frequency increases,  $X_L$  increases and  $X_C$  decreases; hence there is always one frequency at which  $X_L$  and  $X_C$  are equal and  $X_L - X_C$  is zero. At this frequency the impedance  $Z = \sqrt{R^2 + (X_L - X_C)^2}$  has its *smallest* value, equal simply to the resistance  $R$ .

#### Circuit Behavior at Resonance

As we vary the angular frequency  $\omega$  of the source, the current amplitude  $I = V/Z$  varies as shown in Fig. 31.18b; the *maximum* value of  $I$  occurs at the frequency at which the impedance  $Z$  is *minimum*. This peaking of the current amplitude at a certain frequency is called **resonance**. The angular frequency  $\omega_0$  at which the resonance peak occurs is called the **resonance angular frequency**. At  $\omega = \omega_0$  the inductive reactance  $X_L$  and capacitive reactance  $X_C$  are equal, so  $\omega_0 L = 1/\omega_0 C$  and

$$\text{Resonance angular frequency of an } L\text{-}R\text{-}C \text{ series circuit} \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad (31.32)$$

Inductance      Capacitance

This is equal to the natural angular frequency of oscillation of an  $L$ - $C$  circuit, which we derived in Section 30.5, Eq. (30.22). The **resonance frequency**  $f_0$  is  $\omega_0/2\pi$ . At this frequency, the greatest current appears in the circuit for a given source voltage amplitude;  $f_0$  is the frequency to which the circuit is “tuned.”

It’s instructive to look at what happens to the *voltages* in an  $L$ - $R$ - $C$  series circuit at resonance. The current at any instant is the same in  $L$  and  $C$ . The voltage across an inductor always *leads* the current by  $90^\circ$ , or  $\frac{1}{4}$  cycle, and the voltage across a capacitor always *lags* the current by  $90^\circ$ . Therefore the instantaneous voltages across  $L$  and  $C$  always differ in phase by  $180^\circ$ , or  $\frac{1}{2}$  cycle; they have opposite signs at each instant. At the resonance frequency, and *only* at the resonance frequency,  $X_L = X_C$  and the voltage amplitudes  $V_L = IX_L$  and  $V_C = IX_C$  are *equal*; then the instantaneous voltages across  $L$  and  $C$  add to zero at each instant, and the *total* voltage  $v_{bd}$  across the  $L$ - $C$  combination in Fig. 31.13a is exactly zero. The voltage across the resistor is then equal to the source voltage. So at the resonance frequency the circuit behaves as if the inductor and capacitor weren’t there at all!

The *phase* of the voltage relative to the current is given by Eq. (31.24). At frequencies below resonance,  $X_C$  is greater than  $X_L$ ; the capacitive reactance dominates, the voltage *lags* the current, and the phase angle  $\phi$  is between  $0^\circ$  and  $-90^\circ$ . Above resonance, the inductive reactance dominates, the voltage *leads* the current, and the phase angle  $\phi$  is between  $0^\circ$  and  $+90^\circ$ . Figure 31.18b shows this variation of  $\phi$  with angular frequency.

#### Tailoring an ac Circuit

If we can vary the inductance  $L$  or the capacitance  $C$  of a circuit, we can also vary the resonance frequency. This is exactly how a radio is “tuned” to receive a particular station. In the early days of radio this was accomplished by the use of capacitors with movable

metal plates whose overlap could be varied to change  $C$ . (This is what is being done with the radio tuning knob shown in the photograph that opens this chapter.) Another approach is to vary  $L$  by using a coil with a ferrite core that slides in or out.

In an  $L$ - $R$ - $C$  series circuit the impedance is a minimum and the current is a maximum at the resonance frequency. The middle curve in **Fig. 31.19** is a graph of current as a function of frequency for such a circuit, with source voltage amplitude  $V = 100$  V,  $L = 2.0$  H,  $C = 0.50$   $\mu$ F, and  $R = 500$   $\Omega$ . This curve, called a *response curve* or *resonance curve*, has a peak at the resonance angular frequency  $\omega_0 = \sqrt{LC} = 1000$  rad/s.

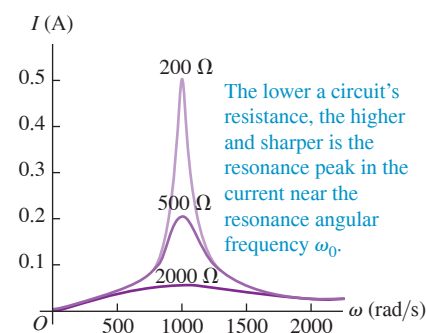
The resonance frequency is determined by  $L$  and  $C$ ; what happens when we change  $R$ ? Figure 31.19 also shows graphs of  $I$  as a function of  $\omega$  for  $R = 200$   $\Omega$  and for  $R = 2000$   $\Omega$ . The curves are similar for frequencies far away from resonance, where the impedance is dominated by  $X_L$  or  $X_C$ . But near resonance, where  $X_L$  and  $X_C$  nearly cancel each other, the curve is higher and more sharply peaked for small values of  $R$  and broader and flatter for large values of  $R$ . At resonance,  $Z = R$  and  $I = V/R$ , so the maximum height of the curve is inversely proportional to  $R$ .

The shape of the response curve is important in the design of radio receiving circuits. The sharply peaked curve is what makes it possible to discriminate between two stations broadcasting on adjacent frequency bands. But if the peak is *too* sharp, some of the information in the received signal is lost, such as the high-frequency sounds in music. The shape of the resonance curve is also related to the overdamped and underdamped oscillations that we described in Section 30.6. A sharply peaked resonance curve corresponds to a small value of  $R$  and a lightly damped oscillating system; a broad, flat curve goes with a large value of  $R$  and a heavily damped system.

In this section we have discussed resonance in an  $L$ - $R$ - $C$  series circuit. Resonance can also occur in an ac circuit in which the inductor, resistor, and capacitor are connected in *parallel*. We leave the details to you (see Problems 31.54 and 31.55).

Resonance phenomena occur not just in ac circuits, but in all areas of physics. We discussed examples of resonance in *mechanical* systems in Sections 14.8 and 16.5. The amplitude of a mechanical oscillation peaks when the driving-force frequency is close to a natural frequency of the system; this is analogous to the peaking of the current in an  $L$ - $R$ - $C$  series circuit.

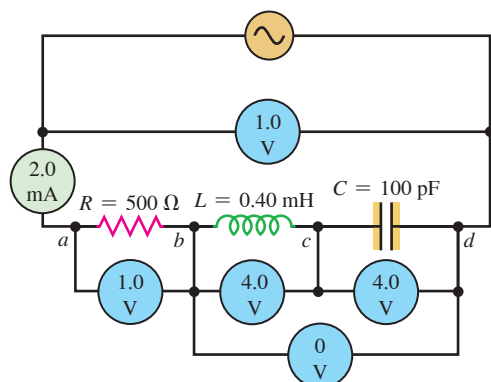
Figure 31.19 Graph of current amplitude  $I$  as a function of angular frequency  $\omega$  for an  $L$ - $R$ - $C$  series circuit with  $V = 100$  V,  $L = 2.0$  H,  $C = 0.50$   $\mu$ F, and three different values of the resistance  $R$ .



### EXAMPLE 31.8 Tuning a radio

The series circuit in **Fig. 31.20** is similar to some radio tuning circuits. It is connected to a variable-frequency ac source with an rms terminal voltage of 1.0 V. (a) Find the resonance frequency. At the resonance frequency, find (b) the inductive reactance  $X_L$ , the capacitive reactance  $X_C$ , and the impedance  $Z$ ; (c) the rms current  $I_{\text{rms}}$ ; (d) the rms voltage across each circuit element.

Figure 31.20 A radio tuning circuit at resonance. The circles denote rms current and voltages.



**IDENTIFY and SET UP** Figure 31.20 shows an  $L$ - $R$ - $C$  series circuit, with ideal meters inserted to measure the rms current and voltages, our target variables. Equation (31.32) gives the formula for the resonance angular frequency  $\omega_0$ , from which we find the resonance frequency  $f_0$ . We use Eqs. (31.12) and (31.18) to find  $X_L$  and  $X_C$ , which are equal at resonance; at resonance, from Eq. (31.23), we have  $Z = R$ . We use Eqs. (31.7), (31.13), and (31.19) to find the voltages across the circuit elements.

**EXECUTE** (a) The values of  $\omega_0$  and  $f_0$  are

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(0.40 \times 10^{-3} \text{ H})(100 \times 10^{-12} \text{ F})}} \\ &= 5.0 \times 10^6 \text{ rad/s} \\ f_0 &= 8.0 \times 10^5 \text{ Hz} = 800 \text{ kHz}\end{aligned}$$

This frequency is in the lower part of the AM radio band.

(b) At this frequency,

$$X_L = \omega L = (5.0 \times 10^6 \text{ rad/s})(0.40 \times 10^{-3} \text{ H}) = 2000 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{(5.0 \times 10^6 \text{ rad/s})(100 \times 10^{-12} \text{ F})} = 2000 \Omega$$

Since  $X_L = X_C$  at resonance as stated above,  $Z = R = 500 \Omega$ .

*Continued*



(c) From Eq. (31.26) the rms current at resonance is

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{V_{\text{rms}}}{R} = \frac{1.0 \text{ V}}{500 \Omega} = 0.0020 \text{ A} = 2.0 \text{ mA}$$

(d) The rms potential difference across the resistor is

$$V_{R-\text{rms}} = I_{\text{rms}}R = (0.0020 \text{ A})(500 \Omega) = 1.0 \text{ V}$$

The rms potential differences across the inductor and capacitor are

$$V_{L-\text{rms}} = I_{\text{rms}}X_L = (0.0020 \text{ A})(2000 \Omega) = 4.0 \text{ V}$$

$$V_{C-\text{rms}} = I_{\text{rms}}X_C = (0.0020 \text{ A})(2000 \Omega) = 4.0 \text{ V}$$

**EVALUATE** The potential differences across the inductor and the capacitor have equal rms values and amplitudes, but are  $180^\circ$  out of phase and so add to zero at each instant. Note also that at resonance,  $V_{R-\text{rms}}$  is equal to the source voltage  $V_{\text{rms}}$ , while in this example,  $V_{L-\text{rms}}$  and  $V_{C-\text{rms}}$  are both considerably *larger* than  $V_{\text{rms}}$ .

**KEYCONCEPT** In an ac circuit, the current amplitude is maximum for a given source voltage amplitude when the source frequency equals the circuit's resonance frequency. For a series  $L$ - $R$ - $C$  circuit the resonance frequency is the same as the circuit's natural oscillation frequency without the resistor or the source.

**TEST YOUR UNDERSTANDING OF SECTION 31.5** How does the resonance frequency of an  $L$ - $R$ - $C$  series circuit change if the plates of the capacitor are brought closer together? (i) It increases; (ii) it decreases; (iii) it is unaffected.

**ANSWER**

(ii) The capacitance  $C$  increases if the plate spacing is decreased (see Section 24.1). Hence the resonance frequency  $f_0 = \omega_0/2\pi = 1/2\pi\sqrt{LC}$  decreases.

## 31.6 TRANSFORMERS

**BIO APPLICATION Dangers of ac Versus dc Voltages** Alternating current at high voltage (above 500 V) is more dangerous than direct current at the same voltage. When a person touches a high-voltage dc source, it usually causes a single muscle contraction that can be strong enough to push the person away from the source. By contrast, touching a high-voltage ac source can cause a continuing muscle contraction that prevents the victim from letting go of the source. Lowering the ac voltage with a transformer reduces the risk of injury.



One advantage of ac over dc for electric-power distribution is that it is much easier to step voltage levels up and down with ac than with dc. For long-distance power transmission it is desirable to use as high a voltage and as small a current as possible; this reduces  $i^2R$  losses in the transmission lines, and smaller wires can be used, saving on material costs. Present-day transmission lines operate at rms voltages of the order of 500 kV. However, safety considerations dictate relatively low voltages in generating equipment and in household and industrial power distribution. The standard voltage for household wiring is 120 V in North and Central America, and 220 V to 240 V in most of the rest of the world. The necessary voltage conversion is accomplished by using **transformers**.

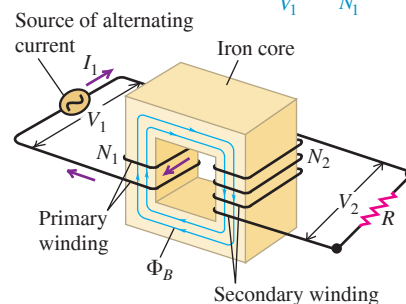
### How Transformers Work

**Figure 31.21** shows an idealized transformer. Its key components are two coils or *windings*, electrically insulated from each other but wound on the same core. The core is typically made of a material, such as iron, with a very large relative permeability  $K_m$ . This keeps the magnetic field lines due to a current in one winding almost completely within the core. Hence almost all of these field lines pass through the other winding, maximizing the *mutual inductance* of the two windings (see Section 30.1). The winding to which power is supplied is called the **primary**; the winding from which power is delivered is called the **secondary**. The circuit symbol for a transformer with an iron core is



**Figure 31.21** Schematic diagram of an idealized step-up transformer. The primary is connected to an ac source; the secondary is connected to a device with resistance  $R$ .

The induced emf *per turn* is the same in both coils, so we adjust the ratio of terminal voltages by adjusting the ratio of turns:  $\frac{V_2}{V_1} = \frac{N_2}{N_1}$



Here's how a transformer works. The ac source causes an alternating current in the primary, which sets up an alternating flux in the core; this induces an emf in each winding, in accordance with Faraday's law. The induced emf in the secondary gives rise to an alternating current in the secondary, and this delivers energy to the device to which the secondary is connected. All currents and emfs have the same frequency as the ac source.

Let's see how the voltage across the secondary can be made larger or smaller in amplitude than the voltage across the primary. We ignore the resistance of the windings and assume that all the magnetic field lines are confined to the iron core, so at any instant the magnetic flux  $\Phi_B$  is the same in each turn of the primary and secondary windings. The primary winding has  $N_1$  turns and the secondary winding has  $N_2$  turns. When the magnetic flux changes because of changing currents in the two coils, the resulting induced emfs are

$$\mathcal{E}_1 = -N_1 \frac{d\Phi_B}{dt} \quad \text{and} \quad \mathcal{E}_2 = -N_2 \frac{d\Phi_B}{dt} \quad (31.33)$$

The flux *per turn*  $\Phi_B$  is the same in both the primary and the secondary, so Eqs. (31.33) show that the induced emf *per turn* is the same in each. The ratio of the secondary emf  $\mathcal{E}_2$  to the primary emf  $\mathcal{E}_1$  is therefore equal at any instant to the ratio of secondary to primary turns:

$$\frac{\mathcal{E}_2}{\mathcal{E}_1} = \frac{N_2}{N_1} \quad (31.34)$$

Since  $\mathcal{E}_1$  and  $\mathcal{E}_2$  both oscillate with the same frequency as the ac source, Eq. (31.34) also gives the ratio of the amplitudes or of the rms values of the induced emfs. If the windings have zero resistance, the induced emfs  $\mathcal{E}_1$  and  $\mathcal{E}_2$  are equal to the terminal voltages across the primary and the secondary, respectively; hence

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} \quad (31.35)$$

Secondary voltage amplitude or rms value  $\rightarrow V_2$   
 Primary voltage amplitude or rms value  $\rightarrow V_1$   
 Number of turns in secondary  $\rightarrow N_2$   
 Number of turns in primary  $\rightarrow N_1$

Terminal voltages in a transformer:

By choosing the appropriate turns ratio  $N_2/N_1$ , we may obtain any desired secondary voltage from a given primary voltage. If  $N_2 > N_1$ , as in Fig. 31.21, then  $V_2 > V_1$  and we have a *step-up* transformer; if  $N_2 < N_1$ , then  $V_2 < V_1$  and we have a *step-down* transformer. At a power-generating station, step-up transformers are used; the primary is connected to the power source and the secondary is connected to the transmission lines, giving the desired high voltage for transmission. Near the consumer, step-down transformers lower the voltage to a value suitable for use in home or industry (**Fig. 31.22**).

Even the relatively low voltage provided by a household wall socket is too high for many electronic devices, so a further step-down transformer is necessary. This is the role of an “ac adapter” such as those used to recharge a mobile phone or laptop computer from line voltage (**Fig. 31.23**).

## Energy Considerations for Transformers

If the secondary circuit is completed by a resistance  $R$ , then the amplitude or rms value of the current in the secondary circuit is  $I_2 = V_2/R$ . From energy considerations, the power delivered to the primary equals that taken out of the secondary (since there is no resistance in the windings), so

$$V_1 I_1 = V_2 I_2 \quad (31.36)$$

Terminal voltages and currents in a transformer:

Primary voltage amplitude or rms value  $\rightarrow V_1$   
 Secondary voltage amplitude or rms value  $\rightarrow V_2$   
 Current in primary  $\rightarrow I_1$   
 Current in secondary  $\rightarrow I_2$

Figure 31.22 The cylindrical can near the top of this power pole is a step-down transformer. It converts the high-voltage ac in the power lines to low-voltage (120 V) ac, which is then distributed to the surrounding homes and businesses.



Figure 31.23 An ac adapter like this one converts household ac into low-voltage dc for use in electronic devices. It contains a step-down transformer to change the line voltage to a lower value, typically 3 to 12 V, as well as diodes to convert alternating current to the direct current that small electronic devices require (see Fig. 31.3).



**APPLICATION When dc Power Transmission Is Better Than ac**

In the cables used for transmitting ac power, eddy currents oppose and cancel the current near the central axis of the cable. Hence current flows only in a reduced area near the cable's surface, increasing the cable's effective resistance and causing the loss of more power. For this and other reasons, high-voltage dc can be more efficient than high-voltage ac for transmitting power over long distances, such as in a planned application to take solar power generated in North Africa to Europe through dc lines at the bottom of the Mediterranean.



We can combine Eqs. (31.35) and (31.36) and the relationship  $I_2 = V_2/R$  to eliminate  $V_2$  and  $I_2$ ; we obtain

$$\frac{V_1}{I_1} = \frac{R}{(N_2/N_1)^2} \quad (31.37)$$

This shows that when the secondary circuit is completed through a resistance  $R$ , the result is the same as if the *source* had been connected directly to a resistance equal to  $R$  divided by the square of the turns ratio,  $(N_2/N_1)^2$ . In other words, the transformer “transforms” not only voltages and currents, but resistances as well. More generally, we can regard a transformer as “transforming” the *impedance* of the network to which the secondary circuit is completed.

Equation (31.37) has many practical consequences. The power supplied by a source to a resistor depends on the resistances of both the resistor and the source. It can be shown that the power transfer is greatest when the two resistances are *equal*. The same principle applies in both dc and ac circuits. When a high-impedance ac source must be connected to a low-impedance circuit, such as an audio amplifier connected to a loudspeaker, the source impedance can be *matched* to that of the circuit by the use of a transformer with an appropriate turns ratio  $N_2/N_1$ .

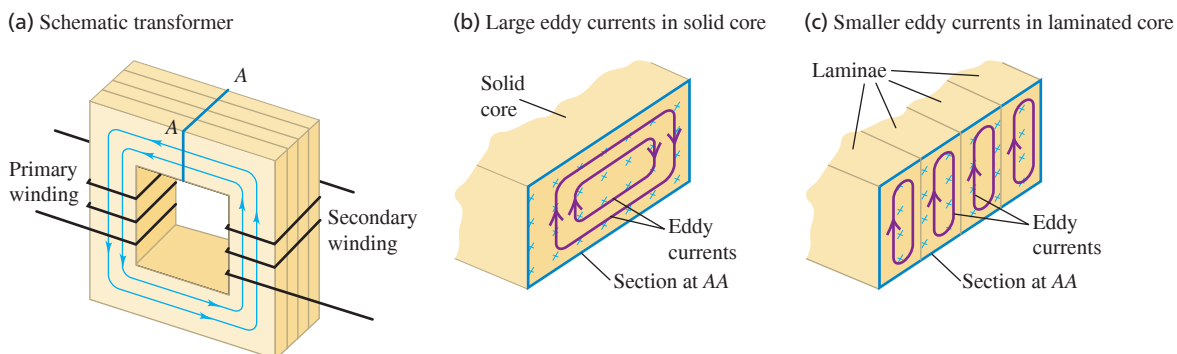
Real transformers always have some energy losses. (That’s why an ac adapter like the one shown in Fig. 31.23 feels warm to the touch after it’s been in use for a while; the transformer is heated by the dissipated energy.) The windings have some resistance, leading to  $i^2R$  losses. There are also energy losses through hysteresis in the core (see Section 28.8). Hysteresis losses are minimized by the use of soft iron with a narrow hysteresis loop.

Eddy currents (Section 29.6) also cause energy loss in transformers. Consider a section AA through an iron transformer core (**Fig. 31.24a**). Since iron is a conductor, any such section can be pictured as several conducting circuits, one within the other (Fig. 31.24b). The flux through each of these circuits is continually changing, so eddy currents circulate in the entire volume of the core, with lines of flow that form planes perpendicular to the flux. These eddy currents waste energy through  $i^2R$  heating and themselves set up an opposing flux.

The effects of eddy currents can be minimized by the use of a *laminated* core—that is, one built up of thin sheets, or laminae. The large electrical surface resistance of each lamina, due either to a natural coating of oxide or to an insulating varnish, effectively confines the eddy currents to individual laminae (Fig. 31.24c). The possible eddy-current paths are narrower, the induced emf in each path is smaller, and the eddy currents are greatly reduced. The alternating magnetic field exerts forces on the current-carrying laminae that cause them to vibrate back and forth; this vibration causes the characteristic “hum” of an operating transformer. You can hear this same “hum” from the magnetic ballast of a fluorescent light fixture (see Section 30.2).

Thanks to the use of soft iron cores and lamination, transformer efficiencies are usually well over 90%; in large installations they may reach 99%.

Figure 31.24 (a) Primary and secondary windings in a transformer. (b) Eddy currents in the iron core, shown in the cross section at AA. (c) Using a laminated core reduces the eddy currents.



**EXAMPLE 31.9 “Wake up and smell the (transformer)!”**

A friend returns to the United States from Europe with a 960 W coffeemaker, designed to operate from a 240 V line. (a) What can she do to operate it at the USA-standard 120 V? (b) What current will the coffeemaker draw from the 120 V line? (c) What is the resistance of the coffeemaker? (The voltages are rms values.)

**IDENTIFY and SET UP** Our friend needs a step-up transformer to convert 120 V ac to the 240 V ac that the coffeemaker requires. We use Eq. (31.35) to determine the transformer turns ratio  $N_2/N_1$ ,  $P_{\text{av}} = V_{\text{rms}}I_{\text{rms}}$  for a resistor to find the current draw, and Eq. (31.37) to calculate the resistance.

**EXECUTE** (a) To get  $V_2 = 240$  V from  $V_1 = 120$  V, the required turns ratio is  $N_2/N_1 = V_2/V_1 = (240 \text{ V})/(120 \text{ V}) = 2$ . That is, the secondary coil (connected to the coffeemaker) should have twice as many turns as the primary coil (connected to the 120 V line).

(b) We find the rms current  $I_1$  in the 120 V primary by using  $P_{\text{av}} = V_1 I_1$ , where  $P_{\text{av}}$  is the average power drawn by the coffeemaker and hence the power supplied by the 120 V line. (We’re assuming that no energy is lost in the transformer.) Hence  $I_1 = P_{\text{av}}/V_1 = (960 \text{ W})/(120 \text{ V}) = 8.0$  A. The secondary current is then  $I_2 = P_{\text{av}}/V_2 = (960 \text{ W})/(240 \text{ V}) = 4.0$  A.

(c) We have  $V_1 = 120$  V,  $I_1 = 8.0$  A, and  $N_2/N_1 = 2$ , so

$$\frac{V_1}{I_1} = \frac{120 \text{ V}}{8.0 \text{ A}} = 15 \Omega$$

From Eq. (31.37),

$$R = 2^2(15 \Omega) = 60 \Omega$$

**EVALUATE** As a check,  $V_2/R = (240 \text{ V})/(60 \Omega) = 4.0 \text{ A} = I_2$ , the same value obtained previously. You can also check this result for  $R$  by using the expression  $P_{\text{av}} = V_2^2/R$  for the power drawn by the coffeemaker.

**KEYCONCEPT** A transformer can be used to raise or lower the voltage amplitude of an alternating current. If there are more turns in the secondary coil than in the primary coil (step-up transformer), the transformer increases the voltage amplitude and decreases the current amplitude; the reverse is true if there are fewer turns in the secondary than in the primary (step-down transformer).

**TEST YOUR UNDERSTANDING OF SECTION 31.6** Each of the following four transformers has 1000 turns in its primary. Rank the transformers from largest to smallest number of turns in the secondary. (i) Converts 120 V ac into 6.0 V ac; (ii) converts 120 V ac into 240 V ac; (iii) converts 240 V ac into 6.0 V ac; (iv) converts 240 V ac into 120 V ac.

**ANSWER**

(ii), (iv), (i), (iii) From Eq. (31.35) the turns ratio is  $N_2/N_1 = V_2/V_1$ , so the number of turns in the secondary is  $N_2 = N_1 V_2/V_1$ . Hence for the four cases we have (i)  $N_2 = (1000)(6.0 \text{ V})/(120 \text{ V}) = 50$  turns; (ii)  $N_2 = (1000)(240 \text{ V})/(120 \text{ V}) = 2000$  turns; (iii)  $N_2 = (1000)(6.0 \text{ V})/(240 \text{ V}) = 25$  turns; and (iv)  $N_2 = (1000)(120 \text{ V})/(240 \text{ V}) = 500$  turns. Note that (i), (iii), and (iv) are step-down transformers with fewer turns in the secondary than in the primary, while (ii) is a step-up transformer with more turns in the secondary than in the primary.



## CHAPTER 31 SUMMARY

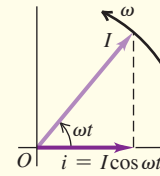
**Phasors and alternating current:** An alternator or ac source produces an emf that varies sinusoidally with time. A sinusoidal voltage or current can be represented by a phasor, a vector that rotates counterclockwise with constant angular velocity  $\omega$  equal to the angular frequency of the sinusoidal quantity. Its projection on the horizontal axis at any instant represents the instantaneous value of the quantity.

For a sinusoidal current, the rectified average and rms (root-mean-square) currents are proportional to the current amplitude  $I$ . Similarly, the rms value of a sinusoidal voltage is proportional to the voltage amplitude  $V$ . (See Example 31.1.)

$$I_{\text{rav}} = \frac{2}{\pi} I = 0.637 I \quad (31.3)$$

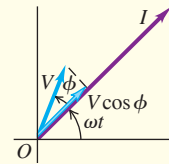
$$I_{\text{rms}} = \frac{I}{\sqrt{2}} \quad (31.4)$$

$$V_{\text{rms}} = \frac{V}{\sqrt{2}} \quad (31.5)$$



**Voltage, current, and phase angle:** In general, the instantaneous voltage  $v = V \cos(\omega t + \phi)$  between two points in an ac circuit is not in phase with the instantaneous current passing through those points. The quantity  $\phi$  is called the phase angle of the voltage relative to the current.

$$i = I \cos \omega t \quad (31.2)$$

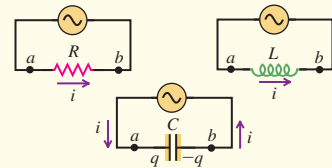


**Resistance and reactance:** The voltage across a resistor  $R$  is in phase with the current. The voltage across an inductor  $L$  leads the current by  $90^\circ$  ( $\phi = +90^\circ$ ), while the voltage across a capacitor  $C$  lags the current by  $90^\circ$  ( $\phi = -90^\circ$ ). The voltage amplitude across each type of device is proportional to the current amplitude  $I$ . An inductor has inductive reactance  $X_L = \omega L$ , and a capacitor has capacitive reactance  $X_C = 1/\omega C$ . (See Examples 31.2 and 31.3.)

$$V_R = IR \quad (31.7)$$

$$V_L = IX_L \quad (31.13)$$

$$V_C = IX_C \quad (31.19)$$

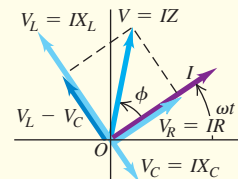


**Impedance and the L-R-C series circuit:** In a general ac circuit, the voltage and current amplitudes are related by the circuit impedance  $Z$ . In an L-R-C series circuit, the values of  $L$ ,  $R$ ,  $C$ , and the angular frequency  $\omega$  determine the impedance and the phase angle  $\phi$  of the voltage relative to the current. (See Examples 31.4 and 31.5.)

$$V = IZ \quad (31.22)$$

$$Z = \sqrt{R^2 + [\omega L - (1/\omega C)]^2} \quad (31.23)$$

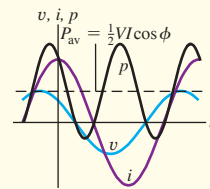
$$\tan \phi = \frac{\omega L - 1/\omega C}{R} \quad (31.24)$$



**Power in ac circuits:** The average power input  $P_{\text{av}}$  to an ac circuit depends on the voltage and current amplitudes (or, equivalently, their rms values) and the phase angle  $\phi$  of the voltage relative to the current. The quantity  $\cos \phi$  is called the power factor. (See Examples 31.6 and 31.7.)

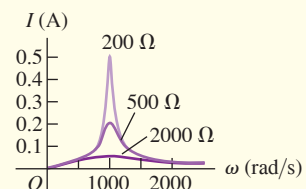
$$P_{\text{av}} = \frac{1}{2} VI \cos \phi \quad (31.31)$$

$$= V_{\text{rms}} I_{\text{rms}} \cos \phi$$



**Resonance in ac circuits:** In an L-R-C series circuit, the current becomes maximum and the impedance becomes minimum at an angular frequency called the resonance angular frequency. This phenomenon is called resonance. At resonance the voltage and current are in phase, and the impedance  $Z$  is equal to the resistance  $R$ . (See Example 31.8.)

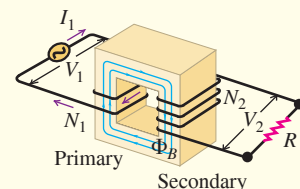
$$\omega_0 = \frac{1}{\sqrt{LC}} \quad (31.32)$$



**Transformers:** A transformer is used to transform the voltage and current levels in an ac circuit. In an ideal transformer with no energy losses, if the primary winding has  $N_1$  turns and the secondary winding has  $N_2$  turns, the amplitudes (or rms values) of the two voltages are related by Eq. (31.35). The amplitudes (or rms values) of the primary and secondary voltages and currents are related by Eq. (31.36). (See Example 31.9.)

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} \quad (31.35)$$

$$V_1 I_1 = V_2 I_2 \quad (31.36)$$





## GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

### KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 31.2 and 31.3** (Section 31.2) before attempting these problems.

**VP31.3.1** An inductor with  $L = 2.50$  mH is part of the circuit for an aviation radio beacon that operates at 108 MHz. Find (a) the inductive reactance of the inductor and (b) the amplitude of the current in the inductor if the amplitude of the voltage across the inductor is 4.20 kV.

**VP31.3.2** An ac circuit includes a resistor with  $R = 125$   $\Omega$  connected in series with a capacitor with  $C = 7.00$   $\mu$ F. The current through the resistor and capacitor is  $i = (2.40 \times 10^{-3} \text{ A})\cos[(1.75 \times 10^3 \text{ rad/s})t]$ . Find (a) the voltage across the resistor as a function of time, (b) the capacitive reactance of the capacitor, and (c) the voltage across the capacitor as a function of time.

**VP31.3.3** An ac circuit includes a 155  $\Omega$  resistor in series with a 8.00  $\mu$ F capacitor. The current in the circuit has amplitude  $4.00 \times 10^{-3}$  A. (a) Find the frequency for which the capacitive reactance equals the resistance. (b) At this frequency, what are the amplitudes of the voltages across the resistor and capacitor? (c) At this frequency, when the voltage across the resistor is maximum, what is the voltage across the capacitor? When the voltage across the capacitor is maximum, what is the voltage across the resistor?

**VP31.3.4** A 116  $\Omega$  resistor and an unknown capacitor are connected in series to an ac source with angular frequency  $5.10 \times 10^3$  rad/s. The amplitude of the resistor voltage is 2.45 V, and the current in the circuit has its maximum positive value at  $t = 0$ . Find (a) the current amplitude, (b) the current at  $t = 3.50$  ms, (c) the maximum voltage across the capacitor, and (d) the capacitance.

Be sure to review **EXAMPLES 31.4 and 31.5** (Section 31.3) before attempting these problems.

**VP31.5.1** In the series circuit of Fig. 31.13a the resistance is 255  $\Omega$ , the inductance is 45.0 mH, the capacitance is 0.400  $\mu$ F, and the source has voltage amplitude 55.0 V. The inductive reactance is equal to the resistance. Find (a) the frequency of the source, (b) the impedance, (c) the current amplitude, (d) the phase angle, and (e) whether the voltage leads or lags the current.

**VP31.5.2** In the series circuit of Fig. 31.13a the source has voltage amplitude 20.0 V and angular frequency  $5.40 \times 10^3$  rad/s, the inductance is 6.50 mH, the capacitance is 0.600  $\mu$ F, and the impedance is 474  $\Omega$ . Find (a) the resistance, (b) the current amplitude, (c) the resistor voltage amplitude, (d) the inductor voltage amplitude, and (e) the capacitor voltage amplitude.

**VP31.5.3** The source in an  $L$ - $R$ - $C$  series circuit has angular frequency  $1.20 \times 10^4$  rad/s. The capacitance is 0.965  $\mu$ F and the resistance is 65.0  $\Omega$ . If you want the source voltage to lead the current by  $15.0^\circ$ , find (a) the required value of  $X_L - X_C$  and (b) the required inductance.

**VP31.5.4** The current in an  $L$ - $R$ - $C$  series circuit has amplitude 0.120 A and angular frequency  $8.00 \times 10^3$  rad/s, and it has its maximum positive value at  $t = 0$ . The resistance is 95.0  $\Omega$ , the inductance is 6.50 mH, and the capacitance is 0.440  $\mu$ F. For the resistor, inductor, and capacitor, find (a) the voltage amplitudes and (b) the instantaneous voltages at  $t = 0.305$  ms.

Be sure to review **EXAMPLES 31.6 and 31.7** (Section 31.4) before attempting these problems.

**VP31.7.1** In operation an electric toaster has an rms current of 3.95 A when plugged into a wall socket with rms voltage  $1.20 \times 10^2$  V. Find the toaster's (a) average power, (b) maximum instantaneous power, and (c) resistance.

**VP31.7.2** An  $L$ - $R$ - $C$  series circuit has a source with voltage amplitude 35.0 V and angular frequency  $1.30 \times 10^3$  rad/s. The resistance is 275  $\Omega$ , the inductance is 82.3 mH, and the capacitance is 1.10  $\mu$ F. Find (a) the inductive and capacitive reactances, (b) the phase angle, and (c) the power factor.

**VP31.7.3** For the circuit in the preceding problem, find (a) the impedance, (b) the current amplitude, and (c) the average power delivered to the resistor.

**VP31.7.4** The power factor of an  $L$ - $R$ - $C$  series circuit is 0.800. The angular frequency is  $1.30 \times 10^4$  rad/s, the inductance is 7.50 mH, and the capacitance is 0.440  $\mu$ F. Find (a) the inductive and capacitive reactances, (b) the phase angle, and (c) the resistance.

### BRIDGING PROBLEM An Alternating-Current Circuit

A series circuit like the circuit in Fig. 31.13a consists of a 1.50 mH inductor, a 125  $\Omega$  resistor, and a 25.0 nF capacitor connected across an ac source having an rms voltage of 35.0 V and variable frequency. (a) At what angular frequencies will the current amplitude be equal to  $\frac{1}{3}$  of its maximum possible value? (b) At the frequencies in part (a), what are the current amplitude and the voltage amplitude across each circuit element (including the ac source)?

#### SOLUTION GUIDE

#### IDENTIFY and SET UP

1. The maximum current amplitude occurs at the resonance angular frequency. This problem concerns the angular frequencies at which the current amplitude is one-third of that maximum.
2. Choose the equation that will allow you to find the angular frequencies in question, and choose the equations that you'll

then use to find the current and voltage amplitudes at each angular frequency.

#### EXECUTE

3. Find the impedance at the angular frequencies in part (a); then solve for the values of angular frequency.
4. Find the voltage amplitude across the source and the current amplitude for each of the angular frequencies in part (a). (*Hint:* Be careful to distinguish between *amplitude* and *rms value*.)
5. Use the results of steps 3 and 4 to find the reactances at each angular frequency. Then calculate the voltage amplitudes for the resistor, inductor, and capacitor.

#### EVALUATE

6. Are any voltage amplitudes greater than the voltage amplitude of the source? If so, does this mean your results are in error?

## PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

### DISCUSSION QUESTIONS

**Q31.1** Household electric power in most of western Europe is supplied at 240 V, rather than the 120 V that is standard in the United States and Canada. What are the advantages and disadvantages of each system?

**Q31.2** The current in an ac power line changes direction 120 times per second, and its average value is zero. Explain how it is possible for power to be transmitted in such a system.

**Q31.3** In an ac circuit, why is the average power for an inductor and a capacitor zero, but not for a resistor?

**Q31.4** Equation (31.14) was derived by using the relationship  $i = dq/dt$  between the current and the charge on the capacitor. In Fig. 31.9a the positive counterclockwise current increases the charge on the capacitor. When the charge on the left plate is positive but decreasing in time, is  $i = dq/dt$  still correct or should it be  $i = -dq/dt$ ? Is  $i = dq/dt$  still correct when the right-hand plate has positive charge that is increasing or decreasing in magnitude? Explain.

**Q31.5** Fluorescent lights often use an inductor, called a ballast, to limit the current through the tubes. Why is it better to use an inductor rather than a resistor for this purpose?

**Q31.6** Equation (31.9) says that  $v_{ab} = L di/dt$  (see Fig. 31.8a). Using Faraday's law, explain why point  $a$  is at higher potential than point  $b$  when  $i$  is in the direction shown in Fig. 31.8a and is increasing in magnitude. When  $i$  is counterclockwise and decreasing in magnitude, is  $v_{ab} = L di/dt$  still correct, or should it be  $v_{ab} = -L di/dt$ ? Is  $v_{ab} = L di/dt$  still correct when  $i$  is clockwise and increasing or decreasing in magnitude? Explain.

**Q31.7** Is it possible for the power factor of an  $L$ - $R$ - $C$  series ac circuit to be zero? Justify your answer on *physical* grounds.

**Q31.8** In an  $L$ - $R$ - $C$  series circuit, can the instantaneous voltage across the capacitor exceed the source voltage at that same instant? Can this be true for the instantaneous voltage across the inductor? Across the resistor? Explain.

**Q31.9** In an  $L$ - $R$ - $C$  series circuit, what are the phase angle  $\phi$  and power factor  $\cos \phi$  when the resistance is much smaller than the inductive or capacitive reactance and the circuit is operated far from resonance? Explain.

**Q31.10** When an  $L$ - $R$ - $C$  series circuit is connected across a 120 V ac line, the voltage rating of the capacitor may be exceeded even if it is rated at 200 or 400 V. How can this be?

**Q31.11** In Example 31.6 (Section 31.4), a hair dryer is treated as a pure resistor. But because there are coils in the heating element and in the motor that drives the blower fan, a hair dryer also has inductance. Qualitatively, does including an inductance increase or decrease the values of  $R$ ,  $I_{\text{rms}}$ , and  $P$ ?

**Q31.12** A light bulb and a parallel-plate capacitor with air between the plates are connected in series to an ac source. What happens to the brightness of the bulb when a dielectric is inserted between the plates of the capacitor? Explain.

**Q31.13** A coil of wire wrapped on a hollow tube and a light bulb are connected in series to an ac source. What happens to the brightness of the bulb when an iron rod is inserted in the tube?

**Q31.14** A circuit consists of a light bulb, a capacitor, and an inductor connected in series to an ac source. What happens to the brightness of the bulb when the inductor is omitted? When the inductor is left in the circuit but the capacitor is omitted? Explain.

**Q31.15** A circuit consists of a light bulb, a capacitor, and an inductor connected in series to an ac source. Is it possible for both the capacitor and the inductor to be removed and the brightness of the bulb to remain the same? Explain.

**Q31.16** Can a transformer be used with dc? Explain. What happens if a transformer designed for 120 V ac is connected to a 120 V dc line?

**Q31.17** An ideal transformer has  $N_1$  windings in the primary and  $N_2$  windings in its secondary. If you double only the number of secondary windings, by what factor does (a) the voltage amplitude in the secondary change, and (b) the effective resistance of the secondary circuit change?

**Q31.18** An inductor, a capacitor, and a resistor are all connected in series across an ac source. If the resistance, inductance, and capacitance are all doubled, by what factor does each of the following quantities change? Indicate whether they increase or decrease: (a) the resonance angular frequency; (b) the inductive reactance; (c) the capacitive reactance. (d) Does the impedance double?

**Q31.19** You want to double the resonance angular frequency of an  $L$ - $R$ - $C$  series circuit by changing only the *pertinent* circuit elements all by the same factor. (a) Which ones should you change? (b) By what factor should you change them?

### EXERCISES

#### Section 31.1 Phasors and Alternating Currents

**31.1 •** You have a special light bulb with a *very* delicate wire filament. The wire will break if the current in it ever exceeds 1.50 A, even for an instant. What is the largest root-mean-square current you can run through this bulb?

**31.2 •** The voltage across the terminals of an ac power supply varies with time according to Eq. (31.1). The voltage amplitude is  $V = 45.0$  V. What are (a) the root-mean-square potential difference  $V_{\text{rms}}$  and (b) the average potential difference  $V_{\text{av}}$  between the two terminals of the power supply?

#### Section 31.2 Resistance and Reactance

**31.3 •** An inductor with  $L = 9.50$  mH is connected across an ac source that has voltage amplitude 45.0 V. (a) What is the phase angle  $\phi$  for the source voltage relative to the current? Does the source voltage lag or lead the current? (b) What value for the frequency of the source results in a current amplitude of 3.90 A?

**31.4 •** A capacitor is connected across an ac source that has voltage amplitude 60.0 V and frequency 80.0 Hz. (a) What is the phase angle  $\phi$  for the source voltage relative to the current? Does the source voltage lag or lead the current? (b) What is the capacitance  $C$  of the capacitor if the current amplitude is 5.30 A?

**31.5 •** (a) What is the reactance of a 3.00 H inductor at a frequency of 80.0 Hz? (b) What is the inductance of an inductor whose reactance is 120  $\Omega$  at 80.0 Hz? (c) What is the reactance of a 4.00  $\mu\text{F}$  capacitor at a frequency of 80.0 Hz? (d) What is the capacitance of a capacitor whose reactance is 120  $\Omega$  at 80.0 Hz?

**31.6 •** A capacitance  $C$  and an inductance  $L$  are operated at the same angular frequency. (a) At what angular frequency will they have the same reactance? (b) If  $L = 5.00$  mH and  $C = 3.50$   $\mu\text{F}$ , what is the numerical value of the angular frequency in part (a), and what is the reactance of each element?

**31.7 •** (a) Compute the reactance of a 0.450 H inductor at frequencies of 60.0 Hz and 600 Hz. (b) Compute the reactance of a 2.50  $\mu\text{F}$  capacitor at the same frequencies. (c) At what frequency is the reactance of a 0.450 H inductor equal to that of a 2.50  $\mu\text{F}$  capacitor?

**31.8 • A Radio Inductor.** You want the current amplitude through a 0.450 mH inductor (part of the circuitry for a radio receiver) to be 1.80 mA when a sinusoidal voltage with amplitude 12.0 V is applied across the inductor. What frequency is required?

**31.9** •• A 0.180 H inductor is connected in series with a 90.0  $\Omega$  resistor and an ac source. The voltage across the inductor is  $v_L = -(12.0 \text{ V})\sin[(480 \text{ rad/s})t]$ . (a) Derive an expression for the voltage  $v_R$  across the resistor. (b) What is  $v_R$  at  $t = 2.00 \text{ ms}$ ?

**31.10** •• A 250  $\Omega$  resistor is connected in series with a 4.80  $\mu\text{F}$  capacitor and an ac source. The voltage across the capacitor is  $v_C = (7.60 \text{ V})\sin[(120 \text{ rad/s})t]$ . (a) Determine the capacitive reactance of the capacitor. (b) Derive an expression for the voltage  $v_R$  across the resistor.

**31.11** •• A 150  $\Omega$  resistor is connected in series with a 0.250 H inductor and an ac source. The voltage across the resistor is  $v_R = (3.80 \text{ V})\cos[(720 \text{ rad/s})t]$ . (a) Derive an expression for the circuit current. (b) Determine the inductive reactance of the inductor. (c) Derive an expression for the voltage  $v_L$  across the inductor.

### Section 31.3 The $L$ - $R$ - $C$ Series Circuit

**31.12** • You have a 200  $\Omega$  resistor, a 0.400 H inductor, and a 6.00  $\mu\text{F}$  capacitor. Suppose you take the resistor and inductor and make a series circuit with a voltage source that has voltage amplitude 30.0 V and an angular frequency of 250 rad/s. (a) What is the impedance of the circuit? (b) What is the current amplitude? (c) What are the voltage amplitudes across the resistor and across the inductor? (d) What is the phase angle  $\phi$  of the source voltage with respect to the current? Does the source voltage lag or lead the current? (e) Construct the phasor diagram.

**31.13** • The resistor, inductor, capacitor, and voltage source described in Exercise 31.12 are connected to form an  $L$ - $R$ - $C$  series circuit. (a) What is the impedance of the circuit? (b) What is the current amplitude? (c) What is the phase angle of the source voltage with respect to the current? Does the source voltage lag or lead the current? (d) What are the voltage amplitudes across the resistor, inductor, and capacitor? (e) Explain how it is possible for the voltage amplitude across the capacitor to be greater than the voltage amplitude across the source.

**31.14** •• A 200  $\Omega$  resistor, 0.900 H inductor, and 6.00  $\mu\text{F}$  capacitor are connected in series across a voltage source that has voltage amplitude 30.0 V and an angular frequency of 250 rad/s. (a) What are  $v$ ,  $v_R$ ,  $v_L$ , and  $v_C$  at  $t = 20.0 \text{ ms}$ ? Compare  $v_R + v_L + v_C$  to  $v$  at this instant. (b) What are  $V_R$ ,  $V_L$ , and  $V_C$ ? Compare  $V$  to  $V_R + V_L + V_C$ . Explain why these two quantities are not equal.

**31.15** • In an  $L$ - $R$ - $C$  series circuit, the rms voltage across the resistor is 30.0 V, across the capacitor it is 90.0 V, and across the inductor it is 50.0 V. What is the rms voltage of the source?

**31.16** •• An  $L$ - $R$ - $C$  series circuit has voltage amplitudes  $V_L = 180 \text{ V}$ ,  $V_C = 120 \text{ V}$ , and  $V_R = 160 \text{ V}$ . At time  $t$  the instantaneous voltage across the inductor is 80.0 V. At this instant, what is the voltage across the capacitor and across the resistor?

**31.17** • An  $L$ - $R$ - $C$  series circuit has source voltage amplitude  $V = 240 \text{ V}$ , and the voltage amplitudes for the inductor and capacitor are  $V_L = 310 \text{ V}$  and  $V_C = 180 \text{ V}$ . What is the phase angle  $\phi$ ?

### Section 31.4 Power in Alternating-Current Circuits

**31.18** •• A resistor with  $R = 300 \Omega$  and an inductor are connected in series across an ac source that has voltage amplitude 500 V. The rate at which electrical energy is dissipated in the resistor is 286 W. What is (a) the impedance  $Z$  of the circuit; (b) the amplitude of the voltage across the inductor; (c) the power factor?

**31.19** • The power of a certain CD player operating at 120 V rms is 20.0 W. Assuming that the CD player behaves like a pure resistor, find (a) the maximum instantaneous power; (b) the rms current; (c) the resistance of this player.

**31.20** •• In an  $L$ - $R$ - $C$  series circuit, the components have the following values:  $L = 20.0 \text{ mH}$ ,  $C = 140 \text{ nF}$ , and  $R = 350 \Omega$ . The generator has an rms voltage of 120 V and a frequency of 1.25 kHz. Determine (a) the power supplied by the generator and (b) the power dissipated in the resistor.

**31.21** •• (a) Use Figs. 31.13b and 31.13c to show that for a series  $L$ - $R$ - $C$  circuit,  $V \cos \phi = IR$ . Use this result in Eq. (31.31) to show that the average power delivered by the source is  $P_{\text{av}} = \frac{1}{2} I^2 R$ . Therefore the average power delivered by the source is equal to the average power consumed by the resistor. This means the average power for the capacitor and for the inductor is zero. These circuit elements consume electrical energy during the part of the current oscillations when they are storing energy, but all the stored energy in each is then released during another part of the current oscillations. (b) In an  $L$ - $R$ - $C$  series circuit the amplitude of the source voltage is 120 V, the source voltage leads the current by  $53.1^\circ$ , and the average power supplied by the source is 80.0 W. What is the resistance  $R$  of the resistor in the circuit?

**31.22** •• A circuit has an ac voltage source and a resistor and capacitor connected in series. There is no inductor. The ac voltage source has voltage amplitude 900 V and angular frequency  $\omega = 20.0 \text{ rad/s}$ . The voltage amplitude across the capacitor is 500 V. The resistor has resistance  $R = 300 \Omega$ . (a) What is the voltage amplitude across the resistor? (b) What is the capacitance  $C$  of the capacitor? (c) Does the source voltage lag or lead the current? (d) What is the average rate at which the ac source supplies electrical energy to the circuit?

**31.23** • An  $L$ - $R$ - $C$  series circuit with  $L = 0.120 \text{ H}$ ,  $R = 240 \Omega$ , and  $C = 7.30 \mu\text{F}$  carries an rms current of 0.450 A with a frequency of 400 Hz. (a) What are the phase angle and power factor for this circuit? (b) What is the impedance of the circuit? (c) What is the rms voltage of the source? (d) What average power is delivered by the source? (e) What is the average rate at which electrical energy is converted to thermal energy in the resistor? (f) What is the average rate at which electrical energy is dissipated (converted to other forms) in the capacitor? (g) In the inductor?

**31.24** •• An  $L$ - $R$ - $C$  series circuit is connected to a 120 Hz ac source that has  $V_{\text{rms}} = 80.0 \text{ V}$ . The circuit has a resistance of 75.0  $\Omega$  and an impedance at this frequency of 105  $\Omega$ . What average power is delivered to the circuit by the source?

**31.25** •• A series ac circuit contains a 250  $\Omega$  resistor, a 15 mH inductor, a 3.5  $\mu\text{F}$  capacitor, and an ac power source of voltage amplitude 45 V operating at an angular frequency of 360 rad/s. (a) What is the power factor of this circuit? (b) Find the average power delivered to the entire circuit. (c) What is the average power delivered to the resistor, to the capacitor, and to the inductor?

### Section 31.5 Resonance in Alternating-Current Circuits

**31.26** •• In an  $L$ - $R$ - $C$  series circuit the source is operated at its resonant angular frequency. At this frequency, the reactance  $X_C$  of the capacitor is 200  $\Omega$  and the voltage amplitude across the capacitor is 600 V. The circuit has  $R = 300 \Omega$ . What is the voltage amplitude of the source?

**31.27** • **Analyzing an  $L$ - $R$ - $C$  Circuit.** You have a 200  $\Omega$  resistor, a 0.400 H inductor, a 5.00  $\mu\text{F}$  capacitor, and a variable-frequency ac source with an amplitude of 3.00 V. You connect all four elements together to form a series circuit. (a) At what frequency will the current in the circuit be greatest? What will be the current amplitude at this frequency? (b) What will be the current amplitude at an angular frequency of 400 rad/s? At this frequency, will the source voltage lead or lag the current?

**31.28** • An  $L$ - $R$ - $C$  series circuit is constructed using a 175  $\Omega$  resistor, a 12.5  $\mu\text{F}$  capacitor, and an 8.00 mH inductor, all connected across an ac source having a variable frequency and a voltage amplitude of 25.0 V. (a) At what angular frequency will the impedance be smallest, and what is the impedance at this frequency? (b) At the angular frequency in part (a), what is the maximum current through the inductor? (c) At the angular frequency in part (a), find the potential difference across the ac source, the resistor, the capacitor, and the inductor at the instant that the current is equal to one-half its greatest positive value. (d) In part (c), how are the potential differences across the resistor, inductor, and capacitor related to the potential difference across the ac source?



**31.29 •** In an  $L$ - $R$ - $C$  series circuit,  $R = 300\ \Omega$ ,  $L = 0.400\ \text{H}$ , and  $C = 6.00 \times 10^{-8}\ \text{F}$ . When the ac source operates at the resonance frequency of the circuit, the current amplitude is  $0.500\ \text{A}$ . (a) What is the voltage amplitude of the source? (b) What is the amplitude of the voltage across the resistor, across the inductor, and across the capacitor? (c) What is the average power supplied by the source?

**31.30 •** An  $L$ - $R$ - $C$  series circuit consists of a source with voltage amplitude  $120\ \text{V}$  and angular frequency  $50.0\ \text{rad/s}$ , a resistor with  $R = 400\ \Omega$ , an inductor with  $L = 3.00\ \text{H}$ , and a capacitor with capacitance  $C$ . (a) For what value of  $C$  will the current amplitude in the circuit be a maximum? (b) When  $C$  has the value calculated in part (a), what is the amplitude of the voltage across the inductor?

**31.31 •** In an  $L$ - $R$ - $C$  series circuit,  $R = 150\ \Omega$ ,  $L = 0.750\ \text{H}$ , and  $C = 0.0180\ \mu\text{F}$ . The source has voltage amplitude  $V = 150\ \text{V}$  and a frequency equal to the resonance frequency of the circuit. (a) What is the power factor? (b) What is the average power delivered by the source? (c) The capacitor is replaced by one with  $C = 0.0360\ \mu\text{F}$  and the source frequency is adjusted to the new resonance value. Then what is the average power delivered by the source?

**31.32 •** An  $L$ - $R$ - $C$  series circuit has  $R = 400\ \Omega$ ,  $L = 0.600\ \text{H}$ , and  $C = 5.00 \times 10^{-8}\ \text{F}$ . The voltage amplitude of the source is  $80.0\ \text{V}$ . When the ac source operates at the resonance frequency of the circuit, what is the average power delivered by the source?

**31.33 ••** In an  $L$ - $R$ - $C$  series circuit,  $L = 0.280\ \text{H}$  and  $C = 4.00\ \mu\text{F}$ . The voltage amplitude of the source is  $120\ \text{V}$ . (a) What is the resonance angular frequency of the circuit? (b) When the source operates at the resonance angular frequency, the current amplitude in the circuit is  $1.70\ \text{A}$ . What is the resistance  $R$  of the resistor? (c) At the resonance angular frequency, what are the peak voltages across the inductor, the capacitor, and the resistor?

### Section 31.6 Transformers

**31.34 • Off to Europe!** You plan to take your hair dryer to Europe, where the electrical outlets put out  $240\ \text{V}$  instead of the  $120\ \text{V}$  seen in the United States. The dryer puts out  $1600\ \text{W}$  at  $120\ \text{V}$ . (a) What could you do to operate your dryer via the  $240\ \text{V}$  line in Europe? (b) What current will your dryer draw from a European outlet? (c) What resistance will your dryer appear to have when operated at  $240\ \text{V}$ ?

**31.35 • A Step-Down Transformer.** A transformer connected to a  $120\ \text{V}$  (rms) ac line is to supply  $12.0\ \text{V}$  (rms) to a portable electronic device. The load resistance in the secondary is  $5.00\ \Omega$ . (a) What should the ratio of primary to secondary turns of the transformer be? (b) What rms current must the secondary supply? (c) What average power is delivered to the load? (d) What resistance connected directly across the  $120\ \text{V}$  line would draw the same power as the transformer? Show that this is equal to  $5.00\ \Omega$  times the square of the ratio of primary to secondary turns.

**31.36 • A Step-Up Transformer.** A transformer connected to a  $120\ \text{V}$  (rms) ac line is to supply  $13,000\ \text{V}$  (rms) for a neon sign. To reduce shock hazard, a fuse is to be inserted in the primary circuit; the fuse is to blow when the rms current in the secondary circuit exceeds  $8.50\ \text{mA}$ . (a) What is the ratio of secondary to primary turns of the transformer? (b) What power must be supplied to the transformer when the rms secondary current is  $8.50\ \text{mA}$ ? (c) What current rating should the fuse in the primary circuit have?

### PROBLEMS

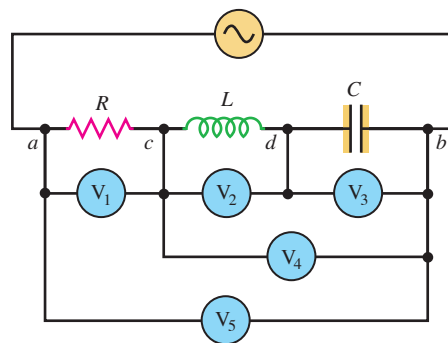
**31.37 •** A coil has a resistance of  $48.0\ \Omega$ . At a frequency of  $80.0\ \text{Hz}$  the voltage across the coil leads the current in it by  $52.3^\circ$ . Determine the inductance of the coil.

**31.38 ••** In an  $L$ - $R$ - $C$  series circuit the phase angle is  $53.0^\circ$  and the source voltage lags the current. The resistance of the resistor is  $300\ \Omega$  and the reactance of the capacitor is  $500\ \Omega$ . The average power delivered by the source is  $80.0\ \text{W}$ . (a) What is the reactance of the inductor? (b) What is the current amplitude in the circuit? (c) What is the voltage amplitude of the source?

**31.39 ••** An  $L$ - $R$ - $C$  series circuit has  $C = 4.80\ \mu\text{F}$ ,  $L = 0.520\ \text{H}$ , and source voltage amplitude  $V = 56.0\ \text{V}$ . The source is operated at the resonance frequency of the circuit. If the voltage across the capacitor has amplitude  $80.0\ \text{V}$ , what is the value of  $R$  for the resistor in the circuit?

**31.40 ••** Five infinite-impedance voltmeters, calibrated to read rms values, are connected as shown in **Fig. P31.40**. Let  $R = 200\ \Omega$ ,  $L = 0.400\ \text{H}$ ,  $C = 6.00\ \mu\text{F}$ , and  $V = 30.0\ \text{V}$ . What is the reading of each voltmeter if (a)  $\omega = 200\ \text{rad/s}$  and (b)  $\omega = 1000\ \text{rad/s}$ ?

Figure P31.40



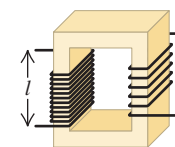
**31.41 •• CP** A parallel-plate capacitor having square plates  $4.50\ \text{cm}$  on each side and  $8.00\ \text{mm}$  apart is placed in series with the following: an ac source of angular frequency  $650\ \text{rad/s}$  and voltage amplitude  $22.5\ \text{V}$ ; a  $75.0\ \Omega$  resistor; and an ideal solenoid that is  $9.00\ \text{cm}$  long, has a circular cross section  $0.500\ \text{cm}$  in diameter, and carries  $125$  coils per centimeter. What is the resonance angular frequency of this circuit? (See Exercise 30.11.)

**31.42 •• CP** A toroidal solenoid has  $2900$  closely wound turns, cross-sectional area  $0.450\ \text{cm}^2$ , mean radius  $9.00\ \text{cm}$ , and resistance  $R = 2.80\ \Omega$ . Ignore the variation of the magnetic field across the cross section of the solenoid. What is the amplitude of the current in the solenoid if it is connected to an ac source that has voltage amplitude  $24.0\ \text{V}$  and frequency  $495\ \text{Hz}$ ?

**31.43 ••** A series circuit has an impedance of  $60.0\ \Omega$  and a power factor of  $0.720$  at  $50.0\ \text{Hz}$ . The source voltage lags the current. (a) What circuit element, an inductor or a capacitor, should be placed in series with the circuit to raise its power factor? (b) What size element will raise the power factor to unity?

**31.44 ••** Consider the ac adapter that bisects the power cord to your laptop computer, cell phone charger, or another electronic device. (a) What dc output voltage  $V$  and current  $I$  are listed on your device? (b) What power  $P$  is listed on your device? Does  $P = IV$ ? (c) Your ac adapter is most likely not a simple transformer; however, it is instructive to analyze the device as if it were: Assume your device is a transformer with a  $120\ \text{V}$  rms voltage on its primary coil and a full-wave rectifier following its secondary coil. If an  $R$ - $C$  circuit with a sufficient time constant followed, the ultimate output would maintain the dc value of the voltage amplitude on the rectified secondary coil. If the primary coil has  $200$  turns, how many turns are needed on the secondary coil to produce the stated dc output according to this scheme? (Give the closest integer value.) (d) Estimate the current amplitude in the primary coil. (e) Based on the size of your device and assuming the transformer geometry shown in **Fig. P31.44**, estimate the length  $l$  of the primary coil. (f) Use Ampere's law to estimate the average strength of the magnetic field inside the core, assuming a relative permeability of  $5000$ .

Figure P31.44

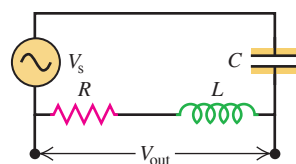


**31.45 ••** In an  $L$ - $R$ - $C$  series circuit,  $R = 300\ \Omega$ ,  $X_C = 300\ \Omega$ , and  $X_L = 500\ \Omega$ . The average electrical power consumed in the resistor is  $60.0\ \text{W}$ . (a) What is the power factor of the circuit? (b) What is the rms voltage of the source?

**31.46 •** At a frequency  $\omega_1$  the reactance of a certain capacitor equals that of a certain inductor. (a) If the frequency is changed to  $\omega_2 = 2\omega_1$ , what is the ratio of the reactance of the inductor to that of the capacitor? Which reactance is larger? (b) If the frequency is changed to  $\omega_3 = \omega_1/3$ , what is the ratio of the reactance of the inductor to that of the capacitor? Which reactance is larger? (c) If the capacitor and inductor are placed in series with a resistor of resistance  $R$  to form an  $L$ - $R$ - $C$  series circuit, what will be the resonance angular frequency of the circuit?

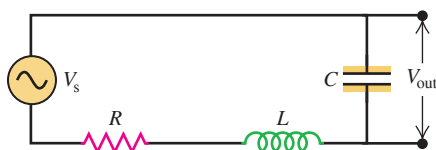
**31.47 •• A High-Pass Filter.** One application of  $L$ - $R$ - $C$  series circuits is to high-pass or low-pass filters, which filter out either the low- or high-frequency components of a signal. A high-pass filter is shown in **Fig. P31.47**, where the output voltage is taken across the  $L$ - $R$  combination. (The  $L$ - $R$  combination represents an inductive coil that also has resistance due to the large length of wire in the coil.) Derive an expression for  $V_{\text{out}}/V_s$ , the ratio of the output and source voltage amplitudes, as a function of the angular frequency  $\omega$  of the source. Show that when  $\omega$  is small, this ratio is proportional to  $\omega$  and thus is small, and show that the ratio approaches unity in the limit of large frequency.

Figure P31.47



**31.48 •• A Low-Pass Filter.** **Figure P31.48** shows a low-pass filter (see Problem 31.47); the output voltage is taken across the capacitor in an  $L$ - $R$ - $C$  series circuit. Derive an expression for  $V_{\text{out}}/V_s$ , the ratio of the output and source voltage amplitudes, as a function of the angular frequency  $\omega$  of the source. Show that when  $\omega$  is large, this ratio is proportional to  $\omega^{-2}$  and thus is very small, and show that the ratio approaches unity in the limit of small frequency.

Figure P31.48



**31.49 •••** An  $L$ - $R$ - $C$  series circuit is connected to an ac source of constant voltage amplitude  $V$  and variable angular frequency  $\omega$ . (a) Show that the current amplitude, as a function of  $\omega$ , is

$$I = \frac{V}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

(b) Show that the average power dissipated in the resistor is

$$P = \frac{V^2 R / 2}{R^2 + (\omega L - 1/\omega C)^2}$$

(c) Show that  $I$  and  $P$  are *both* maximum when  $\omega = 1/\sqrt{LC}$ , the resonance frequency of the circuit. (d) Graph  $P$  as a function of  $\omega$  for  $V = 100$  V,  $R = 200$   $\Omega$ ,  $L = 2.0$  H, and  $C = 0.50$   $\mu$ F. Compare to the light purple curve in Fig. 31.19. Discuss the behavior of  $I$  and  $P$  in the limits  $\omega = 0$  and  $\omega \rightarrow \infty$ .

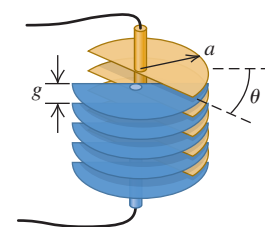
**31.50 •••** An  $L$ - $R$ - $C$  series circuit is connected to an ac source of constant voltage amplitude  $V$  and variable angular frequency  $\omega$ . Using the results of Problem 31.49, find an expression for (a) the amplitude  $V_L$  of the voltage across the inductor as a function of  $\omega$  and (b) the amplitude  $V_C$  of the voltage across the capacitor as a function of  $\omega$ . (c) Graph  $V_L$  and  $V_C$  as functions of  $\omega$  for  $V = 100$  V,  $R = 200$   $\Omega$ ,  $L = 2.0$  H, and  $C = 0.50$   $\mu$ F. (d) Discuss the behavior of  $V_L$  and  $V_C$  in the limits  $\omega = 0$  and  $\omega \rightarrow \infty$ . For what value of  $\omega$  is  $V_L = V_C$ ? What is the significance of this value of  $\omega$ ?

**31.51 ••** In an  $L$ - $R$ - $C$  series circuit the magnitude of the phase angle is  $54.0^\circ$ , with the source voltage lagging the current. The reactance of the capacitor is  $350$   $\Omega$ , and the resistor resistance is  $180$   $\Omega$ . The average power delivered by the source is  $140$  W. Find (a) the reactance of the inductor; (b) the rms current; (c) the rms voltage of the source.

**31.52 •• CP** Cell phones that use 4G technology receive signals broadcast between 2 GHz and 8 GHz. (a) If you want to create a simple  $L$ - $R$ - $C$  series circuit to detect a 4.0 GHz cell phone signal, what is the relevant value of the product  $LC$ , where  $L$  is the inductance and  $C$  is the capacitance? (b) If you choose a capacitor that has  $C = 1.0 \times 10^{-15}$  F, what inductance do you need? (c) Suppose you want to wind your own toroidal inductor and fit it inside a box as thin as your cell phone. Based on the size of your phone, estimate the largest cross-sectional area possible for this. (d) Assume the largest allowable radius of the toroid is 1.0 cm and estimate the lowest number of windings needed to create your inductor, assuming the material inside has a relative permeability of 1.

**31.53 ••• CP** Five semicircular conducting plates, each with radius  $a$ , are attached to a conducting rod along their common axis such that the plates are parallel with a separation gap  $g$  between adjacent plates. A similar arrangement involving five additional semicircular conducting plates is attached to a second conducting rod symmetrically intertwined, upside down, with the first arrangement such that the plates can rotate freely, as shown in **Fig. P31.53**. The two five-plate constructions, with an adjacent plate separation of  $g/2$ , are insulated from each other, and wire leads are attached to the separate axes. This is a common type of variable capacitor, with a capacitance proportional to the overlap angle  $\theta$ .

Figure P31.53



(a) Determine a formula for the capacitance  $C$  in terms of  $a$ ,  $g$ , and  $\theta$ , where  $\theta$  is in radians. (b) The plate radius is  $a = 6.00$  cm while the gap distance is  $g = 2.00$  mm. This variable capacitor is placed in series with a  $100$   $\mu$ H inductor and a  $10.0$   $\Omega$  resistor. A voltage source is supplied by an amplified signal from an antenna, and the output voltage is taken across the capacitor. The circuit is used as a radio receiver that receives a signal frequency equal to the resonance frequency of the  $L$ - $R$ - $C$  circuit. To receive an AM radio signal at  $1180$  kHz, what angle  $\theta$ , in degrees, should be selected? (c) If the input signal is a sinusoidal voltage with amplitude  $100.0$  mV and frequency  $1180$  kHz, what is the amplitude of the output voltage? (d) If the knob on the capacitor is turned so that  $\theta = 120^\circ$ , to what frequency is the circuit tuned? (e) If the input signal has the frequency resonant with this setting and voltage amplitude  $100$  mV, what is the voltage amplitude of the output signal?

**31.54 •• The  $L$ - $R$ - $C$  Parallel Circuit.** A resistor, an inductor, and a capacitor are connected in parallel to an ac source with voltage amplitude  $V$  and angular frequency  $\omega$ . Let the source voltage be given by  $v = V \cos \omega t$ . (a) Show that each of the instantaneous voltages  $v_R$ ,  $v_L$ , and  $v_C$  at any instant is equal to  $v$  and that  $i = i_R + i_L + i_C$ , where  $i$  is the current through the source and  $i_R$ ,  $i_L$ , and  $i_C$  are the currents through the resistor, inductor, and capacitor, respectively. (b) What are the phases of  $i_R$ ,  $i_L$ , and  $i_C$  with respect to  $v$ ? Use current phasors to represent  $i$ ,  $i_R$ ,  $i_L$ , and  $i_C$ . In a phasor diagram, show the phases of these four currents with respect to  $v$ . (c) Use the phasor diagram of part (b) to show that the current amplitude  $I$  for the current  $i$  through the source is  $I = \sqrt{I_R^2 + (I_C - I_L)^2}$ . (d) Show that the result of part (c) can be written as  $I = V/Z$ , with  $1/Z = \sqrt{(1/R^2) + [\omega C - (1/\omega L)]^2}$ .

**31.55 ••** The impedance of an  $L$ - $R$ - $C$  parallel circuit was derived in Problem 31.54. (a) Show that at the resonance angular frequency  $\omega_0 = 1/\sqrt{LC}$ , the impedance  $Z$  is a maximum and therefore the current through the ac source is a minimum. (b) A  $100\ \Omega$  resistor, a  $0.100\ \mu\text{F}$  capacitor, and a  $0.300\ \text{H}$  inductor are connected in parallel to a voltage source with amplitude  $240\ \text{V}$ . What is the resonance angular frequency? For this circuit, what is (c) the maximum current through the source at the resonance frequency; (d) the maximum current in the resistor at resonance; (e) the maximum current in the inductor at resonance; (f) the maximum current in the branch containing the capacitor at resonance?

**31.56 ••** A  $400\ \Omega$  resistor and a  $6.00\ \mu\text{F}$  capacitor are connected in parallel to an ac generator that supplies an rms voltage of  $180\ \text{V}$  at an angular frequency of  $360\ \text{rad/s}$ . Use the results of Problem 31.54. Note that since there is no inductor in this circuit, the  $1/\omega L$  term is not present in the expression for  $1/Z$ . Find (a) the current amplitude in the resistor; (b) the current amplitude in the capacitor; (c) the phase angle of the source current with respect to the source voltage; (d) the amplitude of the current through the generator. (e) Does the source current lag or lead the source voltage?

**31.57 •••** An  $L$ - $R$ - $C$  series circuit consists of a  $2.50\ \mu\text{F}$  capacitor, a  $5.00\ \text{mH}$  inductor, and a  $75.0\ \Omega$  resistor connected across an ac source of voltage amplitude  $15.0\ \text{V}$  having variable frequency. (a) Under what circumstances is the average power delivered to the circuit equal to  $\frac{1}{2}V_{\text{rms}}I_{\text{rms}}$ ? (b) Under the conditions of part (a), what is the average power delivered to each circuit element and what is the maximum current through the capacitor?

**31.58 ••** An  $L$ - $R$ - $C$  series circuit has  $R = 60.0\ \Omega$ ,  $L = 0.800\ \text{H}$ , and  $C = 3.00 \times 10^{-4}\ \text{F}$ . The ac source has voltage amplitude  $90.0\ \text{V}$  and angular frequency  $120\ \text{rad/s}$ . (a) What is the maximum energy stored in the inductor? (b) When the energy stored in the inductor is a maximum, how much energy is stored in the capacitor? (c) What is the maximum energy stored in the capacitor?

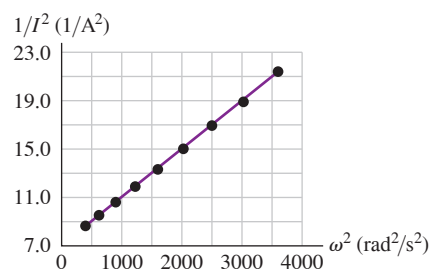
**31.59 •** In an  $L$ - $R$ - $C$  series circuit, the source has a voltage amplitude of  $120\ \text{V}$ ,  $R = 80.0\ \Omega$ , and the reactance of the capacitor is  $480\ \Omega$ . The voltage amplitude across the capacitor is  $360\ \text{V}$ . (a) What is the current amplitude in the circuit? (b) What is the impedance? (c) What two values can the reactance of the inductor have? (d) For which of the two values found in part (c) is the angular frequency less than the resonance angular frequency? Explain.

**31.60 ••** In an  $L$ - $R$ - $C$  series ac circuit, the source has a voltage amplitude of  $240\ \text{V}$ ,  $R = 90.0\ \Omega$ , and the reactance of the inductor is  $320\ \Omega$ . The voltage amplitude across the resistor is  $135\ \text{V}$ . (a) What is the current amplitude in the circuit? (b) What is the voltage amplitude across the inductor? (c) What two values can the reactance of the capacitor have? (d) For which of the two values found in part (c) is the angular frequency less than the resonance angular frequency? Explain.

**31.61 •** A resistance  $R$ , capacitance  $C$ , and inductance  $L$  are connected in series to a voltage source with amplitude  $V$  and variable angular frequency  $\omega$ . If  $\omega = \omega_0$ , the resonance angular frequency, find (a) the maximum current in the resistor; (b) the maximum voltage across the capacitor; (c) the maximum voltage across the inductor; (d) the maximum energy stored in the capacitor; (e) the maximum energy stored in the inductor. Give your answers in terms of  $R$ ,  $C$ ,  $L$ , and  $V$ .

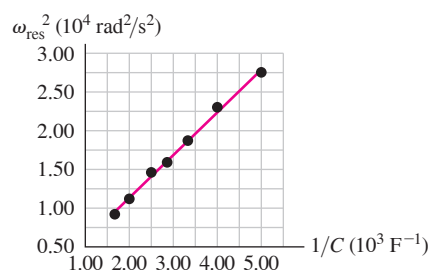
**31.62 •• DATA** A coworker of yours was making measurements of a large solenoid that is connected to an ac voltage source. Unfortunately, she left for vacation before she completed the analysis, and your boss has asked you to finish it. You are given a graph of  $1/I^2$  versus  $\omega^2$  (Fig. P31.62), where  $I$  is the current in the circuit and  $\omega$  is the angular frequency of the source. A note attached to the graph says that the voltage amplitude of the source was kept constant at  $12.0\ \text{V}$ . Calculate the resistance and inductance of the solenoid.

Figure P31.62



**31.63 •• DATA** You are analyzing an ac circuit that contains a solenoid and a capacitor in series with an ac source that has voltage amplitude  $90.0\ \text{V}$  and angular frequency  $\omega$ . For different capacitors in the circuit, each with known capacitance, you measure the value of the frequency  $\omega_{\text{res}}$  for which the current in the circuit is a maximum. You plot your measured values on a graph of  $\omega_{\text{res}}^2$  versus  $1/C$  (Fig. P31.63). The maximum current for each value of  $C$  is the same, you note, and equal to  $4.50\ \text{A}$ . Calculate the resistance and inductance of the solenoid.

Figure P31.63



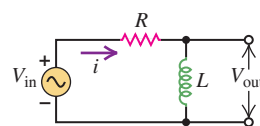
**31.64 •• DATA** You are given this table of data recorded for a circuit that has a resistor, an inductor with negligible resistance, and a capacitor, all in series with an ac voltage source:

|                      |     |     |
|----------------------|-----|-----|
| $f\ (\text{Hz})$     | 80  | 160 |
| $Z\ (\Omega)$        | 15  | 13  |
| $\phi\ (\text{deg})$ | -71 | 67  |

Here  $f$  is the frequency of the voltage source,  $Z$  is the impedance of the circuit, and  $\phi$  is the phase angle. (a) Use the data at both frequencies to calculate the resistance of the resistor. Calculate the average of these two values of the resistance, and use the result as the value of  $R$  in the rest of the analysis. (b) Use the data at  $80\ \text{Hz}$  and  $160\ \text{Hz}$  to calculate the inductance  $L$  and capacitance  $C$  of the circuit. (c) What is the resonance frequency for the circuit, and what are the impedance and phase angle at the resonance frequency?

**31.65 ••• CP** The frequency response of a filter circuit is commonly specified, in decibels, as the attenuation  $G = (20\ \text{dB})\log_{10}(V_{\text{out}}/V_{\text{in}})$ , where  $V_{\text{in}}$  is the input voltage amplitude and  $V_{\text{out}}$  is the output voltage amplitude. This is analogous to the definition of decibels given in Eq. (16.15). For the  $R$ - $L$  circuit shown in Fig. P31.65, the input signal is  $v_{\text{in}}(t) = V_{\text{in}}\cos(\omega t)$ . (a) The current is  $i(t) = I\cos(\omega t - \phi)$ . What is the current amplitude  $I$ ? (b) What is the phase angle  $\phi$ ? (c) What is the ratio  $V_{\text{out}}/V_{\text{in}}$ ? (d) For what frequency  $f = f_{3\text{dB}}$  does this circuit have an attenuation of  $-3.0\ \text{dB}$ ? (The frequency  $f_{3\text{dB}}$  is commonly cited as the nominal boundary demarcating the “pass band.”) (e) If  $R = 100\ \Omega$ , what inductance  $L$  will “pass” frequencies higher than  $10.0\ \text{kHz}$  and block frequencies lower than  $10.0\ \text{kHz}$ ?

Figure P31.65

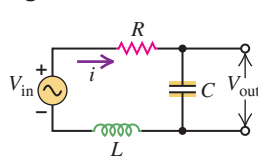




## CHALLENGE PROBLEMS

**31.66 ••• CALC** The  $L$ - $R$ - $C$  series circuit shown in **Fig. P31.66** has an ac voltage source  $v_{\text{in}}(t) = V_{\text{in}} \cos(\omega t)$ , where  $V_{\text{in}}$  is the input voltage amplitude and  $\omega$  is its angular frequency. The values of the resistance, inductance, and capacitance of the components are  $R$ ,  $L$ , and  $C$ . The capacitance is variable. The output voltage is  $v_{\text{out}}(t) = V_{\text{out}} \cos(\omega t + \theta)$ . (a) What is the output voltage amplitude  $V_{\text{out}}$ ? (b) What is the phase angle  $\theta$  of the output voltage? (c) What is the value of  $\theta$  at resonance? (d) If  $R = 100 \, \Omega$ ,  $L = 1.00 \, \mu\text{H}$ , and  $V_{\text{in}} = 10.0 \, \text{V}$ , what value of  $C$  would result in a resonance frequency of  $100 \, \text{kHz}$ ? (e) In that case what would be the output voltage amplitude at resonance?

Figure P31.66



**31.67 •• CALC** In an  $L$ - $R$ - $C$  series circuit the current is given by  $i = I \cos \omega t$ . The voltage amplitudes for the resistor, inductor, and capacitor are  $V_R$ ,  $V_L$ , and  $V_C$ . (a) Show that the instantaneous power into the resistor is  $p_R = V_R I \cos^2 \omega t = \frac{1}{2} V_R I (1 + \cos 2\omega t)$ . What does this expression give for the average power into the resistor? (b) Show that the instantaneous power into the inductor is  $p_L = -V_L I \sin \omega t \cos \omega t = -\frac{1}{2} V_L I \sin 2\omega t$ . What does this expression give for the average power into the inductor? (c) Show that the instantaneous power into the capacitor is  $p_C = V_C I \sin \omega t \cos \omega t = \frac{1}{2} V_C I \sin 2\omega t$ . What does this expression give for the average power into the capacitor? (d) The instantaneous power delivered by the source is shown in Section 31.4 to be  $p = VI \cos \omega t (\cos \phi \cos \omega t - \sin \phi \sin \omega t)$ . Show that  $p_R + p_L + p_C$  equals  $p$  at each instant of time.

**31.68 ••• CALC** (a) At what angular frequency is the voltage amplitude across the resistor in an  $L$ - $R$ - $C$  series circuit at maximum value? (b) At what angular frequency is the voltage amplitude across the inductor at maximum value? (c) At what angular frequency is the voltage amplitude across the capacitor at maximum value? (You may want to refer to the results of Problem 31.49.)

## MCAT-STYLE PASSAGE PROBLEMS

**BIO Converting dc to ac.** An individual cell such as an egg cell (an ovum, produced in the ovaries) is commonly organized spatially, as manifested in part by asymmetries in the cell membrane. These asymmetries include nonuniform distributions of ion transport mechanisms,

which result in a net electric current entering one region of the membrane and leaving another. These steady cellular currents may regulate cell polarity, leading (in the case of eggs) to embryonic polarity; therefore scientists are interested in measuring them.

These cellular currents move in loops through extracellular fluid. Ohm's law requires that there be voltage differences between any two points in this current-carrying fluid surrounding cells. Although the currents may be significant, the extracellular voltage differences are tiny—on the order of nanovolts. If we can map the voltage differences in the fluid outside a cell, we can calculate the current density by using Ohm's law, assuming that the resistivity of the fluid is known. We cannot measure these voltage differences by spacing two electrodes  $10$  or  $20 \, \mu\text{m}$  apart, because the dc impedance (the resistance) of such electrodes is high and the inherent noise in signals detected at the electrodes far exceeds the cellular voltages.

One successful method of measurement uses an electrode with a ball-shaped end made of platinum that is moved sinusoidally between two points in the fluid outside a cell. The electric potential that the electrode measures, with respect to a distant reference electrode, also varies sinusoidally. The dc potential difference between the two extremes (the two points in the fluid) is then converted to a sine-wave ac potential difference. The platinum electrode behaves as a capacitor in series with the resistance of the extracellular fluid. This resistance, called the *access resistance* ( $R_A$ ), has a value of about  $\rho/10a$ , where  $\rho$  is the resistivity of the fluid (usually expressed in  $\Omega \cdot \text{cm}$ ) and  $a$  is the radius of the ball electrode. The platinum ball typically has a diameter of  $20 \, \mu\text{m}$  and a capacitance of  $10 \, \text{nF}$ ; the resistivity of many biological fluids is  $100 \, \Omega \cdot \text{cm}$ .

**31.69** What is the dc impedance of the electrode, assuming that it behaves as an ideal capacitor? (a)  $0$ ; (b) infinite; (c)  $\sqrt{2} \times 10^4 \, \Omega$ ; (d)  $\sqrt{2} \times 10^6 \, \Omega$ .

**31.70** If the electrode oscillates between two points  $20 \, \mu\text{m}$  apart at a frequency of  $(5000/\pi) \, \text{Hz}$ , what is the electrode's impedance? (a)  $0$ ; (b) infinite; (c)  $\sqrt{2} \times 10^4 \, \Omega$ ; (d)  $\sqrt{2} \times 10^6 \, \Omega$ .

**31.71** The signal from the oscillating electrode is fed into an amplifier, which reports the measured voltage as an rms value,  $1.5 \, \text{nV}$ . What is the potential difference between the two extremes? (a)  $1.5 \, \text{nV}$ ; (b)  $3.0 \, \text{nV}$ ; (c)  $2.1 \, \text{nV}$ ; (d)  $4.2 \, \text{nV}$ .

**31.72** If the frequency at which the electrode is oscillated is increased to a very large value, the electrode's impedance (a) approaches infinity; (b) approaches zero; (c) approaches a constant but nonzero value; (d) does not change.

## ANSWERS

## Chapter Opening Question ?

(iv) A radio simultaneously detects transmissions at *all* frequencies. However, a radio is an  $L$ - $R$ - $C$  series circuit, and at any given time it is tuned to have a resonance at just one frequency. Hence the response of the radio to that frequency is much greater than its response to any other frequency, which is why you hear only one broadcasting station through the radio's speaker. (You can sometimes hear a second station if its frequency is sufficiently close to the tuned frequency.)

## Key Example VARIATION Problems

**VP31.3.1** (a)  $1.70 \times 10^6 \, \Omega$  (b)  $2.48 \times 10^{-3} \, \text{A}$

**VP31.3.2** (a)  $(0.300 \, \text{V}) \cos[(1.75 \times 10^3 \, \text{rad/s})t]$  (b)  $81.6 \, \Omega$   
(c)  $(0.196 \, \text{V}) \cos[(1.75 \times 10^3 \, \text{rad/s})t - \pi/2 \, \text{rad}]$

**VP31.3.3** (a)  $128 \, \text{Hz}$  (b) both are  $0.620 \, \text{V}$  (c) zero; zero

**VP31.3.4** (a)  $0.0215 \, \text{A}$  (b)  $0.0116 \, \text{A}$  (c)  $2.45 \, \text{V}$  (d)  $1.72 \, \mu\text{F}$

**VP31.5.1** (a)  $902 \, \text{Hz}$  (b)  $316 \, \Omega$  (c)  $0.174 \, \text{A}$  (d)  $-36.1^\circ$  (e) lags

**VP31.5.2** (a)  $387 \, \Omega$  (b)  $0.0422 \, \text{A}$  (c)  $16.3 \, \text{V}$  (d)  $1.48 \, \text{V}$  (e)  $13.0 \, \text{V}$

**VP31.5.3** (a)  $17.4 \, \Omega$  (b)  $8.65 \, \text{mH}$

**VP31.5.4** (a)  $V_R = 11.4 \, \text{V}$ ,  $V_L = 6.24 \, \text{V}$ ,  $V_C = 34.1 \, \text{V}$

(b)  $v_R = -8.71 \, \text{V}$ ,  $v_L = -4.03 \, \text{V}$ ,  $v_C = +22.0 \, \text{V}$

**VP31.7.1** (a)  $474 \, \text{W}$  (b)  $948 \, \text{W}$  (c)  $30.4 \, \text{W}$

**VP31.7.2** (a)  $X_L = 107 \, \Omega$ ,  $X_C = 699 \, \Omega$  (b)  $-65.1^\circ$  (c)  $0.421$

**VP31.7.3** (a)  $653 \, \Omega$  (b)  $0.0536 \, \text{A}$  (c)  $0.395 \, \text{W}$

**VP31.7.4** (a)  $X_L = 97.5 \, \Omega$ ,  $X_C = 85.5 \, \Omega$  (b)  $36.9^\circ$  (c)  $16.0 \, \Omega$

## Bridging Problem

(a)  $8.35 \times 10^4 \, \text{rad/s}$  and  $3.19 \times 10^5 \, \text{rad/s}$

(b) At  $8.35 \times 10^4 \, \text{rad/s}$ :  $V_{\text{source}} = 49.5 \, \text{V}$ ,  $I = 0.132 \, \text{A}$ ,  $V_R = 16.5 \, \text{V}$ ,  
 $V_L = 16.5 \, \text{V}$ ,  $V_C = 63.2 \, \text{V}$

At  $3.19 \times 10^5 \, \text{rad/s}$ :  $V_{\text{source}} = 49.5 \, \text{V}$ ,  $I = 0.132 \, \text{A}$ ,  $V_R = 16.5 \, \text{V}$ ,  
 $V_L = 63.2 \, \text{V}$ ,  $V_C = 16.5 \, \text{V}$



? Metal objects reflect not only visible light but also radio waves. This is because at the surface of a metal, (i) the electric-field component parallel to the surface must be zero; (ii) the electric-field component perpendicular to the surface must be zero; (iii) the magnetic-field component parallel to the surface must be zero; (iv) the magnetic-field component perpendicular to the surface must be zero; (v) more than one of these.



# 32 Electromagnetic Waves

What is light? This question has been asked by humans for centuries, but there was no answer until electricity and magnetism were unified into *electromagnetism*, as described by Maxwell's equations. These equations show that a time-varying magnetic field acts as a source of electric field and that a time-varying electric field acts as a source of magnetic field. These  $\vec{E}$  and  $\vec{B}$  fields can sustain each other, forming an *electromagnetic wave* that propagates through space. Visible light emitted by the glowing filament of a light bulb is one example of an electromagnetic wave; other kinds of electromagnetic waves are produced by wi-fi base stations, x-ray machines, and radioactive nuclei.

In this chapter we'll use Maxwell's equations as the theoretical basis for understanding electromagnetic waves. We'll find that these waves carry both energy and momentum. In sinusoidal electromagnetic waves, the  $\vec{E}$  and  $\vec{B}$  fields are sinusoidal functions of time and position, with a definite frequency and wavelength. Visible light, radio, x rays, and other types of electromagnetic waves differ only in their frequency and wavelength. Our study of optics in the following chapters will be based in part on the electromagnetic nature of light.

Unlike waves on a string or sound waves in a fluid, electromagnetic waves do not require a material medium; the light that you see coming from the stars at night has traveled without difficulty across tens or hundreds of light-years of (nearly) empty space. Nonetheless, electromagnetic waves and mechanical waves have much in common and are described in much the same language. Before reading further in this chapter, you should review the properties of mechanical waves as discussed in Chapters 15 and 16.

## 32.1 MAXWELL'S EQUATIONS AND ELECTROMAGNETIC WAVES

In the last several chapters we studied various aspects of electric and magnetic fields. We learned that when the fields don't vary with time, such as an electric field produced by charges at rest or the magnetic field of a steady current, we can analyze the electric and magnetic fields independently without considering interactions between them. But when the fields vary with time, they are no longer independent. Faraday's law (see

### LEARNING OUTCOMES

#### In this chapter, you'll learn...

- 32.1 How electromagnetic waves are generated.
- 32.2 How and why the speed of light is related to the fundamental constants of electricity and magnetism.
- 32.3 How to describe the propagation of a sinusoidal electromagnetic wave.
- 32.4 What determines the amount of energy and momentum carried by an electromagnetic wave.
- 32.5 How to describe standing electromagnetic waves.

#### You'll need to review...

- 8.1 Momentum.
- 15.3, 15.7 Traveling waves and standing waves on a string.
- 16.4 Standing sound waves.
- 23.4 Electric field in a conductor.
- 24.3, 24.4 Electric energy density; permittivity of a dielectric.
- 28.1, 28.8 Magnetic field of a moving charge; permeability of a dielectric.
- 29.2, 29.7 Faraday's law and Maxwell's equations.
- 30.3, 30.5 Magnetic energy density; L-C circuits.

Section 29.2) tells us that a time-varying magnetic field acts as a source of electric field, as shown by induced emfs in inductors and transformers. Ampere's law, including the displacement current discovered by James Clerk Maxwell (see Section 29.7), shows that a time-varying electric field acts as a source of magnetic field. This mutual interaction between the two fields is summarized in Maxwell's equations, presented in Section 29.7.

Thus, when *either* an electric or a magnetic field is changing with time, a field of the other kind is induced in adjacent regions of space. We are led (as Maxwell was) to consider the possibility of an electromagnetic disturbance, consisting of time-varying electric and magnetic fields, that can propagate through space from one region to another, even when there is no matter in the intervening region. Such a disturbance, if it exists, will have the properties of a *wave*, and an appropriate term is **electromagnetic wave**.

Such waves do exist; radio and television transmission, light, x rays, and many other kinds of radiation are examples of electromagnetic waves. Our goal in this chapter is to see how such waves are explained by the principles of electromagnetism that we have studied thus far and to examine the properties of these waves.

## Electricity, Magnetism, and Light

The theoretical understanding of electromagnetic waves actually evolved along a considerably more devious path than the one just outlined. In the early days of electromagnetic theory (the early 19th century), two different units of electric charge were used: one for electrostatics and the other for magnetic phenomena involving currents. In the system of units used at that time, these two units of charge had different physical dimensions. Their *ratio* had units of velocity, and measurements showed that the ratio had a numerical value that was precisely equal to the speed of light,  $3.00 \times 10^8$  m/s. At the time, physicists regarded this as an extraordinary coincidence and had no idea how to explain it.

In searching to understand this result, Maxwell (**Fig. 32.1**) proved in 1865 that an electromagnetic disturbance should propagate in free space with a speed equal to that of light and hence that light waves were likely to be electromagnetic in nature. At the same time, he discovered that the basic principles of electromagnetism can be expressed in terms of the four equations that we now call **Maxwell's equations**, which we discussed in Section 29.7. These four equations are (1) Gauss's law for electric fields; (2) Gauss's law for magnetic fields, showing the absence of magnetic monopoles; (3) Faraday's law; and (4) Ampere's law, including displacement current:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (\text{Gauss's law}) \quad (29.18)$$

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (\text{Gauss's law for magnetism}) \quad (29.19)$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad (\text{Faraday's law}) \quad (29.20)$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left( i_C + \epsilon_0 \frac{d\Phi_E}{dt} \right)_{\text{encl}} \quad (\text{Ampere's law}) \quad (29.21)$$

These equations apply to electric and magnetic fields *in vacuum*. If a material is present, the electric constant  $\epsilon_0$  and magnetic constant  $\mu_0$  are replaced by the permittivity  $\epsilon$  and permeability  $\mu$  of the material. If the values of  $\epsilon$  and  $\mu$  are different at different points in the regions of integration, then  $\epsilon$  and  $\mu$  have to be transferred to the left sides of Eqs. (29.18) and (29.21), respectively, and placed inside the integrals. The  $\epsilon$  in Eq. (29.21) also has to be included in the integral that gives  $d\Phi_E/dt$ .

Figure 32.1 The Scottish physicist James Clerk Maxwell (1831–1879) was the first person to truly understand the fundamental nature of light. He also made major contributions to thermodynamics, optics, astronomy, and color photography. Albert Einstein described Maxwell's accomplishments as “the most profound and the most fruitful that physics has experienced since the time of Newton.”



Figure 32.2 (a) When your mobile phone sends a text message or photo, the information is transmitted in the form of electromagnetic waves produced by electrons accelerating within the phone's circuits. (b) Power lines carry a strong alternating current, which means that a substantial amount of charge is accelerating back and forth and generating electromagnetic waves. These waves can produce a buzzing sound from your car radio when you drive near the lines.

(a)



(b)



According to Maxwell's equations, a point charge at rest produces a static  $\vec{E}$  field but no  $\vec{B}$  field, whereas a point charge moving with a constant velocity (see Section 28.1) produces both  $\vec{E}$  and  $\vec{B}$  fields. Maxwell's equations can also be used to show that in order for a point charge to produce electromagnetic waves, the charge must *accelerate*. In fact, in *every* situation where electromagnetic energy is radiated, the source is accelerated charges (**Fig. 32.2**).

## Generating Electromagnetic Radiation

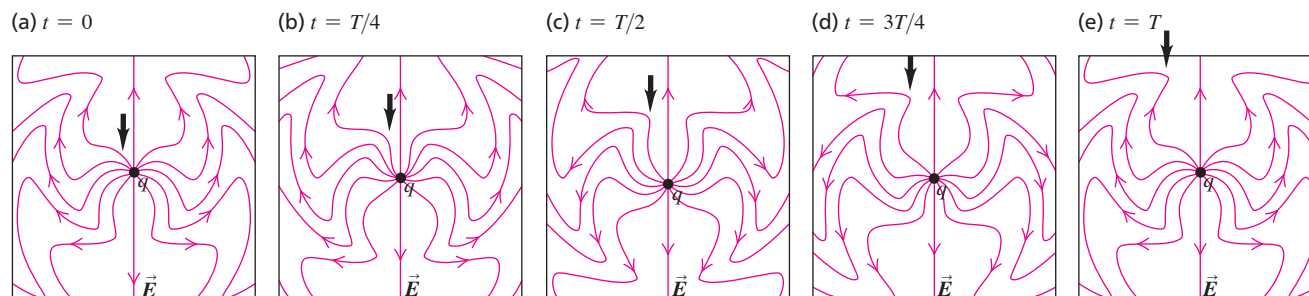
One way in which a point charge can be made to emit electromagnetic waves is by making it oscillate in simple harmonic motion, so that it has an acceleration at almost every instant (the exception is when the charge is passing through its equilibrium position). **Figure 32.3** shows some of the electric field lines produced by such an oscillating point charge. Field lines are *not* material objects, but you may nonetheless find it helpful to think of them as behaving somewhat like strings that extend from the point charge off to infinity. Oscillating the charge up and down makes waves that propagate outward from the charge along these "strings." Note that the charge does not emit waves equally in all directions; the waves are strongest at  $90^\circ$  to the axis of motion of the charge, while there are *no* waves along this axis. This is just what the "string" picture would lead you to conclude. There is also a *magnetic* disturbance that spreads outward from the charge; this is not shown in Fig. 32.3. Because the electric and magnetic disturbances spread or radiate away from the source, the name **electromagnetic radiation** is used interchangeably with the phrase "electromagnetic waves."

Electromagnetic waves with macroscopic wavelengths were first produced in the laboratory in 1887 by the German physicist Heinrich Hertz (for whom the SI unit of frequency is named). As a source of waves, he used charges oscillating in  $L$ - $C$  circuits (see Section 30.5); he detected the resulting electromagnetic waves with other circuits tuned to the same frequency. Hertz also produced electromagnetic *standing waves* and measured the distance between adjacent nodes (one half-wavelength) to determine the wavelength. Knowing the resonant frequency of his circuits, he then found the speed of the waves from the wavelength–frequency relationship  $v = \lambda f$ . He established that their speed was the same as that of light; this verified Maxwell's theoretical prediction directly.

The modern value of the speed of light,  $c$ , is 299,792,458 m/s. (Recall from Section 1.3 that this value is the basis of our standard of length: One meter is defined to be the distance that light travels in  $1/299,792,458$  second.) For our purposes,  $c = 3.00 \times 10^8$  m/s is sufficiently accurate.

In the wake of Hertz's discovery, Guglielmo Marconi and others made radio communication a familiar household experience. In a radio *transmitter*, electric charges are made to oscillate along the length of the conducting antenna, producing oscillating field disturbances like those shown in Fig. 32.3. Since many charges oscillate together in the antenna,

Figure 32.3 Electric field lines of a point charge oscillating in simple harmonic motion, seen at five instants during an oscillation period  $T$ . The charge's trajectory is in the plane of the drawings. At  $t = 0$  the point charge is at its maximum upward displacement. The arrow shows one "kink" in the lines of  $\vec{E}$  as it propagates outward from the point charge. The magnetic field (not shown) contains circles that lie in planes perpendicular to these figures and concentric with the axis of oscillation.





the disturbances are much stronger than those of a single oscillating charge and can be detected at a much greater distance. In a radio *receiver* the antenna is also a conductor; the fields of the wave emanating from a distant transmitter exert forces on free charges within the receiver antenna, producing an oscillating current that is detected and amplified by the receiver circuitry.

For the remainder of this chapter our concern will be with electromagnetic waves themselves, not with the rather complex problem of how they are produced.

## The Electromagnetic Spectrum

The **electromagnetic spectrum** encompasses electromagnetic waves of all frequencies and wavelengths. **Figure 32.4** shows approximate wavelength and frequency ranges for the most commonly encountered portion of the spectrum. Despite vast differences in their uses and means of production, these are all electromagnetic waves with the same propagation speed (in vacuum)  $c = 299,792,458$  m/s. Electromagnetic waves may differ in frequency  $f$  and wavelength  $\lambda$ , but the relationship  $c = \lambda f$  in vacuum holds for each.

We can detect only a very small segment of this spectrum directly through our sense of sight. We call this range **visible light**. Its wavelengths range from about 380 to 750 nm ( $380$  to  $750 \times 10^{-9}$  m), with corresponding frequencies from about 790 to 400 THz ( $7.9$  to  $4.0 \times 10^{14}$  Hz). Different parts of the visible spectrum evoke in humans the sensations of different colors. **Table 32.1** gives the approximate wavelengths for colors in the visible spectrum.

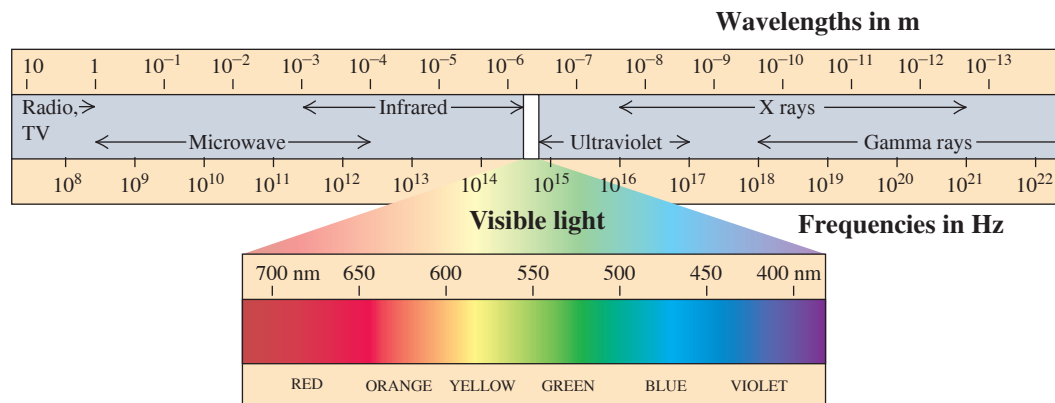
Ordinary white light includes all visible wavelengths. However, by using special sources or filters, we can select a narrow band of wavelengths within a range of a few nm. Such light is approximately *monochromatic* (single-color) light. Absolutely monochromatic light with only a single wavelength is an unattainable idealization. When we say “monochromatic light with  $\lambda = 550$  nm” with reference to a laboratory experiment, we really mean a small band of wavelengths *around* 550 nm. Light from a *laser* is much more nearly monochromatic than is light obtainable in any other way.

Invisible forms of electromagnetic radiation are no less important than visible light. Our system of global communication, for example, depends on radio waves: AM radio uses waves with frequencies from  $5.4 \times 10^5$  Hz to  $1.6 \times 10^6$  Hz, and FM radio broadcasts are at frequencies from  $8.8 \times 10^7$  Hz to  $1.08 \times 10^8$  Hz. Microwaves are also used for communication (for example, by mobile phones and wireless networks) and for weather radar (at frequencies near  $3 \times 10^9$  Hz). Many cameras have a device that emits a

**TABLE 32.1** Wavelengths of Visible Light

|            |        |
|------------|--------|
| 380–450 nm | Violet |
| 450–495 nm | Blue   |
| 495–570 nm | Green  |
| 570–590 nm | Yellow |
| 590–620 nm | Orange |
| 620–750 nm | Red    |

**Figure 32.4** The electromagnetic spectrum. The frequencies and wavelengths found in nature extend over such a wide range that we have to use a logarithmic scale to show all important bands. The boundaries between bands are somewhat arbitrary.





**BIO APPLICATION Ultraviolet**

**Vision** Many insects and birds can see ultraviolet wavelengths that humans cannot. As an example, the left-hand photo shows how black-eyed Susans (genus *Rudbeckia*) look to us. The right-hand photo (in false color), taken with an ultraviolet-sensitive camera, shows how these same flowers appear to the bees that pollinate them. Note the prominent central spot that is invisible to humans. Similarly, many birds with ultraviolet vision—including budgies, parrots, and peacocks—have ultraviolet patterns on their bodies that make them even more vivid to each other than they appear to us.



beam of infrared radiation; by analyzing the properties of the infrared radiation reflected from the subject, the camera determines the distance to the subject and automatically adjusts the focus. X rays are able to penetrate through flesh, which makes them invaluable in dentistry and medicine. Gamma rays, the shortest-wavelength type of electromagnetic radiation, are used in medicine to destroy cancer cells.

**TEST YOUR UNDERSTANDING OF SECTION 32.1** (a) Is it possible to have a purely electric wave propagate through empty space—that is, a wave made up of an electric field but no magnetic field? (b) What about a purely magnetic wave, with a magnetic field but no electric field?

**ANSWER**

(a) **no**, (b) **no** A purely electric wave would have a varying electric field. Such a field necessarily generates a magnetic field through Ampere's law, Eq. (29.21), so a purely electric wave is impossible. In the same way, a purely magnetic wave is impossible: The varying magnetic field in such a wave would automatically give rise to an electric field through Faraday's law, Eq. (29.20).

## 32.2 PLANE ELECTROMAGNETIC WAVES AND THE SPEED OF LIGHT

We are now ready to develop the basic ideas of electromagnetic waves and their relationship to the principles of electromagnetism. Our procedure will be to postulate a simple field configuration that has wavelike behavior. We'll assume an electric field  $\vec{E}$  that has only a  $y$ -component and a magnetic field  $\vec{B}$  with only a  $z$ -component, and we'll assume that both fields move together in the  $+x$ -direction with a speed  $c$  that is initially unknown. (As we go along, it will become clear why we choose  $\vec{E}$  and  $\vec{B}$  to be perpendicular to the direction of propagation as well as to each other.) Then we'll test whether these fields are physically possible by asking whether they are consistent with Maxwell's equations, particularly Ampere's law and Faraday's law. We'll find that the answer is yes, provided that  $c$  has a particular value. We'll also show that the *wave equation*, which we encountered during our study of mechanical waves in Chapter 15, can be derived from Maxwell's equations.

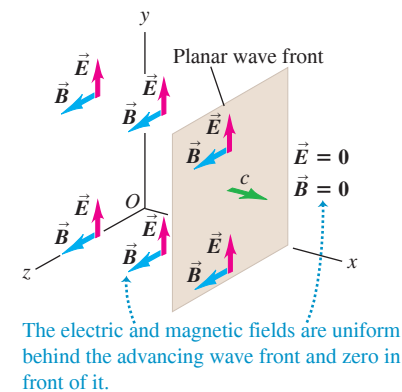
### A Simple Plane Electromagnetic Wave

Using an  $xyz$ -coordinate system (**Fig. 32.5**), we imagine that all space is divided into two regions by a plane perpendicular to the  $x$ -axis (parallel to the  $yz$ -plane). At every point to the left of this plane there are a uniform electric field  $\vec{E}$  in the  $+y$ -direction and a uniform magnetic field  $\vec{B}$  in the  $+z$ -direction, as shown. Furthermore, we suppose that the boundary plane, which we call the *wave front*, moves to the right in the  $+x$ -direction with a constant speed  $c$ , the value of which we'll leave undetermined for now. Thus the  $\vec{E}$  and  $\vec{B}$  fields travel to the right into previously field-free regions with a definite speed. This is a rudimentary electromagnetic wave. Such a wave, in which at any instant the fields are uniform over any plane perpendicular to the direction of propagation, is called a **plane wave**. In the case shown in Fig. 32.5, the fields are zero for planes to the right of the wave front and have the same values on all planes to the left of the wave front; later we'll consider more complex plane waves.

We won't concern ourselves with the problem of actually *producing* such a field configuration. Instead, we simply ask whether it is consistent with the laws of electromagnetism—that is, with all four of Maxwell's equations.

Let us first verify that our wave satisfies Maxwell's first and second equations—that is, Gauss's laws for electric and magnetic fields. To do this, we take as our Gaussian surface

**Figure 32.5** An electromagnetic wave front. The plane representing the wave front moves to the right (in the positive  $x$ -direction) with speed  $c$ .



a rectangular box with sides parallel to the  $xy$ -,  $xz$ -, and  $yz$ -coordinate planes (**Fig. 32.6**). The box encloses no electric charge. The total electric flux and magnetic flux through the box are both zero, even if part of the box is in the region where  $E = B = 0$ . This would *not* be the case if  $\vec{E}$  or  $\vec{B}$  had an  $x$ -component, parallel to the direction of propagation; if the wave front were inside the box, there would be flux through the left-hand side of the box (at  $x = 0$ ) but not the right-hand side (at  $x > 0$ ). Thus to satisfy Maxwell's first and second equations, the electric and magnetic fields must be perpendicular to the direction of propagation; that is, the wave must be **transverse**.

The next of Maxwell's equations that we'll consider is Faraday's law:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} \quad (32.1)$$

To test whether our wave satisfies Faraday's law, we apply this law to a rectangle  $efgh$  that is parallel to the  $xy$ -plane (**Fig. 32.7a**). As shown in Fig. 32.7b, a cross section in the  $xy$ -plane, this rectangle has height  $a$  and width  $\Delta x$ . At the time shown, the wave front has progressed partway through the rectangle, and  $\vec{E}$  is zero along the side  $ef$ . In applying Faraday's law we take the vector area  $d\vec{A}$  of rectangle  $efgh$  to be in the  $+z$ -direction. With this choice the right-hand rule requires that we integrate  $\vec{E} \cdot d\vec{l}$  *counterclockwise* around the rectangle. At every point on side  $ef$ ,  $\vec{E}$  is zero. At every point on sides  $fg$  and  $he$ ,  $\vec{E}$  is either zero or perpendicular to  $d\vec{l}$ . Only side  $gh$  contributes to the integral. On this side,  $\vec{E}$  and  $d\vec{l}$  are opposite, and we find that the left-hand side of Eq. (32.1) is nonzero:

$$\oint \vec{E} \cdot d\vec{l} = -Ea \quad (32.2)$$

To satisfy Faraday's law, Eq. (32.1), there must be a component of  $\vec{B}$  in the  $z$ -direction (perpendicular to  $\vec{E}$ ) so that there can be a nonzero magnetic flux  $\Phi_B$  through the rectangle  $efgh$  and a nonzero derivative  $d\Phi_B/dt$ . Indeed, in our wave,  $\vec{B}$  has *only* a  $z$ -component. We have assumed that this component is in the *positive*  $z$ -direction; let's see whether this assumption is consistent with Faraday's law. During a time interval  $dt$  the wave front (traveling at speed  $c$ ) moves a distance  $c dt$  to the right in Fig. 32.7b, sweeping out an area  $ac dt$  of the rectangle  $efgh$ . During this interval the magnetic flux  $\Phi_B$  through the rectangle  $efgh$  increases by  $d\Phi_B = B(ac dt)$ , so the rate of change of magnetic flux is

$$\frac{d\Phi_B}{dt} = Bac \quad (32.3)$$

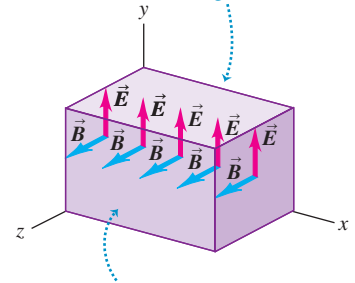
Now we substitute Eqs. (32.2) and (32.3) into Faraday's law, Eq. (32.1); we get  $-Ea = -Bac$ , so

$$\text{Electromagnetic wave in vacuum:} \quad \begin{array}{l} \text{Electric-field magnitude} \\ E \end{array} = \begin{array}{l} \text{Magnetic-field magnitude} \\ cB \end{array} \quad \begin{array}{l} \text{Speed of light} \\ \text{in vacuum} \end{array} \quad (32.4)$$

Our wave is consistent with Faraday's law only if the wave speed  $c$  and the magnitudes of  $\vec{E}$  and  $\vec{B}$  are related as in Eq. (32.4). If we had assumed that  $\vec{B}$  was in the *negative*  $z$ -direction, there would have been an additional minus sign in Eq. (32.4); since  $E$ ,  $c$ , and  $B$  are all positive magnitudes, no solution would then have been possible. Furthermore, any component of  $\vec{B}$  in the  $y$ -direction (parallel to  $\vec{E}$ ) would not contribute to the changing magnetic flux  $\Phi_B$  through the rectangle  $efgh$  (which is parallel to the  $xy$ -plane) and so would not be part of the wave.

Figure 32.6 Gaussian surface for a transverse plane electromagnetic wave.

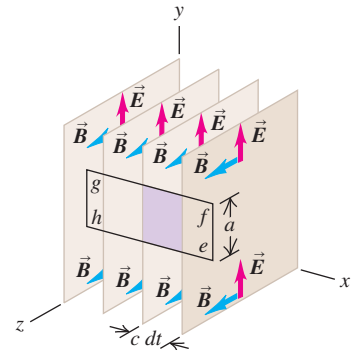
The electric field is the same on the top and bottom sides of the Gaussian surface, so the total electric flux through the surface is zero.



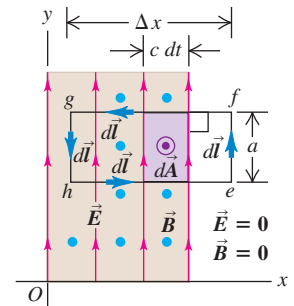
The magnetic field is the same on the left and right sides of the Gaussian surface, so the total magnetic flux through the surface is zero.

Figure 32.7 (a) Applying Faraday's law to a plane wave. (b) In a time  $dt$ , the magnetic flux through the rectangle in the  $xy$ -plane increases by an amount  $d\Phi_B$ . This increase equals the flux through the shaded rectangle with area  $ac dt$ ; that is,  $d\Phi_B = Bac dt$ . Thus  $d\Phi_B/dt = Bac$ .

(a) In time  $dt$ , the wave front moves a distance  $c dt$  in the  $+x$ -direction.



(b) Side view of situation in (a)

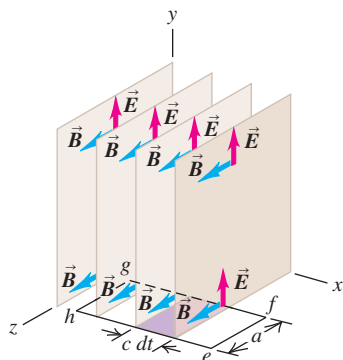


Finally, let's do a similar calculation with Ampere's law, the last of Maxwell's equations. There is no conduction current ( $i_C = 0$ ), so Ampere's law is

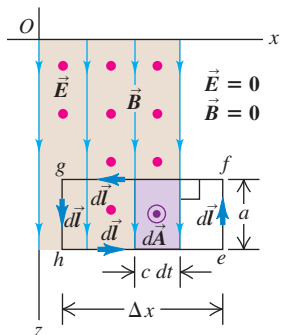
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} \quad (32.5)$$

Figure 32.8 (a) Applying Ampere's law to a plane wave. (Compare to Fig. 32.7a.) (b) In a time  $dt$ , the electric flux through the rectangle in the  $xz$ -plane increases by an amount  $d\Phi_E$ . This increase equals the flux through the shaded rectangle with area  $ac\,dt$ ; that is,  $d\Phi_E = Eac\,dt$ . Thus  $d\Phi_E/dt = Eac$ .

(a) In time  $dt$ , the wave front moves a distance  $c\,dt$  in the  $+x$ -direction.



(b) Top view of situation in (a)



To check whether our wave is consistent with Ampere's law, we move our rectangle so that it lies in the  $xz$ -plane (Fig. 32.8), and we again look at the situation at a time when the wave front has traveled partway through the rectangle. We take the vector area  $d\vec{A}$  in the  $+y$ -direction, and so the right-hand rule requires that we integrate  $\vec{B} \cdot d\vec{l}$  counterclockwise around the rectangle. The  $\vec{B}$  field is zero at every point along side  $ef$ , and at each point on sides  $fg$  and  $he$  it is either zero or perpendicular to  $d\vec{l}$ . Only side  $gh$ , where  $\vec{B}$  and  $d\vec{l}$  are parallel, contributes to the integral, and

$$\oint \vec{B} \cdot d\vec{l} = Ba \quad (32.6)$$

Hence the left-hand side of Eq. (32.5) is nonzero; the right-hand side must be nonzero as well. Thus  $\vec{E}$  must have a  $y$ -component (perpendicular to  $\vec{B}$ ) so that the electric flux  $\Phi_E$  through the rectangle and the time derivative  $d\Phi_E/dt$  can be nonzero. Just as we inferred from Faraday's law, we conclude that in an electromagnetic wave,  $\vec{E}$  and  $\vec{B}$  must be mutually perpendicular.

In a time interval  $dt$  the electric flux  $\Phi_E$  through the rectangle increases by  $d\Phi_E = E(ac\,dt)$ . Since we chose  $d\vec{A}$  to be in the  $+y$ -direction, this flux change is positive; the rate of change of electric flux is

$$\frac{d\Phi_E}{dt} = Eac \quad (32.7)$$

Substituting Eqs. (32.6) and (32.7) into Ampere's law, Eq. (32.5), we find  $Ba = \epsilon_0 \mu_0 Eac$ , so

$$\text{Electromagnetic wave in vacuum:} \quad B = \epsilon_0 \mu_0 c E \quad (32.8)$$

Magnetic-field magnitude Electric-field magnitude  
Electric constant Magnetic constant Speed of light in vacuum

Our assumed wave obeys Ampere's law only if  $B$ ,  $c$ , and  $E$  are related as in Eq. (32.8). The wave must also obey Faraday's law, so Eq. (32.4) must be satisfied as well. This can happen only if  $\epsilon_0 \mu_0 c = 1/c$ , or

$$\text{Speed of electromagnetic waves in vacuum} \quad c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \quad (32.9)$$

Electric constant Magnetic constant

Inserting the numerical values of these quantities to four significant figures, we find

$$c = \frac{1}{\sqrt{(8.854 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(1.257 \times 10^{-6} \text{ N/A}^2)}} = 2.998 \times 10^8 \text{ m/s}$$

Our assumed wave is consistent with all of Maxwell's equations, provided that the wave front moves with the speed given above, which is the speed of light! Recall that the *exact* value of  $c$  is defined to be 299,792,458 m/s; both  $\epsilon_0$  and  $\mu_0$  have small uncertainties (see Sections 21.3 and 28.4), but their product in Eq. (32.9) has *zero* uncertainty.

## Key Properties of Electromagnetic Waves

We chose a simple wave for our study in order to avoid mathematical complications, but this special case illustrates several important features of *all* electromagnetic waves:

1. The wave is *transverse*; both  $\vec{E}$  and  $\vec{B}$  are perpendicular to the direction of propagation of the wave. The electric and magnetic fields are also perpendicular to each other. The direction of propagation is the direction of the vector product  $\vec{E} \times \vec{B}$  (Fig. 32.9).
2. There is a definite ratio between the magnitudes of  $\vec{E}$  and  $\vec{B}$ :  $E = cB$ .
3. The wave travels in vacuum with a definite and unchanging speed.
4. Unlike mechanical waves, which need the particles of a medium such as air to transmit a wave, electromagnetic waves require no medium.

We can generalize this discussion to a more realistic situation. Suppose we have several wave fronts in the form of parallel planes perpendicular to the  $x$ -axis, all of which are moving to the right with speed  $c$ . Suppose that the  $\vec{E}$  and  $\vec{B}$  fields are the same at all points within a single region between two planes, but that the fields differ from region to region. The overall wave is a plane wave, but one in which the fields vary in steps along the  $x$ -axis. Such a wave could be constructed by superposing several of the simple step waves we have just discussed (shown in Fig. 32.5). This is possible because the  $\vec{E}$  and  $\vec{B}$  fields obey the superposition principle in waves just as in static situations: When two waves are superposed, the total  $\vec{E}$  field at each point is the vector sum of the  $\vec{E}$  fields of the individual waves, and similarly for the total  $\vec{B}$  field.

We can extend the above development to show that a wave with fields that vary in steps is also consistent with Ampere's and Faraday's laws, provided that the wave fronts all move with the speed  $c$  given by Eq. (32.9). In the limit that we make the individual steps infinitesimally small, we have a wave in which the  $\vec{E}$  and  $\vec{B}$  fields at any instant vary *continuously* along the  $x$ -axis. The entire field pattern moves to the right with speed  $c$ . In Section 32.3 we'll consider waves in which  $\vec{E}$  and  $\vec{B}$  are *sinusoidal* functions of  $x$  and  $t$ . Because at each point the magnitudes of  $\vec{E}$  and  $\vec{B}$  are related by  $E = cB$ , the periodic variations of the two fields in any periodic traveling wave must be *in phase*.

Electromagnetic waves have the property of **polarization**. In the above discussion the choice of the  $y$ -direction for  $\vec{E}$  was arbitrary. We could instead have specified the  $z$ -axis for  $\vec{E}$ ; then  $\vec{B}$  would have been in the  $-y$ -direction. A wave in which  $\vec{E}$  is always parallel to a certain axis is said to be **linearly polarized** along that axis. More generally, *any* wave traveling in the  $x$ -direction can be represented as a superposition of waves linearly polarized in the  $y$ - and  $z$ -directions. We'll study polarization in greater detail in Chapter 33.

## Derivation of the Electromagnetic Wave Equation

Here is an alternative derivation of Eq. (32.9) for the speed of electromagnetic waves. It is more mathematical than our other treatment, but it includes a derivation of the wave equation for electromagnetic waves. This part of the section can be omitted without loss of continuity in the chapter.

During our discussion of mechanical waves in Section 15.3, we showed that a function  $y(x, t)$  that represents the displacement of any point in a mechanical wave traveling along the  $x$ -axis must satisfy a differential equation, Eq. (15.12):

$$\frac{\partial^2 y(x, t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y(x, t)}{\partial t^2} \quad (32.10)$$

This equation is called the **wave equation**, and  $v$  is the speed of propagation of the wave.

To derive the corresponding equation for an electromagnetic wave, we again consider a plane wave. That is, we assume that at each instant,  $E_y$  and  $B_z$  are uniform over any plane

Figure 32.9 A right-hand rule for electromagnetic waves relates the directions of  $\vec{E}$  and  $\vec{B}$  and the direction of propagation.

**Right-hand rule for an electromagnetic wave:**

- ① Point the thumb of your right hand in the wave's direction of propagation.
- ② Imagine rotating the  $\vec{E}$ -field vector  $90^\circ$  in the sense your fingers curl. That is the direction of the  $\vec{B}$  field.

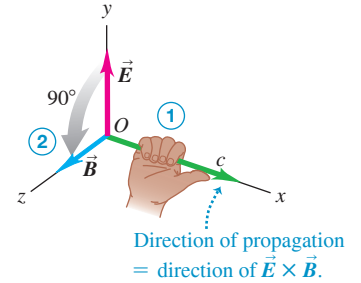
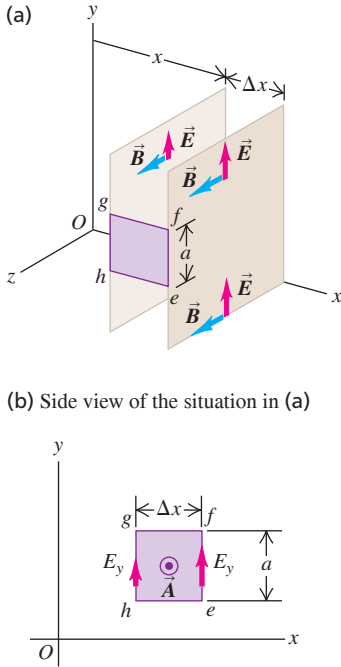




Figure 32.10 Faraday's law applied to a rectangle with height  $a$  and width  $\Delta x$  parallel to the  $xy$ -plane.



perpendicular to the  $x$ -axis, the direction of propagation. But now we let  $E_y$  and  $B_z$  vary continuously as we go along the  $x$ -axis; then each is a function of  $x$  and  $t$ . We consider the values of  $E_y$  and  $B_z$  on two planes perpendicular to the  $x$ -axis, one at  $x$  and one at  $x + \Delta x$ .

Following the same procedure as previously, we apply Faraday's law to a rectangle lying parallel to the  $xy$ -plane, as in **Fig. 32.10**. This figure is similar to Fig. 32.7. Let the left end  $gh$  of the rectangle be at position  $x$ , and let the right end  $ef$  be at position  $(x + \Delta x)$ . At time  $t$ , the values of  $E_y$  on these two sides are  $E_y(x, t)$  and  $E_y(x + \Delta x, t)$ , respectively. When we apply Faraday's law to this rectangle, we find that instead of  $\oint \vec{E} \cdot d\vec{l} = -Ea$  as before, we have

$$\begin{aligned} \oint \vec{E} \cdot d\vec{l} &= -E_y(x, t)a + E_y(x + \Delta x, t)a \\ &= a[E_y(x + \Delta x, t) - E_y(x, t)] \end{aligned} \quad (32.11)$$

To find the magnetic flux  $\Phi_B$  through this rectangle, we assume that  $\Delta x$  is small enough that  $B_z$  is nearly uniform over the rectangle. In that case,  $\Phi_B = B_z(x, t)A = B_z(x, t)a \Delta x$ , and

$$\frac{d\Phi_B}{dt} = \frac{\partial B_z(x, t)}{\partial t} a \Delta x$$

We use partial-derivative notation because  $B_z$  is a function of both  $x$  and  $t$ . When we substitute this expression and Eq. (32.11) into Faraday's law, Eq. (32.1), we get

$$\begin{aligned} a[E_y(x + \Delta x, t) - E_y(x, t)] &= -\frac{\partial B_z}{\partial t} a \Delta x \\ \frac{E_y(x + \Delta x, t) - E_y(x, t)}{\Delta x} &= -\frac{\partial B_z}{\partial t} \end{aligned}$$

Finally, imagine shrinking the rectangle down to a sliver so that  $\Delta x$  approaches zero. When we take the limit of this equation as  $\Delta x \rightarrow 0$ , we get

$$\frac{\partial E_y(x, t)}{\partial x} = -\frac{\partial B_z(x, t)}{\partial t} \quad (32.12)$$

This equation shows that if there is a time-varying component  $B_z$  of magnetic field, there must also be a component  $E_y$  of electric field that varies with  $x$ , and conversely. We put this relationship on the shelf for now; we'll return to it soon.

Next we apply Ampere's law to the rectangle shown in **Fig. 32.11**. The line integral  $\oint \vec{B} \cdot d\vec{l}$  becomes

$$\oint \vec{B} \cdot d\vec{l} = -B_z(x + \Delta x, t)a + B_z(x, t)a \quad (32.13)$$

Again assuming that the rectangle is narrow, we approximate the electric flux  $\Phi_E$  through it as  $\Phi_E = E_y(x, t)A = E_y(x, t)a \Delta x$ . The rate of change of  $\Phi_E$ , which we need for Ampere's law, is then

$$\frac{d\Phi_E}{dt} = \frac{\partial E_y(x, t)}{\partial t} a \Delta x$$

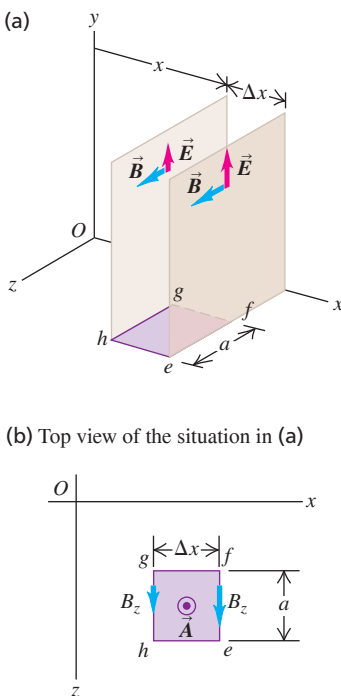
Now we substitute this expression and Eq. (32.13) into Ampere's law, Eq. (32.5):

$$-B_z(x + \Delta x, t)a + B_z(x, t)a = \epsilon_0 \mu_0 \frac{\partial E_y(x, t)}{\partial t} a \Delta x$$

Again we divide both sides by  $a \Delta x$  and take the limit as  $\Delta x \rightarrow 0$ . We find

$$-\frac{\partial B_z(x, t)}{\partial x} = \epsilon_0 \mu_0 \frac{\partial E_y(x, t)}{\partial t} \quad (32.14)$$

Figure 32.11 Ampere's law applied to a rectangle with height  $a$  and width  $\Delta x$  parallel to the  $xz$ -plane.



Now comes the final step. We take the partial derivatives of both sides of Eq. (32.12) with respect to  $x$ , and we take the partial derivatives of both sides of Eq. (32.14) with respect to  $t$ . The results are

$$\begin{aligned} -\frac{\partial^2 E_y(x, t)}{\partial x^2} &= \frac{\partial^2 B_z(x, t)}{\partial x \partial t} \\ -\frac{\partial^2 B_z(x, t)}{\partial x \partial t} &= \epsilon_0 \mu_0 \frac{\partial^2 E_y(x, t)}{\partial t^2} \end{aligned}$$

Combining these two equations to eliminate  $B_z$ , we finally find

$$\frac{\partial^2 E_y(x, t)}{\partial x^2} = \epsilon_0 \mu_0 \frac{\partial^2 E_y(x, t)}{\partial t^2} \quad \text{(electromagnetic wave equation in vacuum)} \quad (32.15)$$

This expression has the same form as the general wave equation, Eq. (32.10). Because the electric field  $E_y$  must satisfy this equation, it behaves as a wave with a pattern that travels through space with a definite speed. Furthermore, comparison of Eqs. (32.15) and (32.10) shows that the wave speed  $v$  is given by

$$\frac{1}{v^2} = \epsilon_0 \mu_0 \quad \text{or} \quad v = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

This agrees with Eq. (32.9) for the speed  $c$  of electromagnetic waves.

We can show that  $B_z$  also must satisfy the same wave equation as  $E_y$ , Eq. (32.15). To prove this, we take the partial derivative of Eq. (32.12) with respect to  $t$  and the partial derivative of Eq. (32.14) with respect to  $x$  and combine the results. We leave this derivation for you to carry out.

**TEST YOUR UNDERSTANDING OF SECTION 32.2** For each of the following electromagnetic waves, state the direction of the magnetic field. (a) The wave is propagating in the positive  $z$ -direction, and  $\vec{E}$  is in the positive  $x$ -direction; (b) the wave is propagating in the positive  $y$ -direction, and  $\vec{E}$  is in the negative  $z$ -direction; (c) the wave is propagating in the negative  $x$ -direction, and  $\vec{E}$  is in the positive  $z$ -direction.

### ANSWER

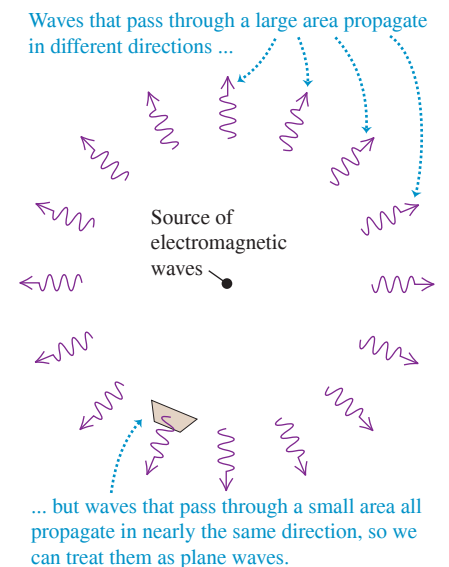
Answers by using the right-hand rule to show that  $\vec{E} \times \vec{B}$  in each case is in the direction of propagation, or by using the rule shown in Fig. 32.9. (a) positive  $y$ -direction, (b) negative  $x$ -direction, (c) positive  $y$ -direction. You can verify these answers by using the right-hand rule to show that  $\vec{E} \times \vec{B}$  in each case is in the direction of propagation.

## 32.3 SINUSOIDAL ELECTROMAGNETIC WAVES

Sinusoidal electromagnetic waves are directly analogous to sinusoidal transverse mechanical waves on a stretched string, which we studied in Section 15.3. In a sinusoidal electromagnetic wave,  $\vec{E}$  and  $\vec{B}$  at any point in space are sinusoidal functions of time, and at any instant of time the *spatial* variation of the fields is also sinusoidal.

Some sinusoidal electromagnetic waves are *plane waves*; they share with the waves described in Section 32.2 the property that at any instant the fields are uniform over any plane perpendicular to the direction of propagation. The entire pattern travels in the direction of propagation with speed  $c$ . The directions of  $\vec{E}$  and  $\vec{B}$  are perpendicular to the direction of propagation (and to each other), so the wave is *transverse*. Electromagnetic waves produced by an oscillating point charge, shown in Fig. 32.3, are an example of sinusoidal waves that are *not* plane waves. But if we restrict our observations to a relatively small region of space at a sufficiently great distance from the source, even these waves are well approximated by plane waves (**Fig. 32.12**). In the same way, the curved surface of the (nearly) spherical earth appears flat to us because of our small size relative to the earth's radius. In this section we'll restrict our discussion to plane waves.

Figure 32.12 Waves passing through a small area at a sufficiently great distance from a source can be treated as plane waves.



The frequency  $f$ , the wavelength  $\lambda$ , and the speed of propagation  $c$  of any periodic wave are related by the usual wavelength–frequency relationship  $c = \lambda f$ . If the frequency  $f$  is  $10^8$  Hz (100 MHz), typical of commercial FM radio broadcasts, the wavelength is

$$\lambda = \frac{3 \times 10^8 \text{ m/s}}{10^8 \text{ Hz}} = 3 \text{ m}$$

Figure 32.4 shows the inverse proportionality between wavelength and frequency.

## Fields of a Sinusoidal Wave

Figure 32.13 Representation of the electric and magnetic fields as functions of  $x$  for a linearly polarized sinusoidal plane electromagnetic wave. One wavelength of the wave is shown at time  $t = 0$ . The fields are shown for only a few points along the  $x$ -axis.

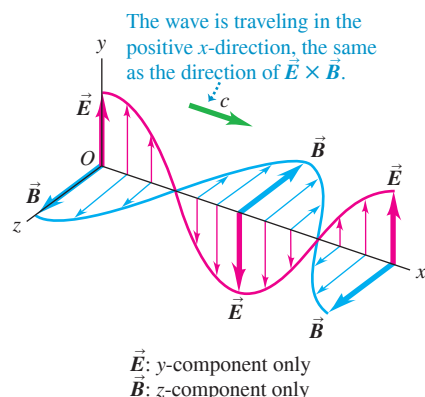


Figure 32.13 shows a linearly polarized sinusoidal electromagnetic wave traveling in the  $+x$ -direction. The electric and magnetic fields oscillate in phase:  $\vec{E}$  is maximum where  $\vec{B}$  is maximum and  $\vec{E}$  is zero where  $\vec{B}$  is zero. Where  $\vec{E}$  is in the  $+y$ -direction,  $\vec{B}$  is in the  $+z$ -direction; where  $\vec{E}$  is in the  $-y$ -direction,  $\vec{B}$  is in the  $-z$ -direction. At all points the vector product  $\vec{E} \times \vec{B}$  is in the direction in which the wave is propagating (the  $+x$ -direction). We mentioned this in Section 32.2 in the list of characteristics of electromagnetic waves.

**CAUTION** In a plane wave,  $\vec{E}$  and  $\vec{B}$  are everywhere Figure 32.13 shows  $\vec{E}$  and  $\vec{B}$  at points on the  $x$ -axis only. But, in fact, in a sinusoidal plane wave there are electric and magnetic fields at *all* points in space. Imagine a plane perpendicular to the  $x$ -axis (that is, parallel to the  $yz$ -plane) at a particular point and time; the fields have the same values at all points in that plane. The values are different on different planes. **|**

We can describe electromagnetic waves by means of *wave functions*, just as we did in Section 15.3 for waves on a string. One form of the wave function for a transverse wave traveling in the  $+x$ -direction along a stretched string is Eq. (15.7):

$$y(x, t) = A \cos(kx - \omega t)$$

where  $y(x, t)$  is the transverse displacement from equilibrium at time  $t$  of a point with coordinate  $x$  on the string. Here  $A$  is the maximum displacement, or *amplitude*, of the wave;  $\omega$  is its *angular frequency*, equal to  $2\pi$  times the frequency  $f$ ; and  $k = 2\pi/\lambda$  is the *wave number*, where  $\lambda$  is the wavelength.

Let  $E_y(x, t)$  and  $B_z(x, t)$  represent the instantaneous values of the  $y$ -component of  $\vec{E}$  and the  $z$ -component of  $\vec{B}$ , respectively, in Fig. 32.13, and let  $E_{\max}$  and  $B_{\max}$  represent the maximum values, or *amplitudes*, of these fields. The wave functions for the wave are then

$$E_y(x, t) = E_{\max} \cos(kx - \omega t) \quad B_z(x, t) = B_{\max} \cos(kx - \omega t) \quad (32.16)$$

We can also write the wave functions in vector form:

$$\begin{aligned} \vec{E}(x, t) &= \hat{j} E_{\max} \cos(kx - \omega t) \\ \vec{B}(x, t) &= \hat{k} B_{\max} \cos(kx - \omega t) \end{aligned} \quad (32.17)$$

Labels in the diagram: Sinusoidal electromagnetic plane wave, propagating in  $+x$ -direction; Electric field; Electric-field magnitude; Wave number; Angular frequency; Magnetic field; Magnetic-field magnitude.

**CAUTION** Two meanings of the symbol  $k$  Note the two different  $k$ 's in Eqs. (32.17): the unit vector  $\hat{k}$  in the  $z$ -direction and the wave number  $k$ . Don't get these confused! **|**

The sine curves in Fig. 32.13 represent the fields as functions of  $x$  at time  $t = 0$ —that is,  $\vec{E}(x, t = 0)$  and  $\vec{B}(x, t = 0)$ . As the wave travels to the right with speed  $c$ , Eqs. (32.16) and (32.17) show that at any point the oscillations of  $\vec{E}$  and  $\vec{B}$  are *in phase*. From Eq. (32.4) the amplitudes must be related by

$$E_{\max} = c B_{\max} \quad (32.18)$$

Labels in the diagram: Sinusoidal electromagnetic wave in vacuum; Electric-field amplitude; Magnetic-field amplitude; Speed of light in vacuum.

### APPLICATION Electromagnetic Plane Waves from Space

The moon, sun, planets, and stars are so distant that the light we receive from them is well approximated by plane waves. This simplifies the problem of designing the optics of astronomical telescopes to collect this light (see Chapter 34). The light from objects in space is not all of a single frequency, however, so the waves are not simple sinusoids as in Eqs. (32.17).



These amplitude and phase relationships are also required for  $E(x, t)$  and  $B(x, t)$  to satisfy Eqs. (32.12) and (32.14), which came from Faraday's law and Ampere's law, respectively. Can you verify this statement? (See Problem 32.30.)

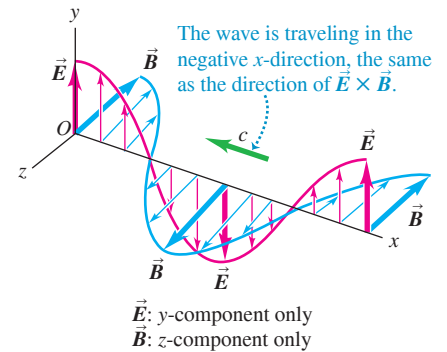
**Figure 32.14** shows the  $\vec{E}$  and  $\vec{B}$  fields of a wave traveling in the *negative*  $x$ -direction. At points where  $\vec{E}$  is in the positive  $y$ -direction,  $\vec{B}$  is in the *negative*  $z$ -direction; where  $\vec{E}$  is in the negative  $y$ -direction,  $\vec{B}$  is in the *positive*  $z$ -direction. As with waves traveling in the  $+x$ -direction, at any point the oscillations of the  $\vec{E}$  and  $\vec{B}$  fields of this wave are in phase, and the vector product  $\vec{E} \times \vec{B}$  points in the propagation direction. The wave functions for this wave are

$$\begin{aligned}\vec{E}(x, t) &= \hat{j}E_{\max} \cos(kx + \omega t) \\ \vec{B}(x, t) &= -\hat{k}B_{\max} \cos(kx + \omega t)\end{aligned}\quad (32.19)$$

(sinusoidal electromagnetic plane wave, propagating in  $-x$ -direction)

The sinusoidal waves shown in both Figs. 32.13 and 32.14 are linearly polarized in the  $y$ -direction; the  $\vec{E}$  field is always parallel to the  $y$ -axis. Example 32.1 concerns a wave that is linearly polarized in the  $z$ -direction.

**Figure 32.14** Representation of one wavelength of a linearly polarized sinusoidal plane electromagnetic wave traveling in the negative  $x$ -direction at  $t = 0$ . The fields are shown only for points along the  $x$ -axis. (Compare with Fig. 32.13.)



### PROBLEM-SOLVING STRATEGY 32.1 Electromagnetic Waves

**IDENTIFY** *the relevant concepts:* Many of the same ideas that apply to mechanical waves apply to electromagnetic waves. One difference is that electromagnetic waves are described by two quantities (in this case, electric field  $\vec{E}$  and magnetic field  $\vec{B}$ ), rather than by a single quantity, such as the displacement of a string.

**SET UP** *the problem* using the following steps:

1. Draw a diagram showing the direction of wave propagation and the directions of  $\vec{E}$  and  $\vec{B}$ .
2. Identify the target variables.

**EXECUTE** *the solution* as follows:

1. Review the treatment of sinusoidal mechanical waves in Chapters 15 and 16, and particularly the four problem-solving strategies suggested there.

2. Keep in mind the basic relationships for periodic waves:  $v = \lambda f$  and  $\omega = vk$ . For electromagnetic waves in vacuum,  $v = c$ . Distinguish between ordinary frequency  $f$ , usually expressed in hertz, and angular frequency  $\omega = 2\pi f$ , expressed in rad/s. Remember that the wave number is  $k = 2\pi/\lambda$ .
3. Concentrate on basic relationships, such as those between  $\vec{E}$  and  $\vec{B}$  (magnitude, direction, and relative phase), how the wave speed is determined, and the transverse nature of the waves.

**EVALUATE** *your answer:* Check that your result is reasonable. For electromagnetic waves in vacuum, the magnitude of the magnetic field in teslas is much smaller (by a factor of  $3.00 \times 10^8$ ) than the magnitude of the electric field in volts per meter. If your answer suggests otherwise, you probably made an error in using the relationship  $E = cB$ . (We'll see later in this section that this relationship is different for electromagnetic waves in a material medium.)

### EXAMPLE 32.1 Electric and magnetic fields of a laser beam

WITH VARIATION PROBLEMS

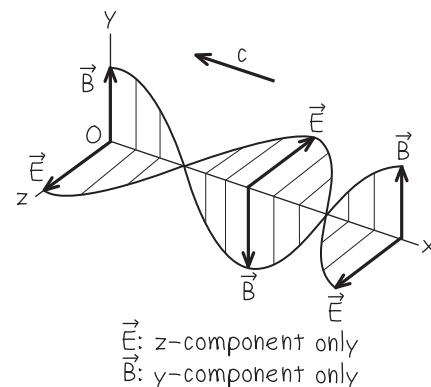
A carbon dioxide laser emits a sinusoidal electromagnetic wave that travels in vacuum in the negative  $x$ -direction. The wavelength is  $10.6 \mu\text{m}$  (in the infrared; see Fig. 32.4) and the  $\vec{E}$  field is parallel to the  $z$ -axis, with  $E_{\max} = 1.5 \text{ MV/m}$ . Write vector equations for  $\vec{E}$  and  $\vec{B}$  as functions of time and position.

**IDENTIFY and SET UP** Equations (32.19) describe a wave traveling in the negative  $x$ -direction with  $\vec{E}$  along the  $y$ -axis—that is, a wave that is linearly polarized along the  $y$ -axis. By contrast, the wave in this example is linearly polarized along the  $z$ -axis. At points where  $\vec{E}$  is in the positive  $z$ -direction,  $\vec{B}$  must be in the positive  $y$ -direction for the vector product  $\vec{E} \times \vec{B}$  to be in the negative  $x$ -direction (the direction of propagation). **Figure 32.15** shows a wave that satisfies these requirements.

**EXECUTE** A possible pair of wave functions that describe the wave shown in Fig. 32.15 are

$$\begin{aligned}\vec{E}(x, t) &= \hat{k}E_{\max} \cos(kx + \omega t) \\ \vec{B}(x, t) &= \hat{j}B_{\max} \cos(kx + \omega t)\end{aligned}$$

**Figure 32.15** Our sketch for this problem.



Continued



The plus sign in the arguments of the cosine functions indicates that the wave is propagating in the negative  $x$ -direction, as it should. Faraday's law requires that  $E_{\max} = cB_{\max}$  [Eq. (32.18)], so

$$B_{\max} = \frac{E_{\max}}{c} = \frac{1.5 \times 10^6 \text{ V/m}}{3.0 \times 10^8 \text{ m/s}} = 5.0 \times 10^{-3} \text{ T}$$

(Recall that  $1 \text{ V} = 1 \text{ Wb/s}$  and  $1 \text{ Wb/m}^2 = 1 \text{ T}$ .)

We have  $\lambda = 10.6 \times 10^{-6} \text{ m}$ , so the wave number and angular frequency are

$$k = \frac{2\pi}{\lambda} = \frac{2\pi \text{ rad}}{10.6 \times 10^{-6} \text{ m}} = 5.93 \times 10^5 \text{ rad/m}$$

$$\begin{aligned}\omega &= ck = (3.00 \times 10^8 \text{ m/s})(5.93 \times 10^5 \text{ rad/m}) \\ &= 1.78 \times 10^{14} \text{ rad/s}\end{aligned}$$

Substituting these values into the above wave functions, we get

$$\begin{aligned}\vec{E}(x, t) &= \hat{k}(1.5 \times 10^6 \text{ V/m}) \\ &\times \cos[(5.93 \times 10^5 \text{ rad/m})x + (1.78 \times 10^{14} \text{ rad/s})t]\end{aligned}$$

$$\begin{aligned}\vec{B}(x, t) &= \hat{j}(5.0 \times 10^{-3} \text{ T}) \\ &\times \cos[(5.93 \times 10^5 \text{ rad/m})x + (1.78 \times 10^{14} \text{ rad/s})t]\end{aligned}$$

**EVALUATE** As we expect, the magnitude  $B_{\max}$  in teslas is much smaller than the magnitude  $E_{\max}$  in volts per meter. To check the directions of  $\vec{E}$  and  $\vec{B}$ , note that  $\vec{E} \times \vec{B}$  is in the direction of  $\hat{k} \times \hat{j} = -\hat{i}$ . This is as it should be for a wave that propagates in the negative  $x$ -direction.

Our expressions for  $\vec{E}(x, t)$  and  $\vec{B}(x, t)$  are not the only possible solutions. We could always add a phase angle  $\phi$  to the arguments of the cosine function, so that  $kx + \omega t$  would become  $kx + \omega t + \phi$ . To determine the value of  $\phi$  we would need to know  $\vec{E}$  and  $\vec{B}$  either as functions of  $x$  at a given time  $t$  or as functions of  $t$  at a given coordinate  $x$ . However, the statement of the problem doesn't include this information.

**KEYCONCEPT** In a sinusoidal electromagnetic wave in vacuum, the electric field  $\vec{E}$  and magnetic field  $\vec{B}$  oscillate in phase with each other, are always perpendicular to each other, and are both perpendicular to the direction of propagation. The magnetic-field amplitude  $B_{\max}$  equals the electric-field amplitude  $E_{\max}$  divided by the speed of light in vacuum.

## Electromagnetic Waves in Matter

So far, our discussion of electromagnetic waves has been restricted to waves in *vacuum*. But electromagnetic waves can also travel in *matter*; think of light traveling through air, water, or glass. In this subsection we extend our analysis to electromagnetic waves in non-conducting materials—that is, *dielectrics*.

In a dielectric the wave speed is not the same as in vacuum, and we denote it by  $v$  instead of  $c$ . Faraday's law is unaltered, but in Eq. (32.4), derived from Faraday's law, the speed  $c$  is replaced by  $v$ . In Ampere's law the displacement current is given not by  $\epsilon_0 d\Phi_E/dt$ , where  $\Phi_E$  is the flux of  $\vec{E}$  through a surface, but by  $\epsilon d\Phi_E/dt = K\epsilon_0 d\Phi_E/dt$ , where  $K$  is the dielectric constant and  $\epsilon$  is the permittivity of the dielectric. (We introduced these quantities in Section 24.4.) Also, the constant  $\mu_0$  in Ampere's law must be replaced by  $\mu = K_m\mu_0$ , where  $K_m$  is the relative permeability of the dielectric and  $\mu$  is its permeability (see Section 28.8). Hence Eqs. (32.4) and (32.8) are replaced by

$$E = vB \quad \text{and} \quad B = \epsilon\mu vE \quad (32.20)$$

Following the same procedure as for waves in vacuum, we find that

$$v = \frac{\text{Speed of electromagnetic waves in a dielectric}}{1} = \frac{1}{\sqrt{\epsilon\mu}} = \frac{1}{\sqrt{KK_m}} \frac{1}{\sqrt{\epsilon_0\mu_0}} = \frac{1}{\sqrt{KK_m}} \frac{c}{1} = \frac{c}{\sqrt{KK_m}} \quad (32.21)$$

Speed of electromagnetic waves in a dielectric    Permeability    Speed of light in vacuum  
Permittivity    Dielectric constant    Relative permeability    Electric constant    Magnetic constant

Figure 32.16 The dielectric constant  $K$  of water is about 1.8 for visible light, so the speed of visible light in water is slower than in vacuum by a factor of  $1/\sqrt{K} = 1/\sqrt{1.8} = 0.75$ .



For most dielectrics the relative permeability  $K_m$  is nearly equal to unity (except for insulating ferromagnetic materials). When  $K_m \approx 1$ ,  $v = c/\sqrt{K}$ . Because  $K$  is always greater than unity, the speed  $v$  of electromagnetic waves in a nonmagnetic dielectric is always *less* than the speed  $c$  in vacuum by a factor of  $1/\sqrt{K}$  (**Fig. 32.16**). The ratio of the speed  $c$  in vacuum to the speed  $v$  in a material is known in optics as the **index of refraction**  $n$  of the material. When  $K_m \approx 1$ ,

$$\frac{c}{v} = n = \sqrt{KK_m} \approx \sqrt{K} \quad (32.22)$$

Usually, we can't use the values of  $K$  in Table 24.1 in this equation because those values are measured in *constant* electric fields. When the fields oscillate rapidly, there is usually not time for the reorientation of electric dipoles that occurs with steady fields. Values of  $K$  with

rapidly varying fields are usually much *smaller* than the values in the table. For example,  $K$  for water is 80.4 for steady fields but only about 1.8 in the frequency range of visible light. Thus the dielectric “constant”  $K$  is actually a function of frequency (the *dielectric function*).

### EXAMPLE 32.2 Electromagnetic waves in different materials

WITH  **ARIATION PROBLEMS**

(a) Visiting a jewelry store one evening, you hold a diamond up to the light of a sodium-vapor street lamp. The heated sodium vapor emits yellow light with a frequency of  $5.09 \times 10^{14}$  Hz. Find the wavelength in vacuum and the wave speed and wavelength in diamond, for which  $K = 5.84$  and  $K_m = 1.00$  at this frequency. (b) A 90.0 MHz radio wave (in the FM radio band) passes from vacuum into an insulating ferrite (a ferromagnetic material used in computer cables to suppress radio interference). Find the wavelength in vacuum and the wave speed and wavelength in the ferrite, for which  $K = 10.0$  and  $K_m = 1000$  at this frequency.

**IDENTIFY and SET UP** In each case we find the wavelength in vacuum by using  $c = \lambda f$ . To use the corresponding equation  $v = \lambda f$  to find the wavelength in a material medium, we find the speed  $v$  of electromagnetic waves in the medium from Eq. (32.21), which relates  $v$  to the values of dielectric constant  $K$  and relative permeability  $K_m$  for the medium.

**EXECUTE** (a) The wavelength in vacuum of the sodium light is

$$\lambda_{\text{vacuum}} = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{5.09 \times 10^{14} \text{ Hz}} = 5.89 \times 10^{-7} \text{ m} = 589 \text{ nm}$$

The wave speed and wavelength in diamond are

$$v_{\text{diamond}} = \frac{c}{\sqrt{KK_m}} = \frac{3.00 \times 10^8 \text{ m/s}}{\sqrt{(5.84)(1.00)}} = 1.24 \times 10^8 \text{ m/s}$$

$$\lambda_{\text{diamond}} = \frac{v_{\text{diamond}}}{f} = \frac{1.24 \times 10^8 \text{ m/s}}{5.09 \times 10^{14} \text{ Hz}} = 2.44 \times 10^{-7} \text{ m} = 244 \text{ nm}$$

(b) Following the same steps as in part (a), we find

$$\lambda_{\text{vacuum}} = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{90.0 \times 10^6 \text{ Hz}} = 3.33 \text{ m}$$

$$v_{\text{ferrite}} = \frac{c}{\sqrt{KK_m}} = \frac{3.00 \times 10^8 \text{ m/s}}{\sqrt{(10.0)(1000)}} = 3.00 \times 10^6 \text{ m/s}$$

$$\lambda_{\text{ferrite}} = \frac{v_{\text{ferrite}}}{f} = \frac{3.00 \times 10^6 \text{ m/s}}{90.0 \times 10^6 \text{ Hz}} = 3.33 \times 10^{-2} \text{ m} = 3.33 \text{ cm}$$

**EVALUATE** The speed of light in transparent materials is typically between  $0.2c$  and  $c$ ; our result in part (a) shows that  $v_{\text{diamond}} = 0.414c$ . As our results in part (b) show, the speed of electromagnetic waves in dense materials like ferrite (for which  $v_{\text{ferrite}} = 0.010c$ ) can be *far* slower than in vacuum.

**KEYCONCEPT** The speed of light in a transparent medium is slower than in vacuum. The greater the dielectric constant  $K$  of the medium and the greater the relative permeability  $K_m$  of the medium, the slower the speed. For waves of a given frequency, the slower the wave speed, the shorter the wavelength.

**TEST YOUR UNDERSTANDING OF SECTION 32.3** The first of Eqs. (32.17) gives the electric field for a plane wave as measured at points along the  $x$ -axis. For this plane wave, how does the electric field at points *off* the  $x$ -axis differ from the expression in Eqs. (32.17)? (i) The amplitude is different; (ii) the phase is different; (iii) both the amplitude and phase are different; (iv) none of these.

#### ANSWER

(iv) In an ideal electromagnetic plane wave, at any instant the fields are the same anywhere in a plane perpendicular to the direction of propagation. The plane wave described by Eqs. (32.17) is propagating in the  $x$ -direction, so the fields depend on the coordinate  $x$  and time  $t$  but do *not* depend on the coordinates  $y$  and  $z$ .

## 32.4 ENERGY AND MOMENTUM IN ELECTROMAGNETIC WAVES

Electromagnetic waves carry energy; the energy in sunlight is a familiar example. Microwave ovens, radio transmitters, and lasers for eye surgery all make use of this wave energy. To understand how to utilize this energy, it's helpful to derive detailed relationships for the energy in an electromagnetic wave.

We begin with the expressions derived in Sections 24.3 and 30.3 for the **energy densities** in electric and magnetic fields; we suggest that you review those derivations now. Equations (24.11) and (30.10) show that in a region of empty space where  $\vec{E}$  and  $\vec{B}$  fields are present, the total energy density  $u$  is

$$u = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2 \mu_0} B^2 \quad (32.23)$$

For electromagnetic waves in vacuum, the magnitudes  $E$  and  $B$  are related by

$$B = \frac{E}{c} = \sqrt{\epsilon_0 \mu_0} E \quad (32.24)$$

Combining Eqs. (32.23) and (32.24), we can also express the energy density  $u$  in a simple electromagnetic wave in vacuum as

$$u = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} (\sqrt{\epsilon_0 \mu_0} E)^2 = \epsilon_0 E^2 \quad (32.25)$$

This shows that in vacuum, the energy density associated with the  $\vec{E}$  field in our simple wave is equal to the energy density of the  $\vec{B}$  field. In general, the electric-field magnitude  $E$  is a function of position and time, as for the sinusoidal wave described by Eqs. (32.16); thus the energy density  $u$  of an electromagnetic wave, given by Eq. (32.25), also depends in general on position and time.

### Electromagnetic Energy Flow and the Poynting Vector

Electromagnetic waves such as those we have described are *traveling* waves that transport energy from one region to another. We can describe this energy transfer in terms of energy transferred *per unit time per unit cross-sectional area*, or *power per unit area*, for an area perpendicular to the direction of wave travel.

To see how the energy flow is related to the fields, consider a stationary plane, perpendicular to the  $x$ -axis, that coincides with the wave front at a certain time. In a time  $dt$  after this, the wave front moves a distance  $dx = c dt$  to the right of the plane. Consider an area  $A$  on this stationary plane (**Fig. 32.17**). The energy in the space to the right of this area had to pass through the area to reach the new location. The volume  $dV$  of the relevant region is the base area  $A$  times the length  $c dt$ , and the energy  $dU$  in this region is the energy density  $u$  times this volume:

$$dU = u dV = (\epsilon_0 E^2)(Ac dt)$$

This energy passes through the area  $A$  in time  $dt$ . The energy flow per unit time per unit area, which we'll call  $S$ , is

$$S = \frac{1}{A} \frac{dU}{dt} = \epsilon_0 c E^2 \quad (\text{in vacuum}) \quad (32.26)$$

Using Eqs. (32.4) and (32.9), you can derive the alternative forms

$$S = \frac{\epsilon_0}{\sqrt{\epsilon_0 \mu_0}} E^2 = \sqrt{\frac{\epsilon_0}{\mu_0}} E^2 = \frac{EB}{\mu_0} \quad (\text{in vacuum}) \quad (32.27)$$

The units of  $S$  are energy per unit time per unit area, or power per unit area. The SI unit of  $S$  is  $1 \text{ J/s} \cdot \text{m}^2$  or  $1 \text{ W/m}^2$ .

We can define a *vector* quantity that describes both the magnitude and direction of the energy flow rate. Introduced by the British physicist John Poynting (1852–1914), this quantity is called the **Poynting vector**:

$$\text{Poynting vector in vacuum} \quad \vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} \quad \begin{array}{l} \text{Electric field} \\ \text{Magnetic field} \\ \text{Magnetic constant} \end{array} \quad (32.28)$$

The vector  $\vec{S}$  points in the direction of propagation of the wave (**Fig. 32.18**). Since  $\vec{E}$  and  $\vec{B}$  are perpendicular, the magnitude of  $\vec{S}$  is  $S = EB/\mu_0$ ; from Eqs. (32.26) and (32.27) this is the energy flow per unit area and per unit time through a cross-sectional area perpendicular to the propagation direction. The total energy flow per unit time (power,  $P$ ) out of any closed surface is the integral of  $\vec{S}$  over the surface:

$$P = \oint \vec{S} \cdot d\vec{A}$$

Figure 32.17 A wave front at a time  $dt$  after it passes through the stationary plane with area  $A$ .

At time  $dt$ , the volume between the stationary plane and the wave front contains an amount of electromagnetic energy  $dU = uAc dt$ .

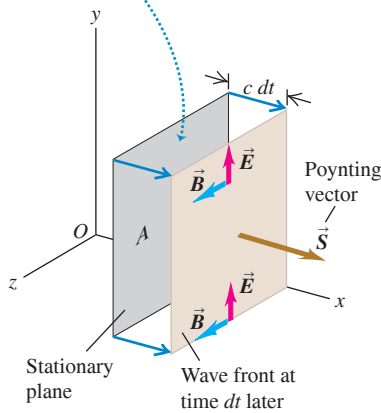


Figure 32.18 These rooftop solar panels are tilted to be face-on to the sun—that is, face-on to the Poynting vector of electromagnetic waves from the sun, so that the panels can absorb the maximum amount of wave energy.



For the sinusoidal waves studied in Section 32.3, as well as for other more complex waves, the electric and magnetic fields at any point vary with time, so the Poynting vector at any point is also a function of time. Because the frequencies of typical electromagnetic waves are very high, the time variation of the Poynting vector is so rapid that it's most appropriate to look at its *average* value. The magnitude of the average value of  $\vec{S}$  at a point is called the **intensity** of the radiation at that point. The SI unit of intensity is the same as for  $S$ ,  $1 \text{ W/m}^2$ .

Let's work out the intensity of the sinusoidal wave described by Eqs. (32.17). We first substitute  $\vec{E}$  and  $\vec{B}$  into Eq. (32.28):

$$\begin{aligned}\vec{S}(x, t) &= \frac{1}{\mu_0} \vec{E}(x, t) \times \vec{B}(x, t) \\ &= \frac{1}{\mu_0} [\hat{j} E_{\max} \cos(kx - \omega t)] \times [\hat{k} B_{\max} \cos(kx - \omega t)]\end{aligned}$$

The vector product of the unit vectors is  $\hat{j} \times \hat{k} = \hat{i}$  and  $\cos^2(kx - \omega t)$  is never negative, so  $\vec{S}(x, t)$  always points in the positive  $x$ -direction (the direction of wave propagation). The  $x$ -component of the Poynting vector is

$$S_x(x, t) = \frac{E_{\max} B_{\max}}{\mu_0} \cos^2(kx - \omega t) = \frac{E_{\max} B_{\max}}{2\mu_0} [1 + \cos 2(kx - \omega t)]$$

The time average value of  $\cos 2(kx - \omega t)$  is zero because at any point, it is positive during one half-cycle and negative during the other half. So the average value of the Poynting vector over a full cycle is  $\vec{S}_{\text{av}} = \hat{i} S_{\text{av}}$ , where

$$S_{\text{av}} = \frac{E_{\max} B_{\max}}{2\mu_0}$$

That is, the magnitude of the average value of  $\vec{S}$  for a sinusoidal wave (the intensity  $I$  of the wave) is  $\frac{1}{2}$  the maximum value. You can verify that by using the relationships  $E_{\max} = B_{\max} c$  and  $\epsilon_0 \mu_0 = 1/c^2$ , we can express the intensity in several equivalent forms:

**Intensity of a sinusoidal electromagnetic wave in vacuum**

$$I = S_{\text{av}} = \frac{E_{\max} B_{\max}}{2\mu_0} = \frac{E_{\max}^2}{2\mu_0 c} = \frac{1}{2} \sqrt{\frac{\epsilon_0}{\mu_0}} E_{\max}^2 = \frac{1}{2} \epsilon_0 c E_{\max}^2 \quad (32.29)$$

Labels in diagram:  
 - Electric-field amplitude (points to  $E_{\max}$ )  
 - Magnetic-field amplitude (points to  $B_{\max}$ )  
 - Electric constant (points to  $\epsilon_0$ )  
 - Magnitude of average Poynting vector (points to  $I = S_{\text{av}}$ )  
 - Magnetic constant (points to  $\mu_0$ )  
 - Speed of light in vacuum (points to  $c$ )

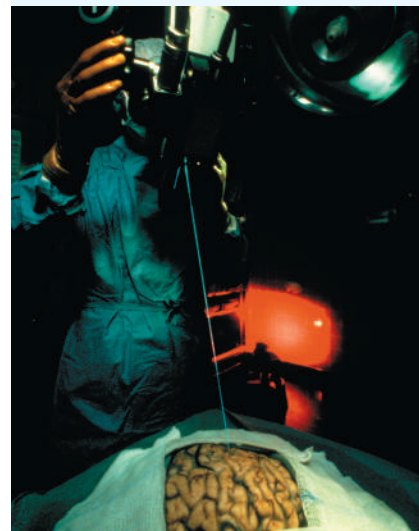
For a wave traveling in the  $-x$ -direction, represented by Eqs. (32.19), the Poynting vector is in the  $-x$ -direction at every point, but its magnitude is the same as for a wave traveling in the  $+x$ -direction. Verifying these statements is left to you.

Throughout this discussion we have considered only electromagnetic waves propagating in vacuum. If the waves are traveling in a dielectric medium, however, the expressions for energy density [Eq. (32.23)], the Poynting vector [Eq. (32.28)], and the intensity of a sinusoidal wave [Eq. (32.29)] must be modified. It turns out that the required modifications are quite simple: Just replace  $\epsilon_0$  with the permittivity  $\epsilon$  of the dielectric, replace  $\mu_0$  with the permeability  $\mu$  of the dielectric, and replace  $c$  with the speed  $v$  of electromagnetic waves in the dielectric. Remarkably, the energy densities in the  $\vec{E}$  and  $\vec{B}$  fields are equal even in a dielectric.

**CAUTION Poynting vector vs. intensity** At any point  $x$ , the magnitude of the Poynting vector varies with time. Hence, the *instantaneous* rate at which electromagnetic energy in a sinusoidal plane wave arrives at a surface is not constant. This may seem to contradict everyday experience; the light from the sun, a light bulb, or the laser in a grocery-store scanner appears steady and unvarying in strength. In fact the Poynting vector from these sources *does* vary in time, but the variation isn't noticeable because the oscillation frequency is so high (around  $5 \times 10^{14} \text{ Hz}$  for visible light). All that you sense is the *average* rate at which energy reaches your eye, which is why we commonly use intensity (the average value of  $S$ ) to describe the strength of electromagnetic radiation. **|**

### BIO APPLICATION Laser Surgery

Lasers are used widely in medicine as ultra-precise, bloodless “scalpels.” They can reach and remove tumors with minimal damage to neighboring healthy tissues, as in the brain surgery shown here. The power output of the laser is typically below 40 W, less than that of a typical light bulb. However, this power is concentrated into a spot from 0.1 to 2.0 mm in diameter, so the intensity of the light (equal to the average value of the Poynting vector) can be as high as  $5 \times 10^9 \text{ W/m}^2$ .





**EXAMPLE 32.3** Energy in a nonsinusoidal wave**WITH VARIATION PROBLEMS**

For the nonsinusoidal wave described in Section 32.2, suppose that  $E = 100 \text{ V/m} = 100 \text{ N/C}$ . Find the value of  $B$ , the energy density  $u$ , and the rate of energy flow per unit area  $S$ .

**IDENTIFY and SET UP** In this wave  $\vec{E}$  and  $\vec{B}$  are uniform behind the wave front (and zero ahead of it). Hence the target variables  $B$ ,  $u$ , and  $S$  must also be uniform behind the wave front. Given the magnitude  $E$ , we use Eq. (32.4) to find  $B$ , Eq. (32.25) to find  $u$ , and Eq. (32.27) to find  $S$ . (We cannot use Eq. (32.29), which applies to sinusoidal waves only.)

**EXECUTE** From Eq. (32.4),

$$B = \frac{E}{c} = \frac{100 \text{ V/m}}{3.00 \times 10^8 \text{ m/s}} = 3.33 \times 10^{-7} \text{ T}$$

From Eq. (32.25),

$$\begin{aligned} u &= \epsilon_0 E^2 = (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(100 \text{ N/C})^2 \\ &= 8.85 \times 10^{-8} \text{ N/m}^2 = 8.85 \times 10^{-8} \text{ J/m}^3 \end{aligned}$$

The magnitude of the Poynting vector is

$$S = \frac{EB}{\mu_0} = \frac{(100 \text{ V/m})(3.33 \times 10^{-7} \text{ T})}{4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}} = 26.5 \text{ V} \cdot \text{A/m}^2 = 26.5 \text{ W/m}^2$$

**EVALUATE** We can check our result for  $S$  by using Eq. (32.26):

$$\begin{aligned} S &= \epsilon_0 c E^2 = (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(3.00 \times 10^8 \text{ m/s})(100 \text{ N/C})^2 \\ &= 26.5 \text{ W/m}^2 \end{aligned}$$

Since  $\vec{E}$  and  $\vec{B}$  have the same values at all points behind the wave front,  $u$  and  $S$  likewise have the same value everywhere behind the wave front. In front of the wave front,  $\vec{E} = \vec{0}$  and  $\vec{B} = \vec{0}$ , and so  $u = 0$  and  $S = 0$ ; where there are no fields, there is no field energy.

**KEYCONCEPT** The Poynting vector  $\vec{S} = (1/\mu_0)\vec{E} \times \vec{B}$  for an electromagnetic wave points in the direction of wave propagation. The average magnitude of  $\vec{S}$  equals the intensity (average power per unit area) of the wave.

**EXAMPLE 32.4** Energy in a sinusoidal wave**WITH VARIATION PROBLEMS**

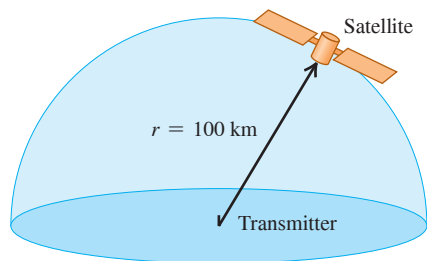
A radio station on the earth's surface emits a sinusoidal wave with average total power 50 kW (**Fig. 32.19**). Assuming that the transmitter radiates equally in all directions above the ground (which is unlikely in real situations), find the electric-field and magnetic-field amplitudes  $E_{\text{max}}$  and  $B_{\text{max}}$  detected by a satellite 100 km from the antenna.

**IDENTIFY and SET UP** We are given the transmitter's average total power  $P$ . The intensity  $I$  is the average power per unit area; to find  $I$  at 100 km from the transmitter we divide  $P$  by the surface area of the hemisphere in **Fig. 32.19**. For a sinusoidal wave,  $I$  is also equal to the magnitude of the average value  $S_{\text{av}}$  of the Poynting vector, so we can use Eq. (32.29) to find  $E_{\text{max}}$ ; Eq. (32.4) yields  $B_{\text{max}}$ .

**EXECUTE** The surface area of a hemisphere of radius  $r = 100 \text{ km} = 1.00 \times 10^5 \text{ m}$  is

$$A = 2\pi R^2 = 2\pi(1.00 \times 10^5 \text{ m})^2 = 6.28 \times 10^{10} \text{ m}^2$$

**Figure 32.19** A radio station radiates waves into the hemisphere shown.



All the radiated power passes through this surface, so the average power per unit area (that is, the intensity) is

$$I = \frac{P}{A} = \frac{P}{2\pi R^2} = \frac{5.00 \times 10^4 \text{ W}}{6.28 \times 10^{10} \text{ m}^2} = 7.96 \times 10^{-7} \text{ W/m}^2$$

From Eq. (32.29),  $I = S_{\text{av}} = E_{\text{max}}^2/2\mu_0 c$ , so

$$\begin{aligned} E_{\text{max}} &= \sqrt{2\mu_0 c S_{\text{av}}} \\ &= \sqrt{2(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(3.00 \times 10^8 \text{ m/s})(7.96 \times 10^{-7} \text{ W/m}^2)} \\ &= 2.45 \times 10^{-2} \text{ V/m} \end{aligned}$$

Then from Eq. (32.4),

$$B_{\text{max}} = \frac{E_{\text{max}}}{c} = 8.17 \times 10^{-11} \text{ T}$$

**EVALUATE** Note that  $E_{\text{max}}$  is comparable to fields commonly seen in the laboratory, but  $B_{\text{max}}$  is extremely small in comparison to  $\vec{B}$  fields we saw in previous chapters. For this reason, most detectors of electromagnetic radiation respond to the effect of the electric field, not the magnetic field. Loop radio antennas are an exception (see the Bridging Problem at the end of this chapter).

**KEYCONCEPT** The magnitude  $S_{\text{av}}$  of the average Poynting vector of a sinusoidal electromagnetic wave depends on the field amplitudes  $E_{\text{max}}$  and  $B_{\text{max}}$ . The value of  $S_{\text{av}}$  is proportional to the product of  $E_{\text{max}}$  and  $B_{\text{max}}$ .

**Electromagnetic Momentum Flow and Radiation Pressure**

We've shown that electromagnetic waves transport energy. It can also be shown that electromagnetic waves carry *momentum*  $p$ , with a corresponding momentum density (momentum  $dp$  per volume  $dV$ ) of magnitude

$$\frac{dp}{dV} = \frac{EB}{\mu_0 c^2} = \frac{S}{c^2} \quad (32.30)$$

This momentum is a property of the field; it is not associated with the mass of a moving particle in the usual sense.

There is also a corresponding momentum flow rate. The volume  $dV$  occupied by an electromagnetic wave (speed  $c$ ) that passes through an area  $A$  in time  $dt$  is  $dV = Ac dt$ . When we substitute this into Eq. (32.30) and rearrange, we find that the momentum flow rate per unit area is

$$\text{Flow rate of electromagnetic momentum} = \frac{1}{A} \frac{dp}{dt} = \frac{S}{c} = \frac{EB}{\mu_0 c} \quad (32.31)$$

Poynting vector magnitude
Electric-field magnitude
Magnetic-field magnitude

Momentum transferred per unit surface area per unit time
Speed of light in vacuum
Magnetic constant

We obtain the *average* rate of momentum transfer per unit area by replacing  $S$  in Eq. (32.31) by  $S_{\text{av}} = I$ .

This momentum is responsible for **radiation pressure**. When an electromagnetic wave is completely absorbed by a surface, the wave's momentum is also transferred to the surface. For simplicity we'll consider a surface perpendicular to the propagation direction. Using the ideas developed in Section 8.1, we see that the rate  $dp/dt$  at which momentum is transferred to the absorbing surface equals the *force* on the surface. The average force per unit area due to the wave, or *radiation pressure*  $p_{\text{rad}}$ , is the average value of  $dp/dt$  divided by the absorbing area  $A$ . (We use the subscript "rad" to distinguish pressure from momentum, for which the symbol  $p$  is also used.) From Eq. (32.31) the radiation pressure is

$$p_{\text{rad}} = \frac{S_{\text{av}}}{c} = \frac{I}{c} \quad (\text{radiation pressure, wave totally absorbed}) \quad (32.32)$$

If the wave is totally reflected, the momentum change is twice as great, and

$$p_{\text{rad}} = \frac{2S_{\text{av}}}{c} = \frac{2I}{c} \quad (\text{radiation pressure, wave totally reflected}) \quad (32.33)$$

For example, the value of  $I$  (or  $S_{\text{av}}$ ) for direct sunlight, before it passes through the earth's atmosphere, is approximately  $1.4 \text{ kW/m}^2$ . From Eq. (32.32) the corresponding average pressure on a completely absorbing surface is

$$p_{\text{rad}} = \frac{I}{c} = \frac{1.4 \times 10^3 \text{ W/m}^2}{3.0 \times 10^8 \text{ m/s}} = 4.7 \times 10^{-6} \text{ Pa}$$

From Eq. (32.33) the average pressure on a totally *reflecting* surface is twice this:  $2I/c$  or  $9.4 \times 10^{-6} \text{ Pa}$ . These are very small pressures, of the order of  $10^{-10} \text{ atm}$ , but they can be measured with sensitive instruments.

The radiation pressure of sunlight is much greater *inside* the sun than at the earth (see Problem 32.35). Inside stars that are much more massive and luminous than the sun, radiation pressure is so great that it substantially augments the gas pressure within the star and so helps to prevent the star from collapsing under its own gravity. In some cases the radiation pressure of stars can have dramatic effects on the material surrounding them (**Fig. 32.20**).

Figure 32.20 At the center of this interstellar gas cloud is a group of intensely luminous stars that exert tremendous radiation pressure on their surroundings. Aided by a "wind" of particles emanating from the stars, over the past million years the radiation pressure has carved out a bubble within the cloud 70 light-years across.

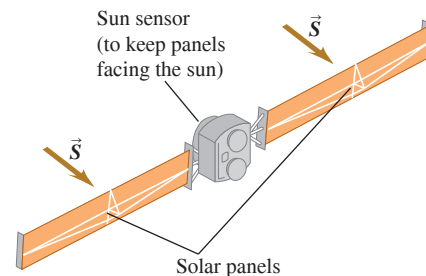


### EXAMPLE 32.5 Power and pressure from sunlight

An earth-orbiting satellite has solar energy-collecting panels with a total area of  $4.0 \text{ m}^2$  (**Fig. 32.21**). If the sun's radiation is perpendicular to the panels and is completely absorbed, find the average solar power absorbed and the average radiation-pressure force.

**IDENTIFY and SET UP** This problem uses the relationships among intensity, power, radiation pressure, and force. In the previous discussion, we used the intensity  $I$  (average power per unit area) of sunlight to find the radiation pressure  $p_{\text{rad}}$  (force per unit area) of sunlight on a completely absorbing surface. (These values are for points above the atmosphere, which is where the satellite orbits.) Multiplying each value by the area of the solar panels gives the average power absorbed and the net radiation force on the panels.

Figure 32.21 Solar panels on a satellite.



Continued

**EXECUTE** The intensity  $I$  (power per unit area) is  $1.4 \times 10^3 \text{ W/m}^2$ . Although the light from the sun is not a simple sinusoidal wave, we can still use the relationship that the average power  $P$  is the intensity  $I$  times the area  $A$ :

$$P = IA = (1.4 \times 10^3 \text{ W/m}^2)(4.0 \text{ m}^2) = 5.6 \times 10^3 \text{ W} = 5.6 \text{ kW}$$

The radiation pressure of sunlight on an absorbing surface is  $p_{\text{rad}} = 4.7 \times 10^{-6} \text{ Pa} = 4.7 \times 10^{-6} \text{ N/m}^2$ . The total force  $F$  is the pressure  $p_{\text{rad}}$  times the area  $A$ :

$$F = p_{\text{rad}}A = (4.7 \times 10^{-6} \text{ N/m}^2)(4.0 \text{ m}^2) = 1.9 \times 10^{-5} \text{ N}$$

**EVALUATE** The absorbed power is quite substantial. Part of it can be used to power the equipment aboard the satellite; the rest goes into heating the panels, either directly or due to inefficiencies in the photocells contained in the panels.

The total radiation force is comparable to the weight (on the earth) of a single grain of salt. Over time, however, this small force can noticeably affect the orbit of a satellite like that in Fig. 32.21, and so radiation pressure must be taken into account.

**KEYCONCEPT** An electromagnetic wave carries both energy and momentum. As a result an electromagnetic wave exerts pressure on any surface that either absorbs or reflects the wave.

**TEST YOUR UNDERSTANDING OF SECTION 32.4** Figure 32.13 shows one wavelength of a sinusoidal electromagnetic wave at time  $t = 0$ . For which of the following four values of  $x$  is (a) the energy density a maximum; (b) the energy density a minimum; (c) the magnitude of the instantaneous (not average) Poynting vector a maximum; (d) the magnitude of the instantaneous (not average) Poynting vector a minimum? (i)  $x = 0$ ; (ii)  $x = \lambda/4$ ; (iii)  $x = \lambda/2$ ; (iv)  $x = 3\lambda/4$ .

### ANSWER

From Fig. 32.13, this occurs at  $x = \lambda/4$  and  $x = 3\lambda/4$ .  
 (a) (i) and (iii), (b) (ii) and (iv), (c) (i) and (iii), (d) (ii) and (iv). Both the energy density  $u$  and the Poynting vector magnitude  $S$  are maximum where the  $\vec{E}$  and  $\vec{B}$  fields have their maximum magnitudes. (The directions of the fields don't matter.) From Fig. 32.13, this occurs at  $x = 0$  and  $x = \lambda/2$ . Both  $u$  and  $S$  have a minimum value of zero; that occurs where  $\vec{E}$  and  $\vec{B}$  are both zero.

## 32.5 STANDING ELECTROMAGNETIC WAVES

Electromagnetic waves can be *reflected* by the surface of a conductor (like a polished sheet of metal) or of a dielectric (such as a sheet of glass). The superposition of an incident wave and a reflected wave forms a **standing wave**. The situation is analogous to standing waves on a stretched string, discussed in Section 15.7.

Suppose a sheet of a perfect conductor (zero resistivity) is placed in the  $yz$ -plane of Fig. 32.22 and a linearly polarized electromagnetic wave, traveling in the negative  $x$ -direction, strikes it. As we discussed in Section 23.4,  $\vec{E}$  cannot have a component parallel to the surface of a perfect conductor. Therefore in the present situation,  $\vec{E}$  must be zero everywhere in the  $yz$ -plane. The electric field of the *incident* electromagnetic wave is *not* zero at all times in the  $yz$ -plane. But this incident wave induces oscillating currents on the surface of the conductor, and these currents give rise to an additional electric field. The *net* electric field, which is the vector sum of this field and the incident  $\vec{E}$ , is zero everywhere inside and on the surface of the conductor.

The currents induced on the surface of the conductor also produce a *reflected* wave that travels out from the plane in the  $+x$ -direction. Suppose the incident wave is described by the wave functions of Eqs. (32.19) (a sinusoidal wave traveling in the  $-x$ -direction) and the reflected wave by the negative of Eqs. (32.16) (a sinusoidal wave traveling in the  $+x$ -direction). We take the *negative* of the wave given by Eqs. (32.16) so that the incident and reflected electric fields cancel at  $x = 0$  (the plane of the conductor, where the total electric field must be zero). The superposition principle states that the total  $\vec{E}$  field at any point is the vector sum of the  $\vec{E}$  fields of the incident and reflected waves, and similarly for the  $\vec{B}$  field. Therefore the wave functions for the superposition of the two waves are

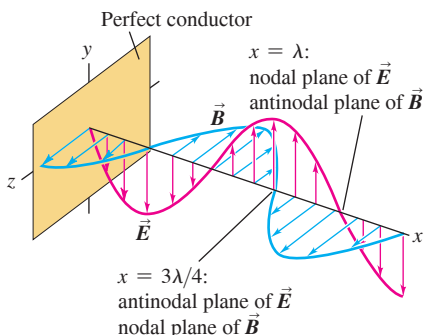
$$E_y(x, t) = E_{\text{max}}[\cos(kx + \omega t) - \cos(kx - \omega t)]$$

$$B_z(x, t) = B_{\text{max}}[-\cos(kx + \omega t) - \cos(kx - \omega t)]$$

We can expand and simplify these expressions by using the identities

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

Figure 32.22 Representation of the electric and magnetic fields of a linearly polarized electromagnetic standing wave when  $\omega t = 3\pi/4$  rad. In any plane perpendicular to the  $x$ -axis,  $E$  is maximum (an antinode) where  $B$  is zero (a node), and vice versa. As time elapses, the pattern does *not* move along the  $x$ -axis; instead, at every point the  $\vec{E}$  and  $\vec{B}$  vectors simply oscillate.



The results are

$$E_y(x, t) = -2E_{\max} \sin kx \sin \omega t \quad (32.34)$$

$$B_z(x, t) = -2B_{\max} \cos kx \cos \omega t \quad (32.35)$$

Equation (32.34) is analogous to Eq. (15.28) for a stretched string. We see that at  $x = 0$  the electric field  $E_y(x = 0, t)$  is *always* zero; this is required by the nature of the ideal conductor, which plays the same role as a fixed point at the end of a string. Furthermore,  $E_y(x, t)$  is zero at *all* times at points in those planes perpendicular to the  $x$ -axis for which  $\sin kx = 0$ —that is,  $kx = 0, \pi, 2\pi, \dots$ . Since  $k = 2\pi/\lambda$ , the positions of these planes are

$$x = 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, \dots \quad (\text{nodal planes of } \vec{E}) \quad (32.36)$$

These planes are called the **nodal planes** of the  $\vec{E}$  field; they are the equivalent of the nodes, or nodal points, of a standing wave on a string. Midway between any two adjacent nodal planes is a plane on which  $\sin kx = \pm 1$ ; on each such plane, the magnitude of  $E(x, t)$  equals the maximum possible value of  $2E_{\max}$  twice per oscillation cycle. These are the **antinodal planes** of  $\vec{E}$ , corresponding to the antinodes of waves on a string.

The total magnetic field is zero at all times at points in planes on which  $\cos kx = 0$ . These are the nodal planes of  $\vec{B}$ , and they occur where

$$x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \dots \quad (\text{nodal planes of } \vec{B}) \quad (32.37)$$

There is an antinodal plane of  $\vec{B}$  midway between any two adjacent nodal planes.

Figure 32.22 shows a standing-wave pattern at one instant of time. The magnetic field is *not* zero at the conducting surface ( $x = 0$ ). The surface currents that must be present to make  $\vec{E}$  exactly zero at the surface cause magnetic fields at the surface. The nodal planes of each field are separated by one half-wavelength. The nodal planes of  $\vec{E}$  are midway between those of  $\vec{B}$ , and vice versa; hence the nodes of  $\vec{E}$  coincide with the antinodes of  $\vec{B}$ , and conversely. Compare this situation to the distinction between pressure nodes and displacement nodes in Section 16.4.

The total electric field is a *sine* function of  $t$ , and the total magnetic field is a *cosine* function of  $t$ . The sinusoidal variations of the two fields are therefore  $90^\circ$  out of phase at each point. At times when  $\sin \omega t = 0$ , the electric field is zero *everywhere*, and the magnetic field is maximum. When  $\cos \omega t = 0$ , the magnetic field is zero everywhere, and the electric field is maximum. This is in contrast to a wave traveling in one direction, as described by Eqs. (32.16) or (32.19) separately, in which the sinusoidal variations of  $\vec{E}$  and  $\vec{B}$  at any particular point are *in phase*. You can show that Eqs. (32.34) and (32.35) satisfy the wave equation, Eq. (32.15). You can also show that they satisfy Eqs. (32.12) and (32.14), the equivalents of Faraday's and Ampere's laws.

## Standing Waves in a Cavity

Let's now insert a second conducting plane, parallel to the first and a distance  $L$  from it, along the  $+x$ -axis. The cavity between the two planes is analogous to a stretched string held at the points  $x = 0$  and  $x = L$ . Both conducting planes must be nodal planes for  $\vec{E}$ ; a standing wave can exist only when the second plane is placed at one of the positions where  $E(x, t) = 0$ , so  $L$  must be an integer multiple of  $\lambda/2$ . The wavelengths that satisfy this condition are

$$\lambda_n = \frac{2L}{n} \quad (n = 1, 2, 3, \dots) \quad (32.38)$$



Figure 32.23 A typical microwave oven sets up a standing electromagnetic wave with  $\lambda = 12.2$  cm, a wavelength that is strongly absorbed by the water in food. Because the wave has nodes spaced  $\lambda/2 = 6.1$  cm apart, the food must be rotated while cooking. Otherwise, the portion that lies at a node—where the electric-field amplitude is zero—will remain cold.



The corresponding frequencies are

$$f_n = \frac{c}{\lambda_n} = n \frac{c}{2L} \quad (n = 1, 2, 3, \dots) \quad (32.39)$$

Thus there is a set of *normal modes*, each with a characteristic frequency, wave shape, and node pattern (Fig. 32.23). By measuring the node positions, we can measure the wavelength. If the frequency is known, the wave speed can be determined. This technique was first used by Hertz in the 1880s in his pioneering investigations of electromagnetic waves.

Conducting surfaces are not the only reflectors of electromagnetic waves. Reflections also occur at an interface between two insulating materials with different dielectric or magnetic properties. The mechanical analog is a junction of two strings with equal tension but different linear mass density. In general, a wave incident on such a boundary surface is partly transmitted into the second material and partly reflected back into the first. For example, light is transmitted through a glass window, but its surfaces also reflect light.

### EXAMPLE 32.6 Intensity in a standing wave

### WITH VARIATION PROBLEMS

Calculate the intensity of the standing wave represented by Eqs. (32.34) and (32.35).

**IDENTIFY and SET UP** The intensity  $I$  of the wave is the time-averaged value  $S_{\text{av}}$  of the magnitude of the Poynting vector  $\vec{S}$ . To find  $S_{\text{av}}$ , we first use Eq. (32.28) to find the instantaneous value of  $\vec{S}$  and then average it over a whole number of cycles of the wave.

**EXECUTE** Using the wave functions of Eqs. (32.34) and (32.35) in Eq. (32.28) for the Poynting vector  $\vec{S}$ , we find

$$\begin{aligned} \vec{S}(x, t) &= \frac{1}{\mu_0} \vec{E}(x, t) \times \vec{B}(x, t) \\ &= \frac{1}{\mu_0} [-2\hat{j}E_{\text{max}} \sin kx \sin \omega t] \times [-2\hat{k}B_{\text{max}} \cos kx \cos \omega t] \\ &= \hat{i} \frac{E_{\text{max}}B_{\text{max}}}{\mu_0} (2 \sin kx \cos kx)(2 \sin \omega t \cos \omega t) = \hat{i} S_x(x, t) \end{aligned}$$

Using the identity  $\sin 2A = 2 \sin A \cos A$ , we can rewrite  $S_x(x, t)$  as

$$S_x(x, t) = \frac{E_{\text{max}}B_{\text{max}} \sin 2kx \sin 2\omega t}{\mu_0}$$

The average value of a sine function over any whole number of cycles is zero. Thus *the time average of  $\vec{S}$  at any point is zero;  $I = S_{\text{av}} = 0$ .*

**EVALUATE** This result is what we should expect. The standing wave is a superposition of two waves with the same frequency and amplitude, traveling in opposite directions. All the energy transferred by one wave is cancelled by an equal amount transferred in the opposite direction by the other wave. When we use electromagnetic waves to transmit power, it is important to avoid reflections that give rise to standing waves.

**KEYCONCEPT** While there is energy flow in an electromagnetic standing wave, the intensity (the magnitude of the average Poynting vector) is zero at any point.

### EXAMPLE 32.7 Standing waves in a cavity

### WITH VARIATION PROBLEMS

Electromagnetic standing waves are set up in a cavity with two parallel, highly conducting walls 1.50 cm apart. (a) Calculate the longest wavelength  $\lambda$  and lowest frequency  $f$  of these standing waves. (b) For a standing wave of this wavelength, where in the cavity does  $\vec{E}$  have maximum magnitude? Where is  $\vec{E}$  zero? Where does  $\vec{B}$  have maximum magnitude? Where is  $\vec{B}$  zero?

**IDENTIFY and SET UP** Only certain normal modes are possible for electromagnetic waves in a cavity, just as only certain normal modes are possible for standing waves on a string. The longest possible wavelength and lowest possible frequency correspond to the  $n = 1$  mode in Eqs. (32.38) and (32.39); we use these to find  $\lambda$  and  $f$ . Equations (32.36) and (32.37) then give the locations of the nodal planes of  $\vec{E}$  and  $\vec{B}$ . The antinodal planes of each field are midway between adjacent nodal planes.

**EXECUTE** (a) From Eqs. (32.38) and (32.39), the  $n = 1$  wavelength and frequency are

$$\begin{aligned} \lambda_1 &= 2L = 2(1.50 \text{ cm}) = 3.00 \text{ cm} \\ f_1 &= \frac{c}{\lambda_1} = \frac{3.00 \times 10^8 \text{ m/s}}{2(1.50 \times 10^{-2} \text{ m})} = 1.00 \times 10^{10} \text{ Hz} = 10 \text{ GHz} \end{aligned}$$

(b) With  $n = 1$  there is a single half-wavelength between the walls. The electric field has nodal planes ( $\vec{E} = 0$ ) at the walls and an antinodal plane (where  $\vec{E}$  has its maximum magnitude) midway between them. The magnetic field has *antinodal* planes at the walls and a nodal plane midway between them.

**EVALUATE** One application of such standing waves is to produce an oscillating  $\vec{E}$  field of definite frequency, which is used to probe the behavior of a small sample of material placed in the cavity. To subject the sample to the strongest possible field, it should be placed near the center of the cavity, at the antinode of  $\vec{E}$ .

**KEYCONCEPT** In a sinusoidal electromagnetic standing wave, the electric field  $\vec{E}$  and magnetic field  $\vec{B}$  are  $90^\circ$  out of phase with each other. The antinodal planes of  $\vec{E}$  are the nodal planes of  $\vec{B}$ , and the antinodal planes of  $\vec{B}$  are the nodal planes of  $\vec{E}$ .

**TEST YOUR UNDERSTANDING OF SECTION 32.5** In the standing wave described in Example 32.7, is there any point in the cavity where the energy density is zero at all times? If so, where? If not, why not?

**ANSWER**

**no** There are places where  $\vec{E} = 0$  at all times (at the walls) and the electric energy density  $\frac{1}{2}\epsilon_0 E^2$  is always zero. There are also places where  $\vec{B} = 0$  at all times (on the plane midway between the walls) and the magnetic energy density  $\frac{1}{2}\mu_0 B^2$  is always zero. However, there are *no* locations where both  $\vec{E}$  and  $\vec{B}$  are always zero. Hence the energy density at any point in the standing wave is always nonzero.

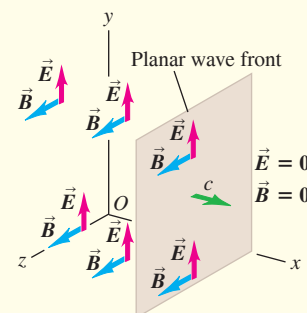
## CHAPTER 32 SUMMARY

**Maxwell's equations and electromagnetic waves:** Maxwell's equations predict the existence of electromagnetic waves that propagate in vacuum at the speed of light,  $c$ . The electromagnetic spectrum covers frequencies from at least 1 to  $10^{24}$  Hz and a correspondingly broad range of wavelengths. Visible light, with wavelengths from 380 to 750 nm, is a very small part of this spectrum. In a plane wave,  $\vec{E}$  and  $\vec{B}$  are uniform over any plane perpendicular to the propagation direction. Faraday's law and Ampere's law give relationships between the magnitudes of  $\vec{E}$  and  $\vec{B}$ ; requiring that both relationships are satisfied gives an expression for  $c$  in terms of  $\epsilon_0$  and  $\mu_0$ . Electromagnetic waves are transverse; the  $\vec{E}$  and  $\vec{B}$  fields are perpendicular to the direction of propagation and to each other. The direction of propagation is the direction of  $\vec{E} \times \vec{B}$ .

$$E = cB \quad (32.4)$$

$$B = \epsilon_0 \mu_0 cE \quad (32.8)$$

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \quad (32.9)$$

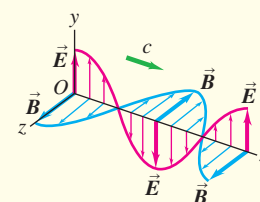


**Sinusoidal electromagnetic waves:** Equations (32.17) and (32.18) describe a sinusoidal plane electromagnetic wave traveling in vacuum in the  $+x$ -direction. If the wave is propagating in the  $-x$ -direction, replace  $kx - \omega t$  by  $kx + \omega t$ . (See Example 32.1.)

$$\vec{E}(x, t) = \hat{j}E_{\max} \cos(kx - \omega t) \quad (32.17)$$

$$\vec{B}(x, t) = \hat{k}B_{\max} \cos(kx - \omega t) \quad (32.18)$$

$$E_{\max} = cB_{\max}$$



**Electromagnetic waves in matter:** When an electromagnetic wave travels through a dielectric, the wave speed  $v$  is less than the speed of light in vacuum  $c$ . (See Example 32.2.)

$$v = \frac{1}{\sqrt{\epsilon\mu}} = \frac{1}{\sqrt{KK_m}} \frac{1}{\sqrt{\epsilon_0\mu_0}} \quad (32.21)$$

$$= \frac{c}{\sqrt{KK_m}}$$

**Energy and momentum in electromagnetic waves:** The energy flow rate (power per unit area) in an electromagnetic wave in vacuum is given by the Poynting vector  $\vec{S}$ . The magnitude of the time-averaged value of the Poynting vector is called the intensity  $I$  of the wave. Electromagnetic waves also carry momentum. When an electromagnetic wave strikes a surface, it exerts a radiation pressure  $p_{\text{rad}}$ . If the surface is perpendicular to the wave propagation direction and is totally absorbing,  $p_{\text{rad}} = I/c$ ; if the surface is a perfect reflector,  $p_{\text{rad}} = 2I/c$ . (See Examples 32.3–32.5.)

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} \quad (32.28)$$

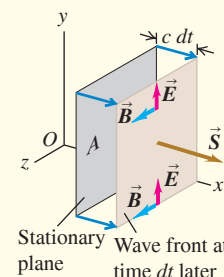
$$I = S_{\text{av}} = \frac{E_{\max} B_{\max}}{2\mu_0} = \frac{E_{\max}^2}{2\mu_0 c}$$

$$= \frac{1}{2} \sqrt{\frac{\epsilon_0}{\mu_0}} E_{\max}^2$$

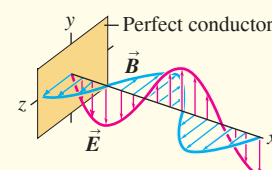
$$= \frac{1}{2} \epsilon_0 c E_{\max}^2 \quad (32.29)$$

$$\frac{1}{A} \frac{dp}{dt} = \frac{S}{c} = \frac{EB}{\mu_0 c} \quad (32.31)$$

(flow rate of electromagnetic momentum)



**Standing electromagnetic waves:** If a perfect reflecting surface is placed at  $x = 0$ , the incident and reflected waves form a standing wave. Nodal planes for  $\vec{E}$  occur at  $kx = 0, \pi, 2\pi, \dots$ , and nodal planes for  $\vec{B}$  at  $kx = \pi/2, 3\pi/2, 5\pi/2, \dots$ . At each point, the sinusoidal variations of  $\vec{E}$  and  $\vec{B}$  with time are  $90^\circ$  out of phase. (See Examples 32.6 and 32.7.)





## GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

## KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 32.1 and 32.2** (Section 32.3) before attempting these problems.

**VP32.2.1** A sinusoidal electromagnetic wave in vacuum has magnetic-field amplitude  $4.30 \times 10^{-3}$  T and wave number  $2.50 \times 10^6$  rad/m. At a certain position and time the electric field  $\vec{E}$  points in the  $-y$ -direction and the magnetic field  $\vec{B}$  points in the  $+x$ -direction. Find (a) the amplitude of  $\vec{E}$ , (b) the wavelength, (c) the frequency, and (d) the direction of wave propagation.

**VP32.2.2** The electric field in a sinusoidal electromagnetic wave in vacuum is given by  $\vec{E}(y,t) = \hat{i}(2.45 \times 10^6 \text{ V/m}) \cos[(6.50 \times 10^6 \text{ rad/m})y - \omega t]$ . Find (a) the wavelength, (b) the angular frequency  $\omega$ , and (c) the direction of wave propagation. (d) Write the wave function for the magnetic field.

**VP32.2.3** A dental laser emits a sinusoidal electromagnetic wave that propagates in vacuum in the positive  $x$ -direction with wavelength  $2.94 \mu\text{m}$ . At  $x = 0$  and  $t = 0$  the electric field  $\vec{E}$  points in the  $+y$ -direction and its magnitude equals the amplitude  $E_{\text{max}} = 4.20$  MV/m. Find the magnitude and direction of (a) the magnetic field  $\vec{B}$  at  $x = 0$  and  $t = 0$ , (b)  $\vec{E}$  at  $x = 1.85 \mu\text{m}$  and  $t = 0$ , and (c)  $\vec{B}$  at  $x = 1.85 \mu\text{m}$  and  $t = 0$ .

**VP32.2.4** The light from a red laser pointer has wavelength  $6.35 \times 10^{-7}$  m in vacuum. When this light is sent into a transparent material with relative permeability 1.00, its wavelength decreases to  $3.47 \times 10^{-7}$  m. Find (a) the frequency of the light in vacuum, (b) the frequency of the light in the material, (c) the speed of the light in the material, and (d) the dielectric constant of the material.

Be sure to review **EXAMPLES 32.3 and 32.4** (Section 32.4) before attempting these problems.

**VP32.4.1** A nonsinusoidal electromagnetic wave like that described in Section 32.2 has uniform electric and magnetic fields. The magnitude of the Poynting vector for this wave is  $11.0 \text{ W/m}^2$ . Find (a) the magnitudes of the electric and magnetic fields and (b) the energy density in the wave.

**VP32.4.2** A sinusoidal electromagnetic wave in vacuum is given by the wave functions in Eqs. (32.17). Find the Poynting vector at (a)  $x = 0$ ,  $t = 0$ ; (b)  $x = \lambda/4$ ,  $t = 0$ ; (c)  $x = \lambda/4$ ,  $t = \pi/4\omega$ .

**VP32.4.3** A transmitter on the earth's surface radiates sinusoidal radio waves equally in all directions above the ground. An airplane flying directly over the radio station at an altitude of 12.5 km measures the wave from the station to have electric-field amplitude 0.360 V/m. Find (a) the magnetic-field amplitude and intensity measured by the airplane and (b) the total power emitted by the transmitter.

**VP32.4.4** The sun emits light (which we can regard as a sinusoidal wave) equally in all directions. The distance from the sun to the earth is  $1.50 \times 10^{11}$  m, and the intensity of the sunlight that reaches the top of earth's atmosphere is  $1.36 \text{ kW/m}^2$ . Find (a) the amplitudes of the electric field and magnetic field at the top of the atmosphere and (b) the total power radiated by the sun.

Be sure to review **EXAMPLES 32.6 and 32.7** (Section 32.5) before attempting these problems.

**VP32.7.1** For the electromagnetic standing wave represented by Eqs. (32.34) and (32.35), find (a) the maximum magnitude of the Poynting vector anywhere in the standing wave; (b) the Poynting vector at  $x = \lambda/8$ ,  $t = 0$ ; (c) the Poynting vector at  $x = \lambda/8$ ,  $t = \pi/3\omega$ ; (d) the Poynting vector at  $x = \lambda/8$ ,  $t = 3\pi/4\omega$ .

**VP32.7.2** In a linearly polarized electromagnetic standing wave like that shown in Fig. 32.22, the amplitude of the magnetic field in the plane of the conductor is  $1.20 \times 10^{-7}$  T. The nodal plane of the magnetic field closest to the conductor is 3.60 mm from the conductor. Find (a) the wavelength, (b) the frequency, and (c) the amplitude of the electric field in the first nodal plane of the magnetic field.

**VP32.7.3** You set up electromagnetic standing waves in a cavity that has two parallel conducting walls, one at  $x = 0$  and one at  $x = 4.56$  cm. Find (a) the three lowest standing-wave frequencies and their corresponding wavelengths, and (b) the positions of the nodal planes of the electric field for each of these frequencies.

**VP32.7.4** A microwave oven sets up a standing wave of wavelength 12.2 cm between two parallel conducting walls 48.8 cm apart. Find (a) the wave frequency and (b) the number of antinodal planes of the electric field between the walls.

## BRIDGING PROBLEM Detecting Electromagnetic Waves

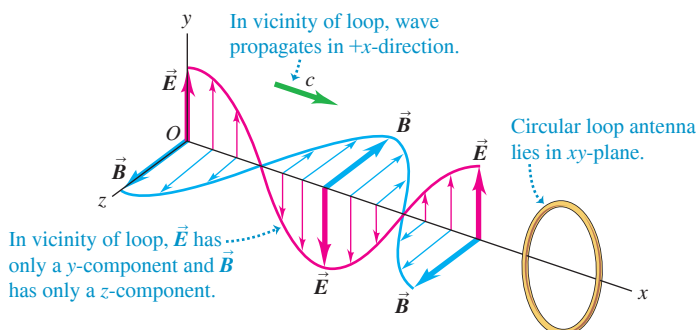
A circular loop of wire can be used as a radio antenna. If an 18.0-cm-diameter antenna is located 2.50 km from a 95.0 MHz source with a total power of 55.0 kW, what is the maximum emf induced in the loop? The orientation of the antenna loop and the polarization of the wave are as shown in Fig. 32.24. Assume that the source radiates uniformly in all directions.

## SOLUTION GUIDE

## IDENTIFY and SET UP

1. The plane of the antenna loop is perpendicular to the direction of the wave's oscillating magnetic field. This causes a magnetic flux through the loop that varies sinusoidally with time. By Faraday's law, this produces an emf equal in magnitude to the rate of change of the flux. The target variable is the magnitude of this emf.

Figure 32.24 Using a circular loop antenna to detect radio waves.



2. Select the equations that you'll need to find (i) the intensity of the wave at the position of the loop, a distance  $r = 2.50$  km from the source of power  $P = 55.0$  kW; (ii) the amplitude of the sinusoidally varying magnetic field at that position; (iii) the magnetic flux through the loop as a function of time; and (iv) the emf produced by the flux.

## EXECUTE

3. Find the wave intensity at the position of the loop.

4. Use your result from step 3 to write expressions for the time-dependent magnetic field at this position and the time-dependent magnetic flux through the loop.
5. Use the results of step 4 to find the time-dependent induced emf in the loop. The amplitude of this emf is your target variable.

## EVALUATE

6. Is the induced emf large enough to detect? (If it is, a receiver connected to this antenna will pick up signals from the source.)

## PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

## DISCUSSION QUESTIONS

**Q32.1** By measuring the electric and magnetic fields at a point in space where there is an electromagnetic wave, can you determine the direction from which the wave came? Explain.

**Q32.2** When driving on the upper level of the Bay Bridge, westbound from Oakland to San Francisco, you can easily pick up a number of radio stations on your car radio. But when driving eastbound on the lower level of the bridge, which has steel girders on either side to support the upper level, the radio reception is much worse. Why is there a difference?

**Q32.3** Give several examples of electromagnetic waves that are encountered in everyday life. How are they all alike? How do they differ?

**Q32.4** Sometimes neon signs located near a powerful radio station are seen to glow faintly at night, even though they are not turned on. What is happening?

**Q32.5** Is polarization a property of all electromagnetic waves, or is it unique to visible light? Can sound waves be polarized? What fundamental distinction in wave properties is involved? Explain.

**Q32.6** Suppose that a positive point charge  $q$  is initially at rest on the  $x$ -axis, in the path of the electromagnetic plane wave described in Section 32.2. Will the charge move after the wave front reaches it? If not, why not? If the charge does move, describe its motion qualitatively. (Remember that  $\vec{E}$  and  $\vec{B}$  have the same value at all points behind the wave front.)

**Q32.7** The light beam from a searchlight may have an electric-field magnitude of  $1000$  V/m, corresponding to a potential difference of  $1500$  V between the head and feet of a  $1.5$ -m-tall person on whom the light shines. Does this cause the person to feel a strong electric shock? Why or why not?

**Q32.8** For a certain sinusoidal wave of intensity  $I$ , the amplitude of the magnetic field is  $B$ . What would be the amplitude (in terms of  $B$ ) in a similar wave of twice the intensity?

**Q32.9** The magnetic-field amplitude of the electromagnetic wave from the laser described in Example 32.1 (Section 32.3) is about 100 times greater than the earth's magnetic field. If you illuminate a compass with the light from this laser, would you expect the compass to deflect? Why or why not?

**Q32.10** Most automobiles have vertical antennas for receiving radio broadcasts. Explain what this tells you about the direction of polarization of  $\vec{E}$  in the radio waves used in broadcasting.

**Q32.11** If a light beam carries momentum, should a person holding a flashlight feel a recoil analogous to the recoil of a rifle when it is fired? Why is this recoil not actually observed?

**Q32.12** A light source radiates a sinusoidal electromagnetic wave uniformly in all directions. This wave exerts an average pressure  $p$  on a perfectly reflecting surface a distance  $R$  away from it. What average pressure (in terms of  $p$ ) would this wave exert on a perfectly absorbing surface that was twice as far from the source?

**Q32.13** Does an electromagnetic *standing* wave have energy? Does it have momentum? Are your answers to these questions the same as for a *traveling* wave? Why or why not?

## EXERCISES

## Section 32.2 Plane Electromagnetic Waves and the Speed of Light

**32.1** • (a) How much time does it take light to travel from the moon to the earth, a distance of  $384,000$  km? (b) Light from the star Sirius takes  $8.61$  years to reach the earth. What is the distance from earth to Sirius in kilometers?

**32.2** • Consider each of the electric- and magnetic-field orientations given next. In each case, what is the direction of propagation of the wave? (a)  $\vec{E}$  in the  $+x$ -direction,  $\vec{B}$  in the  $+y$ -direction; (b)  $\vec{E}$  in the  $-y$ -direction,  $\vec{B}$  in the  $+x$ -direction; (c)  $\vec{E}$  in the  $+z$ -direction,  $\vec{B}$  in the  $-x$ -direction; (d)  $\vec{E}$  in the  $+y$ -direction,  $\vec{B}$  in the  $-z$ -direction.

**32.3** • A sinusoidal electromagnetic wave is propagating in vacuum in the  $+z$ -direction. If at a particular instant and at a certain point in space the electric field is in the  $+x$ -direction and has magnitude  $4.00$  V/m, what are the magnitude and direction of the magnetic field of the wave at this same point in space and instant in time?

**32.4** • Consider each of the following electric- and magnetic-field orientations. In each case, what is the direction of propagation of the wave? (a)  $\vec{E} = E\hat{i}$ ,  $\vec{B} = -B\hat{j}$ ; (b)  $\vec{E} = E\hat{j}$ ,  $\vec{B} = B\hat{i}$ ; (c)  $\vec{E} = -E\hat{k}$ ,  $\vec{B} = -B\hat{i}$ ; (d)  $\vec{E} = E\hat{i}$ ,  $\vec{B} = -B\hat{k}$ .

## Section 32.3 Sinusoidal Electromagnetic Waves

**32.5** • **BIO Medical X Rays.** Medical x rays are taken with electromagnetic waves having a wavelength of around  $0.10$  nm in air. What are the frequency, period, and wave number of such waves?

**32.6** • **BIO Ultraviolet Radiation.** There are two categories of ultraviolet light. Ultraviolet A (UVA) has a wavelength ranging from  $320$  nm to  $400$  nm. It is necessary for the production of vitamin D. UVB, with a wavelength in vacuum between  $280$  nm and  $320$  nm, is more dangerous because it is much more likely to cause skin cancer. (a) Find the frequency ranges of UVA and UVB. (b) What are the ranges of the wave numbers for UVA and UVB?

**32.7** • Consider electromagnetic waves propagating in air. (a) Determine the frequency of a wave with a wavelength of (i)  $5.0$  km, (ii)  $5.0$   $\mu\text{m}$ , (iii)  $5.0$  nm. (b) What is the wavelength (in meters and nanometers) of (i) gamma rays of frequency  $6.50 \times 10^{21}$  Hz and (ii) an AM station radio wave of frequency  $590$  kHz?



**32.8 •** An electromagnetic wave of wavelength 435 nm is traveling in vacuum in the  $-z$ -direction. The electric field has amplitude  $2.70 \times 10^{-3}$  V/m and is parallel to the  $x$ -axis. What are (a) the frequency and (b) the magnetic-field amplitude? (c) Write the vector equations for  $\vec{E}(z, t)$  and  $\vec{B}(z, t)$ .

**32.9 •** An electromagnetic wave has an electric field given by  $\vec{E}(y, t) = (3.10 \times 10^5 \text{ V/m})\hat{k} \cos[ky - (12.65 \times 10^{12} \text{ rad/s})t]$ . (a) In which direction is the wave traveling? (b) What is the wavelength of the wave? (c) Write the vector equation for  $\vec{B}(y, t)$ .

**32.10 •** The electric field of a sinusoidal electromagnetic wave obeys the equation  $E = (375 \text{ V/m}) \cos[(1.99 \times 10^7 \text{ rad/m})x + (5.97 \times 10^{15} \text{ rad/s})t]$ . (a) What is the speed of the wave? (b) What are the amplitudes of the electric and magnetic fields of this wave? (c) What are the frequency, wavelength, and period of the wave? Is this light visible to humans?

**32.11 •** Radio station WCCO in Minneapolis broadcasts at a frequency of 830 kHz. At a point some distance from the transmitter, the magnetic-field amplitude of the electromagnetic wave from WCCO is  $4.82 \times 10^{-11}$  T. Calculate (a) the wavelength; (b) the wave number; (c) the angular frequency; (d) the electric-field amplitude.

**32.12 •** An electromagnetic wave with frequency 65.0 Hz travels in an insulating magnetic material that has dielectric constant 3.64 and relative permeability 5.18 at this frequency. The electric field has amplitude  $7.20 \times 10^{-3}$  V/m. (a) What is the speed of propagation of the wave? (b) What is the wavelength of the wave? (c) What is the amplitude of the magnetic field?

**32.13 •** An electromagnetic wave with frequency  $5.70 \times 10^{14}$  Hz propagates with a speed of  $2.17 \times 10^8$  m/s in a certain piece of glass. Find (a) the wavelength of the wave in the glass; (b) the wavelength of a wave of the same frequency propagating in air; (c) the index of refraction  $n$  of the glass for an electromagnetic wave with this frequency; (d) the dielectric constant for glass at this frequency, assuming that the relative permeability is unity.

### Section 32.4 Energy and Momentum in Electromagnetic Waves

**32.14 • BIO High-Energy Cancer Treatment.** Scientists are working on a new technique to kill cancer cells by zapping them with ultrahigh-energy (in the range of  $10^{12}$  W) pulses of light that last for an extremely short time (a few nanoseconds). These short pulses scramble the interior of a cell without causing it to explode, as long pulses would do. We can model a typical such cell as a disk  $5.0 \mu\text{m}$  in diameter, with the pulse lasting for 4.0 ns with an average power of  $2.0 \times 10^{12}$  W. We shall assume that the energy is spread uniformly over the faces of 100 cells for each pulse. (a) How much energy is given to the cell during this pulse? (b) What is the intensity (in  $\text{W/m}^2$ ) delivered to the cell? (c) What are the maximum values of the electric and magnetic fields in the pulse?

**32.15 •• Fields from a Light Bulb.** We can reasonably model a 75 W incandescent light bulb as a sphere 6.0 cm in diameter. Typically, only about 5% of the energy goes to visible light; the rest goes largely to non-visible infrared radiation. (a) What is the visible-light intensity (in  $\text{W/m}^2$ ) at the surface of the bulb? (b) What are the amplitudes of the electric and magnetic fields at this surface, for a sinusoidal wave with this intensity?

**32.16 ••** A sinusoidal electromagnetic wave from a radio station passes perpendicularly through an open window that has area  $0.500 \text{ m}^2$ . At the window, the electric field of the wave has rms value  $0.0400 \text{ V/m}$ . How much energy does this wave carry through the window during a 30.0 s commercial?

**32.17 •** A space probe  $2.0 \times 10^{10}$  m from a star measures the total intensity of electromagnetic radiation from the star to be  $5.0 \times 10^3 \text{ W/m}^2$ . If the star radiates uniformly in all directions, what is its total average power output?

**32.18 ••** The energy flow to the earth from sunlight is about  $1.4 \text{ kW/m}^2$ . (a) Find the maximum values of the electric and magnetic fields for a sinusoidal wave of this intensity. (b) The distance from the earth to the sun is about  $1.5 \times 10^{11}$  m. Find the total power radiated by the sun.

**32.19 ••** A point source emits monochromatic electromagnetic waves into air uniformly in all directions. You measure the amplitude  $E_{\text{max}}$  of the electric field at several distances from the source. After graphing your results as  $E_{\text{max}}$  versus  $1/r$ , you find that the data lie close to a straight line that has slope  $75.0 \text{ N} \cdot \text{m/C}$ . What is the average power output of the source?

**32.20 •** A sinusoidal electromagnetic wave emitted by a mobile phone has a wavelength of 35.4 cm and an electric-field amplitude of  $5.40 \times 10^{-2}$  V/m at a distance of 250 m from the phone. Calculate (a) the frequency of the wave; (b) the magnetic-field amplitude; (c) the intensity of the wave.

**32.21 •** A monochromatic light source with power output 60.0 W radiates light of wavelength 700 nm uniformly in all directions. Calculate  $E_{\text{max}}$  and  $B_{\text{max}}$  for the 700 nm light at a distance of 5.00 m from the source.

**32.22 • Television Broadcasting.** Public television station KQED in San Francisco broadcasts a sinusoidal radio signal at a power of 777 kW. Assume that the wave spreads out uniformly into a hemisphere above the ground. At a home 5.00 km away from the antenna, (a) what average pressure does this wave exert on a totally reflecting surface, (b) what are the amplitudes of the electric and magnetic fields of the wave, and (c) what is the average density of the energy this wave carries? (d) For the energy density in part (c), what percentage is due to the electric field and what percentage is due to the magnetic field?

**32.23 •• BIO Laser Safety.** If the eye receives an average intensity greater than  $1.0 \times 10^2 \text{ W/m}^2$ , damage to the retina can occur. This quantity is called the *damage threshold* of the retina. (a) What is the largest average power (in mW) that a laser beam 1.5 mm in diameter can have and still be considered safe to view head-on? (b) What are the maximum values of the electric and magnetic fields for the beam in part (a)? (c) How much energy would the beam in part (a) deliver per second to the retina? (d) Express the damage threshold in  $\text{W/cm}^2$ .

**32.24 •** In the 25 ft Space Simulator facility at NASA's Jet Propulsion Laboratory, a bank of overhead arc lamps can produce light of intensity  $2500 \text{ W/m}^2$  at the floor of the facility. (This simulates the intensity of sunlight near the planet Venus.) Find the average radiation pressure (in pascals and in atmospheres) on (a) a totally absorbing section of the floor and (b) a totally reflecting section of the floor. (c) Find the average momentum density (momentum per unit volume) in the light at the floor.

**32.25 • Laboratory Lasers.** He-Ne lasers are often used in physics demonstrations. They produce light of wavelength 633 nm and a power of 0.500 mW spread over a cylindrical beam 1.00 mm in diameter (although these quantities can vary). (a) What is the intensity of this laser beam? (b) What are the maximum values of the electric and magnetic fields? (c) What is the average energy density in the laser beam?

**32.26 • CP** A totally reflecting disk has radius  $8.00 \mu\text{m}$  and average density  $600 \text{ kg/m}^3$ . A laser has an average power output  $P_{\text{av}}$  spread uniformly over a cylindrical beam of radius 2.00 mm. When the laser beam shines upward on the disk in a direction perpendicular to its flat surface, the radiation pressure produces a force equal to the weight of the disk. (a) What value of  $P_{\text{av}}$  is required? (b) What average laser power is required if the radius of the disk is doubled?

### Section 32.5 Standing Electromagnetic Waves

**32.27 • Microwave Oven.** The microwaves in a certain microwave oven have a wavelength of 12.2 cm. (a) How wide must this oven be so that it will contain five antinodal planes of the electric field along its width in the standing-wave pattern? (b) What is the frequency of these microwaves? (c) Suppose a manufacturing error occurred and the oven was made 5.0 cm longer than specified in part (a). In this case, what would have to be the frequency of the microwaves for there still to be five antinodal planes of the electric field along the width of the oven?

**32.28** • An electromagnetic standing wave in air has frequency 75.0 MHz. (a) What is the distance between nodal planes of the  $\vec{E}$  field? (b) What is the distance between a nodal plane of  $\vec{E}$  and the closest nodal plane of  $\vec{B}$ ?

**32.29** •• An air-filled cavity for producing electromagnetic standing waves has two parallel, highly conducting walls separated by a distance  $L$ . One standing-wave pattern in the cavity produces nodal planes of the electric field with a spacing of 1.50 cm. The next-higher-frequency standing wave in the cavity produces nodal planes with a spacing of 1.25 cm. What is the distance  $L$  between the walls of the cavity?

## PROBLEMS

**32.30** •• **CALC** Consider a sinusoidal electromagnetic wave with fields  $\vec{E} = E_{\max} \hat{j} \cos(kx - \omega t)$  and  $\vec{B} = B_{\max} \hat{k} \cos(kx - \omega t + \phi)$ , with  $-\pi \leq \phi \leq \pi$ . Show that if  $\vec{E}$  and  $\vec{B}$  are to satisfy Eqs. (32.12) and (32.14), then  $E_{\max} = cB_{\max}$  and  $\phi = 0$ . (The result  $\phi = 0$  means the  $\vec{E}$  and  $\vec{B}$  fields oscillate in phase.)

**32.31** •• **BIO Laser Surgery.** Very short pulses of high-intensity laser beams are used to repair detached portions of the retina of the eye. The brief pulses of energy absorbed by the retina weld the detached portions back into place. In one such procedure, a laser beam has a wavelength of 810 nm and delivers 250 mW of power spread over a circular spot 510  $\mu\text{m}$  in diameter. The vitreous humor (the transparent fluid that fills most of the eye) has an index of refraction of 1.34. (a) If the laser pulses are each 1.50 ms long, how much energy is delivered to the retina with each pulse? (b) What average pressure would the pulse of the laser beam exert at normal incidence on a surface in air if the beam is fully absorbed? (c) What are the wavelength and frequency of the laser light inside the vitreous humor of the eye? (d) What are the maximum values of the electric and magnetic fields in the laser beam?

**32.32** •• **CP** (a) Estimate how long it takes to bring 1 cup (237 mL) of room-temperature water to 100°C in a microwave oven. (b) How much heat is required to heat this water? (c) Divide the required heat by the time in part (a) to estimate the power supplied to the water by the microwaves. (d) If this energy were supplied by a plane electromagnetic wave incident from one direction, and if half of the wave energy was absorbed by the water, then the average wave intensity would be the twice power estimate divided by the cross-sectional area of the cup seen by the wave. Use this model to estimate the average intensity of the microwaves in the oven. (e) Use Eqs. (32.29) to estimate the amplitude of the electric field in a microwave oven.

**32.33** • A satellite 575 km above the earth's surface transmits sinusoidal electromagnetic waves of frequency 92.4 MHz uniformly in all directions, with a power of 25.0 kW. (a) What is the intensity of these waves as they reach a receiver at the surface of the earth directly below the satellite? (b) What are the amplitudes of the electric and magnetic fields at the receiver? (c) If the receiver has a totally absorbing panel measuring 15.0 cm by 40.0 cm oriented with its plane perpendicular to the direction the waves travel, what average force do these waves exert on the panel? Is this force large enough to cause significant effects?

**32.34** •• **CP** If we move a refrigerator magnet back and forth, we generate an electromagnetic wave that propagates away from us. Assume we move the magnet along the  $z$ -axis centered on the origin with an amplitude of 10 cm and with a frequency of 2.0 Hz. Consider a loop in the  $xy$ -plane centered on the origin with a radius of 2.0 cm. Assume that when the magnet is closest to the loop, the magnetic field within the loop has a spatially uniform value of 0.010 T. When the magnet has moved 10 cm in the direction away from the loop, the field in the loop becomes negligible. (a) Calculate the average rate at which the magnetic flux through the loop changes in time during each half-cycle of the motion of the magnet. (b) Use Faraday's law to estimate the average magnitude of the induced electric field at points on the loop. (c) Use Eq. (32.28) to estimate the average intensity of the electromagnetic wave in the  $xy$ -plane as it propagates radially away from the  $z$ -axis a distance of 2.0 cm. (d) Assume the

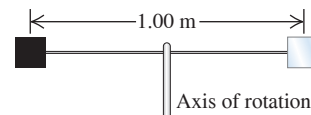
intensity has the same value along the surface of a cylinder centered on the  $z$ -axis that extends 5.0 cm above and 5.0 cm below the  $xy$ -plane, and is negligible above or below this. (We are crudely postulating that the wave is emitted perpendicular to the axis of the motion in a narrow band.) Use this assumption to estimate the total power dissipated by the wave.

**32.35** • The sun emits energy in the form of electromagnetic waves at a rate of  $3.9 \times 10^{26}$  W. This energy is produced by nuclear reactions deep in the sun's interior. (a) Find the intensity of electromagnetic radiation and the radiation pressure on an absorbing object at the surface of the sun (radius  $r = R = 6.96 \times 10^5$  km) and at  $r = R/2$ , in the sun's interior. Ignore any scattering of the waves as they move radially outward from the center of the sun. Compare to the values given in Section 32.4 for sunlight just before it enters the earth's atmosphere. (b) The gas pressure at the sun's surface is about  $1.0 \times 10^4$  Pa; at  $r = R/2$ , the gas pressure is calculated from solar models to be about  $4.7 \times 10^{13}$  Pa. Comparing with your results in part (a), would you expect that radiation pressure is an important factor in determining the structure of the sun? Why or why not?

**32.36** • A small helium-neon laser emits red visible light with a power of 5.80 mW in a beam of diameter 2.50 mm. (a) What are the amplitudes of the electric and magnetic fields of this light? (b) What are the average energy densities associated with the electric field and with the magnetic field? (c) What is the total energy contained in a 1.00 m length of the beam?

**32.37** •• **CP** Two square reflectors, each 1.50 cm on a side and of mass 4.00 g, are located at opposite ends of a thin, extremely light, 1.00 m rod that can rotate without friction and in vacuum about an axle perpendicular to it through its center (**Fig. P32.37**). These reflectors are small enough to be treated as point masses in moment-of-inertia calculations. Both reflectors are illuminated on one face by a sinusoidal light wave having an electric field of amplitude 1.25 N/C that falls uniformly on both surfaces and always strikes them perpendicular to the plane of their surfaces. One reflector is covered with a perfectly absorbing coating, and the other is covered with a perfectly reflecting coating. What is the angular acceleration of this device?

Figure P32.37



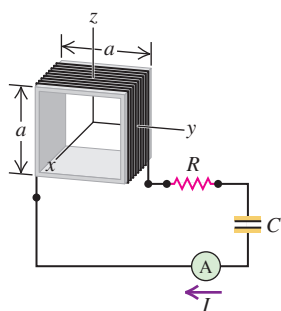
**32.38** •• A source of sinusoidal electromagnetic waves radiates uniformly in all directions. At a distance of 10.0 m from this source, the amplitude of the electric field is measured to be 3.50 N/C. What is the electric-field amplitude 20.0 cm from the source?

**32.39** • **CP CALC** A cylindrical conductor with a circular cross section has a radius  $a$  and a resistivity  $\rho$  and carries a constant current  $I$ . (a) What are the magnitude and direction of the electric-field vector  $\vec{E}$  at a point just inside the wire at a distance  $a$  from the axis? (b) What are the magnitude and direction of the magnetic-field vector  $\vec{B}$  at the same point? (c) What are the magnitude and direction of the Poynting vector  $\vec{S}$  at the same point? (The direction of  $\vec{S}$  is the direction in which electromagnetic energy flows into or out of the conductor.) (d) Use the result in part (c) to find the rate of flow of energy into the volume occupied by a length  $l$  of the conductor. (*Hint:* Integrate  $\vec{S}$  over the surface of this volume.) Compare your result to the rate of generation of thermal energy in the same volume. Discuss why the energy dissipated in a current-carrying conductor, due to its resistance, can be thought of as entering through the cylindrical sides of the conductor.

**32.40** •• **CP** A circular wire loop has a radius of 7.50 cm. A sinusoidal electromagnetic plane wave traveling in air passes through the loop, with the direction of the magnetic field of the wave perpendicular to the plane of the loop. The intensity of the wave at the location of the loop is  $0.0275 \text{ W/m}^2$ , and the wavelength of the wave is 6.90 m. What is the maximum emf induced in the loop?

**32.41 •** In a certain experiment, a radio transmitter emits sinusoidal electromagnetic waves of frequency 110.0 MHz in opposite directions inside a narrow cavity with reflectors at both ends, causing a standing-wave pattern to occur. (a) How far apart are the nodal planes of the magnetic field? (b) If the standing-wave pattern is determined to be in its eighth harmonic, how long is the cavity?

**32.42 •• CP CALC** An antenna is created by wrapping a square frame with side length  $a$  by a wire  $N$  times and then connecting the leads to a resistor  $R$  and a capacitor  $C$ , as shown in **Fig. P32.42**. The loop itself has self-inductance  $L$ . A plane electromagnetic wave with electric field  $\vec{E} = E_{\max} \cos(kz - \omega t) \hat{j}$  propagates in the  $+z$ -direction. The origin is at the center of the frame. (a) What is the magnetic flux through the coil in the direction of the  $+x$ -axis? (*Hint:* The identity  $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$  may prove helpful.) (b) What is the emf generated in the coil? (c) The electromagnetic wave has frequency 4.00 Mz and intensity 100 W/m<sup>2</sup>. The coil has  $N = 50$  windings and side length  $a = 10.0$  cm. It follows that its self-inductance is  $L = 78.0 \mu\text{H}$ . If the resistance in the circuit is  $R = 100 \Omega$ , what value of the capacitance  $C$  results in the resonance frequency of the  $L$ - $R$ - $C$  circuit being equal to the frequency of the wave? (d) What rms value of the current  $I_{\text{rms}}$  flows in that case?



**32.43 •• CP Global Positioning System (GPS).** The GPS network consists of 24 satellites, each of which makes two orbits around the earth per day. Each satellite transmits a 50.0 W (or even less) sinusoidal electromagnetic signal at two frequencies, one of which is 1575.42 MHz. Assume that a satellite transmits half of its power at each frequency and that the waves travel uniformly in a downward hemisphere. (a) What average intensity does a GPS receiver on the ground, directly below the satellite, receive? (*Hint:* First use Newton's laws to find the altitude of the satellite.) (b) What are the amplitudes of the electric and magnetic fields at the GPS receiver in part (a), and how long does it take the signal to reach the receiver? (c) If the receiver is a square panel 1.50 cm on a side that absorbs all of the beam, what average pressure does the signal exert on it? (d) What wavelength must the receiver be tuned to?

**32.44 •• CP Solar Sail.** NASA is giving serious consideration to the concept of *solar sailing*. A solar sailcraft uses a large, low-mass sail and the energy and momentum of sunlight for propulsion. (a) Should the sail be absorbing or reflective? Why? (b) The total power output of the sun is  $3.9 \times 10^{26}$  W. How large a sail is necessary to propel a 10,000 kg spacecraft against the gravitational force of the sun? Express your result in square kilometers. (c) Explain why your answer to part (b) is independent of the distance from the sun. (d) What wavelength must the receiver be tuned to?

**32.45 •• CP Interplanetary space** contains many small particles referred to as *interplanetary dust*. Radiation pressure from the sun sets a lower limit on the size of such dust particles. To see the origin of this limit, consider a spherical dust particle of radius  $R$  and mass density  $\rho$ . (a) Write an expression for the gravitational force exerted on this particle by the sun (mass  $M$ ) when the particle is a distance  $r$  from the sun. (b) Let  $L$  represent the luminosity of the sun, equal to the rate at which it emits energy in electromagnetic radiation. Find the force exerted on the (totally absorbing) particle due to solar radiation pressure, remembering that the intensity of the sun's radiation also depends on the distance  $r$ . The relevant area is the cross-sectional area of the particle, *not* the total surface area of the particle. As part of your answer, explain why this is so. (c) The mass density of a typical interplanetary dust particle is about 3000 kg/m<sup>3</sup>. Find the particle radius  $R$  such that the gravitational

and radiation forces acting on the particle are equal in magnitude. The luminosity of the sun is  $3.9 \times 10^{26}$  W. Does your answer depend on the distance of the particle from the sun? Why or why not? (d) Explain why dust particles with a radius less than that found in part (c) are unlikely to be found in the solar system. [*Hint:* Construct the ratio of the two force expressions found in parts (a) and (b).]

**32.46 •• DATA** The company where you work has obtained and stored five lasers in a supply room. You have been asked to determine the intensity of the electromagnetic radiation produced by each laser. The lasers are marked with specifications, but unfortunately different information is given for each laser:

Laser A: power = 2.6 W; diameter of cylindrical beam = 2.6 mm

Laser B: amplitude of electric field = 480 V/m

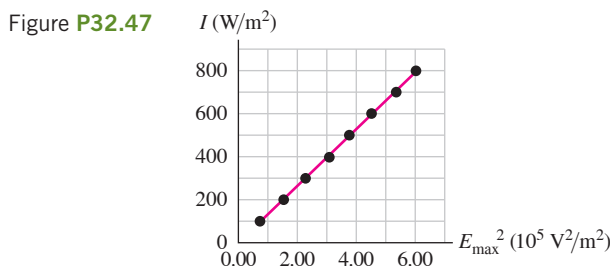
Laser C: amplitude of magnetic field =  $8.7 \times 10^{-6}$  T

Laser D: diameter of cylindrical beam = 1.8 mm; force on totally reflecting surface =  $6.0 \times 10^{-8}$  N

Laser E: average energy density in beam =  $3.0 \times 10^{-7}$  J/m<sup>3</sup>

Calculate the intensity for each laser, and rank the lasers in order of increasing intensity. Assume that the laser beams have uniform intensity distributions over their cross sections.

**32.47 •• DATA** Because the speed of light in vacuum (or air) has such a large value, it is very difficult to measure directly. To measure this speed, you conduct an experiment in which you measure the amplitude of the electric field in a laser beam as you change the intensity of the beam. **Figure P32.47** is a graph of the intensity  $I$  that you measured versus the square of the amplitude  $E_{\max}$  of the electric field. The best-fit straight line for your data has a slope of  $1.33 \times 10^{-3}$  J/(V<sup>2</sup>·s). (a) Explain why the data points plotted this way lie close to a straight line. (b) Use this graph to calculate the speed of light in air.



**32.48 •• DATA** As a physics lab instructor, you conduct an experiment on standing waves of microwaves, similar to the standing waves produced in a microwave oven. A transmitter emits microwaves of frequency  $f$ . The waves are reflected by a flat metal reflector, and a receiver measures the waves' electric-field amplitude as a function of position in the standing-wave pattern that is produced between the transmitter and reflector (**Fig. P32.48**). You measure the distance  $d$  between points of maximum amplitude (antinodes) of the electric field as a function of the frequency of the waves emitted by the transmitter. You obtain the data given in the table.

Figure P32.48



| $f$ (10 <sup>9</sup> Hz) | 1.0  | 1.5 | 2.0 | 2.5 | 3.0 | 3.5 | 4.0 | 5.0 | 6.0 | 8.0 |
|--------------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| $d$ (cm)                 | 15.2 | 9.7 | 7.7 | 5.8 | 5.2 | 4.1 | 3.8 | 3.1 | 2.3 | 1.7 |

Use the data to calculate  $c$ , the speed of the electromagnetic waves in air. Because each measured value has some experimental error, plot the data in such a way that the data points will lie close to a straight line, and use the slope of that straight line to calculate  $c$ .



**32.49 ••• CP** When electromagnetic radiation strikes perpendicular to a flat surface, a totally absorbing surface feels radiation pressure  $I_0/c$ , where  $I_0$  is the intensity of incident electromagnetic radiation. A totally reflecting surface feels twice that pressure. More generally, a surface absorbs a proportion  $e$  of the incident radiation and reflects a complementary proportion,  $1 - e$ , where  $e$  is the emissivity of the surface, as introduced in Chapter 17. Note that  $0 \leq e \leq 1$ . (a) Determine the radiation pressure  $p_{\text{rad}}$  in terms of  $I_0$  and  $e$ . (b) Consider cosmic dust particles in outer space at a distance of  $1.5 \times 10^{11}$  m from the sun, where  $I_{\text{sun}} = 1.4 \text{ kW/m}^2$ . We can model these particles as tiny disks with  $e = 0.61$ , diameter  $8.0 \text{ }\mu\text{m}$  and mass  $1.0 \times 10^{-10}$  grams, all oriented perpendicular to the sun's rays. What is the force on one of these particles that is exerted by the radiation from the sun? (c) What is the ratio of this force to the attractive force of gravity exerted by the sun on the particle?

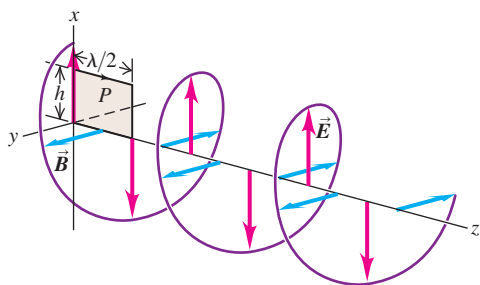
### CHALLENGE PROBLEMS

**32.50 ••• CALC** An electromagnetic wave is specified by the following electric and magnetic fields:

$$\begin{aligned}\vec{E}(z, t) &= E[\cos(kz - \omega t)\hat{i} + \sin(kz - \omega t)\hat{j}] \\ \vec{B}(z, t) &= B[-\sin(kz - \omega t)\hat{i} + \cos(kz - \omega t)\hat{j}]\end{aligned}$$

where  $E$  and  $B$  are constant. (a) What is the value of the line integral  $\oint \vec{E} \cdot d\vec{l}$  over the rectangular loop  $P$  shown in **Fig. P32.50**, which extends a distance  $h$  in the  $x$ -direction and half a wavelength in the  $z$ -direction, and passes through the origin? (b) What is the magnetic flux  $\Phi_B$  through the loop  $P$ ? (c) Use Faraday's law and the relationship  $\omega/k = c$  to determine  $B$  in terms of  $E$  and  $c$ . (d) The wave specified above is "right circularly polarized." We can describe "left circularly polarized" light by swapping the signs on the two terms that involve sine functions. Superpose the above wave with such a reverse polarized analog. Write expressions for the electric and magnetic fields in that sum. Note that the result is a linearly polarized wave. (e) Consider right circularly polarized green light with wavelength  $500 \text{ nm}$  and intensity  $100 \text{ W/m}^2$ . What are the values of  $E$  and  $B$ ? (*Hint:* Use Eq. (32.28) to determine the Poynting vector. Note that this is constant; its magnitude is the intensity of the light.)

Figure P32.50



**32.51 ••• CP** Electromagnetic radiation is emitted by accelerating charges. The rate at which energy is emitted from an accelerating charge that has charge  $q$  and acceleration  $a$  is given by

$$\frac{dE}{dt} = \frac{q^2 a^2}{6\pi\epsilon_0 c^3}$$

where  $c$  is the speed of light. (a) Verify that this equation is dimensionally correct. (b) If a proton with a kinetic energy of  $6.0 \text{ MeV}$  is traveling in a particle accelerator in a circular orbit of radius  $0.750 \text{ m}$ , what fraction of its energy does it radiate per second? (c) Consider an electron orbiting with the same speed and radius. What fraction of its energy does it radiate per second?

**32.52 ••• CP The Classical Hydrogen Atom.** The electron in a hydrogen atom can be considered to be in a circular orbit with a radius of  $0.0529 \text{ nm}$  and a kinetic energy of  $13.6 \text{ eV}$ . If the electron behaved classically, how much energy would it radiate per second (see Challenge Problem 32.51)? What does this tell you about the use of classical physics in describing the atom?

**32.53 ••• CALC** Electromagnetic waves propagate much differently in *conductors* than they do in dielectrics or in vacuum. If the resistivity of the conductor is sufficiently low (that is, if it is a sufficiently good conductor), the oscillating electric field of the wave gives rise to an oscillating conduction current that is much larger than the displacement current. In this case, the wave equation for an electric field  $\vec{E}(x, t) = E_y(x, t)\hat{j}$  propagating in the  $+x$ -direction within a conductor is

$$\frac{\partial^2 E_y(x, t)}{\partial x^2} = \frac{\mu}{\rho} \frac{\partial E_y(x, t)}{\partial t}$$

where  $\mu$  is the permeability of the conductor and  $\rho$  is its resistivity. (a) A solution to this wave equation is  $E_y(x, t) = E_{\text{max}} e^{-k_C x} \cos(k_C x - \omega t)$ , where  $k_C = \sqrt{\omega\mu/2\rho}$ . Verify this by substituting  $E_y(x, t)$  into the above wave equation. (b) The exponential term shows that the electric field decreases in amplitude as it propagates. Explain why this happens. (*Hint:* The field does work to move charges within the conductor. The current of these moving charges causes  $i^2 R$  heating within the conductor, raising its temperature. Where does the energy to do this come from?) (c) Show that the electric-field amplitude decreases by a factor of  $1/e$  in a distance  $1/k_C = \sqrt{2\rho/\omega\mu}$ , and calculate this distance for a radio wave with frequency  $f = 1.0 \text{ MHz}$  in copper (resistivity  $1.72 \times 10^{-8} \text{ }\Omega \cdot \text{m}$ ; permeability  $\mu = \mu_0$ ). Since this distance is so short, electromagnetic waves of this frequency can hardly propagate at all into copper. Instead, they are reflected at the surface of the metal. This is why radio waves cannot penetrate through copper or other metals, and why radio reception is poor inside a metal structure.

### MCAT-STYLE PASSAGE PROBLEMS

**BIO Safe Exposure to Electromagnetic Waves.** There have been many studies of the effects on humans of electromagnetic waves of various frequencies. Using these studies, the International Commission on Non-Ionizing Radiation Protection (ICNIRP) produced guidelines for limiting exposure to electromagnetic fields, with the goal of protecting against known adverse health effects. At frequencies of  $1 \text{ Hz}$  to  $25 \text{ Hz}$ , the maximum exposure level of electric-field amplitude  $E_{\text{max}}$  for the general public is  $14 \text{ kV/m}$ . (Different guidelines were created for people who have occupational exposure to radiation.) At frequencies of  $25 \text{ Hz}$  to  $3 \text{ kHz}$ , the corresponding  $E_{\text{max}}$  is  $350/f \text{ V/m}$ , where  $f$  is the frequency in  $\text{kHz}$ . (Source: "ICNIRP Statement on the 'Guidelines for Limiting Exposure to Time-Varying Electric, Magnetic, and Electromagnetic Fields (up to  $300 \text{ GHz}$ )'"; 2009; *Health Physics* 97(3): 257–258.)

**32.54** In the United States, household electrical power is provided at a frequency of  $60 \text{ Hz}$ , so electromagnetic radiation at that frequency is of particular interest. On the basis of the ICNIRP guidelines, what is the maximum intensity of an electromagnetic wave at this frequency to which the general public should be exposed? (a)  $7.7 \text{ W/m}^2$ ; (b)  $160 \text{ W/m}^2$ ; (c)  $45 \text{ kW/m}^2$ ; (d)  $260 \text{ kW/m}^2$ .

**32.55** Doubling the frequency of a wave in the range of  $25 \text{ Hz}$  to  $3 \text{ kHz}$  represents what change in the maximum allowed electromagnetic-wave intensity? (a) A factor of 2; (b) a factor of  $1/\sqrt{2}$ ; (c) a factor of  $\frac{1}{2}$ ; (d) a factor of  $\frac{1}{4}$ .



**32.56** The ICNIRP also has guidelines for magnetic-field exposure for the general public. In the frequency range of 25 Hz to 3 kHz, this guideline states that the maximum allowed magnetic-field amplitude is  $5/f$  T, where  $f$  is the frequency in kHz. Which is a more stringent limit on allowable electromagnetic-wave intensity in this frequency range: the electric-field guideline or the magnetic-field guideline? (a) The magnetic-field guideline, because at a given

frequency the allowed magnetic field is smaller than the allowed electric field. (b) The electric-field guideline, because at a given frequency the allowed intensity calculated from the electric-field guideline is smaller. (c) It depends on the particular frequency chosen (both guidelines are frequency dependent). (d) Neither—for any given frequency, the guidelines represent the same electromagnetic-wave intensity.

## ANSWERS

### Chapter Opening Question ?

(i) Metals are reflective because they are good conductors of electricity. When an electromagnetic wave strikes a conductor, the electric field of the wave sets up currents on the conductor surface that generate a reflected wave. For a perfect conductor, the requirement that the electric-field component parallel to the surface must be zero implies that this reflected wave is just as intense as the incident wave. Tarnished metals are less shiny because their surface is oxidized and less conductive; polishing the metal removes the oxide and exposes the conducting metal.

### Key Example VARIATION Problems

**VP32.2.1** (a)  $1.29 \times 10^6$  V/m (b)  $2.51 \times 10^{-6}$  m (c)  $1.19 \times 10^{14}$  Hz (d) +z-direction

**VP32.2.2** (a)  $9.67 \times 10^{-7}$  m (b)  $1.95 \times 10^{15}$  rad/s (c) +y-direction (d)  $\vec{B}(y, t) = -\hat{k}(8.17 \times 10^{-3} \text{ T}) \cos[(6.50 \times 10^6 \text{ rad/m})y - (1.95 \times 10^{15} \text{ rad/s})t]$

**VP32.2.3** (a)  $1.40 \times 10^{-2}$  T, +z-direction (b) 2.89 MV/m, -y-direction (c)  $9.64 \times 10^{-3}$  T, -z-direction

**VP32.2.4** (a)  $4.72 \times 10^{14}$  Hz (b)  $4.72 \times 10^{14}$  Hz (c)  $1.64 \times 10^8$  m/s (d) 3.35

**VP32.4.1** (a)  $E = 64.4$  V/m,  $B = 2.15 \times 10^{-7}$  T (b)  $3.67 \times 10^{-8}$  J/m<sup>3</sup>

**VP32.4.2** (a)  $\hat{i}(E_{\max}B_{\max}/\mu_0)$  (b) zero (c)  $\hat{i}(E_{\max}B_{\max}/2\mu_0)$

**VP32.4.3** (a)  $B_{\max} = 1.20 \times 10^{-9}$  T,  $I = 1.72 \times 10^{-4}$  W/m<sup>2</sup> (b)  $1.69 \times 10^5$  W

**VP32.4.4** (a)  $E_{\max} = 1.01 \times 10^3$  V/m,  $B_{\max} = 3.38 \times 10^{-6}$  T (b)  $3.85 \times 10^{26}$  W

**VP32.7.1** (a)  $E_{\max}B_{\max}/\mu_0$  (b) zero (c)  $\hat{i}\sqrt{3}E_{\max}B_{\max}/2\mu_0$  (d)  $-\hat{i}E_{\max}B_{\max}/\mu_0$

**VP32.7.2** (a) 1.44 cm (b)  $2.08 \times 10^{10}$  Hz (c) 36.0 V/m

**VP32.7.3** (a)  $f_1 = 3.29 \times 10^9$  Hz,  $\lambda_1 = 9.12$  cm;  $f_2 = 6.58 \times 10^9$  Hz,  $\lambda_2 = 4.56$  cm;  $f_3 = 9.87 \times 10^9$  Hz,  $\lambda_3 = 3.04$  cm (b)  $x = 0$  and 4.56 cm;  $x = 0$ , 2.28 cm, and 4.56 cm;  $x = 0$ , 1.52 cm, 3.04 cm, and 4.56 cm

**VP32.7.4** (a)  $2.46 \times 10^9$  Hz (b) 8

### Bridging Problem

0.0368 V



**?** When a cut diamond is illuminated with white light, it sparkles brilliantly with a spectrum of vivid colors. These distinctive visual features are a result of (i) light traveling much slower in diamond than in air; (ii) light of different colors traveling at different speeds in diamond; (iii) diamond absorbing light of certain colors; (iv) both (i) and (ii); (v) all of (i), (ii), and (iii).

# 33 The Nature and Propagation of Light

**B**lue lakes, ochre deserts, green forests, and multicolored rainbows can be enjoyed by anyone who has eyes with which to see them. But by studying the branch of physics called **optics**, which deals with the behavior of light and other electromagnetic waves, we can reach a deeper appreciation of the visible world. A knowledge of the properties of light allows us to understand the blue color of the sky and the design of optical devices such as telescopes, microscopes, cameras, eyeglasses, and the human eye. The same basic principles of optics also lie at the heart of modern developments such as the laser, optical fibers, holograms, and new techniques in medical imaging.

The importance of optics to physics, and to science and engineering in general, is so great that we'll devote the next four chapters to its study. In this chapter we begin with a study of the laws of reflection and refraction and the concepts of dispersion, polarization, and scattering of light. Along the way we compare the various possible descriptions of light in terms of particles, rays, or waves, and we introduce Huygens's principle, an important link that connects the ray and wave viewpoints. In Chapter 34 we'll use the ray description of light to understand how mirrors and lenses work, and we'll see how mirrors and lenses are used in optical instruments such as cameras, microscopes, and telescopes. We'll explore the wave characteristics of light further in Chapters 35 and 36.

## 33.1 THE NATURE OF LIGHT

Until the time of Isaac Newton (1642–1727), most scientists thought that light consisted of streams of particles (called *corpuscles*) emitted by light sources. Galileo and others tried (unsuccessfully) to measure the speed of light. Around 1665, evidence of *wave* properties of light began to be discovered. By the early 19th century, evidence that light is a wave had grown very persuasive.

## LEARNING OUTCOMES

### In this chapter, you'll learn...

- 33.1** What light rays are, and how they are related to wave fronts.
- 33.2** The laws that govern the reflection and refraction of light.
- 33.3** The circumstances under which light is totally reflected at an interface.
- 33.4** The consequences of the speed of light in a material being different for different wavelengths.
- 33.5** How to make polarized light out of ordinary light.
- 33.6** How the scattering of light explains the blue color of the sky.
- 33.7** How Huygens's principle helps us analyze reflection and refraction.

### You'll need to review...

- 1.3** Speed of light in vacuum.
- 21.2** Polarization of an object by an electric field.
- 29.7** Maxwell's equations.
- 32.1–32.4** Electromagnetic radiation; plane waves; wave fronts; index of refraction; electromagnetic wave intensity.

In 1873, James Clerk Maxwell predicted the existence of electromagnetic waves and calculated their speed of propagation, as we learned in Section 32.2. This development, along with the experimental work of Heinrich Hertz starting in 1887, showed conclusively that light is indeed an electromagnetic wave.

### The Two Personalities of Light

The wave picture of light is not the whole story, however. Several effects associated with emission and absorption of light reveal a particle aspect, in that the energy carried by light waves is packaged in discrete bundles called *photons* or *quanta*. These apparently contradictory wave and particle properties have been reconciled since 1930 with the development of quantum electrodynamics, a comprehensive theory that includes *both* wave and particle properties. The *propagation* of light is best described by a wave model, but understanding emission and absorption requires a particle approach.

The fundamental sources of all electromagnetic radiation are electric charges in accelerated motion. All objects emit electromagnetic radiation as a result of thermal motion of their molecules; this radiation, called *thermal radiation*, is a mixture of different wavelengths. At sufficiently high temperatures, all matter emits enough visible light to be self-luminous; a very hot object appears “red-hot” (**Fig. 33.1**) or “white-hot.” Thus hot matter in any form is a light source. Familiar examples are a candle flame, hot coals in a campfire, and the coils in an electric toaster oven or room heater.

Light is also produced during electrical discharges through ionized gases. The bluish light of mercury-arc lamps, the orange-yellow of sodium-vapor lamps, and the various colors of “neon” signs are familiar. A variation of the mercury-arc lamp is the *fluorescent* lamp (see Fig. 30.7). This light source uses a material called a *phosphor* to convert the ultraviolet radiation from a mercury arc into visible light. This conversion makes fluorescent lamps more efficient than incandescent lamps in transforming electrical energy into light.

In most light sources, light is emitted independently by different atoms within the source; in a *laser*, by contrast, atoms are induced to emit light in a cooperative, coherent fashion. The result is a very narrow beam of radiation that can be enormously intense and that is much more nearly *monochromatic*, or single-frequency, than light from any other source. Lasers are used by physicians for microsurgery, in a DVD or Blu-ray player to scan the information recorded on a video disc, in industry to cut through steel and to fuse high-melting-point materials, and in many other applications (**Fig. 33.2**).

No matter what its source, electromagnetic radiation travels in vacuum at the same speed  $c$ . As we saw in Sections 1.3 and 32.1, this speed is defined to be

$$c = 2.99792458 \times 10^8 \text{ m/s}$$

or  $3.00 \times 10^8 \text{ m/s}$  to three significant figures. The duration of one second is defined by the cesium clock (see Section 1.3), so one meter is defined to be the distance that light travels in  $1/299,792,458 \text{ s}$ .

### Waves, Wave Fronts, and Rays

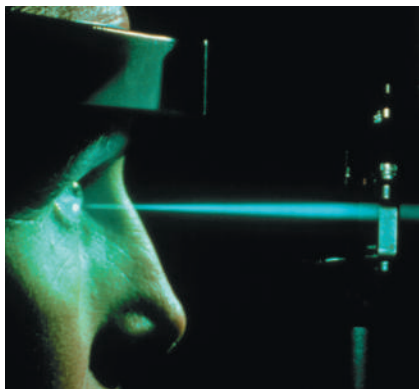
We often use the concept of a **wave front** to describe wave propagation. We introduced this concept in Section 32.2 to describe the leading edge of a wave. More generally, we define a wave front as *the locus of all adjacent points at which the phase of vibration of a physical quantity associated with the wave is the same*. That is, at any instant, all points on a wave front are at the same part of the cycle of their variation.

When we drop a pebble into a calm pool, the expanding circles formed by the wave crests, as well as the circles formed by the wave troughs between them, are wave fronts. Similarly, when sound waves spread out in still air from a pointlike source, or when

Figure 33.1 An electric heating element emits primarily infrared radiation. But if its temperature is high enough, it also emits a discernible amount of visible light.



Figure 33.2 Ophthalmic surgeons use lasers for repairing detached retinas and for cauterizing blood vessels in retinopathy. Pulses of blue-green light from an argon laser are ideal for this purpose, since they pass harmlessly through the transparent part of the eye but are absorbed by red pigments in the retina.



electromagnetic radiation spreads out from a pointlike emitter, any spherical surface that is concentric with the source is a wave front, as shown in **Fig. 33.3**. In diagrams of wave motion we usually draw only parts of a few wave fronts, often choosing consecutive wave fronts that have the same phase and thus are one wavelength apart, such as crests of water waves. Similarly, a diagram for sound waves might show only the “pressure crests,” the surfaces over which the pressure is maximum, and a diagram for electromagnetic waves might show only the “crests” on which the electric or magnetic field is maximum.

We’ll often use diagrams that show the shapes of the wave fronts or their cross sections in some reference plane. For example, when electromagnetic waves are radiated by a small light source, we can represent the wave fronts as spherical surfaces concentric with the source or, as in **Fig. 33.4a**, by the circular intersections of these surfaces with the plane of the diagram. Far away from the source, where the radii of the spheres have become very large, a section of a spherical surface can be considered as a plane, and we have a *plane* wave like those discussed in Sections 32.2 and 32.3 (Fig. 33.4b).

To describe the directions in which light propagates, it’s often convenient to represent a light wave by **rays** rather than by wave fronts. In a particle theory of light, rays are the paths of the particles. From the wave viewpoint *a ray is an imaginary line along the direction of travel of the wave*. In Fig. 33.4a the rays are the radii of the spherical wave fronts, and in Fig. 33.4b they are straight lines perpendicular to the wave fronts. When waves travel in a homogeneous isotropic material (a material with the same properties in all regions and in all directions), the rays are always straight lines normal to the wave fronts. At a boundary surface between two materials, such as the surface of a glass plate in air, the wave speed and the direction of a ray may change, but the ray segments in the air and in the glass are straight lines.

The next several chapters will give you many opportunities to see the interplay of the ray, wave, and particle descriptions of light. The branch of optics for which the ray description is adequate is called **geometric optics**; the branch dealing specifically with wave behavior is called **physical optics**. This chapter and the following one are concerned mostly with geometric optics. In Chapters 35 and 36 we’ll study wave phenomena and physical optics.

**TEST YOUR UNDERSTANDING OF SECTION 33.1** Some crystals are *not* isotropic: Light travels through the crystal at a higher speed in some directions than in others. In a crystal in which light travels at the same speed in the  $x$ - and  $z$ -directions but faster in the  $y$ -direction, what would be the shape of the wave fronts produced by a light source at the origin? (i) Spherical, like those shown in Fig. 33.3; (ii) ellipsoidal, flattened along the  $y$ -axis; (iii) ellipsoidal, stretched out along the  $y$ -axis.

### ANSWER

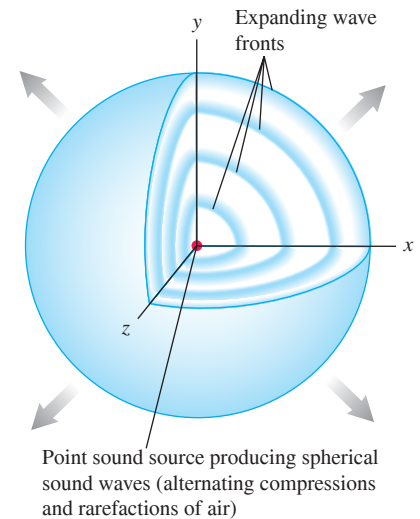
(iii) The waves go farther in the  $y$ -direction in a given amount of time than in the other directions, so the wave fronts are elongated in the  $y$ -direction.

## 33.2 REFLECTION AND REFRACTION

In this section we’ll use the *ray* model of light to explore two of the most important aspects of light propagation: **reflection** and **refraction**. When a light wave strikes a smooth interface separating two transparent materials (such as air and glass or water and glass), the wave is in general partly *reflected* and partly *refracted* (transmitted) into the second material, as shown in **Fig. 33.5a** (next page). For example, when you look into a restaurant window from the street, you see a reflection of the street scene, but a person inside the restaurant can look out through the window at the same scene as light reaches him by refraction.

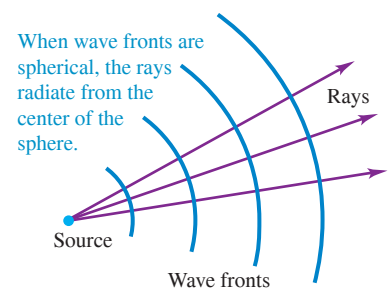
The segments of plane waves shown in Fig. 33.5a can be represented by bundles of rays forming *beams* of light (Fig. 33.5b). For simplicity we often draw only one ray in each beam (Fig. 33.5c). Representing these waves in terms of rays is the basis of geometric optics. We begin our study with the behavior of an individual ray.

**Figure 33.3** Spherical wave fronts of sound spread out uniformly in all directions from a point source in a motionless medium, such as still air, that has the same properties in all regions and in all directions. Electromagnetic waves in vacuum also spread out as shown here.



**Figure 33.4** Wave fronts (blue) and rays (purple).

(a)



(b)

When wave fronts are planar, the rays are perpendicular to the wave fronts and parallel to each other.

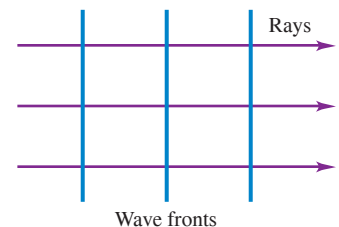
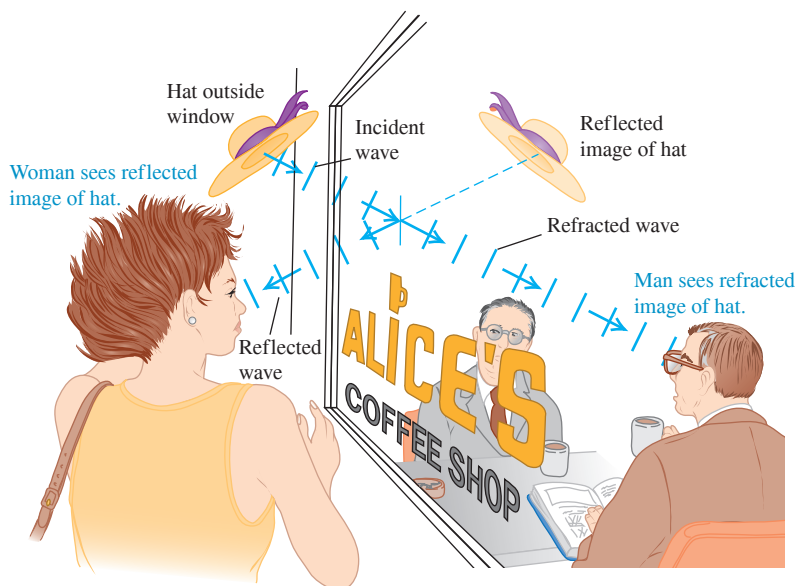


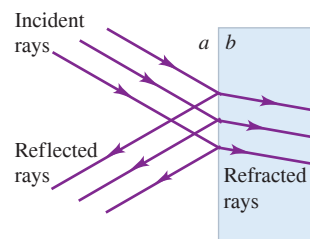


Figure 33.5 (a) A plane wave is in part reflected and in part refracted at the boundary between two media (in this case, air and glass). The light that reaches the inside of the coffee shop is refracted twice, once entering the glass and once exiting the glass. (b), (c) How light behaves at the interface between the air outside the coffee shop (material  $a$ ) and the glass (material  $b$ ). For the case shown here, material  $b$  has a larger index of refraction than material  $a$  ( $n_b > n_a$ ) and the angle  $\theta_b$  is smaller than  $\theta_a$ .

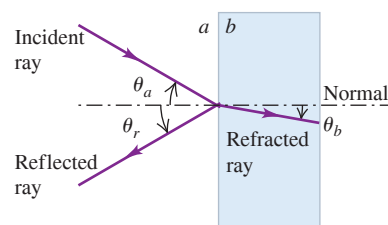
(a) Plane waves reflected and refracted from a window



(b) The waves in the outside air and glass represented by rays



(c) The representation simplified to show just one set of rays



We describe the directions of the incident, reflected, and refracted (transmitted) rays at a smooth interface between two optical materials in terms of the angles they make with the *normal* (perpendicular) to the surface at the point of incidence, as shown in Fig. 33.5c. If the interface is rough, both the transmitted light and the reflected light are scattered in various directions, and there is no single angle of transmission or reflection. Reflection at a definite angle from a very smooth surface is called **specular reflection** (from the Latin word for “mirror”); scattered reflection from a rough surface is called **diffuse reflection** (Fig. 33.6). Both kinds of reflection can occur with either transparent materials or *opaque* materials that do not transmit light. The vast majority of objects in your environment (including plants, other people, and this book) are visible to you because they reflect light in a diffuse manner from their surfaces. Our primary concern, however, will be with specular reflection from a very smooth surface such as highly polished glass or metal. Unless stated otherwise, when referring to “reflection” we’ll always mean *specular* reflection.

The **index of refraction** of an optical material (also called the **refractive index**), denoted by  $n$ , plays a central role in geometric optics:

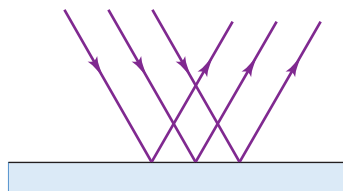
$$\text{Index of refraction of an optical material } n = \frac{c}{v} \quad \text{Speed of light in vacuum} \quad \text{Speed of light in the material} \quad (33.1)$$

Light always travels *more slowly* in a material than in vacuum, so the value of  $n$  in anything other than vacuum is always greater than unity. For vacuum,  $n = 1$ . Since  $n$  is a ratio of two speeds, it is a pure number without units. (In Section 32.3 we described the relationship of the value of  $n$  to the electric and magnetic properties of a material.)

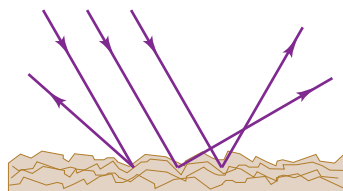
**CAUTION Wave speed and index of refraction** Keep in mind that the wave speed  $v$  is *inversely* proportional to the index of refraction  $n$ . The larger the index of refraction in a material, the *slower* the wave speed in that material. **I**

Figure 33.6 Two types of reflection.

(a) Specular reflection



(b) Diffuse reflection



## The Laws of Reflection and Refraction

Experimental studies of reflection and refraction at a smooth interface between two optical materials lead to the following conclusions (Fig. 33.7):

1. **The incident, reflected, and refracted rays and the normal to the surface all lie in the same plane.** This plane, called the **plane of incidence**, is perpendicular to the plane of the boundary surface between the two materials. We always draw ray diagrams so that the incident, reflected, and refracted rays are in the plane of the diagram.
2. **The angle of reflection  $\theta_r$  is equal to the angle of incidence  $\theta_a$  for all wavelengths and for any pair of materials.** That is, in Fig. 33.5c,

**Law of reflection:**

$$\theta_r = \theta_a \quad (33.2)$$

Angle of reflection (measured from normal)      Angle of incidence (measured from normal)

This relationship, together with the observation that the incident and reflected rays and the normal all lie in the same plane, is called the **law of reflection**.

3. For monochromatic light and for a given pair of materials,  $a$  and  $b$ , on opposite sides of the interface, **the ratio of the sines of the angles  $\theta_a$  and  $\theta_b$ , where both angles are measured from the normal to the surface, is equal to the inverse ratio of the two indexes of refraction:**

$$\frac{\sin \theta_a}{\sin \theta_b} = \frac{n_b}{n_a} \quad (33.3)$$

or

**Law of refraction:**

$$n_a \sin \theta_a = n_b \sin \theta_b \quad (33.4)$$

Angle of incidence (measured from normal)      Angle of refraction (measured from normal)

Index of refraction for material with incident light      Index of refraction for material with refracted light

This result, together with the observation that the incident and refracted rays and the normal all lie in the same plane, is called the **law of refraction** or **Snell's law**, after the Dutch scientist Willebrord Snell (1591–1626). This law was actually first discovered in the 10th century by the Persian scientist Ibn Sahl. The discovery that  $n = c/v$  came much later.

While these results were first observed experimentally, they can be derived theoretically from a wave description of light. We do this in Section 33.7.

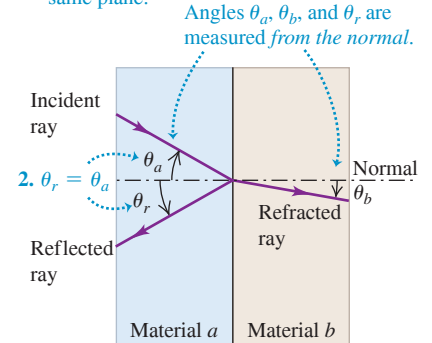
Equations (33.3) and (33.4) show that when a ray passes from one material ( $a$ ) into another material ( $b$ ) having a larger index of refraction ( $n_b > n_a$ ) and hence a slower wave speed, the angle  $\theta_b$  with the normal is *smaller* in the second material than the angle  $\theta_a$  in the first; hence the ray is bent *toward* the normal (Fig. 33.8a). When the second material has a *smaller* index of refraction than the first material ( $n_b < n_a$ ) and hence a faster wave speed, the ray is bent *away from* the normal (Fig. 33.8b).

No matter what the materials on either side of the interface, in the case of *normal* incidence the transmitted ray is not bent at all (Fig. 33.8c). In this case  $\theta_a = 0$  and  $\sin \theta_a = 0$ , so from Eq. (33.4)  $\theta_b$  is also equal to zero; the transmitted ray is also normal to the interface. From Eq. (33.2),  $\theta_r$  is also equal to zero, so the reflected ray travels back along the same path as the incident ray.

The law of refraction explains why a partially submerged ruler or drinking straw appears bent; light rays coming from below the surface change in direction at the

Figure 33.7 The laws of reflection and refraction.

1. The incident, reflected, and refracted rays and the normal to the surface all lie in the same plane.

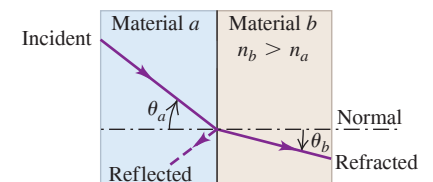


2.  $\theta_r = \theta_a$
3. When a monochromatic light ray crosses the interface between two given materials  $a$  and  $b$ , the angles  $\theta_a$  and  $\theta_b$  are related to the indexes of refraction of  $a$  and  $b$  by

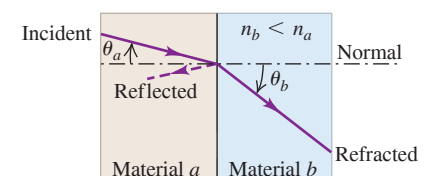
$$\frac{\sin \theta_a}{\sin \theta_b} = \frac{n_b}{n_a}$$

Figure 33.8 Refraction and reflection in three cases. (a) Material  $b$  has a larger index of refraction than material  $a$ . (b) Material  $b$  has a smaller index of refraction than material  $a$ . (c) The incident light ray is normal to the interface between the materials.

- (a) A ray entering a material of *larger* index of refraction bends *toward* the normal.



- (b) A ray entering a material of *smaller* index of refraction bends *away from* the normal.



- (c) A ray oriented along the normal does not bend, regardless of the materials.

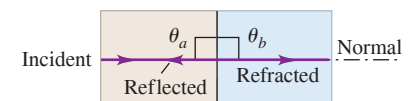
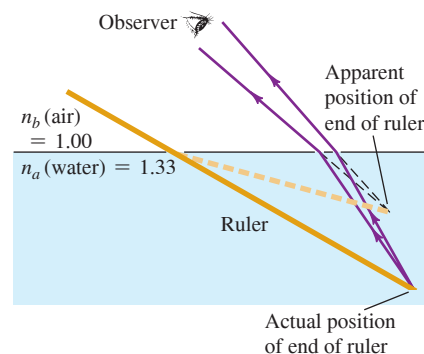


Figure 33.9 (a) This ruler is actually straight, but it appears to bend at the surface of the water. (b) Light rays from any submerged object bend away from the normal when they emerge into the air. As seen by an observer above the surface of the water, the object appears to be much closer to the surface than it actually is.

(a) A straight ruler half-immersed in water

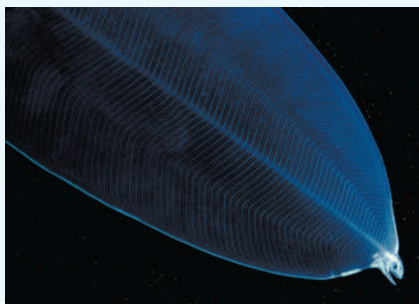


(b) Why the ruler appears bent



### BIO APPLICATION Transparency and Index of Refraction

An eel in its larval stage is nearly as transparent as the seawater in which it swims. The larva in this photo is nonetheless easy to see because its index of refraction is higher than that of seawater, so that some of the light striking it is reflected instead of transmitted. The larva appears particularly shiny around its edges because the light reaching the camera from those points struck the larva at near-grazing incidence ( $\theta_a = 90^\circ$ ), resulting in almost 100% reflection.



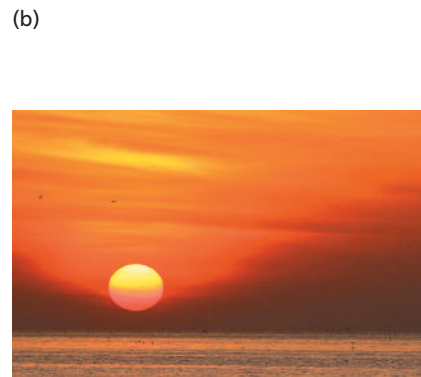
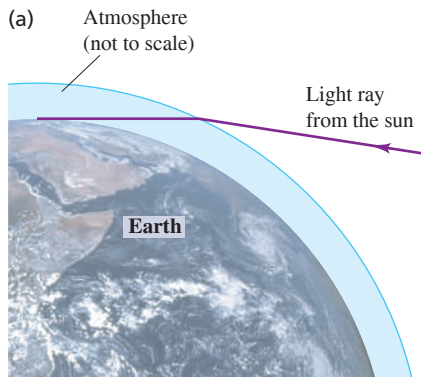
air–water interface, so the rays appear to be coming from a position above their actual point of origin (Fig. 33.9). A similar effect explains the appearance of the setting sun (Fig. 33.10).

An important special case is refraction that occurs at an interface between vacuum, for which the index of refraction is unity by definition, and a material. When a ray passes from vacuum into a material ( $b$ ), so that  $n_a = 1$  and  $n_b > 1$ , the ray is always bent *toward* the normal. When a ray passes from a material into vacuum, so that  $n_a > 1$  and  $n_b = 1$ , the ray is always bent *away from* the normal.

The laws of reflection and refraction apply regardless of which side of the interface the incident ray comes from. If a ray of light approaches the interface in Fig. 33.8a or 33.8b from the right rather than from the left, there are again reflected and refracted rays, and they lie in the same plane as the incident ray and the normal to the surface. Furthermore, the path of a refracted ray is *reversible*; it follows the same path when going from  $b$  to  $a$  as when going from  $a$  to  $b$ . [You can verify this by using Eq. (33.4).] Since the reflected and incident angles are the same, the path of a reflected ray is also reversible. That's why when you see someone's eyes in a mirror, that person can also see you.

The *intensities* of the reflected and refracted rays depend on the angle of incidence, the two indexes of refraction, and the polarization (that is, the direction of the electric-field vector) of the incident ray. The fraction reflected is smallest at normal incidence ( $\theta_a = 0^\circ$ ), where it is about 4% for an air–glass interface. This fraction increases with increasing angle of incidence to 100% at grazing incidence, when  $\theta_a = 90^\circ$ . (It's possible to use Maxwell's equations to predict the amplitude, intensity, phase, and polarization states of the reflected and refracted waves. Such an analysis is beyond our scope, however.)

Figure 33.10 (a) The index of refraction of air is slightly greater than 1, so light rays from the setting sun bend downward when they enter our atmosphere. (The effect is exaggerated in this figure.) (b) Stronger refraction occurs for light coming from the lower limb of the sun (the part that appears closest to the horizon), which passes through denser air in the lower atmosphere. As a result, the setting sun appears flattened vertically. (See Problem 33.51.)



The index of refraction depends not only on the substance but also on the wavelength of the light. The dependence on wavelength is called *dispersion*; we'll consider it in Section 33.4. Indexes of refraction for several solids and liquids are given in **Table 33.1** for a particular wavelength of yellow light.

The index of refraction of air at standard temperature and pressure is about 1.0003, and we'll usually take it to be exactly unity. The index of refraction of a gas increases as its density increases. Most glasses used in optical instruments have indexes of refraction between about 1.5 and 2.0. A few substances have larger indexes; one example is diamond, with 2.417 (see Table 33.1).

### Index of Refraction and the Wave Aspects of Light

We have discussed how the direction of a light ray changes when it passes from one material to another material with a different index of refraction. What aspects of the *wave* characteristics of the light change when this happens?

First, the frequency  $f$  of the wave does *not* change when passing from one material to another. That is, the number of wave cycles arriving per unit time must equal the number leaving per unit time; this is a statement that the boundary surface cannot create or destroy waves.

Second, the wavelength  $\lambda$  of the wave *is* different in general in different materials. This is because in any material,  $v = \lambda f$ ; since  $f$  is the same in any material as in vacuum and  $v$  is always less than the wave speed  $c$  in vacuum,  $\lambda$  is also correspondingly reduced. Thus the wavelength  $\lambda$  of light in a material is *less than* the wavelength  $\lambda_0$  of the same light in vacuum. From the above discussion,  $f = c/\lambda_0 = v/\lambda$ . Combining this with Eq. (33.1),  $n = c/v$ , we find

$$\text{Wavelength of light in a material} \quad \lambda = \frac{\lambda_0}{n} \quad \begin{array}{l} \text{Wavelength of light in vacuum} \\ \text{Index of refraction of the material} \end{array} \quad (33.5)$$

When a wave passes from one material into a second material with larger index of refraction, so  $n_b > n_a$ , the wave speed decreases. The wavelength  $\lambda_b = \lambda_0/n_b$  in the second material is then shorter than the wavelength  $\lambda_a = \lambda_0/n_a$  in the first material. If instead the second material has a smaller index of refraction than the first material, so  $n_b < n_a$ , then the wave speed increases. Then the wavelength  $\lambda_b$  in the second material is longer than the wavelength  $\lambda_a$  in the first material. This makes intuitive sense; the waves get “squeezed” (the wavelength gets shorter) if the wave speed decreases and get “stretched” (the wavelength gets longer) if the wave speed increases.

**TABLE 33.1** Index of Refraction for Yellow Sodium Light,  $\lambda_0 = 589 \text{ nm}$

| Substance                                       | Index of Refraction, $n$ |
|-------------------------------------------------|--------------------------|
| <b>Solids</b>                                   |                          |
| Ice ( $\text{H}_2\text{O}$ )                    | 1.309                    |
| Fluorite ( $\text{CaF}_2$ )                     | 1.434                    |
| Polystyrene                                     | 1.49                     |
| Rock salt ( $\text{NaCl}$ )                     | 1.544                    |
| Quartz ( $\text{SiO}_2$ )                       | 1.544                    |
| Zircon ( $\text{ZrO}_2 \cdot \text{SiO}_2$ )    | 1.923                    |
| Diamond (C)                                     | 2.417                    |
| Fabulite ( $\text{SrTiO}_3$ )                   | 2.409                    |
| Rutile ( $\text{TiO}_2$ )                       | 2.62                     |
| <b>Glasses (typical values)</b>                 |                          |
| Crown                                           | 1.52                     |
| Light flint                                     | 1.58                     |
| Medium flint                                    | 1.62                     |
| Dense flint                                     | 1.66                     |
| Lanthanum flint                                 | 1.80                     |
| <b>Liquids at <math>20^\circ\text{C}</math></b> |                          |
| Methanol ( $\text{CH}_3\text{OH}$ )             | 1.329                    |
| Water ( $\text{H}_2\text{O}$ )                  | 1.333                    |
| Ethanol ( $\text{C}_2\text{H}_5\text{OH}$ )     | 1.36                     |
| Carbon tetrachloride ( $\text{CCl}_4$ )         | 1.460                    |
| Turpentine                                      | 1.472                    |
| Glycerine                                       | 1.473                    |
| Benzene                                         | 1.501                    |
| Carbon disulfide ( $\text{CS}_2$ )              | 1.628                    |

### PROBLEM-SOLVING STRATEGY 33.1 Reflection and Refraction

**IDENTIFY** *the relevant concepts:* Use geometric optics, discussed in this section, whenever light (or electromagnetic radiation of *any* frequency and wavelength) encounters a boundary between materials. In general, part of the light is reflected back into the first material and part is refracted into the second material.

**SET UP** *the problem* using the following steps:

1. In problems involving rays and angles, start by drawing a large, neat diagram. Label all known angles and indexes of refraction.
2. Identify the target variables.

**EXECUTE** *the solution* as follows:

1. Apply the laws of reflection, Eq. (33.2), and refraction, Eq. (33.4). Measure angles of incidence, reflection, and refraction with respect to the *normal* to the surface, *never* from the surface itself.

2. Apply geometry or trigonometry in working out angular relationships. Remember that the sum of the acute angles of a right triangle is  $90^\circ$  (they are *complementary*) and the sum of the interior angles in any triangle is  $180^\circ$ .
3. The frequency of the electromagnetic radiation does not change when it moves from one material to another; the wavelength changes in accordance with Eq. (33.5),  $\lambda = \lambda_0/n$ .

**EVALUATE** *your answer:* In problems that involve refraction, check that your results are consistent with Snell's law ( $n_a \sin \theta_a = n_b \sin \theta_b$ ). If the second material has a higher index of refraction than the first, the angle of refraction must be *smaller* than the angle of incidence: The refracted ray bends toward the normal. If the first material has the higher index of refraction, the refracted angle must be larger than the incident angle: The refracted ray bends away from the normal.

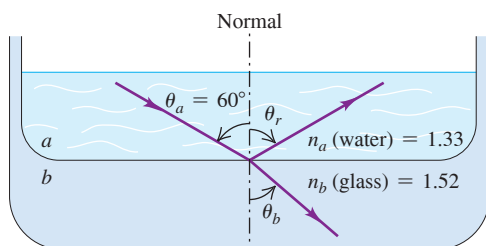


**EXAMPLE 33.1** Reflection and refraction**WITH VARIATION PROBLEMS**

In **Fig. 33.11**, material *a* is water and material *b* is glass with index of refraction 1.52. The incident ray makes an angle of  $60.0^\circ$  with the normal; find the directions of the reflected and refracted rays.

**IDENTIFY and SET UP** This is a problem in geometric optics. We are given the angle of incidence  $\theta_a = 60.0^\circ$  and the indexes of refraction  $n_a = 1.33$  and  $n_b = 1.52$ . We must find the angles of reflection and refraction  $\theta_r$  and  $\theta_b$ ; to do this we use Eqs. (33.2) and (33.4), respectively. Figure 33.11 shows the rays and angles;  $n_b$  is slightly larger than  $n_a$ , so by Snell's law [Eq. (33.4)]  $\theta_b$  is slightly smaller than  $\theta_a$ .

Figure 33.11 Reflection and refraction of light passing from water to glass.



**EXECUTE** According to Eq. (33.2), the angle the reflected ray makes with the normal is the same as that of the incident ray, so  $\theta_r = \theta_a = 60.0^\circ$ .

To find the direction of the refracted ray we use Snell's law, Eq. (33.4):

$$\begin{aligned} n_a \sin \theta_a &= n_b \sin \theta_b \\ \sin \theta_b &= \frac{n_a}{n_b} \sin \theta_a = \frac{1.33}{1.52} \sin 60.0^\circ = 0.758 \\ \theta_b &= \arcsin(0.758) = 49.3^\circ \end{aligned}$$

**EVALUATE** The second material has a larger refractive index than the first, as in Fig. 33.8a. Hence the refracted ray is bent toward the normal and  $\theta_b < \theta_a$ .

**KEYCONCEPT** When light strikes a smooth interface between two transparent materials, it can be both *reflected* back into the first material and *refracted* into the second material. The angles of incidence and reflection are always equal. The angle of refraction is determined by the angle of incidence and the indexes of refraction of the two materials (Snell's law). Always measure these angles from the normal to the interface.

**EXAMPLE 33.2** Index of refraction in the eye**WITH VARIATION PROBLEMS**

The wavelength of the red light from a helium-neon laser is 633 nm in air but 474 nm in the aqueous humor inside your eyeball. Calculate the index of refraction of the aqueous humor and the speed and frequency of the light in it.

**IDENTIFY and SET UP** The key ideas here are (i) the definition of index of refraction  $n$  in terms of the wave speed  $v$  in a medium and the speed  $c$  in vacuum, and (ii) the relationship between wavelength  $\lambda_0$  in vacuum and wavelength  $\lambda$  in a medium of index  $n$ . We use Eq. (33.1),  $n = c/v$ ; Eq. (33.5),  $\lambda = \lambda_0/n$ ; and  $v = \lambda f$ .

**EXECUTE** The index of refraction of air is very close to unity, so we assume that the wavelength  $\lambda_0$  in vacuum is the same as that in air, 633 nm. Then from Eq. (33.5),

$$\lambda = \frac{\lambda_0}{n} \quad n = \frac{\lambda_0}{\lambda} = \frac{633 \text{ nm}}{474 \text{ nm}} = 1.34$$

This is about the same index of refraction as for water. Then, using  $n = c/v$  and  $v = \lambda f$ , we find

$$\begin{aligned} v &= \frac{c}{n} = \frac{3.00 \times 10^8 \text{ m/s}}{1.34} = 2.25 \times 10^8 \text{ m/s} \\ f &= \frac{v}{\lambda} = \frac{2.25 \times 10^8 \text{ m/s}}{474 \times 10^{-9} \text{ m}} = 4.74 \times 10^{14} \text{ Hz} \end{aligned}$$

**EVALUATE** Although the speed and wavelength have different values in air and in the aqueous humor, the *frequency* in air,  $f_0$ , is the same as the frequency  $f$  in the aqueous humor:

$$f_0 = \frac{c}{\lambda_0} = \frac{3.00 \times 10^8 \text{ m/s}}{633 \times 10^{-9} \text{ m}} = 4.74 \times 10^{14} \text{ Hz}$$

**KEYCONCEPT** When a light wave travels from one material into another with a different index of refraction  $n$ , the wave speed and wavelength both change but the wave frequency remains the same. A larger  $n$  means a slower speed and a shorter wavelength; a smaller  $n$  means a faster speed and a longer wavelength.

**EXAMPLE 33.3** A twice-reflected ray

Two mirrors are perpendicular to each other. A ray traveling in a plane perpendicular to both mirrors is reflected from one mirror at *P*, then the other at *Q*, as shown in **Fig. 33.12**. What is the ray's final direction relative to its original direction?

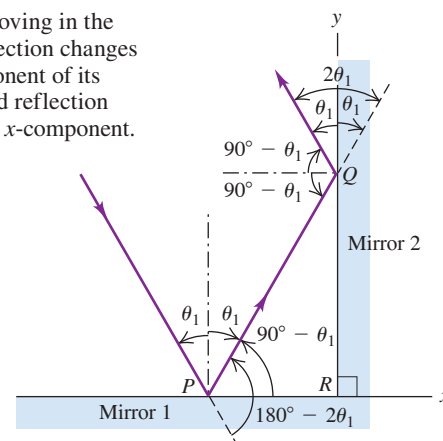
**IDENTIFY and SET UP** This problem involves the law of reflection, which we must apply twice (once for each mirror).

**EXECUTE** For mirror 1 the angle of incidence is  $\theta_1$ , and this equals the angle of reflection. The sum of interior angles in the triangle *PQR* is  $180^\circ$ , so we see that the angles of both incidence and reflection for mirror 2 are  $90^\circ - \theta_1$ . The total change in direction of the ray after both reflections is therefore  $2(90^\circ - \theta_1) + 2\theta_1 = 180^\circ$ . That is, the ray's final direction is opposite to its original direction.

**EVALUATE** An alternative viewpoint is that reflection reverses the sign of the component of light velocity perpendicular to the surface but leaves the other components unchanged. We invite you to verify this in detail. You should also be able to use this result to show that when a ray of light is successively reflected by three mirrors forming a corner of a cube (a “corner reflector”), its final direction is again opposite to its original direction. This principle is widely used in tail-light lenses and bicycle reflectors to improve their night-time visibility. *Apollo* astronauts placed arrays of corner reflectors on the moon. By use of laser beams reflected from these arrays, the earth–moon distance has been measured to within 0.15 m.

**KEYCONCEPT** When light undergoes multiple reflections, multiple refractions, or a combination of reflections and refractions, apply the law of reflection or the law of refraction separately at each interface.

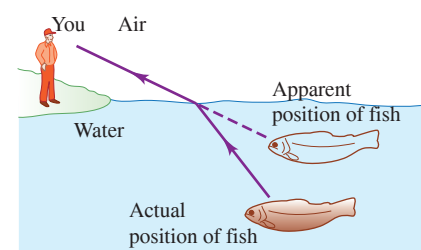
Figure 33.12 A ray moving in the  $xy$ -plane. The first reflection changes the sign of the  $y$ -component of its velocity, and the second reflection changes the sign of the  $x$ -component.



**TEST YOUR UNDERSTANDING OF SECTION 33.2** You are standing on the shore of a lake. You spot a tasty fish swimming some distance below the lake surface. (a) If you want to spear the fish, should you aim the spear (i) above, (ii) below, or (iii) directly at the apparent position of the fish? (b) If instead you use a high-power laser to simultaneously kill and cook the fish, should you aim the laser (i) above, (ii) below, or (iii) directly at the apparent position of the fish?

**ANSWER**

(a) (iii) (b) (ii) (c) (iii) As shown in the figure, light rays coming from the fish bend away from the normal when they pass from the water ( $n = 1.33$ ) into the air ( $n = 1.00$ ). As a result, the fish appears to be higher in the water than it actually is. Hence you should aim a spear *below* the apparent position of the fish. If you use a laser beam, you should aim *at* the apparent position of the fish. The beam of laser light takes the same path from you to the fish as ordinary light takes from the fish to you (though in the opposite direction).

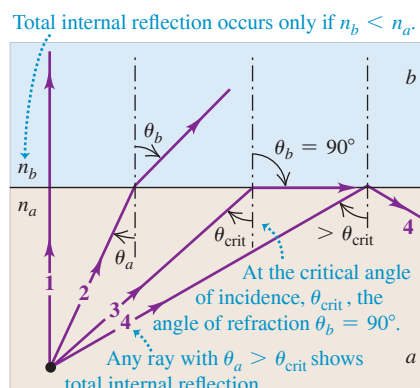


### 33.3 TOTAL INTERNAL REFLECTION

We have described how light is partially reflected and partially transmitted at an interface between two materials with different indexes of refraction. Under certain circumstances, however, *all* of the light can be reflected back from the interface, with none of it being transmitted, even though the second material is transparent. **Figure 33.13a** shows how this can occur. Several rays are shown radiating from a point source in material  $a$  with index of refraction  $n_a$ . The rays strike the surface of a second material  $b$  with index  $n_b$ , where  $n_a > n_b$ . (Materials  $a$  and  $b$  could be water and air, respectively.) From Snell's law of refraction,

$$\sin \theta_b = \frac{n_a}{n_b} \sin \theta_a$$

(a) Total internal reflection



(b) A light beam enters the top left of the tank, then reflects at the bottom from mirrors tilted at different angles. One beam undergoes total internal reflection at the air–water interface.

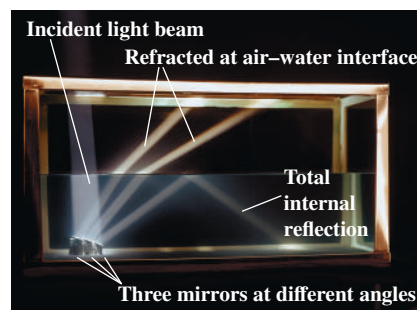


Figure 33.13 (a) Total internal reflection. The angle of incidence for which the angle of refraction is  $90^\circ$  is called the critical angle: This is the case for ray 3. The reflected portions of rays 1, 2, and 3 are omitted for clarity. (b) Rays of laser light enter the water in the fishbowl from above; they are reflected at the bottom by mirrors tilted at slightly different angles. One ray undergoes total internal reflection at the air–water interface.

Because  $n_a/n_b$  is greater than unity,  $\sin \theta_b$  is larger than  $\sin \theta_a$ ; the ray is bent *away from* the normal. Thus there must be some value of  $\theta_a$  *less than*  $90^\circ$  for which  $\sin \theta_b = 1$  and  $\theta_b = 90^\circ$ . This is shown by ray 3 in the diagram, which emerges just grazing the surface at an angle of refraction of  $90^\circ$ . Compare Fig. 33.13a to the photograph of light rays in Fig. 33.13b.

The angle of incidence for which the refracted ray emerges tangent to the surface is called the **critical angle**, denoted by  $\theta_{\text{crit}}$ . (A more detailed analysis using Maxwell's equations shows that as the incident angle approaches the critical angle, the transmitted intensity approaches zero.) If the angle of incidence is *larger* than the critical angle,  $\sin \theta_b$  would have to be greater than unity, which is impossible. Beyond the critical angle, the ray *cannot* pass into the upper material; it is trapped in the lower material and is completely reflected at the boundary surface. This situation, called **total internal reflection**, occurs only when a ray in material *a* is incident on a second material *b* whose index of refraction is *smaller* than that of material *a* (that is,  $n_b < n_a$ ).

We can find the critical angle for two given materials *a* and *b* by setting  $\theta_b = 90^\circ$  ( $\sin \theta_b = 1$ ) in Snell's law. We then have

$$\text{Critical angle for total internal reflection} \rightarrow \sin \theta_{\text{crit}} = \frac{n_b}{n_a} \quad (33.6)$$

Index of refraction of second material  
Index of refraction of first material

Total internal reflection will occur if the angle of incidence  $\theta_a$  is larger than or equal to  $\theta_{\text{crit}}$ .

### Applications of Total Internal Reflection

Total internal reflection finds numerous uses in optical technology. As an example, consider glass with index of refraction  $n = 1.52$ . If light propagating within this glass encounters a glass–air interface, the critical angle is

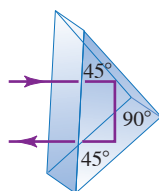
$$\sin \theta_{\text{crit}} = \frac{1}{1.52} = 0.658 \quad \theta_{\text{crit}} = 41.1^\circ$$

The light will be *totally reflected* if it strikes the glass–air surface at an angle of  $41.1^\circ$  or larger. Because the critical angle is slightly smaller than  $45^\circ$ , it is possible to use a prism with angles of  $45^\circ$ – $45^\circ$ – $90^\circ$  as a totally reflecting surface. As reflectors, totally reflecting prisms have some advantages over metallic surfaces such as ordinary coated-glass mirrors. While no metallic surface reflects 100% of the light incident on it, light can be *totally* reflected by a prism. These reflecting properties of a prism are unaffected by tarnishing.

A  $45^\circ$ – $45^\circ$ – $90^\circ$  prism, used as in **Fig. 33.14a**, is called a *Porro* prism. Light enters and leaves at right angles to the hypotenuse and is totally reflected at each of the shorter faces. The total change of direction of the rays is  $180^\circ$ . Binoculars often use combinations of two Porro prisms, as in Fig. 33.14b.

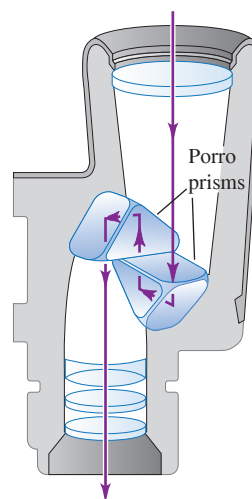
Figure 33.14 (a) Total internal reflection in a Porro prism. (b) A combination of two Porro prisms in binoculars.

(a) Total internal reflection in a Porro prism



If the incident beam is oriented as shown, total internal reflection occurs on the  $45^\circ$  faces (because, for a glass–air interface,  $\theta_{\text{crit}} = 41.1^\circ$ ).

(b) Binoculars use Porro prisms to reflect the light to each eyepiece.



When a beam of light enters at one end of a transparent rod (**Fig. 33.15**), the light can be totally reflected internally if the index of refraction of the rod is greater than that of the surrounding material. The light is “trapped” within even a curved rod, provided that the curvature is not too great. A bundle of fine glass or plastic fibers behaves in the same way and has the advantage of being flexible. A bundle may consist of thousands of individual fibers, each of the order of 0.002 to 0.01 mm in diameter. If the fibers are assembled in the bundle so that the relative positions of the ends are the same (or mirror images) at both ends, the bundle can transmit an image.

Fiber-optic devices have found a wide range of medical applications in instruments called *endoscopes*, which can be inserted directly into the bronchial tubes, the bladder, the colon, and other organs for direct visual examination (**Fig. 33.16**). A bundle of fibers can even be enclosed in a hypodermic needle for studying tissues and blood vessels far beneath the skin.

Fiber optics also have applications in communication systems. The rate at which information can be transmitted by a wave (light, radio, or whatever) is proportional to the frequency. To see qualitatively why this is so, consider modulating (modifying) the wave by chopping off some of the wave crests. Suppose each crest represents a binary digit, with a chopped-off crest representing a zero and an unmodified crest representing a one. The number of binary digits we can transmit per unit time is thus proportional to the frequency of the wave. Infrared and visible-light waves have much higher frequency than do radio waves, so a modulated laser beam can transmit an enormous amount of information through a single fiber-optic cable.

Another advantage of optical fibers is that they can be made thinner than conventional copper wire, so more fibers can be bundled together in a cable of a given diameter. Hence more distinct signals (for instance, different phone lines) can be sent over the same cable. Because fiber-optic cables are electrical insulators, they are immune to electrical interference from lightning and other sources, and they don’t allow unwanted currents between source and receiver. For these and other reasons, fiber-optic cables play an important role in long-distance telephone, television, and Internet communication.

Total internal reflection also plays an important role in the design of jewelry. The brilliance of diamond is due in large measure to its very high index of refraction ( $n = 2.417$ ) and correspondingly small critical angle. Light entering a cut diamond is totally internally reflected from facets on its back surface and then emerges from its front surface (see the photograph that opens this chapter). “Imitation diamond” gems, such as cubic zirconia, are made from less expensive crystalline materials with comparable indexes of refraction.

Figure 33.15 A transparent rod with refractive index greater than that of the surrounding material.

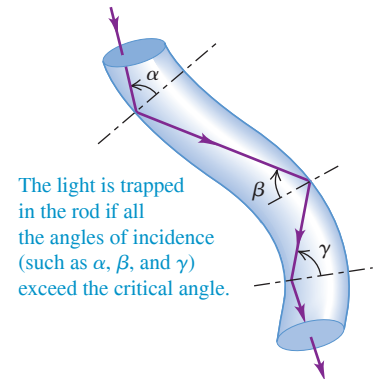
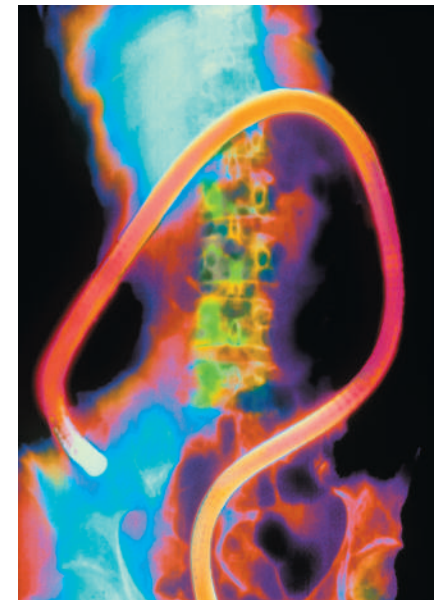


Figure 33.16 This colored x-ray image of a patient’s abdomen shows an endoscope winding through the colon.



### CONCEPTUAL EXAMPLE 33.4 A leaky periscope

A submarine periscope uses two totally reflecting  $45^\circ\text{--}45^\circ\text{--}90^\circ$  prisms with total internal reflection on the sides adjacent to the  $45^\circ$  angles. Explain why the periscope will no longer work if it springs a leak and the bottom prism is covered with water.

**SOLUTION** The critical angle for water ( $n_b = 1.33$ ) on glass ( $n_a = 1.52$ ) is

$$\theta_{\text{crit}} = \arcsin \frac{1.33}{1.52} = 61.0^\circ$$

The  $45^\circ$  angle of incidence for a totally reflecting prism is *smaller* than this new  $61^\circ$  critical angle, so total internal reflection does not occur at the glass–water interface. Most of the light is transmitted into the water, and very little is reflected back into the prism.

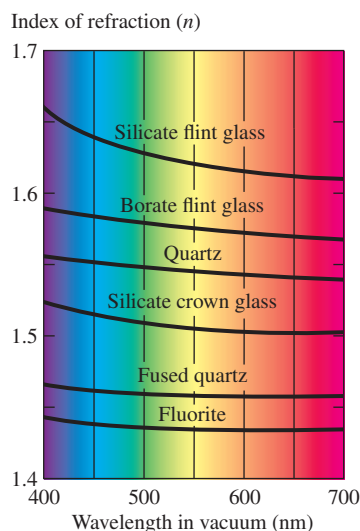
**KEYCONCEPT** For total internal reflection to occur when light in one medium strikes an interface with a second medium, two conditions must be met: The index of refraction  $n_b$  of the second medium must be smaller than the index of refraction  $n_a$  of the first medium, and the angle of incidence must be larger than the critical angle (whose value depends on the ratio  $n_b/n_a$ ).



**TEST YOUR UNDERSTANDING OF SECTION 33.3** In which of the following situations is there total internal reflection? (i) Light propagating in water ( $n = 1.33$ ) strikes a water–air interface at an incident angle of  $70^\circ$ ; (ii) light propagating in glass ( $n = 1.52$ ) strikes a glass–water interface at an incident angle of  $70^\circ$ ; (iii) light propagating in water strikes a water–glass interface at an incident angle of  $70^\circ$ .

**ANSWER** (i), (ii) Total internal reflection can occur only if two conditions are met:  $n_b$  must be less than  $n_a$ , and the critical angle  $\theta_{\text{crit}} = \sin^{-1}(n_b/n_a)$  must be smaller than the angle of incidence  $\theta_a$ . In the first two cases both conditions are met: The critical angles are (i)  $\theta_{\text{crit}} = \sin^{-1}(1/1.33) = 48.8^\circ$  and (ii)  $\theta_{\text{crit}} = \sin^{-1}(1.33/1.52) = 61.0^\circ$ , both of which are smaller than  $\theta_a = 70^\circ$ . In the third case  $n_b = 1.52$  is greater than  $n_a = 1.33$ , so total internal reflection cannot occur for any incident angle.

Figure 33.17 Variation of index of refraction  $n$  with wavelength for different transparent materials. The horizontal axis shows the wavelength  $\lambda_0$  of the light in vacuum; the wavelength in the material is equal to  $\lambda = \lambda_0/n$ .



## 33.4 DISPERSION

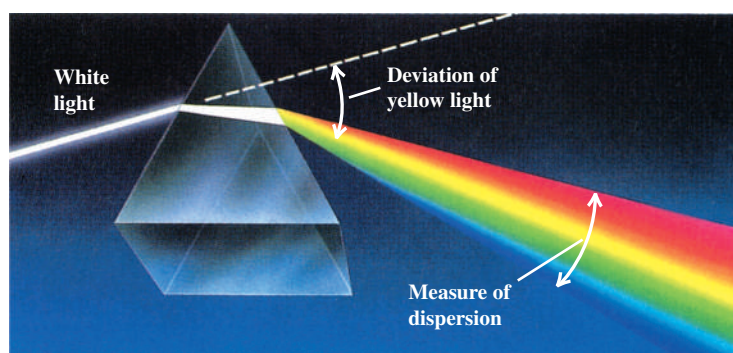
Ordinary white light is a superposition of waves with all visible wavelengths. The speed of light in vacuum is the same for all wavelengths, but the speed in a material substance is different for different wavelengths. Therefore the index of refraction of a material depends on wavelength. The dependence of wave speed and index of refraction on wavelength is called **dispersion**.

Figure 33.17 shows the variation of index of refraction  $n$  with wavelength for some common optical materials. Note that the horizontal axis of this figure is the wavelength of the light in vacuum,  $\lambda_0$ ; the wavelength in the material is given by Eq. (33.5),  $\lambda = \lambda_0/n$ . In most materials the value of  $n$  decreases with increasing wavelength and decreasing frequency, and thus  $n$  increases with decreasing wavelength and increasing frequency. In such a material, light of longer wavelength has greater speed than light of shorter wavelength.

Figure 33.18 shows a ray of white light incident on a prism. The deviation (change of direction) produced by the prism increases with increasing index of refraction and frequency and decreasing wavelength. So violet light is deviated most, and red is deviated least. When it comes out of the prism, the light is spread out into a fan-shaped beam, as shown. The light is said to be *dispersed* into a spectrum. The amount of dispersion depends on the *difference* between the refractive indexes for violet light and for red light. From Fig. 33.17 we can see that for fluorite, the difference between the indexes for red and violet is small, and the dispersion will also be small. A better choice of material for a prism whose purpose is to produce a spectrum would be silicate flint glass, for which there is a larger difference in the value of  $n$  between red and violet.

As we mentioned in Section 33.3, the brilliance of diamond is due in part to its unusually large refractive index; another important factor is its large dispersion, which causes white light entering a diamond to emerge as a multicolored spectrum. Crystals of rutile and of strontium titanate, which can be produced synthetically, have about eight times the dispersion of diamond.

Figure 33.18 Dispersion of light by a prism. The band of colors is called a spectrum.

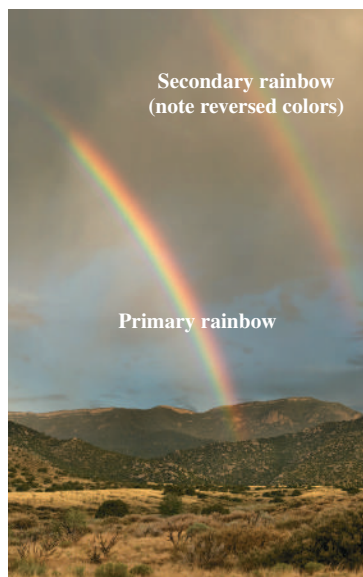


## Rainbows

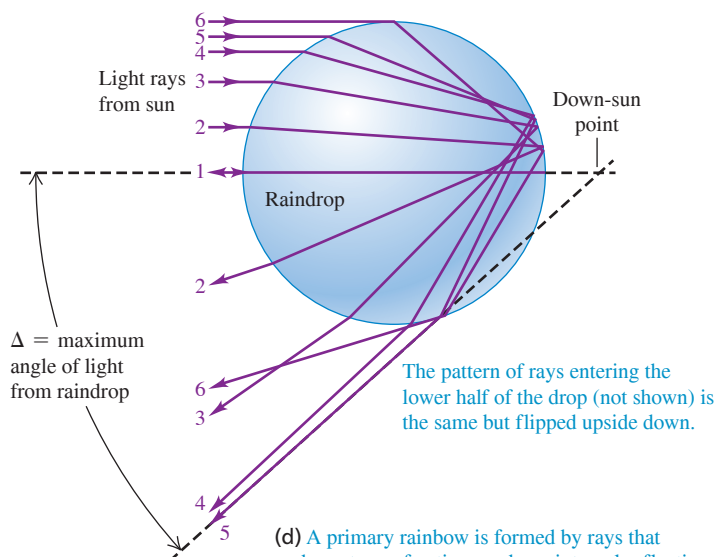
When you experience the beauty of a rainbow, as in **Fig. 33.19a**, you are seeing the combined effects of dispersion, refraction, and reflection. Sunlight comes from behind you, enters a water droplet, is (partially) reflected from the back surface of the droplet, and is refracted again upon exiting the droplet (Fig. 33.19b). A light ray that enters the middle of the raindrop is reflected straight back. All other rays exit the raindrop within an angle  $\Delta$  of that middle ray, with many rays “piling up” at the angle  $\Delta$ . What you see is a disk of light of angular radius  $\Delta$  centered on the down-sun point (the point in the sky opposite the sun); due to the “piling up” of light rays, the disk is brightest around its rim, which we see as a rainbow (Fig. 33.19c). Because no light reaches your eye from angles larger than  $\Delta$ , the sky looks dark outside the rainbow (see Fig. 33.19a). The value of the angle  $\Delta$  depends on the index of refraction of the water that makes up the raindrops, which in turn depends on the wavelength (Fig. 33.19d). The bright disk of red light is slightly larger than that for

Figure 33.19 How rainbows form.

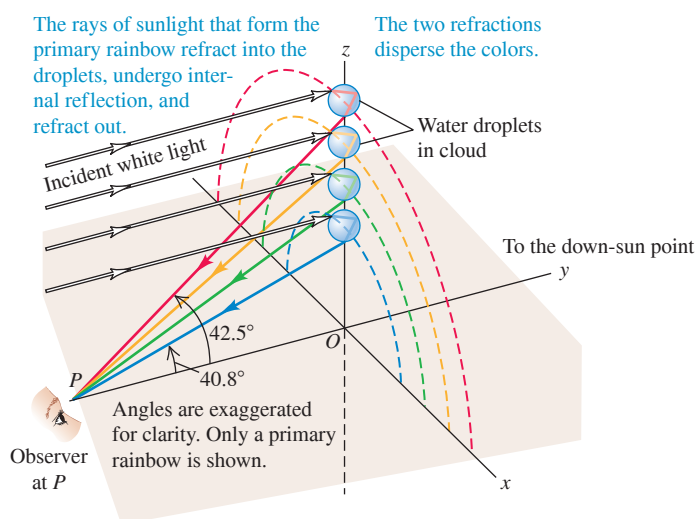
(a) A double rainbow



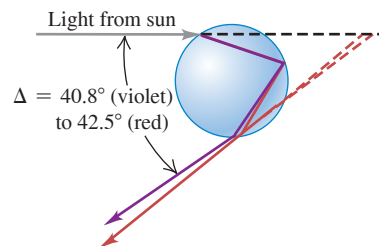
(b) The paths of light rays entering the upper half of a raindrop



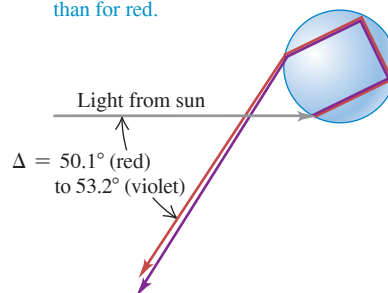
(c) Forming a rainbow. The sun in this illustration is directly behind the observer at  $P$ .



(d) A primary rainbow is formed by rays that undergo two refractions and one internal reflection. The angle  $\Delta$  is larger for red light than for violet.



(e) A secondary rainbow is formed by rays that undergo two refractions and two internal reflections. The angle  $\Delta$  is larger for violet light than for red.



orange light, which in turn is slightly larger than that for yellow light, and so on. As a result, you see the rainbow as a band of colors.

In many cases you can see a second, larger rainbow. It is the result of dispersion, refraction, and *two* reflections from the back surface of the droplet (Fig. 33.19e). Each time a light ray hits the back surface, part of the light is refracted out of the drop (not shown in Fig. 33.19); after two such hits, relatively little light is left inside the drop, which is why the secondary rainbow is noticeably fainter than the primary rainbow. Just as a mirror held up to a book reverses the printed letters, so the second reflection reverses the sequence of colors in the secondary rainbow. You can see this effect in Fig. 33.19a.

### 33.5 POLARIZATION

*Polarization* is a characteristic of all transverse waves. This chapter is about light, but to introduce some basic polarization concepts, let's go back to the transverse waves on a string that we studied in Chapter 15. For a string that in equilibrium lies along the  $x$ -axis, the displacements may be along the  $y$ -direction, as in **Fig. 33.20a**. In this case the string always lies in the  $xy$ -plane. But the displacements might instead be along the  $z$ -axis, as in Fig. 33.20b; then the string always lies in the  $xz$ -plane.

When a wave has only  $y$ -displacements, we say that it is **linearly polarized** in the  $y$ -direction; a wave with only  $z$ -displacements is linearly polarized in the  $z$ -direction. For mechanical waves we can build a **polarizing filter**, or **polarizer**, that permits only waves with a certain polarization direction to pass. In Fig. 33.20c the string can slide vertically in the slot without friction, but no horizontal motion is possible. This filter passes waves that are polarized in the  $y$ -direction but blocks those that are polarized in the  $z$ -direction.

This same language can be applied to electromagnetic waves, which also have polarization. As we learned in Chapter 32, an electromagnetic wave is a transverse wave; the fluctuating electric and magnetic fields are perpendicular to each other and to the direction of propagation. We always define the direction of polarization of an electromagnetic wave to be the direction of the *electric*-field vector  $\vec{E}$ , not the magnetic field, because many common electromagnetic-wave detectors respond to the electric forces on electrons in materials, not the magnetic forces. Thus the electromagnetic wave described by Eq. (32.17),

$$\vec{E}(x, t) = \hat{j}E_{\max} \cos(kx - \omega t)$$

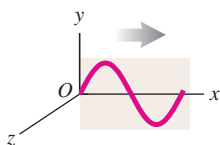
$$\vec{B}(x, t) = \hat{k}B_{\max} \cos(kx - \omega t)$$

is said to be polarized in the  $y$ -direction because the electric field has only a  $y$ -component.

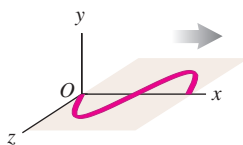
**CAUTION The meaning of “polarization”** It's unfortunate that the same word “polarization” that is used to describe the direction of  $\vec{E}$  in an electromagnetic wave is also used to describe the shifting of electric charge within an object, such as in response to a nearby charged object; we described this latter kind of polarization in Section 21.2 (see Fig. 21.7). Don't confuse these two concepts! ■

Figure 33.20 (a), (b) Polarized waves on a string. (c) Making a polarized wave on a string from an unpolarized one using a polarizing filter.

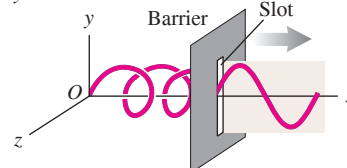
(a) Transverse wave linearly polarized in the  $y$ -direction



(b) Transverse wave linearly polarized in the  $z$ -direction



(c) The slot functions as a polarizing filter, passing only components polarized in the  $y$ -direction.



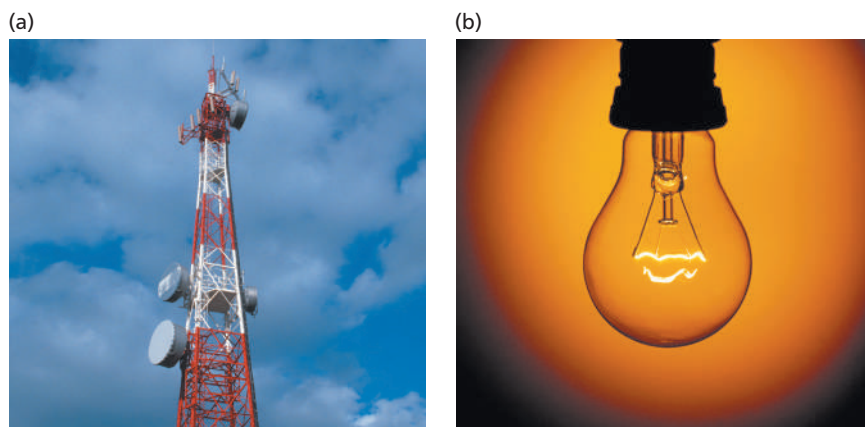


Figure 33.21 (a) Electrons in the red and white broadcast antenna oscillate vertically, producing vertically polarized electromagnetic waves that propagate away from the antenna in the horizontal direction. (The small gray antennas are for relaying cellular phone signals.) (b) No matter how this light bulb is oriented, the random motion of electrons in the filament produces unpolarized light waves.

## Polarizing Filters

Waves emitted by a radio transmitter are usually linearly polarized. The vertical antennas that are used for radio broadcasting emit waves that, in a horizontal plane around the antenna, are polarized in the vertical direction (parallel to the antenna) (Fig. 33.21a).

The situation is different for visible light. Light from incandescent light bulbs and fluorescent light fixtures is *not* polarized (Fig. 33.21b). The “antennas” that radiate light waves are the molecules that make up the sources. The waves emitted by any one molecule may be linearly polarized, like those from a radio antenna. But any actual light source contains a tremendous number of molecules with random orientations, so the emitted light is a random mixture of waves linearly polarized in all possible transverse directions. Such light is called **unpolarized light** or **natural light**. To create polarized light from unpolarized natural light requires a filter that is analogous to the slot for mechanical waves in Fig. 33.20c.

Polarizing filters for electromagnetic waves have different details of construction, depending on the wavelength. For microwaves with a wavelength of a few centimeters, a good polarizer is an array of closely spaced, parallel conducting wires that are insulated from each other. (Think of a barbecue grill with the outer metal ring replaced by an insulating one.) Electrons are free to move along the length of the conducting wires and will do so in response to a wave whose  $\vec{E}$  field is parallel to the wires. The resulting currents in the wires dissipate energy by  $I^2R$  heating; the dissipated energy comes from the wave, so whatever wave passes through the grid is greatly reduced in amplitude. Waves with  $\vec{E}$  oriented perpendicular to the wires pass through almost unaffected, since electrons cannot move through the air between the wires. Hence a wave that passes through such a filter will be predominantly polarized in the direction perpendicular to the wires.

The most common polarizing filter for visible light is a material known by the trade name Polaroid, widely used for sunglasses and polarizing filters for camera lenses. This material incorporates substances that have **dichroism**, a selective absorption in which one of the polarized components is absorbed much more strongly than the other (Fig. 33.22). A Polaroid filter transmits 80% or more of the intensity of a wave that is polarized parallel to the **polarizing axis** of the material, but only 1% or less for waves that are polarized perpendicular to this axis. In one type of Polaroid filter, long-chain molecules within the filter are oriented with their axis perpendicular to the polarizing axis; these molecules preferentially absorb light that is polarized along their length, much like the conducting wires in a polarizing filter for microwaves.

## Using Polarizing Filters

An *ideal* polarizing filter (polarizer) passes 100% of the incident light that is polarized parallel to the filter’s polarizing axis but completely blocks all light that is polarized perpendicular to this axis. Such a device is an unattainable idealization, but the concept is useful in clarifying the basic ideas. In the following discussion we’ll assume that all

Figure 33.22 A Polaroid filter is illuminated by unpolarized natural light (shown by  $\vec{E}$  vectors that point in all directions perpendicular to the direction of propagation). The transmitted light is linearly polarized along the polarizing axis (shown by  $\vec{E}$  vectors along the polarization direction only).

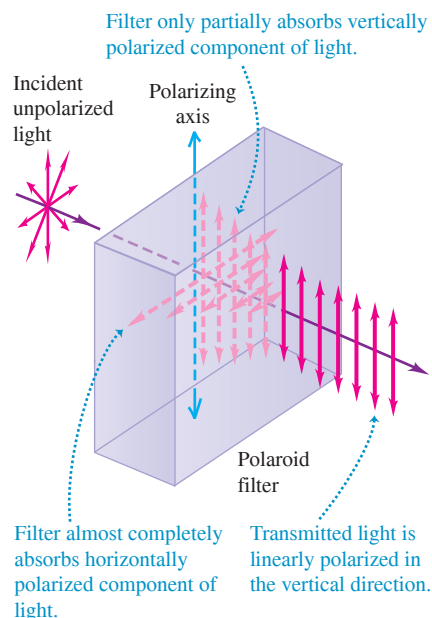
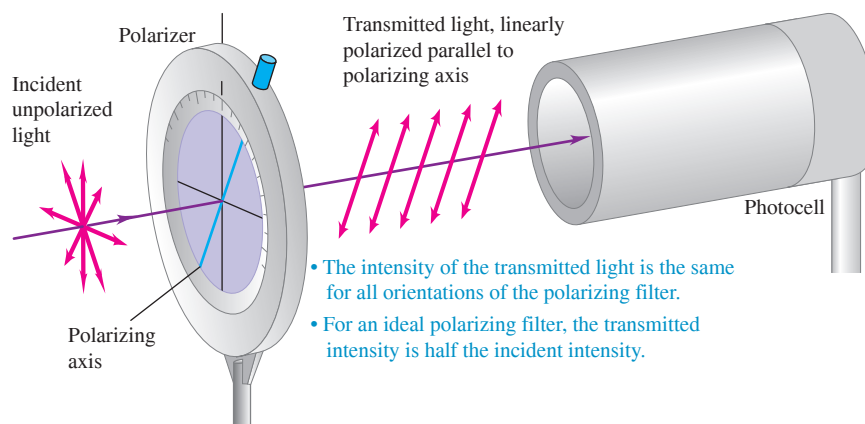




Figure 33.23 Unpolarized natural light is incident on the polarizing filter. The photocell measures the intensity of the transmitted linearly polarized light.



polarizing filters are ideal. In **Fig. 33.23** unpolarized light is incident on a flat polarizing filter. The  $\vec{E}$  vector of the incident wave can be represented in terms of components parallel and perpendicular to the polarizer axis (shown in blue); only the component of  $\vec{E}$  parallel to the polarizing axis is transmitted. Hence the light emerging from the polarizer is linearly polarized parallel to the polarizing axis.

When unpolarized light is incident on an ideal polarizer as in Fig. 33.23, the intensity of the transmitted light is *exactly half* that of the incident unpolarized light, no matter how the polarizing axis is oriented. Here's why: We can resolve the  $\vec{E}$  field of the incident wave into a component parallel to the polarizing axis and a component perpendicular to it. Because the incident light is a random mixture of all states of polarization, these two components are, on average, equal. The ideal polarizer transmits only the component that is parallel to the polarizing axis, so half the incident intensity is transmitted.

What happens when the linearly polarized light emerging from a polarizer passes through a second polarizer, or *analyzer*, as in **Fig. 33.24**? Suppose the polarizing axis of the analyzer makes an angle  $\phi$  with the polarizing axis of the first polarizer. We can resolve the linearly polarized light that is transmitted by the first polarizer into two components, as shown in Fig. 33.24, one parallel and the other perpendicular to the axis of the analyzer. Only the parallel component, with amplitude  $E \cos \phi$ , is transmitted by the analyzer. The transmitted intensity is greatest when  $\phi = 0$ , and it is zero when the polarizer and analyzer are *crossed* so that  $\phi = 90^\circ$  (**Fig. 33.25**). To determine the direction of polarization of the light transmitted by the first polarizer, rotate the analyzer until the photocell in Fig. 33.24 measures zero intensity; the polarization axis of the first polarizer is then perpendicular to that of the analyzer.

To find the transmitted intensity at intermediate values of the angle  $\phi$ , we recall from Section 32.4 that the intensity of an electromagnetic wave is proportional to the *square* of the

Figure 33.24 An ideal analyzer transmits only the electric field component parallel to its transmission direction (that is, its polarizing axis).

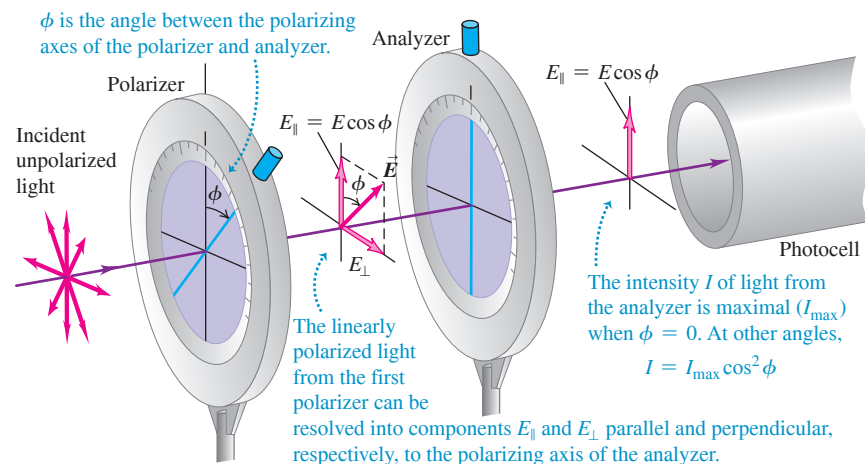


Figure 33.25 These photos show the view through Polaroid sunglasses whose polarizing axes are (left) perpendicular to each other ( $\phi = 90^\circ$ ) and (right) aligned with each other ( $\phi = 0$ ). The transmitted intensity is greatest when the axes are aligned; it is zero when the axes are perpendicular.



amplitude of the wave [see Eq. (32.29)]. The ratio of transmitted to incident *amplitude* is  $\cos \phi$ , so the ratio of transmitted to incident *intensity* is  $\cos^2 \phi$ . Thus the intensity transmitted is

**Malus's law:**

$$I = I_{\max} \cos^2 \phi \quad (33.7)$$

Intensity of polarized light passed through an analyzer  
 Maximum transmitted intensity  
 Angle between polarization axis of light and polarizing axis of analyzer

This relationship, discovered experimentally by Étienne-Louis Malus in 1809, is called **Malus's law**. Malus's law applies *only* if the incident light passing through the analyzer is already linearly polarized.

### PROBLEM-SOLVING STRATEGY 33.2 Linear Polarization

**IDENTIFY** the relevant concepts: In all electromagnetic waves, including light waves, the direction of polarization is the direction of the  $\vec{E}$  field and is perpendicular to the propagation direction. Problems about polarizers are therefore about the components of  $\vec{E}$  parallel and perpendicular to the polarizing axis.

**SET UP** the problem using the following steps:

1. Start by drawing a large, neat diagram. Label all known angles, including the angles of all polarizing axes.
2. Identify the target variables.

**EXECUTE** the solution as follows:

1. Remember that a polarizer lets pass only electric-field components parallel to its polarizing axis.
2. If the incident light is linearly polarized and has amplitude  $E$  and intensity  $I_{\max}$ , the light that passes through an ideal polarizer has amplitude  $E \cos \phi$  and intensity  $I_{\max} \cos^2 \phi$ , where  $\phi$  is the angle

between the incident polarization direction and the filter's polarizing axis.

3. Unpolarized light is a random mixture of all possible polarization states, so on the average it has equal components in any two perpendicular directions. When passed through an ideal polarizer, unpolarized light becomes linearly polarized light with half the incident intensity. Partially linearly polarized light is a superposition of linearly polarized and unpolarized light.
4. The intensity (average power per unit area) of a wave is proportional to the *square* of its amplitude. If you find that two waves differ in amplitude by a certain factor, their intensities differ by the square of that factor.

**EVALUATE** your answer: Check your answer for any obvious errors. If your results say that light emerging from a polarizer has greater intensity than the incident light, something's wrong: A polarizer can't add energy to a light wave.

### EXAMPLE 33.5 Two polarizers in combination

#### WITH VARIATION PROBLEMS

In Fig. 33.24 the incident unpolarized light has intensity  $I_0$ . Find the intensities transmitted by the first and second polarizers if the angle between the axes of the two filters is  $30^\circ$ .

**IDENTIFY and SET UP** This problem involves a polarizer (a polarizing filter on which unpolarized light shines, producing polarized light) and an analyzer (a second polarizing filter on which the polarized light shines). We are given the intensity  $I_0$  of the incident light and the angle  $\phi = 30^\circ$  between the axes of the polarizers. We use Malus's law, Eq. (33.7), to solve for the intensities of the light emerging from each polarizer.

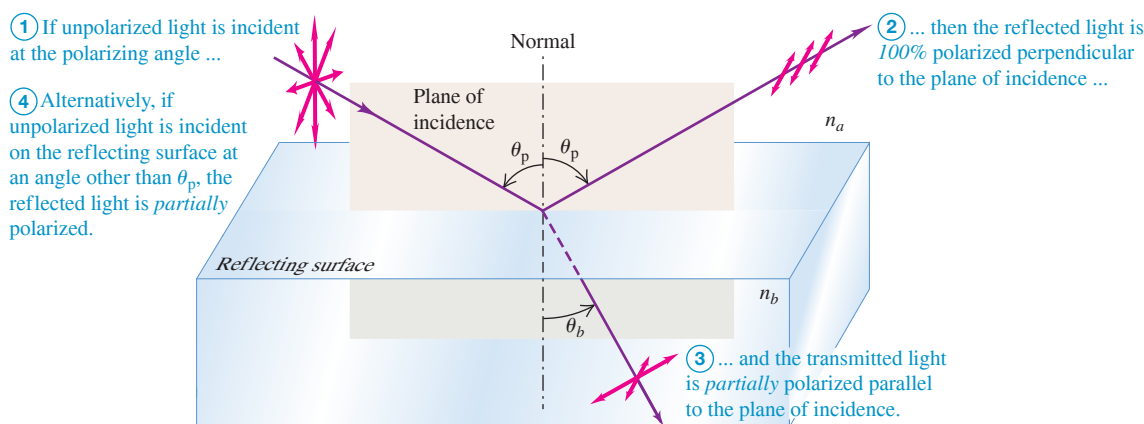
**EXECUTE** The incident light is unpolarized, so the intensity of the linearly polarized light transmitted by the first polarizer is  $I_0/2$ . From Eq. (33.7) with  $\phi = 30^\circ$ , the second polarizer reduces the intensity by a further factor of  $\cos^2 30^\circ = \frac{3}{4}$ . Thus the intensity transmitted by the second polarizer is

$$\left(\frac{I_0}{2}\right)\left(\frac{3}{4}\right) = \frac{3}{8}I_0$$

**EVALUATE** Note that the intensity decreases after each passage through a polarizer. The only situation in which the transmitted intensity does *not* decrease is if the polarizer is ideal (so it absorbs none of the light that passes through it) and if the incident light is linearly polarized along the polarizing axis, so  $\phi = 0$ .

**KEYCONCEPT** When light enters a polarizing filter, the light that emerges is polarized along the polarizing axis of the filter. If the incident light is unpolarized, the intensity of the light that emerges is one-half that of the incident light; if the incident light is polarized at an angle  $\theta$  to the polarization of the filter, the intensity of the light that emerges is  $\cos^2 \theta$  times that of the incident light.

Figure 33.26 When light is incident on a reflecting surface at the polarizing angle, the reflected light is linearly polarized.



### Polarization by Reflection

Unpolarized light can be polarized, either partially or totally, by *reflection*. In **Fig. 33.26**, unpolarized natural light is incident on a reflecting surface between two transparent optical materials. For most angles of incidence, waves for which the electric-field vector  $\vec{E}$  is perpendicular to the plane of incidence (that is, parallel to the reflecting surface) are reflected more strongly than those for which  $\vec{E}$  lies in this plane. In this case the reflected light is *partially polarized* in the direction perpendicular to the plane of incidence.

But at one particular angle of incidence, called the **polarizing angle**  $\theta_p$ , the light for which  $\vec{E}$  lies in the plane of incidence is *not reflected at all* but is completely refracted. At this same angle of incidence the light for which  $\vec{E}$  is perpendicular to the plane of incidence is partially reflected and partially refracted. The *reflected* light is therefore *completely* polarized perpendicular to the plane of incidence, as shown in Fig. 33.26. The *refracted* (transmitted) light is *partially* polarized parallel to this plane; the refracted light is a mixture of the component parallel to the plane of incidence, all of which is refracted, and the remainder of the perpendicular component.

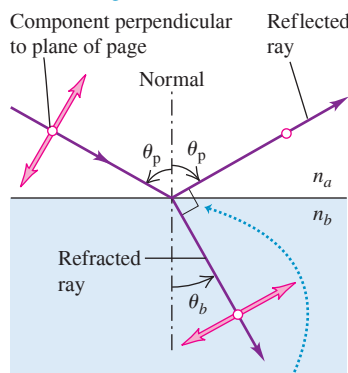
In 1812 the British scientist Sir David Brewster discovered that when the angle of incidence is equal to the polarizing angle  $\theta_p$ , the reflected ray and the refracted ray are perpendicular to each other (**Fig. 33.27**). In this case the angle of refraction  $\theta_b$  equals  $90^\circ - \theta_p$ . From the law of refraction,

$$n_a \sin \theta_p = n_b \sin \theta_b = n_b \sin(90^\circ - \theta_p) = n_b \cos \theta_p$$

Since  $(\sin \theta_p)/(\cos \theta_p) = \tan \theta_p$ , we can rewrite this equation as

Figure 33.27 The significance of the polarizing angle. The open circles represent a component of  $\vec{E}$  that is perpendicular to the plane of the figure (the plane of incidence) and parallel to the surface between the two materials.

Note: This is a side view of the situation shown in Fig. 33.26.



When light strikes a surface at the polarizing angle, the reflected and refracted rays are perpendicular to each other and

$$\tan \theta_p = \frac{n_b}{n_a}$$

**Brewster's law for the polarizing angle:**

Polarizing angle (angle of incidence for which reflected light is 100% polarized)

$$\tan \theta_p = \frac{n_b}{n_a} \quad (33.8)$$

Index of refraction of second material  
Index of refraction of first material

This relationship is known as **Brewster's law**. Although discovered experimentally, it can also be *derived* from a wave model by using Maxwell's equations.

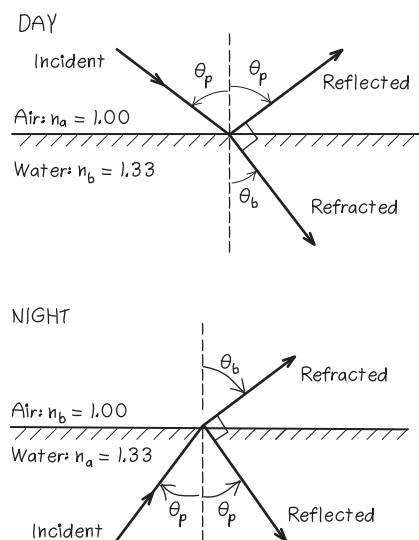
Polarization by reflection is the reason polarizing filters are widely used in sunglasses (Fig. 33.25). When sunlight is reflected from a horizontal surface, the plane of incidence is vertical, and the reflected light contains a preponderance of light that is polarized in the horizontal direction. When the reflection occurs at a smooth asphalt road surface, it causes unwanted glare. To eliminate this glare, the polarizing axis of the lens material is made vertical, so very little of the horizontally polarized light reflected from the road is transmitted to the eyes. The glasses also reduce the overall intensity of the transmitted light to somewhat less than 50% of the intensity of the unpolarized incident light.

**EXAMPLE 33.6** Reflection from a swimming pool's surface**WITH VARIATION PROBLEMS**

Sunlight reflects off the smooth surface of a swimming pool. (a) For what angle of reflection is the reflected light completely polarized? (b) What is the corresponding angle of refraction? (c) At night, an underwater floodlight is turned on in the pool. Repeat parts (a) and (b) for rays from the floodlight that strike the surface from below.

**IDENTIFY and SET UP** This problem involves polarization by reflection at an air–water interface in parts (a) and (b) and at a water–air interface in part (c). **Figure 33.28** shows our sketches. For both cases our first target variable is the polarizing angle  $\theta_p$ , which we find from Brewster's law, Eq. (33.8). For this angle of reflection, the angle of refraction  $\theta_b$  is the complement of  $\theta_p$  (that is,  $\theta_b = 90^\circ - \theta_p$ ).

Figure 33.28 Our sketches for this problem.



**EXECUTE** (a) During the day (shown in the upper part of Fig. 33.28) the light moves in air toward water, so  $n_a = 1.00$  (air) and  $n_b = 1.33$  (water). From Eq. (33.8),

$$\theta_p = \arctan \frac{n_b}{n_a} = \arctan \frac{1.33}{1.00} = 53.1^\circ$$

(b) The incident light is at the polarizing angle, so the reflected and refracted rays are perpendicular; hence

$$\theta_b = 90^\circ - \theta_p = 90^\circ - 53.1^\circ = 36.9^\circ$$

(c) At night (shown in the lower part of Fig. 33.28) the light moves in water toward air, so now  $n_a = 1.33$  and  $n_b = 1.00$ . Again using Eq. (33.8), we have

$$\theta_p = \arctan \frac{1.00}{1.33} = 36.9^\circ$$

$$\theta_b = 90^\circ - 36.9^\circ = 53.1^\circ$$

**EVALUATE** We check our answer in part (b) by using Snell's law,  $n_a \sin \theta_a = n_b \sin \theta_b$ , to solve for  $\theta_b$ :

$$\sin \theta_b = \frac{n_a \sin \theta_p}{n_b} = \frac{1.00 \sin 53.1^\circ}{1.33} = 0.601$$

$$\theta_b = \arcsin(0.601) = 36.9^\circ$$

Note that the two polarizing angles found in parts (a) and (c) add to  $90^\circ$ . This is *not* an accident; can you see why?

**KEYCONCEPT** When light strikes an interface between two materials, how much is reflected depends on the polarization of the incident light. If the angle of incidence equals the polarizing angle, light polarized in the plane of incidence is completely refracted into the second material; none is reflected. Light polarized perpendicular to the plane of incidence is partially refracted and partially reflected, so the reflected light is completely polarized in the perpendicular direction.

**Circular and Elliptical Polarization**

Light and other electromagnetic radiation can also have *circular* or *elliptical* polarization. To introduce these concepts, let's return once more to mechanical waves on a stretched string. In Fig. 33.20, suppose the two linearly polarized waves in parts (a) and (b) are in phase and have equal amplitude. When they are superposed, each point in the string has simultaneous  $y$ - and  $z$ -displacements of equal magnitude. A little thought shows that the resultant wave lies in a plane oriented at  $45^\circ$  to the  $y$ - and  $z$ -axes (i.e., in a plane making a  $45^\circ$  angle with the  $xy$ - and  $xz$ -planes). The amplitude of the resultant wave is larger by a factor of  $\sqrt{2}$  than that of either component wave, and the resultant wave is linearly polarized.

But now suppose the two equal-amplitude waves differ in phase by a quarter-cycle. Then the resultant motion of each point corresponds to a superposition of two simple harmonic motions at right angles, with a quarter-cycle phase difference. The  $y$ -displacement at a point is greatest at times when the  $z$ -displacement is zero, and vice versa. The string as a whole then no longer moves in a single plane. It can be shown that each point on the rope moves in a *circle* in a plane parallel to the  $yz$ -plane. Successive points on the rope have successive phase differences, and the overall motion of the string has the appearance of a rotating helix, as shown to the left of the polarizing filter in Fig. 33.20c. Such a superposition of two linearly polarized waves is called **circular polarization**.



**APPLICATION Circular Polarization and 3-D Movies**

The lenses of the special glasses you wear to see a 3-D movie are circular polarizing filters. The lens over one eye allows only right circularly polarized light to pass; the other lens allows only left circularly polarized light to pass. The projector alternately projects the images intended for the left eye and those intended for the right eye. A special filter synchronized with the projector and in front of its lens circularly polarizes the projected light, with alternate polarization for each frame. Hence alternate images go to your left and right eyes, with such a short time interval between them that they produce the illusion of viewing with both eyes simultaneously.



**Figure 33.29** shows the analogous situation for an electromagnetic wave. Two sinusoidal waves of equal amplitude, polarized in the  $y$ - and  $z$ -directions and with a quarter-cycle phase difference, are superposed. The result is a wave in which the  $\vec{E}$  vector at each point has a constant magnitude but *rotates* around the direction of propagation. The wave in Fig. 33.29 is propagating toward you and the  $\vec{E}$  vector appears to be rotating clockwise, so it is called a *right circularly polarized* electromagnetic wave. If instead the  $\vec{E}$  vector of a wave coming toward you appears to be rotating counterclockwise, it is called a *left circularly polarized* electromagnetic wave.

If the phase difference between the two component waves is something other than a quarter-cycle, or if the two component waves have different amplitudes, then each point on the string traces out not a circle but an *ellipse*. The resulting wave is said to be **elliptically polarized**.

For electromagnetic waves with radio frequencies, circular or elliptical polarization can be produced by using two antennas at right angles, fed from the same transmitter but with a phase-shifting network that introduces the appropriate phase difference. For light, the phase shift can be introduced by use of a material that exhibits *birefringence*—that is, has different indexes of refraction for different directions of polarization. A common example is calcite ( $\text{CaCO}_3$ ). When a calcite crystal is oriented appropriately in a beam of unpolarized light, its refractive index (for a wavelength in vacuum of 589 nm) is 1.658 for one direction of polarization and 1.486 for the perpendicular direction. When two waves with equal amplitude and with perpendicular directions of polarization enter such a material, they travel with different speeds. If they are in phase when they enter the material, then in general they are no longer in phase when they emerge. If the crystal is just thick enough to introduce a quarter-cycle phase difference, then the crystal converts linearly polarized light to circularly polarized light. Such a crystal is called a *quarter-wave plate*. Such a plate also converts circularly polarized light to linearly polarized light. Can you prove this?

**Photoelasticity**

Some optical materials that are not normally birefringent become so when they are subjected to mechanical stress. This is the basis of the science of *photoelasticity*. Stresses in girders, boiler plates, gear teeth, and cathedral pillars can be analyzed by constructing a transparent model of the object, usually of a plastic material, subjecting it to stress, and examining it between a polarizer and an analyzer in the crossed position. Very complicated stress distributions can be studied by these optical methods.

**Figure 33.29** Circular polarization of an electromagnetic wave moving toward you parallel to the  $x$ -axis. The  $y$ -component of  $\vec{E}$  lags the  $z$ -component by a quarter-cycle. This phase difference results in right circular polarization.

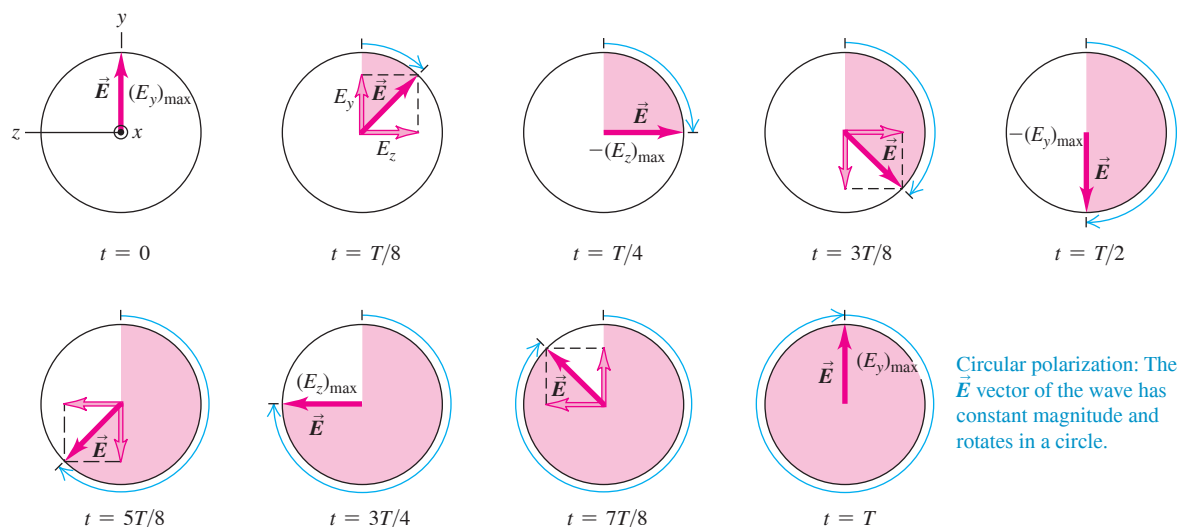
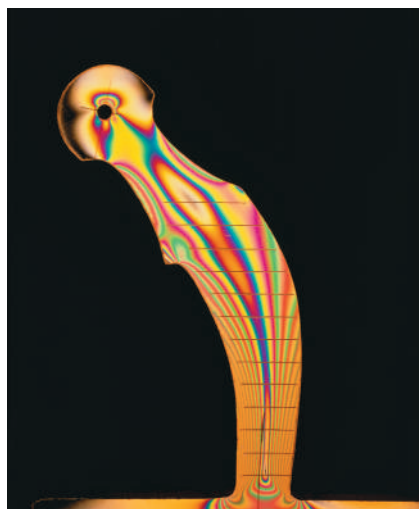


Figure 33.30 This plastic model of an artificial hip joint was photographed between two polarizing filters (a polarizer and an analyzer) with perpendicular polarizing axes. The colored interference pattern reveals the direction and magnitude of stresses in the model. Engineers use these results to help design the actual hip joint (used in hip replacement surgery), which is made of metal.



**Figure 33.30** is a photograph of a photoelastic model under stress. The polarized light that enters the model can be thought of as having a component along each of the two directions of the birefringent plastic. Since these two components travel through the plastic at different speeds, the light that emerges from the other side of the model can have a different overall direction of polarization. Hence some of this transmitted light will be able to pass through the analyzer even though its polarization axis is at a  $90^\circ$  angle to the polarizer's axis, and the stressed areas in the plastic will appear as bright spots. The amount of birefringence is different for different wavelengths and hence different colors of light; the color that appears at each location in Fig. 33.30 is that for which the transmitted light is most nearly polarized along the analyzer's polarization axis.

**TEST YOUR UNDERSTANDING OF SECTION 33.5** You are taking a photograph of a sunlit office building at sunrise, so the plane of incidence is nearly horizontal. In order to minimize the reflections from the building's windows, you place a polarizing filter on the camera lens. How should you orient the filter? (i) With the polarizing axis vertical; (ii) with the polarizing axis horizontal; (iii) either orientation will minimize the reflections just as well; (iv) neither orientation will have any effect.

#### ANSWER

(iii) The sunlight reflected from the windows of the high-rise building is partially polarized in the vertical direction, perpendicular to the horizontal plane of incidence. The Polaroid filter in front of the lens is oriented with its polarizing axis perpendicular to the dominant direction of polarization of the reflected light.

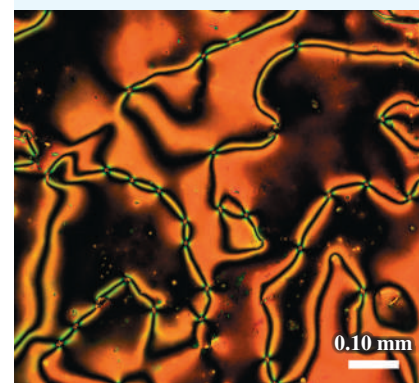
## 33.6 SCATTERING OF LIGHT

The sky is blue. Sunsets are red. Skylight is partially polarized; that's why the sky looks darker from some angles than from others when it is viewed through Polaroid sunglasses. As we'll see, a single phenomenon is responsible for all of these effects.

When you look at the daytime sky, the light that you see is sunlight that has been absorbed and then re-radiated in a variety of directions. This process is called **scattering**. (If the earth had no atmosphere, the sky would appear as black in the daytime as it does at night, just as it does to an astronaut in space or on the moon.) **Figure 33.31** (next page)

**APPLICATION Birefringence and Liquid Crystal Displays** In each pixel of an LCD computer screen is a birefringent material called a liquid crystal. This material is composed of rod-shaped molecules that align to produce a fluid with two different indexes of refraction. The liquid crystal is placed between linear polarizing filters with perpendicular polarizing axes, and the sandwich of filters and liquid crystal is backlit. The two polarizers by themselves would not transmit light, but like the birefringent object in Fig. 33.30, the liquid crystal allows light to pass through. Varying the voltage across a pixel turns the birefringence effect on and off, changing the pixel from bright to dark and back again.

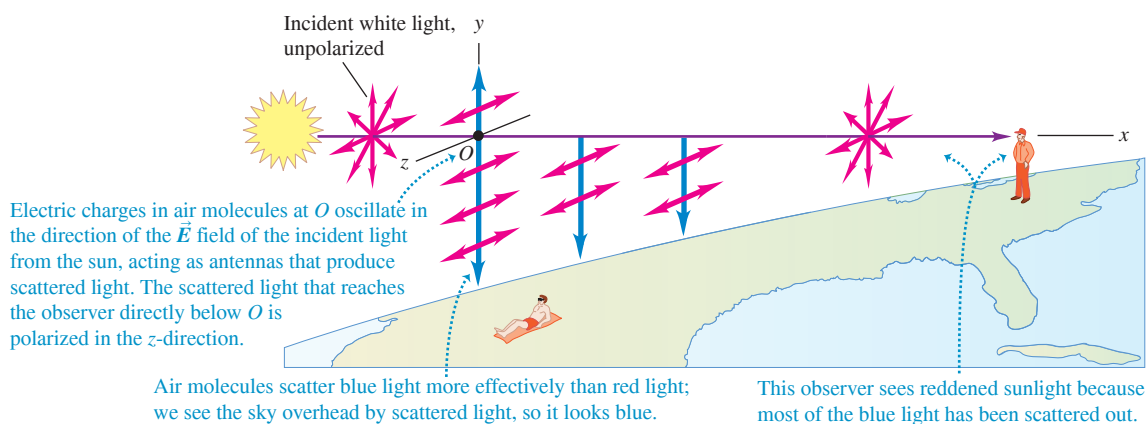
Microscope image of a liquid crystal



Liquid crystal display



Figure 33.31 When the sunbathing observer on the left looks up, he sees blue, polarized sunlight that has been scattered by air molecules. The observer on the right sees reddened, unpolarized light when he looks at the sun.



### BIO APPLICATION Bee Vision and Polarized Light from the Sky

The eyes of a bee can detect the polarization of light. Bees use this ability when they navigate between the hive and food sources. As Fig. 33.31 would suggest, a bee sees unpolarized light if it looks directly toward the sun and sees completely polarized light if it looks  $90^\circ$  away from the sun. These polarizations are unaffected by the presence of clouds, so a bee can navigate relative to the sun even on an overcast day.



Figure 33.32 Clouds are white because they efficiently scatter sunlight of all wavelengths.



shows some of the details of the scattering process. Sunlight, which is unpolarized, comes from the left along the  $x$ -axis and passes over an observer looking vertically upward along the  $y$ -axis. (We are viewing the situation from the side.) Consider the molecules of the earth's atmosphere located at point  $O$ . The electric field in the beam of sunlight sets the electric charges in these molecules into vibration. Since light is a transverse wave, the direction of the electric field in any component of the sunlight lies in the  $yz$ -plane, and the motion of the charges takes place in this plane. There is no field, and hence no motion of charges, in the direction of the  $x$ -axis.

An incident light wave sets the electric charges in the molecules at point  $O$  vibrating along the line of  $\vec{E}$ . We can resolve this vibration into two components, one along the  $y$ -axis and the other along the  $z$ -axis. Each component in the incident light produces the equivalent of two molecular "antennas," oscillating with the same frequency as the incident light and lying along the  $y$ - and  $z$ -axes.

We mentioned in Chapter 32 that an oscillating charge, like those in an antenna, does not radiate in the direction of its oscillation. (See Fig. 32.3 in Section 32.1.) Thus the "antenna" along the  $y$ -axis does not send any light to the observer directly below it, although it does emit light in other directions. Therefore the only light reaching this observer comes from the other molecular "antenna," corresponding to the oscillation of charge along the  $z$ -axis. This light is linearly polarized, with its electric field along the  $z$ -axis (parallel to the "antenna"). The red vectors on the  $y$ -axis below point  $O$  in Fig. 33.31 show the direction of polarization of the light reaching the observer.

As the original beam of sunlight passes through the atmosphere, its intensity decreases as its energy goes into the scattered light. Detailed analysis of the scattering process shows that the intensity of the light scattered from air molecules increases in proportion to the fourth power of the frequency (inversely to the fourth power of the wavelength). Thus the intensity ratio for the two ends of the visible spectrum is  $(750 \text{ nm}/380 \text{ nm})^4 = 15$ . Roughly speaking, scattered light contains 15 times as much blue light as red, and that's why the sky is blue.

Clouds contain a high concentration of suspended water droplets or ice crystals, which also scatter light. Because of this high concentration, light passing through the cloud has many more opportunities for scattering than does light passing through a clear sky. Thus light of *all* wavelengths is eventually scattered out of the cloud, so the cloud looks white (Fig. 33.32). Milk looks white for the same reason; the scattering is due to fat globules suspended in the milk.

Near sunset, when sunlight has to travel a long distance through the earth's atmosphere, a substantial fraction of the blue light is removed by scattering. White light minus blue light looks yellow or red. This explains the yellow or red hue that we so often see from the setting sun (and that is seen by the observer at the far right of Fig. 33.31).



### 33.7 HUYGENS'S PRINCIPLE

The laws of reflection and refraction of light rays, as introduced in Section 33.2, were discovered experimentally long before the wave nature of light was firmly established. However, we can *derive* these laws from wave considerations and show that they are consistent with the wave nature of light.

We begin with a principle called **Huygens's principle**. This principle, stated originally by the Dutch scientist Christiaan Huygens in 1678, is a geometrical method for finding, from the known shape of a wave front at some instant, the shape of the wave front at some later time. Huygens assumed that **every point of a wave front may be considered the source of secondary wavelets that spread out in all directions with a speed equal to the speed of propagation of the wave**. The new wave front at a later time is then found by constructing a surface *tangent* to the secondary wavelets or, as it is called, the *envelope* of the wavelets. All the results that we obtain from Huygens's principle can also be obtained from Maxwell's equations, but Huygens's simple model is easier to use.

**Figure 33.33** illustrates Huygens's principle. The original wave front  $AA'$  is traveling outward from a source, as indicated by the arrows. We want to find the shape of the wave front after a time interval  $t$ . We assume that  $v$ , the speed of propagation of the wave, is the same at all points. Then in time  $t$  the wave front travels a distance  $vt$ . We construct several circles (traces of spherical wavelets) with radius  $r = vt$ , centered at points along  $AA'$ . The trace of the envelope of these wavelets, which is the new wave front, is the curve  $BB'$ .

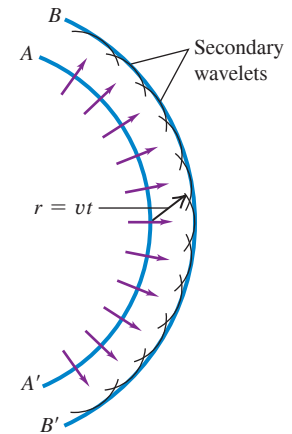


Figure 33.33 Applying Huygens's principle to wave front  $AA'$  to construct a new wave front  $BB'$ .

### Reflection and Huygens's Principle

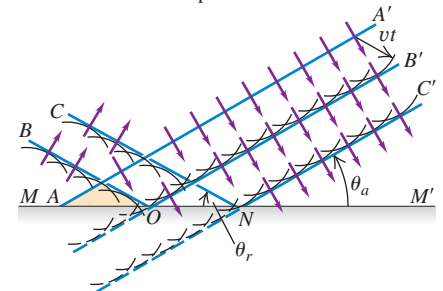
To derive the law of reflection from Huygens's principle, we consider a plane wave approaching a plane reflecting surface. In **Fig. 33.34a** the lines  $AA'$ ,  $OB'$ , and  $NC'$  represent successive positions of a wave front approaching the surface  $MM'$ . Point  $A$  on the wave front  $AA'$  has just arrived at the reflecting surface. We can use Huygens's principle to find the position of the wave front after a time interval  $t$ . With points on  $AA'$  as centers, we draw several secondary wavelets with radius  $vt$ . The wavelets that originate near the upper end of  $AA'$  spread out unhindered, and their envelope gives the portion  $OB'$  of the new wave front. If the reflecting surface were not there, the wavelets originating near the lower end of  $AA'$  would similarly reach the positions shown by the broken circular arcs. Instead, these wavelets strike the reflecting surface.

The effect of the reflecting surface is to *change the direction* of travel of those wavelets that strike it, so the part of a wavelet that would have penetrated the surface actually lies to the left of it, as shown by the full lines. The first such wavelet is centered at point  $A$ ; the envelope of all such reflected wavelets is the portion  $OB$  of the wave front. The trace of the entire wave front at this instant is the bent line  $BOB'$ . A similar construction gives the line  $CNC'$  for the wave front after another interval  $t$ .

From plane geometry the angle  $\theta_a$  between the incident wave front and the surface is the same as that between the incident ray and the normal to the surface and is therefore the angle of incidence. Similarly,  $\theta_r$  is the angle of reflection. To find the relationship between these angles, we consider **Fig. 33.34b**. From  $O$  we draw  $OP = vt$ , perpendicular to  $AA'$ . Now  $OB$ , by construction, is tangent to a circle of radius  $vt$  with center at  $A$ . If we draw  $AQ$  from  $A$  to the point of tangency, the triangles  $APO$  and  $OQA$  are congruent because they are right triangles with the side  $AO$  in common and with  $AQ = OP = vt$ . The angle  $\theta_a$  therefore equals the angle  $\theta_r$ , and we have the law of reflection.

Figure 33.34 Using Huygens's principle to derive the law of reflection.

(a) Successive positions of a plane wave  $AA'$  as it is reflected from a plane surface



(b) Magnified portion of (a)

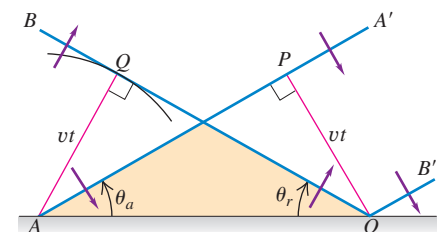
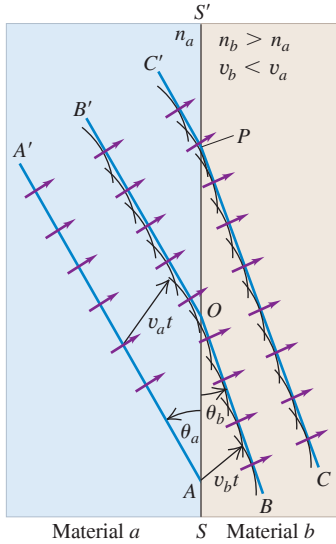


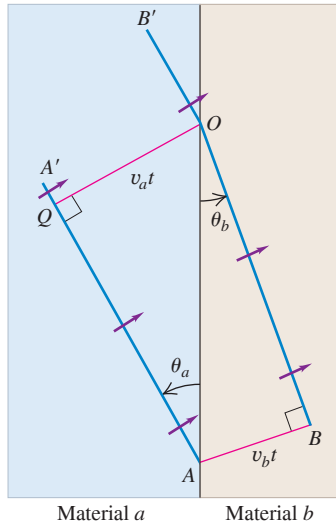


Figure 33.35 Using Huygens's principle to derive the law of refraction. The case  $v_b < v_a$  ( $n_b > n_a$ ) is shown.

(a) Successive positions of a plane wave  $AA'$  as it is refracted by a plane surface



(b) Magnified portion of (a)



## Refraction and Huygens's Principle

We can derive the law of *refraction* by a similar procedure. In **Fig. 33.35a** we consider a wave front, represented by line  $AA'$ , for which point  $A$  has just arrived at the boundary surface  $SS'$  between two transparent materials  $a$  and  $b$ , with indexes of refraction  $n_a$  and  $n_b$  and wave speeds  $v_a$  and  $v_b$ . (The *reflected* waves are not shown; they proceed as in **Fig. 33.34**.) We can apply Huygens's principle to find the position of the refracted wave fronts after a time  $t$ .

With points on  $AA'$  as centers, we draw several secondary wavelets. Those originating near the upper end of  $AA'$  travel with speed  $v_a$  and, after a time interval  $t$ , are spherical surfaces of radius  $v_a t$ . The wavelet originating at point  $A$ , however, is traveling in the second material  $b$  with speed  $v_b$  and at time  $t$  is a spherical surface of radius  $v_b t$ . The envelope of the wavelets from the original wave front is the plane whose trace is the bent line  $BOB'$ . A similar construction leads to the trace  $CPC'$  after a second interval  $t$ .

The angles  $\theta_a$  and  $\theta_b$  between the surface and the incident and refracted wave fronts are the angle of incidence and the angle of refraction, respectively. To find the relationship between these angles, refer to **Fig. 33.35b**. We draw  $OQ = v_a t$ , perpendicular to  $AQ$ , and we draw  $AB = v_b t$ , perpendicular to  $BO$ . From the right triangle  $AOQ$ ,

$$\sin \theta_a = \frac{v_a t}{AO}$$

and from the right triangle  $AOB$ ,

$$\sin \theta_b = \frac{v_b t}{AO}$$

Combining these, we find

$$\frac{\sin \theta_a}{\sin \theta_b} = \frac{v_a}{v_b} \quad (33.9)$$

We have defined the index of refraction  $n$  of a material as the ratio of the speed of light  $c$  in vacuum to its speed  $v$  in the material:  $n_a = c/v_a$  and  $n_b = c/v_b$ . Thus

$$\frac{n_b}{n_a} = \frac{c/v_b}{c/v_a} = \frac{v_a}{v_b}$$

and we can rewrite **Eq. (33.9)** as

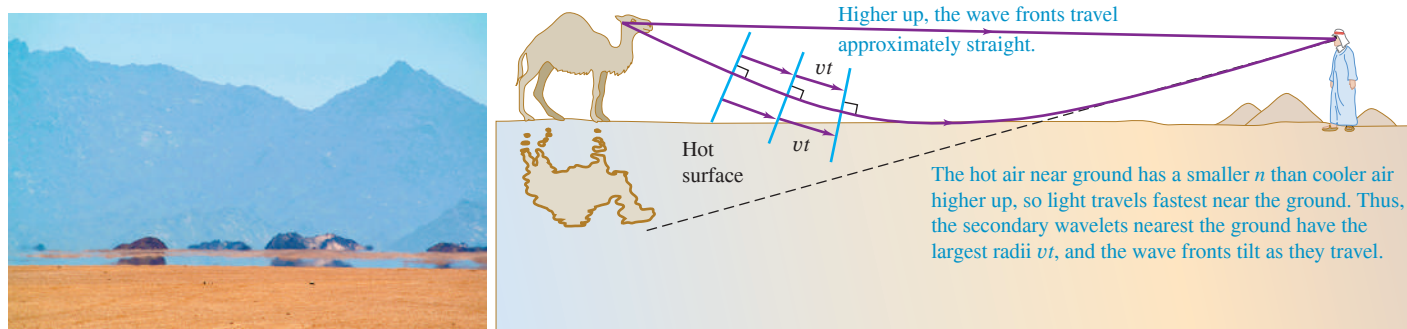
$$\frac{\sin \theta_a}{\sin \theta_b} = \frac{n_b}{n_a} \quad \text{or} \quad n_a \sin \theta_a = n_b \sin \theta_b$$

which we recognize as Snell's law, **Eq. (33.4)**. So we have derived Snell's law from a wave theory. Alternatively, we can regard Snell's law as an experimental result that defines the index of refraction of a material; in that case this analysis helps confirm the relationship  $v = c/n$  for the speed of light in a material.

Mirages are an example of Huygens's principle in action. When the surface of pavement or desert sand is heated intensely by the sun, a hot, less dense, smaller- $n$  layer of air forms near the surface. The speed of light is slightly greater in the hotter air near the ground, the Huygens wavelets have slightly larger radii, the wave fronts tilt slightly, and rays that were headed toward the surface with a large angle of incidence (near  $90^\circ$ ) can be bent up as shown in **Fig. 33.36**. Light farther from the ground is bent less and travels nearly in a straight line. The observer sees the object in its natural position, with an inverted image below it, as though seen in a horizontal reflecting surface. A thirsty traveler can interpret the apparent reflecting surface as a sheet of water.

It is important to keep in mind that Maxwell's equations are the fundamental relationships for electromagnetic wave propagation. But Huygens's principle provides a convenient way to visualize this propagation.

Figure 33.36 How mirages are formed.



**TEST YOUR UNDERSTANDING OF SECTION 33.7** Sound travels faster in warm air than in cold air. Imagine a weather front that runs north-south, with warm air to the west of the front and cold air to the east. A sound wave traveling in a northeast direction in the warm air encounters this front. How will the direction of this sound wave change when it passes into the cold air? (i) The wave direction will deflect toward the north; (ii) the wave direction will deflect toward the east; (iii) the wave direction will be unchanged.

### ANSWER

(ii) Huygens's principle applies to waves of all kinds, including sound waves. Hence this situation is exactly like that shown in Fig. 33.35, with material  $a$  representing the warm air, material  $b$  representing the cold air in which the waves travel more slowly, and the interface between the materials representing the weather front. North is toward the top of the figure and east is toward the right, so Fig. 33.35 shows that the rays (which indicate the direction of propagation) deflect toward the east.

## CHAPTER 33 SUMMARY

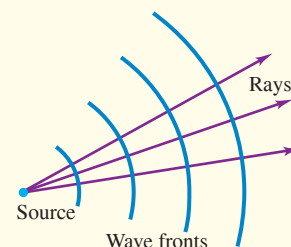
**Light and its properties:** Light is an electromagnetic wave. When emitted or absorbed, it also shows particle properties. It is emitted by accelerated electric charges.

A wave front is a surface of constant phase; wave fronts move with a speed equal to the propagation speed of the wave. A ray is a line along the direction of propagation, perpendicular to the wave fronts.

When light is transmitted from one material to another, the frequency of the light is unchanged, but the wavelength and wave speed can change. The index of refraction  $n$  of a material is the ratio of the speed of light in vacuum  $c$  to the speed  $v$  in the material. If  $\lambda_0$  is the wavelength in vacuum, the same wave has a shorter wavelength  $\lambda$  in a medium with index of refraction  $n$ . (See Example 33.2.)

$$n = \frac{c}{v} \quad (33.1)$$

$$\lambda = \frac{\lambda_0}{n} \quad (33.5)$$



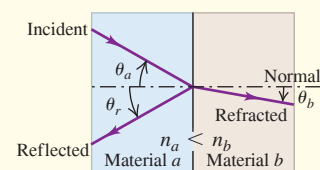
**Reflection and refraction:** At a smooth interface between two optical materials, the incident, reflected, and refracted rays and the normal to the interface all lie in a single plane called the plane of incidence. The law of reflection states that the angles of incidence and reflection are equal. The law of refraction relates the angles of incidence and refraction to the indexes of refraction of the materials. (See Examples 33.1 and 33.3.)

$$\theta_r = \theta_a \quad (33.2)$$

(law of reflection)

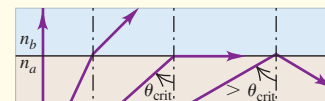
$$n_a \sin \theta_a = n_b \sin \theta_b \quad (33.4)$$

(law of refraction)



**Total internal reflection:** When a ray travels in a material of index of refraction  $n_a$  toward a material of index  $n_b < n_a$ , total internal reflection occurs at the interface when the angle of incidence equals or exceeds a critical angle  $\theta_{\text{crit}}$ . (See Example 33.4.)

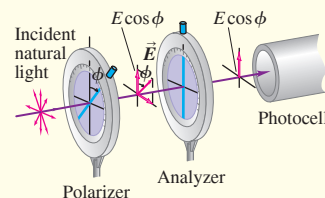
$$\sin \theta_{\text{crit}} = \frac{n_b}{n_a} \quad (33.6)$$



**Polarization of light:** The direction of polarization of a linearly polarized electromagnetic wave is the direction of the  $\vec{E}$  field. A polarizing filter passes waves that are linearly polarized along its polarizing axis and blocks waves polarized perpendicularly to that axis. When polarized light of intensity  $I_{\max}$  is incident on a polarizing filter used as an analyzer, the intensity  $I$  of the light transmitted through the analyzer depends on the angle  $\phi$  between the polarization direction of the incident light and the polarizing axis of the analyzer. (See Example 33.5.)

$$I = I_{\max} \cos^2 \phi \quad (33.7)$$

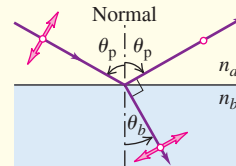
(Malus's law)



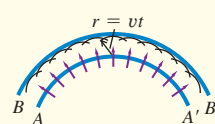
**Polarization by reflection:** When unpolarized light strikes an interface between two materials, Brewster's law states that the reflected light is completely polarized perpendicular to the plane of incidence (parallel to the interface) if the angle of incidence equals the polarizing angle  $\theta_p$ . (See Example 33.6.)

$$\tan \theta_p = \frac{n_b}{n_a} \quad (33.8)$$

(Brewster's law)



**Huygens's principle:** Huygens's principle states that if the position of a wave front at one instant is known, then the position of the front at a later time can be constructed by imagining the front as a source of secondary wavelets. Huygens's principle can be used to derive the laws of reflection and refraction.



Chapter 33 Media Assets



## GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

### KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 33.1 and 33.2** (Section 33.2) before attempting these problems.

**VP33.2.1** A block of glass with index of refraction 1.80 has a smooth surface. Light in air strikes this surface at an angle of incidence of  $70.0^\circ$  measured from the normal to the surface of the glass. Find the angles measured relative to this normal of (a) the reflected ray and (b) the refracted ray.

**VP33.2.2** A ray of light in water ( $n = 1.33$ ) strikes a submerged glass block at an angle of incidence of  $55.0^\circ$ . The angle of refraction for the light that enters the glass is  $37.0^\circ$ . Find (a) the index of refraction of the glass and (b) the speed of light in the glass.

**VP33.2.3** The light from a red laser pointer has wavelength 635 nm in air and 508 nm in a transparent liquid. You point the laser in air so that the beam strikes the surface of the liquid at an angle of  $35.0^\circ$  from the normal. Find (a) the index of refraction of the liquid, (b) the angle of refraction, (c) the frequency of the light in air, and (d) the frequency of the light in the liquid.

**VP33.2.4** A glass of ethanol ( $n = 1.36$ ) has an ice cube ( $n = 1.309$ ) floating in it. A light beam in the ethanol goes into the ice cube at an angle of refraction of  $85.0^\circ$ . Find (a) the angle of incidence in the ethanol and (b) the ratio of the wavelength of the light in ice to its wavelength in ethanol.

Be sure to review **EXAMPLE 33.5** (Section 33.5) before attempting these problems.

**VP33.5.1** A polarized laser beam of intensity  $255 \text{ W/m}^2$  shines on an ideal polarizer. The angle between the polarization direction of the laser beam and the polarizing axis of the polarizer is  $15.0^\circ$ . What is the intensity of the light that emerges from the polarizer?

**VP33.5.2** You shine unpolarized light with intensity  $54.0 \text{ W/m}^2$  on an ideal polarizer, and then the light that emerges from this polarizer falls on a second ideal polarizer. The light that emerges from the second polarizer has intensity  $19.0 \text{ W/m}^2$ . Find (a) the intensity of the light that emerges from the first polarizer and (b) the angle between the polarizing axes of the two polarizers.

**VP33.5.3** A beam of polarized light of intensity  $60.0 \text{ W/m}^2$  propagates in the  $+x$ -direction. The light is polarized in the  $+y$ -direction. The beam strikes an ideal polarizer whose plane is parallel to the  $yz$ -plane and has its polarizing axis at  $25.0^\circ$  clockwise from the  $y$ -direction. Then the beam that emerges from this polarizer strikes a second ideal polarizer whose plane is also parallel to the  $yz$ -plane but has its polarizing axis at  $50.0^\circ$  clockwise from the  $y$ -direction. Find the intensity of the light that emerges (a) from the first polarizer, (b) from the second polarizer, and (c) from the second polarizer if the first polarizer is removed.

**VP33.5.4** You simultaneously shine two light beams, each of intensity  $I_0$ , on an ideal polarizer. One beam is unpolarized, and the other beam is polarized at an angle of exactly  $30^\circ$  to the polarizing axis of the polarizer. Find the intensity of the light that emerges from the polarizer.

Be sure to review **EXAMPLE 33.6** (Section 33.5) before attempting these problems.

**VP33.6.1** Unpolarized sunlight in air shines on a block of a transparent solid with index of refraction 1.73. (a) For what angle of incidence is the reflected light completely polarized? (b) For this angle of incidence, is the light refracted into the solid completely polarized, partially polarized, or unpolarized?

**VP33.6.2** You shine a beam of unpolarized light in air on a block of glass. You find that if the angle of incidence is  $57.0^\circ$ , the reflected light is completely polarized. Find (a) the index of refraction of the glass and (b) the angle of refraction.

**VP33.6.3** You shine a beam of polarized light in air on a piece of dense flint glass ( $n = 1.66$ ). (a) If the polarization direction is perpendicular to the plane of incidence, is there an angle of incidence for which no light is reflected from the glass? If so, what is this angle? (b) Repeat part (a) if the polarization direction is in the plane of incidence.

**VP33.6.4** A glass container holds water ( $n = 1.33$ ). If unpolarized light propagating in the glass strikes the glass-water interface, the light reflected back into the glass will be completely polarized if the angle of refraction is  $53.5^\circ$ . Find (a) the polarizing angle in this situation and (b) the index of refraction of the glass.

## BRIDGING PROBLEM Reflection and Refraction

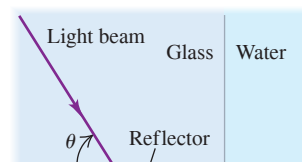
**Figure 33.37** shows a rectangular glass block that has a metal reflector on one face and water on an adjoining face. A light beam strikes the reflector as shown. You gradually increase the angle  $\theta$  of the light beam. If  $\theta \geq 59.2^\circ$ , no light enters the water. What is the speed of light in this glass?

### SOLUTION GUIDE

#### IDENTIFY and SET UP

1. Specular reflection occurs where the light ray in the glass strikes the reflector. If no light is to enter the water, we require that there be reflection only and no refraction where this ray strikes the glass–water interface—that is, there must be total internal reflection.
2. The target variable is the speed of light  $v$  in the glass, which you can determine from the index of refraction  $n$  of the glass. (Table 33.1 gives the index of refraction of water.) Write down the equations you'll use to find  $n$  and  $v$ .

**Figure 33.37** Glass bounded by water and a metal reflector.



#### EXECUTE

3. Use the figure to find the angle of incidence of the ray at the glass–water interface.
4. Use the result of step 3 to find  $n$ .
5. Use the result of step 4 to find  $v$ .

#### EVALUATE

6. How does the speed of light in the glass compare to the speed in water? Does this make sense?

## PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

### DISCUSSION QUESTIONS

**Q33.1** Light requires about 8 minutes to travel from the sun to the earth. Is it delayed appreciably by the earth's atmosphere? Explain.

**Q33.2** Sunlight or starlight passing through the earth's atmosphere is always bent toward the vertical. Why? Does this mean that a star is not really where it appears to be? Explain.

**Q33.3** A beam of light goes from one material into another. On *physical* grounds, explain *why* the wavelength changes but the frequency and period do not.

**Q33.4** A student claimed that, because of atmospheric refraction (see Discussion Question Q33.2), the sun can be seen after it has set and that the day is therefore longer than it would be if the earth had no atmosphere. First, what does she mean by saying that the sun can be seen after it has set? Second, comment on the validity of her conclusion.

**Q33.5** When hot air rises from a radiator or heating duct, objects behind it appear to shimmer or waver. What causes this?

**Q33.6** Devise straightforward experiments to measure the speed of light in a given glass using (a) Snell's law; (b) total internal reflection; (c) Brewster's law.

**Q33.7** Sometimes when looking at a window, you see two reflected images slightly displaced from each other. What causes this?

**Q33.8** If you look up from underneath toward the surface of the water in your aquarium, you may see an upside-down reflection of your pet fish in the surface of the water. Explain how this can happen.

**Q33.9** A ray of light in air strikes a glass surface. Is there a range of angles for which total internal reflection occurs? Explain.

**Q33.10** When light is incident on an interface between two materials, the angle of the refracted ray depends on the wavelength, but the angle of the reflected ray does not. Why should this be?

**Q33.11** A salesperson at a bargain counter claims that a certain pair of sunglasses has Polaroid filters; you suspect that the glasses are just tinted plastic. How could you find out for sure?

**Q33.12** Does it make sense to talk about the polarization of a *longitudinal* wave, such as a sound wave? Why or why not?

**Q33.13** How can you determine the direction of the polarizing axis of a single polarizer?

**Q33.14** It has been proposed that automobile windshields and headlights should have polarizing filters to reduce the glare of oncoming lights during night driving. Would this work? How should the polarizing axes be arranged? What advantages would this scheme have? What disadvantages?

**Q33.15** When a sheet of plastic food wrap is placed between two crossed polarizers, no light is transmitted. When the sheet is stretched in one direction, some light passes through the crossed polarizers. What is happening?

**Q33.16** If you sit on the beach and look at the ocean through Polaroid sunglasses, the glasses help to reduce the glare from sunlight reflecting off the water. But if you lie on your side on the beach, there is little reduction in the glare. Explain why there is a difference.

**Q33.17** When unpolarized light is incident on two crossed polarizers, no light is transmitted. A student asserted that if a third polarizer is inserted between the other two, some transmission will occur. Does this make sense? How can adding a third filter *increase* transmission?

**Q33.18** For the old "rabbit-ear" style TV antennas, it's possible to alter the quality of reception considerably simply by changing the orientation of the antenna. Why?

**Q33.19** In Fig. 33.31, since the light that is scattered out of the incident beam is polarized, why is the transmitted beam not also partially polarized?

**Q33.20** You are sunbathing in the late afternoon when the sun is relatively low in the western sky. You are lying flat on your back, looking straight up through Polaroid sunglasses. To minimize the amount of sky light reaching your eyes, how should you lie: with your feet pointing north, east, south, west, or in some other direction? Explain your reasoning.

**Q33.21** Light scattered from blue sky is strongly polarized because of the nature of the scattering process described in Section 33.6. But light scattered from white clouds is usually *not* polarized. Why not?

**Q33.22** Atmospheric haze is due to water droplets or smoke particles ("smog"). Such haze reduces visibility by scattering light, so that the light from distant objects becomes randomized and images become indistinct. Explain why visibility through haze can be improved by wearing red-tinted sunglasses, which filter out blue light.



**Q33.23** The explanation given in Section 33.6 for the color of the setting sun should apply equally well to the *rising* sun, since sunlight travels the same distance through the atmosphere to reach your eyes at either sunrise or sunset. Typically, however, sunsets are redder than sunrises. Why? (Hint: Particles of all kinds in the atmosphere contribute to scattering.)

**Q33.24** Huygens's principle also applies to sound waves. During the day, the temperature of the atmosphere decreases with increasing altitude above the ground. But at night, when the ground cools, there is a layer of air just above the surface in which the temperature *increases* with altitude. Use this to explain why sound waves from distant sources can be heard more clearly at night than in the daytime. (Hint: The speed of sound increases with increasing temperature. Use the ideas displayed in Fig. 33.36 for light.)

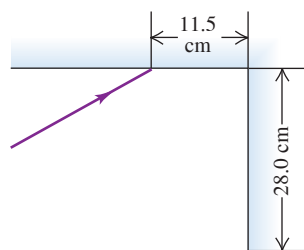
**Q33.25** Can water waves be reflected and refracted? Give examples. Does Huygens's principle apply to water waves? Explain.

## EXERCISES

### Section 33.2 Reflection and Refraction

**33.1** • Two plane mirrors intersect at right angles. A laser beam strikes the first of them at a point 11.5 cm from their point of intersection, as shown in Fig. E33.1. For what angle of incidence at the first mirror will this ray strike the midpoint of the second mirror (which is 28.0 cm long) after reflecting from the first mirror?

Figure E33.1



**33.2** • **BIO Light Inside the Eye.** The vitreous humor, a transparent, gelatinous fluid that fills most of the eyeball, has an index of refraction of 1.34. Visible light ranges in wavelength from 380 nm (violet) to 750 nm (red), as measured in air. This light travels through the vitreous humor and strikes the rods and cones at the surface of the retina. What are the ranges of (a) the wavelength, (b) the frequency, and (c) the speed of the light just as it approaches the retina within the vitreous humor?

**33.3** • A beam of light has a wavelength of 650 nm in vacuum. (a) What is the speed of this light in a liquid whose index of refraction at this wavelength is 1.47? (b) What is the wavelength of these waves in the liquid?

**33.4** • Light with a frequency of  $5.80 \times 10^{14}$  Hz travels in a block of glass that has an index of refraction of 1.52. What is the wavelength of the light (a) in vacuum and (b) in the glass?

**33.5** • A liquid lies on top of the horizontal surface of a block of glass. A ray of light traveling in the glass has speed  $1.85 \times 10^8$  m/s, wavelength 365 nm, and frequency  $5.07 \times 10^{14}$  Hz. The ray is incident on the surface of the glass at an angle of  $38.0^\circ$  with respect to the normal to the surface. The ray that refracts into the liquid makes an angle of  $44.7^\circ$  with the normal to the interface between the two materials. What are the speed, wavelength, and frequency of the light when it is traveling in the liquid?

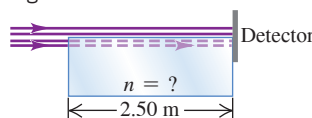
**33.6** • Light of a certain frequency has a wavelength of 526 nm in water. What is the wavelength of this light in benzene?

**33.7** • A parallel beam of light in air makes an angle of  $47.5^\circ$  with the surface of a glass plate having a refractive index of 1.66. (a) What is the angle between the reflected part of the beam and the surface of the glass? (b) What is the angle between the refracted beam and the surface of the glass?

**33.8** • A laser beam shines along the surface of a block of transparent material (see Fig. E33.8). Half of the beam goes straight to a detector, while the other half travels through the block and then hits the detector.

The time delay between the arrival of the two light beams at the detector is 6.25 ns. What is the index of refraction of this material?

Figure E33.8

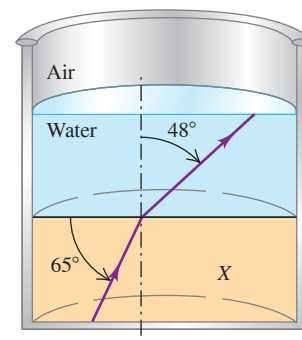


**33.9** • Light traveling in air is incident on the surface of a block of plastic at an angle of  $62.7^\circ$  to the normal and is bent so that it makes a  $48.1^\circ$  angle with the normal in the plastic. Find the speed of light in the plastic.

**33.10** • (a) A tank containing methanol has walls 2.50 cm thick made of glass of refractive index 1.550. Light from the outside air strikes the glass at a  $41.3^\circ$  angle with the normal to the glass. Find the angle the light makes with the normal in the methanol. (b) The tank is emptied and refilled with an unknown liquid. If light incident at the same angle as in part (a) enters the liquid in the tank at an angle of  $20.2^\circ$  from the normal, what is the refractive index of the unknown liquid?

**33.11** • As shown in Fig. E33.11, a layer of water covers a slab of material X in a beaker. A ray of light traveling upward follows the path indicated. Using the information on the figure, find (a) the index of refraction of material X and (b) the angle the light makes with the normal in the air.

Figure E33.11



**33.12** • A horizontal, parallel-sided plate of glass having a refractive index of 1.52 is in contact with the surface of water in a tank. A ray coming from above in air makes an angle of incidence of  $35.0^\circ$  with the normal to the top surface of the glass. (a) What angle does the ray refract into the water make with the normal to the surface?

(b) What is the dependence of this angle on the refractive index of the glass?

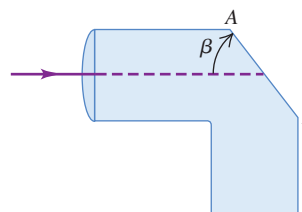
**33.13** • A block of a transparent solid sits on top of the horizontal surface of a block of glass. A ray of light traveling in the glass is incident on the top surface of the glass at an angle of  $62.0^\circ$  with respect to the normal to the surface. The light has wavelength 447 nm in the glass and 315 nm in the transparent solid. What angle does the ray that refracts into the transparent solid make with the normal to the surface?

**33.14** • A ray of light traveling in water is incident on an interface with a flat piece of glass. The wavelength of the light in the water is 726 nm, and its wavelength in the glass is 544 nm. If the ray in water makes an angle of  $56.0^\circ$  with respect to the normal to the interface, what angle does the refracted ray in the glass make with respect to the normal?

### Section 33.3 Total Internal Reflection

**33.15** • **Light Pipe.** Light enters a solid pipe made of plastic having an index of refraction of 1.60. The light travels parallel to the upper part of the pipe (Fig. E33.15). You want to cut the face AB so that all the light will reflect back into the pipe after it first strikes that face. (a) What is the largest that  $\beta$  can be if the pipe is in air? (b) If the pipe is immersed in water of refractive index 1.33, what is the largest that  $\beta$  can be?

Figure E33.15



**33.16** • A flat piece of glass covers the top of a vertical cylinder that is completely filled with water. If a ray of light traveling in the glass is incident on the interface with the water at an angle of  $\theta_a = 36.2^\circ$ , the ray refracted into the water makes an angle of  $49.8^\circ$  with the normal to the interface. What is the smallest value of the incident angle  $\theta_a$  for which none of the ray refracts into the water?

**33.17** • The critical angle for total internal reflection at a liquid–air interface is  $42.5^\circ$ . (a) If a ray of light traveling in the liquid has an angle of incidence at the interface of  $35.0^\circ$ , what angle does the refracted ray in the air make with the normal? (b) If a ray of light traveling in air has an angle of incidence at the interface of  $35.0^\circ$ , what angle does the refracted ray in the liquid make with the normal?

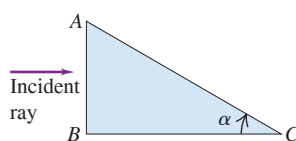
**33.18** • A beam of light is traveling inside a solid glass cube that has index of refraction 1.62. It strikes the surface of the cube from the inside. (a) If the cube is in air, at what minimum angle with the normal inside the glass will this light *not* enter the air at this surface? (b) What would be the minimum angle in part (a) if the cube were immersed in water?

**33.19** • A ray of light is traveling in a glass cube that is totally immersed in water. You find that if the ray is incident on the glass–water interface at an angle to the normal larger than  $48.7^\circ$ , no light is refracted into the water. What is the refractive index of the glass?

**33.20** • At the very end of Wagner's series of operas *Ring of the Nibelung*, Brünnhilde takes the golden ring from the finger of the dead Siegfried and throws it into the Rhine, where it sinks to the bottom of the river. Assuming that the ring is small enough compared to the depth of the river to be treated as a point and that the Rhine is 10.0 m deep where the ring goes in, what is the area of the largest circle at the surface of the water over which light from the ring could escape from the water?

**33.21** • Light is incident along the normal on face  $AB$  of a glass prism of refractive index 1.52, as shown in **Fig. E33.21**. Find the largest value the angle  $\alpha$  can have without any light refracted out of the prism at face  $AC$  if (a) the prism is immersed in air and (b) the prism is immersed in water.

Figure E33.21

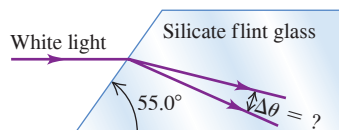


**33.22** • A thick sheet of transparent plastic sits on top of the horizontal surface of a block of glass. For a ray traveling in the glass, the critical angle at the glass–plastic interface is  $48.6^\circ$  and the wavelength of the light in the glass is 350 nm. For an incident angle of  $40.0^\circ$  in the glass, what is the wavelength of the light after it refracts into the plastic?

### Section 33.4 Dispersion

**33.23** •• A narrow beam of white light strikes one face of a slab of silicate flint glass. The light is traveling parallel to the two adjoining faces, as shown in **Fig. E33.23**. For the transmitted light inside the glass, through what angle  $\Delta\theta$  is the portion of the visible spectrum between 400 nm and 700 nm dispersed? (Consult the graph in Fig. 33.17.)

Figure E33.23



**33.24** • The indexes of refraction for violet light ( $\lambda = 400$  nm) and red light ( $\lambda = 700$  nm) in diamond are 2.46 and 2.41, respectively. A ray of light traveling through air strikes the diamond surface at an angle of  $53.5^\circ$  to the normal. Calculate the angular separation between these two colors of light in the refracted ray.

**33.25** • A beam of light strikes a sheet of glass at an angle of  $57.0^\circ$  with the normal in air. You observe that red light makes an angle of  $38.1^\circ$  with the normal in the glass, while violet light makes a  $36.7^\circ$  angle. (a) What are the indexes of refraction of this glass for these colors of light? (b) What are the speeds of red and violet light in the glass?

### Section 33.5 Polarization

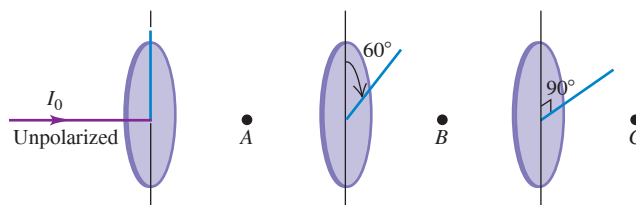
**33.26** •• Birefringence is discussed in Section 33.5 and the refractive indexes for the two perpendicular polarization directions in calcite are given. A crystal of calcite serves as a quarter-wave plate; it converts linearly polarized light to circularly polarized light if the numbers of wavelengths within the crystal differ by one-fourth for the two polarization components. For light with wavelength 589 nm in air, what is the minimum thickness of a quarter-wave plate made of calcite?

**33.27** • Unpolarized light with intensity  $I_0$  is incident on two polarizing filters. The axis of the first filter makes an angle of  $60.0^\circ$  with the vertical, and the axis of the second filter is horizontal. What is the intensity of the light after it has passed through the second filter?

**33.28** • A layer of liquid sits on top of the horizontal surface of a transparent solid. For a ray traveling in the solid and incident on the interface of the two materials, the critical angle is  $38.7^\circ$ . (a) For a ray traveling in the solid and reflecting at the interface with the liquid, for what incident angle with respect to the normal is the reflected ray 100% polarized? (b) What is the polarizing angle if the ray is traveling in the liquid?

**33.29** •• A beam of unpolarized light of intensity  $I_0$  passes through a series of ideal polarizing filters with their polarizing axes turned to various angles as shown in **Fig. E33.29**. (a) What is the light intensity (in terms of  $I_0$ ) at points A, B, and C? (b) If we remove the middle filter, what will be the light intensity at point C?

Figure E33.29



**33.30** •• Light of original intensity  $I_0$  passes through two ideal polarizing filters having their polarizing axes oriented as shown in **Fig. E33.30**. You want to adjust the angle  $\phi$  so that the intensity at point P is equal to  $I_0/10$ . (a) If the original light is unpolarized, what should  $\phi$  be? (b) If the original light is linearly polarized in the same direction as the polarizing axis of the first polarizer the light reaches, what should  $\phi$  be?

Figure E33.30



**33.31** • A parallel beam of unpolarized light in air is incident at an angle of  $54.5^\circ$  (with respect to the normal) on a plane glass surface. The reflected beam is completely linearly polarized. (a) What is the refractive index of the glass? (b) What is the angle of refraction of the transmitted beam?

**33.32** • The refractive index of a certain glass is 1.66. For what incident angle is light reflected from the surface of this glass completely polarized if the glass is immersed in (a) air and (b) water?

**33.33** •• Unpolarized light of intensity  $20.0 \text{ W/cm}^2$  is incident on two polarizing filters. The axis of the first filter is at an angle of  $25.0^\circ$  counterclockwise from the vertical (viewed in the direction the light is traveling), and the axis of the second filter is at  $62.0^\circ$  counterclockwise from the vertical. What is the intensity of the light after it has passed through the second polarizer?

**33.34** •• Three polarizing filters are stacked, with the polarizing axis of the second and third filters at  $23.0^\circ$  and  $62.0^\circ$ , respectively, clockwise to that of the first. If unpolarized light is incident on the stack, the light has intensity  $55.0 \text{ W/cm}^2$  after it passes through the stack. If the incident intensity is kept constant but the second polarizer is removed, what is the intensity of the light after it has passed through the stack?

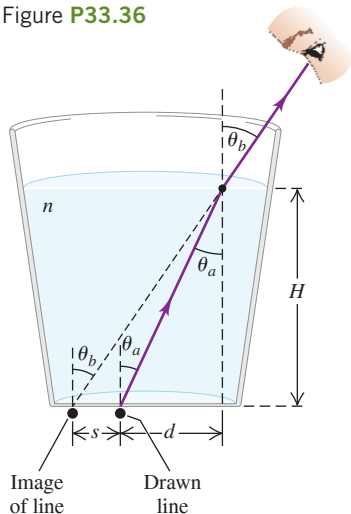
## Section 33.6 Scattering of Light

**33.35 •** A beam of white light passes through a uniform thickness of air. If the intensity of the scattered light in the middle of the green part of the visible spectrum is  $I$ , find the intensity (in terms of  $I$ ) of scattered light in the middle of (a) the red part of the spectrum and (b) the violet part of the spectrum. Consult Table 32.1.

## PROBLEMS

**33.36 ••** Draw a straight line on a piece of paper. Fill a transparent drinking glass with water and place it on top of the line you have drawn, making sure the line extends outward from both sides of the glass. Position your head above the glass, but not directly above, such that you look downward at an angle to see the image of the line through the water, as shown in Fig. P33.36. The image of the line beneath the glass will appear shifted away from the line you have drawn. (a) Mark a spot on the paper beside the glass where the line seen through the water would extend. Estimate the distance  $s$  between that mark and the original line. (b) Keeping your eye in the same position, hold a pen along the line of sight from your eye to the surface of the water, where the image of the line appears. The tip of the pen should just touch the surface of the water. Now move your head above the glass and sight downward from directly above the tip of the pen. Mark a spot on the paper beside the glass indicating the horizontal distance between the drawn line and the tip of the pen. Estimate that distance  $d$ . (c) Let  $\theta_a$  and  $\theta_b$  represent the angle of incidence and the angle of refraction, respectively, of the upward light ray traveling toward your eye, as shown in the figure, and let  $H$  be the depth of the water. Write expressions for  $\tan \theta_a$  and  $\tan \theta_b$  in terms of  $s$ ,  $d$ , and  $H$ . (d) Assume  $H$  is large so the angles are small and we can approximate  $\tan \theta_a \approx \sin \theta_a$  and  $\tan \theta_b \approx \sin \theta_b$ . Using Snell's law and these approximations, write an expression for the index of refraction  $n$  of the water in terms of  $s$  and  $d$ . (e) Use your values of  $s$  and  $d$  to estimate the index of refraction of the water.

Figure P33.36

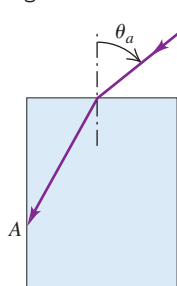


**33.37 •• BIO Heart Sonogram.** Physicians use high-frequency ( $f = 1\text{--}5\text{ MHz}$ ) sound waves, called ultrasound, to image internal organs. The speed of these ultrasound waves is  $1480\text{ m/s}$  in muscle and  $344\text{ m/s}$  in air. We define the index of refraction of a material for sound waves to be the ratio of the speed of sound in air to the speed of sound in the material. Snell's law then applies to the refraction of sound waves. (a) At what angle from the normal does an ultrasound beam enter the heart if it leaves the lungs at an angle of  $9.73^\circ$  from the normal to the heart wall? (Assume that the speed of sound in the lungs is  $344\text{ m/s}$ .) (b) What is the critical angle for sound waves in air incident on muscle?

**33.38 •••** In a physics lab, light with wavelength  $490\text{ nm}$  travels in air from a laser to a photocell in  $17.0\text{ ns}$ . When a slab of glass  $0.840\text{ m}$  thick is placed in the light beam, with the beam incident along the normal to the parallel faces of the slab, it takes the light  $21.2\text{ ns}$  to travel from the laser to the photocell. What is the wavelength of the light in the glass?

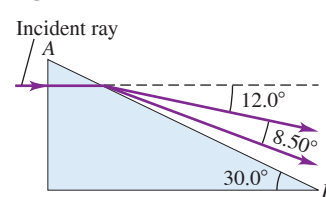
**33.39 ••** A ray of light is incident in air on a block of a transparent solid whose index of refraction is  $n$ . If  $n = 1.38$ , what is the *largest* angle of incidence  $\theta_a$  for which total internal reflection will occur at the vertical face (point A shown in Fig. P33.39)?

Figure P33.39



**33.40 •** A light ray in air strikes the right-angle prism shown in Fig. P33.40. The prism angle at B is  $30.0^\circ$ . This ray consists of two different wavelengths. When it emerges at face AB, it has been split into two different rays that diverge from each other by  $8.50^\circ$ . Find the index of refraction of the prism for each of the two wavelengths.

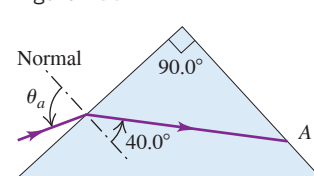
Figure P33.40



**33.41 ••** A ray of light traveling in a block of glass ( $n = 1.52$ ) is incident on the top surface at an angle of  $57.2^\circ$  with respect to the normal in the glass. If a layer of oil is placed on the top surface of the glass, the ray is totally reflected. What is the maximum possible index of refraction of the oil?

**33.42 ••** A ray of light traveling in air is incident at angle  $\theta_a$  on one face of a  $90.0^\circ$  prism made of glass. Part of the light refracts into the prism and strikes the opposite face at point A (Fig. P33.42). If the ray at A is at the critical angle, what is the value of  $\theta_a$ ?

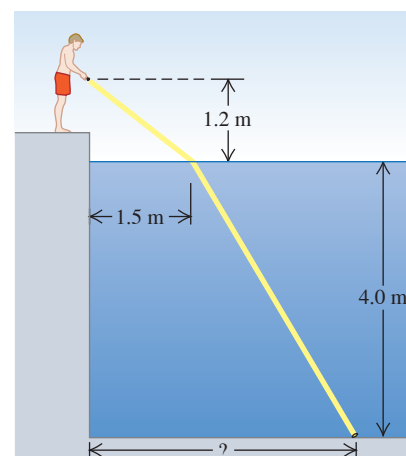
Figure P33.42



**33.43 •••** A glass plate  $2.50\text{ mm}$  thick, with an index of refraction of  $1.40$ , is placed between a point source of light with wavelength  $540\text{ nm}$  (in vacuum) and a screen. The distance from source to screen is  $1.80\text{ cm}$ . How many wavelengths are there between the source and the screen?

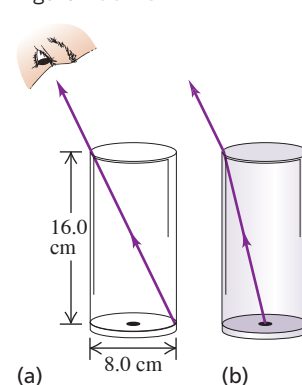
**33.44 •** After a long day of driving you take a late-night swim in a motel swimming pool. When you go to your room, you realize that you have lost your room key in the pool. You borrow a powerful flashlight and walk around the pool, shining the light into it. The light shines on the key, which is lying on the bottom of the pool, when the flashlight is held  $1.2\text{ m}$  above the water surface and is directed at the surface a horizontal distance of  $1.5\text{ m}$  from the edge (Fig. P33.44). If the water here is  $4.0\text{ m}$  deep, how far is the key from the edge of the pool?

Figure P33.44



**33.45 •** You sight along the rim of a glass with vertical sides so that the top rim is lined up with the opposite edge of the bottom (Fig. P33.45a). The glass is a thin-walled, hollow cylinder  $16.0\text{ cm}$  high. The diameter of the top and bottom of the glass is  $8.0\text{ cm}$ . While you keep your eye in the same position, a friend fills the glass with a transparent liquid, and you then see a dime that is lying at the center of the bottom of the glass (Fig. P33.45b). What is the index of refraction of the liquid?

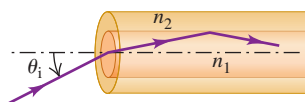
Figure P33.45





**33.46 ••** Optical fibers are constructed with a cylindrical core surrounded by a sheath of cladding material. Common materials used are pure silica ( $n_2 = 1.450$ ) for the cladding and silica doped with germanium ( $n_1 = 1.465$ ) for the core. (a) What is the critical angle  $\theta_{\text{crit}}$  for light traveling in the core and reflecting at the interface with the cladding material? (b) The numerical aperture (NA) is defined as the angle of incidence  $\theta_i$  at the flat end of the cable for which light is incident on the core-cladding interface at angle  $\theta_{\text{crit}}$  (Fig. P33.46). Show that  $\sin \theta_i = \sqrt{n_1^2 - n_2^2}$ . (c) What is the value of  $\theta_i$  for  $n_1 = 1.465$  and  $n_2 = 1.450$ ?

Figure P33.46

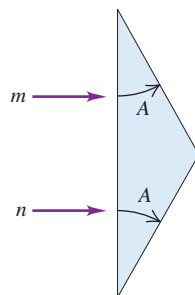


**33.47 •** A thin layer of ice ( $n = 1.309$ ) floats on the surface of water ( $n = 1.333$ ) in a bucket. A ray of light from the bottom of the bucket travels upward through the water. (a) What is the largest angle with respect to the normal that the ray can make at the ice-water interface and still pass out into the air above the ice? (b) What is this angle after the ice melts?

**33.48 ••** When linearly polarized light passes through a polarizer, its polarizing axis may be rotated by any angle  $\phi < 90^\circ$  at the expense of a loss of intensity, as determined by Malus's law. By using sequential polarizers, you can achieve a similar axis rotation but retain greater intensity. In fact, if you use many intermediate polarizers, the polarization axis can be rotated by  $90^\circ$  with virtually undiminished intensity. (a) Derive an equation for the resulting intensity if linearly polarized light passes through successive  $N$  polarizers, each with the polarizing axis rotated by an angle  $90^\circ/2N$  larger than the preceding polarizer. (b) By making a table of the resulting intensity for various values of  $N$ , estimate the minimum number  $N$  of polarizers needed so that the light will have its polarization axis rotated by  $90^\circ$  while maintaining more than 90% of its intensity. (c) Estimate the minimum number of polarizers needed to maintain more than 95% and 99% intensity.

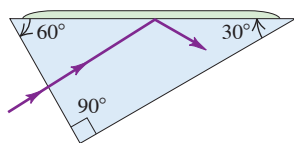
**33.49 ••** The prism shown in Fig. P33.49 has a refractive index of 1.66, and the angles  $A$  are  $25.0^\circ$ . Two light rays  $m$  and  $n$  are parallel as they enter the prism. What is the angle between them after they emerge?

Figure P33.49



**33.50 ••** Light is incident normally on the short face of a  $30^\circ$ - $60^\circ$ - $90^\circ$  prism (Fig. P33.50). A drop of liquid is placed on the hypotenuse of the prism. If the index of refraction of the prism is 1.56, find the maximum index that the liquid may have for the light to be totally reflected.

Figure P33.50



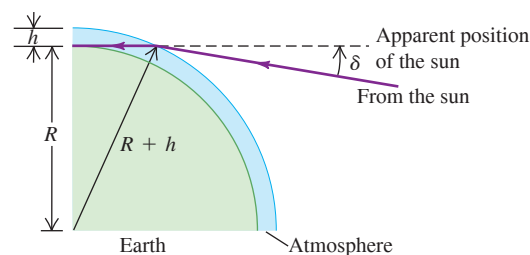
**33.51 ••** When the sun is either rising or setting and appears to be just on the horizon, it is in fact *below* the horizon. The explanation for this seeming paradox is that light from the sun bends slightly when entering the earth's atmosphere, as shown in Fig. P33.51. Since our perception is based on the idea that light travels in straight lines, we perceive the light to be coming from an apparent position that is an angle  $\delta$  above the sun's true position. (a) Make the simplifying assumptions that the atmosphere has uniform density, and hence uniform index of refraction

$n$ , and extends to a height  $h$  above the earth's surface, at which point it abruptly stops. Show that the angle  $\delta$  is given by

$$\delta = \arcsin\left(\frac{nR}{R+h}\right) - \arcsin\left(\frac{R}{R+h}\right)$$

where  $R = 6378$  km is the radius of the earth. (b) Calculate  $\delta$  using  $n = 1.0003$  and  $h = 20$  km. How does this compare to the angular radius of the sun, which is about one quarter of a degree? (In actuality a light ray from the sun bends gradually, not abruptly, since the density and refractive index of the atmosphere change gradually with altitude.)

Figure P33.51

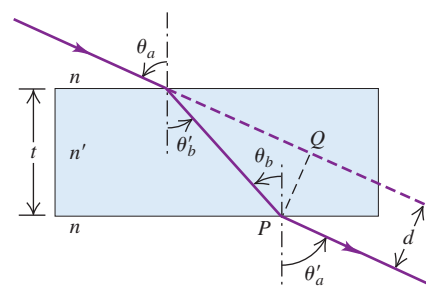


**33.52 ••** Light is incident in air at an angle  $\theta_a$  (Fig. P33.52) on the upper surface of a transparent plate, the surfaces of the plate being plane and parallel to each other. (a) Prove that  $\theta_a = \theta'_a$ . (b) Show that this is true for any number of different parallel plates. (c) Prove that the lateral displacement  $d$  of the emergent beam is given by the relationship

$$d = t \frac{\sin(\theta_a - \theta'_b)}{\cos \theta'}$$

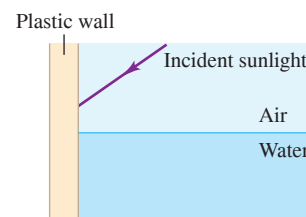
where  $t$  is the thickness of the plate. (d) A ray of light is incident at an angle of  $66.0^\circ$  on one surface of a glass plate 2.40 cm thick with an index of refraction of 1.80. The medium on either side of the plate is air. Find the lateral displacement between the incident and emergent rays.

Figure P33.52



**33.53 ••** A beam of unpolarized sunlight strikes the vertical plastic wall of a water tank at an unknown angle. Some of the light reflects from the wall and enters the water (Fig. P33.53). The refractive index of the plastic wall is 1.61. If the light that has been reflected from the wall into the water is observed to be completely polarized, what angle does this beam make with the normal inside the water?

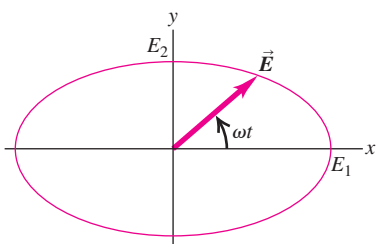
Figure P33.53





**33.54 •• CP** Figure P33.54

A circularly polarized electromagnetic wave propagating in air has an electric field given by  $\vec{E} = E[\cos(kz - \omega t)\hat{i} + \sin(kz - \omega t)\hat{j}]$ . This wave is incident with an intensity of  $150 \text{ W/m}^2$  at the polarizing angle  $\theta_p$  onto a flat interface perpendicular to the  $xz$ -plane with a material that has index of refraction  $n = 1.62$ . (a) What is the angle of refraction  $\theta_b$ ? (b) The ratio between the electric-field amplitude in the reflected wave and the electric-field amplitude of the component of the incident wave polarized parallel to the interface is  $\sin(\theta_p - \theta_b)/\sin(\theta_p + \theta_b)$ . (This result, known as the Fresnel equation, may be derived using deeper analysis.) From this relationship, determine the intensity  $I_{\text{reflected}}$  of the reflected wave. (c) Determine the intensity  $I_{\parallel}$  of the component of the refracted wave polarized parallel to the interface. (d) Determine the intensity  $I_{\perp}$  of the component of the refracted wave polarized in the  $x$ -direction. (e) The reflected wave is linearly polarized and the refracted wave is elliptically polarized, such that its electric field is characterized as shown in **Fig. P33.54**. Determine the elliptical eccentricity  $e = \sqrt{1 - (E_1/E_2)^2}$ .



**33.55 •• DATA** In physics lab, you are studying the properties of four transparent liquids. You shine a ray of light (in air) onto the surface of each liquid—A, B, C, and D—one at a time, at a  $60.0^\circ$  angle of incidence; you then measure the angle of refraction. The table gives your data:

| Liquid                  | A    | B    | C    | D    |
|-------------------------|------|------|------|------|
| $\theta_a$ ( $^\circ$ ) | 36.4 | 40.5 | 32.1 | 35.2 |

The wavelength of the light when it is traveling in air is  $589 \text{ nm}$ . (a) Find the refractive index of each liquid at this wavelength. Use Table 33.1 to identify each liquid, assuming that all four are listed in the table. (b) For each liquid, what is the dielectric constant  $K$  at the frequency of the  $589 \text{ nm}$  light? For each liquid, the relative permeability ( $K_m$ ) is very close to unity. (c) What is the frequency of the light in air and in each liquid?

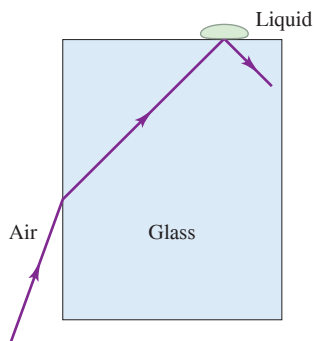
**33.56 •• DATA** Figure P33.56

Given small samples of three liquids, you are asked to determine their refractive indexes. However, you do not have enough of each liquid to measure the angle of refraction for light refracting from air into the liquid. Instead, for each liquid, you take a rectangular block of glass ( $n = 1.52$ ) and place a drop of the liquid on the top surface of the block. You shine a laser beam with wavelength  $638 \text{ nm}$  in vacuum at one side of the block and measure the largest angle of incidence  $\theta_a$  for which there is total internal reflection at the interface between the glass and the liquid (**Fig. P33.56**). Your results are given in the table:

| Liquid                  | A    | B    | C    |
|-------------------------|------|------|------|
| $\theta_a$ ( $^\circ$ ) | 52.0 | 44.3 | 36.3 |

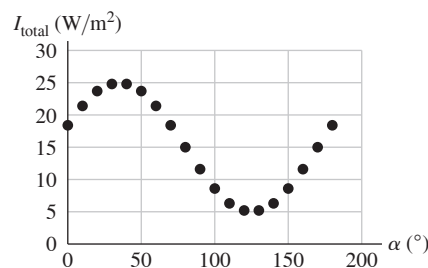
What is the refractive index of each liquid at this wavelength?

Figure P33.56

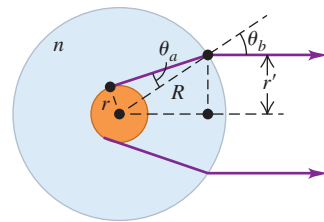


**33.57 •• DATA** A beam of light traveling horizontally is made of an unpolarized component with intensity  $I_0$  and a polarized component with intensity  $I_p$ . The plane of polarization of the polarized component is oriented at an angle  $\theta$  with respect to the vertical. **Figure P33.57** is a graph of the total intensity  $I_{\text{total}}$  after the light passes through a polarizer versus the angle  $\alpha$  that the polarizer's axis makes with respect to the vertical. (a) What is the orientation of the polarized component? (That is, what is  $\theta$ ?) (b) What are the values of  $I_0$  and  $I_p$ ?

Figure P33.57

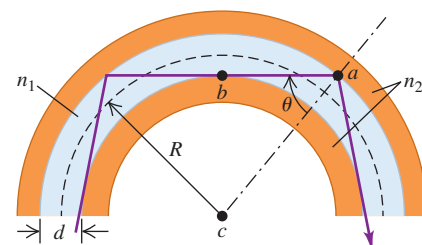
**33.58 •• CP** Figure P33.58

An object surrounded by a translucent material with index of refraction greater than unity appears enlarged. This includes items such as plants or shells inside plastic or resin paperweights, as well as ancient bugs trapped in amber. Consider a spherical object in air with radius  $r$  fixed at the center of a sphere with radius  $R > r$  with refractive index  $n$ , as shown in **Fig. P33.58**. When viewed from outside, the spherical object appears to have radius  $r' > r$ . (a) Using right triangles found in the figure, write an expression for  $\sin \theta_a$  in terms of  $r$  and  $R$ . (b) Similarly, write an expression for  $\sin \theta_b$  in terms of  $r'$  and  $R$ . (c) Using Snell's law, derive an expression for  $r'$  in terms of  $r$ ,  $R$ , and  $n$ . (d) If the object is a spherical dandelion seed head with diameter  $45.0 \text{ mm}$  fixed at the center of a solidified resin sphere with radius  $80.0 \text{ mm}$  and index of refraction  $1.53$ , what diameter will the seed head appear to have when viewed from outside?

**CHALLENGE PROBLEMS**

**33.59 •••** A fiber-optic cable consists of a thin cylindrical core with thickness  $d$  made of a material with index of refraction  $n_1$ , surrounded by cladding made of a material with index of refraction  $n_2 < n_1$ . Light rays traveling within the core remain trapped in the core provided they do not strike the core-cladding interface at an angle larger than the critical angle for total internal reflection. (a) **Figure P33.59** shows a light ray traveling within a cable bent to its critical extent, beyond which light will leak out of the core. The radius of curvature of the center of the core is  $R$ . Derive an expression for the angle of incidence  $\theta$  at which the indicated light ray strikes the outer edge of the core. (Hint: Construct a right triangle with vertices  $a$ ,  $b$ , and  $c$  as in the figure.) (b) Determine the

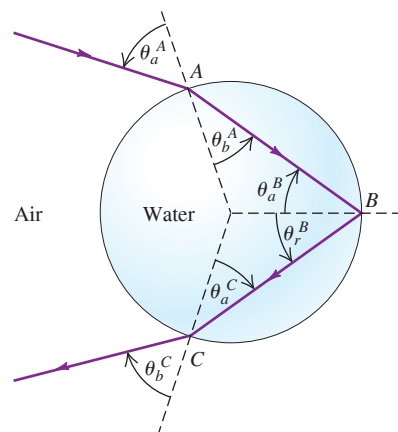
Figure P33.59



critical radius  $R$  in terms of  $n_1$ ,  $n_2$ , and  $d$ . (c) A typical fiber-optic cable has a  $50.0\text{-}\mu\text{m}$ -thick core with index of refraction 1.4475 and has a cladding with index of refraction 1.4440. What is the minimum radius of curvature for this cable? (d) If a light ray traversed a  $1.00\text{ km}$  length of such a cable bent to its critical extent, and if the ray were maximally angled within the core as shown in the figure, what would be the extra distance, beyond  $1.00\text{ km}$ , traveled by the ray owing to its multiple reflections? Keep extra digits in intermediate steps to reduce rounding errors. (e) How much longer would it take light to travel  $1.00\text{ km}$  through the cable than  $1.00\text{ km}$  in air?

**33.60 ••• CALC** A rainbow is produced by the reflection of sunlight by spherical drops of water in the air. **Figure P33.60** shows a ray that refracts into a drop at point A, is reflected from the back surface of the drop at point B, and refracts back into the air at point C. The angles of incidence and refraction,  $\theta_a$  and  $\theta_b$ , are shown at points A and C, and the angles of incidence and reflection,  $\theta_a$  and  $\theta_r$ , are shown at point B.

Figure P33.60



(a) Show that  $\theta_a^B = \theta_b^A$ ,  $\theta_a^C = \theta_b^A$ , and  $\theta_b^C = \theta_a^A$ . (b) Show that the angle in radians between the ray before it enters the drop at A and after it exits at C (the total angular deflection of the ray) is  $\Delta = 2\theta_a^A - 4\theta_b^A + \pi$ . (Hint: Find the angular deflections that occur at A, B, and C, and add them to get  $\Delta$ .) (c) Use Snell's law to write  $\Delta$  in terms of  $\theta_a^A$  and  $n$ , the refractive index of the water in the drop. (d) A rainbow will form when the angular deflection  $\Delta$  is *stationary* in the incident angle  $\theta_a^A$ —that is, when  $d\Delta/d\theta_a^A = 0$ . If this condition is satisfied, all the rays with incident angles close to  $\theta_a^A$  will be sent back in the same direction, producing a bright zone in the sky. Let  $\theta_1$  be the value of  $\theta_a^A$  for which this occurs. Show that  $\cos^2 \theta_1 = \frac{1}{3}(n^2 - 1)$ . [Hint: You may find the derivative formula  $d(\arcsin u(x))/dx = (1 - u^2)^{-1/2}(du/dx)$  helpful.] (e) The index of refraction in water is 1.342 for violet light and 1.330 for red light. Use the results of parts (c) and (d) to find  $\theta_1$  and  $\Delta$  for violet and red light. Do your results agree with the angles shown in Fig. 33.19d? When you view the rainbow, which color, red or violet, is higher above the horizon?

**33.61 ••• CALC** A *secondary rainbow* is formed when the incident light undergoes two internal reflections in a spherical drop of water as shown in Fig. 33.19e. (See Challenge Problem 33.60.) (a) In terms of the incident angle  $\theta_a^A$  and the refractive index  $n$  of the drop, what is the angular deflection  $\Delta$  of the ray? That is, what is the angle between the ray before it enters the drop and after it exits? (b) What is the incident angle  $\theta_2$  for which the derivative of  $\Delta$  with respect to the incident angle  $\theta_a^A$  is zero? (c) The indexes of refraction for red and violet light in water are given in part (e) of Challenge Problem 33.60. Use the results of parts (a) and (b) to find  $\theta_2$  and  $\Delta$  for violet and red light. Do your results agree with the angles shown in Fig. 33.19e? When you view a secondary rainbow, is red or violet higher above the horizon? Explain.

## MCAT-STYLE PASSAGE PROBLEMS

**BIO Seeing Polarized Light.** Some insect eyes have two types of cells that are sensitive to the plane of polarization of light. In a simple model, one cell type (type H) is sensitive to horizontally polarized light only, and the other cell type (type V) is sensitive to vertically polarized light only. To study the responses of these cells, researchers fix the insect in a normal, upright position so that one eye is illuminated by a light source. Then several experiments are carried out.

**33.62** First, light with a plane of polarization at  $45^\circ$  to the horizontal shines on the insect. Which statement is true about the two types of cells? (a) Both types detect this light. (b) Neither type detects this light. (c) Only type H detects the light. (d) Only type V detects the light.

**33.63** Next, unpolarized light is reflected off a smooth horizontal piece of glass, and the reflected light shines on the insect. Which statement is true about the two types of cells? (a) When the light is directly above the glass, only type V detects the reflected light. (b) When the light is directly above the glass, only type H detects the reflected light. (c) When the light is about  $35^\circ$  above the horizontal, type V responds much more strongly than type H does. (d) When the light is about  $35^\circ$  above the horizontal, type H responds much more strongly than type V does.

**33.64** To vary the angle as well as the intensity of polarized light, ordinary unpolarized light is passed through one polarizer with its transmission axis vertical, and then a second polarizer is placed between the first polarizer and the insect. When the light leaving the second polarizer has half the intensity of the original unpolarized light, which statement is true about the two types of cells? (a) Only type H detects this light. (b) Only type V detects this light. (c) Both types detect this light, but type H detects more light. (d) Both types detect this light, but type V detects more light.

## ANSWERS

### Chapter Opening Question ?

(iv) The brilliance and color of a diamond are due to total internal reflection from its surfaces (Section 33.3) and to dispersion, which spreads this light into a spectrum (Section 33.4).

### Key Example VARIATION Problems

**VP33.2.1** (a)  $70.0^\circ$  (b)  $31.5^\circ$

**VP33.2.2** (a) 1.81 (b)  $1.66 \times 10^8\text{ m/s}$

**VP33.2.3** (a) 1.25 (b)  $27.3^\circ$  (c)  $4.72 \times 10^{14}\text{ Hz}$  (d)  $4.72 \times 10^{14}\text{ Hz}$

**VP33.2.4** (a)  $73.5^\circ$  (b) 1.04

**VP33.5.1**  $238\text{ W/m}^2$

**VP33.5.2** (a)  $27.0\text{ W/m}^2$  (b)  $33.0^\circ$

**VP33.5.3** (a)  $49.3\text{ W/m}^2$  (b)  $40.5\text{ W/m}^2$  (c)  $24.8\text{ W/m}^2$

**VP33.5.4**  $5I_0/4$

**VP33.6.1** (a)  $60.0^\circ$  (b) partially polarized

**VP33.6.2** (a) 1.54 (b)  $33.0^\circ$

**VP33.6.3** (a) no (b) yes,  $58.9^\circ$

**VP33.6.4** (a)  $36.5^\circ$  (b) 1.80

### Bridging Problem

$1.93 \times 10^8\text{ m/s}$

? This surgeon performing microsurgery needs a sharp, magnified view of the surgical site. To obtain this, she's wearing glasses with magnifying lenses that must be (i) at a particular distance from her eye; (ii) at a particular distance from the object being magnified; (iii) both (i) and (ii); (iv) neither (i) nor (ii).



# 34 Geometric Optics

## LEARNING OUTCOMES

### In this chapter, you'll learn...

- 34.1 How a plane mirror forms an image.
- 34.2 Why concave and convex mirrors form images of different kinds.
- 34.3 How images can be formed by a curved interface between two transparent materials.
- 34.4 What aspects of a lens determine the type of image that it produces.
- 34.5 What determines the field of view of a camera lens.
- 34.6 What causes various defects in human vision, and how they can be corrected.
- 34.7 The principle of the simple magnifier.
- 34.8 How microscopes and telescopes work.

### You'll need to review...

- 33.2 Reflection and refraction.

Your reflection in the bathroom mirror, the view of the moon through a telescope, an insect seen through a magnifying lens—all of these are examples of *images*. In each case the object that you're looking at appears to be in a different place than its actual position: Your reflection is on the other side of the mirror, the moon appears to be much closer when seen through a telescope, and an insect seen through a magnifying lens appears *more distant* (so your eye can focus on it easily). In each case, light rays that come from a point on an object are deflected by reflection or refraction (or a combination of the two), so they converge toward or appear to diverge from a point called an *image point*. Our goal in this chapter is to see how this is done and to explore the different kinds of images that can be made with simple optical devices.

To understand images and image formation, all we need are the ray model of light, the laws of reflection and refraction (Section 33.2), and some simple geometry and trigonometry. The key role played by geometry in our analysis explains why we give the name *geometric optics* to the study of how light rays form images. We'll begin our analysis with one of the simplest image-forming optical devices, a plane mirror. We'll go on to study how images are formed by curved mirrors, by refracting surfaces, and by thin lenses. Our results will lay the foundation for understanding many familiar optical instruments, including camera lenses, magnifiers, the human eye, microscopes, and telescopes.

## 34.1 REFLECTION AND REFRACTION AT A PLANE SURFACE

Before discussing what is meant by an image, we first need the concept of **object** as it is used in optics. By an *object* we mean anything from which light rays radiate. This light could be emitted by the object itself if it is *self-luminous*, like the glowing filament of a light bulb. Alternatively, the light could be emitted by another source (such as a lamp or the sun) and then reflected from the object; an example is the light you see coming from



the pages of this book. **Figure 34.1** shows light rays radiating in all directions from an object at a point  $P$ . Note that light rays from the object reach the observer's left and right eyes at different angles; these differences are processed by the observer's brain to infer the *distance* from the observer to the object.

The object in Fig. 34.1 is a **point object** that has no physical extent. Real objects with length, width, and height are called **extended objects**. To start with, we'll consider only an idealized point object, since we can always think of an extended object as being made up of a very large number of point objects.

Suppose some of the rays from the object strike a smooth, plane reflecting surface (**Fig. 34.2**). This could be the surface of a material with a different index of refraction, which reflects part of the incident light, or a polished metal surface that reflects almost 100% of the light that strikes it. We'll always draw the reflecting surface as a black line with a shaded area behind it, as in Fig. 34.2. Bathroom mirrors have a thin sheet of glass that lies in front of and protects the reflecting surface; we'll ignore the effects of this thin sheet.

According to the law of reflection, all rays striking the surface are reflected at an angle from the normal equal to the angle of incidence. Since the surface is plane, the normal is in the same direction at all points on the surface, and we have *specular* reflection. After the rays are reflected, their directions are the same as though they had come from point  $P'$ . We call point  $P$  an *object point* and point  $P'$  the corresponding *image point*, and we say that the reflecting surface forms an **image** of point  $P$ . An observer who can see only the rays reflected from the surface, and who doesn't know that he's seeing a reflection, *thinks* that the rays originate from the image point  $P'$ . The image point is therefore a convenient way to describe the directions of the various reflected rays, just as the object point  $P$  describes the directions of the rays arriving at the surface *before* reflection.

If the surface in Fig. 34.2 were *not* smooth, the reflection would be *diffuse*. Rays reflected from different parts of the surface would go in uncorrelated directions (see Fig. 33.6b), and there would be no definite image point  $P'$  from which all reflected rays seem to emanate. You can't see your reflection in a tarnished piece of metal because its surface is rough; polishing the metal smoothes the surface so that specular reflection occurs and a reflected image becomes visible.

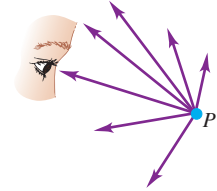
A plane *refracting* surface also forms an image (**Fig. 34.3**). Rays coming from point  $P$  are refracted at the interface between two optical materials. When the angles of incidence are small, the final directions of the rays after refraction are the same as though they had come from an *image point*  $P'$  as shown. In Section 33.2 we described how this effect makes underwater objects appear closer to the surface than they really are (see Fig. 33.9).

In both Figs. 34.2 and 34.3 the rays do not actually pass through the image point  $P'$ . Indeed, if the mirror in Fig. 34.2 is opaque, there is no light at all on its right side. If the outgoing rays don't actually pass through the image point, we call the image a **virtual image**. Later we'll see cases in which the outgoing rays really *do* pass through an image point, and we'll call the resulting image a **real image**. The images that are formed on a projection screen, on the electronic sensor in a camera, and on the retina of your eye are real images.

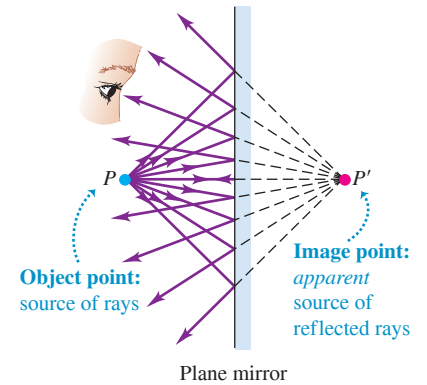
### Image Formation by a Plane Mirror

Let's concentrate for now on images produced by *reflection*; we'll return to refraction later in the chapter. **Figure 34.4** (next page) shows how to find the precise location of the virtual image  $P'$  that a plane mirror forms of an object at  $P$ . The diagram shows two rays diverging from an object point  $P$  at a distance  $s$  to the left of a plane mirror. We call  $s$  the **object distance**. The ray  $PV$  is perpendicular to the mirror surface, and it returns along its original path.

**Figure 34.1** Light rays radiate from a point object  $P$  in all directions. For an observer to see this object directly, there must be no obstruction between the object and the observer's eyes.



**Figure 34.2** Light rays from the object at point  $P$  are reflected from a plane mirror. The reflected rays entering the eye look as though they had come from image point  $P'$ .



**Figure 34.3** Light rays from the object at point  $P$  are refracted at the plane interface. The refracted rays entering the eye look as though they had come from image point  $P'$ .

When  $n_a > n_b$ ,  $P'$  is closer to the surface than  $P$ ; for  $n_a < n_b$ , the reverse is true.

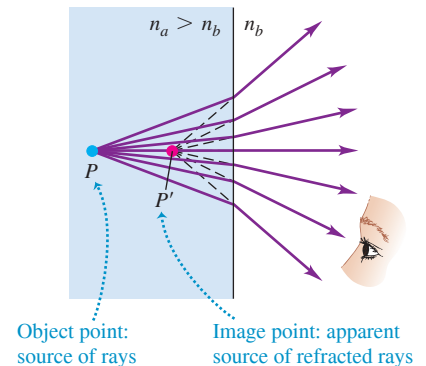




Figure 34.4 Construction for determining the location of the image formed by a plane mirror. The image point  $P'$  is as far behind the mirror as the object point  $P$  is in front of it.

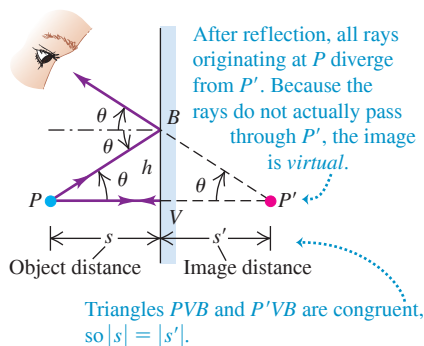


Figure 34.5 For both of these situations, the object distance  $s$  is positive (rule 1) and the image distance  $s'$  is negative (rule 2).

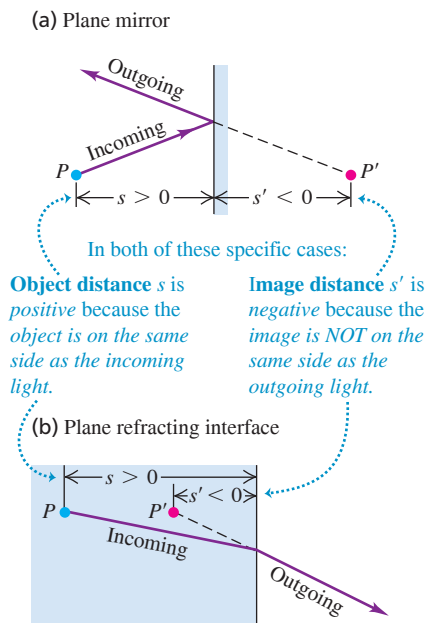
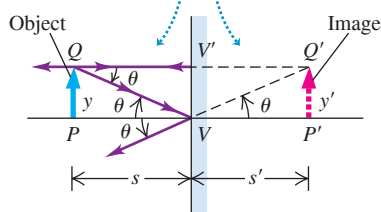


Figure 34.6 Construction for determining the height of an image formed by reflection at a plane reflecting surface.

For a plane mirror,  $PQV$  and  $P'Q'V$  are congruent, so  $y = y'$  and the object and image are the same size (the lateral magnification is 1).



The ray  $PB$  makes an angle  $\theta$  with  $PV$ . It strikes the mirror at an angle of incidence  $\theta$  and is reflected at an equal angle with the normal. When we extend the two reflected rays backward, they intersect at point  $P'$ , at a distance  $s'$  behind the mirror. We call  $s'$  the **image distance**. The line between  $P$  and  $P'$  is perpendicular to the mirror. The two triangles  $PVB$  and  $P'VB$  are congruent, so  $P$  and  $P'$  are at equal distances from the mirror, and  $s$  and  $s'$  have equal magnitudes. The image point  $P'$  is located exactly opposite the object point  $P$  as far *behind* the mirror as the object point is from the front of the mirror.

We can repeat the construction of Fig. 34.4 for each ray diverging from  $P$ . The directions of *all* the outgoing reflected rays are the same as though they had originated at point  $P'$ , confirming that  $P'$  is the *image* of  $P$ . No matter where the observer is located, she will always see the image at the point  $P'$ .

## Sign Rules

Before we go further, let's introduce some general sign rules. These may seem unnecessarily complicated for the simple case of an image formed by a plane mirror, but we want to state the rules in a form that will be applicable to *all* the situations we'll encounter later. These will include image formation by a plane or spherical reflecting or refracting surface, or by a pair of refracting surfaces forming a lens. Here are the rules:

1. **Sign rule for the object distance:** When the object is on the same side of the reflecting or refracting surface as the incoming light, object distance  $s$  is positive; otherwise, it is negative.
2. **Sign rule for the image distance:** When the image is on the same side of the reflecting or refracting surface as the outgoing light, image distance  $s'$  is positive; otherwise, it is negative.
3. **Sign rule for the radius of curvature of a spherical surface:** When the center of curvature  $C$  is on the same side as the outgoing light, the radius of curvature is positive; otherwise, it is negative.

Figure 34.5 illustrates rules 1 and 2 for two different situations. For a mirror the incoming and outgoing sides are always the same; for example, in Figs. 34.2, 34.4, and 34.5a they are both on the left side. For the refracting surfaces in Figs. 34.3 and 34.5b the incoming and outgoing sides are on the left and right sides, respectively, of the interface between the two materials. (Note that other textbooks may use different rules.)

In Figs. 34.4 and 34.5a the object distance  $s$  is *positive* because the object point  $P$  is on the incoming side (the left side) of the reflecting surface. The image distance  $s'$  is *negative* because the image point  $P'$  is *not* on the outgoing side (the left side) of the surface. The object and image distances  $s$  and  $s'$  are related by

$$s = -s' \quad (\text{plane mirror}) \quad (34.1)$$

For a plane reflecting or refracting surface, the radius of curvature is infinite and not a particularly interesting or useful quantity; in these cases we really don't need sign rule 3. But this rule will be of great importance when we study image formation by *curved* reflecting and refracting surfaces later in the chapter.

## Image of an Extended Object: Plane Mirror

Next we consider an *extended* object with finite size. For simplicity we often consider an object that has only one dimension, like a slender arrow, oriented parallel to the reflecting surface; an example is the arrow  $PQ$  in Fig. 34.6. The distance from the head to the tail of an arrow oriented in this way is called its *height*. In Fig. 34.6 the height is  $y$ . The image formed by such an extended object is an extended image; to each point on the object, there corresponds a point on the image. Two of the rays from  $Q$  are shown; *all* the rays from  $Q$

appear to diverge from its image point  $Q'$  after reflection. The image of the arrow is the line  $P'Q'$ , with height  $y'$ . Other points of the object  $PQ$  have image points between  $P'$  and  $Q'$ . The triangles  $PQV$  and  $P'Q'V$  are congruent, so the object  $PQ$  and image  $P'Q'$  have the same size and orientation, and  $y = y'$ .

The ratio of image height to object height,  $y'/y$ , in *any* image-forming situation is called the **lateral magnification**  $m$ ; that is,

$$\text{Lateral magnification in an image-forming situation } m = \frac{y'}{y} \quad (34.2)$$

Image height  
Object height

For a plane mirror  $y = y'$ , so the lateral magnification  $m$  is unity. When you look at yourself in a plane mirror, your image is the same size as the real you.

In Fig. 34.6 the image arrow points in the *same* direction as the object arrow; we say that the image is **erect**. In this case,  $y$  and  $y'$  have the same sign, and the lateral magnification  $m$  is positive. The image formed by a plane mirror is always erect, so  $y$  and  $y'$  have both the same magnitude and the same sign; from Eq. (34.2) the lateral magnification of a plane mirror is always  $m = +1$ . Later we'll encounter situations in which the image is **inverted**; that is, the image arrow points in the direction *opposite* to that of the object arrow. For an inverted image,  $y$  and  $y'$  have *opposite* signs, and the lateral magnification  $m$  is *negative*.

The object in Fig. 34.6 has only one dimension. **Figure 34.7** shows a *three-dimensional* object and its three-dimensional virtual image formed by a plane mirror. The object and image are related in the same way as a left hand and a right hand.

**CAUTION Reflections in a plane mirror** At this point, you may be asking, “Why does a plane mirror reverse images left and right but not top and bottom?” This question is quite misleading! As Fig. 34.7 shows, the up-down image  $P'Q'$  and the left-right image  $P'S'$  are parallel to their objects and are not reversed at all. Only the front-back image  $P'R'$  is reversed relative to  $PR$ . Hence it's most correct to say that a plane mirror reverses *back to front*. When an object and its image are related in this way, the image is said to be **reversed**; this means that *only* the front-back dimension is reversed. ■

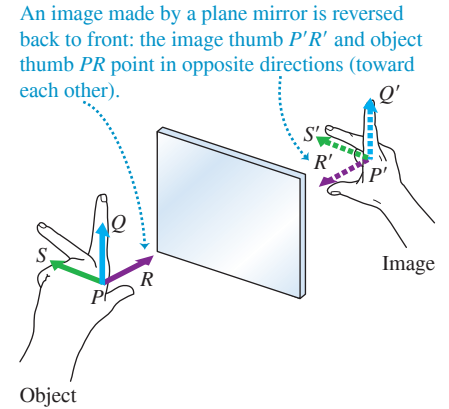
The reversed image of a three-dimensional object formed by a plane mirror is the same *size* as the object in all its dimensions. When the transverse dimensions of object and image are in the same direction, the image is erect. Thus a plane mirror always forms an erect but reversed image (**Fig. 34.8**).

An important property of all images formed by reflecting or refracting surfaces is that an *image* formed by one surface or optical device can serve as the *object* for a second surface or device. **Figure 34.9** shows a simple example. Mirror 1 forms an image  $P'_1$  of the object point  $P$ , and mirror 2 forms another image  $P'_2$ , each in the way we have just discussed. But in addition, the image  $P'_1$  formed by mirror 1 serves as an object for mirror 2, which then forms an image of this object at point  $P'_3$  as shown. Similarly, mirror 1 uses the image  $P'_2$  formed by mirror 2 as an object and forms an image of it. We leave it to you to show that this image point is also at  $P'_3$ . The idea that an image formed by one device can act as the object for a second device is of great importance. We'll use it later in this chapter to locate the image formed by two successive curved-surface refractions in a lens. We'll also use it to understand image formation by combinations of lenses, as in a microscope or a refracting telescope.

**TEST YOUR UNDERSTANDING OF SECTION 34.1** If you walk directly toward a plane mirror at a speed  $v$ , at what speed does your image approach you? (i) Slower than  $v$ ; (ii)  $v$ ; (iii) faster than  $v$  but slower than  $2v$ ; (iv)  $2v$ ; (v) faster than  $2v$ .

**ANSWER** (iv) When you are a distance  $s$  from the mirror, your image is a distance  $s$  on the other side of the mirror and the distance from you to your image is  $2s$ . As you move toward the mirror, the distance  $2s$  changes at twice the rate of the distance  $s$ , so your image moves toward you at speed  $2v$ . ■

**Figure 34.7** The image formed by a plane mirror is virtual, erect, and reversed. It is the same size as the object.



**Figure 34.8** The image formed by a plane mirror is reversed; the image of a right hand is a left hand, and so on. (The hand is resting on a horizontal mirror.) Are images of the letters I, H, and T reversed?



**Figure 34.9** Images  $P'_1$  and  $P'_2$  are formed by a single reflection of each ray from the object at  $P$ . Image  $P'_3$ , located by treating either of the other images as an object, is formed by a double reflection of each ray.

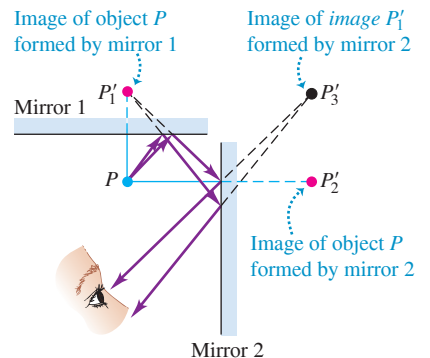
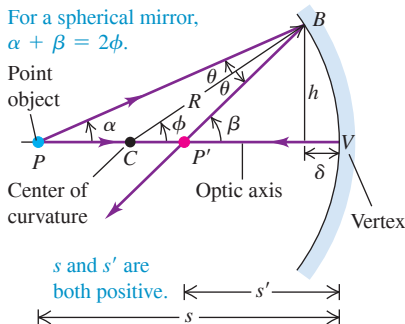


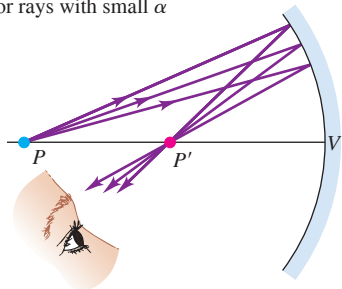
Figure 34.10 (a) A concave spherical mirror forms a real image of a point object  $P$  on the mirror's optic axis. (b) The eye sees some of the outgoing rays and perceives them as having come from  $P'$ .

(a) Construction for finding the position  $P'$  of an image formed by a concave spherical mirror

For a spherical mirror,  
 $\alpha + \beta = 2\phi$ .

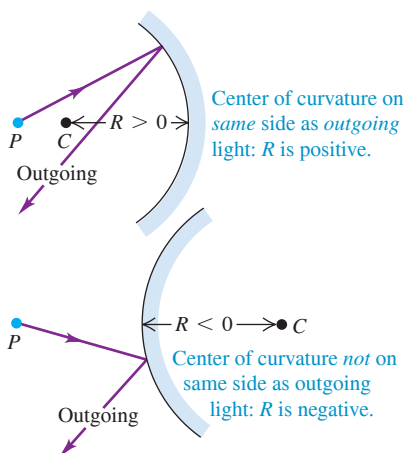


(b) The paraxial approximation, which holds for rays with small  $\alpha$



All rays from  $P$  that have a small angle  $\alpha$  pass through  $P'$ , forming a real image.

Figure 34.11 The sign rule for the radius of a spherical mirror.



## 34.2 REFLECTION AT A SPHERICAL SURFACE

A plane mirror produces an image that is the same size as the object. But there are many applications for mirrors in which the image and object must be of different sizes. A magnifying mirror used when applying makeup gives an image that is *larger* than the object, and surveillance mirrors (used in stores to help spot shoplifters) give an image that is *smaller* than the object. There are also applications of mirrors in which a *real* image is desired, so light rays do indeed pass through the image point  $P'$ . A plane mirror by itself cannot perform any of these tasks. Instead, *curved* mirrors are used.

### Image of a Point Object: Spherical Mirror

We'll consider the special (and easily analyzed) case of image formation by a *spherical* mirror. Figure 34.10a shows a spherical mirror with radius of curvature  $R$ , with its concave side facing the incident light. The **center of curvature** of the surface (the center of the sphere of which the surface is a part) is at  $C$ , and the **vertex** of the mirror (the center of the mirror surface) is at  $V$ . The line  $CV$  is called the **optic axis**. Point  $P$  is an object point that lies on the optic axis; for the moment, we assume that the distance from  $P$  to  $V$  is greater than  $R$ .

Ray  $PV$ , passing through  $C$ , strikes the mirror normally and is reflected back on itself. Ray  $PB$ , at an angle  $\alpha$  with the axis, strikes the mirror at  $B$ , where the angles of incidence and reflection are  $\theta$ . The reflected ray intersects the axis at point  $P'$ . We'll show shortly that *all* rays from  $P$  intersect the axis at the *same* point  $P'$ , as in Fig. 34.10b, provided that the angle  $\alpha$  is small. Point  $P'$  is therefore the *image* of object point  $P$ . Unlike the reflected rays in Fig. 34.1, the reflected rays in Fig. 34.10b actually do intersect at point  $P'$ , then diverge from  $P'$  as if they had originated at this point. Thus  $P'$  is a *real* image.

To see the usefulness of having a real image, suppose that the mirror is in a darkened room in which the only source of light is a self-luminous object at  $P$ . If you place a small piece of photographic film at  $P'$ , all the rays of light coming from point  $P$  that reflect off the mirror will strike the same point  $P'$  on the film; when developed, the film will show a single bright spot, representing a sharply focused image of the object at point  $P$ . This principle is at the heart of most astronomical telescopes, which use large concave mirrors to make photographs of celestial objects. With a *plane* mirror like that in Fig. 34.2, the light rays never actually pass through the image point, and the image can't be recorded on film. Real images are *essential* for photography.

Let's now locate the real image point  $P'$  in Fig. 34.10a and prove that all rays from  $P$  intersect at  $P'$  (provided that their angle with the optic axis is small). The object distance, measured from the vertex  $V$ , is  $s$ ; the image distance, also measured from  $V$ , is  $s'$ . The signs of  $s$ ,  $s'$ , and the radius of curvature  $R$  are determined by the sign rules given in Section 34.1. The object point  $P$  is on the same side as the incident light, so according to sign rule 1,  $s$  is positive. The image point  $P'$  is on the same side as the reflected light, so according to sign rule 2, the image distance  $s'$  is also positive. The center of curvature  $C$  is on the same side as the reflected light, so according to sign rule 3,  $R$ , too, is positive;  $R$  is always positive when reflection occurs at the *concave* side of a surface (Fig. 34.11).

We now use the following theorem from plane geometry: An exterior angle of a triangle equals the sum of the two opposite interior angles. Applying this theorem to triangles  $PBC$  and  $P'BC$  in Fig. 34.10a, we have

$$\phi = \alpha + \theta \quad \beta = \phi + \theta$$

Eliminating  $\theta$  between these equations gives

$$\alpha + \beta = 2\phi \quad (34.3)$$



We may now compute the image distance  $s'$ . Let  $h$  represent the height of point  $B$  above the optic axis, and let  $\delta$  represent the short distance from  $V$  to the foot of this vertical line. We now write expressions for the tangents of  $\alpha$ ,  $\beta$ , and  $\phi$ , remembering that  $s$ ,  $s'$ , and  $R$  are all positive quantities:

$$\tan \alpha = \frac{h}{s - \delta} \quad \tan \beta = \frac{h}{s' - \delta} \quad \tan \phi = \frac{h}{R - \delta}$$

These trigonometric equations cannot be solved as simply as the corresponding algebraic equations for a plane mirror. However, *if the angle  $\alpha$  is small*, the angles  $\beta$  and  $\phi$  are also small. The tangent of an angle that is much less than one radian is nearly equal to the angle itself (measured in radians), so we can replace  $\tan \alpha$  by  $\alpha$ , and so on, in the equations above. Also, if  $\alpha$  is small, we can ignore the distance  $\delta$  compared with  $s'$ ,  $s$ , and  $R$ . So for small angles we have the following approximate relationships:

$$\alpha = \frac{h}{s} \quad \beta = \frac{h}{s'} \quad \phi = \frac{h}{R}$$

Substituting these into Eq. (34.3) and dividing by  $h$ , we obtain a general relationship among  $s$ ,  $s'$ , and  $R$ :

$$\frac{1}{s} + \frac{1}{s'} = \frac{2}{R} \quad (\text{object-image relationship, spherical mirror}) \quad (34.4)$$

This equation does not contain the angle  $\alpha$ . Hence *all* rays from  $P$  that make sufficiently small angles with the axis intersect at  $P'$  after they are reflected; this verifies our earlier assertion. Such rays, nearly parallel to the axis and close to it, are called **paraxial rays**. (The term **paraxial approximation** is often used for the approximations we have just described.) Since all such reflected light rays converge on the image point, a concave mirror is also called a *converging mirror*.

Be sure you understand that Eq. (34.4), as well as many similar relationships that we'll derive later in this chapter and the next, is only *approximately* correct. It results from a calculation containing approximations, and it is valid only for paraxial rays. If we increase the angle  $\alpha$  that a ray makes with the optic axis, the point  $P'$  where the ray intersects the optic axis moves somewhat closer to the vertex than for a paraxial ray. As a result, a spherical mirror, unlike a plane mirror, does not form a precise point image of a point object; the image is “smeared out.” This property of a spherical mirror is called **spherical aberration**. When the primary mirror of the Hubble Space Telescope (Fig. 34.12a) was manufactured, tiny errors were made in its shape that led to an unacceptable amount of spherical aberration (Fig. 34.12b). The performance of the telescope improved dramatically after the installation of corrective optics (Fig. 34.12c).

If the radius of curvature becomes infinite ( $R = \infty$ ), the mirror becomes *plane*, and Eq. (34.4) reduces to Eq. (34.1) for a plane reflecting surface.

## Focal Point and Focal Length

When the object point  $P$  is very far from the spherical mirror ( $s = \infty$ ), the incoming rays are parallel. (The star shown in Fig. 34.12c is an example of such a distant object.) From Eq. (34.4) the image distance  $s'$  in this case is given by

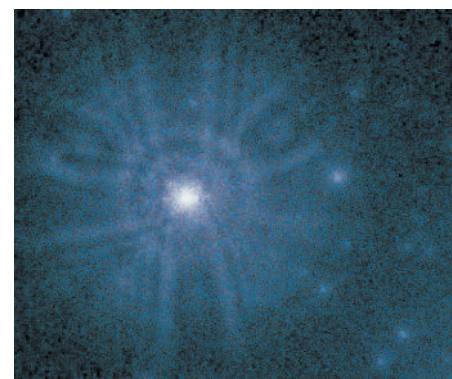
$$\frac{1}{\infty} + \frac{1}{s'} = \frac{2}{R} \quad s' = \frac{R}{2}$$

Figure 34.12 (a), (b) Soon after the Hubble Space Telescope (HST) was placed in orbit in 1990, it was discovered that the concave primary mirror (also called the *objective mirror*) was too shallow by about  $\frac{1}{50}$  the width of a human hair, leading to spherical aberration of the star's image. (c) After corrective optics were installed in 1993, the effects of spherical aberration were almost completely eliminated.

(a) The 2.4-m-diameter primary mirror of the Hubble Space Telescope



(b) A star seen with the original mirror



(c) The same star with corrective optics

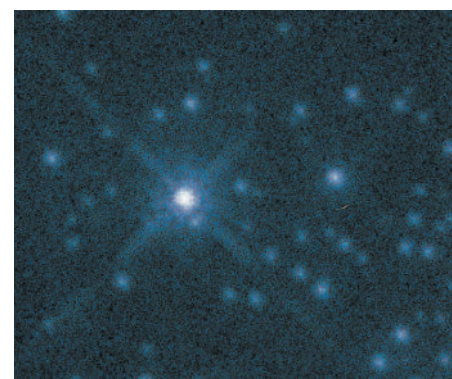
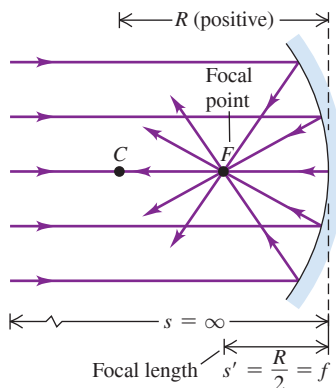


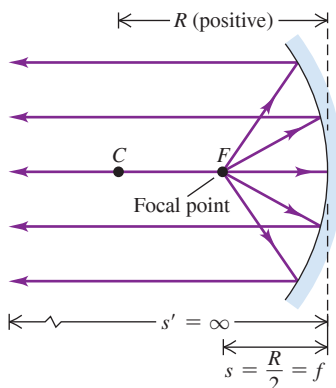


Figure 34.13 The focal point and focal length of a concave mirror.

(a) All parallel rays incident on a spherical mirror reflect through the focal point.



(b) Rays diverging from the focal point reflect to form parallel outgoing rays.



The situation is shown in Fig. 34.13a. The beam of incident parallel rays converges, after reflection from the mirror, to a point  $F$  at a distance  $R/2$  from the vertex of the mirror. The point  $F$  at which the incident parallel rays converge is called the **focal point**; we say that these rays are brought to a focus. The distance from the vertex to the focal point, denoted by  $f$ , is called the **focal length**. We see that  $f$  is related to the radius of curvature  $R$  by

$$f = \frac{R}{2} \quad (\text{focal length of a spherical mirror}) \quad (34.5)$$

Figure 34.13b shows the opposite situation. Now the *object* is placed at the focal point  $F$ , so the object distance is  $s = f = R/2$ . The image distance  $s'$  is again given by Eq. (34.4):

$$\frac{2}{R} + \frac{1}{s'} = \frac{2}{R} \quad \frac{1}{s'} = 0 \quad s' = \infty$$

With the object at the focal point, the reflected rays in Fig. 34.13b are parallel to the optic axis; they meet only at a point infinitely far from the mirror, so the image is at infinity.

Thus the focal point  $F$  of a spherical mirror has the properties that (1) any incoming ray parallel to the optic axis is reflected through the focal point and (2) any incoming ray that passes through the focal point is reflected parallel to the optic axis. For spherical mirrors these statements are true only for paraxial rays. For parabolic mirrors these statements are *exactly* true. Spherical or parabolic mirrors are used in flashlights and headlights to form the light from the bulb into a parallel beam. Some solar-power plants use an array of plane mirrors to simulate an approximately spherical concave mirror; sunlight is collected by the mirrors and directed to the focal point, where a steam boiler is placed. (The concepts of focal point and focal length also apply to lenses, as we'll see in Section 34.4.)

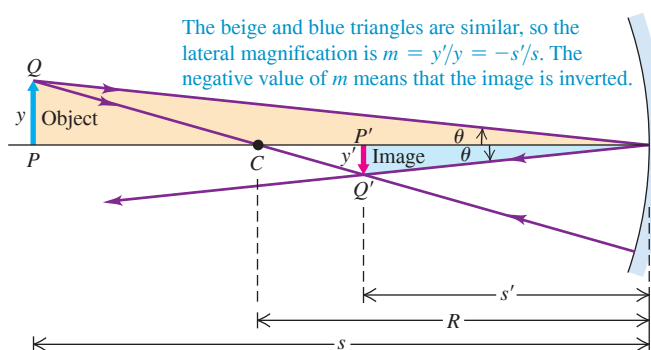
We'll usually express the relationship between object and image distances for a mirror, Eq. (34.4), in terms of the focal length  $f$ :

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f} \quad (\text{object-image relationship, spherical mirror}) \quad (34.6)$$

### Image of an Extended Object: Spherical Mirror

Now suppose we have an object with *finite* size, represented by the arrow  $PQ$  in Fig. 34.14, perpendicular to the optic axis  $CV$ . The image of  $P$  formed by paraxial rays is at  $P'$ . The object distance for point  $Q$  is very nearly equal to that for point  $P$ , so the image  $P'Q'$  is nearly straight and perpendicular to the axis. Note that the object and image arrows have different sizes,  $y$  and  $y'$ , respectively, and that they have opposite orientation.

Figure 34.14 Construction for determining the position, orientation, and height of an image formed by a concave spherical mirror.



In Eq. (34.2) we defined the *lateral magnification*  $m$  as the ratio of image size  $y'$  to object size  $y$ :

$$m = \frac{y'}{y}$$

Because triangles  $PVQ$  and  $P'VQ'$  in Fig. 34.14 are *similar*, we also have the relationship  $y/s = -y'/s'$ . The negative sign is needed because object and image are on opposite sides of the optic axis; if  $y$  is positive,  $y'$  is negative. Therefore

$$m = \frac{y'}{y} = -\frac{s'}{s} \quad (\text{lateral magnification, spherical mirror}) \quad (34.7)$$

If  $m$  is positive, the image is erect in comparison to the object; if  $m$  is negative, the image is *inverted* relative to the object, as in Fig. 34.14. For a *plane* mirror,  $s = -s'$ , so  $y' = y$  and  $m = +1$ ; since  $m$  is positive, the image is erect, and since  $|m| = 1$ , the image is the same size as the object.

**CAUTION** **Lateral magnification can be less than 1** Although the ratio of image size to object size is called the *lateral magnification*, the image formed by a mirror or lens may be larger than, smaller than, or the same size as the object. If it is smaller, then the lateral magnification is less than unity in absolute value:  $|m| < 1$ . The image formed by an astronomical telescope mirror or a camera lens is usually *much* smaller than the object. For example, the image of the bright star shown in Fig. 34.12c is just a few millimeters across, while the star itself is hundreds of thousands of kilometers in diameter. ■

In our discussion of concave mirrors we have so far considered only objects that lie *outside* or at the focal point, so that the object distance  $s$  is greater than or equal to the (positive) focal length  $f$ . In this case the image point is on the same side of the mirror as the outgoing rays, and the image is real and inverted. If an object is *inside* the focal point of a concave mirror, so that  $s < f$ , the resulting image is *virtual* (that is, the image point is on the opposite side of the mirror from the object), *erect*, and *larger* than the object. Mirrors used when you apply makeup (referred to at the beginning of this section) are concave mirrors; in use, the distance from the face to the mirror is less than the focal length, and you see an enlarged, erect image. You can prove these statements about concave mirrors by applying Eqs. (34.6) and (34.7). We'll also be able to verify these results later in this section, after we've learned some graphical methods for relating the positions and sizes of the object and the image.

### EXAMPLE 34.1 Image formation by a concave mirror I

A concave mirror forms an image, on a wall 3.00 m in front of the mirror, of a headlamp filament 10.0 cm in front of the mirror. (a) What are the radius of curvature and focal length of the mirror? (b) What is the lateral magnification? What is the image height if the object height is 5.00 mm?

**IDENTIFY and SET UP** Figure 34.15 shows our sketch. Our target variables are the radius of curvature  $R$ , focal length  $f$ , lateral magnification  $m$ , and image height  $y'$ . We are given the distances from the mirror to the object ( $s$ ) and from the mirror to the image ( $s'$ ). We solve Eq. (34.4) for  $R$ , and then use Eq. (34.5) to find  $f$ . Equation (34.7) yields both  $m$  and  $y'$ .

**EXECUTE** (a) Both the object and the image are on the concave (reflective) side of the mirror, so both  $s$  and  $s'$  are positive; we have  $s = 10.0$  cm and  $s' = 300$  cm. We solve Eq. (34.4) for  $R$ :

$$\frac{1}{10.0 \text{ cm}} + \frac{1}{300 \text{ cm}} = \frac{2}{R}$$

$$R = 2 \left( \frac{1}{10.0 \text{ cm}} + \frac{1}{300 \text{ cm}} \right)^{-1} = 19.4 \text{ cm}$$

### APPLICATION Satellite Television

**Dishes** A dish antenna used to receive satellite TV broadcasts is actually a concave parabolic mirror. The waves are of much lower frequency than visible light ( $1.2$  to  $1.8 \times 10^{10}$  Hz compared with  $4.0$  to  $7.9 \times 10^{14}$  Hz), but the laws of reflection are the same. The transmitter in orbit is so far away that the arriving waves have essentially parallel rays, as in Fig. 34.13a. The dish reflects the waves and brings them to a focus at a feed horn, from which they are “piped” to a decoder that extracts the signal.



Dish = segment of a curved mirror. Only a segment away from the optic axis is used so that the feed horn does not block incoming waves.

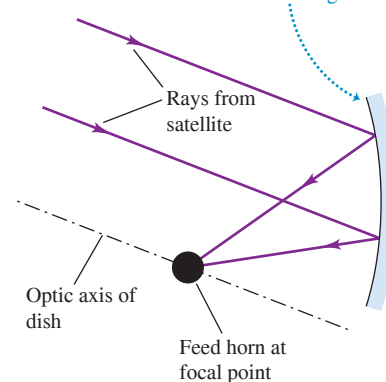
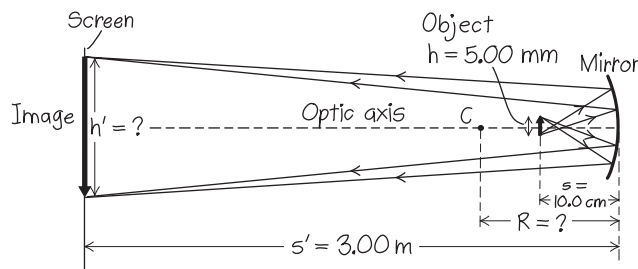


Figure 34.15 Our sketch for this problem.



The focal length of the mirror is  $f = R/2 = 9.68$  cm.

(b) From Eq. (34.7) the lateral magnification is

$$m = -\frac{s'}{s} = -\frac{300 \text{ cm}}{10.0 \text{ cm}} = -30.0$$

Continued

Because  $m$  is negative, the image is inverted. The height of the image is 30.0 times the height of the object, or  $(30.0)(5.00 \text{ mm}) = 150 \text{ mm}$ .

**EVALUATE** Our sketch indicates that the image is inverted; our calculations agree. Note that the object (at  $s = 10.0 \text{ cm}$ ) is just outside the focal point ( $f = 9.68 \text{ cm}$ ). This is very similar to what is done in automobile headlights. With the filament close to the focal point, the concave mirror produces a beam of nearly parallel rays.

**KEYCONCEPT** For a concave spherical mirror, the focal length—the distance from the mirror to the point where incoming parallel light rays come to a focus—equals one-half the radius of the mirror. The distance from the mirror to an object, the distance from the mirror to the resulting image, and the focal length are related by Eq. (34.6). The magnification of the image (which is negative if the image is inverted) equals the ratio of the image height to the object height.

### CONCEPTUAL EXAMPLE 34.2 Image formation by a concave mirror II

In Example 34.1, suppose that the lower half of the mirror's reflecting surface is covered with nonreflective soot. What effect will this have on the image of the filament?

**SOLUTION** It would be natural to guess that the image would now show only half of the filament. But in fact the image will still show the *entire* filament. You can see why by examining Fig. 34.10b. Light rays coming from any object point  $P$  are reflected from *all* parts of the mirror and converge on the corresponding image point  $P'$ . If part of the mirror surface is made nonreflective (or is removed altogether), rays from the remaining reflective surface still form an image of every part of the object.

Reducing the reflecting area reduces the light energy reaching the image point, however: The image becomes *dimmer*. If the area is reduced by one-half, the image will be one-half as bright. Conversely, *increasing* the reflective area makes the image brighter. To make reasonably bright images of faint stars, astronomical telescopes use mirrors that are up to several meters in diameter (see Fig. 34.12a).

**KEYCONCEPT** A concave mirror of any size makes a complete image of an object. The larger the mirror, the more light is incorporated into the image and the brighter the image is.

### Convex Mirrors

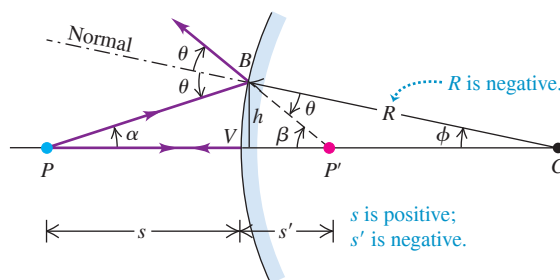
In Fig. 34.16a the *convex* side of a spherical mirror faces the incident light. The center of curvature is on the side opposite to the outgoing rays; according to sign rule 3 in Section 34.1,  $R$  is negative (see Fig. 34.11). Ray  $PB$  is reflected, with the angles of incidence and reflection both equal to  $\theta$ . The reflected ray, projected backward, intersects the axis at  $P'$ . As with a concave mirror, *all* rays from  $P$  that are reflected by the mirror diverge from the same point  $P'$ , provided that the angle  $\alpha$  is small. Therefore  $P'$  is the image of  $P$ . The object distance  $s$  is positive, the image distance  $s'$  is negative, and the radius of curvature  $R$  is *negative* for a *convex* mirror.

Figure 34.16b shows two rays diverging from the head of the arrow  $PQ$  and the virtual image  $P'Q'$  of this arrow. The same procedure that we used for a concave mirror can be used to show that for a convex mirror, the expressions for the object–image relationship and the lateral magnification are

$$\frac{1}{s} + \frac{1}{s'} = \frac{2}{R} \quad \text{and} \quad m = \frac{y'}{y} = -\frac{s'}{s}$$

Figure 34.16 Image formation by a convex mirror.

(a) Construction for finding the position of an image formed by a convex mirror



(b) Construction for finding the magnification of an image formed by a convex mirror

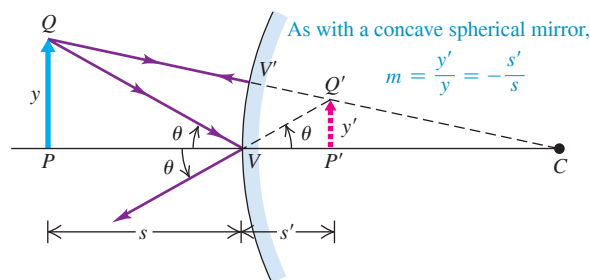
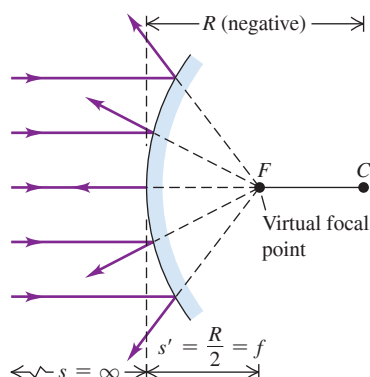
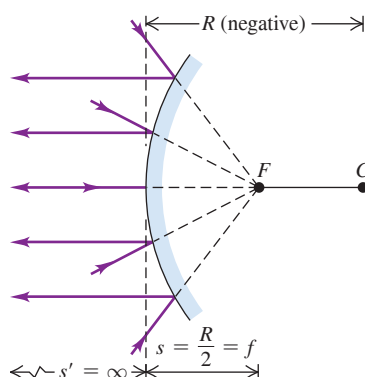


Figure 34.17 The focal point and focal length of a convex mirror.

(a) Paraxial rays incident on a convex spherical mirror diverge from a virtual focal point.



(b) Rays aimed at the virtual focal point are parallel to the axis after reflection.



These expressions are exactly the same as Eqs. (34.4) and (34.7) for a concave mirror. Thus when we use our sign rules consistently, Eqs. (34.4) and (34.7) are valid for both concave and convex mirrors.

When  $R$  is negative (convex mirror), incoming rays that are parallel to the optic axis are not reflected through the focal point  $F$ . Instead, they diverge as though they had come from the point  $F$  at a distance  $f$  behind the mirror, as shown in **Fig. 34.17a**. In this case,  $f$  is the focal length, and  $F$  is called a *virtual focal point*. The corresponding image distance  $s'$  is negative, so both  $f$  and  $R$  are negative, and Eq. (34.5),  $f = R/2$ , holds for convex as well as concave mirrors. In **Fig. 34.17b** the incoming rays are converging as though they would meet at the virtual focal point  $F$ , and they are reflected parallel to the optic axis.

In summary, Eqs. (34.4) through (34.7), the basic relationships for image formation by a spherical mirror, are valid for *both* concave and convex mirrors, provided that we use the sign rules consistently.

### EXAMPLE 34.3 Santa's image problem

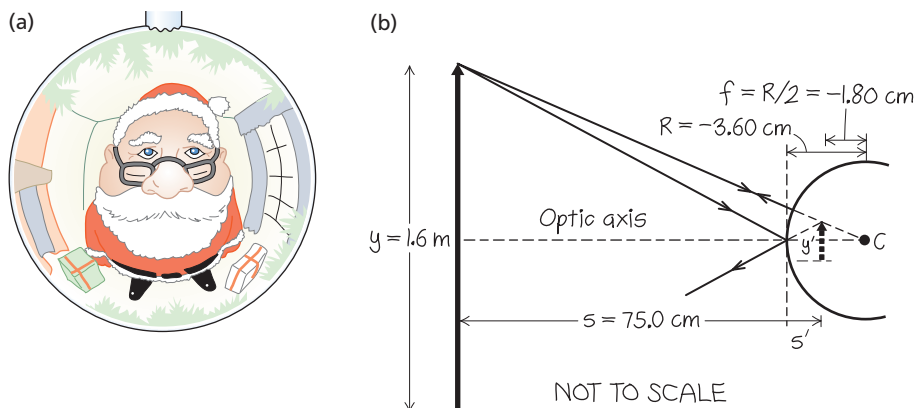
WITH **ARIATION PROBLEMS**

Santa checks himself for soot, using his reflection in a silvered Christmas tree ornament 0.750 m away (**Fig. 34.18a**). The diameter of the ornament is 7.20 cm. Standard reference texts state that he is a “right jolly old elf,” so we estimate his height to be 1.6 m. Where and how tall is the image of Santa formed by the ornament? Is it erect or inverted?

**IDENTIFY and SET UP** Figure 34.18b shows the situation. Santa is the object, and the surface of the ornament closest to him acts as a convex

mirror. The relationships among object distance, image distance, focal length, and magnification are the same as for concave mirrors, provided we use the sign rules consistently. The radius of curvature and the focal length of a convex mirror are *negative*. The object distance is  $s = 0.750 \text{ m} = 75.0 \text{ cm}$ , and Santa's height is  $y = 1.6 \text{ m}$ . We solve Eq. (34.6) to find the image distance  $s'$ , and then use Eq. (34.7) to find the lateral magnification  $m$  and the image height  $y'$ . The sign of  $m$  tells us whether the image is erect or inverted.

Figure 34.18 (a) The ornament forms a virtual, reduced, erect image of Santa. (b) Our sketch of two of the rays forming the image.



Continued



**EXECUTE** The radius of the mirror (half the diameter) is  $R = -(7.20 \text{ cm})/2 = -3.60 \text{ cm}$ , and the focal length is  $f = R/2 = -1.80 \text{ cm}$ . From Eq. (34.6),

$$\frac{1}{s'} = \frac{1}{f} - \frac{1}{s} = \frac{1}{-1.80 \text{ cm}} - \frac{1}{75.0 \text{ cm}}$$

$$s' = -1.76 \text{ cm}$$

Because  $s'$  is negative, the image is behind the mirror—that is, on the side opposite to the outgoing light (Fig. 34.18b)—and it is virtual. The image is about halfway between the front surface of the ornament and its center.

From Eq. (34.7), the lateral magnification and the image height are

$$m = \frac{y'}{y} = -\frac{s'}{s} = -\frac{-1.76 \text{ cm}}{75.0 \text{ cm}} = 0.0234$$

$$y' = my = (0.0234)(1.6 \text{ m}) = 3.8 \times 10^{-2} \text{ m} = 3.8 \text{ cm}$$

**EVALUATE** Our sketch indicates that the image is erect so both  $m$  and  $y'$  are positive; our calculations agree. When the object distance  $s$  is positive, a convex mirror *always* forms an erect, virtual, reduced, reversed image. For this reason, convex mirrors are used at blind intersections, for surveillance in stores, and as wide-angle rear-view mirrors for cars and trucks. (Many such mirrors read “Objects in mirror are closer than they appear.”)

**KEYCONCEPT** A convex spherical mirror has a negative focal length: Incoming parallel light rays reflect from the mirror as though they were coming from a focal point behind the mirror. The relationships among object distance, image distance, and focal length, and among magnification, object distance, and image distance are the same as for a concave mirror.

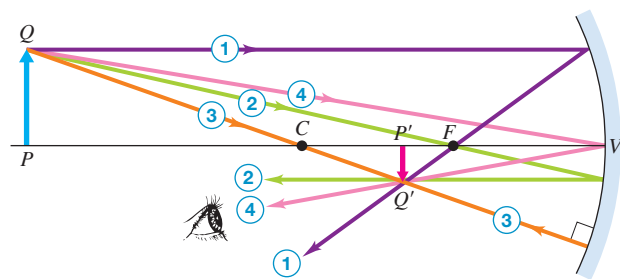
## Graphical Methods for Mirrors

In Examples 34.1 and 34.3, we used Eqs. (34.6) and (34.7) to find the position and size of the image formed by a mirror. We can also determine the properties of the image by a simple *graphical* method. This method consists of finding the point of intersection of a few particular rays that diverge from a point of the object (such as point  $Q$  in Fig. 34.19) and are reflected by the mirror. Then (ignoring aberrations) *all* rays from this object point that strike the mirror will intersect at the same point. For this construction we always choose an object point that is *not* on the optic axis. Four rays that we can usually draw easily are shown in Fig. 34.19. These are called **principal rays**.

1. A ray *parallel to the axis*, after reflection, passes through the focal point  $F$  of a concave mirror or appears to come from the (virtual) focal point of a convex mirror.
2. A ray *through (or proceeding toward) the focal point  $F$*  is reflected parallel to the axis.
3. A ray *along the radius* through or away from the center of curvature  $C$  intersects the surface normally and is reflected back along its original path.
4. A ray *to the vertex  $V$*  is reflected forming equal angles with the optic axis.

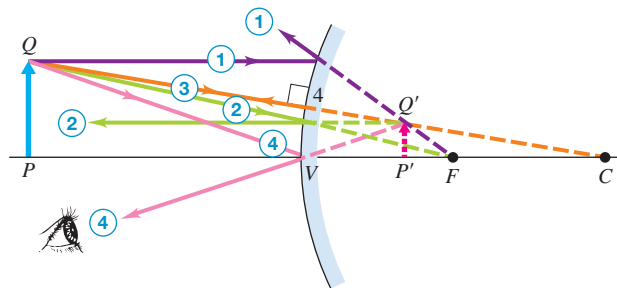
Figure 34.19 The graphical method of locating an image formed by a spherical mirror. The colors of the rays are for identification only; they do not refer to specific colors of light.

(a) Principal rays for concave mirror



- ① Ray parallel to axis reflects through focal point.
- ② Ray through focal point reflects parallel to axis.
- ③ Ray through center of curvature intersects the surface normally and reflects along its original path.
- ④ Ray to vertex reflects symmetrically around optic axis.

(b) Principal rays for convex mirror



- ① Reflected parallel ray appears to come from focal point.
- ② Ray toward focal point reflects parallel to axis.
- ③ As with concave mirror: Ray radial to center of curvature intersects the surface normally and reflects along its original path.
- ④ As with concave mirror: Ray to vertex reflects symmetrically around optic axis.

Once we have found the position of the image point by means of the intersection of any two of these principal rays (1, 2, 3, 4), we can draw the path of any other ray from the object point to the same image point.

**CAUTION** **Principal rays are not the only rays** Although we've emphasized the principal rays, in fact *any* ray from the object that strikes the mirror will pass through the image point (for a real image) or appear to originate from the image point (for a virtual image). Usually, you need to draw only the principal rays in order to locate the image. 

### PROBLEM-SOLVING STRATEGY 34.1 Image Formation by Mirrors

**IDENTIFY** *the relevant concepts:* Problems involving image formation by mirrors can be solved in two ways: using principal-ray diagrams and using equations. A successful problem solution uses *both* approaches.

**SET UP** *the problem:* Identify the target variables. One of them is likely to be the focal length, the object distance, or the image distance, with the other two quantities given.

**EXECUTE** *the solution* as follows:

1. Draw a large, clear principal-ray diagram if you have enough information.
2. Orient your diagram so that incoming rays go from left to right. Draw only the principal rays; color-code them as in Fig. 34.19. If possible, use graph paper or quadrille-ruled paper. Use a ruler and measure distances carefully! A freehand sketch will *not* give good results.
3. If your principal rays don't converge at a real image point, you may have to extend them straight backward to locate a virtual

image point, as in Fig. 34.19b. We recommend drawing the extensions with broken lines.

4. Measure the resulting diagram to obtain the magnitudes of the target variables.
5. Solve for the target variables by using Eq. (34.6),  $1/s + 1/s' = 1/f$ , and the lateral magnification equation, Eq. (34.7), as appropriate. Apply the sign rules given in Section 34.1 to object and image distances, radii of curvature, and object and image heights.
6. Use the sign rules to interpret the results that you deduced from your ray diagram and calculations. Note that the *same* sign rules (given in Section 34.1) work for all four cases in this chapter: reflection and refraction from plane and spherical surfaces.

**EVALUATE** *your answer:* Check that the results of your calculations agree with your ray-diagram results for image position, image size, and whether the image is real or virtual.

### EXAMPLE 34.4 Concave mirror with various object distances

#### WITH VARIATION PROBLEMS

A concave mirror has a radius of curvature with absolute value 20 cm. Find graphically the image of an object in the form of an arrow perpendicular to the axis of the mirror at object distances of (a) 30 cm, (b) 20 cm, (c) 10 cm, and (d) 5 cm. Check the construction by *computing* the size and lateral magnification of each image.

**IDENTIFY and SET UP** We must use graphical methods *and* calculations to analyze the image made by a mirror. The mirror is concave, so its radius of curvature is  $R = +20$  cm and its focal length is  $f = R/2 = +10$  cm. Our target variables are the image distances  $s'$  and lateral magnifications  $m$  corresponding to four cases with successively smaller object distances  $s$ . In each case we solve Eq. (34.6) for  $s'$  and use  $m = -s'/s$  to find  $m$ .

**EXECUTE** **Figure 34.20** (next page) shows the principal-ray diagrams for the four cases. Study each of these diagrams carefully and confirm that each numbered ray is drawn in accordance with the rules given earlier (under “Graphical Methods for Mirrors”). Several points are worth noting. First, in case (b) the object and image distances are equal. Ray 3 cannot be drawn in this case because a ray from  $Q$  through the center of curvature  $C$  does not strike the mirror. In case (c), ray 2 cannot be drawn because a ray from  $Q$  through  $F$  does not strike the mirror. In this case the outgoing rays are parallel, corresponding to an infinite image distance. In case (d), the outgoing rays diverge; they have been extended backward to the *virtual image point*  $Q'$ , from which they appear to diverge. Case (d) illustrates the general observation that an object placed inside the focal point of a concave mirror produces a virtual image.

Measurements of the figures, with appropriate scaling, give the following approximate image distances: (a) 15 cm; (b) 20 cm; (c)  $\infty$  or  $-\infty$  (because the outgoing rays are parallel and do not converge at any finite distance); (d)  $-10$  cm. To *compute* these distances, we solve Eq. (34.6) for  $s'$  and insert  $f = 10$  cm:

$$\begin{aligned} \text{(a)} \quad \frac{1}{30 \text{ cm}} + \frac{1}{s'} &= \frac{1}{10 \text{ cm}} & s' &= 15 \text{ cm} \\ \text{(b)} \quad \frac{1}{20 \text{ cm}} + \frac{1}{s'} &= \frac{1}{10 \text{ cm}} & s' &= 20 \text{ cm} \\ \text{(c)} \quad \frac{1}{10 \text{ cm}} + \frac{1}{s'} &= \frac{1}{10 \text{ cm}} & s' &= \infty \text{ (or } -\infty) \\ \text{(d)} \quad \frac{1}{5 \text{ cm}} + \frac{1}{s'} &= \frac{1}{10 \text{ cm}} & s' &= -10 \text{ cm} \end{aligned}$$

The signs of  $s'$  tell us that the image is real in cases (a) and (b) and virtual in case (d).

The lateral magnifications measured from the figures are approximately (a)  $-\frac{1}{2}$ ; (b)  $-1$ ; (c)  $\infty$  or  $-\infty$ ; (d)  $+2$ . From Eq. (34.7),

$$\begin{aligned} \text{(a)} \quad m &= -\frac{15 \text{ cm}}{30 \text{ cm}} = -\frac{1}{2} \\ \text{(b)} \quad m &= -\frac{20 \text{ cm}}{20 \text{ cm}} = -1 \\ \text{(c)} \quad m &= -\frac{\infty \text{ cm}}{10 \text{ cm}} = -\infty \text{ (or } +\infty) \end{aligned}$$

*Continued*

$$(d) m = -\frac{-10 \text{ cm}}{5 \text{ cm}} = +2$$

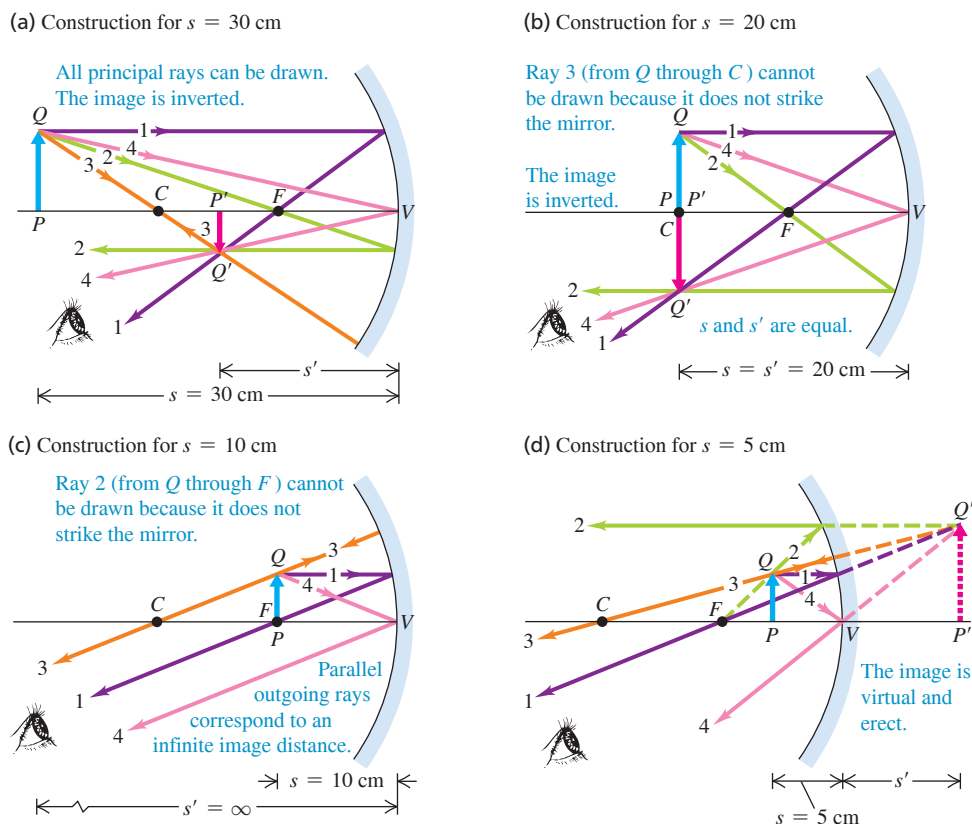
The signs of  $m$  tell us that the image is inverted in cases (a) and (b) and erect in case (d).

**EVALUATE** Notice the trend of the results in the four cases. When the object is far from the mirror, as in Fig. 34.20a, the image is smaller than the object, inverted, and real. As the object distance  $s$  decreases, the image moves farther from the mirror and gets larger (Fig. 34.20b). When the object is at the focal point, the image is at infinity (Fig. 34.20c). When the

object is inside the focal point, the image becomes larger than the object, erect, and virtual (Fig. 34.20d). You can confirm these conclusions by looking at objects reflected in the concave bowl of a shiny metal spoon.

**KEYCONCEPT** If an object is *outside* the focal point of a concave mirror, the reflected rays converge to a real, inverted image. If an object is *at* the focal point, the reflected rays are parallel and form an image at infinity. If an object is *inside* the focal point, the reflected rays appear to be coming from behind the mirror and so form a virtual, erect image.

Figure 34.20 Using principal-ray diagrams to locate the image  $P'Q'$  made by a concave mirror.



**TEST YOUR UNDERSTANDING OF SECTION 34.2** A cosmetics mirror is designed so that your reflection appears right-side up and enlarged. (a) Is the mirror concave or convex? (b) To see an enlarged image, what should be the distance from the mirror (of focal length  $f$ ) to your face? (i)  $|f|$ ; (ii) less than  $|f|$ ; (iii) greater than  $|f|$ .

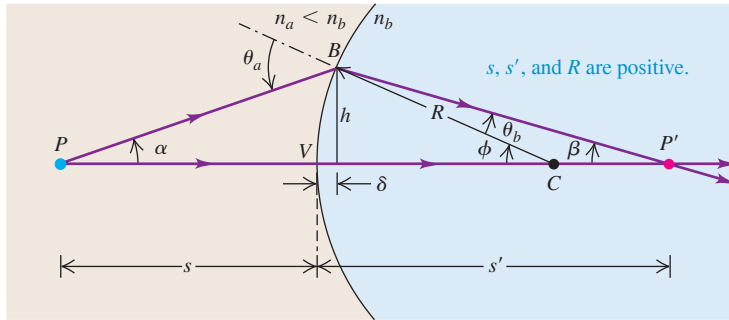
### ANSWER

(a) **concave**, (b) **(ii)** A convex mirror always produces an erect image, but that image is smaller than the object (see Fig. 34.16b). Hence a concave mirror must be used. The image will be erect and enlarged only if the distance from the object (your face) to the mirror is less than the focal length of the mirror, as in Fig. 34.20d.

## 34.3 REFRACTION AT A SPHERICAL SURFACE

As we mentioned in Section 34.1, images can be formed by refraction as well as by reflection. To begin with, let's consider refraction at a spherical surface—that is, at a spherical interface between two optical materials with different indexes of refraction. This analysis is directly applicable to some real optical systems, such as the human eye. It also provides a stepping-stone for the analysis of lenses, which usually have *two* spherical (or nearly spherical) surfaces.

Figure 34.21 Construction for finding the position of the image point  $P'$  of a point object  $P$  formed by refraction at a spherical surface. The materials to the left and right of the interface have indexes of refraction  $n_a$  and  $n_b$ , respectively. In the case shown here,  $n_a < n_b$ .



### Image of a Point Object: Spherical Refracting Surface

In **Fig. 34.21** a spherical surface with radius  $R$  forms an interface between two materials with different indexes of refraction  $n_a$  and  $n_b$ . The surface forms an image  $P'$  of an object point  $P$ ; we want to find how the object and image distances ( $s$  and  $s'$ ) are related. We'll use the same sign rules that we used for spherical mirrors. The center of curvature  $C$  is on the outgoing side of the surface, so  $R$  is positive. Ray  $PV$  strikes the vertex  $V$  and is perpendicular to the surface (that is, to the plane that is tangent to the surface at the point of incidence  $V$ ). It passes into the second material without deviation. Ray  $PB$ , making an angle  $\alpha$  with the axis, is incident at an angle  $\theta_a$  with the normal and is refracted at an angle  $\theta_b$ . These rays intersect at  $P'$ , a distance  $s'$  to the right of the vertex. The figure is drawn for the case  $n_a < n_b$ . Both the object and image distances are positive.

We are going to prove that if the angle  $\alpha$  is small, *all* rays from  $P$  intersect at the same point  $P'$ , so  $P'$  is the *real image* of  $P$ . We use much the same approach as we did for spherical mirrors in Section 34.2. We again use the theorem that an exterior angle of a triangle equals the sum of the two opposite interior angles; applying this to the triangles  $PBC$  and  $P'BC$  gives

$$\theta_a = \alpha + \phi \quad \phi = \beta + \theta_b \quad (34.8)$$

From the law of refraction,

$$n_a \sin \theta_a = n_b \sin \theta_b$$

Also, the tangents of  $\alpha$ ,  $\beta$ , and  $\phi$  are

$$\tan \alpha = \frac{h}{s + \delta} \quad \tan \beta = \frac{h}{s' - \delta} \quad \tan \phi = \frac{h}{R - \delta} \quad (34.9)$$

For paraxial rays,  $\theta_a$  and  $\theta_b$  are both small in comparison to a radian, and we may approximate both the sine and tangent of either of these angles by the angle itself (measured in radians). The law of refraction then gives

$$n_a \theta_a = n_b \theta_b$$

Combining this with the first of Eqs. (34.8), we obtain

$$\theta_b = \frac{n_a}{n_b}(\alpha + \phi)$$

When we substitute this into the second of Eqs. (34.8), we get

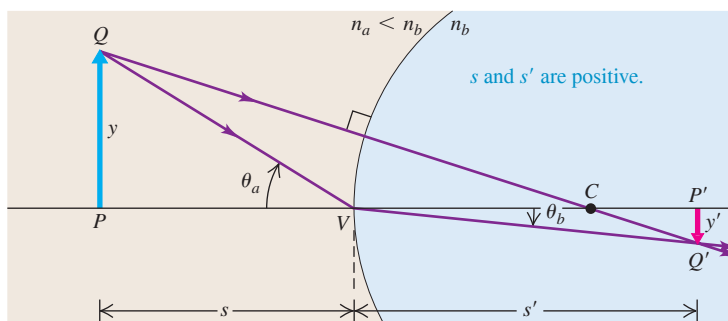
$$n_a \alpha + n_b \beta = (n_b - n_a) \phi \quad (34.10)$$

Now we use the approximations  $\tan \alpha = \alpha$ , and so on, in Eqs. (34.9) and also ignore the small distance  $\delta$ ; those equations then become

$$\alpha = \frac{h}{s} \quad \beta = \frac{h}{s'} \quad \phi = \frac{h}{R}$$



Figure 34.22 Construction for determining the height of an image formed by refraction at a spherical surface. In the case shown here,  $n_a < n_b$ .



Finally, we substitute these into Eq. (34.10) and divide out the common factor  $h$ . We obtain

$$\frac{n_a}{s} + \frac{n_b}{s'} = \frac{n_b - n_a}{R} \quad (\text{object-image relationship, spherical refracting surface}) \quad (34.11)$$

This equation does not contain the angle  $\alpha$ , so the image distance is the same for *all* paraxial rays emanating from  $P$ ; this proves that  $P'$  is the image of  $P$ .

To obtain the lateral magnification  $m$  for this situation, we use the construction in **Fig. 34.22**. We draw two rays from point  $Q$ , one through the center of curvature  $C$  and the other incident at the vertex  $V$ . From the triangles  $PQV$  and  $P'Q'V$ ,

$$\tan \theta_a = \frac{y}{s} \quad \tan \theta_b = \frac{-y'}{s'}$$

and from the law of refraction,

$$n_a \sin \theta_a = n_b \sin \theta_b$$

For small angles,

$$\tan \theta_a = \sin \theta_a \quad \tan \theta_b = \sin \theta_b$$

so finally

$$\frac{n_a y}{s} = -\frac{n_b y'}{s'} \quad \text{or} \quad m = \frac{y'}{y} = -\frac{n_a s'}{n_b s} \quad (\text{lateral magnification, spherical refracting surface}) \quad (34.12)$$

Equations (34.11) and (34.12) apply to both convex and concave refracting surfaces, provided that you use the sign rules consistently. It doesn't matter whether  $n_b$  is greater or less than  $n_a$ . To verify these statements, construct diagrams like Figs. 34.21 and 34.22 for the following three cases: (i)  $R > 0$  and  $n_a > n_b$ , (ii)  $R < 0$  and  $n_a < n_b$ , and (iii)  $R < 0$  and  $n_a > n_b$ . Then in each case, use your diagram to again derive Eqs. (34.11) and (34.12).

Here's a final note on the sign rule for the radius of curvature  $R$  of a surface. For the convex reflecting surface in Fig. 34.16, we considered  $R$  negative, but the convex *refracting* surface in Fig. 34.21 has a *positive* value of  $R$ . This may seem inconsistent, but it isn't. The rule is that  $R$  is positive if the center of curvature  $C$  is on the outgoing side of the surface and negative if  $C$  is on the other side. For the convex reflecting surface in Fig. 34.16,  $R$  is negative because point  $C$  is to the right of the surface but outgoing rays are to the left. For the convex refracting surface in Fig. 34.21,  $R$  is positive because both  $C$  and the outgoing rays are to the right of the surface.

Refraction at a curved surface is one reason gardeners avoid watering plants at midday. As sunlight enters a water drop resting on a leaf (**Fig. 34.23**), the light rays are refracted toward each other as in Figs. 34.21 and 34.22. The sunlight that strikes the leaf is therefore more concentrated and able to cause damage.

An important special case of a spherical refracting surface is a *plane* surface between two optical materials. This corresponds to setting  $R = \infty$  in Eq. (34.11). In this case,

$$\frac{n_a}{s} + \frac{n_b}{s'} = 0 \quad (\text{plane refracting surface}) \quad (34.13)$$

Figure 34.23 Light rays refract as they pass through the curved surfaces of these water droplets.



To find the lateral magnification  $m$  for this case, we combine this equation with the general relationship, Eq. (34.12), obtaining the simple result

$$m = 1$$

That is, the image formed by a *plane* refracting surface always has the same lateral size as the object, and it is always erect.

An example of image formation by a plane refracting surface is the appearance of a partly submerged drinking straw or canoe paddle. When viewed from some angles, the submerged part appears to be only about three-quarters of its actual distance below the surface. (We commented on the appearance of a submerged object in Section 33.2; see Fig. 33.9.)

### EXAMPLE 34.5 Image formation by refraction I

A cylindrical glass rod (**Fig. 34.24**) has index of refraction 1.52. It is surrounded by air. One end is ground to a hemispherical surface with radius  $R = 2.00$  cm. A small object is placed on the axis of the rod, 8.00 cm to the left of the vertex. Find (a) the image distance and (b) the lateral magnification.

**IDENTIFY and SET UP** This problem uses the ideas of refraction at a curved surface. Our target variables are the image distance  $s'$  and the lateral magnification  $m$ . Here material  $a$  is air ( $n_a = 1.00$ ) and material  $b$  is the glass of which the rod is made ( $n_b = 1.52$ ). We are given  $s = 8.00$  cm. The center of curvature of the spherical surface is on the outgoing side of the surface, so the radius is positive:  $R = +2.00$  cm. We solve Eq. (34.11) for  $s'$ , and we use Eq. (34.12) to find  $m$ .

**EXECUTE** (a) From Eq. (34.11),

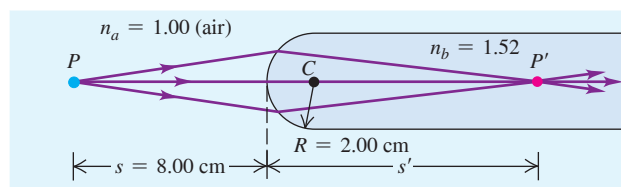
$$\frac{1.00}{8.00 \text{ cm}} + \frac{1.52}{s'} = \frac{1.52 - 1.00}{+2.00 \text{ cm}}$$

$$s' = +11.3 \text{ cm}$$

(b) From Eq. (34.12),

$$m = -\frac{n_a s'}{n_b s} = -\frac{(1.00)(11.3 \text{ cm})}{(1.52)(8.00 \text{ cm})} = -0.929$$

Figure 34.24 The glass rod in air forms a real image.



**EVALUATE** Because the image distance  $s'$  is positive, the image is formed 11.3 cm to the *right* of the vertex (on the outgoing side), as Fig. 34.24 shows. The value of  $m$  tells us that the image is somewhat smaller than the object and that it is inverted. If the object is an arrow 1.000 mm high, pointing upward, the image is an arrow 0.929 mm high, pointing downward.

**KEYCONCEPT** The position of the image formed by a spherical refracting surface depends on the position of the object, the radius of curvature of the surface, and the indexes of refraction of the materials on either side of the surface.

### EXAMPLE 34.6 Image formation by refraction II

The glass rod of Example 34.5 is immersed in water, which has index of refraction  $n = 1.33$  (**Fig. 34.25**). The object distance is again 8.00 cm. Find the image distance and lateral magnification.

**IDENTIFY and SET UP** The situation is the same as in Example 34.5 except that now  $n_a = 1.33$ . We again use Eqs. (34.11) and (34.12) to determine  $s'$  and  $m$ , respectively.

**EXECUTE** Our solution of Eq. (34.11) in Example 34.5 yields

$$\frac{1.33}{8.00 \text{ cm}} + \frac{1.52}{s'} = \frac{1.52 - 1.33}{+2.00 \text{ cm}}$$

$$s' = -21.3 \text{ cm}$$

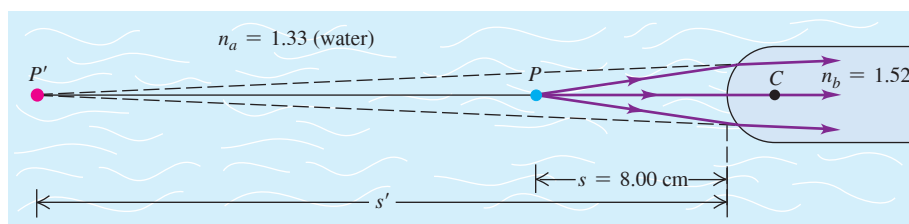
The lateral magnification in this case is

$$m = \frac{n_a s'}{n_b s} = -\frac{(1.33)(-21.3 \text{ cm})}{(1.52)(8.00 \text{ cm})} = +2.33$$

**EVALUATE** The negative value of  $s'$  means that the refracted rays do not converge, but appear to diverge from a point 21.3 cm to the *left* of the vertex. We saw a similar case in the reflection of light from a convex mirror; in both cases we call the result a *virtual image*. The vertical image is erect (because  $m$  is positive) and 2.33 times as large as the object.

**KEYCONCEPT** A spherical refracting surface forms a real image if the refracted rays converge but forms a virtual image if the refracted rays diverge.

Figure 34.25 When immersed in water, the glass rod forms a virtual image.



**EXAMPLE 34.7** Apparent depth of a swimming pool

If you look straight down into a swimming pool where it is 2.00 m deep, how deep does it appear to be?

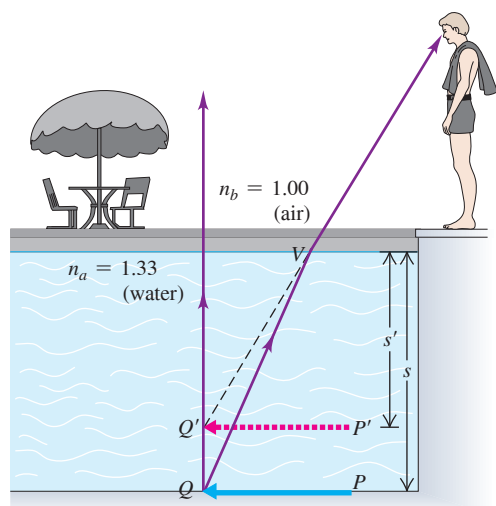
**IDENTIFY and SET UP** Figure 34.26 shows the situation. The surface of the water acts as a plane refracting surface. To determine the pool's apparent depth, we imagine an arrow  $PQ$  painted on the bottom. The pool's refracting surface forms a virtual image  $P'Q'$  of this arrow. We solve Eq. (34.13) to find the image distance  $s'$ ; that's the pool's apparent depth.

**EXECUTE** The object distance is the actual depth of the pool,  $s = 2.00$  m. Material  $a$  is water ( $n_a = 1.33$ ) and material  $b$  is air ( $n_b = 1.00$ ). From Eq. (34.13),

$$\frac{n_a}{s} + \frac{n_b}{s'} = \frac{1.33}{2.00 \text{ m}} + \frac{1.00}{s'} = 0$$

$$s' = -1.50 \text{ m}$$

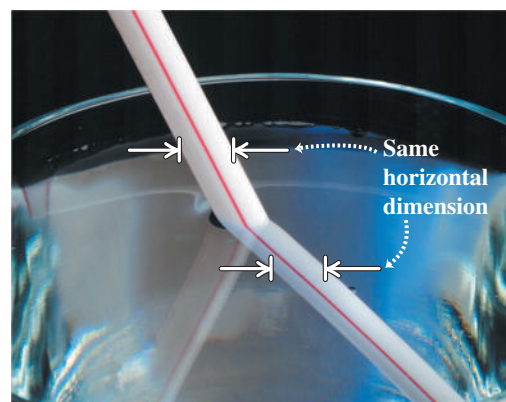
Figure 34.26 Arrow  $P'Q'$  is the virtual image of underwater arrow  $PQ$ . The angles of the ray with the vertical are exaggerated for clarity.



The image distance is negative. By the sign rules in Section 34.1, this means that the image is virtual and on the incoming side of the refracting surface—that is, on the same side as the object, namely underwater. The pool's apparent depth is 1.50 m, or just 75% of its true depth.

**EVALUATE** Recall that the lateral magnification for a plane refracting surface is  $m = 1$ . Hence the image  $P'Q'$  of the arrow has the same horizontal length as the actual arrow  $PQ$  (Fig. 34.27). Only its depth is different from that of  $PQ$ .

Figure 34.27 The submerged portion of this straw appears to be at a shallower depth (closer to the surface) than it actually is.



**KEYCONCEPT** A plane refracting surface always forms a virtual image. An object on the other side of the refracting surface appears to be at a different distance from the surface than its actual distance.

**TEST YOUR UNDERSTANDING OF SECTION 34.3** The water droplets in Fig. 34.23 have radius of curvature  $R$  and index of refraction  $n = 1.33$ . Can they form an image of the sun on the leaf?

**ANSWER** The rays are nonetheless concentrated and can cause damage to the leaf. The rays of sunlight never reach the image point, and the drops do not form an image of the sun. Thus the rays of sunlight never reach the image point, and the drops do not form an image of the sun. The image would be formed 4.0 drop radii from the front surface of the drop. But since each drop is only a part of a complete sphere, the distance from the front to the back of the drop is less than  $2R$ .

$$\frac{n_a}{s} + \frac{n_b}{s'} = \frac{n_b}{R} \quad \text{or} \quad 0 + \frac{s'}{1.33} = \frac{R}{1.33 - 1.00}$$

$$s' = \frac{0.33}{1.33} R = 4.0R$$

**no** The sun is very far away, so the object distance is essentially infinite:  $s = \infty$  and  $1/s = 0$ . Material  $a$  is air ( $n_a = 1.00$ ) and material  $b$  is water ( $n_b = 1.33$ ), so the image position  $s'$  is

## 34.4 THIN LENSES

The most familiar and widely used optical device (after the plane mirror) is the *lens*. A lens is an optical system with two refracting surfaces. The simplest lens has two *spherical* surfaces close enough together that we can ignore the distance between them (the thickness of the lens); we call this a **thin lens**. If you wear eyeglasses or contact lenses while reading, you are viewing these words through a pair of thin lenses. Later in this section we'll analyze thin lenses in detail by using the results of Section 34.3 for refraction by a single spherical surface. However, let's first discuss the properties of thin lenses.

### Properties of a Lens

A lens of the shape shown in **Fig. 34.28** has an important property: When a beam of rays parallel to the axis passes through the lens, the rays converge to a point  $F_2$  (Fig. 34.28a) and form a real image at that point. Such a lens is called a **converging lens**. Similarly, rays passing through point  $F_1$  emerge from the lens as a beam of parallel rays (Fig. 34.28b). The points  $F_1$  and  $F_2$  are called the first and second *focal points*, and the distance  $f$  (measured from the center of the lens) is called the *focal length*. Note the similarities between the two focal points of a converging lens and the single focal point of a concave mirror (see Fig. 34.13). As for a concave mirror, the focal length of a converging lens is defined to be a *positive* quantity, and such a lens is also called a *positive lens*.

The central horizontal line in Fig. 34.28 is called the *optic axis*, as with spherical mirrors. The centers of curvature of the two spherical surfaces lie on and define the optic axis. The two focal lengths in Fig. 34.28, both labeled  $f$ , are *always equal* for a thin lens, even when the two sides have different curvatures. We'll show this result later in the section, when we derive the relationship of  $f$  to the index of refraction of the lens and the radii of curvature of its surfaces.

### Image of an Extended Object: Converging Lens

Like a concave mirror, a converging lens can form an image of an extended object. **Figure 34.29** shows how to find the position and lateral magnification of an image made by a thin converging lens. Using the same notation and sign rules as before, we let  $s$  and  $s'$  be the object and image distances, respectively, and let  $y$  and  $y'$  be the object and image heights. Ray  $QA$ , parallel to the optic axis before refraction, passes through the second focal point  $F_2$  after refraction. Ray  $QOQ'$  passes undeflected straight through the center of the lens because at the center the two surfaces are parallel and (we have assumed) very close together. There is refraction where the ray enters and leaves the material but no net change in direction.

The two angles labeled  $\alpha$  in Fig. 34.29 are equal, so the two right triangles  $PQO$  and  $P'Q'O$  are *similar* and ratios of corresponding sides are equal. Thus

$$\frac{y}{s} = -\frac{y'}{s'} \quad \text{or} \quad \frac{y'}{y} = -\frac{s'}{s} \quad (34.14)$$

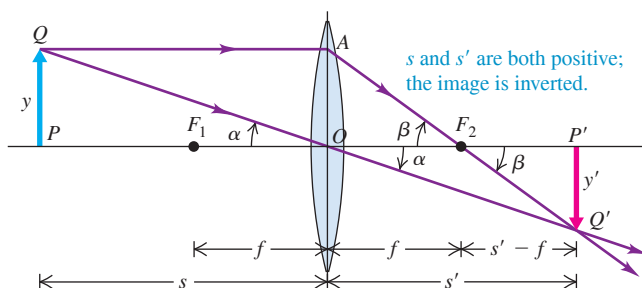
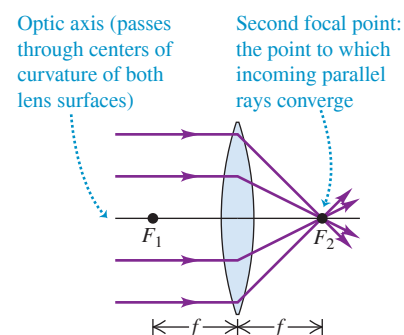


Figure 34.28  $F_1$  and  $F_2$  are the first and second focal points of a converging thin lens. The numerical value of  $f$  is positive.

(a)



- Focal length
- Measured from lens center
  - Always the same on both sides of the lens
  - Positive for a converging thin lens

(b)

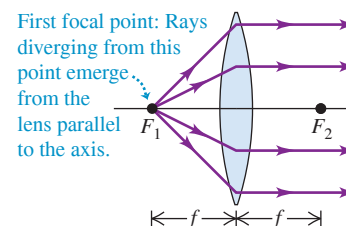


Figure 34.29 Construction used to find image position for a thin lens. To emphasize that the lens is assumed to be very thin, the ray  $QAQ'$  is shown as bent at the midplane of the lens rather than at the two surfaces and ray  $QOQ'$  is shown as a straight line.



(The reason for the negative sign is that the image is below the optic axis and  $y'$  is negative.) Also, the two angles labeled  $\beta$  are equal, and the two right triangles  $OAF_2$  and  $P'Q'F_2$  are similar, so

$$\frac{y}{f} = -\frac{y'}{s' - f} \quad \text{or}$$

$$\frac{y'}{y} = -\frac{s' - f}{f} \quad (34.15)$$

We now equate Eqs. (34.14) and (34.15), divide by  $s'$ , and rearrange to obtain

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f} \quad (34.16)$$

Object-image relationship, thin lens:      Object distance      Image distance      Focal length of lens

Equation (34.14) also gives the lateral magnification for the lens:

$$m = y'/y = -\frac{s'}{s} \quad (\text{lateral magnification, thin lens}) \quad (34.17)$$

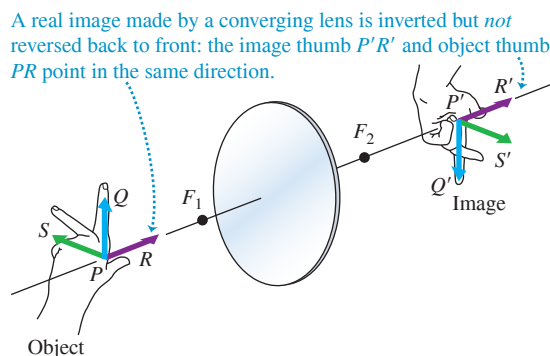
The negative sign tells us that when  $s$  and  $s'$  are both positive, as in Fig. 34.29, the image is *inverted*, and  $y$  and  $y'$  have opposite signs.

Equations (34.16) and (34.17) are the basic equations for thin lenses. They are *exactly* the same as the corresponding equations for spherical mirrors, Eqs. (34.6) and (34.7). As we'll see, the same sign rules that we used for spherical mirrors are also applicable to lenses. In particular, consider a lens with a positive focal length (a converging lens). When an object is outside the first focal point  $F_1$  of this lens (that is, when  $s > f$ ), the image distance  $s'$  is positive (that is, the image is on the same side as the outgoing rays); this image is real and inverted, as in Fig. 34.29. An object placed inside the first focal point of a converging lens, so that  $s < f$ , produces an image with a negative value of  $s'$ ; this image is located on the same side of the lens as the object and is virtual, erect, and larger than the object. You can verify these statements algebraically by using Eqs. (34.16) and (34.17); we'll also verify them in the next section, using graphical methods analogous to those introduced for mirrors in Section 34.2.

**Figure 34.30** shows how a lens forms a three-dimensional image of a three-dimensional object. Point  $R$  is nearer the lens than point  $P$ . From Eq. (34.16), image point  $R'$  is farther from the lens than is image point  $P'$ , and the image  $P'R'$  points in the same direction as the object  $PR$ . Arrows  $P'S'$  and  $P'Q'$  are reversed relative to  $PS$  and  $PQ$ .

Let's compare Fig. 34.30 with Fig. 34.7, which shows the image formed by a plane *mirror*. We note that the image formed by the lens is inverted, but it is *not* reversed front to back along the optic axis. That is, if the object is a left hand, its image is also a left hand. You can verify this by pointing your left thumb along  $PR$ , your left forefinger along  $PQ$ ,

Figure 34.30 The image  $S'P'Q'R'$  of a three-dimensional object  $SPQR$  is not reversed by a lens.



and your left middle finger along  $PS$ . Then rotate your hand  $180^\circ$ , using your thumb as an axis; this brings the fingers into coincidence with  $P'Q'$  and  $P'S'$ . In other words, an *inverted* image is equivalent to an image that has been rotated by  $180^\circ$  about the lens axis.

## Diverging Lenses

So far we have been discussing *converging* lenses. **Figure 34.31** shows a **diverging lens**; the beam of parallel rays incident on this lens *diverges* after refraction. The focal length of a diverging lens is a negative quantity, and the lens is also called a *negative lens*. The focal points of a negative lens are reversed, relative to those of a positive lens. The second focal point,  $F_2$ , of a negative lens is the point from which rays that are originally parallel to the axis *appear to diverge* after refraction, as in Fig. 34.31a. Incident rays converging toward the first focal point  $F_1$ , as in Fig. 34.31b, emerge from the lens parallel to its axis. Comparing with Section 34.2, you can see that a diverging lens has the same relationship to a converging lens as a convex mirror has to a concave mirror.

Equations (34.16) and (34.17) apply to *both* positive and negative lenses. **Figure 34.32** shows various types of lenses, both converging and diverging. Here's an important observation: *Any lens that is thicker at its center than at its edges is a converging lens with positive  $f$ , and any lens that is thicker at its edges than at its center is a diverging lens with negative  $f$*  (provided that the lens has a greater index of refraction than the surrounding material). We can prove this by using the *lensmaker's equation*, which it is our next task to derive.

## The Lensmaker's Equation

We'll now derive Eq. (34.16) in more detail and at the same time derive the *lensmaker's equation*, which is a relationship among the focal length  $f$ , the index of refraction  $n$  of the lens, and the radii of curvature  $R_1$  and  $R_2$  of the lens surfaces. We use the principle that an image formed by one reflecting or refracting surface can serve as the object for a second reflecting or refracting surface.

We begin with the somewhat more general problem of two spherical interfaces separating three materials with indexes of refraction  $n_a$ ,  $n_b$ , and  $n_c$ , as shown in **Fig. 34.33** (next page). The object and image distances for the first surface are  $s_1$  and  $s'_1$ , and those for the second surface are  $s_2$  and  $s'_2$ . We assume that the lens is thin, so that the distance  $t$  between the two surfaces is small in comparison with the object and image distances and can therefore be ignored. This is usually the case with eyeglass lenses (**Fig. 34.34**). Then  $s_2$  and  $s'_1$  have the same magnitude but opposite sign. For example, if the first image is on the outgoing side of the first surface,  $s'_1$  is positive. But when viewed as an object for the second surface, the first image is *not* on the incoming side of that surface. So we can say that  $s_2 = -s'_1$ .

We need to use the single-surface equation, Eq. (34.11), twice, once for each surface. The two resulting equations are

$$\frac{n_a}{s_1} + \frac{n_b}{s'_1} = \frac{n_b - n_a}{R_1}$$

$$\frac{n_b}{s_2} + \frac{n_c}{s'_2} = \frac{n_c - n_b}{R_2}$$

Ordinarily, the first and third materials are air or vacuum, so we set  $n_a = n_c = 1$ . The second index  $n_b$  is that of the lens, which we can call simply  $n$ . Substituting these values and the relationship  $s_2 = -s'_1$ , we get

$$\frac{1}{s_1} + \frac{n}{s'_1} = \frac{n - 1}{R_1}$$

$$-\frac{n}{s'_1} + \frac{1}{s'_2} = \frac{1 - n}{R_2}$$

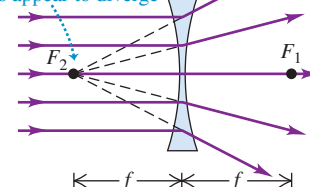
To get a relationship between the initial object position  $s_1$  and the final image position  $s'_2$ , we add these two equations. This eliminates the term  $n/s'_1$ :

$$\frac{1}{s_1} + \frac{1}{s'_2} = (n - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right)$$

**Figure 34.31**  $F_2$  and  $F_1$  are the second and first focal points of a diverging thin lens, respectively. The numerical value of  $f$  is negative.

(a)

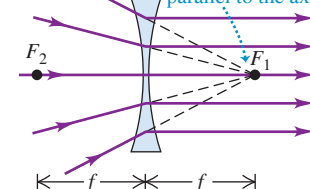
Second focal point: The point from which parallel incident rays appear to diverge



For a diverging thin lens,  $f$  is negative.

(b)

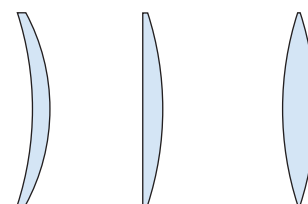
First focal point: Rays converging on this point emerge from the lens parallel to the axis.



**Figure 34.32** Various types of lenses.

(a)

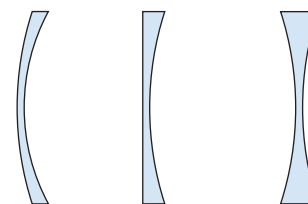
**Converging lenses**



Meniscus    Planoconvex    Double convex

(b)

**Diverging lenses**



Meniscus    Planoconcave    Double concave

Figure 34.33 The image formed by the first surface of a lens serves as the object for the second surface. The distances  $s'_1$  and  $s_2$  are taken to be equal; this is a good approximation if the lens thickness  $t$  is small.

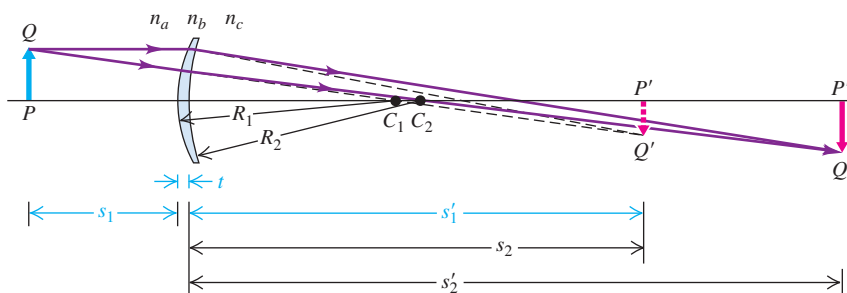


Figure 34.34 These eyeglass lenses satisfy the thin-lens approximation: Their thickness is small compared to the object and image distances.



Finally, thinking of the lens as a single unit, we rename the object distance simply  $s$  instead of  $s_1$ , and we rename the final image distance  $s'$  instead of  $s'_2$ :

$$\frac{1}{s} + \frac{1}{s'} = (n - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right) \quad (34.18)$$

Now we compare this with the other thin-lens equation, Eq. (34.16). We see that the object and image distances  $s$  and  $s'$  appear in exactly the same places in both equations and that the focal length  $f$  is given by the **lensmaker's equation**:

Lensmaker's equation for a thin lens:

$$\frac{1}{f} = (n - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right) \quad (34.19)$$

Index of refraction of lens material

Focal length

Radius of curvature of first surface

Radius of curvature of second surface

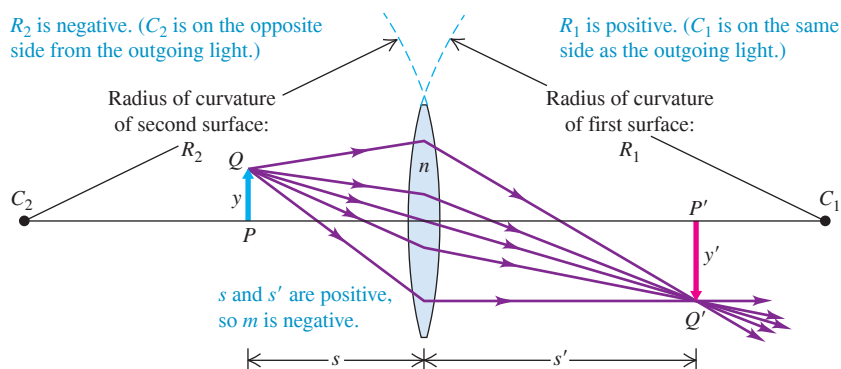
In the process of rederiving the relationship among object distance, image distance, and focal length for a thin lens, we have also derived Eq. (34.19), an expression for the focal length  $f$  of a lens in terms of its index of refraction  $n$  and the radii of curvature  $R_1$  and  $R_2$  of its surfaces. This can be used to show that all the lenses in Fig. 34.32a are converging lenses with  $f > 0$  and that all the lenses in Fig. 34.32b are diverging lenses with  $f < 0$ .

We use all our sign rules from Section 34.1 with Eqs. (34.18) and (34.19). For example, in Fig. 34.35,  $s$ ,  $s'$ , and  $R_1$  are positive, but  $R_2$  is negative.

It is not hard to generalize Eq. (34.19) to the situation in which the lens is immersed in a material with an index of refraction greater than unity. We invite you to work out the lensmaker's equation for this more general situation.

We stress that the paraxial approximation is indeed an approximation! Rays that are at sufficiently large angles to the optic axis of a spherical lens will not be brought to the same focus as paraxial rays; this is the same problem of spherical aberration that plagues spherical *mirrors* (see Section 34.2). To avoid this and other limitations of thin spherical lenses, lenses of more complicated shape are used in precision optical instruments.

Figure 34.35 A converging thin lens with a positive focal length  $f$ .



**EXAMPLE 34.8 Determining the focal length of a lens****WITH VARIATION PROBLEMS**

(a) Suppose the absolute values of the radii of curvature of the lens surfaces in Fig. 34.35 are both equal to 10 cm and the index of refraction of the glass is  $n = 1.52$ . What is the focal length  $f$  of the lens?  
 (b) Suppose the lens in Fig. 34.31 also has  $n = 1.52$  and the absolute values of the radii of curvature of its lens surfaces are also both equal to 10 cm. What is the focal length of this lens?

**IDENTIFY and SET UP** We are asked to find the focal length  $f$  of (a) a lens that is convex on both sides (Fig. 34.35) and (b) a lens that is concave on both sides (Fig. 34.31). In both cases we solve the lensmaker's equation, Eq. (34.19), to determine  $f$ . We apply the sign rules given in Section 34.1 to the radii of curvature  $R_1$  and  $R_2$  to take account of whether the surfaces are convex or concave.

**EXECUTE** (a) The lens in Fig. 34.35 is *double convex*: The center of curvature of the first surface ( $C_1$ ) is on the outgoing side of the lens, so  $R_1$  is positive, and the center of curvature of the second surface ( $C_2$ ) is on the *incoming* side, so  $R_2$  is negative. Hence  $R_1 = +10$  cm and  $R_2 = -10$  cm. Then from Eq. (34.19),

$$\frac{1}{f} = (1.52 - 1) \left( \frac{1}{+10 \text{ cm}} - \frac{1}{-10 \text{ cm}} \right) \quad \text{or} \quad f = 9.6 \text{ cm}$$

(b) The lens in Fig. 34.31 is *double concave*: The center of curvature of the first surface is on the *incoming* side, so  $R_1$  is negative, and the center of curvature of the second surface is on the outgoing side, so  $R_2$  is positive. Hence in this case  $R_1 = -10$  cm and  $R_2 = +10$  cm. Again using Eq. (34.19), we get

$$\frac{1}{f} = (1.52 - 1) \left( \frac{1}{-10 \text{ cm}} - \frac{1}{+10 \text{ cm}} \right) \quad \text{or} \quad f = -9.6 \text{ cm}$$

**EVALUATE** In part (a) the focal length is *positive*, so this is a converging lens; this makes sense, since the lens is thicker at its center than at its edges. In part (b) the focal length is *negative*, so this is a diverging lens; this also makes sense, since the lens is thicker at its edges than at its center.

**KEYCONCEPT** The lensmaker's equation, Eq. (34.19), describes how the focal length of a thin lens surrounded by air depends on the radii of curvature of its surfaces and the index of refraction of the material of which it is made. A radius is positive if the center of curvature is on the side of the outgoing light, and negative if the center is on the side of the incoming light.

**Graphical Methods for Lenses**

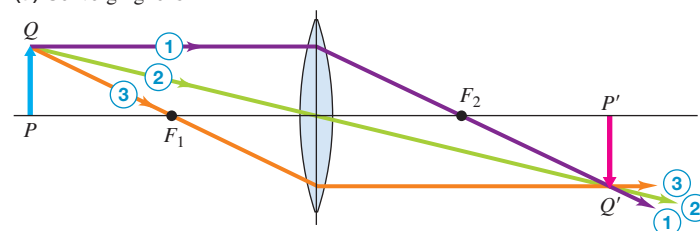
We can determine the position and size of an image formed by a thin lens by using a graphical method very similar to the one we used in Section 34.2 for spherical mirrors. Again we draw a few special rays called *principal rays* that diverge from a point of the object that is *not* on the optic axis. The intersection of these rays, after they pass through the lens, determines the position and size of the image. In using this graphical method, we'll consider the entire deviation of a ray as occurring at the midplane of the lens, as shown in Fig. 34.36. This is consistent with the assumption that the distance between the lens surfaces is negligible.

The three principal rays whose paths are usually easy to trace for lenses are shown in Fig. 34.36:

1. A ray *parallel to the axis* emerges from the lens in a direction that passes through the second focal point  $F_2$  of a converging lens, or appears to come from the second focal point of a diverging lens.
2. A ray *through the center of the lens* is not appreciably deviated; at the center of the lens the two surfaces are parallel, so this ray emerges at essentially the same angle at which it enters and along essentially the same line.
3. A ray *through (or proceeding toward) the first focal point  $F_1$*  emerges parallel to the axis.

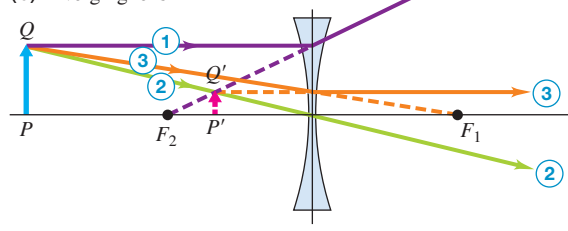
Figure 34.36 The graphical method of locating an image formed by a thin lens. The colors of the rays are for identification only; they do not refer to specific colors of light. (Compare Fig. 34.19 for spherical mirrors.)

(a) Converging lens



- ① Parallel incident ray refracts to pass through second focal point  $F_2$ .
- ② Ray through center of lens does not deviate appreciably.
- ③ Ray through the first focal point  $F_1$  emerges parallel to the axis.

(b) Diverging lens



- ① Parallel incident ray appears after refraction to have come from the second focal point  $F_2$ .
- ② Ray through center of lens does not deviate appreciably.
- ③ Ray aimed at the first focal point  $F_1$  emerges parallel to the axis.



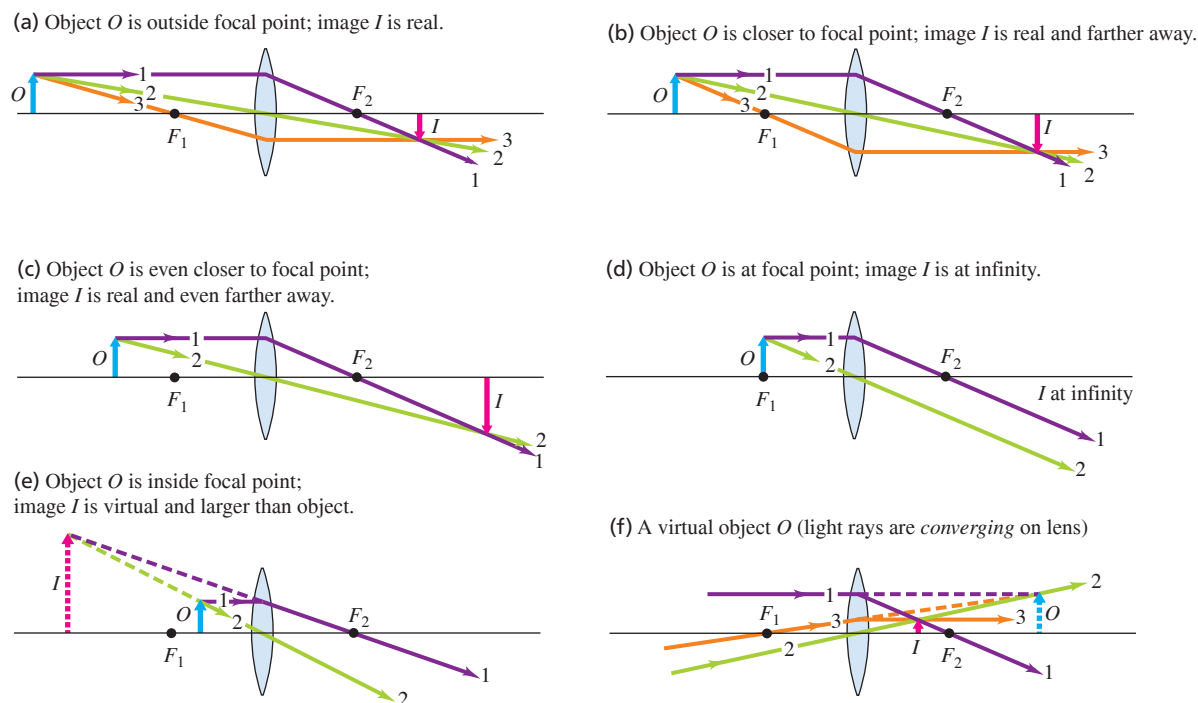
When the image is real, the position of the image point is determined by the intersection of any two rays 1, 2, and 3 (Fig. 34.36a). When the image is virtual, we extend the diverging outgoing rays backward to their intersection point to find the image point (Fig. 34.36b).

**CAUTION** **Principal rays are not the only rays** Keep in mind that *any* ray from the object that strikes the lens will pass through the image point (for a real image) or appear to originate from the image point (for a virtual image). (We made a similar comment about image formation by mirrors in Section 34.2.) We've emphasized the principal rays because they're the only ones you need to draw to locate the image. ■

**Figure 34.37** shows principal-ray diagrams for a converging lens for several object distances. We suggest you study each of these diagrams very carefully, comparing each numbered ray with the above description.

Parts (a), (b), and (c) of Fig. 34.37 help explain what happens in focusing a camera. For a photograph to be in sharp focus, the electronic sensor or film must be at the position of the real image made by the camera's lens. The image distance increases as the object is brought closer, so the sensor is moved farther behind the lens (i.e., the lens is moved farther in front of the sensor). In Fig. 34.37d the object is at the focal point; ray 3 can't be drawn because it doesn't pass through the lens. In Fig. 34.37e the object distance is less than the focal length. The outgoing rays are divergent, and the image is *virtual*; its position is located by extending the outgoing rays backward, so the image distance  $s'$  is negative. Note also that the image is erect and larger than the object. (We'll see the usefulness of this in Section 34.6.) Figure 34.37f corresponds to a *virtual object*. The incoming rays do not diverge from a real object, but are *converging* as though they would meet at the tip of the virtual object  $O$  on the right side; the object distance  $s$  is negative in this case. The image is real and is located between the lens and the second focal point. This situation can arise if the rays that strike the lens in Fig. 34.37f emerge from another converging lens (not shown) to the left of the figure.

Figure 34.37 Formation of images by a thin converging lens for various object distances. The principal rays are numbered. (Compare Fig. 34.20 for a concave spherical mirror.)



### PROBLEM-SOLVING STRATEGY 34.2 Image Formation by Thin Lenses

**IDENTIFY** *the relevant concepts:* Review Problem-Solving Strategy 34.1 (Section 34.2) for mirrors, which is equally applicable here. As for mirrors, you should use *both* principal-ray diagrams and equations to solve problems that involve image formation by lenses.

**SET UP** *the problem:* Identify the target variables.

**EXECUTE** *the solution* as follows:

1. Draw a large principal-ray diagram if you have enough information, using graph paper or quadrille-ruled paper. Orient your diagram so that incoming rays go from left to right. Draw the rays with a ruler, and measure distances carefully.
2. Draw the principal rays so they change direction at the midplane of the lens, as in Fig. 34.36. For a lens there are only three principal rays (compared to four for a mirror). Draw all three whenever

possible; the intersection of any two rays determines the image location, but the third ray should pass through the same point.

3. If the outgoing principal rays diverge, extend them backward to find the virtual image point on the *incoming* side of the lens, as in Fig. 34.27e.
4. Solve Eqs. (34.16) and (34.17), as appropriate, for the target variables. Carefully use the sign rules given in Section 34.1.
5. The *image* from a first lens or mirror may serve as the *object* for a second lens or mirror. In finding the object and image distances for this intermediate image, be sure you include the distance between the two elements (lenses and/or mirrors) correctly.

**EVALUATE** *your answer:* Your calculated results must be consistent with your ray-diagram results. Check that they give the same image position and image size, and that they agree on whether the image is real or virtual.

### EXAMPLE 34.9 Image position and magnification with a converging lens

#### WITH VARIATION PROBLEMS

Use ray diagrams to find the image position and magnification for an object at each of the following distances from a converging lens with a focal length of 20 cm: (a) 50 cm; (b) 20 cm; (c) 15 cm; (d) -40 cm. Check your results by calculating the image position and lateral magnification by using Eqs. (34.16) and (34.17), respectively.

**IDENTIFY and SET UP** We are given the focal length  $f = 20$  cm and four object distances  $s$ . Our target variables are the corresponding image distances  $s'$  and lateral magnifications  $m$ . We solve Eq. (34.16) for  $s'$ , and find  $m$  from Eq. (34.17),  $m = -s'/s$ .

**EXECUTE** Figures 34.37a, d, e, and f, respectively, show the appropriate principal-ray diagrams. You should be able to reproduce these without referring to the figures. Measuring these diagrams yields the approximate results:  $s' = 35$  cm,  $-\infty$ , -40 cm, and 15 cm, and  $m = -\frac{2}{3}$ ,  $+\infty$ ,  $+3$ , and  $+\frac{1}{3}$ , respectively.

Calculating the image distances from Eq. (34.16), we find

$$(a) \frac{1}{50 \text{ cm}} + \frac{1}{s'} = \frac{1}{20 \text{ cm}} \quad s' = 33.3 \text{ cm}$$

$$(b) \frac{1}{20 \text{ cm}} + \frac{1}{s'} = \frac{1}{20 \text{ cm}} \quad s' = \pm\infty$$

$$(c) \frac{1}{15 \text{ cm}} + \frac{1}{s'} = \frac{1}{20 \text{ cm}} \quad s' = -60 \text{ cm}$$

$$(d) \frac{1}{-40 \text{ cm}} + \frac{1}{s'} = \frac{1}{20 \text{ cm}} \quad s' = 13.3 \text{ cm}$$

The graphical results are fairly close to these except for part (c); the accuracy of the diagram in Fig. 34.37e is limited because the rays extended backward have nearly the same direction.

From Eq. (34.17),

$$(a) m = -\frac{33.3 \text{ cm}}{50 \text{ cm}} = -\frac{2}{3} \quad (b) m = -\frac{\pm\infty \text{ cm}}{20 \text{ cm}} = \pm\infty$$

$$(c) m = -\frac{-60 \text{ cm}}{15 \text{ cm}} = +4 \quad (d) m = -\frac{13.3 \text{ cm}}{-40 \text{ cm}} = +\frac{1}{3}$$

**EVALUATE** Note that the image distance  $s'$  is positive in parts (a) and (d) but negative in part (c). This makes sense: The image is real in parts (a) and (d) but virtual in part (c). The light rays that emerge from the lens in part (b) are parallel and never converge, so the image can be regarded as being at either  $+\infty$  or  $-\infty$ .

The values of magnification  $m$  tell us that the image is inverted in part (a) and erect in parts (c) and (d), in agreement with the principal-ray diagrams. The infinite value of magnification in part (b) is another way of saying that the image is formed infinitely far away.

**KEYCONCEPT** If an object is *outside* the focal point of a converging lens, the outgoing rays converge to a real, inverted image. If an object is *at* the focal point, the outgoing rays are parallel and form an image at infinity. If an object is *inside* the focal point, the outgoing rays appear to be coming from in front of the lens and so form a virtual, erect image.

### EXAMPLE 34.10 Image formation by a diverging lens

#### WITH VARIATION PROBLEMS

A beam of parallel rays spreads out after passing through a thin diverging lens, as if the rays all came from a point 20.0 cm from the center of the lens. You want to use this lens to form an erect, virtual image that is  $\frac{1}{3}$  the height of the object. (a) Where should the object be placed? Where will the image be? (b) Draw a principal-ray diagram.

**IDENTIFY and SET UP** The result with parallel rays shows that the focal length is  $f = -20$  cm. We want the lateral magnification to be  $m = +\frac{1}{3}$  (positive because the image is to be erect). Our target variables are the object distance  $s$  and the image distance  $s'$ . In part (a), we solve the magnification equation, Eq. (34.17), for  $s'$  in terms of  $s$ ; we then use the object-image relationship, Eq. (34.16), to find  $s$  and  $s'$  individually.

*Continued*

**EXECUTE** (a) From Eq. (34.17),  $m = +\frac{1}{3} = -s'/s$ , so  $s' = -s/3$ . We insert this result into Eq. (34.16) and solve for the object distance  $s$ :

$$\frac{1}{s} + \frac{1}{-s/3} = \frac{1}{s} - \frac{3}{s} = -\frac{2}{s} = \frac{1}{f}$$

$$s = -2f = -2(-20.0 \text{ cm}) = 40.0 \text{ cm}$$

The object should be 40.0 cm from the lens. The image distance will be

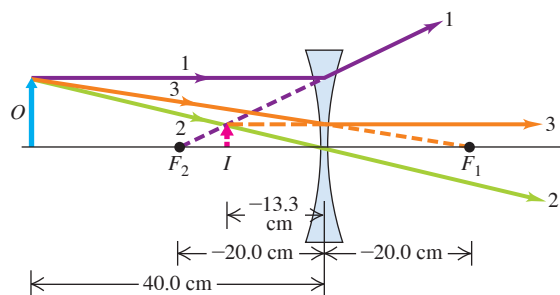
$$s' = -\frac{s}{3} = -\frac{40.0 \text{ cm}}{3} = -13.3 \text{ cm}$$

The image distance is negative, so the object and image are on the same side of the lens.

(b) **Figure 34.38** is a principal-ray diagram for this problem, with the rays numbered as in Fig. 34.36b.

**EVALUATE** You should be able to draw a principal-ray diagram like Fig. 34.38 without referring to the figure. From your diagram, you can confirm our results in part (a) for the object and image distances. You can also check our results for  $s$  and  $s'$  by substituting them back into Eq. (34.16).

**Figure 34.38** Principal-ray diagram for an image formed by a thin diverging lens.



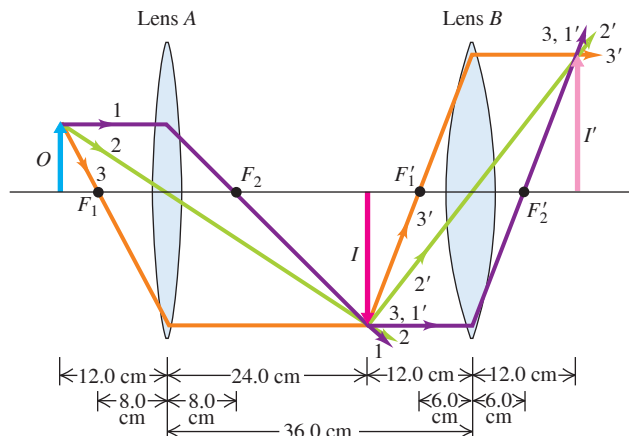
**KEYCONCEPT** No matter where an object is placed in front of a diverging lens, the outgoing rays appear to be coming from in front of the lens and so form a virtual, erect image. This image is always smaller than the object.

### EXAMPLE 34.11 An image of an image

Converging lenses *A* and *B*, of focal lengths 8.0 cm and 6.0 cm, respectively, are placed 36.0 cm apart. Both lenses have the same optic axis. An object 8.0 cm high is placed 12.0 cm to the left of lens *A*. Find the position, size, and orientation of the image produced by the lenses in combination. (Such combinations are used in telescopes and microscopes, which we'll discuss in Section 34.7.)

**IDENTIFY and SET UP** **Figure 34.39** shows the situation. The object *O* lies outside the first focal point  $F_1$  of lens *A*, which therefore produces a real image *I*. The light rays that strike lens *B* diverge from this real image just as if *I* was a material object; image *I* therefore acts as an *object* for lens *B*. Our goal is to determine the properties of the image *I'* made by lens *B*. We use both ray-diagram and computational methods to do this.

**Figure 34.39** Principal-ray diagram for a combination of two converging lenses. The first lens (*A*) makes a real image of the object. This real image acts as an object for the second lens (*B*).



**EXECUTE** In Fig. 34.39 we have drawn principal rays 1, 2, and 3 from the head of the object arrow *O* to find the position of the image *I* made by lens *A*, and principal rays 1', 2', and 3' from the head of *I* to find the position of the image *I'* made by lens *B* (even though rays 2' and 3' don't actually exist in this case). The image is inverted *twice*, once by each lens, so the second image *I'* has the same orientation as the original object.

We first find the position and size of the first image *I*. Applying Eq. (34.16),  $1/s + 1/s' = 1/f$ , to lens *A* gives

$$\frac{1}{12.0 \text{ cm}} + \frac{1}{s'_{I,A}} = \frac{1}{8.0 \text{ cm}} \quad s'_{I,A} = +24.0 \text{ cm}$$

Image *I* is 24.0 cm to the right of lens *A*. The lateral magnification is  $m_A = -(24.0 \text{ cm})/(12.0 \text{ cm}) = -2.00$ , so image *I* is inverted and twice as tall as object *O*.

Image *I* is  $36.0 \text{ cm} - 24.0 \text{ cm} = 12.0 \text{ cm}$  to the left of lens *B*, so the object distance for lens *B* is  $+12.0 \text{ cm}$ . Applying Eq. (34.16) to lens *B* then gives

$$\frac{1}{12.0 \text{ cm}} + \frac{1}{s'_{I',B}} = \frac{1}{6.0 \text{ cm}} \quad s'_{I',B} = +12.0 \text{ cm}$$

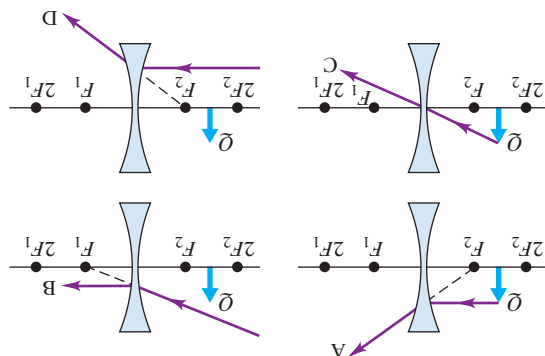
The final image *I'* is 12.0 cm to the right of lens *B*. The magnification produced by lens *B* is  $m_B = -(12.0 \text{ cm})/(12.0 \text{ cm}) = -1.00$ .

**EVALUATE** The value of  $m_B$  means that the final image *I'* is just as large as the first image *I* but has the opposite orientation. The *overall* magnification is  $m_A m_B = (-2.00)(-1.00) = +2.00$ . Hence the final image *I'* is  $(2.00)(8.0 \text{ cm}) = 16 \text{ cm}$  tall and has the same orientation as the original object *O*, just as Fig. 34.39 shows.

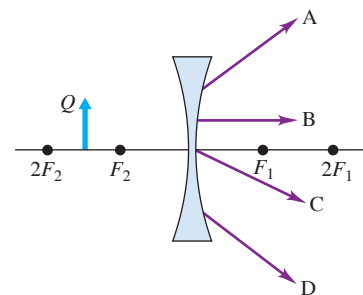
**KEYCONCEPT** To analyze an optical system that has more than one lens, first find the position of the image made by the first lens. This first image then acts as an object for the second lens, which in turn makes an image of this object.

**TEST YOUR UNDERSTANDING OF SECTION 34.4** A diverging lens and an object are positioned as shown in the figure at right. Which of the rays A, B, C, and D could emanate from point  $Q$  at the top of the object?

**ANSWER**



**I A and C** When rays A and D are extended backward, they pass through focal point  $F_2$ ; thus, before they passed through the lens, they were parallel to the optic axis. The figures show that ray A emanated from point  $Q$ , but ray D did not. Ray B is parallel to the optic axis, so before it passed through the lens, it was directed toward focal point  $F_1$ . Hence it cannot have come from point  $Q$ . Ray C passes through the center of the lens and hence is not deflected by its passage; tracing the ray backward shows that it emanates from point  $Q$ .



## 34.5 CAMERAS

The concept of *image*, which is so central to understanding simple mirror and lens systems, plays an equally important role in the analysis of optical instruments (also called *optical devices*). Among the most common optical devices are cameras, which make an image of an object and record it either electronically or on film.

The basic elements of a **camera** are a light-tight box (“camera” is a Latin word meaning “a room or enclosure”), a converging lens, a shutter to open the lens for a prescribed length of time, and a light-sensitive recording medium (**Fig. 34.40**). In digital cameras (including mobile-phone cameras), this is an electronic sensor; in older cameras, it is photographic film. The lens forms an inverted real image on the recording medium of the object being photographed. High-quality camera lenses have several elements, permitting partial correction of various *aberrations*, including the dependence of index of refraction on wavelength and the limitations imposed by the paraxial approximation.

When the camera is in proper *focus*, the position of the recording medium coincides with the position of the real image formed by the lens. The resulting photograph will then be as sharp as possible. With a converging lens, the image distance increases as the

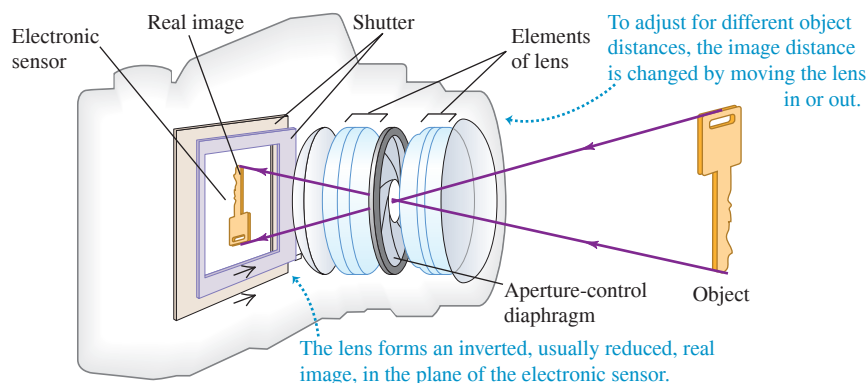


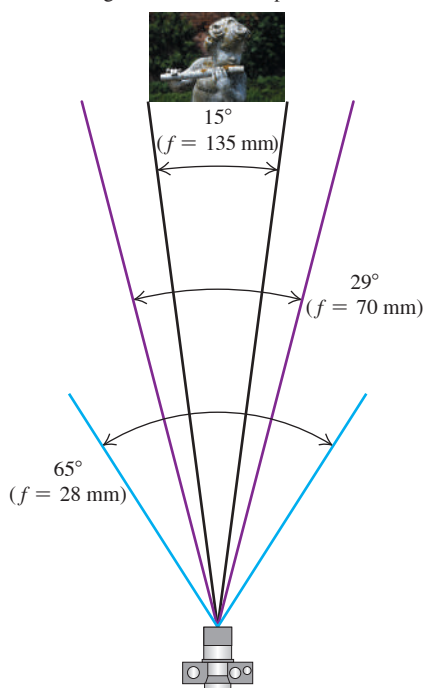
Figure 34.40 Key elements of a digital camera.



Figure 34.41 (a), (b), (c) Three photographs taken with the same camera from the same position, using lenses with focal lengths  $f = 28$  mm, 70 mm, and 135 mm. Increasing the focal length increases the image size proportionately. (d) The larger the value of  $f$ , the narrower the angle of view. The angles shown here are for a camera with image area  $24$  mm  $\times$   $36$  mm (corresponding to 35 mm film) and refer to the angle of view along the 36 mm width of the film.

(a)  $f = 28$  mm(b)  $f = 70$  mm(c)  $f = 135$  mm

(d) The angles of view for the photos in (a)–(c)



object distance decreases (see Figs. 34.41a, 34.41b, and 34.41c, and the discussion in Section 34.4). Hence in “focusing” the camera, we move the lens closer to the sensor or film for a distant object and farther from the sensor or film for a nearby object.

### Camera Lenses: Focal Length

The choice of the focal length  $f$  for a camera lens depends on the size of the electronic sensor or film and the desired angle of view. Figure 34.41 shows three photographs taken on 35 mm film with the same camera at the same position, but with lenses of different focal lengths. A lens of long focal length, called a *telephoto* lens, gives a narrow angle of view and a large image of a distant object (such as the statue in Fig. 34.41c); a lens of short focal length gives a small image and a wide angle of view (as in Fig. 34.41a) and is called a *wide-angle* lens. To understand this behavior, recall that the focal length is the distance from the lens to the image when the object is infinitely far away. In general, for *any* object distance, using a lens of longer focal length gives a greater image distance. This also increases the height of the image; as discussed in Section 34.4, the ratio of the image height  $y'$  to the object height  $y$  (the *lateral magnification*) is equal in absolute value to the ratio of the image distance  $s'$  to the object distance  $s$  [Eq. (34.17)]:

$$m = \frac{y'}{y} = -\frac{s'}{s}$$

With a lens of short focal length, the ratio  $s'/s$  is small, and a distant object gives only a small image. When a lens with a long focal length is used, the image of this same object may entirely cover the area of the electronic sensor or film. Hence the longer the focal length, the narrower the angle of view (Fig. 34.41d).

### Camera Lenses: $f$ -Number

For a camera to record the image properly, the total light energy per unit area reaching the electronic sensor or film (the “exposure”) must fall within certain limits. This is controlled by the *shutter* and the *lens aperture*. The shutter controls the time interval during which light enters the lens. This is usually adjustable in steps corresponding to factors of about 2, often from 1 s to  $\frac{1}{1000}$  s.

The intensity of light reaching the sensor or film is proportional to the area viewed by the camera lens and to the effective area of the lens. The size of the area that the lens “sees” is proportional to the square of the angle of view of the lens, and so is roughly proportional to  $1/f^2$ . The effective area of the lens is controlled by means of an adjustable lens aperture, or *diaphragm*, a nearly circular hole with variable diameter  $D$ ; hence the effective area is proportional to  $D^2$ . Putting these factors together, we see that the intensity of light reaching the sensor or film with a particular lens is proportional to  $D^2/f^2$ .

Photographers commonly express the light-gathering capability of a lens in terms of the ratio  $f/D$ , called the ***f*-number** of the lens:

$$f\text{-number of a lens} = \frac{f}{D} \quad (34.20)$$

$\leftarrow$  Focal length of lens  
 $\leftarrow$  Aperture diameter

For example, a lens with a focal length  $f = 50$  mm and an aperture diameter  $D = 25$  mm has an  $f$ -number of 2, or “an aperture of  $f/2$ .” The light intensity reaching the sensor or film is *inversely* proportional to the square of the  $f$ -number.

For a lens with a variable-diameter aperture, increasing the diameter by a factor of  $\sqrt{2}$  changes the  $f$ -number by  $1/\sqrt{2}$  and increases the intensity at the sensor or film by a factor of 2. Adjustable apertures usually have scales labeled with successive numbers (often called *f*-stops) related by factors of  $\sqrt{2}$ , such as

$$f/2 \quad f/2.8 \quad f/4 \quad f/5.6 \quad f/8 \quad f/11 \quad f/16$$

and so on. The larger numbers represent smaller apertures and exposures, and each step corresponds to a factor of 2 in intensity (**Fig. 34.42**). The actual *exposure* (total amount of light reaching the sensor or film) is proportional to both the aperture area and the time of exposure. Thus  $f/4$  and  $\frac{1}{500}$  s,  $f/5.6$  and  $\frac{1}{250}$  s, and  $f/8$  and  $\frac{1}{125}$  s all correspond to the same exposure.

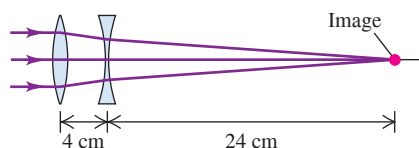
## Zoom Lenses and Projectors

Many photographers use a *zoom lens*, which is not a single lens but a complex collection of several lens elements that give a continuously variable focal length, often over a range as great as 15 to 1. **Figures 34.43a** and 34.43b show a simple system with variable focal length, and Fig. 34.43c shows a typical zoom lens for a digital single-lens reflex camera. Zoom lenses give a range of image sizes of a given object. It is an enormously complex problem in optical design to keep the image in focus and maintain a constant  $f$ -number while the focal length changes. When you vary the focal length of a typical zoom lens, two groups of elements move within the lens and a diaphragm opens and closes.

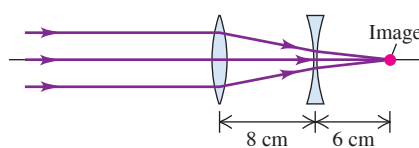
A *digital projector* for viewing lecture slides, photos, or movies operates very much like a digital camera in reverse. In the most common type of digital projector, the pixels of data to be projected are shown on a small, transparent liquid-crystal-display (LCD) screen inside the projector and behind the projection lens. A lamp illuminates the LCD screen, which acts as an object for the lens. The lens forms a real, enlarged, inverted image of the LCD screen. Because the image is inverted, the pixels shown on the LCD screen are upside down so that the image on the projection screen appears right-side up.

**Figure 34.43** A simple zoom lens uses a converging lens and a diverging lens in tandem. (a) When the two lenses are close together, the combination behaves like a single lens of long focal length. (b) If the two lenses are moved farther apart, the combination behaves like a short-focal-length lens. (c) This zoom lens contains twelve elements arranged in four groups.

(a) Zoom lens set for long focal length



(b) Zoom lens set for short focal length



(c) A practical zoom lens



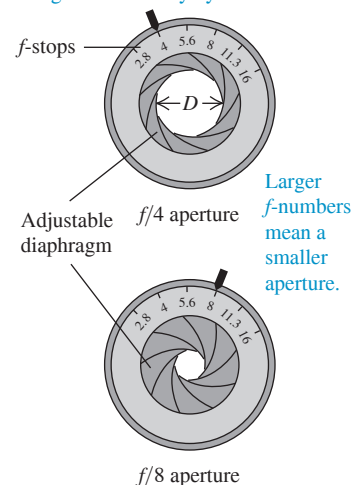
## APPLICATION Inverting an Inverted Image

A camera lens makes an inverted image on the camera's light-sensitive electronic detector. The internal software of the camera then inverts the image again so it appears the right way around on the camera's display. A similar thing happens with your vision: The image formed on the retina of your eye is inverted, but your brain's “software” erects the image so you see the world right-side up.



**Figure 34.42** A camera lens with an adjustable diaphragm.

Changing the diameter by a factor of  $\sqrt{2}$  changes the intensity by a factor of 2.



**EXAMPLE 34.12** Photographic exposures

A common telephoto lens for a 35 mm film camera has a focal length of 200 mm; its  $f$ -stops range from  $f/2.8$  to  $f/22$ . (a) What is the corresponding range of aperture diameters? (b) What is the corresponding range of image intensities on the film?

**IDENTIFY and SET UP** Part (a) of this problem uses the relationship among lens focal length  $f$ , aperture diameter  $D$ , and  $f$ -number. Part (b) uses the relationship between intensity and aperture diameter. We use Eq. (34.20) to relate  $D$  (the target variable) to the  $f$ -number and the focal length  $f = 200$  mm. The intensity of the light reaching the film is proportional to  $D^2/f^2$ ; since  $f$  is the same in each case, we conclude that the intensity in this case is proportional to  $D^2$ , the square of the aperture diameter.

**EXECUTE** (a) From Eq. (34.20), the diameter ranges from

$$D = \frac{f}{f\text{-number}} = \frac{200 \text{ mm}}{2.8} = 71 \text{ mm}$$

to

$$D = \frac{200 \text{ mm}}{22} = 9.1 \text{ mm}$$

(b) Because the intensity is proportional to  $D^2$ , the ratio of the intensity at  $f/2.8$  to the intensity at  $f/22$  is

$$\left(\frac{71 \text{ mm}}{9.1 \text{ mm}}\right)^2 = \left(\frac{22}{2.8}\right)^2 = 62 \quad (\text{about } 2^6)$$

**EVALUATE** If the correct exposure time at  $f/2.8$  is  $\frac{1}{1000}$  s, then the exposure at  $f/22$  is  $(62)\left(\frac{1}{1000} \text{ s}\right) = \frac{1}{16}$  s to compensate for the lower intensity. In general, the smaller the aperture and the larger the  $f$ -number, the longer the required exposure. Nevertheless, many photographers prefer to use small apertures so that only the central part of the lens is used to make the image. This minimizes aberrations that occur near the edges of the lens and gives the sharpest possible image.

**KEYCONCEPT** The  $f$ -number of a photographic lens is the ratio of the focal length to the aperture diameter. The larger the  $f$ -number, the lower the intensity of the image produced by the lens and the longer the exposure time needed to record the image.

**TEST YOUR UNDERSTANDING OF SECTION 34.5** When used with 35 mm film (image area  $24 \text{ mm} \times 36 \text{ mm}$ ), a lens with  $f = 50 \text{ mm}$  gives a  $45^\circ$  angle of view and is called a “normal lens.” When used with an electronic sensor that measures  $5 \text{ mm} \times 5 \text{ mm}$ , this same lens is (i) a wide-angle lens; (ii) a normal lens; (iii) a telephoto lens.

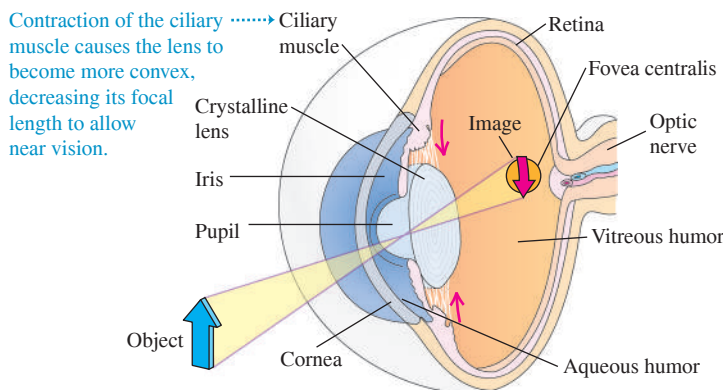
**ANSWER** (iii) The smaller image area of the electronic sensor means that the angle of view is decreased for a given focal length. Individual objects make images of the same size in either case; when a smaller light-sensitive area is used, fewer images fit into the area and the field of view is narrower.

**34.6 THE EYE**

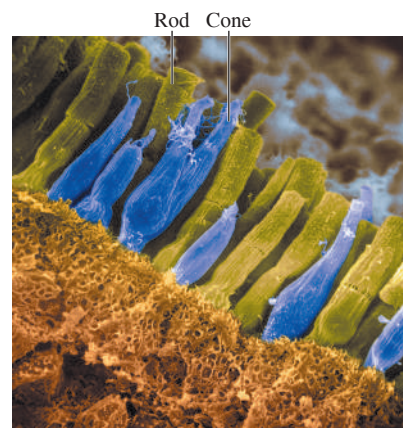
The optical behavior of the eye is similar to that of a camera. **Figure 34.44** shows the essential parts of the human eye, considered as an optical system. The eye is nearly spherical and about 2.5 cm in diameter. The front portion is somewhat more sharply curved and is covered by a tough, transparent membrane called the *cornea*. The region behind

**Figure 34.44** (a) The eye. (b) There are two types of light-sensitive cells on the retina. Rods are more sensitive to light than cones, but only the cones are sensitive to differences in color. A typical human eye contains about  $1.3 \times 10^8$  rods and about  $7 \times 10^6$  cones.

(a) Diagram of the eye



(b) Scanning electron micrograph showing retinal rods and cones in different colors





the cornea contains a liquid called the *aqueous humor*. Next comes the *crystalline lens*, a capsule containing a fibrous jelly, hard at the center and progressively softer at the outer portions. The crystalline lens is held in place by ligaments that attach it to the ciliary muscle, which encircles it. Behind the lens, the eye is filled with a thin watery jelly called the *vitreous humor*. The indexes of refraction of both the aqueous humor and the vitreous humor are about 1.336, nearly equal to that of water. The crystalline lens, while not homogeneous, has an average index of 1.437. This is not very different from the indexes of the aqueous and vitreous humors. As a result, most of the refraction of light entering the eye occurs at the outer surface of the cornea.

Refraction at the cornea and the surfaces of the lens produces a *real image* of the object being viewed. This image is formed on the light-sensitive *retina*, lining the rear inner surface of the eye. The retina plays the same role as the electronic sensor in a digital camera. The *rods* and *cones* in the retina act like an array of miniature photocells (Fig. 34.44b); they sense the image and transmit it via the *optic nerve* to the brain. Vision is most acute in a small central region called the *fovea centralis*, about 0.25 mm in diameter.

In front of the lens is the *iris*. It contains an aperture with variable diameter called the *pupil*, which opens and closes to adapt to changing light intensity. The receptors of the retina also have intensity adaptation mechanisms.

For an object to be seen sharply, the image must be formed exactly at the location of the retina. The eye adjusts to different object distances  $s$  by changing the focal length  $f$  of its lens; the lens-to-retina distance, corresponding to  $s'$ , does not change. (Contrast this with focusing a camera, in which the focal length is fixed and the lens-to-sensor distance is changed.) For the normal eye, an object at infinity is sharply focused when the ciliary muscle is relaxed. To focus sharply on a closer object, the tension in the ciliary muscle surrounding the lens increases, the ciliary muscle contracts, the lens bulges, and the radii of curvature of its surfaces decrease; this decreases  $f$ . This process is called *accommodation*.

The extremes of the range over which distinct vision is possible are known as the *far point* and the *near point* of the eye. The far point of a normal eye is at infinity. The position of the near point depends on the amount by which the ciliary muscle can increase the curvature of the crystalline lens. The range of accommodation gradually diminishes with age because the crystalline lens becomes less flexible throughout a person's life and the ciliary muscles are less able to distort the lens. For this reason, the near point gradually recedes as one grows older. This recession of the near point is called *presbyopia*. **Table 34.1** shows the approximate position of the near point for an average person at various ages. For example, an average person 50 years of age cannot focus on an object that is closer than about 40 cm.

## Defects of Vision

Several common defects of vision result from incorrect distance relationships in the eye. A normal eye forms an image on the retina of an object at infinity when the eye is relaxed (**Fig. 34.45a**). In the *myopic* (nearsighted) eye, the eyeball is too long from front to back in comparison with the radius of curvature of the cornea (or the cornea is too sharply curved), and rays from an object at infinity are focused in front of the retina (**Fig. 34.45b**). The most distant object for which an image can be formed on the retina is then nearer than infinity. In the *hyperopic* (farsighted) eye, the eyeball is too short or the cornea is not curved enough, and the image of an infinitely distant object is behind the retina (**Fig. 34.45c**). The myopic eye produces *too much* convergence in a parallel bundle of rays for an image to be formed on the retina; the hyperopic eye, *not enough* convergence.

All of these defects can be corrected by the use of corrective lenses (eyeglasses or contact lenses). The near point of either a presbyopic or a hyperopic eye is *farther* from the eye than normal. To see clearly an object at normal reading distance (often assumed to be 25 cm), we need a lens that forms a virtual image of the object at or beyond the near point. This can be accomplished by a converging (positive) lens (**Fig. 34.46**, next page). In effect the lens moves the object farther away from the eye to a point where a sharp retinal image

## BIO APPLICATION Focusing in the Animal Kingdom

The crystalline lens and ciliary muscle found in humans and other mammals are among a number of focusing mechanisms used by animals. Birds can change the shape not only of their lens but also of the corneal surface. In aquatic animals the corneal surface is not very useful for focusing because its index of refraction is close to that of water. Thus, focusing is accomplished entirely by the lens, which is nearly spherical. Fish focus by using a muscle to move the lens either inward or outward. Whales and dolphins achieve the same effect by filling or emptying a fluid chamber behind the lens to move the lens in or out.

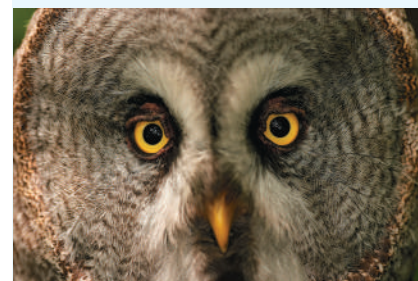
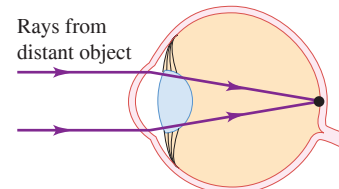
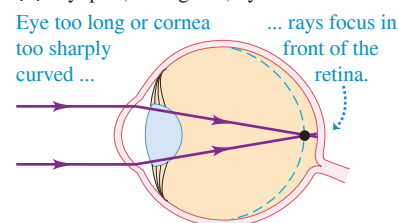


Figure 34.45 Refractive errors for (a) a normal eye, (b) a myopic (nearsighted) eye, and (c) a hyperopic (farsighted) eye viewing a very distant object. In each case, the eye is shown with the ciliary muscle relaxed. The dashed blue curve indicates the required position of the retina.

(a) Normal eye



(b) Myopic (nearsighted) eye



(c) Hyperopic (farsighted) eye

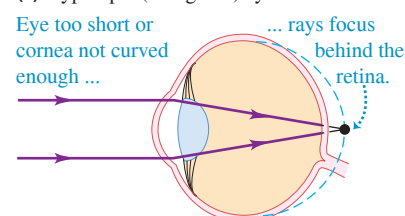




Figure 34.46 (a) An uncorrected hyperopic (farsighted) eye. (b) A positive (converging) lens gives the extra convergence needed for a hyperopic eye to focus the image on the retina.

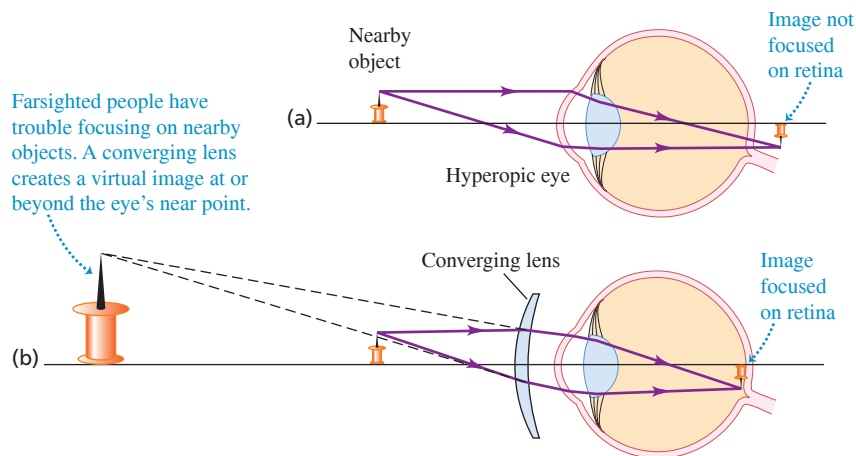


Figure 34.47 (a) An uncorrected myopic (nearsighted) eye. (b) A negative (diverging) lens spreads the rays farther apart to compensate for the excessive convergence of the myopic eye.

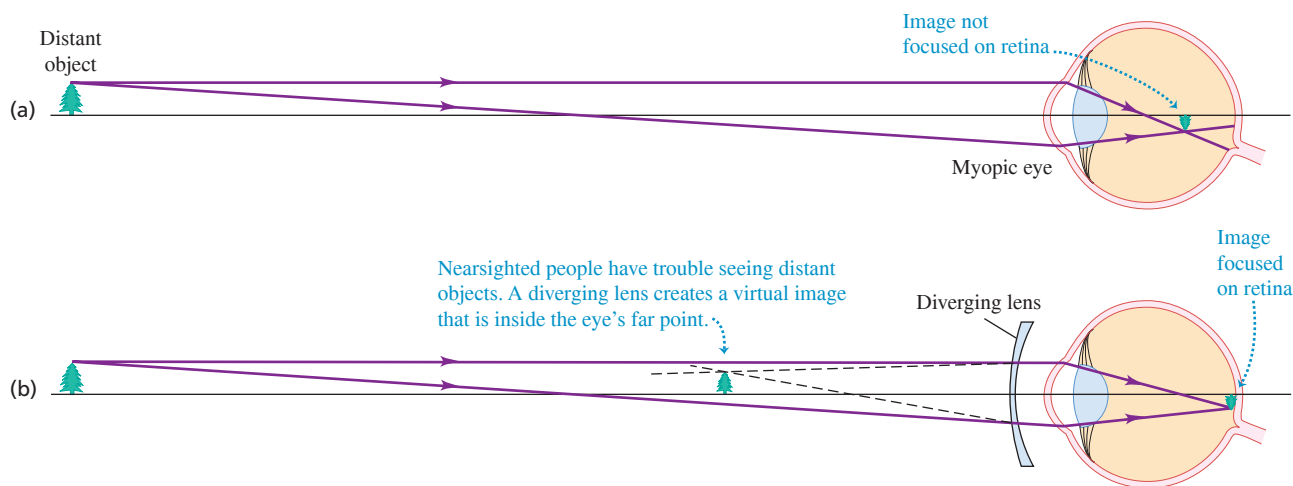


TABLE 34.1 Receding of Near Point with Age

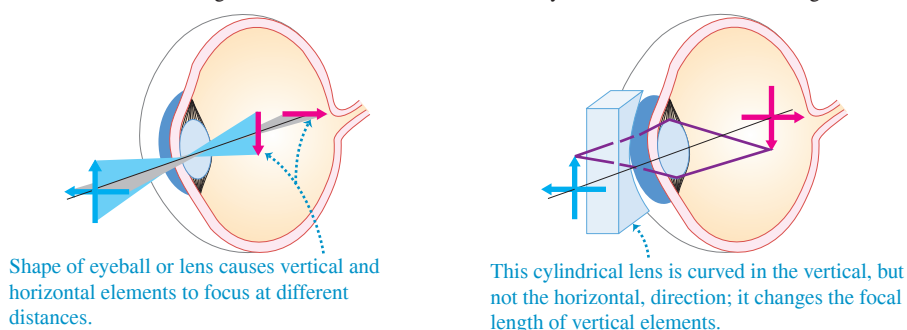
| Age (years) | Near Point (cm) |
|-------------|-----------------|
| 10          | 7               |
| 20          | 10              |
| 30          | 14              |
| 40          | 22              |
| 50          | 40              |
| 60          | 200             |

can form. Similarly, correcting the myopic eye involves the use of a diverging (negative) lens to move the image closer to the eye than the actual object is (Fig. 34.47).

*Astigmatism* is a different type of defect in which the surface of the cornea is not spherical but rather more sharply curved in one plane than in another. As a result, horizontal lines may be imaged in a different plane from vertical lines (Fig. 34.48a). Astigmatism may make it impossible, for example, to focus clearly on both the horizontal and vertical bars of a window at the same time.

Figure 34.48 One type of astigmatism and how it is corrected.

(a) Vertical lines are imaged in front of the retina. (b) A cylindrical lens corrects for astigmatism.



Astigmatism can be corrected by use of a lens with a *cylindrical* surface. For example, suppose the curvature of the cornea in a horizontal plane is correct to focus rays from infinity on the retina but the curvature in the vertical plane is too great to form a sharp retinal image. When a cylindrical lens with its axis horizontal is placed before the eye, the rays in a horizontal plane are unaffected, but the additional divergence of the rays in a vertical plane causes these to be sharply imaged on the retina (Fig. 34.48b).

Lenses for vision correction are usually described in terms of the **power**, defined as the reciprocal of the focal length expressed in meters. The unit of power is the **diopter**. Thus a lens with  $f = 0.50$  m has a power of 2.0 diopters,  $f = -0.25$  m corresponds to  $-4.0$  diopters, and so on. The numbers on a prescription for glasses are usually powers expressed in diopters. When the correction involves both astigmatism and myopia or hyperopia, there are three numbers: one for the spherical power, one for the cylindrical power, and an angle to describe the orientation of the cylinder axis.

### EXAMPLE 34.13 Correcting for farsightedness

The near point of a certain hyperopic eye is 100 cm in front of the eye. Find the focal length and power of the contact lens that will permit the wearer to see clearly an object that is 25 cm in front of the eye.

**IDENTIFY and SET UP** Figure 34.49 shows the situation. We want the lens to form a virtual image of the object at the near point of the eye, 100 cm from it. The contact lens (which we treat as having negligible thickness) is at the surface of the cornea, so the object distance is  $s = 25$  cm. The virtual image is on the incoming side of the contact lens, so the image distance is  $s' = -100$  cm. We use Eq. (34.16) to determine the required focal length  $f$  of the contact lens; the corresponding power is  $1/f$ .

**EXECUTE** From Eq. (34.16),

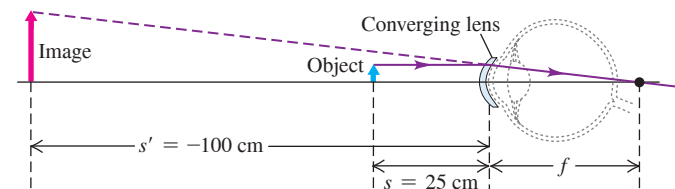
$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'} = \frac{1}{+25 \text{ cm}} + \frac{1}{-100 \text{ cm}}$$

$$f = +33 \text{ cm}$$

We need a converging lens with focal length  $f = 33$  cm and power  $1/(0.33 \text{ m}) = +3.0$  diopters.

**EVALUATE** In this example we used a contact lens to correct hyperopia. Had we used eyeglasses, we would have had to account for the separation between the eye and the eyeglass lens, and a somewhat different power would have been required (see Example 34.14).

Figure 34.49 Using a contact lens to correct for farsightedness. For clarity, the eye and contact lens are shown much larger than the scale of the figure; the 2.5 cm diameter of the eye is actually much smaller than the focal length  $f$  of the contact lens.



**KEYCONCEPT** In a hyperopic (farsighted) eye, the cornea and lens cause too little convergence of light rays. Distant objects can form sharp images on the retina, but nearby objects cannot. Hyperopia can be corrected by placing a converging lens in front of the eye; the virtual, erect image made by this lens acts as a more distant object on which the eye can focus.

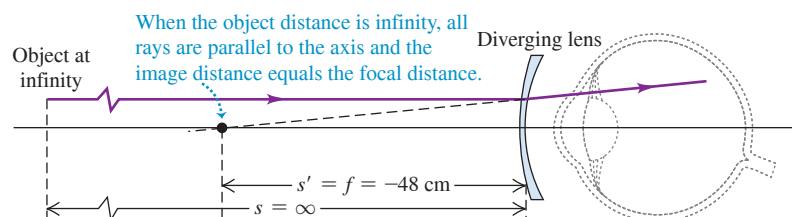
### EXAMPLE 34.14 Correcting for nearsightedness

The far point of a certain myopic eye is 50 cm in front of the eye. Find the focal length and power of the eyeglass lens that will permit the wearer to see clearly an object at infinity. Assume that the lens is worn 2 cm in front of the eye.

**IDENTIFY and SET UP** Figure 34.50 shows the situation. The far point of a myopic eye is nearer than infinity. To see clearly objects beyond the

far point, we need a lens that forms a virtual image of such objects no farther from the eye than the far point. Assume that the virtual image of the object at infinity is formed at the far point, 50 cm in front of the eye (48 cm in front of the eyeglass lens). Then when the object distance is  $s = \infty$ , we want the image distance to be  $s' = -48$  cm. As in Example 34.13, we use the values of  $s$  and  $s'$  to calculate the required focal length.

Figure 34.50 Using an eyeglass lens to correct for nearsightedness. For clarity, the eye and eyeglass lens are shown much larger than the scale of the figure.



Continued

**EXECUTE** From Eq. (34.16),

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'} = \frac{1}{\infty} + \frac{1}{-48 \text{ cm}}$$

$$f = -48 \text{ cm}$$

We need a *diverging* lens with focal length  $f = -48 \text{ cm}$  and power  $1/(-0.48 \text{ m}) = -2.1$  diopters.

**EVALUATE** If a *contact* lens were used to correct this myopia, we would need  $f = -50 \text{ cm}$  and a power of  $-2.0$  diopters. Can you see why?

**KEYCONCEPT** In a myopic (nearsighted) eye, the cornea and lens cause too much convergence of light rays. Nearby objects can form sharp images on the retina, but distant objects cannot. Myopia can be corrected by placing a diverging lens in front of the eye; the virtual, erect image made by this lens acts as a closer object on which the eye can focus.

### BIO APPLICATION The Telephoto Eyes of Chameleons

The crystalline lens of a human eye can change shape but is always a converging (positive) lens. The lens in the eye of a chameleon lizard (family Chamaeleonidae) is different: It can change shape to be either a converging or a *diverging* (negative) lens. When it acts as a diverging lens just behind the cornea (which acts as a converging lens), the combination is like the long-focal-length zoom lens shown in Fig. 34.43a. This “telephoto vision” gives the chameleon a sharp view of potential insect prey.



**TEST YOUR UNDERSTANDING OF SECTION 34.6** A certain eyeglass lens is thin at its center, even thinner at its top and bottom edges, and relatively thick at its left and right edges. What defects of vision is this lens intended to correct? (i) Hyperopia for objects oriented both vertically and horizontally; (ii) myopia for objects oriented both vertically and horizontally; (iii) hyperopia for objects oriented vertically and myopia for objects oriented horizontally; (iv) hyperopia for objects oriented horizontally and myopia for objects oriented vertically.

### ANSWER

(iii) This lens is designed to correct for a type of astigmatism. Along the vertical axis, the lens is configured as a converging lens; along the horizontal axis, the lens is configured as a diverging lens. Hence the eye is hyperopic (see Fig. 34.46) for objects that are oriented vertically but myopic for objects that are oriented horizontally (see Fig. 34.47). Without correction, the eye focuses vertical objects behind the retina but horizontal objects in front of the retina.

## 34.7 THE MAGNIFIER

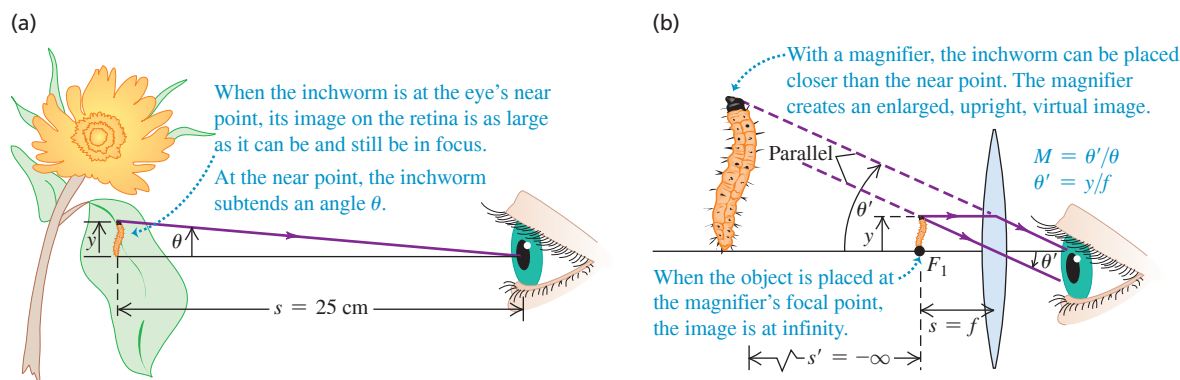
The apparent size of an object is determined by the size of its image on the retina. If the eye is unaided, this size depends on the *angle*  $\theta$  subtended by the object at the eye, called its **angular size** (Fig. 34.51a).

To look closely at a small object such as an insect or a crystal, you bring it close to your eye, making the subtended angle and the retinal image as large as possible. But your eye cannot focus sharply on objects that are closer than the near point, so an object subtends the largest possible viewing angle when it is placed at the near point. In the following discussion we'll assume an average viewer for whom the near point is  $25 \text{ cm}$  from the eye.

A converging lens can be used to form a virtual image that is larger and farther **?** from the eye than the object itself, as shown in Fig. 34.51b. Then the object can be moved closer to the eye, and the angular size of the image may be substantially larger than the angular size of the object at  $25 \text{ cm}$  without the lens. A lens used in this way is called a **magnifier**, otherwise known as a *magnifying glass* or a *simple magnifier*. The virtual image is most comfortable to view when it is placed at infinity, so that the ciliary muscle of the eye is relaxed; this means that the object is placed at the focal point  $F_1$  of the magnifier. In the following discussion we assume that this is done.

In Fig. 34.51a the object is at the near point, where it subtends an angle  $\theta$  at the eye. In Fig. 34.51b a magnifier in front of the eye forms an image at infinity, and the angle

Figure 34.51 (a) The angular size  $\theta$  is largest when the object is at the near point. (b) The magnifier gives a virtual image at infinity. This virtual image appears to the eye to be a real object subtending a larger angle  $\theta'$  at the eye.



subtended at the magnifier is  $\theta'$ . The usefulness of the magnifier is given by the ratio of the angle  $\theta'$  (with the magnifier) to the angle  $\theta$  (without the magnifier). This ratio is called the **angular magnification**  $M$ :

$$M = \frac{\theta'}{\theta} \quad (\text{angular magnification}) \quad (34.21)$$

**CAUTION Angular magnification vs. lateral magnification** Don't confuse *angular* magnification  $M$  with *lateral* magnification  $m$ . Angular magnification is the ratio of the *angular* size of an image to the angular size of the corresponding object; lateral magnification refers to the ratio of the *height* of an image to the height of the corresponding object. For the situation shown in Fig. 34.51b, the angular magnification is about  $3\times$ , since the inchworm subtends an angle about three times larger than that in Fig. 34.51a; hence the inchworm will look about three times larger to the eye. The *lateral* magnification  $m = -s'/s$  in Fig. 34.51b is *infinite* because the virtual image is at infinity, but that doesn't mean that the inchworm looks infinitely large through the magnifier! When you deal with a magnifier,  $M$  is useful but  $m$  is not. **|**

To find the value of  $M$ , we first assume that the angles are small enough that each angle (in radians) is equal to its sine and its tangent. Using Fig. 34.51a and drawing the ray in Fig. 34.51b that passes undeviated through the center of the lens, we find that  $\theta$  and  $\theta'$  (in radians) are

$$\theta = \frac{y}{25 \text{ cm}} \quad \theta' = \frac{y}{f}$$

Combining these expressions with Eq. (34.21), we find

$$\text{Angular magnification for a simple magnifier} \quad M = \frac{\theta'}{\theta} = \frac{y/f}{y/25 \text{ cm}} = \frac{25 \text{ cm}}{f} \quad (34.22)$$

Angular size of object seen with magnifier  $\theta'$  Object height  
 Angular size of object seen without magnifier  $\theta$  Near point  
Focal length

It may seem that we can make the angular magnification as large as we like by decreasing the focal length  $f$ . In fact, the aberrations of a simple double-convex lens set a limit to  $M$  of about  $3\times$  to  $4\times$ . If these aberrations are corrected, the angular magnification may be made as great as  $20\times$ . A compound microscope, discussed in the next section, provides even greater magnification.

**TEST YOUR UNDERSTANDING OF SECTION 34.7** You are using a magnifier to examine a gem. If you change to a different magnifier with twice the focal length of the first one, you'll have to hold the object at (i) twice the distance and the angular magnification will be twice as great; (ii) twice the distance and the angular magnification will be  $\frac{1}{2}$  as great; (iii)  $\frac{1}{2}$  the distance and the angular magnification will be twice as great; (iv)  $\frac{1}{2}$  the distance and the angular magnification will be  $\frac{1}{2}$  as great.

#### ANSWER

(ii) The object must be held at the focal point, which is twice as far away if the focal length  $f$  is twice as great. Equation (34.24) shows that the angular magnification  $M$  is inversely proportional to  $f$ , so doubling the focal length makes  $M \frac{1}{2}$  as great. To improve the magnification, you should use a magnifier with a *shorter* focal length.

## 34.8 MICROSCOPES AND TELESCOPES

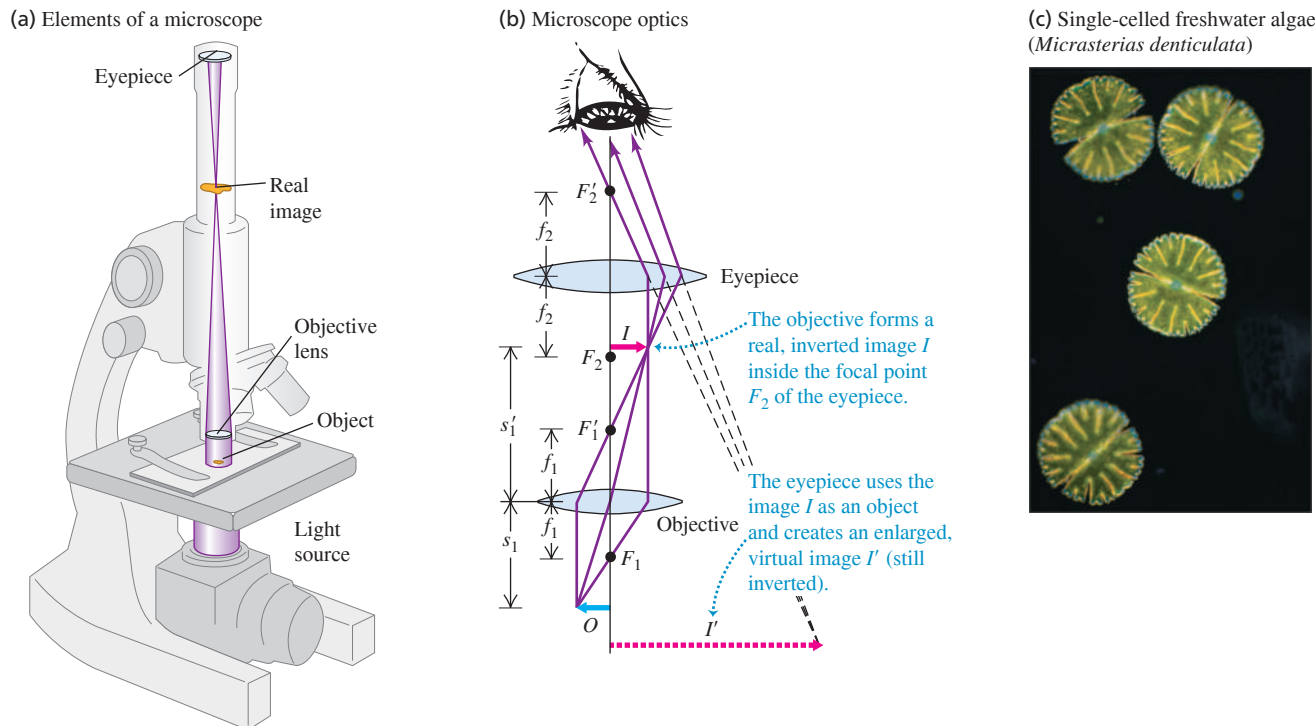
Cameras, eyeglasses, and magnifiers use a single lens to form an image. Two important optical devices that use *two* lenses are the microscope and the telescope. In each device a primary lens, or *objective*, forms a real image, and a second lens, or *eyepiece*, is used as a magnifier to make an enlarged, virtual image.

### Microscopes

**Figure 34.52a** (next page) shows the essential features of a **microscope**, sometimes called a *compound microscope*. To analyze this system, we use the principle that an image formed by one optical element such as a lens or mirror can serve as the object for a second



Figure 34.52 (a) Elements of a microscope. (b) The object  $O$  is placed just outside the first focal point of the objective (the distance  $s_1$  has been exaggerated for clarity). (c) This microscope image shows single-celled organisms about  $2 \times 10^{-4}$  m (0.2 mm) across. Typical light microscopes can resolve features as small as  $2 \times 10^{-7}$  m, comparable to the wavelength of light.



element. We used this principle in Section 34.4 when we derived the lensmaker's equation by repeated application of the single-surface refraction equation; we used this principle again in Example 34.11 (Section 34.4), in which the image formed by a lens was used as the object of a second lens.

The object  $O$  to be viewed is placed just beyond the first focal point  $F_1$  of the **objective**, a converging lens that forms a real and enlarged image  $I$  (Fig. 34.52b). In a properly designed instrument this image lies just inside the first focal point  $F_2$  of a second converging lens called the **eyepiece** or **ocular**. (The reason the image should lie just *inside*  $F_2$  is left for you to discover.) The eyepiece acts as a simple magnifier, as discussed in Section 34.7, and forms a final virtual image  $I'$  of  $I$ . The position of  $I'$  may be anywhere between the near and far points of the eye. Both the objective and the eyepiece of an actual microscope are highly corrected compound lenses with several optical elements, but for simplicity we show them here as simple thin lenses.

As for a simple magnifier, what matters when viewing through a microscope is the *angular* magnification  $M$ . The overall angular magnification of the compound microscope is the product of two factors. The first factor is the *lateral* magnification  $m_1$  of the objective, which determines the linear size of the real image  $I$ ; the second factor is the *angular* magnification  $M_2$  of the eyepiece, which relates the angular size of the virtual image seen through the eyepiece to the angular size that the real image  $I$  would have if you viewed it *without* the eyepiece. The first of these factors is given by

$$m_1 = -\frac{s'_1}{s_1} \quad (34.23)$$

where  $s_1$  and  $s'_1$  are the object and image distances, respectively, for the objective lens. Ordinarily, the object is very close to the focal point, and the resulting image distance  $s'_1$  is very great in comparison to the focal length  $f_1$  of the objective lens. Thus  $s_1$  is approximately equal to  $f_1$ , and we can write  $m_1 = -s'_1/f_1$ .

The real image  $I$  is close to the focal point  $F_2$  of the eyepiece, so to find the eyepiece angular magnification, we can use Eq. (34.22):  $M_2 = (25 \text{ cm})/f_2$ , where  $f_2$  is the focal length of the eyepiece (considered as a simple lens). The overall angular magnification  $M$  of the compound microscope (apart from a negative sign, which is customarily ignored) is the product of the two magnifications:

$$M = m_1 M_2 = \frac{(25 \text{ cm})s'_1}{f_1 f_2} \quad \begin{array}{l} \text{(angular magnification} \\ \text{for a microscope)} \end{array} \quad (34.24)$$

where  $s'_1$ ,  $f_1$ , and  $f_2$  are measured in centimeters. The final image is inverted with respect to the object. Microscope manufacturers usually specify the values of  $m_1$  and  $M_2$  rather than the focal lengths of the objective and eyepiece.

Equation (34.24) shows that the angular magnification of a microscope can be increased by using an objective of shorter focal length  $f_1$ , thereby increasing  $m_1$  and the size of the real image  $I$ . Most optical microscopes have a rotating “turret” with three or more objectives of different focal lengths so that the same object can be viewed at different magnifications. The eyepiece should also have a short focal length  $f_2$  to help to maximize the value of  $M$ .

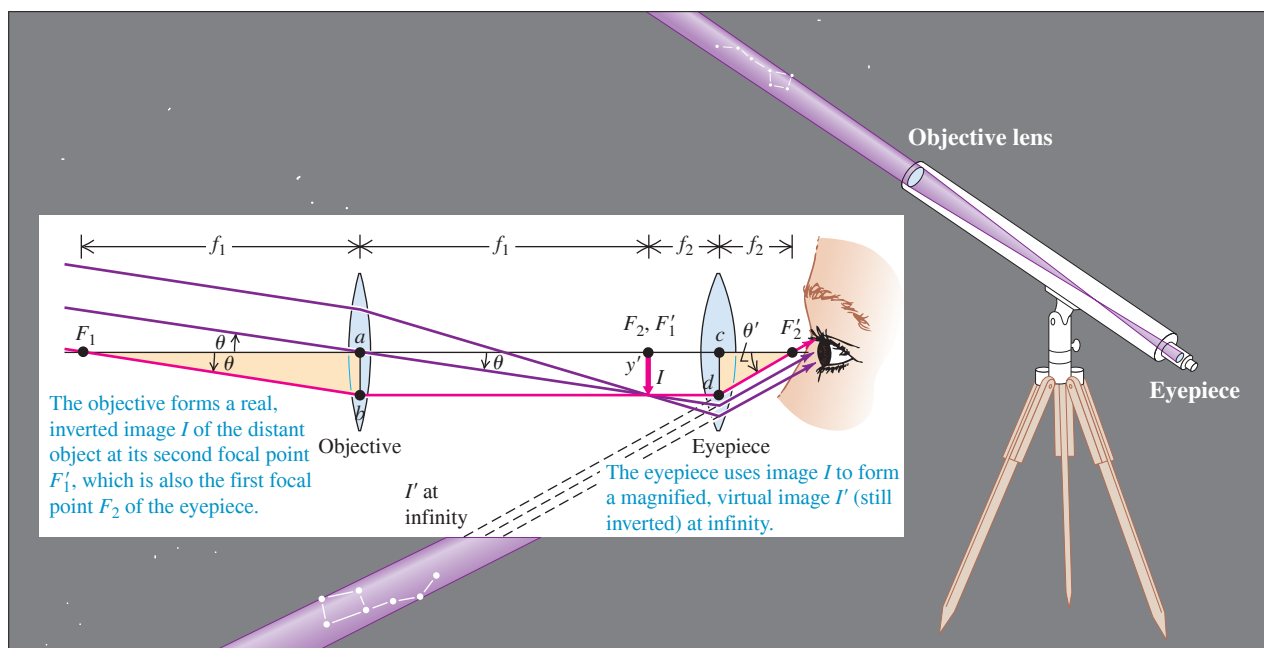
To use a microscope to take a photograph (called a *photomicrograph* or *micrograph*), the eyepiece is removed and a camera placed so that the real image  $I$  falls on the camera's electronic sensor or film. Figure 34.52c shows such a photograph. In this case what matters is the *lateral* magnification of the microscope as given by Eq. (34.23).

## Telescopes

The optical system of a **telescope** is similar to that of a compound microscope. In both instruments the image formed by an objective is viewed through an eyepiece. The key difference is that the telescope is used to view large objects at large distances and the microscope is used to view small objects close at hand. Another difference is that many telescopes use a curved mirror, not a lens, as an objective.

**Figure 34.53** shows an *astronomical telescope*. Because this telescope uses a lens as an objective, it is called a *refracting telescope* or *refractor*. The objective lens forms a

Figure 34.53 Optical system of an astronomical refracting telescope.



real, reduced image  $I$  of the object. This image is the object for the eyepiece lens, which forms an enlarged, virtual image of  $I$ . Objects that are viewed with a telescope are usually so far away from the instrument that the first image  $I$  is formed very nearly at the second focal point of the objective lens. If the final image  $I'$  formed by the eyepiece is at infinity (for most comfortable viewing by a normal eye), the first image must also be at the first focal point of the eyepiece. The distance between objective and eyepiece, which is the length of the telescope, is therefore the *sum* of the focal lengths of objective and eyepiece,  $f_1 + f_2$ .

The angular magnification  $M$  of a telescope is defined as the ratio of the angle subtended at the eye by the final image  $I'$  to the angle subtended at the (unaided) eye by the object. We can express this ratio in terms of the focal lengths of objective and eyepiece. In Fig. 34.53 the ray passing through  $F_1$ , the first focal point of the objective, and through  $F'_2$ , the second focal point of the eyepiece, is shown in red. The object (not shown) subtends an angle  $\theta$  at the objective and would subtend essentially the same angle at the unaided eye. Also, since the observer's eye is placed just to the right of the focal point  $F'_2$ , the angle subtended at the eye by the final image is very nearly equal to the angle  $\theta'$ . Because  $bd$  is parallel to the optic axis, the distances  $ab$  and  $cd$  are equal to each other and also to the height  $y'$  of the real image  $I$ . Because the angles  $\theta$  and  $\theta'$  are small, they may be approximated by their tangents. From the right triangles  $F_1ab$  and  $F'_2cd$ ,

$$\theta = \frac{-y'}{f_1}$$

$$\theta' = \frac{y'}{f_2}$$

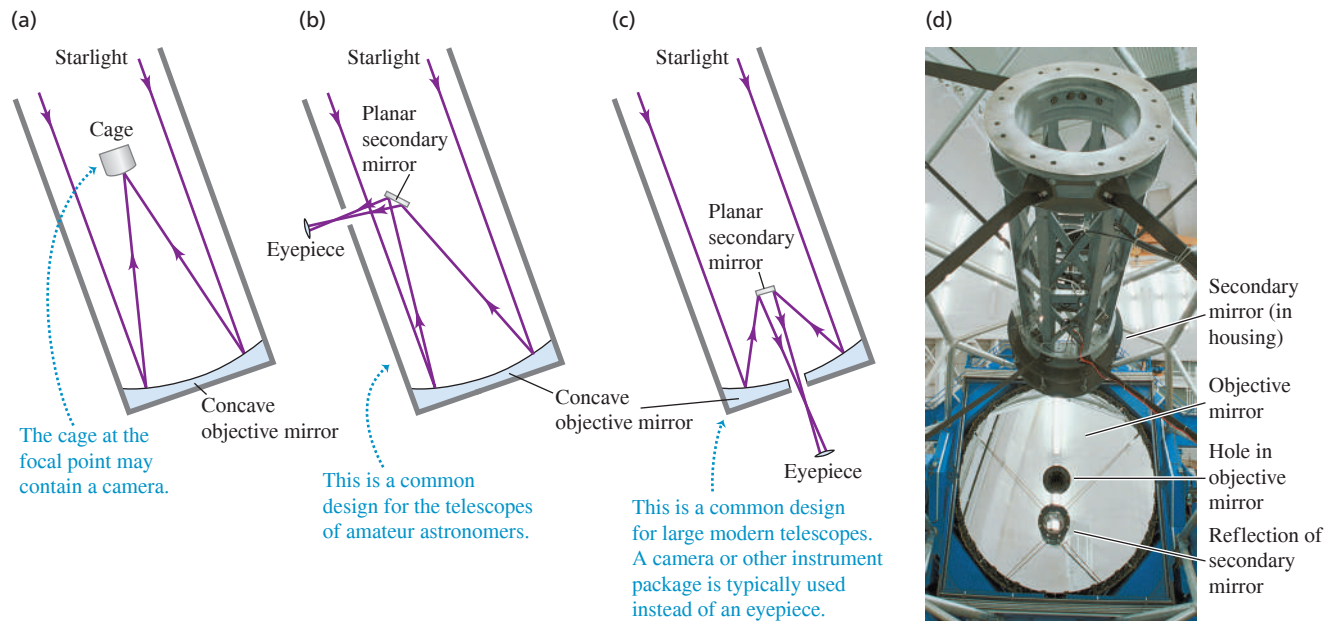
and the angular magnification  $M$  is

$$M = \frac{\theta'}{\theta} = -\frac{y'/f_2}{y'/f_1} = -\frac{f_1}{f_2} \quad \begin{array}{l} \text{(angular magnification} \\ \text{for a telescope)} \end{array} \quad (34.25)$$

The angular magnification  $M$  of a telescope is equal to the ratio of the focal length of the objective to that of the eyepiece. The negative sign shows that the final image is inverted. Equation (34.25) shows that to achieve good angular magnification, a *telescope* should have a *long* objective focal length  $f_1$ . By contrast, Eq. (34.24) shows that a *microscope* should have a *short* objective focal length. However, a telescope objective with a long focal length should also have a large diameter  $D$  so that the  $f$ -number  $f_1/D$  will not be too large; as described in Section 34.5, a large  $f$ -number means a dim, low-intensity image. Telescopes typically do not have interchangeable objectives; instead, the magnification is varied by using different eyepieces with different focal lengths  $f_2$ . Just as for a microscope, smaller values of  $f_2$  give larger angular magnifications.

An inverted image is no particular disadvantage for astronomical observations. When we use a telescope or binoculars—essentially a pair of telescopes mounted side by side—to view objects on the earth, though, we want the image to be right-side up. In prism binoculars, this is accomplished by reflecting the light several times along the path from the objective to the eyepiece. The combined effect of the reflections is to flip the image both horizontally and vertically. Binoculars are usually described by two numbers separated by a multiplication sign, such as  $7 \times 50$ . The first number is the angular magnification  $M$ , and the second is the diameter of the objective lenses (in millimeters). The diameter helps to determine the light-gathering capacity of the objective lenses and thus the brightness of the image.

Figure 34.54 (a), (b), (c) Three designs for reflecting telescopes. (d) This photo shows the interior of the Gemini North telescope, which uses the design shown in (c). The objective mirror is 8 meters in diameter.



In the *reflecting telescope* (**Fig. 34.54a**) the objective lens is replaced by a concave mirror. In large telescopes this scheme has many advantages. Mirrors are inherently free of chromatic aberrations (dependence of focal length on wavelength), and spherical aberrations (associated with the paraxial approximation) are easier to correct than with a lens. The reflecting surface is sometimes nonspherical. The material of the mirror need not be transparent, and it can be made more rigid than a lens, which has to be supported only at its edges.

The largest reflecting telescope in the world, the Gran Telescopio Canarias in the Canary Islands, has an objective mirror of overall diameter 10.4 m made up of 36 separate hexagonal reflecting elements.

One challenge in designing reflecting telescopes is that the image is formed in front of the objective mirror, in a region traversed by incoming rays. Isaac Newton devised one solution to this problem. A flat secondary mirror oriented at  $45^\circ$  to the optic axis causes the image to be formed in a hole on the side of the telescope, where it can be magnified with an eyepiece (**Fig. 34.54b**). Another solution uses a secondary mirror that causes the focused light to pass through a hole in the objective mirror (**Fig. 34.54c**). Large research telescopes, as well as many amateur telescopes, use this design (**Fig. 34.54d**).

Like a microscope, when a telescope is used for photography the eyepiece is removed and an electronic sensor is placed at the position of the real image formed by the objective. (Some long-focal-length “lenses” for photography are actually reflecting telescopes used in this way.) Most telescopes used for astronomical research are never used with an eyepiece.

**TEST YOUR UNDERSTANDING OF SECTION 34.8** Which gives a lateral magnification of greater absolute value? (i) The objective lens in a microscope (**Fig. 34.52**); (ii) the objective lens in a refracting telescope (**Fig. 34.53**); or (iii) not enough information is given to decide.

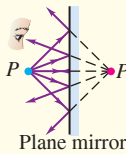
### ANSWER

(i) The objective lens of a microscope makes enlarged images of small objects, so the absolute value of its lateral magnification  $m$  is greater than 1. By contrast, the objective lens of a refracting telescope makes *reduced* images. For example, the moon is thousands of kilometers in diameter, but its image may fit on an electronic sensor a few centimeters across. Thus  $|m|$  is much less than 1 for a refracting telescope. (In both cases  $m$  is negative because the objective makes an inverted image, which is why the question asks about the absolute value of  $m$ .)



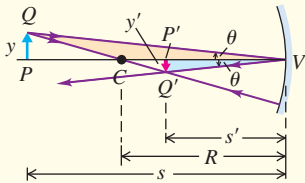
CHAPTER 34 SUMMARY

**Reflection or refraction at a plane surface:** When rays diverge from an object point  $P$  and are reflected or refracted, the directions of the outgoing rays are the same as though they had diverged from a point  $P'$  called the image point. If they actually converge at  $P'$  and diverge again beyond it,  $P'$  is a real image of  $P$ ; if they only appear to have diverged from  $P'$ , it is a virtual image. Images can be either erect or inverted.

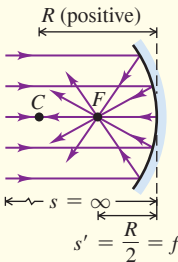


**Lateral magnification:** The lateral magnification  $m$  in any reflecting or refracting situation is defined as the ratio of image height  $y'$  to object height  $y$ . When  $m$  is positive, the image is erect; when  $m$  is negative, the image is inverted.

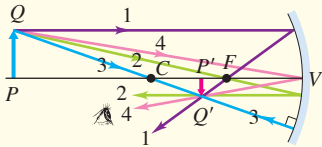
$$m = \frac{y'}{y} \tag{34.2}$$



**Focal point and focal length:** The focal point of a mirror is the point where parallel rays converge after reflection from a concave mirror, or the point from which they appear to diverge after reflection from a convex mirror. Rays diverging from the focal point of a concave mirror are parallel after reflection; rays converging toward the focal point of a convex mirror are parallel after reflection. The distance from the focal point to the vertex is called the focal length, denoted as  $f$ . The focal points of a lens are defined similarly.



**Relating object and image distances:** The formulas for object distance  $s$  and image distance  $s'$  for plane and spherical mirrors and single refracting surfaces are summarized in the table. The equation for a plane surface can be obtained from the corresponding equation for a spherical surface by setting  $R = \infty$ . (See Examples 34.1–34.7.)



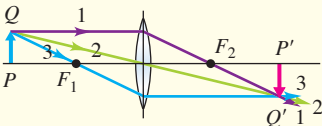
|                            | Plane Mirror                     | Spherical Mirror                                         | Plane Refracting Surface             | Spherical Refracting Surface                           |
|----------------------------|----------------------------------|----------------------------------------------------------|--------------------------------------|--------------------------------------------------------|
| Object and image distances | $\frac{1}{s} + \frac{1}{s'} = 0$ | $\frac{1}{s} + \frac{1}{s'} = \frac{2}{R} = \frac{1}{f}$ | $\frac{n_a}{s} + \frac{n_b}{s'} = 0$ | $\frac{n_a}{s} + \frac{n_b}{s'} = \frac{n_b - n_a}{R}$ |
| Lateral magnification      | $m = -\frac{s'}{s} = 1$          | $m = -\frac{s'}{s}$                                      | $m = -\frac{n_a s'}{n_b s} = 1$      | $m = -\frac{n_a s'}{n_b s}$                            |

Object–image relationships derived in this chapter are valid for only rays close to and nearly parallel to the optic axis; these are called paraxial rays. Nonparaxial rays do not converge precisely to an image point. This effect is called spherical aberration.

**Thin lenses:** The object–image relationship, given by Eq. (34.16), is the same for a thin lens as for a spherical mirror. Equation (34.19), the lensmaker’s equation, relates the focal length of a lens to its index of refraction and the radii of curvature of its surfaces. (See Examples 34.8–34.11.)

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f} \tag{34.16}$$

$$\frac{1}{f} = (n - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right) \tag{34.19}$$

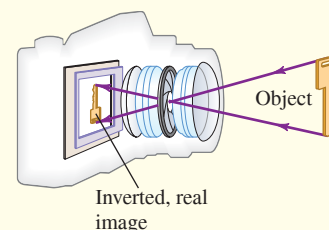


**Sign rules:** The following sign rules are used with all plane and spherical reflecting and refracting surfaces:

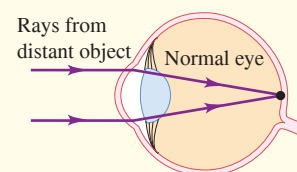
- $s > 0$  when the object is on the incoming side of the surface (a real object);  $s < 0$  otherwise.
- $s' > 0$  when the image is on the outgoing side of the surface (a real image);  $s' < 0$  otherwise.
- $R > 0$  when the center of curvature is on the outgoing side of the surface;  $R < 0$  otherwise.
- $m > 0$  when the image is erect;  $m < 0$  when inverted.

**Cameras:** A camera forms a real, inverted image of the object being photographed on a light-sensitive surface. The amount of light striking this surface is controlled by the shutter speed and the aperture. The intensity of this light is inversely proportional to the square of the  $f$ -number of the lens. (See Example 34.12.)

$$\begin{aligned} f\text{-number} &= \frac{\text{Focal length}}{\text{Aperture diameter}} \\ &= \frac{f}{D} \end{aligned} \quad (34.20)$$

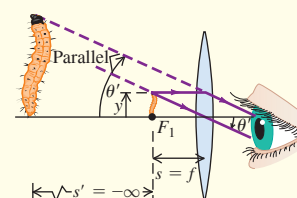


**The eye:** In the eye, refraction at the surface of the cornea forms a real image on the retina. Adjustment for various object distances is made by squeezing the lens, thereby making it bulge and decreasing its focal length. A nearsighted eye is too long for its lens; a farsighted eye is too short. The power of a corrective lens, in diopters, is the reciprocal of the focal length in meters. (See Examples 34.13 and 34.14.)

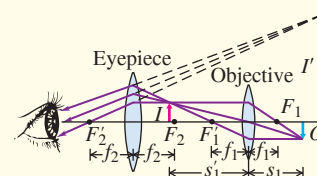


**The simple magnifier:** The simple magnifier creates a virtual image whose angular size  $\theta'$  is larger than the angular size  $\theta$  of the object itself at a distance of 25 cm, the nominal closest distance for comfortable viewing. The angular magnification  $M$  of a simple magnifier is the ratio of the angular size of the virtual image to that of the object at this distance.

$$M = \frac{\theta'}{\theta} = \frac{25 \text{ cm}}{f} \quad (34.22)$$



**Microscopes and telescopes:** In a compound microscope, the objective lens forms a first image in the barrel of the instrument, and the eyepiece forms a final virtual image, often at infinity, of the first image. The telescope operates on the same principle, but the object is far away. In a reflecting telescope, the objective lens is replaced by a concave mirror, which eliminates chromatic aberrations.



Chapter 34 Media Assets



## GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

### KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 34.1, 34.3, and 34.4** (Section 34.2) before attempting these problems.

**VP34.4.1** You place a light bulb 15.0 cm in front of a concave spherical mirror. The mirror forms an image of the bulb on a wall 4.50 m in front of the mirror. Find (a) the mirror's focal length, (b) the mirror's radius of curvature, and (c) the lateral magnification of the image.

**VP34.4.2** A concave spherical mirror has radius of curvature 37.0 cm. Find the image distance and lateral magnification for each of the following object distances. In each case state whether the image is real or virtual, whether it is erect or inverted, and whether it is larger or smaller than the object. (a) 11.0 cm; (b) 31.0 cm; (c) 55.0 cm.

**VP34.4.3** A spherical mirror has radius of curvature  $-44.0$  cm. When you look at your eye in the mirror, your eye's reflection appears to be 18.0 cm behind the mirror's surface. (a) Is the mirror concave or convex? (b) How far is your eye from the mirror? (c) What is the lateral magnification of the image of your eye? Is the image real or virtual? Erect or inverted? Larger or smaller than your eye?

**VP34.4.4** A convex spherical mirror has radius of curvature  $-37.0$  cm. Find the image distance and lateral magnification for each of the following object distances. In each case state whether the image is real or virtual, whether it is erect or inverted, and whether it is larger or smaller than the object. (a) 11.0 cm; (b) 31.0 cm; (c) 55.0 cm.

Be sure to review **EXAMPLE 34.8** (Section 34.4) before attempting these problems.

**VP34.8.1** Both sides of a double convex thin lens have radii of curvature of the same magnitude. The lens is made of glass with index of refraction 1.65, and the focal length of the lens is  $+30.0$  cm. Find the radius of curvature of (a) the front surface and (b) the back surface.

**VP34.8.2** The side of a thin lens that faces the object is convex and has radius of curvature 15.0 cm. The other side is concave and has radius of curvature 25.0 cm. The lens is made of glass with index of refraction 1.55. (a) Is this lens thicker at its center or at its edges? (b) What is the focal length of the lens? (c) Is the lens converging or diverging?

**VP34.8.3** The side of a thin lens that faces the object is convex and has radius of curvature 25.0 cm. The other side is concave and has radius of curvature 15.0 cm. The lens is made of glass with index of refraction 1.55. (a) Is this lens thicker at its center or at its edges? (b) What is the focal length of the lens? (c) Is the lens converging or diverging?

**VP34.8.4** You are designing a lens to be made of glass with index of refraction 1.70. The first surface (the surface toward the object) is to be convex with radius of curvature 28.0 cm, and the focal length of the lens is to be 14.0 cm. (a) What must be the radius of curvature of the second surface (the surface away from the object)? (b) Will the second surface be concave or convex?

Be sure to review **EXAMPLES 34.9 and 34.10** (Section 34.4) before attempting these problems.

**VP34.10.1** You place a strawberry 15.0 cm in front of a converging lens with focal length +25.0 cm. (a) What is the image position? Is it on the same side of the lens as the strawberry or on the opposite side? (b) What is the lateral magnification of the image? (c) Is the image real or virtual? Erect or inverted? Larger or smaller than the strawberry?

**VP34.10.2** If you put an eraser 28.0 cm in front of a lens, an image of the eraser is formed 42.0 cm behind the lens. (a) Find the focal length of the lens. Is the lens converging or diverging? (b) What is the lateral magnification of the image? (c) Is the image real or virtual? Erect or inverted? Larger or smaller than the eraser?

**VP34.10.3** When you place a thumbtack 48.0 cm in front of a lens, the resulting image is real, inverted, and the same size as the thumbtack. (a) Is the lens converging or diverging? (b) What is the image distance? (c) What is the focal length of the lens?

**VP34.10.4** The front door of many houses has a small “peephole” lens that gives a wide-angle view of the outside. To provide this wide-angle view, the lens must produce an image of an object outside that is erect and smaller than the object. For one lens, if an object 20.0 cm tall is 2.50 m in front of the door, its erect image is only 4.00 cm tall. (a) What is the lateral magnification of the image? (b) What is the image distance? Is the image on the same side of the lens as the object or on the opposite side? (c) What is the focal length of the lens? Is the lens converging or diverging?

## BRIDGING PROBLEM Image Formation by a Wine Goblet

A thick-walled wine goblet can be considered to be a hollow glass sphere with an outer radius of 4.00 cm and an inner radius of 3.40 cm. The index of refraction of the goblet glass is 1.50. (a) A beam of parallel light rays enters the side of the empty goblet along a horizontal radius. Where, if anywhere, will an image be formed? (b) The goblet is filled with white wine ( $n = 1.37$ ). Where is the image formed?

### SOLUTION GUIDE

#### IDENTIFY and SET UP

1. The goblet is *not* a thin lens, so you cannot use the thin-lens formula. Instead, you must think of the inner and outer surfaces of the goblet walls as spherical refracting surfaces. The image formed by one surface serves as the object for the next surface. Draw a diagram that shows the goblet and the light rays that enter it.
2. Choose the appropriate equation that relates the image and object distances for a spherical refracting surface.

#### EXECUTE

3. For the empty goblet, each refracting surface has glass on one side and air on the other. Find the position of the image formed by the first surface, the outer wall of the goblet. Use this as the object for the second surface (the inner wall of the same side of the goblet) and find the position of the second image. (*Hint:* Be sure to account for the thickness of the goblet wall.)
4. Continue the process of step 3. Consider the refractions at the inner and outer surfaces of the glass on the opposite side of the goblet, and find the position of the final image. (*Hint:* Be sure to account for the distance between the two sides of the goblet.)
5. Repeat steps 3 and 4 for the case in which the goblet is filled with wine.

#### EVALUATE

6. Are the images real or virtual? How can you tell?

## PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

### DISCUSSION QUESTIONS

**Q34.1** A spherical mirror is cut in half horizontally. Will an image be formed by the bottom half of the mirror? If so, where will the image be formed?

**Q34.2** For the situation shown in Fig. 34.3, is the image distance  $s'$  positive or negative? Is the image real or virtual? Explain your answers.

**Q34.3** The laws of optics also apply to electromagnetic waves invisible to the eye. A satellite TV dish is used to detect radio waves coming from orbiting satellites. Why is a curved reflecting surface (a “dish”) used? The dish is always concave, never convex; why? The actual radio receiver is placed on an arm and suspended in front of the dish. How far in front of the dish should it be placed?

**Q34.4** Explain why the focal length of a *plane* mirror is infinite, and explain what it means for the focal point to be at infinity.

**Q34.5** If a spherical mirror is immersed in water, does its focal length change? Explain.

**Q34.6** For what range of object positions does a concave spherical mirror form a real image? What about a convex spherical mirror?

**Q34.7** When a room has mirrors on two opposite walls, an infinite series of reflections can be seen. Discuss this phenomenon in terms of images. Why do the distant images appear fainter?

**Q34.8** For a spherical mirror, if  $s = f$ , then  $s' = \infty$ , and the lateral magnification  $m$  is infinite. Does this make sense? If so, what does it mean?

**Q34.9** You may have noticed a small convex mirror next to your bank’s ATM. Why is this mirror convex, as opposed to flat or concave? What considerations determine its radius of curvature?

**Q34.10** A student claims that she can start a fire on a sunny day using just the sun’s rays and a concave mirror. How is this done? Is the concept of image relevant? Can she do the same thing with a convex mirror? Explain.

**Q34.11** A person looks at his reflection in the concave side of a shiny spoon. Is it right side up or inverted? Does it matter how far his face is from the spoon? What if he looks in the convex side? (Try this yourself!)

**Q34.12** In Example 34.4 (Section 34.2), there appears to be an ambiguity for the case  $s = 10\text{ cm}$  as to whether  $s'$  is  $+\infty$  or  $-\infty$  and whether the image is erect or inverted. How is this resolved? Or is it?

**Q34.13** Suppose that in the situation of Example 34.7 of Section 34.3 (see Fig. 34.26) a vertical arrow 2.00 m tall is painted on the side of the pool beneath the water line. According to the calculations in the example, this arrow would appear to the person shown in Fig. 34.26 to be 1.50 m long. But the discussion following Eq. (34.13) states that the magnification for a plane refracting surface is  $m = 1$ , which suggests that the arrow would appear to the person to be 2.00 m long. How can you resolve this apparent contradiction?

**Q34.14** The bottom of the passenger-side mirror on your car notes, "Objects in mirror are closer than they appear." Is this true? Why?

**Q34.15** How could you very quickly make an approximate measurement of the focal length of a converging lens? Could the same method be applied if you wished to use a diverging lens? Explain.

**Q34.16** The focal length of a simple lens depends on the color (wavelength) of light passing through it. Why? Is it possible for a lens to have a positive focal length for some colors and negative for others? Explain.

**Q34.17** When a converging lens is immersed in water, does its focal length increase or decrease in comparison with the value in air? Explain.

**Q34.18** A spherical air bubble in water can function as a lens. Is it a converging or diverging lens? How is its focal length related to its radius?

**Q34.19** Can an image formed by one reflecting or refracting surface serve as an object for a second reflection or refraction? Does it matter whether the first image is real or virtual? Explain.

**Q34.20** If a piece of photographic film is placed at the location of a real image, the film will record the image. Can this be done with a virtual image? How might one record a virtual image?

**Q34.21** According to the discussion in Section 34.2, light rays are reversible. Are the formulas in the table in this chapter's Summary still valid if object and image are interchanged? What does reversibility imply with respect to the *forms* of the various formulas?

**Q34.22** You've entered a survival contest that will include building a crude telescope. You are given a large box of lenses. Which two lenses do you pick? How do you quickly identify them?

**Q34.23 BIO** You can't see clearly underwater with the naked eye, but you *can* if you wear a face mask or goggles (with air between your eyes and the mask or goggles). Why is there a difference? Could you instead wear eyeglasses (with water between your eyes and the eyeglasses) in order to see underwater? If so, should the lenses be converging or diverging? Explain.

**Q34.24** You take a lens and mask it so that light can pass through only the bottom half of the lens. How does the image formed by the masked lens compare to the image formed before masking?

## EXERCISES

### Section 34.1 Reflection and Refraction at a Plane Surface

**34.1** • A candle 4.85 cm tall is 39.2 cm to the left of a plane mirror. Where is the image formed by the mirror, and what is the height of this image?

**34.2** • The image of a tree just covers the length of a plane mirror 4.00 cm tall when the mirror is held 35.0 cm from the eye. The tree is 28.0 m from the mirror. What is its height?

**34.3** • A pencil that is 9.0 cm long is held perpendicular to the surface of a plane mirror with the tip of the pencil lead 12.0 cm from the mirror surface and the end of the eraser 21.0 cm from the mirror surface. What is the length of the image of the pencil that is formed by the mirror? Which end of the image is closer to the mirror surface: the tip of the lead or the end of the eraser?

### Section 34.2 Reflection at a Spherical Surface

**34.4** • A concave mirror has a radius of curvature of 34.0 cm. (a) What is its focal length? (b) If the mirror is immersed in water (refractive index 1.33), what is its focal length?

**34.5** • An object 0.600 cm tall is placed 16.5 cm to the left of the vertex of a concave spherical mirror having a radius of curvature of 22.0 cm. (a) Draw a principal-ray diagram showing the formation of the image. (b) Determine the position, size, orientation, and nature (real or virtual) of the image.

**34.6** • Repeat Exercise 34.5 for the case in which the mirror is convex.

**34.7** •• The diameter of Mars is 6794 km, and its minimum distance from the earth is  $5.58 \times 10^7$  km. When Mars is at this distance, find the diameter of the image of Mars formed by a spherical, concave telescope mirror with a focal length of 1.75 m.

**34.8** •• An object is 18.0 cm from the center of a spherical silvered-glass Christmas tree ornament 6.00 cm in diameter. What are the position and magnification of its image?

**34.9** • A coin is placed next to the convex side of a thin spherical glass shell having a radius of curvature of 18.0 cm. Reflection from the surface of the shell forms an image of the 1.5-cm-tall coin that is 6.00 cm behind the glass shell. Where is the coin located? Determine the size, orientation, and nature (real or virtual) of the image.

**34.10** • You hold a spherical salad bowl 60 cm in front of your face with the bottom of the bowl facing you. The bowl is made of polished metal with a 35 cm radius of curvature. (a) Where is the image of your 5.0-cm-tall nose located? (b) What are the image's size, orientation, and nature (real or virtual)?

**34.11** • A spherical, concave shaving mirror has a radius of curvature of 32.0 cm. (a) What is the magnification of a person's face when it is 12.0 cm to the left of the vertex of the mirror? (b) Where is the image? Is the image real or virtual? (c) Draw a principal-ray diagram showing the formation of the image.

**34.12** • For a concave spherical mirror that has focal length  $f = +18.0$  cm, what is the distance of an object from the mirror's vertex if the image is real and has the same height as the object?

**34.13** • **Dental Mirror.** A dentist uses a curved mirror to view teeth on the upper side of the mouth. Suppose she wants an erect image with a magnification of 2.00 when the mirror is 1.25 cm from a tooth. (Treat this problem as though the object and image lie along a straight line.) (a) What kind of mirror (concave or convex) is needed? Use a ray diagram to decide, without performing any calculations. (b) What must be the focal length and radius of curvature of this mirror? (c) Draw a principal-ray diagram to check your answer in part (b).

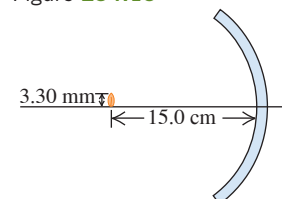
**34.14** • For a convex spherical mirror that has focal length  $f = -12.0$  cm, what is the distance of an object from the mirror's vertex if the height of the image is half the height of the object?

**34.15** • The thin glass shell shown in

Fig. E34.15 has a spherical shape with a radius of curvature of 12.0 cm, and both of its surfaces can act as mirrors. A seed 3.30 mm high is placed 15.0 cm from the center of the mirror along the optic axis, as shown in the figure. (a) Calculate the location and height of the image of this seed. (b) Suppose now that the shell is reversed. Find the location and height of the seed's image.

**34.16** • An object 0.600 cm tall is placed 24.0 cm to the left of the vertex of a concave spherical mirror. The image of the object is inverted and is 2.50 cm tall. What is the radius of curvature of the mirror?

Figure E34.15





**34.17 •** A shiny spoon provides both a concave mirror and a convex mirror, one on each side. (a) Hold a shiny spoon about 25 cm from your face. Observe your image in the concave side. Is your image up-side down or right-side up? (b) Does that mean that your image is real or virtual? (c) Estimate the height of your head. (d) Estimate the height of the image of your head. (Use the size of the spoon as a guide.) (e) Using these estimates, determine the magnification. (f) Use Eqs. (34.6) and (34.5) to estimate the radius of curvature of the spoon. (g) Now observe your image in the convex side of the spoon. Is it up-side down or right-side up? (h) Is your image real or virtual?

### Section 34.3 Refraction at a Spherical Surface

**34.18 ••** A tank whose bottom is a mirror is filled with water to a depth of 20.0 cm. A small fish floats motionless 7.0 cm under the surface of the water. (a) What is the apparent depth of the fish when viewed at normal incidence? (b) What is the apparent depth of the image of the fish when viewed at normal incidence?

**34.19 •** A speck of dirt is embedded 3.50 cm below the surface of a sheet of ice ( $n = 1.309$ ). What is its apparent depth when viewed at normal incidence?

**34.20 •** Parallel rays from a distant object are traveling in air and then are incident on the concave end of a glass rod with a radius of curvature of 15.0 cm. The refractive index of the glass is 1.50. What is the distance between the vertex of the glass surface and the image formed by the refraction at the concave surface of the rod? Is the image in the air or in the glass?

**34.21 •** A person swimming 0.80 m below the surface of the water in a swimming pool looks at the diving board that is directly overhead and sees the image of the board that is formed by refraction at the surface of the water. This image is a height of 5.20 m above the swimmer. What is the actual height of the diving board above the surface of the water?

**34.22 •** A person is lying on a diving board 3.00 m above the surface of the water in a swimming pool. She looks at a penny that is on the bottom of the pool directly below her. To her, the penny appears to be a distance of 7.00 m from her. What is the depth of the water at this point?

**34.23 •• A Spherical Fish Bowl.** A small tropical fish is at the center of a water-filled, spherical fish bowl 28.0 cm in diameter. (a) Find the apparent position and magnification of the fish to an observer outside the bowl. The effect of the thin walls of the bowl may be ignored. (b) A friend advised the owner of the bowl to keep it out of direct sunlight to avoid blinding the fish, which might swim into the focal point of the parallel rays from the sun. Is the focal point actually within the bowl?

**34.24 •** The left end of a long glass rod 6.00 cm in diameter has a convex hemispherical surface 3.00 cm in radius. The refractive index of the glass is 1.60. Determine the position of the image if an object is placed in air on the axis of the rod at the following distances to the left of the vertex of the curved end: (a) infinitely far; (b) 12.0 cm; (c) 2.00 cm.

**34.25 ••** The glass rod of Exercise 34.24 is immersed in oil ( $n = 1.45$ ). An object placed to the left of the rod on the rod's axis is to be imaged 1.20 m inside the rod. How far from the left end of the rod must the object be located to form the image?

**34.26 ••** The left end of a long glass rod 8.00 cm in diameter, with an index of refraction of 1.60, is ground and polished to a convex hemispherical surface with a radius of 4.00 cm. An object in the form of an arrow 1.50 mm tall, at right angles to the axis of the rod, is located on the axis 24.0 cm to the left of the vertex of the convex surface. Find the position and height of the image of the arrow formed by paraxial rays incident on the convex surface. Is the image erect or inverted?

**34.27 •** A 3.20-mm-tall object is 20.0 cm from the vertex of a spherical mirror. The mirror forms an image 60.0 cm from the mirror. (a) If the image is real, what is the radius of curvature of the mirror? What is the height of the image? Is it upright or inverted? (b) Repeat part (a) for the case where the image is virtual.

### Section 34.4 Thin Lenses

**34.28 ••** To determine the focal length  $f$  of a converging thin lens, you place a 4.00-mm-tall object a distance  $s$  to the left of the lens and measure the height  $h'$  of the real image that is formed to the right of the lens. You repeat this process for several values of  $s$  that produce a real image. After graphing your results as  $1/h'$  versus  $s$ , both in cm, you find that they lie close to a straight line that has slope  $0.208 \text{ cm}^{-2}$ . What is the focal length of the lens?

**34.29 •** An insect 3.75 mm tall is placed 22.5 cm to the left of a thin planoconvex lens. The left surface of this lens is flat, the right surface has a radius of curvature of magnitude 13.0 cm, and the index of refraction of the lens material is 1.70. (a) Calculate the location and size of the image this lens forms of the insect. Is it real or virtual? Erect or inverted? (b) Repeat part (a) if the lens is reversed.

**34.30 •** A lens forms an image of an object. The object is 16.0 cm from the lens. The image is 12.0 cm from the lens on the same side as the object. (a) What is the focal length of the lens? Is the lens converging or diverging? (b) If the object is 8.50 mm tall, how tall is the image? Is it erect or inverted? (c) Draw a principal-ray diagram.

**34.31 •** A converging meniscus lens (see Fig. 34.32a) with a refractive index of 1.52 has spherical surfaces whose radii are 7.00 cm and 4.00 cm. What is the position of the image if an object is placed 24.0 cm to the left of the lens? What is the magnification?

**34.32 •** A converging lens with a focal length of 70.0 cm forms an image of a 3.20-cm-tall real object that is to the left of the lens. The image is 4.50 cm tall and inverted. Where are the object and image located in relation to the lens? Is the image real or virtual?

**34.33 ••** A converging lens forms an image of an 8.00-mm-tall real object. The image is 12.0 cm to the left of the lens, 3.40 cm tall, and erect. What is the focal length of the lens? Where is the object located?

**34.34 •** A photographic slide is to the left of a lens. The lens projects an image of the slide onto a wall 6.00 m to the right of the slide. The image is 80.0 times the size of the slide. (a) How far is the slide from the lens? (b) Is the image erect or inverted? (c) What is the focal length of the lens? (d) Is the lens converging or diverging?

**34.35 ••** A double-convex thin lens has surfaces with equal radii of curvature of magnitude 2.50 cm. Using this lens, you observe that it forms an image of a very distant tree at a distance of 1.87 cm from the lens. What is the index of refraction of the lens?

**34.36 •** A converging lens with a focal length of 9.00 cm forms an image of a 4.00-mm-tall real object that is to the left of the lens. The image is 1.30 cm tall and erect. Where are the object and image located? Is the image real or virtual?

**34.37 •** A thin lens is made of glass that has refractive index  $n = 1.50$ . The lens is surrounded by air. The left-hand surface of the lens is flat and the right-hand spherical surface is convex with radius of curvature 20.0 cm, so the lens is thicker in the middle than at its edges. What is the height of the image formed by the lens for a 6.00-mm-tall object that is placed 20.0 cm to the left of the center of the lens?

**34.38 ••** A lensmaker wants to make a magnifying glass from glass that has an index of refraction  $n = 1.55$  and a focal length of 20.0 cm. If the two surfaces of the lens are to have equal radii, what should that radius be?

**34.39 ••** A thin lens is made of glass that has refractive index  $n = 1.50$ . The lens is surrounded by air. The left-hand spherical surface of the lens is concave with radius of curvature  $R$  and the right-hand side is flat, so the lens is thinner in the middle than at its edges. An object is placed 12.0 cm to the left of the center of the lens. What is the value of  $R$  if the image formed by the lens is 8.00 cm to the left of the center of the lens?

**34.40 •** A converging lens with a focal length of 12.0 cm forms a virtual image 8.00 mm tall, 17.0 cm to the right of the lens. Determine the position and size of the object. Is the image erect or inverted? Are the object and image on the same side or opposite sides of the lens? Draw a principal-ray diagram for this situation.

**34.41 •** Repeat Exercise 34.40 for the case in which the lens is diverging, with a focal length of  $-48.0$  cm.

**34.42 •** An object is 16.0 cm to the left of a lens. The lens forms an image 36.0 cm to the right of the lens. (a) What is the focal length of the lens? Is the lens converging or diverging? (b) If the object is 8.00 mm tall, how tall is the image? Is it erect or inverted? (c) Draw a principal-ray diagram.

**34.43 •• Combination of Lenses I.** A 1.20-cm-tall object is 50.0 cm to the left of a converging lens of focal length 40.0 cm. A second converging lens, this one having a focal length of 60.0 cm, is located 300.0 cm to the right of the first lens along the same optic axis. (a) Find the location and height of the image (call it  $I_1$ ) formed by the lens with a focal length of 40.0 cm. (b)  $I_1$  is now the object for the second lens. Find the location and height of the image produced by the second lens. This is the final image produced by the combination of lenses.

**34.44 •• Combination of Lenses II.** Repeat Exercise 34.43 using the same lenses except for the following changes: (a) The second lens is a *diverging* lens having a focal length of magnitude 60.0 cm. (b) The first lens is a *diverging* lens having a focal length of magnitude 40.0 cm. (c) Both lenses are diverging lenses having focal lengths of the same *magnitudes* as in Exercise 34.43.

**34.45 •• Combination of Lenses III.** Two thin lenses with a focal length of magnitude 12.0 cm, the first diverging and the second converging, are located 9.00 cm apart. An object 2.50 mm tall is placed 20.0 cm to the left of the first (diverging) lens. (a) How far from this first lens is the final image formed? (b) Is the final image real or virtual? (c) What is the height of the final image? Is it erect or inverted? (*Hint:* See the preceding two exercises.)

**34.46 •** The focal points of a thin diverging lens are 25.0 cm from the center of the lens. An object is placed to the left of the lens, and the lens forms an image of the object that is 18.0 cm from the lens. (a) Is the image to the left or right of the lens? (b) How far is the object from the center of the lens? (c) Is the height of the image less than, greater than, or the same as the height of the object?

**34.47 •** Figure 34.37e shows the principal-ray diagram of the image formation by a converging lens with focal length  $f$  and object distance  $s = 2f/3$  to the left of the lens. (a) Use Eq. (34.17) to find the image distance  $s'$  in terms of  $f$ . Based on your result, is the image real or virtual? Is it to the left or right of the lens? (b) If the height of the object is  $h$ , use Eq. (34.17) to find the height of the image. Does that mean the image is upright or inverted? (c) Do your results agree with what is shown in Fig. 34.37e?

**34.48 ••** An object is to the left of a thin lens. The lens forms an image on a screen that is 2.60 m to the right of the object. The height of the image is 2.50 times the height of the object. (a) Is the image upright or inverted? (b) What is the focal length of the lens? Is the lens converging or diverging?

### Section 34.5 Cameras

**34.49 •** When a camera is focused, the lens is moved away from or toward the digital image sensor. If you take a picture of your friend, who is standing 3.90 m from the lens, using a camera with a lens with an 85 mm focal length, how far from the sensor is the lens? Will the whole image of your friend, who is 175 cm tall, fit on a sensor that is 24 mm  $\times$  36 mm?

**34.50 •** You wish to project the image of a slide on a screen 9.00 m from the lens of a slide projector. (a) If the slide is placed 15.0 cm from the lens, what focal length lens is required? (b) If the dimensions of the picture on a 35 mm color slide are 24 mm  $\times$  36 mm, what is the minimum size of the projector screen required to accommodate the image?

**34.51 • Zoom Lens.** Consider the simple model of the zoom lens shown in Fig. 34.43a. The converging lens has focal length  $f_1 = 12$  cm, and the diverging lens has focal length  $f_2 = -12$  cm. The lenses are separated by 4 cm as shown in Fig. 34.43a. (a) For a distant object, where is the image of the converging lens? (b) The image of the converging lens serves as the object for the diverging lens. What is the object distance for the diverging lens? (c) Where is the final image? Compare your answer to Fig. 34.43a. (d) Repeat parts (a), (b), and (c) for the situation shown in Fig. 34.43b, in which the lenses are separated by 8 cm.

### Section 34.6 The Eye

**34.52 •• BIO Curvature of the Cornea.** In a simplified model of the human eye, the aqueous and vitreous humors and the lens all have a refractive index of 1.40, and all the bending occurs at the cornea, whose vertex is 2.60 cm from the retina. What should be the radius of curvature of the cornea such that the image of an object 40.0 cm from the cornea's vertex is focused on the retina?

**34.53 •• BIO** (a) Where is the near point of an eye for which a contact lens with a power of  $+2.75$  diopters is prescribed? (b) Where is the far point of an eye for which a contact lens with a power of  $-1.30$  diopters is prescribed for distant vision?

**34.54 • BIO Contact Lenses.** Contact lenses are placed right on the eyeball, so the distance from the eye to an object (or image) is the same as the distance from the lens to that object (or image). A certain person can see distant objects well, but his near point is 45.0 cm from his eyes instead of the usual 25.0 cm. (a) Is this person nearsighted or farsighted? (b) What type of lens (converging or diverging) is needed to correct his vision? (c) If the correcting lenses will be contact lenses, what focal length lens is needed and what is its power in diopters?

**34.55 •• BIO Ordinary Glasses.** Ordinary glasses are worn in front of the eye and usually 2.0 cm in front of the eyeball. Suppose that the person in Exercise 34.54 prefers ordinary glasses to contact lenses. What focal length lenses are needed to correct his vision, and what is their power in diopters?

**34.56 • BIO** A person can see clearly up close but cannot focus on objects beyond 75.0 cm. She opts for contact lenses to correct her vision. (a) Is she nearsighted or farsighted? (b) What type of lens (converging or diverging) is needed to correct her vision? (c) What focal length contact lens is needed, and what is its power in diopters?

**34.57 ••** A woman wears bifocal glasses with the lenses 2.0 cm in front of her eyes. The upper half of each lens has power  $-0.500$  diopter and corrects her far vision so that she can focus clearly on distant objects when looking through that half. The lower half of each lens has power  $+2.00$  diopters and corrects her near vision when she looks through that half. (a) What are the far point and near point of her eyes? (b) While the woman is repairing a leaky pipe under her kitchen sink, she looks at close objects through the upper half of her bifocal lenses. What is the closest object that she can see clearly?

### Section 34.7 The Magnifier

**34.58 ••** A thin lens with a focal length of 6.00 cm is used as a simple magnifier. (a) What angular magnification is obtainable with the lens if the object is at the focal point? (b) When an object is examined through the lens, how close can it be brought to the lens? Assume that the image viewed by the eye is at the near point, 25.0 cm from the eye, and that the lens is very close to the eye.

**34.59 •** The focal length of a simple magnifier is 8.00 cm. Assume the magnifier is a thin lens placed very close to the eye. (a) How far in front of the magnifier should an object be placed if the image is formed at the observer's near point, 25.0 cm in front of her eye? (b) If the object is 1.00 mm high, what is the height of its image formed by the magnifier?

**34.60 •** You want to view through a magnifier an insect that is 2.00 mm long. If the insect is to be at the focal point of the magnifier, what focal length will give the image of the insect an angular size of 0.032 radian?

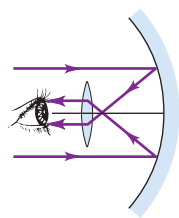
### Section 34.8 Microscopes and Telescopes

**34.61 ••** A telescope is constructed from two lenses with focal lengths of 95.0 cm and 15.0 cm, the 95.0 cm lens being used as the objective. Both the object being viewed and the final image are at infinity. (a) Find the angular magnification for the telescope. (b) Find the height of the image formed by the objective of a building 60.0 m tall, 3.00 km away. (c) What is the angular size of the final image as viewed by an eye very close to the eyepiece?

**34.62 ••** The eyepiece of a refracting telescope (see Fig. 34.53) has a focal length of 9.00 cm. The distance between objective and eyepiece is 1.20 m, and the final image is at infinity. What is the angular magnification of the telescope?

**34.63 ••** A reflecting telescope (Fig. E34.63) is to be made by using a spherical mirror with a radius of curvature of 1.30 m and an eyepiece with a focal length of 1.10 cm. The final image is at infinity. (a) What should the distance between the eyepiece and the mirror vertex be if the object is taken to be at infinity? (b) What will the angular magnification be?

Figure E34.63



**34.64 •** A compound microscope has an objective lens with focal length 14.0 mm and an eyepiece with focal length 20.0 mm. The final image is at infinity. The object to be viewed is placed 2.0 mm beyond the focal point of the objective lens. (a) What is the distance between the two lenses? (b) Without making the approximation  $s_1 \approx f_1$ , use  $M = m_1 m_2$  with  $m_1 = -s'_1/s_1$  to find the overall angular magnification of the microscope. (c) What is the percentage difference between your result and the result obtained if the approximation  $s_1 \approx f_1$  is used to find  $M$ ?

**34.65 ••** The overall angular magnification of a microscope is  $M = -178$ . The eyepiece has focal length 15.0 mm and the final image is at infinity. The separation between the two lenses is 202 mm. What is the focal length of the objective? Do not use the approximation  $s_1 \approx f_1$  in the expression for  $M$ .

### PROBLEMS

**34.66 •** Where must you place an object in front of a concave mirror with radius  $R$  so that the image is erect and  $2\frac{1}{2}$  times the size of the object? Where is the image?

**34.67 ••** A concave mirror is to form an image of the filament of a headlight lamp on a screen 8.00 m from the mirror. The filament is 6.00 mm tall, and the image is to be 24.0 cm tall. (a) How far in front of the vertex of the mirror should the filament be placed? (b) What should be the radius of curvature of the mirror?

**34.68 •** A light bulb is 3.00 m from a wall. You are to use a concave mirror to project an image of the bulb on the wall, with the image 3.50 times the size of the object. How far should the mirror be from the wall? What should its radius of curvature be?

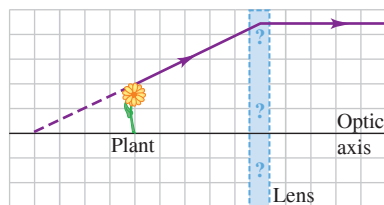
**34.69 •• CP CALC** You are in your car driving on a highway at 25 m/s when you glance in the passenger-side mirror (a convex mirror with radius of curvature 150 cm) and notice a truck approaching. If the image of the truck is approaching the vertex of the mirror at a speed of 1.9 m/s when the truck is 2.0 m from the mirror, what is the speed of the truck relative to the highway?

**34.70 ••** A layer of benzene ( $n = 1.50$ ) that is 4.20 cm deep floats on water ( $n = 1.33$ ) that is 5.70 cm deep. What is the apparent distance from the upper benzene surface to the bottom of the water when you view these layers at normal incidence?

**34.71 •• Rear-View Mirror.** A mirror on the passenger side of your car is convex and has a radius of curvature with magnitude 18.0 cm. (a) Another car is behind your car, 9.00 m from the mirror, and this car is viewed in the mirror by your passenger. If this car is 1.5 m tall, what is the height of the image? (b) The mirror has a warning attached that objects viewed in it are closer than they appear. Why is this so?

**34.72 •• Figure P34.72** shows a small plant near a thin lens. The ray shown is one of the principal rays for the lens. Each square is 2.0 cm along the horizontal direction, but the vertical direction is not to the same scale. Use information from the diagram for the following: (a) Using only the ray shown, decide what type of lens (converging or diverging) this is. (b) What is the focal length of the lens? (c) Locate the image by drawing the other two principal rays. (d) Calculate where the image should be, and compare this result with the graphical solution in part (c).

Figure P34.72



**34.73 •• Pinhole Camera.** A pinhole camera is just a rectangular box with a tiny hole in one face. The film is on the face opposite this hole, and that is where the image is formed. The camera forms an image *without* a lens. (a) Make a clear ray diagram to show how a pinhole camera can form an image on the film without using a lens. (*Hint:* Put an object outside the hole, and then draw rays passing through the hole to the opposite side of the box.) (b) A certain pinhole camera is a box that is 25 cm square and 20.0 cm deep, with the hole in the middle of one of the 25 cm  $\times$  25 cm faces. If this camera is used to photograph a fierce chicken that is 18 cm high and 1.5 m in front of the camera, how large is the image of this bird on the film? What is the lateral magnification of this camera?

**34.74 ••** An object with height 4.00 mm is placed 28.0 cm to the left of a converging lens that has focal length 8.40 cm. A second lens is placed 8.00 cm to the right of the converging lens. (a) What is the focal length of the second lens if the final image is inverted relative to the 4.00-mm-tall object and has height 5.60 mm? (b) What is the distance between the original object and the final image?

**34.75 ••** What should be the index of refraction of a transparent sphere in order for paraxial rays from an infinitely distant object to be brought to a focus at the vertex of the surface opposite the point of incidence?

**34.76 ••** Fill a transparent cylindrical drinking glass with water. Hold a small coin between two fingers submerged in the water immediately inside the glass, and then view the coin from outside the glass. Watch the coin as you move it steadily toward the back of the glass. (a) Does the coin appear to widen as you move it backward in the glass? (b) Estimate the apparent horizontal magnification  $m_{\text{back}}$  when the coin is flush with the back side of the glass. (c) Since the curvature is in a horizontal plane, this horizontal magnification is the same as the lateral magnification analyzed in Section 34.3. Use Eqs. (34.11) and (34.12) to derive an expression for the horizontal magnification  $m$  as a function of the object position  $s$ , the magnitude of the radius of curvature of the glass  $|R|$ , and the index of refraction  $n$  of the water. (d) Use your result to derive an expression for  $n$  as a function of  $m_{\text{back}}$ . (e) Substitute your estimate for  $m_{\text{back}}$  into your equation for  $n$  to estimate the index of refraction of the water.



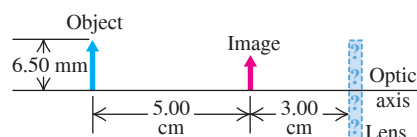
**34.77 ••** (a) You want to use a lens with a focal length of 35.0 cm to produce a real image of an object, with the height of the image twice the height of the object. What kind of lens do you need, and where should the object be placed? (b) Suppose you want a virtual image of the same object, with the same magnification—what kind of lens do you need, and where should the object be placed?

**34.78 •• Autocollimation.** You place an object alongside a white screen, and a plane mirror is 60.0 cm to the right of the object and the screen, with the surface of the mirror tilted slightly from the perpendicular to the line from object to mirror. You then place a converging lens between the object and the mirror. Light from the object passes through the lens, reflects from the mirror, and passes back through the lens; the image is projected onto the screen. You adjust the distance of the lens from the object until a sharp image of the object is focused on the screen. The lens is then 22.0 cm from the object. Because the screen is alongside the object, the distance from object to lens is the same as the distance from screen to lens. (a) Draw a sketch that shows the locations of the object, lens, plane mirror, and screen. (b) What is the focal length of the lens?

**34.79 ••** A lens forms a real image that is 214 cm away from the object and  $1\frac{2}{3}$  times its height. What kind of lens is this, and what is its focal length?

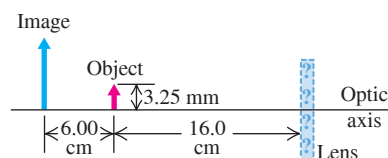
**34.80 • Figure P34.80** shows an object and its image formed by a thin lens. (a) What is the focal length of the lens, and what type of lens (converging or diverging) is it? (b) What is the height of the image? Is it real or virtual?

Figure P34.80



**34.81 • Figure P34.81** shows an object and its image formed by a thin lens. (a) What is the focal length of the lens, and what type of lens (converging or diverging) is it? (b) What is the height of the image? Is it real or virtual?

Figure P34.81



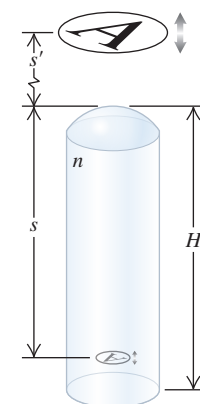
**34.82 •••** A transparent rod 30.0 cm long is cut flat at one end and rounded to a hemispherical surface of radius 10.0 cm at the other end. A small object is embedded within the rod along its axis and halfway between its ends, 15.0 cm from the flat end and 15.0 cm from the vertex of the curved end. When the rod is viewed from its flat end, the apparent depth of the object is 8.20 cm from the flat end. What is its apparent depth when the rod is viewed from its curved end?

**34.83 • BIO Focus of the Eye.** The cornea of the eye has a radius of curvature of approximately 0.50 cm, and the aqueous humor behind it has an index of refraction of 1.35. The thickness of the cornea itself is small enough that we shall neglect it. The depth of a typical human eye is around 25 mm. (a) What would have to be the radius of curvature of the cornea so that it alone would focus the image of a distant mountain on the retina, which is at the back of the eye opposite the cornea? (b) If the cornea focused the mountain correctly on the retina as described in part (a), would it also focus the text from a computer screen on the retina if that screen were 25 cm in front of the eye? If not, where would it focus that text: in front of or behind the retina? (c) Given that the cornea has a radius of curvature of about 5.0 mm, where does it actually focus the mountain? Is this in front of or behind the retina? Does this help you see why the eye needs help from a lens to complete the task of focusing?

**34.84 •** The radii of curvature of the surfaces of a thin converging meniscus lens are  $R_1 = +12.0$  cm and  $R_2 = +28.0$  cm. The index of refraction is 1.60. (a) Compute the position and size of the image of an object in the form of an arrow 5.00 mm tall, perpendicular to the lens axis, 45.0 cm to the left of the lens. (b) A second converging lens with the same focal length is placed 3.15 m to the right of the first. Find the position and size of the final image. Is the final image erect or inverted with respect to the original object? (c) Repeat part (b) except with the second lens 45.0 cm to the right of the first.

**34.85 ••• CP CALC** A transparent cylindrical tube with radius  $r = 1.50$  cm has a flat circular bottom and a top that is convex as seen from above, with radius of curvature of magnitude 2.50 cm. The cylinder is filled with quinine, a colorless highly refractive liquid with index of refraction  $n = 1.627$ . Near the bottom of the tube, immersed in the liquid, is a luminescent LED display mounted on a platform whose height may be varied. The display is the letter A inside a circle that has a diameter of 1.00 cm. A real image of this display is formed at a height  $s'$  above the top of the tube, as shown in **Fig. P34.85**. (a) What is the minimum tube height  $H$  for which this display apparatus can function? (b) The luminescent object is moved up and down periodically so that the real image moves up and down in the air above the tube. A mist in the air renders this display visible and dramatic. If we want the image to move from 50.0 cm above the top of the cylinder to 1.00 m above the top of the cylinder during this motion, what is the corresponding range of motion for the object distance  $s$ ? (c) What is the height  $s'$  of the image when the object is halfway through its motion? (d) The object in part (b) is moved sinusoidally according to  $s = A\sin(\omega t)$ , where  $A$  is the amplitude and  $\omega$  is the angular frequency. The frequency of the motion is 1.00 Hz. What is the velocity of the image when the object is at its midpoint and traveling upward? (e) What is the diameter of the image when it is at its largest size?

Figure P34.85



**34.86 •** An object is placed 22.0 cm from a screen. (a) At what two points between object and screen may a converging lens with a 3.00 cm focal length be placed to obtain an image on the screen? (b) What is the magnification of the image for each position of the lens?

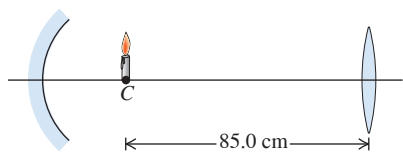
**34.87 •• BIO** A person with a near point of 85 cm, but excellent distant vision, normally wears corrective glasses. But he loses them while traveling. Fortunately, he has his old pair as a spare. (a) If the lenses of the old pair have a power of +2.25 diopters, what is his near point (measured from his eye) when he is wearing the old glasses if they rest 2.0 cm in front of his eye? (b) What would his near point be if his old glasses were contact lenses with the same power instead?

**34.88 ••** A screen is placed a distance  $d$  to the right of an object. A converging lens with focal length  $f$  is placed between the object and the screen. In terms of  $f$ , what is the smallest value  $d$  can have for an image to be in focus on the screen?

**34.89 ••** As shown in **Fig. P34.89** (next page), the candle is at the center of curvature of the concave mirror, whose focal length is 10.0 cm. The converging lens has a focal length of 32.0 cm and is 85.0 cm to the right of the candle. The candle is viewed looking through the lens from the right. The lens forms two images of the candle. The first is formed by light passing directly through the lens. The second image is formed from the light that goes from the candle to the mirror, is reflected, and then passes through the lens. (a) For each image, draw a principal-ray diagram that locates the image. (b) For each image, answer the following questions: (i) Where is the image? (ii) Is the image real or virtual? (iii) Is the image erect or inverted with respect to the original object?



Figure P34.89



**34.90 •• Two Lenses in Contact.** (a) Prove that when two thin lenses with focal lengths  $f_1$  and  $f_2$  are placed *in contact*, the focal length of the combination is given by the relationship

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

(b) A converging meniscus lens (see Fig. 34.32a) has an index of refraction of 1.55 and radii of curvature for its surfaces of magnitudes 4.50 cm and 9.00 cm. The concave surface is placed upward and filled with carbon tetrachloride ( $\text{CCl}_4$ ), which has  $n = 1.46$ . What is the focal length of the  $\text{CCl}_4$ -glass combination?

**34.91 •••** When an object is placed at the proper distance to the left of a converging lens, the image is focused on a screen 30.0 cm to the right of the lens. A diverging lens is now placed 15.0 cm to the right of the converging lens, and it is found that the screen must be moved 19.2 cm farther to the right to obtain a sharp image. What is the focal length of the diverging lens?

**34.92 ••** (a) Repeat the derivation of Eq. (34.19) for the case in which the lens is totally immersed in a liquid of refractive index  $n_{\text{liq}}$ . (b) A lens is made of glass that has refractive index 1.60. In air, the lens has focal length +18.0 cm. What is the focal length of this lens if it is totally immersed in a liquid that has refractive index 1.42?

**34.93 •••** A convex spherical mirror with a focal length of magnitude 24.0 cm is placed 20.0 cm to the left of a plane mirror. An object 0.250 cm tall is placed midway between the surface of the plane mirror and the vertex of the spherical mirror. The spherical mirror forms multiple images of the object. Where are the two images of the object formed by the spherical mirror that are closest to the spherical mirror, and how tall is each image?

**34.94 •• BIO What Is the Smallest Thing We Can See?** The smallest object we can resolve with our eye is limited by the size of the light receptor cells in the retina. In order for us to distinguish any detail in an object, its image cannot be any smaller than a single retinal cell. Although the size depends on the type of cell (rod or cone), a diameter of a few microns ( $\mu\text{m}$ ) is typical near the center of the eye. We shall model the eye as a sphere 2.50 cm in diameter with a single thin lens at the front and the retina at the rear, with light receptor cells 5.0  $\mu\text{m}$  in diameter. (a) What is the smallest object you can resolve at a near point of 25 cm? (b) What angle is subtended by this object at the eye? Express your answer in units of minutes ( $1^\circ = 60 \text{ min}$ ), and compare it with the typical experimental value of about 1.0 min. (Note: There are other limitations, such as the bending of light as it passes through the pupil, but we shall ignore them here.)

**34.95 •** Three thin lenses, each with a focal length of 40.0 cm, are aligned on a common axis; adjacent lenses are separated by 52.0 cm. Find the position of the image of a small object on the axis, 80.0 cm to the left of the first lens.

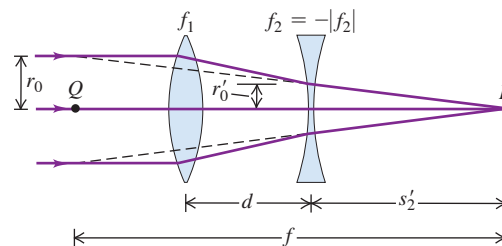
**34.96 ••** A camera with a 90-mm-focal-length lens is focused on an object 1.30 m from the lens. To refocus on an object 6.50 m from the lens, by how much must the distance between the lens and the sensor be changed? To refocus on the more distant object, is the lens moved toward or away from the sensor?

**34.97 •• BIO** In one form of cataract surgery the person's natural lens, which has become cloudy, is replaced by an artificial lens. The refracting properties of the replacement lens can be chosen so that the person's eye focuses on distant objects. But there is no accommodation, and glasses or contact lenses are needed for close vision. What is the power, in diopters, of the corrective contact lenses that will enable a person who has had such surgery to focus on the page of a book at a distance of 24 cm?

**34.98 •• BIO A Nearsighted Eye.** A certain very nearsighted person cannot focus on anything farther than 36.0 cm from the eye. Consider the simplified model of the eye described in Exercise 34.52. If the radius of curvature of the cornea is 0.75 cm when the eye is focusing on an object 36.0 cm from the cornea vertex and the indexes of refraction are as described in Exercise 34.52, what is the distance from the cornea vertex to the retina? What does this tell you about the shape of the nearsighted eye?

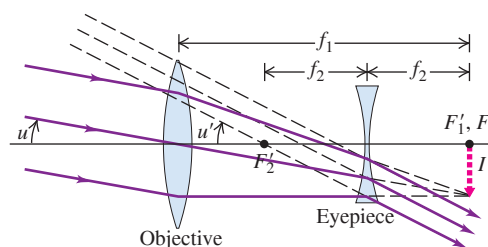
**34.99 ••• Focal Length of a Zoom Lens.** Figure P34.99 shows a simple version of a zoom lens. The converging lens has focal length  $f_1$  and the diverging lens has focal length  $f_2 = -|f_2|$ . The two lenses are separated by a variable distance  $d$  that is always less than  $f_1$ . Also, the magnitude of the focal length of the diverging lens satisfies the inequality  $|f_2| > (f_1 - d)$ . To determine the effective focal length of the combination lens, consider a bundle of parallel rays of radius  $r_0$  entering the converging lens. (a) Show that the radius of the ray bundle decreases to  $r'_0 = r_0(f_1 - d)/f_1$  at the point that it enters the diverging lens. (b) Show that the final image  $I'$  is formed a distance  $s'_2 = |f_2|(f_1 - d)/(|f_2| - f_1 + d)$  to the right of the diverging lens. (c) If the rays that emerge from the diverging lens and reach the final image point are extended backward to the left of the diverging lens, they will eventually expand to the original radius  $r_0$  at some point  $Q$ . The distance from the final image  $I'$  to the point  $Q$  is the *effective focal length*  $f$  of the lens combination; if the combination were replaced by a single lens of focal length  $f$  placed at  $Q$ , parallel rays would still be brought to a focus at  $I'$ . Show that the effective focal length is given by  $f = f_1|f_2|/(|f_2| - f_1 + d)$ . (d) If  $f_1 = 12.0 \text{ cm}$ ,  $f_2 = -18.0 \text{ cm}$ , and the separation  $d$  is adjustable between 0 and 4.0 cm, find the maximum and minimum focal lengths of the combination. What value of  $d$  gives  $f = 30.0 \text{ cm}$ ?

Figure P34.99



**34.100 •• The Galilean Telescope.** Figure P34.100 is a diagram of a *Galilean telescope*, or *opera glass*, with both the object and its final image at infinity. The image  $I$  serves as a virtual object for the eyepiece. The final image is virtual and erect. (a) Prove that the angular magnification is  $M = -f_1/f_2$ . (b) A Galilean telescope is to be constructed with the same objective lens as in Exercise 34.61. What focal length should the eyepiece have if this telescope is to have the same magnitude of angular magnification as the one in Exercise 34.61? (c) Compare the lengths of the telescopes.

Figure P34.100



**34.101 •• DATA** It is your first day at work as a summer intern at an optics company. Your supervisor hands you a diverging lens and asks you to measure its focal length. You know that with a *converging* lens, you can measure the focal length by placing an object a distance  $s$  to the left of the lens, far enough from the lens for the image to be real, and viewing the image on a screen that is to the right of the lens. By adjusting the position of the screen until the image is in sharp focus, you can determine the image distance  $s'$  and then use Eq. (34.16) to calculate the focal length  $f$  of the lens. But this procedure won't work with a diverging lens—by itself, a diverging lens produces only virtual images, which can't be projected onto a screen. Therefore, to determine the focal length of a diverging lens, you do the following: First you take a *converging* lens and measure that, for an object 20.0 cm to the left of the lens, the image is 29.7 cm to the right of the lens. You then place a *diverging* lens 20.0 cm to the right of the converging lens and measure the final image to be 42.8 cm to the right of the converging lens. Suspecting some inaccuracy in measurement, you repeat the lens-combination measurement with the same object distance for the converging lens but with the diverging lens 25.0 cm to the right of the converging lens. You measure the final image to be 31.6 cm to the right of the converging lens. (a) Use both lens-combination measurements to calculate the focal length of the diverging lens. Take as your best experimental value for the focal length the average of the two values. (b) Which position of the diverging lens, 20.0 cm to the right or 25.0 cm to the right of the converging lens, gives the tallest image?

**34.102 •• DATA** In setting up an experiment for a high school biology lab, you use a concave spherical mirror to produce real images of a 4.00-mm-tall firefly. The firefly is to the right of the mirror, on the mirror's optic axis, and serves as a real object for the mirror. You want to determine how far the object must be from the mirror's vertex (that is, object distance  $s$ ) to produce an image of a specified height. First you place a square of white cardboard to the right of the object and find what its distance from the vertex needs to be so that the image is sharply focused on it. Next you measure the height of the sharply focused images for five values of  $s$ . For each  $s$  value, you then calculate the lateral magnification  $m$ . You find that if you graph your data with  $s$  on the vertical axis and  $1/m$  on the horizontal axis, then your measured points fall close to a straight line. (a) Explain why the data plotted this way should fall close to a straight line. (b) Use the graph in **Fig. P34.102** to calculate the focal length of the mirror. (c) How far from the mirror's vertex should you place the object in order for the image to be real, 8.00 mm tall, and inverted? (d) According to **Fig. P34.102**, starting from the position that you calculated in part (c), should you move the object closer to the mirror or farther from it to increase the height of the inverted, real image? What distance should you move the object in order to increase the image height from 8.00 mm to 12.00 mm? (e) Explain why  $1/m$  approaches zero as  $s$  approaches 25 cm. Can you produce a sharp image on the cardboard when  $s = 25$  cm? (f) Explain why you can't see sharp images on the cardboard when  $s < 25$  cm (and  $m$  is positive).

**34.103 •• DATA** The science museum where you work is constructing a new display. You are given a glass rod that is surrounded by air and was ground on its left-hand end to form a hemispherical surface there. You must determine the radius of curvature of that surface and the index of refraction of the glass. Remembering the optics portion of your physics course, you place a small object to the left of the rod, on the rod's optic axis, at a distance  $s$  from the vertex of the hemispherical

surface. You measure the distance  $s'$  of the image from the vertex of the surface, with the image being to the right of the vertex. Your measurements are as follows:

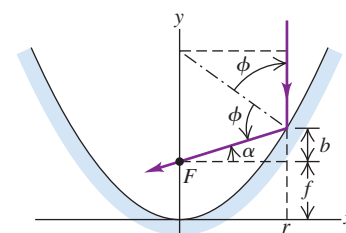
| $s$ (cm)  | 22.5  | 25.0  | 30.0 | 35.0 | 40.0 | 45.0 |
|-----------|-------|-------|------|------|------|------|
| $s'$ (cm) | 271.6 | 148.3 | 89.4 | 71.1 | 60.8 | 53.2 |

Recalling that the object–image relationships for thin lenses and spherical mirrors include reciprocals of distances, you plot your data as  $1/s'$  versus  $1/s$ . (a) Explain why your data points plotted this way lie close to a straight line. (b) Use the slope and y-intercept of the best-fit straight line to your data to calculate the index of refraction of the glass and the radius of curvature of the hemispherical surface of the rod. (c) Where is the image if the object distance is 15.0 cm?

## CHALLENGE PROBLEMS

**34.104 •••** In a spherical mirror, rays parallel to but relatively distant from the optic axis do not reflect precisely to the focal point, and this causes spherical aberration of images. In parabolic mirrors, by contrast, all paraxial rays that enter the mirror, arbitrarily distant from the optic axis, converge to a focal point on the optic axis without any approximation whatsoever. We can prove this and determine the

Figure P34.104



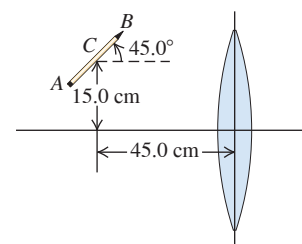
location of the focal point by considering **Fig. P34.104**. The mirror is a parabola defined by  $y = ax^2$ , with  $a$  in units of  $(\text{distance})^{-1}$ , rotated around the  $y$ -axis. Consider the ray that enters the mirror a distance  $r$  from the axis. (a) What is the slope of the parabola at the point where the ray strikes the mirror, in terms of  $a$  and  $r$ ? (b) What is the slope of the dashed line that is normal to the curve at that point? (c) What is the angle  $\phi$ ? (d) Using trigonometry, find the angle  $\alpha$  in terms of  $\phi$ . (e) Using trigonometry, find the distance  $b$  in terms of  $a$  and  $r$ . [Hint: The identities  $\tan(\theta - \pi/2) = -\cot\theta$  and  $\cot(2\theta) = (\cot^2\theta - 1)/2\cot\theta$  may be useful.] (f) Find the distance  $f$  in terms of  $a$ . Note that your answer does not depend on the distance  $r$ ; this proves our assertion.

**34.105 ••• CALC** (a) For a lens with focal length  $f$ , find the smallest distance possible between the object and its real image. (b) Graph the distance between the object and the real image as a function of the distance of the object from the lens. Does your graph agree with the result you found in part (a)?

**34.106 ••• An Object at an Angle.**

A 16.0-cm-long pencil is placed at a  $45.0^\circ$  angle, with its center 15.0 cm above the optic axis and 45.0 cm from a lens with a 20.0 cm focal length as shown in **Fig. P34.106**. (Note that the figure is not drawn to scale.) Assume that the diameter of the lens is large enough for the paraxial approximation to be valid. (a) Where is the image of the pencil? (Give the location of the images of the points A, B, and C on the object, which are located at the eraser, point, and center of the pencil, respectively.) (b) What is the length of the image (that is, the distance between the images of points A and B)? (c) Show the orientation of the image in a sketch.

Figure P34.106



**34.107** ••• **BIO** People with normal vision cannot focus their eyes underwater if they aren't wearing a face mask or goggles and there is water in contact with their eyes (see Discussion Question Q34.23). (a) Why not? (b) With the simplified model of the eye described in Exercise 34.52, what corrective lens (specified by focal length as measured in air) would be needed to enable a person underwater to focus an infinitely distant object? (Be careful—the focal length of a lens underwater is *not* the same as in air! See Problem 34.92. Assume that the corrective lens has a refractive index of 1.62 and that the lens is used in eyeglasses, not goggles, so there is water on both sides of the lens. Assume that the eyeglasses are 2.00 cm in front of the eye.)

### MCAT-STYLE PASSAGE PROBLEMS

**BIO Amphibian Vision.** The eyes of amphibians such as frogs have a much flatter cornea but a more strongly curved (almost spherical) lens than do the eyes of air-dwelling mammals. In mammalian eyes, the shape (and therefore the focal length) of the lens changes to enable the eye to focus at different distances. In amphibian eyes, the shape of the lens doesn't change. Amphibians focus on objects at different distances by using specialized muscles to move the lens closer to or farther from the retina, like the focusing mechanism of a camera. In air, most frogs are nearsighted; correcting the distance vision of a typical frog in air would require contact lenses with a power of about  $-6.0$  D.

**34.108** A frog can see an insect clearly at a distance of 10 cm. At that point the effective distance from the lens to the retina is 8 mm. If the insect moves 5 cm farther from the frog, by how much and in which direction does the lens of the frog's eye have to move to keep the insect in focus? (a) 0.02 cm, toward the retina; (b) 0.02 cm, away from the retina; (c) 0.06 cm, toward the retina; (d) 0.06 cm, away from the retina.

**34.109** What is the farthest distance at which a typical “nearsighted” frog can see clearly in air? (a) 12 m; (b) 6.0 m; (c) 80 cm; (d) 17 cm.

**34.110** Given that frogs are nearsighted in air, which statement is most likely to be true about their vision in water? (a) They are even more nearsighted; because water has a higher index of refraction than air, a frog's ability to focus light increases in water. (b) They are less nearsighted, because the cornea is less effective at refracting light in water than in air. (c) Their vision is no different, because only structures that are internal to the eye can affect the eye's ability to focus. (d) The images projected on the retina are no longer inverted, because the eye in water functions as a diverging lens rather than a converging lens.

**34.111** To determine whether a frog can judge distance by means of the amount its lens must move to focus on an object, researchers covered one eye with an opaque material. An insect was placed in front of the frog, and the distance that the frog snapped its tongue out to catch the insect was measured with high-speed video. The experiment was repeated with a contact lens over the eye to determine whether the frog could correctly judge the distance under these conditions. If such an experiment is performed twice, once with a lens of power  $-9$  D and once with a lens of power  $-15$  D, in which case does the frog have to focus at a shorter distance, and why? (a) With the  $-9$  D lens; because the lenses are diverging, the lens with the longer focal length creates an image that is closer to the frog. (b) With the  $-15$  D lens; because the lenses are diverging, the lens with the shorter focal length creates an image that is closer to the frog. (c) With the  $-9$  D lens; because the lenses are converging, the lens with the longer focal length creates a larger real image. (d) With the  $-15$  D lens; because the lenses are converging, the lens with the shorter focal length creates a larger real image.

## ANSWERS

### Chapter Opening Question ?

(ii) A magnifying lens (simple magnifier) produces a virtual image with a large angular size that is infinitely far away, so you can see it in sharp focus with your eyes relaxed. (A surgeon would not appreciate having to strain her eyes while working.) The object should be at the focal point of the lens, so the object and lens are separated by one focal length. The distance from magnifier to eye is not crucial.

### Key Example VARIATION Problems

**VP34.4.1** (a) 14.5 cm (b) 29.0 cm (c)  $-30.0$

**VP34.4.2** (a)  $s' = -27.1$  cm,  $m = +2.47$ , virtual, erect, larger

(b)  $s' = +45.9$  cm,  $m = -1.48$ , real, inverted, larger

(c)  $s' = +27.9$  cm,  $m = -0.507$ , real, inverted, smaller

**VP34.4.3** (a) convex (b) 99.0 cm (c)  $+0.182$ , virtual, erect, smaller

**VP34.4.4** (a)  $s' = -6.90$  cm,  $m = +0.627$ , virtual, erect, smaller

(b)  $s' = -11.6$  cm,  $m = +0.374$ , virtual, erect, smaller

(c)  $s' = -13.8$  cm,  $m = +0.252$ , virtual, erect, smaller

**VP34.8.1** (a)  $+39.0$  cm (b)  $-39.0$  cm

**VP34.8.2** (a) center (b)  $+68$  cm (c) converging

**VP34.8.3** (a) edges (b)  $-68$  cm (c) diverging

**VP34.8.4** (a)  $-15$  cm (b) convex

**VP34.10.1** (a)  $-37.5$  cm, same side (b)  $+2.50$  (c) virtual, erect, larger

**VP34.10.2** (a)  $+16.8$  cm, converging (b)  $-1.50$  (c) real, inverted, larger

**VP34.10.3** (a) converging (b)  $+48.0$  cm (c)  $+24.0$  cm

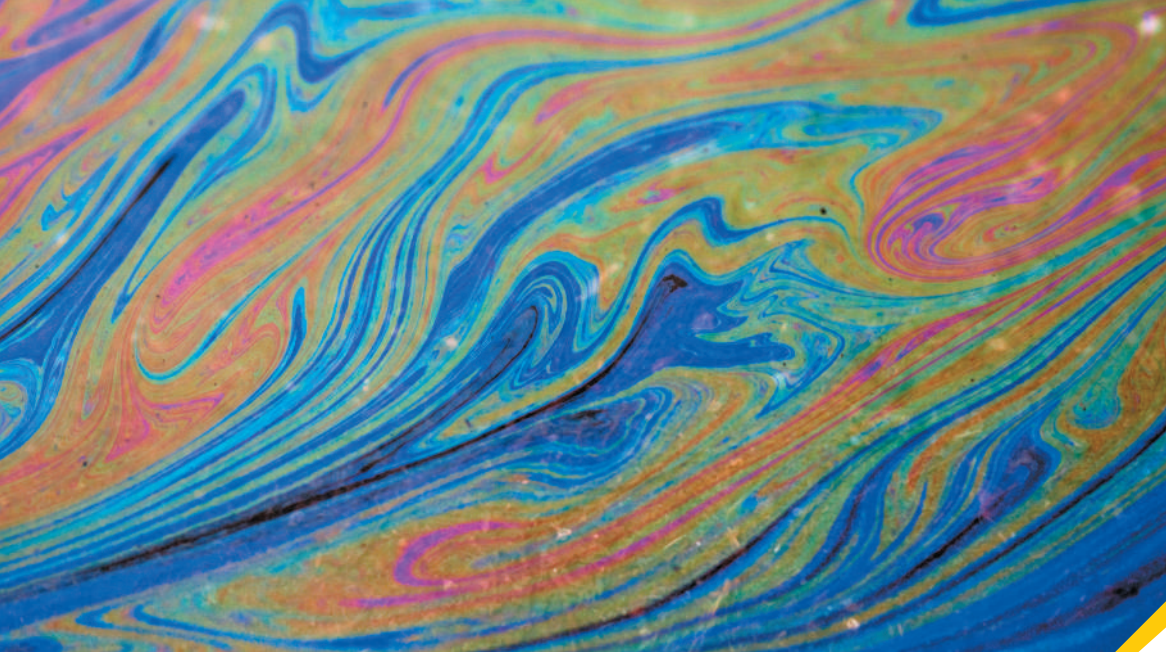
**VP34.10.4** (a)  $+0.200$  (b)  $-50.0$  cm, same side (c)  $-62.5$  cm, diverging

### Bridging Problem

(a) 29.9 cm to the left of the goblet

(b) 3.73 cm to the right of the goblet





**?** When white light shines downward on a thin, horizontal layer of oil, light waves reflected from the upper and lower surfaces of the film of oil interfere, producing vibrant colors. The color that you see reflected from a certain spot on the film depends on (i) the film thickness at that spot; (ii) the index of refraction of the oil; (iii) the index of refraction of the material below the oil; (iv) both (i) and (ii); (v) all of (i), (ii), and (iii).

# 35 Interference

**A**n ugly black oil spot on the pavement can become a thing of beauty after a rain, when the oil reflects a rainbow of colors. Multicolored reflections can also be seen from the surfaces of soap bubbles and DVDs. How is it possible for colorless objects to produce these remarkable colors?

In our discussion of lenses, mirrors, and optical instruments we used the model of *geometric optics*, in which we represent light as *rays*, straight lines that are bent at a reflecting or refracting surface. But light is fundamentally a *wave*, and in some situations we have to consider its wave properties explicitly. If two or more light waves of the same frequency overlap at a point, the total effect depends on the *phases* of the waves as well as their amplitudes. The resulting patterns of light are a result of the *wave* nature of light and cannot be understood on the basis of rays. Optical effects that depend on the wave nature of light are grouped under the heading **physical optics**.

In this chapter we'll look at *interference* phenomena that occur when two waves combine. Effects that occur when *many* sources of waves are present are called *diffraction* phenomena; we'll study these in Chapter 36. In that chapter we'll see that diffraction effects occur whenever a wave passes through an aperture or around an obstacle. They are important in practical applications of physical optics such as diffraction gratings, x-ray diffraction, and holography.

While our primary concern is with light, interference and diffraction can occur with waves of *any* kind. As we go along, we'll point out applications to other types of waves such as sound and water waves.

## LEARNING OUTCOMES

### In this chapter, you'll learn...

- 35.1** What happens when two waves combine, or interfere, in space.
- 35.2** How to understand the interference pattern formed by the interference of two coherent light waves.
- 35.3** How to calculate the intensity at various points in an interference pattern.
- 35.4** How interference occurs when light reflects from the two surfaces of a thin film.
- 35.5** How interference makes it possible to measure extremely small distances.

### You'll need to review...

- 14.2, 31.1** Phasors.
- 15.3, 15.6, 15.7** Wave number, wave superposition, standing waves on a string.
- 16.4** Standing sound waves.
- 32.1, 32.4, 32.5** Electromagnetic spectrum, wave intensity, standing electromagnetic waves.

## 35.1 INTERFERENCE AND COHERENT SOURCES

As we discussed in Chapter 15, the term **interference** refers to any situation in which two or more waves overlap in space. When this occurs, the total wave at any point at any instant of time is governed by the **principle of superposition**, which we introduced in



Section 15.6 in the context of waves on a string. This principle also applies to electromagnetic waves and is the most important principle in all of physical optics.

**PRINCIPLE OF SUPERPOSITION** When two or more waves overlap, the resultant displacement at any point and at any instant is found by adding the instantaneous displacements that would be produced at the point by the individual waves if each were present alone.

(In some special situations, such as electromagnetic waves propagating in a crystal, this principle may not apply. A discussion of these situations is beyond our scope.)

We use the term “displacement” in a general sense. With waves on the surface of a liquid, we mean the actual displacement of the surface above or below its normal level. With sound waves, the term refers to the excess or deficiency of pressure. For electromagnetic waves, we usually mean a specific component of electric or magnetic field.

### Interference in Two or Three Dimensions

We have already discussed one important case of interference, in which two identical waves propagating in opposite directions combine to produce a *standing wave*. We saw this in Section 15.7 for transverse waves on a string and in Section 16.4 for longitudinal waves in a fluid filling a pipe; we described the same phenomenon for electromagnetic waves in Section 32.5. In all of these cases the waves propagated along only a single axis: along a string, along the length of a fluid-filled pipe, or along the propagation direction of an electromagnetic plane wave. But light waves can (and do) travel in *two* or *three* dimensions, as can any kind of wave that propagates in a two- or three-dimensional medium. In this section we’ll see what happens when we combine waves that spread out in two or three dimensions from a pair of identical wave sources.

Interference effects are most easily seen when we combine *sinusoidal* waves with a single frequency  $f$  and wavelength  $\lambda$ . **Figure 35.1** shows a “snapshot” of a *single* source  $S_1$  of sinusoidal waves and some of the wave fronts produced by this source. The figure shows only the wave fronts corresponding to wave *crests*, so the spacing between successive wave fronts is one wavelength. The material surrounding  $S_1$  is uniform, so the wave speed is the same in all directions, and there is no refraction (and hence no bending of the wave fronts). If the waves are two-dimensional, like waves on the surface of a liquid, the circles in Fig. 35.1 represent circular wave fronts; if the waves propagate in three dimensions, the circles represent spherical wave fronts spreading away from  $S_1$ .

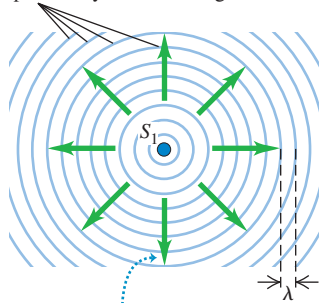
In optics, sinusoidal waves are characteristic of **monochromatic light** (light of a single color). While it’s fairly easy to make water waves or sound waves of a single frequency, common sources of light *do not* emit monochromatic (single-frequency) light. For example, incandescent light bulbs and flames emit a continuous distribution of wavelengths. By far the most nearly monochromatic light source is the *laser*. An example is the helium–neon laser, which emits red light at 632.8 nm with a wavelength range of the order of  $\pm 0.000001$  nm, or about one part in  $10^9$ . In this chapter and the next, we’ll assume that we are working with monochromatic waves (unless we explicitly state otherwise).

### Constructive and Destructive Interference

**Figure 35.2a** shows two identical sources of monochromatic waves,  $S_1$  and  $S_2$ . The two sources produce waves of the same amplitude and the same wavelength  $\lambda$ . In addition, the two sources are permanently *in phase*; they vibrate in unison. They might be two loudspeakers driven by the same amplifier, two radio antennas powered by the same transmitter, or two small slits in an opaque screen, illuminated by the same monochromatic light source. We’ll see that if there were not a constant phase relationship between the two sources, the phenomena we are about to discuss would not occur. Two monochromatic sources of the same frequency and with a constant phase relationship (not necessarily in phase) are said to be **coherent**. We also use the term *coherent waves* (or, for light waves, *coherent light*) to refer to the waves emitted by two such sources.

Figure 35.1 A “snapshot” of sinusoidal waves of frequency  $f$  and wavelength  $\lambda$  spreading out from source  $S_1$  in all directions.

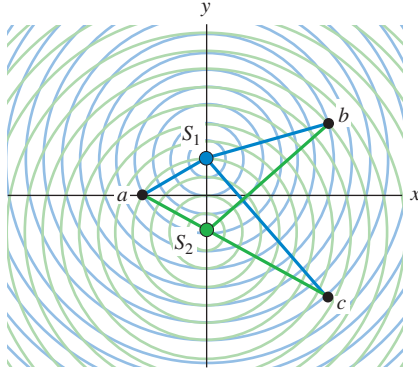
Wave fronts: crests of the wave (frequency  $f$ ) separated by one wavelength  $\lambda$



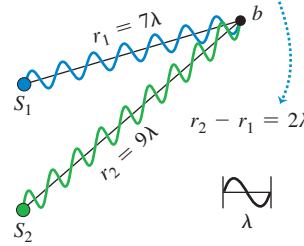
The wave fronts move outward from source  $S_1$  at the wave speed  $v = f\lambda$ .

Figure 35.2 (a) A “snapshot” of sinusoidal waves spreading out from two coherent sources  $S_1$  and  $S_2$ . Constructive interference occurs at point  $a$  (equidistant from the two sources) and (b) at point  $b$ . (c) Destructive interference occurs at point  $c$ .

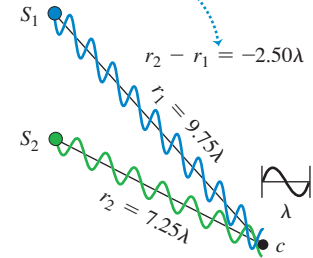
(a) Two coherent wave sources separated by a distance  $4\lambda$



(b) Conditions for constructive interference: Waves interfere constructively if their path lengths differ by an integer number of wavelengths:  $r_2 - r_1 = m\lambda$ .



(c) Conditions for destructive interference: Waves interfere destructively if their path lengths differ by a half-integer number of wavelengths:  $r_2 - r_1 = (m + \frac{1}{2})\lambda$ .



If the waves emitted by the two coherent sources are *transverse*, like electromagnetic waves, then we'll also assume that the wave disturbances produced by both sources have the same *polarization* (that is, they lie along the same line). For example, the sources  $S_1$  and  $S_2$  in Fig. 35.2a could be two radio antennas in the form of long rods oriented parallel to the  $z$ -axis (perpendicular to the plane of the figure); at any point in the  $xy$ -plane the waves produced by both antennas have  $\vec{E}$  fields with only a  $z$ -component. Then we need only a single scalar function to describe each wave; this makes the analysis much easier.

We position the two sources of equal amplitude, equal wavelength, and (if the waves are transverse) the same polarization along the  $z$ -axis in Fig. 35.2a, equidistant from the origin. Consider a point  $a$  on the  $x$ -axis. From symmetry the two distances from  $S_1$  to  $a$  and from  $S_2$  to  $a$  are *equal*; waves from the two sources thus require equal times to travel to  $a$ . Hence waves that leave  $S_1$  and  $S_2$  in phase arrive at  $a$  in phase, and the total amplitude at  $a$  is *twice* the amplitude of each individual wave. This is true for *any* point on the  $x$ -axis.

Similarly, the distance from  $S_2$  to point  $b$  is exactly two wavelengths *greater* than the distance from  $S_1$  to  $b$ . A wave crest from  $S_1$  arrives at  $b$  exactly two cycles earlier than a crest emitted at the same time from  $S_2$ , and again the two waves arrive in phase. As at point  $a$ , the total amplitude is the sum of the amplitudes of the waves from  $S_1$  and  $S_2$ .

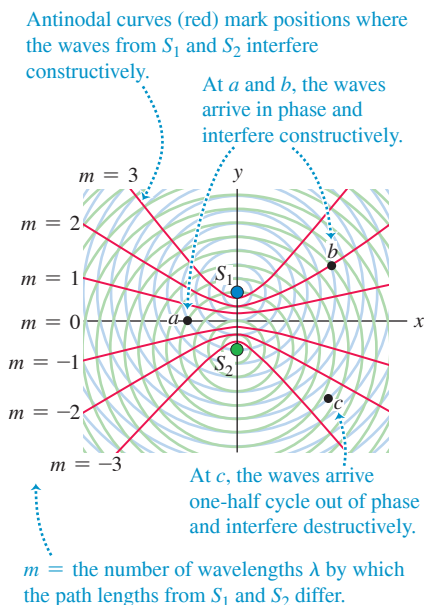
In general, when waves from two or more sources arrive at a point *in phase*, they reinforce each other: The amplitude of the resultant wave is the *sum* of the amplitudes of the individual waves. This is called **constructive interference** (Fig. 35.2b). Let the distance from  $S_1$  to any point  $P$  be  $r_1$ , and let the distance from  $S_2$  to  $P$  be  $r_2$ . For constructive interference to occur at  $P$ , the path difference  $r_2 - r_1$  for the two sources must be an integer multiple of the wavelength  $\lambda$ :

$$r_2 - r_1 = m\lambda \quad (m = 0, \pm 1, \pm 2, \pm 3, \dots) \quad \begin{array}{l} \text{(constructive} \\ \text{interference,} \\ \text{sources in phase)} \end{array} \quad (35.1)$$

In Fig. 35.2a, points  $a$  and  $b$  satisfy Eq. (35.1) with  $m = 0$  and  $m = +2$ , respectively.

Something different occurs at point  $c$  in Fig. 35.2a. At this point, the path difference  $r_2 - r_1 = -2.50\lambda$ , which is a *half-integer* number of wavelengths. Waves from the two sources arrive at point  $c$  exactly a half-cycle out of phase. A crest of one wave arrives at the same time as a crest in the opposite direction (a “trough”) of the other wave (Fig. 35.2c). The resultant amplitude is the *difference* between the two individual amplitudes. If the individual amplitudes are equal, then the total amplitude is *zero*! This cancellation or partial

Figure 35.3 The same as Fig. 35.2a, but with red antinodal curves (curves of maximum amplitude) superimposed. All points on each curve satisfy Eq. (35.1) with the value of  $m$  shown. The nodal curves (not shown) lie between each adjacent pair of antinodal curves.



**BIO APPLICATION** **Phase Difference, Path Difference, and Localization in Human Hearing** Your auditory system uses the phase differences between sounds received by your left and your right ears for *localization*—determining the direction from which the sounds are coming. For sound waves with frequencies lower than about 800 Hz (which are important in speech and music), the distance between your ears is less than a half-wavelength and the phase difference between the waves detected by each ear is less than a half cycle. Remarkably, your brain can detect this phase difference, determine the corresponding path difference, and use this information to localize the direction of the sound source.



cancellation of the individual waves is called **destructive interference**. The condition for destructive interference in the situation shown in Fig. 35.2a is

$$r_2 - r_1 = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, \pm 1, \pm 2, \pm 3, \dots) \quad \begin{array}{l} \text{(destructive} \\ \text{interference,} \\ \text{sources in phase)} \end{array} \quad (35.2)$$

The path difference at point  $c$  in Fig. 35.2a satisfies Eq. (35.2) with  $m = -3$ .

**Figure 35.3** shows the same situation as in Fig. 35.2a, but with red curves that show all positions where *constructive* interference occurs. On each curve, the path difference  $r_2 - r_1$  is equal to an integer  $m$  times the wavelength, as in Eq. (35.1). These curves are called **antinodal curves**. They are directly analogous to *antinodes* in the standing-wave patterns described in Chapters 15 and 16 and Section 32.5. In a standing wave formed by interference between waves propagating in opposite directions, the antinodes are points at which the amplitude is maximum; likewise, the wave amplitude in the situation of Fig. 35.3 is maximum along the antinodal curves. Not shown in Fig. 35.3 are the **nodal curves**, which are the curves that show where *destructive* interference occurs in accordance with Eq. (35.2); these are analogous to the *nodes* in a standing-wave pattern. A nodal curve lies between each two adjacent antinodal curves in Fig. 35.3; one such curve, corresponding to  $r_2 - r_1 = -2.50\lambda$ , passes through point  $c$ .

In some cases, such as two loudspeakers or two radio-transmitter antennas, the interference pattern is three-dimensional. Think of rotating the color curves of Fig. 35.3 around the  $y$ -axis; then maximum constructive interference occurs at all points on the resulting surfaces of revolution.

**CAUTION** **Interference patterns are not standing waves** In the standing waves described in Sections 15.7, 16.4, and 32.5, the interference is between two waves propagating in opposite directions; there is *no* net energy flow in either direction (the energy in the wave is left “standing”). In the situations shown in Figs. 35.2a and 35.3, there is likewise a stationary pattern of antinodal and nodal curves, but there is a net flow of energy *outward* from the two sources. All that interference does is to “channel” the energy flow so that it is greatest along the antinodal curves and least along the nodal curves. ■

For Eqs. (35.1) and (35.2) to hold, the two sources must be monochromatic, must have the same wavelength, and must *always* be in phase. These conditions are rather easy to satisfy for sound waves. But with *light* waves there is no practical way to achieve a constant phase relationship (coherence) with two independent sources. This is because of the way light is emitted. In ordinary light sources, atoms gain excess energy by thermal agitation or by impact with accelerated electrons. Such an “excited” atom begins to radiate energy and continues until it has lost all the energy it can, typically in a time of the order of  $10^{-8}$  s. The many atoms in a source ordinarily radiate in an unsynchronized and random phase relationship, and the light that is emitted from *two* such sources has no definite phase relationship.

However, the light from a single source can be split so that parts of it emerge from two or more regions of space, forming two or more *secondary sources*. Then any random phase change in the source affects these secondary sources equally and does not change their *relative* phase.

The distinguishing feature of light from a *laser* is that the emission of light from many atoms is synchronized in frequency and phase. As a result, the random phase changes mentioned above occur much less frequently. Definite phase relationships are preserved over correspondingly much greater lengths in the beam, and laser light is much more coherent than ordinary light.

**TEST YOUR UNDERSTANDING OF SECTION 35.1** Consider a point in Fig. 35.3 on the positive  $y$ -axis above  $S_1$ . Does this point lie on (i) an antinodal curve; (ii) a nodal curve; or (iii) neither? (*Hint:* The distance between  $S_1$  and  $S_2$  is  $4\lambda$ .)

**ANSWER** ■ (i) At any point  $P$  on the positive  $y$ -axis above  $S_1$ , the distance  $r_2$  from  $S_2$  to  $P$  is greater than the distance  $r_1$  from  $S_1$  to  $P$  by  $4\lambda$ . This corresponds to  $m = 4$  in Eq. (35.1), the equation for constructive interference. Hence all such points make up an antinodal curve.

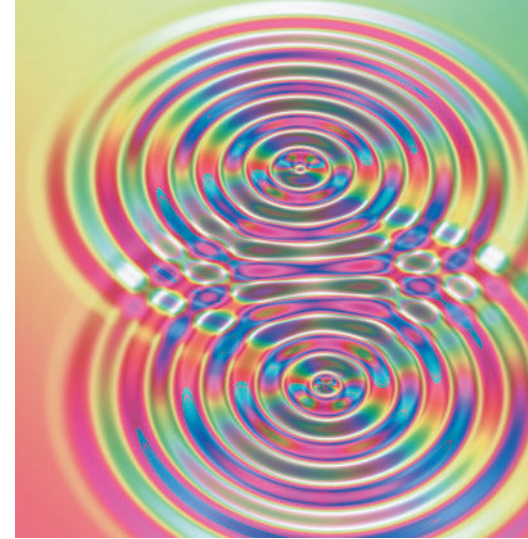
## 35.2 TWO-SOURCE INTERFERENCE OF LIGHT

The interference pattern produced by two coherent sources of *water* waves of the same wavelength can be readily seen in a ripple tank with a shallow layer of water (**Fig. 35.4**). This pattern is not directly visible when the interference is between *light* waves, since light traveling in a uniform medium cannot be seen. (Sunlight in a room is made visible by scattering from airborne dust.)

**Figure 35.5a** shows one of the earliest quantitative experiments to reveal the interference of light from two sources, first performed in 1800 by the English scientist Thomas Young. Let's examine this important experiment in detail. A light source (not shown) emits monochromatic light; however, this light is not suitable for use in an interference experiment because emissions from different parts of an ordinary source are not synchronized. To remedy this, the light is directed at a screen with a narrow slit  $S_0$ ,  $1\text{ }\mu\text{m}$  or so wide. The light emerging from the slit originated from only a small region of the light source; thus slit  $S_0$  behaves more nearly like the idealized source shown in Fig. 35.1. (In modern versions of the experiment, a laser is used as a source of coherent light, and the slit  $S_0$  isn't needed.) The light from slit  $S_0$  falls on a screen with two other narrow slits  $S_1$  and  $S_2$ , each  $1\text{ }\mu\text{m}$  or so wide and a few tens or hundreds of micrometers apart. Cylindrical wave fronts spread out from slit  $S_0$  and reach slits  $S_1$  and  $S_2$  *in phase* because they travel equal distances from  $S_0$ . The waves *emerging* from slits  $S_1$  and  $S_2$  are therefore also always in phase, so  $S_1$  and  $S_2$  are *coherent* sources. The interference of waves from  $S_1$  and  $S_2$  produces a pattern in space like that to the right of the sources in Figs. 35.2a and 35.3.

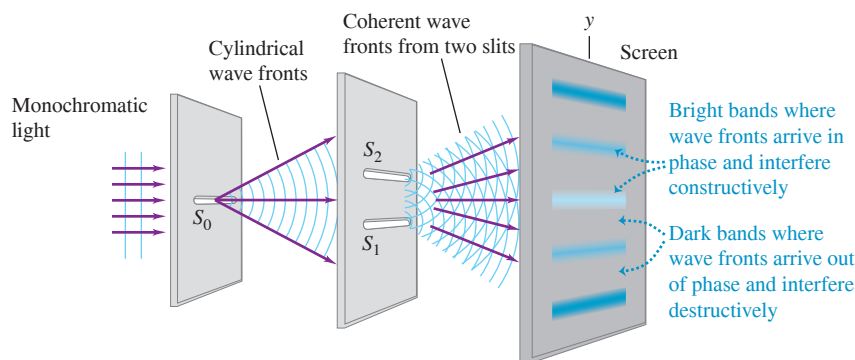
To visualize the interference pattern, a screen is placed so that the light from  $S_1$  and  $S_2$  falls on it (Fig. 35.5b). The screen will be most brightly illuminated at points  $P$ , where the light waves from the slits interfere constructively, and will be darkest at points where the interference is destructive.

**Figure 35.4** The concepts of constructive interference and destructive interference apply to these water waves as well as to light waves and sound waves.

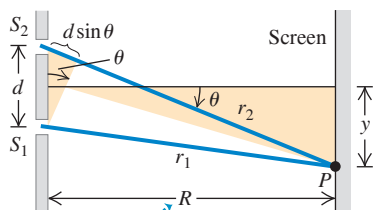


**Figure 35.5** (a) Young's experiment to show interference of light passing through two slits. A pattern of bright and dark areas appears on the screen (see Fig. 35.6). (b) Geometrical analysis of Young's experiment. For the case shown,  $r_2 > r_1$  and both  $y$  and  $\theta$  are positive. If point  $P$  is on the other side of the screen's center,  $r_2 < r_1$  and both  $y$  and  $\theta$  are negative. (c) Approximate geometry when the distance  $R$  to the screen is much greater than the distance  $d$  between the slits.

(a) Interference of light waves passing through two slits

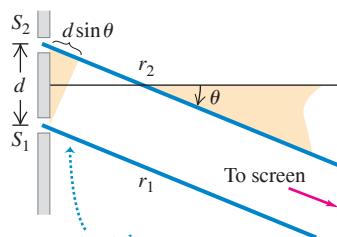


(b) Actual geometry (seen from the side)



In real situations, the distance  $R$  to the screen is usually very much greater than the distance  $d$  between the slits ...

(c) Approximate geometry



... so we can treat the rays as parallel, in which case the path difference is simply  $r_2 - r_1 = d \sin \theta$ .



To simplify the analysis of Young's experiment, we assume that the distance  $R$  from the slits to the screen is so large in comparison to the distance  $d$  between the slits that the lines from  $S_1$  and  $S_2$  to  $P$  are very nearly parallel, as in Fig. 35.5c. This is usually the case for experiments with light; the slit separation is typically a few millimeters, while the screen may be a meter or more away. The difference in path length is then given by

$$r_2 - r_1 = d \sin \theta \quad (35.3)$$

where  $\theta$  is the angle between a line from slits to screen (shown in blue in Fig. 35.5c) and the normal to the plane of the slits (shown as a thin black line).

### Constructive and Destructive Two-Slit Interference

We found in Section 35.1 that constructive interference (reinforcement) occurs at points where the path difference is an integer number of wavelengths,  $m\lambda$ , where  $m = 0, \pm 1, \pm 2, \pm 3, \dots$ . So the bright regions on the screen in Fig. 35.5a occur at angles  $\theta$  for which

**Constructive interference, two slits:**

$$d \sin \theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \quad (35.4)$$

Distance between slits  
Wavelength  
Angle of line from slits to  $m$ th bright region on screen

Similarly, destructive interference (cancellation) occurs, forming dark regions on the screen, at points for which the path difference is a half-integer number of wavelengths,  $(m + \frac{1}{2})\lambda$ :

**Destructive interference, two slits:**

$$d \sin \theta = (m + \frac{1}{2})\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \quad (35.5)$$

Distance between slits  
Wavelength  
Angle of line from slits to  $m$ th dark region on screen

Thus the pattern on the screen of Figs. 35.5a and 35.5b is a succession of bright and dark bands, or **interference fringes**, parallel to the slits  $S_1$  and  $S_2$ . A photograph of such a pattern is shown in Fig. 35.6.

We can derive an expression for the positions of the centers of the bright bands on the screen. In Fig. 35.5b,  $y$  is measured from the center of the pattern, corresponding to the distance from the center of Fig. 35.6. Let  $y_m$  be the distance from the center of the pattern ( $\theta = 0$ ) to the center of the  $m$ th bright band. Let  $\theta_m$  be the corresponding value of  $\theta$ ; then

$$y_m = R \tan \theta_m$$

In such experiments, the distances  $y_m$  are often much smaller than the distance  $R$  from the slits to the screen. Hence  $\theta_m$  is very small,  $\tan \theta_m \approx \sin \theta_m$ , and

$$y_m = R \sin \theta_m$$

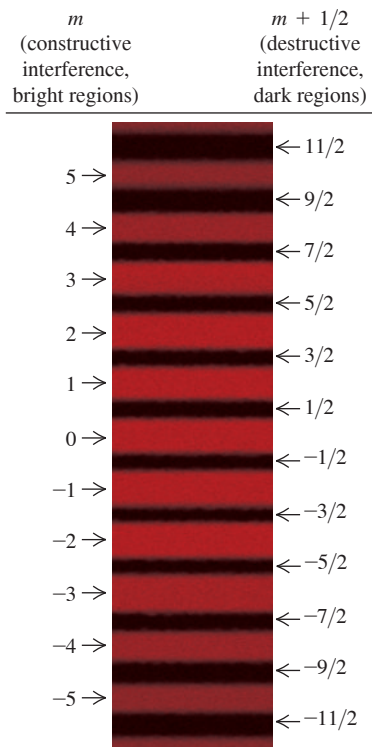
Combining this with Eq. (35.4), we find that for *small angles only*,

**Constructive interference, Young's experiment (small angles only):**

$$y_m = R \frac{m\lambda}{d} \quad (m = 0, \pm 1, \pm 2, \dots) \quad (35.6)$$

Position of  $m$ th bright band  
Wavelength  
Distance from slits to screen  
Distance between slits

Figure 35.6 Photograph of interference fringes produced on a screen in Young's double-slit experiment. The center of the pattern is a bright band corresponding to  $m = 0$  in Eq. (35.4); this point on the screen is equidistant from the two slits.



**CAUTION** Equation (35.6) is for **small angles only**. While Eqs. (35.4) and (35.5) are valid for any angle, Eq. (35.6) is valid for *small* angles only. It can be used *only* if the distance  $R$  from slits to screen is much greater than the slit separation  $d$  and if  $R$  is much greater than the distance  $y_m$  from the center of the interference pattern to the  $m$ th bright band. ■

We can measure  $R$  and  $d$ , as well as the positions  $y_m$  of the bright fringes, so this experiment provides a direct measurement of the wavelength  $\lambda$ . Young's experiment was in fact the first direct measurement of wavelengths of light.

The distance between adjacent bright bands in the pattern is *inversely* proportional to the distance  $d$  between the slits. The closer together the slits are, the more the pattern spreads out. When the slits are far apart, the bands in the pattern are closer together.

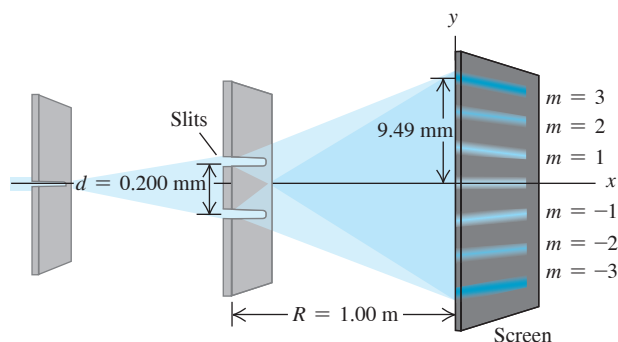
While we have described the experiment that Young performed with visible light, the results given in Eqs. (35.4) and (35.5) are valid for *any* type of wave, provided that the resultant wave from two coherent sources is detected at a point that is far away in comparison to the separation  $d$ .

### EXAMPLE 35.1 Two-slit interference

WITH  **ARIATION PROBLEMS**

**Figure 35.7** shows a two-slit interference experiment in which the slits are 0.200 mm apart and the screen is 1.00 m from the slits. The  $m = 3$  bright fringe in the figure is 9.49 mm from the central bright fringe. Find the wavelength of the light.

Figure 35.7 Using a two-slit interference experiment to measure the wavelength of light.



**IDENTIFY and SET UP** Our target variable in this two-slit interference problem is the wavelength  $\lambda$ . We are given the slit separation  $d = 0.200$  mm, the distance from slits to screen  $R = 1.00$  m, and the distance  $y_3 = 9.49$  mm on the screen from the center of the interference pattern to the  $m = 3$  bright fringe. We may use Eq. (35.6) to find  $\lambda$ , since the value of  $R$  is so much greater than the value of  $d$  or  $y_3$ .

**EXECUTE** We solve Eq. (35.6) for  $\lambda$  for the case  $m = 3$ :

$$\lambda = \frac{y_m d}{m R} = \frac{(9.49 \times 10^{-3} \text{ m})(0.200 \times 10^{-3} \text{ m})}{(3)(1.00 \text{ m})} = 633 \times 10^{-9} \text{ m} = 633 \text{ nm}$$

**EVALUATE** This bright fringe could also correspond to  $m = -3$ . Can you show that this gives the same result for  $\lambda$ ?

**KEYCONCEPT** In two-slit interference, each wave front of a monochromatic wave enters the two slits simultaneously. Hence the waves that emerge from the slits always have the same phase relationship and so produce a steady interference pattern. Equation (35.6) gives the positions of the bright fringes where constructive interference occurs.

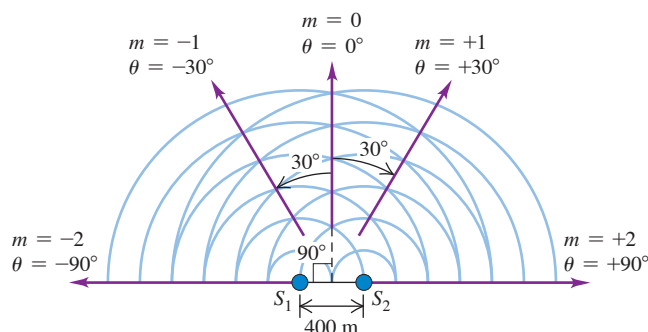
### EXAMPLE 35.2 Broadcast pattern of a radio station

WITH  **ARIATION PROBLEMS**

It is often desirable to radiate most of the energy from a radio transmitter in particular directions rather than uniformly in all directions. Pairs or rows of antennas are often used to produce the desired radiation pattern. As an example, consider two identical vertical antennas 400 m apart, operating at 1500 kHz =  $1.5 \times 10^6$  Hz (near the top end of the AM broadcast band) and oscillating in phase. At distances much greater than 400 m, in what directions is the intensity from the two antennas greatest?

**IDENTIFY and SET UP** The antennas, shown in Fig. 35.8, correspond to sources  $S_1$  and  $S_2$  in Fig. 35.5. Hence we can apply the ideas of two-slit interference to this problem. Since the resultant wave is detected at distances much greater than  $d = 400$  m, we may use Eq. (35.4) to give the directions of the intensity maxima, the values of  $\theta$  for which the path difference is zero or a whole number of wavelengths.

Figure 35.8 Two radio antennas broadcasting in phase. The purple arrows indicate the directions of maximum intensity. The waves that are emitted toward the lower half of the figure are not shown.



**EXECUTE** The wavelength is  $\lambda = c/f = 200$  m. From Eq. (35.4) with  $m = 0, \pm 1$ , and  $\pm 2$ , the intensity maxima are given by

$$\sin \theta = \frac{m \lambda}{d} = \frac{m(200 \text{ m})}{400 \text{ m}} = \frac{m}{2} \quad \theta = 0, \pm 30^\circ, \pm 90^\circ$$

In this example, values of  $m$  greater than 2 or less than  $-2$  give values of  $\sin \theta$  greater than 1 or less than  $-1$ , which is impossible. There is *no* direction for which the path difference is three or more wavelengths, so values of  $m$  of  $\pm 3$  or beyond have no meaning in this example.

**EVALUATE** We can check our result by calculating the angles for *minimum* intensity, using Eq. (35.5). There should be one intensity minimum between each pair of intensity maxima, just as in Fig. 35.6. From Eq. (35.5), with  $m = -2, -1, 0$ , and  $1$ ,

$$\sin \theta = \frac{(m + \frac{1}{2}) \lambda}{d} = \frac{m + \frac{1}{2}}{2} \quad \theta = \pm 14.5^\circ, \pm 48.6^\circ$$

These angles fall between the angles for intensity maxima, as they should. The angles are not small, so the angles for the minima are *not* exactly halfway between the angles for the maxima.

**KEYCONCEPT** Equations (35.4) and (35.5) give the angles in two-slit interference for which constructive interference (bright regions) and destructive interference (dark regions) occur. These equations are also valid for any two monochromatic sources that emit in phase.

**TEST YOUR UNDERSTANDING OF SECTION 35.2** You shine a tunable laser (whose wavelength can be adjusted by turning a knob) on a pair of closely spaced slits. The light emerging from the two slits produces an interference pattern on a screen like that shown in Fig. 35.6. If you adjust the wavelength so that the laser light changes from red to blue, how will the spacing between bright fringes change? (i) The spacing increases; (ii) the spacing decreases; (iii) the spacing is unchanged; (iv) not enough information is given to decide.

### ANSWER

(ii) Blue light has a shorter wavelength than red light (see Section 32.1). Equation (35.6) tells us that the distance  $y_m$  from the center of the pattern to the  $m$ th bright fringe is proportional to the wavelength  $\lambda$ . Hence all of the fringes will move toward the center of the pattern as the wavelength decreases, and the spacing between fringes will decrease.

## 35.3 INTENSITY IN INTERFERENCE PATTERNS

In Section 35.2 we found the positions of maximum and minimum intensity in a two-source interference pattern. Let's now see how to find the intensity at *any* point in the pattern. To do this, we have to combine the two sinusoidally varying fields (from the two sources) at a point  $P$  in the radiation pattern, taking proper account of the phase difference of the two waves at point  $P$ , which results from the path difference. The intensity is then proportional to the square of the resultant electric-field amplitude, as we learned in Section 32.4.

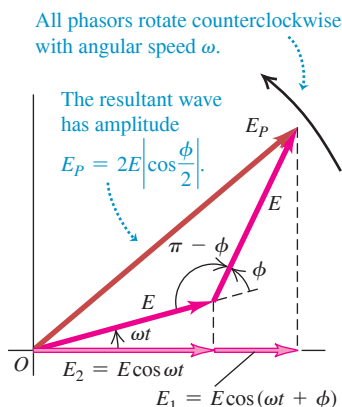
To calculate the intensity, we'll assume, as in Section 35.2, that the waves from the two sources have equal amplitude  $E$  and the same polarization. This assumes that the sources are identical and ignores the slight amplitude difference caused by the unequal path lengths (the amplitude decreases with increasing distance from the source). From Eq. (32.29), each source by itself would give an intensity  $\frac{1}{2}\epsilon_0 c E^2$  at point  $P$ . If the two sources are in phase, then the waves that arrive at  $P$  differ in phase by an amount  $\phi$  that is proportional to the difference in their path lengths,  $(r_2 - r_1)$ . Then we can use the following expressions for the two electric fields superposed at  $P$ :

$$E_1(t) = E \cos(\omega t + \phi)$$

$$E_2(t) = E \cos \omega t$$

The superposition of the two fields at  $P$  is a sinusoidal function with some amplitude  $E_P$  that depends on  $E$  and the phase difference  $\phi$ . First we'll work on finding the amplitude  $E_P$  if  $E$  and  $\phi$  are known. Then we'll find the intensity  $I$  of the resultant wave, which is proportional to  $E_P^2$ . Finally, we'll relate the phase difference  $\phi$  to the path difference, which is determined by the geometry of the situation.

Figure 35.9 Phasor diagram for the superposition at a point  $P$  of two waves of equal amplitude  $E$  with a phase difference  $\phi$ .



### Amplitude in Two-Source Interference

To add the two sinusoidal functions with a phase difference, we use the same *phasor* representation that we used for simple harmonic motion (see Section 14.2) and for voltages and currents in ac circuits (see Section 31.1). We suggest that you review these sections now. Each sinusoidal function is represented by a rotating vector (phasor) whose projection on the horizontal axis at any instant represents the instantaneous value of the sinusoidal function.

In Fig. 35.9,  $E_1$  is the horizontal component of the phasor representing the wave from source  $S_1$ , and  $E_2$  is the horizontal component of the phasor for the wave from  $S_2$ . As shown in the diagram, both phasors have the same magnitude  $E$ , but  $E_1$  is *ahead* of  $E_2$  in phase by an angle  $\phi$ . Both phasors rotate counterclockwise with constant angular speed  $\omega$ , and the sum of the projections on the horizontal axis at any time gives the instantaneous value of the total  $E$  field at point  $P$ . Thus the amplitude  $E_P$  of the resultant sinusoidal wave at  $P$  is the magnitude of the dark red phasor in the diagram (labeled  $E_P$ ); this is the *vector sum*

of the other two phasors. To find  $E_P$ , we use the law of cosines and the trigonometric identity  $\cos(\pi - \phi) = -\cos \phi$ :

$$\begin{aligned} E_P^2 &= E^2 + E^2 - 2E^2 \cos(\pi - \phi) \\ &= E^2 + E^2 + 2E^2 \cos \phi \end{aligned}$$

Then, using the identity  $1 + \cos \phi = 2\cos^2(\phi/2)$ , we obtain

$$E_P^2 = 2E^2(1 + \cos \phi) = 4E^2 \cos^2\left(\frac{\phi}{2}\right)$$

$$E_P = 2E \left| \cos \frac{\phi}{2} \right| \quad (35.7)$$

You can also obtain this result without using phasors.

When the two waves are in phase,  $\phi = 0$  and  $E_P = 2E$ . When they are exactly a half-cycle out of phase,  $\phi = \pi \text{ rad} = 180^\circ$ ,  $\cos(\phi/2) = \cos(\pi/2) = 0$ , and  $E_P = 0$ . Thus the superposition of two sinusoidal waves with the same frequency and amplitude but with a phase difference yields a sinusoidal wave with the same frequency and an amplitude between zero and twice the individual amplitudes, depending on the phase difference.

### Intensity in Two-Source Interference

To obtain the intensity  $I$  at point  $P$ , we recall from Section 32.4 that  $I$  is equal to the average magnitude of the Poynting vector,  $S_{\text{av}}$ . For a sinusoidal wave with electric-field amplitude  $E_P$ , this is given by Eq. (32.29) with  $E_{\text{max}}$  replaced by  $E_P$ . Thus we can express the intensity in several equivalent forms:

$$I = S_{\text{av}} = \frac{E_P^2}{2\mu_0 c} = \frac{1}{2} \sqrt{\frac{\epsilon_0}{\mu_0}} E_P^2 = \frac{1}{2} \epsilon_0 c E_P^2 \quad (35.8)$$

The essential content of these expressions is that  $I$  is proportional to  $E_P^2$ . When we substitute Eq. (35.7) into the last expression in Eq. (35.8), we get

$$I = \frac{1}{2} \epsilon_0 c E_P^2 = 2\epsilon_0 c E^2 \cos^2 \frac{\phi}{2} \quad (35.9)$$

In particular, the *maximum* intensity  $I_0$ , which occurs at points where the phase difference is zero ( $\phi = 0$ ), is

$$I_0 = 2\epsilon_0 c E^2$$

Note that the maximum intensity  $I_0$  is *four times* (not twice) as great as the intensity  $\frac{1}{2} \epsilon_0 c E^2$  from each individual source. Substituting the expression for  $I_0$  into Eq. (35.9), we find

$$I = I_0 \cos^2 \frac{\phi}{2} \quad (35.10)$$

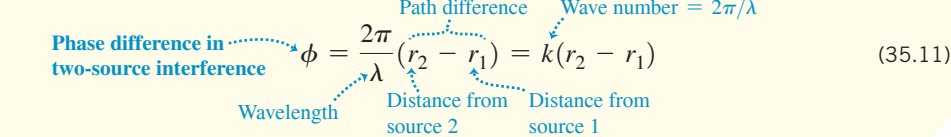
The intensity depends on the phase difference  $\phi$  and varies between  $I_0$  and zero. If we average Eq. (35.10) over all possible phase differences, the result is  $I_0/2 = \epsilon_0 c E^2$  [the average of  $\cos^2(\phi/2)$  is  $\frac{1}{2}$ ]. This is just twice the intensity from each individual source, as we should expect. The total energy output from the two sources isn't changed by the interference effects, but the energy is redistributed (see Section 35.1).



## Phase Difference and Path Difference

Our next task is to find the phase difference  $\phi$  between the two fields at any point  $P$ . We know that  $\phi$  is proportional to the difference in path length from the two sources to point  $P$ . When the path difference is one wavelength, the phase difference is one cycle, and  $\phi = 2\pi \text{ rad} = 360^\circ$ . When the path difference is  $\lambda/2$ ,  $\phi = \pi \text{ rad} = 180^\circ$ , and so on. That is, the ratio of the phase difference  $\phi$  to  $2\pi$  is equal to the ratio of the path difference  $r_2 - r_1$  to  $\lambda$ :

$$\frac{\phi}{2\pi} = \frac{r_2 - r_1}{\lambda}$$



$$\phi = \frac{2\pi}{\lambda}(r_2 - r_1) = k(r_2 - r_1) \quad (35.11)$$

We introduced the wave number  $k = 2\pi/\lambda$  in Section 15.3.

If the material in the space between the sources and  $P$  is anything other than vacuum, we must use the wavelength *in the material* in Eq. (35.11). If  $\lambda_0$  and  $k_0$  are the wavelength and wave number, respectively, in vacuum and the material has index of refraction  $n$ , then

$$\lambda = \frac{\lambda_0}{n} \quad \text{and} \quad k = nk_0 \quad (35.12)$$

Finally, if the point  $P$  is far away from the sources in comparison to their separation  $d$ , the path difference is given by Eq. (35.3):

$$r_2 - r_1 = d \sin \theta$$

Combining this with Eq. (35.11), we find

$$\phi = k(r_2 - r_1) = kd \sin \theta = \frac{2\pi d}{\lambda} \sin \theta \quad (35.13)$$

When we substitute this into Eq. (35.10), we find

$$I = I_0 \cos^2\left(\frac{1}{2}kd \sin \theta\right) = I_0 \cos^2\left(\frac{\pi d}{\lambda} \sin \theta\right) \quad \text{(intensity far from two sources)} \quad (35.14)$$

*Maximum* intensity occurs when the cosine has the values  $\pm 1$ —that is, when

$$\frac{\pi d}{\lambda} \sin \theta = m\pi \quad (m = 0, \pm 1, \pm 2, \dots)$$

or

$$d \sin \theta = m\lambda$$

in agreement with Eq. (35.4). You can also derive Eq. (35.5) for the zero-intensity directions from Eq. (35.14).

As we noted in Section 35.2, in experiments with light we visualize the interference pattern due to two slits by using a screen placed at a distance  $R$  from the slits. We can describe positions on the screen with the coordinate  $y$ ; the positions of the bright fringes are given by Eq. (35.6), where ordinarily  $y \ll R$ . In that case,  $\sin \theta$  is approximately equal to  $y/R$ , and we obtain the following expressions for the intensity at *any* point on the screen as a function of  $y$ :

$$I = I_0 \cos^2\left(\frac{kdy}{2R}\right) = I_0 \cos^2\left(\frac{\pi dy}{\lambda R}\right) \quad \text{(intensity in two-slit interference)} \quad (35.15)$$

Figure 35.10 Intensity distribution in the interference pattern from two identical slits.

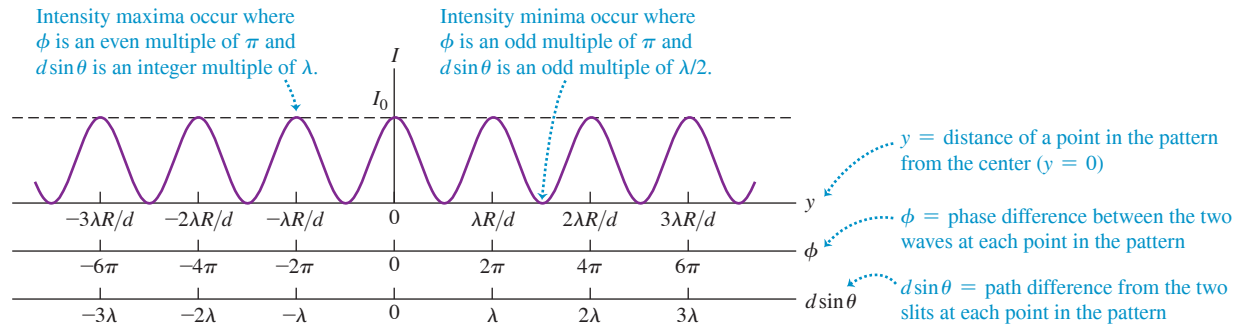


Figure 35.10 shows a graph of Eq. (35.15); we can compare this with the photographically recorded pattern of Fig. 35.6. All peaks in Fig. 35.10 have the same intensity, while those in Fig. 35.6 fade off as we go away from the center. We'll explore the reasons for this variation in peak intensity in Chapter 36.

### EXAMPLE 35.3 A directional transmitting antenna array

### WITH VARIATION PROBLEMS

Suppose the two identical radio antennas of Fig. 35.8 are moved to be only 10.0 m apart and the broadcast frequency is increased to  $f = 60.0$  MHz. At a distance of 700 m from the point midway between the antennas and in the direction  $\theta = 0$  (see Fig. 35.8), the intensity is  $I_0 = 0.020$  W/m<sup>2</sup>. At this same distance, find (a) the intensity in the direction  $\theta = 4.0^\circ$ ; (b) the direction near  $\theta = 0$  for which the intensity is  $I_0/2$ ; and (c) the directions in which the intensity is zero.

**IDENTIFY and SET UP** This problem involves the intensity distribution as a function of angle. Because the 700 m distance from the antennas to the point at which the intensity is measured is much greater than the distance  $d = 10.0$  m between the antennas, the amplitudes of the waves from the two antennas are very nearly equal. Hence we can use Eq. (35.14) to relate intensity  $I$  and angle  $\theta$ .

**EXECUTE** The wavelength is  $\lambda = c/f = 5.00$  m. The spacing  $d = 10.0$  m between the antennas is just twice the wavelength (as was the case in Example 35.2), so  $d/\lambda = 2.00$  and Eq. (35.14) becomes

$$I = I_0 \cos^2 \left( \frac{\pi d}{\lambda} \sin \theta \right) = I_0 \cos^2 [(2.00\pi \text{ rad}) \sin \theta]$$

(a) When  $\theta = 4.0^\circ$ ,

$$\begin{aligned} I &= I_0 \cos^2 [(2.00\pi \text{ rad}) \sin 4.0^\circ] = 0.82 I_0 \\ &= (0.82)(0.020 \text{ W/m}^2) = 0.016 \text{ W/m}^2 \end{aligned}$$

(b) The intensity  $I$  equals  $I_0/2$  when the cosine in Eq. (35.14) has the value  $\pm 1/\sqrt{2}$ . The smallest angles at which this occurs correspond to  $2.00\pi \sin \theta = \pm \pi/4$  rad, so  $\sin \theta = \pm (1/8.00) = \pm 0.125$  and  $\theta = \pm 7.2^\circ$ .

(c) The intensity is zero when  $\cos[(2.00\pi \text{ rad}) \sin \theta] = 0$ . This occurs for  $2.00\pi \sin \theta = \pm \pi/2, \pm 3\pi/2, \pm 5\pi/2, \dots$ , or  $\sin \theta = \pm 0.250, \pm 0.750, \pm 1.25, \dots$ . Values of  $\sin \theta$  greater than 1 have no meaning, so the answers are

$$\theta = \pm 14.5^\circ, \pm 48.6^\circ$$

**EVALUATE** The condition in part (b) that  $I = I_0/2$ , so that  $(2.00\pi \text{ rad}) \sin \theta = \pm \pi/4$  rad, is also satisfied when  $\sin \theta = \pm 0.375, \pm 0.625$ , or  $\pm 0.875$  so that  $\theta = \pm 22.0^\circ, \pm 38.7^\circ$ , or  $\pm 61.0^\circ$ . (Can you verify this?) It would be incorrect to include these angles in the solution, however, because the problem asked for the angle near  $\theta = 0$  at which  $I = I_0/2$ . These additional values of  $\theta$  aren't the ones we're looking for.

**KEYCONCEPT** The intensity at a given point in a two-source interference pattern depends on the phase difference  $\phi$  between the waves that arrive at that point from the two sources [Eqs. (35.10) and (35.11)]. The intensity is *maximum* at points where  $\phi$  is an *even* multiple of  $\pi$  ( $0, \pm 2\pi, \pm 4\pi, \dots$ ) and *minimum* at point where  $\phi$  is an *odd* multiple of  $\pi$  ( $\pm \pi, \pm 3\pi, \dots$ ).

**TEST YOUR UNDERSTANDING OF SECTION 35.3** A two-slit interference experiment uses coherent light of wavelength  $5.00 \times 10^{-7}$  m. Rank the following points in the interference pattern according to the intensity at each point, from highest to lowest. (i) A point that is closer to one slit than the other by  $4.00 \times 10^{-7}$  m; (ii) a point where the light waves received from the two slits are out of phase by  $4.00$  rad; (iii) a point that is closer to one slit than the other by  $7.50 \times 10^{-7}$  m; (iv) a point where the light waves received by the two slits are out of phase by  $2.00$  rad.

### ANSWER

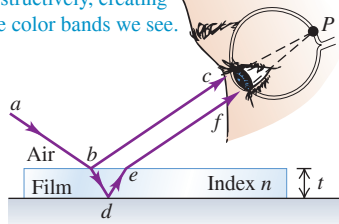
(i)  $I = I_0 \cos^2(\pi d \sin \theta / \lambda) = I_0 \cos^2(\pi (4.00 \times 10^{-7} \text{ m}) / (5.00 \times 10^{-7} \text{ m})) = I_0 \cos^2(\pi/2) = 0$ .  
(ii)  $I = I_0 \cos^2(\pi d \sin \theta / \lambda) = I_0 \cos^2(\pi (7.50 \times 10^{-7} \text{ m}) / (5.00 \times 10^{-7} \text{ m})) = I_0 \cos^2(3\pi/2) = 0$ .  
(iii)  $I = I_0 \cos^2(\pi d \sin \theta / \lambda) = I_0 \cos^2(\pi (4.00 \times 10^{-7} \text{ m}) / (5.00 \times 10^{-7} \text{ m})) = I_0 \cos^2(\pi/2) = 0$ .  
(iv)  $I = I_0 \cos^2(\pi d \sin \theta / \lambda) = I_0 \cos^2(\pi (7.50 \times 10^{-7} \text{ m}) / (5.00 \times 10^{-7} \text{ m})) = I_0 \cos^2(3\pi/2) = 0$ .  
Hence we use Eq. (35.14),  $I = I_0 \cos^2(\phi/2)$ , and we find  $\phi$  and use Eq. (35.10),  $\phi = 2\pi d \sin \theta / \lambda$ . In parts (ii) and (iii) we are given the phase difference  $\phi$  and we use Eq. (35.14),  $I = I_0 \cos^2(\phi/2)$ . In cases (i) and (iv) we are given the wavelength  $\lambda$  and path difference  $d \sin \theta$ . In cases (i), (ii), (iii), and (iv),  $I = I_0 \cos^2(\pi d \sin \theta / \lambda)$ .

Figure 35.11 (a) A diagram and (b) a photograph showing interference of light reflected from a thin film.

(a) Interference between rays reflected from the two surfaces of a thin film

Light reflected from the upper and lower surfaces of the film comes together in the eye at  $P$  and undergoes interference.

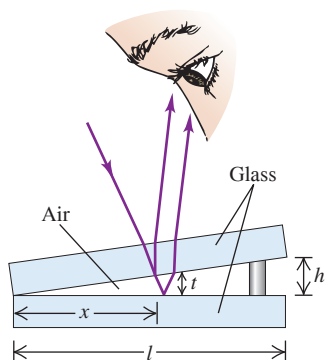
Some colors interfere constructively and others destructively, creating the color bands we see.



(b) Colorful reflections from a soap bubble



Figure 35.12 Interference between light waves reflected from the two sides of an air wedge separating two glass plates. The angles and the thickness of the air wedge have been exaggerated for clarity; in the text we assume that the light strikes the upper plate at normal incidence and that the distances  $h$  and  $t$  are much less than  $l$ .



## 35.4 INTERFERENCE IN THIN FILMS

You often see bright bands of color when light reflects from a thin layer of oil floating on water or from a soap bubble (see the photograph that opens this chapter). These are the results of interference. Light waves are reflected from the front and back surfaces of such thin films, and constructive interference between the two reflected waves (with different path lengths) occurs in different places for different wavelengths. **Figure 35.11a** shows the situation. Light shining on the upper surface of a thin film with thickness  $t$  is partly reflected at the upper surface (path  $abc$ ). Light *transmitted* through the upper surface is partly reflected at the lower surface (path  $abdef$ ). The two reflected waves come together at point  $P$  on the retina of the eye. Depending on the phase relationship, they may interfere constructively or destructively. Different colors have different wavelengths, so the interference may be constructive for some colors and destructive for others. That's why we see colored patterns in the photograph that opens this chapter (which shows a thin film of oil floating on water) and in Fig. 35.11b (which shows thin films of soap solution that make up the bubble walls). The complex shapes of the colored patterns result from variations in the thickness of the film.

### Thin-Film Interference and Phase Shifts During Reflection

Let's look at a simplified situation in which *monochromatic* light reflects from two nearly parallel surfaces at nearly normal incidence. **Figure 35.12** shows two plates of glass separated by a thin wedge, or film, of air. We want to consider interference between the two light waves reflected from the surfaces adjacent to the air wedge. (Reflections also occur at the top surface of the upper plate and the bottom surface of the lower plate; to keep our discussion simple, we won't include these.) The situation is the same as in Fig. 35.11a except that the film (wedge) thickness is not uniform. The path difference between the two waves is just twice the thickness  $t$  of the air wedge at each point. At points where  $2t$  is an integer number of wavelengths, we expect to see constructive interference and a bright area; where it is a half-integer number of wavelengths, we expect to see destructive interference and a dark area. Where the plates are in contact, there is practically *no* path difference, and we expect a bright area.

When we carry out the experiment, the bright and dark fringes appear, but they are interchanged! Along the line where the plates are in contact, we find a *dark* fringe, not a bright one. This suggests that one or the other of the reflected waves has undergone a half-cycle phase shift during its reflection. In that case the two waves that are reflected at the line of contact are a half-cycle out of phase even though they have the same path length.

In fact, this phase shift can be predicted from Maxwell's equations and the electromagnetic nature of light. The details of the derivation are beyond our scope, but here is the result. Suppose a light wave with electric-field amplitude  $E_i$  is traveling in an optical material with index of refraction  $n_a$ . It strikes, at normal incidence, an interface with another optical material with index  $n_b$ . The amplitude  $E_r$  of the wave reflected from the interface is given by

$$E_r = \frac{n_a - n_b}{n_a + n_b} E_i \quad (\text{normal incidence}) \quad (35.16)$$

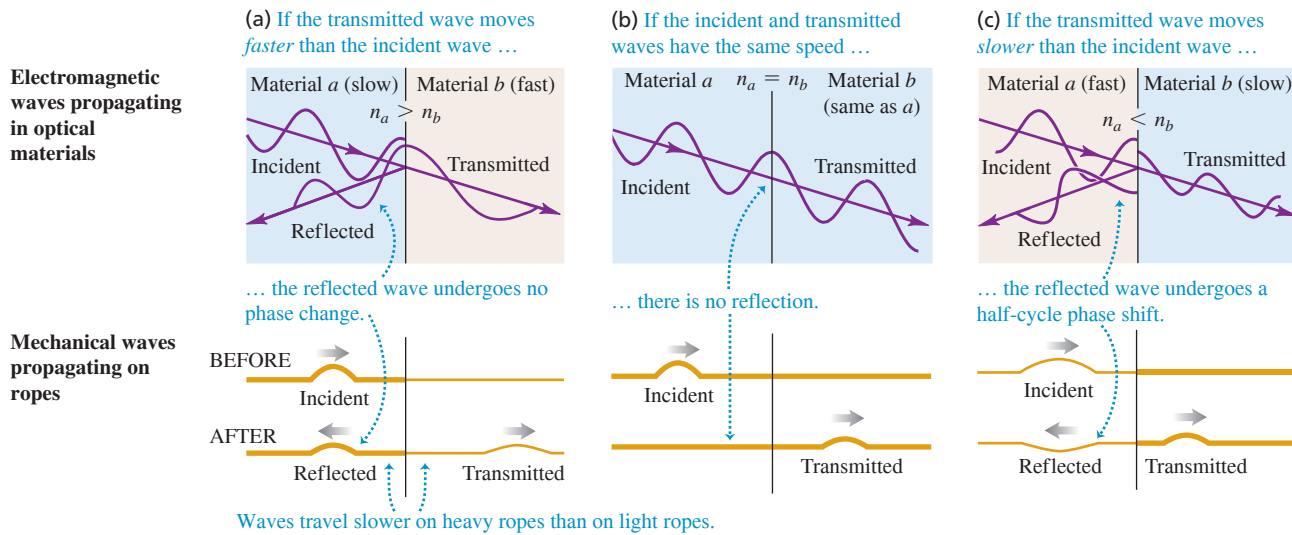
This result shows that the incident and reflected amplitudes have the same sign when  $n_a$  is larger than  $n_b$  and opposite signs when  $n_b$  is larger than  $n_a$ . Because amplitudes must always be positive or zero, a *negative* value means that the wave actually undergoes a half-cycle ( $180^\circ$ ) phase shift. **Figure 35.13** shows three possibilities:

**Figure 35.13a:** When  $n_a > n_b$ , light travels more slowly in the first material than in the second. In this case,  $E_r$  and  $E_i$  have the same sign, and the phase shift of the reflected wave relative to the incident wave is zero. This is analogous to reflection of a transverse mechanical wave on a heavy rope at a point where it is tied to a lighter rope.

**Figure 35.13b:** When  $n_a = n_b$ , the amplitude  $E_r$  of the reflected wave is zero. In effect there is no interface, so there is *no* reflected wave.

**Figure 35.13c:** When  $n_a < n_b$ , light travels more slowly in the second material than in the first. In this case,  $E_r$  and  $E_i$  have opposite signs, and the phase shift of the reflected wave relative to the incident wave is  $\pi$  rad (a half-cycle). This is analogous to reflection (with inversion) of a transverse mechanical wave on a light rope at a point where it is tied to a heavier rope.

Figure 35.13 Upper figures: electromagnetic waves striking an interface between optical materials at normal incidence (shown as a small angle for clarity). Lower figures: mechanical wave pulses on ropes.



Let's compare with the situation of Fig. 35.12. For the wave reflected from the upper surface of the air wedge,  $n_a$  (glass) is greater than  $n_b$ , so this wave has zero phase shift. For the wave reflected from the lower surface,  $n_a$  (air) is less than  $n_b$  (glass), so this wave has a half-cycle phase shift. Waves that are reflected from the line of contact have no path difference to give additional phase shifts, and they interfere destructively; this is what we observe. You can use the above principle to show that for normal incidence, the wave reflected at point  $b$  in Fig. 35.11a is shifted by a half-cycle, while the wave reflected at  $d$  is not (if there is air below the film).

We can summarize this discussion mathematically. If the film has thickness  $t$ , the light is at normal incidence and has wavelength  $\lambda$  in the film; if neither or both of the reflected waves from the two surfaces have a half-cycle reflection phase shift, the conditions for constructive and destructive interference are

$$\text{Constructive reflection} \quad 2t = m\lambda \quad (m = 0, 1, 2, \dots) \quad (35.17a)$$

(From thin film,  
no relative phase shift)

Thickness of film      Wavelength

$$\text{Destructive reflection} \quad 2t = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, 1, 2, \dots) \quad (35.17b)$$

If *one* of the two waves has a half-cycle reflection phase shift, the conditions for constructive and destructive interference are reversed:

$$\text{Constructive reflection} \quad 2t = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, 1, 2, \dots) \quad (35.18a)$$

(From thin film,  
half-cycle phase shift)

Thickness of film      Wavelength

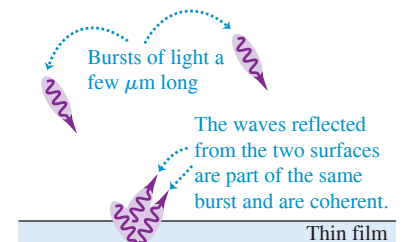
$$\text{Destructive reflection} \quad 2t = m\lambda \quad (m = 0, 1, 2, \dots) \quad (35.18b)$$

## Thin and Thick Films

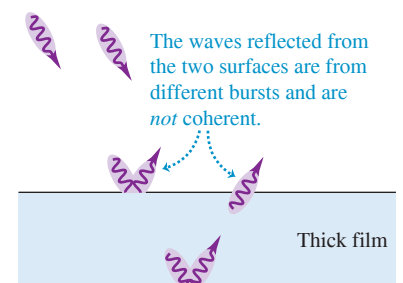
We have emphasized *thin* films in our discussion because of a principle we introduced in Section 35.1: In order for two waves to cause a steady interference pattern, the waves must be *coherent*, with a definite and constant phase relationship. The sun and light bulbs emit light in a stream of short bursts, each of which is only a few micrometers long (1 micrometer =  $1 \mu\text{m} = 10^{-6} \text{ m}$ ). If light reflects from the two surfaces of a thin film, the two reflected waves are part of the same burst (Fig. 35.14a). Hence these waves are coherent and interference occurs as we have described. If the film is too thick, however,

Figure 35.14 (a) Light reflecting from a thin film produces a steady interference pattern, but (b) light reflecting from a thick film does not.

(a) Light reflecting from a thin film



(b) Light reflecting from a thick film





the two reflected waves will belong to different bursts (Fig. 35.14b). There is no definite phase relationship between different light bursts, so the two waves are incoherent and there is no fixed interference pattern. That's why you see interference colors in light reflected from a soap bubble a few micrometers thick (see Fig. 35.11b), but you do *not* see such colors in the light reflected from a pane of window glass with a thickness of a few millimeters (a thousand times greater).

### PROBLEM-SOLVING STRATEGY 35.1 Interference in Thin Films

**IDENTIFY** *the relevant concepts:* Problems with thin films involve interference of two waves, one reflected from the film's front surface and one reflected from the back surface. Typically you'll be asked to relate the wavelength, the film thickness, and the index of refraction of the film.

**SET UP** *the problem* using the following steps:

1. Make a drawing showing the geometry of the film. Identify the materials that adjoin the film; their properties determine whether one or both of the reflected waves have a half-cycle phase shift.
2. Identify the target variable.

**EXECUTE** *the solution* as follows:

1. Apply the rule for phase changes to each reflected wave: There is a half-cycle phase shift when  $n_b > n_a$  and none when  $n_b < n_a$ .

2. If *neither* reflected wave undergoes a phase shift, or if *both* do, use Eqs. (35.17). If only one reflected wave undergoes a phase shift, use Eqs. (35.18).
3. Solve the resulting equation for the target variable. Use the wavelength  $\lambda = \lambda_0/n$  of light *in the film* in your calculations, where  $n$  is the index of refraction of the film. (For air,  $n = 1.000$  to four-figure precision.)
4. If you are asked about the wave that is transmitted through the film, remember that minimum intensity in the reflected wave corresponds to maximum *transmitted* intensity, and vice versa.

**EVALUATE** *your answer:* Interpret your results by examining what would happen if the wavelength were changed or if the film had a different thickness.

### EXAMPLE 35.4 Thin-film interference I

#### WITH VARIATION PROBLEMS

Suppose the two glass plates in Fig. 35.12 are two microscope slides 10.0 cm long. At one end they are in contact; at the other end they are separated by a piece of paper 0.0200 mm thick. What is the spacing of the interference fringes seen by reflection? Is the fringe at the line of contact bright or dark? Assume monochromatic light with a wavelength in air of  $\lambda = \lambda_0 = 500$  nm.

**IDENTIFY and SET UP** Figure 35.15 depicts the situation. We'll consider only interference between the light reflected from the upper and lower surfaces of the air wedge between the microscope slides. [The top slide has a relatively great thickness, about 1 mm, so we can ignore interference between the light reflected from its upper and lower surfaces (see Fig. 35.14b).] Light travels more slowly in the glass of the slides than it does in air. Hence the wave reflected from the upper surface of the air wedge has no phase shift (see Fig. 35.13a), while the wave reflected from the lower surface has a half-cycle phase shift (see Fig. 35.13c).

**EXECUTE** Since only one of the reflected waves undergoes a phase shift, the condition for *destructive* interference (a dark fringe) is Eq. (35.18b):

$$2t = m\lambda_0 \quad (m = 0, 1, 2, \dots)$$

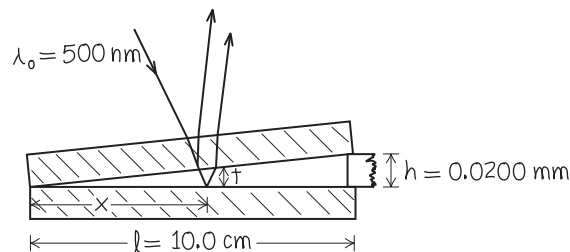
From similar triangles in Fig. 35.15 the thickness  $t$  of the air wedge at each point is proportional to the distance  $x$  from the line of contact:

$$\frac{t}{x} = \frac{h}{l}$$

Combining this with Eq. (35.18b), we find

$$\begin{aligned} \frac{2xh}{l} &= m\lambda_0 \\ x &= m \frac{l\lambda_0}{2h} = m \frac{(0.100 \text{ m})(500 \times 10^{-9} \text{ m})}{(2)(0.0200 \times 10^{-3} \text{ m})} = m(1.25 \text{ mm}) \end{aligned}$$

Figure 35.15 Our sketch for this problem.



Successive dark fringes, corresponding to  $m = 1, 2, 3, \dots$ , are spaced 1.25 mm apart. Substituting  $m = 0$  into this equation gives  $x = 0$ , which is where the two slides touch (at the left-hand side of Fig. 35.15). Hence there is a dark fringe at the line of contact.

**EVALUATE** Our result shows that the fringe spacing is proportional to the wavelength of the light used; the fringes would be farther apart with red light (larger  $\lambda_0$ ) than with blue light (smaller  $\lambda_0$ ). If we use white light, the reflected light at any point is a mixture of wavelengths for which constructive interference occurs; the wavelengths that interfere destructively are weak or absent in the reflected light. (This same effect explains the colors seen when a soap bubble is illuminated by white light, as in Fig. 35.11b.)

**KEYCONCEPT** In thin-film interference, light waves reflected from the front surface of a thin film interfere with light waves reflected from the back surface. The phase difference between these waves depends on the thickness of the film, the wavelength of the light in the film, and whether the light undergoes a phase change on reflecting from each surface.

**EXAMPLE 35.5 Thin-film interference II****WITH VARIATION PROBLEMS**

Suppose the glass plates of Example 35.4 have  $n = 1.52$  and the space between plates contains water ( $n = 1.33$ ) instead of air. What happens now?

**IDENTIFY and SET UP** The index of refraction of the water wedge is still less than that of the glass on either side of it, so the phase shifts are the same as in Example 35.4. Once again we use Eq. (35.18b) to find the positions of the dark fringes; the only difference is that the wavelength  $\lambda$  in this equation is now the wavelength in water instead of in air.

**EXECUTE** In the film of water ( $n = 1.33$ ), the wavelength is  $\lambda = \lambda_0/n = (500 \text{ nm})/(1.33) = 376 \text{ nm}$ . When we replace  $\lambda_0$  by  $\lambda$  in the expression from Example 35.4 for the position  $x$  of the  $m$ th dark fringe, we find that the fringe spacing is reduced by the same factor of

1.33 and is equal to 0.940 mm. There is still a dark fringe at the line of contact.

**EVALUATE** Can you see that to obtain the same fringe spacing as in Example 35.4, the dimension  $h$  in Fig. 35.15 would have to be reduced to  $(0.0200 \text{ mm})/1.33 = 0.0150 \text{ mm}$ ? This shows that what matters in thin-film interference is the *ratio*  $t/\lambda$  between film thickness and wavelength. [Consider Eqs. (35.17) and (35.18).]

**KEYCONCEPT** The wavelength of interest in thin-film interference is the wavelength of light in the material of which the film is composed. This equals the wavelength in vacuum divided by the index of refraction of the material.

**EXAMPLE 35.6 Thin-film interference III****WITH VARIATION PROBLEMS**

Suppose the upper of the two plates of Example 35.4 is a plastic with  $n = 1.40$ , the wedge is filled with a silicone grease with  $n = 1.50$ , and the bottom plate is a dense flint glass with  $n = 1.60$ . What happens now?

**IDENTIFY and SET UP** The geometry is again the same as shown in Fig. 35.15, but now half-cycle phase shifts occur at *both* surfaces of the grease wedge (see Fig. 35.13c). Hence there is no *relative* phase shift and we must use Eq. (35.17b) to find the positions of the dark fringes.

**EXECUTE** The value of  $\lambda$  to use in Eq. (35.17b) is the wavelength in the silicone grease,  $\lambda = \lambda_0/n = (500 \text{ nm})/1.50 = 333 \text{ nm}$ . You can readily show that the fringe spacing is 0.833 mm. Note that the two reflected waves from the line of contact are in phase (they both undergo the same phase shift), so the line of contact is at a *bright* fringe.

**EVALUATE** What would happen if you carefully removed the upper microscope slide so that the grease wedge retained its shape? There would still be half-cycle phase changes at the upper and lower surfaces of the wedge, so the pattern of fringes would be the same as with the upper slide present.

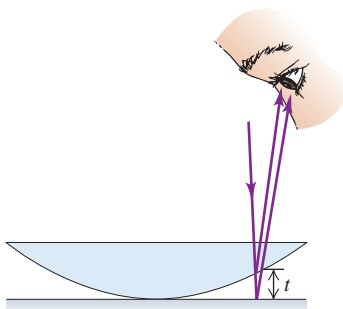
**KEYCONCEPT** When a light wave in one medium reflects at the surface of a second medium, it undergoes a half-cycle phase shift only if the speed of light is slower in the second medium than in the first; otherwise, there is no phase shift. If both reflected waves in thin-film interference undergo the same phase shift on reflection, there is no *relative* phase shift.

**Newton's Rings**

**Figure 35.16a** shows the convex surface of a lens in contact with a plane glass plate. A thin film of air is formed between the two surfaces. When you view the setup with monochromatic light, you see circular interference fringes (Fig. 35.16b). These were studied by Newton and are called **Newton's rings**.

**Figure 35.16** (a) Air film between a convex lens and a plane surface. The thickness of the film  $t$  increases from zero as we move out from the center, giving (b) a series of alternating dark and bright rings for monochromatic light.

(a) A convex lens in contact with a glass plane



(b) Newton's rings: circular interference fringes

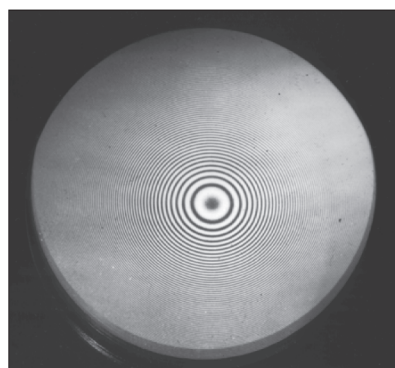


Figure 35.17 The surface of a telescope objective lens under inspection during manufacture.

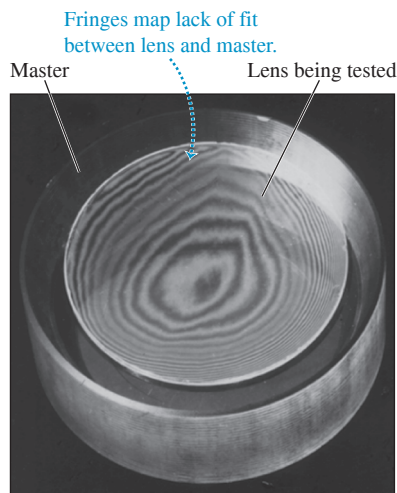
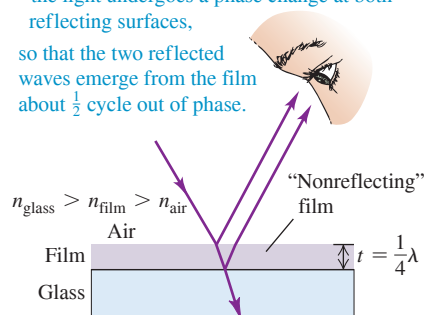


Figure 35.18 A nonreflective coating has an index of refraction intermediate between those of glass and air.

Destructive interference occurs when

- the film is about  $\frac{1}{4}\lambda$  thick and
- the light undergoes a phase change at both reflecting surfaces,

so that the two reflected waves emerge from the film about  $\frac{1}{2}$  cycle out of phase.



We can use interference fringes to compare the surfaces of two optical parts by placing the two in contact and observing the interference fringes. **Figure 35.17** is a photograph made during the grinding of a telescope objective lens. The lower, larger-diameter, thicker disk is the correctly shaped master, and the smaller, upper disk is the lens under test. The “contour lines” are Newton’s interference fringes; each one indicates an additional distance between the specimen and the master of one half-wavelength. At 10 lines from the center spot the distance between the two surfaces is five wavelengths, or about 0.003 mm. This isn’t very good; high-quality lenses are routinely ground with a precision of less than one wavelength. The surface of the primary mirror of the Hubble Space Telescope was ground to a precision of better than  $\frac{1}{50}$  wavelength. Unfortunately, it was ground to incorrect specifications, creating one of the most precise errors in the history of optical technology (see Section 34.2).

## Nonreflective and Reflective Coatings

**Nonreflective coatings** for lens surfaces make use of thin-film interference. A thin layer or film of hard transparent material with an index of refraction smaller than that of the glass is deposited on the lens surface, as in **Fig. 35.18**. Light is reflected from both surfaces of the layer. In both reflections the light is reflected from a medium of greater index than that in which it is traveling, so the same phase change occurs in both reflections. If the film thickness is a quarter (one-fourth) of the wavelength *in the film* (assuming normal incidence), the total path difference is a half-wavelength. Light reflected from the first surface is then a half-cycle out of phase with light reflected from the second, and there is destructive interference.

The thickness of the nonreflective coating can be a quarter-wavelength for only one particular wavelength. This is usually chosen in the central yellow-green portion of the spectrum ( $\lambda = 550$  nm), where the eye is most sensitive. Then there is somewhat more reflection at both longer (red) and shorter (blue) wavelengths, and the reflected light has a purple hue. The overall reflection from a lens or prism surface can be reduced in this way from 4–5% to less than 1%. This also increases the net amount of light that is *transmitted* through the lens, since light that is not reflected will be transmitted. The same principle is used to minimize reflection from silicon photovoltaic solar cells ( $n = 3.5$ ) by use of a thin surface layer of silicon monoxide ( $\text{SiO}$ ,  $n = 1.45$ ); this helps to increase the amount of light that actually reaches the solar cells.

If a quarter-wavelength thickness of a material with an index of refraction *greater* than that of glass is deposited on glass, then the reflectivity is *increased*, and the deposited material is called a **reflective coating**. In this case there is a half-cycle phase shift at the air–film interface but none at the film–glass interface, and reflections from the two sides of the film interfere constructively. For example, a coating with refractive index 2.5 causes 38% of the incident energy to be reflected, compared with 4% or so with no coating. By use of multiple-layer coatings, it is possible to achieve nearly 100% reflection (or, if desired, 100% transmission) for particular wavelengths. Some practical applications of reflective coatings are for color separation in television cameras and for infrared “heat reflectors” in motion-picture projectors, mirrored sunglasses, and astronauts’ visors.

### EXAMPLE 35.7 A nonreflective coating

A common lens coating material is magnesium fluoride ( $\text{MgF}_2$ ), with  $n = 1.38$ . What thickness should a nonreflective coating have for 550 nm light if it is applied to glass with  $n = 1.52$ ?

**IDENTIFY and SET UP** This coating is of the sort shown in Fig. 35.18. The thickness must be one-quarter of the wavelength of this light *in the coating*.

**EXECUTE** The wavelength in air is  $\lambda_0 = 550$  nm, so its wavelength in the  $\text{MgF}_2$  coating is  $\lambda = \lambda_0/n = (550 \text{ nm})/1.38 = 400$  nm. The coating thickness should be one-quarter of this, or  $\lambda/4 = 100$  nm.

**EVALUATE** This is a very thin film, no more than a few hundred molecules thick. Note that this coating is *reflective* for light whose wavelength is *twice* the coating thickness; light of that wavelength reflected

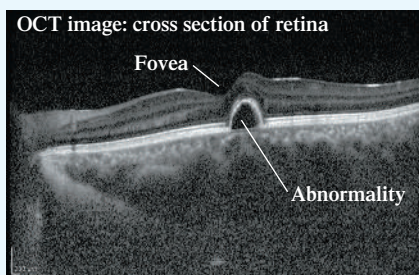
from the coating’s lower surface travels one wavelength farther than light reflected from the upper surface, so the two waves are in phase and interfere constructively. This occurs for light with a wavelength in  $\text{MgF}_2$  of 200 nm and a wavelength in air of  $(200 \text{ nm})(1.38) = 276$  nm. This is an ultraviolet wavelength that isn’t visible to the eye (see Section 32.1), so a lens designer need not worry about such enhanced reflection.

**KEYCONCEPT** A thin-film coating *minimizes* reflection for certain wavelengths by causing *destructive* interference of the reflected waves. The same coating *maximizes* reflection for other wavelengths by causing *constructive* interference of the reflected waves.





**BIO APPLICATION Seeing Below the Surface with Interferometry** Visible light can penetrate up to half a millimeter below the surface of biological tissue. However, only a very small fraction of the light is reflected back. To capture this light and image the subsurface features, a technique called *optical coherence tomography* (OCT) is used. OCT uses the same setup as a Michelson interferometer (Fig. 35.19), but with one of the mirrors replaced by the tissue to be studied. Only the light that reflects directly off the subsurface tissue gives constructive interference, which makes it possible to isolate that reflected light and form an image. OCT is widely used by ophthalmologists to get a cross-sectional view of the subsurface structure of the retina, which makes it easier to diagnose and monitor abnormalities of the eye.



The whole apparatus in Fig. 35.19 is mounted on a very rigid frame, and the position of mirror  $M_2$  can be adjusted with a fine, very accurate micrometer screw. If the distances  $L_1$  and  $L_2$  are exactly equal and the mirrors  $M_1$  and  $M_2$  are exactly at right angles, the virtual image of  $M_1$  formed by reflection at the silvered surface of plate  $C$  coincides with mirror  $M_2$ . If  $L_1$  and  $L_2$  are *not* exactly equal, the image of  $M_1$  is displaced slightly from  $M_2$ ; and if the mirrors are not exactly perpendicular, the image of  $M_1$  makes a slight angle with  $M_2$ . Then the mirror  $M_2$  and the virtual image of  $M_1$  play the same roles as the two surfaces of a wedge-shaped thin film (see Section 35.4), and light reflected from these surfaces forms the same sort of interference fringes.

Suppose the angle between mirror  $M_2$  and the virtual image of  $M_1$  is just large enough that five or six vertical fringes are present in the field of view. If we now move the mirror  $M_2$  slowly either backward or forward a distance  $\lambda/2$ , the difference in path length between rays 1 and 2 changes by  $\lambda$ , and each fringe moves to the left or right a distance equal to the fringe spacing. If we observe the fringe positions through a telescope with a crosshair eyepiece and  $m$  fringes cross the crosshairs when we move the mirror a distance  $y$ , then

$$y = m \frac{\lambda}{2} \quad \text{or} \quad \lambda = \frac{2y}{m} \quad (35.19)$$

If  $m$  is several thousand, the distance  $y$  is large enough that it can be measured with good accuracy, and we can obtain an accurate value for the wavelength  $\lambda$ . Alternatively, if the wavelength is known, a distance  $y$  can be measured by simply counting fringes when  $M_2$  is moved by this distance. In this way, distances that are comparable to a wavelength of light can be measured with relative ease.

### The Michelson-Morley Experiment

The original application of the Michelson interferometer was to the historic **Michelson-Morley experiment**. Before the electromagnetic theory of light became established, most physicists thought that the propagation of light waves occurred in a medium called the **ether**, which was believed to permeate all space. In 1887 the American scientists Albert Michelson and Edward Morley used the Michelson interferometer in an attempt to detect the motion of the earth through the ether. Suppose the interferometer in Fig. 35.19 is moving from left to right relative to the ether. According to the ether theory, this would lead to changes in the speed of light in the portions of the path shown as horizontal lines in the figure. There would be fringe shifts relative to the positions that the fringes would have if the instrument were at rest in the ether. Then when the entire instrument was rotated  $90^\circ$ , the other portions of the paths would be similarly affected, giving a fringe shift in the opposite direction.

Michelson and Morley expected that the motion of the earth through the ether would cause a shift of about four-tenths of a fringe when the instrument was rotated. The shift that was actually observed was less than a hundredth of a fringe and, within the limits of experimental uncertainty, appeared to be exactly zero. Despite its orbital motion around the sun, the earth appeared to be *at rest* relative to the ether. This negative result baffled physicists until 1905, when Albert Einstein developed the special theory of relativity (which we'll study in detail in Chapter 37). Einstein postulated that the speed of a light wave in vacuum has the same magnitude  $c$  relative to *all* inertial reference frames, no matter what their velocity may be relative to each other. The presumed ether then plays no role, and the concept of an ether has been abandoned.

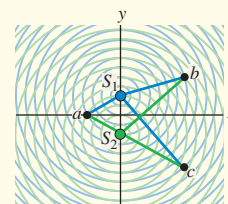
**TEST YOUR UNDERSTANDING OF SECTION 35.5** You are observing the pattern of fringes in a Michelson interferometer like that shown in Fig. 35.19. If you change the index of refraction (but not the thickness) of the compensator plate, will the pattern change?

### ANSWER

Yes Changing the index of refraction changes the wavelength of the light inside the compensator plate, and so changes the number of wavelengths within the thickness of the plate. Hence this has the same effect as changing the distance  $L_1$  from the beam splitter to mirror  $M_1$ , which would change the interference pattern.

## CHAPTER 35 SUMMARY

**Interference and coherent sources:** Monochromatic light is light with a single frequency. Coherence is a definite, unchanging phase relationship between two waves. The overlap of waves from two coherent sources of monochromatic light forms an interference pattern. The principle of superposition states that the total wave disturbance at any point is the sum of the disturbances from the separate waves.



**Two-source interference of light:** When two sources are in phase, constructive interference occurs where the difference in path length from the two sources is zero or an integer number of wavelengths; destructive interference occurs where the path difference is a half-integer number of wavelengths. If two sources separated by a distance  $d$  are both very far from a point  $P$ , and the line from the sources to  $P$  makes an angle  $\theta$  with the line perpendicular to the line of the sources, then the condition for constructive interference at  $P$  is Eq. (35.4). The condition for destructive interference is Eq. (35.5). When  $\theta$  is very small, the position  $y_m$  of the  $m$ th bright fringe on a screen located a distance  $R$  from the sources is given by Eq. (35.6). (See Examples 35.1 and 35.2.)

$$d \sin \theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \quad (35.4)$$

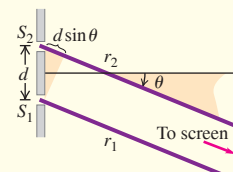
(constructive interference)

$$d \sin \theta = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \quad (35.5)$$

(destructive interference)

$$y_m = R \frac{m\lambda}{d} \quad (m = 0, \pm 1, \pm 2, \dots) \quad (35.6)$$

(bright fringes)

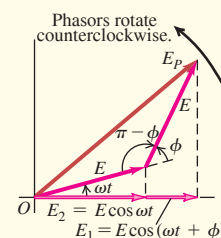


**Intensity in interference patterns:** When two sinusoidal waves with equal amplitude  $E$  and phase difference  $\phi$  are superimposed, the resultant amplitude  $E_P$  and intensity  $I$  are given by Eqs. (35.7) and (35.10), respectively. If the two sources emit in phase, the phase difference  $\phi$  at a point  $P$  (located a distance  $r_1$  from source 1 and a distance  $r_2$  from source 2) is directly proportional to the path difference  $r_2 - r_1$ . (See Example 35.3.)

$$E_P = 2E \left| \cos \frac{\phi}{2} \right| \quad (35.7)$$

$$I = I_0 \cos^2 \frac{\phi}{2} \quad (35.10)$$

$$\phi = \frac{2\pi}{\lambda}(r_2 - r_1) = k(r_2 - r_1) \quad (35.11)$$



**Interference in thin films:** When light is reflected from both sides of a thin film of thickness  $t$  and no phase shift occurs at either surface, constructive interference between the reflected waves occurs when  $2t$  is equal to an integer number of wavelengths. If a half-cycle phase shift occurs at one surface, this is the condition for destructive interference. A half-cycle phase shift occurs during reflection whenever the index of refraction in the second material is greater than that in the first. (See Examples 35.4–35.7.)

$$2t = m\lambda \quad (m = 0, 1, 2, \dots) \quad (35.17a)$$

(constructive reflection from thin film, no relative phase shift)

$$2t = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, 1, 2, \dots) \quad (35.17b)$$

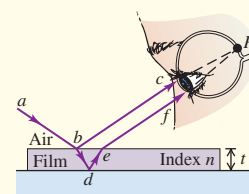
(destructive reflection from thin film, no relative phase shift)

$$2t = \left(m + \frac{1}{2}\right)\lambda \quad (m = 0, 1, 2, \dots) \quad (35.18a)$$

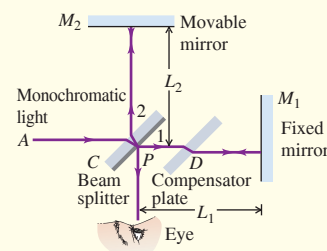
(constructive reflection from thin film, half-cycle phase shift)

$$2t = m\lambda \quad (m = 0, 1, 2, \dots) \quad (35.18b)$$

(destructive reflection from thin film, half-cycle phase shift)



**Michelson interferometer:** The Michelson interferometer uses a monochromatic light source and can be used for high-precision measurements of wavelengths. Its original purpose was to detect motion of the earth relative to a hypothetical ether, the supposed medium for electromagnetic waves. The ether has never been detected, and the concept has been abandoned; the speed of light is the same relative to all observers. This is part of the foundation of the special theory of relativity.





## GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE  VARIATION PROBLEMS

Be sure to review Examples **EXAMPLES 35.1 and 35.2** (Section 35.2) before attempting these problems.

**VP35.2.1** In a two-slit interference experiment like that shown in Fig. 35.7, the slits are 0.180 mm apart and the screen is 1.50 m from the slits. The  $m = 2$  bright fringe is 11.4 mm from the central bright fringe. Find (a) the wavelength of the light and (b) the distance from the central bright fringe to the  $m = -3$  bright fringe.

**VP35.2.2** Two identical vertical radio antennas are 0.810 km apart. Both emit in phase at a frequency of  $1.05 \text{ MHz} = 1.05 \times 10^6 \text{ Hz}$ . Find the angles in the range from  $0^\circ$  and  $90^\circ$  at which the intensity measured several kilometers away from these antennas is (a) maximum and (b) minimum.

**VP35.2.3** You shine red laser light of wavelength 685 nm on two closely spaced slits. This creates a pattern on a screen that is 2.10 m from the double slit. The  $m = 3$  bright fringe in this pattern is 9.15 mm from the central bright fringe. (a) What is the distance between the slits? (b) What would be the distance between the central and  $m = 3$  bright fringes if you used green laser light of wavelength 515 nm?

**VP35.2.4** In a two-slit interference experiment the distance between the slits is 0.370 mm and the angle of the line from the slits to the  $m = 4$  bright region on the screen is  $0.407^\circ$ . Find (a) the wavelength of the light used and (b) the angle of the line from the slits to the  $m = 2$  dark region on the screen.

Be sure to review **EXAMPLE 35.3** (Section 35.3) before attempting these problems.

**VP35.3.1** Two identical radio antennas are 13.0 m apart and transmit in phase at frequency 59.3 MHz. At a distance of 0.850 km from the midpoint between the antennas in the direction  $\theta = 0$ , the intensity is  $I_0 = 0.0330 \text{ W/m}^2$ . Find (a) the intensity at this same distance in the direction  $\theta = 5.00^\circ$  and (b) the smallest angle for which the intensity is  $I_0/2$ .

**VP35.3.2** A two-slit interference experiment uses laser light of wavelength 655 nm. The slits are 0.230 mm apart and 1.75 m from the screen on which the interference pattern appears. The intensity of the light at the central bright fringe is  $I_0 = 0.0520 \text{ W/m}^2$ . Find (a) the intensity at a point on the screen 6.50 mm from the central bright fringe and (b) the distance on the screen from the central bright fringe to the nearest point where the intensity is  $I_0/4$ .

**VP35.3.3** You walk in a circle of radius 1.20 km around the midpoint of a pair of identical radio antennas that are 8.00 m apart. The intensity decreases continuously from  $0.0540 \text{ W/m}^2$  in the direction  $\theta = 0^\circ$  to

$0.0303 \text{ W/m}^2$  in the direction  $\theta = 6.00^\circ$ . Find (a) the wavelength of the radio waves and (b) the intensity in the direction  $\theta = 12.0^\circ$ .

**VP35.3.4** A laser of unknown wavelength illuminates a pair of slits separated by 0.110 mm, producing an interference pattern on a wall 5.00 m from the slits. A point on the wall 10.5 mm from the central bright fringe is between that fringe and the  $m = 1$  bright fringe, and the intensity there is 9.00% as great as at the central bright fringe. Find (a) the wavelength of the laser and (b) the distance from the central bright fringe to the nearest point on the wall where the intensity is 30.0% as great as at the central bright fringe.

Be sure to review **EXAMPLES 35.4, 35.5, and 35.6** (Section 35.4) before attempting these problems.

**VP35.6.1** Two identical microscope slides made of glass ( $n = 1.50$ ) are stacked on top of each other. The slides are 8.00 cm long and are in contact at one end; at the other end they are separated by a piece of paper. You illuminate the slides from above with monochromatic light that has a wavelength in air of 565 nm. The spacing between the interference fringes seen by reflection is 1.30 mm. (a) Find the thickness of the piece of paper. (b) What would be the spacing between fringes if the space between the slides were filled with ethanol ( $n = 1.36$ ) rather than air?

**VP35.6.2** A thin, flat slide of glass ( $n = 1.50$ ) lies on a table, and a second, identical glass slide is suspended 146 nm above the first one. The surfaces of the slides are parallel to each other. You illuminate the two slides from above. Find the wavelengths in air, if any, in the visible range (380 to 750 nm) for which there will be constructive reflection and destructive reflection if the space between the slides is filled with (a) air or (b) carbon tetrachloride ( $n = 1.46$ ).

**VP35.6.3** A thin, flat sheet of plastic ( $n = 1.40$ ) lies on a table, and a second sheet of the same dimensions but made of flint glass ( $n = 1.60$ ) is suspended 172 nm above the first sheet. The surfaces of the sheets are parallel to each other. You illuminate the two sheets from above. Find the wavelengths in air, if any, in the visible range (380 to 750 nm) for which there will be constructive reflection and destructive reflection if the space between the sheets is filled with (a) air or (b) ethanol ( $n = 1.36$ ).

**VP35.6.4** A bowl of water ( $n = 1.33$ ) has a thin layer of grease ( $n = 1.60$ ) floating on top of the water. When you illuminate the grease from above with monochromatic light, you get constructive reflection if the wavelength in air is 565 nm. Find (a) the smallest possible thickness of the grease layer and (b) the second smallest possible thickness.

## BRIDGING PROBLEM Thin-Film Interference in an Oil Slick

An oil tanker spills a large amount of oil ( $n = 1.45$ ) into the sea ( $n = 1.33$ ). (a) If you look down onto the oil spill from overhead, what predominant wavelength of light do you see at a point where the oil is 380 nm thick? What color is the light? (*Hint:* See Table 32.1.) (b) In the water under the slick, what visible wavelength (as measured in air) is predominant in the transmitted light at the same place in the slick as in part (a)?

## SOLUTION GUIDE

## IDENTIFY and SET UP

1. The oil layer acts as a thin film, so we must consider interference between light that reflects from the top and bottom surfaces of the oil. If a wavelength is prominent in the *transmitted* light, there is destructive interference for that wavelength in the *reflected* light.

2. Choose the appropriate interference equations that relate the thickness of the oil film and the wavelength of light. Take account of the indexes of refraction of the air, oil, and water.

## EXECUTE

3. For part (a), find the wavelengths for which there is constructive interference as seen from above the oil film. Which of these are in the visible spectrum?
4. For part (b), find the visible wavelength for which there is destructive interference as seen from above the film. (This will ensure that there is substantial transmitted light at the wavelength.)

## EVALUATE

5. If a diver below the water's surface shines a light up at the bottom of the oil film, at what wavelengths would there be constructive interference in the light that reflects back downward?

## PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

### DISCUSSION QUESTIONS

**Q35.1** A two-slit interference experiment is set up, and the fringes are displayed on a screen. Then the whole apparatus is immersed in the nearest swimming pool. How does the fringe pattern change?

**Q35.2** Could an experiment similar to Young's two-slit experiment be performed with sound? How might this be carried out? Does it matter that sound waves are longitudinal and electromagnetic waves are transverse? Explain.

**Q35.3** Monochromatic coherent light passing through two thin slits is viewed on a distant screen. Are the bright fringes equally spaced on the screen? If so, why? If not, which ones are closest to being equally spaced?

**Q35.4** In a two-slit interference pattern on a distant screen, are the bright fringes midway between the dark fringes? Is this ever a good approximation?

**Q35.5** Would the headlights of a distant car form a two-source interference pattern? If so, how might it be observed? If not, why not?

**Q35.6** The two sources  $S_1$  and  $S_2$  shown in Fig. 35.3 emit waves of the same wavelength  $\lambda$  and are in phase with each other. Suppose  $S_1$  is a weaker source, so that the waves emitted by  $S_1$  have half the amplitude of the waves emitted by  $S_2$ . How would this affect the positions of the antinodal lines and nodal lines? Would there be total reinforcement at points on the antinodal curves? Would there be total cancellation at points on the nodal curves? Explain your answers.

**Q35.7** Could the Young two-slit interference experiment be performed with gamma rays? If not, why not? If so, discuss differences in the experimental design compared to the experiment with visible light.

**Q35.8** Coherent red light illuminates two narrow slits that are 25 cm apart. Will a two-slit interference pattern be observed when the light from the slits falls on a screen? Explain.

**Q35.9** Coherent light with wavelength  $\lambda$  falls on two narrow slits separated by a distance  $d$ . If  $d$  is less than some minimum value, no dark fringes are observed. Explain. In terms of  $\lambda$ , what is this minimum value of  $d$ ?

**Q35.10** A fellow student, who values memorizing equations above understanding them, combines Eqs. (35.4) and (35.13) to "prove" that  $\phi$  can *only* equal  $2\pi m$ . How would you explain to this student that  $\phi$  can have values other than  $2\pi m$ ?

**Q35.11** If the monochromatic light shown in Fig. 35.5a were replaced by white light, would a two-slit interference pattern be seen on the screen? Explain.

**Q35.12** In using the superposition principle to calculate intensities in interference patterns, could you add the intensities of the waves instead of their amplitudes? Explain.

**Q35.13** A glass windowpane with a thin film of water on it reflects less than when it is perfectly dry. Why?

**Q35.14** A *very* thin soap film ( $n = 1.33$ ), whose thickness is much less than a wavelength of visible light, looks black; it appears to reflect no light at all. Why? By contrast, an equally thin layer of soapy water ( $n = 1.33$ ) on glass ( $n = 1.50$ ) appears quite shiny. Why is there a difference?

**Q35.15** Interference can occur in thin films. Why is it important that the films be *thin*? Why don't you get these effects with a relatively *thick* film? Where should you put the dividing line between "thin" and "thick"? Explain your reasoning.

**Q35.16** If we shine white light on an air wedge like that shown in Fig. 35.12, the colors that are weak in the light *reflected* from any point along the wedge are strong in the light *transmitted* through the wedge. Explain why this should be so.

**Q35.17** Monochromatic light is directed at normal incidence on a thin film. There is destructive interference for the reflected light, so the intensity of the reflected light is very low. What happened to the energy of the incident light?

**Q35.18** When a thin oil film spreads out on a puddle of water, the thinnest part of the film looks dark in the resulting interference pattern. What does this tell you about the relative magnitudes of the refractive indexes of oil and water?

### EXERCISES

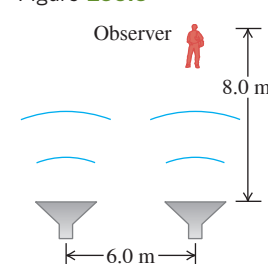
#### Section 35.1 Interference and Coherent Sources

**35.1 ••** A radio transmitting station operating at a frequency of 120 MHz has two identical antennas that radiate in phase. Antenna  $B$  is 9.00 m to the right of antenna  $A$ . Consider point  $P$  between the antennas and along the line connecting them, a horizontal distance  $x$  to the right of antenna  $A$ . For what values of  $x$  will constructive interference occur at point  $P$ ?

**35.2 • Radio Interference.** Two radio antennas  $A$  and  $B$  radiate in phase. Antenna  $B$  is 120 m to the right of antenna  $A$ . Consider point  $Q$  along the extension of the line connecting the antennas, a horizontal distance of 40 m to the right of antenna  $B$ . The frequency, and hence the wavelength, of the emitted waves can be varied. (a) What is the longest wavelength for which there will be destructive interference at point  $Q$ ? (b) What is the longest wavelength for which there will be constructive interference at point  $Q$ ?

**35.3 •** Two speakers, emitting identical sound waves of wavelength 2.0 m in phase with each other, and an observer are located as shown in Fig. E35.3. (a) At the observer's location, what is the path difference for waves from the two speakers? (b) Will the sound waves interfere constructively or destructively at the observer's location—or something in between constructive and destructive? (c) Suppose the observer now increases her distance from the closest speaker to 17.0 m, staying directly in front of the same speaker as initially. Answer the questions of parts (a) and (b) for this new situation.

Figure E35.3



**35.4 •** Two light sources can be adjusted to emit monochromatic light of any visible wavelength. The two sources are coherent,  $2.04 \mu\text{m}$  apart, and in line with an observer, so that one source is  $2.04 \mu\text{m}$  farther from the observer than the other. (a) For what visible wavelengths (380 to 750 nm) will the observer see the brightest light, owing to constructive interference? (b) How would your answers to part (a) be affected if the two sources were not in line with the observer, but were still arranged so that one source is  $2.04 \mu\text{m}$  farther away from the observer than the other? (c) For what visible wavelengths will there be *destructive* interference at the location of the observer?

**35.5 ••** Antenna  $B$  is 40.0 m to the right of antenna  $A$ . The two antennas emit electromagnetic waves that are in phase and have wavelength 7.00 m. (a) At how many points along the line connecting  $A$  and  $B$  is the interference constructive? (b) What is the smallest distance to the right of antenna  $A$  for which is there a point of constructive interference?



## Section 35.2 Two-Source Interference of Light

**35.6 ••** Coherent light of wavelength 500 nm is incident on two very narrow and closely spaced slits. The interference pattern is observed on a very tall screen that is 2.00 m from the slits. Near the center of the screen the separation between two adjacent interference maxima is 3.53 cm. What is the distance on the screen between the  $m = 49$  and  $m = 50$  maxima?

**35.7 •** Young's experiment is performed with light from excited helium atoms ( $\lambda = 502$  nm). Fringes are measured carefully on a screen 1.20 m away from the double slit, and the center of the 20th fringe (not counting the central bright fringe) is found to be 10.6 mm from the center of the central bright fringe. What is the separation of the two slits?

**35.8 ••** Coherent light with wavelength 450 nm falls on a pair of slits. On a screen 1.80 m away, the distance between dark fringes is 3.90 mm. What is the slit separation?

**35.9 ••** Two slits spaced 0.450 mm apart are placed 75.0 cm from a screen. What is the distance between the second and third dark lines of the interference pattern on the screen when the slits are illuminated with coherent light with a wavelength of 500 nm?

**35.10 ••** Coherent light of frequency  $f$  travels in air and is incident on two narrow slits. The interference pattern is observed on a distant screen that is directly opposite the slits. The frequency of light  $f$  can be varied. For  $f = 5.60 \times 10^{12}$  Hz there is an interference maximum for  $\theta = 60.0^\circ$ . The next higher frequency for which there is an interference maximum at this angle is  $7.47 \times 10^{12}$  Hz. What is the separation  $d$  between the two slits?

**35.11 ••** Two thin parallel slits that are 0.0116 mm apart are illuminated by a laser beam of wavelength 585 nm. (a) On a very large distant screen, what is the *total* number of bright fringes (those indicating complete constructive interference), including the central fringe and those on both sides of it? Solve this problem without calculating all the angles! (*Hint:* What is the largest that  $\sin \theta$  can be? What does this tell you is the largest value of  $m$ ?) (b) At what angle, relative to the original direction of the beam, will the fringe that is most distant from the central bright fringe occur?

**35.12 •** Coherent light with wavelength 400 nm passes through two very narrow slits that are separated by 0.200 mm, and the interference pattern is observed on a screen 4.00 m from the slits. (a) What is the width (in mm) of the central interference maximum? (b) What is the width of the first-order bright fringe?

**35.13 ••** Two very narrow slits are spaced  $1.80 \mu\text{m}$  apart and are placed 35.0 cm from a screen. What is the distance between the first and second dark lines of the interference pattern when the slits are illuminated with coherent light with  $\lambda = 550$  nm? (*Hint:* The angle  $\theta$  in Eq. (35.5) is *not* small.)

**35.14 ••** Coherent light that contains two wavelengths, 660 nm (red) and 470 nm (blue), passes through two narrow slits that are separated by 0.300 mm. Their interference pattern is observed on a screen 4.00 m from the slits. What is the distance on the screen between the first-order bright fringes for the two wavelengths?

**35.15 ••** Coherent light with wavelength 600 nm passes through two very narrow slits and the interference pattern is observed on a screen 3.00 m from the slits. The first-order bright fringe is at 4.84 mm from the center of the central bright fringe. For what wavelength of light will the first-order dark fringe be observed at this same point on the screen?

**35.16 ••** Coherent light of frequency  $6.32 \times 10^{14}$  Hz passes through two thin slits and falls on a screen 85.0 cm away. You observe that the third bright fringe occurs at  $\pm 3.11$  cm on either side of the central bright fringe. (a) How far apart are the two slits? (b) At what distance from the central bright fringe will the third dark fringe occur?

## Section 35.3 Intensity in Interference Patterns

**35.17 ••** In a two-slit interference pattern, the intensity at the peak of the central maximum is  $I_0$ . (a) At a point in the pattern where the phase difference between the waves from the two slits is  $60.0^\circ$ , what is the intensity? (b) What is the path difference for 480 nm light from the two slits at a point where the phase difference is  $60.0^\circ$ ?

**35.18 •** Two slits spaced 0.260 mm apart are 0.900 m from a screen and illuminated by coherent light of wavelength 660 nm. The intensity at the center of the central maximum ( $\theta = 0^\circ$ ) is  $I_0$ . What is the distance on the screen from the center of the central maximum (a) to the first minimum; (b) to the point where the intensity has fallen to  $I_0/2$ ?

**35.19 •** Coherent light with wavelength 500 nm passes through narrow slits separated by 0.340 mm. At a distance from the slits large compared to their separation, what is the phase difference (in radians) in the light from the two slits at an angle of  $23.0^\circ$  from the centerline?

**35.20 ••** Two slits spaced 0.0720 mm apart are 0.800 m from a screen. Coherent light of wavelength  $\lambda$  passes through the two slits. In their interference pattern on the screen, the distance from the center of the central maximum to the first minimum is 3.00 mm. If the intensity at the peak of the central maximum is  $0.0600 \text{ W/m}^2$ , what is the intensity at points on the screen that are (a) 2.00 mm and (b) 1.50 mm from the center of the central maximum?

**35.21 •** Consider two antennas separated by 9.00 m that radiate in phase at 120 MHz, as described in Exercise 35.1. A receiver placed 150 m from both antennas measures an intensity  $I_0$ . The receiver is moved so that it is 1.8 m closer to one antenna than to the other. (a) What is the phase difference  $\phi$  between the two radio waves produced by this path difference? (b) In terms of  $I_0$ , what is the intensity measured by the receiver at its new position?

## Section 35.4 Interference in Thin Films

**35.22 ••** Two identical horizontal sheets of glass have a thin film of air of thickness  $t$  between them. The glass has refractive index 1.40. The thickness  $t$  of the air layer can be varied. Light with wavelength  $\lambda$  in air is at normal incidence onto the top of the air film. There is constructive interference between the light reflected at the top and bottom surfaces of the air film when its thickness is 650 nm. For the same wavelength of light the next larger thickness for which there is constructive interference is 910 nm. (a) What is the wavelength  $\lambda$  of the light when it is traveling in air? (b) What is the smallest thickness  $t$  of the air film for which there is constructive interference for this wavelength of light?

**35.23 •** What is the thinnest film of a coating with  $n = 1.42$  on glass ( $n = 1.52$ ) for which destructive interference of the red component (650 nm) of an incident white light beam in air can take place by reflection?

**35.24 •• Nonglare Glass.** When viewing a piece of art that is behind glass, one often is affected by the light that is reflected off the front of the glass (called *glare*), which can make it difficult to see the art clearly. One solution is to coat the outer surface of the glass with a film to cancel part of the glare. (a) If the glass has a refractive index of 1.62 and you use  $\text{TiO}_2$ , which has an index of refraction of 2.62, as the coating, what is the minimum film thickness that will cancel light of wavelength 505 nm? (b) If this coating is too thin to stand up to wear, what other thickness would also work? Find only the three thinnest ones.

**35.25 ••** A uniform film of  $\text{TiO}_2$ , 1036 nm thick and having index of refraction 2.62, is spread uniformly over the surface of crown glass of refractive index 1.52. Light of wavelength 520.0 nm falls at normal incidence onto the film from air. You want to increase the thickness of this film so that the reflected light cancels. (a) What is the *minimum* thickness of  $\text{TiO}_2$  that you must *add* so the reflected light cancels as desired? (b) After you make the adjustment in part (a), what is the path difference between the light reflected off the top of the film and the light that cancels it after traveling through the film? Express your answer in (i) nanometers and (ii) wavelengths of the light in the  $\text{TiO}_2$  film.

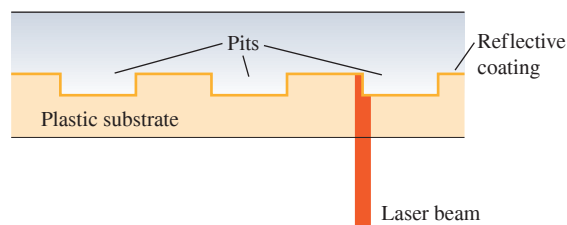
**35.26 •** A plastic film with index of refraction 1.70 is applied to the surface of a car window to increase the reflectivity and thus to keep the car's interior cooler. The window glass has index of refraction 1.52. (a) What minimum thickness is required if light of wavelength 550 nm in air reflected from the two sides of the film is to interfere constructively? (b) Coatings as thin as that calculated in part (a) are difficult to manufacture and install. What is the next greater thickness for which constructive interference will also occur?

**35.27 •** The walls of a soap bubble have about the same index of refraction as that of plain water,  $n = 1.33$ . There is air both inside and outside the bubble. (a) What wavelength (in air) of visible light is most strongly reflected from a point on a soap bubble where its wall is 290 nm thick? To what color does this correspond (see Fig. 32.4 and Table 32.1)? (b) Repeat part (a) for a wall thickness of 340 nm.

**35.28 ••** A researcher measures the thickness of a layer of benzene ( $n = 1.50$ ) floating on water by shining monochromatic light onto the film and varying the wavelength of the light. She finds that light of wavelength 575 nm is reflected most strongly from the film. What does she calculate for the minimum thickness of the film?

**35.29 •• Compact Disc Player.** A compact disc (CD) is read from the bottom by a semiconductor laser with wavelength 790 nm passing through a plastic substrate of refractive index 1.8. When the beam encounters a pit, part of the beam is reflected from the pit and part from the flat region between the pits, so these two beams interfere with each other (Fig. E35.29). What must the minimum pit depth be so that the part of the beam reflected from a pit cancels the part of the beam reflected from the flat region? (It is this cancellation that allows the player to recognize the beginning and end of a pit.)

Figure E35.29



### Section 35.5 The Michelson Interferometer

**35.30 •** Jan first uses a Michelson interferometer with the 606 nm light from a krypton-86 lamp. He displaces the movable mirror away from him, counting 818 fringes moving across a line in his field of view. Then Linda replaces the krypton lamp with filtered 502 nm light from a helium lamp and displaces the movable mirror toward her. She also counts 818 fringes, but they move across the line in her field of view opposite to the direction they moved for Jan. Assume that both Jan and Linda counted to 818 correctly. (a) What distance did each person move the mirror? (b) What is the resultant displacement of the mirror?

**35.31 •** How far must the mirror  $M_2$  (see Fig. 35.19) of the Michelson interferometer be moved so that 1800 fringes of He-Ne laser light ( $\lambda = 633$  nm) move across a line in the field of view?

## PROBLEMS

**35.32 ••** The LIGO experiment, which historically detected gravitational waves for the first time in September 2015, uses a pair of highly sensitive Michelson interferometers. These have arms that are 4.00 km long and use powerful Nd:Yag lasers with 1064 nm wavelength. The beams traverse the arms both ways 280 times before recombining, which effectively lengthens the arm length to 1120 km. The devices are tuned so that the beams destructively interfere when they recombine if no gravitational wave is present. (a) The beam has a power of 100 kW, concentrated into an area of a square centimeter. Calculate the amplitude of the electric field in the beam. (b) LIGO can detect a gravitational wave that temporarily lengthens one arm by the minuscule amount of  $10^{-18}$  m! When this happens, the beams combine with a phase difference of  $\pi + \delta$ . Estimate the shift  $\delta$  in radians. Note that the phase difference accumulates during both traversals of each round trip. (c) Use Eq. (35.7) to estimate the sensitivity of the photodetector in terms of the minimal electric field strength needed to detect a gravitational wave.

**35.33 ••** One round face of a 3.25 m, solid, cylindrical plastic pipe is covered with a thin black coating that completely blocks light. The opposite face is covered with a fluorescent coating that glows when it is struck by light. Two straight, thin, parallel scratches, 0.225 mm apart, are made in the center of the black face. When laser light of wavelength 632.8 nm shines through the slits perpendicular to the black face, you find that the central bright fringe on the opposite face is 5.82 mm wide, measured between the dark fringes that border it on either side. What is the index of refraction of the plastic?

**35.34 ••** Newton's rings are visible when a planoconvex lens is placed on a flat glass surface. For a particular lens with an index of refraction of  $n = 1.50$  and a glass plate with an index of  $n = 1.80$ , the diameter of the third bright ring is 0.640 mm. If water ( $n = 1.33$ ) now fills the space between the lens and the glass plate, what is the new diameter of this ring? Assume the radius of curvature of the lens is much greater than the wavelength of the light.

**35.35 • BIO Coating Eyeglass Lenses.** Eyeglass lenses can be coated on the *inner* surfaces to reduce the reflection of stray light to the eye. If the lenses are medium flint glass of refractive index 1.62 and the coating is fluorite of refractive index 1.432, (a) what minimum thickness of film is needed on the lenses to cancel light of wavelength 550 nm reflected toward the eye at normal incidence? (b) Will any other wavelengths of visible light be cancelled or enhanced in the reflected light?

**35.36 •• BIO Sensitive Eyes.** After an eye examination, you put some eyedrops on your sensitive eyes. The cornea (the front part of the eye) has an index of refraction of 1.38, while the eyedrops have a refractive index of 1.45. After you put in the drops, your friends notice that your eyes look red, because red light of wavelength 600 nm has been reinforced in the reflected light. (a) What is the minimum thickness of the film of eyedrops on your cornea? (b) Will any other wavelengths of visible light be reinforced in the reflected light? Will any be cancelled? (c) Suppose you had contact lenses, so that the eyedrops went on them instead of on your corneas. If the refractive index of the lens material is 1.50 and the layer of eyedrops has the same thickness as in part (a), what wavelengths of visible light will be reinforced? What wavelengths will be cancelled?

**35.37 ••** Two flat plates of glass with parallel faces are on a table, one plate on the other. Each plate is 11.0 cm long and has a refractive index of 1.55. A very thin sheet of metal foil is inserted under the end of the upper plate to raise it slightly at that end, in a manner similar to that discussed in Example 35.4. When you view the glass plates from above with reflected white light, you observe that, at 1.15 mm from the line where the sheets are in contact, the violet light of wavelength 400.0 nm is enhanced in this reflected light, but no visible light is enhanced closer to the line of contact. (a) How far from the line of contact will green light (of wavelength 550.0 nm) and orange light (of wavelength 600.0 nm) first be enhanced? (b) How far from the line of contact will the violet, green, and orange light again be enhanced in the reflected light? (c) How thick is the metal foil holding the ends of the plates apart?

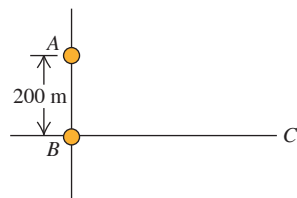
**35.38 ••** A typical red laser pointer has a wavelength of 650 nm. Suppose we wanted to test the wave nature of light by carefully cutting two parallel slits in a dark plastic sheet. We would then shine the laser through the slits onto a wall located 1 m beyond the sheet. (a) Determine the slit spacing needed so that the bright spots on the wall would be discernible with 1 cm spacing. (b) Is this a feasible "home experiment"? Is it possible to cut slits with that separation using typical household tools? (c) Suppose we wanted to test the wave nature of sound in a similar manner, by placing two small speakers driven in-phase near each other, separated by 40 cm, both facing a wall 2 m distant. If we used a 1.0 kHz tone, determine the distance between points along the wall that would exhibit enhanced sound. (d) Suppose we wanted the sound-enhanced points to be separated by only 1.75 m to render this experiment feasible. Estimate an audible frequency  $f$  and use it to determine a speaker separation distance  $d$  that would accomplish this. (e) Is this a feasible "home experiment"?

**35.39 ••** Suppose you illuminate two thin slits by monochromatic coherent light in air and find that they produce their first interference *minima* at  $\pm 35.20^\circ$  on either side of the central bright spot. You then immerse these slits in a transparent liquid and illuminate them with the same light. Now you find that the first minima occur at  $\pm 19.46^\circ$  instead. What is the index of refraction of this liquid?

**35.40 •• CP CALC** A very thin sheet of brass contains two thin parallel slits. When a laser beam shines on these slits at normal incidence and room temperature ( $20.0^\circ\text{C}$ ), the first interference dark fringes occur at  $\pm 26.6^\circ$  from the original direction of the laser beam when viewed from some distance. If this sheet is now slowly heated to  $135^\circ\text{C}$ , by how many degrees do these dark fringes change position? Do they move closer together or farther apart? See Table 17.1 for pertinent information, and ignore any effects that might occur due to a change in the thickness of the slits. (*Hint:* Thermal expansion normally produces very small changes in length, so you can use differentials to find the change in the angle.)

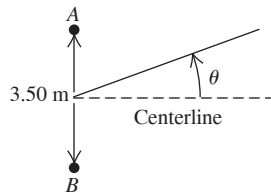
**35.41 ••** Two radio antennas radiating in phase are located at points A and B, 200 m apart (Fig. P35.41). The radio waves have a frequency of 5.80 MHz. A radio receiver is moved out from point B along a line perpendicular to the line connecting A and B (line BC shown in Fig. P35.41). At what distances from B will there be *destructive* interference? (*Note:* The distance of the receiver from the sources is not large in comparison to the separation of the sources, so Eq. (35.5) does not apply.)

Figure P35.41



**35.42 ••** Two speakers A and B are 3.50 m apart, and each one is emitting a frequency of 444 Hz. However, because of signal delays in the cables, speaker A is one-fourth of a period *ahead* of speaker B. For points far from the speakers, find all the angles relative to the centerline (Fig. P35.42) at which the sound from these speakers cancels. Include angles on *both* sides of the centerline. The speed of sound is 340 m/s.

Figure P35.42



**35.43 •• CP** A thin uniform film of refractive index 1.750 is placed on a sheet of glass of refractive index 1.50. At room temperature ( $20.0^\circ\text{C}$ ), this film is just thick enough for light with wavelength 582.4 nm reflected off the top of the film to be cancelled by light reflected from the top of the glass. After the glass is placed in an oven and slowly heated to  $170^\circ\text{C}$ , you find that the film cancels reflected light with wavelength 588.5 nm. What is the coefficient of linear expansion of the film? (Ignore any changes in the refractive index of the film due to the temperature change.)

**35.44 ••• GPS Transmission.** The GPS (Global Positioning System) satellites are approximately 5.18 m across and transmit two low-power signals, one of which is at 1575.42 MHz (in the UHF band). In a series of laboratory tests on the satellite, you put two 1575.42 MHz UHF transmitters at opposite ends of the satellite. These broadcast in phase uniformly in all directions. You measure the intensity at points on a circle that is several hundred meters in radius and centered on the satellite. You measure angles on this circle relative to a point that lies along the centerline of the satellite (that is, the perpendicular bisector of a line that extends from one transmitter to the other). At this point on the circle, the measured intensity is  $2.00 \text{ W/m}^2$ . (a) At how many other angles in the range  $0^\circ < \theta < 90^\circ$  is the intensity also  $2.00 \text{ W/m}^2$ ? (b) Find the four smallest angles in the range  $0^\circ < \theta < 90^\circ$  for which the intensity is  $2.00 \text{ W/m}^2$ . (c) What is the intensity at a point on the circle at an angle of  $4.65^\circ$  from the centerline?

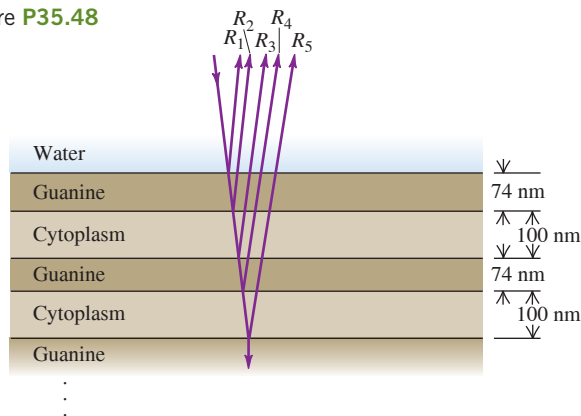
**35.45 ••** White light reflects at normal incidence from the top and bottom surfaces of a glass plate ( $n = 1.52$ ). There is air above and below the plate. Constructive interference is observed for light whose wavelength in air is 477.0 nm. What is the thickness of the plate if the next longer wavelength for which there is constructive interference is 540.6 nm?

**35.46 ••** Laser light of wavelength 510 nm is traveling in air and shines at normal incidence onto the flat end of a transparent plastic rod that has  $n = 1.30$ . The end of the rod has a thin coating of a transparent material that has refractive index 1.65. What is the minimum (nonzero) thickness of the coating (a) for which there is maximum transmission of the light into the rod; (b) for which transmission into the rod is minimized?

**35.47 ••** Red light with wavelength 700 nm is passed through a two-slit apparatus. At the same time, monochromatic visible light with another wavelength passes through the same apparatus. As a result, most of the pattern that appears on the screen is a mixture of two colors; however, the center of the third bright fringe ( $m = 3$ ) of the red light appears pure red, with none of the other color. What are the possible wavelengths of the second type of visible light? Do you need to know the slit spacing to answer this question? Why or why not?

**35.48 •• BIO Reflective Coatings and Herring.** Herring and related fish have a brilliant silvery appearance that camouflages them while they are swimming in a sunlit ocean. The silveriness is due to *platelets* attached to the surfaces of these fish. Each platelet is made up of several alternating layers of crystalline guanine ( $n = 1.80$ ) and of cytoplasm ( $n = 1.333$ , the same as water), with a guanine layer on the outside in contact with the surrounding water (Fig. P35.48). In one typical platelet, the guanine layers are 74 nm thick and the cytoplasm layers are 100 nm thick. (a) For light striking the platelet surface at normal incidence, for which vacuum wavelengths of visible light will all of the reflections  $R_1, R_2, R_3, R_4$ , and  $R_5$ , shown in Fig. P35.48, be approximately in phase? If white light is shone on this platelet, what color will be most strongly reflected (see Fig. 32.4)? The surface of a herring has very many platelets side by side with layers of different thickness, so that *all* visible wavelengths are reflected. (b) Explain why such a “stack” of layers is more reflective than a single layer of guanine with cytoplasm underneath. (A stack of five guanine layers separated by cytoplasm layers reflects more than 80% of incident light at the wavelength for which it is “tuned.”) (c) The color that is most strongly reflected from a platelet depends on the angle at which it is viewed. Explain why this should be so. (You can see these changes in color by examining a herring from different angles. Most of the platelets on these fish are oriented in the same way, so that they are vertical when the fish is swimming.)

Figure P35.48

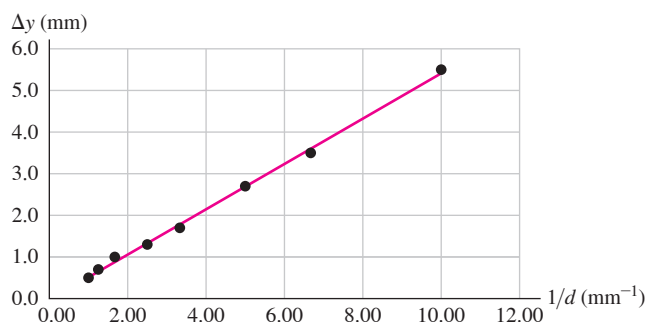


**35.49 ••** After a laser beam passes through two thin parallel slits, the first completely dark fringes occur at  $\pm 19.0^\circ$  with the original direction of the beam, as viewed on a screen far from the slits. (a) What is the ratio of the distance between the slits to the wavelength of the light illuminating the slits? (b) What is the smallest angle, relative to the original direction of the laser beam, at which the intensity of the light is  $\frac{1}{10}$  the maximum intensity on the screen?



**35.50 •• DATA** In your summer job at an optics company, you are asked to measure the wavelength  $\lambda$  of the light that is produced by a laser. To do so, you pass the laser light through two narrow slits that are separated by a distance  $d$ . You observe the interference pattern on a screen that is 0.900 m from the slits and measure the separation  $\Delta y$  between adjacent bright fringes in the portion of the pattern that is near the center of the screen. Using a microscope, you measure  $d$ . But both  $\Delta y$  and  $d$  are small and difficult to measure accurately, so you repeat the measurements for several pairs of slits, each with a different value of  $d$ . Your results are shown in **Fig. P35.50**, where you have plotted  $\Delta y$  versus  $1/d$ . The line in the graph is the best-fit straight line for the data. (a) Explain why the data points plotted this way fall close to a straight line. (b) Use **Fig. P35.50** to calculate  $\lambda$ .

Figure P35.50

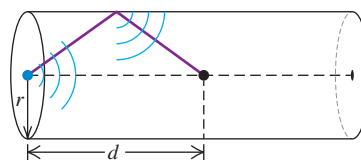


**35.51 •• DATA** Short-wave radio antennas  $A$  and  $B$  are connected to the same transmitter and emit coherent waves in phase and with the same frequency  $f$ . You must determine the value of  $f$  and the placement of the antennas that produce a maximum intensity through constructive interference at a receiving antenna that is located at point  $P$ , which is at the corner of your garage. First you place antenna  $A$  at a point 240.0 m due east of  $P$ . Next you place antenna  $B$  on the line that connects  $A$  and  $P$ , a distance  $x$  due east of  $P$ , where  $x < 240.0$  m. Then you measure that a maximum in the total intensity from the two antennas occurs when  $x = 210.0$  m, 216.0 m, and 222.0 m. You don't investigate smaller or larger values of  $x$ . (Treat the antennas as point sources.) (a) What is the frequency  $f$  of the waves that are emitted by the antennas? (b) What is the greatest value of  $x$ , with  $x < 240.0$  m, for which the interference at  $P$  is destructive?

**35.52 •• DATA** In your research lab, a very thin, flat piece of glass with refractive index 1.40 and uniform thickness covers the opening of a chamber that holds a gas sample. The refractive indexes of the gases on either side of the glass are very close to unity. To determine the thickness of the glass, you shine coherent light of wavelength  $\lambda_0$  in vacuum at normal incidence onto the surface of the glass. When  $\lambda_0 = 496$  nm, constructive interference occurs for light that is reflected at the two surfaces of the glass. You find that the next shorter wavelength in vacuum for which there is constructive interference is 386 nm. (a) Use these measurements to calculate the thickness of the glass. (b) What is the longest wavelength in vacuum for which there is constructive interference for the reflected light?

**35.53 •• CP** An acoustic waveguide consists of a long cylindrical tube with radius  $r$  designed to channel sound waves, as shown in **Fig. P35.53**. A tone with frequency  $f$  is emitted from a small source at the center of one end of this tube. Depending on the radius of the tube and the frequency of the tone, pressure nodes can develop along the tube axis where rays reflected from the periphery constructively interfere with direct rays. (a) For sound waves with wavelength  $\lambda$ , what is the minimum tube radius for which at least one such node exists? (b) The tube has radius 25.0 cm and the temperature is 20°C. If the tone has frequency 2.50 kHz, how many nodes exist? (c) At what distance  $d$  are these nodes located? (d) If the tube were filled with helium rather than air, how many nodes would exist, and at what value of  $d$ ?

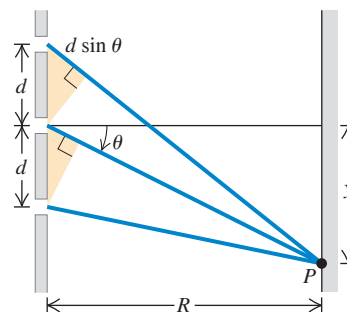
Figure P35.53



## CHALLENGE PROBLEMS

**35.54 ••• Three-Slit Interference.** Monochromatic light with wavelength  $\lambda$  is incident on a screen with three narrow slits with separation  $d$ , as shown in **Fig. P35.54**. Light from the middle slit reaches point  $P$  with electric field  $E \cos(\omega t)$ . From the small-angle approximation, light from the upper and lower slits reaches point  $P$  with electric fields  $E \cos(\omega t + \phi)$  and  $E \cos(\omega t - \phi)$ , respectively, where  $\phi = (2\pi d \sin \theta)/\lambda$  is the phase lag and phase lead associated with the different path lengths. (a) Using either a phasor analysis similar to **Fig. 35.9** or trigonometric identities, determine the electric-field amplitude  $E_P$  associated with the net field at point  $P$ , in terms of  $\phi$ . (b) Determine the intensity  $I$  at point  $P$  in terms of the maximum net intensity  $I_0$  and the phase angle  $\phi$ . (c) There are two sets of relative maxima: one with intensity  $I_0$  when  $\phi = 2\pi m$ , where  $m$  is an integer, and another with a smaller intensity at other values of  $\phi$ . What values of  $\phi$  exhibit the "lesser" maxima? (d) What is the intensity at the lesser maxima, in terms of  $I_0$ ? (e) What values of  $\phi$  correspond to the dark fringes closest to the center? (f) If the incident light has wavelength 650 nm, the slits are separated by  $d = 0.200$  mm, and the distance to the far screen is  $R = 1.00$  m, what is the distance from the central maximum to the first lesser maximum? (g) What is the distance from the central maximum to the closest absolute maximum?

Figure P35.54

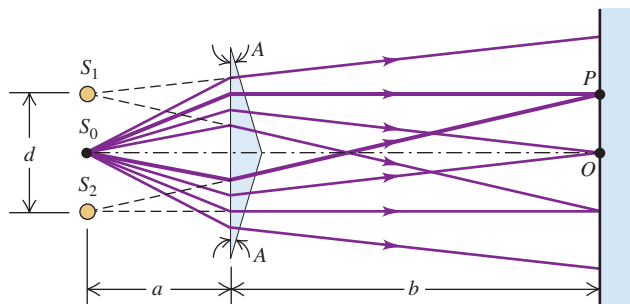


**35.55 ••• CP** The index of refraction of a glass rod is 1.48 at  $T = 20.0^\circ\text{C}$  and varies linearly with temperature, with a coefficient of  $2.50 \times 10^{-5}/^\circ\text{C}$ . The coefficient of linear expansion of the glass is  $5.00 \times 10^{-6}/^\circ\text{C}$ . At  $20.0^\circ\text{C}$  the length of the rod is 3.00 cm. A Michelson interferometer has this glass rod in one arm, and the rod is being heated so that its temperature increases at a rate of  $5.00^\circ\text{C}/\text{min}$ . The light source has wavelength  $\lambda = 589$  nm, and the rod initially is at  $T = 20.0^\circ\text{C}$ . How many fringes cross the field of view each minute?



**35.56 ••• CP** Figure P35.56 shows an interferometer known as *Fresnel's biprism*. The magnitude of the prism angle  $A$  is extremely small. (a) If  $S_0$  is a very narrow source slit, show that the separation of the two virtual coherent sources  $S_1$  and  $S_2$  is given by  $d = 2aA(n - 1)$ , where  $n$  is the index of refraction of the material of the prism. (b) Calculate the spacing of the fringes of green light with wavelength 500 nm on a screen 2.00 m from the biprism. Take  $a = 0.200$  m,  $A = 3.50$  mrad, and  $n = 1.50$ .

Figure P35.56



### MCAT-STYLE PASSAGE PROBLEMS

**Interference and Sound Waves.** Interference occurs with not only light waves but also all frequencies of electromagnetic waves and all other types of waves, such as sound and water waves. Suppose that your physics professor sets up two sound speakers in the front of your classroom and uses an electronic oscillator to produce sound waves of a single frequency. When she turns the oscillator on (take this to be its original setting), you and many students hear a loud tone while other students hear nothing. (The speed of sound in air is 340 m/s.)

**35.57** The professor then adjusts the apparatus. The frequency that you hear does not change, but the loudness decreases. Now all of your fellow students can hear the tone. What did the professor do? (a) She turned off the oscillator. (b) She turned down the volume of the speakers. (c) She changed the phase relationship of the speakers. (d) She disconnected one speaker.

**35.58** The professor returns the apparatus to the original setting. She then adjusts the speakers again. All of the students who had heard nothing originally now hear a loud tone, while you and the others who had originally heard the loud tone hear nothing. What did the professor do? (a) She turned off the oscillator. (b) She turned down the volume of the speakers. (c) She changed the phase relationship of the speakers. (d) She disconnected one speaker.

**35.59** The professor again returns the apparatus to its original setting, so you again hear the original loud tone. She then slowly moves one speaker away from you until it reaches a point at which you can no longer hear the tone. If she has moved the speaker by 0.34 m (farther from you), what is the frequency of the tone? (a) 1000 Hz; (b) 2000 Hz; (c) 500 Hz; (d) 250 Hz.

**35.60** The professor once again returns the apparatus to its original setting, but now she adjusts the oscillator to produce sound waves of half the original frequency. What happens? (a) The students who originally heard a loud tone again hear a loud tone, and the students who originally heard nothing still hear nothing. (b) The students who originally heard a loud tone now hear nothing, and the students who originally heard nothing now hear a loud tone. (c) Some of the students who originally heard a loud tone again hear a loud tone, but others in that group now hear nothing. (d) Among the students who originally heard nothing, some still hear nothing but others now hear a loud tone.

## ANSWERS

### Chapter Opening Question ?

(v) The colors appear as a result of constructive interference between light waves reflected from the upper and lower surfaces of the oil film. The wavelength of light for which the most constructive interference occurs at a point, and hence the color that appears the brightest at that point, depends on (1) the thickness of the film (which determines the path difference between light waves that reflect off the two surfaces), (2) the oil's index of refraction (which gives the wavelength of light in the oil a different value than in air), and (3) the index of refraction of the material below the oil (which determines whether the wave that reflects from the lower surface undergoes a half-cycle phase shift). (See Examples 35.4, 35.5, and 35.6 in Section 35.4.)

### Key Example VARIATION Problems

**VP35.2.1** (a) 684 nm (b) 17.1 mm

**VP35.2.2** (a)  $0^\circ$ ,  $20.7^\circ$ ,  $44.9^\circ$  (b)  $10.2^\circ$ ,  $31.9^\circ$ ,  $61.9^\circ$

**VP35.2.3** (a) 0.472 mm (b) 6.88 mm

**VP35.2.4** (a) 657 nm (b)  $0.254^\circ$

**VP35.3.1** (a)  $0.0192 \text{ W/m}^2$  (b)  $5.58^\circ$

**VP35.3.2** (a)  $0.0173 \text{ W/m}^2$  (b) 1.66 mm

**VP35.3.3** (a) 3.63 m (b)  $9.14 \times 10^{-4} \text{ W/m}^2$

**VP35.3.4** (a) 573 nm (b) 8.22 mm

**VP35.6.1** (a) 0.0174 mm (b) 0.956 mm

**VP35.6.2** (a) constructive: 584 nm; destructive: none (b) constructive: none; destructive: 426 nm

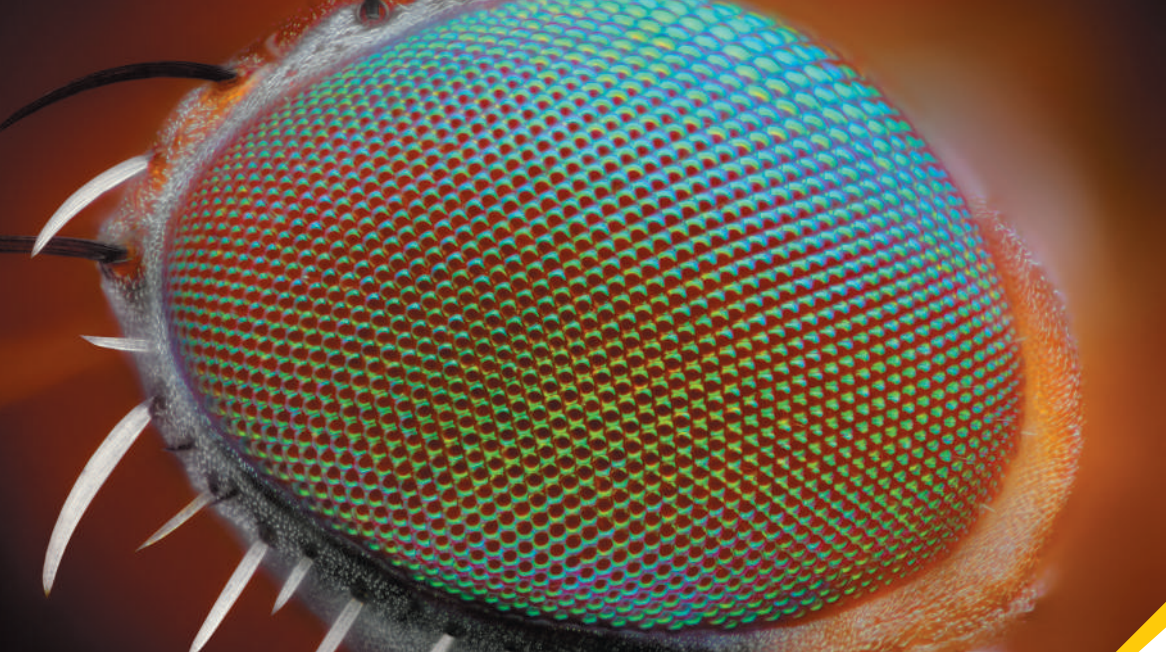
**VP35.6.3** (a) constructive: none; destructive: 688 nm (b) constructive: 468 nm; destructive: none

**VP35.6.4** (a) 88.3 nm (b) 265 nm

### Bridging Problem

(a) 441 nm; violet

(b) 551 nm



? Flies have *compound* eyes with thousands of miniature lenses. The overall diameter of the eye is about 1 mm, but each lens is only about  $20\ \mu\text{m}$  in diameter and produces an individual image of a small region in the fly's field of view. Compared to the resolving power of the human eye (in which the light-gathering region is about 16 mm across), the ability of a fly's eye to resolve small details is (i) worse because the lenses are so small; (ii) worse because the eye as a whole is so small; (iii) better because the lenses are so small; (iv) better because the eye as a whole is so small; (v) about the same.

# 36 Diffraction

Everyone is used to the idea that sound bends around corners. If sound didn't behave this way, you couldn't hear a police siren that's out of sight or the speech of a person whose back is turned to you. But *light* can bend around corners as well. When light from a point source falls on a straightedge and casts a shadow, the edge of the shadow is never perfectly sharp. Some light appears in the area that we expect to be in the shadow, and we find alternating bright and dark fringes in the illuminated area. In general, light emerging from apertures doesn't precisely follow the predictions of the straight-line ray model of geometric optics.

The reason for these effects is that light, like sound, has wave characteristics. In Chapter 35 we studied the interference patterns that can arise when two light waves are combined. In this chapter we'll investigate interference effects due to combining *many* light waves. Such effects are referred to as *diffraction*. The behavior of waves after they pass through an aperture is an example of diffraction; each infinitesimal part of the aperture acts as a source of waves, and these waves interfere, producing a pattern of bright and dark fringes.

Similar patterns appear when light emerges from *arrays* of apertures. The nature of these patterns depends on the color of the light and the size and spacing of the apertures. Examples of this effect include the colors of iridescent butterflies and the "rainbow" you see reflected from the surface of a compact disc. We'll explore similar effects with x rays that are used to study the atomic structure of solids and liquids. Finally, we'll look at the physics of a *hologram*, a special kind of interference pattern used to form three-dimensional images.

## 36.1 FRESNEL AND FRAUNHOFER DIFFRACTION

According to geometric optics, when an opaque object is placed between a point light source and a screen, as in (Fig. 36.1, next page) the shadow of the object forms a perfectly sharp line. No light at all strikes the screen at points within the shadow, and the area outside the shadow is illuminated nearly uniformly. But as we saw in Chapter 35, the *wave* nature of light causes effects that can't be understood with geometric optics. An important class of such effects occurs when light strikes a barrier that has an aperture or an edge. The interference patterns formed in such a situation are grouped under the heading **diffraction**.

## LEARNING OUTCOMES

### In this chapter, you'll learn...

- 36.1 What happens when coherent light shines on an object with an edge or aperture.
- 36.2 How to understand the diffraction pattern formed when coherent light passes through a narrow slit.
- 36.3 How to calculate the intensity at various points in a single-slit diffraction pattern.
- 36.4 What happens when coherent light shines on an array of narrow, closely spaced slits.
- 36.5 How scientists use diffraction gratings for precise measurements of wavelength.
- 36.6 How x-ray diffraction reveals the arrangement of atoms in a crystal.
- 36.7 How diffraction sets limits on the smallest details that can be seen with an optical system.
- 36.8 How holograms work.

### You'll need to review...

- 33.4, 33.7 Prisms and dispersion; Huygens's principle.
- 34.4, 34.5 Image formation by a lens;  $f$ -number.
- 35.1–35.3 Coherent light, two-slit interference, and phasors.

Figure 36.1 A point source of light illuminates a straightedge.

Geometric optics predicts that this situation should produce a sharp boundary between illumination and solid shadow.

That's NOT what really happens!

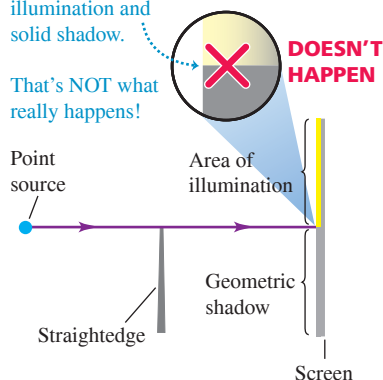


Figure 36.2 shows an example of diffraction. The photograph in Fig. 36.2a was made by placing a razor blade halfway between a pinhole, illuminated by monochromatic light, and a photographic film. The film recorded the shadow cast by the blade. Figure 36.2b is an enlargement of a region near the shadow of the right edge of the blade. The position of the *geometric* shadow line is indicated by arrows. The area outside the geometric shadow is bordered by alternating bright and dark bands. There is some light in the shadow region, although this is not very visible in the photograph. The first bright band in Fig. 36.2b, just to the right of the geometric shadow, is considerably brighter than in the region of uniform illumination to the extreme right. This simple experiment gives us some idea of the richness and complexity of diffraction.

We don't often observe diffraction patterns such as Fig. 36.2 in everyday life because most ordinary light sources are neither monochromatic nor point sources. If we use a white frosted light bulb instead of a point source to illuminate the razor blade in Fig. 36.2, each wavelength of the light from every point of the bulb forms its own diffraction pattern, but the patterns overlap so much that we can't see any individual pattern.

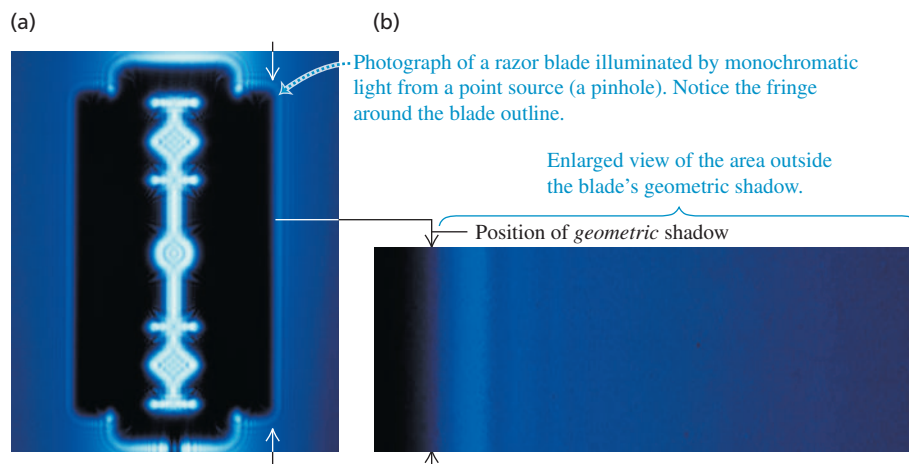
## Diffraction and Huygens's Principle

We can use Huygens's principle (see Section 33.7) to analyze diffraction patterns. This principle states that we can consider every point of a wave front as a source of secondary wavelets. These spread out in all directions with a speed equal to the speed of propagation of the wave. The position of the wave front at any later time is the *envelope* of the secondary wavelets at that time. To find the resultant displacement at any point, we use the superposition principle to combine all the individual displacements produced by these secondary waves.

In Fig. 36.1, both the point source and the screen are relatively close to the obstacle forming the diffraction pattern. This situation is described as *near-field diffraction* or **Fresnel diffraction**, pronounced “Freh-nell” (after the French scientist Augustin Jean Fresnel, 1788–1827). By contrast, we use the term **Fraunhofer diffraction** (after the German physicist Joseph von Fraunhofer, 1787–1826) for situations in which the source, obstacle, and screen are far enough apart that we can consider all lines from the source to the obstacle to be parallel, and can likewise consider all lines from the obstacle to a given point on the screen to be parallel. We'll restrict the following discussion to Fraunhofer diffraction, which is usually simpler to analyze in detail than Fresnel diffraction.

Diffraction is sometimes described as “the bending of light around an obstacle.” But the process that causes diffraction is present in the propagation of *every* wave. When part of the wave is cut off by some obstacle, we observe diffraction effects that result from interference of the remaining parts of the wave fronts. Optical instruments typically use only a limited portion of a wave; for example, a telescope uses only the part of a wave that is admitted by its objective lens or mirror. Thus diffraction plays a role in nearly all optical phenomena.

Figure 36.2 An example of diffraction.



Finally, we emphasize that there is no fundamental distinction between *interference* and *diffraction*. In Chapter 35 we used the term *interference* for effects involving waves from a small number of sources, usually two. *Diffraction* usually involves a *continuous* distribution of Huygens's wavelets across the area of an aperture, or a very large number of sources or apertures. But both interference and diffraction are consequences of superposition and Huygens's principle.

**TEST YOUR UNDERSTANDING OF SECTION 36.1** Can *sound* waves undergo diffraction around an edge?

**ANSWER**

**I yes** When you hear the voice of someone standing around a corner, you are hearing sound waves that underwent diffraction. If there were no diffraction or reflection of sound, you could hear sounds only from objects that were in plain view.

## 36.2 DIFFRACTION FROM A SINGLE SLIT

In this section we'll discuss the diffraction pattern formed by plane-wave (parallel-ray) monochromatic light when it emerges from a long, narrow slit, as shown in **Fig. 36.3**. We call the narrow dimension the *width*, even though in this figure it is a vertical dimension.

According to geometric optics, the transmitted beam should have the same cross section as the slit, as in Fig. 36.3a. What is *actually* observed is the pattern shown in Fig. 36.3b. The beam spreads out vertically after passing through the slit. The diffraction pattern consists of a central bright band, which may be much broader than the width of the slit, bordered by alternating dark and bright bands with rapidly decreasing intensity. About 85% of the power in the transmitted beam is in the central bright band, whose width is *inversely* proportional to the slit width. In general, the narrower the slit, the broader the entire diffraction pattern. (The *horizontal* spreading of the beam in Fig. 36.3b is negligible because the horizontal dimension of the slit is relatively large.) You can observe a similar diffraction pattern by looking at a point source, such as a distant street light, through a narrow slit formed between your two thumbs held in front of your eye; the retina of your eye acts as the screen.

### Single-Slit Diffraction: Locating the Dark Fringes

**Figure 36.4** (next page) shows a side view of the same setup; the long sides of the slit are perpendicular to the figure, and plane waves are incident on the slit from the left. According to Huygens's principle, each element of area of the slit opening can be considered as a source of secondary waves. In particular, imagine dividing the slit into several narrow strips of equal width, parallel to the long edges and perpendicular to the page. Figure 36.4a shows two such strips. Cylindrical secondary wavelets, shown in cross section, spread out from each strip.

Figure 36.3 (a) The “shadow” of a horizontal slit as incorrectly predicted by geometric optics. (b) A horizontal slit actually produces a diffraction pattern. The slit width has been greatly exaggerated.

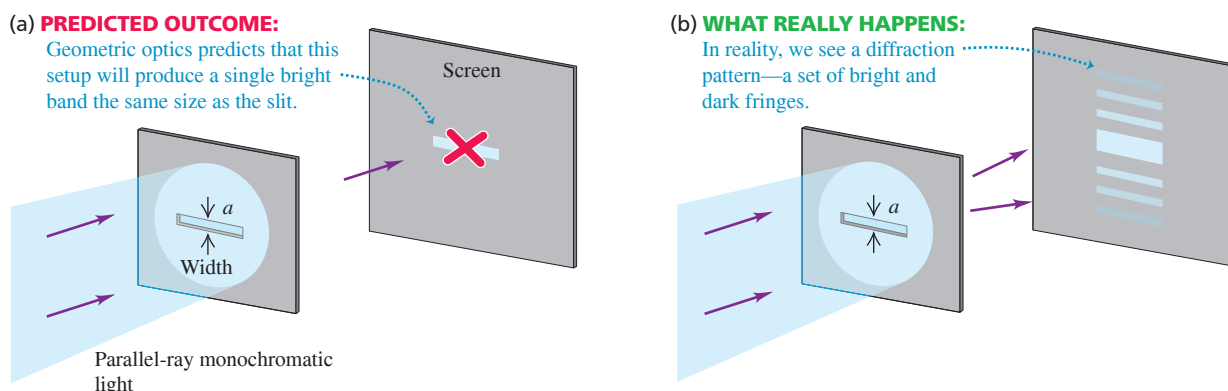
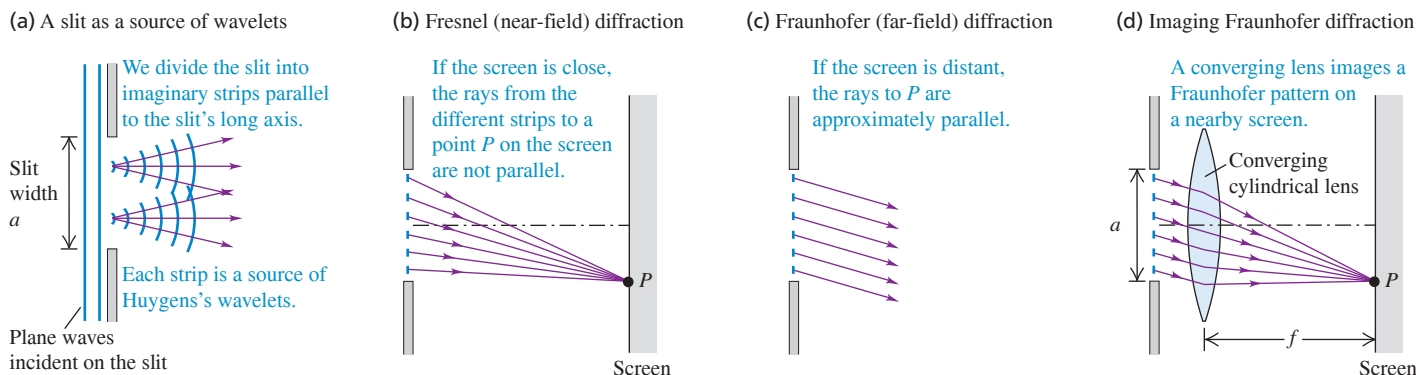




Figure 36.4 Diffraction by a single rectangular slit. The long sides of the slit are perpendicular to the figure.

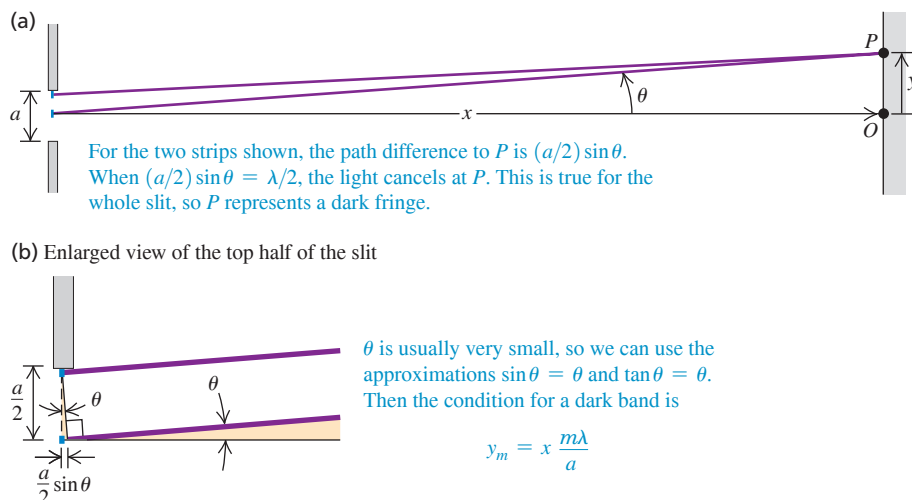


In Fig. 36.4b a screen is placed to the right of the slit. We can calculate the resultant intensity at a point  $P$  on the screen by adding the contributions from the individual wavelets, taking proper account of their various phases and amplitudes. It's easiest to do this calculation if we assume that the screen is far enough away that all the rays from various parts of the slit to a particular point  $P$  on the screen are parallel, as in Fig. 36.4c. An equivalent situation is Fig. 36.4d, in which the rays to the lens are parallel and the lens forms a reduced image of the same pattern that would be formed on an infinitely distant screen without the lens. We might expect that the various light paths through the lens would introduce additional phase shifts, but in fact it can be shown that all the paths have *equal* phase shifts, so this is not a problem.

The situation of Fig. 36.4b is Fresnel diffraction; those in Figs. 36.4c and 36.4d, where the outgoing rays are considered parallel, are Fraunhofer diffraction. We can derive quite simply the most important characteristics of the Fraunhofer diffraction pattern from a single slit. First consider two narrow strips, one just below the top edge of the drawing of the slit and one at its center, shown in end view in **Fig. 36.5**. The difference in path length to point  $P$  is  $(a/2)\sin\theta$ , where  $a$  is the slit width and  $\theta$  is the angle between the perpendicular to the slit and a line from the center of the slit to  $P$ . Suppose this path difference happens to be equal to  $\lambda/2$ ; then light from these two strips arrives at point  $P$  with a half-cycle phase difference, and cancellation occurs.

Similarly, light from two strips immediately *below* the two in the figure also arrives at  $P$  a half-cycle out of phase. In fact, the light from *every* strip in the top half of the slit cancels out the light from a corresponding strip in the bottom half. Hence the combined

Figure 36.5 Side view of a horizontal slit. When the distance  $x$  to the screen is much greater than the slit width  $a$ , the rays from a distance  $a/2$  apart may be considered parallel.



light from the entire slit completely cancels at  $P$ , giving a dark fringe in the interference pattern. A dark fringe occurs whenever

$$\frac{a}{2} \sin \theta = \pm \frac{\lambda}{2} \quad \text{or} \quad \sin \theta = \pm \frac{\lambda}{a} \quad (36.1)$$

The plus-or-minus ( $\pm$ ) sign in Eq. (36.1) says that there are symmetric dark fringes above and below point  $O$  in Fig. 36.5a. The upper fringe ( $\theta > 0$ ) occurs at a point  $P$  where light from the bottom half of the slit travels  $\lambda/2$  farther to  $P$  than does light from the top half; the lower fringe ( $\theta < 0$ ) occurs where light from the *top* half travels  $\lambda/2$  farther than light from the *bottom* half.

We may also divide the slit into quarters, sixths, and so on, and use the above argument to show that a dark fringe occurs whenever  $\sin \theta = \pm 2\lambda/a$ ,  $\pm 3\lambda/a$ , and so on. Thus the condition for a *dark* fringe is

**Dark fringes, single-slit diffraction:**

Angle of line from center of slit to  $m$ th dark fringe on screen

$$\sin \theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \dots) \quad (36.2)$$

Slit width  $a$       Wavelength

For example, if the slit width is equal to ten wavelengths ( $a = 10\lambda$ ), dark fringes occur at  $\sin \theta = \pm \frac{1}{10}$ ,  $\pm \frac{2}{10}$ ,  $\pm \frac{3}{10}$ ,  $\dots$ . Between the dark fringes are bright fringes. Note that  $\sin \theta = 0$  corresponds to a *bright* band; in this case, light from the entire slit arrives at  $P$  in phase. Thus it would be wrong to put  $m = 0$  in Eq. (36.2).

With light, the wavelength  $\lambda$  is of the order of  $500 \text{ nm} = 5 \times 10^{-7} \text{ m}$ . This is often much smaller than the slit width  $a$ ; a typical slit width is  $10^{-2} \text{ cm} = 10^{-4} \text{ m}$ . Therefore the values of  $\theta$  in Eq. (36.2) are often so small that the approximation  $\sin \theta \approx \theta$  (where  $\theta$  is in radians) is a very good one. In that case we can rewrite this equation as

$$\theta = \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \pm 3, \dots) \quad (\text{for small angles } \theta \text{ in radians})$$

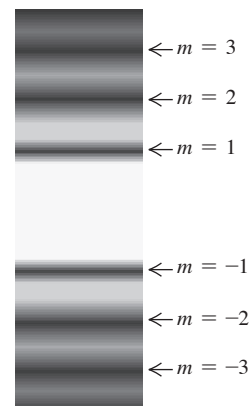
Also, if the distance from slit to screen is  $x$ , as in Fig. 36.5a, and the vertical distance of the  $m$ th dark band from the center of the pattern is  $y_m$ , then  $\tan \theta = y_m/x$ . For small  $\theta$  we may also approximate  $\tan \theta$  by  $\theta$  (in radians). We then find

$$y_m = x \frac{m\lambda}{a} \quad (\text{for } y_m \ll x) \quad (36.3)$$

**Figure 36.6** is a photograph of a single-slit diffraction pattern with the  $m = \pm 1$ ,  $\pm 2$ , and  $\pm 3$  minima labeled. The central bright fringe is wider than the other bright fringes; in the small-angle approximation used in Eq. (36.3), it is exactly twice as wide.

**CAUTION Single-slit diffraction vs. two-slit interference** Equation (36.3) has the same form as the equation for the two-slit pattern, Eq. (35.6), except that in Eq. (36.3) we use  $x$  rather than  $R$  for the distance to the screen. But Eq. (36.3) gives the positions of the *dark* fringes in a *single-slit* pattern rather than the *bright* fringes in a *double-slit* pattern. Also,  $m = 0$  in Eq. (36.2) is *not* a dark fringe. Finally, note that Eq. (36.2) gives the angle  $\theta$  from the  $m$ th dark fringe to the center of the diffraction pattern—not the angle from a dark fringe on one side of the pattern to the corresponding dark fringe on the other side. Be careful! ■

Figure 36.6 Photograph of the Fraunhofer diffraction pattern of a single horizontal slit.



### EXAMPLE 36.1 Single-slit diffraction

### WITH VARIATION PROBLEMS

You pass  $633 \text{ nm}$  laser light through a narrow slit and observe the diffraction pattern on a screen  $6.0 \text{ m}$  away. The distance on the screen between the centers of the first minima on either side of the central bright fringe is  $32 \text{ mm}$  (**Fig. 36.7**, next page). How wide is the slit?

**IDENTIFY and SET UP** This problem involves the relationship between the positions of dark fringes in a single-slit diffraction pattern and the slit width  $a$  (our target variable). The distances between fringes on the screen are much smaller than the slit-to-screen distance, so the angle  $\theta$  shown in Fig. 36.5a is very small and we can use Eq. (36.3) to solve for  $a$ .

**EXECUTE** The first minimum corresponds to  $m = 1$  in Eq. (36.3). The distance  $y_1$  from the central maximum to the first minimum on either side is half the distance between the two first minima, so  $y_1 = (32 \text{ mm})/2 = 16 \text{ mm}$ . Solving Eq. (36.3) for  $a$ , we find

$$a = \frac{x\lambda}{y_1} = \frac{(6.0 \text{ m})(633 \times 10^{-9} \text{ m})}{16 \times 10^{-3} \text{ m}} = 2.4 \times 10^{-4} \text{ m} = 0.24 \text{ mm}$$

*Continued*



$E_P$  of the resultant electric field at  $P$  is equal to the chord  $AB$ , which is  $2(E_0/\beta) \sin(\beta/2)$ . (Note that  $\beta$  *must* be in radians!) We then have

$$E_P = E_0 \frac{\sin(\beta/2)}{\beta/2} \quad (\text{amplitude in single-slit diffraction}) \quad (36.4)$$

The intensity at each point on the screen is proportional to the square of the amplitude given by Eq. (36.4). If  $I_0$  is the intensity in the straight-ahead direction where  $\theta = 0$  and  $\beta = 0$ , then the intensity  $I$  at any point is

$$I = I_0 \left[ \frac{\sin(\beta/2)}{\beta/2} \right]^2 \quad (\text{intensity in single-slit diffraction}) \quad (36.5)$$

We can express the phase difference  $\beta$  in terms of geometric quantities, as we did for the two-slit pattern. From Eq. (35.11) the phase difference is  $2\pi/\lambda$  times the path difference. Figure 36.5 shows that the path difference between the ray from the top of the slit and the ray from the middle of the slit is  $(a/2) \sin \theta$ . The path difference between the rays from the top of the slit and the bottom of the slit is twice this, so

$$\beta = \frac{2\pi}{\lambda} a \sin \theta \quad (36.6)$$

and Eq. (36.5) becomes

$$I = I_0 \left\{ \frac{\sin \left[ \frac{\pi a (\sin \theta)}{\lambda} \right]}{\frac{\pi a (\sin \theta)}{\lambda}} \right\}^2 \quad (36.7)$$

Angle of line from center of slit to position on screen  
Intensity in single-slit diffraction  
Intensity at  $\theta = 0$   
Slit width  
Wavelength

This equation expresses the intensity directly in terms of the angle  $\theta$ . In many calculations it is easier first to calculate the phase angle  $\beta$ , from Eq. (36.6), and then to use Eq. (36.5).

Equation (36.7) is plotted in **Fig. 36.9a**. Note that the central intensity peak is much larger than any of the others. This means that most of the power in the wave remains within an angle  $\theta$  from the perpendicular to the slit, where  $\sin \theta = \lambda/a$  (the first diffraction minimum). You can see this easily in Fig. 36.9b, which is a photograph of water waves undergoing single-slit diffraction. Note also that the peak intensities in Fig. 36.9a decrease rapidly as we go away from the center of the pattern. (Compare Fig. 36.6, which shows a single-slit diffraction pattern for light.)

The dark fringes in the pattern are the places where  $I = 0$ . These occur at points for which the numerator of Eq. (36.5) is zero so that  $\beta$  is a multiple of  $2\pi$ . From Eq. (36.6) this corresponds to

$$\begin{aligned} \frac{a \sin \theta}{\lambda} &= m \quad (m = \pm 1, \pm 2, \dots) \\ \sin \theta &= \frac{m\lambda}{a} \quad (m = \pm 1, \pm 2, \dots) \end{aligned} \quad (36.8)$$

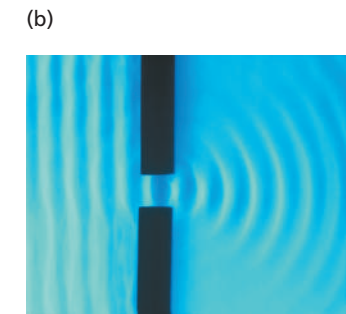
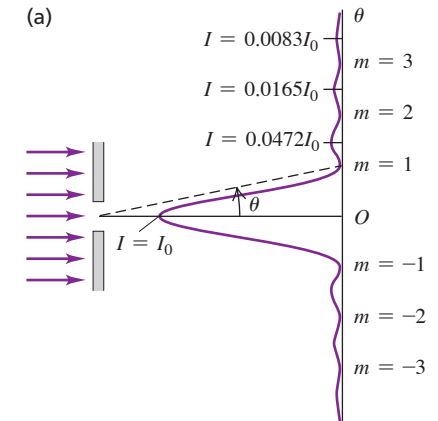
This agrees with our previous result, Eq. (36.2). Note again that  $\beta = 0$  (corresponding to  $\theta = 0$ ) is *not* a minimum. Equation (36.5) is indeterminate at  $\beta = 0$ , but we can evaluate the limit as  $\beta \rightarrow 0$  by using L'Hôpital's rule. We find that at  $\beta = 0$ ,  $I = I_0$ , as we should expect.

### Intensity Maxima in the Single-Slit Pattern

We can also use Eq. (36.5) to calculate the positions of the peaks, or *intensity maxima*, and the intensities at these peaks. This is not quite as simple as it may appear. We might expect the peaks to occur where the sine function reaches the value  $\pm 1$ —namely, where  $\beta = \pm \pi, \pm 3\pi, \pm 5\pi$ , or in general,

$$\beta \approx \pm (2m + 1)\pi \quad (m = 0, 1, 2, \dots) \quad (36.9)$$

Figure 36.9 (a) Intensity versus angle in single-slit diffraction. The values of  $m$  label intensity minima given by Eq. (36.8). Most of the wave power goes into the central intensity peak (between the  $m = 1$  and  $m = -1$  intensity minima). (b) These water waves passing through a small aperture behave exactly like light waves in single-slit diffraction. Only the diffracted waves within the central intensity peak are visible; the waves at larger angles are too faint to see.





This is *approximately* correct, but because of the factor  $(\beta/2)^2$  in the denominator of Eq. (36.5), the maxima don't occur precisely at these points. When we take the derivative of Eq. (36.5) with respect to  $\beta$  and set it equal to zero to try to find the maxima and minima, we get a transcendental equation that has to be solved numerically. In fact there is *no* maximum near  $\beta = \pm\pi$ . The first maxima on either side of the central maximum, near  $\beta = \pm 3\pi$ , actually occur at  $\pm 2.860\pi$ . The second side maxima, near  $\beta = \pm 5\pi$ , are actually at  $\pm 4.918\pi$ , and so on. The error in Eq. (36.9) vanishes in the limit of large  $m$ —that is, for intensity maxima far from the center of the pattern.

To find the intensities at the side maxima, we substitute these values of  $\beta$  back into Eq. (36.5). Using the approximate expression in Eq. (36.9), we get

$$I_m \approx \frac{I_0}{\left(m + \frac{1}{2}\right)^2 \pi^2} \quad (36.10)$$

where  $I_m$  is the intensity of the  $m$ th side maximum and  $I_0$  is the intensity of the central maximum. Equation (36.10) gives the series of intensities

$$0.0450I_0 \quad 0.0162I_0 \quad 0.0083I_0$$

and so on. As we have pointed out, this equation is only approximately correct. The actual intensities of the side maxima turn out to be

$$0.0472I_0 \quad 0.0165I_0 \quad 0.0083I_0 \quad \dots$$

These intensities decrease very rapidly, as Fig. 36.9a also shows. Even the first side maxima have less than 5% of the intensity of the central maximum.

### Width of the Single-Slit Pattern

For small angles the angular spread of the diffraction pattern is inversely proportional to the ratio of the slit width  $a$  to the wavelength  $\lambda$ . **Figure 36.10** shows graphs of intensity  $I$  as a function of the angle  $\theta$  for three values of the ratio  $a/\lambda$ .

With light waves, the wavelength  $\lambda$  is often much smaller than the slit width  $a$ , and the values of  $\theta$  in Eqs. (36.6) and (36.7) are so small that the approximation  $\sin\theta = \theta$  is very good. With this approximation the position  $\theta_1$  of the first ( $m = 1$ ) minimum, corresponding to  $\beta/2 = \pi$ , is, from Eq. (36.7),

$$\theta_1 = \frac{\lambda}{a} \quad (36.11)$$

This characterizes the width (angular spread) of the central maximum, and we see that it is *inversely* proportional to the slit width  $a$ . When the small-angle approximation is valid, the central maximum is exactly twice as wide as each side maximum. When  $a$  is of the order of a centimeter or more,  $\theta_1$  is so small that we can consider practically all the light to be concentrated at the geometrical focus. But when  $a$  is less than  $\lambda$ , the central maximum spreads over  $180^\circ$ , and the fringe pattern is not seen at all.

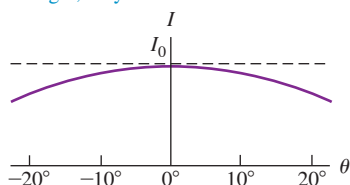
Figure 36.10 The single-slit diffraction pattern depends on the ratio of the slit width  $a$  to the wavelength  $\lambda$ .

(a)  $a = \lambda$

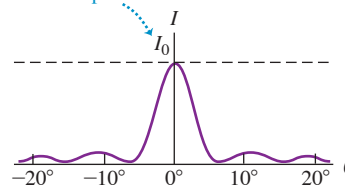
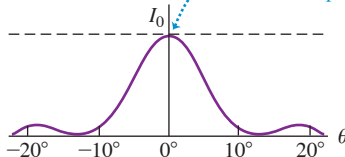
(b)  $a = 5\lambda$

(c)  $a = 8\lambda$

If the slit width is equal to or narrower than the wavelength, only one broad maximum forms.



The wider the slit (or the shorter the wavelength), the narrower and sharper is the central peak.



It's important to keep in mind that diffraction occurs for *all* kinds of waves, not just light. Sound waves undergo diffraction when they pass through a slit or aperture such as an ordinary doorway. The sound waves used in speech have wavelengths of about a meter or greater, and a typical doorway is less than 1 m wide; in this situation,  $a$  is less than  $\lambda$ , and the central intensity maximum extends over  $180^\circ$ . This is why the sounds coming through an open doorway can easily be heard by an eavesdropper hiding out of sight around the corner. In the same way, sound waves can bend around the head of an instructor who faces the blackboard while lecturing (**Fig. 36.11**). By contrast, there is essentially no diffraction of visible light through a doorway because the width  $a$  is very much greater than the wavelength  $\lambda$  (of order  $5 \times 10^{-7}$  m). You can *hear* around corners because typical sound waves have relatively long wavelengths; you cannot *see* around corners because the wavelength of visible light is very short.

**Figure 36.11** The sound waves used in speech have a long wavelength (about 1 m) and can easily bend around this instructor's head. By contrast, light waves have very short wavelengths and undergo very little diffraction. Hence you can't *see* around his head!



### EXAMPLE 36.2 Single-slit diffraction: Intensity I

#### WITH VARIATION PROBLEMS

(a) The intensity at the center of a single-slit diffraction pattern is  $I_0$ . What is the intensity at a point in the pattern where there is a 66 radian phase difference between wavelets from the two edges of the slit? (b) If this point is  $7.0^\circ$  away from the central maximum, how many wavelengths wide is the slit?

**IDENTIFY and SET UP** In our analysis of Fig. 36.8 we used the symbol  $\beta$  for the phase difference between wavelets from the two edges of the slit. In part (a) we use Eq. (36.5) to find the intensity  $I$  at the point in the pattern where  $\beta = 66$  rad. In part (b) we need to find the slit width  $a$  as a multiple of the wavelength  $\lambda$  so our target variable is  $a/\lambda$ . We are given the angular position  $\theta$  of the point where  $\beta = 66$  rad, so we can use Eq. (36.6) to solve for  $a/\lambda$ .

**EXECUTE** (a) We have  $\beta/2 = 33$  rad, so from Eq. (36.5),

$$I = I_0 \left[ \frac{\sin(33 \text{ rad})}{33 \text{ rad}} \right]^2 = (9.2 \times 10^{-4}) I_0$$

(b) From Eq. (36.6),

$$\frac{a}{\lambda} = \frac{\beta}{2\pi \sin \theta} = \frac{66 \text{ rad}}{(2\pi \text{ rad}) \sin 7.0^\circ} = 86$$

For example, for 550 nm light the slit width is  $a = (86)(550 \text{ nm}) = 4.7 \times 10^{-5} \text{ m} = 0.047 \text{ mm}$ , or roughly  $\frac{1}{20} \text{ mm}$ .

**EVALUATE** To what point in the diffraction pattern does this value of  $\beta$  correspond? To find out, note that  $\beta = 66$  rad is approximately equal to  $21\pi$ . This is an odd multiple of  $\pi$ , corresponding to the form  $(2m + 1)\pi$  found in Eq. (36.9) for the intensity *maxima*. Hence  $\beta = 66$  rad corresponds to a point near the tenth ( $m = 10$ ) maximum. This is well beyond the range shown in Fig. 36.9a, which shows only maxima out to  $m = \pm 3$ .

**KEYCONCEPT** The intensity at a given point in a single-slit diffraction pattern depends on the phase difference  $\beta$  between the wavelets that arrive at that point from the two edges of the slit [Eqs. (36.6) and (36.7)]. The intensity is greatest at the center of the pattern where  $\beta = 0$ ; the intensity is zero at points where  $\beta = \pm 2\pi, \pm 4\pi, \pm 6\pi, \dots$

### EXAMPLE 36.3 Single-slit diffraction: Intensity II

#### WITH VARIATION PROBLEMS

In the experiment described in Example 36.1 (Section 36.2), the intensity at the center of the pattern is  $I_0$ . What is the intensity at a point on the screen 3.0 mm from the center of the pattern?

**IDENTIFY and SET UP** This is similar to Example 36.2, except that we are not given the value of the phase difference  $\beta$  at the point in question. We use geometry to determine the angle  $\theta$  for our point and then use Eq. (36.7) to find the intensity  $I$  (the target variable).

**EXECUTE** Referring to Fig. 36.5a, we have  $y = 3.0 \text{ mm}$  and  $x = 6.0 \text{ m}$ , so  $\tan \theta = y/x = (3.0 \times 10^{-3} \text{ m})/(6.0 \text{ m}) = 5.0 \times 10^{-4}$ . This is so small that the values of  $\tan \theta$ ,  $\sin \theta$ , and  $\theta$  (in radians) are all nearly the same. Then, from Eq. (36.7),

$$\frac{\pi a \sin \theta}{\lambda} = \frac{\pi(2.4 \times 10^{-4} \text{ m})(5.0 \times 10^{-4})}{6.33 \times 10^{-7} \text{ m}} = 0.60$$

$$I = I_0 \left( \frac{\sin 0.60}{0.60} \right)^2 = 0.89 I_0$$

**EVALUATE** Figure 36.9a shows that an intensity this high can occur only within the central intensity maximum. This checks out; from Example 36.1, the first intensity minimum ( $m = 1$  in Fig. 36.9a) is  $(32 \text{ mm})/2 = 16 \text{ mm}$  from the center of the pattern, so the point in question here at  $y = 3 \text{ mm}$  does, indeed, lie within the central maximum.

**KEYCONCEPT** When light of wavelength  $\lambda$  passes through a single slit of width  $a$ , you can write the intensity of the diffracted light at angle  $\theta$  [Eq. (36.7)] in terms of the ratio  $(\pi a \sin \theta)/\lambda$ , which is one-half of the phase difference  $\beta$  between wavelets coming from the two edges of the slit. The intensity is greatest at the center of the pattern ( $\theta = 0$ ); the intensity is zero where  $(\pi a \sin \theta)/\lambda = \pm \pi, \pm 2\pi, \pm 3\pi, \dots$

**TEST YOUR UNDERSTANDING OF SECTION 36.3** Coherent electromagnetic radiation is sent through a slit of width  $0.0100\text{ mm}$ . For which of the following wavelengths will there be *no* points in the diffraction pattern where the intensity is zero? (i) Blue light of wavelength  $500\text{ nm}$ ; (ii) infrared light of wavelength  $10.6\text{ }\mu\text{m}$ ; (iii) microwaves of wavelength  $1.00\text{ mm}$ ; (iv) ultraviolet light of wavelength  $50.0\text{ nm}$ .

**ANSWER**

(i) and (ii) If the slit width  $a$  is less than the wavelength  $\lambda$ , there are no points in the diffraction pattern at which the intensity is zero (see Fig. 36.10a). The slit width is  $0.0100\text{ mm}$ . (iii)  $1.00\text{ mm} > 0.0100\text{ mm}$ , so this condition is satisfied for (iii) and (iv)  $10.6\text{ }\mu\text{m} = 0.0106\text{ mm} > 0.0100\text{ mm}$ , so this condition is satisfied for (iii) and (iv). (iv)  $10.6\text{ }\mu\text{m} = 0.0106\text{ mm} > 0.0100\text{ mm}$ , so this condition is satisfied for (iii) and (iv). (v)  $50.0\text{ nm} = 0.00005\text{ mm} < 0.0100\text{ mm}$ , so this condition is not satisfied for (v).

**36.4 MULTIPLE SLITS**

In Sections 35.2 and 35.3 we analyzed interference from two point sources and from two very narrow slits; in this analysis we ignored effects due to the finite (that is, nonzero) slit width. In Sections 36.2 and 36.3 we considered the diffraction effects that occur when light passes through a single slit of finite width. Additional interesting effects occur when we have two slits with finite width or when there are several very narrow slits.

**Two Slits of Finite Width**

Let's take another look at the two-slit pattern in the more realistic case in which the slits have finite width. If the slits are narrow in comparison to the wavelength, we can assume that light from each slit spreads out uniformly in all directions to the right of the slit. We used this assumption in Section 35.3 to calculate the interference pattern described by Eq. (35.10) or (35.15), consisting of a series of equally spaced, equally intense maxima. However, when the slits have finite width, the peaks in the two-slit interference pattern are modulated by the single-slit diffraction pattern characteristic of the width of each slit.

**Figure 36.12a** shows the intensity in a single-slit diffraction pattern with slit width  $a$ . The *diffraction minima* are labeled by the integer  $m_d = \pm 1, \pm 2, \dots$  (“d” for “diffraction”). Figure 36.12b shows the pattern formed by two very narrow slits with distance  $d$  between slits, where  $d$  is four times as great as the single-slit width  $a$  in Fig. 36.12a; that is,  $d = 4a$ . The *interference maxima* are labeled by the integer  $m_i = 0, \pm 1, \pm 2, \dots$  (“i” for “interference”). We note that the spacing between adjacent minima in the single-slit pattern is four times as great as in the two-slit pattern. Now suppose we widen each of the narrow slits to the same width  $a$  as that of the single slit in Fig. 36.12a. Figure 36.12c shows the pattern from two slits with width  $a$ , separated by a distance (between centers)  $d = 4a$ . The effect of the finite width of the slits is to superimpose the two patterns—that is, to multiply the two intensities at each point. The two-slit peaks are in the same positions as before, but their intensities are modulated by the single-slit pattern, which acts as an “envelope” for the intensity function. The expression for the intensity shown in Fig. 36.12c is proportional to the product of the two-slit and single-slit expressions, Eqs. (35.10) and (36.5):

$$I = I_0 \cos^2 \frac{\phi}{2} \left[ \frac{\sin(\beta/2)}{\beta/2} \right]^2 \quad (\text{two slits of finite width}) \quad (36.12)$$

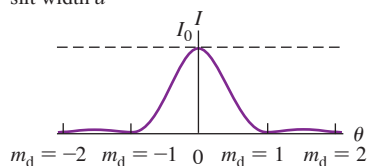
where, as before,

$$\phi = \frac{2\pi d}{\lambda} \sin \theta \quad \beta = \frac{2\pi a}{\lambda} \sin \theta$$

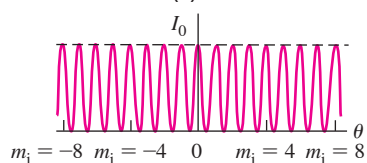
In Fig. 36.12c, every fourth interference maximum at the sides is *missing* because these interference maxima ( $m_i = \pm 4, \pm 8, \dots$ ) coincide with diffraction minima ( $m_d = \pm 1, \pm 2, \dots$ ). This can also be seen in Fig. 36.12d, which is a photograph of an actual pattern with  $d = 4a$ . You should be able to convince yourself that there will be “missing” maxima whenever  $d$  is an integer multiple of  $a$ .

Figure 36.12 Finding the intensity pattern for two slits of finite width.

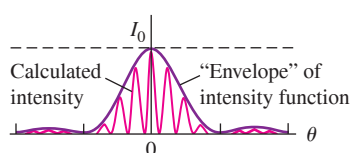
(a) Single-slit diffraction pattern for a slit width  $a$



(b) Two-slit interference pattern for narrow slits whose separation  $d$  is four times the width of the slit in (a)



(c) Calculated intensity pattern for two slits of width  $a$  and separation  $d = 4a$ , including both interference and diffraction effects



(d) Photograph of the pattern calculated in (c)



For  $d = 4a$ , every fourth interference maximum at the sides ( $m_i = \pm 4, \pm 8, \dots$ ) is missing.

Figures 36.12c and 36.12d show that as you move away from the central bright maximum of the two-slit pattern, the intensity of the maxima decreases. This is a result of the single-slit modulating pattern shown in Fig. 36.12a; mathematically, the decrease in intensity arises from the factor  $(\beta/2)^2$  in the denominator of Eq. (36.12). You can also see this decrease in Fig. 35.6 (Section 35.2). The narrower the slits, the broader the single-slit pattern (as in Fig. 36.10) and the slower the decrease in intensity from one interference maximum to the next.

Shall we call the pattern in Fig. 36.12d *interference* or *diffraction*? It's really both, since it results from the superposition of waves coming from various parts of the two apertures.

## Several Slits

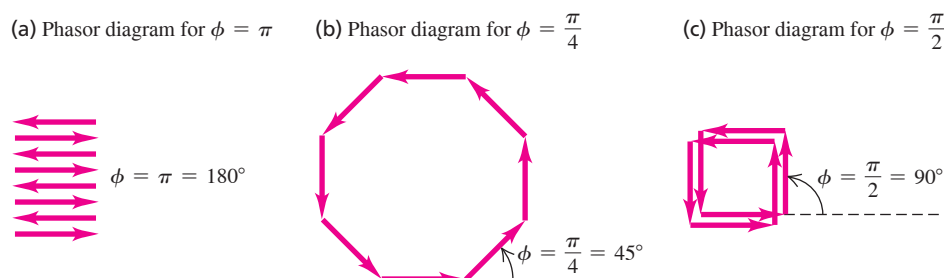
Next let's consider patterns produced by *several* very narrow slits. As we'll see, systems of narrow slits are of tremendous practical importance in *spectroscopy*, the determination of the particular wavelengths of light coming from a source. Assume that each slit is narrow in comparison to the wavelength, so its diffraction pattern spreads out nearly uniformly. **Figure 36.13** shows an array of eight narrow slits, with distance  $d$  between adjacent slits. Constructive interference occurs for rays at angle  $\theta$  to the normal that arrive at point  $P$  with a path difference between adjacent slits equal to an integer number of wavelengths:

$$d \sin \theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots)$$

This means that reinforcement occurs when the phase difference  $\phi$  at  $P$  for light from adjacent slits is an integer multiple of  $2\pi$ . That is, the maxima in the pattern occur at the *same* positions as for two slits with the same spacing.

What happens *between* the maxima is different with multiple slits, however. In the two-slit pattern, there is exactly one intensity minimum located midway between each pair of maxima, corresponding to angles for which the phase difference between waves from the two sources is  $\pi$ ,  $3\pi$ ,  $5\pi$ , and so on. In the eight-slit pattern these are also minima because the light from adjacent slits cancels out in pairs, corresponding to the phasor diagram in **Fig. 36.14a**. But these are not the only minima in the eight-slit pattern. For example, when the phase difference  $\phi$  from adjacent sources is  $\pi/4$ , the phasor diagram is as shown in Fig. 36.14b; the total (resultant) phasor is zero, and the intensity is zero. When  $\phi = \pi/2$ , we get the phasor diagram of Fig. 36.14c, and again both the total phasor and the intensity are zero. More generally, the intensity with eight slits is zero whenever  $\phi$  is an integer multiple of  $\pi/4$ , *except* when  $\phi$  is a multiple of  $2\pi$ . Thus there are seven minima for every maximum.

**Figure 36.14** Phasor diagrams for light passing through eight narrow slits. Intensity maxima occur when the phase difference  $\phi = 0, 2\pi, 4\pi, \dots$ . Between the maxima at  $\phi = 0$  and  $\phi = 2\pi$  are seven minima, corresponding to  $\phi = \pi/4, \pi/2, 3\pi/4, \pi, 5\pi/4, 3\pi/2$ , and  $7\pi/4$ . Can you draw phasor diagrams for the other minima?



**Figure 36.13** Multiple-slit diffraction. Here a lens is used to give a Fraunhofer pattern on a nearby screen, as in Fig. 36.4d.

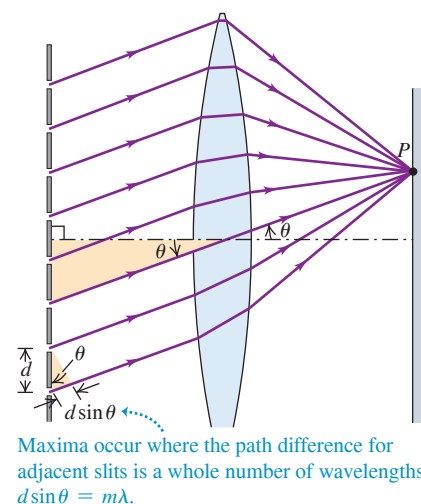
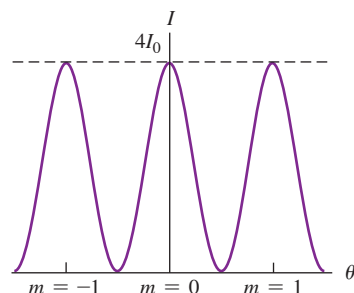


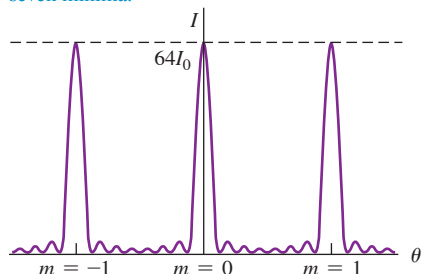


Figure 36.15 Interference patterns for  $N$  equally spaced, very narrow slits. (a) Two slits. (b) Eight slits. (c) Sixteen slits. The vertical scales are different for each graph; the maximum intensity is  $I_0$  for a single slit and  $N^2 I_0$  for  $N$  slits. The width of each peak is proportional to  $1/N$ .

(a)  $N = 2$ : two slits produce one minimum between adjacent maxima.



(b)  $N = 8$ : eight slits produce taller, narrower maxima in the same locations, separated by seven minima.



(c)  $N = 16$ : with 16 slits, the maxima are even taller and narrower, with more intervening minima.

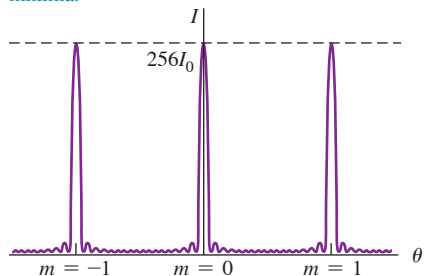


Figure 36.15b shows the result of a detailed calculation of the eight-slit pattern. The large maxima, called *principal maxima*, are in the same positions as for the two-slit pattern of Fig. 36.15a but are much narrower. If the phase difference  $\phi$  between adjacent slits is slightly different from a multiple of  $2\pi$ , the waves from slits 1 and 2 will be only a little out of phase; however, the phase difference between slits 1 and 3 will be greater, that between slits 1 and 4 will be greater still, and so on. This leads to a partial cancellation for angles that are only slightly different from the angle for a maximum, giving the narrow maxima in Fig. 36.15b. The maxima are even narrower with 16 slits (Fig. 36.15c).

You should show that when there are  $N$  slits, there are  $(N - 1)$  minima between each pair of principal maxima and a minimum occurs whenever  $\phi$  is an integer multiple of  $2\pi/N$  (except when  $\phi$  is an integer multiple of  $2\pi$ , which gives a principal maximum). There are small *secondary* intensity maxima between the minima; these become smaller in comparison to the principal maxima as  $N$  increases. The greater the value of  $N$ , the narrower the principal maxima become. From an energy standpoint the total power in the entire pattern is proportional to  $N$ . The height of each principal maximum is proportional to  $N^2$ , so from energy conservation the width of each principal maximum must be proportional to  $1/N$ . In the next section we'll see why the details of the multiple-slit pattern are of great practical importance.

**TEST YOUR UNDERSTANDING OF SECTION 36.4** Suppose two slits, each of width  $a$ , are separated by a distance  $d = 2.5a$ . Are there any missing maxima in the interference pattern produced by these slits? If so, which are missing? If not, why not?

**ANSWER** Yes;  $m = \pm 5, \pm 10, \dots$ . A "missing maximum" satisfies both  $d \sin \theta = m\lambda$  (the condition for an interference maximum) and  $a \sin \theta = m'\lambda$  (the condition for a diffraction minimum). Substituting  $d = 2.5a$ , we can combine these two conditions into the relationship  $m = 2.5m'$ . This is satisfied for  $m = \pm 5$  and  $m = \pm 10$  (the fifth interference maximum is missing because it coincides with the second diffraction minimum),  $m = \pm 15$  and  $m = \pm 20$  (the tenth interference maximum is missing because it coincides with the fourth diffraction minimum), and so on.

## 36.5 THE DIFFRACTION GRATING

We have just seen that increasing the number of slits in an interference experiment (while keeping the spacing of adjacent slits constant) gives interference patterns in which the maxima are in the same positions, but progressively narrower, than with two slits. Because these maxima are so narrow, their angular position, and hence the wavelength, can be measured to very high precision. As we'll see, this effect has many important applications.

An array of a large number of parallel slits, all with the same width  $a$  and spaced equal distances  $d$  between centers, is called a **diffraction grating**. The first one was constructed by Fraunhofer using fine wires. Gratings can be made by using a diamond point to scratch many equally spaced grooves on a glass or metal surface, or by photographic reduction of a pattern of black and white stripes on paper. For a grating, what we have been calling *slits* are often called *rulings* or *lines*.

In Fig. 36.16,  $GG'$  is a cross section of a *transmission grating*; the slits are perpendicular to the plane of the page, and an interference pattern is formed by the light that is transmitted through the slits. The diagram shows only six slits; an actual grating may contain several thousand. The spacing  $d$  between centers of adjacent slits is called the *grating spacing*. A plane monochromatic wave is incident normally on the grating from the left side. We assume far-field (Fraunhofer) conditions; that is, the pattern is formed on a screen that is far enough away that all rays emerging from the grating and going to a particular point on the screen can be considered to be parallel.

We found in Section 36.4 that the principal intensity maxima with multiple slits occur in the same directions as for the two-slit pattern. These are the directions for which the path difference for adjacent slits is an integer number of wavelengths. So the positions of the maxima are once again

$$d \sin \theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots) \quad (36.13)$$

Intensity maxima, multiple slits:   
 Distance between slits  $d$    
 Wavelength  $\lambda$    
 Angle of line from center of slit array to  $m$ th bright region on screen

The intensity patterns for two, eight, and 16 slits displayed in Fig. 36.15 show the progressive increase in sharpness of the maxima as the number of slits increases.

When a grating containing hundreds or thousands of slits is illuminated by a beam of parallel rays of monochromatic light, the pattern is a series of very sharp lines at angles determined by Eq. (36.13). The  $m = \pm 1$  lines are called the *first-order lines*, the  $m = \pm 2$  lines the *second-order lines*, and so on. If the grating is illuminated by white light with a continuous distribution of wavelengths, each value of  $m$  corresponds to a continuous spectrum in the pattern. The angle for each wavelength is determined by Eq. (36.13); for a given value of  $m$ , long wavelengths (the red end of the spectrum) lie at larger angles (that is, are deviated more from the straight-ahead direction) than do the shorter wavelengths at the violet end of the spectrum.

As Eq. (36.13) shows, the sines of the deviation angles of the maxima are proportional to the ratio  $\lambda/d$ . For substantial deviation to occur, the grating spacing  $d$  should be of the same order of magnitude as the wavelength  $\lambda$ . Gratings for use with visible light ( $\lambda$  from 400 to 700 nm) usually have about 1000 slits per millimeter; the value of  $d$  is the *reciprocal* of the number of slits per unit length, so  $d$  is of the order of  $\frac{1}{1000} \text{ mm} = 1000 \text{ nm}$ .

In a *reflection grating*, the array of equally spaced slits shown in Fig. 36.16 is replaced by an array of equally spaced ridges or grooves on a reflective screen. The reflected light has maximum intensity at angles where the phase difference between light waves reflected from adjacent ridges or grooves is an integer multiple of  $2\pi$ . If light of wavelength  $\lambda$  is incident normally on a reflection grating with a spacing  $d$  between adjacent ridges or grooves, the *reflected* angles at which intensity maxima occur are given by Eq. (36.13).

The rainbow-colored reflections from the surface of a DVD are a reflection-grating effect (Fig. 36.17). The “grooves” are tiny pits  $0.12 \mu\text{m}$  deep in the surface of the disc, with a uniform radial spacing of  $0.74 \mu\text{m} = 740 \text{ nm}$ . Information is coded on the DVD by varying the *length* of the pits. The reflection-grating aspect of the disc is merely an aesthetic side benefit.

Figure 36.16 A portion of a transmission diffraction grating. The separation between the centers of adjacent slits is  $d$ .

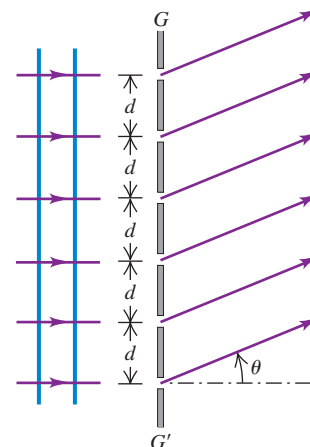


Figure 36.17 Microscopic pits on the surface of this DVD act as a reflection grating, splitting white light into its component colors.



### EXAMPLE 36.4 Width of a grating spectrum

WITH VARIATION PROBLEMS

The wavelengths of the visible spectrum are approximately 380 nm (violet) to 750 nm (red). (a) Find the angular limits of the first-order visible spectrum produced by a plane grating with 600 slits per millimeter when white light falls normally on the grating. (b) Do the first-order and second-order spectra overlap? What about the second-order and third-order spectra? Do your answers depend on the grating spacing?

**IDENTIFY and SET UP** We must find the angles spanned by the visible spectrum in the first-, second-, and third-order spectra. These correspond to  $m = 1, 2$ , and  $3$  in Eq. (36.13).

**EXECUTE** (a) The grating spacing is

$$d = \frac{1}{600 \text{ slits/mm}} = 1.67 \times 10^{-6} \text{ m}$$

We solve Eq. (36.13) for  $\theta$ :

$$\theta = \arcsin \frac{m\lambda}{d}$$

Then for  $m = 1$ , the angular deviations  $\theta_{v1}$  and  $\theta_{r1}$  for violet and red light, respectively, are

$$\theta_{v1} = \arcsin \left( \frac{380 \times 10^{-9} \text{ m}}{1.67 \times 10^{-6} \text{ m}} \right) = 13.2^\circ$$

$$\theta_{r1} = \arcsin \left( \frac{750 \times 10^{-9} \text{ m}}{1.67 \times 10^{-6} \text{ m}} \right) = 26.7^\circ$$

That is, the first-order visible spectrum appears with deflection angles from  $\theta_{v1} = 13.2^\circ$  (violet) to  $\theta_{r1} = 26.7^\circ$  (red).

Continued

(b) With  $m = 2$  and  $m = 3$ , our equation  $\theta = \arcsin(m\lambda/d)$  for 380 nm violet light yields

$$\theta_{v2} = \arcsin\left(\frac{2(380 \times 10^{-9} \text{ m})}{1.67 \times 10^{-6} \text{ m}}\right) = 27.1^\circ$$

$$\theta_{v3} = \arcsin\left(\frac{3(380 \times 10^{-9} \text{ m})}{1.67 \times 10^{-6} \text{ m}}\right) = 43.0^\circ$$

For 750 nm red light, this same equation gives

$$\theta_{r2} = \arcsin\left(\frac{2(750 \times 10^{-9} \text{ m})}{1.67 \times 10^{-6} \text{ m}}\right) = 63.9^\circ$$

$$\theta_{r3} = \arcsin\left(\frac{3(750 \times 10^{-9} \text{ m})}{1.67 \times 10^{-6} \text{ m}}\right) = \arcsin(1.35) = \text{undefined}$$

Hence the second-order spectrum extends from  $27.1^\circ$  to  $63.9^\circ$  and the third-order spectrum extends from  $43.0^\circ$  to  $90^\circ$  (the largest possible value of  $\theta$ ). The undefined value of  $\theta_{r3}$  means that the third-order spectrum

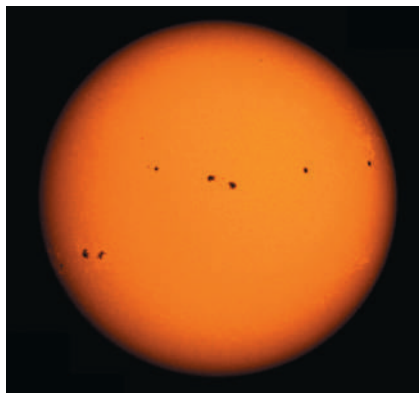
reaches  $\theta = 90^\circ = \arcsin(1)$  at a wavelength shorter than 750 nm; you should be able to show that this happens for  $\lambda = 557 \text{ nm}$ . Hence the first-order spectrum (from  $13.2^\circ$  to  $26.7^\circ$ ) does not overlap with the second-order spectrum, but the second- and third-order spectra do overlap. You can convince yourself that this is true for any value of the grating spacing  $d$ .

**EVALUATE** The fundamental reason the first-order and second-order visible spectra don't overlap is that the human eye is sensitive to only a narrow range of wavelengths. Can you show that if the eye could detect wavelengths from 380 nm to 900 nm (in the near-infrared range), the first and second orders *would* overlap?

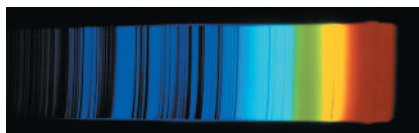
**KEYCONCEPT** In the interference pattern produced by a transmission or reflection grating, the intensity maxima occur at angles that are determined by the grating spacing and the wavelength of the light used. These angles are the same as in the interference pattern produced in the double-slit experiment; the difference is that the maxima are much narrower for a grating.

Figure 36.18 (a) A visible-light photograph of the sun. (b) Sunlight is dispersed into a spectrum by a diffraction grating. Specific wavelengths are absorbed as sunlight passes through the sun's atmosphere, leaving dark lines in the spectrum.

(a)



(b)



## Grating Spectrographs

Diffraction gratings are widely used to measure the spectrum of light emitted by a source, a process called *spectroscopy* or *spectrometry*. Light incident on a grating of known spacing is dispersed into a spectrum. The angles of deviation of the maxima are then measured, and Eq. (36.13) is used to compute the wavelength. With a grating that has many slits, very sharp maxima are produced, and the angle of deviation (and hence the wavelength) can be measured very precisely.

An important application of this technique is to astronomy. As light generated within the sun passes through the sun's atmosphere, certain wavelengths are selectively absorbed. The result is that the spectrum of sunlight produced by a diffraction grating has dark *absorption lines* (Fig. 36.18). Experiments in the laboratory show that different types of atoms and ions absorb light of different wavelengths. By comparing these laboratory results with the wavelengths of absorption lines in the spectrum of sunlight, astronomers can deduce the chemical composition of the sun's atmosphere. The same technique is used to make chemical assays of galaxies that are millions of light-years away.

Figure 36.19 shows one design for a *grating spectrograph* used in astronomy. A transmission grating is used in the figure; in other setups, a reflection grating is used. In older designs a prism was used rather than a grating, and a spectrum was formed by dispersion (see Section 33.4) rather than diffraction. However, there is no simple relationship between wavelength and angle of deviation for a prism, prisms absorb some of the light that passes through them, and they are less effective for many nonvisible wavelengths that are important in astronomy. For these and other reasons, gratings are preferred in precision applications.

## Resolution of a Grating Spectrograph

In spectroscopy it is often important to distinguish slightly differing wavelengths. The minimum wavelength difference  $\Delta\lambda$  that can be distinguished by a spectrograph is described by the **chromatic resolving power**  $R$ , defined as

$$R = \frac{\lambda}{\Delta\lambda} \quad (\text{chromatic resolving power}) \quad (36.14)$$

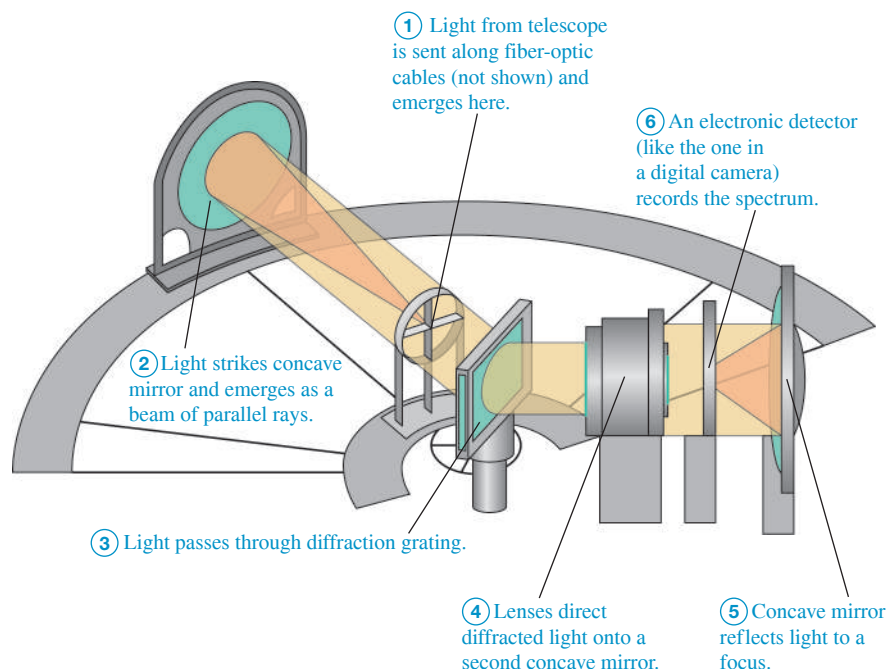


Figure 36.19 A schematic diagram of a diffraction-grating spectrograph for use in astronomy. Note that the light does not strike the grating normal to its surface, so the intensity maxima are given by a somewhat different expression than Eq. (36.13).

As an example, when sodium atoms are heated, they emit strongly at the yellow wavelengths 589.00 nm and 589.59 nm. A spectrograph that can barely distinguish these two lines in the spectrum (called the *sodium doublet*) has a chromatic resolving power  $R = (589.00 \text{ nm})/(0.59 \text{ nm}) = 1000$ . (You can see these wavelengths when boiling water with table salt in it on a gas range. If the water boils over onto the flame, dissolved sodium from the salt emits a burst of yellow light.)

We can derive an expression for the resolving power of a diffraction grating used in a spectrograph. Two different wavelengths give diffraction maxima at slightly different angles. As a reasonable (though arbitrary) criterion, let's assume that we can distinguish them as two separate peaks if the maximum of one coincides with the first minimum of the other.

From our discussion in Section 36.4 the  $m$ th-order maximum occurs when the phase difference  $\phi$  for adjacent slits is  $\phi = 2\pi m$ . The first minimum beside that maximum occurs when  $\phi = 2\pi m + 2\pi/N$ , where  $N$  is the number of slits. The phase difference is also given by  $\phi = (2\pi d \sin \theta)/\lambda$ , so the angular interval  $d\theta$  corresponding to a small increment  $d\phi$  in the phase shift can be obtained from the differential of this equation:

$$d\phi = \frac{2\pi d \cos \theta d\theta}{\lambda}$$

When  $d\phi = 2\pi/N$ , this corresponds to the angular interval  $d\theta$  between a maximum and the first adjacent minimum. Thus  $d\theta$  is given by

$$\frac{2\pi}{N} = \frac{2\pi d \cos \theta d\theta}{\lambda} \quad \text{or} \quad d \cos \theta d\theta = \frac{\lambda}{N}$$

Now we need to find the angular spacing  $d\theta$  between maxima for two slightly different wavelengths. The positions of these maxima are given by  $d \sin \theta = m\lambda$ , and the differential of this equation gives

$$d \cos \theta d\theta = m d\lambda$$

According to our criterion, the limit or resolution is reached when these two angular spacings are equal. Equating the two expressions for the quantity  $(d \cos \theta d\theta)$ , we find

$$\frac{\lambda}{N} = m d\lambda \quad \text{and} \quad \frac{\lambda}{d\lambda} = Nm$$

### BIO APPLICATION Detecting DNA with Diffraction

Diffraction gratings are used in a common piece of laboratory equipment known as a spectrophotometer. Light shining across a diffraction grating is dispersed into its component wavelengths. A slit is used to block all but a very narrow range of wavelengths, producing a beam of almost perfectly monochromatic light. The instrument then measures how much of that light is absorbed by a solution of biological molecules. For example, the sample tube shown here contains a solution of DNA, which is transparent to visible light but which strongly absorbs ultraviolet light with a wavelength of exactly 260 nm. Therefore, by illuminating the sample with 260 nm light and measuring the amount absorbed, we can determine the concentration of DNA in the solution.



**CAUTION** Watch out for different uses of the symbol  $d$  Don't confuse the slit spacing  $d$  with the differential " $d$ " in the angular interval  $d\theta$  or in the phase shift increment  $d\phi$ !



If  $\Delta\lambda$  is small, we can replace  $d\lambda$  by  $\Delta\lambda$ , and the resolving power  $R$  is

$$R = \frac{\lambda}{\Delta\lambda} = Nm \quad (36.15)$$

The greater the number of slits  $N$ , the better the resolution; also, the higher the order  $m$  of the diffraction-pattern maximum that we use, the better the resolution.

**TEST YOUR UNDERSTANDING OF SECTION 36.5** What minimum number of slits would be required in a grating to resolve the sodium doublet in the fourth order? (i) 250; (ii) 400; (iii) 1000; (iv) 4000.

### ANSWER

(i) As described in the text, the resolving power needed is  $R = Nm = 1000$ . In the first order ( $m = 1$ ) we need  $N = 1000$  slits, but in the fourth order ( $m = 4$ ) we need only  $N = 1000/4 = 250$  slits. (These numbers are only approximate because of the arbitrary nature of our criterion for resolution and because real gratings always have slight imperfections in the shapes and spacings of the slits.)

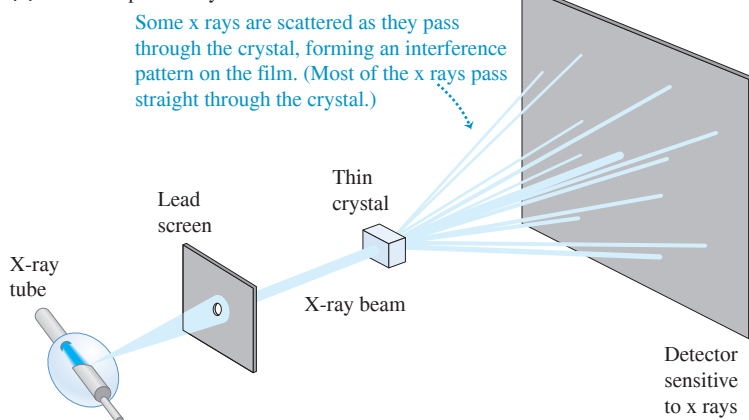
## 36.6 X-RAY DIFFRACTION

X rays were discovered by Wilhelm Röntgen (1845–1923) in 1895, and early experiments suggested that they were electromagnetic waves with wavelengths of the order of  $10^{-10}$  m. At about the same time, the idea began to emerge that in a crystalline solid the atoms are arranged in a regular repeating pattern, with spacing between adjacent atoms also of the order of  $10^{-10}$  m. Putting these two ideas together, Max von Laue (1879–1960) proposed in 1912 that a crystal might serve as a kind of three-dimensional diffraction grating for x rays. That is, a beam of x rays might be scattered (that is, absorbed and re-emitted) by the individual atoms in a crystal, and the scattered waves might interfere just like waves from a diffraction grating.

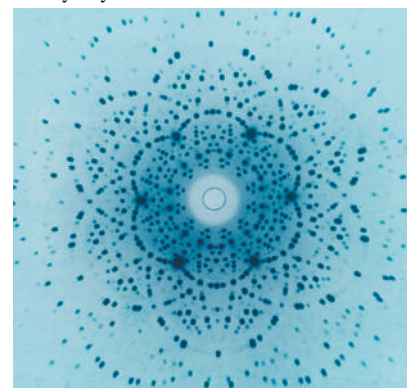
The first **x-ray diffraction** experiments were performed in 1912 by Friedrich, Knipping, and von Laue, using the experimental setup shown in **Fig. 36.20a**. The scattered x rays *did* form an interference pattern, which they recorded on photographic film. Figure 36.20b is a photograph of such a pattern. These experiments verified that x rays *are* waves, or at least have wavelike properties, and also that the atoms in a crystal *are* arranged in a regular pattern (**Fig. 36.21**). Since that time, x-ray diffraction has proved to be an invaluable research tool, both for measuring x-ray wavelengths and for studying the structure of crystals and complex molecules.

Figure 36.20 (a) An x-ray diffraction experiment. (b) Diffraction pattern (or *Laue pattern*) formed by directing a beam of x rays at a thin section of beryl crystal.

(a) Basic setup for x-ray diffraction



(b) Laue diffraction pattern for a thin section of beryl crystal



## A Simple Model of X-Ray Diffraction

To better understand x-ray diffraction, we consider first a two-dimensional scattering situation, as shown in Fig. 36.22a, in which a plane wave is incident on a rectangular array of scattering centers. The situation might be a ripple tank with an array of small posts or x rays incident on an array of atoms. In the case of electromagnetic waves, the wave induces an oscillating electric dipole moment in each scatterer. These dipoles act like little antennas, emitting scattered waves. The resulting interference pattern is the superposition of all these scattered waves. The situation is different from that with a diffraction grating, in which the waves from all the slits are emitted *in phase* (for a plane wave at normal incidence). Here the scattered waves are *not* all in phase because their distances from the *source* are different. To compute the interference pattern, we have to consider the *total* path differences for the scattered waves, including the distances from source to scatterer and from scatterer to observer.

As Fig. 36.22b shows, the path length from source to observer is the same for all the scatterers in a single row if the two angles  $\theta_a$  and  $\theta_r$  are equal. Scattered radiation from *adjacent* rows is *also* in phase if the path difference for adjacent rows is an integer number of wavelengths. Figure 36.22c shows that this path difference is  $2d\sin\theta$ , where  $\theta$  is the common value of  $\theta_a$  and  $\theta_r$ . Therefore the conditions for radiation from the *entire* array to reach the observer in phase are (1) the angle of incidence must equal the angle of scattering and (2) the path difference for adjacent rows must equal  $m\lambda$ , where  $m$  is an integer. We can express the second condition, called the **Bragg condition** in honor of x-ray diffraction pioneers Sir William Bragg and his son Laurence Bragg, as

**Bragg condition for constructive interference from an array:**

$$2d\sin\theta = m\lambda \quad (m = 1, 2, 3, \dots) \quad (36.16)$$

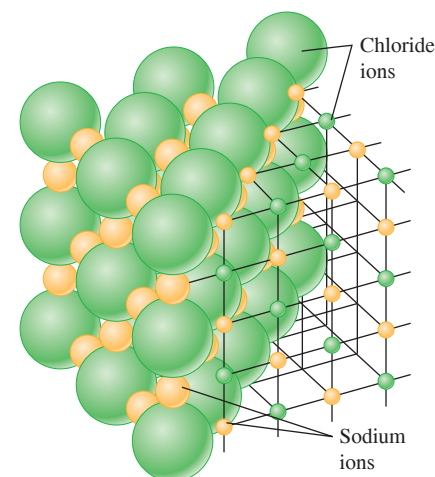
Distance between adjacent rows in array

Wavelength

Angle of line from surface of array to  $m$ th bright region on screen

In directions for which Eq. (36.16) is satisfied, we see a strong maximum in the interference pattern. We can describe this interference in terms of *reflections* of the wave from the horizontal rows of scatterers in Fig. 36.22a. Strong reflection (constructive interference) occurs at angles such that the incident and scattered angles are equal and Eq. (36.16) is satisfied. Since  $\sin\theta$  can never be greater than 1, Eq. (36.16) says that to have constructive interference the quantity  $m\lambda$  must be less than  $2d$  and so  $\lambda$  must be less than  $2d/m$ . For example, the value of  $d$  in an NaCl crystal (see Fig. 36.21) is only 0.282 nm. Hence to have the  $m$ th-order maximum present in the diffraction pattern,  $\lambda$  must be less than  $2(0.282 \text{ nm})/m$ ; that is,  $\lambda < 0.564 \text{ nm}$  for  $m = 1$ ,  $\lambda < 0.282 \text{ nm}$  for  $m = 2$ ,  $\lambda < 0.188 \text{ nm}$  for  $m = 3$ , and so on. These are all x-ray wavelengths (see Fig. 32.4), which is why x rays are used for studying crystal structure.

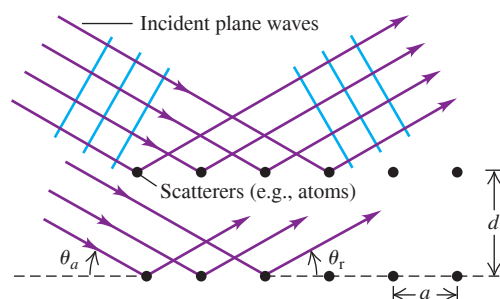
Figure 36.21 Model of the arrangement of ions in a crystal of NaCl (table salt). The spacing of adjacent atoms is 0.282 nm. (The electron clouds of the atoms actually overlap slightly.)



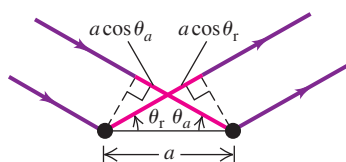
**CAUTION** Scattering from an array In Eq. (36.16) the angle  $\theta$  is measured with respect to the *surface* of the crystal rather than with respect to the *normal* to the plane of an array of slits or a grating. Also, note that the path difference in Eq. (36.16) is  $2d\sin\theta$ , not  $d\sin\theta$  as in Eq. (36.13) for a diffraction grating. ■

Figure 36.22 A two-dimensional model of scattering from a rectangular array. The distance between adjacent atoms in a horizontal row is  $a$ ; the distance between adjacent rows is  $d$ . The angles in (b) are measured from the *surface* of the array, not from its normal.

(a) Scattering of waves from a rectangular array



(b) Scattering from adjacent atoms in a row is constructive when the path lengths  $a\cos\theta_a$  and  $a\cos\theta_r$  are equal, so that the angle of incidence  $\theta_a$  equals the angle of reflection (scattering)  $\theta_r$ .



(c) Scattering from atoms in adjacent rows is constructive when the path difference  $2d\sin\theta$  is an integer number of wavelengths, as in Eq. (36.16).

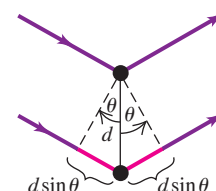
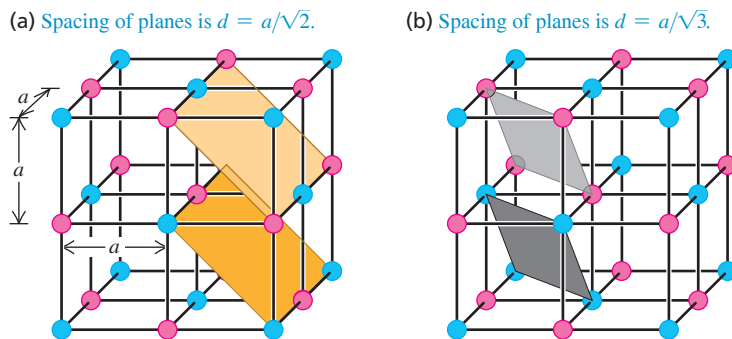
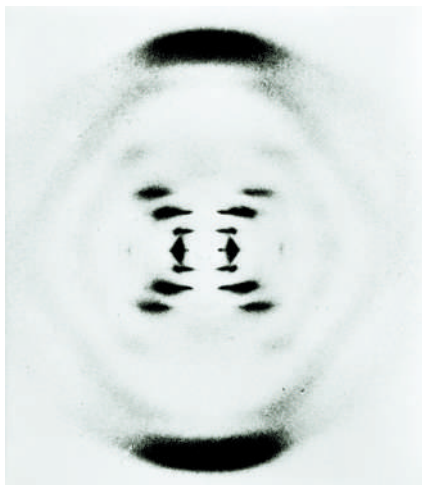


Figure 36.23 A cubic crystal and two different families of crystal planes. There are also three sets of planes parallel to the cube faces, with spacing  $a$ .

**CAUTION** Bragg reflection is really Bragg interference. While we are using the term *reflection*, remember that we are dealing with an *interference* effect. The reflections from various planes are closely analogous to interference effects in thin films (see Section 35.4).

Figure 36.24 The British scientist Rosalind Franklin made this groundbreaking x-ray diffraction image of DNA in 1953. The dark bands arranged in a cross provided the first evidence of the helical structure of the DNA molecule.



We can extend this discussion to a three-dimensional array by considering *planes* of scatterers instead of *rows*. Figure 36.23 shows two different sets of parallel planes that pass through all the scatterers. Waves from all the scatterers in a given plane interfere constructively if the angles of incidence and scattering are equal. There is also constructive interference between planes when Eq. (36.16) is satisfied, where  $d$  is now the distance between adjacent planes. Because there are many different sets of parallel planes, there are also many values of  $d$  and many sets of angles that give constructive interference for the whole crystal lattice. This phenomenon is called **Bragg reflection**.

As Fig. 36.20b shows, in x-ray diffraction there is nearly complete cancellation in all but certain very specific directions in which constructive interference occurs and forms bright spots. Such a pattern is usually called an *x-ray diffraction pattern*, although *interference pattern* might be more appropriate.

We can determine the wavelength of x rays by examining the diffraction pattern for a crystal of known structure and known spacing between atoms, just as we determined wavelengths of visible light by measuring patterns from slits or gratings. (The spacing between atoms in simple crystals of known structure, such as sodium chloride, can be found from the density of the crystal and Avogadro's number.) Then, once we know the x-ray wavelength, we can use x-ray diffraction to explore the structure and determine the spacing between atoms in crystals with unknown structure.

X-ray diffraction is by far the most important experimental tool in the investigation of crystal structure of solids. X-ray diffraction also plays an important role in studies of the structures of liquids and of organic molecules. It has been one of the chief experimental techniques in working out the double-helix structure of DNA (Fig. 36.24) and subsequent advances in molecular genetics.

### EXAMPLE 36.5 X-ray diffraction

### WITH VARIATION PROBLEMS

You direct a beam of 0.154 nm x rays at certain planes of a silicon crystal. As you increase the angle of incidence of the beam from zero, the first strong interference maximum occurs when the beam makes an angle of  $34.5^\circ$  with the planes. (a) How far apart are the planes? (b) Will you find other interference maxima from these planes at larger angles of incidence?

**IDENTIFY and SET UP** This problem involves Bragg reflection of x rays from the planes of a crystal. In part (a) we use the Bragg condition, Eq. (36.16), to find the distance  $d$  between adjacent planes from the known wavelength  $\lambda = 0.154$  nm and angle of incidence  $\theta = 34.5^\circ$  for the  $m = 1$  interference maximum. Given the value of  $d$ , we use the Bragg condition again in part (b) to find the values of  $\theta$  for interference maxima corresponding to other values of  $m$ .

**EXECUTE** (a) We solve Eq. (36.16) for  $d$  and set  $m = 1$ :

$$d = \frac{m\lambda}{2 \sin \theta} = \frac{(1)(0.154 \text{ nm})}{2 \sin 34.5^\circ} = 0.136 \text{ nm}$$

This is the distance between adjacent planes.

(b) To calculate other angles, we solve Eq. (36.16) for  $\sin \theta$ :

$$\sin \theta = \frac{m\lambda}{2d} = m \frac{0.154 \text{ nm}}{2(0.136 \text{ nm})} = m(0.566)$$

Values of  $m$  of 2 or greater give values of  $\sin \theta$  greater than unity, which is impossible. Hence there are *no* other angles for interference maxima for this particular set of crystal planes.

**EVALUATE** Our result in part (b) shows that there *would* be a second interference maximum if the quantity  $2\lambda/2d = \lambda/d$  were less than 1. This would be the case if the wavelength of the x rays were less than  $d = 0.136$  nm. How short would the wavelength need to be to have *three* interference maxima?

**KEYCONCEPT** In x-ray diffraction, waves that pass through or reflect from a crystal are scattered by rows of atoms within the crystal. The pattern of scattered light has maxima at points where waves that interact with different rows arrive in phase and interfere constructively [Eq. (36.16)].

**TEST YOUR UNDERSTANDING OF SECTION 36.6** You are doing an x-ray diffraction experiment with a crystal in which the atomic planes are 0.200 nm apart. You are using x rays of wavelength 0.0900 nm. What is the highest-order maximum present in the diffraction pattern? (i) Third; (ii) fourth; (iii) fifth; (iv) sixth; (v) seventh.

**ANSWER**

(ii) The angular position of the  $m$ th maximum is given by Eq. (36.16),  $2d \sin \theta = m\lambda$ . This gives  $m = (2d \sin \theta)/\lambda$ . The sine function can never be greater than 1, so the largest value of  $m$  in the pattern can be no greater than  $2d/\lambda = 2(0.200 \text{ nm})/(0.0900 \text{ nm}) = 4.44$ . Since  $m$  must be an integer, the highest-order maximum in the pattern is  $m = 4$  (fourth order). The  $m = 5, 6, 7, \dots$  maxima do not appear.

## 36.7 CIRCULAR APERTURES AND RESOLVING POWER

We have studied in detail the diffraction patterns formed by long, thin slits or arrays of slits. But an aperture of *any* shape forms a diffraction pattern. The diffraction pattern formed by a *circular* aperture is of special interest because of its role in limiting how well an optical instrument can resolve fine details. In principle, we could compute the intensity at any point  $P$  in the diffraction pattern by dividing the area of the aperture into small elements, finding the resulting wave amplitude and phase at  $P$ , and then integrating over the aperture area to find the resultant amplitude and intensity at  $P$ . In practice, the integration cannot be carried out in terms of elementary functions. We'll simply *describe* the pattern and quote a few relevant numbers.

The diffraction pattern formed by a circular aperture consists of a central bright spot surrounded by a series of bright and dark rings, as **Fig. 36.25** shows. We can describe the pattern in terms of the angle  $\theta$ , representing the angular radius of each ring. The angular radius  $\theta_1$  of the first *dark* ring is given by

$$\text{Diffraction by a circular aperture:} \quad \sin \theta_1 = 1.22 \frac{\lambda}{D} \quad \text{Angular radius of first dark ring = angular radius of Airy disk} \quad (36.17)$$

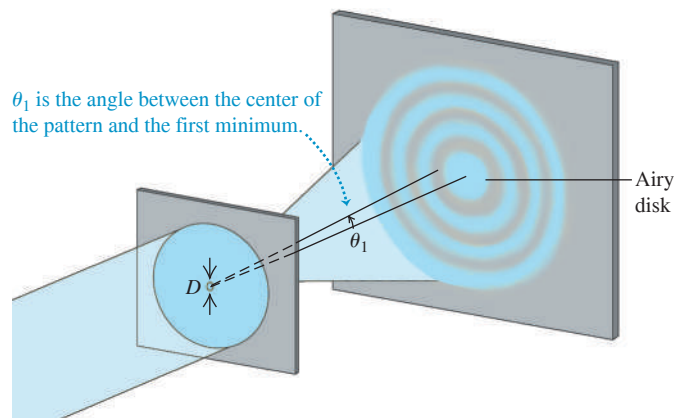
$\lambda$  ← Wavelength  
 $D$  ← Aperture diameter

The angular radii of the next two dark rings are given by

$$\sin \theta_2 = 2.23 \frac{\lambda}{D} \quad \sin \theta_3 = 3.24 \frac{\lambda}{D} \quad (36.18)$$

The central bright spot is called the **Airy disk**, in honor of Sir George Airy (1801–1892), who first derived the expression for the intensity in the pattern. The angular radius of the Airy disk is that of the first dark ring, given by Eq. (36.17). The angular radii of the first three *bright* rings outside the Airy disk are

$$\sin \theta = 1.63 \frac{\lambda}{D}, \quad 2.68 \frac{\lambda}{D}, \quad 3.70 \frac{\lambda}{D} \quad (36.19)$$



**APPLICATION Bigger Telescope, Better Resolving Power** The large aperture diameter of very large telescopes minimizes diffraction effects. The effective diameter of a telescope can be increased by using arrays of smaller telescopes. The Very Large Array (VLA) in New Mexico is a collection of 27 radio telescopes, each 25 m in diameter, that can be spread out in a Y-shaped arrangement 36 km across. Hence the effective aperture diameter is 36 km, giving the VLA a limit of resolution of  $5 \times 10^{-8}$  rad at a radio wavelength of 1.5 cm. If your eye had this resolving power, you could read the “20/20” line on an eye chart more than 30 km away!



**Figure 36.25** Diffraction pattern formed by a circular aperture of diameter  $D$ . The pattern consists of a central bright spot and alternating dark and bright rings. The angular radius  $\theta_1$  of the first dark ring is shown. (This diagram is not drawn to scale.)



Figure 36.26 Photograph of the diffraction pattern formed by a circular aperture.

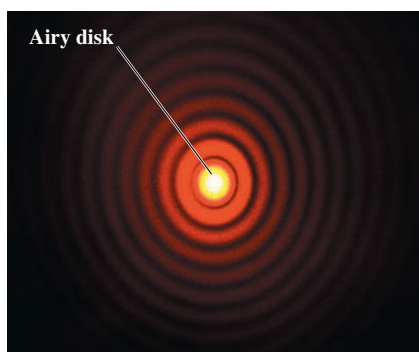
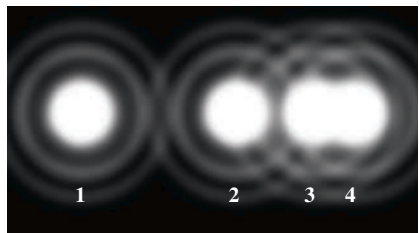
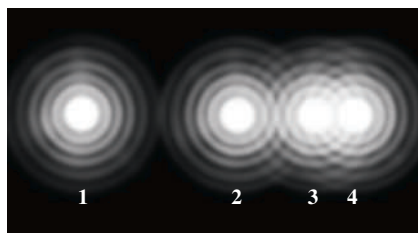


Figure 36.27 These images show the diffraction patterns produced when four very small (“point”) sources of light are viewed through a lens with a circular aperture placed in front of the lens. The photographs were made with a circular aperture in front of the lens. (a) The aperture is so small that the patterns of sources 3 and 4 overlap and are barely resolved by Rayleigh’s criterion. Increasing the size of the aperture decreases the size of the diffraction patterns, as shown in (b) and (c).

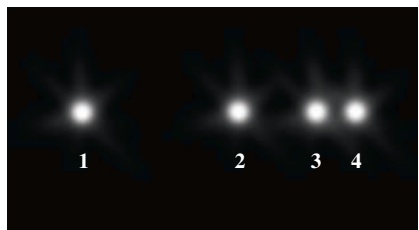
(a) Small aperture



(b) Medium aperture



(c) Large aperture



The intensities in the bright rings drop off very quickly with increasing angle. When  $D$  is much larger than the wavelength  $\lambda$ , the usual case for optical instruments, the peak intensity in the first ring is only 1.7% of the value at the center of the Airy disk, and the peak intensity of the second ring is only 0.4%. Most (85%) of the light energy falls within the Airy disk. **Figure 36.26** shows a diffraction pattern from a circular aperture.

## Diffraction and Image Formation

Diffraction has far-reaching implications for image formation by lenses and mirrors. In our study of optical instruments in Chapter 34 we assumed that a lens with focal length  $f$  focuses a parallel beam (plane wave) to a *point* at a distance  $f$  from the lens. We now see that what we get is *not* a point but the diffraction pattern just described. If we have two point objects, their images are not two points but two diffraction patterns. When the objects are close together, their diffraction patterns overlap; if they are close enough, their patterns overlap almost completely and cannot be distinguished. The effect is shown in **Fig. 36.27**, which presents the patterns for four very small “point” sources of light. In Fig. 36.27a the image of source 1 is well separated from the others, but the images of sources 3 and 4 have merged. In Fig. 36.27b, with a larger aperture diameter and hence smaller Airy disks, the images of sources 3 and 4 are better resolved. In Fig. 36.27c, with a still larger aperture, they are well resolved.

A widely used criterion for resolution of two point objects, proposed by the English physicist Lord Rayleigh (1842–1919) and called **Rayleigh’s criterion**, is that the objects are just barely resolved (that is, distinguishable) if the center of one diffraction pattern coincides with the first minimum of the other. In that case the angular separation of the image centers is given by Eq. (36.17). The angular separation of the *objects* is the same as that of the *images* made by a telescope, microscope, or other optical device. So two point objects are barely resolved when their angular separation is given by Eq. (36.17).

The minimum separation of two objects that can just be resolved by an optical instrument is called the **limit of resolution** of the instrument. The smaller the limit of resolution, the greater the *resolution*, or **resolving power**, of the instrument. Diffraction sets the ultimate limits on resolution of lenses. *Geometric* optics may make it seem that we can make images as large as we like. Eventually, though, we always reach a point at which the image becomes larger but does not gain in detail. The images in Fig. 36.27 would not become sharper if enlarged.

**CAUTION Resolving power vs. chromatic resolving power** Don’t confuse the resolving power of an optical instrument with the *chromatic* resolving power of a grating (Section 36.5). Resolving power refers to the ability to distinguish the images of objects that appear close to each other, when looking either through an optical instrument or at a photograph made with the instrument. Chromatic resolving power describes how well different wavelengths can be distinguished in a spectrum formed by a diffraction grating. **I**

Rayleigh’s criterion combined with Eq. (36.17) shows that resolution (resolving power) improves with larger diameter; it also improves with shorter wavelengths. **?** Ultraviolet microscopes have higher resolution than visible-light microscopes. In electron microscopes the resolution is limited by the wavelengths associated with the electrons, which have wavelike aspects (to be discussed further in Chapter 39). These wavelengths can be made 100,000 times smaller than wavelengths of visible light, with a corresponding gain in resolution. Resolving power also explains the difference in storage capacity between DVDs (introduced in 1995) and Blu-ray discs (introduced in 2003). Information is stored in both of these in a series of tiny pits. In order not to lose information in the scanning process, the scanning optics must be able to resolve two adjacent pits so that they do not seem to blend into a single pit (see sources 3 and 4 in Fig. 36.27). The blue scanning laser used in a Blu-ray player has a shorter wavelength (405 nm) and hence better resolving power than the 650 nm red laser in a DVD player. Hence pits can be spaced closer together in a Blu-ray disc than in a DVD, and more information can be stored on a disc of the same size (50 gigabytes on a Blu-ray disc versus 4.7 gigabytes on a DVD).

**BIO APPLICATION The Airy Disk in an Eagle's Eye** Diffraction by the pupil of an eye limits resolving power. In a human's eye, the maximum pupil diameter  $D$  is about 5 mm; in an eagle's eye,  $D$  is about 9 mm. From Eq. (36.17) this means that an eagle's eye has superior resolution: A distant point source of light produces an Airy disk on an eagle's retina that is only about  $\frac{5}{9}$  the angular size of the disk produced on the retina of the human eye. (If our eye produces an image like Fig. 36.27b, an eagle's eye produces one like Fig. 36.27c.) To record the fine details of this high-resolution image, the light-sensitive cones in an eagle's retina are smaller and more closely packed than those in a human retina.



### EXAMPLE 36.6 Resolving power of a camera lens

### WITH VARIATION PROBLEMS

A camera lens with focal length  $f = 50$  mm and maximum aperture  $f/2$  forms an image of an object 9.0 m away. (a) If the resolution is limited by diffraction, what is the minimum distance between two points on the object that are barely resolved? What is the corresponding distance between image points? (b) How does the situation change if the lens is “stopped down” to  $f/16$ ? Use  $\lambda = 500$  nm in both cases.

**IDENTIFY and SET UP** This example uses the ideas about resolving power, image formation by a lens (Section 34.4), and  $f$ -number (Section 34.5). From Eq. (34.20), the  $f$ -number of a lens is its focal length  $f$  divided by the aperture diameter  $D$ . We use this equation to determine  $D$  and then use Eq. (36.17) (the Rayleigh criterion) to find the angular separation  $\theta$  between two barely resolved points on the object. We then use the geometry of image formation by a lens to determine the distance  $y$  between those points and the distance  $y'$  between the corresponding image points.

**EXECUTE** (a) The aperture diameter is  $D = f/(f\text{-number}) = (50 \text{ mm})/2 = 25 \text{ mm} = 25 \times 10^{-3} \text{ m}$ . From Eq. (36.17) the angular separation  $\theta$  of two object points that are barely resolved is

$$\theta \approx \sin \theta = 1.22 \frac{\lambda}{D} = 1.22 \frac{500 \times 10^{-9} \text{ m}}{25 \times 10^{-3} \text{ m}} = 2.4 \times 10^{-5} \text{ rad}$$

We know from our thin-lens analysis in Section 34.4 that, apart from sign,  $y/s = y'/s'$  [see Eq. (34.14)]. Thus the angular separations of the object points and the corresponding image points are both equal to  $\theta$ . Because the object distance  $s$  is much greater than the focal length  $f = 50$  mm, the image distance  $s'$  is approximately equal to  $f$ . Thus

$$\begin{aligned} \frac{y}{9.0 \text{ m}} &= 2.4 \times 10^{-5} & y &= 2.2 \times 10^{-4} \text{ m} = 0.22 \text{ mm} \\ \frac{y'}{50 \text{ mm}} &= 2.4 \times 10^{-5} & y' &= 1.2 \times 10^{-3} \text{ mm} \\ & & &= 0.0012 \text{ mm} \approx \frac{1}{800} \text{ mm} \end{aligned}$$

(b) The aperture diameter is now  $(50 \text{ mm})/16$ , or one-eighth as large as before. The angular separation between barely resolved points is eight times as great, and the values of  $y$  and  $y'$  are also eight times as great as before:

$$y = 1.8 \text{ mm} \quad y' = 0.0096 \text{ mm} = \frac{1}{100} \text{ mm}$$

Only the best camera lenses can approach this resolving power.

**EVALUATE** Many photographers use the smallest possible aperture for maximum sharpness, since lens aberrations cause light rays that are far from the optic axis to converge to a different image point than do rays near the axis. But as this example shows, diffraction effects become more significant at small apertures. One cause of fuzzy images has to be balanced against another.

**KEYCONCEPT** Like waves that pass through a narrow slit, waves that pass through a circular aperture undergo diffraction and spread out. This sets a limit on the ability of a lens or telescope to resolve objects that are close to each other.

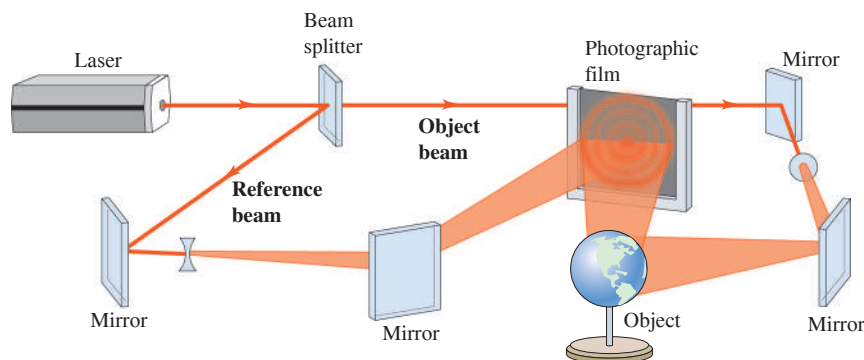
**TEST YOUR UNDERSTANDING OF SECTION 36.7** You have been asked to compare four proposals for telescopes to be placed in orbit above the blurring effects of the earth's atmosphere. Rank the proposed telescopes in order of their ability to resolve small details, from best to worst. (i) A radio telescope 100 m in diameter observing at a wavelength of 21 cm; (ii) an optical telescope 2.0 m in diameter observing at a wavelength of 500 nm; (iii) an ultraviolet telescope 1.0 m in diameter observing at a wavelength of 100 nm; (iv) an infrared telescope 2.0 m in diameter observing at a wavelength of  $10 \mu\text{m}$ .

### ANSWER

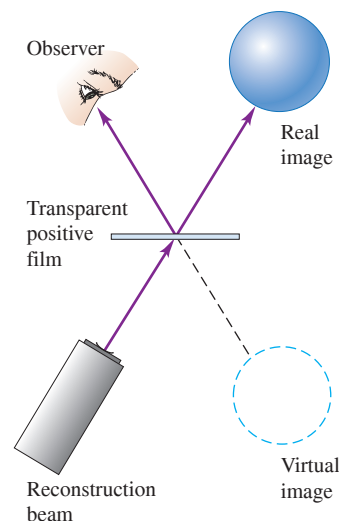
Rayleigh's criterion combined with Eq. (36.17) shows that the smaller the value of the ratio  $\lambda/D$ , the better the resolving power of a telescope of diameter  $D$ . For the four telescopes, this ratio is equal to (i)  $(21 \text{ cm})/(100 \text{ m}) = (0.21 \text{ m})/(100 \text{ m}) = 2.1 \times 10^{-3}$ ; (ii)  $(500 \text{ nm})/(2.0 \text{ m}) = (5.0 \times 10^{-7} \text{ m})/(2.0 \text{ m}) = 2.5 \times 10^{-7}$ ; (iii)  $(100 \text{ nm})/(1.0 \text{ m}) = (1.0 \times 10^{-7} \text{ m})/(1.0 \text{ m}) = 1.0 \times 10^{-7}$ ; (iv)  $(10 \mu\text{m})/(2.0 \text{ m}) = (1.0 \times 10^{-5} \text{ m})/(2.0 \text{ m}) = 5.0 \times 10^{-6}$ . Ranking these ratios from smallest to largest gives (iii), (ii), (iv), (i). **I**

Figure 36.28 (a) A hologram is the record on film of the interference pattern formed with light from the coherent source and light scattered from the object. (b) Images are formed when light is projected through the hologram. The observer sees the virtual image formed behind the hologram.

(a) Recording a hologram



(b) Viewing the hologram



## 36.8 HOLOGRAPHY

**Holography** is a technique for recording and reproducing an image of an object through the use of interference effects. Unlike the two-dimensional images recorded by an ordinary photograph or television system, a holographic image is truly three-dimensional. Such an image can be viewed from different directions to reveal different sides and from various distances to reveal changing perspective. If you had never seen a hologram, you wouldn't believe it was possible!

Figure 36.28a shows the basic procedure for making a hologram. We illuminate the object to be holographed with monochromatic light, and we place a photographic film so that it is struck by scattered light from the object and also by direct light from the source. In practice, the light source must be a laser, for reasons we'll discuss later. Interference between the direct and scattered light forms a complex interference pattern that is recorded on the film.

To form the images, we simply project light through the developed film (Fig. 36.28b). Two images are formed: a virtual image on the side of the film nearer the source and a real image on the opposite side.

### Holography and Interference Patterns

A complete analysis of holography is beyond our scope, but we can gain some insight into the process by looking at how a single point is holographed and imaged. Consider the interference pattern that is formed on a sheet of photographic negative film by the superposition of an incident plane wave and a spherical wave, as shown in Fig. 36.29a. The spherical wave originates at a point source  $P$  at a distance  $b_0$  from the film;  $P$  may in fact be a small object that scatters part of the incident plane wave. We assume that the two waves are monochromatic and coherent and that the phase relationship is such that constructive interference occurs at point  $O$  on the diagram. Then constructive interference will *also* occur at any point  $Q$  on the film that is farther from  $P$  than  $O$  is by an integer number of wavelengths. That is, if  $b_m - b_0 = m\lambda$ , where  $m$  is an integer, then constructive interference occurs. The points where this condition is satisfied form circles on the film centered at  $O$ , with radii  $r_m$  given by

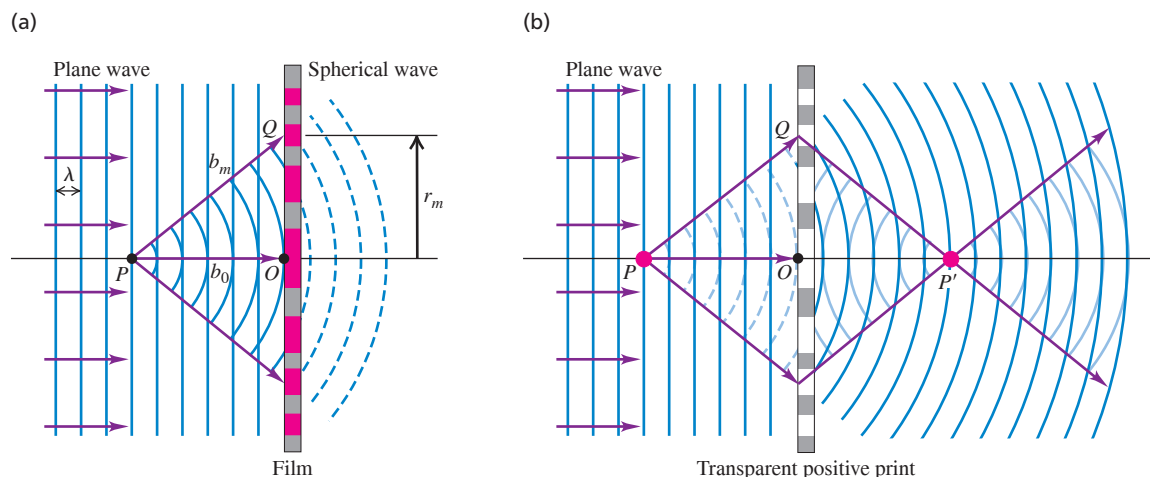
$$b_m - b_0 = \sqrt{b_0^2 + r_m^2} - b_0 = m\lambda \quad (m = 1, 2, 3, \dots) \quad (36.20)$$

Solving this for  $r_m^2$ , we find

$$r_m^2 = \lambda(2mb_0 + m^2\lambda)$$



Figure 36.29 (a) Constructive interference of the plane and spherical waves occurs in the plane of the film at every point  $Q$  for which the distance  $b_m$  from  $P$  is greater than the distance  $b_0$  from  $P$  to  $O$  by an integer number of wavelengths  $m\lambda$ . For the point  $Q$  shown,  $m = 2$ . (b) When a plane wave strikes a transparent positive print of the developed film, the diffracted wave consists of a wave converging to  $P'$  and then diverging again and a diverging wave that appears to originate at  $P$ . These waves form the real and virtual images, respectively.



Ordinarily,  $b_0$  is very much larger than  $\lambda$ , so we ignore the second term in parentheses and obtain

$$r_m = \sqrt{2m\lambda b_0} \quad (m = 1, 2, 3, \dots) \quad (36.21)$$

The interference pattern consists of a series of concentric bright circular fringes with radii given by Eq. (36.21). Between these bright fringes are dark fringes.

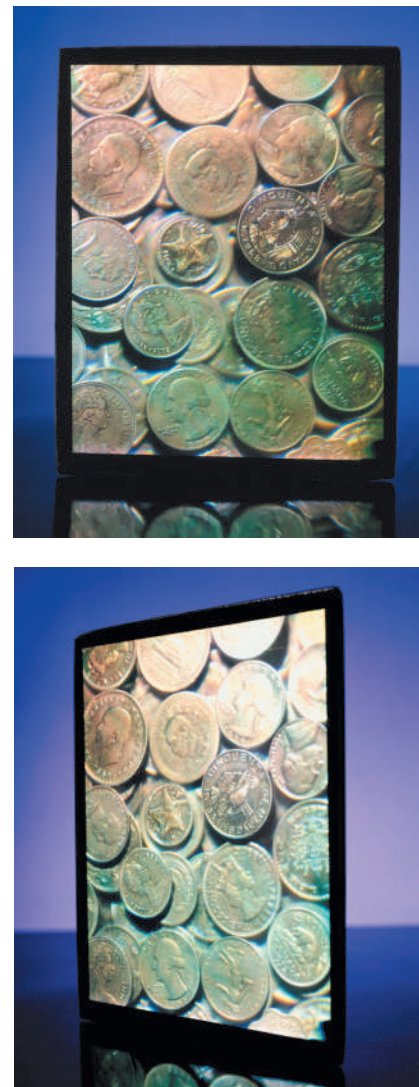
Now we develop the film and make a transparent positive print, so the bright-fringe areas have the greatest transparency on the film. Then we illuminate it with monochromatic plane-wave light of the same wavelength  $\lambda$  that we used initially. In Fig. 36.29b, consider a point  $P'$  at a distance  $b_0$  along the axis from the film. The centers of successive bright fringes differ in their distances from  $P'$  by an integer number of wavelengths, and therefore a strong *maximum* in the diffracted wave occurs at  $P'$ . That is, light converges to  $P'$  and then diverges from it on the opposite side. Therefore  $P'$  is a *real image* of point  $P$ .

This is not the entire diffracted wave, however. The interference of the wavelets that spread out from all the transparent areas forms a second spherical wave that is diverging rather than converging. When this wave is traced back behind the film in Fig. 36.29b, it appears to be spreading out from point  $P$ . Thus the total diffracted wave from the hologram is a superposition of a spherical wave converging to form a real image at  $P'$  and a spherical wave that diverges as though it had come from the virtual image point  $P$ .

Because of the principle of superposition for waves, what is true for the imaging of a single point is also true for the imaging of any number of points. The film records the superposed interference pattern from the various points, and when light is projected through the film, the various image points are reproduced simultaneously. Thus the images of an extended object can be recorded and reproduced just as for a single point object. **Figure 36.30** shows photographs of a holographic image from two different angles, showing the changing perspective in this three-dimensional image.

In making a hologram, we have to overcome two practical problems. First, the light used must be *coherent* over distances that are large in comparison to the dimensions of the object and its distance from the film. Ordinary light sources *do not* satisfy this requirement, for reasons that we discussed in Section 35.1. Therefore laser light is essential for making a hologram. (Ordinary white light can be used for *viewing* certain types of hologram, such as those used on credit cards.) Second, extreme mechanical stability is needed. If any relative motion of source, object, or film occurs during exposure, even by as much as a quarter of a wavelength, the interference pattern on the film is blurred enough to prevent satisfactory image formation. These obstacles are not insurmountable, however, and holography has become important in research, entertainment, and a wide variety of technological applications.

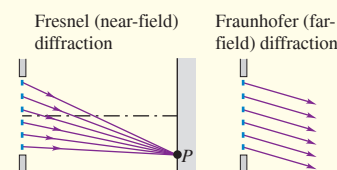
Figure 36.30 Two views of the same hologram seen from different angles.





## CHAPTER 36 SUMMARY

**Fresnel and Fraunhofer diffraction:** Diffraction occurs when light passes through an aperture or around an edge. When the source and the observer are so far away from the obstructing surface that the outgoing rays can be considered parallel, it is called Fraunhofer diffraction. When the source or the observer is relatively close to the obstructing surface, it is Fresnel diffraction.

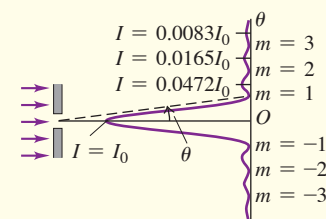


**Single-slit diffraction:** Monochromatic light sent through a narrow slit of width  $a$  produces a diffraction pattern on a distant screen. Equation (36.2) gives the condition for destructive interference (a dark fringe) at a point  $P$  in the pattern at angle  $\theta$ . Equation (36.7) gives the intensity in the pattern as a function of  $\theta$ . (See Examples 36.1–36.3.)

$$\sin \theta = \frac{m\lambda}{a} \quad (36.2)$$

$$(m = \pm 1, \pm 2, \pm 3, \dots)$$

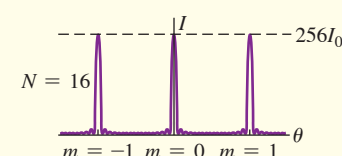
$$I = I_0 \left\{ \frac{\sin[\pi a(\sin \theta)/\lambda]}{\pi a(\sin \theta)/\lambda} \right\}^2 \quad (36.7)$$



**Diffraction gratings:** A diffraction grating consists of a large number of thin parallel slits, spaced a distance  $d$  apart. The condition for maximum intensity in the interference pattern is the same as for the two-source pattern, but the maxima for the grating are very sharp and narrow. (See Example 36.4.)

$$d \sin \theta = m\lambda \quad (36.13)$$

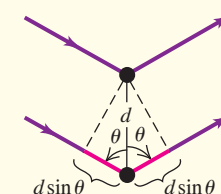
$$(m = 0, \pm 1, \pm 2, \pm 3, \dots)$$



**X-ray diffraction:** A crystal serves as a three-dimensional diffraction grating for x rays with wavelengths of the same order of magnitude as the spacing between atoms in the crystal. For a set of crystal planes spaced a distance  $d$  apart, constructive interference occurs when the angles of incidence and scattering (measured from the crystal planes) are equal and when the Bragg condition [Eq. (36.16)] is satisfied. (See Example 36.5.)

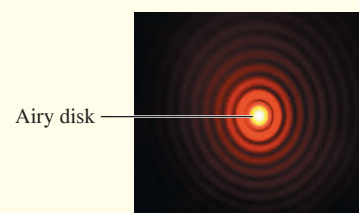
$$2d \sin \theta = m\lambda \quad (36.16)$$

$$(m = 1, 2, 3, \dots)$$



**Circular apertures and resolving power:** The diffraction pattern from a circular aperture of diameter  $D$  consists of a central bright spot, called the Airy disk, and a series of concentric dark and bright rings. Equation (36.17) gives the angular radius  $\theta_1$  of the first dark ring, equal to the angular size of the Airy disk. Diffraction sets the ultimate limit on resolution (image sharpness) of optical instruments. According to Rayleigh's criterion, two point objects are just barely resolved when their angular separation  $\theta$  is given by Eq. (36.17). (See Example 36.6.)

$$\sin \theta_1 = 1.22 \frac{\lambda}{D} \quad (36.17)$$



Chapter 36 Media Assets



## GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

### KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLE 36.1** (Section 36.2) and **EXAMPLES 36.2 and 36.3** (Section 36.3) before attempting these problems.

**VP36.3.1** You pass laser light of wavelength 645 nm through a slit 0.250 mm in width and observe the diffraction pattern on a screen a large distance away. On the screen, the centers of the second minima on either side of the central bright fringe are 28.0 mm apart. (a) How far away is the screen? (b) What would the distance be between these minima if the wavelength were 525 nm?

**VP36.3.2** You shine a laser on a narrow slit 0.221 mm in width. In the diffraction pattern that appears on a screen 5.00 m from the slit, the third minimum is 45.7 mm from the middle of the central bright fringe. Find (a) the wavelength of the laser light and (b) the angle of a line from the center of the slit to the second dark fringe on the screen.

**VP36.3.3** At a certain point in a single-slit diffraction pattern there is a phase difference of 35.0 radians between wavelets arriving at the point from the two edges of the slit. The slit is 0.250 mm wide, and the light used has wavelength 545 nm. (a) What is the angle of a line from the center of the slit to this point in the diffraction pattern? (b) If the intensity at the center of the diffraction pattern is  $I_0$ , what is the intensity at this point?

**VP36.3.4** In a single-slit diffraction experiment, the slit width is 0.185 mm and the wavelength of the light used is 576 nm. (a) Find the angle of a line from the center of the slit to the first dark fringe. (b) You might expect the intensity at one-half the angle found in part (a), which is halfway between the middle of the central bright fringe (where the intensity is  $I_0$ ) and the first dark fringe (where the intensity is zero), to be  $I_0/2$ . What is the actual intensity at this angle?

**Be sure to review EXAMPLE 36.4 (Section 36.5) and EXAMPLE 36.5 (Section 36.6) before attempting these problems.**

**VP36.5.1** You shine laser light so that it falls normally on a transmission grating with 825 slits per millimeter. The second-order bright region occurs at an angle of  $41.0^\circ$  from the central maximum. Find (a) the wavelength of the light and (b) the angles for the first-order and third-order bright regions.

**VP36.5.2** The first-order bright region created by a diffraction grating occurs at an angle of  $24.0^\circ$  from the central maximum. The wavelength of light that falls normally onto the grating is 625 nm. Find (a) the number of slits per mm on the grating and (b) the angles of the second-order bright region and the third-order bright region (if any).

**VP36.5.3** In a certain crystal the spacing between crystal planes is 0.165 nm. (a) If you shine a beam of x rays of wavelength 0.124 nm on this crystal, for what angle between the beam and the crystal planes does the first strong interference maximum occur? (b) For what larger angles (if any) do strong interference maxima occur?

**VP36.5.4** In an x-ray diffraction experiment there is only one strong interference maximum, and this occurs when the x-ray beam makes an angle of  $36.0^\circ$  with the crystal planes. The spacing between the crystal planes is 0.158 nm. (a) What is the wavelength of the x rays? (b) What

would the wavelength have to be to have three strong interference maxima, with the third at an angle of  $88.0^\circ$ ?

**Be sure to review EXAMPLE 36.6 (Section 36.7) before attempting these problems.**

**VP36.6.1** You wish to study the radio emission from the sun at wavelength 1.70 cm. In order to see details on the sun's surface no larger than the diameter of the earth, the limit of resolution of the telescope must be  $9.00 \times 10^{-5}$  rad (about  $0.005^\circ$ ). Using the Rayleigh criterion, find (a) the minimum diameter your radio telescope must have and (b) the limit of resolution of this telescope at wavelength 21.1 cm.

**VP36.6.2** A telephoto lens for a camera has focal length 0.250 m and maximum aperture  $f/4.0$ . Take the wavelength of visible light to be  $5.50 \times 10^{-7}$  m. (a) Assuming that the resolution is limited by diffraction, how far away can an object be for you to be able to resolve two points on the object that are 5.00 mm apart? (b) How far apart are the corresponding points on the image made by the lens?

**VP36.6.3** A distant point source of light (like a star) emits light of wavelength 575 nm. When this light enters a camera whose lens has a focal length of 135 mm, the diffraction pattern formed on the camera's detector has an Airy disk of radius 0.0112 mm. Find (a) the diameter of the lens aperture and (b) the  $f$ -number.

**VP36.6.4** One important goal of astronomers is to have a telescope in space that can resolve planets like the earth orbiting other stars. If a planet orbits its star at a distance of  $1.5 \times 10^{11}$  m (the radius of the earth's orbit around the sun) and the telescope has a mirror of diameter 8.0 m, how far from the telescope could the star and its planet be if the wavelength used was (a) 690 nm and (b) 1400 nm? Use the Rayleigh criterion and give your answers in light-years ( $1 \text{ ly} = 9.46 \times 10^{15} \text{ m}$ ).

## BRIDGING PROBLEM Observing the Expanding Universe

An astronomer who is studying the light from a galaxy has identified the spectrum of hydrogen but finds that the wavelengths are somewhat shifted from those found in the laboratory. In the lab, the  $H_\alpha$  line in the hydrogen spectrum has a wavelength of 656.3 nm. The astronomer is using a transmission diffraction grating having 5758 lines/cm in the first order and finds that the first bright fringe for the  $H_\alpha$  line occurs at  $\pm 23.41^\circ$  from the central spot. How fast is the galaxy moving? Express your answer in m/s and as a percentage of the speed of light. Is the galaxy moving toward us or away from us?

### SOLUTION GUIDE

#### IDENTIFY and SET UP

1. You can use the information about the grating to find the wavelength of the  $H_\alpha$  line in the galaxy's spectrum.
2. In Section 16.8 we learned about the Doppler effect for electromagnetic radiation: The frequency that we receive from a

moving source, such as the galaxy, is different from the frequency that is emitted. Equation (16.30) relates the emitted frequency, the received frequency, and the velocity of the source (the target variable). The equation  $c = f\lambda$  relates the frequency  $f$  and wavelength  $\lambda$  through the speed of light  $c$ .

#### EXECUTE

3. Find the wavelength of the  $H_\alpha$  spectral line in the received light.
4. Rewrite Eq. (16.30) as a formula for the velocity  $v$  of the galaxy in terms of the received wavelength and the wavelength emitted by the source.
5. Solve for  $v$ . Express it in m/s and as a percentage of  $c$ , and decide whether the galaxy is moving toward us or moving away.

#### EVALUATE

6. Is your answer consistent with the relative sizes of the received wavelength and the emitted wavelength?

## PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

### DISCUSSION QUESTIONS

**Q36.1** Why can we readily observe diffraction effects for sound waves and water waves, but not for light? Is this because light travels so much faster than these other waves? Explain.

**Q36.2** What is the difference between Fresnel and Fraunhofer diffraction? Are they different *physical* processes? Explain.

**Q36.3** You use a lens of diameter  $D$  and light of wavelength  $\lambda$  and frequency  $f$  to form an image of two closely spaced and distant objects. Which of the following will increase the resolving power? (a) Use a lens with a smaller diameter; (b) use light of higher frequency; (c) use light of longer wavelength. In each case justify your answer.

**Q36.4** Light of wavelength  $\lambda$  and frequency  $f$  passes through a single slit of width  $a$ . The diffraction pattern is observed on a screen a distance  $x$  from the slit. Which of the following will *decrease* the width of the central maximum? (a) Decrease the slit width; (b) decrease the frequency  $f$  of the light; (c) decrease the wavelength  $\lambda$  of the light; (d) decrease the distance  $x$  of the screen from the slit. In each case justify your answer.

**Q36.5** In a diffraction experiment with waves of wavelength  $\lambda$ , there will be *no* intensity minima (that is, no dark fringes) if the slit width is small enough. What is the maximum slit width for which this occurs? Explain your answer.

**Q36.6** An interference pattern is produced by four parallel and equally spaced narrow slits. By drawing appropriate phasor diagrams, explain why there is an interference minimum when the phase difference  $\phi$  from adjacent slits is (a)  $\pi/2$ ; (b)  $\pi$ ; (c)  $3\pi/2$ . In each case, for which pairs of slits is there totally destructive interference?

**Q36.7 Phasor Diagram for Eight Slits.** An interference pattern is produced by eight equally spaced narrow slits. The caption for Fig. 36.14 claims that minima occur for  $\phi = 3\pi/4$ ,  $\pi/4$ ,  $3\pi/2$ , and  $7\pi/4$ . Draw the phasor diagram for each of these four cases, and explain why each diagram proves that there is in fact a minimum. In each case, for which pairs of slits is there totally destructive interference?

**Q36.8** A rainbow ordinarily shows a range of colors (see Section 33.4). But if the water droplets that form the rainbow are small enough, the rainbow will appear white. Explain why, using diffraction ideas. How small do you think the raindrops would have to be for this to occur?

**Q36.9** Some loudspeaker horns for outdoor concerts (at which the entire audience is seated on the ground) are wider vertically than horizontally. Use diffraction ideas to explain why this is more efficient at spreading the sound uniformly over the audience than either a square speaker horn or a horn that is wider horizontally than vertically. Would this still be the case if the audience were seated at different elevations, as in an amphitheater? Why or why not?

**Q36.10** Figure 31.12 (Section 31.2) shows a loudspeaker system. Low-frequency sounds are produced by the *woofer*, which is a speaker with large diameter; the *tweeter*, a speaker with smaller diameter, produces high-frequency sounds. Use diffraction ideas to explain why the tweeter is more effective for distributing high-frequency sounds uniformly over a room than is the woofer.

**Q36.11** Information is stored on an audio compact disc, CD-ROM, or DVD disc in a series of pits on the disc. These pits are scanned by a laser beam. An important limitation on the amount of information that can be stored on such a disc is the width of the laser beam. Explain why this should be, and explain how using a shorter-wavelength laser allows more information to be stored on a disc of the same size.

**Q36.12** With which color of light can the Hubble Space Telescope see finer detail in a distant astronomical object: red, blue, or ultraviolet? Explain your answer.

**Q36.13** At the end of Section 36.4, the following statements were made about an array of  $N$  slits. Explain, using phasor diagrams, why each statement is true. (a) A minimum occurs whenever  $\phi$  is an integer multiple of  $2\pi/N$  (except when  $\phi$  is an integer multiple of  $2\pi$ , which gives a principal maximum). (b) There are  $(N - 1)$  minima between each pair of principal maxima.

**Q36.14** Could x-ray diffraction effects with crystals be observed by using visible light instead of x rays? Why or why not?

**Q36.15** Why is a diffraction grating better than a two-slit setup for measuring wavelengths of light?

**Q36.16** One sometimes sees rows of evenly spaced radio antenna towers. A student remarked that these act like diffraction gratings. What did she mean? Why would one want them to act like a diffraction grating?

**Q36.17** If a hologram is made using 600 nm light and then viewed with 500 nm light, how will the images look compared to those observed when viewed with 600 nm light? Explain.

**Q36.18** A hologram is made using 600 nm light and then viewed by using white light from an incandescent bulb. What will be seen? Explain.

**Q36.19** Ordinary photographic film reverses black and white, in the sense that the most brightly illuminated areas become blackest upon development (hence the term *negative*). Suppose a hologram negative is viewed directly, without making a positive transparency. How will the resulting images differ from those obtained with the positive? Explain.

## EXERCISES

### Section 36.2 Diffraction from a Single Slit

**36.1 ••** Monochromatic light from a distant source is incident on a slit 0.750 mm wide. On a screen 2.00 m away, the distance from the central maximum of the diffraction pattern to the first minimum is measured to be 1.35 mm. Calculate the wavelength of the light.

**36.2 •** Coherent electromagnetic waves with wavelength  $\lambda$  pass through a narrow slit of width  $a$ . The diffraction pattern is observed on a tall screen that is 2.00 m from the slit. When  $\lambda = 500$  nm, the width on the screen of the central maximum in the diffraction pattern is 8.00 mm. For the same slit and screen, what is the width of the central maximum when  $\lambda = 0.125$  mm?

**36.3 ••** Light of wavelength 585 nm falls on a slit 0.0666 mm wide. (a) On a very large and distant screen, how many *totally* dark fringes (indicating complete cancellation) will there be, including both sides of the central bright spot? Solve this problem *without* calculating all the angles! (*Hint:* What is the largest that  $\sin\theta$  can be? What does this tell you is the largest that  $m$  can be?) (b) At what angle will the dark fringe that is most distant from the central bright fringe occur?

**36.4 •** Light of wavelength 633 nm from a distant source is incident on a slit 0.750 mm wide, and the resulting diffraction pattern is observed on a screen 3.50 m away. What is the distance between the two dark fringes on either side of the central bright fringe?

**36.5 ••** Diffraction occurs for all types of waves, including sound waves. High-frequency sound from a distant source with wavelength 9.00 cm passes through a slit 12.0 cm wide. A microphone is placed 8.00 m directly in front of the center of the slit, corresponding to point  $O$  in Fig. 36.5a. The microphone is then moved in a direction perpendicular to the line from the center of the slit to point  $O$ . At what distances from  $O$  will the intensity detected by the microphone be zero?

**36.6 • CP Tsunami!** On December 26, 2004, a violent earthquake of magnitude 9.1 occurred off the coast of Sumatra. This quake triggered a huge tsunami (similar to a tidal wave) that killed more than 150,000 people. Scientists observing the wave on the open ocean measured the time between crests to be 1.0 h and the speed of the wave to be 800 km/h. Computer models of the evolution of this enormous wave showed that it bent around the continents and spread to all the oceans of the earth. When the wave reached the gaps between continents, it diffracted between them as through a slit. (a) What was the wavelength of this tsunami? (b) The distance between the southern tip of Africa and northern Antarctica is about 4500 km, while the distance between the southern end of Australia and Antarctica is about 3700 km. As an approximation, we can model this wave's behavior by using Fraunhofer diffraction. Find the smallest angle away from the central maximum for which the waves would cancel after going through each of these continental gaps.

**36.7 •• CP** A series of parallel linear water wave fronts are traveling directly toward the shore at 15.0 cm/s on an otherwise placid lake. A long concrete barrier that runs parallel to the shore at a distance of 3.20 m away has a hole in it. You count the wave crests and observe that 75.0 of them pass by each minute, and you also observe that no waves reach the shore at  $\pm 61.3$  cm from the point directly opposite the hole, but waves do reach the shore everywhere within this distance. (a) How wide is the hole in the barrier? (b) At what other angles do you find no waves hitting the shore?



**36.8 •** Monochromatic electromagnetic radiation with wavelength  $\lambda$  from a distant source passes through a slit. The diffraction pattern is observed on a screen 2.50 m from the slit. If the width of the central maximum is 6.00 mm, what is the slit width  $a$  if the wavelength is (a) 500 nm (visible light); (b)  $50.0\ \mu\text{m}$  (infrared radiation); (c) 0.500 nm (x rays)?

**36.9 •• Doorway Diffraction.** Sound of frequency 1250 Hz leaves a room through a 1.00-m-wide doorway (see Exercise 36.5). At which angles relative to the centerline perpendicular to the doorway will someone outside the room hear no sound? Use 344 m/s for the speed of sound in air and assume that the source and listener are both far enough from the doorway for Fraunhofer diffraction to apply. You can ignore effects of reflections.

**36.10 • CP** Light waves, for which the electric field is given by  $E_y(x, t) = E_{\text{max}} \sin[(1.40 \times 10^7\ \text{m}^{-1})x - \omega t]$ , pass through a slit and produce the first dark bands at  $\pm 28.6^\circ$  from the center of the diffraction pattern. (a) What is the frequency of this light? (b) How wide is the slit? (c) At which angles will other dark bands occur?

**36.11 ••** Red light of wavelength 633 nm from a helium–neon laser passes through a slit 0.350 mm wide. The diffraction pattern is observed on a screen 3.00 m away. Define the width of a bright fringe as the distance between the minima on either side. (a) What is the width of the central bright fringe? (b) What is the width of the first bright fringe on either side of the central one?

**36.12 •** When coherent electromagnetic waves with wavelength  $\lambda = 120\ \mu\text{m}$  are incident on a single slit of width  $a$ , the width of the central maximum on a tall screen 1.50 m from the slit is 90.0 cm. For the same slit and screen, for what wavelength of the incident waves is the width of the central maximum 180.0 cm, double the value when  $\lambda = 120\ \mu\text{m}$ ?

### Section 36.3 Intensity in the Single-Slit Pattern

**36.13 ••** Monochromatic light of wavelength 580 nm passes through a single slit and the diffraction pattern is observed on a screen. Both the source and screen are far enough from the slit for Fraunhofer diffraction to apply. (a) If the first diffraction minima are at  $\pm 90.0^\circ$ , so the central maximum completely fills the screen, what is the width of the slit? (b) For the width of the slit as calculated in part (a), what is the ratio of the intensity at  $\theta = 45.0^\circ$  to the intensity at  $\theta = 0^\circ$ ?

**36.14 ••** Monochromatic light of wavelength  $\lambda = 620\ \text{nm}$  from a distant source passes through a slit 0.450 mm wide. The diffraction pattern is observed on a screen 3.00 m from the slit. In terms of the intensity  $I_0$  at the peak of the central maximum, what is the intensity of the light at the screen the following distances from the center of the central maximum: (a) 1.00 mm; (b) 3.00 mm; (c) 5.00 mm?

**36.15 ••** Public Radio station KXPR-FM in Sacramento broadcasts at 88.9 MHz. The radio waves pass between two tall skyscrapers that are 15.0 m apart along their closest walls. (a) At what horizontal angles, relative to the original direction of the waves, will a distant antenna not receive any signal from this station? (b) If the maximum intensity is  $3.50\ \text{W/m}^2$  at the antenna, what is the intensity at  $\pm 5.00^\circ$  from the center of the central maximum at the distant antenna?

**36.16 •** Monochromatic light of wavelength 592 nm from a distant source passes through a slit that is 0.0290 mm wide. In the resulting diffraction pattern, the intensity at the center of the central maximum ( $\theta = 0^\circ$ ) is  $4.00 \times 10^{-5}\ \text{W/m}^2$ . What is the intensity at a point on the screen that corresponds to  $\theta = 1.20^\circ$ ?

### Section 36.4 Multiple Slits

**36.17 •** Nearly monochromatic coherent light waves leave two rectangular slits in phase and at an angle of  $\theta = 22.0^\circ$  with the normal. When the light reaches a distant screen, the waves from the center of one slit are  $344^\circ$  out of phase with the waves from the center of the other slit, and the waves from the top of either slit are  $172^\circ$  out of phase with the waves from the bottom of that slit. (a) How is the center-to-center distance between the slits related to the width of either slit? (b) Calculate the intensity at the screen for  $\theta = 22.0^\circ$  if the intensity at  $\theta = 0^\circ$  is  $0.234\ \text{W/m}^2$ .

**36.18 •** Parallel rays of monochromatic light with wavelength 568 nm illuminate two identical slits and produce an interference pattern on a screen that is 75.0 cm from the slits. The centers of the slits are 0.640 mm apart and the width of each slit is 0.434 mm. If the intensity at the center of the central maximum is  $5.00 \times 10^{-4}\ \text{W/m}^2$ , what is the intensity at a point on the screen that is 0.900 mm from the center of the central maximum?

**36.19 • Number of Fringes in a Diffraction Maximum.** In Fig. 36.12c the central diffraction maximum contains exactly seven interference fringes, and in this case  $d/a = 4$ . (a) What must the ratio  $d/a$  be if the central maximum contains exactly five fringes? (b) In the case considered in part (a), how many fringes are contained within the first diffraction maximum on one side of the central maximum?

**36.20 •• Diffraction and Interference Combined.** Consider the interference pattern produced by two parallel slits of width  $a$  and separation  $d$ , in which  $d = 3a$ . The slits are illuminated by normally incident light of wavelength  $\lambda$ . (a) First we ignore diffraction effects due to the slit width. At what angles  $\theta$  from the central maximum will the next four maxima in the two-slit interference pattern occur? Your answer will be in terms of  $d$  and  $\lambda$ . (b) Now we include the effects of diffraction. If the intensity at  $\theta = 0^\circ$  is  $I_0$ , what is the intensity at each of the angles in part (a)? (c) Which double-slit interference maxima are missing in the pattern? (d) Compare your results to those illustrated in Fig. 36.12c. In what ways are your results different?

**36.21 ••** An interference pattern is produced by light of wavelength 580 nm from a distant source incident on two identical parallel slits separated by a distance (between centers) of 0.530 mm. (a) If the slits are very narrow, what would be the angular positions of the first-order and second-order, two-slit interference maxima? (b) Let the slits have width 0.320 mm. In terms of the intensity  $I_0$  at the center of the central maximum, what is the intensity at each of the angular positions in part (a)?

**36.22 ••** Laser light of wavelength 500.0 nm illuminates two identical slits, producing an interference pattern on a screen 90.0 cm from the slits. The bright bands are 1.00 cm apart, and the third bright bands on either side of the central maximum are missing in the pattern. Find the width and the separation of the two slits.

**36.23 •** Coherent electromagnetic waves with wavelength  $\lambda = 500\ \text{nm}$  pass through two identical slits. The width of each slit is  $a$ , and the distance between the centers of the slits is  $d = 9.00\ \text{mm}$ . (a) What is the smallest possible width  $a$  of the slits if the  $m = 3$  maximum in the interference pattern is not present? (b) What is the next larger value of the slit width for which the  $m = 3$  maximum is absent?

**36.24 ••** Coherent light with wavelength 200 nm passes through two identical slits. The width of each slit is  $a$ , and the distance between the centers of the slits is  $d = 1.00\ \text{mm}$ . The  $m = 5$  maximum in the two-slit interference pattern is absent, but the maxima for  $m = 0$  through  $m = 4$  are present. What is the ratio of the intensities for the  $m = 1$  and  $m = 2$  maxima in the two-slit pattern?

### Section 36.5 The Diffraction Grating

**36.25 •** When laser light of wavelength 632.8 nm passes through a diffraction grating, the first bright spots occur at  $\pm 17.8^\circ$  from the central maximum. (a) What is the line density (in lines/cm) of this grating? (b) How many additional bright spots are there beyond the first bright spots, and at what angles do they occur?

**36.26 ••** Monochromatic light is at normal incidence on a plane transmission grating. The first-order maximum in the interference pattern is at an angle of  $11.3^\circ$ . What is the angular position of the fourth-order maximum?

**36.27 •** You send coherent 550 nm light through a diffraction grating that has slits of equal widths and constant separation between adjacent slits. You expect to see the fourth-order interference maximum at an angle of  $66.6^\circ$  with respect to the normal to the grating. However, that order is missing because  $66.6^\circ$  is also the angle for the third diffraction minimum (as measured from the central diffraction maximum) for each slit. (a) Find the center-to-center distance between adjacent slits. (b) Find the number of slits per mm. (c) Find the width of each slit.



**36.28 •** If a diffraction grating produces a third-order bright spot for red light (of wavelength 700 nm) at  $65.0^\circ$  from the central maximum, at what angle will the second-order bright spot be for violet light (of wavelength 400 nm)?

**36.29 •** Two Fraunhofer lines in the solar absorption spectrum have wavelengths of 430.790 nm and 430.774 nm. A diffraction grating has 12,800 slits. (a) What is the minimum chromatic resolving power needed to resolve these two spectral lines? (b) What is the lowest order required to resolve these two lines?

**36.30 •** The wavelength range of the visible spectrum is approximately 380–750 nm. White light falls at normal incidence on a diffraction grating that has 350 slits/mm. Find the angular width of the visible spectrum in (a) the first order and (b) the third order. (*Note:* An advantage of working in higher orders is the greater angular spread and better resolution. A disadvantage is the overlapping of different orders, as shown in Example 36.4.)

**36.31 •** (a) What is the wavelength of light that is deviated in the first order through an angle of  $13.5^\circ$  by a transmission grating having 5000 slits/cm? (b) What is the second-order deviation of this wavelength? Assume normal incidence.

**36.32 •• CDs and DVDs as Diffraction Gratings.** A laser beam of wavelength  $\lambda = 632.8$  nm shines at normal incidence on the reflective side of a compact disc. (a) The tracks of tiny pits in which information is coded onto the CD are  $1.60\text{ }\mu\text{m}$  apart. For what angles of reflection (measured from the normal) will the intensity of light be maximum? (b) On a DVD, the tracks are only  $0.740\text{ }\mu\text{m}$  apart. Repeat the calculation of part (a) for the DVD.

**36.33 • Identifying Isotopes by Spectra.** Different isotopes of the same element emit light at slightly different wavelengths. A wavelength in the emission spectrum of a hydrogen atom is 656.45 nm; for deuterium, the corresponding wavelength is 656.27 nm. (a) What minimum number of slits is required to resolve these two wavelengths in second order? (b) If the grating has 500.00 slits/mm, find the angles and angular separation of these two wavelengths in the second order.

### Section 36.6 X-Ray Diffraction

**36.34 •** If the planes of a crystal are  $3.50\text{ }\text{\AA}$  ( $1\text{ }\text{\AA} = 10^{-10}\text{ m} = 1\text{ }\text{\AA}$  angstrom unit) apart, (a) what wavelength of electromagnetic waves is needed so that the first strong interference maximum in the Bragg reflection occurs when the waves strike the planes at an angle of  $22.0^\circ$ , and in what part of the electromagnetic spectrum do these waves lie? (See Fig. 32.4.) (b) At what other angles will strong interference maxima occur?

**36.35 •** X rays of wavelength 0.0850 nm are scattered from the atoms of a crystal. The second-order maximum in the Bragg reflection occurs when the angle  $\theta$  in Fig. 36.22 is  $21.5^\circ$ . What is the spacing between adjacent atomic planes in the crystal?

**36.36 •** Monochromatic x rays are incident on a crystal for which the spacing of the atomic planes is 0.440 nm. The first-order maximum in the Bragg reflection occurs when the incident and reflected x rays make an angle of  $39.4^\circ$  with the crystal planes. What is the wavelength of the x rays?

### Section 36.7 Circular Apertures and Resolving Power

**36.37 ••** Monochromatic light with wavelength 620 nm passes through a circular aperture with diameter  $7.4\text{ }\mu\text{m}$ . The resulting diffraction pattern is observed on a screen that is 4.5 m from the aperture. What is the diameter of the Airy disk on the screen?

**36.38 ••** Monochromatic light with wavelength 490 nm passes through a circular aperture, and a diffraction pattern is observed on a screen that is 1.20 m from the aperture. If the distance on the screen between the first and second dark rings is 1.65 mm, what is the diameter of the aperture?

**36.39 •** Two satellites at an altitude of 1200 km are separated by 28 km. If they broadcast 3.6 cm microwaves, what minimum receiving-dish diameter is needed to resolve (by Rayleigh's criterion) the two transmissions?

**36.40 • BIO** If you can read the bottom row of your doctor's eye chart, your eye has a resolving power of 1 arcminute, equal to  $\frac{1}{60}$  degree. If this resolving power is diffraction limited, to what effective diameter of your eye's optical system does this correspond? Use Rayleigh's criterion and assume  $\lambda = 550$  nm.

**36.41 ••** The VLBA (Very Long Baseline Array) uses a number of individual radio telescopes to make one unit having an equivalent diameter of about 8000 km. When this radio telescope is focusing radio waves of wavelength 2.0 cm, what would have to be the diameter of the mirror of a visible-light telescope focusing light of wavelength 550 nm so that the visible-light telescope has the same resolution as the radio telescope?

**36.42 • Photography.** A wildlife photographer uses a moderate telephoto lens of focal length 135 mm and maximum aperture  $f/4.00$  to photograph a bear that is 11.5 m away. Assume the wavelength is 550 nm. (a) What is the width of the smallest feature on the bear that this lens can resolve if it is opened to its maximum aperture? (b) If, to gain depth of field, the photographer stops the lens down to  $f/22.0$ , what would be the width of the smallest resolvable feature on the bear?

**36.43 •• Hubble Versus Arecibo.** The Hubble Space Telescope has an aperture of 2.4 m and focuses visible light (380–750 nm). The Arecibo radio telescope in Puerto Rico is 305 m (1000 ft) in diameter (it is built in a mountain valley) and focuses radio waves of wavelength 75 cm. (a) Under optimal viewing conditions, what is the smallest crater that each of these telescopes could resolve on our moon? (b) If the Hubble Space Telescope were to be converted to surveillance use, what is the highest orbit above the surface of the earth it could have and still be able to resolve the license plate (not the letters, just the plate) of a car on the ground? Assume optimal viewing conditions, so that the resolution is diffraction limited.

**36.44 • Observing Jupiter.** You are asked to design a space telescope for earth orbit. When Jupiter is  $5.93 \times 10^8$  km away (its closest approach to the earth), the telescope is to resolve, by Rayleigh's criterion, features on Jupiter that are 250 km apart. What minimum-diameter mirror is required? Assume a wavelength of 500 nm.

### PROBLEMS

**36.45 •• BIO Thickness of Human Hair.** Although we have discussed single-slit diffraction only for a slit, a similar result holds when light bends around a straight, thin object, such as a strand of hair. In that case,  $a$  is the width of the strand. From actual laboratory measurements on a human hair, it was found that when a beam of light of wavelength 632.8 nm was shone on a single strand of hair, and the diffracted light was viewed on a screen 1.25 m away, the first dark fringes on either side of the central bright spot were 5.22 cm apart. How thick was this strand of hair?

**36.46 •• CP** A loudspeaker with a diaphragm that vibrates at 960 Hz is traveling at 80.0 m/s directly toward a pair of holes in a very large wall. The speed of sound in the region is 344 m/s. Far from the wall, you observe that the sound coming through the openings first cancels at  $\pm 11.4^\circ$  with respect to the direction in which the speaker is moving. (a) How far apart are the two openings? (b) At what angles would the sound first cancel if the source stopped moving?

**36.47 •••** Laser light of wavelength 632.8 nm falls normally on a slit that is 0.0250 mm wide. The transmitted light is viewed on a distant screen where the intensity at the center of the central bright fringe is  $8.50\text{ W/m}^2$ . (a) Find the maximum number of totally dark fringes on the screen, assuming the screen is large enough to show them all. (b) At what angle does the dark fringe that is most distant from the center occur? (c) What is the maximum intensity of the bright fringe that occurs immediately before the dark fringe in part (b)? Approximate the angle at which this fringe occurs by assuming it is midway between the angles to the dark fringes on either side of it.

**36.48 • Grating Design.** Your boss asks you to design a diffraction grating that will disperse the first-order visible spectrum through an angular range of  $27.0^\circ$ . (See Example 36.4 in Section 36.5.) (a) What must be the number of slits per centimeter for this grating? (b) At what angles will the first-order visible spectrum begin and end?

**36.49 • Measuring Refractive Index.** A thin slit illuminated by light of frequency  $f$  produces its first dark band at  $\pm 38.2^\circ$  in air. When the entire apparatus (slit, screen, and space in between) is immersed in an unknown transparent liquid, the slit's first dark bands occur instead at  $\pm 21.6^\circ$ . Find the refractive index of the liquid.

**36.50 • Underwater Photography.** An underwater camera has a lens with focal length in air of 35.0 mm and a maximum aperture of  $f/2.80$ . The film it uses has an emulsion that is sensitive to light of frequency  $6.00 \times 10^{14}$  Hz. If the photographer takes a picture of an object 2.75 m in front of the camera with the lens wide open, what is the width of the smallest resolvable detail on the subject if the object is (a) a fish underwater with the camera in the water and (b) a person on the beach with the camera out of the water?

**36.51 ••• CALC** The intensity of light in the Fraunhofer diffraction pattern of a single slit is given by Eq. (36.5). Let  $\gamma = \beta/2$ . (a) Show that the equation for the values of  $\gamma$  at which  $I$  is a maximum is  $\tan \gamma = \gamma$ . (b) Determine the two smallest positive values of  $\gamma$  that are solutions of this equation. (*Hint:* You can use a trial-and-error procedure. Guess a value of  $\gamma$  and adjust your guess to bring  $\tan \gamma$  closer to  $\gamma$ . A graphical solution of the equation is very helpful in locating the solutions approximately, to get good initial guesses.) (c) What are the positive values of  $\gamma$  for the first, second, and third minima on one side of the central maximum? Are the  $\gamma$  values in part (b) precisely halfway between the  $\gamma$  values for adjacent minima? (d) If  $a = 12\lambda$ , what are the angles  $\theta$  (in degrees) that locate the first minimum, the first maximum beyond the central maximum, and the second minimum?

**36.52 ••** A slit 0.360 mm wide is illuminated by parallel rays of light that have a wavelength of 540 nm. The diffraction pattern is observed on a screen that is 1.20 m from the slit. The intensity at the center of the central maximum ( $\theta = 0^\circ$ ) is  $I_0$ . (a) What is the distance on the screen from the center of the central maximum to the first minimum? (b) What is the distance on the screen from the center of the central maximum to the point where the intensity has fallen to  $I_0/2$ ?

**36.53 •• CP CALC** In a large vacuum chamber, monochromatic laser light passes through a narrow slit in a thin aluminum plate and forms a diffraction pattern on a screen that is 0.620 m from the slit. When the aluminum plate has a temperature of  $20.0^\circ\text{C}$ , the width of the central maximum in the diffraction pattern is 2.75 mm. What is the change in the width of the central maximum when the temperature of the plate is raised to  $520.0^\circ\text{C}$ ? Does the width of the central diffraction maximum increase or decrease when the temperature is increased?

**36.54 •• CP** In a laboratory, light from a particular spectrum line of helium passes through a diffraction grating and the second-order maximum is at  $18.9^\circ$  from the center of the central bright fringe. The same grating is then used for light from a distant galaxy that is moving away from the earth with a speed of  $2.65 \times 10^7$  m/s. For the light from the galaxy, what is the angular location of the second-order maximum for the same spectral line as was observed in the lab? (See Section 16.8.)

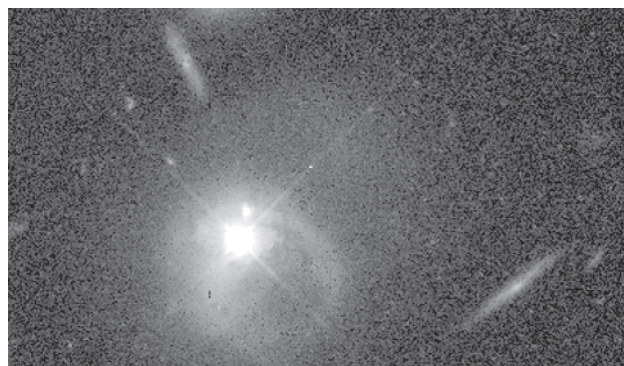
**36.55 •** What is the longest wavelength that can be observed in the third order for a transmission grating having 9200 slits/cm? Assume normal incidence.

**36.56 ••** Humans perceive sound with frequencies in the range 20 to 20,000 Hz. Speech lies in the middle of this range, 400 to 1000 Hz, while bells, sirens, and the knock of silverware on plates all extend above the range for speech. Musical instruments that keep rhythm, such as drums and bass guitars, supply sounds with lower frequencies. Doorways act as natural filters, so noises sound different when they come from around corners. (a) Estimate the width of a typical doorway in your house. (b) Determine the sound frequency that corresponds to a wavelength comparable to the width of a doorway. (Such waves freely diffract around the corner of the door.) (c) Determine the frequency of sound that would diffract only  $\pm 20^\circ$  from directly forward when passing through a doorway. (d) Based on your estimates, does a doorway act as a noticeable filter for sounds we can hear? (e) If so, which frequency range (low, middle, high) is harder to hear from around the corner of a doorway? (f) Experiment to see if you can notice this effect. It is more readily perceived from outside a house as you listen to sounds from inside. Why is this so?

**36.57 •** A diffraction grating has 650 slits/mm. What is the highest order that contains the entire visible spectrum? (The wavelength range of the visible spectrum is approximately 380–750 nm.)

**36.58 ••** *Quasars*, an abbreviation for *quasi-stellar radio sources*, are distant objects that look like stars through a telescope but that emit far more electromagnetic radiation than an entire normal galaxy of stars. An example is the bright object below and to the left of center in **Fig. P36.58**; the other elongated objects in this image are normal galaxies. The leading model for the structure of a quasar is a galaxy with a supermassive black hole at its center. In this model, the radiation is emitted by interstellar gas and dust within the galaxy as this material falls toward the black hole. The radiation is thought to emanate from a region just a few light-years in diameter. (The diffuse glow surrounding the bright quasar shown in **Fig. P36.58** is thought to be this quasar's host galaxy.) To investigate this model of quasars and to study other exotic astronomical objects, the Russian Space Agency has placed a radio telescope in a large orbit around the earth. When this telescope is 77,000 km from earth and the signals it receives are combined with signals from the ground-based telescopes of the VLBA, the resolution is that of a single radio telescope 77,000 km in diameter. What is the size of the smallest detail that this arrangement can resolve in quasar 3C 405, which is  $7.2 \times 10^8$  light-years from earth, using radio waves at a frequency of 1665 MHz? (*Hint:* Use Rayleigh's criterion.) Give your answer in light-years and in kilometers.

Figure P36.58

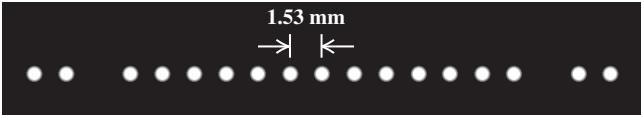


**36.59 ••** In the 1920s Clinton Davisson and Lester Germer accidentally observed diffraction when electrons with 54 eV of energy were scattered off crystalline nickel. The diffraction peak occurred when the angle between the incident beam and the scattered beam was  $50^\circ$ . (a) What is the corresponding angle  $\theta$  relevant for Eq. (36.16)? (b) The planes in crystalline nickel are separated by 0.091 nm, as determined by x-ray scattering experiments. According to the Bragg condition, what wavelength do the electrons in these experiments have? (c) Given the mass of an electron as  $9.11 \times 10^{-31}$  kg, what is the corresponding classical speed  $v_{cl}$  of the diffracted electrons? (d) Assuming the electrons correspond to a wave with speed  $v_{cl}$  and wavelength  $\lambda$ , what is the frequency  $f$  of the diffracted waves? (e) Quantum mechanics postulates that the energy  $E$  and the frequency  $f$  of a particle are related by  $E = hf$ , where  $h$  is known as Planck’s constant. Estimate  $h$  from these observations. (f) Our analysis has a small flaw: The relevant wave velocity, known as a quantum phase velocity, is *half* the classical particle velocity, for reasons explained by deeper aspects of quantum physics. Re-estimate the value of  $h$  using this modification. (g) The established value of Planck’s constant is  $6.6 \times 10^{-34}$  J·s. Does this agree with your estimate?

**36.60 •• BIO Resolution of the Eye.** The maximum resolution of the eye depends on the diameter of the opening of the pupil (a diffraction effect) and the size of the retinal cells. The size of the retinal cells (about  $5.0 \mu\text{m}$  in diameter) limits the size of an object at the near point (25 cm) of the eye to a height of about  $50 \mu\text{m}$ . (To get a reasonable estimate without having to go through complicated calculations, we shall ignore the effect of the fluid in the eye.) (a) Given that the diameter of the human pupil is about 2.0 mm, does the Rayleigh criterion allow us to resolve a  $50\text{-}\mu\text{m}$ -tall object at 25 cm from the eye with light of wavelength 550 nm? (b) According to the Rayleigh criterion, what is the shortest object we could resolve at the 25 cm near point with light of wavelength 550 nm? (c) What angle would the object in part (b) subtend at the eye? Express your answer in minutes ( $60 \text{ min} = 1^\circ$ ), and compare it with the experimental value of about 1 min. (d) Which effect is more important in limiting the resolution of our eyes: diffraction or the size of the retinal cells?

**36.61 •• DATA** While researching the use of laser pointers, you conduct a diffraction experiment with two thin parallel slits. Your result is the pattern of closely spaced bright and dark fringes shown in **Fig. P36.61**. (Only the central portion of the pattern is shown.) You measure that the bright spots are equally spaced at 1.53 mm center to center (except for the missing spots) on a screen that is 2.50 m from the slits. The light source was a helium–neon laser producing a wavelength of 632.8 nm. (a) How far apart are the two slits? (b) How wide is each one?

Figure P36.61



**36.62 •• DATA** Your physics study partner tells you that the width of the central bright band in a single-slit diffraction pattern is inversely proportional to the width of the slit. This means that the width of the central maximum increases when the width of the slit decreases. The claim seems counterintuitive to you, so you make measurements to test it. You shine monochromatic laser light with wavelength  $\lambda$  onto a very narrow slit of width  $a$  and measure the width  $w$  of the central maximum in the diffraction pattern that is produced on a screen 1.50 m from the slit. (By “width,” you mean the distance on the screen between the two minima on either side of the central maximum.) Your measurements are given in the table.

|                           |      |      |      |      |      |      |      |       |       |
|---------------------------|------|------|------|------|------|------|------|-------|-------|
| $a \text{ (}\mu\text{m)}$ | 0.78 | 0.91 | 1.04 | 1.82 | 3.12 | 5.20 | 7.80 | 10.40 | 15.60 |
| $w \text{ (m)}$           | 2.68 | 2.09 | 1.73 | 0.89 | 0.51 | 0.30 | 0.20 | 0.15  | 0.10  |

(a) If  $w$  is inversely proportional to  $a$ , then the product  $aw$  is constant, independent of  $a$ . For the data in the table, graph  $aw$  versus  $a$ . Explain why  $aw$  is not constant for smaller values of  $a$ . (b) Use your graph in part (a) to calculate the wavelength  $\lambda$  of the laser light. (c) What is the angular position of the first minimum in the diffraction pattern for (i)  $a = 0.78 \mu\text{m}$  and (ii)  $a = 15.60 \mu\text{m}$ ?

**36.63 •• DATA** At the metal fabrication company where you work, you are asked to measure the diameter  $D$  of a very small circular hole in a thin, vertical metal plate. To do so, you pass coherent monochromatic light with wavelength 562 nm through the hole and observe the diffraction pattern on a screen that is a distance  $x$  from the hole. You measure the radius  $r$  of the first dark ring in the diffraction pattern (see Fig. 36.26). You make the measurements for four values of  $x$ . Your results are given in the table.

|                  |      |      |      |      |
|------------------|------|------|------|------|
| $x \text{ (m)}$  | 1.00 | 1.50 | 2.00 | 2.50 |
| $r \text{ (cm)}$ | 5.6  | 8.5  | 11.6 | 14.1 |

(a) Use each set of measurements to calculate  $D$ . Because the measurements contain some error, calculate the average of the four values of  $D$  and take that to be your reported result. (b) For  $x = 1.00$  m, what are the radii of the second and third dark rings in the diffraction pattern?

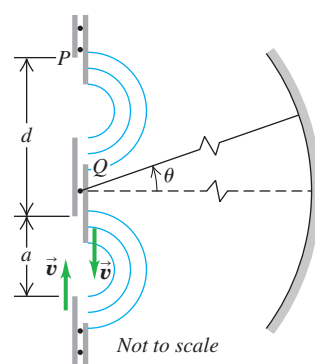
**36.64 ••** A glass sheet is covered by a very thin opaque coating. In the middle of this sheet there is a thin scratch  $0.00125$  mm thick. The sheet is totally immersed beneath the surface of a liquid. Parallel rays of monochromatic coherent light with wavelength 612 nm in air strike the sheet perpendicular to its surface and pass through the scratch. A screen is placed in the liquid a distance of 30.0 cm away from the sheet and parallel to it. You observe that the first dark fringes on either side of the central bright fringe on the screen are 22.4 cm apart. What is the refractive index of the liquid?



## CHALLENGE PROBLEMS

**36.65** ••• An opaque barrier has an inner membrane and an outer membrane that slide past each other, as shown in **Fig. P36.65**. Each membrane includes parallel slits of width  $a$  separated by a distance  $d$ . A screen forms a circular arc subtending  $60^\circ$  at the fixed midpoint between the slits. A green 532 nm laser impinges on the slits from the left. The outer membrane moves upward with speed  $v$  while the inner membrane moves downward with the same speed, propelled by nanomotors. At time  $t = 0$ , point  $P$

Figure P36.65



on the outer membrane is adjacent to point  $Q$  on the inner membrane so that the effective aperture width is zero. The aperture is fully closed again at  $t = 3.00$  s. (a) At  $t = 1.00$  s, there are 19 evenly spaced bright spots on the screen, each of approximately the same intensity. At the edges of the screen the first diffraction minimum and a two-slit interference maximum coincide. What is the slit distance  $d$ ? (Note: The screen does not encompass the entire diffraction pattern.) (b) What is the speed  $v$ ? (c) What is the maximum aperture width  $a$ ? (d) At a certain time, the outermost spots (the  $m = \pm 9$  spots) disappear. What is that time? (e) At  $t = 1.50$  s, what is the intensity of the  $m = \pm 1$  spots in terms of the  $m = 0$  central spot? (f) What are the angular positions of these spots?

**36.66** ••• **Intensity Pattern of  $N$  Slits.** (a) Consider an arrangement of  $N$  slits with a distance  $d$  between adjacent slits. The slits emit coherently and in phase at wavelength  $\lambda$ . Show that at a time  $t$ , the electric field at a distant point  $P$  is

$$E_P(t) = E_0 \cos(kR - \omega t) + E_0 \cos(kR - \omega t + \phi) \\ + E_0 \cos(kR - \omega t + 2\phi) + \cdots \\ + E_0 \cos(kR - \omega t + (N-1)\phi)$$

where  $E_0$  is the amplitude at  $P$  of the electric field due to an individual slit,  $\phi = (2\pi d \sin \theta)/\lambda$ ,  $\theta$  is the angle of the rays reaching  $P$  (as measured from the perpendicular bisector of the slit arrangement), and  $R$  is the distance from  $P$  to the most distant slit. In this problem, assume that  $R$  is much larger than  $d$ . (b) To carry out the sum in part (a), it is convenient to use the complex-number relationship  $e^{iz} = \cos z + i \sin z$ , where  $i = \sqrt{-1}$ . In this expression,  $\cos z$  is the *real part* of the complex number  $e^{iz}$ , and  $\sin z$  is its *imaginary part*. Show that the electric field  $E_P(t)$  is equal to the real part of the complex quantity

$$\sum_{n=0}^{N-1} E_0 e^{i(kR - \omega t + n\phi)}$$

(c) Using the properties of the exponential function that  $e^A e^B = e^{(A+B)}$  and  $(e^A)^n = e^{nA}$ , show that the sum in part (b) can be written as

$$E_0 \left( \frac{e^{iN\phi} - 1}{e^{i\phi} - 1} \right) e^{i(kR - \omega t)} \\ = E_0 \left( \frac{e^{iN\phi/2} - e^{-iN\phi/2}}{e^{i\phi/2} - e^{-i\phi/2}} \right) e^{i[kR - \omega t + (N-1)\phi/2]}$$

Then, using the relationship  $e^{iz} = \cos z + i \sin z$ , show that the (real) electric field at point  $P$  is

$$E_P(t) = \left[ E_0 \frac{\sin(N\phi/2)}{\sin(\phi/2)} \right] \cos[kR - \omega t + (N-1)\phi/2]$$

The quantity in the first square brackets in this expression is the amplitude of the electric field at  $P$ . (d) Use the result for the electric-field amplitude in part (c) to show that the intensity at an angle  $\theta$  is

$$I = I_0 \left[ \frac{\sin(N\phi/2)}{\sin(\phi/2)} \right]^2$$

where  $I_0$  is the maximum intensity for an individual slit. (e) Check the result in part (d) for the case  $N = 2$ . It will help to recall that  $\sin 2A = 2 \sin A \cos A$ . Explain why your result differs from Eq. (35.10), the expression for the intensity in two-source interference, by a factor of 4. (Hint: Is  $I_0$  defined in the same way in both expressions?)

**36.67** ••• **CALC Intensity Pattern of  $N$  Slits, Continued.** Part (d) of Challenge Problem 36.66 gives an expression for the intensity in the interference pattern of  $N$  identical slits. Use this result to verify the following statements. (a) The maximum intensity in the pattern is  $N^2 I_0$ . (b) The principal maximum at the center of the pattern extends from  $\phi = -2\pi/N$  to  $\phi = 2\pi/N$ , so its width is inversely proportional to  $1/N$ . (c) A minimum occurs whenever  $\phi$  is an integer multiple of  $2\pi/N$ , except when  $\phi$  is an integer multiple of  $2\pi$  (which gives a principal maximum). (d) There are  $(N-1)$  minima between each pair of principal maxima. (e) Halfway between two principal maxima, the intensity can be no greater than  $I_0$ ; that is, it can be no greater than  $1/N^2$  times the intensity at a principal maximum.

**36.68** ••• **CALC** It is possible to calculate the intensity in the single-slit Fraunhofer diffraction pattern *without* using the phasor method of Section 36.3. Let  $y'$  represent the position of a point within the slit of width  $a$  in Fig. 36.5a, with  $y' = 0$  at the center of the slit so that the slit extends from  $y' = -a/2$  to  $y' = a/2$ . We imagine dividing the slit up into infinitesimal strips of width  $dy'$ , each of which acts as a source of secondary wavelets. (a) The amplitude of the total wave at the point  $O$  on the distant screen in Fig. 36.5a is  $E_0$ . Explain why the amplitude of the wavelet from each infinitesimal strip within the slit is  $E_0(dy'/a)$ , so that the electric field of the wavelet a distance  $x$  from the infinitesimal strip is  $dE = E_0(dy'/a) \sin(kx - \omega t)$ . (b) Explain why the wavelet from each strip as detected at point  $P$  in Fig. 36.5a can be expressed as

$$dE = E_0 \frac{dy'}{a} \sin[k(D - y' \sin \theta) - \omega t]$$

where  $D$  is the distance from the center of the slit to point  $P$  and  $k = 2\pi/\lambda$ . (c) By integrating the contributions  $dE$  from all parts of the slit, show that the total wave detected at point  $P$  is

$$E = E_0 \sin(kD - \omega t) \frac{\sin[ka(\sin \theta)/2]}{ka(\sin \theta)/2} \\ = E_0 \sin(kD - \omega t) \frac{\sin[\pi a(\sin \theta)/\lambda]}{\pi a(\sin \theta)/\lambda}$$

(The trigonometric identities in Appendix B will be useful.) Show that at  $\theta = 0^\circ$ , corresponding to point  $O$  in Fig. 36.5a, the wave is  $E = E_0 \sin(kD - \omega t)$  and has amplitude  $E_0$ , as stated in part (a). (d) Use the result of part (c) to show that if the intensity at point  $O$  is  $I_0$ , then the intensity at a point  $P$  is given by Eq. (36.7).



## MCAT-STYLE PASSAGE PROBLEMS

**Bragg Reflection on a Different Scale.** A *colloid* consists of particles of one type of substance dispersed in another substance. Suspensions of electrically charged microspheres (microscopic spheres, such as polystyrene) in a liquid such as water can form a colloidal crystal when the microspheres arrange themselves in a regular repeating pattern under the influence of the electrostatic force. Colloidal crystals can selectively manipulate different wavelengths of visible light. Just as we can study crystalline solids by using Bragg reflection of x rays, we can study colloidal crystals through Bragg scattering of visible light from the regular arrangement of charged microspheres. Because the light is traveling through a liquid when it experiences the path differences that lead to constructive interference, it is the wavelength in the liquid that determines the angles at which Bragg reflections are seen. In one experiment, laser light with a wavelength in vacuum of 650 nm is passed through a sample of charged polystyrene spheres in water. A strong interference maximum is then observed when the incident and reflected beams make an angle of  $39^\circ$  with the colloidal crystal planes.

**36.69** Why is visible light, which has much longer wavelengths than x rays do, used for Bragg reflection experiments on colloidal crystals? (a) The microspheres are suspended in a liquid, and it is more difficult for x rays to penetrate liquid than it is for visible light. (b) The irregular

spacing of the microspheres allows the longer-wavelength visible light to produce more destructive interference than can x rays. (c) The microspheres are much larger than atoms in a crystalline solid, and in order to get interference maxima at reasonably large angles, the wavelength must be much longer than the size of the individual scatterers. (d) The microspheres are spaced more widely than atoms in a crystalline solid, and in order to get interference maxima at reasonably large angles, the wavelength must be comparable to the spacing between scattering planes.

**36.70** What plane spacing in the colloidal crystal could produce the maximum in this experiment? (a) 390 nm; (b) 520 nm; (c) 650 nm; (d) 780 nm.

**36.71** When the light is passed through the bottom of the sample container, the interference maximum is observed to be at  $41^\circ$ ; when it is passed through the top, the corresponding maximum is at  $37^\circ$ . What is the best explanation for this observation? (a) The microspheres are more tightly packed at the bottom, because they tend to settle in the suspension. (b) The microspheres are more tightly packed at the top, because they tend to float to the top of the suspension. (c) The increased pressure at the bottom makes the microspheres smaller there. (d) The maximum at the bottom corresponds to  $m = 2$ , whereas the maximum at the top corresponds to  $m = 1$ .

## ANSWERS

## Chapter Opening Question ?

(i) For an optical system that uses a lens, the ability to resolve fine details—its resolving power, or resolution—improves as the lens diameter  $D$  increases (Section 36.7). Each miniature lens in a fly's eye produces its own image, so these images have very poor resolution, compared to those produced by a human eye, because the lens is so small. However, a fly's eye is much better than a human eye at detecting movement.

Key Example  VARIATION Problems

**VP36.3.1** (a) 2.30 m (b) 22.8 mm

**VP36.3.2** (a) 673 nm (b)  $0.349^\circ$

**VP36.3.3** (a)  $0.696^\circ$  (b)  $(3.11 \times 10^{-3})I_0$

**VP36.3.4** (a)  $0.178^\circ$  (b)  $0.405I_0$

**VP36.5.1** (a) 398 nm (a)  $19.1^\circ$  for  $m = 1$ ,  $79.8^\circ$  for  $m = 3$

**VP36.5.2** (a) 651 slits/mm (b)  $54.4^\circ$  for  $m = 2$ ; there is no  $m = 3$  bright region

**VP36.5.3** (a)  $22.1^\circ$  (b)  $48.7^\circ$

**VP36.5.4** (a) 0.186 nm (b) 0.105 nm

**VP36.6.1** (a)  $2.50 \times 10^2$  m (b)  $1.11 \times 10^{-3}$  rad =  $0.0640^\circ$

**VP36.6.2** (a) 466 m (b) 0.00268 mm

**VP36.6.3** (a) 8.46 mm (b)  $f/16.0$

**VP36.6.4** (a) 150 ly (b) 74 ly

## Bridging Problem

$1.501 \times 10^7$  m/s or 5.00% of  $c$ ; away from us



? At Brookhaven National Laboratory in New York, atomic nuclei are accelerated to 99.995% of the ultimate speed limit of the universe—the speed of light,  $c$ . Compared to the kinetic energy of a nucleus moving at 99.000% of  $c$ , the kinetic energy of the same nucleus moving at 99.995% of  $c$  is about (i) 0.001% greater; (ii) 0.1% greater; (iii) 1% greater; (iv) 2% greater; (v) 16 times greater.

# 37 Relativity

In 1905, Albert Einstein—then an unknown 25-year-old assistant in the Swiss patent office—published four papers of extraordinary importance. One was an analysis of Brownian motion; a second (for which he was awarded the Nobel Prize) was on the photoelectric effect. In the last two, Einstein introduced his **special theory of relativity**, proposing drastic revisions in the Newtonian concepts of space and time.

Einstein based the special theory of relativity on two postulates. One states that the laws of physics are the same in all inertial frames of reference; the other states that the speed of light in vacuum is the same in all inertial frames. These innocent-sounding propositions have far-reaching implications. Here are three: (1) Events that are simultaneous for one observer may not be simultaneous for another. (2) When two observers moving relative to each other measure a time interval or a length, they may not get the same results. (3) For the conservation principles for momentum and energy to be valid in all inertial systems, Newton's second law and the equations for momentum and kinetic energy have to be revised.

Relativity has important consequences in *all* areas of physics, including electromagnetism, atomic and nuclear physics, and high-energy physics. Although many of the results derived in this chapter may run counter to your intuition, the theory is in solid agreement with experimental observations.

## 37.1 INVARIANCE OF PHYSICAL LAWS

Let's take a look at the two postulates that make up the special theory of relativity. Both postulates describe what is seen by an observer in an *inertial frame of reference*, which we introduced in Section 4.2. The theory is “special” in the sense that it applies to observers in such special reference frames.

### Einstein's First Postulate

Einstein's first postulate, called the **principle of relativity**, states: **The laws of physics are the same in every inertial frame of reference.** If the laws differed, that difference could distinguish one inertial frame from the others or make one frame more “correct” than another. As an example, suppose you watch two children playing catch with a ball while the three of you are aboard a train moving with constant velocity. Your observations

## LEARNING OUTCOMES

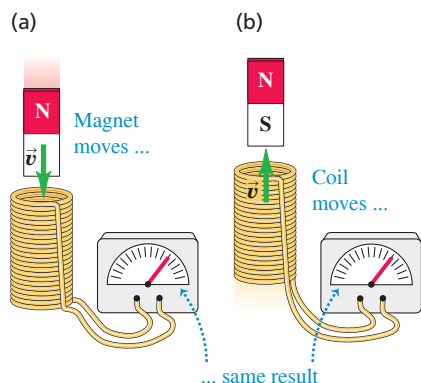
*In this chapter, you'll learn...*

- 37.1 The two postulates of Einstein's special theory of relativity, and what motivates these postulates.
- 37.2 Why different observers can disagree about whether two events are simultaneous.
- 37.3 How relativity predicts that moving clocks run slow, and what experimental evidence confirms this.
- 37.4 How the length of an object changes due to the object's motion.
- 37.5 How the velocity of an object depends on the frame of reference from which it is observed.
- 37.6 How the frequency of a light wave has different values for different observers.
- 37.7 How the theory of relativity modifies the relationship between velocity and momentum.
- 37.8 How to solve problems involving work and kinetic energy for particles moving at relativistic speeds.
- 37.9 Some of the key concepts of Einstein's general theory of relativity.

*You'll need to review...*

- 3.4, 3.5 Motion in a circle, relative velocity.
- 4.2 Inertial frames of reference.
- 16.8 Doppler effect for sound.
- 29.1 Electromagnetic induction.
- 32.2 Maxwell's equations and the speed of light.
- 35.5 Michelson-Morley experiment.

Figure 37.1 The same emf is induced in the coil whether (a) the magnet moves relative to the coil or (b) the coil moves relative to the magnet.



of the motion of the ball, no matter how carefully done, can't tell you how fast (or whether) the train is moving. This is because Newton's laws of motion are the same in every inertial frame.

Another example is the electromotive force (emf) induced in a coil of wire by a nearby moving permanent magnet. In the frame of reference in which the coil is stationary (Fig. 37.1a), the moving magnet causes a change of magnetic flux through the coil, and this induces an emf. In a different frame of reference in which the magnet is stationary (Fig. 37.1b), the motion of the coil through a magnetic field induces the emf. According to the principle of relativity, both of these frames of reference are equally valid. Hence the same emf must be induced in both situations shown in Fig. 37.1. As we saw in Section 29.1, this is indeed the case, so Faraday's law is consistent with the principle of relativity. Indeed, all of the laws of electromagnetism are the same in every inertial frame of reference.

Equally significant is the prediction of the speed of electromagnetic radiation, derived from Maxwell's equations (see Section 32.2). According to this analysis, light and all other electromagnetic waves travel in vacuum with a constant speed, now defined to equal exactly 299,792,458 m/s. (We often use the approximate value  $c = 3.00 \times 10^8$  m/s.) As we'll see, the speed of light in vacuum plays a central role in the theory of relativity.

## Einstein's Second Postulate

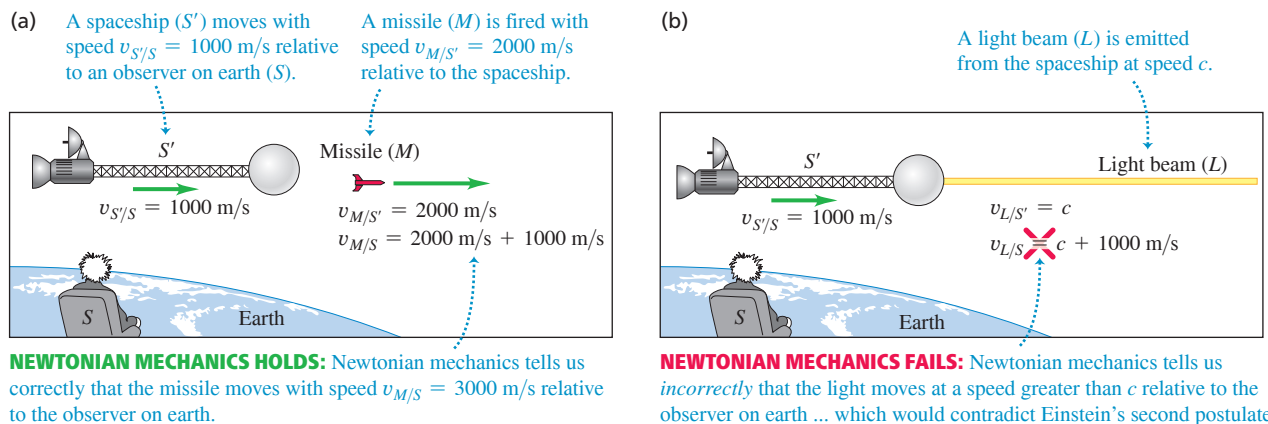
During the 19th century, most physicists believed that light traveled through a hypothetical medium called the *ether*, just as sound waves travel through air. If so, the speed of light measured by observers would depend on their motion relative to the ether and would therefore be different in different directions. The Michelson-Morley experiment, described in Section 35.5, was an effort to detect motion of the earth relative to the ether.

Einstein's conceptual leap was to recognize that if Maxwell's equations are valid in all inertial frames, then the speed of light in vacuum should also be the same in all frames and in all directions. In fact, Michelson and Morley detected *no* ether motion across the earth, and the ether concept has been discarded. Although Einstein may not have known about this negative result, it supported his bold hypothesis. We call this **Einstein's second postulate: The speed of light in vacuum is the same in all inertial frames of reference and is independent of the motion of the source.**

Let's think about what this means. Suppose two observers measure the speed of light in vacuum. One is at rest with respect to the light source, and the other is moving away from it. Both are in inertial frames of reference. According to the principle of relativity, the two observers must obtain the same result, despite the fact that one is moving with respect to the other.

If this seems too easy, consider the following situation. A spacecraft moving past the earth at 1000 m/s fires a missile straight ahead with a speed of 2000 m/s (relative to the spacecraft) (Fig. 37.2). What is the missile's speed relative to the earth? Simple, you say; this is an elementary problem in relative velocity (see Section 3.5). The correct answer,

Figure 37.2 (a) Newtonian mechanics makes correct predictions about relatively slow-moving objects; (b) it makes incorrect predictions about the behavior of light.



according to Newtonian mechanics, is 3000 m/s. But now suppose the spacecraft turns on a searchlight, pointing in the same direction in which the missile was fired. An observer on the spacecraft measures the speed of light emitted by the searchlight and obtains the value  $c$ . According to Einstein's second postulate, the motion of the light after it has left the source cannot depend on the motion of the source. So the observer on earth who measures the speed of this same light must also obtain the value  $c$ , *not*  $c + 1000$  m/s. This result contradicts our elementary notion of relative velocities, and it may not appear to agree with common sense. But "common sense" is intuition based on everyday experience, and this does not usually include measurements of the speed of light.

## The Ultimate Speed Limit

Einstein's second postulate immediately implies the following result: **It is impossible for an inertial observer to travel at  $c$ , the speed of light in vacuum.**

We can prove this by showing that travel at  $c$  implies a logical contradiction. Suppose that the spacecraft  $S'$  in Fig. 37.2b is moving at the speed of light relative to an observer on the earth, so that  $v_{S'/S} = c$ . If the spacecraft turns on a headlight, the second postulate now asserts that the earth observer  $S$  measures the headlight beam to be also moving at  $c$ . Thus this observer measures that the headlight beam and the spacecraft move together and are always at the same point in space. But Einstein's second postulate also asserts that the headlight beam moves at a speed  $c$  relative to the spacecraft, so they *cannot* be at the same point in space. This contradictory result can be avoided only if it is impossible for an inertial observer, such as a passenger on the spacecraft, to move at  $c$ . As we go through our discussion of relativity, you may find yourself asking the question Einstein asked himself as a 16-year-old student, "What would I see if I were traveling at the speed of light?" Einstein realized only years later that his question's basic flaw was that he could *not* travel at  $c$ .

## The Galilean Coordinate Transformation

Let's restate this argument symbolically, using two inertial frames of reference, labeled  $S$  for the observer on earth and  $S'$  for the moving spacecraft, as shown in **Fig. 37.3**. To keep things as simple as possible, we have omitted the  $z$ -axes. The  $x$ -axes of the two frames lie along the same line, but the origin  $O'$  of frame  $S'$  moves relative to the origin  $O$  of frame  $S$  with constant velocity  $u$  along the common  $x$ - $x'$ -axis. We on earth set our clocks so that the two origins coincide at time  $t = 0$ , so their separation at a later time  $t$  is  $ut$ .

**CAUTION Choose your inertial frame coordinates wisely** Many of the equations derived in this chapter are true *only* if you define your inertial reference frames as stated in the preceding paragraph. The positive  $x$ -direction must be the direction in which the origin  $O'$  moves relative to the origin  $O$ . In Fig. 37.3 this direction is to the right; if instead  $O'$  moves to the left relative to  $O$ , you must define the positive  $x$ -direction to be to the left. When solving problems with reference frames, you should *always* draw a diagram showing which reference frame goes with which observer. ■

Now think about how we describe the motion of a particle  $P$ . This might be an exploratory vehicle launched from the spacecraft or a pulse of light from a laser. We can describe the *position* of this particle by using the earth coordinates  $(x, y, z)$  in  $S$  or the spacecraft coordinates  $(x', y', z')$  in  $S'$ . Figure 37.3 shows that these are simply related by

$$x = x' + ut \quad y = y' \quad z = z' \quad (\text{Galilean coordinate transformation}) \quad (37.1)$$

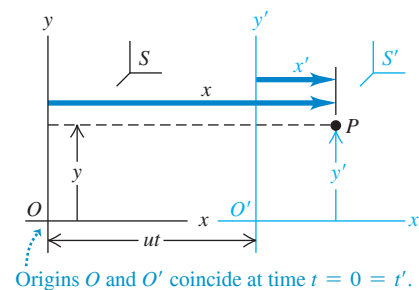
These equations, based on the familiar Newtonian notions of space and time, are called the **Galilean coordinate transformation**.

If particle  $P$  moves in the  $x$ -direction, its instantaneous velocity  $v_x$  as measured by an observer stationary in  $S$  is  $v_x = dx/dt$ . Its velocity  $v'_x$  as measured by an observer stationary in  $S'$  is  $v'_x = dx'/dt$ . We can derive a relationship between  $v_x$  and  $v'_x$  by taking the derivative with respect to  $t$  of the first of Eqs. (37.1):

$$\frac{dx}{dt} = \frac{dx'}{dt} + u$$

Figure 37.3 The position of particle  $P$  can be described by the coordinates  $x$  and  $y$  in frame of reference  $S$  or by  $x'$  and  $y'$  in frame  $S'$ .

Frame  $S'$  moves relative to frame  $S$  with constant velocity  $u$  along the common  $x$ - $x'$ -axis.





Now  $dx/dt$  is the velocity  $v_x$  measured in  $S$ , and  $dx'/dt'$  is the velocity  $v'_x$  measured in  $S'$ , so we get the *Galilean velocity transformation* for one-dimensional motion:

$$v_x = v'_x + u \quad (\text{Galilean velocity transformation}) \quad (37.2)$$

Although the notation differs, this result agrees with our discussion of relative velocities in Section 3.5.

Now here's the fundamental problem. Applied to the speed of light in vacuum, Eq. (37.2) says that  $c = c' + u$ . Einstein's second postulate, supported subsequently by a wealth of experimental evidence, says that  $c = c'$ . This is a genuine inconsistency, not an illusion, and it demands resolution. If we accept this postulate, we are forced to conclude that Eqs. (37.1) and (37.2) *cannot* be precisely correct, despite our convincing derivation. These equations have to be modified to bring them into harmony with this principle.

The resolution involves some very fundamental modifications in our kinematic concepts. The first idea to be changed is the seemingly obvious assumption that the observers in frames  $S$  and  $S'$  use the same *time scale*, formally stated as  $t = t'$ . Alas, we are about to show that this everyday assumption cannot be correct; the two observers *must* have different time scales. We must define the velocity  $v'$  in frame  $S'$  as  $v' = dx'/dt'$ , not as  $dx'/dt$ ; the two quantities are not the same. The difficulty lies in the concept of *simultaneity*, which is our next topic. A careful analysis of simultaneity will help us develop the appropriate modifications of our notions about space and time.

**TEST YOUR UNDERSTANDING OF SECTION 37.1** As a high-speed spaceship flies past you, it fires a strobe light that sends out a pulse of light in all directions. An observer aboard the spaceship measures a spherical wave front that spreads away from the spaceship with the same speed  $c$  in all directions. (a) What is the shape of the wave front that you measure? (i) Spherical; (ii) ellipsoidal, with the longest axis of the ellipsoid along the direction of the spaceship's motion; (iii) ellipsoidal, with the shortest axis of the ellipsoid along the direction of the spaceship's motion; (iv) not enough information is given to decide. (b) As measured by you, does the wave front remain centered on the spaceship?

**ANSWER** (a) (i), (b) no. You, too, will measure a spherical wave front that expands at the same speed  $c$  in all directions. This is a consequence of Einstein's second postulate. The wave front that you measure does *not* stay centered on the position of the moving spaceship; rather, it is centered on the point  $P$  where the spaceship was located at the instant that it emitted the light pulse. For example, suppose the spaceship is moving at speed  $c/2$ . When your watch shows that a time  $t$  has elapsed since the pulse of light was emitted, your measurements will show that the wave front is a sphere of radius  $ct$  centered on  $P$  and that the spaceship is a distance  $ct/2$  from  $P$ .

Figure 37.4 An event has a definite position and time—for instance, on the pavement directly below the center of the Eiffel Tower at midnight on New Year's Eve.



## 37.2 RELATIVITY OF SIMULTANEITY

Measuring times and time intervals involves the concept of **simultaneity**. In a given frame of reference, an **event** is an occurrence that has a definite position and time (**Fig. 37.4**). When you say that you awoke at seven o'clock, you mean that two events (your awakening and your clock showing 7:00) occurred *simultaneously*. The fundamental problem in measuring time intervals is this: In general, two events that are simultaneous in one frame of reference are *not* simultaneous in a second frame that is moving relative to the first, even if both are inertial frames.

### A Thought Experiment in Simultaneity

This may seem to be contrary to common sense. To illustrate the point, here is a version of one of Einstein's *thought experiments*—mental experiments that follow concepts to their logical conclusions. Imagine a train moving with a speed comparable to  $c$ , with uniform velocity (**Fig. 37.5**). Two lightning bolts strike a passenger car, one near each end. Each bolt leaves a mark on the car and one on the ground at the instant the bolt hits. The points on the ground are labeled  $A$  and  $B$  in the figure, and the corresponding points on the car are  $A'$  and  $B'$ . Stanley is stationary on the ground at  $O$ , midway between  $A$  and  $B$ .

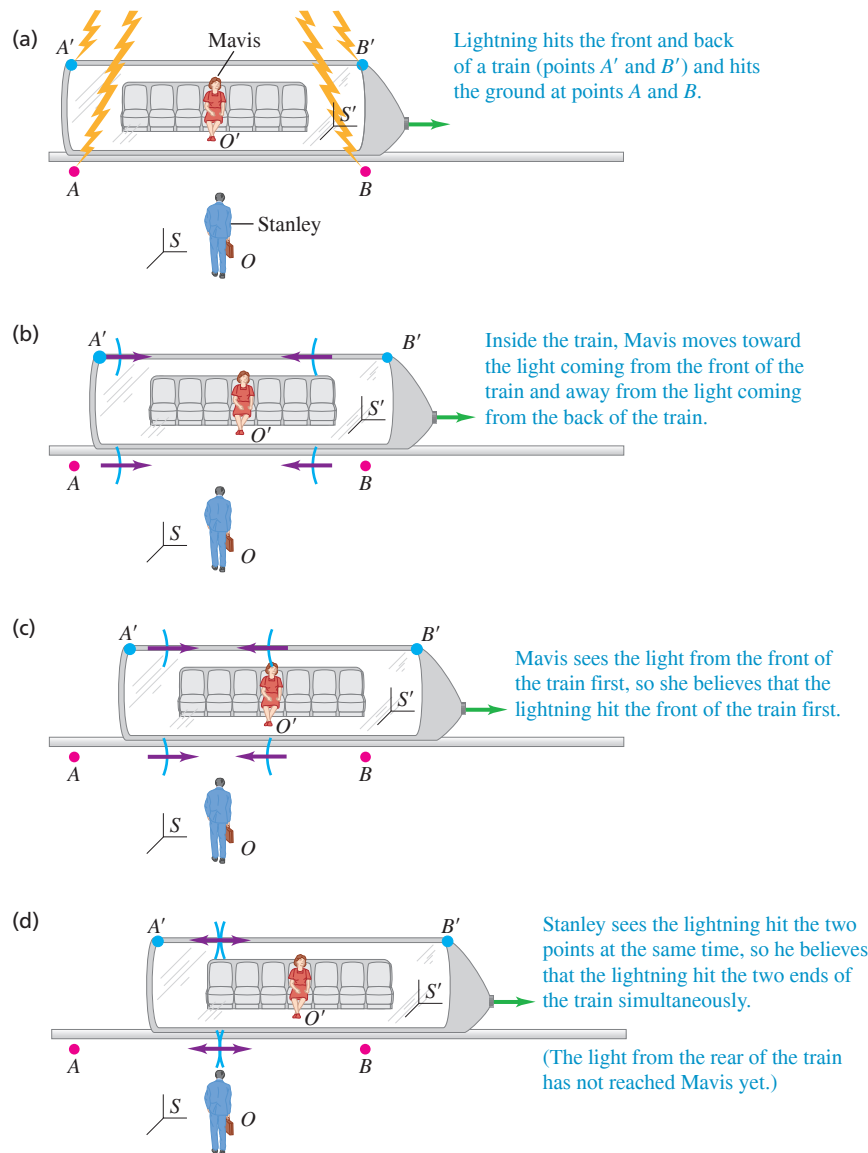


Figure 37.5 A thought experiment in simultaneity.

Mavis is moving with the train at  $O'$  in the middle of the passenger car, midway between  $A'$  and  $B'$ . Both Stanley and Mavis see both light flashes emitted from the points where the lightning strikes.

Suppose the two wave fronts from the lightning strikes reach Stanley at  $O$  simultaneously. He knows that he is the same distance from  $B$  and  $A$ , so Stanley concludes that the two bolts struck  $B$  and  $A$  simultaneously. Mavis agrees that the two wave fronts reached Stanley at the same time, but she disagrees that the flashes were emitted simultaneously.

Stanley and Mavis agree that the two wave fronts do not reach Mavis at the same time. Mavis at  $O'$  is moving to the right with the train, so she runs into the wave front from  $B'$  *before* the wave front from  $A'$  catches up to her. However, because she is in the middle of the passenger car equidistant from  $A'$  and  $B'$ , her observation is that both wave fronts took the same time to reach her because both moved the same distance at the same speed  $c$ . (Recall that the speed of each wave front with respect to *either* observer is  $c$ .) Thus she concludes that the lightning bolt at  $B'$  struck *before* the one at  $A'$ . Stanley at  $O$  measures the two events to be simultaneous, but Mavis at  $O'$  does not! *Whether or not two events at different  $x$ -axis locations are simultaneous depends on the state of motion of the observer.*

You may want to argue that in this example the lightning bolts really *are* simultaneous and that if Mavis at  $O'$  could communicate with the distant points without the time delay

caused by the finite speed of light, she would realize this. But that would be erroneous; the finite speed of information transmission is not the real issue. If  $O'$  is midway between  $A'$  and  $B'$ , then in her frame of reference the time for a signal to travel from  $A'$  to  $O'$  is the same as that from  $B'$  to  $O'$ . Two signals arrive simultaneously at  $O'$  only if they were emitted simultaneously at  $A'$  and  $B'$ . In this example they *do not* arrive simultaneously at  $O'$ , and so Mavis must conclude that the events at  $A'$  and  $B'$  were *not* simultaneous.

Furthermore, there is no basis for saying that Stanley is right and Mavis is wrong, or vice versa. According to the principle of relativity, no inertial frame of reference is more correct than any other in the formulation of physical laws. Each observer is correct *in his or her own frame of reference*. In other words, simultaneity is not an absolute concept. Whether two events are simultaneous depends on the frame of reference. As we mentioned at the beginning of this section, simultaneity plays an essential role in measuring time intervals. It follows that *the time interval between two events may be different in different frames of reference*. So our next task is to learn how to compare time intervals in different frames of reference.

**TEST YOUR UNDERSTANDING OF SECTION 37.2** Stanley, who works for the rail system shown in Fig. 37.5, has carefully synchronized the clocks at all of the rail stations. At the moment that Stanley measures all of the clocks striking noon, Mavis is on a high-speed passenger car traveling from Ogdenville toward North Haverbrook. According to Mavis, when the Ogdenville clock strikes noon, what time is it in North Haverbrook? (i) Noon; (ii) before noon; (iii) after noon.

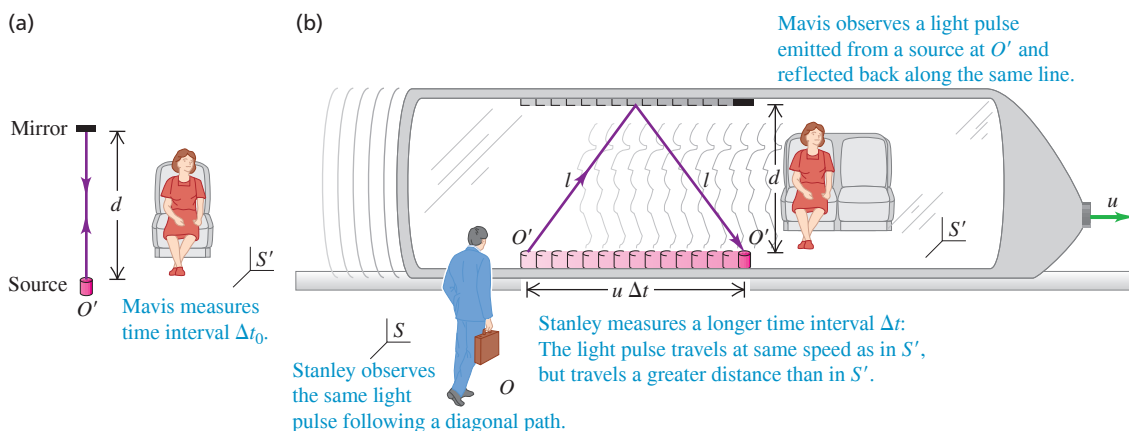
### ANSWER

(iii) In Mavis's frame of reference, the two events (the Ogdenville clock striking noon and the North Haverbrook clock striking noon) are not simultaneous. Figure 37.5 shows that the event toward the front of the rail car occurs first. Since the rail car is moving toward North Haverbrook, that clock struck noon before the one on Ogdenville. So, according to Mavis, it is after noon in North Haverbrook.

## 37.3 RELATIVITY OF TIME INTERVALS

We can derive a quantitative relationship between time intervals in different coordinate systems. To do this, let's consider another thought experiment. As before, a frame of reference  $S'$  moves along the common  $x$ - $x'$ -axis with constant speed  $u$  relative to a frame  $S$ . As discussed in Section 37.1,  $u$  must be less than the speed of light  $c$ . Mavis, who is riding along with frame  $S'$ , measures the time interval between two events that occur at the *same* point in space. Event 1 is when a flash of light from a light source leaves  $O'$ . Event 2 is when the flash returns to  $O'$ , having been reflected from a mirror a distance  $d$  away, as shown in **Fig. 37.6a**. We label the time interval  $\Delta t_0$ , using the subscript zero as a reminder

Figure 37.6 (a) Mavis, in frame of reference  $S'$ , observes a light pulse emitted from a source at  $O'$  and reflected back along the same line. (b) How Stanley (in frame of reference  $S$ ) and Mavis observe the same light pulse. The positions of  $O'$  at the times of departure and return of the pulse are shown.



that the apparatus is at rest, with zero velocity, in frame  $S'$ . The flash of light moves a total distance  $2d$ , so the time interval is

$$\Delta t_0 = \frac{2d}{c} \quad (37.3)$$

The round-trip time measured by Stanley in frame  $S$  is a different interval  $\Delta t$ ; in his frame of reference the two events occur at *different* points in space. During the time  $\Delta t$ , the source moves relative to  $S$  a distance  $u \Delta t$  (Fig. 37.6b). In  $S'$  the round-trip distance is  $2d$  perpendicular to the relative velocity, but the round-trip distance in  $S$  is the longer distance  $2l$ , where

$$l = \sqrt{d^2 + \left(\frac{u \Delta t}{2}\right)^2}$$

In writing this expression, we have assumed that both observers measure the same distance  $d$ . We'll justify this assumption in the next section. The speed of light is the same for both observers, so the round-trip time measured in  $S$  is

$$\Delta t = \frac{2l}{c} = \frac{2}{c} \sqrt{d^2 + \left(\frac{u \Delta t}{2}\right)^2} \quad (37.4)$$

We would like to have a relationship between  $\Delta t$  and  $\Delta t_0$  that is independent of  $d$ . To get this, we solve Eq. (37.3) for  $d$  and substitute the result into Eq. (37.4):

$$\Delta t = \frac{2}{c} \sqrt{\left(\frac{c \Delta t_0}{2}\right)^2 + \left(\frac{u \Delta t}{2}\right)^2} \quad (37.5)$$

Now we square this and solve for  $\Delta t$ ; the result is

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}}$$

Since the quantity  $\sqrt{1 - u^2/c^2}$  is less than 1,  $\Delta t$  is greater than  $\Delta t_0$ : Thus Stanley measures a *longer* round-trip time for the light pulse than does Mavis.

## Time Dilation and Proper Time

We may generalize this important result. Suppose that in a particular frame of reference, two events occur at the same point in space. If these events are two ticks of a clock, then this is the frame of reference at which the clock is at rest. We call this the *rest frame* of the clock. There is only one frame of reference in which a clock is at rest, and there are infinitely many in which it is moving. Therefore the time interval measured between two events (such as two ticks of the clock) that occur at the same point in a particular frame is a more fundamental quantity than the interval between events at different points. We use the term **proper time** to describe the time interval between two events that occur *at the same point*.

Let  $\Delta t_0$  be the proper time between the two events—that is, the time as measured by an observer at rest in the frame in which the events occur at the same point. Then our above result says that an observer in a second frame moving with constant speed  $u$  relative to the rest frame will measure the time interval to be  $\Delta t$ , where

**Time dilation:**

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}} \quad (37.6)$$

Proper time between two events (measured in rest frame)

Speed of light in vacuum

Speed of second frame relative to rest frame

Time interval between same events measured in second frame of reference



Figure 37.7 This image shows an exploding star, called a *supernova*, within a distant galaxy. The brightness of a typical supernova decays at a certain rate. But supernovae that are moving away from us at a substantial fraction of the speed of light decay more slowly, in accordance with Eq. (37.6). The decaying supernova is a moving “clock” that runs slow.

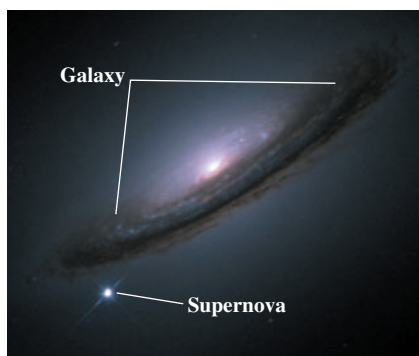


Figure 37.8 The Lorentz factor  $\gamma = 1/\sqrt{1 - u^2/c^2}$  as a function of the relative speed  $u$  of two frames of reference.

As speed  $u$  approaches the speed of light  $c$ ,  $\gamma$  approaches infinity.

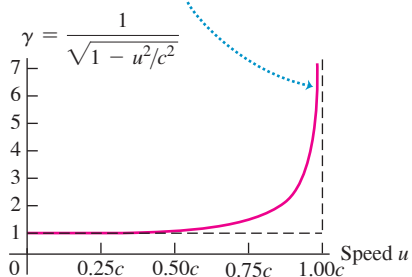
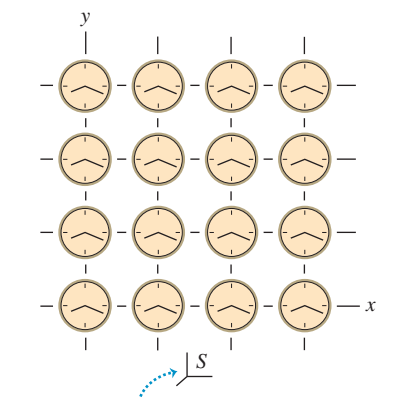


Figure 37.9 A frame of reference pictured as a coordinate system with a grid of synchronized clocks.



The grid is three dimensional; identical planes of clocks lie in front of and behind the page, connected by grid lines perpendicular to the page.

We recall that no inertial observer can travel at  $u = c$  and we note that  $\sqrt{1 - u^2/c^2}$  is imaginary for  $u > c$ . Thus Eq. (37.6) gives sensible results only when  $u < c$ . The denominator of Eq. (37.6) is always smaller than 1, so  $\Delta t$  is always *larger* than  $\Delta t_0$ . Thus we call this effect **time dilation**.

Think of an old-fashioned pendulum clock that has one second between ticks, as measured by Mavis in the clock’s rest frame; this is  $\Delta t_0$ . If the clock’s rest frame is moving relative to Stanley, he measures a time between ticks  $\Delta t$  that is longer than one second. In brief, *observers measure any clock to run slow if it moves relative to them (Fig. 37.7)*. Note that this conclusion is a direct result of the fact that the speed of light in vacuum is the same in both frames of reference.

The quantity  $1/\sqrt{1 - u^2/c^2}$  in Eq. (37.6) is called the **Lorentz factor**. It appears often in relativity and is denoted by the symbol  $\gamma$  (the Greek letter gamma):

$$\text{Lorentz factor } \gamma = \frac{1}{\sqrt{1 - u^2/c^2}} \quad (37.7)$$

Speed of light in vacuum  
Speed of one frame of reference relative to another

In terms of this symbol, we can express the time dilation formula, Eq. (37.6), as

$$\text{Time dilation: } \Delta t = \gamma \Delta t_0 \quad (37.8)$$

Proper time between two events (measured in rest frame)  
Lorentz factor relating the two frames  
Time interval between same events measured in second frame of reference

As a further simplification,  $u/c$  is sometimes given the symbol  $\beta$  (the Greek letter beta); then  $\gamma = 1/\sqrt{1 - \beta^2}$ .

Figure 37.8 shows a graph of  $\gamma$  as a function of the relative speed  $u$  of two frames of reference. When  $u$  is very small compared to  $c$ ,  $u^2/c^2$  is much smaller than 1 and  $\gamma$  is very nearly *equal* to 1. In that limit, Eqs. (37.6) and (37.8) approach the Newtonian relationship  $\Delta t = \Delta t_0$ , corresponding to the same time interval in all frames of reference.

If the relative speed  $u$  is great enough that  $\gamma$  is appreciably greater than 1, the speed is said to be *relativistic*; if the difference between  $\gamma$  and 1 is negligibly small, the speed  $u$  is called *nonrelativistic*. Thus  $u = 6.00 \times 10^7 \text{ m/s} = 0.200c$  (for which  $\gamma = 1.02$ ) is a relativistic speed, but  $u = 6.00 \times 10^4 \text{ m/s} = 0.000200c$  (for which  $\gamma = 1.00000002$ ) is a nonrelativistic speed.

**CAUTION Measuring time intervals** It is important to note that the time interval  $\Delta t$  in Eq. (37.6) involves events that occur *at different space points* in the frame of reference  $S$ . Note also that any differences between  $\Delta t$  and the proper time  $\Delta t_0$  are *not* caused by differences in the times required for light to travel from those space points to an observer at rest in  $S$ . We assume that our observer can correct for differences in light transit times, just as an astronomer who’s observing the sun understands that an event seen now on earth actually occurred 500 s ago on the sun’s surface. Alternatively, we can use *two* observers, one stationary at the location of the first event and the other at the second, each with his or her own clock. We can synchronize these two clocks without difficulty, as long as they are at rest in the same frame of reference. For example, we could send a light pulse simultaneously to the two clocks from a point midway between them. When the pulses arrive, the observers set their clocks to a prearranged time. (But clocks that are synchronized in one frame of reference *are not* in general synchronized in any other frame.)

In thought experiments, it’s often helpful to imagine many observers with synchronized clocks at rest at various points in a particular frame of reference. We can picture a frame of reference as a coordinate grid with lots of synchronized clocks distributed around it, as suggested by Fig. 37.9. Only when a clock is moving relative to a given frame of reference do we have to watch for ambiguities of synchronization or simultaneity.

Throughout this chapter we’ll frequently use phrases like “Stanley *observes* that Mavis passes the point  $x = 5.00 \text{ m}$ ,  $y = 0$ ,  $z = 0$  at time  $2.00 \text{ s}$ .” This means that Stanley is using

a grid of clocks in his frame of reference, like the grid shown in Fig. 37.9, to record the time of an event. We could restate the phrase as “When Mavis passes the point at  $x = 5.00$  m,  $y = 0$ ,  $z = 0$ , the clock at that location in Stanley’s frame of reference reads 2.00 s.” We’ll avoid using phrases like “Stanley *sees* that Mavis is at a certain point at a certain time,” because there is a time delay for light to travel to Stanley’s eye from the position of an event.

### PROBLEM-SOLVING STRATEGY 37.1 Time Dilation

**IDENTIFY** *the relevant concepts:* The concept of time dilation is used whenever we compare the time intervals between events as measured by observers in different inertial frames of reference.

**SET UP** *the problem* using the following steps:

1. First decide what two events define the beginning and the end of the time interval. Then identify the two frames of reference in which the time interval is measured.
2. Identify the target variable.

**EXECUTE** *the solution* as follows:

1. In many problems, the time interval as measured in one frame of reference is the *proper* time  $\Delta t_0$ . This is the time interval

between two events in a frame of reference in which the two events occur at the same point in space. In a second frame of reference that has a speed  $u$  relative to that first frame, there is a longer time interval  $\Delta t$  between the same two events. In this second frame the two events occur at different points. You’ll need to decide in which frame the time interval is  $\Delta t_0$  and in which frame it is  $\Delta t$ .

2. Use Eq. (37.6) or (37.8) to relate  $\Delta t_0$  and  $\Delta t$ , and then solve for the target variable.

**EVALUATE** *your answer:* Note that  $\Delta t$  is never smaller than  $\Delta t_0$ , and  $u$  is never greater than  $c$ . If your results suggest otherwise, you need to rethink your calculation.

### EXAMPLE 37.1 Time dilation at 0.990c

#### WITH VARIATION PROBLEMS

High-energy subatomic particles coming from space interact with atoms in the earth’s upper atmosphere, in some cases producing unstable particles called *muons*. A muon decays into other particles with a mean lifetime of  $2.20 \mu\text{s} = 2.20 \times 10^{-6}$  s as measured in a reference frame in which it is at rest. If a muon is moving at  $0.990c$  relative to the earth, what will an observer on earth measure its mean lifetime to be?

**IDENTIFY and SET UP** The muon’s lifetime is the time interval between two events: the production of the muon and its subsequent decay. Our target variable is the lifetime in your frame of reference on earth, which we call frame  $S$ . We are given the lifetime in a frame  $S'$  in which the muon is at rest; this is its *proper* lifetime,  $\Delta t_0 = 2.20 \mu\text{s}$ . The relative speed of these two frames is  $u = 0.990c$ . We use Eq. (37.6) to relate the lifetimes in the two frames.

**EXECUTE** The muon moves relative to the earth between the two events, so the two events occur at different positions as measured in

$S$  and the time interval in that frame is  $\Delta t$  (the target variable). From Eq. (37.6),

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}} = \frac{2.20 \mu\text{s}}{\sqrt{1 - (0.990)^2}} = 15.6 \mu\text{s}$$

**EVALUATE** Our result predicts that the mean lifetime of the muon in the earth frame ( $\Delta t$ ) is about seven times longer than in the muon’s frame ( $\Delta t_0$ ). This prediction has been verified experimentally; indeed, this was the first experimental confirmation of the time dilation formula, Eq. (37.6).

**KEYCONCEPT** The time interval between two events depends on the frame of reference of the observer. The time interval is shortest in an inertial frame  $S$  where the two events occur at the same point; this is the proper time  $\Delta t_0$ . In any other inertial frame that is moving with respect to  $S$ , the time interval is longer (it is *dilated*) [Eq. (37.6)].

### EXAMPLE 37.2 Time dilation at airliner speeds

#### WITH VARIATION PROBLEMS

An airplane flies from San Francisco to New York (about 4800 km, or  $4.80 \times 10^6$  m) at a steady speed of 300 m/s (about 670 mi/h). How much time does the trip take, as measured by an observer on the ground? By an observer in the plane?

**IDENTIFY and SET UP** Here we’re interested in the time interval between the airplane departing from San Francisco and landing in New York. The target variables are the time intervals as measured in the frame of reference of the ground  $S$  and in the frame of reference of the airplane  $S'$ .

**EXECUTE** As measured in  $S$  the two events occur at different positions (San Francisco and New York), so the time interval measured by ground observers corresponds to  $\Delta t$  in Eq. (37.6). To find it, we simply divide the distance by the speed  $u = 300$  m/s:

$$\Delta t = \frac{4.80 \times 10^6 \text{ m}}{300 \text{ m/s}} = 1.60 \times 10^4 \text{ s} \quad (\text{about } 4\frac{1}{2} \text{ hours})$$

In the airplane’s frame  $S'$ , San Francisco and New York passing under the plane occur at the same point (the position of the plane). Hence the time interval in the airplane is a proper time, corresponding to  $\Delta t_0$  in Eq. (37.6). We have

$$\frac{u^2}{c^2} = \frac{(300 \text{ m/s})^2}{(3.00 \times 10^8 \text{ m/s})^2} = 1.00 \times 10^{-12}$$

From Eq. (37.6),

$$\Delta t_0 = (1.60 \times 10^4 \text{ s})\sqrt{1 - 1.00 \times 10^{-12}}$$

The square root can’t be evaluated with adequate precision with an ordinary calculator. But we can approximate it using the binomial theorem (see Appendix B):

$$(1 - 1.00 \times 10^{-12})^{1/2} = 1 - \left(\frac{1}{2}\right)(1.00 \times 10^{-12}) + \dots$$

*Continued*

The remaining terms are of the order of  $10^{-24}$  or smaller and can be discarded. The approximate result for  $\Delta t_0$  is

$$\Delta t_0 = (1.60 \times 10^4 \text{ s})(1 - 0.50 \times 10^{-12})$$

The proper time  $\Delta t_0$ , measured in the airplane, is very slightly less (by less than one part in  $10^{12}$ ) than the time measured on the ground.

**EVALUATE** We don't notice such effects in everyday life. But present-day atomic clocks (see Section 1.3) can attain a precision of about one

part in  $10^{13}$ . A cesium clock traveling a long distance in an airliner has been used to measure this effect and thereby verify Eq. (37.6) even at speeds much less than  $c$ .

**KEYCONCEPT** If two events take place at the same point in an inertial frame  $S$ , an observer in a frame  $S'$  moving relative to  $S$  observes a time interval between these events that is greater than in  $S$  by the Lorentz factor  $\gamma$  [Eqs. (37.7) and (37.8)]. This factor is substantially different from 1 only if the relative speed of  $S'$  and  $S$  is a substantial fraction of the speed of light.

### EXAMPLE 37.3 Just when is it proper?

### WITH VARIATION PROBLEMS

Mavis boards a spaceship and then zips past Stanley on earth at a relative speed of  $0.600c$ . At the instant she passes him, they both start timers. (a) A short time later Stanley measures that Mavis has traveled  $9.00 \times 10^7 \text{ m}$  beyond him and is passing a space station. What does Stanley's timer read as she passes the space station? What does Mavis's timer read? (b) Stanley starts to blink just as Mavis flies past him, and Mavis measures that the blink takes  $0.400 \text{ s}$  from beginning to end. According to Stanley, what is the duration of his blink?

**IDENTIFY and SET UP** This problem involves time dilation for two *different* sets of events measured in Stanley's frame of reference (which we call  $S$ ) and in Mavis's frame of reference (which we call  $S'$ ). The two events of interest in part (a) are when Mavis passes Stanley and when Mavis passes the space station; the target variables are the time intervals between these two events as measured in  $S$  and in  $S'$ . The two events in part (b) are the start and finish of Stanley's blink; the target variable is the time interval between these two events as measured in  $S$ .

**EXECUTE** (a) The two events, Mavis passing the earth and Mavis passing the space station, occur at different positions in Stanley's frame but at the same position in Mavis's frame. Hence Stanley measures time interval  $\Delta t$ , while Mavis measures the *proper* time  $\Delta t_0$ . As measured by Stanley, Mavis moves at  $0.600c = 0.600(3.00 \times 10^8 \text{ m/s}) = 1.80 \times 10^8 \text{ m/s}$  and travels  $9.00 \times 10^7 \text{ m}$  in time  $\Delta t = (9.00 \times 10^7 \text{ m})/(1.80 \times 10^8 \text{ m/s}) = 0.500 \text{ s}$ . From Eq. (37.6), Mavis's timer reads an elapsed time of

$$\Delta t_0 = \Delta t \sqrt{1 - u^2/c^2} = 0.500 \text{ s} \sqrt{1 - (0.600)^2} = 0.400 \text{ s}$$

Since Stanley measures  $0.500 \text{ s}$  elapsed on his timer but only  $0.400 \text{ s}$  elapsed on Mavis's timer, Stanley concludes that Mavis's timer runs slow.

(b) It is tempting to answer that Stanley's blink lasts  $0.500 \text{ s}$  in his frame. But this is wrong, because we are now considering a *different* pair of events than in part (a). The start and finish of Stanley's blink occur at the same point in his frame  $S$  but at different positions in Mavis's frame  $S'$ , so the time interval of  $0.400 \text{ s}$  that she measures between these events is equal to  $\Delta t$ . The duration of the blink measured on Stanley's timer is the proper time  $\Delta t_0$ :

$$\Delta t_0 = \Delta t \sqrt{1 - u^2/c^2} = 0.400 \text{ s} \sqrt{1 - (0.600)^2} = 0.320 \text{ s}$$

Since Mavis measures  $0.400 \text{ s}$  elapsed on her timer for the duration of Stanley's blink but only  $0.320 \text{ s}$  elapsed on Stanley's timer, Mavis concludes that Stanley's timer runs slow. *Both* observers measure that a moving clock runs slow!

**EVALUATE** This example illustrates both time dilation and the relativity of simultaneity. In Mavis's frame she passes the space station at the same instant that Stanley finishes his blink,  $0.400 \text{ s}$  after she passed Stanley. Hence these two events are simultaneous to Mavis in frame  $S'$ . But these two events are *not* simultaneous to Stanley in his frame  $S$ : According to his timer, he finishes his blink after  $0.320 \text{ s}$  and Mavis passes the space station after  $0.500 \text{ s}$ .

**KEYCONCEPT** No matter which inertial frame of reference you are in, you observe dilated time intervals for timekeepers (clocks) of any kind that move relative to you. If  $S$  and  $S'$  are two inertial frames in relative motion, an observer in  $S$  measures clocks in  $S'$  to run slow, and an observer in  $S'$  measures clocks in  $S$  to run slow.

## The Twin Paradox

Equations (37.6) and (37.8) for time dilation suggest an apparent paradox called the **twin paradox**. Consider identical twin astronauts named Eartha and Astrid. Eartha remains on earth while her twin Astrid takes off on a high-speed trip through the galaxy. Because of time dilation, Eartha observes Astrid's heartbeat and all other life processes proceeding more slowly than her own. Thus to Eartha, Astrid ages more slowly; when Astrid returns to earth she is younger (has aged less) than Eartha.

Here is the paradox: All inertial frames are equivalent. Can't Astrid make exactly the same arguments to conclude that Eartha is in fact the younger? Then each twin measures the other to be younger when they're back together, and that's a paradox.

To resolve the paradox, note that the twins are *not* identical in all respects. While Eartha remains in an approximately inertial frame at all times, Astrid must *accelerate* with respect to that frame during parts of her trip in order to leave, turn around, and return to earth. Eartha's reference frame is always approximately inertial; Astrid's is often far from inertial. Thus there is a real physical difference between the circumstances of the two twins. Careful analysis shows that Eartha is correct; when Astrid returns, she *is* younger than Eartha.

**TEST YOUR UNDERSTANDING OF SECTION 37.3** Samir (who is standing on the ground) starts his stopwatch at the instant that Maria flies past him in her spaceship at a speed of  $0.600c$ . At the same instant, Maria starts her stopwatch. (a) As measured in Samir's frame of reference, what is the reading on Maria's stopwatch at the instant that Samir's stopwatch reads  $10.0$  s? (i)  $10.0$  s; (ii) less than  $10.0$  s; (iii) more than  $10.0$  s. (b) As measured in Maria's frame of reference, what is the reading on Samir's stopwatch at the instant that Maria's stopwatch reads  $10.0$  s? (i)  $10.0$  s; (ii) less than  $10.0$  s; (iii) more than  $10.0$  s.

**ANSWER** (a) (ii), (b) (iii) The statement that moving clocks run slow refers to any clock that is moving relative to an observer. Maria and her stopwatch are moving relative to Samir, so Samir measures Maria's stopwatch to be running slow and to have ticked off fewer seconds than his own stopwatch. Samir and his stopwatch are moving relative to Maria, so she likewise measures Samir's stopwatch to be running slow. Each observer's measurement is correct for his or her own frame of reference. Both observers conclude that a moving stopwatch runs slow. This is consistent with the principle of relativity (see Section 37.1), which states that the laws of physics are the same in all inertial frames of reference.

## 37.4 RELATIVITY OF LENGTH

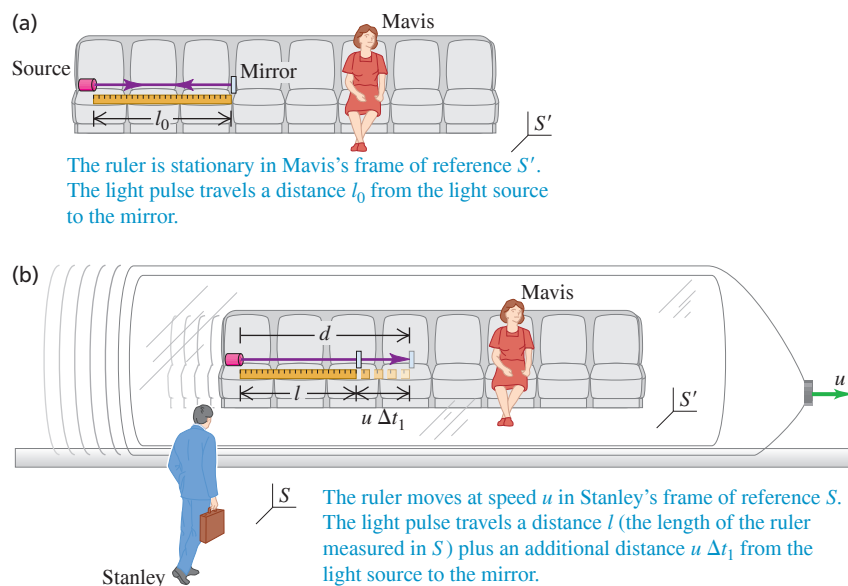
Not only does the time interval between two events depend on the observer's frame of reference, but the *distance* between two points may also depend on the observer's frame of reference. The concept of simultaneity is involved. Suppose you want to measure the length of a moving car. One way is to have two assistants make marks on the pavement at the positions of the front and rear bumpers. Then you measure the distance between the marks. But your assistants have to make their marks *at the same time*. If one marks the position of the front bumper at one time and the other marks the position of the rear bumper half a second later, you won't get the car's true length. Since we've learned that simultaneity isn't an absolute concept, we have to proceed with caution.

### Lengths Parallel to the Relative Motion

To develop a relationship between lengths that are measured parallel to the direction of motion in various coordinate systems, we consider another thought experiment. We attach a light source to one end of a ruler and a mirror to the other end. The ruler is at rest in reference frame  $S'$ , and its length in this frame is  $l_0$  (**Fig. 37.10a**). Then the time  $\Delta t_0$  required for a light pulse to make the round trip from source to mirror and back is

$$\Delta t_0 = \frac{2l_0}{c} \quad (37.9)$$

This is a *proper* time interval because departure and return occur at the same point in  $S'$ .



**APPLICATION Which One's the Grandmother?** The answer to this question may seem obvious, but it could depend on which person had traveled to a distant destination at relativistic speeds. Imagine that a 20-year-old woman had given birth to a child and then immediately left on a 100 light-year trip (50 light-years out and 50 light-years back) at 99.5% the speed of light. Because of time dilation for the traveler, only 10 years would pass for her, and she would be 30 years old when she returned, even though 100 years had passed by for people on earth. Meanwhile, the child she left behind at home could have had a baby 20 years after her departure, and this grandchild would now be 80 years old!



Figure 37.10 (a) A ruler is at rest in Mavis's frame  $S'$ . A light pulse is emitted from a source at one end of the ruler, reflected by a mirror at the other end, and returned to the source position. (b) Motion of the light pulse as measured in Stanley's frame  $S$ .



In reference frame  $S$  the ruler is moving to the right with speed  $u$  during this travel of the light pulse (Fig. 37.10b). The length of the ruler in  $S$  is  $l$ , and the time of travel from source to mirror, as measured in  $S$ , is  $\Delta t_1$ . During this interval the ruler, with source and mirror attached, moves a distance  $u \Delta t_1$ . The total length of path  $d$  from source to mirror is not  $l$ , but rather

$$d = l + u \Delta t_1 \quad (37.10)$$

The light pulse travels with speed  $c$ , so it is also true that

$$d = c \Delta t_1 \quad (37.11)$$

Combining Eqs. (37.10) and (37.11) to eliminate  $d$ , we find

$$\begin{aligned} c \Delta t_1 &= l + u \Delta t_1 \quad \text{or} \\ \Delta t_1 &= \frac{l}{c - u} \end{aligned} \quad (37.12)$$

(Dividing the distance  $l$  by  $c - u$  does *not* mean that light travels with speed  $c - u$ , but rather that the distance the pulse travels in  $S$  is greater than  $l$ .)

In the same way we can show that the time  $\Delta t_2$  for the return trip from mirror to source is

$$\Delta t_2 = \frac{l}{c + u} \quad (37.13)$$

The *total* time  $\Delta t = \Delta t_1 + \Delta t_2$  for the round trip, as measured in  $S$ , is

$$\Delta t = \frac{l}{c - u} + \frac{l}{c + u} = \frac{2l}{c(1 - u^2/c^2)} \quad (37.14)$$

We also know that  $\Delta t$  and  $\Delta t_0$  are related by Eq. (37.6) because  $\Delta t_0$  is a proper time in  $S'$ . Thus Eq. (37.9) for the round-trip time in the rest frame  $S'$  of the ruler becomes

$$\Delta t \sqrt{1 - \frac{u^2}{c^2}} = \frac{2l_0}{c} \quad (37.15)$$

Finally, we combine Eqs. (37.14) and (37.15) to eliminate  $\Delta t$  and simplify:

$$\begin{aligned} \text{Length contraction:} \quad l &= l_0 \sqrt{1 - \frac{u^2}{c^2}} = \frac{l_0}{\gamma} \end{aligned} \quad (37.16)$$

Proper length of object (measured in rest frame)   
 Speed of second frame relative to rest frame   
 Lorentz factor relating the two frames   
 Speed of light in vacuum

Figure 37.11 The speed at which electrons traverse the 3 km beam line of the SLAC National Accelerator Laboratory is slower than  $c$  by only 10 cm/s. As measured in the reference frame of such an electron, the beam line (which extends from the top to the bottom of this photograph) is only about 8 cm long!



**CAUTION** Length contraction is real. This is *not* an optical illusion! The ruler really is shorter in reference frame  $S$  than it is in  $S'$ . ■

[We have used the Lorentz factor  $\gamma$  defined in Eq. (37.7).] Thus the length  $l$  measured in  $S$ , in which the ruler is moving, is *shorter* than the length  $l_0$  measured in its rest frame  $S'$ .

A length measured in the frame in which the object is at rest (the rest frame of the object) is called a **proper length**; thus  $l_0$  is a proper length in  $S'$ , and the length measured in any other frame moving relative to  $S'$  is *less than*  $l_0$ . This effect is called **length contraction**.

When  $u$  is very small in comparison to  $c$ ,  $\gamma$  approaches 1. Thus in the limit of small speeds we approach the Newtonian relationship  $l = l_0$ . This and the corresponding result for time dilation show that Eqs. (37.1), the Galilean coordinate transformation, are usually sufficiently accurate for relative speeds much smaller than  $c$ . If  $u$  is a reasonable fraction of  $c$ , however, the quantity  $\sqrt{1 - u^2/c^2}$  can be appreciably less than 1. Then  $l$  can be substantially smaller than  $l_0$ , and the effects of length contraction can be substantial (Fig. 37.11).

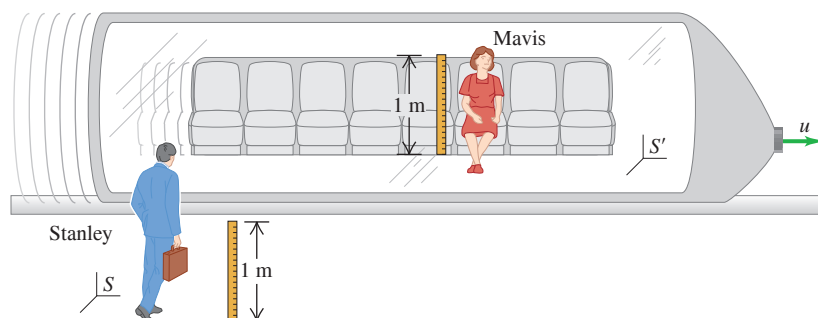


Figure 37.12 The meter sticks are perpendicular to the relative velocity. For any value of  $u$ , both Stanley and Mavis measure either meter stick to have a length of 1 meter.

## Lengths Perpendicular to the Relative Motion

We have derived Eq. (37.16) for lengths measured in the direction *parallel* to the relative motion of the two frames of reference. Lengths that are measured *perpendicular* to the direction of motion are *not* contracted. To prove this, consider two identical meter sticks. One stick is at rest in frame  $S$  and lies along the positive  $y$ -axis with one end at  $O$ , the origin of  $S$ . The other is at rest in frame  $S'$  and lies along the positive  $y'$ -axis with one end at  $O'$ , the origin of  $S'$ . Frame  $S'$  moves in the positive  $x$ -direction relative to frame  $S$ . Observers Stanley and Mavis, at rest in  $S$  and  $S'$  respectively, station themselves at the 50 cm mark of their sticks. At the instant the two origins coincide, the two sticks lie along the same line. At this instant, Mavis makes a mark on Stanley's stick at the point that coincides with her own 50 cm mark, and Stanley does the same to Mavis's stick.

Suppose for the sake of argument that Stanley observes Mavis's stick as longer than his own. Then the mark Stanley makes on her stick is *below* its center. In that case, Mavis will think Stanley's stick has become shorter, since half of its length coincides with *less* than half her stick's length. So Mavis observes moving sticks getting shorter and Stanley observes them getting longer. But this implies an asymmetry between the two frames that contradicts the basic postulate of relativity that tells us all inertial frames are equivalent. We conclude that consistency with the postulates of relativity requires that both observers measure the rulers as having the *same* length, even though to each observer one of them is stationary and the other is moving (**Fig. 37.12**). So *there is no length contraction perpendicular to the direction of relative motion of the coordinate systems*. We used this result in our derivation of Eq. (37.6) in assuming that the distance  $d$  is the same in both frames of reference.

For example, suppose a moving rod of length  $l_0$  makes an angle  $\theta_0$  with the direction of relative motion (the  $x$ -axis) as measured in its rest frame. Its length component in that frame parallel to the motion,  $l_0 \cos \theta_0$ , is contracted to  $(l_0 \cos \theta_0)/\gamma$ . However, its length component perpendicular to the motion,  $l_0 \sin \theta_0$ , remains the same.

## PROBLEM-SOLVING STRATEGY 37.2 Length Contraction

**IDENTIFY** the relevant concepts: The concept of length contraction is used whenever we compare the length of an object as measured by observers in different inertial frames of reference.

**SET UP** the problem using the following steps:

1. Decide what defines the length in question. If the problem describes an object such as a ruler, it is just the distance between the ends of the object. If the problem is about a distance between two points in space, it helps to envision an object like a ruler that extends from one point to the other.
2. Identify the target variable.

**EXECUTE** the solution as follows:

1. Determine the reference frame in which the object in question is at rest. In this frame, the length of the object is its proper length  $l_0$ .

In a second reference frame moving at speed  $u$  relative to the first frame, the object has contracted length  $l$ .

2. Keep in mind that length contraction occurs only for lengths parallel to the direction of relative motion of the two frames. Any length that is perpendicular to the relative motion is the same in both frames.
3. Use Eq. (37.16) to relate  $l$  and  $l_0$ , and then solve for the target variable.

**EVALUATE** your answer: Check that your answers make sense:  $l$  is never larger than  $l_0$ , and  $u$  is never greater than  $c$ .

**EXAMPLE 37.4** How long is the spaceship?**WITH VARIATION PROBLEMS**

A spaceship flies past earth at a speed of  $0.990c$ . A crew member on board the spaceship measures its length, obtaining the value 400 m. What length do observers measure on earth?

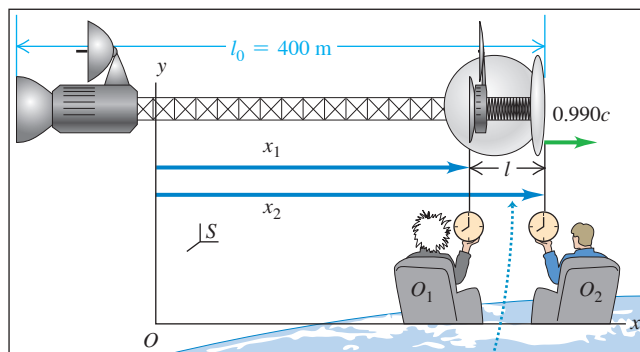
**IDENTIFY and SET UP** This problem is about the nose-to-tail length of the spaceship as measured on the spaceship and on earth. This length is along the direction of relative motion (**Fig. 37.13**), so there will be length contraction. The spaceship's 400 m length is the *proper* length  $l_0$  because it is measured in the frame in which the spaceship is at rest. Our target variable is the length  $l$  measured in the earth frame, relative to which the spaceship is moving at  $u = 0.990c$ .

**EXECUTE** From Eq. (37.16), the length in the earth frame is

$$l = l_0 \sqrt{1 - \frac{u^2}{c^2}} = (400 \text{ m}) \sqrt{1 - (0.990)^2} = 56.4 \text{ m}$$

**EVALUATE** The spaceship is shorter in a frame in which it is in motion than in a frame in which it is at rest. To measure the length  $l$ , two earth observers with synchronized clocks could measure the positions of the two ends of the spaceship simultaneously in the earth's reference frame, as shown in **Fig. 37.13**. (These two measurements will *not* appear simultaneous to an observer in the spaceship.)

**Figure 37.13** Measuring the length of a moving spaceship.



The two observers on earth ( $S$ ) must measure  $x_2$  and  $x_1$  simultaneously to obtain the correct length  $l = x_2 - x_1$  in their frame of reference.

**KEYCONCEPT** The dimensions of an object depend on the frame of reference of the observer. If the object is moving at constant velocity relative to your inertial frame, its length is contracted *along* the direction of relative motion [Eq. (37.16)]. There is no change in length *perpendicular* to the direction of relative motion.

**EXAMPLE 37.5** How far apart are the observers?**WITH VARIATION PROBLEMS**

Observers  $O_1$  and  $O_2$  in **Fig. 37.13** are 56.4 m apart on the earth. How far apart does the spaceship crew measure them to be?

**IDENTIFY and SET UP** In this example the 56.4 m distance is the *proper* length  $l_0$ . It represents the length of a ruler that extends from  $O_1$  to  $O_2$  and is at rest in the earth frame in which the observers are at rest. Our target variable is the length  $l$  of this ruler measured in the spaceship frame, in which the earth and ruler are moving at  $u = 0.990c$ .

**EXECUTE** As in Example 37.4, but with  $l_0 = 56.4 \text{ m}$ ,

$$l = l_0 \sqrt{1 - \frac{u^2}{c^2}} = (56.4 \text{ m}) \sqrt{1 - (0.990)^2} = 7.96 \text{ m}$$

**EVALUATE** This answer does *not* say that the crew measures their spaceship to be both 400 m long and 7.96 m long. As measured on earth,

the tail of the spacecraft is at the position of  $O_1$  at the same instant that the nose of the spacecraft is at the position of  $O_2$ . Hence the length of the spaceship measured on earth equals the 56.4 m distance between  $O_1$  and  $O_2$ . But in the spaceship frame  $O_1$  and  $O_2$  are only 7.96 m apart, and the nose (which is 400 m in front of the tail) passes  $O_2$  before the tail passes  $O_1$ .

**KEYCONCEPT** No matter which inertial frame of reference you are in, you observe objects that move relative to you to be contracted along the direction of relative motion. If  $S$  and  $S'$  are two inertial frames in relative motion, an observer in  $S$  measures objects in  $S'$  to have contracted lengths, and an observer in  $S'$  measures objects in  $S$  to have contracted lengths.

**How an Object Moving Near  $c$  Would Appear**

Let's think a little about the visual appearance of a moving three-dimensional object. If we could see the positions of all points of the object simultaneously, it would appear to shrink only in the direction of motion. But we *don't* see all the points simultaneously; light from points farther from us takes longer to reach us than does light from points near to us, so we see the farther points at the positions they had at earlier times.

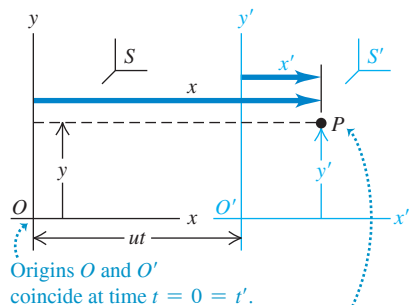
Suppose we have a rectangular rod with its faces parallel to the coordinate planes. When we look end-on at the center of the closest face of such a rod at rest, we see only that face. (See the center rod in computer-generated **Fig. 37.14a**.) But when that rod is moving past us toward the right at an appreciable fraction of the speed of light, we may also see its left side because of the earlier-time effect just described. That is, we can see some points that we couldn't see when the rod was at rest because the rod moves out of the way of the light rays from those points to us. Conversely, some light that can get to us when the rod is at rest is blocked by the moving rod. Because of all this, the rods in **Figs. 37.14b** and **37.14c** appear rotated and distorted.





Figure 37.15 As measured in frame of reference  $S$ ,  $x'$  is contracted to  $x'/\gamma$ , so  $x = ut + (x'/\gamma)$  and  $x' = \gamma(x - ut)$ .

Frame  $S'$  moves relative to frame  $S$  with constant velocity  $u$  along the common  $x$ - $x'$ -axis.



The Lorentz coordinate transformation relates the spacetime coordinates of an event as measured in the two frames:  $(x, y, z, t)$  in frame  $S$  and  $(x', y', z', t')$  in frame  $S'$ .

We now equate Eqs. (37.18) and (37.19) to eliminate  $x'$ . This gives us an equation for  $t'$  in terms of  $x$  and  $t$ . You can do the algebra to show that

$$t' = \frac{t - ux/c^2}{\sqrt{1 - u^2/c^2}} \quad (37.20)$$

As we discussed previously, lengths perpendicular to the direction of relative motion are not affected by the motion, so  $y' = y$  and  $z' = z$ .

Collecting our results, we have the *Lorentz coordinate transformation*:

**Lorentz coordinate transformation:**  
Spacetime coordinates of an event are  $x, y, z, t$  in frame  $S$  and  $x', y', z', t'$  in frame  $S'$ .

Velocity of  $S'$  relative to  $S$  in positive direction along  $x$ - $x'$ -axis

$$\begin{aligned} x' &= \frac{x - ut}{\sqrt{1 - u^2/c^2}} = \gamma(x - ut) \\ y' &= y \\ z' &= z \\ t' &= \frac{t - ux/c^2}{\sqrt{1 - u^2/c^2}} = \gamma(t - ux/c^2) \end{aligned} \quad (37.21)$$

Lorentz factor relating the two frames  
Speed of light in vacuum

These equations are the relativistic generalization of the Galilean coordinate transformation, Eqs. (37.1) and  $t = t'$ . For values of  $u$  that approach zero,  $\sqrt{1 - u^2/c^2}$  and  $\gamma$  approach 1, and the  $ux/c^2$  term approaches zero. In this limit, Eqs. (37.21) become identical to Eqs. (37.1) along with  $t = t'$ . In general, though, both the coordinates and time of an event in one frame depend on its coordinates and time in another frame. *Space and time have become intertwined; we can no longer say that length and time have absolute meanings independent of the frame of reference.* For this reason, we refer to time and the three dimensions of space collectively as a four-dimensional entity called **spacetime**, and we call  $(x, y, z, t)$  together the **spacetime coordinates** of an event.

## The Lorentz Velocity Transformation

We can use Eqs. (37.21) to derive the relativistic generalization of the Galilean velocity transformation, Eq. (37.2). We consider only one-dimensional motion along the  $x$ -axis and use the term “velocity” as being short for the “ $x$ -component of the velocity.” Suppose that in a time  $dt$  a particle moves a distance  $dx$ , as measured in frame  $S$ . We obtain the corresponding distance  $dx'$  and time  $dt'$  in  $S'$  by taking differentials of Eqs. (37.21):

$$\begin{aligned} dx' &= \gamma(dx - u dt) \\ dt' &= \gamma(dt - u dx/c^2) \end{aligned}$$

We divide the first equation by the second and then divide the numerator and denominator of the result by  $dt$  to obtain

$$\frac{dx'}{dt'} = \frac{\frac{dx}{dt} - u}{1 - \frac{u}{c^2} \frac{dx}{dt}}$$

Now  $dx/dt$  is the velocity  $v_x$  in  $S$ , and  $dx'/dt'$  is the velocity  $v'_x$  in  $S'$ , so

**Lorentz velocity transformation**  
(velocity in  $S'$  in terms of velocity in  $S$ ):

$$v'_x = \frac{v_x - u}{1 - uv_x/c^2} \quad (37.22)$$

Velocity of  $S'$  relative to  $S$  in positive direction along  $x$ - $x'$ -axis  
Speed of light in vacuum

When  $u$  and  $v_x$  are much smaller than  $c$ , the denominator in Eq. (37.22) approaches 1, and we approach the nonrelativistic result  $v'_x = v_x - u$ . The opposite extreme is the case  $v_x = c$ ; then we find

$$v'_x = \frac{c - u}{1 - uc/c^2} = \frac{c(1 - u/c)}{1 - u/c} = c$$

This says that anything moving with velocity  $v_x = c$  measured in  $S$  also has velocity  $v'_x = c$  measured in  $S'$ , despite the relative motion of the two frames. So Eq. (37.22) is consistent with Einstein's postulate that the speed of light in vacuum is the same in all inertial frames of reference.

The principle of relativity tells us there is no fundamental distinction between the two frames  $S$  and  $S'$ . Thus the expression for  $v_x$  in terms of  $v'_x$  must have the same form as Eq. (37.22), with  $v_x$  changed to  $v'_x$ , and vice versa, and the sign of  $u$  reversed. Carrying out these operations with Eq. (37.22), we find

$$v_x = \frac{v'_x + u}{1 + uv'_x/c^2} \quad (37.23)$$

Lorentz velocity transformation (velocity in  $S$  in terms of velocity in  $S'$ ):
x-velocity of object in frame  $S'$ 
x-velocity of object in frame  $S$ 
Velocity of  $S'$  relative to  $S$  in positive direction along  $x$ - $x'$ -axis
Speed of light in vacuum

You can also obtain this equation by solving Eq. (37.22) for  $v_x$ . Both Eqs. (37.22) and (37.23) are *Lorentz velocity transformations* for one-dimensional motion.

When  $u$  is less than  $c$ , the Lorentz velocity transformations show us that an object moving with a speed less than  $c$  in one frame of reference always has a speed less than  $c$  in *every other* frame of reference. This is one reason for concluding that no material object may travel with a speed equal to or greater than  $c$  relative to *any* inertial frame of reference. Later we'll see that the relativistic generalizations of energy and momentum give further support to this hypothesis.

**CAUTION Use the correct reference frame coordinates** The Lorentz transformation equations given by Eqs. (37.21), (37.22), and (37.23) assume that frame  $S'$  is moving in the positive  $x$ -direction with velocity  $u$  relative to frame  $S$ . Always set up your coordinate system to follow this convention. ■

**APPLICATION Relative Velocity and Reference Frames** A relay race illustrates the frames of reference used in Eqs. (37.22) and (37.23). The runner in purple is the particle, and the two frames of reference in which the particle's motion is observed are our rest frame  $S$  (we are spectators standing next to the track) and the rest frame  $S'$  of the runner in red, who has speed  $u$  relative to us. Since the runner in red is moving to the left relative to us, we must take the positive  $x$ -direction to the left. The runner in purple has positive velocity  $v_x$  relative to us (she is moving to the left); if the runner in red is moving faster ( $u > v_x$ ), then from Eq. (37.22) the runner in purple has a *negative* velocity  $v'_x$  relative to the runner in red.

Reference frame  $S$ : Our rest frame as we stand next to the track

Reference frame  $S'$ : Rest frame of runner in red. She moves with speed  $u$  relative to us.



Runner in purple has velocity  $v_x$  in  $S$  and velocity  $v'_x$  in  $S'$ .

### PROBLEM-SOLVING STRATEGY 37.3 Lorentz Transformations

**IDENTIFY** *the relevant concepts:* The Lorentz coordinate-transformation equations relate the spacetime coordinates of an event in one inertial reference frame to the coordinates of the same event in a second inertial frame. The Lorentz velocity-transformation equations relate the velocity of an object in one inertial reference frame to its velocity in a second inertial frame.

**SET UP** *the problem* using the following steps:

1. Identify the target variable.
2. Define the two inertial frames  $S$  and  $S'$ . Remember that  $S'$  moves relative to  $S$  at a constant velocity  $u$  in the  $+x$ -direction.
3. If the coordinate-transformation equations are needed, make a list of spacetime coordinates in the two frames, such as  $x_1, x'_1, t_1, t'_1$ , and so on. Label carefully which of these you know and which you don't.
4. In velocity-transformation problems, clearly identify  $u$  (the relative velocity of the two frames of reference),  $v_x$  (the velocity of the object relative to  $S$ ), and  $v'_x$  (the velocity of the object relative to  $S'$ ).

**EXECUTE** *the solution* as follows:

1. In a coordinate-transformation problem, use Eqs. (37.21) to solve for the spacetime coordinates of the event as measured in  $S'$  in terms of the corresponding values in  $S$ . (If you need to solve for the spacetime coordinates in  $S$  in terms of the corresponding values in  $S'$ , you can easily convert the expressions in Eqs. (37.21): Replace all of the primed quantities with unprimed ones, and vice versa, and replace  $u$  with  $-u$ .)
2. In a velocity-transformation problem, use either Eq. (37.22) or Eq. (37.23), as appropriate, to solve for the target variable.

**EVALUATE** *your answer:* Don't be discouraged if some of your results don't seem to make sense or if they disagree with "common sense." It takes time to develop intuition about relativity; you'll gain it with experience.

**EXAMPLE 37.6** Was it received before it was sent?**WITH VARIATION PROBLEMS**

Winning an interstellar race, Mavis pilots her spaceship across a finish line in space at a speed of  $0.600c$  relative to that line. A “hooray” message is sent from the back of her ship (event 2) at the instant (in her frame of reference) that the front of her ship crosses the line (event 1). She measures the length of her ship to be 300 m. Stanley is at the finish line and is at rest relative to it. When and where does he measure events 1 and 2 to occur?

**IDENTIFY and SET UP** This example involves the Lorentz coordinate transformation. Our derivation of this transformation assumes that the origins of frames  $S$  and  $S'$  coincide at  $t = 0 = t'$ . Thus for simplicity we fix the origin of  $S$  at the finish line and the origin of  $S'$  at the front of the spaceship so that Stanley and Mavis measure event 1 to be at  $x = 0 = x'$  and  $t = 0 = t'$ .

Mavis in  $S'$  measures her spaceship to be 300 m long, so she has the “hooray” sent from 300 m behind her spaceship’s front at the instant she measures the front to cross the finish line. That is, she measures event 2 at  $x' = -300$  m and  $t' = 0$ .

Our target variables are the coordinate  $x$  and time  $t$  of event 2 that Stanley measures in  $S$ .

**EXECUTE** To solve for the target variables, we modify the first and last of Eqs. (37.21) to give  $x$  and  $t$  as functions of  $x'$  and  $t'$ . We do so in the same way that we obtained Eq. (37.23) from Eq. (37.22). We remove

the primes from  $x'$  and  $t'$ , add primes to  $x$  and  $t$ , and replace each  $u$  with  $-u$ . The results are

$$x = \gamma(x' + ut') \quad \text{and} \quad t = \gamma(t' + ux'/c^2)$$

From Eq. (37.7),  $\gamma = 1.25$  for  $u = 0.600c = 1.80 \times 10^8$  m/s. We also substitute  $x' = -300$  m,  $t' = 0$ ,  $c = 3.00 \times 10^8$  m/s, and  $u = 1.80 \times 10^8$  m/s in the equations for  $x$  and  $t$  to find  $x = -375$  m at  $t = -7.50 \times 10^{-7}$  s =  $-0.750 \mu\text{s}$  for event 2.

**EVALUATE** Mavis says that the events are simultaneous, but Stanley says that the “hooray” was sent *before* Mavis crossed the finish line. This does not mean that the effect preceded the cause. The fastest that Mavis can send a signal the length of her ship is  $300 \text{ m} / (3.00 \times 10^8 \text{ m/s}) = 1.00 \mu\text{s}$ . She cannot send a signal from the front at the instant it crosses the finish line that would cause a “hooray” to be broadcast from the back at the same instant. She would have to send that signal from the front at least  $1.00 \mu\text{s}$  before then, so she had to slightly anticipate her success.

**KEYCONCEPT** Given the position and time of an event as measured in one inertial frame of reference, you can use the Lorentz coordinate transformation to find the position and time of that same event as measured in a second inertial frame moving relative to the first one.

**EXAMPLE 37.7** Relative velocities**WITH VARIATION PROBLEMS**

(a) A spaceship moving away from the earth at  $0.900c$  fires a robot space probe in the same direction as its motion at  $0.700c$  relative to the spaceship. What is the probe’s velocity relative to the earth? (b) A scoutship is sent to catch up with the spaceship by traveling at  $0.950c$  relative to the earth. What is the velocity of the scoutship relative to the spaceship?

**IDENTIFY and SET UP** This example uses the Lorentz velocity transformation. Let the earth and spaceship reference frames be  $S$  and  $S'$ , respectively (**Fig. 37.16**); their relative velocity is  $u = 0.900c$ . In part (a) we are given the probe velocity  $v'_x = 0.700c$  with respect to  $S'$ , and the target variable is the velocity  $v_x$  of the probe relative to  $S$ . In part (b) we are given the velocity  $v_x = 0.950c$  of the scoutship relative to  $S$ , and the target variable is its velocity  $v'_x$  relative to  $S'$ .

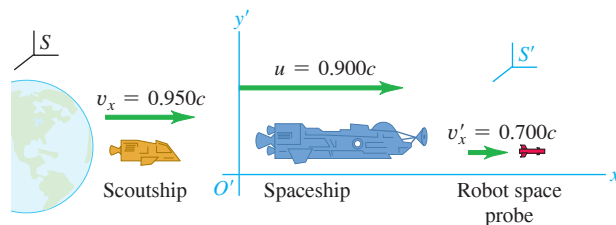
**EXECUTE** (a) We use Eq. (37.23) to find the probe velocity relative to the earth:

$$v_x = \frac{v'_x + u}{1 + uv'_x/c^2} = \frac{0.700c + 0.900c}{1 + (0.900c)(0.700c)/c^2} = 0.982c$$

(b) We use Eq. (37.22) to find the scoutship velocity relative to the spaceship:

$$v'_x = \frac{v_x - u}{1 - uv_x/c^2} = \frac{0.950c - 0.900c}{1 - (0.900c)(0.950c)/c^2} = 0.345c$$

Figure 37.16 The spaceship, robot space probe, and scoutship.



**EVALUATE** What would the Galilean velocity-transformation formula, Eq. (37.2), say? In part (a) we would have found the probe’s velocity relative to the earth to be  $v_x = v'_x + u = 0.700c + 0.900c = 1.600c$ , which is greater than  $c$  and hence impossible. In part (b), we would have found the scoutship’s velocity relative to the spaceship to be  $v'_x = v_x - u = 0.950c - 0.900c = 0.050c$ ; the relativistically correct value,  $v'_x = 0.345c$ , is almost seven times greater than the incorrect Galilean value.

**KEYCONCEPT** Given the velocity of an object as measured in one inertial frame of reference, you can use the Lorentz velocity transformation to find the velocity of that same object as measured in a second inertial frame moving relative to the first one. If an object is moving slower than  $c$  as measured in one inertial frame, its speed is less than  $c$  as measured in any other inertial frame.

**TEST YOUR UNDERSTANDING OF SECTION 37.5** (a) In frame  $S$  events  $P_1$  and  $P_2$  occur at the same  $x$ -,  $y$ -, and  $z$ -coordinates, but event  $P_1$  occurs before event  $P_2$ . In frame  $S'$ , which event occurs first? (b) In frame  $S$  events  $P_3$  and  $P_4$  occur at the same time  $t$  and the same  $y$ - and  $z$ -coordinates, but event  $P_3$  occurs at a less positive  $x$ -coordinate than event  $P_4$ . In frame  $S'$ , which event occurs first?





Using  $c^2 - u^2 = (c - u)(c + u)$  gives

**Doppler effect, electromagnetic waves, source approaching observer:**

$$f = \sqrt{\frac{c + u}{c - u}} f_0 \quad (37.25)$$

Frequency measured by observer  
Frequency measured in rest frame of source  
Speed of light in vacuum  
Speed of source relative to observer

**Figure 37.18** This handheld radar gun emits a radio beam of frequency  $f_0$ , which in the frame of reference of an approaching car has a higher frequency  $f$  given by Eq. (37.25). The reflected beam also has frequency  $f$  in the car's frame, but has an even higher frequency  $f'$  in the police officer's frame. The radar gun calculates the car's speed by comparing the frequencies of the emitted beam and the doubly Doppler-shifted reflected beam. (Compare Example 16.18 in Section 16.8.)



This shows that when the source moves *toward* the observer, the observed frequency  $f$  is *greater* than the emitted frequency  $f_0$ . The difference  $f - f_0 = \Delta f$  is called the Doppler frequency shift. When  $u/c$  is much smaller than 1, the fractional shift  $\Delta f/f$  is also small and is approximately equal to  $u/c$ :

$$\frac{\Delta f}{f} = \frac{u}{c}$$

When the source moves *away from* the observer, we change the sign of  $u$  in Eq. (37.25) to get

$$f = \sqrt{\frac{c - u}{c + u}} f_0 \quad (\text{Doppler effect, electromagnetic waves, source moving away from observer}) \quad (37.26)$$

This agrees with Eq. (16.30) with minor notation changes.

With light, unlike sound, there is no distinction between motion of source and motion of observer; only the *relative* velocity of the two is significant. The last four paragraphs of Section 16.8 discuss several practical applications of the Doppler effect with light and other electromagnetic radiation; we suggest you review those paragraphs now. **Figure 37.18** shows one common application.

### EXAMPLE 37.8 A jet from a black hole

Many galaxies have supermassive black holes at their centers (see Section 13.8). As material swirls around such a black hole, it is heated, becomes ionized, and generates strong magnetic fields. The resulting magnetic forces steer some of the material into high-speed jets that blast out of the galaxy and into intergalactic space (**Fig. 37.19**). The light we observe from the jet in Fig. 37.19 has a frequency of  $6.66 \times 10^{14}$  Hz (in the far ultraviolet; see Fig. 32.4), but in the reference frame of the jet material the light has a frequency of  $5.55 \times 10^{13}$  Hz (in the infrared). What is the speed of the jet material with respect to us?

**IDENTIFY and SET UP** This problem involves the Doppler effect for electromagnetic waves. The frequency we observe is  $f = 6.66 \times 10^{14}$  Hz, and the frequency in the frame of the source is  $f_0 = 5.55 \times 10^{13}$  Hz. Since  $f > f_0$ , the jet is approaching us and we use Eq. (37.25) to find the target variable  $u$ .

**EXECUTE** We need to solve Eq. (37.25) for  $u$ . We'll leave it as an exercise for you to show that the result is

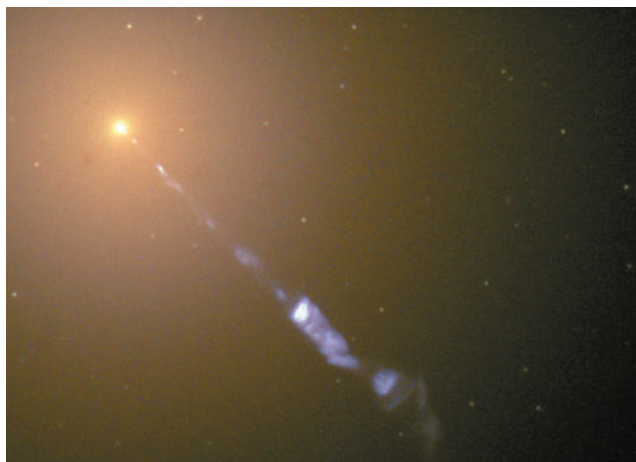
$$u = \frac{(f/f_0)^2 - 1}{(f/f_0)^2 + 1} c$$

We have  $f/f_0 = (6.66 \times 10^{14} \text{ Hz}) / (5.55 \times 10^{13} \text{ Hz}) = 12.0$ , so

$$u = \frac{(12.0)^2 - 1}{(12.0)^2 + 1} c = 0.986c$$

**EVALUATE** Because the frequency shift is quite substantial, it would have been erroneous to use the approximate expression  $\Delta f/f = u/c$ . Had you done so, you would have found that  $u = c(\Delta f/f_0) = c(6.66 \times 10^{14} \text{ Hz} - 5.55 \times 10^{13} \text{ Hz}) / (5.55 \times 10^{13} \text{ Hz}) = 11.0c$ .

**Figure 37.19** This image shows a fast-moving jet 5000 light-years in length emanating from the center of the galaxy M87. The light from the jet is emitted by fast-moving electrons spiraling around magnetic field lines (see Fig. 27.18).



This result cannot be correct because the jet material cannot travel faster than light.

**KEYCONCEPT** The frequency of electromagnetic waves as measured by an observer moving relative to their source is different from the frequency measured in the rest frame of the source. The measured frequency is greater than the source frequency if the observer and source are approaching, and less than the source frequency if they are receding [Eq. (37.25)].

### 37.7 RELATIVISTIC MOMENTUM

Newton's laws of motion have the same form in all inertial frames of reference. When we use transformations to change from one inertial frame to another, the laws should be *invariant* (unchanging). But we have just learned that the principle of relativity forces us to replace the Galilean transformations with the more general Lorentz transformations. As we'll see, this requires corresponding generalizations in the laws of motion and the definitions of momentum and energy.

The principle of conservation of momentum states that *when two objects interact, the total momentum is constant*, provided that the net external force acting on the objects in an inertial reference frame is zero (for example, if they form an isolated system, interacting only with each other). If conservation of momentum is a valid physical law, it must be valid in *all* inertial frames of reference. Now, here's the problem: Suppose we look at a collision in one inertial coordinate system  $S$  and find that momentum is conserved. Then we use the Lorentz transformation to obtain the velocities in a second inertial system  $S'$ . We find that if we use the Newtonian definition of momentum ( $\vec{p} = m\vec{v}$ ), momentum is *not* conserved in the second system! The only way to make momentum conservation consistent with relativity is to generalize the *definition* of momentum.

We won't derive the correct relativistic generalization of momentum, but here is the result. Suppose we measure the mass of a particle to be  $m$  when it is at rest relative to us: We often call  $m$  the **rest mass**. We'll use the term "material particle" for a particle that has a nonzero rest mass. When such a particle has a velocity  $\vec{v}$ , its **relativistic momentum**  $\vec{p}$  is

$$\text{Relativistic momentum } \vec{p} = \frac{\text{Rest mass of particle } m \times \text{Velocity of particle } \vec{v}}{\sqrt{1 - \text{Speed of particle}^2 / \text{Speed of light in vacuum}^2}} \quad (37.27)$$

When the particle's speed  $v$  is much less than  $c$ , this is approximately equal to the Newtonian expression  $\vec{p} = m\vec{v}$ , but in general the momentum is greater in magnitude than  $mv$  (Fig. 37.20). In fact, as  $v$  approaches  $c$ , the momentum approaches infinity.

#### Relativity, Newton's Second Law, and Relativistic Mass

What about the relativistic generalization of Newton's second law? In Newtonian mechanics the most general form of the second law is

$$\vec{F} = \frac{d\vec{p}}{dt} \quad (37.28)$$

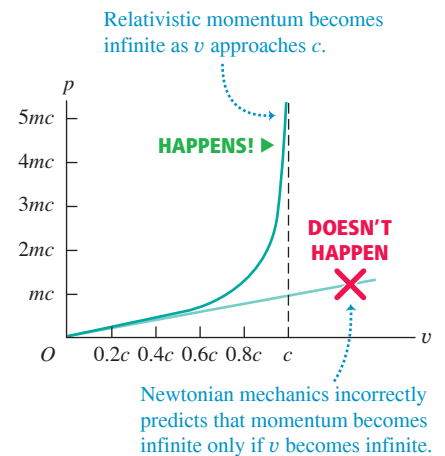
That is, the net force  $\vec{F}$  on a material particle equals the time rate of change of its momentum. Experiments show that this result is still valid in relativistic mechanics, provided that we use the relativistic momentum given by Eq. (37.27). That is, the relativistically correct generalization of Newton's second law is

$$\vec{F} = \frac{d}{dt} \frac{m\vec{v}}{\sqrt{1 - v^2/c^2}} \quad (37.29)$$

Because momentum is no longer directly proportional to velocity, the rate of change of momentum is no longer directly proportional to the acceleration. As a result, *constant force does not cause constant acceleration*. For example, when the net force and the velocity are both along the  $x$ -axis, Eq. (37.29) gives

$$F = \frac{m}{(1 - v^2/c^2)^{3/2}} a \quad (\vec{F} \text{ and } \vec{v} \text{ along the same line}) \quad (37.30)$$

Figure 37.20 Graph of the magnitude of the momentum of a particle of rest mass  $m$  as a function of speed  $v$ . Also shown is the Newtonian prediction, which gives correct results only at speeds much less than  $c$ .



where  $a$  is the acceleration, also along the  $x$ -axis. Solving Eq. (37.30) for the acceleration  $a$  gives

$$a = \frac{F}{m} \left( 1 - \frac{v^2}{c^2} \right)^{3/2}$$

We see that as a particle's speed increases, the acceleration caused by a given force continuously *decreases*. As the speed approaches  $c$ , the acceleration approaches zero, no matter how great a force is applied. Thus it is impossible to accelerate a particle with nonzero rest mass to a speed equal to or greater than  $c$ . We again see that the speed of light in vacuum represents an ultimate speed limit.

Equation (37.27) for relativistic momentum is sometimes interpreted to mean that a rapidly moving material particle undergoes an increase in mass. If the mass at zero velocity (the rest mass) is denoted by  $m$ , then the “relativistic mass”  $m_{\text{rel}}$  is

$$m_{\text{rel}} = \frac{m}{\sqrt{1 - v^2/c^2}}$$

Indeed, when we consider the motion of a system of particles (such as rapidly moving ideal-gas molecules in a stationary container), the total rest mass of the system is the sum of the relativistic masses of the particles, not the sum of their rest masses.

However, if blindly applied, the concept of relativistic mass has its pitfalls. As Eq. (37.29) shows, the relativistic generalization of Newton's second law is *not*  $\vec{F} = m_{\text{rel}}\vec{a}$ , and we'll show in Section 37.8 that the relativistic kinetic energy of a particle is *not*  $K = \frac{1}{2}m_{\text{rel}}v^2$ . The use of relativistic mass has its supporters and detractors, some quite strong in their opinions. We'll mostly deal with individual particles, so we'll sidestep the controversy and use Eq. (37.27) as the generalized definition of momentum with  $m$  as a constant for each particle, independent of its state of motion.

The quantity  $1/\sqrt{1 - v^2/c^2}$  in Eqs. (37.27) and (37.29) is the Lorentz factor  $\gamma$  from Eq. (37.7) (Section 37.3), but with a difference: We've replaced  $u$ , the relative speed of two coordinate systems, by  $v$ , the speed of a particle in a particular coordinate system—that is, the speed of the particle's *rest frame* with respect to that system. In terms of  $\gamma$ , Eqs. (37.27) and (37.30) become

$$\text{Relativistic momentum } \vec{p} = \frac{\text{Rest mass of particle } m}{\text{Lorentz factor relating rest frame of particle and frame of observer}} \times \text{Velocity of particle } \vec{v} \quad (37.31)$$

$$F = \gamma^3 m a \quad (\vec{F} \text{ and } \vec{v} \text{ along the same line}) \quad (37.32)$$

In linear accelerators (used in medicine as well as nuclear and elementary-particle physics; see Fig. 37.11) the net force  $\vec{F}$  and the velocity  $\vec{v}$  of the accelerated particle are along the same straight line. But for much of the path in most *circular* accelerators the particle moves in uniform circular motion at constant speed  $v$ . Then the net force and velocity are perpendicular, so the force can do no work on the particle and the kinetic energy and speed remain constant. Thus the denominator in Eq. (37.29) is constant, and we obtain

$$F = \frac{m}{(1 - v^2/c^2)^{1/2}} a = \gamma m a \quad (\vec{F} \text{ and } \vec{v} \text{ perpendicular}) \quad (37.33)$$

Recall from Section 3.4 that if the particle moves in a circle, the net force and acceleration are directed inward along the radius  $r$ , and  $a = v^2/r$ .

What about the general case in which  $\vec{F}$  and  $\vec{v}$  are neither along the same line nor perpendicular? Then we can resolve the net force  $\vec{F}$  at any instant into components parallel to and perpendicular to  $\vec{v}$ . The resulting acceleration will have corresponding components obtained from Eqs. (37.32) and (37.33). Because of the different  $\gamma^3$  and  $\gamma$  factors, the acceleration components will not be proportional to the net force components. That is, *unless the net force on a relativistic particle is either along the same line as the particle's velocity or perpendicular to it, the net force and acceleration vectors are not parallel.*

**EXAMPLE 37.9 Relativistic dynamics of an electron****WITH VARIATION PROBLEMS**

An electron (rest mass  $9.11 \times 10^{-31}$  kg, charge  $-1.60 \times 10^{-19}$  C) is moving opposite to an electric field of magnitude  $E = 5.00 \times 10^5$  N/C. All other forces are negligible in comparison to the electric-field force. (a) Find the magnitudes of momentum and of acceleration at the instants when  $v = 0.010c$ ,  $0.90c$ , and  $0.99c$ . (b) Find the corresponding accelerations if a net force of the same magnitude is perpendicular to the velocity.

**IDENTIFY and SET UP** In addition to the expressions from this section for relativistic momentum and acceleration, we need the relationship between electric force and electric field from Chapter 21. In part (a) we use Eq. (37.31) to determine the magnitude of momentum; the force acts along the same line as the velocity, so we use Eq. (37.32) to determine the magnitude of acceleration. In part (b) the force is perpendicular to the velocity, so we use Eq. (37.33) rather than Eq. (37.32).

**EXECUTE** (a) For  $v = 0.010c$ ,  $0.90c$ , and  $0.99c$  we have  $\gamma = \sqrt{1 - v^2/c^2} = 1.00$ ,  $2.29$ , and  $7.09$ , respectively. The values of the momentum magnitude  $p = \gamma mv$  are

$$\begin{aligned} p_1 &= (1.00)(9.11 \times 10^{-31} \text{ kg})(0.010)(3.00 \times 10^8 \text{ m/s}) \\ &= 2.7 \times 10^{-24} \text{ kg} \cdot \text{m/s at } v_1 = 0.010c \\ p_2 &= (2.29)(9.11 \times 10^{-31} \text{ kg})(0.90)(3.00 \times 10^8 \text{ m/s}) \\ &= 5.6 \times 10^{-22} \text{ kg} \cdot \text{m/s at } v_2 = 0.90c \\ p_3 &= (7.09)(9.11 \times 10^{-31} \text{ kg})(0.99)(3.00 \times 10^8 \text{ m/s}) \\ &= 1.9 \times 10^{-21} \text{ kg} \cdot \text{m/s at } v_3 = 0.99c \end{aligned}$$

From Eq. (21.4), the magnitude of the force on the electron is

$$F = |q|E = (1.60 \times 10^{-19} \text{ C})(5.00 \times 10^5 \text{ N/C}) = 8.00 \times 10^{-14} \text{ N}$$

From Eq. (37.32),  $a = F/\gamma^3 m$ . For  $v = 0.010c$  and  $\gamma = 1.00$ ,

$$a_1 = \frac{8.00 \times 10^{-14} \text{ N}}{(1.00)^3(9.11 \times 10^{-31} \text{ kg})} = 8.8 \times 10^{16} \text{ m/s}^2$$

The accelerations at the two higher speeds are smaller than the nonrelativistic value by factors of  $\gamma^3 = 12.0$  and  $356$ , respectively:

$$a_2 = 7.3 \times 10^{15} \text{ m/s}^2 \quad a_3 = 2.5 \times 10^{14} \text{ m/s}^2$$

(b) From Eq. (37.33),  $a = F/\gamma m$  if  $\vec{F}$  and  $\vec{v}$  are perpendicular. When  $v = 0.010c$  and  $\gamma = 1.00$ ,

$$a_1 = \frac{8.00 \times 10^{-14} \text{ N}}{(1.00)(9.11 \times 10^{-31} \text{ kg})} = 8.8 \times 10^{16} \text{ m/s}^2$$

Now the accelerations at the two higher speeds are smaller by factors of  $\gamma = 2.29$  and  $7.09$ , respectively:

$$a_2 = 3.8 \times 10^{16} \text{ m/s}^2 \quad a_3 = 1.2 \times 10^{16} \text{ m/s}^2$$

These accelerations are larger than the corresponding ones in part (a) by factors of  $\gamma^2$ .

**EVALUATE** Our results in part (a) show that at higher speeds, the relativistic values of momentum differ more and more from the nonrelativistic values calculated from  $p = mv$ . The momentum at  $0.99c$  is more than three times as great as at  $0.90c$  because of the increase in the factor  $\gamma$ . Our results also show that the acceleration drops off very quickly as  $v$  approaches  $c$ .

**KEYCONCEPT** The expression for momentum [Eq. (37.31)] and the relationship between net force and acceleration [Newton's second law, Eqs. (37.32) and (37.33)] are both modified in the special theory of relativity. As the speed of a material particle approaches  $c$ , its momentum approaches infinity and the acceleration caused by a net force approaches zero.

**TEST YOUR UNDERSTANDING OF SECTION 37.7** According to relativistic mechanics, when you double the speed of a particle, the magnitude of its momentum increases by (i) a factor of 2; (ii) a factor greater than 2; (iii) a factor between 1 and 2 that depends on the mass of the particle.

**ANSWER**

(ii) Equation (37.27) tells us that the magnitude of momentum of a particle with mass  $m$  and speed  $v$  is  $p = mv/\sqrt{1 - v^2/c^2}$ . If  $v$  increases by a factor of 2, the numerator  $mv$  increases by a factor of 2 and the denominator  $\sqrt{1 - v^2/c^2}$  decreases. Hence  $p$  increases by a factor greater than 2. (Note that in order to double the speed, the initial speed must be less than  $c/2$ . That's because the speed of light is the ultimate speed limit.)

**37.8 RELATIVISTIC WORK AND ENERGY**

When we developed the relationship between work and kinetic energy in Chapter 6, we used Newton's laws of motion. Since we have generalized these laws according to the principle of relativity, we need a corresponding generalization of the equation for kinetic energy.

**Relativistic Kinetic Energy**

We use the work–energy theorem, beginning with the definition of work. When the net force and displacement are in the same direction, the work done by that force is  $W = \int F dx$ . We substitute the expression for  $F$  from Eq. (37.30), the relativistic version of Newton's second law for straight-line motion. In moving a particle of rest mass  $m$  from point  $x_1$  to point  $x_2$ ,

$$W = \int_{x_1}^{x_2} F dx = \int_{x_1}^{x_2} \frac{ma dx}{(1 - v_x^2/c^2)^{3/2}} \quad (37.34)$$



We've replaced  $v$  in Eq. (37.34) with  $v_x$  because the motion is along the  $x$ -axis only. So  $v_x$  is the varying  $x$ -component of the particle's velocity as the net force accelerates it. To derive the generalized expression for kinetic energy  $K$ , first remember that the kinetic energy of a particle equals the net work done on it in moving it from rest to speed  $v$ :  $K = W$ . Thus we let the speeds be zero at point  $x_1$  and  $v$  at point  $x_2$ . It's useful to convert Eq. (37.34) to an integral on  $v_x$ . To do this, note that  $dx$  and  $dv_x$  are the infinitesimal changes in  $x$  and  $v_x$ , respectively, in the time interval  $dt$ . Because  $v_x = dx/dt$  and  $a = dv_x/dt$ , we can rewrite  $a dx$  in Eq. (37.34) as

$$a dx = \frac{dv_x}{dt} dx = dx \frac{dv_x}{dt} = \frac{dx}{dt} dv_x = v_x dv_x$$

Making these substitutions gives us

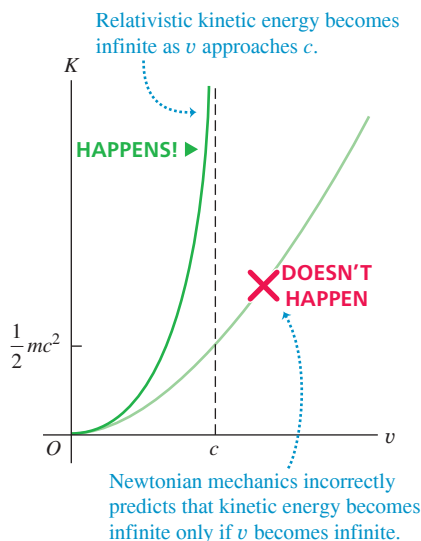
$$K = W = \int_0^v \frac{mv_x dv_x}{(1 - v_x^2/c^2)^{3/2}} \quad (37.35)$$

We can evaluate this integral by a simple change of variable; the final result is

$$\text{Relativistic kinetic energy } K = \frac{\text{Rest mass of particle } mc^2}{\sqrt{1 - \frac{\text{Speed of particle } v^2}{\text{Speed of light in vacuum } c^2}}} - mc^2 = (\gamma - 1)mc^2 \quad (37.36)$$

Lorentz factor relating rest frame of particle and frame of observer

Figure 37.21 Graph of the kinetic energy of a particle of rest mass  $m$  as a function of speed  $v$ . Also shown is the Newtonian prediction, which gives correct results only at speeds much less than  $c$ .



As  $v$  approaches  $c$ , the kinetic energy approaches infinity. If Eq. (37.36) is correct, it must also approach the Newtonian expression  $K = \frac{1}{2}mv^2$  when  $v$  is much smaller than  $c$  (Fig. 37.21). To verify this, we expand the radical, using the binomial theorem in the form

$$(1 + x)^n = 1 + nx + n(n-1)x^2/2 + \cdots$$

In our case,  $n = -\frac{1}{2}$  and  $x = -v^2/c^2$ , and we get

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} = 1 + \frac{1}{2}\frac{v^2}{c^2} + \frac{3}{8}\frac{v^4}{c^4} + \cdots$$

Combining this with  $K = (\gamma - 1)mc^2$ , we find

$$K = \left(1 + \frac{1}{2}\frac{v^2}{c^2} + \frac{3}{8}\frac{v^4}{c^4} + \cdots - 1\right)mc^2 = \frac{1}{2}mv^2 + \frac{3}{8}\frac{mv^4}{c^2} + \cdots \quad (37.37)$$

When  $v$  is much smaller than  $c$ , all the terms in the series in Eq. (37.37) except the first are negligibly small, and we obtain the Newtonian expression  $\frac{1}{2}mv^2$ .

## Rest Energy and $E = mc^2$

Equation (37.36) for the kinetic energy of a moving particle includes a term  $mc^2/\sqrt{1 - v^2/c^2}$  that depends on the motion and a second energy term  $mc^2$  that is independent of the motion. It seems that the kinetic energy of a particle is the difference between some **total energy**  $E$  and an energy  $mc^2$  that it has even when it is at rest. Thus we can rewrite Eq. (37.36) as

$$\text{Total energy of a particle } E = K + mc^2 = \frac{\text{Kinetic energy } K + \text{Rest energy } mc^2}{\sqrt{1 - \frac{\text{Speed of particle } v^2}{\text{Speed of light in vacuum } c^2}}} = \gamma mc^2 \quad (37.38)$$

Lorentz factor relating rest frame of particle and frame of observer

For a particle at rest ( $K = 0$ ), we see that  $E = mc^2$ . The energy  $mc^2$  associated with rest mass  $m$  rather than motion is called the **rest energy** of the particle.

There is abundant experimental evidence that rest energy really does exist. The simplest example is the decay of a neutral *pion*. This is an unstable subatomic particle of rest mass  $m_\pi$ ; when it decays, it disappears and electromagnetic radiation appears. If a neutral pion has no kinetic energy before its decay, the total energy of the radiation after its decay is found to equal exactly  $m_\pi c^2$ . In many other fundamental particle transformations the sum of the rest masses of the particles changes. In every case there is a corresponding energy change, consistent with the assumption of a rest energy  $mc^2$  associated with a rest mass  $m$ .

Historically, the principles of conservation of mass and of energy developed quite independently. The theory of relativity shows that they are actually two special cases of a single broader conservation principle, the *principle of conservation of mass and energy*. In some physical phenomena, neither the sum of the rest masses of the particles nor the total energy other than rest energy is separately conserved, but there is a more general conservation principle: In an isolated system, when the sum of the rest masses changes, there is always a change in  $1/c^2$  times the total energy other than the rest energy. This change is equal in magnitude but opposite in sign to the change in the sum of the rest masses.

This more general mass-energy conservation law is the fundamental principle involved in the generation of nuclear power. When a uranium nucleus undergoes fission in a nuclear reactor, the sum of the rest masses of the resulting fragments is *less than* the rest mass of the parent nucleus. An amount of energy is released that equals the mass decrease multiplied by  $c^2$ . Most of this energy can be used to produce steam to operate turbines for electric power generators.

We can also relate the total energy  $E$  of a particle (kinetic energy plus rest energy) directly to its momentum by combining Eq. (37.27) for relativistic momentum and Eq. (37.38) for total energy to eliminate the particle's velocity. The simplest procedure is to rewrite these equations in the following forms:

$$\left(\frac{E}{mc^2}\right)^2 = \frac{1}{1 - v^2/c^2} \quad \text{and} \quad \left(\frac{p}{mc}\right)^2 = \frac{v^2/c^2}{1 - v^2/c^2}$$

Subtracting the second of these from the first and rearranging, we find

$$\begin{array}{l} \text{Total energy,} \\ \text{rest energy,} \\ \text{and momentum:} \end{array} \quad \begin{array}{l} \text{Total} \\ \text{energy} \end{array} \quad \begin{array}{l} \text{Rest} \\ \text{energy} \end{array} \quad \begin{array}{l} \text{Magnitude of} \\ \text{momentum} \end{array} \quad \begin{array}{l} E^2 = (mc^2)^2 + (pc)^2 \\ \text{Rest mass} \quad \text{Speed of light in vacuum} \end{array} \quad (37.39)$$

Again we see that for a particle at rest ( $p = 0$ ),  $E = mc^2$ .

Equation (37.39) also suggests that a particle may have energy and momentum even when it has no rest mass. In such a case,  $m = 0$  and

$$E = pc \quad (\text{zero rest mass}) \quad (37.40)$$

In fact, zero rest mass particles do exist. Such particles always travel at the speed of light in vacuum. One example is the *photon*, the quantum of electromagnetic radiation (to be discussed in Chapter 38). Photons are emitted and absorbed during changes of state of an atomic or nuclear system when the energy and momentum of the system change.

### EXAMPLE 37.10 Energetic electrons

### WITH VARIATION PROBLEMS

(a) Find the rest energy of an electron ( $m = 9.109 \times 10^{-31}$  kg,  $q = -e = -1.602 \times 10^{-19}$  C) in joules and in electron volts. (b) Find the speed of an electron that has been accelerated by an electric field, from rest, through a potential increase of 20.0 kV or of 5.00 MV (typical of a high-voltage x-ray machine).

**IDENTIFY and SET UP** This problem uses the ideas of rest energy, relativistic kinetic energy, and (from Chapter 23) electric potential energy. We use  $E = mc^2$  to find the rest energy and Eqs. (37.7) and (37.38) to find the speed that gives the stated total energy.

**EXECUTE** (a) The rest energy is

$$mc^2 = (9.109 \times 10^{-31} \text{ kg})(2.998 \times 10^8 \text{ m/s})^2 = 8.187 \times 10^{-14} \text{ J}$$

From the definition of the electron volt in Section 23.2,  $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ . Using this, we find

$$\begin{aligned} mc^2 &= (8.187 \times 10^{-14} \text{ J}) \frac{1 \text{ eV}}{1.602 \times 10^{-19} \text{ J}} \\ &= 5.11 \times 10^5 \text{ eV} = 0.511 \text{ MeV} \end{aligned}$$

*Continued*

**APPLICATION Monitoring Mass-Energy Conversion** Although the control room of a nuclear power plant is very complex, the physical principle on which such a plant operates is a simple one: Part of the rest energy of atomic nuclei is converted to thermal energy, which in turn is used to produce steam to drive electric generators.



(b) In calculations such as this, it is often convenient to work with the quantity  $\gamma = 1/\sqrt{1 - v^2/c^2}$  from Eq. (37.38). Solving this for  $v$ , we find

$$v = c \sqrt{1 - (1/\gamma)^2}$$

The total energy  $E$  of the accelerated electron is the sum of its rest energy  $mc^2$  and the kinetic energy  $eV_{ba}$  that it gains from the work done on it by the electric field in moving from point  $a$  to point  $b$ :

$$E = \gamma mc^2 = mc^2 + eV_{ba} \quad \text{or} \quad \gamma = 1 + \frac{eV_{ba}}{mc^2}$$

An electron accelerated through a potential increase of  $V_{ba} = 20.0$  kV gains 20.0 keV of energy, so for this electron

$$\gamma = 1 + \frac{20.0 \times 10^3 \text{ eV}}{0.511 \times 10^6 \text{ eV}} = 1.039$$

and

$$v = c \sqrt{1 - (1/1.039)^2} = 0.272c = 8.15 \times 10^7 \text{ m/s}$$

Repeating the calculation for  $V_{ba} = 5.00$  MV, we find  $eV_{ba}/mc^2 = 9.78$ ,  $\gamma = 10.78$ , and  $v = 0.996c$ .

**EVALUATE** With  $V_{ba} = 20.0$  kV, the added kinetic energy of 20.0 keV is less than 4% of the rest energy of 0.511 MeV, and the final speed is about one-fourth the speed of light. With  $V_{ba} = 5.00$  MV, the added kinetic energy of 5.00 MeV is much greater than the rest energy and the speed is close to  $c$ .

**CAUTION Three electron energies** All electrons have *rest* energy 0.511 MeV. An electron accelerated from rest through a 5.00 MeV potential increase has *kinetic* energy 5.00 MeV (we call it a “5.00 MeV electron”) and *total* energy 5.51 MeV. Be careful to distinguish these energies from one another. I

**KEYCONCEPT** The expression for kinetic energy [Eq. (37.36)] is modified in the special theory of relativity. As the speed of a material particle approaches  $c$ , its kinetic energy approaches infinity. The total energy of a particle of rest mass  $m$  is the sum of its kinetic energy and its rest energy  $mc^2$ .

### EXAMPLE 37.11 A relativistic collision

### WITH VARIATION PROBLEMS

Two protons (each with mass  $m_p = 1.67 \times 10^{-27}$  kg) are initially moving with equal speeds in opposite directions. They continue to exist after a head-on collision that also produces a neutral pion of mass  $m_\pi = 2.40 \times 10^{-28}$  kg (**Fig. 37.22**). If all three particles are at rest after the collision, find the initial speed of the protons. Energy is conserved in the collision.

**IDENTIFY and SET UP** Relativistic total energy is conserved in the collision, so we can equate the (unknown) total energy of the two protons before

the collision to the combined rest energies of the two protons and the pion after the collision. We then use Eq. (37.38) to find the speed of each proton.

**EXECUTE** The total energy of each proton before the collision is  $\gamma m_p c^2$ . By conservation of energy,

$$2(\gamma m_p c^2) = 2(m_p c^2) + m_\pi c^2$$

$$\gamma = 1 + \frac{m_\pi}{2m_p} = 1 + \frac{2.40 \times 10^{-28} \text{ kg}}{2(1.67 \times 10^{-27} \text{ kg})} = 1.072$$

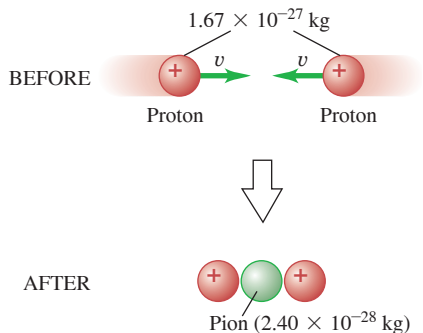
From Eq. (37.38), the initial proton speed is

$$v = c \sqrt{1 - (1/\gamma)^2} = 0.360c$$

**EVALUATE** The proton rest energy is 938 MeV, so the initial kinetic energy of each proton is  $(\gamma - 1)m_p c^2 = 0.072m_p c^2 = (0.072)(938 \text{ MeV}) = 67.5 \text{ MeV}$ . You can verify that the rest energy  $m_\pi c^2$  of the pion is twice this, or 135 MeV. All the kinetic energy “lost” in this completely inelastic collision is transformed into the rest energy of the pion.

**KEYCONCEPT** In a relativistic collision, momentum and total energy are both conserved. Particles can be created or destroyed in such collisions by converting kinetic energy to rest energy, or vice versa.

Figure 37.22 In this collision the kinetic energy of two protons is transformed into the rest energy of a new particle, a pion.



**TEST YOUR UNDERSTANDING OF SECTION 37.8** A proton is accelerated from rest by a constant force that always points in the direction of the particle's motion. Compared to the amount of kinetic energy that the proton gains during the first meter of its travel, how much kinetic energy does the proton gain during one meter of travel while it is moving at 99% of the speed of light? (i) The same amount; (ii) a greater amount; (iii) a smaller amount.

#### ANSWER

(i) As the proton moves a distance  $s$ , the constant force of magnitude  $F$  does work  $W = Fs$  and increases the kinetic energy by an amount  $\Delta K = W = Fs$ . This is true no matter what the speed of the proton before moving this distance. Thus the constant force increases the proton's kinetic energy by the same amount during the first meter of travel as during any subsequent meter of travel. (As the proton approaches the ultimate speed limit of  $c$ , the increase in the proton's speed is less and less with each subsequent meter of travel. That's not what the question asks, however.) I

## 37.9 NEWTONIAN MECHANICS AND RELATIVITY

The sweeping changes required by the principle of relativity go to the very roots of Newtonian mechanics, including the concepts of length and time, the equations of motion, and the conservation principles. Thus it may appear that we have destroyed the foundations on which Newtonian mechanics is built. In one sense this is true, yet the Newtonian formulation is still accurate whenever speeds are small in comparison with the speed of light in vacuum. In such cases, time dilation, length contraction, and the modifications of the laws of motion are so small that they are unobservable. In fact, every one of the principles of Newtonian mechanics survives as a special case of the more general relativistic formulation.

The laws of Newtonian mechanics are not *wrong*; they are *incomplete*. They are a limiting case of relativistic mechanics. They are *approximately* correct when all speeds are small in comparison to  $c$ , and they become exactly correct in the limit when all speeds approach zero. Thus relativity does not completely destroy the laws of Newtonian mechanics but *generalizes* them. This is a common pattern in the development of physical theory. Whenever a new theory is in partial conflict with an older, established theory, the new must yield the same predictions as the old in areas in which the old theory is supported by experimental evidence. Every new physical theory must pass this test, called the **correspondence principle**.

### The General Theory of Relativity

At this point we may ask whether the special theory of relativity gives the final word on mechanics or whether *further* generalizations are possible or necessary. For example, inertial frames have occupied a privileged position in our discussion. Can the principle of relativity be extended to noninertial frames as well?

Here's an example that illustrates some implications of this question. A student decides to go over Niagara Falls while enclosed in a large wooden box. During her free fall she doesn't fall to the floor of the box because both she and the box are in free fall with a downward acceleration of  $9.8 \text{ m/s}^2$ . But an alternative interpretation, from her point of view, is that she doesn't fall to the floor because her gravitational interaction with the earth has suddenly been turned off. As long as she remains in the box and it remains in free fall, she cannot tell whether she is indeed in free fall or whether the gravitational interaction has vanished.

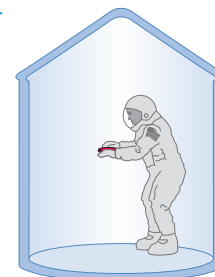
A similar problem occurs in a space station in orbit around the earth. Objects in the space station *seem* to be weightless, but without looking outside the station there is no way to determine whether gravity has been turned off or whether the station and all its contents are accelerating toward the center of the earth. **Figure 37.23** makes a similar point for a spaceship that is not in free fall but may be accelerating relative to an inertial frame or be at rest on the earth's surface.

These considerations form the basis of Einstein's **general theory of relativity**. If we cannot distinguish experimentally between a uniform gravitational field at a particular location and a uniformly accelerated reference frame, then there cannot be any real distinction between the two. Pursuing this concept, we may try to represent *any* gravitational field in terms of special characteristics of the coordinate system. This turns out to require even more sweeping revisions of our spacetime concepts than did the special theory of relativity. In the general theory of relativity the geometric properties of space are affected by the presence of matter (**Fig. 37.24**, next page).

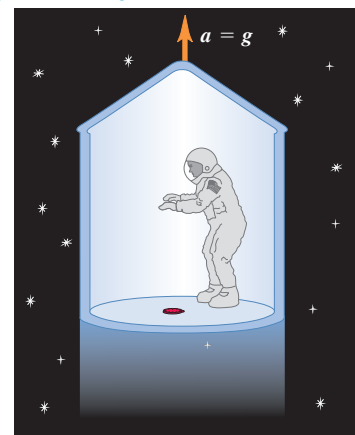
The general theory of relativity has passed several experimental tests, including three proposed by Einstein. One test has to do with understanding the rotation of the axes of the planet Mercury's elliptical orbit, called the *precession of the perihelion*. (The perihelion is the point of closest approach to the sun.) A second test concerns the apparent bending of light rays from distant stars when they pass near the sun. The third test is the *gravitational red shift*, the increase in wavelength of light proceeding outward from a massive source. Some details of the general theory are more difficult to test, but this theory has played a central role in investigations of the formation and evolution of stars, black holes, and studies of the evolution of the universe.

Figure 37.23 Without information from outside the spaceship, the astronaut cannot distinguish situation (b) from situation (c).

(a) An astronaut is about to drop her watch in a spaceship.



(b) In gravity-free space, the floor accelerates upward at  $a = g$  and hits the watch.



(c) On the earth's surface, the watch accelerates downward at  $a = g$  and hits the floor.

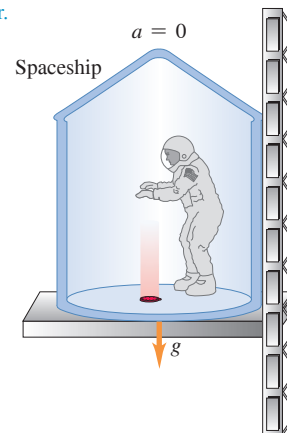




Figure 37.24 A two-dimensional representation of curved space. We imagine the space (a plane) as being distorted as shown by a massive object (the sun). Light from a distant star (solid line) follows the distorted surface on its way to the earth. The dashed line shows the direction from which the light *appears* to be coming. The effect is greatly exaggerated; for the sun, the maximum deviation is only  $0.00048^\circ$ .

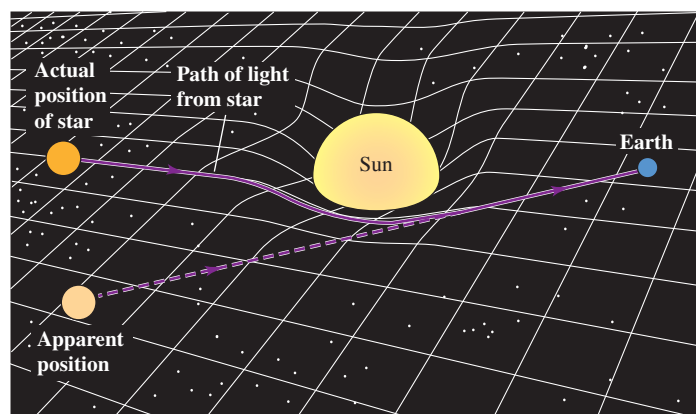


Figure 37.25 A GPS receiver uses radio signals from the orbiting GPS satellites to determine its position. To account for the effects of relativity, the receiver must be tuned to a slightly higher frequency (10.23 MHz) than the frequency emitted by the satellites (10.22999999543 MHz).

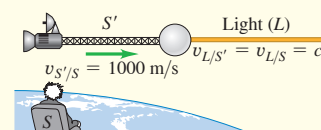


The general theory of relativity may seem to be an exotic bit of knowledge with little practical application. In fact, this theory plays an essential role in the Global Positioning System (GPS), which makes it possible to determine your position on the earth's surface to within a few meters using a handheld receiver (Fig. 37.25). The heart of the GPS system is a collection of more than two dozen satellites in very precise orbits. Each satellite emits carefully timed radio signals, and a GPS receiver simultaneously detects the signals from several satellites. The receiver then calculates the time delay between when each signal was emitted and when it was received, and uses this information to calculate the receiver's position.

To ensure the proper timing of the signals, it's necessary to include corrections due to the special theory of relativity (because the satellites are moving relative to the receiver on earth) as well as the general theory (because the satellites are higher in the earth's gravitational field than the receiver). The corrections due to relativity are small—less than one part in  $10^9$ —but are crucial to the superb precision of the GPS system.

## CHAPTER 37 SUMMARY

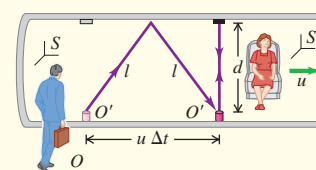
**Invariance of physical laws, simultaneity:** All of the fundamental laws of physics have the same form in all inertial frames of reference. The speed of light in vacuum is the same in all inertial frames and is independent of the motion of the source. Simultaneity is not an absolute concept; events that are simultaneous in one frame are not necessarily simultaneous in a second frame moving relative to the first.



**Time dilation:** If two events occur at the same space point in a particular frame of reference, the time interval  $\Delta t_0$  between the events as measured in that frame is called a proper time interval. If this frame moves with constant velocity  $u$  relative to a second frame, the time interval  $\Delta t$  between the events as observed in the second frame is longer than  $\Delta t_0$ . (See Examples 37.1–37.3.)

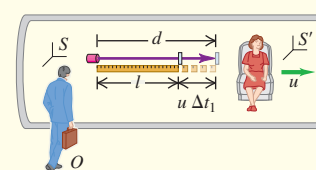
$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}} = \gamma \Delta t_0 \quad (37.6), (37.8)$$

$$\gamma = \frac{1}{\sqrt{1 - u^2/c^2}} \quad (37.7)$$



**Length contraction:** If two points are at rest in a particular frame of reference, the distance  $l_0$  between the points as measured in that frame is called a proper length. If this frame moves with constant velocity  $u$  relative to a second frame and the distances are measured parallel to the motion, the distance  $l$  between the points as measured in the second frame is shorter than  $l_0$ . (See Examples 37.4 and 37.5.)

$$l = l_0 \sqrt{1 - \frac{u^2}{c^2}} = \frac{l_0}{\gamma} \quad (37.16)$$

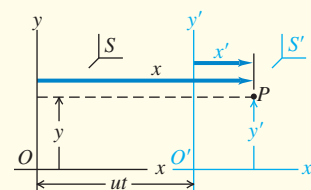


**The Lorentz transformations:** The Lorentz coordinate transformations relate the coordinates and time of an event in an inertial frame  $S$  to the coordinates and time of the same event as observed in a second inertial frame  $S'$  moving at velocity  $u$  relative to the first. For one-dimensional motion, a particle's velocities  $v_x$  in  $S$  and  $v'_x$  in  $S'$  are related by the Lorentz velocity transformation. (See Examples 37.6 and 37.7.)

$$\begin{aligned}x' &= \frac{x - ut}{\sqrt{1 - u^2/c^2}} = \gamma(x - ut) \\y' &= y \quad z' = z\end{aligned}\quad (37.21)$$

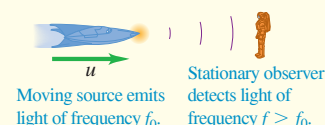
$$\begin{aligned}t' &= \frac{t - ux/c^2}{\sqrt{1 - u^2/c^2}} = \gamma(t - ux/c^2) \\v'_x &= \frac{v_x - u}{1 - uv_x/c^2}\end{aligned}\quad (37.22)$$

$$v_x = \frac{v'_x + u}{1 + uv'_x/c^2}\quad (37.23)$$



**The Doppler effect for electromagnetic waves:** The Doppler effect is the frequency shift in light from a source due to the relative motion of source and observer. For a source moving toward the observer with speed  $u$ , Eq. (37.25) gives the received frequency  $f$  in terms of the emitted frequency  $f_0$ . (See Example 37.8.)

$$f = \sqrt{\frac{c + u}{c - u}} f_0\quad (37.25)$$



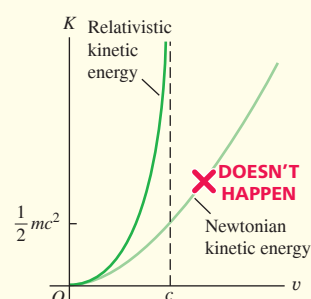
**Relativistic momentum and energy:** For a particle of rest mass  $m$  moving with velocity  $\vec{v}$ , the relativistic momentum  $\vec{p}$  is given by Eq. (37.27) or (37.31) and the relativistic kinetic energy  $K$  is given by Eq. (37.36). The total energy  $E$  is the sum of the kinetic energy and the rest energy  $mc^2$ . The total energy can also be expressed in terms of the magnitude of momentum  $p$  and rest mass  $m$ . (See Examples 37.9–37.11.)

$$\vec{p} = \frac{m\vec{v}}{\sqrt{1 - v^2/c^2}} = \gamma m\vec{v}\quad (37.27), (37.31)$$

$$K = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2 = (\gamma - 1)mc^2\quad (37.36)$$

$$\begin{aligned}E &= K + mc^2 = \frac{mc^2}{\sqrt{1 - v^2/c^2}} \\&= \gamma mc^2\end{aligned}\quad (37.38)$$

$$E^2 = (mc^2)^2 + (pc)^2\quad (37.39)$$



Chapter 37 Media Assets



## GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

### KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 37.1, 37.2, and 37.3** (Section 37.3) and **EXAMPLES 37.4 and 37.5** (Section 37.4) before attempting these problems.

**VP37.5.1** A positive K meson ( $K^+$ ) is an unstable subatomic particle that decays into other particles. At rest, its mean lifetime is  $1.23 \times 10^{-8}$  s. (a) What do you measure the mean lifetime of a  $K^+$  to be if it is moving at  $0.800c$  relative to you? (b) As measured from the reference frame of this  $K^+$ , what is the mean lifetime of a second  $K^+$  that is at rest relative to you?

**VP37.5.2** Alia flies in her spacecraft at  $0.600c$  relative to the planet Arrakis. As she passes Paul, at rest on Arrakis, they both start timers. (a) According to Alia's timer, 20.0 s elapses from when Paul starts his timer to when he stops his timer. What does Paul's timer read when he stops it? (b) Alia stops her timer when it reads 24.0 s. As measured by Paul, how much time elapses from when Alia starts her timer to when she stops it?

**VP37.5.3** At one point in its orbit the earth is  $1.50 \times 10^{11}$  m from the sun. A spacecraft flies along a line from the earth to the sun at  $0.950c$ . (a) As measured by you on the earth, how much time does it take the spacecraft to travel the distance from the earth to the sun? (b) As measured by an astronaut on the spacecraft, what is the distance from the earth to the sun, and how much time does it take her to travel that distance?

**VP37.5.4** A spacecraft flies past the earth in the direction toward the moon, a distance of  $3.84 \times 10^5$  km. As measured by your clock on the earth, it takes 6.00 s for the clock on the spacecraft to tick off 2.00 s. You measure the length of the spacecraft in the direction of its motion to be 24.0 m. As measured by an astronaut at rest on the spacecraft, what are (a) the distance from the earth to the moon and (b) the length of the spacecraft?

Be sure to review **EXAMPLES 37.6 and 37.7** (Section 37.5) before attempting these problems.

**VP37.7.1** Gamora flies her spacecraft in the  $+x$ -direction past the planet Xandar at  $0.750c$  while her sister Nebula is at rest on the surface of the planet. As Gamora passes Nebula, both sisters set their clocks to zero. Each chooses the zero of the  $x$ -axis to be at her position. (a) Nebula sets off fireworks next to her 2.50 s after Gamora passes her. What are the coordinates of this event as measured by Gamora? (b) Gamora detects an explosion in space that occurs  $4.00 \times 10^8$  m in front of her 2.50 s after she passes Nebula. What are the coordinates of this event as measured by Nebula?

**VP37.7.2** Two events occur at different  $x$ -coordinates but at the same time as measured by Kamala. Doreen, who is moving at  $0.800c$  relative to Kamala in the  $+x$ -direction, measures that one event takes place 0.600 s before the other. What is the distance between the two events as measured by (a) Kamala and (b) Doreen?

**VP37.7.3** The spaceship *Nostramo* flies away from the earth in the  $+x$ -direction at  $0.800c$ . It fires a probe in the  $-x$ -direction at  $0.600c$  relative to *Nostramo*. A scoutship is sent from the earth to recover the probe; the scoutship travels in the  $+x$ -direction at  $0.700c$  relative to the earth. Find the velocity (in terms of  $c$ ) of (a) the probe relative to the earth and (b) the scoutship relative to *Nostramo*.

**VP37.7.4** As measured from the earth, the spaceship *Macross* is flying directly toward the earth at  $0.750c$  while on the other side of the earth the spaceship *Yamato* is flying at  $0.650c$  along the same line as *Macross*. Find the speed (in terms of  $c$ ) of *Yamato* relative to *Macross* if *Yamato* is flying (a) toward the earth and (b) away from the earth.

**Be sure to review** **EXAMPLE 37.9** (Section 37.7) and **EXAMPLES 37.10 and 37.11** (Section 37.8) before attempting these problems.

**VP37.11.1** A proton (rest mass  $1.67 \times 10^{-27}$  kg) is moving at  $0.925c$ . Find (a) the momentum of the proton, (b) its acceleration if a force of

$9.00 \times 10^{-14}$  N acts on the proton in the direction of its motion, and (c) its acceleration if a force of  $9.00 \times 10^{-14}$  N acts on the proton perpendicular to the direction of its motion.

**VP37.11.2** A moving electron (rest mass  $9.11 \times 10^{-31}$  kg) has total energy  $4.00 \times 10^{-13}$  J. Find (a) the kinetic energy of the electron, (b) its Lorentz factor  $\gamma$ , and (c) its speed in terms of  $c$ .

**VP37.11.3** A moving electron (rest mass  $9.11 \times 10^{-31}$  kg) has momentum  $2.60 \times 10^{-22}$  kg  $\cdot$  m/s. Find (a) the total energy of the electron, (b) its Lorentz factor  $\gamma$ , and (c) its speed in terms of  $c$ .

**VP37.11.4** The  $\psi(2S)$  meson is an unstable particle with rest energy 3686 MeV. One way in which a  $\psi(2S)$  can decay is into a positive K meson ( $K^+$ ) and a negative K meson ( $K^-$ ), each with rest energy 495 MeV. Assume the  $\psi(2S)$  is at rest before it decays. Find (a) the kinetic energy in MeV, (b) the Lorentz factor  $\gamma$ , and (c) the speed (in terms of  $c$ ) of each K meson.

## BRIDGING PROBLEM Colliding Protons

In an experiment, two protons are shot directly toward each other. Their speeds are such that in the frame of reference of each proton, the other proton is moving at  $0.500c$ . (a) What does an observer in the laboratory measure for the speed of each proton? (b) What is the kinetic energy of each proton as measured by an observer in the laboratory? (c) What is the kinetic energy of each proton as measured by the other proton?

### SOLUTION GUIDE

#### IDENTIFY and SET UP

1. This problem uses the Lorentz velocity transformation, which allows us to relate the velocity  $v_x$  of a proton in one frame to its velocity  $v'_x$  in a different frame. It also uses the idea of relativistic kinetic energy.
2. Draw a sketch of the situation. Take the  $x$ -axis to be the line of motion of the protons, and take the  $+x$ -direction to be to the right. In the frame in which the left-hand proton is at rest, the right-hand proton has velocity  $-0.500c$ . In the laboratory frame the two protons have velocities  $-ac$  and  $+ac$ , where  $a$  (each proton's laboratory speed as a fraction of  $c$ ) is our first

target variable. Given this we can find the laboratory kinetic energy of each proton.

#### EXECUTE

3. Write a Lorentz velocity-transformation equation that relates the velocity of the right-hand proton in the laboratory frame to its velocity in the frame of the left-hand proton. Solve this equation for  $a$ . (Hint: Remember that  $a$  cannot be greater than 1. Why?)
4. Use your result from step 3 to find the laboratory kinetic energy of each proton.
5. Find the kinetic energy of the right-hand proton as measured in the frame of the left-hand proton.

#### EVALUATE

6. How much total kinetic energy must be imparted to the protons by a scientist in the laboratory? If the experiment were to be repeated with one proton stationary, what kinetic energy would have to be given to the other proton for the collision to be equivalent?

## PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

## DISCUSSION QUESTIONS

**Q37.1** You are standing on a train platform watching a high-speed train pass by. A light inside one of the train cars is turned on and then a little later it is turned off. (a) Who can measure the proper time interval for the duration of the light: you or a passenger on the train? (b) Who can measure the proper length of the train car: you or a passenger on the train? (c) Who can measure the proper length of a sign attached to a post on the train platform: you or a passenger on the train? In each case explain your answer.

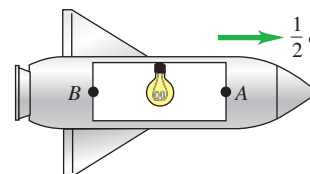
**Q37.2** If simultaneity is not an absolute concept, does that mean that we must discard the concept of causality? If event  $A$  is to *cause* event  $B$ ,  $A$  must occur first. Is it possible that in some frames  $A$  appears to be the cause of  $B$ , and in others  $B$  appears to be the cause of  $A$ ? Explain.

**Q37.3** A rocket is moving to the right at  $\frac{1}{2}$  the speed of light relative to the earth. A light bulb in the center of a room inside the rocket suddenly

turns on. Call the light hitting the front end of the room event  $A$  and the light hitting the back of the room event  $B$  (**Fig. Q37.3**). Which event occurs first,  $A$  or  $B$ , or are they simultaneous, as viewed by (a) an astronaut riding in the rocket and (b) a person at rest on the earth?

**Q37.4** A spaceship is traveling toward the earth from the space colony on Asteroid 1040A. The ship is at the halfway point of the trip, passing Mars at a speed of  $0.9c$  relative to the Mars frame of reference. At the same instant, a passenger on the spaceship receives a radio message from her boyfriend on 1040A and another from her sister on earth. According to the passenger on the ship, were these messages sent simultaneously or at different times? If at different times, which one was sent first? Explain your reasoning.

Figure Q37.3



**Q37.5** The average life span in the United States is about 70 years. Does this mean that it is impossible for an average person to travel a distance greater than 70 light-years away from the earth? (A light-year is the distance light travels in a year.) Explain.

**Q37.6** You are holding an elliptical serving platter. How would you need to travel for the serving platter to appear round to another observer?

**Q37.7** Two events occur at the same space point in a particular inertial frame of reference and are simultaneous in that frame. Is it possible that they may not be simultaneous in a different inertial frame? Explain.

**Q37.8** A high-speed train passes a train platform. Larry is a passenger on the train, Adam is standing on the train platform, and David is riding a bicycle toward the platform in the same direction as the train is traveling. Compare the length of a train car as measured by Larry, Adam, and David.

**Q37.9** The theory of relativity sets an upper limit on the speed that a particle can have. Are there also limits on the energy and momentum of a particle? Explain.

**Q37.10** A student asserts that a material particle must always have a speed slower than that of light, and a massless particle must always move at exactly the speed of light. Is she correct? If so, how do massless particles such as photons and neutrinos acquire this speed? Can't they start from rest and accelerate? Explain.

**Q37.11** The speed of light relative to still water is  $2.25 \times 10^8$  m/s. If the water is moving past us, the speed of light we measure depends on the speed of the water. Do these facts violate Einstein's second postulate? Explain.

**Q37.12** When a monochromatic light source moves toward an observer, its wavelength appears to be shorter than the value measured when the source is at rest. Does this contradict the hypothesis that the speed of light is the same for all observers? Explain.

**Q37.13** In principle, does a hot gas have more mass than the same gas when it is cold? Explain. In practice, would this be a measurable effect? Explain.

**Q37.14** Why do you think the development of Newtonian mechanics preceded the more refined relativistic mechanics by so many years?

**Q37.15** What do you think would be different in everyday life if the speed of light were 10 m/s instead of  $3.00 \times 10^8$  m/s?

## EXERCISES

### Section 37.2 Relativity of Simultaneity

**37.1** • Suppose the two lightning bolts shown in Fig. 37.5a are simultaneous to an observer on the train. Show that they are *not* simultaneous to an observer on the ground. Which lightning strike does the ground observer measure to come first?

### Section 37.3 Relativity of Time Intervals

**37.2** • The positive muon ( $\mu^+$ ), an unstable particle, lives on average  $2.20 \times 10^{-6}$  s (measured in its own frame of reference) before decaying. (a) If such a particle is moving, with respect to the laboratory, with a speed of  $0.900c$ , what average lifetime is measured in the laboratory? (b) What average distance, measured in the laboratory, does the particle move before decaying?

**37.3** • How fast must a rocket travel relative to the earth so that time in the rocket "slows down" to half its rate as measured by earth-based observers? Do present-day jet planes approach such speeds?

**37.4** • A spaceship flies past Mars with a speed of  $0.985c$  relative to the surface of the planet. When the spaceship is directly overhead, a signal light on the Martian surface blinks on and then off. An observer on Mars measures that the signal light was on for  $75.0 \mu\text{s}$ . (a) Does the observer on Mars or the pilot on the spaceship measure the proper time? (b) What is the duration of the light pulse measured by the pilot of the spaceship?

**37.5** • The negative pion ( $\pi^-$ ) is an unstable particle with an average lifetime of  $2.60 \times 10^{-8}$  s (measured in the rest frame of the pion). (a) If the pion is made to travel at very high speed relative to a laboratory, its average lifetime is measured in the laboratory to be  $4.20 \times 10^{-7}$  s. Calculate the speed of the pion expressed as a fraction of  $c$ . (b) What distance, measured in the laboratory, does the pion travel during its average lifetime?

**37.6** • As you pilot your space utility vehicle at a constant speed toward the moon, a race pilot flies past you in her spaceracer at a constant speed of  $0.800c$  relative to you. At the instant the spaceracer passes you, both of you start timers at zero. (a) At the instant when you measure that the spaceracer has traveled  $1.20 \times 10^8$  m past you, what does the race pilot read on her timer? (b) When the race pilot reads the value calculated in part (a) on her timer, what does she measure to be your distance from her? (c) At the instant when the race pilot reads the value calculated in part (a) on her timer, what do you read on yours?

**37.7** • An alien spacecraft is flying overhead at a great distance as you stand in your backyard. You see its searchlight blink on for  $0.150$  s. The first officer on the spacecraft measures that the searchlight is on for  $12.0$  ms. (a) Which of these two measured times is the proper time? (b) What is the speed of the spacecraft relative to the earth, expressed as a fraction of the speed of light  $c$ ?

### Section 37.4 Relativity of Length

**37.8** • At  $x = x' = 0$  and  $t = t' = 0$  a clock ticks aboard an extremely fast spaceship moving past us in the  $+x$ -direction with a Lorentz factor of 100 so  $v \approx c$ . The captain hears the clock tick again  $1.00$  s later. Where and when do we measure the second tick to occur?

**37.9** • A spacecraft of the Trade Federation flies past the planet Coruscant at a speed of  $0.600c$ . A scientist on Coruscant measures the length of the moving spacecraft to be  $74.0$  m. The spacecraft later lands on Coruscant, and the same scientist measures the length of the now stationary spacecraft. What value does she get?

**37.10** • A meter stick moves past you at great speed. Its motion relative to you is parallel to its long axis. If you measure the length of the moving meter stick to be  $1.00$  ft ( $1 \text{ ft} = 0.3048 \text{ m}$ )—for example, by comparing it to a 1 foot ruler that is at rest relative to you—at what speed is the meter stick moving relative to you?

**37.11** • **Why Are We Bombarded by Muons?** Muons are unstable subatomic particles that decay to electrons with a mean lifetime of  $2.2 \mu\text{s}$ . They are produced when cosmic rays bombard the upper atmosphere about  $10$  km above the earth's surface, and they travel very close to the speed of light. The problem we want to address is why we see any of them at the earth's surface. (a) What is the greatest distance a muon could travel during its  $2.2 \mu\text{s}$  lifetime? (b) According to your answer in part (a), it would seem that muons could never make it to the ground. But the  $2.2 \mu\text{s}$  lifetime is measured in the frame of the muon, and muons are moving very fast. At a speed of  $0.999c$ , what is the mean lifetime of a muon as measured by an observer at rest on the earth? How far would the muon travel in this time? Does this result explain why we find muons in cosmic rays? (c) From the point of view of the muon, it still lives for only  $2.2 \mu\text{s}$ , so how does it make it to the ground? What is the thickness of the  $10$  km of atmosphere through which the muon must travel, as measured by the muon? Is it now clear how the muon is able to reach the ground?

**37.12** • A rocket ship flies past the earth at  $91.0\%$  of the speed of light. Inside, an astronaut who is undergoing a physical examination is having his height measured while he is lying down parallel to the direction in which the ship is moving. (a) If his height is measured to be  $2.00$  m by his doctor inside the ship, what height would a person watching this from the earth measure? (b) If the earth-based person had measured  $2.00$  m, what would the doctor in the spaceship have measured for the astronaut's height? Is this a reasonable height? (c) Suppose the astronaut in part (a) gets up after the examination and stands with his body perpendicular to the direction of motion. What would the doctor in the rocket and the observer on earth measure for his height now?



**37.13 •** As measured by an observer on the earth, a spacecraft runway on earth has a length of 3600 m. (a) What is the length of the runway as measured by a pilot of a spacecraft flying past at a speed of  $4.00 \times 10^7$  m/s relative to the earth? (b) An observer on earth measures the time interval from when the spacecraft is directly over one end of the runway until it is directly over the other end. What result does she get? (c) The pilot of the spacecraft measures the time it takes him to travel from one end of the runway to the other end. What value does he get?

### Section 37.5 The Lorentz Transformations

**37.14 •** The Lorentz coordinate transformation assumes that  $t = t'$  at  $x = x' = 0$ . At what other values of  $x$  and  $x'$  does  $t = t'$ ?

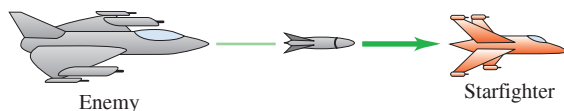
**37.15 •** An observer in frame  $S'$  is moving to the right ( $+x$ -direction) at speed  $u = 0.600c$  away from a stationary observer in frame  $S$ . The observer in  $S'$  measures the speed  $v'$  of a particle moving to the right away from her. What speed  $v$  does the observer in  $S$  measure for the particle if (a)  $v' = 0.400c$ ; (b)  $v' = 0.900c$ ; (c)  $v' = 0.990c$ ?

**37.16 •** Space pilot Mavis zips past Stanley at a constant speed relative to him of  $0.800c$ . Mavis and Stanley start timers at zero when the front of Mavis's ship is directly above Stanley. When Mavis reads 5.00 s on her timer, she turns on a bright light under the front of her spaceship. (a) Use the Lorentz coordinate transformation derived in Example 37.6 to calculate  $x$  and  $t$  as measured by Stanley for the event of turning on the light. (b) Use the time dilation formula, Eq. (37.6), to calculate the time interval between the two events (the front of the spaceship passing overhead and turning on the light) as measured by Stanley. Compare to the value of  $t$  you calculated in part (a). (c) Multiply the time interval by Mavis's speed, both as measured by Stanley, to calculate the distance she has traveled as measured by him when the light turns on. Compare to the value of  $x$  you calculated in part (a).

**37.17 ••** A pursuit spacecraft from the planet Tatooine is attempting to catch up with a Trade Federation cruiser. As measured by an observer on Tatooine, the cruiser is traveling away from the planet with a speed of  $0.600c$ . The pursuit ship is traveling at a speed of  $0.800c$  relative to Tatooine, in the same direction as the cruiser. (a) For the pursuit ship to catch the cruiser, should the velocity of the cruiser relative to the pursuit ship be directed toward or away from the pursuit ship? (b) What is the speed of the cruiser relative to the pursuit ship?

**37.18 ••** An enemy spaceship is moving toward your starfighter with a speed, as measured in your frame, of  $0.400c$ . The enemy ship fires a missile toward you at a speed of  $0.700c$  relative to the enemy ship (**Fig. E37.18**). (a) What is the speed of the missile relative to you? Express your answer in terms of the speed of light. (b) If you measure that the enemy ship is  $8.00 \times 10^6$  km away from you when the missile is fired, how much time, measured in your frame, will it take the missile to reach you?

Figure E37.18



**37.19 ••** Two particles are created in a high-energy accelerator and move off in opposite directions. The speed of one particle, as measured in the laboratory, is  $0.650c$ , and the speed of each particle relative to the other is  $0.950c$ . What is the speed of the second particle, as measured in the laboratory?

**37.20 ••** Two particles in a high-energy accelerator experiment are approaching each other head-on, each with a speed of  $0.9380c$  as measured in the laboratory. What is the magnitude of the velocity of one particle relative to the other?

**37.21 ••** Two particles in a high-energy accelerator experiment approach each other head-on with a relative speed of  $0.890c$ . Both particles travel at the same speed as measured in the laboratory. What is the speed of each particle, as measured in the laboratory?

### Section 37.6 The Doppler Effect for Electromagnetic Waves

**37.22 •** Electromagnetic radiation from a star is observed with an earth-based telescope. The star is moving away from the earth at a speed of  $0.520c$ . If the radiation has a frequency of  $8.64 \times 10^{14}$  Hz in the rest frame of the star, what is the frequency measured by an observer on earth?

**37.23 • Tell It to the Judge.** (a) How fast must you be approaching a red traffic light ( $\lambda = 675$  nm) for it to appear yellow ( $\lambda = 575$  nm)? Express your answer in terms of the speed of light. (b) If you used this as a reason not to get a ticket for running a red light, how much of a fine would you get for speeding? Assume that the fine is \$1.00 for each kilometer per hour that your speed exceeds the posted limit of 90 km/h.

**37.24 •** A source of electromagnetic radiation is moving in a radial direction relative to you. The frequency you measure is 1.25 times the frequency measured in the rest frame of the source. What is the speed of the source relative to you? Is the source moving toward you or away from you?

### Section 37.7 Relativistic Momentum

**37.25 •** A particle zips by us with a Lorentz factor of 1.12. Then another particle zips by us moving at twice the speed of the first particle. (a) What is the Lorentz factor of the second particle? (b) If the particles were moving with a speed much less than  $c$ , the magnitude of the momentum of the second particle would be twice that of the first. However, what is the ratio of the magnitudes of momentum for these relativistic particles?

**37.26 • Relativistic Baseball.** Calculate the magnitude of the force required to give a 0.145 kg baseball an acceleration  $a = 1.00$  m/s<sup>2</sup> in the direction of the baseball's initial velocity when this velocity has a magnitude of (a) 10.0 m/s; (b)  $0.900c$ ; (c)  $0.990c$ . (d) Repeat parts (a), (b), and (c) if the force and acceleration are perpendicular to the velocity.

**37.27 •** A proton has momentum with magnitude  $p_0$  when its speed is  $0.400c$ . In terms of  $p_0$ , what is the magnitude of the proton's momentum when its speed is doubled to  $0.800c$ ?

**37.28 •** A spaceship has length 120 m, diameter 25 m, and mass  $4.0 \times 10^3$  kg as measured by its crew. As the spaceship moves parallel to its cylindrical axis and passes us, we measure its length to be 90 m. (a) What do we measure its diameter to be? (b) What do we measure the magnitude of its momentum to be?

**37.29 •** (a) At what speed is the momentum of a particle twice as great as the result obtained from the nonrelativistic expression  $mv$ ? Express your answer in terms of the speed of light. (b) A force is applied to a particle along its direction of motion. At what speed is the magnitude of force required to produce a given acceleration twice as great as the force required to produce the same acceleration when the particle is at rest? Express your answer in terms of the speed of light.

### Section 37.8 Relativistic Work and Energy

**37.30 •** The sun produces energy by nuclear fusion reactions, in which matter is converted into energy. By measuring the amount of energy we receive from the sun, we know that it is producing energy at a rate of  $3.8 \times 10^{26}$  W. (a) How many kilograms of matter does the sun lose each second? Approximately how many tons of matter is this (1 ton = 2000 lb)? (b) At this rate, how long would it take the sun to use up all its mass?

**37.31 ••** What is the speed of a particle whose kinetic energy is equal to (a) its rest energy and (b) five times its rest energy?

**37.32 •** If a muon is traveling at  $0.999c$ , what are its momentum and kinetic energy? (The mass of such a muon at rest in the laboratory is 207 times the electron mass.)

**37.33 •** A proton (rest mass  $1.67 \times 10^{-27}$  kg) has total energy that is 4.00 times its rest energy. What are (a) the kinetic energy of the proton; (b) the magnitude of the momentum of the proton; (c) the proton's speed?

**37.34 ••** (a) How much work must be done on a particle with mass  $m$  to accelerate it (a) from rest to a speed of  $0.090c$  and (b) from a speed of  $0.900c$  to a speed of  $0.990c$ ? (Express the answers in terms of  $mc^2$ .) (c) How do your answers in parts (a) and (b) compare?

**37.35 •** A particle has rest mass  $6.64 \times 10^{-27}$  kg and momentum  $2.10 \times 10^{-18}$  kg  $\cdot$  m/s. (a) What is the total energy (kinetic plus rest energy) of the particle? (b) What is the kinetic energy of the particle? (c) What is the ratio of the kinetic energy to the rest energy of the particle?

**37.36 ••** Electrons are accelerated through a potential difference of 750 kV, so that their kinetic energy is  $7.50 \times 10^5$  eV. (a) What is the ratio of the speed  $v$  of an electron having this energy to the speed of light,  $c$ ? (b) What would the speed be if it were computed from the principles of classical mechanics?

**37.37 •** Compute the kinetic energy of a proton (mass  $1.67 \times 10^{-27}$  kg) using both the nonrelativistic and relativistic expressions, and compute the ratio of the two results (relativistic divided by nonrelativistic) for speeds of (a)  $8.00 \times 10^7$  m/s and (b)  $2.85 \times 10^8$  m/s.

**37.38 • Creating a Particle.** Two protons (each with rest mass  $M = 1.67 \times 10^{-27}$  kg) are initially moving with equal speeds in opposite directions. The protons continue to exist after a collision that also produces an  $\eta^0$  particle (see Chapter 44). The rest mass of the  $\eta^0$  is  $m = 9.75 \times 10^{-28}$  kg. (a) If the two protons and the  $\eta^0$  are all at rest after the collision, find the initial speed of the protons, expressed as a fraction of the speed of light. (b) What is the kinetic energy of each proton? Express your answer in MeV. (c) What is the rest energy of the  $\eta^0$ , expressed in MeV? (d) Discuss the relationship between the answers to parts (b) and (c).

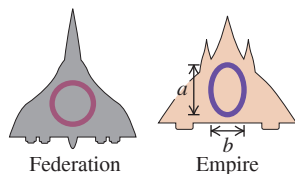
**37.39 •** (a) Through what potential difference does an electron have to be accelerated, starting from rest, to achieve a speed of  $0.980c$ ? (b) What is the kinetic energy of the electron at this speed? Express your answer in joules and in electron volts.

## PROBLEMS

**37.40 •** Inside a spaceship flying past the earth at three-fourths the speed of light, a pendulum is swinging. (a) If each swing takes 1.80 s as measured by an astronaut performing an experiment inside the spaceship, how long will the swing take as measured by a person at mission control (on earth) who is watching the experiment? (b) If each swing takes 1.80 s as measured by a person at mission control, how long will it take as measured by the astronaut in the spaceship?

**37.41 ••** The starships of the Solar Federation are marked with the symbol of the federation, a circle, while starships of the Denebian Empire are marked with the empire's symbol, an ellipse whose major axis is 1.40 times longer than its minor axis ( $a = 1.40b$  in Fig. P37.41). How fast, relative to an observer, does an empire ship have to travel for its marking to be confused with the marking of a federation ship?

Figure P37.41



**37.42 ••** A cube of metal with sides of length  $a$  sits at rest in a frame  $S$  with one edge parallel to the  $x$ -axis. Therefore, in  $S$  the cube has volume  $a^3$ . Frame  $S'$  moves along the  $x$ -axis with a speed  $u$ . As measured by an observer in frame  $S'$ , what is the volume of the metal cube?

**37.43 ••** A space probe is sent to the vicinity of the star Capella, which is 42.2 light-years from the earth. (A light-year is the distance light travels in a year.) The probe travels with a speed of  $0.9930c$ . An astronaut recruit on board is 19 years old when the probe leaves the earth. What is her biological age when the probe reaches Capella?

**37.44 ••** A  $\Sigma^+$  particle has a mean lifetime of 80.2 ps. A physicist measures that mean lifetime to be 403 ps as the particle moves in his lab. The rest mass of the particle is  $2.12 \times 10^{-27}$  kg. (a) How fast is the particle moving? (b) How far does it travel, as measured in the lab frame, over one mean lifetime? (c) What are its rest, kinetic, and total energies in the lab frame of reference? (d) What are its rest, kinetic, and total energies in the particle's frame?

**37.45 ••** Spaceship  $A$  moves past the earth at  $0.80c$  to the west. Spaceship  $B$  approaches  $A$ , moving to the east. Both spaceship crews measure their relative speed of approach to be  $0.98c$ . What mass would the crews of both spaceships measure for the standard kilogram, kept at rest on the earth, (a) according to classical physics and (b) according to the special theory of relativity?

**37.46 ••** One way to strictly enforce a speed limit would be to alter the laws of nature. Suppose the speed of light were 65 mph and your workplace was 30 miles from your home. Assume you travel to work at a typical driving speed of 60 mph. (a) If you drove at that speed for the round trip to and from work, light, how much would your wrist-watch lag your kitchen clock each day? (b) Estimate the length of your car. (c) If you were driving at your estimated driving speed, how long would your car be when viewed from the roadside? (d) What would be the speed relative to you of similar cars traveling toward you in the opposite lane with the same ground speed as you? (e) How long would you measure those cars to be? (f) If the total mass of you and your car was 2000 kg, how much work would be required to get you up to speed? (Note: Your rest mass energy in this world is  $mc^2$ , where  $c = 65$  mph.) (g) How much work would be required in the real world, where the speed of light is  $3.0 \times 10^8$  m/s, to get you up to speed?

**37.47 • The Large Hadron Collider (LHC).** Physicists and engineers from around the world came together to build the largest accelerator in the world, the Large Hadron Collider (LHC) at the CERN Laboratory in Geneva, Switzerland. The machine accelerates protons to high kinetic energies in an underground ring 27 km in circumference. (a) What is the speed  $v$  of a proton in the LHC if the proton's kinetic energy is 7.0 TeV? (Because  $v$  is very close to  $c$ , write  $v = (1 - \Delta)c$  and give your answer in terms of  $\Delta$ .) (b) Find the relativistic mass,  $m_{\text{rel}}$ , of the accelerated proton in terms of its rest mass.

**37.48 ••** The net force  $\vec{F}$  on a particle of mass  $m$  is directed at  $30.0^\circ$  counterclockwise from the  $+x$ -axis. At one instant of time, the particle is traveling in the  $+x$ -direction with a speed (measured relative to the earth) of  $0.700c$ . At this instant, what is the direction of the particle's acceleration?

**37.49 •• Everyday Time Dilation.** Two atomic clocks are carefully synchronized. One remains in New York, and the other is loaded on an airliner that travels at an average speed of 250 m/s and then returns to New York. When the plane returns, the elapsed time on the clock that stayed behind is 4.00 h. By how much will the readings of the two clocks differ, and which clock will show the shorter elapsed time? (Hint: Since  $u \ll c$ , you can simplify  $\sqrt{1 - u^2/c^2}$  by a binomial expansion.)

**37.50 ••** The distance to a particular star, as measured in the earth's frame of reference, is 7.11 light-years (1 light-year is the distance that light travels in 1 y). A spaceship leaves the earth and takes 3.35 y to arrive at the star, as measured by passengers on the ship. (a) How long does the trip take, according to observers on earth? (b) What distance for the trip do passengers on the spacecraft measure?

**37.51 • CP Čerenkov Radiation.** The Russian physicist P. A. Čerenkov discovered that a charged particle traveling in a solid with a speed exceeding the speed of light in that material radiates electromagnetic radiation. (This is analogous to the sonic boom produced by an aircraft moving faster than the speed of sound in air; see Section 16.9. Čerenkov shared the 1958 Nobel Prize for this discovery.) What is the minimum kinetic energy (in electron volts) that an electron must have while traveling inside a slab of crown glass ( $n = 1.52$ ) in order to create this Čerenkov radiation?

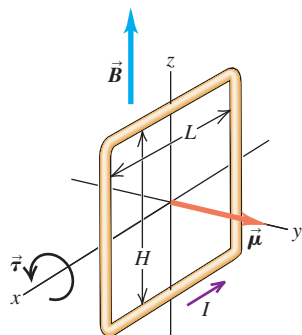
**37.52 ••** Quarks and gluons are fundamental particles that will be discussed in Chapter 44. A proton, which is a bound state of two up quarks and a down quark, has a rest mass of  $m_p = 1.67 \times 10^{-27}$  kg. This is significantly greater than the sum of the rest mass of the up quarks, which is  $m_u = 4.12 \times 10^{-30}$  kg each, and the rest mass of the down quark, which is  $m_d = 8.59 \times 10^{-30}$  kg. Suppose we (incorrectly) model the rest energy of the proton  $m_p c^2$  as derived from the kinetic energy of the three quarks, and we split that energy equally among them. (a) Estimate the Lorentz factor  $\gamma = (1 - v^2/c^2)^{-1/2}$  for each of the up quarks using Eq. (37.36). (b) Similarly estimate the Lorentz factor  $\gamma$  for the down quark. (c) Are the corresponding speeds  $v_u$  and  $v_d$  greater than 99% of the speed of light? (d) More realistically, the quarks are held together by massless gluons, which mediate the strong nuclear interaction. Suppose we model the proton as the three quarks, each with a speed of  $0.90c$ , with the remainder of the proton rest energy supplied by gluons. In this case, estimate the percentage of the proton rest energy associated with gluons. (e) Model a quark as oscillating with an average speed of  $0.90c$  across the diameter of a proton,  $1.7 \times 10^{-15}$  m. Estimate the frequency of that motion.

**37.53 • CP** A nuclear bomb containing 12.0 kg of plutonium explodes. The sum of the rest masses of the products of the explosion is less than the original rest mass by one part in  $10^4$ . (a) How much energy is released in the explosion? (b) If the explosion takes place in  $4.00 \mu\text{s}$ , what is the average power developed by the bomb? (c) What mass of water could the released energy lift to a height of 1.00 km?

**37.54 ••** In the earth's rest frame, two protons are moving away from each other at equal speed. In the frame of each proton, the other proton has a speed of  $0.700c$ . What does an observer in the rest frame of the earth measure for the speed of each proton?

**37.55 •• CP** In a laboratory, a rectangular loop of wire surrounds the origin in the  $xz$ -plane, with extent  $H$  in the  $z$ -direction and extent  $L$  in the  $x$ -direction (Fig. P37.55). The loop carries current  $I$  in the counterclockwise direction as viewed from the positive  $y$ -axis. A magnetic field  $\vec{B} = B\hat{k}$  is present. (a) What is the magnetic dipole moment  $\vec{\mu}$  as seen in the frame  $S$  of the laboratory? (b) What is the torque  $\vec{\tau}$  felt by the current loop? (c) The same loop is viewed from a passing alien spaceship moving with velocity  $\vec{v} = v\hat{i}$  as seen from the laboratory. From the point of view of the aliens, the loop is length contracted in the direction of this motion. What is the magnetic dipole moment  $\vec{\mu}'$  according to the aliens if their coordinate axes are aligned with those of the humans? (d) The torque on the loop is the same when viewed from the laboratory frame and from the spaceship frame. Accordingly, the magnetic field must be frame-dependent. What is the magnetic field  $\vec{B}'$  in the spaceship frame  $S'$ ? (e) If the electromagnetic field is  $(\vec{E}, \vec{B}) = (0, \vec{B})$  in an inertial frame  $S$ , then in another frame  $S'$  moving at velocity  $\vec{v}$  relative to  $S$ , what is the component of the magnetic field  $\vec{B}'_{\perp}$  perpendicular to  $\vec{v}$ ?

Figure P37.55



**37.56 •• CP** A small sphere with charge  $Q$  is resting motionless on an insulated pedestal in a laboratory. In the frame of reference  $S$  of the laboratory, the  $z$ -axis points upward, there is a magnetic field  $\vec{B} = -B\hat{j}$ , and there is no electric field. (a) What is the net force on the sphere? (b) The same sphere is viewed from a passing spacecraft moving with velocity  $\vec{v} = v\hat{i}$  as seen from the laboratory. Using the result of Problem 37.55, what is the magnetic field  $\vec{B}'$  seen by observers in the spacecraft if their coordinate axes are aligned with those of the laboratory? (c) From the perspective of the spacecraft, the sphere has a nonzero velocity and therefore feels a nonzero magnetic force. What is that

magnetic force? (d) The net electromagnetic force on the sphere is necessarily the same when viewed from the laboratory frame and from the spaceship frame. This can be true only if there is a nonzero electric field  $\vec{E}'$  in the spaceship frame. Determine  $\vec{E}'$ . (e) Generalize your conclusion: If the electromagnetic field is  $(\vec{E}, \vec{B}) = (0, \vec{B})$  in inertial frame  $S$ , then in another frame  $S'$  moving at velocity  $\vec{v}$  relative to  $S$ , what is the component of the electric field  $\vec{E}'_{\perp}$  perpendicular to  $\vec{v}$ ?

**37.57 •** One of the wavelengths of light emitted by hydrogen atoms under normal laboratory conditions is  $\lambda = 656.3$  nm, in the red portion of the electromagnetic spectrum. In the light emitted from a distant galaxy this same spectral line is observed to be Doppler-shifted to  $\lambda = 953.4$  nm, in the infrared portion of the spectrum. How fast are the emitting atoms moving relative to the earth? Are they approaching the earth or receding from it?

**37.58 ••** Two events are observed in a frame of reference  $S$  to occur at the same space point, the second occurring 1.80 s after the first. In a frame  $S'$  moving relative to  $S$ , the second event is observed to occur 2.15 s after the first. What is the difference between the positions of the two events as measured in  $S'$ ?

**37.59 •• Measuring Speed by Radar.** A baseball coach uses a radar device to measure the speed of an approaching pitched baseball. This device sends out electromagnetic waves with frequency  $f_0$  and then measures the shift in frequency  $\Delta f$  of the waves reflected from the moving baseball. If the fractional frequency shift produced by a baseball is  $\Delta f/f_0 = 2.86 \times 10^{-7}$ , what is the baseball's speed in km/h? (Hint: Are the waves Doppler-shifted a second time when reflected off the ball?)

**37.60 •• Albert in Wonderland.** Einstein and Lorentz, avid tennis players, play a fast-paced game on a court where they stand 20.0 m from each other. They play without a net. The tennis ball has mass 0.0580 kg. Ignore gravity and assume that the ball travels parallel to the ground as it travels between the two players. Unless otherwise specified, all measurements are made by the two men. (a) Lorentz serves the ball at  $80.0$  m/s. What is the ball's kinetic energy? (b) Einstein slams a return at  $1.80 \times 10^8$  m/s. What is the ball's kinetic energy? (c) During Einstein's return of the ball in part (a), a white rabbit runs beside the court in the direction from Einstein to Lorentz. The rabbit has a speed of  $2.20 \times 10^8$  m/s relative to the two men. What is the speed of the rabbit relative to the ball? (d) What does the rabbit measure as the distance from Einstein to Lorentz? (e) How much time does it take for the rabbit to run 20.0 m, according to the players? (f) The white rabbit uses his pocket watch to measure the time (as he sees it) for the distance from Einstein to Lorentz to pass by under him. What time does he measure?

**37.61 • CP** In a particle accelerator a proton moves at constant speed  $0.750c$  in a circle of radius 628 m. What is the net force on the proton?

**37.62 ••** A spaceship moving at constant speed  $u$  relative to us broadcasts a radio signal at constant frequency  $f_0$ . As the spaceship approaches us, we receive a higher frequency  $f$ ; after it has passed, we receive a lower frequency. (a) As the spaceship passes by, so it is instantaneously moving neither toward nor away from us, show that the frequency we receive is not  $f_0$ , and derive an expression for the frequency we do receive. Is the frequency we receive higher or lower than  $f_0$ ? (Hint: In this case, successive wave crests move the same distance to the observer and so they have the same transit time. Thus  $f$  equals  $1/T$ . Use the time dilation formula to relate the periods in the stationary and moving frames.) (b) A spaceship emits electromagnetic waves of frequency  $f_0 = 345$  MHz as measured in a frame moving with the ship. The spaceship is moving at a constant speed  $0.758c$  relative to us. What frequency  $f$  do we receive when the spaceship is approaching us? When it is moving away? In each case what is the shift in frequency,  $f - f_0$ ? (c) Use the result of part (a) to calculate the frequency  $f$  and the frequency shift  $(f - f_0)$  we receive at the instant that the ship passes by us. How does the shift in frequency calculated here compare to the shifts calculated in part (b)?



**37.63 •• DATA** As a research scientist at a linear accelerator, you are studying an unstable particle. You measure its mean lifetime  $\Delta t$  as a function of the particle's speed relative to your laboratory equipment. You record the speed of the particle  $u$  as a fraction of the speed of light in vacuum  $c$ . The table gives the results of your measurements.

| $u/c$                     | 0.70 | 0.80 | 0.85 | 0.88 | 0.90 | 0.92 | 0.94 |
|---------------------------|------|------|------|------|------|------|------|
| $\Delta t$ ( $10^{-8}$ s) | 3.57 | 4.41 | 5.02 | 5.47 | 6.05 | 6.58 | 7.62 |

(a) Your team leader suggests that if you plot your data as  $(\Delta t)^2$  versus  $(1 - u^2/c^2)^{-1}$ , the data points will be fit well by a straight line. Construct this graph and verify the team leader's prediction. Use the best-fit straight line to your data to calculate the mean lifetime of the particle in its rest frame. (b) What is the speed of the particle relative to your lab equipment (expressed as  $u/c$ ) if the lifetime that you measure is four times its rest-frame lifetime?

**37.64 •• CP** The French physicist Armand Fizeau was the first to measure the speed of light accurately. He also found experimentally that the speed, relative to the lab frame, of light traveling in a tank of water that is itself moving at a speed  $V$  relative to the lab frame is

$$v = \frac{c}{n} + kV$$

where  $n = 1.333$  is the index of refraction of water. Fizeau called  $k$  the dragging coefficient and obtained an experimental value of  $k = 0.44$ . What value of  $k$  do you calculate from relativistic transformations?

**37.65 •• DATA** You are a scientist studying small aerosol particles that are contained in a vacuum chamber. The particles carry a net charge, and you use a uniform electric field to exert a constant force of  $8.00 \times 10^{-14}$  N on one of them. That particle moves in the direction of the exerted force. Your instruments measure the acceleration of the particle as a function of its speed  $v$ . The table gives the results of your measurements for this particular particle.

| $v/c$                           | 0.60 | 0.65 | 0.70 | 0.75 | 0.80 | 0.85 |
|---------------------------------|------|------|------|------|------|------|
| $a$ ( $10^3$ m/s <sup>2</sup> ) | 20.3 | 17.9 | 14.8 | 11.2 | 8.5  | 5.9  |

(a) Graph your data so that the data points are well fit by a straight line. Use the slope of this line to calculate the mass  $m$  of the particle. (b) What magnitude of acceleration does the exerted force produce if the speed of the particle is 100 m/s?

**37.66 •• DATA** You are an astronomer investigating four astronomical sources of infrared radiation. You have identified the nature of each source, so you know the frequency  $f_0$  of each when it is at rest relative to you. Your detector, which is at rest relative to the earth, measures the frequency  $f$  of the moving source. Your results are given in the table.

| Source      | A   | B   | C   | D   |
|-------------|-----|-----|-----|-----|
| $f$ (THz)   | 7.1 | 5.4 | 6.1 | 8.1 |
| $f_0$ (THz) | 9.2 | 8.6 | 7.9 | 8.9 |

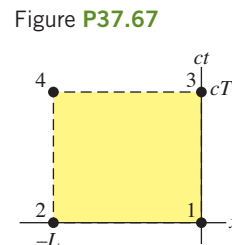
(a) Which source is moving at the highest speed relative to your detector? What is its speed? Is that source moving toward or away from the detector? (b) Which source is moving at the lowest speed relative to your detector? What is its speed? Is that source moving toward or away from the detector? (c) For source B, what frequency would your detector measure if the source were moving at the same speed relative to the detector but toward it rather than away from it?

## CHALLENGE PROBLEMS

**37.67 •••** A fluorescent tube with length  $L$  is fixed horizontally above a ticket window in a train station. At time  $t = 0$  the tube lights up,

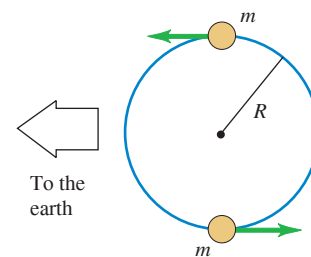
changing color from white to bright yellow. After a duration  $T$  the tube turns off, reverting to its white color. The tube lies along the  $x$ -axis of frame  $S$  fixed with respect to the station, with its right end at the origin. We can represent the history of the tube using a “spacetime” diagram, as shown in

**Fig. P37.67**. Events 1 and 2 correspond, respectively, to the right and left ends of the tube at time  $t = 0$ , while events 3 and 4 correspond to the right and left ends of the tube at time  $t = T$ . The totality of events corresponding to positions on the tube when it is lit correspond to the yellow region on the diagram. The tube is observed from the window of a rocket train passing the station from left to right at speed  $v$ . The frame  $S'$  is fixed with respect to the train such that the  $x'$ -axis coincides with the  $x$ -axis of frame  $S$ , and such that event 1 occurs at the origin  $(x', t') = (0, 0)$ . (a) Use the Lorentz transformations to determine the spacetime coordinates  $(x', t')$  for events 1, 2, 3, and 4 in the train frame  $S'$ . (b) Construct a spacetime diagram analogous to Figure P37.67 showing the tube in the train frame  $S'$ . Carefully label relevant points, using the economical definition  $\gamma = (1 - v^2/c^2)^{-1/2}$  when needed. (c) We use the product  $ct$  rather than  $t$  on the vertical axis of our figure so that both axes have the dimension of distance. Thus the yellow region has a well-defined area. What is the area of the yellow region in the  $S$  frame plot? (d) Compute the area of the yellow region in the  $S'$  frame plot you constructed in part (b). (Hint: The area of a parallelogram is given by the magnitude of the vector product of two vectors describing adjacent edges.) (e) Is the area of the yellow region the same when we change the frame of reference from  $S$  to  $S'$ ?



**37.68 ••• CP Determining the Masses of Stars.** Many of the stars in the sky are actually *binary stars*, in which two stars orbit about their common center of mass. If the orbital speeds of the stars are high enough, the motion of the stars can be detected by the Doppler shifts of the light they emit. Stars for which this is the case are called *spectroscopic binary stars*.

Figure P37.68

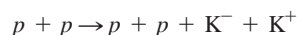


**Figure P37.68** shows the simplest case of a spectroscopic binary star:

two identical stars, each with mass  $m$ , orbiting their center of mass in a circle of radius  $R$ . The plane of the stars' orbits is edge-on to the line of sight of an observer on the earth. (a) The light produced by heated hydrogen gas in a laboratory on the earth has a frequency of  $4.568110 \times 10^{14}$  Hz. In the light received from the stars by a telescope on the earth, hydrogen light is observed to vary in frequency between  $4.567710 \times 10^{14}$  Hz and  $4.568910 \times 10^{14}$  Hz. Determine whether the binary star system as a whole is moving toward or away from the earth, the speed of this motion, and the orbital speeds of the stars. (Hint: The speeds involved are much less than  $c$ , so you may use the approximate result  $\Delta f/f = u/c$  given in Section 37.6.) (b) The light from each star in the binary system varies from its maximum frequency to its minimum frequency and back again in 11.0 days. Determine the orbital radius  $R$  and the mass  $m$  of each star. Give your answer for  $m$  in kilograms and as a multiple of the mass of the sun,  $1.99 \times 10^{30}$  kg. Compare the value of  $R$  to the distance from the earth to the sun,  $1.50 \times 10^{11}$  m. (This technique is actually used in astronomy to determine the masses of stars. In practice, the problem is more complicated because the two stars in a binary system are usually not identical, the orbits are usually not circular, and the plane of the orbits is usually tilted with respect to the line of sight from the earth.)



**37.69 ••• CP Kaon Production.** In high-energy physics, new particles can be created by collisions of fast-moving projectile particles with stationary particles. Some of the kinetic energy of the incident particle is used to create the mass of the new particle. A proton–proton collision can result in the creation of a negative kaon ( $K^-$ ) and a positive kaon ( $K^+$ ):



(a) Calculate the minimum kinetic energy of the incident proton that will allow this reaction to occur if the second (target) proton is initially at rest. The rest energy of each kaon is 493.7 MeV, and the rest energy of each proton is 938.3 MeV. (*Hint:* It is useful here to work in the frame in which the total momentum is zero. But note that the Lorentz transformation must be used to relate the velocities in the laboratory frame to those in the zero-total-momentum frame.) (b) How does this calculated minimum kinetic energy compare with the total rest mass energy of the created kaons? (c) Suppose that instead the two protons are both in motion with velocities of equal magnitude and opposite direction. Find the minimum combined kinetic energy of the two protons that will allow the reaction to occur. How does this calculated minimum kinetic energy compare with the total rest mass energy of the created kaons? (This example shows that when colliding beams of particles are used instead of a stationary target, the energy requirements for producing new particles are reduced substantially.)

**37.70 ••• CP CALC Relativity and the Wave Equation.** (a) Consider the Galilean transformation along the  $x$ -direction:  $x' = x - vt$  and  $t' = t$ . In frame  $S$  the wave equation for electromagnetic waves in a vacuum is

$$\frac{\partial^2 E(x, t)}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 E(x, t)}{\partial t^2} = 0$$

where  $E$  represents the electric field in the wave. Show that by using the Galilean transformation the wave equation in frame  $S'$  is found to be

$$\left(1 - \frac{v^2}{c^2}\right) \frac{\partial^2 E(x', t')}{\partial x'^2} + \frac{2v}{c^2} \frac{\partial^2 E(x', t')}{\partial x' \partial t'} - \frac{1}{c^2} \frac{\partial^2 E(x', t')}{\partial t'^2} = 0$$

This has a different form than the wave equation in  $S$ . Hence the Galilean transformation *violates* the first relativity postulate that all physical laws have the same form in all inertial reference frames. (*Hint:* Express the derivatives  $\partial/\partial x$  and  $\partial/\partial t$  in terms of  $\partial/\partial x'$  and  $\partial/\partial t'$  by use of the chain rule.) (b) Repeat the analysis of part (a), but use the Lorentz coordinate transformations, Eqs. (37.21), and show that in frame  $S'$  the wave equation has the same form as in frame  $S$ :

$$\frac{\partial^2 E(x', t')}{\partial x'^2} - \frac{1}{c^2} \frac{\partial^2 E(x', t')}{\partial t'^2} = 0$$

Explain why this shows that the speed of light in vacuum is  $c$  in both frames  $S$  and  $S'$ .

## MCAT-STYLE PASSAGE PROBLEMS

**Speed of Light.** Our universe has properties that are determined by the values of the fundamental physical constants, and it would be a much different place if the charge of the electron, the mass of the proton, or the speed of light was substantially different from its actual value. For instance, the speed of light is so great that the effects of relativity usually go unnoticed in everyday events. Let's imagine an alternate universe where the speed of light is 1,000,000 times less than it is in our universe to see what would happen.

**37.71** An airplane has a length of 60 m when measured at rest. When the airplane is moving at 180 m/s (400 mph) in the alternate universe, how long would the plane appear to be to a stationary observer? (a) 24 m; (b) 36 m; (c) 48 m; (d) 60 m; (e) 75 m.

**37.72** If the airplane of Passage Problem 37.71 has a rest mass of 20,000 kg, what is its relativistic mass when the plane is moving at 180 m/s? (a) 8000 kg; (b) 12,000 kg; (c) 16,000 kg; (d) 25,000 kg; (e) 33,300 kg.

**37.73** In our universe, the rest energy of an electron is approximately  $8.2 \times 10^{-14}$  J. What would it be in the alternate universe? (a)  $8.2 \times 10^{-8}$  J; (b)  $8.2 \times 10^{-26}$  J; (c)  $8.2 \times 10^{-2}$  J; (d) 0.82 J.

**37.74** In the alternate universe, how fast must an object be moving for it to have a kinetic energy equal to its rest mass? (a) 225 m/s; (b) 260 m/s; (c) 300 m/s; (d) The kinetic energy could not be equal to the rest mass.

## ANSWERS

### Chapter Opening Question ?

(v) From Eq. (37.36), the relativistic expression for the kinetic energy of a particle of mass  $m$  moving at speed  $v$  is  $K = (\gamma - 1)mc^2$ , where  $\gamma = 1/\sqrt{1 - v^2/c^2}$ . If  $v = 0.99000c$ ,  $\gamma - 1 = 6.08881$ ; if  $v = 0.99995c$ ,  $\gamma - 1 = 99.001$ , which is 16.260 times greater than the value at  $v = 0.99000c$ . As the speed approaches  $c$ , a relatively small increase in  $v$  corresponds to a large increase in kinetic energy (see Fig. 37.21).

### Key Example VARIATION Problems

**VP37.5.1** (a)  $2.05 \times 10^{-8}$  s (b)  $2.05 \times 10^{-8}$  s

**VP37.5.2** (a) 16.0 s (b) 30.0 s

**VP37.5.3** (a) 526 s (b)  $4.68 \times 10^{10}$  m, 164 s

**VP37.5.4** (a)  $1.28 \times 10^5$  km (b) 72.0 m

**VP37.7.1** (a)  $x' = -8.50 \times 10^8$  m,  $t' = 3.78$  s

(b)  $x = 1.46 \times 10^9$  m,  $t = 5.29$  s

**VP37.7.2** (a)  $1.35 \times 10^8$  m (b)  $2.25 \times 10^8$  m

**VP37.7.3** (a)  $+0.385c$  (b)  $-0.227c$

**VP37.7.4** (a)  $0.941c$  (b)  $0.195c$

**VP37.11.1** (a)  $1.22 \times 10^{-18}$  kg  $\cdot$  m/s (b)  $2.96 \times 10^{12}$  m/s<sup>2</sup>

(c)  $2.05 \times 10^{13}$  m/s<sup>2</sup>

**VP37.11.2** (a)  $3.18 \times 10^{-13}$  J (b) 4.88 (c)  $0.979c$

**VP37.11.3** (a)  $1.13 \times 10^{-13}$  J (b) 1.38 (c)  $0.689c$

**VP37.11.4** (a) 1348 MeV (b) 3.72 (c)  $0.963c$

### Bridging Problem

(a)  $0.268c$

(b) 35.6 MeV

(c) 145 MeV



? This plastic surgeon is using two light sources: a headlamp that emits a beam of visible light and a handheld laser that emits infrared light. The light from both sources is emitted in the form of packets of energy called photons. For which source are the individual photons more energetic? (i) The headlamp; (ii) the laser; (iii) both are equally energetic; (iv) not enough information is given.

# 38 Photons: Light Waves Behaving as Particles

In Chapter 32 we saw how Maxwell, Hertz, and others established firmly that light is an electromagnetic wave. Interference, diffraction, and polarization, discussed in Chapters 35 and 36, further demonstrate this *wave nature* of light.

When we look more closely at the emission, absorption, and scattering of electromagnetic radiation, however, we discover a completely different aspect of light. We find that the energy of an electromagnetic wave is *quantized*; it is emitted and absorbed in particle-like packages of definite energy, called *photons*. The energy of a single photon is proportional to the frequency of the radiation.

We'll find that light and other electromagnetic radiation exhibit *wave-particle duality*: Light acts sometimes like waves and sometimes like particles. Interference and diffraction demonstrate wave behavior, while emission and absorption of photons demonstrate the particle behavior. This radical reinterpretation of light will lead us in the next chapter to no less radical changes in our views of the nature of matter.

## 38.1 LIGHT ABSORBED AS PHOTONS: THE PHOTOELECTRIC EFFECT

A phenomenon that gives insight into the nature of light is the **photoelectric effect**, in which a material emits electrons from its surface when illuminated (**Fig. 38.1**, next page). To escape from the surface, an electron must absorb enough energy from the incident light to overcome the attraction of positive ions in the material. These attractions constitute a potential-energy barrier; the light supplies the “kick” that enables the electron to escape.

The photoelectric effect has a number of applications. Digital cameras and night-vision scopes use it to convert light energy into an electric signal that is reconstructed into an

### LEARNING OUTCOMES

*In this chapter, you'll learn...*

- 38.1** How Einstein's photon picture of light explains the photoelectric effect.
- 38.2** How experiments with x-ray production provided evidence that light is emitted in the form of photons.
- 38.3** How the scattering of gamma rays helped confirm the photon picture of light.
- 38.4** How the Heisenberg uncertainty principle imposes fundamental limits on what can be measured.

*You'll need to review...*

- 8.5** Center of mass.
- 16.7** Beats.
- 23.2** Electron volts.
- 32.1, 32.4** Light as an electromagnetic wave.
- 33.6** Light scattering.
- 36.2, 36.3, 36.6** Single-slit diffraction, x-ray diffraction.
- 37.8** Relativistic energy and momentum.

Figure 38.1 The photoelectric effect.

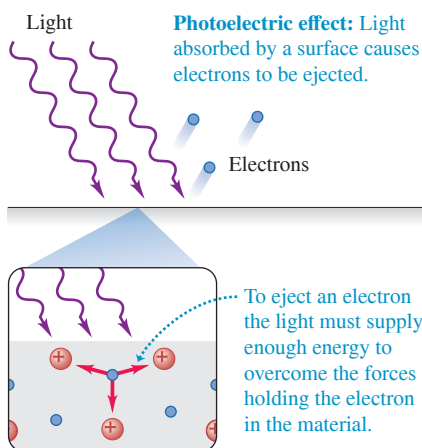


Figure 38.2 (a) A night-vision scope makes use of the photoelectric effect. Photons entering the scope strike a plate, ejecting electrons that pass through a thin disk in which there are millions of tiny channels. The current through each channel is amplified electronically and then directed toward a screen that glows when hit by electrons. (b) The image formed on the screen, which is a combination of these millions of glowing spots, is thousands of times brighter than the naked-eye view.

(a)



(b)



image (Fig. 38.2). Sunlight striking the moon causes surface dust to eject electrons, leaving the dust particles with a positive charge. The mutual electric repulsion of these charged dust particles causes them to rise above the moon's surface, a phenomenon that was observed from lunar orbit by the *Apollo* astronauts.

## Threshold Frequency and Stopping Potential

In Section 32.1 we explored the wave model of light, which Maxwell formulated two decades before the photoelectric effect was observed. Is the photoelectric effect consistent with this model? Figure 38.3a shows a modern version of one of the experiments that explored this question. Two conducting electrodes are enclosed in an evacuated glass tube and connected by a battery, and the cathode is illuminated. Depending on the potential difference  $V_{AC}$  between the two electrodes, electrons emitted by the illuminated cathode (called *photoelectrons*) may travel across to the anode, producing a *photocurrent* in the external circuit. (The tube is evacuated to a pressure of 0.01 Pa or less to minimize collisions between the electrons and gas molecules.)

The illuminated cathode emits photoelectrons with various kinetic energies. If the electric field points toward the cathode, as in Fig. 38.3a, all the electrons are accelerated toward the anode and contribute to the photocurrent. But by reversing the field and adjusting its strength as in Fig. 38.3b, we can prevent the less energetic electrons from reaching the anode. In fact, we can determine the *maximum* kinetic energy  $K_{\max}$  of the emitted electrons by making the potential of the anode relative to the cathode,  $V_{AC}$ , just negative enough so that the current stops. This occurs for  $V_{AC} = -V_0$ , where  $V_0$  is called the **stopping potential**. As an electron moves from the cathode to the anode, the potential decreases by  $V_0$  and negative work  $-eV_0$  is done on the (negatively charged) electron. The most energetic electron leaves the cathode with kinetic energy  $K_{\max} = \frac{1}{2}mv_{\max}^2$  and has zero kinetic energy at the anode. Using the work-energy theorem, we have

$$\begin{aligned} W_{\text{tot}} &= -eV_0 = \Delta K = 0 - K_{\max} && \text{(maximum kinetic energy of photoelectrons)} \\ K_{\max} &= \frac{1}{2}mv_{\max}^2 = eV_0 \end{aligned} \quad (38.1)$$

Hence by measuring the stopping potential  $V_0$ , we can determine the maximum kinetic energy with which electrons leave the cathode. (We are ignoring any effects due to differences in the materials of the cathode and anode.)

In this experiment, how does the photocurrent depend on the voltage across the electrodes and on the frequency and intensity of the light? Based on Maxwell's picture of light as an electromagnetic wave, here is what we would predict:

**Wave-Model Prediction 1:** We saw in Section 32.4 that the intensity of an electromagnetic wave depends on its amplitude but not on its frequency. So the photoelectric effect should occur for light of any frequency, and *the magnitude of the photocurrent should not depend on the frequency of the light.*

**Wave-Model Prediction 2:** It takes a certain minimum amount of energy, called the **work function**, to eject a single electron from a particular surface (see Fig. 38.1). If the light falling on the surface is very faint, some time may elapse before the total energy absorbed by the surface equals the work function. Hence, for faint illumination, *we expect a time delay between when we switch on the light and when photoelectrons appear.*

**Wave-Model Prediction 3:** Because the energy delivered to the cathode surface depends on the intensity of illumination, *we expect the stopping potential to increase with increasing light intensity.* Since intensity does not depend on frequency, we further expect that *the stopping potential should not depend on the frequency of the light.*

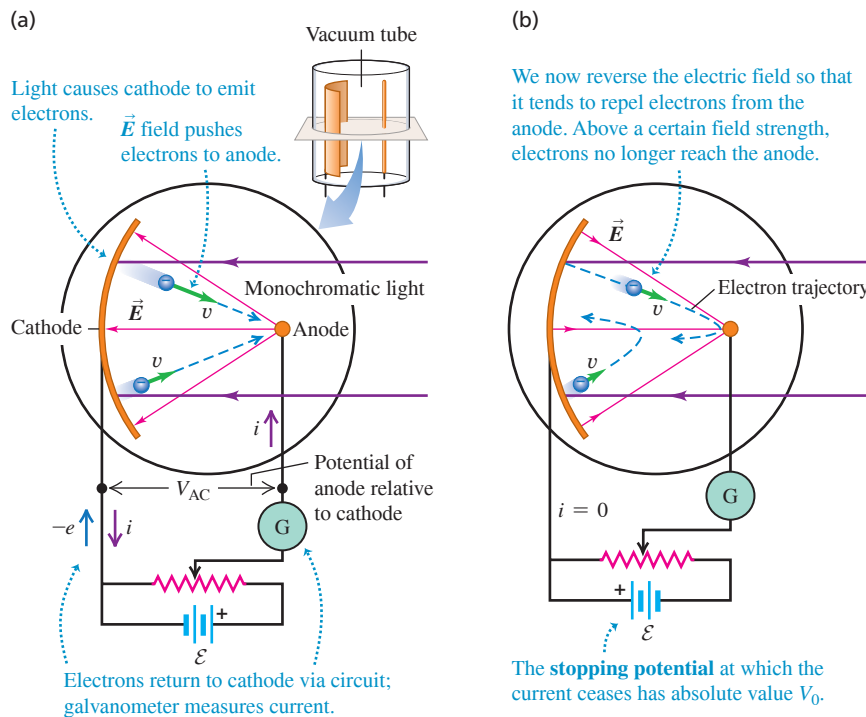


Figure 38.3 An experiment testing whether the photoelectric effect is consistent with the wave model of light.

The experimental results proved to be *very* different from these predictions. Here is what was found in the years between 1877 and 1905:

**Experimental Result 1:** The photocurrent depends on the light frequency. For a given material, monochromatic light with a frequency below a minimum **threshold frequency** produces *no* photocurrent, regardless of intensity. For most metals the threshold frequency is in the ultraviolet (corresponding to wavelengths  $\lambda$  between 200 and 300 nm), but for other materials like potassium oxide and cesium oxide it is in the visible spectrum ( $\lambda$  between 380 and 750 nm).

**Experimental Result 2:** There is no measurable time delay between when the light is turned on and when the cathode emits photoelectrons (assuming the frequency of the light exceeds the threshold frequency). This is true no matter how faint the light is.

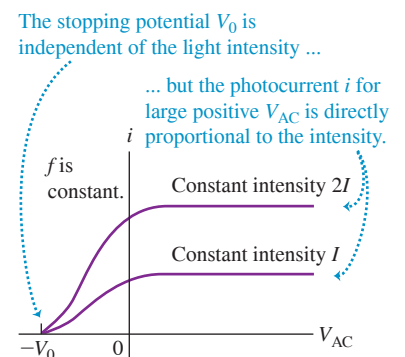
**Experimental Result 3:** The stopping potential does not depend on intensity, but does depend on frequency. **Figure 38.4** shows graphs of photocurrent as a function of potential difference  $V_{AC}$  for light of a given frequency and two different intensities. The reverse potential difference  $-V_0$  needed to reduce the current to zero is the same for both intensities. The only effect of increasing the intensity is to increase the number of electrons per second and hence the photocurrent  $i$ . (The curves level off when  $V_{AC}$  is large and positive because at that point all the emitted electrons are being collected by the anode.) If the intensity is held constant but the frequency is increased, the stopping potential also increases. In other words, the greater the light frequency, the higher the energy of the ejected photoelectrons.

These results directly contradict Maxwell's description of light as an electromagnetic wave. A solution to this dilemma was provided by Albert Einstein in 1905. His proposal involved nothing less than a new picture of the nature of light.

## Einstein's Photon Explanation

Einstein made the radical postulate that a beam of light consists of small packages of energy called **photons** or *quanta*. This postulate was an extension of an idea developed five years earlier by Max Planck to explain the properties of blackbody radiation, which we discussed in Section 17.7. (We'll explore Planck's ideas in Section 39.5.) In Einstein's

Figure 38.4 Photocurrent  $i$  for light frequency  $f$  as a function of the potential  $V_{AC}$  of the anode with respect to the cathode.





**CAUTION** Photons are not “particles” in the usual sense. It’s common, but inaccurate, to envision photons as miniature billiard balls. Billiard balls have a rest mass and travel slower than the speed of light  $c$ , while photons travel at the speed of light and have *zero* rest mass. Furthermore, unlike billiard balls, photons have wave aspects (frequency and wavelength) that are easy to observe. The photon concept is a very strange one, and the true nature of photons is difficult to visualize in a simple way. We’ll discuss this in more detail in Section 38.4. ■

picture, the energy  $E$  of an individual photon is equal to a constant times the photon frequency  $f$ . From the relationship  $f = c/\lambda$  for electromagnetic waves in vacuum, we have

$$\text{Energy of a photon} \rightarrow E = hf = \frac{hc}{\lambda} \quad (38.2)$$

Planck's constant
Speed of light in vacuum  
Frequency
Wavelength

The greater the frequency of a photon, the greater the photon energy and the shorter its wavelength; the longer the wavelength of a photon, the smaller the photon energy and the lower its frequency. Here **Planck’s constant**,  $h$ , is a universal constant. In the modern SI system of units (Section 1.3), its value is *defined* to be

$$h = 6.62607015 \times 10^{-34} \text{ J} \cdot \text{s}$$

In Einstein’s picture, all the energy of an individual photon arriving at the surface in Fig. 38.1 is absorbed by a single electron. The electron can escape from the surface only if the energy it acquires is greater than the work function  $\phi$ . Thus photoelectrons will be ejected only if  $hf > \phi$ , or  $f > \phi/h$ . Einstein’s postulate therefore explains why the photoelectric effect occurs only for frequencies greater than a minimum threshold frequency. This postulate is also consistent with the observation that greater intensity causes a greater photocurrent (Fig. 38.4). Greater intensity at a particular frequency means more photons absorbed per second, and thus more electrons emitted per second and a greater photocurrent.

Einstein’s postulate also explains why there is no delay between illumination and the emission of photoelectrons. As soon as photons of sufficient energy strike the surface, electrons can absorb them and be ejected.

Finally, Einstein’s postulate explains why the stopping potential for a given surface depends only on the light frequency. Recall that  $\phi$  is the *minimum* energy needed to remove an electron from the surface. Einstein applied conservation of energy to find that the *maximum* kinetic energy  $K_{\text{max}} = \frac{1}{2}mv_{\text{max}}^2$  for an emitted electron is the energy  $hf$  gained from a photon minus the work function  $\phi$ :

$$K_{\text{max}} = \frac{1}{2}mv_{\text{max}}^2 = hf - \phi \quad (38.3)$$

Substituting  $K_{\text{max}} = eV_0$  from Eq. (38.1), we find

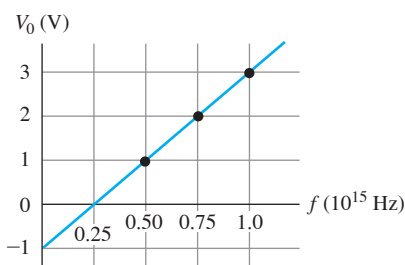
$$\text{Photoelectric effect:} \quad eV_0 = hf - \phi \quad (38.4)$$

Maximum kinetic energy of photoelectron
Energy of absorbed photon  
Magnitude of electron charge
Stopping potential
Work function
Light frequency  

Planck's constant

**CAUTION** Greater work function means smaller photoelectron energy. Note what Eq. (38.3) tells us: The *greater* the work function of the material, the *smaller* the kinetic energy of the electrons emitted when photons of a given frequency shine on the material. ■

Figure 38.5 Stopping potential as a function of frequency for a particular cathode material.



Equation (38.4) shows that the stopping potential  $V_0$  increases with increasing frequency  $f$ . The intensity doesn’t appear in Eq. (38.4), so  $V_0$  is independent of intensity. As a check of Eq. (38.4), we can measure the stopping potential  $V_0$  for each of several values of frequency  $f$  for a given cathode material (**Fig. 38.5**). A graph of  $V_0$  as a function of  $f$  turns out to be a straight line, verifying Eq. (38.4), and from such a graph we can determine both the work function  $\phi$  for the material and the value of the quantity  $h/e$ . After the electron charge  $-e$  was measured by Robert Millikan in 1909, Planck’s constant  $h$  could also be determined from these measurements.

Electron energies and work functions are usually expressed in electron volts (eV), defined in Section 23.2. To four significant figures,

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

To this accuracy, Planck's constant is

$$h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s} = 4.136 \times 10^{-15} \text{ eV} \cdot \text{s}$$

**Table 38.1** lists the work functions of several elements. These values are approximate because they are very sensitive to surface impurities. The greater the work function, the higher the minimum frequency needed to emit photoelectrons (**Fig. 38.6**).

The photon picture also explains other phenomena in which light is absorbed. A *sun-tan* is caused when the energy in sunlight triggers a chemical reaction in skin cells that leads to increased production of the pigment melanin. This reaction can occur only if a specific molecule in the cell absorbs a certain minimum amount of energy. A short-wavelength ultraviolet photon has enough energy to trigger the reaction, but a longer-wavelength visible-light photon does not. Hence ultraviolet light causes tanning, while visible light cannot.

## Photon Momentum

Einstein's photon concept applies to *all* regions of the electromagnetic spectrum, including radio waves, x rays, and so on. A photon of any frequency  $f$  and wavelength  $\lambda$  has energy  $E$  given by Eq. (38.2). Furthermore, according to the special theory of relativity, every particle that has energy must have momentum. Photons have zero rest mass, and a particle with zero rest mass and energy  $E$  has momentum with magnitude  $p$  given by  $E = pc$  [Section 37.8; see Eq. (37.40)]. Thus the magnitude  $p$  of the momentum of a photon is

$$\text{Momentum of a photon } p = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda} \quad (38.5)$$

Photon energy
Planck's constant
Wavelength
Frequency

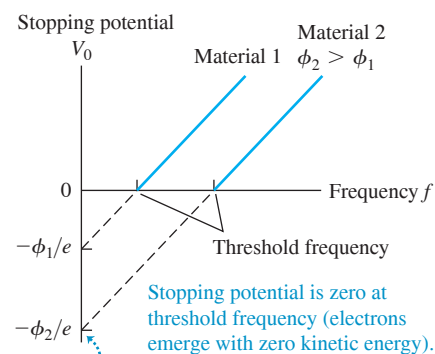
Speed of light in vacuum

The greater the frequency and the shorter the wavelength of a photon, the greater the magnitude of the photon's momentum. The *direction* of the photon's momentum is simply the direction in which the electromagnetic wave is moving.

**TABLE 38.1** Work Functions of Several Elements

| Element  | Work Function (eV) |
|----------|--------------------|
| Aluminum | 4.3                |
| Carbon   | 5.0                |
| Copper   | 4.7                |
| Gold     | 5.1                |
| Nickel   | 5.1                |
| Silicon  | 4.8                |
| Silver   | 4.3                |
| Sodium   | 2.7                |

**Figure 38.6** Stopping potential as a function of frequency for two cathode materials having different work functions  $\phi$ .



For each material,

$$eV_0 = hf - \phi \quad \text{or} \quad V_0 = \frac{hf}{e} - \frac{\phi}{e}$$

so the plots have same slope  $h/e$  but different intercepts  $-\phi/e$  on the vertical axis.

## PROBLEM-SOLVING STRATEGY 38.1 Photons

**IDENTIFY** the relevant concepts: The energy and momentum of an individual photon are proportional to the frequency and inversely proportional to the wavelength. Einstein's interpretation of the photoelectric effect is that energy is conserved as a photon ejects an electron from a material surface.

**SET UP** the problem: Identify the target variable. It could be the photon's wavelength  $\lambda$ , frequency  $f$ , energy  $E$ , or momentum  $p$ . If the problem involves the photoelectric effect, the target variable could be the maximum kinetic energy of photoelectrons  $K_{\max}$ , the stopping potential  $V_0$ , or the work function  $\phi$ .

**EXECUTE** the solution as follows:

1. Use Eqs. (38.2) and (38.5) to relate the energy and momentum of a photon to its wavelength and frequency. If the problem involves the photoelectric effect, use Eqs. (38.1), (38.3), and (38.4)

to relate the photon frequency, stopping potential, work function, and maximum photoelectron kinetic energy.

2. The electron volt (eV), which we introduced in Section 23.2, is a convenient unit. It is the kinetic energy gained by an electron when it moves freely through an increase of potential of one volt:  $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ . If the photon energy  $E$  is given in electron volts, use  $h = 4.136 \times 10^{-15} \text{ eV} \cdot \text{s}$ ; if  $E$  is in joules, use  $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$ .

**EVALUATE** your answer: In problems involving photons, at first the numbers will be unfamiliar to you and errors will not be obvious. It helps to remember that a visible-light photon with  $\lambda = 600 \text{ nm}$  and  $f = 5 \times 10^{14} \text{ Hz}$  has an energy  $E$  of about 2 eV, or about  $3 \times 10^{-19} \text{ J}$ .

**EXAMPLE 38.1** Laser-pointer photons**WITH VARIATION PROBLEMS**

A laser pointer with a power output of 5.00 mW emits red light ( $\lambda = 650$  nm). (a) What is the magnitude of the momentum of each photon? (b) How many photons does the laser pointer emit each second?

**IDENTIFY and SET UP** This problem involves the ideas of (a) photon momentum and (b) photon energy. In part (a) we'll use Eq. (38.5) and the given wavelength to find the magnitude of each photon's momentum. In part (b), Eq. (38.5) gives the energy per photon, and the power output tells us the energy emitted per second. We can combine these quantities to calculate the number of photons emitted per second.

**EXECUTE** (a) We have  $\lambda = 650$  nm  $= 6.50 \times 10^{-7}$  m, so from Eq. (38.5) the photon momentum is

$$p = \frac{h}{\lambda} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{6.50 \times 10^{-7} \text{ m}} \\ = 1.02 \times 10^{-27} \text{ kg} \cdot \text{m/s}$$

(Recall that  $1 \text{ J} = 1 \text{ kg} \cdot \text{m}^2/\text{s}^2$ .)

(b) From Eq. (38.5), the energy of a single photon is

$$E = pc = (1.02 \times 10^{-27} \text{ kg} \cdot \text{m/s})(3.00 \times 10^8 \text{ m/s}) \\ = 3.06 \times 10^{-19} \text{ J} = 1.91 \text{ eV}$$

The laser pointer emits energy at the rate of  $5.00 \times 10^{-3} \text{ J/s}$ , so it emits photons at the rate of

$$\frac{5.00 \times 10^{-3} \text{ J/s}}{3.06 \times 10^{-19} \text{ J/photon}} = 1.63 \times 10^{16} \text{ photons/s}$$

**EVALUATE** The result in part (a) is very small; a typical oxygen molecule in room-temperature air has 2500 times more momentum. As a check on part (b), we can calculate the photon energy from Eq. (38.2):

$$E = hf = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(3.00 \times 10^8 \text{ m/s})}{6.50 \times 10^{-7} \text{ m}} \\ = 3.06 \times 10^{-19} \text{ J} = 1.91 \text{ eV}$$

Our result in part (b) shows that a huge number of photons leave the laser pointer each second, each of which has an infinitesimal amount of energy. Hence the discreteness of the photons isn't noticed, and the radiated energy appears to be a continuous flow.

**KEYCONCEPT** Photons have both particle properties (energy and momentum) and wave properties (frequency and wavelength). The energy  $E$  and the magnitude of momentum  $p$  of a photon are both proportional to the frequency and inversely proportional to the wavelength; furthermore,  $E$  is directly proportional to  $p$ .

**EXAMPLE 38.2** A photoelectric-effect experiment**WITH VARIATION PROBLEMS**

While conducting a photoelectric-effect experiment with light of a certain frequency, you find that a reverse potential difference of 1.25 V is required to reduce the current to zero. Find (a) the maximum kinetic energy and (b) the maximum speed of the emitted photoelectrons.

**IDENTIFY and SET UP** The value of 1.25 V is the stopping potential  $V_0$  for this experiment. We'll use this in Eq. (38.1) to find the maximum photoelectron kinetic energy  $K_{\text{max}}$ , and from this we'll find the maximum photoelectron speed.

**EXECUTE** (a) From Eq. (38.1),

$$K_{\text{max}} = eV_0 = (1.60 \times 10^{-19} \text{ C})(1.25 \text{ V}) = 2.00 \times 10^{-19} \text{ J}$$

(Recall that  $1 \text{ V} = 1 \text{ J/C}$ .) In terms of electron volts,

$$K_{\text{max}} = eV_0 = e(1.25 \text{ V}) = 1.25 \text{ eV}$$

because the electron volt (eV) is the magnitude of the electron charge  $e$  times one volt (1 V).

(b) From  $K_{\text{max}} = \frac{1}{2}mv_{\text{max}}^2$  we get

$$v_{\text{max}} = \sqrt{\frac{2K_{\text{max}}}{m}} = \sqrt{\frac{2(2.00 \times 10^{-19} \text{ J})}{9.11 \times 10^{-31} \text{ kg}}} \\ = 6.63 \times 10^5 \text{ m/s}$$

**EVALUATE** The value of  $v_{\text{max}}$  is about 0.2% of the speed of light, so we are justified in using the nonrelativistic expression for kinetic energy. (An equivalent justification is that the electron's 1.25 eV kinetic energy is much less than its rest energy  $mc^2 = 0.511 \text{ MeV} = 5.11 \times 10^5 \text{ eV}$ .)

**KEYCONCEPT** In an experiment with the photoelectric effect, not all photoelectrons have the same kinetic energy. The *stopping voltage*—the voltage required to stop photoelectrons emitted at the cathode from reaching the anode—tells you the *maximum* kinetic energy of the photoelectrons.

**EXAMPLE 38.3** Determining  $\phi$  and  $h$  experimentally**WITH VARIATION PROBLEMS**

For a particular cathode material in a photoelectric-effect experiment, you measure stopping potentials  $V_0 = 1.0$  V for light of wavelength  $\lambda = 600$  nm, 2.0 V for 400 nm, and 3.0 V for 300 nm. Determine the work function  $\phi$  for this material and the implied value of Planck's constant  $h$ .

**IDENTIFY and SET UP** This example uses the relationship among stopping potential  $V_0$ , frequency  $f$ , and work function  $\phi$  in the photoelectric effect. According to Eq. (38.4), a graph of  $V_0$  versus  $f$  should be a

straight line as in Fig. 38.5 or 38.6. Such a graph is completely determined by its slope and the value at which it intercepts the vertical axis; we'll use these to determine the values of the target variables  $\phi$  and  $h$ .

**EXECUTE** We rewrite Eq. (38.4) as

$$V_0 = \frac{h}{e}f - \frac{\phi}{e}$$

In this form we see that the slope of the line is  $h/e$  and the vertical-axis intercept (corresponding to  $f = 0$ ) is  $-\phi/e$ . The frequencies, obtained from  $f = c/\lambda$  and  $c = 3.00 \times 10^8$  m/s, are  $0.50 \times 10^{15}$  Hz,  $0.75 \times 10^{15}$  Hz, and  $1.0 \times 10^{15}$  Hz, respectively. From a graph of these data (see Fig. 38.6), we find

$$-\frac{\phi}{e} = \text{vertical intercept} = -1.0 \text{ V}$$

$$\phi = 1.0 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

and

$$\text{Slope} = \frac{\Delta V_0}{\Delta f} = \frac{3.0 \text{ V} - (-1.0 \text{ V})}{1.00 \times 10^{15} \text{ s}^{-1} - 0} = 4.0 \times 10^{-15} \text{ J} \cdot \text{s/C}$$

$$h = \text{slope} \times e = (4.0 \times 10^{-15} \text{ J} \cdot \text{s/C})(1.60 \times 10^{-19} \text{ C})$$

$$= 6.4 \times 10^{-34} \text{ J} \cdot \text{s}$$

**EVALUATE** The value of Planck's constant  $h$  determined from your experiment differs from the accepted value by only about 3%. The small value  $\phi = 1.0$  eV tells us that the cathode surface is not composed solely of one of the elements in Table 38.1.

**KEYCONCEPT** To analyze the results of a photoelectric-effect experiment, graph the stopping potential as a function of the frequency of the light used. You can calculate the work function of the material used from the vertical intercept of this graph, and you can calculate the value of Planck's constant from the slope of this graph.

**TEST YOUR UNDERSTANDING OF SECTION 38.1** Silicon films become better electrical conductors when illuminated by photons with energies of 1.14 eV or greater, an effect called *photoconductivity*. Which of the following wavelengths of electromagnetic radiation can cause photoconductivity in silicon films? (i) Ultraviolet light with  $\lambda = 300$  nm; (ii) red light with  $\lambda = 600$  nm; (iii) infrared light with  $\lambda = 1200$  nm; (iv) both (i) and (ii); (v) all of (i), (ii), and (iii).

**ANSWER** This is in the infrared part of the spectrum. Since wavelength is inversely proportional to photon energy, the *minimum* photon energy of 1.14 eV corresponds to the *maximum* wavelength that causes photoconductivity in silicon. Thus the wavelength must be 1090 nm or less.

$$\lambda = hc/E = (4.136 \times 10^{-15} \text{ eV} \cdot \text{s})(3.00 \times 10^8 \text{ m/s})/(1.14 \text{ eV})$$

$$= 1.09 \times 10^3 \text{ nm} = 1090 \text{ nm}$$

(iv) From Eq. (38.2), a photon of energy  $E = 1.14$  eV has wavelength

## 38.2 LIGHT EMITTED AS PHOTONS: X-RAY PRODUCTION

The photoelectric effect provides solid evidence that light is *absorbed* in the form of photons. For physicists to accept Einstein's radical photon concept, however, it was also necessary to show that light is *emitted* as photons. An experiment that demonstrates this convincingly is the inverse of the photoelectric effect: Instead of releasing electrons from a surface by shining electromagnetic radiation on it, we cause a surface to emit radiation—specifically, *x rays*—by bombarding it with fast-moving electrons.

### X-Ray Photons

X rays were first produced in 1895 by the German physicist Wilhelm Röntgen, using an apparatus similar in principle to the setup shown in Fig. 38.7. When the cathode is heated to a very high temperature, it releases electrons in a process called *thermionic emission*. (As in the photoelectric effect, the minimum energy that an individual electron must be given to escape from the cathode's surface is equal to the work function for the surface. In this case the energy is provided to the electrons by heat rather than by light.) The electrons are then accelerated toward the anode by a potential difference  $V_{AC}$ . The bulb is evacuated (residual pressure  $10^{-7}$  atm or less), so the electrons can travel from the cathode to the anode without colliding with air molecules. When  $V_{AC}$  is a few thousand volts or more, x rays are emitted from the anode surface.

### BIO APPLICATION Sterilizing with High-Energy Photons

One technique for killing harmful microorganisms is to illuminate them with ultraviolet light with a wavelength shorter than 254 nm. If a photon of such short wavelength strikes a DNA molecule within a microorganism, the energy of the photon is great enough to break the bonds within the molecule. This renders the microorganism unable to grow or reproduce. Such ultraviolet germicidal irradiation is used for medical sanitation, to keep laboratories sterile (as shown here), and to treat both drinking water and wastewater.

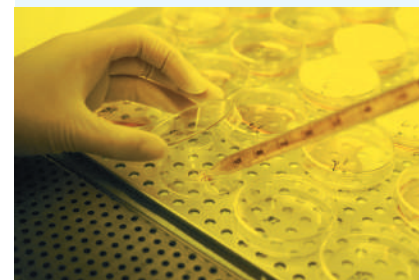
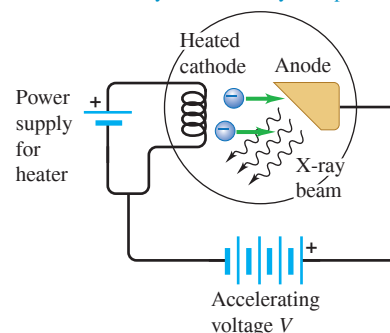


Figure 38.7 An apparatus used to produce x rays, similar to Röntgen's 1895 apparatus.

Electrons are emitted thermionically from the heated cathode and are accelerated toward the anode; when they strike it, x rays are produced.





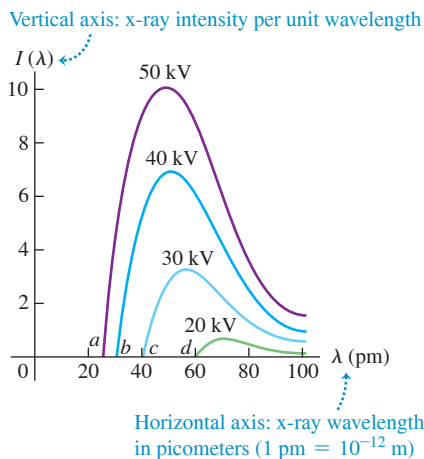
The anode produces x rays in part simply by slowing the electrons abruptly. (Recall from Section 32.1 that accelerated charges emit electromagnetic waves.) This process is called *bremsstrahlung* (German for “braking radiation”). Because the electrons undergo accelerations of very great magnitude, they emit much of their radiation at short wavelengths in the x-ray range, about  $10^{-9}$  to  $10^{-12}$  m (1 nm to 1 pm). (X-ray wavelengths can be measured quite precisely by crystal diffraction techniques, which we discussed in Section 36.6.) Most electrons are braked by a series of collisions and interactions with anode atoms, so bremsstrahlung produces a continuous spectrum of electromagnetic radiation.

Just as we did for the photoelectric effect in Section 38.1, let’s compare what Maxwell’s wave theory of electromagnetic radiation would predict about this radiation to what is observed experimentally.

**Wave-Model Prediction:** The electromagnetic waves produced when an electron slams into the anode should be analogous to the sound waves produced by crashing cymbals together. These waves include sounds of all frequencies. By analogy, the x rays produced by bremsstrahlung should have a spectrum that includes *all* frequencies and hence *all* wavelengths.

**Experimental Result:** Figure 38.8 shows bremsstrahlung spectra obtained when the same cathode and anode are used with four different accelerating voltages  $V_{AC}$ . *Not* all x-ray frequencies and wavelengths are emitted: Each spectrum has a maximum frequency  $f_{\max}$  and a corresponding minimum wavelength  $\lambda_{\min}$ . The greater the value of  $V_{AC}$ , the higher the maximum frequency and the shorter the minimum wavelength.

Figure 38.8 The continuous spectrum of x rays produced when a tungsten target is struck by electrons accelerated through a voltage  $V_{AC}$ . The curves represent different values of  $V_{AC}$ ; points *a*, *b*, *c*, and *d* show the minimum wavelength  $\lambda_{\min}$  for each voltage.



The wave model of electromagnetic radiation cannot explain these experimental results. But we can readily understand them by using the photon model. An electron has charge  $-e$  and gains kinetic energy  $eV_{AC}$  when accelerated through a potential increase  $V_{AC}$ . The most energetic photon (highest frequency and shortest wavelength) is produced if the electron is braked to a stop all at once when it hits the anode, so that all of its kinetic energy goes to produce one photon; that is,

**Bremsstrahlung:**

$$eV_{AC} = hf_{\max} = \frac{hc}{\lambda_{\min}} \quad (38.6)$$

Annotations for Equation (38.6):

- $eV_{AC}$ : Kinetic energy lost by electron; Magnitude of electron charge  $e$  times Accelerating voltage  $V_{AC}$ .
- $hf_{\max}$ : Maximum energy of an emitted photon; Planck's constant  $h$  times Maximum photon frequency  $f_{\max}$ .
- $\frac{hc}{\lambda_{\min}}$ : Planck's constant  $h$  times Speed of light in vacuum  $c$  divided by Minimum photon wavelength  $\lambda_{\min}$ .

(In this equation we ignore the work function of the target anode and the initial kinetic energy of the electrons “boiled off” from the cathode. These energies are very small compared to the kinetic energy  $eV_{AC}$  gained due to the potential difference.) If only a portion of an electron’s kinetic energy goes into producing a photon, the photon energy will be less than  $eV_{AC}$  and the wavelength will be longer than  $\lambda_{\min}$ . Experiment shows that the measured values for  $\lambda_{\min}$  for different values of  $eV_{AC}$  (see Fig. 38.8) agree with Eq. (38.6). Note that according to Eq. (38.6), the maximum frequency and minimum wavelength in the bremsstrahlung process do not depend on the target material; this also agrees with experiment. So we can conclude that the photon picture of electromagnetic radiation is valid for the *emission* as well as the absorption of radiation.

The apparatus shown in Fig. 38.7 can also produce x rays by a second process in which electrons transfer their kinetic energy partly or completely to individual atoms within the target. It turns out that this process not only is consistent with the photon model of electromagnetic radiation, but also provides insight into the structure of atoms. We’ll return to this process in Section 41.7.

### EXAMPLE 38.4 Producing x rays

Electrons in an x-ray tube accelerate through a potential difference of 10.0 kV before striking a target. If an electron produces one photon on impact with the target, what is the minimum wavelength of the resulting x rays? Find the answer by expressing energies in both SI units and electron volts.

**IDENTIFY and SET UP** To produce an x-ray photon with minimum wavelength and hence maximum energy, all of the electron's kinetic energy must go into producing a single x-ray photon. We'll use Eq. (38.6) to determine the wavelength.

**EXECUTE** From Eq. (38.6), using SI units we have

$$\begin{aligned}\lambda_{\min} &= \frac{hc}{eV_{\text{AC}}} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(3.00 \times 10^8 \text{ m/s})}{(1.602 \times 10^{-19} \text{ C})(10.0 \times 10^3 \text{ V})} \\ &= 1.24 \times 10^{-10} \text{ m} = 0.124 \text{ nm}\end{aligned}$$

Using electron volts, we have

$$\begin{aligned}\lambda_{\min} &= \frac{hc}{eV_{\text{AC}}} = \frac{(4.136 \times 10^{-15} \text{ eV} \cdot \text{s})(3.00 \times 10^8 \text{ m/s})}{e(10.0 \times 10^3 \text{ V})} \\ &= 1.24 \times 10^{-10} \text{ m} = 0.124 \text{ nm}\end{aligned}$$

In the second calculation, the “ $e$ ” for the magnitude of the electron charge cancels the “ $e$ ” in the unit “eV,” because the electron volt (eV) is the magnitude of the electron charge  $e$  times one volt (1 V).

**EVALUATE** To check our result, recall from Example 38.1 that a 1.91 eV photon has a wavelength of 650 nm. Here the electron energy, and therefore the x-ray photon energy, is  $10.0 \times 10^3 \text{ eV} = 10.0 \text{ keV}$ , about 5000 times greater than in Example 38.1, and the wavelength is about  $\frac{1}{5000}$  as great as in Example 38.1. This makes sense, since wavelength and photon energy are inversely proportional.

**KEYCONCEPT** In bremsstrahlung, a photon of maximum energy and minimum wavelength  $\lambda_{\min}$  is produced when all of the kinetic energy of an electron goes into producing that photon. The value of  $\lambda_{\min}$  is inversely proportional to the voltage used to accelerate the electron.

## Applications of X Rays

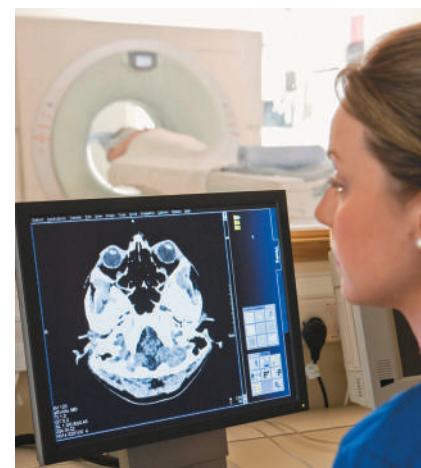
X rays have many practical applications in medicine and industry. Because x-ray photons are of such high energy, they can penetrate several centimeters of solid matter. Hence they can be used to visualize the interiors of materials that are opaque to ordinary light, such as broken bones or defects in structural steel. The object to be visualized is placed between an x-ray source and an electronic detector (like that used in a digital camera). The darker an area in the image recorded by such a detector, the greater the radiation exposure. Bones are much more effective x-ray absorbers than soft tissue, so bones appear as light areas. A crack or air bubble allows greater transmission and shows as a dark area.

A widely used and vastly improved x-ray technique is *computed tomography*; the corresponding instrument is called a *CT scanner*. The x-ray source produces a thin, fan-shaped beam that is detected on the opposite side of the subject by an array of several hundred detectors in a line. Each detector measures absorption along a thin line through the subject. The entire apparatus is rotated around the subject in the plane of the beam, and the changing photon-counting rates of the detectors are recorded digitally. A computer processes this information and reconstructs a picture of absorption over an entire cross section of the subject (see **Fig. 38.9**). Differences in absorption as small as 1% or less can be detected with CT scans, and tumors and other anomalies that are much too small to be seen with older x-ray techniques can be detected.

X rays cause damage to living tissues. As x-ray photons are absorbed in tissues, their energy breaks molecular bonds and creates highly reactive free radicals (such as neutral H and OH), which in turn can disturb the molecular structure of proteins and especially genetic material. Young and rapidly growing cells are particularly susceptible, which is why x rays are useful for selective destruction of cancer cells. Conversely, however, a cell may be damaged by radiation but survive, continue dividing, and produce generations of defective cells; thus x rays can *cause* cancer.

Even when the organism itself shows no apparent damage, excessive exposure to x rays can cause changes in the organism's reproductive system that will affect its offspring. A careful assessment of the balance between risks and benefits of radiation exposure is essential in each individual case.

Figure 38.9 This radiologist is operating a CT scanner (seen through the window) from a separate room to avoid repeated exposure to x rays.



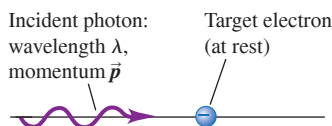
**BIO APPLICATION X-Ray Absorption and Medical Imaging**

Atomic electrons can absorb x rays. Hence materials with many electrons per atom tend to be better x-ray absorbers than materials with few electrons. In this x-ray image the lighter areas show where x rays are absorbed as they pass through the body; the darker areas indicate regions that are relatively transparent to x rays. Bones contain large amounts of elements such as phosphorus and calcium, with 15 and 20 electrons per atom, respectively. In soft tissue, the predominant elements are hydrogen, carbon, and oxygen, with only 1, 6, and 8 electrons per atom, respectively. Hence x rays are absorbed by bone but pass relatively easily through soft tissue.

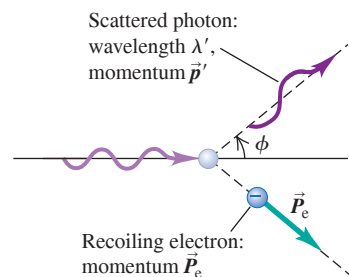


Figure 38.10 The photon model of light scattering by an electron.

(a) Before collision: The target electron is at rest.



(b) After collision: The angle between the directions of the scattered photon and the incident photon is  $\phi$ .



**TEST YOUR UNDERSTANDING OF SECTION 38.2** In the apparatus shown in Fig. 38.7, suppose you increase the number of electrons that are emitted from the cathode per second while keeping the potential difference  $V_{AC}$  the same. How will this affect the intensity  $I$  and minimum wavelength  $\lambda_{\min}$  of the emitted x rays? (i)  $I$  and  $\lambda_{\min}$  will both increase; (ii)  $I$  will increase but  $\lambda_{\min}$  will be unchanged; (iii)  $I$  will increase but  $\lambda_{\min}$  will decrease; (iv)  $I$  will remain the same but  $\lambda_{\min}$  will decrease; (v) none of these.

**ANSWER**

(ii) Equation (38.6) shows that the minimum wavelength of x rays produced by bremsstrahlung depends on the potential difference  $V_{AC}$  but does not depend on the rate at which electrons strike the anode. Increasing the number of electrons per second will only cause an increase in the number of x-ray photons emitted per second (that is, the x-ray intensity  $I$ ).

### 38.3 LIGHT SCATTERED AS PHOTONS: COMPTON SCATTERING AND PAIR PRODUCTION

The final aspect of light that we must test against Einstein's photon model is its behavior after the light is produced and before it is eventually absorbed. We can do this by considering the *scattering* of light. As we discussed in Section 33.6, scattering is what happens when light bounces off particles such as molecules in the air.

**Compton Scattering**

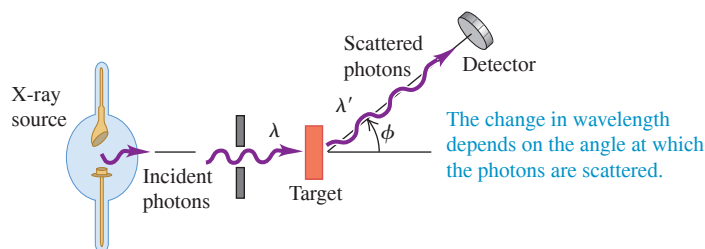
Let's see what Maxwell's wave model and Einstein's photon model would predict for how light behaves when it undergoes scattering by a single electron, such as an individual electron within an atom.

**Wave-Model Prediction:** In the wave description, scattering would be a process of absorption and re-radiation. Part of the energy of the light wave would be absorbed by the electron, which would oscillate in response to the oscillating electric field of the wave. The oscillating electron would act like a miniature antenna (see Section 32.1), re-radiating its acquired energy as *scattered* waves in a variety of directions. The frequency at which the electron oscillates would be the same as the frequency of the incident light, and the re-radiated light would have the same frequency as the oscillations of the electron. So, *in the wave model, the scattered light and incident light have the same frequency and same wavelength.*

**Photon-Model Prediction:** In the photon model we imagine the scattering process as a collision of two *particles*, the incident photon and an electron that is initially at rest (**Fig. 38.10a**). The incident photon would give up part of its energy and momentum to the electron, which recoils as a result of this impact. The scattered photon that remains can fly off at a variety of angles  $\phi$  with respect to the incident direction, but it has less energy and less momentum than the incident photon (**Fig. 38.10b**). The energy and momentum of a photon are given by  $E = hf = hc/\lambda$  [Eq. (38.2)] and  $p = hf/c = h/\lambda$  [Eq. (38.5)]. Therefore, *in the photon model, the scattered light has a lower frequency  $f$  and longer wavelength  $\lambda$  than the incident light.*

The definitive experiment that tested these predictions was carried out in 1922 by the American physicist Arthur H. Compton. He aimed a beam of x rays at a solid target and measured the wavelength of the radiation scattered from the target (**Fig. 38.11**). Compton

Figure 38.11 A Compton-effect experiment.



discovered that some of the scattered radiation has smaller frequency (longer wavelength) than the incident radiation and that the change in wavelength depends on the angle through which the radiation is scattered. This is precisely what the photon model predicts for light scattered from electrons in the target, a process that is now called **Compton scattering**.

Specifically, Compton found that if the scattered radiation emerges at an angle  $\phi$  with respect to the incident direction, as shown in Fig. 38.11, then

$$\text{Compton scattering:} \quad \lambda' - \lambda = \frac{h}{mc} (1 - \cos \phi) \quad (38.7)$$

Wavelength of scattered radiation
Wavelength of incident radiation
Planck's constant
Scattering angle  
 $\lambda'$ 
 $\lambda$ 
 $h$ 
 $(1 - \cos \phi)$   
Electron rest mass
 $mc$ 
Speed of light in vacuum

In other words,  $\lambda'$  is greater than  $\lambda$ . The quantity  $h/mc$  that appears in Eq. (38.7) has units of length. Its numerical value is

$$\frac{h}{mc} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{(9.109 \times 10^{-31} \text{ kg})(2.998 \times 10^8 \text{ m/s})} = 2.426 \times 10^{-12} \text{ m}$$

Compton showed that Einstein's photon theory, combined with the principles of conservation of energy and conservation of momentum, provides a beautifully clear explanation of his experimental results. We outline the derivation below. The electron recoil energy may be in the relativistic range, so we have to use the relativistic energy–momentum relationships, Eqs. (37.39) and (37.40). The incident photon has momentum  $\vec{p}$ , with magnitude  $p$  and energy  $pc$ . The scattered photon has momentum  $\vec{p}'$ , with magnitude  $p'$  and energy  $p'c$ . The electron is initially at rest, so its initial momentum is zero and its initial energy is its rest energy  $mc^2$ . The final electron momentum  $\vec{P}_e$  has magnitude  $P_e$ , and the final electron energy is  $E_e^2 = (mc^2)^2 + (P_e c)^2$ . Then energy conservation gives us the relationship

$$pc + mc^2 = p'c + E_e$$

Rearranging, we find

$$(pc - p'c + mc^2)^2 = E_e^2 = (mc^2)^2 + (P_e c)^2 \quad (38.8)$$

We can eliminate the electron momentum  $\vec{P}_e$  from Eq. (38.8) by using momentum conservation. From Fig. 38.12 we see that  $\vec{p} = \vec{p}' + \vec{P}_e$ , or

$$\vec{P}_e = \vec{p} - \vec{p}' \quad (38.9)$$

By taking the scalar product of each side of Eq. (38.9) with itself, we find

$$P_e^2 = p^2 + p'^2 - 2pp' \cos \phi \quad (38.10)$$

We now substitute this expression for  $P_e^2$  into Eq. (38.8) and multiply out the left side. We divide out a common factor  $c^2$ ; several terms cancel, and when the resulting equation is divided through by  $(pp')$ , the result is

$$\frac{mc}{p'} - \frac{mc}{p} = 1 - \cos \phi \quad (38.11)$$

Finally, we substitute  $p' = h/\lambda'$  and  $p = h/\lambda$ , then multiply by  $h/mc$  to obtain Eq. (38.7).

When the wavelengths of x rays scattered at a certain angle are measured, the curve of intensity per unit wavelength as a function of wavelength has two peaks (Fig. 38.13). The longer-wavelength peak represents Compton scattering. The shorter-wavelength peak,  $\lambda_0$ , is at the wavelength of the incident x rays and corresponds to x-ray scattering from tightly bound electrons. In such scattering processes the entire atom must recoil, so  $m$  in Eq. (38.7) is the mass of the entire atom rather than of a single electron. The resulting wavelength shifts are negligible.

Figure 38.12 Vector diagram showing conservation of momentum in Compton scattering.

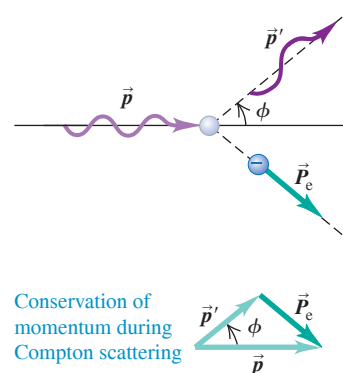
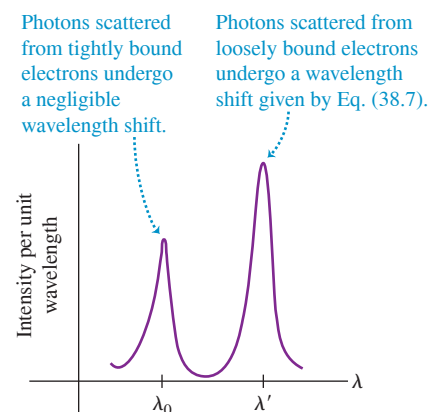


Figure 38.13 Intensity as a function of wavelength for photons scattered at an angle of  $135^\circ$  in a Compton-scattering experiment.





## EXAMPLE 38.5 Compton scattering

## WITH VARIATION PROBLEMS

You use 0.124 nm x-ray photons in a Compton-scattering experiment. (a) At what angle is the wavelength of the scattered x rays 1.0% longer than that of the incident x rays? (b) At what angle is it 0.050% longer?

**IDENTIFY and SET UP** We'll use the relationship between scattering angle and wavelength shift in the Compton effect. In each case our target variable is the angle  $\phi$  (see Fig. 38.10b). We solve for  $\phi$  by using Eq. (38.7).

**EXECUTE** (a) In Eq. (38.7) we want  $\Delta\lambda = \lambda' - \lambda$  to be 1.0% of 0.124 nm, so  $\Delta\lambda = 0.00124 \text{ nm} = 1.24 \times 10^{-12} \text{ m}$ . Using the value  $h/mc = 2.426 \times 10^{-12} \text{ m}$ , we find

$$\Delta\lambda = \frac{h}{mc}(1 - \cos\phi)$$

$$\cos\phi = 1 - \frac{\Delta\lambda}{h/mc} = 1 - \frac{1.24 \times 10^{-12} \text{ m}}{2.426 \times 10^{-12} \text{ m}} = 0.4889$$

$$\phi = 60.7^\circ$$

(b) For  $\Delta\lambda$  to be 0.050% of 0.124 nm, or  $6.2 \times 10^{-14} \text{ m}$ ,

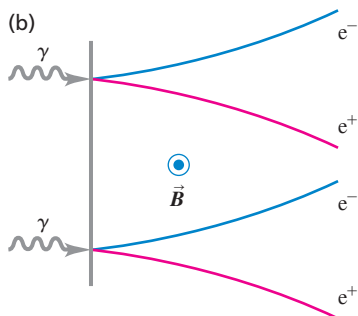
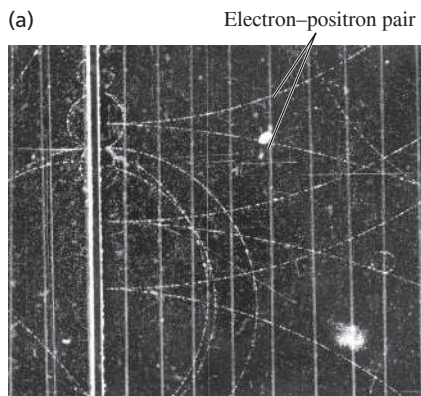
$$\cos\phi = 1 - \frac{6.2 \times 10^{-14} \text{ m}}{2.426 \times 10^{-12} \text{ m}} = 0.9744$$

$$\phi = 13.0^\circ$$

**EVALUATE** Our results show that smaller scattering angles give smaller wavelength shifts. Thus in a grazing collision the photon energy loss and the electron recoil energy are smaller than when the scattering angle is larger. This is just what we would expect for an elastic collision, whether between a photon and an electron or between two billiard balls.

**KEYCONCEPT** In Compton scattering, a photon collides with an electron; both energy and momentum are conserved in the collision. If the electron is initially at rest, the photon loses both energy and momentum to the electron. The scattered photon wavelength is longer than the wavelength of the incident photon by an amount  $\Delta\lambda$  that depends on the photon scattering angle.

Figure 38.14 (a) Photograph of bubble-chamber tracks of electron-positron pairs that are produced when 300 MeV photons strike a lead sheet. A magnetic field directed out of the photograph made the electrons ( $e^-$ ) and positrons ( $e^+$ ) curve in opposite directions. (b) Diagram showing the pair-production process for two of the gamma-ray photons ( $\gamma$ ).



## Pair Production

Another effect that can be explained only with the photon picture involves *gamma rays*, the shortest-wavelength and highest-frequency variety of electromagnetic radiation. If a gamma-ray photon of sufficiently short wavelength is fired at a target, it may not scatter. Instead, as depicted in **Fig. 38.14**, it may disappear completely and be replaced by two new particles: an electron and a **positron** (a particle that has the same rest mass  $m$  as an electron but has a positive charge  $+e$  rather than the negative charge  $-e$  of the electron). This process, called **pair production**, was first observed by the physicists Patrick Blackett and Giuseppe Occhialini in 1933. The electron and positron have to be produced in pairs in order to conserve electric charge: The incident photon has zero charge, and the electron-positron pair has net charge  $(-e) + (+e) = 0$ . Enough energy must be available to account for the rest energy  $2mc^2$  of the two particles. To four significant figures, this minimum energy is

$$E_{\min} = 2mc^2 = 2(9.109 \times 10^{-31} \text{ kg})(2.998 \times 10^8 \text{ m/s})^2$$

$$= 1.637 \times 10^{-13} \text{ J} = 1.022 \text{ MeV}$$

Thus the photon must have at least this much energy to produce an electron-positron pair. From Eq. (38.2),  $E = hc/\lambda$ , the photon wavelength has to be shorter than

$$\lambda_{\max} = \frac{hc}{E_{\min}} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(2.998 \times 10^8 \text{ m/s})}{1.637 \times 10^{-13} \text{ J}}$$

$$= 1.213 \times 10^{-12} \text{ m} = 1.213 \times 10^{-3} \text{ nm} = 1.213 \text{ pm}$$

This is a very short wavelength, about  $\frac{1}{1000}$  as large as the x-ray wavelengths that Compton used in his scattering experiments. (The requisite minimum photon energy is actually a bit higher than 1.022 MeV, so the photon wavelength must be a bit shorter than 1.213 pm. The reason is that when the incident photon encounters an atomic nucleus in the target, some of the photon energy goes into the kinetic energy of the recoiling nucleus.) Just as for the photoelectric effect, the wave model of electromagnetic radiation cannot explain why pair production occurs only when very short wavelengths are used.

The inverse process, *electron-positron pair annihilation*, occurs when a positron and an electron collide. Both particles disappear, and two (or occasionally three) photons can appear, with total energy of at least  $2m_e c^2 = 1.022 \text{ MeV}$ . Decay into a *single* photon is